

# The Dual-Scale String

T-Duality,  $K3 \times T^2$ , and Their Formalization in Lean 4

*A Student's Companion*

Xavier Callens

SocrateAI Scientific Agora Collaboration

SocrateAI Research Laboratory for Mathematical Physics & Formal Verification

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Companion to the Lean 4 repository `SocrateAI-Scientific-Agora-LeanMaster`

**How this book was made.** The text was drafted by AI language models (Anthropic’s Claude family) working as chapter authors under the direction of the first author, from a fixed library of open-access sources that the models were required to open and cite by file and line range, and from the Lean 4 source code of the companion repository. Every chapter was then examined by an independent reviewer model that re-opened the cited sources and the Lean files. The Lean theorems quoted in this book were checked by the Lean 4 kernel; the prose was not, and cannot be. Readers should treat the exposition as they would any first edition of lecture notes: check derivations, and report errors.

**What the word “checked” means here.** A claim marked **[A]** is a theorem whose proof has been verified by the Lean 4 kernel and which depends on no axioms beyond Lean’s three standard ones (`propext`, `Classical.choice`, `Quot.sound`). A claim marked **[L]** is taken from the cited literature and has not been formalized here. A claim marked **[C]** is a conjecture, or a proposal of the dual-scale programme. That a formal object in Lean faithfully represents a physical system is never a **[A]** claim.

**Sources.** The source library (arXiv identifiers and SHA-256 hashes) is listed in the bibliography and in `papers/foundations/MANIFEST.md`. We thank the authors of those lecture notes and reviews; any error introduced in paraphrasing them is ours.

# Preface

A point particle moving on a circle cannot tell you the size of the circle twice. A string can: it carries momentum around the circle, with energy quantized in units of  $1/R$ , and it can wind around the circle, with energy proportional to  $R/\alpha'$ . Exchange the two kinds of quantum number and replace  $R$  by  $\alpha'/R$ , and every energy level is where it was. This is T-duality. It says that for a string the geometry of radius  $R$  and the geometry of radius  $\alpha'/R$  are one and the same, and therefore that a length shorter than  $\sqrt{\alpha'}$  has no independent meaning. A theory of strings has two scales in every compact direction,  $R$  and  $\alpha'/R$ , and physical statements must respect both. That is the sense of the title.

This book has three aims, and tries to keep them apart.

The first is to *teach*: to take a student from the relativistic string to toroidal compactification, the duality group  $O(d, d; \mathbb{Z})$ , double field theory, the lattice theory of  $K3$  surfaces, compactification on  $K3 \times T^2$  with fluxes, and Mathieu moonshine. The derivations are written out. Where we follow a source we say which, down to the line range of the text we read.

The second is to *record exactly what has been formalized*. Alongside the physics, each chapter has a section called “What the machine has checked”, which quotes statements from the companion Lean 4 libraries and says in plain words what each one asserts. Much of string theory is far beyond what any proof assistant can currently express: there is no formal path integral here, no worldsheet conformal field theory, no Calabi–Yau metric. What *can* be formalized today is the algebraic skeleton: lattices and their signatures, the matrices of  $O(d, d; \mathbb{Z})$  and the identities they satisfy, the generalized metric, inequalities such as  $\text{tr } G + \text{tr } G^{-1} \geq 2d$ , tadpole arithmetic, and the dimension counts of moonshine. The older libraries in the repository go less far and model physical quantities by integers or rationals; we call these *arithmetic shadows* and describe them as such.

The third is to *propose*: the dual-scale programme treats the pair  $(G, G^{-1})$  as the basic geometric datum and asks what follows for moduli stabilization and early-universe cosmology. These proposals are conjectural and are marked **[C]** throughout. The inequalities they rest on are theorems; the physics is not.

## How to read this book

Parts I–II are a self-contained first course on the bosonic string and T-duality. Part III develops double field theory. Part IV is an introduction to lattices and  $K3$  surfaces for physicists; Part V puts the two together on  $K3 \times T^2$ . Part VI treats Mathieu moonshine. Part VII states the dual-scale proposal. Part VIII is about the formalization itself: a Lean primer for physicists, the architecture of the libraries, a graph-and-retrieval atlas of all theorems, the AI-assisted proof pipeline that produced them, and open problems sized for student projects. A reader interested mainly in Lean can start with Chapter 34 and then read the “machine” sections of Parts II–IV. Exercises are graded  $\star$  (routine),  $\star\star$  (needs an idea),  $\star\star\star$  (small project); many ask for a Lean statement or proof.

## **What this book is not**

It is not a replacement for the standard textbooks, and it does not cover the superstring, string perturbation theory beyond the spectrum, or AdS/CFT. It does not claim that string theory, or the dual-scale proposal, describes nature.

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## **Part I**

# **Foundations of the String**



# Chapter 1

## Why Strings, Why Two Scales, Why a Proof Assistant

This chapter answers three questions before any machinery is built. Why would one replace point particles by strings? Why does a string, unlike a particle, treat a circle of radius  $R$  and a circle of radius  $\alpha'/R$  as the same space, so that every length comes with a partner? And why does this book accompany the physics with statements checked by a proof assistant? A student should care about the first two because they contain, in a form that needs only quantum mechanics and special relativity, the idea that the rest of the book develops with increasing generality: from one circle to tori, to the doubled geometry of double field theory, to K3 surfaces and their lattices. A student should care about the third because the subject mixes theorems, established physics and open conjectures, and it is easy to lose track of which is which. The chapter ends with one small result carried through every stage: derived by hand, checked with numbers, and stated as a theorem of Lean 4.

### 1.1 Why strings

#### 1.1.1 Gravity resists the usual quantization

Units with  $\hbar = c = 1$  are used throughout, so that energy, mass and inverse length are the same dimension. Newton's constant  $G_N$  then defines a mass, the *Planck mass*. In the normalization of the Einstein–Hilbert action  $S_{EH} = \frac{1}{16\pi G_N} \int d^4x \sqrt{-g} \mathcal{R}$  one writes

$$8\pi G_N = \frac{1}{M_{\text{pl}}^2}, \quad M_{\text{pl}} \approx 2 \times 10^{18} \text{ GeV}, \quad (1.1)$$

the *reduced* Planck mass; without the  $8\pi$  one finds  $1/\sqrt{G_N} \approx 10^{19} \text{ GeV}$  [Tong §0.1, ll. 465–493]. Here  $g$  is the determinant of the spacetime metric  $g_{\mu\nu}$  and  $\mathcal{R}$  its Ricci scalar.

To see what Equation (1.1) means for a quantum theory, expand the metric around flat space,  $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}/M_{\text{pl}}$ . The factor  $1/M_{\text{pl}}$  is chosen so that the kinetic term of the fluctuation  $h$  carries no power of  $M_{\text{pl}}$ . The action then has the schematic form

$$S_{EH} = \int d^4x \left[ (\partial h)^2 + \frac{1}{M_{\text{pl}}} h(\partial h)^2 + \frac{1}{M_{\text{pl}}^2} h^2(\partial h)^2 + \dots \right] \quad (1.2)$$

[Tong §0.1, ll. 494–523]. Every interaction is suppressed by a power of  $M_{\text{pl}}$ . A process at energy  $E$  is therefore controlled by the dimensionless ratio  $E^2/M_{\text{pl}}^2$ : gravity is weak at low energy, and perturbation theory fails when  $E$  approaches  $M_{\text{pl}}$ . A coupling with the dimension of an inverse mass is the signature of a *non-renormalizable* theory: the divergences of its loop

diagrams cannot be absorbed in a finite number of counterterms. For pure gravity the first such divergence appears at two loops; with matter, at one loop [Tong §0.1, ll. 531–560].

This does not make general relativity useless. It makes it an *effective field theory*, accurate to order  $(E/M_{\text{pl}})^2$ , in the same way that Fermi’s four-fermion theory of the weak interaction is accurate below about 100 GeV. Fermi’s theory fails at its cutoff because it lacks the degrees of freedom that tame the high-energy behaviour, the  $W$  and  $Z$  bosons. The question of quantum gravity, put in this language, is: what are the missing degrees of freedom of gravity? [Tong §0.1, ll. 572–587]

### 1.1.2 The string hypothesis and its one parameter

String theory answers with a change of the elementary object: matter consists of tiny loops of string, not of points [Tong §0, ll. 373–378]. A relativistic string is characterized by a single dimensionful number, its tension  $T$ , the energy per unit length: a static string of length  $L$  has potential energy  $TL$ , growing linearly with  $L$  (unlike a Hooke spring, where it grows quadratically) [Tong §1.2, ll. 1090–1096]. For historical reasons the tension is written

$$T = \frac{1}{2\pi\alpha'}, \quad \alpha' = \ell_s^2, \quad (1.3)$$

where  $\alpha'$  is the *Regge slope* and  $\ell_s$  the *string length* [Tong §1.2, eqs. (1.17)–(1.18), ll. 1097–1120]. A dimensional check:  $T$  is energy per length, that is, mass squared in natural units, so  $\alpha'$  is a length squared and  $\ell_s = \sqrt{\alpha'}$  is a length. These are the conventions of the whole book (mostly-plus signature,  $\alpha'$  of dimension length<sup>2</sup>).

For the fundamental strings of this book, which have no thickness and no further structure, the tension is taken far above any laboratory scale, “typically an order of magnitude or so below the Planck scale” [Tong §1.2, ll. 1146–1150]. To convert to a length one uses  $\hbar c \approx 1.97 \times 10^{-14}$  GeV cm (a standard value; not checked against a source in this book’s library): an energy of  $10^{19}$  GeV corresponds to  $1.97 \times 10^{-14} / 10^{19} \approx 2 \times 10^{-33}$  cm, in agreement with the figure  $10^{-33}$  cm quoted for the Planck length in [GPR §1, ll. 209–213]. By the same conversion the Fermi scale of 100 GeV is  $2 \times 10^{-16}$  cm. Seventeen orders of magnitude separate the two, and this is the practical reason why no experiment tests the ideas of this book directly. Tong says so without hedging: string theory is “speculative science”, with no experimental evidence that it describes our world [Tong §0, ll. 380–387].

#### Physical picture: what the new length does

A point particle can probe arbitrarily short distances if it is given enough energy. A string cannot: it is itself an extended, fluctuating object of size about  $\ell_s$ , so that spacetime looks “fuzzy” to it at the scale  $\sqrt{\alpha'}$ . One consequence is that string scattering amplitudes fall off exponentially at high energy, where those of general relativity grow; this is the first sign that the ultraviolet problem of Equation (1.2) may be cured [Tong §6, ll. 8055–8080]. The string length is a *minimum length*, but “not in any crude simple manner”: it shows up differently depending on the question asked. Section 1.2 describes the manifestation that organizes this book.

Why study such a theory? The source gives four reasons that a student should weigh independently [Tong §0, ll. 388–463]: it is *a* consistent framework for quantum gravity, in which sharp questions about short-distance spacetime can be posed; it may be *the* theory of quantum gravity, since general relativity, gauge theories, scalars and chiral fermions all emerge at low energy; it gives new tools for ordinary gauge theories; and it produces new mathematics, mirror symmetry being the best-known case. This book lives mostly in the

fourth reason. Its subject is a part of string theory where the mathematics is rigid enough — lattices, integer matrices, modular forms, finite groups — for a proof assistant to check it.

#### Historical note

String theory “was born from attempts to understand the strong force” [Tong §0, ll. 437–438]; the name “Regge slope” for  $\alpha'$  is a remnant of that period [Tong §3.1.3, ll. 3391–3394]. The word *duality* has a longer record in physics: wave–particle duality, the high/low temperature duality of the Ising model, and electric–magnetic duality all precede the target-space duality of strings; according to the standard review each shares features with it and “none is identical to it” [GPR §1, ll. 237–255].

## 1.2 Why two scales

The cleanest way to see what is new about a string is to put it on the simplest space that is not  $\mathbb{R}^n$ : one spatial direction curled into a circle,

$$X \sim X + 2\pi R, \quad (1.4)$$

with  $R$  the radius. Chapters 7 and 8 treat this carefully; here only the two facts that need no string quantization are derived.

### 1.2.1 A particle on a circle sees one scale

A quantum particle with momentum  $p$  along the circle has the wavefunction  $\psi(X) = e^{ipX}$ . The points  $X$  and  $X + 2\pi R$  are the same point, so  $\psi$  must take the same value there:  $e^{2\pi ipR} = 1$ , hence

$$p = \frac{n}{R}, \quad n \in \mathbb{Z} \quad (1.5)$$

[Tong §8.2, ll. 11352–11358]. Suppose the particle is massless in the full spacetime,  $-p_\mu p^\mu - p^2 = 0$ , where  $\mu$  runs over the non-compact directions. An observer who cannot resolve the circle measures the mass  $M^2 = -p_\mu p^\mu$ , and finds a tower

$$M_{\text{particle}}^2 = \frac{n^2}{R^2}, \quad n \in \mathbb{Z}, \quad (1.6)$$

the *Kaluza–Klein tower*. The spacing  $1/R$  of the tower measures the radius: a small circle gives heavy states, a large circle gives light ones, and two different radii are never confused. As  $R \rightarrow 0$  every state with  $n \neq 0$  becomes infinitely heavy and the circle direction disappears from low-energy physics.

### 1.2.2 A string on a circle sees two

A closed string can do what a particle does: move around the circle with momentum (1.5). It can also do something a particle cannot: wrap around the circle. Parametrize the string by  $\sigma \in [0, 2\pi]$ . Closing up after one turn of  $\sigma$  only requires returning to the same *point of the circle*, so by (1.4)

$$X(\sigma + 2\pi) = X(\sigma) + 2\pi w R, \quad w \in \mathbb{Z}, \quad (1.7)$$

where  $w$  is the *winding number* [Tong §8.2, ll. 11359–11371; Tong writes  $m$  for our  $w$ ]. A string wound  $w$  times has length at least  $2\pi|w|R$ , and a string costs energy  $T$  per unit length. With (1.3) the winding contributes

$$M_{\text{winding}} = 2\pi|w|RT = \frac{|w|R}{\alpha'} \quad (1.8)$$

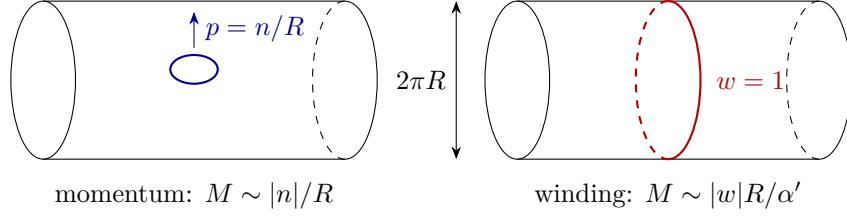


Figure 1.1: One compact direction, drawn as the circumference of a cylinder; the axis of the cylinder stands for the non-compact directions. Left: a small closed string moving around the circle carries quantized momentum  $n/R$ , as a particle would. Right: a string wrapped around the circle carries winding number  $w$ ; its minimal length  $2\pi|w|R$  costs the energy (1.8). A particle has no analogue of the second state.

to the mass [Tong §8.2, ll. 11480–11487]. Figure 1.1 shows the two kinds of excitation.

Quantizing the string (Chapters 3 and 7) adds the oscillator levels  $N, \tilde{N} \in \{0, 1, 2, \dots\}$  of the right- and left-moving vibrations and a zero-point term. The result, for the bosonic string and in the conventions of this book, is

$$M^2 = \frac{n^2}{R^2} + \frac{w^2 R^2}{\alpha'^2} + \frac{2}{\alpha'}(N + \tilde{N} - 2), \quad N - \tilde{N} = nw, \quad (1.9)$$

[Tong §8.2, eqs. (8.5)–(8.6), ll. 11425–11475]. The first term is the particle’s (1.6); the second is the square of (1.8); the third is the vibrational energy, in units set by the tension. The condition  $N - \tilde{N} = nw$  is *level matching*: it expresses that no point of a closed string is preferred as the origin of  $\sigma$ . Equation (1.9) can be packaged with the combinations  $p_{L,R} = n/R \pm wR/\alpha'$ : a line of algebra gives  $\frac{1}{2}(p_L^2 + p_R^2) = n^2/R^2 + w^2 R^2/\alpha'^2$  and  $p_L^2 - p_R^2 = 4nw/\alpha'$ , so that both the mass and the level-matching condition are functions of  $(p_L, p_R)$  alone. This is the seed of the Narain lattice of Chapter 10.

Now compare a string on a circle of radius  $R$  with a string on a circle of radius

$$\tilde{R} = \frac{\alpha'}{R}. \quad (1.10)$$

Substituting in (1.9),  $n^2/\tilde{R}^2 = n^2 R^2/\alpha'^2$  and  $w^2 \tilde{R}^2/\alpha'^2 = w^2/R^2$ : the first two terms are exchanged. If the labels are exchanged as well,  $n \leftrightarrow w$ , the formula returns to itself, and level matching, which contains only the product  $nw$ , is untouched. Hence

$$M^2(R; n, w, N, \tilde{N}) = M^2(\alpha'/R; w, n, N, \tilde{N}) : \quad (1.11)$$

the two spectra coincide state by state [Tong §8.3, eqs. (8.7)–(8.8), ll. 11626–11640; BW §III.B, eq. (24), ll. 720–737]. This is *T-duality*. The statement extends from the spectrum to the full conformal field theory and hence to interactions [Tong §8.3, ll. 11655–11658] [L]; Chapters 8 and 9 explain why. The standard review puts it this way: a string “cannot tell” whether it propagates on a circle of radius  $R$  or  $1/R$  in string units [GPR §1, ll. 229–234].

#### Physical picture: two rulers

A particle owns one ruler: the spacing  $1/R$  of its momentum tower. A string owns two: the momentum spacing  $1/R$  and the winding spacing  $R/\alpha'$ . When  $R \gg \sqrt{\alpha'}$  the momentum states are light and nearly continuous — which is what “a large dimension” means operationally — and the winding states are too heavy to matter. When  $R \ll \sqrt{\alpha'}$  the roles are reversed: the *winding* states form a near-continuum, and the spectrum



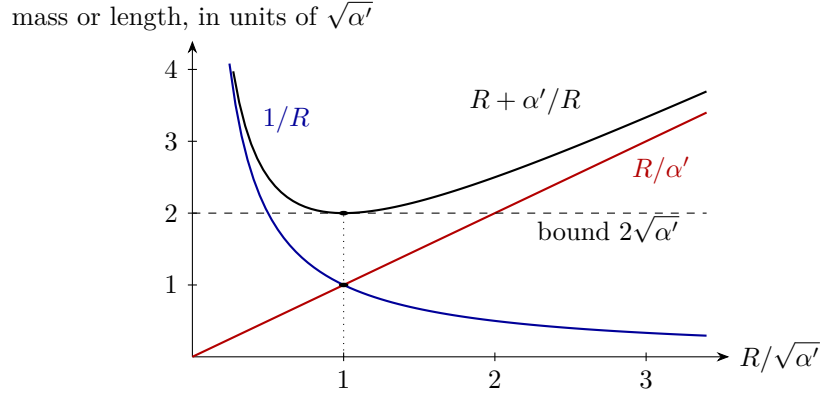


Figure 1.2: The two rulers of a string on a circle, in units  $\sqrt{\alpha'} = 1$ : the mass  $1/R$  of the first momentum state (blue) and the mass  $R/\alpha'$  of the first winding state (red) cross at the self-dual radius  $R = \sqrt{\alpha'}$ . The black curve is the effective scale  $R + \alpha'/R$  of Section 1.3; it is symmetric under  $R \mapsto \alpha'/R$  and never drops below  $2\sqrt{\alpha'}$  (dashed).

looks “as if another dimension of space is opening up” [Tong §8.3, ll. 11641–11654]. An observer made of strings who measures the size of the circle by the lightest tower reads off  $\max(R, \alpha'/R)$ , never less than  $\sqrt{\alpha'}$ . This is the sense in which the theory has two scales,  $R$  and  $\alpha'/R$ , of equal standing.

The fixed point of (1.10),  $R = \sqrt{\alpha'} = \ell_s$ , is the *self-dual radius*. Shrinking a circle below it is the same as growing the dual circle, so  $\sqrt{\alpha'}$  acts as a minimal radius [Tong §8.3, ll. 11659–11663]. Figure 1.2 plots the two rulers and their sum.

**Example 1.1** (The same spectrum at  $R = 2$  and  $R = \frac{1}{2}$ ). Set  $\alpha' = 1$ , so that lengths are measured in string lengths, and compare  $R = 2$  with its dual  $\tilde{R} = \frac{1}{2}$ . Equation (1.9) reads  $M^2 = n^2/R^2 + w^2R^2 + 2(N + \tilde{N} - 2)$ . Each row of the table is a state allowed by level matching ( $N - \tilde{N} = nw$  is checked in the second column).

| $(n, w)$ | $(N, \tilde{N})$ | $M^2$ at $R = 2$ | $M^2$ at $R = \frac{1}{2}$ | particle, $n^2/R^2$ at $R = 2$ / $R = \frac{1}{2}$ |
|----------|------------------|------------------|----------------------------|----------------------------------------------------|
| (0, 0)   | (1, 1)           | 0                | 0                          | 0 / 0                                              |
| (1, 0)   | (1, 1)           | $\frac{1}{4}$    | 4                          | $\frac{1}{4}$ / 4                                  |
| (0, 1)   | (1, 1)           | 4                | $\frac{1}{4}$              | —                                                  |
| (2, 0)   | (1, 1)           | 1                | 16                         | 1 / 16                                             |
| (0, 2)   | (1, 1)           | 16               | 1                          | —                                                  |
| (1, 1)   | (1, 0)           | $\frac{9}{4}$    | $\frac{9}{4}$              | —                                                  |

For instance, for  $(n, w) = (1, 1)$ ,  $(N, \tilde{N}) = (1, 0)$  at  $R = 2$ :  $M^2 = \frac{1}{4} + 4 + 2(1 + 0 - 2) = \frac{9}{4}$ . Read as *sets*, the third and fourth columns agree:  $\{0, \frac{1}{4}, 4, 1, 16, \frac{9}{4}\}$  in both cases. Row by row they agree after the exchange  $n \leftrightarrow w$ : the state (1, 0) at  $R = 2$  has the mass of the state (0, 1) at  $R = \frac{1}{2}$ . The last column shows what a particle sees: the towers  $\{0, \frac{1}{4}, 1, \dots\}$  and  $\{0, 4, 16, \dots\}$  are different, so a particle distinguishes the two circles and a string does not. (The level  $N = \tilde{N} = 0$ ,  $n = w = 0$  gives  $M^2 = -4/\alpha'$ , the tachyon of the bosonic string; it is discussed in Chapter 3 and plays no role here.)

**Common pitfall**

T-duality does *not* say that the function  $M^2(R)$  of a given state is symmetric under  $R \mapsto \alpha'/R$ ; the table shows that it is not. It says that the two *theories* have the same set of states, under a relabelling that exchanges momentum and winding. A second trap: the conclusion depends on having both towers. Any approximation that keeps only the momentum states — ordinary field theory on the circle, for example — loses the duality. Double field theory (Chapter 14) is the attempt to write a field theory that keeps both.

**1.3 A toy theorem, three ways:  $R + \alpha'/R \geq 2\sqrt{\alpha'}$** **1.3.1 By hand**

A quantity that treats  $R$  and  $\alpha'/R$  on an equal footing must be invariant under the exchange (1.10). The simplest one with the dimension of a length is the sum

$$R_{\text{eff}}(R) = R + \frac{\alpha'}{R}, \quad R > 0, \quad (1.12)$$

which this book calls the *effective scale*. Invariance is immediate:  $R_{\text{eff}}(\alpha'/R) = \alpha'/R + \alpha'/(\alpha'/R) = \alpha'/R + R$ .

**Proposition 1.2.** *For all real  $R > 0$  and  $\alpha' > 0$ ,  $R + \alpha'/R \geq 2\sqrt{\alpha'}$ , with equality if and only if  $R = \sqrt{\alpha'}$ .*

*First proof (completing the square).* Since  $R > 0$  we may write  $R = (\sqrt{R})^2$  and  $\alpha'/R = (\sqrt{\alpha'}/\sqrt{R})^2$ , and the product of the two square roots is  $\sqrt{\alpha'}$ . Therefore

$$R + \frac{\alpha'}{R} - 2\sqrt{\alpha'} = \left(\sqrt{R} - \frac{\sqrt{\alpha'}}{\sqrt{R}}\right)^2 = \frac{(R - \sqrt{\alpha'})^2}{R} \geq 0. \quad (1.13)$$

The middle expression is a square, the right-hand one is a square divided by a positive number; either way it is non-negative and vanishes exactly when  $R = \sqrt{\alpha'}$ .  $\square$

*Second proof (calculus).*  $R'_{\text{eff}}(R) = 1 - \alpha'/R^2$  vanishes on  $R > 0$  only at  $R = \sqrt{\alpha'}$ , and  $R''_{\text{eff}}(R) = 2\alpha'/R^3 > 0$ , so  $R_{\text{eff}}$  is convex with a single minimum,  $R_{\text{eff}}(\sqrt{\alpha'}) = 2\sqrt{\alpha'}$ .  $\square$

*Third proof (arithmetic and geometric means).* For  $a, b \geq 0$ ,  $(a + b)/2 \geq \sqrt{ab}$ , the AM–GM inequality, which is itself  $(\sqrt{a} - \sqrt{b})^2 \geq 0$ . With  $a = R$  and  $b = \alpha'/R$  the geometric mean is  $\sqrt{R \cdot \alpha'/R} = \sqrt{\alpha'}$ , independent of  $R$ : *T-dual radii have a fixed geometric mean, so their arithmetic mean is bounded below by it.*  $\square$

The identities in (1.13) were also confirmed with a computer-algebra system (a symbolic check, which is not a proof-assistant check). It is useful to remove the units. Put  $x = R/\sqrt{\alpha'}$ , the radius in string lengths. Then

$$R + \frac{\alpha'}{R} = \sqrt{\alpha'} \left(x + \frac{1}{x}\right), \quad \text{so} \quad R + \frac{\alpha'}{R} \geq 2\sqrt{\alpha'} \iff x + \frac{1}{x} \geq 2. \quad (1.14)$$

The dimensionless statement  $x + 1/x \geq 2$  for  $x > 0$  is the one that appears in Lean below.

**Example 1.3 (Numbers).** In units  $\sqrt{\alpha'} = 1$ : at  $R = 2$ ,  $R_{\text{eff}} = 2 + \frac{1}{2} = 2.5$ ; at the dual radius  $R = \frac{1}{2}$ ,  $R_{\text{eff}} = \frac{1}{2} + 2 = 2.5$  again; at  $R = 1$ ,  $R_{\text{eff}} = 2$ , the minimum. The excess over the bound, from (1.13), is  $(R - 1)^2/R$ : for  $R = 2$  this is  $\frac{1}{2}$ , for  $R = \frac{1}{2}$  it is  $(\frac{1}{4})/(\frac{1}{2}) = \frac{1}{2}$  as well, as it must be. For a circle ten times the string length,  $R_{\text{eff}} = 10.1$ : the correction to the naive size is one per cent, and for a macroscopic circle it is unobservably small,  $\alpha'/R^2$  in relative terms. The effective scale differs from  $R$  only near or below the string length.

### 1.3.2 In Lean

The same statement, in the dimensionless form (1.14), is a theorem of this book's Lean library. It is the first `leanbox` of the book, so it is annotated in more detail than later ones.

#### In Lean: the circle bound

```
-- DualScaleStream2/DualScale/TraceBound.lean
theorem circle_effective_scale_ge_two (R : ℝ) (hR : 0 < R) :
  2 ≤ R + R-1 := by
  have h : R + R-1 - 2 = (R - 1) ^ 2 / R := by field_simp; ring
  have h_nonneg : 0 ≤ (R - 1) ^ 2 / R := by
    apply div_nonneg
    · exact sq_nonneg (R - 1)
    · linarith
  linarith
```

**Reading.** For every real number  $R$  and every proof  $hR$  that  $0 < R$ , one has  $2 \leq R + R^{-1}$ . Nothing in the statement mentions strings,  $\alpha'$  or a circle:  $R$  is a real number, and the identification  $R = (\text{radius})/\sqrt{\alpha'}$  is made by the reader.

**Proof idea.** Exactly the first proof above with  $\alpha' = 1$ : the tactic `field_simp` clears the denominator and `ring` verifies the polynomial identity  $R + R^{-1} - 2 = (R - 1)^2/R$ ; a square divided by a positive number is non-negative (`div_nonneg`, `sq_nonneg`); `linarith` combines the two facts.

**How to read a `leanbox`.** Every box of this kind in the book has the same four parts.

1. *The statement, verbatim.* Text before the colon declares the data and the hypotheses; a hypothesis is a named proof, here `hR : 0 < R`. Text after the colon is the conclusion. The part after `:=` is the proof; from now on it is usually abbreviated to `:= by ...`, because for the reader the *statement* is what matters — the proof has been checked by the Lean kernel and need not be trusted or even read.
2. *A plain-language reading* that says what the statement literally asserts and no more.
3. *The proof idea*, two or three sentences, so that the result can be connected to the derivation in the text.
4. *A tier* (Section 1.5): **[A]** for the Lean statement, and separately the tier of the physical statement that it supports.

The gap between the Lean statement and Proposition 1.2 is small but real, and is typical. The Lean theorem is the case  $\alpha' = 1$ ; the general case follows from the rescaling (1.14), which was done on this page and not in Lean. The equality case ( $R = 1$  is the only minimizer) is not part of the Lean statement either; Exercise 1.5 asks for it.

#### Common pitfall: the hypothesis is not decoration

In Lean's mathematical library division is a total function:  $0^{-1}$  is *defined* to be 0. Without `hR` the claimed inequality at  $R = 0$  would read  $2 \leq 0 + 0$ , which is false, and for  $R < 0$  it fails as well ( $R = -1$  gives  $2 \leq -2$ ). Lean would refuse the proof. When reading any formal statement, first find the hypotheses and ask what each one excludes; a theorem with a forgotten hypothesis is unprovable, and a theorem with an *extra*, unsatisfiable hypothesis is provable and empty.

### 1.3.3 What the inequality means, and with what confidence

Proposition 1.2 is elementary mathematics, and its Lean form is Tier A. Its *interpretation* is a different kind of statement. That  $\sqrt{\alpha'}$  acts as a minimal radius for a string on a circle is a result of the literature [L] [Tong §8.3, ll. 11659–11663]. That the same mechanism matters in the early universe is the proposal of string gas cosmology, which starts from the observation that strings at small scale factor behave like strings at large scale factor and therefore act on a cosmological background differently from point particles [BW §III.B, ll. 720–737]; the same review describes scenarios in which compact dimensions are driven towards the self-dual radius, and their fine-tuning problems [L] [BW §I, ll. 160–176; §IV, ll. 1108–1137]. That  $R_{\text{eff}}$  itself is *the* physically measured length, or that the bound removes the initial singularity of cosmology, is a proposal of this programme [C] and is examined as such in Chapters 31 and 32. The general version for a  $d$ -dimensional torus with metric  $G$ ,  $\text{tr } G + \text{tr } G^{-1} \geq 2d$ , is proved in Chapter 31; in Lean it is `dualScale_ge`, in the same file as the circle bound [A].

## 1.4 The programme of this book

The circle is the first case of a pattern. Replace the circle by a  $d$ -dimensional torus: the pair  $(n, w)$  becomes a vector of  $2d$  integers, the exchange  $n \leftrightarrow w$  becomes the group  $O(d, d; \mathbb{Z})$  of integer matrices preserving the pairing  $2n \cdot w$ , and the radius becomes a metric  $G$  and a two-form  $B$  assembled in the generalized metric  $\mathcal{H}$  (Parts II and III). Replace the torus by a K3 surface: the lattice of charges becomes the even unimodular lattice of signature  $(4, 20)$ , whose automorphisms again act as dualities (Part IV). Compactify on  $K3 \times T^2$  and add fluxes: integrality conditions and the number  $\chi/24$  appear (Part V). Count states on K3: the coefficients organize into dimensions of representations of the sporadic group  $M_{24}$  (Part VI). In each case the physics rests on a piece of exact, discrete mathematics.

This observation defines the scope of the formalization. Its goal, in the words of the project’s workflow document, is a kernel-checked Lean 4 layer for the *mathematical skeleton* of string theory on  $K3 \times T^2$ : charge lattices, their duality groups, the doubled algebra, flux and tadpole counting, and the moonshine numerics [docs/STREAM2\_WORKFLOW.md §1]. The same document states what is not a goal and is never to be claimed: formalizing string theory as physics. Moduli-space quotients, the classification of even unimodular lattices, Fourier–Mukai transforms and similar deep results remain Tier L unless a specific work package proves them. A student should read every chapter with this division in mind: *the physics is explained and sourced; the skeleton is proved; the bridge between them is argued, never machine-checked*.

*Remark 1.4* (Older and newer libraries). The repository contains libraries of different vintage. The newest, `DualScaleStream2`, works with real and integer matrices of arbitrary size and proves statements that are recognizably the mathematical ones (the circle bound above is an example). Several older libraries model physical quantities by integers or rationals and prove arithmetic consequences. This book calls such results *arithmetic shadows* of the physical statement, says so each time, and explains what a faithful formalization would need. Chapter 35 maps the libraries.

## 1.5 Three tiers of confidence

Every claim that matters in this book carries one of three badges.

**[A]** *Kernel-checked*. A named declaration in the Lean 4 libraries of this project, compiled without `sorry`. It may depend only on the three standard axioms of Lean, namely

`propext`, `Classical.choice` and `Quot.sound`.

[L] *Literature*. Established in a published source, cited with the line range that was read; not formalized here.

[C] *Conjecture*. An open conjecture of the field, or a proposal of this programme.

Two rules govern the badges. First, a computer-algebra verification is a “symbolic check”, useful but not Tier A. Second, and more important: *the identification of a Lean object with a physical system is never Tier A*. The theorem of Section 1.3 is about a real number; the sentence “the radius of a compact dimension, measured by strings, is bounded below” is Tier L. The second rule is not a matter of taste. It follows from a simple principle about how confidence propagates, and that principle has itself been formalized.

### 1.5.1 The tier order and the ledger, in Lean

The module `DualScaleM24Formalization/Epistemic/Tier.lean` defines a type of tiers. It is finer than the book’s: it has five values, in increasing order of strength

$$X < C < L < B < A,$$

where  $X$  (exploratory: floating-point numerics, sampling, machine-generated text) and  $B$  (a deterministic exact-arithmetic check accompanied by a negative control that fails) are used in the project’s internal ledgers. This book does not print  $X$  material and files its symbolic checks in the running text, so three badges suffice. The order is defined by sending each tier to a natural number and comparing (`Tier.toNat`, `Tier.le`); the module proves that  $\leq$  is reflexive, transitive and antisymmetric (`tier_refl`, `tier_trans`, `tier_antisymm`), that  $A$  is the top and  $X$  the bottom element (`tier_A_is_maximal`, `tier_X_is_minimal`), and the small lemma on which everything else turns:

```
| theorem eq_A_of_A_le {s : Tier} (h : Tier.A ≤ s) : s = Tier.A := by ...
```

if something is at least as strong as  $A$ , it *is*  $A$ .

The module `DualScaleM24Formalization/Epistemic/Ledger.lean` models the book-keeping. A *claim* is a tier together with the list of claims it cites; a *ledger* assigns a claim to each identifier of some index type  $\iota$ ; a ledger is *sound* if no claim is filed above anything it directly cites.

```
structure Claim (ι : Type) where
  tier : Tier
  supports : List ι

abbrev Ledger (ι : Type) := ι → Claim ι

def Sound {ι : Type} (L : Ledger ι) : Prop :=
  ∀ a b, b ∈ (L a).supports → (L a).tier ≤ (L b).tier
```

Soundness is a *local* condition: it looks one citation deep. What one wants is a *global* guarantee, along chains of citations of any length. The relation `Depends L a b` (“ $a$  rests on  $b$ , directly or through a chain”) is defined inductively, and the main theorem states that the local condition implies the global one.

#### In Lean: confidence is a weakest-link quantity

```
| theorem tier_le_of_depends {ι : Type} {L : Ledger ι} (hL : Sound L) :
  ∀ {a b : ι}, Depends L a b → (L a).tier ≤ (L b).tier := by ...
```

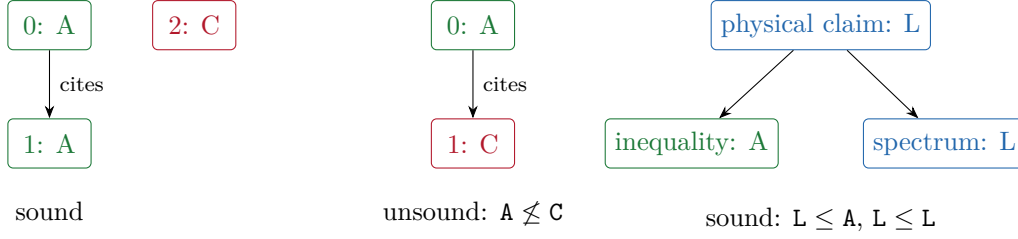


Figure 1.3: Ledgers as directed graphs; an arrow points from a claim to a claim it cites. Left and middle: the two test ledgers of the Lean module (Example 1.5). Right: the way this book files a physical statement that uses a kernel-checked inequality: at the tier of its weakest support.

```
theorem no_kernel_claim_rests_on_weaker {ι : Type} {L : Ledger ι}
  (hL : Sound L) {a b : ι} (ha : (L a).tier = Tier.A)
  (hab : Depends L a b) : (L b).tier = Tier.A := by ...
```

**Reading.** In a sound ledger, a claim is never filed above anything in its transitive support; in particular everything a Tier-A claim rests on, at any depth, is Tier A. The contrapositive is `not_A_of_weak_support`: one weaker link anywhere in the chain forbids the badge A.

**Proof idea.** Induction on the chain. A chain of length one is the soundness hypothesis; a longer chain is a first citation followed by a shorter chain, and the two inequalities compose by `tier_trans`. The corollary rewrites the tier of  $a$  as A and applies `eq_A_of_A_le`.

**What it does not say.** The theorem is about an abstract function  $\iota \rightarrow \text{Claim } \iota$ . It does not inspect this book. Whether the badges printed in these pages form a sound ledger is an editorial discipline, which the theorem motivates but cannot enforce.

**Example 1.5** (A sound and an unsound ledger). The Lean module contains two test ledgers, drawn in Figure 1.3. In `witnessSound`, claim 0 is Tier A and cites claim 1, also Tier A; claim 2 is a free-standing conjecture. The only citation to check is  $0 \rightarrow 1$ , and  $A \leq A$  holds: `witnessSound_sound` proves `Sound witnessSound`. In `witnessUnsound`, claim 0 is filed at A while citing a Tier-C claim; in numbers, soundness would need  $4 \leq 1$ . The theorem `witnessUnsound_not_sound` proves  $\neg \text{Sound witnessUnsound}$ . The second test is as important as the first: a definition of “sound” that every ledger satisfied would make the main theorem true and useless, and the negative control rules this out.

Now apply the principle to this chapter. The physical sentence “a string cannot resolve a radius below  $\sqrt{\alpha'}$ ” cites (i) the inequality, Tier A; (ii) the mass formula (1.9) and the extension of T-duality to the full theory, Tier L; (iii) the identification of the real number  $R$  in Lean with a radius in string units, which no kernel can check. By `tier_le_of_depends` the sentence can be filed at Tier L at best. This is the rule stated above, now seen as a consequence of soundness.

## 1.6 Why a proof assistant

A proof assistant is a program that checks a proof down to the axioms of a fixed logical foundation. In Lean 4 a small trusted component, the *kernel*, accepts a theorem only if every step reduces to the rules of the underlying type theory; the tactics that produced the proof (`ring`, `linarith`, a language model, a student) need not be trusted, because their output is

re-checked. Chapter 34 teaches enough Lean to read and write the statements in this book. Three properties make the tool suited to this subject.

*It removes a specific kind of error.* Much of the mathematics in later chapters is finite but unforgiving: an  $8 \times 8$  integer matrix must have determinant exactly 1; the squares of 26 character degrees must add up to a group order; a block-matrix identity must hold with every sign right. Such statements are rarely doubted and often mistyped. The project's own record has an instructive case: a table of the dimensions of the irreducible representations of  $M_{24}$  entered in an early library had two entries duplicated and two missing, so that the squares summed to 351 061 029 instead of  $|M_{24}| = 244\,823\,040 = 2^{10} \cdot 3^3 \cdot 5 \cdot 7 \cdot 11 \cdot 23$ . The table was replaced by the one in the literature and is now guarded by a theorem, `M24RepDim_sum_sq`, which states the sum rule [docs/STREAM2\_WORKFLOW.md §6, "Side findings"]; see Chapter 29.

*It makes the hypotheses explicit*, as the pitfall box of Section 1.3 showed. In physics texts, conditions such as " $G$  positive definite", " $B$  antisymmetric" or " $R > 0$ " are often left to context; a formal statement cannot leave them out.

*It separates the statement from the proof.* Once the kernel accepts a proof, the only thing a reader has to examine is whether the *statement* says what the text claims. This is the one check that no kernel can perform, and the project's workflow reserves a separate, recorded statement-review step for it [docs/STREAM2\_WORKFLOW.md §4, step S3]. The same document fixes the acceptance rule used for every Tier-A badge in this book: the library builds, no `sorry` (Lean's placeholder for a missing proof) remains, and the command `#print axioms` reports only the three standard axioms; the producer of a proof never grades it [docs/STREAM2\_WORKFLOW.md §2, items 1–2].

#### Common pitfall: what a green build does not mean

A kernel-checked theorem is true *as stated, from its hypotheses*. It is not evidence that the statement models nature, nor that a toy statement about integers captures a statement about quantum fields, nor that the definitions are the intended ones. A library can compile perfectly and prove only trivialities. The habit to acquire is to read a formal statement the way one reads a contract: literally, and with attention to what is absent.

## 1.7 A reader's guide

The book has eight parts. Parts I–VI build the physics and mathematics, Part VII presents the programme's own proposals with their tiers, and Part VIII documents the formalization.

| Part | Title                                                        | What it does                                                                                                                                 |
|------|--------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| I    | Foundations of the String<br>(Chapter 1–Chapter 6)           | Classical string, quantization and critical dimension, the conformal field theory needed later, background fields and the low-energy action. |
| II   | Compactification and<br>T-Duality (Chapter 7–<br>Chapter 13) | The circle in full, Buscher rules, the Narain lattice, the group $O(d, d; \mathbb{Z})$ , the two-torus, D-branes.                            |
| III  | Double Field Theory (Chap-<br>ter 14–Chapter 17)             | Doubled coordinates, the strong constraint, the generalized metric $\mathcal{H}$ , Courant brackets, the action.                             |
| IV   | K3 Surfaces and Lattices<br>(Chapter 18–Chapter 23)          | Even unimodular lattices, $E_8$ and $U$ ; K3 surfaces, periods and Torelli; Kummer surfaces; the Mukai lattice; strings on K3.               |
| V    | $K3 \times T^2$ and Fluxes (Chap-<br>ter 24–Chapter 27)      | Compactification on $K3 \times T^2$ , tadpoles and $\chi/24$ , flux superpotentials, swampland constraints.                                  |
| VI   | Mathieu Moonshine (Chap-<br>ter 28–Chapter 30)               | The elliptic genus of K3, the group $M_{24}$ , and the observed decompositions.                                                              |



|      |                                                  |                                                                                                                      |
|------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| VII  | The Dual-Scale Theory<br>(Chapter 31–Chapter 33) | The inequality of this chapter in general form; string gas cosmology; the programme’s conjectures, labelled as such. |
| VIII | The Formalization (Chapter 34–Chapter 38)        | A Lean primer, the architecture of the libraries, an atlas of the theorems, the proof pipeline, open problems.       |

**Routes through the book.** A physics student can read Parts I–III in order, take Chapter 18 as a mathematical interlude and continue. A mathematics student who accepts the mass formula (1.9) as a definition can start at Chapter 10, read Parts III–IV and VI, and return to Part I for motivation. A reader mainly interested in formalization should read this chapter, then Chapter 34, and then the section “What the machine has checked” of each chapter.

**Conventions for every chapter.** Each chapter opens with the question it answers, derives its results, works at least one example with numbers, and ends with the machine-checked section, a summary, graded exercises (★ routine, ★★ requires an idea, ★★★ a small project; some are to be done in Lean) and further reading. Sources are cited as [key §section, ll. lines]; the line numbers refer to the plain-text copies of the papers in the project repository, so that every citation can be located mechanically.

## 1.8 What the machine has checked

This chapter used two groups of declarations. The first is the circle bound of Section 1.3, in `DualScaleStream2.DualScale`. The file that contains it proves the general torus bound first. The circle bound is then proved directly, as shown above, and not by specialization; what links the two is the following identity.

### In Lean: from the torus to the circle

```
theorem dualScale_circle (R : ℝ) (hR : R ≠ 0) :
  dualScale (!! [R ^ 2] : Matrix (Fin 1) (Fin 1) ℝ)
    = (R + R⁻¹) ^ 2 - 2 := by ...
```

**Reading.** Here `dualScale G` is defined as the trace of the generalized metric at  $B = 0$ , and `dualScale_eq` shows it equals  $\text{tr } G + \text{tr } G^{-1}$ . For the  $1 \times 1$  metric  $G = (R^2)$  — a circle of radius  $R$  in units  $\alpha' = 1$  — this is  $R^2 + R^{-2}$ , and the theorem is the identity  $R^2 + R^{-2} = (R + R^{-1})^2 - 2$ .

**Proof idea.** Compute the inverse of a  $1 \times 1$  matrix explicitly, reduce both traces to their single entry, clear denominators (`field_simp`) and finish with `ring`.

**Why it matters.** With  $d = 1$  the torus bound `dualScale_ge` gives  $(R + R^{-1})^2 - 2 \geq 2$ , that is  $R + R^{-1} \geq 2$  for  $R > 0$ : the circle bound is the  $d = 1$  case of a statement about all positive-definite metrics, taken up in Chapters 15 and 31.

The second group is the tier order and the ledger of Section 1.5, in the namespace `SocrateAI.Epistemic`. The table collects what was cited, with the tier of the Lean statement and the tier of the use made of it in the text.

| Declaration                                | Literal content                      | Tier       |
|--------------------------------------------|--------------------------------------|------------|
| <code>circle_effective_scale_ge_two</code> | $2 \leq R + R^{-1}$ for real $R > 0$ | <b>[A]</b> |



|                                                                                                                                                                    |                                                                                           |     |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-----|
| <code>dualScale_circle</code>                                                                                                                                      | $\text{tr } G + \text{tr } G^{-1} = (R + R^{-1})^2 - 2$ for $G = (R^2)$ , $R \neq 0$      | [A] |
| <code>dualScale_ge</code> (preview)                                                                                                                                | $2d \leq \text{tr } G + \text{tr } G^{-1}$ for real positive-definite $G$                 | [A] |
| <code>tier_refl</code> , <code>tier_trans</code> , <code>tier_antisymm</code>                                                                                      | the five tiers are partially ordered (in fact linearly, through <code>Tier.toNat</code> ) | [A] |
| <code>tier_A_is_maximal</code> ,<br><code>tier_X_is_minimal</code> , <code>eq_A_of_A_le</code><br><code>tier_le_of_depends</code>                                  | A is the top element, X the bottom                                                        | [A] |
| <code>no_kernel_claim_rests_on_weaker</code> ,<br><code>not_A_of_weak_support</code><br><code>witnessSound_sound</code> ,<br><code>witnessUnsound_not_sound</code> | sound ledger $\Rightarrow$ tier is monotone along citation chains                         | [A] |
| <code>M24RepDim_sum_sq</code> (mentioned)                                                                                                                          | support of an A claim is all A; contrapositive                                            | [A] |
|                                                                                                                                                                    | positive and negative control ledgers                                                     | [A] |
|                                                                                                                                                                    | $\sum_i (\dim \rho_i)^2 = 244\,823\,040$ for the 26 tabulated dimensions                  | [A] |
| T-duality of the circle spectrum (1.11)                                                                                                                            | Tong §8.3; formalized versions in Chapter 8                                               | [L] |
| $\sqrt{\alpha'}$ as a minimal radius                                                                                                                               | Tong §8.3                                                                                 | [L] |
| $R_{\text{eff}}$ as the physical length; singularity avoidance                                                                                                     | proposal of this programme                                                                | [C] |

For each theorem in the table the command `#print axioms` reports only axioms among `propext`, `Classical.choice` and `Quot.sound`.

#### Chapter summary

- Perturbative gravity is an expansion in  $E^2/M_{\text{pl}}^2$  and is non-renormalizable; it is an effective theory that lacks its short-distance degrees of freedom. String theory proposes that these are strings, characterized by one scale,  $\ell_s = \sqrt{\alpha'}$ , with tension  $T = 1/(2\pi\alpha')$ .
- On a circle  $X \sim X + 2\pi R$  a particle has momentum  $n/R$  only; a closed string also has winding  $w$  with energy  $|w|R/\alpha'$ . The spectrum (1.9) is invariant under  $R \mapsto \alpha'/R$  with  $n \leftrightarrow w$ : T-duality. A string has two rulers,  $R$  and  $\alpha'/R$ .
- The T-duality-invariant effective scale satisfies  $R + \alpha'/R \geq 2\sqrt{\alpha'}$ , with equality only at the self-dual radius; the proof is  $(R - \sqrt{\alpha'})^2/R \geq 0$ . In Lean: `circle_effective_scale_ge_two` (the case  $\alpha' = 1$ ).
- Claims carry tiers: [A] kernel-checked, [L] literature, [C] conjecture. Confidence is a weakest-link quantity (`tier_le_of_depends`); identifying a Lean object with a physical system is never Tier A.
- A `leanbox` gives a verbatim statement, a literal reading, a proof idea and a tier. Read the hypotheses first.

## Exercises

**Exercise 1.1** (\*). Using  $\hbar c \approx 1.97 \times 10^{-14}$  GeV cm, convert to centimetres (a) the reduced Planck mass (1.1); (b) a string scale one order of magnitude below  $10^{19}$  GeV. Express the tension  $T = 1/(2\pi\alpha')$  of case (b) in GeV<sup>2</sup>.

**Exercise 1.2** (\*). Extend the table of Example 1.1 by all states with  $|n|, |w| \leq 2$  and  $N + \tilde{N} \leq 3$  allowed by level matching, at  $R = 3$  and at  $R = \frac{1}{3}$  (with  $\alpha' = 1$ ). Verify that the two lists of masses coincide and exhibit the bijection between states.

**Exercise 1.3** (★). With  $p_L = n/R + wR/\alpha'$  and  $p_R = n/R - wR/\alpha'$ , verify that  $\frac{1}{2}(p_L^2 + p_R^2) = n^2/R^2 + w^2R^2/\alpha'^2$  and  $p_L^2 - p_R^2 = 4nw/\alpha'$ . Show that under  $R \mapsto \alpha'/R$ ,  $n \leftrightarrow w$  one has  $p_L \mapsto p_L$  and  $p_R \mapsto -p_R$ , and deduce (1.11) without expanding the mass formula.

**Exercise 1.4** (★★). Find all states of (1.9) with  $M^2 = 0$  (a) at a generic radius, (b) at  $R = \sqrt{\alpha'}$ . You should find four additional solutions with  $n, w \in \{\pm 1\}$  in case (b). Why can they exist only at the self-dual radius? (Their meaning, an enhanced gauge symmetry, is the subject of Chapter 8.)

**Exercise 1.5** (★★). State and prove in Lean the equality case missing from the circle bound:

```
example (R : ℝ) (hR : 0 < R) (h : R + R-1 = 2) : R = 1 := by
  sorry
```

Hint: from the identity  $R + R^{-1} - 2 = (R - 1)^2/R$  obtain  $(R - 1)^2 = 0$ ; `field_simp` at `h` followed by `nlinarith [sq_nonneg (R - 1)]` is one route.

**Exercise 1.6** (★★). State in Lean the inequality with a general  $\alpha'$ , in a form that avoids square roots: for real  $R > 0$  and  $a > 0$ ,  $4a \leq (R + a/R)^2$ . Prove it on paper from Proposition 1.2, then in Lean. Explain why the square-root-free form is equivalent to  $R + a/R \geq 2\sqrt{a}$  when  $R, a > 0$ . Lean’s linter will report that the hypothesis  $a > 0$  is never used: why is the square-root-free statement true for every real  $a$ , and why is the original one not?

**Exercise 1.7** (★★). In the ledger of Figure 1.3 (right), suppose an author refiles the physical claim at Tier A. Which instance of the condition `Sound` fails? Write the four-element ledger (the claim, the inequality, the spectrum, and a Tier-C interpretation cited by nobody) as a Lean definition `Ledger` (Fin 4) in the style of `witnessSound`, and prove it sound with the same three-line proof.

**Exercise 1.8** (★★★). (a) Show that every function of  $R > 0$  invariant under  $R \mapsto \alpha'/R$  can be written as a function of  $R_{\text{eff}} = R + \alpha'/R$  alone. Hint: for  $s \geq 2\sqrt{\alpha'}$  the equation  $R + \alpha'/R = s$  has exactly the two solutions  $R$  and  $\alpha'/R$ . (b) Deduce that  $\max(R, \alpha'/R)$ , the size read off from the lightest tower, is an increasing function of  $R_{\text{eff}}$ , and find it explicitly. (c) Discuss: does (a) single out  $R_{\text{eff}}$  as “the” physical length? Which tier does your answer carry?

## Further reading

- Motivation for string theory and the problem of quantum gravity: [Tong §0, ll. 369–680]. The tension,  $\alpha'$  and the string length: [Tong §1.2, ll. 1090–1150]. The string length as a minimum length in scattering: [Tong §6, ll. 8055–8072].
- The circle, winding and T-duality at the level used here: [Tong §8.2–8.3, ll. 11345–11660]; continued in Chapters 7 and 8.
- The scope of target-space duality, from circles to curved backgrounds, and the history of the word “duality”: [GPR §1, ll. 207–330].
- Strings as a gas in the early universe and the role of T-duality there: [BW §I, ll. 128–190; §III.B, ll. 720–737]; continued in Chapter 32.
- The project’s rules for what counts as checked, and the measured record of the proof pipeline: [docs/STREAM2\_WORKFLOW.md §§1–2, 4, 6]; continued in Chapter 37.

## Chapter 2

# The Relativistic String: Nambu–Goto and Polyakov

What is the simplest relativistic dynamics that a one-dimensional object can have, and how does one solve it? This chapter answers that question for the classical closed string. The answer has three layers. First, an action: the area of the surface that the string sweeps out in spacetime, in the same way that the action of a free particle is the length of its worldline. Second, a reformulation, the Polyakov action, which removes the square root at the price of a new field and a new gauge symmetry. Third, a solution: in a suitable gauge the string obeys the free wave equation, its general motion is a superposition of left-moving and right-moving waves, and what remains of the gauge symmetry appears as an infinite set of quadratic constraints on the Fourier modes. A student should care because every later chapter starts from the end point of this one. The momentum and winding of Chapter 7, the duality  $R \mapsto \alpha'/R$  of Chapter 8, the Narain lattice of Chapter 10 are all statements about the zero modes and the constraints  $L_0 = \tilde{L}_0 = 0$  derived here. The chapter also marks a boundary of this book's formal content: almost nothing in it is machine-checked, and Section 2.7 explains why that is the expected state of affairs and not an omission.

Conventions are those of the whole book:  $\hbar = c = 1$ , mostly-plus signature  $\eta_{\mu\nu} = \text{diag}(-1, +1, \dots, +1)$  in  $D$  spacetime dimensions [Tong §1.1, ll. 688–693], and  $A \cdot B = A^\mu B^\nu \eta_{\mu\nu}$ ,  $A^2 = A \cdot A$ . A timelike vector has  $A^2 < 0$ .

### 2.1 Warm-up: the relativistic point particle

Every feature of the string action has a simpler counterpart for the particle. It pays to see each one there first.

#### 2.1.1 From a frame-dependent to a covariant action

Fix an inertial frame with coordinates  $X^\mu = (t, \vec{x})$ . A free particle of mass  $m$  has the action

$$S = -m \int dt \sqrt{1 - \dot{\vec{x}} \cdot \dot{\vec{x}}}, \quad \dot{\vec{x}} = d\vec{x}/dt \quad (2.1)$$

[Tong §1.1, ll. 694–712]. One checks it by computing the conjugate momentum and the energy. With  $L = -m\sqrt{1 - \dot{\vec{x}}^2}$ ,

$$\vec{p} = \frac{\partial L}{\partial \dot{\vec{x}}} = \frac{m\dot{\vec{x}}}{\sqrt{1 - \dot{\vec{x}}^2}}, \quad E = \vec{p} \cdot \dot{\vec{x}} - L = \frac{m\dot{\vec{x}}^2}{\sqrt{1 - \dot{\vec{x}}^2}} + m\sqrt{1 - \dot{\vec{x}}^2} = \frac{m}{\sqrt{1 - \dot{\vec{x}}^2}}, \quad (2.2)$$

so that  $E^2 - \vec{p}^2 = m^2(1 - \dot{\vec{x}}^2)/(1 - \dot{\vec{x}}^2) = m^2$ , the relativistic dispersion relation  $E = \sqrt{m^2 + \vec{p}^2}$ .

The action (2.1) is correct but treats  $t$  and  $\vec{x}$  differently:  $\vec{x}$  is a dynamical variable,  $t$  a label. Lorentz transformations mix them, so Lorentz invariance is hidden. The remedy is to let  $t$  become a dynamical variable too, and to pay for it with a gauge symmetry [Tong §1.1, ll. 713–738]. Introduce an arbitrary parameter  $\tau$  along the worldline and write

$$S = -m \int d\tau \sqrt{-\dot{X}^\mu \dot{X}^\nu \eta_{\mu\nu}}, \quad \dot{X}^\mu = dX^\mu/d\tau, \quad \mu = 0, \dots, D-1 \quad (2.3)$$

[Tong §1.1, ll. 739–760]. Since  $ds = \sqrt{-dX \cdot dX}$  is the proper time elapsed along an element of a timelike curve, the action is  $-m$  times the total proper time, that is,  $-m$  times the Lorentzian length of the worldline. It is manifestly invariant under the Poincaré transformations  $X^\mu \rightarrow \Lambda^\mu_\nu X^\nu + c^\mu$ , because only the invariant  $\dot{X}^2$  enters [Tong §1.1, ll. 815–822].

### 2.1.2 Reparametrization invariance is a gauge symmetry

The action (2.3) now has  $D$  functions  $X^\mu(\tau)$ , one more than the  $D-1$  functions  $\vec{x}(t)$  we started with. The extra one is not physical, because the action does not depend on the choice of parameter. Let  $\tilde{\tau}(\tau)$  be any monotonic function. Then  $d\tau = d\tilde{\tau} |d\tau/d\tilde{\tau}|$  and  $dX^\mu/d\tau = (dX^\mu/d\tilde{\tau})(d\tilde{\tau}/d\tau)$ . The square root is homogeneous of degree one in the velocities, so the two Jacobian factors cancel:

$$S = -m \int d\tilde{\tau} \sqrt{-\frac{dX^\mu}{d\tilde{\tau}} \frac{dX^\nu}{d\tilde{\tau}} \eta_{\mu\nu}} \quad (2.4)$$

[Tong §1.1, ll. 765–782]. This *reparametrization invariance* is a redundancy of the description, not a symmetry relating different physical states. We can use it to choose  $\tau = X^0(\tau) = t$ ; then  $-\dot{X}^2 = 1 - \dot{\vec{x}}^2$  and (2.3) reduces to (2.1) [Tong §1.1, ll. 783–797].

The redundancy is also visible in the momenta. From (2.3),

$$p_\mu = \frac{\partial L}{\partial \dot{X}^\mu} = \frac{m \dot{X}^\nu \eta_{\mu\nu}}{\sqrt{-\dot{X}^2}} \implies p_\mu p^\mu = \frac{m^2 \dot{X}^2}{-\dot{X}^2} = -m^2. \quad (2.5)$$

The  $D$  momenta are not independent; they satisfy the *mass-shell constraint*  $p^2 + m^2 = 0$  identically, for every trajectory, before any equation of motion is used [Tong §1.1, ll. 798–814]. A second consequence of the same homogeneity: the canonical Hamiltonian vanishes,

$$H = \dot{X}^\mu p_\mu - L = \frac{m \dot{X}^2}{\sqrt{-\dot{X}^2}} + m \sqrt{-\dot{X}^2} = -m \sqrt{-\dot{X}^2} + m \sqrt{-\dot{X}^2} = 0 \quad (2.6)$$

[Tong §1.1.1, ll. 826–840]. There is no evolution in  $\tau$  because  $\tau$  means nothing; all the dynamics sits in the constraint. In the quantum theory the constraint is imposed on wavefunctions,  $(p^2 + m^2)\Psi = 0$  with  $p_\mu = -i\partial/\partial X^\mu$ , which is the Klein–Gordon equation [Tong §1.1.1, ll. 848–857].

#### Physical picture: a constraint for every gauge symmetry

A gauge symmetry with one arbitrary function of  $\tau$  removes one function from the physical data and produces one constraint on the momenta. Remember this bookkeeping. The string has worldsheet reparametrizations with *two* arbitrary functions of *two* variables; it will have two constraints at every point of the string, and their Fourier modes will be the  $L_n$  and  $\tilde{L}_n$  of Section 2.5.

### 2.1.3 The einbein: removing the square root

There is a second action for the particle. Introduce an additional worldline field  $e(\tau) > 0$ , the *einbein*, and set

$$S = \frac{1}{2} \int d\tau \left( e^{-1} \dot{X}^2 - e m^2 \right) \quad (2.7)$$

[Tong §1.1.2, ll. 885–897]. No derivative of  $e$  appears, so its equation of motion is algebraic:  $\partial L / \partial e = -\frac{1}{2} e^{-2} \dot{X}^2 - \frac{1}{2} m^2 = 0$ , that is

$$\dot{X}^2 + e^2 m^2 = 0, \quad e = \frac{\sqrt{-\dot{X}^2}}{m}. \quad (2.8)$$

Substituting back,  $\frac{1}{2}(e^{-1} \dot{X}^2 - e m^2) = \frac{1}{2}(-m\sqrt{-\dot{X}^2} - m\sqrt{-\dot{X}^2}) = -m\sqrt{-\dot{X}^2}$ , which is (2.3) [Tong §1.1.2, ll. 913–918]. The two actions are classically equivalent. The new one has two advantages: it makes sense for  $m = 0$ , and it is quadratic in  $X$ , which is what a path integral needs [Tong §1.1.2, ll. 919–922]. Writing  $e = \sqrt{-g_{\tau\tau}}$  shows that  $e$  is a metric on the one-dimensional worldline, so (2.7) is “ $D$  scalar fields coupled to one-dimensional gravity” [Tong §1.1.2, ll. 898–912]. Under  $\tau \rightarrow \tau - \eta(\tau)$  the fields transform as  $\delta X^\mu = \eta \dot{X}^\mu$  (a scalar) and  $\delta e = \frac{d}{d\tau}(\eta e)$  (a density) [Tong §1.1.2, ll. 923–938].

The step from (2.3) to (2.7) is the model for the step from the Nambu–Goto action to the Polyakov action below.

## 2.2 The Nambu–Goto action

### 2.2.1 Worldsheet, induced metric, area

A particle sweeps out a worldline; a string sweeps out a two-dimensional surface, the *worldsheet*. Parametrize it by a timelike coordinate  $\tau$  and a spacelike coordinate  $\sigma$ , collected as  $\sigma^\alpha = (\tau, \sigma)$ ,  $\alpha = 0, 1$ . The string is described by  $D$  functions  $X^\mu(\tau, \sigma)$ , a map from the worldsheet into spacetime, which in this context is called the *target space*. This chapter treats closed strings:  $\sigma$  is periodic,

$$\sigma \in [0, 2\pi), \quad X^\mu(\tau, \sigma + 2\pi) = X^\mu(\tau, \sigma) \quad (2.9)$$

[Tong §1.2, ll. 945–968]; see Figure 2.1. Open strings appear in Chapter 13.

We want an action that depends on the surface and not on the coordinates  $\sigma^\alpha$  used to describe it. For the particle the answer was the length of the worldline. For the string it is the area of the worldsheet. To compute it, note that the flat metric of spacetime induces a metric on the surface: two nearby worldsheet points separated by  $d\sigma^\alpha$  are separated in spacetime by  $dX^\mu = \partial_\alpha X^\mu d\sigma^\alpha$ , hence by the invariant interval  $ds^2 = \eta_{\mu\nu} \partial_\alpha X^\mu \partial_\beta X^\nu d\sigma^\alpha d\sigma^\beta$ . The *induced metric* (or pull-back of  $\eta$ ) is therefore

$$\gamma_{\alpha\beta} = \frac{\partial X^\mu}{\partial \sigma^\alpha} \frac{\partial X^\nu}{\partial \sigma^\beta} \eta_{\mu\nu} = \begin{pmatrix} \dot{X}^2 & \dot{X} \cdot X' \\ \dot{X} \cdot X' & X'^2 \end{pmatrix}, \quad \dot{X}^\mu = \partial_\tau X^\mu, \quad X'^\mu = \partial_\sigma X^\mu, \quad (2.10)$$

and the *Nambu–Goto action* is

$$S_{\text{NG}} = -T \int d^2\sigma \sqrt{-\det \gamma} = -T \int d^2\sigma \sqrt{(\dot{X} \cdot X')^2 - \dot{X}^2 X'^2} \quad (2.11)$$

[Tong §1.2, ll. 969–1016]. The second form follows from  $\det \gamma = \dot{X}^2 X'^2 - (\dot{X} \cdot X')^2$ . For a physical worldsheet  $\dot{X}$  is timelike and  $X'$  spacelike, so  $\det \gamma < 0$  and the square root is real. The constant  $T$  is fixed below.

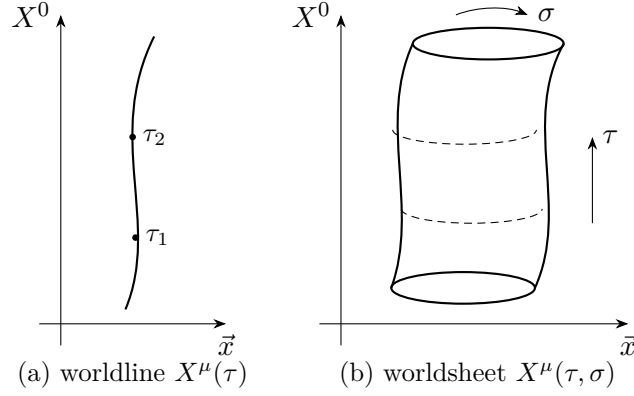


Figure 2.1: (a) A particle traces a worldline; the parameter  $\tau$  labels its points and has no physical meaning. (b) A closed string traces a tube, the worldsheet. Dashed curves are lines of constant  $\tau$ , one snapshot of the string each;  $\sigma \sim \sigma + 2\pi$  runs around the tube.

Why is  $\sqrt{|\det \gamma|} d^2\sigma$  an area? Consider first a surface in Euclidean space,  $\vec{X}(\tau, \sigma)$ . The coordinate cell  $d\tau d\sigma$  is mapped to the parallelogram spanned by  $\vec{u} = \partial_\tau \vec{X} d\tau$  and  $\vec{v} = \partial_\sigma \vec{X} d\sigma$ , with angle  $\theta$  between them. Its area is

$$|\vec{u}||\vec{v}| \sin \theta = \sqrt{\vec{u}^2 \vec{v}^2 (1 - \cos^2 \theta)} = \sqrt{\vec{u}^2 \vec{v}^2 - (\vec{u} \cdot \vec{v})^2} = \sqrt{\det \gamma} d\tau d\sigma \quad (2.12)$$

[Tong §1.2, ll. 1018–1060]. The quantity under the root is non-negative by the Cauchy–Schwarz inequality. In Minkowski signature the sign under the root flips, which is the origin of the minus sign in (2.11). The area is also manifestly independent of coordinates: under  $\sigma^\alpha \rightarrow \tilde{\sigma}^\alpha(\sigma)$  with Jacobian matrix  $J^\gamma_\alpha = \partial \sigma^\gamma / \partial \tilde{\sigma}^\alpha$ , the induced metric becomes  $\tilde{\gamma} = J^\top \gamma J$ , so  $\sqrt{-\det \tilde{\gamma}} = |\det J| \sqrt{-\det \gamma}$ , while  $d^2 \tilde{\sigma} = |\det J|^{-1} d^2 \sigma$ . This is the two-dimensional version of the cancellation that led to (2.4).

### 2.2.2 Tension, $\alpha'$ and the string length

To interpret  $T$ , take a string momentarily at rest. Choose the parametrization so that  $X^0 = t = R\tau$  with  $R$  a constant of dimension length, and suppose  $\dot{\vec{x}} = 0$  at the instant considered. Then  $\dot{X}^2 = -R^2$ ,  $\dot{X} \cdot X' = 0$ ,  $X'^2 = (d\vec{x}/d\sigma)^2$ , and

$$S_{\text{NG}} = -T \int d\tau d\sigma R \sqrt{(d\vec{x}/d\sigma)^2} = -T \int dt \times (\text{length of the string}) \quad (2.13)$$

[Tong §1.2, ll. 1061–1086]. For a system at rest the Lagrangian is minus the potential energy, hence

$$\text{potential energy} = T \times \text{length}.$$

$T$  is the energy per unit length, the *tension*. The energy grows linearly with the length, unlike a Hooke spring, whose energy grows quadratically; a classical string lowers its energy by shrinking [Tong §1.2, ll. 1087–1096].

The book-wide parametrization of the tension is

$$T = \frac{1}{2\pi\alpha'}, \quad \alpha' = \ell_s^2 \quad (2.14)$$

[Tong §1.2, ll. 1097–1123], where  $\alpha'$  is called the universal Regge slope for historical reasons and  $\ell_s$  is the string length. Dimensional analysis confirms the powers. With  $\hbar = 1$  the action

is dimensionless;  $X^\mu$  has dimension of length and  $\sigma^\alpha$  is dimensionless (it is an angle, Equation (2.9)). The integrand of (2.11) then has dimension  $\text{length}^2$ , so  $T$  has dimension  $\text{length}^{-2}$ , which for  $c = \hbar = 1$  is energy per length, and  $\alpha'$  is a length squared. For fundamental strings  $\ell_s$  is the only dimensionful parameter of the theory. Its value is not known; the tension is typically taken an order of magnitude or so below the Planck scale,  $T \lesssim (10^{18} \text{ GeV})^2$  [Tong §1.2, ll. 1143–1156].

**Example 2.1** (The tension of a QCD flux tube in everyday units). The Nambu–Goto action also describes, approximately, string-like objects of finite width such as the chromo-electric flux tubes of QCD, as long as they are nearly straight on the scale of their width; for those  $T \sim (1 \text{ GeV})^2$  [Tong §1.2, ll. 1124–1142]. Take  $T = 1 \text{ GeV}^2$  exactly to see the sizes involved. Restoring units with  $\hbar c = 0.1973 \text{ GeV fm}$  (standard value; not taken from this book’s library),

$$T = \frac{1 \text{ GeV}^2}{\hbar c} = 5.07 \frac{\text{GeV}}{\text{fm}} = 5.07 \times \frac{1.602 \times 10^{-10} \text{ J}}{10^{-15} \text{ m}} \approx 8 \times 10^5 \text{ N},$$

the weight of about eighty tonnes, acting between two quarks. The corresponding string length is  $\ell_s = \sqrt{\alpha'} = (2\pi T)^{-1/2} = 0.399 \text{ GeV}^{-1} = 0.079 \text{ fm}$ . A fundamental string with  $T$  near  $(10^{18} \text{ GeV})^2$  has a tension larger by a factor  $10^{36}$  and a length  $\ell_s$  smaller by a factor  $10^{18}$ .

### 2.2.3 Symmetries, momenta and equations of motion

The Nambu–Goto action has two symmetries of different nature [Tong §1.2.1, ll. 1159–1170]:

- *Poincaré invariance*  $X^\mu \rightarrow \Lambda^\mu_\nu X^\nu + c^\mu$ . From the worldsheet point of view this is a *global* symmetry:  $\Lambda$  and  $c$  do not depend on  $\sigma^\alpha$ . By Noether’s theorem it yields conserved momentum and angular momentum.
- *Reparametrization invariance*  $\sigma^\alpha \rightarrow \tilde{\sigma}^\alpha(\sigma)$ , a *gauge* symmetry, shown above.

Define the worldsheet momentum densities, with  $\mathcal{L}$  the integrand of (2.11),

$$\Pi_\mu^\tau = \frac{\partial \mathcal{L}}{\partial \dot{X}^\mu} = -T \frac{(\dot{X} \cdot X') X'_\mu - X'^2 \dot{X}_\mu}{\sqrt{(\dot{X} \cdot X')^2 - \dot{X}^2 X'^2}}, \quad \Pi_\mu^\sigma = \frac{\partial \mathcal{L}}{\partial X'^\mu} = -T \frac{(\dot{X} \cdot X') \dot{X}_\mu - \dot{X}^2 X'_\mu}{\sqrt{(\dot{X} \cdot X')^2 - \dot{X}^2 X'^2}}. \quad (2.15)$$

The Euler–Lagrange equations are

$$\partial_\tau \Pi_\mu^\tau + \partial_\sigma \Pi_\mu^\sigma = 0 \quad \Longleftrightarrow \quad \partial_\alpha (\sqrt{-\det \gamma} \gamma^{\alpha\beta} \partial_\beta X^\mu) = 0 \quad (2.16)$$

[Tong §1.2.2, ll. 1171–1208]. The second form uses  $\delta \sqrt{-\gamma} = \frac{1}{2} \sqrt{-\gamma} \gamma^{\alpha\beta} \delta \gamma_{\alpha\beta}$  and  $\delta \gamma_{\alpha\beta} = 2 \partial_\alpha X \cdot \partial_\beta \delta X$  (by symmetry in  $\alpha\beta$ ), followed by an integration by parts. Both forms are non-linear in  $X$ .

As for the particle, the momenta are constrained. Contracting (2.15) with  $X'^\mu$  and with itself gives, identically,

$$\Pi^\tau \cdot X' = 0, \quad (\Pi^\tau)^2 + T^2 X'^2 = 0. \quad (2.17)$$

The first is immediate: the numerator becomes  $(\dot{X} \cdot X') X'^2 - X'^2 (\dot{X} \cdot X')$ . For the second, the square of the numerator is  $X'^2 [X'^2 \dot{X}^2 - (\dot{X} \cdot X')^2]$ , which is  $-X'^2$  times the expression under the root. (Both identities were also confirmed by a symbolic check for  $D = 4$ .) There are two constraints at each point of the string, as the counting in the box of Section 2.1 predicted. They are the string’s version of  $p^2 + m^2 = 0$ , and they will reappear in a simpler gauge as (2.31).

## 2.3 The Polyakov action

### 2.3.1 Trading the square root for a worldsheet metric

The square root makes the Nambu–Goto action awkward, in particular in a path integral. The cure is the one that worked for the particle: introduce an independent metric  $g_{\alpha\beta}(\tau, \sigma)$  on the worldsheet, of signature  $(-, +)$ , and write

$$S_P = -\frac{1}{4\pi\alpha'} \int d^2\sigma \sqrt{-g} g^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu \eta_{\mu\nu}, \quad g = \det g_{\alpha\beta}, \quad (2.18)$$

the *Polyakov action* [Tong §1.3, ll. 1209–1227]. From the worldsheet point of view it describes  $D$  massless scalar fields  $X^\mu$  coupled to two-dimensional gravity. Note the prefactor:  $1/(4\pi\alpha') = T/2$ .

#### Historical note

The action (2.18) was not discovered by Polyakov; it carries his name because he understood how to use it in the path integral [Tong §1.3, ll. 1220–1223]. That use is the subject of Chapter 5.

**Equation of motion for  $X$ .** Varying  $X^\mu$  with  $g_{\alpha\beta}$  fixed gives

$$\partial_\alpha (\sqrt{-g} g^{\alpha\beta} \partial_\beta X^\mu) = 0, \quad (2.19)$$

the same as (2.16) but with  $g$  in place of  $\gamma$  [Tong §1.3, ll. 1228–1235].

**Equation of motion for  $g$ .** We need  $\delta\sqrt{-g}$ . From  $\delta \det g = \det g g^{\alpha\beta} \delta g_{\alpha\beta}$  and  $g^{\alpha\beta} \delta g_{\alpha\beta} = -g_{\alpha\beta} \delta g^{\alpha\beta}$  (vary  $g^{\alpha\beta} g_{\beta\gamma} = \delta^\alpha_\gamma$ ),

$$\delta\sqrt{-g} = \frac{1}{2}\sqrt{-g} g^{\alpha\beta} \delta g_{\alpha\beta} = -\frac{1}{2}\sqrt{-g} g_{\alpha\beta} \delta g^{\alpha\beta}. \quad (2.20)$$

Therefore

$$\delta S_P = -\frac{T}{2} \int d^2\sigma \sqrt{-g} \delta g^{\alpha\beta} \left( \partial_\alpha X \cdot \partial_\beta X - \frac{1}{2} g_{\alpha\beta} g^{\rho\sigma} \partial_\rho X \cdot \partial_\sigma X \right) = 0 \quad (2.21)$$

[Tong §1.3, ll. 1236–1256]. The bracket must vanish. Recognizing  $\partial_\alpha X \cdot \partial_\beta X = \gamma_{\alpha\beta}$ , this says

$$g_{\alpha\beta} = 2f(\tau, \sigma) \gamma_{\alpha\beta}, \quad f^{-1} = g^{\rho\sigma} \gamma_{\rho\sigma} \quad (2.22)$$

[Tong §1.3, ll. 1257–1268]: the worldsheet metric equals the induced metric up to a positive function  $f$  that the equation leaves undetermined. (Insert  $g = 2f\gamma$  into the definition of  $f$ :  $g^{\rho\sigma} \gamma_{\rho\sigma} = (2f)^{-1} \gamma^{\rho\sigma} \gamma_{\rho\sigma} = (2f)^{-1} \cdot 2 = f^{-1}$ , an identity;  $f > 0$  is needed for  $g$  to have the signature of  $\gamma$ .)

**Classical equivalence.** Substitute (2.22) into (2.18). In two dimensions  $\sqrt{-\det(2f\gamma)} = 2f\sqrt{-\det\gamma}$  and  $g^{\alpha\beta} = (2f)^{-1}\gamma^{\alpha\beta}$ , so

$$\sqrt{-g} g^{\alpha\beta} \gamma_{\alpha\beta} = 2f\sqrt{-\gamma} \cdot \frac{1}{2f} \gamma^{\alpha\beta} \gamma_{\alpha\beta} = 2\sqrt{-\gamma}, \quad S_P \rightarrow -\frac{2}{4\pi\alpha'} \int d^2\sigma \sqrt{-\gamma} = S_{NG} \quad (2.23)$$

[Tong §1.3, ll. 1269–1281]. The function  $f$  cancels, both here and in (2.19). This cancellation signals a new symmetry.



### 2.3.2 Symmetries: Poincaré, diffeomorphisms and Weyl

The Polyakov action has [Tong §1.3.1, ll. 1283–1327]:

1. *Poincaré invariance*, global on the worldsheet, as before.
2. *Diffeomorphism invariance*  $\sigma^\alpha \rightarrow \tilde{\sigma}^\alpha(\sigma)$ , under which  $X^\mu$  is a scalar,  $\tilde{X}^\mu(\tilde{\sigma}) = X^\mu(\sigma)$ , and  $g_{\alpha\beta}$  a tensor,  $\tilde{g}_{\alpha\beta}(\tilde{\sigma}) = \frac{\partial \sigma^\gamma}{\partial \tilde{\sigma}^\alpha} \frac{\partial \sigma^\delta}{\partial \tilde{\sigma}^\beta} g_{\gamma\delta}(\sigma)$ . Infinitesimally, for  $\tilde{\sigma}^\alpha = \sigma^\alpha - \eta^\alpha(\sigma)$ :  $\delta X^\mu = \eta^\alpha \partial_\alpha X^\mu$  and  $\delta g_{\alpha\beta} = \nabla_\alpha \eta_\beta + \nabla_\beta \eta_\alpha$ , with  $\nabla$  the Levi-Civita connection of  $g$ . Compare with the einbein transformations of Section 2.1.3.
3. *Weyl invariance*, which is new:

$$X^\mu(\sigma) \rightarrow X^\mu(\sigma), \quad g_{\alpha\beta}(\sigma) \rightarrow \Omega^2(\sigma) g_{\alpha\beta}(\sigma). \quad (2.24)$$

A Weyl transformation is not a change of coordinates. It is a position-dependent change of the unit of length on the worldsheet which preserves angles. The Polyakov string regards two metrics related by (2.24) as the same physical configuration.

To verify the invariance, and to see why it is special to strings, do the computation on a worldvolume of dimension  $n$ . Under (2.24),  $\det g \rightarrow \Omega^{2n} \det g$  and  $g^{\alpha\beta} \rightarrow \Omega^{-2} g^{\alpha\beta}$ , so

$$\sqrt{-g} g^{\alpha\beta} \rightarrow \Omega^{n-2} \sqrt{-g} g^{\alpha\beta}. \quad (2.25)$$

The combination that enters (2.18) is invariant if and only if  $n = 2$  [Tong §1.3.1, ll. 1328–1338]. A particle ( $n = 1$ ) or a membrane ( $n = 3$ ) has no such symmetry. Even for  $n = 2$  the symmetry is fragile: a potential  $\int d^2\sigma \sqrt{-g} V(X)$  or a worldsheet cosmological constant  $\mu \int d^2\sigma \sqrt{-g}$  scales with  $\Omega^2$  and breaks it [Tong §1.3.1, ll. 1339–1356]. In the quantum theory the requirement of Weyl invariance becomes much stronger; it fixes the spacetime dimension (Chapter 3, Chapter 5) and yields the equations of motion of the background fields (Chapter 6).

#### Common pitfall: three metrics, one symbol

Three different metrics are in play and sources often write them all as “ $g$ ” or “ $\eta$ ”. (i) The spacetime metric  $\eta_{\mu\nu}$ , fixed, with indices  $\mu, \nu$ . (ii) The induced metric  $\gamma_{\alpha\beta}$  of (2.10), a function of the  $X^\mu$ , not an independent field. (iii) The worldsheet metric  $g_{\alpha\beta}$  of the Polyakov action, an independent field. On solutions  $g = 2f\gamma$ , and only the conformal class of  $g$  is determined. When conformal gauge sets  $g_{\alpha\beta} = \eta_{\alpha\beta}$  below, that  $\eta$  is a two-by-two worldsheet object and has nothing to do with (i); in particular  $\gamma_{\alpha\beta}$  is then *not* equal to  $\eta_{\alpha\beta}$  but to  $\eta_{\alpha\beta}/(2f)$ .

## 2.4 Conformal gauge: wave equation plus constraints

### 2.4.1 Counting and fixing

A symmetric two-by-two metric has three independent components. Diffeomorphisms provide two arbitrary functions, a Weyl transformation a third. One therefore expects to be able to remove the metric completely [Tong §1.3.2, ll. 1357–1380]. In two steps: first use the two reparametrizations to bring the metric to the *conformally flat* form

$$g_{\alpha\beta} = e^{2\phi(\tau,\sigma)} \eta_{\alpha\beta}, \quad \eta_{\alpha\beta} = \text{diag}(-1, +1), \quad (2.26)$$

which is called *conformal gauge*; then use a Weyl transformation with  $\Omega = e^{-\phi}$  to set  $\phi = 0$ , so that  $g_{\alpha\beta} = \eta_{\alpha\beta}$ . That the first step is possible, at least locally, is an existence statement for a system of partial differential equations. A quick argument in Euclidean signature: under  $g'_{\alpha\beta} = e^{2\phi} g_{\alpha\beta}$  the Ricci scalar obeys  $\sqrt{g'} R' = \sqrt{g} (R - 2\nabla^2 \phi)$ ; choose  $\phi$  to solve  $2\nabla^2 \phi = R$ , so that  $R' = 0$ ; in two dimensions the Riemann tensor is  $R_{\alpha\beta\gamma\delta} = \frac{R}{2} (g_{\alpha\gamma} g_{\beta\delta} - g_{\alpha\delta} g_{\beta\gamma})$ , so  $R' = 0$  implies that  $g'$  is flat [Tong §1.3.2, ll. 1381–1411] [L].

### 2.4.2 Equations of motion and the stress tensor

With  $g_{\alpha\beta} = \eta_{\alpha\beta}$  the Polyakov action becomes the action of  $D$  free massless scalar fields in two dimensions,

$$S = -\frac{1}{4\pi\alpha'} \int d^2\sigma \partial_\alpha X \cdot \partial^\alpha X = \frac{1}{4\pi\alpha'} \int d\tau d\sigma (\dot{X}^2 - X'^2), \quad (2.27)$$

and (2.19) becomes the wave equation

$$\partial_\alpha \partial^\alpha X^\mu = 0 \quad \Longleftrightarrow \quad \ddot{X}^\mu - X''^\mu = 0 \quad (2.28)$$

[Tong §1.3.2, ll. 1412–1426]. (In fact any conformal gauge gives the same, since  $\sqrt{-g}g^{\alpha\beta}$  does not depend on  $\phi$ .) The non-linear equations (2.16) have become linear. This cannot be the whole story, and it is not: fixing  $g$  by a gauge choice does not remove its equation of motion. The equation (2.21) must still be imposed. Define the worldsheet *stress-energy tensor*

$$T_{\alpha\beta} = -\frac{2}{T} \frac{1}{\sqrt{-g}} \frac{\partial S_P}{\partial g^{\alpha\beta}} \xrightarrow{g=\eta} T_{\alpha\beta} = \partial_\alpha X \cdot \partial_\beta X - \frac{1}{2} \eta_{\alpha\beta} \eta^{\rho\sigma} \partial_\rho X \cdot \partial_\sigma X \quad (2.29)$$

[Tong §1.3.2, ll. 1427–1447]. (The stress tensor always carries indices; the bare letter  $T$  remains the tension.) The equation of motion of the metric is  $T_{\alpha\beta} = 0$ . Write out the components with  $\eta^{\rho\sigma} \partial_\rho X \cdot \partial_\sigma X = -\dot{X}^2 + X'^2$ :

$$\begin{aligned} T_{01} &= \dot{X} \cdot X', \\ T_{00} &= \dot{X}^2 + \frac{1}{2}(-\dot{X}^2 + X'^2) = \frac{1}{2}(\dot{X}^2 + X'^2), \\ T_{11} &= X'^2 - \frac{1}{2}(-\dot{X}^2 + X'^2) = \frac{1}{2}(\dot{X}^2 + X'^2). \end{aligned} \quad (2.30)$$

The trace  $\eta^{\alpha\beta} T_{\alpha\beta} = -T_{00} + T_{11}$  vanishes identically. That is a consequence of Weyl invariance: a Weyl variation is  $\delta g^{\alpha\beta} = -2\phi g^{\alpha\beta}$ , and invariance of  $S_P$  under it for arbitrary  $\phi(\sigma)$  forces  $g^{\alpha\beta} T_{\alpha\beta} = 0$ . Only two of the three equations  $T_{\alpha\beta} = 0$  are therefore independent:

$$\dot{X} \cdot X' = 0, \quad \dot{X}^2 + X'^2 = 0 \quad (2.31)$$

[Tong §1.3.2, ll. 1448–1460]. *The classical string is the free wave equation (2.28) subject to the two constraints (2.31).* All the non-linearity of the Nambu–Goto equations has been moved into the constraints.

These are the constraints (2.17) in a new guise. In conformal gauge the momentum density is  $\Pi_\mu^\tau = \partial \mathcal{L} / \partial \dot{X}^\mu = T \dot{X}_\mu$ , so  $\Pi^\tau \cdot X' = 0$  is  $\dot{X} \cdot X' = 0$  and  $(\Pi^\tau)^2 + T^2 X'^2 = 0$  is  $\dot{X}^2 + X'^2 = 0$ . Moreover the canonical Hamiltonian of (2.27) is  $H = \frac{T}{2} \int d\sigma (\dot{X}^2 + X'^2) = T \int d\sigma T_{00}$ . It vanishes on the second constraint, exactly as  $H = 0$  for the particle in (2.6).

### 2.4.3 What the constraints mean

Conformal gauge leaves some freedom, described in Section 2.4.4, which is enough to reach *static gauge*  $X^0 = t = R\tau$  for a constant  $R$ . With  $X^\mu = (t, \vec{x})$  we have  $\dot{X}^2 = -R^2 + \dot{\vec{x}}^2$  and  $X'^2 = \vec{x}'^2$ , and the system becomes

$$\ddot{\vec{x}} - \vec{x}'' = 0, \quad \dot{\vec{x}} \cdot \vec{x}' = 0, \quad \dot{\vec{x}}^2 + \vec{x}'^2 = R^2 \quad (2.32)$$

[Tong §1.3.2, ll. 1461–1489]. The first constraint says that the velocity is perpendicular to the string. A motion of the string along itself would only relabel its points, which is a gauge transformation; the physical oscillations are *transverse*. The second constraint fixes the meaning of  $R$ : at an instant when  $\dot{\vec{x}} = 0$ ,  $|\vec{x}'| = R$ , so the length of the string is  $\int_0^{2\pi} d\sigma |\vec{x}'| = 2\pi R$  [Tong §1.3.2, ll. 1490–1498]. When the string moves, the same equation trades stretching  $|\vec{x}'|$  against speed: since  $d\vec{x}/dt = \dot{\vec{x}}/R$ , it reads  $|d\vec{x}/dt|^2 + |\vec{x}'|^2/R^2 = 1$ . A point where the string is momentarily unstretched,  $\vec{x}' = 0$ , moves at the speed of light.

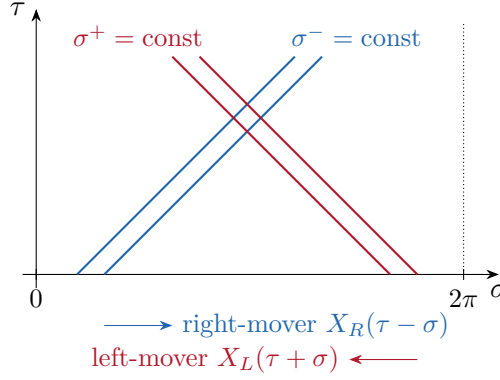


Figure 2.2: The worldsheet cylinder cut open along  $\sigma = 0 \sim 2\pi$ . A right-moving pulse is constant along lines  $\sigma^- = \tau - \sigma = \text{const}$  and travels towards increasing  $\sigma$  at unit coordinate speed; a left-moving pulse travels the other way. The two pass through each other unchanged.

#### 2.4.4 Residual symmetry

Introduce worldsheet *lightcone coordinates*

$$\sigma^\pm = \tau \pm \sigma, \quad \partial_\pm = \frac{1}{2}(\partial_\tau \pm \partial_\sigma), \quad ds^2 = -d\tau^2 + d\sigma^2 = -d\sigma^+ d\sigma^-. \quad (2.33)$$

A reparametrization of the special form  $\sigma^+ \rightarrow \tilde{\sigma}^+(\sigma^+)$ ,  $\sigma^- \rightarrow \tilde{\sigma}^-(\sigma^-)$  multiplies the flat metric by a function,  $d\tilde{\sigma}^+ d\tilde{\sigma}^- = (\partial_+ \tilde{\sigma}^+)(\partial_- \tilde{\sigma}^-) d\sigma^+ d\sigma^-$ , and this factor can be removed by a compensating Weyl transformation. The combination preserves  $g_{\alpha\beta} = \eta_{\alpha\beta}$  [Tong §2.2, ll. 1937–1957]. The counting of Section 2.4 was not wrong: these transformations depend on functions of one variable only, a negligible subset of the functions of two variables that we used up. They are the *conformal transformations* of the flat worldsheet and the reason why conformal field theory (Chapter 4) is the natural language of the quantum string. For the classical string they complete the count of degrees of freedom: the general solution of (2.28) contains  $2D$  functions of one variable; the two constraints remove two; the residual symmetry removes two more;  $2(D-2)$  remain, the left- and right-moving waves in the  $D-2$  transverse directions [Tong §2.2, ll. 1966–1985].

## 2.5 Mode expansion: left- and right-movers

### 2.5.1 General solution

In lightcone coordinates  $\partial_\alpha \partial^\alpha = -\partial_\tau^2 + \partial_\sigma^2 = -4\partial_+ \partial_-$ , so the wave equation is  $\partial_+ \partial_- X^\mu = 0$  and its general solution is

$$X^\mu(\tau, \sigma) = X_L^\mu(\sigma^+) + X_R^\mu(\sigma^-) \quad (2.34)$$

with arbitrary functions  $X_L$ ,  $X_R$  [Tong §1.4, ll. 1500–1512]. A function of  $\sigma^+ = \tau + \sigma$  keeps its shape and moves towards decreasing  $\sigma$  as  $\tau$  grows; it is called *left-moving*. A function of  $\sigma^-$  is *right-moving*. The two sets of waves pass through each other without interacting; see Figure 2.2. This decoupling is the single most useful fact about the closed string, and it survives quantization: the quantum closed string is, to a first approximation, two copies of one structure, tied together only by the zero modes and by one constraint (level matching, below).

Now impose periodicity (2.9). A periodic function is a Fourier series; but  $X_L$  and  $X_R$  need not be separately periodic, only their sum. The derivative  $\partial_+ X_L$  must be periodic (it equals

$\partial_+ X$ ), so  $X_L$  is a Fourier series plus a term linear in  $\sigma^+$ , and likewise  $X_R$ . The standard parametrization is

$$\begin{aligned} X_L^\mu(\sigma^+) &= \frac{1}{2}x^\mu + \frac{1}{2}\alpha' p^\mu \sigma^+ + i\sqrt{\frac{\alpha'}{2}} \sum_{n \neq 0} \frac{1}{n} \tilde{\alpha}_n^\mu e^{-in\sigma^+}, \\ X_R^\mu(\sigma^-) &= \frac{1}{2}x^\mu + \frac{1}{2}\alpha' p^\mu \sigma^- + i\sqrt{\frac{\alpha'}{2}} \sum_{n \neq 0} \frac{1}{n} \alpha_n^\mu e^{-in\sigma^-} \end{aligned} \quad (2.35)$$

[Tong §1.4, ll. 1513–1538]. The normalizations are chosen for later convenience. Three remarks make the formula concrete.

(i) *The linear terms.* Their sum is  $\frac{1}{2}\alpha' p^\mu (\sigma^+ + \sigma^-) = \alpha' p^\mu \tau$ , which is periodic in  $\sigma$  although neither piece is. Had we allowed different coefficients  $p_L^\mu \neq p_R^\mu$ , the sum would contain  $\frac{1}{2}\alpha' (p_L - p_R)^\mu \sigma$ , which is not periodic. In flat, non-compact spacetime periodicity therefore forces equal left and right zero modes. This is exactly the point that changes in Chapter 7; see Remark 2.2.

(ii) *Centre of mass.* Averaging over the string,  $\frac{1}{2\pi} \int_0^{2\pi} d\sigma X^\mu = x^\mu + \alpha' p^\mu \tau$ : the oscillators drop out and the centre of mass moves on a straight line. The constant  $p^\mu$  is the total spacetime momentum. Indeed, the Noether current of the translation symmetry  $X^\mu \rightarrow X^\mu + c^\mu$  of (2.27) is  $P_\mu^\alpha = -T \partial^\alpha X_\mu$ , conserved by (2.28), and its charge is

$$\int_0^{2\pi} d\sigma P_\mu^\tau = T \int_0^{2\pi} d\sigma \dot{X}_\mu = \frac{1}{2\pi\alpha'} \cdot 2\pi \cdot \alpha' p_\mu = p_\mu \quad (2.36)$$

[Tong §1.4, ll. 1545–1549]. This is why the factor  $\alpha'$  accompanies  $p^\mu$  in (2.35).

(iii) *Reality.*  $X^\mu$  is real, hence

$$\alpha_{-n}^\mu = (\alpha_n^\mu)^*, \quad \tilde{\alpha}_{-n}^\mu = (\tilde{\alpha}_n^\mu)^* \quad (2.37)$$

[Tong §1.4, ll. 1550–1563]. Each pair  $(\alpha_n^\mu, \alpha_{-n}^\mu)$ ,  $n > 0$ , is the complex amplitude of one right-moving harmonic of frequency  $n$ . In Chapter 3 these amplitudes become annihilation and creation operators.

### 2.5.2 The constraints in modes: $L_n$ and $\tilde{L}_n$

From (2.33),  $\partial_\pm X = \frac{1}{2}(\dot{X} \pm X')$ , so

$$(\partial_\pm X)^2 = \frac{1}{4}(\dot{X}^2 + X'^2 \pm 2\dot{X} \cdot X') = \frac{1}{2}(T_{00} \pm T_{01}), \quad (2.38)$$

and the two constraints (2.31) are equivalent to  $(\partial_+ X)^2 = (\partial_- X)^2 = 0$  [Tong §1.4.1, ll. 1565–1570]: one constraint on the left-movers alone, one on the right-movers alone. Differentiate (2.35); each oscillator term contributes  $i\sqrt{\alpha'/2} \cdot \frac{1}{n} \cdot (-in) = \sqrt{\alpha'/2}$  times  $\alpha_n e^{-in\sigma^-}$ :

$$\partial_- X^\mu = \frac{\alpha'}{2} p^\mu + \sqrt{\frac{\alpha'}{2}} \sum_{n \neq 0} \alpha_n^\mu e^{-in\sigma^-} = \sqrt{\frac{\alpha'}{2}} \sum_{n \in \mathbb{Z}} \alpha_n^\mu e^{-in\sigma^-}, \quad \alpha_0^\mu \equiv \sqrt{\frac{\alpha'}{2}} p^\mu. \quad (2.39)$$

Squaring and collecting the coefficient of  $e^{-in\sigma^-}$ ,

$$(\partial_- X)^2 = \frac{\alpha'}{2} \sum_{m,k} \alpha_m \cdot \alpha_k e^{-i(m+k)\sigma^-} = \alpha' \sum_{n \in \mathbb{Z}} L_n e^{-in\sigma^-}, \quad L_n = \frac{1}{2} \sum_{m \in \mathbb{Z}} \alpha_{n-m} \cdot \alpha_m, \quad (2.40)$$

and in the same way  $(\partial_+ X)^2 = \alpha' \sum_n \tilde{L}_n e^{-in\sigma^+}$  with  $\tilde{L}_n = \frac{1}{2} \sum_m \tilde{\alpha}_{n-m} \cdot \tilde{\alpha}_m$  and  $\tilde{\alpha}_0^\mu \equiv \sqrt{\alpha'/2} p^\mu = \alpha_0^\mu$  [Tong §1.4.1, ll. 1571–1650]. A Fourier series vanishes only if every coefficient

does. A classical closed string is therefore a set of amplitudes obeying the infinitely many quadratic conditions

$$L_n = \tilde{L}_n = 0 \quad \text{for all } n \in \mathbb{Z}. \quad (2.41)$$

The  $L_n$  are the Fourier modes of the stress tensor. In Chapter 4 they return as generators of the Virasoro algebra; here they are just numbers.

### 2.5.3 The mass formula and level matching

The conditions with  $n = 0$  contain  $\alpha_0^2 = \frac{\alpha'}{2} p^2$ , and  $p^2 = -M^2$  is minus the squared rest mass of the string regarded as a particle. Separate the  $m = 0$  term in  $L_0$  and use  $\alpha_{-m} \cdot \alpha_m = \alpha_m \cdot \alpha_{-m}$  (classical amplitudes commute) to combine  $m$  and  $-m$ :

$$L_0 = \frac{1}{2} \alpha_0^2 + \sum_{n>0} \alpha_{-n} \cdot \alpha_n = -\frac{\alpha'}{4} M^2 + \sum_{n>0} \alpha_{-n} \cdot \alpha_n. \quad (2.42)$$

Setting  $L_0 = 0$  and  $\tilde{L}_0 = 0$  gives two expressions for the same mass,

$$M^2 = \frac{4}{\alpha'} \sum_{n>0} \alpha_{-n} \cdot \alpha_n = \frac{4}{\alpha'} \sum_{n>0} \tilde{\alpha}_{-n} \cdot \tilde{\alpha}_n \quad (2.43)$$

[Tong §1.4.1, ll. 1651–1679] [L]. The equality of the right-moving and the left-moving sums is *level matching*. It holds because  $\alpha_0 = \tilde{\alpha}_0$ , which in turn came from periodicity in remark (i). The mass of a string is the energy stored in its oscillations: by (2.37),  $\alpha_{-n} \cdot \alpha_n = \sum_{\mu} \eta_{\mu\mu} |\alpha_n^{\mu}|^2$ , a sum of squared amplitudes (with a minus sign for  $\mu = 0$ , which the constraints  $L_{n \neq 0} = 0$  keep under control). Classically  $M^2$  is any non-negative real number. Quantization (Chapter 3) makes the sums integer-valued,  $\sum_{n>0} \alpha_{-n} \cdot \alpha_n \rightarrow N$ , and adds a constant from normal ordering, giving the book's convention  $M^2 = \frac{2}{\alpha'} (N + \tilde{N} - 2)$  with  $N = \tilde{N}$  in non-compact flat space.

*Remark 2.2* (What changes on a circle: a preview). Suppose one coordinate is periodic,  $X \sim X + 2\pi R$ . Then a closed string may close only up to this identification,  $X(\tau, \sigma + 2\pi) = X(\tau, \sigma) + 2\pi R w$  with  $w \in \mathbb{Z}$  the *winding number*. Remark (i) is then modified: allow  $p_L \neq p_R$  in (2.35). The zero-mode part is  $\frac{\alpha'}{2} (p_L + p_R) \tau + \frac{\alpha'}{2} (p_L - p_R) \sigma$ , and matching it to  $\alpha' p \tau + w R \sigma$  gives

$$p_L = p + \frac{wR}{\alpha'}, \quad p_R = p - \frac{wR}{\alpha'}, \quad \alpha_0 = \sqrt{\frac{\alpha'}{2}} p_R, \quad \tilde{\alpha}_0 = \sqrt{\frac{\alpha'}{2}} p_L. \quad (2.44)$$

Now  $\alpha_0 \neq \tilde{\alpha}_0$ , the two sums in (2.43) differ by  $\frac{\alpha'}{4} (p_L^2 - p_R^2) = p w R$ , and after quantization, where  $p = n/R$ , level matching becomes  $N - \tilde{N} = n w$ . This short computation (derived here, not quoted) is the seed of Chapter 7 and of T-duality in Chapter 8: the exchange  $p_R \rightarrow -p_R$ ,  $p_L \rightarrow p_L$  is the exchange  $p \leftrightarrow wR/\alpha'$ .

## 2.6 Worked example: a pulsating loop

Take  $D \geq 3$  and look for a circular string in the  $(x^1, x^2)$ -plane that starts at rest with radius  $R$ . In static gauge  $X^0 = R\tau$  try

$$\vec{x}(\tau, \sigma) = R \cos \tau (\cos \sigma, \sin \sigma). \quad (2.45)$$

**Equations of motion and constraints.**  $\ddot{\vec{x}} = -\vec{x} = \vec{x}''$ , so the wave equation holds. Further  $\dot{\vec{x}} = -R \sin \tau (\cos \sigma, \sin \sigma)$  is radial and  $\vec{x}' = R \cos \tau (-\sin \sigma, \cos \sigma)$  is tangential, so  $\dot{\vec{x}} \cdot \vec{x}' = 0$ ; and  $\dot{\vec{x}}^2 + \vec{x}'^2 = R^2 \sin^2 \tau + R^2 \cos^2 \tau = R^2$ . All of (2.32) is satisfied. Note that the constant in static gauge had to be the initial radius: the constraint ties the unit of worldsheet time to the size of the string.

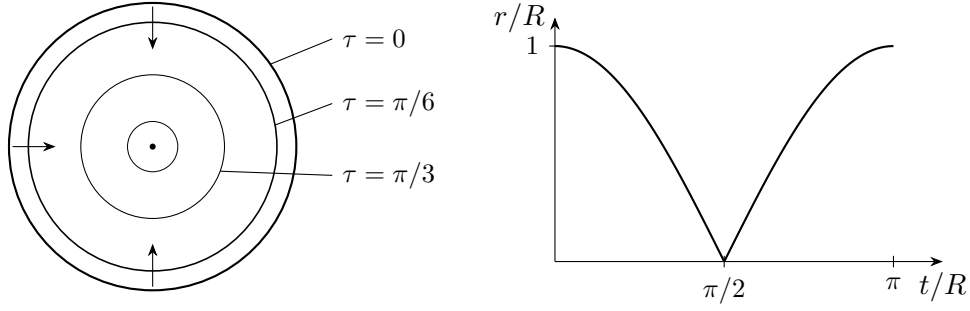


Figure 2.3: The pulsating loop (2.45). Left: snapshots of the string at  $\tau = 0, \pi/6, \pi/3$  and shortly before collapse. Right: radius against time; the slope at  $t = \pi R/2$  is  $\mp 1$ , the speed of light.

**Motion.** In terms of target-space time  $t = R\tau$  the radius is  $r(t) = R|\cos(t/R)|$  (Figure 2.3). The tension pulls the loop inward; it collapses to a point at  $t = \pi R/2$ . Its radial speed is  $|dr/dt| = |\sin(t/R)|$ , which reaches the speed of light at the moment of collapse, in agreement with the remark after (2.32), since there  $\vec{x}' = 0$ . The loop then re-expands to radius  $R$  at  $t = \pi R$ . (At the collapse point the Nambu–Goto description is singular, since  $\det \gamma = 0$ ; the conformal-gauge solution passes through it smoothly.)

**Energy and mass.** From (2.36),  $E = p^0 = T \int d\sigma \dot{X}^0 = 2\pi RT = R/\alpha'$ ; consistently, comparing  $X^0 = R\tau$  with the zero-mode term  $\alpha' p^0 \tau$  gives  $p^0 = R/\alpha'$ . This is the tension times the initial length  $2\pi R$ , as it must be for a string initially at rest, and it is conserved although the length is not: potential energy is converted into kinetic energy. The total spatial momentum vanishes, so  $M = E = R/\alpha'$ .

**The same mass from the oscillators.** Use  $2 \cos \tau \cos \sigma = \cos \sigma^+ + \cos \sigma^-$  and  $2 \cos \tau \sin \sigma = \sin \sigma^+ - \sin \sigma^-$  to split (2.45) into movers:

$$X_R^1 = \frac{R}{2} \cos \sigma^-, \quad X_R^2 = -\frac{R}{2} \sin \sigma^-, \quad X_L^1 = \frac{R}{2} \cos \sigma^+, \quad X_L^2 = +\frac{R}{2} \sin \sigma^+.$$

Only  $n = \pm 1$  appear. For  $n = \pm 1$  the oscillator part of (2.35) is  $i\sqrt{\alpha'/2}(\alpha_1 e^{-is} - \alpha_1^* e^{is}) = -2\sqrt{\alpha'/2} \operatorname{Im}(\alpha_1 e^{-is})$ . With  $c \equiv R/(2\sqrt{2\alpha'})$  one finds by inspection

$$\alpha_1^1 = -ic, \quad \alpha_1^2 = -c, \quad \tilde{\alpha}_1^1 = -ic, \quad \tilde{\alpha}_1^2 = +c. \quad (2.46)$$

(Check the first:  $\operatorname{Im}(-ic e^{-is}) = -c \cos s$ , so  $X_R^1 = 2c\sqrt{\alpha'/2} \cos s = \frac{R}{2} \cos s$ .) Then

$$\sum_{n>0} \alpha_{-n} \cdot \alpha_n = |\alpha_1^1|^2 + |\alpha_1^2|^2 = 2c^2 = \frac{R^2}{4\alpha'}, \quad M^2 = \frac{4}{\alpha'} \cdot \frac{R^2}{4\alpha'} = \frac{R^2}{\alpha'^2},$$

in agreement with  $M = R/\alpha'$ ; the left-movers give the same, so the level-matching condition holds. The remaining constraints hold too:  $L_{\pm 1} \propto \alpha_0 \cdot \alpha_{\pm 1} = 0$  because  $\alpha_0$  points in the time direction and  $\alpha_{\pm 1}$  in space, and  $L_2 = \frac{1}{2} \alpha_1 \cdot \alpha_1 = \frac{1}{2}((-ic)^2 + (-c)^2) = 0$ , likewise  $\tilde{L}_2 = \frac{1}{2}((-ic)^2 + c^2) = 0$ . The vanishing of  $L_2$  is the statement that the right-mover is circularly polarized; a linearly polarized mode, say  $\alpha_1^2 = 0$ , would violate it. (All formulas of this example were confirmed by a symbolic check.)

**Numbers.** Take  $R = 10\ell_s$ . Then  $M = 10/\ell_s$ , the loop collapses after  $t = \pi R/2 \approx 15.7\ell_s$ , and the oscillator sum is  $R^2/(4\alpha') = 25$ . In the quantum theory the corresponding level is  $N = \tilde{N} \approx 25$  (the normal-ordering constant of Chapter 3 shifts the precise value by one). For comparison with Chapter 7: a string wound once around a circle of the same radius,  $X = R\sigma$ , has  $M = 2\pi R T = R/\alpha'$  as well, but it cannot collapse, because its winding number is conserved.

## 2.7 What the machine has checked

### 2.7.1 The honest inventory

The short answer for this chapter is: nothing about strings. The Lean module assigned to this chapter, `StringTheoryFormalization/Foundations/MathlibCore.lean`, contains exactly one declaration.

#### In Lean: the foundation layer

```
import Mathlib.Algebra.Category.ModuleCat.Basic
import Mathlib.Analysis.SpecialFunctions.Complex.Circle
import Mathlib.Analysis.InnerProductSpace.Basic
import Mathlib.Topology.Algebra.Module.FiniteDimension
import Mathlib.CategoryTheory.Functor.Basic
import Mathlib.MeasureTheory.Function.L2Space

namespace StringTheory.Foundations

/-- Re-export summary: all Mathlib primitives used by the pipeline.
    This namespace is 'open'ed in every downstream block. -/
def mathlib_version : String := "Mathlib4 @ v4.33.1"

end StringTheory.Foundations
```

**Plain-language reading.** The file imports six Mathlib modules (categories of modules, the unit circle in  $\mathbb{C}$ , inner product spaces, finite-dimensional topological vector spaces, functors,  $L^2$  spaces) and defines a constant of type `String`. The definition `StringTheory.Foundations.mathlib_version` is a label, not a proposition; it has no proof because there is nothing to prove. The file’s header comment describes it as “verified” with no `sorry`; that is true and vacuous. The comment also says that the file is consumed by all downstream blocks; a text search of the repository finds its name in eight `.lean` files including itself. Comments are not checked by the kernel, and neither is the agreement of the string with the toolchain (the repository’s `lake-manifest.json` does pin Mathlib at the revision `v4.33.1`).

**What it contributes.** Importing a module makes its theorems available. One of them is relevant to this chapter: Mathlib’s `real_inner_mul_inner_self_le`, in `Mathlib/Analysis/InnerProductSpace/Basic.lean`, is the Cauchy–Schwarz inequality  $\langle u, v \rangle^2 \leq \langle u, u \rangle \langle v, v \rangle$ , the fact that makes the Euclidean area element (2.12) real. It is a theorem of Mathlib, not of this project.

### 2.7.2 Why an action functional is harder to formalize than its consequences

This is not an accident of scheduling. Compare what is needed to *state* the two kinds of result in this chapter.

*The end products are algebra.* The mass formula (2.43), the zero modes (2.44), and everything built on them in Chapter 7 and Chapter 8 are identities and inequalities among

finitely many integers, rationals or reals. Lean states them in one line, and tactics such as `ring`, `linarith` and `nlinarith` often prove them in one more. The Tier A content of this book lives there.

*The starting point is analysis on infinite-dimensional spaces.* To state “the Polyakov action is Weyl invariant and its critical points are the solutions of the wave equation with constraints” faithfully, one needs at least the following.

1. A type of field configurations: smooth maps  $X : \Sigma \rightarrow \mathbb{R}^{1,D-1}$  from a two-manifold, together with Lorentzian metrics on  $\Sigma$ . These are not Banach spaces, so the Fréchet derivative for normed spaces, which Mathlib has, does not directly apply.
2. The action as an integral over a non-compact  $\Sigma$ . It diverges for most configurations; “stationary” must be defined through compactly supported variations, and every integration by parts that we performed silently (from (2.21) or to (2.16)) needs a version of Stokes’ theorem with its hypotheses.
3. Lorentzian signature. A text search of the Mathlib revision pinned by this repository finds Riemannian (positive-definite) metrics on manifolds and lengths of paths, but no file mentioning pseudo-Riemannian metrics or Euler–Lagrange equations. The expression  $\sqrt{-\det g}$  needs the signature hypothesis merely to be well defined.
4. Gauge redundancy. The physical configuration space is a quotient by diffeomorphisms and Weyl transformations. “Every metric can be brought to conformal gauge” is an existence theorem for a partial differential equation (isothermal coordinates), true only locally.
5. A dictionary. Even with 1–4 in place, that the resulting mathematical object *is* a string is a modelling statement, never Tier A.

None of this is out of reach in principle, but each item is a project of its own, while the algebra downstream is accessible today. The pointwise algebraic cores of this chapter’s derivations can be isolated and proved: the determinant identity in (2.11), the cancellation (2.25), the einbein elimination, the tracelessness in (2.30). Exercises 2.8–2.9 ask for them. The authors checked that the statements given there compile against the repository’s Mathlib, but they are not declarations of the project and carry no tier.

| Claim                                                             | Tier | Basis                                                   |
|-------------------------------------------------------------------|------|---------------------------------------------------------|
| Particle: reparametrization invariance, $p^2 + m^2 = 0$ , $H = 0$ | [L]  | Tong §1.1; derived on the page                          |
| Nambu–Goto action = $-T \times$ area; $T$ is energy per length    | [L]  | Tong §1.2; derived on the page                          |
| Constraints (2.17)                                                | [L]  | derived; symbolic check                                 |
| Polyakov $\Leftrightarrow$ Nambu–Goto classically                 | [L]  | Tong §1.3; derived on the page                          |
| Weyl invariance only for $n = 2$                                  | [L]  | Tong §1.3.1; derived on the page                        |
| Conformal gauge exists locally                                    | [L]  | Tong §1.3.2 (sketch only)                               |
| Mode expansion, $L_n = \tilde{L}_n = 0$ , mass formula            | [L]  | Tong §1.4; derived on the page                          |
| Pulsating loop, $M = R/\alpha'$                                   | [L]  | derived; symbolic check                                 |
| $p_{L,R} = p \pm wR/\alpha'$ (preview)                            | [L]  | derived; see Chapter 7                                  |
| Cauchy–Schwarz (area is real)                                     | —    | Mathlib theorem, kernel-checked there                   |
| <code>mathlib_version</code>                                      | [A]  | a <code>String</code> constant; no mathematical content |

### Chapter summary

- The covariant particle action is  $-m \times$  proper time. Reparametrization invariance makes one of the  $D$  coordinates unphysical and produces the constraint  $p^2 + m^2 = 0$  and  $H = 0$ . An einbein removes the square root.



- The Nambu–Goto action is  $-T \times$  the area of the worldsheet, computed with the induced metric  $\gamma_{\alpha\beta} = \partial_\alpha X \cdot \partial_\beta X$ .  $T = 1/(2\pi\alpha')$  is the energy per unit length, and  $\ell_s = \sqrt{\alpha'}$  is the only scale.
- The Polyakov action introduces an independent worldsheet metric. Its equation of motion gives  $g = 2f\gamma$ ; substituting back returns Nambu–Goto. It has Poincaré, diffeomorphism and Weyl invariance; the last exists only for two-dimensional worldvolumes.
- In conformal gauge  $g_{\alpha\beta} = \eta_{\alpha\beta}$  the string obeys  $\ddot{X} = X''$  with constraints  $\dot{X} \cdot X' = 0$ ,  $\dot{X}^2 + X'^2 = 0$ , i.e.  $T_{\alpha\beta} = 0$ . Physical oscillations are transverse;  $2(D-2)$  functions of one variable remain.
- $X = X_L(\sigma^+) + X_R(\sigma^-)$ . Periodicity forces equal zero modes  $\alpha_0 = \tilde{\alpha}_0 = \sqrt{\alpha'/2}p$  in non-compact space. The constraints are  $L_n = \tilde{L}_n = 0$ ;  $n=0$  gives  $M^2 = \frac{4}{\alpha'} \sum_{n>0} \alpha_{-n} \cdot \alpha_n$  and level matching.
- On a circle the zero modes split,  $p_{L,R} = p \pm wR/\alpha'$ : the entry point of Chapter 7.
- Tier status: everything here is Tier L. The assigned Lean file contains a version string and imports. Formalizing an action principle needs global analysis that is not yet available in the library; the algebra that follows from it is what later chapters check.

## Exercises

**Exercise 2.1** (★). Expand (2.1) for  $|\dot{x}| \ll 1$  to second order and identify the two terms. Then show directly from (2.2) that  $\vec{p}/E$  is the velocity.

**Exercise 2.2** (★). Set  $m = 0$  in the einbein action (2.7). Derive the equations of motion of  $X^\mu$  and of  $e$  and show that the particle moves on a null straight line. Verify that the action is invariant under the transformations  $\delta X^\mu = \eta \dot{X}^\mu$ ,  $\delta e = \frac{d}{d\tau}(\eta e)$  up to a total derivative.

**Exercise 2.3** (★). Verify the two identities (2.17) from (2.15). Then show that in static gauge with  $\dot{x} = 0$  the density  $-\Pi_0^\tau$  integrates to  $T \times$  length, confirming (2.13) from the Hamiltonian side.

**Exercise 2.4** (★★). Consider a  $p$ -dimensional extended object with worldvolume dimension  $n = p+1$  and the action  $-\frac{T_p}{2} \int d^n \sigma \sqrt{-g} (g^{\alpha\beta} \partial_\alpha X \cdot \partial_\beta X - (n-2))$ . Show that the equation of motion of  $g_{\alpha\beta}$  is solved by  $g_{\alpha\beta} = \gamma_{\alpha\beta}$  for  $n \neq 2$ , with no free function, and that substituting back gives  $-T_p \int d^n \sigma \sqrt{-\gamma}$ . Explain the role of (2.25) in the difference between  $n = 2$  and  $n \neq 2$ .

**Exercise 2.5** (★★). Show, for an arbitrary worldsheet metric and without using the equations of motion, that  $g^{\alpha\beta} T_{\alpha\beta} = 0$  for the tensor in the bracket of (2.21). Which property of two dimensions did you use?

**Exercise 2.6** (★★ A rotating folded string and the Regge slope). In static gauge  $X^0 = R\tau$  consider  $\vec{x} = R \cos \sigma (\cos \tau, \sin \tau)$ . (a) Verify (2.32). (b) Describe the motion; which points move at the speed of light? (c) Compute  $E = T \int d\sigma \dot{X}^0$  and the angular momentum  $J = T \int d\sigma (x^1 \dot{x}^2 - x^2 \dot{x}^1)$  and show  $J = \frac{1}{2} \alpha' E^2$ . This linear relation between spin and mass squared is the origin of the name “Regge slope” for  $\alpha'$ . (d) Find the non-zero  $\alpha_n^\mu$ ,  $\tilde{\alpha}_n^\mu$  and check  $L_2 = \tilde{L}_2 = 0$ .

**Exercise 2.7** (★★). Repeat the derivation of Remark 2.2 in full: starting from (2.35) with independent  $p_L, p_R$  in one compact direction, compute  $L_0 - \tilde{L}_0$  and show that the classical level-matching condition reads  $\sum_{n>0} \alpha_{-n} \cdot \alpha_n - \sum_{n>0} \tilde{\alpha}_{-n} \cdot \tilde{\alpha}_n = p w R$ . Show that under  $R \rightarrow \alpha'/R$  together with  $p \leftrightarrow wR/\alpha'$  the pair  $(p_L, p_R)$  goes to  $(p_L, -p_R)$  and that both  $L_0$  and  $\tilde{L}_0$  are unchanged.

**Exercise 2.8** (★★ Lean). Prove in Lean 4 with Mathlib the pointwise core of (2.25) for  $n = 2$ : a Weyl rescaling multiplies the determinant of a symmetric two-by-two matrix by  $\Omega^4$ .

```
example (Ω a b c : ℝ) :
  (Ω^2 * a) * (Ω^2 * c) - (Ω^2 * b)^2 = Ω^4 * (a * c - b^2) := by
  sorry
```

Then state and prove the einbein elimination of Section 2.1.3: for  $m > 0$  and  $s > 0$  (standing for  $\sqrt{-\dot{X}^2}$ ),  $\frac{1}{2}((s/m)^{-1}(-s^2) - (s/m)m^2) = -ms$ . Hint: `field_simp` then `ring`.

**Exercise 2.9** (★★★ Lean). Let  $F$  be a real inner product space. Prove that the Gram determinant of two vectors is non-negative, so that the Euclidean area element (2.12) is real. Use the Mathlib theorem `real_inner_mul_inner_self_le`.

```
example {F : Type*} [NormedAddCommGroup F] [InnerProductSpace ℝ F]
  (u v : F) :
  0 ≤ inner ℝ u u * inner ℝ v v - inner ℝ u v * inner ℝ u v := by
  sorry
```

Then explain in a paragraph why the analogous statement in Minkowski signature,  $(\dot{X} \cdot X')^2 - \dot{X}^2 X'^2 \geq 0$ , is *not* a theorem about arbitrary pairs of vectors, and formulate a hypothesis on  $\dot{X}$  and  $X'$  under which it holds.

**Exercise 2.10** (★★★). Show that a reparametrization  $\sigma^\pm \rightarrow \tilde{\sigma}^\pm(\sigma^\pm)$  maps solutions of (2.28) and (2.31) to solutions. Use it to show that any solution can be brought to static gauge  $X^0 = R\tau$ . (Hint:  $X^0$  itself solves the wave equation, so  $X^0 = X_L^0(\sigma^+) + X_R^0(\sigma^-)$ ; choose the new coordinates proportional to  $X_L^0$  and  $X_R^0$ . What must be assumed about these functions?)

## Further reading

- The chapter follows [Tong §1, ll. 681–1680] throughout: particle §1.1 (ll. 688–938), Nambu–Goto §1.2 (ll. 945–1208), Polyakov §1.3 (ll. 1209–1498), mode expansions §1.4 (ll. 1500–1680).
- Residual conformal symmetry and the counting of  $2(D - 2)$  transverse functions: [Tong §2.2, ll. 1932–1990]; lightcone gauge, which solves the constraints explicitly, is taken up in Chapter 3.
- Strings of finite width (flux tubes, cosmic strings) and corrections to the Nambu–Goto action: [Tong §1.2, ll. 1124–1142].
- For the state of the formal library: Chapter 34 and Chapter 35; for the open problem of formalizing variational principles, Chapter 38.

## Chapter 3

# Quantization, the Spectrum and the Critical Dimension

Chapter 2 described a classical relativistic string:  $D$  free fields  $X^\mu(\sigma, \tau)$  on the worldsheet, subject to constraints. This chapter asks what that system becomes in quantum mechanics. Which particles does one quantized string describe? What are their masses and spins? And why does the answer depend on the number  $D$  of spacetime dimensions, so that a consistent bosonic string exists only for  $D = 26$ ? A student should care for two reasons. First, the spectrum derived here contains a massless spin-two particle, and this is the reason string theory is a theory of gravity. It also contains the two other massless fields,  $B_{\mu\nu}$  and the dilaton  $\Phi$ , from which the whole of T-duality and double field theory in Parts II and III is built. Second, the mass formula obtained here,  $M^2 = \frac{4}{\alpha'}(N - 1)$ , is the  $R \rightarrow \infty$  limit of the formula with momentum and winding in Chapter 7; every later statement about dual scales is a statement about how this formula changes on a circle. The chapter closes by stating exactly which part of the “ $D = 26$ ” story has been checked in Lean 4: the arithmetic of central charges, and nothing more.

Conventions are those of Chapter 2:  $\hbar = c = 1$ , mostly-plus Minkowski metric  $\eta_{\mu\nu}$ ,  $\alpha'$  of dimension  $\text{length}^2$ , tension  $T = 1/(2\pi\alpha')$ , closed strings with  $\sigma \sim \sigma + 2\pi$ . A dot between two vectors is the Minkowski product,  $A \cdot B = \eta_{\mu\nu} A^\mu B^\nu$ .

### 3.1 Canonical quantization: from fields to oscillators

#### 3.1.1 Two roads

The string action has a gauge symmetry (reparametrizations and Weyl rescalings of the worldsheet), and a gauge theory can be quantized in two ways [Tong §2, ll. 1683–1707].

- *Covariant quantization*: quantize all  $D$  fields  $X^\mu$ , including the time  $X^0$ , and afterwards impose the constraints as conditions on states. This is the analogue of the Gupta–Bleuler treatment of electrodynamics in Lorentz gauge. Lorentz invariance is manifest; positivity of the Hilbert space is not.
- *Lightcone quantization*: solve the constraints classically, find the independent degrees of freedom, and quantize only those. This is the analogue of Coulomb gauge. Positivity is manifest; Lorentz invariance is not.

If both are carried out correctly they agree. We follow Tong: a short look at the first road, to see where the difficulties are, and then the second road to the end.

### 3.1.2 Commutation relations for the modes

In conformal gauge the action is that of  $D$  free scalar fields,  $S = -\frac{1}{4\pi\alpha'} \int d^2\sigma \partial_\alpha X \cdot \partial^\alpha X$ , and the constraints are

$$\dot{X} \cdot X' = 0, \quad \dot{X}^2 + X'^2 = 0, \quad (3.1)$$

where a dot is  $\partial_\tau$  and a prime is  $\partial_\sigma$  [Tong §2.1, ll. 1708–1713]. With worldsheet lightcone coordinates  $\sigma^\pm = \tau \pm \sigma$  the general periodic solution of the wave equation  $\partial_+ \partial_- X^\mu = 0$  is  $X^\mu = X_L^\mu(\sigma^+) + X_R^\mu(\sigma^-)$  with

$$\begin{aligned} X_L^\mu(\sigma^+) &= \frac{1}{2}x^\mu + \frac{1}{2}\alpha' p^\mu \sigma^+ + i\sqrt{\frac{\alpha'}{2}} \sum_{n \neq 0} \frac{1}{n} \tilde{\alpha}_n^\mu e^{-in\sigma^+}, \\ X_R^\mu(\sigma^-) &= \frac{1}{2}x^\mu + \frac{1}{2}\alpha' p^\mu \sigma^- + i\sqrt{\frac{\alpha'}{2}} \sum_{n \neq 0} \frac{1}{n} \alpha_n^\mu e^{-in\sigma^-}, \end{aligned} \quad (3.2)$$

[Tong §1.4, ll. 1502–1563]. Here  $x^\mu$  and  $p^\mu$  are the position and momentum of the centre of mass,  $\alpha_n^\mu$  are the amplitudes of right-moving waves,  $\tilde{\alpha}_n^\mu$  those of left-moving waves, and reality of  $X^\mu$  requires  $\alpha_{-n}^\mu = (\alpha_n^\mu)^*$ . One also defines  $\alpha_0^\mu = \tilde{\alpha}_0^\mu = \sqrt{\alpha'/2} p^\mu$ .

The momentum conjugate to  $X^\mu$  is  $\Pi_\mu = \partial\mathcal{L}/\partial\dot{X}^\mu = \dot{X}_\mu/(2\pi\alpha')$ . Quantization replaces fields by operators with the equal-time commutators

$$[X^\mu(\sigma, \tau), \Pi_\nu(\sigma', \tau)] = i\delta(\sigma - \sigma') \delta^\mu_\nu, \quad [X^\mu, X^\nu] = [\Pi_\mu, \Pi_\nu] = 0, \quad (3.3)$$

and these are equivalent to

$$[x^\mu, p_\nu] = i\delta^\mu_\nu, \quad [\alpha_n^\mu, \alpha_m^\nu] = [\tilde{\alpha}_n^\mu, \tilde{\alpha}_m^\nu] = n\eta^{\mu\nu} \delta_{n+m,0}, \quad (3.4)$$

with all other commutators zero [Tong §2.1, ll. 1714–1742].

It is worth checking once that Equation (3.4) reproduces Equation (3.3), because the check fixes every normalization in Equation (3.2). At  $\tau = 0$  one has  $e^{-in\sigma^-} = e^{in\sigma}$  and  $e^{-in\sigma^+} = e^{-in\sigma}$ , and

$$\dot{X}^\nu(\sigma') = \alpha' p^\nu + \sqrt{\frac{\alpha'}{2}} \sum_{m \neq 0} (\alpha_m^\nu e^{im\sigma'} + \tilde{\alpha}_m^\nu e^{-im\sigma'}).$$

The zero modes give  $[x^\mu, \alpha' p^\nu] = i\alpha' \eta^{\mu\nu}$ . In the oscillator part the commutator  $[\alpha_n^\mu, \alpha_m^\nu] = n\eta^{\mu\nu} \delta_{n+m,0}$  cancels the factor  $1/n$  and sets  $m = -n$ :

$$i\frac{\alpha'}{2} \sum_{n \neq 0} \frac{1}{n} n \eta^{\mu\nu} (e^{in(\sigma-\sigma')} + e^{-in(\sigma-\sigma')}) = i\alpha' \eta^{\mu\nu} \sum_{n \neq 0} e^{in(\sigma-\sigma')}.$$

Add the two pieces and use the Fourier series of the periodic delta function,  $\sum_{n \in \mathbb{Z}} e^{in(\sigma-\sigma')} = 2\pi\delta(\sigma - \sigma')$ . The result is

$$[X^\mu(\sigma), \dot{X}^\nu(\sigma')] = 2\pi i \alpha' \eta^{\mu\nu} \delta(\sigma - \sigma'),$$

and dividing by  $2\pi\alpha'$  gives Equation (3.3). The factor  $n$  on the right of Equation (3.4) is therefore forced by the  $1/n$  in the mode expansion.

### 3.1.3 Harmonic oscillators and the Fock space

For a single direction  $\mu$  and  $n > 0$  define

$$a_n = \frac{\alpha_n}{\sqrt{n}}, \quad a_n^\dagger = \frac{\alpha_{-n}}{\sqrt{n}} \quad \implies \quad [a_n, a_m^\dagger] = \delta_{nm} \quad (3.5)$$

[Tong §2.1, ll. 1743–1757]. Each spatial field  $X^i$  is therefore two infinite towers of harmonic oscillators:  $\alpha_n$  with  $n > 0$  annihilates a right-moving quantum of worldsheet frequency  $n$  and  $\alpha_{-n}$  creates one; the  $\tilde{\alpha}_n$  do the same for left-movers.

**Physical picture: a violin string with infinitely many notes**

A classical closed string vibrates in normal modes labelled by an integer  $n$  (the number of wavelengths around the string), a direction  $\mu$  and a sense of travel (left or right). Quantization turns each normal mode into a harmonic oscillator whose excitations come in integer quanta. Seen from far away, a small vibrating loop looks like a point particle, and the vibrational energy appears as the rest mass of that particle. One string, excited in different ways, is therefore an infinite list of particle species [Tong §2.1, ll. 1790–1800].

The vacuum of the oscillators is defined by

$$\alpha_n^\mu |0; p\rangle = \tilde{\alpha}_n^\mu |0; p\rangle = 0 \quad (n > 0), \quad \hat{p}^\mu |0; p\rangle = p^\mu |0; p\rangle, \quad (3.6)$$

where the label  $p$  is needed because the zero modes  $x^\mu, p^\mu$  act as in the quantum mechanics of a free particle. This state is the ground state of *one string* with centre-of-mass momentum  $p$ . It is not the vacuum of spacetime, which contains no string at all [Tong §2.1, ll. 1758–1782]. The Fock space is spanned by the states obtained by acting on  $|0; p\rangle$  with any finite number of creation operators  $\alpha_{-n}^\mu, \tilde{\alpha}_{-n}^\mu$ ,  $n > 0$ .

## 3.2 Negative norms and the Virasoro constraints

### 3.2.1 The problem with the time direction

The Fock space just built is not a Hilbert space. Because  $\eta^{00} = -1$ , the commutator in Equation (3.4) for the time direction reads  $[\alpha_n^0, \alpha_{-n}^0] = -n$ , so

$$\|\alpha_{-1}^0 |0; p\rangle\|^2 = \langle 0; p' | \alpha_1^0 \alpha_{-1}^0 |0; p\rangle = \langle 0; p' | [\alpha_1^0, \alpha_{-1}^0] |0; p\rangle = -\delta^D(p - p'). \quad (3.7)$$

States with an odd number of timelike oscillators have negative norm; such states are called *ghosts* [Tong §2.1.1, ll. 1802–1822]. A negative norm would mean a negative probability, so these states must not appear in any physical process. In electrodynamics the same problem occurs for the timelike photon, and the gauge-fixing condition removes it. In string theory the constraints of Equation (3.1) must do this job.

**Common pitfall: two meanings of “ghost”**

In this chapter a ghost is a *state of negative norm*. In Chapter 5 the word also denotes the anticommuting Faddeev–Popov fields  $b, c$  introduced when the path integral is gauge-fixed. The two are related, since the Faddeev–Popov fields are what cancels the unphysical polarizations, but they are different objects. The Lean file discussed in Section 3.8 is about the central charge of the second kind.

### 3.2.2 The constraints in modes

In lightcone worldsheet coordinates the constraints of Equation (3.1) read  $(\partial_+ X)^2 = (\partial_- X)^2 = 0$ . From Equation (3.2),  $\partial_- X^\mu = \sqrt{\alpha'/2} \sum_{n \in \mathbb{Z}} \alpha_n^\mu e^{-in\sigma^-}$ , where the  $n = 0$  term is the definition of  $\alpha_0^\mu$ . Squaring,

$$(\partial_- X)^2 = \alpha' \sum_{n \in \mathbb{Z}} L_n e^{-in\sigma^-}, \quad L_n = \frac{1}{2} \sum_{m \in \mathbb{Z}} \alpha_{n-m} \cdot \alpha_m, \quad (3.8)$$

and similarly  $\tilde{L}_n$  from the left-movers [Tong §1.4.1, ll. 1565–1630]. Classically the constraints are  $L_n = \tilde{L}_n = 0$  for all  $n$ . The  $L_n$  are the *Virasoro generators*; their algebra, and the central charge that appears in it, are the subject of Chapter 4.

The  $n = 0$  constraint is special because it contains  $\frac{1}{2}\alpha_0^2 = \frac{\alpha'}{4}p^2 = -\frac{\alpha'}{4}M^2$ , where  $M$  is the rest mass of the string seen as a particle,  $p_\mu p^\mu = -M^2$ . Classically  $L_0 = \frac{\alpha'}{4}p^2 + \sum_{n>0} \alpha_{-n} \cdot \alpha_n = 0$  gives

$$M^2 = \frac{4}{\alpha'} \sum_{n>0} \alpha_{-n} \cdot \alpha_n = \frac{4}{\alpha'} \sum_{n>0} \tilde{\alpha}_{-n} \cdot \tilde{\alpha}_n \quad (3.9)$$

[Tong §1.4.1, ll. 1650–1680]. The mass is the vibrational energy. The second equality, from  $\tilde{L}_0 = 0$ , holds because  $\alpha_0 = \tilde{\alpha}_0$ : left- and right-movers share the same centre-of-mass momentum. It is called *level matching*.

### 3.2.3 Imposing the constraints on states

In the quantum theory one cannot demand  $L_n|\text{phys}\rangle = 0$  for all  $n$ ; as in Gupta–Bleuler quantization one demands only that matrix elements between physical states vanish. Since  $L_n^\dagger = L_{-n}$ , it suffices that

$$L_n|\text{phys}\rangle = \tilde{L}_n|\text{phys}\rangle = 0 \quad (n > 0) \quad (3.10)$$

[Tong §2.1.2, ll. 1823–1846]. For  $n \neq 0$  the two factors in each term of Equation (3.8) commute, so  $L_n$  is unambiguous. For  $n = 0$  they do not:  $\alpha_{-m}$  and  $\alpha_m$  fail to commute, and different orderings differ by a constant. One fixes an ordering, the normal ordering with annihilation operators on the right,

$$L_0 = \frac{1}{2}\alpha_0^2 + \sum_{m=1}^{\infty} \alpha_{-m} \cdot \alpha_m, \quad \tilde{L}_0 = \frac{1}{2}\tilde{\alpha}_0^2 + \sum_{m=1}^{\infty} \tilde{\alpha}_{-m} \cdot \tilde{\alpha}_m, \quad (3.11)$$

and parametrizes the ignorance by a constant  $a$ :

$$(L_0 - a)|\text{phys}\rangle = (\tilde{L}_0 - a)|\text{phys}\rangle = 0. \quad (3.12)$$

Inserting  $\frac{1}{2}\alpha_0^2 = -\frac{\alpha'}{4}M^2$  into Equation (3.11), these two conditions read

$$M^2 = \frac{4}{\alpha'} \left( \sum_{m \geq 1} \alpha_{-m} \cdot \alpha_m - a \right) = \frac{4}{\alpha'} \left( \sum_{m \geq 1} \tilde{\alpha}_{-m} \cdot \tilde{\alpha}_m - a \right) \quad (3.13)$$

[Tong §2.1.2, ll. 1847–1912]. The constant  $a$  is not a harmless convention: it shifts the mass of every state. Determining  $a$ , together with  $D$ , is the main task of this chapter.

**Example 3.1** (The first excited level, covariantly). This example shows why  $a = 1$  is special before we derive it. For brevity keep only the right-movers (the left-movers behave identically) and consider  $|\zeta\rangle = \zeta_\mu \alpha_{-1}^\mu |0; p\rangle$  with a polarization vector  $\zeta$ . The calculation is a short exercise with Equation (3.4); the argument is standard but is not spelled out in this book’s source library, where Tong refers to Green, Schwarz and Witten for it [Tong §2.1.2, ll. 1920–1931].

*Mass shell.*  $\sum_{m \geq 1} \alpha_{-m} \cdot \alpha_m |\zeta\rangle = |\zeta\rangle$ , so Equation (3.13) gives  $M^2 = -p^2 = 4(1 - a)/\alpha'$ .

*Constraint.* In  $L_1 = \alpha_0 \cdot \alpha_1 + \alpha_{-1} \cdot \alpha_2 + \dots$  only the first term acts non-trivially:  $L_1|\zeta\rangle = \alpha_0^\mu \zeta^\nu [\alpha_{1\mu}, \alpha_{-1\nu}] |0; p\rangle = \sqrt{\alpha'/2} (p \cdot \zeta) |0; p\rangle$ . So Equation (3.10) requires  $p \cdot \zeta = 0$ : the polarization is transverse to the momentum.  $L_n$  with  $n \geq 2$  annihilates  $|\zeta\rangle$  automatically.

*Norm.*  $\langle \zeta | \zeta \rangle \propto \zeta^* \cdot \zeta$ . Three cases:

- $a < 1$ :  $M^2 > 0$ . In the rest frame  $p = (M, 0, \dots, 0)$ , so  $p \cdot \zeta = 0$  removes  $\zeta^0$ . The  $D - 1$  remaining states have positive norm: a massive vector.
- $a > 1$ :  $M^2 < 0$ ,  $p$  is spacelike, say  $p = (0, \dots, 0, |M|)$ . Then  $\zeta = (1, 0, \dots, 0)$  is allowed and has  $\zeta \cdot \zeta = -1$ : a physical ghost. The theory is inconsistent.

- $a = 1$ :  $M^2 = 0$ . Take  $p = (k, 0, \dots, 0, k)$ . The condition  $p \cdot \zeta = 0$  allows the  $D-2$  transverse  $\zeta$ , of positive norm, and also  $\zeta \propto p$ , of norm  $p \cdot p = 0$ . This null state is orthogonal to all physical states and decouples. It expresses the gauge invariance  $\zeta_\mu \rightarrow \zeta_\mu + \lambda p_\mu$  of a massless vector.

The boundary value  $a = 1$  is where the theory is as rich as possible without ghosts: a massless particle with  $D-2$  polarizations and a gauge symmetry. For the closed string, with both  $\alpha_{-1}$  and  $\tilde{\alpha}_{-1}$  acting, the same computation gives a polarization tensor  $\zeta_{\mu\nu}$  with  $p^\mu \zeta_{\mu\nu} = \zeta_{\mu\nu} p^\nu = 0$  and the invariance  $\zeta_{\mu\nu} \rightarrow \zeta_{\mu\nu} + p_\mu \xi_\nu + \xi'_\mu p_\nu$ , which in its symmetric part is the linearized diffeomorphism  $h_{\mu\nu} \rightarrow h_{\mu\nu} + \partial_\mu \xi_\nu + \partial_\nu \xi_\mu$  [Tong §2.3.2, ll. 2725–2770].

The full covariant analysis shows that the absence of ghosts at all levels requires  $a = 1$  and  $D = 26$  [L] [Tong §2.1.2, ll. 1920–1931]. We shall obtain the same values on the second road.

### 3.3 Lightcone gauge: solving the constraints

#### 3.3.1 The residual gauge symmetry

Conformal gauge  $g_{\alpha\beta} = \eta_{\alpha\beta}$  does not use up all the gauge freedom. The flat worldsheet metric is  $ds^2 = -d\sigma^+ d\sigma^-$ , so a reparametrization

$$\sigma^+ \rightarrow \tilde{\sigma}^+(\sigma^+), \quad \sigma^- \rightarrow \tilde{\sigma}^-(\sigma^-) \quad (3.14)$$

only multiplies the metric by a function, which a Weyl rescaling removes [Tong §2.2, ll. 1932–1962]. This has a consequence for the counting of degrees of freedom. The solutions  $X_L^\mu(\sigma^+) + X_R^\mu(\sigma^-)$  are  $2D$  functions of one variable. The constraints  $(\partial_\pm X)^2 = 0$  remove two, and the freedom of Equation (3.14) shows that two more are pure gauge. What remains is

$$2D - 2 - 2 = 2(D - 2) \text{ functions,} \quad (3.15)$$

the left- and right-moving *transverse* oscillations of the string [Tong §2.2, ll. 1963–1985]. A relativistic string has no longitudinal vibrations: a motion of the string along itself is a relabelling of  $\sigma$  and not a physical motion.

#### 3.3.2 The gauge choice

Introduce spacetime lightcone coordinates

$$X^\pm = \frac{1}{\sqrt{2}}(X^0 \pm X^{D-1}), \quad ds^2 = -2dX^+ dX^- + \sum_{i=1}^{D-2} dX^i dX^i, \quad (3.16)$$

so that  $A \cdot B = -A^+ B^- - A^- B^+ + A^i B^i$  and  $A_\pm = -A^\mp$  [Tong §2.2.1, ll. 1992–2035]. The choice singles out one spatial direction, and Lorentz invariance is no longer manifest; we return to this in Section 3.6. The field  $X^+$  solves the wave equation, so  $X^+ = X_L^+(\sigma^+) + X_R^+(\sigma^-)$ , and the residual freedom of Equation (3.14) is exactly a choice of what the functions  $X_L^+$  and  $X_R^+$  are. One uses it to remove all oscillators from  $X^+$ :

$$X^+ = x^+ + \alpha' p^+ \tau \quad (\text{lightcone gauge}) \quad (3.17)$$

[Tong §2.2.1, ll. 2036–2066]. The zero modes  $x^+$  and  $p^+$  survive, because a reparametrization must respect the periodicity in  $\sigma$ . In this gauge the worldsheet time  $\tau$  is, up to a shift and a scale, the spacetime coordinate  $X^+$ : every point of the string carries the same “lightcone clock”. The gauge is good as long as  $p^+ \neq 0$ .

### 3.3.3 Solving for $X^-$

The benefit of the gauge choice in Equation (3.17) is that the constraints become *linear* in  $X^-$ . Written with Equation (3.16), the constraint  $(\partial_+ X)^2 = 0$  is  $2\partial_+ X^- \partial_+ X^+ = \sum_i \partial_+ X^i \partial_+ X^i$ , and Equation (3.17) gives  $\partial_+ X^+ = \frac{1}{2}\alpha' p^+$ . Hence

$$\partial_+ X^- = \frac{1}{\alpha' p^+} \sum_{i=1}^{D-2} \partial_+ X^i \partial_+ X^i, \quad \partial_- X^- = \frac{1}{\alpha' p^+} \sum_{i=1}^{D-2} \partial_- X^i \partial_- X^i \quad (3.18)$$

[Tong §2.2.1, ll. 2068–2113]. The field  $X^-$  is determined by the transverse fields up to one integration constant  $x^-$ . In modes, insert  $\partial_- X^- = \sqrt{\alpha'/2} \sum_n \alpha_n^- e^{-in\sigma^-}$  and  $(\partial_- X^i)^2 = \frac{\alpha'}{2} \sum_n \sum_m \alpha_{n-m}^i \alpha_m^i e^{-in\sigma^-}$  and compare coefficients:

$$\alpha_n^- = \frac{1}{\sqrt{2\alpha'} p^+} \sum_{m \in \mathbb{Z}} \sum_{i=1}^{D-2} \alpha_{n-m}^i \alpha_m^i \quad (3.19)$$

[Tong §2.2.1, ll. 2114–2141]. The case  $n = 0$ , with  $\alpha_0^- = \sqrt{\alpha'/2} p^-$  and  $\alpha_0^i \alpha_0^i = \frac{\alpha'}{2} p^i p^i$ , is an equation for  $p^-$ :

$$\frac{\alpha' p^-}{2} = \frac{1}{2p^+} \sum_{i=1}^{D-2} \left( \frac{\alpha'}{2} p^i p^i + \sum_{n \neq 0} \alpha_{-n}^i \alpha_n^i \right), \quad (3.20)$$

and the left-movers give the same equation with  $\tilde{\alpha}$  [Tong §2.2.1, ll. 2142–2200]. Now compute the mass. Since  $p^2 = -2p^+ p^- + p^i p^i$ ,

$$M^2 = 2p^+ p^- - p^i p^i = \frac{2}{\alpha'} \sum_i \sum_{n \neq 0} \alpha_{-n}^i \alpha_n^i = \frac{4}{\alpha'} \sum_{i=1}^{D-2} \sum_{n>0} \alpha_{-n}^i \alpha_n^i = \frac{4}{\alpha'} \sum_{i=1}^{D-2} \sum_{n>0} \tilde{\alpha}_{-n}^i \tilde{\alpha}_n^i, \quad (3.21)$$

where the third step uses that classical amplitudes commute. This is Equation (3.9) again, with the essential difference that only the  $D - 2$  transverse directions appear [Tong §2.2.1, ll. 2201–2238]. The independent variables are  $x^i, p^i, x^-, p^+$  and the transverse oscillators  $\alpha_n^i, \tilde{\alpha}_n^i$ . Note that  $p^-$  generates translations of  $x^+$ , that is, of  $\tau$ : Equation (3.20) is the lightcone Hamiltonian of the string.

## 3.4 Lightcone quantization and the normal-ordering constant

### 3.4.1 A positive Hilbert space

Quantize the independent variables:

$$[x^i, p^j] = i\delta^{ij}, \quad [x^-, p^+] = -i, \quad [\alpha_n^i, \alpha_m^j] = [\tilde{\alpha}_n^i, \tilde{\alpha}_m^j] = n\delta^{ij}\delta_{n+m,0} \quad (3.22)$$

[Tong §2.2.2, ll. 2239–2256]. (That the naive commutators survive the elimination of  $X^\pm$  requires an argument, because  $\alpha_n^-$  is a function of the  $\alpha_n^i$ . It holds because  $X^+$  in Equation (3.17) has no oscillators, so the symplectic form receives no contribution from  $\alpha_n^-$  [Tong §2.2.2, ll. 2294–2313].) The Fock space is built on  $|0; p\rangle$  with the transverse creation operators only. Since  $\delta^{ij}$  is positive definite, every state has positive norm. There are no ghosts, by construction [Tong §2.2.2, ll. 2276–2290].

The constraint of Equation (3.20) must be imposed on states, and it has the ordering ambiguity met before. With the *level operators*

$$N = \sum_{i=1}^{D-2} \sum_{n>0} \alpha_{-n}^i \alpha_n^i, \quad \tilde{N} = \sum_{i=1}^{D-2} \sum_{n>0} \tilde{\alpha}_{-n}^i \tilde{\alpha}_n^i, \quad (3.23)$$



the quantum mass formula is

$$M^2 = \frac{4}{\alpha'}(N - a) = \frac{4}{\alpha'}(\tilde{N} - a) \quad (3.24)$$

[Tong §2.2.2, ll. 2318–2387]. In terms of Equation (3.5),  $\alpha_{-n}\alpha_n = n a_n^\dagger a_n$ , so  $N$  is not the number of quanta but the sum of their mode numbers:  $\alpha_{-2}^i|0; p\rangle$  and  $\alpha_{-1}^j\alpha_{-1}^j|0; p\rangle$  both have  $N = 2$ . Indeed, from  $[\alpha_{-m}^i\alpha_m^i, \alpha_{-n}^j] = n\alpha_{-n}^j\delta_{mn}$  one finds  $[N, \alpha_{-n}^j] = n\alpha_{-n}^j$ : each  $\alpha_{-n}$  raises  $N$  by  $n$ .

### 3.4.2 Level matching as a symmetry

The second equality in Equation (3.24),  $N = \tilde{N}$ , has a simple meaning, which we derive because it will be modified in Chapter 7. No point on a closed string is marked, so a rigid shift  $\sigma \rightarrow \sigma + \delta$  must act trivially on physical states. In Equation (3.2) the shift sends  $e^{-in\sigma^-} \rightarrow e^{in\delta}e^{-in\sigma^-}$  and  $e^{-in\sigma^+} \rightarrow e^{-in\delta}e^{-in\sigma^+}$ , that is,

$$\alpha_n \rightarrow e^{in\delta}\alpha_n, \quad \tilde{\alpha}_n \rightarrow e^{-in\delta}\tilde{\alpha}_n.$$

Since  $[N, \alpha_n] = -n\alpha_n$ , the operator  $U_\delta = e^{-i\delta(N - \tilde{N})}$  implements exactly this:  $U_\delta\alpha_n U_\delta^{-1} = e^{in\delta}\alpha_n$  and  $U_\delta\tilde{\alpha}_n U_\delta^{-1} = e^{-in\delta}\tilde{\alpha}_n$ . A state is invariant under all  $U_\delta$  if and only if  $(N - \tilde{N})|\psi\rangle = 0$ . Level matching is the statement that the origin of  $\sigma$  is unobservable. On a circle the zero modes also transform, and the condition becomes  $N - \tilde{N} = nw$  (Chapter 7).

### 3.4.3 The constant $a$ as a Casimir energy

Here is Tong’s heuristic determination of  $a$  [Tong §2.2.2, ll. 2391–2498]. Take the classical expression  $\frac{1}{2}\sum_{n \neq 0}\alpha_{-n}^i\alpha_n^i$  from Equation (3.20) at face value and bring it to normal-ordered form. The terms with  $n < 0$  are in the wrong order; relabel  $n \rightarrow -n$  and use  $\alpha_n^i\alpha_{-n}^i = \alpha_{-n}^i\alpha_n^i + n(D - 2)$  (summed over  $i$ ):

$$\frac{1}{2}\sum_{n \neq 0}\alpha_{-n}^i\alpha_n^i = \frac{1}{2}\sum_{n > 0}\alpha_{-n}^i\alpha_n^i + \frac{1}{2}\sum_{n > 0}\alpha_n^i\alpha_{-n}^i = \sum_{n > 0}\alpha_{-n}^i\alpha_n^i + \frac{D - 2}{2}\sum_{n > 0}n. \quad (3.25)$$

The last term is the sum of the zero-point energies  $\frac{1}{2}\omega_n = \frac{1}{2}n$  of  $D - 2$  towers of oscillators. It diverges, as the vacuum energy does in any field theory. In a finite box, however, the *finite part* of such a sum is physical: it is the Casimir energy. Regulate with a cutoff  $\epsilon \ll 1$  that suppresses high frequencies,

$$\sum_{n=1}^{\infty} n e^{-\epsilon n} = -\frac{\partial}{\partial \epsilon} \sum_{n=1}^{\infty} e^{-\epsilon n} = -\frac{\partial}{\partial \epsilon} \frac{1}{e^\epsilon - 1} = \frac{1}{\epsilon^2} - \frac{1}{12} + \frac{\epsilon^2}{240} + O(\epsilon^4). \quad (3.26)$$

The expansion follows from  $1/(e^\epsilon - 1) = 1/\epsilon - \frac{1}{2} + \frac{\epsilon}{12} - \frac{\epsilon^3}{720} + \dots$ ; we have confirmed it with a symbolic check in sympy. The divergent term  $1/\epsilon^2$  is independent of the state and has the form of a cosmological constant on the worldsheet. It must be cancelled by a counterterm, because such a term would violate the Weyl invariance of the Polyakov action. The finite remainder is

$$\sum_{n=1}^{\infty} n \longrightarrow -\frac{1}{12} \quad \implies \quad a = \frac{D - 2}{24}, \quad M^2 = \frac{4}{\alpha'}\left(N - \frac{D - 2}{24}\right). \quad (3.27)$$

The same number is obtained from *zeta-function regularization*: the series  $\zeta(s) = \sum_{n \geq 1} n^{-s}$  converges for  $\text{Re } s > 1$ , has a unique analytic continuation, and  $\zeta(-1) = -\frac{1}{12}$  [Tong §2.2.2, ll. 2507–2530].

**Example 3.2** (The cutoff sum with numbers). Take  $\epsilon = 0.1$ . Summing the series numerically gives  $\sum_n n e^{-0.1n} = 99.916708\dots$ , while  $1/\epsilon^2 - \frac{1}{12} = 99.916667$  and the next term  $\epsilon^2/240 = 0.000042$  accounts for the difference. With  $\epsilon = 0.01$  the sum is 9999.9166671 against  $10^4 - \frac{1}{12} = 9999.9166667$ . The divergent part changes by a factor 100; the constant  $-\frac{1}{12} = -0.08333\dots$  does not move. This stability is what makes the finite part meaningful.

**Common pitfall:** “ $1 + 2 + 3 + \dots = -\frac{1}{12}$ ”

The divergent series does not equal  $-\frac{1}{12}$  in any ordinary sense, and nothing in the argument uses such an equality. What is claimed is: (i) the regulated sum has the structure (divergent, state-independent) + (finite) + (vanishing with the cutoff); (ii) the divergent part is removed by a local counterterm that the symmetries require anyway; (iii) the finite part does not depend on how the cutoff is chosen. Point (iii) is not proved by Equation (3.26), which uses one particular cutoff. The derivation that does not discard infinities uses conformal field theory: a theory of central charge  $c$  on a cylinder of circumference  $2\pi$  has ground-state energy  $-(c + \tilde{c})/24$ , and a free boson has  $c = \tilde{c} = 1$ , so each transverse direction contributes  $-\frac{1}{24}$  to  $L_0$  and to  $\tilde{L}_0$  [Tong §4.4.1, ll. 5018–5074]; see Chapter 4.

## 3.5 The spectrum

We can now list the states of the closed bosonic string level by level, using  $M^2 = \frac{4}{\alpha'}(N - \frac{D-2}{24})$  with  $N = \tilde{N}$ .

### 3.5.1 The ground state: a tachyon

The state  $|0; p\rangle$  has  $N = \tilde{N} = 0$  and

$$M^2 = -\frac{1}{\alpha'} \frac{D-2}{6} \quad (3.28)$$

[Tong §2.3.1, ll. 2535–2544]. A particle with  $M^2 < 0$  is a *tachyon*. The right reading is field-theoretic. If  $T(X)$  is the spacetime field whose quantum this particle is, then  $M^2 = \partial^2 V / \partial T^2$  at  $T = 0$ , so  $M^2 < 0$  says that  $T = 0$  is a *maximum* of the potential  $V(T)$  (Figure 3.1). The expansion is carried out around an unstable point, just as the Higgs field at  $H = 0$  is tachyonic. Whether  $V(T)$  for the closed bosonic string has a stable minimum is not known: the cubic term produces one, the quartic term removes it again, and  $T$  mixes with other scalars [Tong §2.3.1, ll. 2546–2580]. The superstring has no tachyon, and the lessons of the bosonic string carry over to it; but the instability of the bosonic vacuum should be kept in mind throughout Part I.

### 3.5.2 The first excited level and $D = 26$

Level matching forbids a single oscillator. The first excited states are

$$\alpha_{-1}^i \tilde{\alpha}_{-1}^j |0; p\rangle, \quad i, j = 1, \dots, D-2, \quad M^2 = \frac{4}{\alpha'} \left(1 - \frac{D-2}{24}\right), \quad (3.29)$$

$(D-2)^2$  states in all [Tong §2.3.2, ll. 2595–2622]. They transform as a two-index tensor of the rotation group  $SO(D-2)$  of the transverse directions, which is all that lightcone gauge leaves manifest. They must nevertheless form a representation of the full Lorentz group  $SO(1, D-1)$ . Wigner’s classification says how [Tong §2.3.2, ll. 2631–2650]:

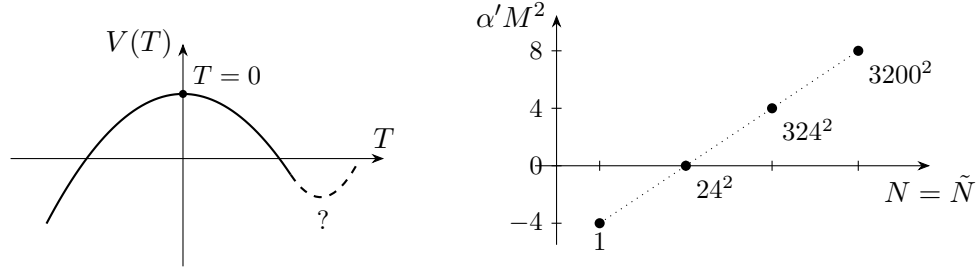


Figure 3.1: Left: a negative  $M^2$  means that the expansion point  $T = 0$  is a maximum of the tachyon potential; whether a minimum exists elsewhere (dashed) is not known. Right: the first four levels of the closed bosonic string in  $D = 26$ , with  $\alpha' M^2 = 4(N - 1)$  and the number of states at each level (Example 3.3).

- a *massive* particle can be brought to rest,  $p^\mu = (M, 0, \dots, 0)$ ; the rotations that preserve this momentum form  $SO(D - 1)$ , and the internal states must fill representations of that group,  $SO(D - 1)$ ;
- a *massless* particle cannot be brought to rest; one takes  $p^\mu = (k, 0, \dots, 0, k)$ , which is preserved by  $SO(D - 2)$ , and the internal states need only fill representations of  $SO(D - 2)$ . This is why a photon in four dimensions has two polarizations while a massive vector has three.

The  $(D - 2)^2$  states of Equation (3.29) form a tensor of  $SO(D - 2)$ . They cannot be completed into a representation of  $SO(D - 1)$ : a two-index tensor of  $SO(D - 1)$  would need  $(D - 1)^2$  states, and nothing else is available at this level. Lorentz invariance therefore requires these states to be massless, and Equation (3.29) then gives

$$1 - \frac{D - 2}{24} = 0 \quad \Longleftrightarrow \quad D = 26, \quad a = 1 \quad (3.30)$$

[L] [Tong §2.3.2, ll. 2643–2653]. This agrees with Example 3.1:  $a = 1$  is the value at which the first level is a massless gauge particle. With  $D = 26$  the mass formula takes the form used in the rest of the book,

$$M^2 = \frac{4}{\alpha'}(N - 1) = \frac{2}{\alpha'}(N + \tilde{N} - 2), \quad N = \tilde{N}. \quad (3.31)$$

### 3.5.3 Graviton, $B$ -field and dilaton

For  $D = 26$  the massless states are a tensor  $\zeta_{ij}$  of  $SO(24)$  with  $24^2 = 576$  components. A two-index tensor decomposes into irreducible parts,

$$\underbrace{\mathbf{24} \otimes \mathbf{24}}_{576} = \underbrace{\text{traceless symmetric}}_{\frac{1}{2} \cdot 24 \cdot 25 - 1 = 299} \oplus \underbrace{\text{antisymmetric}}_{\frac{1}{2} \cdot 24 \cdot 23 = 276} \oplus \underbrace{\text{trace}}_1, \quad (3.32)$$

and to each part corresponds a massless field in spacetime [Tong §2.3.2, ll. 2655–2684]:

| $SO(24)$ representation | states | field           | name                                |
|-------------------------|--------|-----------------|-------------------------------------|
| traceless symmetric     | 299    | $G_{\mu\nu}(X)$ | metric; its quantum is the graviton |
| antisymmetric           | 276    | $B_{\mu\nu}(X)$ | Kalb–Ramond field, or 2-form        |
| singlet                 | 1      | $\Phi(X)$       | dilaton                             |

These three fields are common to all string theories. The fields carry spacetime indices  $\mu, \nu = 0, \dots, 25$  and not just  $i, j$ , because  $G_{\mu\nu}$  and  $B_{\mu\nu}$  possess gauge symmetries that remove the extra components; lightcone gauge shows only the physical polarizations, and Example 3.1 showed where the gauge symmetry comes from [Tong §2.3.2, ll. 2760–2770].

#### Why a massless spin-two particle means gravity

A massless spin-two field  $h_{\mu\nu}$  quantized covariantly has negative-norm states, created by operators with one timelike index, for the same reason as Equation (3.7). They decouple only if the theory has the gauge symmetry  $h_{\mu\nu} \rightarrow h_{\mu\nu} + \partial_\mu \xi_\nu + \partial_\nu \xi_\mu$ . An argument due to Feynman and Weinberg shows that this symmetry survives interactions only if the full theory is invariant under diffeomorphisms, that is, if it is general relativity, possibly with higher-derivative corrections [Tong §2.3.2, ll. 2686–2757]. The string was not designed to contain gravity. The graviton appears because the constraint algebra forces  $a = 1$ . How Einstein's equations themselves emerge is the subject of Chapter 6; the roles of  $B_{\mu\nu}$  and  $\Phi$  in T-duality begin in Chapter 9.

### 3.5.4 Massive levels

At higher levels all states are massive and must fill representations of  $SO(D-1) = SO(25)$ , although only  $SO(24)$  is manifest. At level  $N = 2$  the right-moving states are  $\alpha_{-1}^i \alpha_{-1}^j |0\rangle$  (symmetric in  $i, j$ ) and  $\alpha_{-2}^i |0\rangle$ . Their number is

$$\frac{1}{2}(D-2)(D-1) + (D-2) = \frac{1}{2}D(D-1) - 1, \quad (3.33)$$

which is exactly the dimension of the traceless symmetric tensor of  $SO(D-1)$  [Tong §2.3.3, ll. 2771–2806]. (A symmetric tensor of  $SO(n)$  has  $\frac{1}{2}n(n+1)$  components, and removing the trace gives  $\frac{1}{2}n(n+1) - 1$ ; put  $n = D-1$ .) The vector  $\alpha_{-2}^i |0\rangle$  of  $SO(24)$  supplies precisely the components that the symmetric tensor of  $SO(24)$  lacks. One can show that all higher levels also assemble into  $SO(D-1)$  representations, so the only condition from this argument is the one at level 1 [Tong §2.3.3, ll. 2807–2812].

**Example 3.3** (Counting states with a generating function). Each right-moving oscillator  $\alpha_{-n}^i$  may be excited  $k = 0, 1, 2, \dots$  times and then contributes  $kn$  to  $N$ . The number  $d_N$  of right-moving states at level  $N$  is therefore the coefficient of  $q^N$  in

$$\prod_{n=1}^{\infty} \left( \frac{1}{1-q^n} \right)^{24} = 1 + 24q + 324q^2 + 3200q^3 + 25650q^4 + \dots$$

(expansion obtained by a symbolic check in sympy). Check the first entries by hand:  $d_1 = 24$ ;  $d_2 = \frac{1}{2} \cdot 24 \cdot 25 + 24 = 324 = \frac{1}{2} \cdot 25 \cdot 26 - 1$ , in agreement with Equation (3.33); and  $d_3 = 24 + 24^2 + \binom{26}{3} = 24 + 576 + 2600 = 3200$  from  $\alpha_{-3}$ ,  $\alpha_{-2}\alpha_{-1}$  and  $\alpha_{-1}\alpha_{-1}\alpha_{-1}$ . For  $SO(25)$  the traceless symmetric three-index tensor has  $\binom{27}{3} - 25 = 2900$  components and the antisymmetric two-index tensor has  $\binom{25}{2} = 300$ ; the sum is 3200, as required for a massive level. The closed string at level  $N = \tilde{N}$  has  $d_N^2$  states:

| $N = \tilde{N}$ | $\alpha' M^2$ | $d_N$ | closed-string states | content                                                       |
|-----------------|---------------|-------|----------------------|---------------------------------------------------------------|
| 0               | -4            | 1     | 1                    | tachyon                                                       |
| 1               | 0             | 24    | 576                  | $G_{\mu\nu}, B_{\mu\nu}, \Phi$                                |
| 2               | 4             | 324   | 104 976              | massive, from $\mathbf{324} \otimes \mathbf{324}$ of $SO(25)$ |
| 3               | 8             | 3200  | 10 240 000           | massive                                                       |

The spacing of the levels is  $\Delta M^2 = 4/\alpha'$ . If  $\sqrt{\alpha'}$  is of the order of the Planck length, the first massive level has  $M = 2/\sqrt{\alpha'}$ , near the Planck mass, far beyond any accelerator [Tong §2.3.3, ll. 2813–2819]. At low energy only the massless level matters; this is why Chapter 6 can describe the string by a field theory of  $G$ ,  $B$  and  $\Phi$ .

The same product appears, with  $q = e^{2\pi i \tau}$ , in the one-loop partition function of the string, and its analogue for K3 in the elliptic genus of Chapter 28.

### 3.6 Lorentz invariance revisited

The argument for  $D = 26$  in the previous section was quick: it fixed  $a$  by a regularization and  $D$  by one representation-theoretic observation. A more systematic test within lightcone quantization is to construct the Lorentz generators and verify their algebra [Tong §2.4, ll. 2820–2835].

Before gauge fixing, the action is invariant under the global Poincaré transformations  $X^\mu \rightarrow \Lambda^\mu_\nu X^\nu + c^\mu$ . The Noether currents are

$$P_\mu^\alpha = T \partial^\alpha X_\mu, \quad J_{\mu\nu}^\alpha = P_\mu^\alpha X_\nu - P_\nu^\alpha X_\mu, \quad (3.34)$$

conserved by the equations of motion [Tong §2.4, ll. 2836–2860]. The charges  $M^{\mu\nu} = \int_0^{2\pi} d\sigma J^{\tau\mu\nu}$  are, in modes,

$$M^{\mu\nu} = \underbrace{(p^\mu x^\nu - p^\nu x^\mu)}_{l^{\mu\nu}} - i \underbrace{\sum_{n=1}^{\infty} \frac{1}{n} (\alpha_{-n}^\nu \alpha_n^\mu - \alpha_{-n}^\mu \alpha_n^\nu)}_{S^{\mu\nu}} + \tilde{S}^{\mu\nu}, \quad (3.35)$$

where  $l^{\mu\nu}$  is the orbital angular momentum of the centre of mass,  $S^{\mu\nu}$  the spin carried by right-moving oscillators and  $\tilde{S}^{\mu\nu}$  the same expression with  $\tilde{\alpha}$  [Tong §2.4, ll. 2861–2880]. In covariant quantization these operators obey the Lorentz algebra

$$[M^{\rho\sigma}, M^{\tau\nu}] = \eta^{\sigma\tau} M^{\rho\nu} - \eta^{\rho\tau} M^{\sigma\nu} + \eta^{\rho\nu} M^{\sigma\tau} - \eta^{\sigma\nu} M^{\rho\tau} \quad (3.36)$$

without difficulty. In lightcone gauge the generators  $M^{i-}$  are dangerous. They contain  $\alpha_n^-$  and  $p^-$ , which by Equations (3.19) and (3.20) are *quadratic* in the transverse oscillators and carry the ordering constant  $a$ . So  $M^{i-}$  is cubic in oscillators, and its commutators can produce anomalous terms. The algebra of Equation (3.36) demands

$$[M^{i-}, M^{j-}] = 0, \quad (3.37)$$

because with  $\rho = i, \tau = j, \sigma = \nu = -$  every term on the right of Equation (3.36) vanishes:  $\eta^{--} = \eta^{i-} = \eta^{-j} = 0$  in the metric of Equation (3.16), and the remaining term is  $-\delta^{ij} M^{--}$ , which is zero by antisymmetry. Keeping  $a$  and  $D$  arbitrary, a long calculation gives instead

$$[M^{i-}, M^{j-}] = \frac{2}{(p^+)^2} \sum_{n>0} \left( \left[ \frac{D-2}{24} - 1 \right] n + \frac{1}{n} \left[ a - \frac{D-2}{24} \right] \right) (\alpha_{-n}^i \alpha_n^j - \alpha_{-n}^j \alpha_n^i) + (\alpha \leftrightarrow \tilde{\alpha}) \quad (3.38)$$

[L] [Tong §2.4, ll. 2905–2945]. The operators  $\alpha_{-n}^i \alpha_n^j - \alpha_{-n}^j \alpha_n^i$  for different  $n$  are independent, so each coefficient must vanish. The coefficient is a combination of  $n$  and  $1/n$ , and it vanishes for all  $n \geq 1$  only if both brackets vanish:

$$\frac{D-2}{24} = 1 \quad \text{and} \quad a = \frac{D-2}{24} \quad \Longleftrightarrow \quad D = 26, \quad a = 1 \quad (3.39)$$

[Tong §2.4, ll. 2953–2956]. Two remarks. First, the second condition is Equation (3.27): the Lorentz algebra confirms the Casimir value of  $a$  without any regularization. Second, this is an *anomaly*: a symmetry of the classical theory that fails in the quantum theory unless a condition is met. We have not carried out the calculation leading to Equation (3.38); neither does Tong, and neither does the Lean library (Section 3.8).

#### Historical note

Lightcone quantization of the string was first carried out by Goddard, Goldstone, Rebbi and Thorn, “Quantum Dynamics of a Massless Relativistic String”, Nucl. Phys. B56 (1973); the commutator in Equation (3.38) is the central computation of that paper [Tong §2.4, footnote, ll. 2949–2951].

### 3.7 The critical dimension as a central charge; the superstring

The number 26 will be derived a third time in Chapter 5, and that derivation is the one formalized in Lean, so we summarize it here. Gauge-fixing the Polyakov path integral introduces anticommuting Faddeev–Popov fields  $b, c$ . They form a conformal field theory, and any conformal field theory has a *central charge*  $c$ , a number that measures its response to a rescaling of the worldsheet metric (Chapter 4). The results needed are [L] [Tong §5.2–5.3, ll. 6832–6850]:

- a free scalar field has  $c = 1$ , so  $D$  fields  $X^\mu$  have  $c = D$ ;
- the  $bc$  ghost system has  $c = -26$ ;
- the Weyl symmetry is anomalous unless the total central charge vanishes.

Since the Weyl symmetry is a gauge symmetry, it must not be anomalous. Hence  $D - 26 = 0$ . In this form the statement is more general than “spacetime has 26 dimensions”: the matter sector may be *any* conformal field theory with  $c = 26$ , for instance four free scalars for Minkowski space together with an “internal” theory of  $c = 22$ . The critical dimension is really a critical central charge [Tong §5.3, ll. 6851–6868].

The lightcone result and the central-charge result are connected by the Casimir energy. A conformal field theory on the cylinder has ground-state energy  $-(c+\tilde{c})/24$  [Tong §4.4.1, ll. 5060–5074], which gives  $a = c_\perp/24$  for a transverse theory of central charge  $c_\perp$ ; with  $c_\perp = D - 2$  this is Equation (3.27). The identity

$$a = \frac{D - 2}{24} = 1 + \frac{D - 26}{24}$$

then shows that the ordering constant takes the ghost-free value  $a = 1$  of Example 3.1 exactly when the total central charge  $D - 26$  vanishes. In both languages two of the  $D$  directions do not contribute physical oscillators: in lightcone gauge they are eliminated by hand, in the covariant path integral the  $bc$  system cancels them.

**The superstring.** Adding fermions  $\psi^\mu$  to the worldsheet, related to the  $X^\mu$  by worldsheet supersymmetry, gives the superstring. Supersymmetry is a further gauge symmetry and requires further Faddeev–Popov fields, the commuting  $\beta\gamma$  system, with  $c = +11$ . The matter sector must then have  $c = 26 - 11 = 15$ . A free fermion has  $c = \frac{1}{2}$ , and supersymmetry pairs each boson with a fermion, so  $D(1 + \frac{1}{2}) = 15$  and

$$D = 10 \tag{3.40}$$

[L] [Tong §5.3.1, ll. 6884–6891]. The quantized superstring has no tachyon, and its massless bosons include  $G_{\mu\nu}$ ,  $B_{\mu\nu}$  and  $\Phi$  again [Tong §2.5, ll. 2957–2975]. There are several superstring theories, distinguished by discrete choices: type II (worldsheet fermions moving in both directions, 32 supercharges in  $D = 10$ ), subdivided into IIA and IIB by their additional Ramond–Ramond fields; heterotic (fermions in one direction only, 16 supercharges) with gauge group  $SO(32)$  or  $E_8 \times E_8$ ; and type I with open strings [Tong §2.5, ll. 2976–3008]. For this book the important consequences are these. In  $D = 10$  one can compactify six dimensions, for instance on  $K3 \times T^2$  (Chapter 24). By the same counting, a supersymmetric sigma model on a space of four real dimensions, such as a K3 surface, has four bosons and four fermions and hence  $c = 4 \cdot 1 + 4 \cdot \frac{1}{2} = 6$ , the value used in Chapter 28. And the appearance of the lattice  $E_8$  in a string theory is one motivation for Chapter 18.

### 3.8 What the machine has checked

The Lean file for this chapter is

StringTheoryFormalization/UseCases/CriticalDimension.lean

and all its declarations live in the namespace `StringTheory.UseCases.CriticalDimension`. The file belongs to the older part of the library, in which a physical quantity is modelled by a rational number and the theorems are arithmetic. It is a good first Lean file to read, because every proof is one line. It is also a good first lesson in reading a formal statement for what it literally says.

#### In Lean: the two ghost central-charge formulas

```
def fermionicGhostCharge (lam : ℚ) : ℚ := 1 - 3 * (2 * lam - 1) ^ 2

def bosonicGhostCharge (lam : ℚ) : ℚ := 3 * (2 * lam - 1) ^ 2 - 1

theorem bosonic_eq_neg_fermionic (lam : ℚ) :
  bosonicGhostCharge lam = - fermionicGhostCharge lam := by ...
```

**Reading.** These are *definitions* of two functions  $\mathbb{Q} \rightarrow \mathbb{Q}$ :  $c_{bc}(\lambda) = 1 - 3(2\lambda - 1)^2$  and  $c_{\beta\gamma}(\lambda) = 3(2\lambda - 1)^2 - 1$ . In the literature they are the central charges of an anticommuting, respectively commuting, first-order system whose fields have conformal weights  $\lambda$  and  $1 - \lambda$  (the file’s header cites Polchinski and Green–Schwarz–Witten for them; these are not in this book’s source library; Tong computes only the case  $\lambda = 2$ , where  $b$  has weight 2 and  $c$  weight  $-1$  [Tong §5.2, ll. 6700–6839]). Lean has not derived these formulas from an operator product expansion; they are input. The theorem states that the second polynomial is the negative of the first. **Proof idea.** Unfold the definitions; the tactic `ring` verifies the polynomial identity.

#### In Lean: evaluation and the two cancellations

```
theorem bc_ghost_charge_eq : fermionicGhostCharge 2 = -26 := by ...

theorem betagamma_ghost_charge_eq : bosonicGhostCharge (3 / 2) = 11 := by ...

theorem bosonic_string_critical_dimension :
  (26 : ℚ) * 1 + fermionicGhostCharge 2 = 0 := by ...

theorem superstring_critical_dimension :
  (10 : ℚ) * 1 + 10 * (1 / 2) + fermionicGhostCharge 2
```



```
| + bosonicGhostCharge (3 / 2) = 0 := by ...
```

**Reading.** The first two theorems evaluate the polynomials:  $1 - 3 \cdot 3^2 = -26$  and  $3 \cdot 2^2 - 1 = 11$ . The third says  $26 \cdot 1 + (-26) = 0$  in  $\mathbb{Q}$ ; the fourth says  $10 \cdot 1 + 10 \cdot \frac{1}{2} - 26 + 11 = 0$  in  $\mathbb{Q}$ . The literals 1 and  $1/2$  stand for the central charges of a free boson and a free fermion, but in the statement they are just rational numbers. **Proof idea.** `norm_num` evaluates the arithmetic after the definitions are unfolded; the last two theorems first rewrite with the first two.

#### In Lean: uniqueness of the solution

```
| theorem bosonic_dimension_unique (D : ℚ)
  | (h : D * 1 + fermionicGhostCharge 2 = 0) :
  | D = 26 := by ...
```

**Reading.** If a rational number  $D$  satisfies  $D \cdot 1 + c_{bc}(2) = 0$ , then  $D = 26$ . This is the solution of one linear equation. Note that  $D$  is a rational number, not a natural number, so the theorem is slightly stronger than “no other integer dimension works”. **Proof idea.** Rewrite the hypothesis with `bc_ghost_charge_eq` to obtain  $D - 26 = 0$ , then `linarith`.

**What this does and does not establish.** The chain of reasoning behind  $D = 26$  in the central-charge form has four links: (1) gauge fixing produces the  $bc$  system with  $\lambda = 2$ ; (2) a first-order system of weight  $\lambda$  has  $c = 1 - 3(2\lambda - 1)^2$ , and a free boson has  $c = 1$ ; (3) consistency requires  $c_{\text{total}} = 0$ ; (4) arithmetic. The Lean file checks link (4) and records link (2) as a definition. Links (1)–(3) are physics and conformal field theory established in the literature. In particular the file does *not*:

- define the Virasoro algebra, an operator product expansion or a central charge as a mathematical object (so it cannot *derive*  $-26$ );
- contain the oscillators  $\alpha_n^\mu$ , the generators  $M^{\mu\nu}$ , or the commutator of Equation (3.38); it does not derive the closure of the Lorentz algebra, which is the content of Section 3.6;
- determine the normal-ordering constant  $a$ , prove a no-ghost theorem, or construct any conformal field theory with  $c = 26$ .

In the language of this book, the file proves an *arithmetic shadow* of the critical-dimension statement. A faithful formalization would need, at least, a Lie-algebra definition of the Virasoro algebra with its central extension, the Fock-space representation built from Equation (3.4), and a proof that the normal-ordered  $L_n$  satisfy the algebra with  $c = D$ ; Exercise 3.9 indicates the first step. All theorems of the file depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`; this was checked with `#print axioms` for each of the six theorems. Table 3.1 collects the status of the claims of this chapter.

The identification of the rational numbers in these theorems with central charges of world-sheet theories is part of the physical modelling and carries no tier A status.

#### Chapter summary

- Canonical quantization turns the modes of  $X^\mu$  into oscillators,  $[\alpha_n^\mu, \alpha_m^\nu] = n\eta^{\mu\nu}\delta_{n+m,0}$ . The timelike oscillators create negative-norm states, which the Virasoro constraints  $L_n|\text{phys}\rangle = 0$  ( $n > 0$ ) and  $(L_0 - a)|\text{phys}\rangle = 0$  must remove.
- In lightcone gauge  $X^+ = x^+ + \alpha' p^+ \tau$  the constraints determine  $X^-$ ; only  $D - 2$



Table 3.1: Epistemic status of the main claims of this chapter.

| Claim                                                                                                   | Tier | Basis                              |
|---------------------------------------------------------------------------------------------------------|------|------------------------------------|
| $1 - 3(2\lambda - 1)^2 = -26$ at $\lambda = 2$                                                          | [A]  | bc_ghost_charge_eq                 |
| $3(2\lambda - 1)^2 - 1 = 11$ at $\lambda = \frac{3}{2}$                                                 | [A]  | betagamma_ghost_charge_eq          |
| $26 \cdot 1 - 26 = 0$ in $\mathbb{Q}$                                                                   | [A]  | bosonic_string_critical_dimension  |
| $10 \cdot 1 + 10 \cdot \frac{1}{2} - 26 + 11 = 0$ in $\mathbb{Q}$                                       | [A]  | superstring_critical_dimension     |
| $D \in \mathbb{Q}$ and $D - 26 = 0$ imply $D = 26$                                                      | [A]  | bosonic_dimension_unique           |
| $c_{\beta\gamma}(\lambda) = -c_{bc}(\lambda)$ as polynomials                                            | [A]  | bosonic_eq_neg_fermionic           |
| The $bc$ ghosts have $c = -26$ , the $\beta\gamma$ ghosts $c = 11$                                      | [L]  | Tong §5.2–5.3.1                    |
| Weyl invariance needs $c_{\text{total}} = 0$ , so $D = 26$ (resp. 10)                                   | [L]  | Tong §5.3, §5.3.1                  |
| Lightcone Lorentz algebra closes iff $D = 26$ , $a = 1$                                                 | [L]  | Tong §2.4                          |
| $\sum_n n e^{-\epsilon n} = \epsilon^{-2} - \frac{1}{12} + O(\epsilon^2)$ ;<br>$d_N = 1, 24, 324, 3200$ | —    | symbolic check (sympy), not Tier A |

transverse oscillators remain and the Hilbert space is positive. The price is that Lorentz invariance must be checked.

- The mass formula is  $M^2 = \frac{4}{\alpha'}(N - a)$  with  $N = \tilde{N}$  (level matching = invariance under shifts of  $\sigma$ ). The ordering constant is a Casimir energy,  $a = (D - 2)/24$ .
- Level 0 is a tachyon,  $M^2 = -4/\alpha'$  for  $D = 26$ . Level 1 has  $(D - 2)^2$  states, which can represent the Lorentz group only if massless:  $D = 26$ ,  $a = 1$ . They are the graviton  $G_{\mu\nu}$ , the Kalb–Ramond field  $B_{\mu\nu}$  and the dilaton  $\Phi$  ( $299 + 276 + 1 = 576$ ).
- The same values follow from closure of the lightcone Lorentz algebra,  $[M^{i-}, M^{j-}] = 0$ , and from cancellation of the central charge,  $D - 26 = 0$ . For the superstring,  $\frac{3}{2}D - 26 + 11 = 0$  gives  $D = 10$ .
- Lean checks the central-charge arithmetic [A]. The origin of the numbers  $-26$ ,  $11$ ,  $1$ ,  $\frac{1}{2}$  and the requirement  $c_{\text{total}} = 0$  are [L].

## Exercises

**Exercise 3.1** (★). Using Equation (3.4), show that  $[N, \alpha_{-n}^j] = n \alpha_{-n}^j$  and  $[N, \alpha_n^j] = -n \alpha_n^j$  for  $n > 0$ . Deduce that  $\alpha_{-2}^i \alpha_{-1}^j \alpha_{-1}^k |0; p\rangle$  has  $N = 4$ , and find the state(s) with which level matching allows it to be combined if the left-moving part contains only oscillators  $\tilde{\alpha}_{-1}$  and  $\tilde{\alpha}_{-3}$ .

**Exercise 3.2** (★). Compute the norm of  $\alpha_{-1}^0 \alpha_{-1}^0 |0; p\rangle$  and of  $\alpha_{-1}^0 \alpha_{-1}^1 |0; p\rangle$  (strip off the momentum delta function). Conclude that the sign of the norm is  $(-1)^k$ , with  $k$  the number of timelike oscillators.

**Exercise 3.3** (★). Verify the decomposition in Equation (3.32) for general  $n = D - 2$ : show that  $[\frac{1}{2}n(n+1) - 1] + \frac{1}{2}n(n-1) + 1 = n^2$ . In  $D = 4$  (not a consistent bosonic string, but a useful check) how many polarizations would the “graviton”, “ $B$ -field” and “dilaton” have? Compare with the two helicities of the four-dimensional graviton.

**Exercise 3.4** (★★). Repeat the regularization of Equation (3.26) with a Gaussian cutoff, that

is, for the sum  $\sum_n n e^{-\epsilon^2 n^2}$ . Use the Euler–Maclaurin formula

$$\sum_{n \geq 0} f(n) = \int_0^\infty f(x) dx + \frac{1}{2}f(0) - \frac{1}{12}f'(0) + \frac{1}{720}f'''(0) - \dots$$

Show that the divergent part changes but the finite part is again  $-\frac{1}{12}$ . Why does every smooth cutoff function  $f(n) = n g(\epsilon n)$  with  $g(0) = 1$  give the same finite part?

**Exercise 3.5** (★★). Open strings with free ends in all directions (Chapter 13) have a single set of oscillators and  $M^2 = \frac{1}{\alpha'}(N - a)$  [Tong §3.1, ll. 3270–3290], with the same  $a = (D - 2)/24$ . Repeat the argument of Section 3.5: how many states are there at  $N = 1$ , which little group must they represent, and what follows for  $D$ ? Identify the massless particle.

**Exercise 3.6** (★★). (a) Show that the number of right-moving states at level 4 in  $D = 26$  is 25650 by listing the partitions of 4 and counting the states for each. (b) For  $SO(25)$  the traceless symmetric tensors of rank  $r$  have dimension  $\binom{24+r}{r} - \binom{22+r}{r-2}$ , and the traceless three-index tensor of mixed symmetry has dimension  $\frac{1}{3}n(n^2 - 1) - n$  with  $n = 25$ . Check that the traceless symmetric tensors of rank 4 and rank 2, the mixed tensor and a singlet have total dimension 25650. (A dimension count does not prove that the states transform in these representations; what else would be needed?)

**Exercise 3.7** (★★). The library proves uniqueness only for the bosonic string. State and prove in Lean the superstring analogue: for a rational number  $D$ , the hypothesis

`| D * 1 + D * (1 / 2) + fermionicGhostCharge 2 + bosonicGhostCharge (3 / 2) = 0`

implies  $D = 10$ . Hint: rewrite the hypothesis with the two evaluation theorems of Section 3.8, then use `linarith`. This statement is an exercise; it is not a declaration of the library.

**Exercise 3.8** (★★). Evaluate `fermionicGhostCharge` at  $\lambda = 1$  and  $\lambda = \frac{1}{2}$  by hand, and prove both values in Lean with `norm_num`. Then prove that `fermionicGhostCharge` is symmetric under  $\lambda \rightarrow 1 - \lambda$ , i.e. `fermionicGhostCharge (1 - lam) = fermionicGhostCharge lam`. What is the physical meaning of this symmetry for a system with fields of weights  $\lambda$  and  $1 - \lambda$ ?

**Exercise 3.9** (★★★). (a) Using only Equation (3.4), show that  $[L_m, \alpha_n^\mu] = -n \alpha_{m+n}^\mu$ . (b) Deduce that  $[L_m, L_n] = (m - n)L_{m+n}$  for  $m + n \neq 0$ . (c) For  $m + n = 0$  the normal ordering of  $L_0$  produces an extra constant  $A(m)$ . Compute  $A(1)$  and  $A(2)$  by evaluating  $\langle 0; 0 | [L_m, L_{-m}] | 0; 0 \rangle$  directly, and check that they agree with  $A(m) = \frac{D}{12}m(m^2 - 1)$ . (d) Write a Lean `structure` whose fields are a family  $L : \mathbb{Z} \rightarrow V$  in a Lie algebra  $V$  over  $\mathbb{Q}$ , a central element, and the commutation relation with central term  $\frac{c}{12}m(m^2 - 1)\delta_{m+n,0}$ . This is the first step towards a formalization in which “ $c = D$ ” could be a theorem and not a definition.

## Further reading

- Covariant quantization, ghosts and the constraints: [Tong §2.1, ll. 1708–1931].
- Lightcone gauge, quantization, the Casimir energy and zeta-function regularization: [Tong §2.2, ll. 1932–2530].
- The spectrum, Wigner’s little-group argument, and why massless spin two means general relativity: [Tong §2.3, ll. 2533–2819].

- The Lorentz algebra in lightcone gauge and the original reference: [Tong §2.4, ll. 2820–2956]. The superstring in outline: [Tong §2.5, ll. 2957–3008].
- The Casimir energy from the Schwarzian derivative, without discarded infinities: [Tong §4.4.1, ll. 5018–5074]. The ghost central charge and the critical central charge: [Tong §5.2–5.3.1, ll. 6832–6891]; both are developed in Chapters 4 and 5.
- The Lean source, including its own account of what is and is not proved:  
`StringTheoryFormalization/UseCases/CriticalDimension.lean`.



## Chapter 4

# Conformal Field Theory for String Theorists

After gauge fixing, the worldsheet theory of a string is a two-dimensional field theory with no intrinsic length scale: a *conformal field theory* (CFT). This chapter answers the question: what can be said about such a theory using symmetry alone, and how much of the string spectrum of Chapter 3 is already fixed by that symmetry? The answer is surprisingly much. The symmetry group is infinite dimensional; it organizes all local operators by two numbers, the weights  $(h, \bar{h})$ ; it has a quantum anomaly measured by one number, the central charge  $c$ ; and it identifies states on a closed string with local operators at a point. A student should care for three reasons. First, the consistency condition  $D = 26$  of the bosonic string is the statement  $c = 26$  (Chapter 5). Second, the vertex operators  $e^{ikX}$  studied here become, on a circle, the momentum and winding operators of Chapter 7, and their weights are the two halves of the mass formula on which T-duality acts. Third, the K3 sigma model of the second half of the book is known almost entirely through its conformal data ( $c = 6$ , weights, characters). The chapter follows [Tong §4, ll. 3553–6190] and uses its conventions (Euclidean worldsheet,  $\alpha'$  of dimension length<sup>2</sup>).

### 4.1 Conformal transformations in two dimensions

#### 4.1.1 Definition and the role of the metric

Let  $\sigma^\alpha$ ,  $\alpha = 1, 2$ , be coordinates on the worldsheet and  $g_{\alpha\beta}$  its metric. A *conformal transformation* is a change of coordinates  $\sigma^\alpha \rightarrow \tilde{\sigma}^\alpha(\sigma)$  under which the metric changes only by a position-dependent positive factor,

$$g_{\alpha\beta}(\sigma) \longrightarrow \Omega^2(\sigma) g_{\alpha\beta}(\sigma) \quad (4.1)$$

[Tong §4, ll. 3560–3566]. Such a map preserves the angle between two curves at each point, because an angle is a ratio of inner products and the factor  $\Omega^2$  cancels; it does not preserve lengths. A *conformal field theory* is a field theory invariant under all maps of the form (4.1).

The interpretation of (4.1) depends on whether the metric is dynamical. In the Polyakov formulation of Chapter 2 it is, and a conformal transformation is a diffeomorphism whose effect on the metric can be undone by a Weyl rescaling  $g \rightarrow e^{2\omega}g$ : the part of the gauge group that survives conformal gauge. If instead the metric is a fixed flat background, the same map is a genuine global symmetry with conserved currents [Tong §4, ll. 3567–3578]. The second point of view is taken in this chapter; the first returns in Chapter 5. Classically the two are equivalent; whether they remain so in the quantum theory is the question behind the critical dimension [Tong §4, ll. 3590–3597].

**Physical picture: no scale, no particles**

Invariance under (4.1) includes uniform dilatations  $\sigma \rightarrow \lambda\sigma$ . A theory with a mass  $m$  has a preferred length  $1/m$  and cannot be dilatation invariant, so a CFT has only massless excitations. The natural questions are not about particles and S-matrices but about correlation functions and the transformation of operators [Tong §4, ll. 3598–3604]. For strings this is no loss: the particles of interest live in spacetime and appear as *operators* of the worldsheet CFT.

**4.1.2 Euclidean worldsheet and complex coordinates**

Continue the worldsheet time to imaginary values,  $(\sigma^1, \sigma^2) = (\sigma^1, i\sigma^0)$ , and define

$$z = \sigma^1 + i\sigma^2, \quad \bar{z} = \sigma^1 - i\sigma^2, \quad \partial \equiv \partial_z = \frac{1}{2}(\partial_1 - i\partial_2), \quad \bar{\partial} \equiv \partial_{\bar{z}} = \frac{1}{2}(\partial_1 + i\partial_2), \quad (4.2)$$

so that  $\partial z = \bar{\partial} \bar{z} = 1$  and  $\partial \bar{z} = \bar{\partial} z = 0$  [Tong §4.0.1, ll. 3605–3630]. These are the Euclidean versions of the light-cone coordinates  $\sigma^\pm$  of Chapter 2; by this analogy holomorphic functions of  $z$  are called left-moving and antiholomorphic functions right-moving. The flat metric is

$$ds^2 = (d\sigma^1)^2 + (d\sigma^2)^2 = dz d\bar{z}, \quad g_{zz} = g_{\bar{z}\bar{z}} = 0, \quad g_{z\bar{z}} = \frac{1}{2}, \quad (4.3)$$

and the measure is  $d^2z \equiv dz d\bar{z} = 2 d\sigma^1 d\sigma^2$  [Tong §4.0.1, ll. 3631–3646].

A warning about factors of two: because  $g_{z\bar{z}} = \frac{1}{2}$ , one has  $g^{z\bar{z}} = 2$ ,  $\partial \bar{\partial} = \frac{1}{4} \partial_\alpha \partial^\alpha$ , and the delta function normalized by  $\int d^2z \delta(z, \bar{z}) = 1$  is half of  $\delta(\sigma)$ . Most numerical disagreements between CFT texts come from these conventions.

**4.1.3 Conformal maps are holomorphic maps**

In these coordinates, take any holomorphic function  $f$  and set

$$z \rightarrow z' = f(z), \quad \bar{z} \rightarrow \bar{z}' = \bar{f}(\bar{z}). \quad (4.4)$$

Then  $dz' d\bar{z}' = |f'(z)|^2 dz d\bar{z}$ , which has the form (4.1) with  $\Omega^2 = |f'|^2$ . Every holomorphic change of coordinates is conformal [Tong §4.0.2, ll. 3651–3660]. There are infinitely many: one for each function  $f$ . This is special to two dimensions; for flat  $\mathbb{R}^{p,q}$  with  $p+q > 2$  the conformal group is the finite-dimensional group  $SO(p+1, q+1)$  [Tong §4.0.2, ll. 3660–3665]. The two simplest examples are the translation  $z \rightarrow z+a$  and  $z \rightarrow \zeta z$ , which is a rotation for  $|\zeta| = 1$  and a dilatation for  $\zeta$  real and positive.

It is convenient to treat  $z$  and  $\bar{z}$  as independent complex variables and to impose  $\bar{z} = z^*$  at the end; this makes contour integration available [Tong §4.0.2, ll. 3668–3673]. Almost every statement below therefore comes in a holomorphic and an antiholomorphic copy, of which only the first will be written.

**4.2 The stress tensor****4.2.1 Definition, conservation and tracelessness**

The stress tensor is the set of Noether currents of translations. A quick way to obtain it: couple the theory to a background metric. A coordinate change  $\delta\sigma^\alpha = \epsilon^\alpha(\sigma)$  together with the metric change  $\delta g_{\alpha\beta} = \partial_\alpha \epsilon_\beta + \partial_\beta \epsilon_\alpha$  is a diffeomorphism and leaves the action  $S$  invariant. Hence the change of  $S$  under the coordinate change alone is minus its change

under the metric variation alone,  $\delta S = -2 \int d^2\sigma (\partial S / \partial g_{\alpha\beta}) \partial_\alpha \epsilon_\beta$ . This has the form  $\int J^\alpha \partial_\alpha \epsilon$  that defines a Noether current. With the normalization standard in string theory,

$$T_{\alpha\beta} = -\frac{4\pi}{\sqrt{g}} \frac{\partial S}{\partial g^{\alpha\beta}}, \quad \partial^\alpha T_{\alpha\beta} = 0 \quad \text{on a flat worldsheet} \quad (4.5)$$

[Tong §4.1.1, ll. 3680–3739]. In a conformal theory a uniform rescaling  $\delta g_{\alpha\beta} = \epsilon g_{\alpha\beta}$  is a symmetry. The corresponding variation of the action is  $\delta S = -\frac{1}{4\pi} \int d^2\sigma \sqrt{g} \epsilon T^\alpha_\alpha$ , and it must vanish for all  $\epsilon$ :

$$T^\alpha_\alpha = 0. \quad (4.6)$$

[Tong §4.1.1, ll. 3740–3770]. Tracelessness holds classically in many theories and is usually lost on quantization, because regularization introduces a scale. Section 4.6 shows how much of it survives in two dimensions.

In complex coordinates (4.6) reads  $T_{z\bar{z}} = 0$  (since  $T^\alpha_\alpha = 4T_{z\bar{z}}$  with the metric (4.3)), and conservation becomes

$$\bar{\partial} T_{zz} = 0, \quad \partial T_{\bar{z}\bar{z}} = 0, \quad (4.7)$$

so that  $T(z) \equiv T_{zz}$  is holomorphic and  $\bar{T}(\bar{z}) \equiv T_{\bar{z}\bar{z}}$  is antiholomorphic [Tong §4.1.1, ll. 3782–3795]. To see (4.7) directly, write  $\partial^\alpha T_{\alpha z} = 2(\bar{\partial} T_{zz} + \partial T_{\bar{z}z})$  and drop the second term using  $T_{z\bar{z}} = 0$ .

#### 4.2.2 One current for each conformal map

For the infinitesimal map  $z' = z + \epsilon(z)$  the same argument, with  $\epsilon$  promoted to a function  $\epsilon(z, \bar{z})$ , gives

$$\delta S = \frac{1}{2\pi} \int d^2z [T_{zz} \partial_{\bar{z}} \epsilon + T_{\bar{z}\bar{z}} \partial_z \bar{\epsilon}] \quad (4.8)$$

[Tong §4.1.2, ll. 3796–3826]. The step from a general  $T_{\alpha\beta}$  to (4.8) uses  $T_{z\bar{z}} = 0$ ; this is where conformal invariance enters. If  $\epsilon$  is holomorphic the variation vanishes: the map is a symmetry. The associated conserved current is read off in the usual way and is

$$J^z = 0, \quad J^{\bar{z}} = T(z) \epsilon(z), \quad (4.9)$$

which is itself holomorphic, a much stronger property than  $\partial_\alpha J^\alpha = 0$  [Tong §4.1.2, ll. 3836–3856]. Since  $\epsilon(z)$  is arbitrary there are infinitely many conserved currents: expanding  $\epsilon(z) = \sum_n \epsilon_n z^{n+1}$  gives one charge for each integer  $n$ , the Virasoro generators  $L_n$  of Section 4.7.

#### 4.2.3 The example that accompanies the chapter: the free scalar

The Euclidean action of one free scalar field  $X(\sigma)$  is

$$S = \frac{1}{4\pi\alpha'} \int d^2\sigma \partial_\alpha X \partial^\alpha X = \frac{1}{2\pi\alpha'} \int d^2z \partial X \bar{\partial} X, \quad (4.10)$$

the Polyakov action of Chapter 2 for a single coordinate  $X$  in conformal gauge, with the overall sign changed by the Euclidean continuation [Tong §4.1.3, ll. 3864–3876]. (The second form follows from  $\partial_\alpha X \partial^\alpha X = 4 \partial X \bar{\partial} X$  and  $d^2\sigma = \frac{1}{2} d^2z$ .) Under  $\sigma \rightarrow \lambda\sigma$  the two derivatives contribute  $\lambda^{-2}$  and the measure  $\lambda^2$ , so the action is scale invariant; any polynomial potential for  $X$  would break this [Tong §4.1.3, ll. 3884–3900]. From (4.5),

$$T_{\alpha\beta} = -\frac{1}{\alpha'} \left( \partial_\alpha X \partial_\beta X - \frac{1}{2} \delta_{\alpha\beta} (\partial X)^2 \right), \quad T = -\frac{1}{\alpha'} \partial X \partial X, \quad \bar{T} = -\frac{1}{\alpha'} \bar{\partial} X \bar{\partial} X \quad (4.11)$$

[Tong §4.1.3, ll. 3903–3925]. The trace vanishes because  $\delta_\alpha^\alpha = 2$ . The equation of motion  $\partial \bar{\partial} X = 0$  has the general solution  $X(z, \bar{z}) = X(z) + \bar{X}(\bar{z})$ , and on it  $T$  is holomorphic, in agreement with (4.7).

### 4.3 Operator products and Ward identities

#### 4.3.1 The operator product expansion

In a CFT the word *field* or *local operator* means any local expression, not only the variables of the path integral:  $X$ ,  $\partial^n X$ ,  $e^{ikX}$ , products of these. Let  $\{\mathcal{O}_i\}$  be the set of all local operators. The *operator product expansion* (OPE) states that two operators at nearby points can be replaced by a sum of single operators,

$$\mathcal{O}_i(z, \bar{z}) \mathcal{O}_j(w, \bar{w}) = \sum_k C_{ij}^k(z-w, \bar{z}-\bar{w}) \mathcal{O}_k(w, \bar{w}) \quad (4.12)$$

[Tong §4.2.1, ll. 3943–3966]. Three conventions must be kept in mind. (i) The equation holds inside time-ordered correlation functions  $\langle \mathcal{O}_i \mathcal{O}_j \dots \rangle$  with arbitrary other insertions that stay farther from  $w$  than  $z$  does; inside a correlator bosonic operators commute. (ii) In a CFT the expansion converges, with radius equal to the distance to the nearest other insertion [Tong §4.2.1, ll. 4006–4012]; the reason is given in Section 4.9. (iii) Only the terms singular as  $z \rightarrow w$  will matter: they carry the same information as commutation relations and as transformation laws. One writes “ $\sim$ ” when the regular terms are omitted.

#### 4.3.2 Ward identities

Noether’s theorem in the quantum theory is expressed by Ward identities. Let  $\phi \rightarrow \phi + \epsilon \delta\phi$  be a symmetry of  $\mathcal{D}\phi e^{-S}$  and promote  $\epsilon$  to a function  $\epsilon(\sigma)$ . To first order the integrand changes by  $-\frac{1}{2\pi} \int J^\alpha \partial_\alpha \epsilon$ , which defines the current  $J^\alpha$ . A change of integration variable cannot change the integral, so  $\langle \partial_\alpha J^\alpha \dots \rangle = 0$  as long as the other insertions lie outside the support of  $\epsilon$ . If instead  $\epsilon$  is constant on a region containing the insertion point  $\sigma_1$  of  $\mathcal{O}_1$  and zero outside, the operator itself changes, and the same manipulation gives

$$-\frac{1}{2\pi} \int_\epsilon d^2\sigma \partial_\alpha \langle J^\alpha(\sigma) \mathcal{O}_1(\sigma_1) \dots \rangle = \langle \delta \mathcal{O}_1(\sigma_1) \dots \rangle, \quad (4.13)$$

the integral running over the region where  $\epsilon \neq 0$  [Tong §4.2.2, ll. 4018–4189].

In two dimensions the left side is a boundary integral by Stokes’ theorem. For the conformal currents (4.9) the integrand  $J_z(z) \mathcal{O}_1(w)$  is holomorphic in  $z$  away from  $w$ , so the contour integral is  $2\pi i$  times a residue, where “residue” means the coefficient of  $(z-w)^{-1}$  in the OPE. The Ward identity for  $\delta z = \epsilon(z)$  becomes

$$\delta \mathcal{O}(w, \bar{w}) = -\text{Res}_{z \rightarrow w} [\epsilon(z) T(z) \mathcal{O}(w, \bar{w})] \quad (4.14)$$

[Tong §4.2.2, ll. 4196–4250], and similarly for  $\bar{\epsilon}$  and  $\bar{T}$ . This is the working tool of the chapter. Read from right to left: the singular part of the OPE of  $T$  with an operator determines how the operator transforms under every conformal map. Read from left to right: if the transformation is known, part of the OPE is known.

### 4.4 Primary operators and their weights

#### 4.4.1 Weights from translations, rotations and dilatations

Under a translation  $\delta z = \epsilon$  (constant) every operator changes by  $\delta \mathcal{O} = -\epsilon \partial \mathcal{O}$ . By (4.14) the  $T\mathcal{O}$  OPE therefore always contains the term  $\partial \mathcal{O}(w)/(z-w)$  [Tong §4.2.3, ll. 4259–4280]. Next take  $\delta z = \epsilon z$ ,  $\delta \bar{z} = \bar{\epsilon} \bar{z}$ : a dilatation for real  $\epsilon$ , a rotation for imaginary  $\epsilon$ . As with commuting observables in quantum mechanics, one can choose a basis of local operators that are eigenvectors of both.



**Definition 4.1.** An operator  $\mathcal{O}$  has *weight*  $(h, \tilde{h})$  if under  $\delta z = \epsilon z$ ,  $\delta \bar{z} = \bar{\epsilon} \bar{z}$  it transforms as  $\delta \mathcal{O} = -\epsilon(h \mathcal{O} + z \partial \mathcal{O}) - \bar{\epsilon}(\tilde{h} \mathcal{O} + \bar{z} \bar{\partial} \mathcal{O})$  [Tong §4.2.3, ll. 4307–4312]. Its *scaling dimension* and *spin* are

$$\Delta = h + \tilde{h}, \quad s = h - \tilde{h}. \quad (4.15)$$

The derivative terms are present for any operator (they come from moving the argument); the terms  $h\mathcal{O}$ ,  $\tilde{h}\mathcal{O}$  are the eigenvalues. For  $\epsilon = \bar{\epsilon} = \lambda$  real the eigenvalue part is  $-\lambda(h + \tilde{h})\mathcal{O}$ : the operator scales like  $(\text{length})^{-\Delta}$ , the usual engineering dimension. For  $\epsilon = -\bar{\epsilon} = i\theta$  it is  $-i\theta(h - \tilde{h})\mathcal{O}$ , the phase of an object of spin  $s$  under a rotation by  $\theta$  [Tong §4.2.3, ll. 4318–4345]. The current for  $\delta z = \epsilon z$  is  $zT(z)$  by (4.9); its residue with  $\mathcal{O}(w)$  picks out the  $(z - w)^{-2}$  term of the  $T\mathcal{O}$  OPE. Therefore

$$T(z) \mathcal{O}(w, \bar{w}) = \dots + \frac{h \mathcal{O}(w, \bar{w})}{(z - w)^2} + \frac{\partial \mathcal{O}(w, \bar{w})}{z - w} + \dots \quad (4.16)$$

#### 4.4.2 Primary operators

**Definition 4.2.** A *primary operator* is an operator for which (4.16) has no term more singular than  $(z - w)^{-2}$ , and likewise for  $\bar{T}$ :

$$T(z) \mathcal{O}(w, \bar{w}) = \frac{h \mathcal{O}(w, \bar{w})}{(z - w)^2} + \frac{\partial \mathcal{O}(w, \bar{w})}{z - w} + \text{non-singular} \quad (4.17)$$

[Tong §4.2.3, ll. 4363–4375].

For a primary operator the whole singular part is known, so (4.14) gives its change under *any* conformal map. Expand  $\epsilon(z) = \epsilon(w) + \epsilon'(w)(z - w) + \dots$  and collect the coefficient of  $(z - w)^{-1}$ :

$$\delta \mathcal{O}(w, \bar{w}) = -h \epsilon'(w) \mathcal{O}(w, \bar{w}) - \epsilon(w) \partial \mathcal{O}(w, \bar{w}). \quad (4.18)$$

Integrating this to a finite map  $z \rightarrow \tilde{z}(z)$  gives

$$\tilde{\mathcal{O}}(\tilde{z}, \tilde{\bar{z}}) = \left( \frac{\partial \tilde{z}}{\partial z} \right)^{-h} \left( \frac{\partial \tilde{\bar{z}}}{\partial \bar{z}} \right)^{-\tilde{h}} \mathcal{O}(z, \bar{z}) \quad (4.19)$$

[Tong §4.2.3, ll. 4376–4425]. A primary operator of weight  $(h, \tilde{h})$  thus transforms like a tensor with  $h$  lower  $z$  indices and  $\tilde{h}$  lower  $\bar{z}$  indices, except that  $h$  and  $\tilde{h}$  need not be integers.

The list of weights of the primary operators is the basic datum of a CFT: for a string it encodes the mass spectrum (Section 4.9), in statistical physics the critical exponents [Tong §4.2.3, ll. 4426–4431].

## 4.5 The free boson

### 4.5.1 Propagator

All the structure above can be computed explicitly for the free scalar (4.10). The path integral of a total functional derivative vanishes:

$$0 = \int \mathcal{D}X \frac{\delta}{\delta X(\sigma)} \left[ e^{-S} X(\sigma') \right] = \int \mathcal{D}X e^{-S} \left[ \frac{1}{2\pi\alpha'} \partial^2 X(\sigma) X(\sigma') + \delta(\sigma - \sigma') \right],$$

where  $\partial^2 = \partial_\alpha \partial^\alpha$  and  $\delta S / \delta X = -\partial^2 X / (2\pi\alpha')$ . Hence  $\langle \partial^2 X(\sigma) X(\sigma') \rangle = -2\pi\alpha' \delta(\sigma - \sigma')$  [Tong §4.3.1, ll. 4463–4484]. The Green function of the two-dimensional Laplacian obeys  $\partial^2 \ln(\sigma -$

$\sigma')^2 = 4\pi \delta(\sigma - \sigma')$ ; one checks this by integrating over a disc and using Stokes' theorem, which gives  $2 \oint r^2 d\theta / r^2 = 4\pi$  [Tong §4.3.1, ll. 4490–4524]. Therefore

$$\langle X(\sigma)X(\sigma') \rangle = -\frac{\alpha'}{2} \ln(\sigma - \sigma')^2, \quad X(z)X(w) \sim -\frac{\alpha'}{2} \ln(z - w), \quad (4.20)$$

the second formula being the OPE of the holomorphic parts. The logarithm means that  $X$  itself has no well-defined weight. Its derivative does:

$$\partial X(z) \partial X(w) \sim -\frac{\alpha'}{2} \frac{1}{(z - w)^2} \quad (4.21)$$

[Tong §4.3.1, ll. 4545–4566]. (Differentiate  $-\frac{\alpha'}{2} \ln(z - w)$  once in  $z$  and once in  $w$ .) The growth of (4.20) at large separation means that the wave function of  $X$  spreads over its whole target; this is why there are no Goldstone bosons in two dimensions [Tong §4.3.2, ll. 4572–4608].

### 4.5.2 Normal ordering and Wick's theorem

The product  $\partial X(z) \partial X(z)$  in (4.11) is singular by (4.21). Define the normal-ordered stress tensor by subtracting the singularity,

$$T(z) = -\frac{1}{\alpha'} : \partial X \partial X : (z) \equiv -\frac{1}{\alpha'} \lim_{z' \rightarrow z} \left( \partial X(z') \partial X(z) - \langle \partial X(z') \partial X(z) \rangle \right), \quad (4.22)$$

so that  $\langle T \rangle = 0$  [Tong §4.3.3, ll. 4609–4636]. Products of normal-ordered operators are evaluated by Wick's theorem: sum over all ways of replacing pairs of fields, one from each factor, by the propagator (a *contraction*), leaving the rest normal ordered.

**Proposition 4.3** ([L]).  $\partial X$  is a primary operator of weight  $(1, 0)$ .

*Proof.* In  $T(z) \partial X(w) = -\frac{1}{\alpha'} : \partial X(z) \partial X(z) : \partial X(w)$  there are two ways to make one contraction, each giving (4.21):

$$T(z) \partial X(w) \sim -\frac{2}{\alpha'} \partial X(z) \left( -\frac{\alpha'}{2} \frac{1}{(z - w)^2} \right) = \frac{\partial X(z)}{(z - w)^2} = \frac{\partial X(w)}{(z - w)^2} + \frac{\partial^2 X(w)}{z - w} + \dots,$$

after Taylor expanding  $\partial X(z)$  about  $w$ . This is (4.17) with  $h = 1$  [Tong §4.3.3, ll. 4638–4690].  $\square$

By contrast  $\partial^2 X$  has weight  $(2, 0)$  but is not primary: differentiating in  $w$  produces a term  $2 \partial X(w)/(z - w)^3$  [Tong §4.3.3, ll. 4691–4705].

**Proposition 4.4** ([L]). The operator  $:e^{ikX}:$  is primary of weight  $h = \tilde{h} = \alpha' k^2/4$ .

*Proof.* First contract  $\partial X(z)$  with the exponential. Expanding the exponential, a single contraction with one of the  $n$  factors of  $X(w)$  in  $:X(w)^n:$  gives  $n :X^{n-1}: \times (-\frac{\alpha'}{2}) \frac{1}{z-w}$ , and resumming,

$$\partial X(z) :e^{ikX(w)}: \sim -\frac{i\alpha' k}{2} \frac{:e^{ikX(w)}:}{z - w}. \quad (4.23)$$

In  $T(z) :e^{ikX(w)}:$  both  $\partial X(z)$  factors can be contracted, giving  $-\frac{1}{\alpha'} \left( -\frac{i\alpha' k}{2} \right)^2 (z - w)^{-2} = \frac{\alpha' k^2}{4} (z - w)^{-2}$ , or only one of the two, giving  $-\frac{2}{\alpha'} \left( -\frac{i\alpha' k}{2} \right) : \partial X(z) e^{ikX(w)} : / (z - w) = ik : \partial X e^{ikX} : (w) / (z - w) + \dots$ , which is  $\partial_w :e^{ikX(w)}: / (z - w)$ . Hence

$$T(z) :e^{ikX(w)}: \sim \frac{\alpha' k^2}{4} \frac{:e^{ikX(w)}:}{(z - w)^2} + \frac{\partial_w :e^{ikX(w)}:}{z - w} \quad (4.24)$$

[Tong §4.3.3, ll. 4719–4800]. The antiholomorphic computation is identical.  $\square$

The weight  $\alpha'k^2/4$  is a quantum effect: classically  $e^{ik\cdot X}$  has dimension zero, and the result is proportional to  $\alpha'$ , which stands in the action where  $\hbar$  would. Because  $k$  is continuous the spectrum of weights is continuous, a consequence of the non-compact range of  $X$  [Tong §4.3.3, ll. 4722–4730].

**Example 4.5** (The tachyon from a weight). Chapter 5 will show that an operator can be integrated over the worldsheet in a conformally invariant way only if its weight is  $(1, 1)$ , because  $d^2z$  has weight  $(-1, -1)$ . Apply this to  $D$  free bosons  $X^\mu$  and the operator  $:e^{ik\cdot X}:$ , whose weight is  $h = \tilde{h} = \alpha'k^2/4$  by Proposition 4.4 applied to each direction. The condition  $h = 1$  is

$$\frac{\alpha'k^2}{4} = 1 \iff M^2 \equiv -k^2 = -\frac{4}{\alpha'}.$$

This is the mass of the closed-string tachyon found in Chapter 3 from  $M^2 = \frac{2}{\alpha'}(N + \tilde{N} - 2)$  at  $N = \tilde{N} = 0$ ; in units  $\alpha' = 1$ ,  $M^2 = -4$ . Similarly  $:\partial X^\mu \bar{\partial} X^\nu e^{ik\cdot X}:$  has weight  $(1 + \alpha'k^2/4, 1 + \alpha'k^2/4)$ , which is  $(1, 1)$  exactly when  $k^2 = 0$ : the massless level (graviton,  $B$ -field, dilaton), once a polarization  $\zeta_{\mu\nu}$  with  $k^\mu \zeta_{\mu\nu} = 0$  is included so that the operator is primary [Tong §5.4, ll. 7050–7095]. The mass-shell condition of spacetime is a statement about conformal weights on the worldsheet.

### 4.5.3 The $TT$ OPE of the free boson

The same technique applied to two stress tensors gives

$$\begin{aligned} T(z)T(w) &= \frac{1}{\alpha'^2} : \partial X(z) \partial X(z) :: \partial X(w) \partial X(w) : \\ &\sim \frac{2}{\alpha'^2} \left( -\frac{\alpha'}{2} \frac{1}{(z-w)^2} \right)^2 + \frac{4}{\alpha'^2} \left( -\frac{\alpha'}{2} \right) \frac{\partial X(z) \partial X(w)}{(z-w)^2}. \end{aligned} \quad (4.25)$$

The factor 2 counts the two ways of making two contractions, the factor 4 the four ways of making one. The first term is  $\frac{1/2}{(z-w)^4}$ . In the second, expand  $\partial X(z) = \partial X(w) + (z-w)\partial^2 X(w) + \dots$  and recognize  $-\frac{2}{\alpha'} : \partial X \partial X := 2T$  and  $-\frac{2}{\alpha'} : \partial^2 X \partial X := \partial T$ :

$$T(z)T(w) \sim \frac{1/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w} \quad (4.26)$$

[Tong §4.3.3, ll. 4802–4853]. So  $T$  has weight  $(2, 0)$  but is *not* primary: the quartic pole is an obstruction. The number multiplying it is the subject of the next section.

## 4.6 The central charge

### 4.6.1 The general form of the $TT$ OPE

In any CFT the stress tensor has dimension  $\Delta = 2$  (its integral over space is an energy) and spin  $s = 2$  (it is a symmetric two-index tensor), hence weight  $(2, 0)$  by (4.15). The terms  $2T/(z-w)^2 + \partial T/(z-w)$  are therefore always present. Every term in the  $TT$  OPE has total dimension 4, so a term  $\mathcal{O}_n/(z-w)^n$  requires an operator of dimension  $4-n$ . In a unitary theory no operator has negative weight (Section 4.7.4), so  $n \leq 4$ , and the operator multiplying  $(z-w)^{-4}$  has dimension zero: it is a constant. One writes

$$T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}, \quad \bar{T}(\bar{z})\bar{T}(\bar{w}) \sim \frac{\bar{c}/2}{(\bar{z}-\bar{w})^4} + \dots \quad (4.27)$$

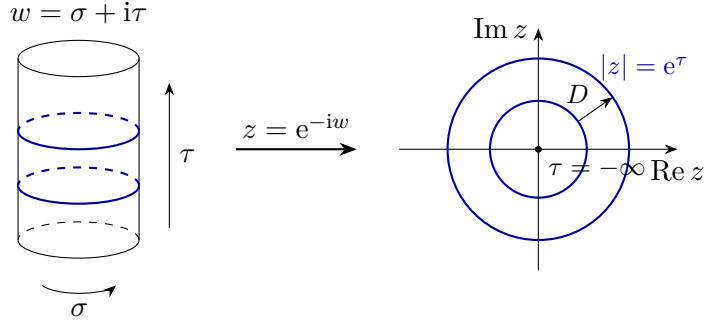


Figure 4.1: The conformal map from the cylinder to the plane. Circles of constant time  $\tau$  on the cylinder (the closed string at a given instant) become circles of constant radius; the infinite past becomes the origin; time translation becomes the dilatation  $D$ .

and calls  $c, \tilde{c}$  the (left and right) *central charges* [Tong §4.4, ll. 4855–4922]. A cubic pole is absent because the OPE must be symmetric under  $z \leftrightarrow w$ . The simple pole, also odd, passes this test only because Taylor expanding  $T(z)$  and  $\partial T(z)$  about  $w$  trades it against the double pole [Tong §4.4, ll. 4923–4956].

Comparing (4.27) with (4.26): *one free boson has  $c = \tilde{c} = 1$* . For  $D$  non-interacting bosons,  $T = \sum_{\mu} T^{(\mu)}$  and cross terms have no contractions, so the quartic poles add and  $c = D$ . More generally, for two decoupled CFTs the central charges add. The central charge counts degrees of freedom, in a weighted sense: it need not be an integer [Tong §4.4, ll. 4915–4922]. A free Majorana fermion has  $c = \frac{1}{2}$  [Tong §5.3.1, ll. 6886–6888], a value this book quotes and does not derive.

#### 4.6.2 How the stress tensor transforms

Inserting (4.27) into the Ward identity (4.14) and expanding  $\epsilon(z)$  to third order about  $w$ , the residue collects  $\epsilon \partial T$  from the simple pole,  $2\epsilon' T$  from the double pole and  $\frac{1}{6}\epsilon''' \cdot \frac{c}{2}$  from the quartic pole:

$$\delta T(w) = -\epsilon(w) \partial T(w) - 2\epsilon'(w) T(w) - \frac{c}{12} \epsilon'''(w). \quad (4.28)$$

The first two terms are the law (4.18) for  $h = 2$ ; the last is new and does not depend on  $T$ . The finite form is

$$\tilde{T}(\tilde{z}) = \left(\frac{\partial \tilde{z}}{\partial z}\right)^{-2} \left[ T(z) - \frac{c}{12} S(\tilde{z}, z) \right], \quad S(\tilde{z}, z) = \frac{\tilde{z}'''}{\tilde{z}'} - \frac{3}{2} \left( \frac{\tilde{z}''}{\tilde{z}'} \right)^2, \quad (4.29)$$

with primes denoting  $\partial_z$ ;  $S$  is the *Schwarzian derivative* [Tong §4.4, ll. 4957–5016]. For  $\tilde{z} = z + \epsilon$  one has  $S = \epsilon''' + O(\epsilon^2)$ , which reproduces (4.28) (symbolic check). Two properties of  $S$  will be used: it vanishes identically for  $\tilde{z} = (az + b)/(cz + d)$  (Section 4.8), and  $S(e^{-iw}, w) = \frac{1}{2}$  (both are symbolic checks; the second is also stated in [Tong §4.4.1, l. 5044]).

#### 4.6.3 Worked example: the Casimir energy and the intercept

A closed string of coordinate length  $2\pi$  sweeps out a cylinder,  $w = \sigma + i\tau$  with  $\sigma \sim \sigma + 2\pi$ . The map  $z = e^{-iw}$  sends it to the plane (Figure 4.1):  $|z| = e^\tau$ , so equal-time circles become concentric circles and  $\tau \rightarrow -\infty$  becomes  $z = 0$  [Tong §4.4.1, ll. 5018–5043]. Apply (4.29) with  $w$  in the role of  $\tilde{z}$ :  $T_{\text{cyl}}(w) = (\partial w / \partial z)^{-2} [T_{\text{plane}}(z) - \frac{c}{12} S(w, z)]$ . With  $\partial w / \partial z = i/z$  one has  $(\partial w / \partial z)^{-2} = -z^2$  and  $S(w, z) = \frac{1}{2z^2}$  (symbolic check), so

$$T_{\text{cyl}}(w) = -z^2 T_{\text{plane}}(z) + \frac{c}{24} \quad (4.30)$$

[Tong §4.4.1, ll. 5044–5050]. Suppose the vacuum energy vanishes on the plane,  $\langle T_{\text{plane}} \rangle = 0$ . On the cylinder the Hamiltonian is  $H = \int d\sigma T_{\tau\tau} = - \int d\sigma (T_{ww} + \bar{T}_{\bar{w}\bar{w}})$ , and (4.30) gives a ground state energy

$$\frac{E_0}{2\pi} = -\frac{c + \tilde{c}}{24} \quad (4.31)$$

[Tong §4.4.1, ll. 5051–5076] (the  $2\pi$  comes from the normalization of (4.5)). This is a Casimir energy: it exists because the cylinder has a finite circumference, and it is fixed by  $c$  alone.

Now put in numbers. For one boson  $c = \tilde{c} = 1$  and  $E_0/2\pi = -\frac{1}{12}$ . In the light-cone quantization of Chapter 3 the closed string has 24 transverse bosons, so  $c = \tilde{c} = 24$  and

$$\frac{E_0}{2\pi} = -\frac{24 + 24}{24} = -2.$$

This is the  $-2$  in  $M^2 = \frac{2}{\alpha'}(N + \tilde{N} - 2)$ . In Chapter 3 it arose from the regularized sum  $\sum_{n \geq 1} n \rightarrow -\frac{1}{12}$ :  $2 \times 24 \times \frac{1}{2} \times (-\frac{1}{12}) = -2$  for two sectors and 24 directions. Here no divergent sum was manipulated; the number follows from the quartic pole of the  $TT$  OPE and the Schwarzian of the exponential map [Tong §4.4.1, ll. 5072–5076].

#### 4.6.4 The Weyl anomaly and why $c$ decides consistency

The central charge has a second, related meaning. On a curved worldsheet with Ricci scalar  $\mathcal{R}$  the trace of the stress tensor no longer vanishes:

$$\langle T^\alpha_\alpha \rangle = -\frac{c}{12} \mathcal{R} \quad (4.32)$$

in every state [Tong §4.4.2, ll. 5094–5110 and 5262–5272]; the coefficient follows from a contact term implied by (4.27). On a fixed flat background this is harmless. For a string the Weyl rescaling is a gauge symmetry, (4.32) says it is anomalous, and the theory is consistent only if the total central charge of all worldsheet fields, including the gauge-fixing ghosts with  $c = -26$ , vanishes. This is the origin of  $D = 26$  and, more accurately, of the statement that the “matter” CFT must have  $c = 26$  whatever its geometric interpretation [Tong §5.3, ll. 6832–6870]; see Chapter 5.

**Example 4.6** (Central charge bookkeeping for K3). For the superstring the corresponding requirement is  $c = 15$  for the matter fields, met by ten bosons and ten Majorana fermions:  $10(1 + \frac{1}{2}) = 15$  [Tong §5.3.1, ll. 6878–6892]. Compactify on a space of four real dimensions, leaving  $\mathbb{R}^{1,5}$ . The six non-compact directions contribute  $6 \times \frac{3}{2} = 9$ ; the internal theory must supply  $15 - 9 = 6$ . If the internal space is a flat torus  $T^4$  this is visibly  $4 \cdot 1 + 4 \cdot \frac{1}{2} = 6$ . The sigma model on K3 (Chapter 23) is interacting, but it has the same  $c = \tilde{c} = 6$ : on the orbifold  $T^4/\mathbb{Z}_2$  of Chapter 21 the fields are still free, and  $c$  cannot change under a continuous deformation preserving conformal invariance (standard; not checked against a source in this book’s library). This arithmetic is what `central_charge_k3_eq_six` records (Section 4.10).

## 4.7 The Virasoro algebra

### 4.7.1 Radial quantization and the generators

States of a closed string live on circles  $\tau = \text{const}$  of the cylinder, hence on circles  $|z| = \text{const}$  of the plane, and time evolution  $H = \partial_\tau$  becomes the dilatation  $D = z\partial + \bar{z}\bar{\partial}$ . This is *radial quantization*; time ordering becomes radial ordering, operators at larger  $|z|$  standing to the left [Tong §4.5.1, ll. 5408–5448].

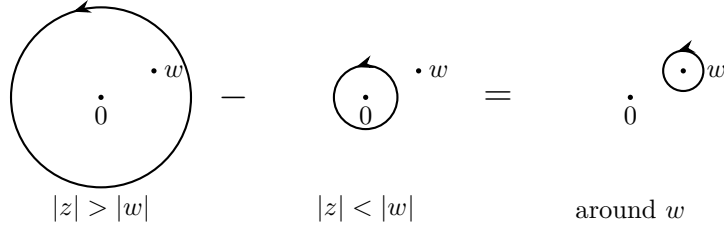


Figure 4.2: The  $z$  contours in  $[L_m, L_n]$  at fixed  $w$ : outside  $w$  for  $L_m L_n$ , inside for  $L_n L_m$ . The difference is a small circle around  $w$ , so only the singular part of the OPE contributes.

On the cylinder one Fourier expands  $T_{\text{cyl}}(w) = -\sum_m L_m e^{imw} + \frac{c}{24}$ . By (4.30) this is a Laurent expansion on the plane,

$$T(z) = \sum_{m \in \mathbb{Z}} \frac{L_m}{z^{m+2}}, \quad L_n = \oint \frac{dz}{2\pi i} z^{n+1} T(z), \quad (4.33)$$

and similarly  $\tilde{L}_n$  from  $\bar{T}$  [Tong §4.5.1, ll. 5450–5482]. By (4.9),  $L_n$  is the conserved charge of the conformal map  $\delta z = z^{n+1}$ ; the contour is a constant-time slice. In particular  $L_{-1}$ ,  $\tilde{L}_{-1}$  generate translations,  $L_0$ ,  $\tilde{L}_0$  dilatations and rotations,  $D = L_0 + \tilde{L}_0$ , and a state with  $L_0 = h$ ,  $\tilde{L}_0 = \tilde{h}$  has energy on the cylinder

$$\frac{E}{2\pi} = h + \tilde{h} - \frac{c + \tilde{c}}{24} \quad (4.34)$$

[Tong §4.5.1, ll. 5498–5514; §4.5.3, ll. 5668–5676]. The  $L_n$  are the quantum versions of the Virasoro constraints of Chapters 2 and 3.

#### 4.7.2 From the OPE to the commutator

Write  $L_m$  as a contour integral over  $z$  and  $L_n$  over  $w$ . In  $L_m L_n$  radial ordering requires  $|z| > |w|$ , in  $L_n L_m$  it requires  $|z| < |w|$ . At fixed  $w$  the difference of the two  $z$  contours can be deformed to a small circle around  $w$  (Figure 4.2), so

$$[L_m, L_n] = \oint \frac{dw}{2\pi i} w^{n+1} \text{Res}_{z \rightarrow w} \left[ z^{m+1} \left( \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w} \right) \right] \quad (4.35)$$

[Tong §4.5.2, ll. 5515–5600]. Expand  $z^{m+1}$  about  $w$ :

$$z^{m+1} = w^{m+1} + (m+1)w^m(z-w) + \frac{1}{2}m(m+1)w^{m-1}(z-w)^2 + \frac{1}{6}m(m+1)m(m-1)w^{m-2}(z-w)^3 + \dots$$

The residue receives  $w^{m+1}\partial T$  from the simple pole,  $2(m+1)w^m T$  from the double pole, and  $\frac{c}{2} \cdot \frac{1}{6}m(m+1)m(m-1)w^{m-2} = \frac{c}{12}m(m^2-1)w^{m-2}$  from the quartic pole. Integrate the first term by parts,  $\oint w^{m+n+2}\partial T = -(m+n+2)\oint w^{m+n+1}T$ , and use (4.33):

$$[L_m, L_n] = [-(m+n+2) + 2(m+1)]L_{m+n} + \frac{c}{12}m(m^2-1)\oint \frac{dw}{2\pi i} w^{m+n-1}.$$

The last integral is  $\delta_{m+n,0}$ . The result is the *Virasoro algebra*

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12}m(m^2-1)\delta_{m+n,0}, \quad [L_m, \tilde{L}_n] = 0, \quad (4.36)$$

with the same algebra and  $\tilde{c}$  for the  $\tilde{L}_n$  [Tong §4.5.2, ll. 5601–5646]. The number  $c$  commutes with everything: it is a *central* term, which is the origin of the name central charge.

Without the central term, (4.36) is the algebra of the vector fields  $l_n = z^{n+1} \partial_z$  generating  $\delta z \propto z^{n+1}$ :  $[l_n, l_m] = (m-n) l_{m+n}$  (Exercise 4.1). The central term arises because a conformal transformation is a coordinate change *followed by a Weyl rescaling*, which is anomalous by (4.32) [Tong §4.5.2, ll. 5647–5661].

**Common pitfall: “the central term is  $\frac{c}{12}m^3$ ”**

The term linear in  $m$  depends on the convention  $\langle T_{\text{plane}} \rangle = 0$  (a shift of  $L_0$  changes it); the cubic term cannot be removed. In the plane convention the central term vanishes for  $m = -1, 0, 1$ , so  $L_{-1}, L_0, L_1$  close without anomaly (Section 4.8). Note also that the identity  $\frac{c}{12}(m^3 - m) = \frac{cm(m^2-1)}{12}$  holds for every number  $c$  and carries no information about any commutator; compare Section 4.10.

### 4.7.3 Representations: primaries and descendants

Let  $|\psi\rangle$  be an eigenstate with  $L_0|\psi\rangle = h|\psi\rangle$ . From (4.36),  $[L_0, L_n] = -nL_n$ , so

$$L_0 L_n |\psi\rangle = (h - n) L_n |\psi\rangle : \quad (4.37)$$

$L_n$  with  $n > 0$  lowers the energy,  $L_{-n}$  raises it. If the energy is bounded below there are states annihilated by all lowering operators,

$$L_n |\psi\rangle = \tilde{L}_n |\psi\rangle = 0 \quad \text{for all } n > 0, \quad (4.38)$$

called *primary* or highest-weight states. Acting on a primary state with  $L_{-n}$ ,  $n > 0$ , gives its *descendants*; at level  $N$  there is one for each partition of  $N$ :  $L_{-1}|\psi\rangle$ ; then  $L_{-1}^2|\psi\rangle$ ,  $L_{-2}|\psi\rangle$ ; and so on. The whole tower is a *Verma module* [Tong §4.5.3, ll. 5662–5707]. The vacuum  $|0\rangle$  has  $h = 0$  and obeys  $L_n|0\rangle = 0$  for all  $n \geq 0$ ; in a unitary theory  $L_{-1}|0\rangle$  has zero norm by (4.39) below and also vanishes. One cannot demand  $L_n|0\rangle = 0$  for all  $n$ , because  $[L_2, L_{-2}]|0\rangle = \frac{c}{2}|0\rangle \neq 0$  [Tong §4.5.3, ll. 5708–5718]. The conditions (4.38) are the ones imposed on physical states in the covariant quantization of Chapter 3, where  $(L_0 - 1)|\text{phys}\rangle = 0$  in addition.

### 4.7.4 Unitarity

Hermiticity of the Hamiltonian on the Minkowski cylinder requires  $L_n^\dagger = L_{-n}$  [Tong §4.5.4, ll. 5743–5760]. Two consequences follow in one line each. For a primary state of weight  $h$ ,

$$\|L_{-1}|\psi\rangle\|^2 = \langle\psi|[L_1, L_{-1}]|\psi\rangle = 2h \langle\psi|\psi\rangle \geq 0 \implies h \geq 0, \quad (4.39)$$

and for the vacuum

$$\|L_{-n}|0\rangle\|^2 = \langle 0|[L_n, L_{-n}]|0\rangle = \frac{c}{12} n(n^2 - 1) \geq 0 \implies c \geq 0, \quad (4.40)$$

with  $c = 0$  only for the trivial theory [Tong §4.5.4, ll. 5764–5777]. These bounds justify the assumption “no negative weights” used to derive (4.27).

## 4.8 The global subgroup $SL(2, \mathbb{C})$ and Möbius maps

### 4.8.1 Which conformal maps are globally defined?

The maps  $\delta z = \epsilon z^{n+1}$  exist for every  $n$  as infinitesimal transformations near a point. On the Riemann sphere  $\mathbb{C} \cup \{\infty\}$ , the worldsheet of a tree-level closed-string amplitude, most of

them are singular. The vector field  $l_n = z^{n+1}\partial_z$  is regular at  $z = 0$  only for  $n \geq -1$ . In the coordinate  $u = 1/z$  near  $z = \infty$  one finds  $l_n = -u^{1-n}\partial_u$ , regular at  $u = 0$  only for  $n \leq 1$ . Only

$$l_{-1} : z \rightarrow z + \alpha, \quad l_0 : z \rightarrow \lambda z, \quad l_1 : z \rightarrow \frac{z}{1 - \beta z}$$

survive, and they combine into the *Möbius transformations*

$$z \mapsto \frac{az + b}{cz + d}, \quad a, b, c, d \in \mathbb{C}, \quad ad - bc = 1 \quad (4.41)$$

[Tong §6.2.1, ll. 7506–7572]. (In this section  $c$  is a matrix entry; the central charge does not appear until Section 4.8.3.) The normalization  $ad - bc = 1$  removes the freedom to rescale all four parameters, leaving three, one for each generator. The group is  $SL(2, \mathbb{C})$ , or more precisely  $SL(2, \mathbb{C})/\mathbb{Z}_2$ , since  $(a, b, c, d)$  and  $(-a, -b, -c, -d)$  give the same map [Tong §6.2.1, ll. 7572–7581]. A Möbius map can move any three points to any three others; this is what is used to fix the residual gauge freedom in string amplitudes.

### 4.8.2 Composition is matrix multiplication

**Proposition 4.7 ([A]).** *Let  $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$  and  $N = \begin{pmatrix} a' & b' \\ c' & d' \end{pmatrix}$  have determinant one and write  $M \cdot z = (az + b)/(cz + d)$ . If  $c'z + d' \neq 0$  and  $c(N \cdot z) + d \neq 0$ , then  $(MN) \cdot z = M \cdot (N \cdot z)$ , and  $\det(MN) = 1$ .*

*Proof.* Put  $N \cdot z = p/q$  with  $p = a'z + b'$ ,  $q = c'z + d' \neq 0$ . Then  $M \cdot (N \cdot z) = \frac{ap/q+b}{cp/q+d} = \frac{ap+bq}{cp+dq}$ , where the denominator  $cp + dq = q(c(N \cdot z) + d)$  is non-zero by hypothesis. Expanding,  $ap + bq = (aa' + bc')z + (ab' + bd')$  and  $cp + dq = (ca' + dc')z + (cb' + dd')$ : the coefficients are the entries of  $MN$ . Finally  $\det(MN) = \det M \det N = 1$ .  $\square$

This is the statement formalized as `mobius_compose` (Section 4.10). The hypotheses are needed: on  $\mathbb{C}$  alone, without the point  $\infty$ , the map is undefined at  $z = -d/c$ . Two further facts are one-line computations:

$$\frac{d}{dz} \frac{az + b}{cz + d} = \frac{ad - bc}{(cz + d)^2} = \frac{1}{(cz + d)^2}, \quad S\left(\frac{az + b}{cz + d}, z\right) = 0. \quad (4.42)$$

By (4.19) and the first of these, a primary operator of weight  $h$  picks up the factor  $(cz + d)^{2h}$  under a Möbius map (or  $(cz + d)^{-2h}$  if one writes the law in the opposite direction, old operator in terms of new). By the second,  $T$  transforms under Möbius maps as if it were primary: the anomaly (4.29) is invisible to the global group. On the algebraic side this is the fact that the central term  $\frac{c}{12}m(m^2 - 1)$  vanishes for  $m \in \{-1, 0, 1\}$ , so that

$$[L_1, L_{-1}] = 2L_0, \quad [L_0, L_{\pm 1}] = \mp L_{\pm 1} \quad (4.43)$$

is an ordinary  $sl(2)$  algebra, and the vacuum, annihilated by all three generators (and by their tilded partners), is  $SL(2, \mathbb{C})$  invariant.

### 4.8.3 Global Ward identities fix the two-point function

Invariance of the vacuum under  $L_{-1}, L_0, L_1$  means that a correlation function does not change when all operators are transformed by  $\epsilon(z) = 1, z, z^2$ . Using (4.18) for primaries  $\mathcal{O}_i$  of weights  $h_i$ , the statement  $\sum_i \langle \mathcal{O}_1 \cdots \delta \mathcal{O}_i \cdots \mathcal{O}_n \rangle = 0$  gives three differential equations for the holomorphic dependence of  $G = \langle \mathcal{O}_1(z_1) \cdots \mathcal{O}_n(z_n) \rangle$ :

$$\sum_i \partial_{z_i} G = 0, \quad \sum_i (z_i \partial_{z_i} + h_i) G = 0, \quad \sum_i (z_i^2 \partial_{z_i} + 2h_i z_i) G = 0. \quad (4.44)$$



These are derived here from (4.18); they are standard, and not checked against a source in this book’s library. For  $n = 2$  the first equation says  $G = G(z_{12})$  with  $z_{12} = z_1 - z_2$ . The second becomes  $z_{12}G' + (h_1 + h_2)G = 0$ , so  $G = C z_{12}^{-(h_1+h_2)}$ . Substituting into the third,

$$-(h_1 + h_2)(z_1 + z_2) + 2h_1z_1 + 2h_2z_2 = (h_1 - h_2)z_{12} = 0,$$

which forces  $h_1 = h_2$  unless  $C = 0$ . Including the antiholomorphic side:

$$\langle \mathcal{O}_1(z_1, \bar{z}_1) \mathcal{O}_2(z_2, \bar{z}_2) \rangle = \frac{C_{12}}{z_{12}^{2h} \bar{z}_{12}^{2\tilde{h}}} \quad \text{if } (h_1, \tilde{h}_1) = (h_2, \tilde{h}_2) = (h, \tilde{h}), \text{ and } 0 \text{ otherwise.} \quad (4.45)$$

The free boson illustrates this: (4.21) is (4.45) with  $h = 1$  and  $C = -\alpha'/2$ . Symmetry fixes the functional form; only the constant  $C_{12}$  is dynamical.

## 4.9 The state–operator map

### 4.9.1 States are local operators at the origin

In a general quantum field theory a state is a functional  $\Psi[\phi(\sigma)]$  on a whole spatial slice, while a local operator lives at a point. In a CFT the two are in one-to-one correspondence, because the infinite past of the cylinder is the single point  $z = 0$  (Figure 4.1).

In path-integral language a state at radius  $r_f$  is obtained by integrating over fields on an annulus  $r_i \leq |z| \leq r_f$ , with boundary value  $\phi_f$  outside and the initial wave functional as a weight inside. As  $r_i \rightarrow 0$  the integral runs over the whole disc, and the only trace of the initial state is a modified weight at  $z = 0$ , which is what a local operator insertion is:

$$\Psi_{\mathcal{O}}[\phi_f; r] = \int^{\phi(r)=\phi_f} \mathcal{D}\phi e^{-S[\phi]} \mathcal{O}(z=0) \quad (4.46)$$

[Tong §4.6, ll. 5779–5875]. The identity operator gives the vacuum. The map requires conformal invariance (it uses the map of Figure 4.1), and it is a bijection between states and *local* operators; the mode operators  $L_n$ ,  $\alpha_n$  are contour integrals, not local operators [Tong §4.6, ll. 5876–5897].

Acting with a Virasoro generator on the state  $|\mathcal{O}\rangle$  means surrounding the origin by the contour in (4.33). For a primary  $\mathcal{O}$  of weight  $h$ , by (4.17),

$$L_n|\mathcal{O}\rangle = \oint \frac{dz}{2\pi i} z^{n+1} \left( \frac{h\mathcal{O}(0)}{z^2} + \frac{\partial\mathcal{O}(0)}{z} + \dots \right), \quad (4.47)$$

from which one reads

$$L_{-1}|\mathcal{O}\rangle = |\partial\mathcal{O}\rangle, \quad L_0|\mathcal{O}\rangle = h|\mathcal{O}\rangle, \quad L_n|\mathcal{O}\rangle = 0 \quad (n > 0) \quad (4.48)$$

[Tong §4.6.1, ll. 5899–5937]. The first holds for every operator, the second for every operator of definite weight, the third exactly for primaries: *primary operators correspond to primary states*, and the weights  $(h, \tilde{h})$  are, by (4.34), the energy and momentum of closed-string states on the cylinder [Tong §4.6.1, ll. 5938–5946]. The map also explains why the OPE converges: the path integral over a disc containing two operators and nothing else yields a state on the boundary circle, and that state is a sum of local operators at the centre [Tong §4.6.1, ll. 5960–5981].

### 4.9.2 The free boson again

On the plane the primary field  $\partial X$  has the mode expansion

$$\partial X(z) = -i\sqrt{\frac{\alpha'}{2}} \sum_{n \in \mathbb{Z}} \frac{\alpha_n}{z^{n+1}}, \quad \alpha_n = i\sqrt{\frac{2}{\alpha'}} \oint \frac{dz}{2\pi i} z^n \partial X(z), \quad \alpha_0 = \sqrt{\frac{\alpha'}{2}} p \quad (4.49)$$

[Tong §4.6.2, ll. 5982–6030], with  $\alpha_0$  related to the momentum  $p$  as in Chapter 3 (the source’s Euclidean convention has an extra factor  $i$ ). Repeating the contour argument of Figure 4.2 with the OPE (4.21),

$$[\alpha_m, \alpha_n] = -\frac{2}{\alpha'} \oint \frac{dw}{2\pi i} \text{Res}_{z \rightarrow w} \left[ z^m w^n \left( -\frac{\alpha'}{2} \right) \frac{1}{(z-w)^2} \right] = m \oint \frac{dw}{2\pi i} w^{m+n-1} = m \delta_{m+n,0}, \quad (4.50)$$

the oscillator algebra obtained by canonical quantization in Chapter 3 [Tong §4.6.2, ll. 6031–6075]. The singular part of an OPE and an equal-time commutator contain the same information.

The dictionary between oscillator states and operators is

$$|0; 0\rangle \leftrightarrow 1, \quad \alpha_{-m}|0; 0\rangle \leftrightarrow \propto \partial^m X(0) \quad (m \geq 1), \quad |0; p\rangle \leftrightarrow :e^{ipX(0)}: \quad (4.51)$$

[Tong §4.6.2, ll. 6076–6189]. For the first entry: fields contributing to the path integral over the disc are smooth, so  $\oint dw w^m \partial X(w) = 0$  for  $m \geq 0$  and the state created by the identity is annihilated by all  $\alpha_{m \geq 0}$ . For the second,  $\alpha_n$  acting on  $\partial^m X(0)$  gives by (4.21) a contour integral of  $w^{n-m-1}$ , which vanishes unless  $n = m$ . For the third, (4.23) shows that  $:e^{ipX}:$  has  $\alpha_0$  eigenvalue  $\sqrt{\alpha'/2} p$ .

#### Physical picture: looking ahead to the circle

On a circle the left- and right-moving zero modes  $p_L, p_R$  of  $X$  differ when the string winds. The operator  $:e^{ip_L X(z) + ip_R \bar{X}(\bar{z})}:$  has weights  $(\alpha' p_L^2/4, \alpha' p_R^2/4)$  by Proposition 4.4 applied on each side. Its dimension gives the mass formula of Chapter 7; its spin  $h - \bar{h}$  must be an integer, which is level matching and, in Chapter 10, the evenness of the Narain lattice.

## 4.10 What the machine has checked

Four modules of the library `StringTheoryFormalization` carry names from this chapter: three in the folder `Frontier/` (`CentralCharge`, `SL2CSymmetry`, `ChiralPrimaries`) and one in `StringDynamics/` (`VertexOperators`). To formalize the chapter one would need local operators, correlation functions, contour integrals of operator-valued functions, and a proof of (4.36) from (4.27). *None of this is in the library.* What is checked is a set of algebraic fragments: one useful and faithful (Möbius composition), some arithmetic shadows in the sense of Chapter 1 (the number 6), and some placeholders whose names promise more than their statements say. Reading a formal statement for what it literally asserts is a skill this book wants to teach (Chapter 34), so the section goes through them one by one. The files contain no `sorry`, although their header comments still say “in progress”. The theorems whose axioms were inspected for this chapter (`mobius_compose`, `central_charge_k3_eq_six`) depend only on `propext`, `Classical.choice` and `Quot.sound`.

### 4.10.1 Möbius transformations: a genuine algebraic fragment

The module `SL2CSymmetry` contains the one faithful piece of mathematics of this section. A Möbius transformation is a quadruple of complex numbers with  $ad - bc = 1$ , and it acts on  $\mathbb{C}$  by (4.41):

```

structure MobiusTransform where
  (a b c d : ℂ)
  det_one : a * d - b * c = 1

noncomputable def MobiusTransform.act (M : MobiusTransform) (z : ℂ) : ℂ :=
  (M.a * z + M.b) / (M.c * z + M.d)

```

In Lean: `StringTheory.Frontier.mobius_compose`

```

theorem mobius_compose (M N : MobiusTransform) (z : ℂ)
  (h : N.c * z + N.d ≠ 0)
  (h' : M.c * (N.act z) + M.d ≠ 0) :
  (MobiusTransform.mk
    (M.a * N.a + M.b * N.c)
    (M.a * N.b + M.b * N.d)
    (M.c * N.a + M.d * N.c)
    (M.c * N.b + M.d * N.d)
  (by
    have key : (M.a * N.a + M.b * N.c) * (M.c * N.b + M.d * N.d)
      - (M.a * N.b + M.b * N.d) * (M.c * N.a + M.d * N.c)
      = (M.a * M.d - M.b * M.c) * (N.a * N.d - N.b * N.c) := by ring
    rw [key, M.det_one, N.det_one, one_mul])).act z =
  M.act (N.act z) := by ...

```

**Reading.** The theorem builds the quadruple of entries of the matrix product  $MN$ ; the inner `by` block proves that its determinant is again 1 (multiplicativity of the  $2 \times 2$  determinant, by `ring`). The conclusion is  $(MN) \cdot z = M \cdot (N \cdot z)$  under the two non-vanishing hypotheses: exactly Proposition 4.7. A companion, `mobius_id_act`, says that the identity matrix acts trivially.

**Proof idea.** Unfold the action, clear denominators with `field_simp` (which consumes  $h$  and  $h'$ ), and close the polynomial identity with `ring`, as on paper.

**What it does not say.** It does not construct the group  $SL(2, \mathbb{C})/\mathbb{Z}_2$ , its action on the Riemann sphere (the point  $\infty$  is absent, hence the side conditions), or any relation to conformal symmetry. In Lean division by zero returns zero, so the value of `act` at the pole is a convention, not geometry.

The same file defines `primaryTransform h M z` as  $(cz + d)^{-2h}$ , a complex power. By (4.42) this is  $(d\bar{z}/dz)^h$ , the factor of (4.19) written in the direction “old operator in terms of new”. No theorem uses it; for non-integer  $2h$  one would have to handle the branch cut. The file also contains

```

theorem sl2c_fixes_two_point (h : ℚ) :
  ∃ (C : ℂ) (f : ℂ → ℂ → ℂ), ∀ (z w : ℂ) (hzw : z ≠ w),
  f z w = twoPointFunction h C z w hzw := by ...

```

and two declarations `ward_identity_translation`, `ward_identity_dilatation` whose conclusion is the proposition `True`: they assert nothing about (4.44). The theorem displayed says: *there exist* a constant  $C$  and a function  $f$  equal to  $C/(z - w)^{2h}$  off the diagonal. It is proved by taking  $C = 1$  and  $f$  to be that very function. The content of Section 4.8.3 is a *uniqueness* statement (every solution of (4.44) has this form), which is a different and harder theorem. Exercise 4.7 asks for a faithful formulation.

#### 4.10.2 The central charge: arithmetic shadows

The module `CentralCharge` records numbers from Section 4.6, not derivations.

**In Lean: StringTheory.Frontier.central\_charge\_k3\_eq\_six**

```

def centralChargeBoson : ℚ := 1
def centralChargeFermion : ℚ := 1/2
def centralChargeK3 : ℚ :=
  4 * centralChargeBoson + 4 * centralChargeFermion

theorem central_charge_k3_eq_six : centralChargeK3 = 6 := by ...

theorem virasoro_ope_central_term (z w : ℂ) (h : z ≠ w) :
  (centralChargeK3 / 2 : ℚ) = 3 := by ...

theorem virasoro_algebra_commutator (c : ℚ) (m n : ℤ) (L : ℤ → ℤ) :
  c / 12 * (m^3 - m) = c * m * (m^2 - 1) / 12 := by ...

```

**Reading.** Three rational constants are *defined*:  $1$ ,  $\frac{1}{2}$  and  $4 \cdot 1 + 4 \cdot \frac{1}{2}$ . The first theorem evaluates the third constant to 6. The second states  $6/2 = 3$ ; the points  $z, w$  and the hypothesis  $z \neq w$  do not occur in the conclusion. The third is the polynomial identity  $\frac{c}{12}(m^3 - m) = \frac{cm(m^2-1)}{12}$  for all rational  $c$  and integer  $m$ ; the arguments  $n$  and  $L$  do not occur in it, and no commutator is mentioned. (A fourth, `central_charge_from_worldsheet_action`, says that some rational number equals both `centralChargeK3` and 6.) Proofs: `norm_num` and `ring`.

**What it does not say.** That a free boson has  $c = 1$  is the result (4.26) of a Wick contraction; here it is an input, typed in as a definition. Nothing in the file refers to a stress tensor, an OPE, the algebra (4.36) or a K3 surface. The file is an arithmetic shadow of Example 4.6: it certifies the bookkeeping  $4 \cdot 1 + 4 \cdot \frac{1}{2} = 6$ , which is Tier A, and nothing else. Identifying `centralChargeK3` with the central charge of the K3 sigma model is Tier L physics, not a formal statement. The theorem names describe goals, not contents; the file's own documentation says so.

A general lesson: a Lean theorem may list hypotheses that its conclusion never uses, and is then true for a trivial reason; ask of every bound variable where it occurs after the colon. A faithful formalization of (4.36) would define a Lie algebra with basis  $\{L_n\}_{n \in \mathbb{Z}} \cup \{c\}$  and prove the Jacobi identity, which is where  $m^3 - m$  is forced (Exercise 4.4).

### 4.10.3 Chiral primaries and vertex operators

The K3 sigma model has more symmetry than the Virasoro algebra: its algebra contains a  $U(1)$  current, and states carry a charge  $q$  in addition to the weight  $h$ . Unitary representations obey  $h \geq |q|/2$ , and the states saturating the bound with  $q \geq 0$ , the *chiral primaries*, are counted by the Dolbeault cohomology  $H^{q,0}$  of the target. (Standard; not checked against a source in this book's library. See Chapter 28.) In `ChiralPrimaries` an `N2State` is a pair of rationals `confWeight`, `u1Charge`; it is not a vector in a representation. The definitions and the main theorem are

```

def bpsBound (s : N2State) : Prop :=
  s.confWeight ≥ |s.u1Charge| / 2
def isChiralPrimary (s : N2State) : Prop :=
  0 ≤ s.u1Charge ∧ s.confWeight = s.u1Charge / 2
theorem chiral_primary_saturates_bps (s : N2State) (h : isChiralPrimary s) :
  bpsBound s := by ...

```

The theorem says: if  $q \geq 0$  and  $h = q/2$  then  $h \geq |q|/2$ . It is one direction, and it is the elementary fact  $|q| = q$  for  $q \geq 0$  (`abs_of_nonneg`); the bound itself is a definition, not a theorem about a superconformal algebra. `K3ChiralPrimaryCount` is a hand-typed table

1, 0, 1, matching the Hodge numbers  $h^{0,0}, h^{1,0}, h^{2,0}$  of K3 (Chapter 19) but not derived from them, and `k3_chiral_primary_total` adds its entries:  $1 + 0 + 1 = 2$ . The declaration `chiral_primary_ring_associativity` has conclusion `True` and must not be quoted as a fact about the chiral ring.

In the module `VertexOperators` a `VertexOperator` is a record of a rational `confWeight`, a natural `level` and a sequence `modeCoeffs` of complex numbers, with no relation imposed between them. The function

```
noncomputable def ope (V1 V2 : VertexOperator) (z w : ℂ) : ℂ :=
  if z = w then 0
  else (z - w)-1 ^ 2 * (V1.confWeight * V2.confWeight : ℚ).num
```

returns  $(z - w)^{-2}$  times the *numerator* of the rational number  $h_1 h_2$ , and the one theorem, `ope_vacuum_left`, says that this number is 0 when the first argument is `vacuumVertex`, whose weight is 0. This `ope` is not the operator product expansion (4.12): an OPE is operator valued, its leading power for two exponentials is  $(z - w)^{\alpha' k_1 k_2 / 2}$  (Exercise 4.2), not  $(z - w)^{-2}$ , and the numerator of a fraction has no physical meaning. Even the theorem needs care in words: the OPE of the identity with  $V$  is  $V$ ; what vanishes is its *singular part*. Proposition 4.4 is not formalized.

#### 4.10.4 Tier table

| Statement                                                    | Status                                                   | Tier |
|--------------------------------------------------------------|----------------------------------------------------------|------|
| Möbius composition (Proposition 4.7)                         | <code>mobius_compose</code> , <code>mobius_id_act</code> | [A]  |
| $4 \cdot 1 + 4 \cdot \frac{1}{2} = 6$                        | <code>central_charge_k3_eq_six</code>                    | [A]  |
| $\frac{c}{12}(m^3 - m) = \frac{cm(m^2 - 1)}{12}$             | <code>virasoro_algebra_commutator</code>                 | [A]  |
| $q \geq 0, h = \frac{q}{2} \Rightarrow h \geq \frac{ q }{2}$ | <code>chiral_primary_saturates_bps</code>                | [A]  |
| $1 + 0 + 1 = 2$                                              | <code>k3_chiral_primary_total</code>                     | [A]  |
| $TT$ OPE, Virasoro algebra, $h \geq 0, c \geq 0$             | Tong §4.4–4.5; not formalized                            | [L]  |
| Casimir energy, Weyl anomaly                                 | Tong §4.4.1–4.4.2; not formalized                        | [L]  |
| Free boson: $c = 1, h(e^{ikX}) = \alpha' k^2 / 4$            | Tong §4.3; not formalized                                | [L]  |
| State-operator map                                           | Tong §4.6; not formalized                                | [L]  |
| $c = 6$ for the K3 sigma model                               | Example 4.6; arithmetic only in Lean                     | [L]  |

Nothing in this chapter is Tier C; the Schwarzian values and the Gram determinant of Exercise 4.3 are symbolic checks (sympy). The modules named after conformal field theory check only the algebra in the upper half of the table; a faithful formalization of CFT does not yet exist in this library (Chapter 38).

##### Chapter summary

- On a flat Euclidean worldsheet the conformal maps are the holomorphic maps  $z \rightarrow f(z)$ ; the stress tensor is traceless and splits into  $T(z)$  and  $\bar{T}(\bar{z})$ . The Ward identity  $\delta\mathcal{O} = -\text{Res}[\epsilon T\mathcal{O}]$  turns OPEs with  $T$  into transformation laws. A primary of weight  $(h, \tilde{h})$  has  $T\mathcal{O} \sim h\mathcal{O}/(z - w)^2 + \partial\mathcal{O}/(z - w)$ , dimension  $h + \tilde{h}$ , spin  $h - \tilde{h}$ .
- Free boson:  $\partial X$  has weight  $(1, 0)$ ,  $:e^{ikX}:$  has weight  $(\alpha' k^2 / 4, \alpha' k^2 / 4)$ , and  $c = 1$ . Requiring weight  $(1, 1)$  gives the tachyon mass  $M^2 = -4/\alpha'$ .
- $T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \dots$  defines the central charge. It gives the Schwarzian term in the transformation of  $T$ , the Casimir energy  $-(c + \bar{c})/24$  (the  $-2$  of the closed-string mass formula for  $c = \bar{c} = 24$ ), and the Weyl anomaly.
- The modes  $L_n$  obey  $[L_m, L_n] = (m - n)L_{m+n} + \frac{c}{12}m(m^2 - 1)\delta_{m+n,0}$ ; unitarity gives

$h \geq 0, c \geq 0$ .  $L_{-1}, L_0, L_1$  generate the anomaly-free Möbius group  $SL(2, \mathbb{C})/\mathbb{Z}_2$ , which fixes two-point functions.

- States on the circle correspond one-to-one to local operators at the origin, primary states to primary operators; weights are energies on the cylinder.
- Machine-checked: Möbius composition and arithmetic identities ( $4 + 2 = 6$ ,  $m^3 - m = m(m^2 - 1)$ ,  $1 + 0 + 1 = 2$ ). The OPE, the Virasoro algebra and the free boson are Tier L.

## Exercises

**Exercise 4.1** (★). Let  $l_n = z^{n+1} \partial_z$ . Show that  $[l_n, l_m] = (m - n) l_{m+n}$ . Which sign must be put in the definition of  $l_n$  for the generators to satisfy  $[l_m, l_n] = (m - n) l_{m+n}$ , the form of (4.36)? Verify that  $l_{-1}, l_0, l_1$  close among themselves.

**Exercise 4.2** (★★). Using Wick's theorem and (4.20), show that the holomorphic parts satisfy  $:e^{ik_1 X(z)} : :e^{ik_2 X(w)} : = (z - w)^{\alpha' k_1 k_2 / 2} :e^{ik_1 X(z) + ik_2 X(w)} :$ . Compare with (4.45): for which  $k_1, k_2$  can the two-point function be non-zero, and what does the condition mean in spacetime? Is the exponent then consistent with Proposition 4.4?

**Exercise 4.3** (★★). Let  $|\psi\rangle$  be a normalized primary state of weight  $h$ . Using (4.36),  $L_n^\dagger = L_{-n}$  and (4.38), show that the Gram matrix of  $L_{-2}|\psi\rangle, L_{-1}^2|\psi\rangle$  has rows  $(4h + \frac{c}{2}, 6h), (6h, 4h(2h + 1))$  and determinant  $2h(16h^2 - 10h + 2hc + c)$ . Deduce that for  $c = 2h(5 - 8h)/(2h + 1)$  the state  $L_{-2}|\psi\rangle - \frac{3}{2(2h+1)} L_{-1}^2|\psi\rangle$  is *null* (zero norm) [Tong §4.5.3, ll. 5724–5735]. Which  $c$  corresponds to  $h = \frac{1}{2}$ ?

**Exercise 4.4** (★★). Suppose  $[L_m, L_n] = (m - n)L_{m+n} + f(m)\delta_{m+n,0}$  with  $f$  unknown and odd. (a) Show that the Jacobi identity for  $L_m, L_n, L_p$ ,  $m + n + p = 0$ , reads  $(m - n)f(m + n) + (2n + m)f(-m) - (2m + n)f(-n) = 0$ . (b) Show that  $f(m) = am^3 + bm$  solves it, and that  $b$  can be changed by  $L_0 \rightarrow L_0 + \text{const}$ . (c) State (b) for  $f(m) = m^3 - m$  as a Lean theorem over  $\mathbb{Q}$ ; `ring` proves it.

**Exercise 4.5** (★★). Verify (4.42) and  $S(w, z) = 1/(2z^2)$  for  $w = i \ln z$ . For a cylinder of circumference  $L$ , mapped to the plane by  $z = e^{-2\pi i w/L}$ , show that  $E_0 = -2\pi(c + \tilde{c})/(24L)$ .

**Exercise 4.6** (★★, Lean). (a) Prove in Lean 4 with Mathlib: for rational  $c$  and an integer  $m \in \{-1, 0, 1\}$  (as a disjunction),  $\frac{c}{12}((m : \mathbb{Q})^3 - m) = 0$  (`rcases`, then `norm_num`). (b) Define the inverse  $(d, -b, -c, a)$  of a `MobiusTransform` (`linear_combination M.det_one` proves `det_one`) and use `mobius_compose` to show that it undoes  $M$  wherever both side conditions hold.

**Exercise 4.7** (★★★, Lean). `sl2c_fixes_two_point` is an existence statement with a trivial witness. Formalize *uniqueness* for integer weight: if  $k \in \mathbb{N}$  and  $f : \mathbb{C} \rightarrow \mathbb{C} \rightarrow \mathbb{C}$  satisfy (i)  $f(z + a, w + a) = f(z, w)$  and (ii)  $f(\lambda z, \lambda w) = \lambda^{-k} f(z, w)$  for  $\lambda \neq 0$ , then  $f(z, w) = f(1, 0)/(z - w)^k$  for  $z \neq w$ . (Hint:  $a = -w$ , then  $\lambda = (z - w)^{-1}$ .) Which equations of (4.44) were used, and what does the third add?

## Further reading

- [Tong §4, ll. 3553–6190] is the source of this chapter; see also [Tong §5.3–5.4, ll. 6832–7110] and [Tong §6.2.1, ll. 7506–7581]. Its first footnote (ll. 3583–3588) names the founding papers of Belavin, Polyakov and Zamolodchikov (1984) and of Friedan, Martinec and Shenker (1986) and the review by Ginsparg; none is in this library.

## Chapter 5

# The Polyakov Path Integral, Ghosts and the Superstring in Brief

Chapter 3 found the critical dimension  $D = 26$  by demanding Lorentz invariance in lightcone gauge, a calculation in which the number 26 appears at the end of a long commutator. This chapter answers a sharper question: *where does 26 come from, and is it really a number of dimensions?* The answer uses the conformal field theory of Chapter 4. The worldsheet metric is a gauge field; removing it from the path integral costs a Jacobian; the Jacobian can be written as a new conformal field theory of anticommuting “ghost” fields; and that theory has central charge  $c = -26$ . Consistency of the gauge symmetry then demands that everything else on the worldsheet has  $c = +26$ . A student should care for two reasons. First, the result is a *budget*: any conformal field theory with the right central charge is a legitimate string background, which is what makes compactification on circles, tori and K3 surfaces (the subject of most of this book) possible at all. Second, the same argument, repeated for the superstring, gives  $D = 10$  and the budget  $c = 6$  for a K3 surface that returns in Chapters 23 and 28. We end with a short, descriptive account of the superstring theories, at the level needed later, and with an exact statement of which parts of the chapter’s arithmetic have been checked in Lean.

### 5.1 The problem: a gauge symmetry with an anomaly

Recall from Chapter 2 that the Polyakov action for  $D$  scalar fields  $X^\mu(\sigma)$  on a worldsheet with metric  $g_{\alpha\beta}$  is, in Euclidean signature,

$$S_{\text{Poly}}[X, g] = \frac{1}{4\pi\alpha'} \int d^2\sigma \sqrt{g} g^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu \delta_{\mu\nu}, \quad (5.1)$$

[Tong §5.1, ll. 6300–6308]. It has two local symmetries. *Diffeomorphisms*  $\sigma^\alpha \rightarrow \sigma'^\alpha(\sigma)$  relabel the points of the worldsheet. *Weyl transformations*  $g_{\alpha\beta} \rightarrow e^{2\omega(\sigma)} g_{\alpha\beta}$  rescale the metric point by point and leave  $X^\mu$  alone; they are a symmetry because  $\sqrt{g} g^{\alpha\beta}$  is unchanged in two dimensions ( $\sqrt{g} \rightarrow e^{2\omega} \sqrt{g}$  and  $g^{\alpha\beta} \rightarrow e^{-2\omega} g^{\alpha\beta}$ ).

Both are *gauge* symmetries. A gauge symmetry is not a symmetry of nature but a redundancy of the description: two configurations related by it are the same physical state. A global symmetry may be broken by quantum effects and the theory survives; if a gauge symmetry is broken, the redundancy no longer cancels and the unphysical degrees of freedom (here, the timelike oscillators with negative norm met in Chapter 3) no longer decouple. A gauge anomaly is therefore fatal [Tong §5, ll. 6274–6290].

Now recall two facts from Chapter 4. First, on a curved worldsheet with Ricci scalar  $R$ , the trace of the stress tensor of a conformal field theory with central charge  $c$  does not vanish:

$$\langle T^\alpha_\alpha \rangle = -\frac{c}{12} R \quad (5.2)$$

[Tong §4.4.2, ll. 5094–5112]. Since  $T^\alpha_\alpha$  is the response of the theory to a Weyl rescaling (it is  $g^{\alpha\beta} \delta S / \delta g^{\alpha\beta}$  up to normalization), Equation (5.2) says that the quantum theory is *not* Weyl invariant unless  $c = 0$ . This is the Weyl anomaly. Second, every non-trivial unitary conformal field theory has  $c > 0$ ; for instance  $D$  free scalars have  $c = D$ . On a fixed background this is harmless. In string theory, where we integrate over the metric, it looks like a contradiction: we need  $c = 0$  and cannot have it [Tong §5, ll. 6290–6299].

The loophole is that we have not yet computed the path integral correctly. Removing the gauge redundancy produces additional fields, and these fields are not unitary: they carry *negative* central charge. The condition  $c = 0$  applies to the total.

## 5.2 The Faddeev–Popov method

### 5.2.1 What “dividing by the volume” means

The quantum string is defined by the path integral

$$Z = \frac{1}{\text{Vol}} \int \mathcal{D}g \mathcal{D}X e^{-S_{\text{Poly}}[X,g]}, \quad (5.3)$$

where “Vol” is the volume of the group of diffeomorphisms and Weyl transformations [Tong §5.1, ll. 6316–6340]. The integrand is constant along each *gauge orbit* (the set of all configurations obtained from one by gauge transformations), so the integral over all metrics overcounts each physical configuration by the volume of its orbit. We want to integrate once over each orbit. To do so we choose a *gauge slice*: a surface in field space that meets every orbit exactly once (Figure 5.1). Changing variables from “all fields” to “position on the slice” and “position along the orbit” costs a Jacobian. The method of Faddeev and Popov computes this Jacobian for any gauge symmetry [Tong §5.1, ll. 6341–6353].

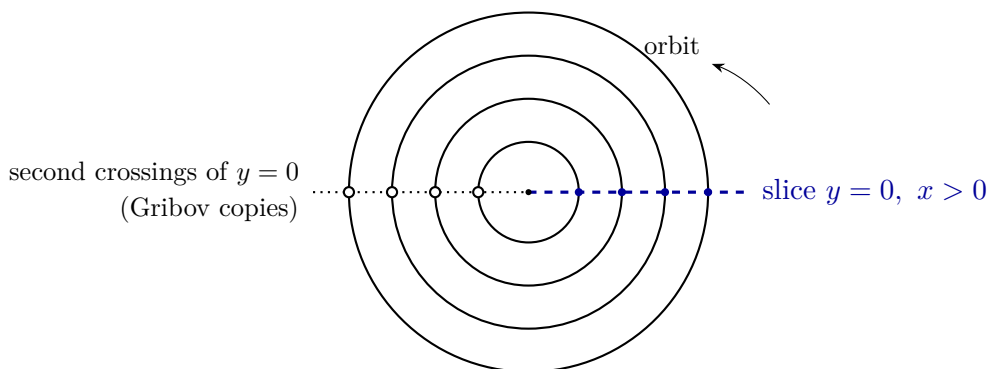


Figure 5.1: Gauge orbits (solid circles) of the rotation group acting on the plane, and a gauge slice (dashed half-line) that meets each orbit once. The full line  $y = 0$  would meet each orbit twice. In string theory the plane is replaced by the space of worldsheet metrics, the rotations by diffeomorphisms and Weyl transformations, and the slice by a choice of fiducial metric  $\hat{g}$ .



### 5.2.2 A warm-up with numbers

**Example 5.1** (Faddeev–Popov in the plane). Let the “fields” be a point  $\vec{x} = (x, y) \in \mathbb{R}^2$ , the “action”  $S = x^2 + y^2 = r^2$ , and the gauge group the rotations  $\vec{x} \rightarrow \vec{x}^\theta$  by an angle  $\theta \in [0, 2\pi)$ , with  $\text{Vol} = 2\pi$ . The quantity to compute is

$$Z = \frac{1}{2\pi} \int_{\mathbb{R}^2} dx dy e^{-(x^2+y^2)} = \frac{1}{2\pi} \cdot \pi = \frac{1}{2}.$$

Now repeat the computation the Faddeev–Popov way. Choose the slice  $y = 0, x > 0$ . Define  $\Delta(\vec{x})$  by

$$\Delta(\vec{x})^{-1} = \int_0^{2\pi} d\theta \delta(y^\theta) \Big|_{x^\theta > 0}, \quad y^\theta = r \sin(\theta + \theta_0),$$

where  $\theta_0$  is the polar angle of  $\vec{x}$ . Using  $\int d\theta \delta(f(\theta)) = 1/|f'(\theta_*)|$  at the zero  $\theta_* = -\theta_0$  gives  $\Delta^{-1} = 1/(r \cos 0) = 1/r$ , so  $\Delta(\vec{x}) = r$ ; it depends only on the orbit, not on the position along it. Insert  $1 = \Delta(\vec{x}) \int d\theta \delta(y^\theta)$  into  $Z$ , change variables  $\vec{x} \rightarrow \vec{x}^{-\theta}$  (the measure and the action are rotation invariant), and the  $\theta$  integral factors out as  $2\pi = \text{Vol}$ :

$$Z = \int dx dy \Delta(\vec{x}) \delta(y) \Big|_{x>0} e^{-(x^2+y^2)} = \int_0^\infty dx x e^{-x^2} = \frac{1}{2}.$$

The two answers agree. The Jacobian  $\Delta = r$  is nothing but the  $r$  in  $dx dy = r dr d\theta$ . Had we forgotten it we would have obtained  $\int_0^\infty e^{-x^2} dx = \sqrt{\pi}/2 \approx 0.886$ , which is wrong. Had we used the whole line  $y = 0$  as the slice we would have counted each orbit twice; such repeated intersections are called *Gribov copies* [Tong §5.1.1, ll. 6396–6401].

### 5.2.3 The same steps for the string

Write  $\zeta$  for a combined diffeomorphism and Weyl transformation, and  $g^\zeta$  for the transformed metric,  $g_{\alpha\beta}^\zeta(\sigma') = e^{2\omega(\sigma)} \frac{\partial \sigma^\gamma}{\partial \sigma'^\alpha} \frac{\partial \sigma^\delta}{\partial \sigma'^\beta} g_{\gamma\delta}(\sigma)$ . A metric in two dimensions has three independent components, and  $\zeta$  contains three arbitrary functions ( $\omega$  and the two components of  $\sigma'$ ). So locally every metric can be brought to a chosen *fiducial metric*  $\hat{g}$ ; this is the slice. Two caveats are postponed: globally this fails on worldsheets of higher genus, where a finite number of parameters (moduli) survive, and even locally a residual symmetry, the conformal group, remains [Tong §5.1.1, ll. 6354–6378].

Define the Faddeev–Popov determinant  $\Delta_{FP}[g]$  by

$$\Delta_{FP}[g]^{-1} = \int \mathcal{D}\zeta \delta(g - \hat{g}^\zeta), \quad (5.4)$$

the exact analogue of  $\Delta^{-1} = \int d\theta \delta(y^\theta)$  [Tong §5.1.1, ll. 6379–6392]. The measure  $\mathcal{D}\zeta$  is assumed invariant under the group,  $\mathcal{D}\zeta = \mathcal{D}(\zeta'\zeta)$ , like  $d\theta$  above.

**Lemma 5.2.**  $\Delta_{FP}$  is gauge invariant:  $\Delta_{FP}[g^\zeta] = \Delta_{FP}[g]$ .

*Proof.*  $\Delta_{FP}[g^\zeta]^{-1} = \int \mathcal{D}\zeta' \delta(g^\zeta - \hat{g}^{\zeta'}) = \int \mathcal{D}\zeta' \delta(g - \hat{g}^{\zeta^{-1}\zeta'}) = \int \mathcal{D}\zeta'' \delta(g - \hat{g}^{\zeta''})$ , where the second step acts with  $\zeta^{-1}$  on both arguments of the delta functional and the third uses the invariance of the measure with  $\zeta'' = \zeta^{-1}\zeta'$  [Tong §5.1.1, ll. 6406–6419].  $\square$

Insert  $1 = \Delta_{FP}[g] \int \mathcal{D}\zeta \delta(g - \hat{g}^\zeta)$  into Equation (5.3). The delta functional performs the integral over  $g$  and sets  $g = \hat{g}^\zeta$ :

$$\begin{aligned} Z[\hat{g}] &= \frac{1}{\text{Vol}} \int \mathcal{D}\zeta \mathcal{D}X \Delta_{FP}[\hat{g}^\zeta] e^{-S_{\text{Poly}}[X, \hat{g}^\zeta]} \\ &= \frac{1}{\text{Vol}} \int \mathcal{D}\zeta \mathcal{D}X \Delta_{FP}[\hat{g}] e^{-S_{\text{Poly}}[X, \hat{g}]} = \int \mathcal{D}X \Delta_{FP}[\hat{g}] e^{-S_{\text{Poly}}[X, \hat{g}]}. \end{aligned} \quad (5.5)$$

The second line uses Lemma 5.2 and the gauge invariance of the action (together with a relabelling of the integration variable  $X$ ); after that nothing depends on  $\zeta$  and  $\int \mathcal{D}\zeta = \text{Vol}$  cancels the prefactor [Tong §5.1.1, ll. 6421–6480].

#### Common pitfall: the hidden assumption

The step “the action and the measure are gauge invariant” is exactly where an anomaly would enter. The combined measure  $\mathcal{D}X \Delta_{FP}$  is invariant under diffeomorphisms, but under a Weyl transformation it changes by the anomaly of Equation (5.2), with  $c$  the *total* central charge. The result  $Z[\hat{g}]$  is independent of the arbitrary choice of  $\hat{g}$ , as a gauge-fixed answer must be, only if that total vanishes. The rest of this chapter computes the total.

## 5.3 From a determinant to ghost fields

### 5.3.1 Evaluating the determinant

Near the identity, a gauge transformation is an infinitesimal Weyl rescaling  $\omega(\sigma)$  and an infinitesimal diffeomorphism  $\delta\sigma^\alpha = v^\alpha(\sigma)$ . The metric changes by

$$\delta\hat{g}_{\alpha\beta} = 2\omega\hat{g}_{\alpha\beta} + \nabla_\alpha v_\beta + \nabla_\beta v_\alpha, \quad (5.6)$$

By Lemma 5.2 we need  $\Delta_{FP}$  only at  $g = \hat{g}$ . There, assuming the slice meets each orbit once, the delta functional in Equation (5.4) is supported at the identity transformation, so it suffices to expand  $\hat{g}^\zeta$  to first order, and the integral over the group becomes an integral over its Lie algebra,  $\Delta_{FP}[\hat{g}]^{-1} = \int \mathcal{D}\omega \mathcal{D}v \delta(2\omega\hat{g}_{\alpha\beta} + \nabla_\alpha v_\beta + \nabla_\beta v_\alpha)$  [Tong §5.1.2, ll. 6482–6500]. A delta functional restricts a symmetric tensor field to vanish, so its Fourier representation needs a symmetric tensor field  $\beta^{\alpha\beta}$  as the conjugate variable:

$$\Delta_{FP}[\hat{g}]^{-1} = \int \mathcal{D}\omega \mathcal{D}v \mathcal{D}\beta \exp\left(2\pi i \int d^2\sigma \sqrt{\hat{g}} \beta^{\alpha\beta} [2\omega\hat{g}_{\alpha\beta} + \nabla_\alpha v_\beta + \nabla_\beta v_\alpha]\right).$$

The field  $\omega$  appears without derivatives and linearly. Integrating over it gives another delta functional, which sets  $\beta^{\alpha\beta}\hat{g}_{\alpha\beta} = 0$ : from now on  $\beta^{\alpha\beta}$  is symmetric and *traceless*. Using the symmetry of  $\beta$  to combine the two remaining terms,

$$\Delta_{FP}[\hat{g}]^{-1} = \int \mathcal{D}v \mathcal{D}\beta \exp\left(4\pi i \int d^2\sigma \sqrt{\hat{g}} \beta^{\alpha\beta} \nabla_\alpha v_\beta\right) \quad (5.7)$$

[Tong §5.1.2, ll. 6501–6527]. The counting is consistent: in two dimensions a symmetric traceless tensor has  $3 - 1 = 2$  components and so does the vector  $v_\beta$ . The operator in the exponent maps vectors to symmetric traceless tensors and can be treated as a square matrix, which is necessary for a determinant to make sense.

### 5.3.2 Commuting versus anticommuting variables

Equation (5.7) is a Gaussian integral over commuting variables, which produces an *inverse* determinant. We want the determinant itself. The tool is integration over Grassmann (anticommuting) variables. For a single pair  $\theta, \bar{\theta}$  with  $\theta^2 = \bar{\theta}^2 = 0$ ,  $\theta\bar{\theta} = -\bar{\theta}\theta$ , and the rules  $\int d\theta 1 = 0$ ,  $\int d\theta \theta = 1$ , one has

$$\int d\bar{\theta} d\theta e^{-\bar{\theta}a\theta} = \int d\bar{\theta} d\theta (1 - a\bar{\theta}\theta) = a \int d\bar{\theta} d\theta \theta\bar{\theta} = a,$$

whereas for a complex commuting variable  $\int d\bar{z} dz e^{-\bar{z}az} \propto 1/a$ . For  $n$  pairs and a matrix  $M$  the same computation gives

$$\int \prod_i d\bar{\theta}_i d\theta_i e^{-\bar{\theta}M\theta} = \det M, \quad \int \prod_i d\bar{z}_i dz_i e^{-\bar{z}Mz} \propto \frac{1}{\det M}, \quad (5.8)$$

because expanding the exponential to order  $n$  and keeping the terms that contain every  $\theta_i$  and every  $\bar{\theta}_i$  once reproduces the signed sum over permutations that defines the determinant (Exercise 5.2 asks for the  $2 \times 2$  case). This is standard material, not taken from a source in this book's library.

So to invert Equation (5.7) we replace the commuting fields by anticommuting fields with the same index structure,

$$\beta_{\alpha\beta} \longrightarrow b_{\alpha\beta}, \quad v^\alpha \longrightarrow c^\alpha,$$

and obtain, after a rescaling of the fields that changes only the overall normalization of  $Z$  and after rotating back to Euclidean signature,

$$\Delta_{FP}[\hat{g}] = \int \mathcal{D}b \mathcal{D}c e^{-S_{\text{ghost}}}, \quad S_{\text{ghost}} = \frac{1}{2\pi} \int d^2\sigma \sqrt{g} b_{\alpha\beta} \nabla^\alpha c^\beta \quad (5.9)$$

[Tong §5.1.3, ll. 6529–6560]. The fields  $b_{\alpha\beta}$  (symmetric, traceless) and  $c^\alpha$  are the Faddeev–Popov ghosts. The gauge-fixed partition function Equation (5.5) is

$$Z[\hat{g}] = \int \mathcal{D}X \mathcal{D}b \mathcal{D}c \exp(-S_{\text{Poly}}[X, \hat{g}] - S_{\text{ghost}}[b, c, \hat{g}]). \quad (5.10)$$

#### Physical picture: what ghosts are for

Ghosts are anticommuting fields of integer spin. They violate the spin–statistics theorem, so they cannot describe physical particles, and they are not meant to: they are a local way of writing a Jacobian. Their effect is to cancel the unphysical, gauge degrees of freedom of  $X^\mu$ , leaving the  $D - 2$  transverse oscillations found in lightcone gauge, but without breaking manifest Lorentz invariance [Tong §5.1.3, ll. 6566–6574]. Tong's one-loop partition function is computed in lightcone gauge with 24 oscillators precisely to avoid carrying the ghosts along [Tong §6.4.2, ll. 8519–8527]; in the covariant computation the ghost oscillators remove the contribution of two of the 26 directions (standard; not checked against a source in this book's library). Do not confuse these ghosts with the negative-norm states of covariant quantization in Chapter 3, which are also called ghosts; the Faddeev–Popov ghosts are the cure, those states are the disease.

### 5.3.3 Conformal gauge

Choose the fiducial metric  $\hat{g}_{\alpha\beta} = e^{2\omega} \delta_{\alpha\beta}$  and complex coordinates  $z = \sigma^1 + i\sigma^2$  as in Chapter 4. Tracelessness,  $\hat{g}^{\alpha\beta} b_{\alpha\beta} \propto b_{zz} = 0$ , leaves the components  $b_{zz}$  and  $b_{\bar{z}\bar{z}}$ . With  $\sqrt{\hat{g}} = e^{2\omega}$ ,  $d^2\sigma = \frac{1}{2} d^2z$  and  $\nabla^z = \hat{g}^{z\bar{z}} \nabla_{\bar{z}} = 2e^{-2\omega} \nabla_{\bar{z}}$ , the factors of  $e^{2\omega}$  cancel:

$$S_{\text{ghost}} = \frac{1}{2\pi} \int d^2z (b_{zz} \nabla_{\bar{z}} c^z + b_{\bar{z}\bar{z}} \nabla_z c^{\bar{z}}).$$

The covariant derivative is  $\nabla_{\bar{z}} c^z = \partial_{\bar{z}} c^z + \Gamma_{\bar{z}\alpha}^z c^\alpha$ , and  $\Gamma_{\bar{z}\alpha}^z = \frac{1}{2} g^{z\bar{z}} (\partial_{\bar{z}} g_{\alpha\bar{z}} + \partial_\alpha g_{\bar{z}\bar{z}} - \partial_{\bar{z}} g_{\bar{z}\alpha}) = 0$ : the first and third terms cancel and  $g_{\bar{z}\bar{z}} = 0$  in conformal gauge. Writing  $b = b_{zz}$ ,  $c = c^z$ ,  $\bar{b} = b_{\bar{z}\bar{z}}$ ,  $\bar{c} = c^{\bar{z}}$ ,  $\partial = \partial_z$  and  $\bar{\partial} = \partial_{\bar{z}}$ ,

$$S_{\text{ghost}} = \frac{1}{2\pi} \int d^2z (b \bar{\partial} c + \bar{b} \partial \bar{c}) \quad (5.11)$$

[Tong §5.1.3–5.2, ll. 6578–6640]. Two observations. The conformal factor  $\omega$  has dropped out, so the ghost action is Weyl invariant with  $b_{\alpha\beta}$  and  $c^\alpha$  inert; this holds for this index placement only, since raising or lowering an index introduces the metric. And the equations of motion  $\bar{\partial}b = \bar{\partial}c = 0$ ,  $\partial\bar{b} = \partial\bar{c} = 0$  say that  $b, c$  are holomorphic and  $\bar{b}, \bar{c}$  antiholomorphic: the ghosts form a free conformal field theory, to which all the tools of Chapter 4 apply.

## 5.4 The ghost conformal field theory

We treat the holomorphic half; the antiholomorphic half is identical. It costs nothing to be slightly more general than the string requires, and the generality pays off in Section 5.6. Consider anticommuting fields  $b$  and  $c$  with action  $\frac{1}{2\pi} \int d^2z b \bar{\partial}c$ , and suppose  $b$  has conformal weight  $\lambda$  and  $c$  has weight  $1 - \lambda$ , so that  $b \bar{\partial}c$  has weight  $(1, 1)$  and the action is conformally invariant. The reparametrization ghosts have  $\lambda = 2$ , because  $b_{zz}$  has two lower holomorphic indices and  $c^z$  one upper index.

### 5.4.1 Propagator

As in Chapter 4, the operator product follows from the statement that the path integral of a total functional derivative vanishes:

$$0 = \int \mathcal{D}b \mathcal{D}c \frac{\delta}{\delta b(\sigma)} \left[ e^{-S_{\text{ghost}}} b(\sigma') \right] = \int \mathcal{D}b \mathcal{D}c e^{-S_{\text{ghost}}} \left[ -\frac{1}{2\pi} \bar{\partial}c(\sigma) b(\sigma') + \delta(\sigma - \sigma') \right],$$

that is,  $\bar{\partial}c(\sigma) b(\sigma') = 2\pi \delta(\sigma - \sigma')$  inside correlation functions. Since  $\bar{\partial}(1/z) = 2\pi \delta(z, \bar{z})$ , this integrates to

$$c(z) b(w) \sim \frac{1}{z - w}, \quad b(z) c(w) \sim \frac{1}{z - w}, \quad (5.12)$$

where  $\sim$  means “equal up to terms regular as  $z \rightarrow w$ ”. The second relation follows from the first and anticommutativity:  $b(z)c(w) = -c(w)b(z) \sim -1/(w - z) = 1/(z - w)$ . The products  $b(z)b(w)$  and  $c(z)c(w)$  are regular [Tong §5.2, ll. 6674–6720].

### 5.4.2 Stress tensor and conformal weights

The stress tensor is

$$T(z) = -\lambda :b \partial c:(z) + (1 - \lambda) : \partial b c:(z). \quad (5.13)$$

For  $\lambda = 2$  this is  $T = -2 :b \partial c: - : \partial b c: = 2 : \partial c b: + : c \partial b:$ , which is the result Tong obtains by varying the covariant action Equation (5.9) with respect to the metric [Tong §5.2, ll. 6650–6672, 6721–6726]. For general  $\lambda$  we do not derive Equation (5.13) from an action; we justify it by checking that it assigns the intended weights. A primary field  $\mathcal{O}$  of weight  $h$  satisfies  $T(z)\mathcal{O}(w) \sim h \mathcal{O}(w)/(z - w)^2 + \partial \mathcal{O}(w)/(z - w)$  (Chapter 4).

*The field  $b$ .* Contract the  $c$  in each term of  $T(z)$  with  $b(w)$ , using  $\partial c(z) b(w) \sim \partial_z(z - w)^{-1} = -(z - w)^{-2}$ :

$$T(z) b(w) \sim -\lambda b(z) \cdot \frac{-1}{(z - w)^2} + (1 - \lambda) \frac{\partial b(z)}{z - w} = \frac{\lambda b(w)}{(z - w)^2} + \frac{[\lambda + (1 - \lambda)] \partial b(w)}{z - w},$$

after the Taylor expansion  $b(z) = b(w) + (z - w)\partial b(w) + \dots$ . So  $b$  is primary of weight  $\lambda$ .

*The field  $c$ .* Now the  $b$  in each term of  $T(z)$  is contracted with  $c(w)$ , and  $c(w)$  must first be moved past one anticommuting field, which costs a sign:  $:b \partial c:(z) c(w) \rightarrow -\partial c(z)/(z - w)$  and  $: \partial b c:(z) c(w) \rightarrow -\partial_z(z - w)^{-1} c(z) = c(z)/(z - w)^2$ . Hence

$$T(z) c(w) \sim \frac{\lambda \partial c(z)}{z - w} + \frac{(1 - \lambda) c(z)}{(z - w)^2} = \frac{(1 - \lambda) c(w)}{(z - w)^2} + \frac{\partial c(w)}{z - w},$$

so  $c$  is primary of weight  $1 - \lambda$ . For  $\lambda = 2$ :  $h_b = 2$  and  $h_c = -1$  [Tong §5.2, ll. 6727–6769]. These are the weights dictated by the index structure  $b_{zz}$ ,  $c^z$ : a field with  $n$  lower  $z$  indices has weight  $n$ , and an upper index counts as  $-1$ . A negative weight is a sign that the theory is not unitary, which is what we need to evade the bound  $c > 0$ .

### 5.4.3 The central charge

The central charge is read off from the most singular term of  $T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}$  [Tong §4.4, ll. 4895–4915]. The  $(z-w)^{-4}$  term comes from contracting both fields of  $T(z)$  with both fields of  $T(w)$ . There are four products of terms, and in each exactly one pairing is possible ( $b$  with  $c$ ). In every case the field being moved passes two anticommuting fields, so the reordering sign is  $+$ . With  $G = (z-w)^{-1}$  and

$$\partial_w G = \frac{1}{(z-w)^2}, \quad \partial_z G = \frac{-1}{(z-w)^2}, \quad \partial_z \partial_w G = \frac{-2}{(z-w)^3},$$

the four double contractions are as follows. In the column “pairings” the first factor is the contraction of the  $b$ -type field at  $z$  with the  $c$ -type field at  $w$ , and the second that of the  $c$ -type field at  $z$  with the  $b$ -type field at  $w$ ; each is a derivative of Equation (5.12).

| product                                                    | pairings                          | value $\times (z-w)^4$ | at $\lambda = 2$ |
|------------------------------------------------------------|-----------------------------------|------------------------|------------------|
| $\lambda^2 :b \partial c:(z) :b \partial c:(w)$            | $\partial_w G \cdot \partial_z G$ | $-\lambda^2$           | $-4$             |
| $-\lambda(1-\lambda) :b \partial c:(z) : \partial b c:(w)$ | $G \cdot \partial_z \partial_w G$ | $+2\lambda(1-\lambda)$ | $-4$             |
| $-\lambda(1-\lambda) : \partial b c:(z) :b \partial c:(w)$ | $\partial_z \partial_w G \cdot G$ | $+2\lambda(1-\lambda)$ | $-4$             |
| $(1-\lambda)^2 : \partial b c:(z) : \partial b c:(w)$      | $\partial_z G \cdot \partial_w G$ | $-(1-\lambda)^2$       | $-1$             |

The sum is  $c/2 = -\lambda^2 + 4\lambda(1-\lambda) - (1-\lambda)^2 = -6\lambda^2 + 6\lambda - 1$ , that is

$$c_{bc}(\lambda) = -2(6\lambda^2 - 6\lambda + 1) = 1 - 3(2\lambda - 1)^2 \quad (5.14)$$

(the two forms agree:  $1 - 3(4\lambda^2 - 4\lambda + 1) = -12\lambda^2 + 12\lambda - 2$ ; the table and this identity were also confirmed by a symbolic check in sympy). For the reparametrization ghosts the last column adds up to  $-13$ , so

$$T(z)T(w) \sim \frac{-13}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}, \quad c_{bc}(2) = -26, \quad (5.15)$$

in agreement with [Tong §5.2, ll. 6770–6839], who carries out the  $\lambda = 2$  case including the less singular terms. The general- $\lambda$  formula is standard; here it is derived on the page rather than quoted.

**Example 5.3** (The formula at the weights that matter). Equation (5.14) is a downward parabola in  $\lambda$ , symmetric under  $\lambda \leftrightarrow 1 - \lambda$  (exchange of the roles of  $b$  and  $c$ ), with maximum  $c = 1$  at  $\lambda = \frac{1}{2}$  (Figure 5.2).

| $\lambda$ | $1 - \lambda$ | $(2\lambda - 1)^2$ | $c_{bc}(\lambda)$ | system                                              |
|-----------|---------------|--------------------|-------------------|-----------------------------------------------------|
| $1/2$     | $1/2$         | $0$                | $+1$              | a complex free fermion = two real fermions          |
| $1$       | $0$           | $1$                | $-2$              |                                                     |
| $3/2$     | $-1/2$        | $4$                | $-11$             | (the commuting version has $+11$ , see Section 5.6) |
| $2$       | $-1$          | $9$                | $-26$             | reparametrization ghosts                            |

The first row is a useful by-product. At  $\lambda = \frac{1}{2}$  the fields  $b$  and  $c$  both have weight  $\frac{1}{2}$  and can be combined into two real (Majorana) fermions  $\psi_{1,2} = (b \pm c)/\sqrt{2}$  up to a factor of  $i$ . So one real free fermion has

$$c_\psi = \frac{1}{2},$$

the value used for the superstring below.

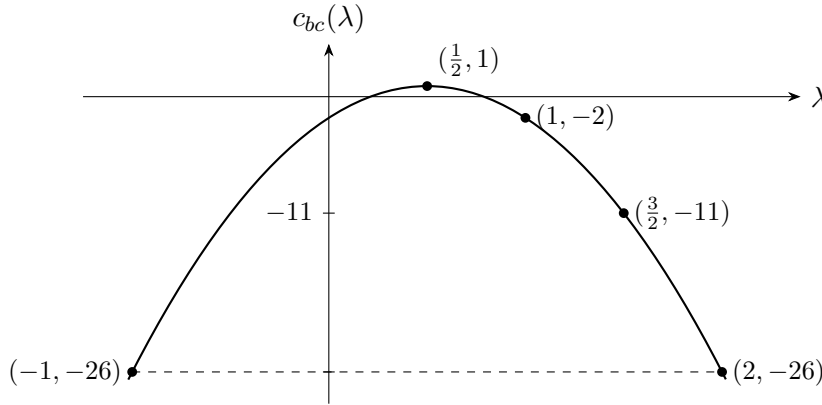


Figure 5.2: The central charge  $c_{bc}(\lambda) = 1 - 3(2\lambda - 1)^2$  of an anticommuting first-order system whose fields have weights  $\lambda$  and  $1 - \lambda$ . The points  $\lambda = 2$  and  $\lambda = -1$  describe the same system with the names of  $b$  and  $c$  exchanged. The commuting system of the same weights has the opposite central charge.

## 5.5 The critical central charge

Collect the pieces. After gauge fixing, the worldsheet theory in Equation (5.10) consists of a “matter” conformal field theory (so far,  $D$  free scalars) and the ghosts. Central charges of decoupled theories add, because their stress tensors add and have no singular product with each other. The Weyl anomaly of the full theory, Equation (5.2), is proportional to

$$c_{\text{tot}} = c_{\text{matter}} + c_{bc}(2) = c_{\text{matter}} - 26, \quad (5.16)$$

and the gauge symmetry we used to fix the metric survives quantization only if  $c_{\text{tot}} = 0$ :

**Proposition 5.4** (Critical central charge [L]). *The gauge-fixed bosonic string is Weyl invariant if and only if the matter conformal field theory has  $c_{\text{matter}} = \tilde{c}_{\text{matter}} = 26$ . If the matter theory consists of  $D$  free scalars, each with  $c = 1$ , this requires  $D = 26$  [Tong §5.3, ll. 6840–6851].*

This agrees with the lightcone result of Chapter 3, but the logic is different and more informative. There,  $D = 26$  was needed to keep a global symmetry (Lorentz invariance) that the gauge choice had hidden. Here, Lorentz invariance is manifest throughout and  $c = 26$  is needed to keep the gauge symmetry itself.

### Physical picture: a budget, not a dimension

Nothing in the argument required the matter theory to be free scalars. *Any* conformal field theory with  $c = \tilde{c} = 26$  gives a consistent gauge-fixed string, and each describes a different background in which the string propagates. “Critical dimension” is thus a misnomer for “critical central charge”: only for special conformal field theories can  $c$

be read as a number of spacetime dimensions. To describe strings in four-dimensional Minkowski space one takes four free scalars and an “internal” conformal field theory with  $c = 26 - 4 = 22$ , which may or may not have a geometric interpretation [Tong §5.3, ll. 6852–6867]. When this book compactifies on a circle (Chapter 7), a torus (Chapter 10) or a K3 surface (Chapter 23), it is choosing the internal theory. T-duality will turn out to be the statement that two different-looking geometries give the *same* internal conformal field theory.

*Remark 5.5 (Time).* One direction must be timelike, i.e. have a wrong-sign kinetic term. The safe construction keeps  $X^0$  as a free scalar and lets the remaining  $c = 25$  be an arbitrary conformal field theory, which excludes backgrounds that evolve in time. The worldsheet description of time-dependent backgrounds has unresolved technical obstacles, and string cosmology is usually studied through the low-energy effective action of Chapter 6 instead [Tong §5.3, ll. 6868–6877]. This caveat applies to the cosmological discussion of Chapter 32.

*Remark 5.6 (Non-critical strings).* If  $c_{\text{matter}} = D \neq 26$  one may give up Weyl invariance from the start. The conformal factor  $\omega$  of the metric then does not decouple: integrating the anomaly equation with  $c = D - 26$  shows that  $\omega$  acquires a kinetic term proportional to  $(26 - D)$  and, if a worldsheet cosmological constant  $\mu$  is present, an exponential potential  $\mu e^{2\omega}$ . It becomes a new dynamical field, the Liouville field [Tong §5.3.2, ll. 6892–6987]. We shall not use non-critical strings; the remark shows that “ $D \neq 26$ ” does not simply fail but turns into a different and harder theory.

### 5.5.1 Physical states without BRST

With the ghosts present the space of states is larger, not smaller, and one needs a criterion for physical states. The systematic answer is BRST quantization, which neither Tong nor this book develops [Tong §5.4, ll. 6988–7001]. The following shortcut is enough for our purposes. By the state–operator map of Chapter 4, a state corresponds to a local operator  $\mathcal{O}(z, \bar{z})$  of the matter theory. An operator at a point is not diffeomorphism invariant; its integral over the worldsheet is:

$$V \sim \int d^2z \mathcal{O}(z, \bar{z}). \quad (5.17)$$

Under a rescaling the measure  $d^2z$  has weight  $(-1, -1)$ , so Weyl invariance of  $V$  requires  $\mathcal{O}$  to be a primary operator of weight  $(h, \tilde{h}) = (1, 1)$ . Such  $V$  are the *vertex operators* [Tong §5.4, ll. 7020–7059].

**Example 5.7** (Masses from weights). The operator  $:e^{ip \cdot X}:$  has weight  $h = \tilde{h} = \alpha' p^2/4$  (Chapter 4). Setting  $h = 1$  gives  $p^2 = 4/\alpha'$ , i.e.  $M^2 = -p^2 = -4/\alpha'$ : the tachyon. The operator  $:e^{ip \cdot X} \partial X^\mu \bar{\partial} X^\nu: \zeta_{\mu\nu}$  has  $h = \tilde{h} = 1 + \alpha' p^2/4$ , so  $h = 1$  gives  $M^2 = 0$ ; demanding that it be primary (no  $(z - w)^{-3}$  term in its product with  $T$ ) imposes the transversality conditions  $p^\mu \zeta_{\mu\nu} = p^\nu \zeta_{\mu\nu} = 0$  [Tong §5.4.1, ll. 7060–7116]. Both masses agree with the book’s mass formula  $M^2 = \frac{2}{\alpha'}(N + \tilde{N} - 2)$  at  $N = \tilde{N} = 0$  and  $N = \tilde{N} = 1$ . The intercept “ $-2$ ” of that formula, which in Chapter 3 came from a regularized sum over zero-point energies, comes here from the requirement of weight  $(1, 1)$ .

Ghosts can also be ignored in tree-level scattering: on the sphere they decouple from the matter fields, whereas on surfaces of higher genus the moduli of the metric couple the two, in analogy with the field-theory statement that ghosts matter only in loops [Tong §6.1.1, ll. 7444–7458].

## 5.6 The superstring in brief

This book works mostly with the bosonic fields  $G$ ,  $B$  and the dilaton, which are common to all string theories, so a full treatment of the superstring is not needed. But the K3 part of the book (Chapters 23, 24 and 28) takes place in superstring theory, and the student needs to know what the words mean. Tong’s notes, our main source, deliberately give “nothing more than a list of facts” about the superstring [Tong §2.5, ll. 2958–2961]; we do the same, and mark what goes beyond that source. Everything in this section is [L].

### 5.6.1 Worldsheet fermions and $D = 10$

In the Neveu–Schwarz–Ramond formulation the worldsheet theory contains, besides the scalars  $X^\mu$ , anticommuting fields  $\psi^\mu$  of weight  $\frac{1}{2}$  (and/or their antiholomorphic partners  $\tilde{\psi}^\mu$ ), related to  $X^\mu$  by a worldsheet supersymmetry. This supersymmetry is a further *gauge* symmetry on the worldsheet, so gauge fixing it produces further ghosts, the  $\beta\gamma$  system, with central charge  $+11$ . The matter theory must therefore have

$$c_{\text{matter}} = 26 - 11 = 15.$$

Supersymmetry pairs each boson ( $c = 1$ ) with a fermion ( $c = \frac{1}{2}$ ; compare Example 5.3), so  $D$  flat dimensions contribute  $\frac{3}{2}D$ , and

$$\frac{3}{2}D = 15 \implies D = 10 \quad (5.18)$$

[Tong §5.3.1, ll. 6878–6891].

Where does  $+11$  come from? Tong states the number without derivation. It follows from Section 5.4 with two changes. (i) The gauge parameter of a local supersymmetry is a spinor, a “square root” of the vector  $c^z$  of weight  $-1$ ; its ghost  $\gamma$  has weight  $-\frac{1}{2}$ , and the conjugate field  $\beta$  has weight  $\lambda = \frac{3}{2}$ . (ii) Ghosts have the opposite statistics to the gauge parameter, so  $\beta, \gamma$  *commute*. For commuting fields with the same action  $\frac{1}{2\pi} \int d^2z \beta \bar{\partial} \gamma$  the derivation of Equation (5.12) still gives  $\gamma(z)\beta(w) \sim 1/(z-w)$ , but now  $\beta(z)\gamma(w) = +\gamma(w)\beta(z) \sim 1/(w-z) = -1/(z-w)$ . In the computation of the weights this sign replaces the reordering sign of the anticommuting case, so the weights are still  $(\lambda, 1-\lambda)$ . In each of the four double contractions of the table in Section 5.4 exactly one of the two pairings is of the type  $\beta(z)\gamma(w)$ , so every entry changes sign:

$$c_{\beta\gamma}(\lambda) = -c_{bc}(\lambda) = 3(2\lambda - 1)^2 - 1, \quad c_{\beta\gamma}(\tfrac{3}{2}) = 3 \cdot 4 - 1 = 11. \quad (5.19)$$

The identification of the weights in (i) is standard but is not checked against a source in this book’s library; the sign argument in (ii) is derived here.

**Example 5.8** (The budget for  $K3 \times T^2$ ). Take the superstring on  $\mathbb{R}^{1,3} \times T^2 \times K3$  (Chapter 24). The ghosts contribute  $-26 + 11 = -15$ . The matter side is

| factor             | real dimensions | $c$ per dimension | $c$ |
|--------------------|-----------------|-------------------|-----|
| $\mathbb{R}^{1,3}$ | 4               | $1 + \frac{1}{2}$ | 6   |
| $T^2$              | 2               | $1 + \frac{1}{2}$ | 3   |
| K3                 | 4               | $1 + \frac{1}{2}$ | 6   |
| total              | 10              |                   | 15  |

and  $15 - 15 = 0$ . The entry for K3 deserves a comment. A K3 surface is curved, and its sigma model is an interacting theory, not four free bosons and four free fermions. The value  $c = 6$



is nevertheless correct, because the anomaly budget leaves exactly  $15 - 9 = 6$  for the internal theory once  $\mathbb{R}^{1,3} \times T^2$  is accounted for, and because a Ricci-flat target preserves conformal invariance to the order discussed in Chapter 6. The free-field count  $4 \cdot 1 + 4 \cdot \frac{1}{2}$  is a mnemonic (it is exact for the orbifold  $T^4/\mathbb{Z}_2$  of Chapter 21). The same number  $c = 6$  labels the  $N = 4$  superconformal algebra whose characters organize the elliptic genus in Chapter 28.

### 5.6.2 The five theories

The bosonic string is unique, but there are discrete choices in adding fermions to the worldsheet. The main one is whether fermions are added to both the left- and right-moving sectors or to one sector only [Tong §2.5, ll. 2969–2985]. In all cases  $D = 10$ , there is no tachyon, and the massless bosonic fields include the metric  $G_{\mu\nu}$ , the two-form  $B_{\mu\nu}$  and the dilaton  $\Phi$  [Tong §2.5, ll. 2958–2968], which is why the results of Chapters 6, 8 and 9 carry over.

| theory                     | worldsheet fermions       | super-charges | additional massless bosons                                                                  |
|----------------------------|---------------------------|---------------|---------------------------------------------------------------------------------------------|
| type IIA                   | both sectors              | 32            | Ramond–Ramond one-form $C_1$ and three-form $C_3$                                           |
| type IIB                   | both sectors              | 32            | Ramond–Ramond $C_0, C_2, C_4$ ; the field strength of $C_4$ is self-dual, $F_5 = \star F_5$ |
| heterotic $SO(32)$         | one sector                | 16            | $SO(32)$ gauge field                                                                        |
| heterotic $E_8 \times E_8$ | one sector                | 16            | $E_8 \times E_8$ gauge field                                                                |
| type I                     | (open and closed strings) |               | mentioned by Tong only in passing                                                           |

[Tong §2.5, ll. 2981–3008]. Three comments connect this table to later chapters.

*Heterotic strings and lattices.* In a heterotic string one sector is that of the superstring, with matter central charge 15, and the other is that of the bosonic string, with matter central charge 26. Ten dimensions use 10 of the 26, and the remaining  $26 - 10 = 16$  units are carried by an internal theory that produces the gauge group; both  $SO(32)$  and  $E_8 \times E_8$  have rank 16. The consistent choices are governed by even unimodular lattices of rank 16, the subject of Chapter 18, and on a torus by the Narain lattices of Chapter 10. (This paragraph is standard; Tong gives only the list of gauge groups, and the counting is not checked against a source in this book’s library.)

*T-duality.* The superstring theories are not individually invariant under  $R \rightarrow \alpha'/R$ ; they are exchanged in pairs. Type IIA on a circle of radius  $R$  is equivalent to type IIB on a circle of radius  $\alpha'/R$ , and the two heterotic strings are likewise exchanged [Tong §8.3.3, ll. 11828–11835]. This is consistent with the D-brane content: type IIA has stable  $Dp$ -branes with  $p$  even, type IIB with  $p$  odd, and T-duality along a circle changes  $p$  by one (Chapter 13).

*K3.* Aspinwall’s lectures, the main source for Chapter 23, state their goal as understanding “the type II string and the heterotic string compactified on a K3 surface and what such models are dual to”, and remark that the heterotic string appears to be intrinsically tied to the geometry of K3 [Asp §1, ll. 112–190]. The student who knows the table above knows enough to follow that discussion.

**Common pitfall: two meanings of “supersymmetry”**

The supersymmetry used in Equation (5.18) lives on the worldsheet and is a gauge redundancy. The “32 supercharges” of the table refer to spacetime supersymmetry of the ten-dimensional theory, which is a physical symmetry. The second does not follow from the first by inspection; it requires a projection on the spectrum that this book does not describe (standard; not covered by the sources in this book’s library).

**Historical note**

The analysis of the string path integral described in this chapter is due to Polyakov, “Quantum geometry of bosonic strings”, Phys. Lett. B **103**, 207 (1981), a paper of four pages, followed by “Quantum geometry of fermionic strings”, Phys. Lett. B **103**, 211 (1981), for the superstring. Polyakov’s main interest was the non-critical string of Remark 5.6, not  $D = 26$  [Tong §5.1, ll. 6312–6315; §5.3.2, ll. 6892–6896].

## 5.7 What the machine has checked

Two files of the library `StringTheoryFormalization` concern this chapter. Both model central charges as rational numbers and prove arithmetic facts about them. In the language of this book they are *arithmetic shadows*: they check the last line of the argument, not the argument. No path integral, no Grassmann field, no operator product and no Virasoro algebra is defined in either file.

**In Lean: UseCases/CriticalDimension.lean**

The two central-charge formulas Equations (5.14) and (5.19) are *definitions*:

```
def fermionicGhostCharge (lam : ℚ) : ℚ := 1 - 3 * (2 * lam - 1) ^ 2
def bosonicGhostCharge (lam : ℚ) : ℚ := 3 * (2 * lam - 1) ^ 2 - 1
```

The theorems proved about them, in the namespace `StringTheory.UseCases.CriticalDimension`, are:

```
theorem bosonic_eq_neg_fermionic (lam : ℚ) :
  bosonicGhostCharge lam = - fermionicGhostCharge lam := by ...

theorem bc_ghost_charge_eq : fermionicGhostCharge 2 = -26 := by ...

theorem betagamma_ghost_charge_eq : bosonicGhostCharge (3 / 2) = 11 := by ...

theorem bosonic_string_critical_dimension :
  (26 : ℚ) * 1 + fermionicGhostCharge 2 = 0 := by ...

theorem superstring_critical_dimension :
  (10 : ℚ) * 1 + 10 * (1 / 2) + fermionicGhostCharge 2
  + bosonicGhostCharge (3 / 2) = 0 := by ...

theorem bosonic_dimension_unique (D : ℚ)
  (h : D * 1 + fermionicGhostCharge 2 = 0) :
  D = 26 := by ...
```

**Plain reading.** The polynomial  $3(2\lambda - 1)^2 - 1$  is the negative of  $1 - 3(2\lambda - 1)^2$  for every rational  $\lambda$ ; the first polynomial equals 11 at  $\lambda = \frac{3}{2}$  and the second equals  $-26$  at  $\lambda = 2$ ;  $26 - 26 = 0$ ;  $10 + 5 - 26 + 11 = 0$ ; and the only rational  $D$  with  $D - 26 = 0$  is 26. The numerals 1 and  $\frac{1}{2}$  multiplying 26 and 10 stand for the central charges of a free boson and a free fermion, but in the statements they are just numerals.

**Proof idea.** Unfold the definitions; `ring` proves the polynomial identity, `norm_num` evaluates at a fixed rational number, and `linarith` solves the linear equation in `bosonic_dimension_unique`.

**What is not proved.** That a *bc* system of weight  $\lambda$  has central charge `fermionicGhostCharge`  $\lambda$  (our derivation of Equation (5.14) is on paper only); that the Faddeev–Popov ghosts have  $\lambda = 2$ ; that the Weyl anomaly is proportional to the total central charge; or that a consistent conformal field theory with  $c = 26$  exists. There is also no uniqueness statement for the superstring in the file (Exercise 5.6).

#### In Lean: `Frontier/CentralCharge.lean`

In the namespace `StringTheory.Frontier`:

```
def centralChargeBoson : ℚ := 1
def centralChargeFermion : ℚ := 1/2
def centralChargeK3 : ℚ :=
  4 * centralChargeBoson + 4 * centralChargeFermion

theorem central_charge_k3_eq_six : centralChargeK3 = 6 := by ...
```

**Plain reading.**  $4 \cdot 1 + 4 \cdot \frac{1}{2} = 6$  in  $\mathbb{Q}$ . This is the K3 row of Example 5.8, with the free-field mnemonic taken as the definition.

The same file contains three theorems whose names promise more than their statements deliver, as the file’s own documentation says explicitly:

```
theorem virasoro_ope_central_term (z w : ℂ) (h : z ≠ w) :
  (centralChargeK3 / 2 : ℚ) = 3 := by ...

theorem virasoro_algebra_commutator (c : ℚ) (m n : ℤ) (L : ℤ → ℤ) :
  c / 12 * (m^3 - m) = c * m * (m^2 - 1) / 12 := by ...

theorem central_charge_from_worldsheet_action :
  ∃ (c : ℚ), c = centralChargeK3 ∧ c = 6 := by ...
```

The first does not use  $z$ ,  $w$  or  $h$  and states that  $6/2 = 3$ . The second does not use  $n$  or  $L$  and is the identity  $\frac{c}{12}(m^3 - m) = \frac{cm(m^2-1)}{12}$ , valid for every  $c$ ; it says nothing about commutators  $[L_m, L_n]$ . The third is witnessed by  $c = 6$  and involves no action. They are recorded here so that a reader who meets these names in the catalogue (Chapter 36) is not misled. One sentence of that file’s documentation should also be read with care: it describes the bookkeeping as “ $26 = 20 + 6$ ”, which mixes the bosonic budget 26 with the superstring value  $c = 6$ . The consistent superstring bookkeeping is  $15 = 9 + 6$ , as in Example 5.8. This affects a comment only, not a theorem.

All declarations above depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`.

| statement                                                                                            | tier | evidence                                                                                          |
|------------------------------------------------------------------------------------------------------|------|---------------------------------------------------------------------------------------------------|
| $1 - 3(2\lambda - 1)^2$ equals $-26$ at $\lambda = 2$ ; its negative equals $11$ at $\frac{3}{2}$    | [A]  | bc_ghost_charge_eq,<br>betagamma_ghost_charge_eq                                                  |
| $26 - 26 = 0$ ; $10 + 5 - 26 + 11 = 0$ ; $D - 26 = 0 \Rightarrow D = 26$                             | [A]  | bosonic_string_critical_dimension,<br>superstring_critical_dimension,<br>bosonic_dimension_unique |
| $4 \cdot 1 + 4 \cdot \frac{1}{2} = 6$                                                                | [A]  | central_charge_k3_eq_six                                                                          |
| $c_{bc}(\lambda) = 1 - 3(2\lambda - 1)^2$ for a $bc$ system of weight $\lambda$                      | [L]  | derived in Section 5.4; sympy check; $\lambda = 2$ in Tong §5.2                                   |
| Gauge fixing yields $bc$ ghosts with $\lambda = 2$ ; Weyl invariance iff $c_{\text{matter}} = 26$    | [L]  | Tong §5.1–5.3                                                                                     |
| $\beta\gamma$ ghosts have $c = 11$ ; superstring has $D = 10$                                        | [L]  | Tong §5.3.1                                                                                       |
| Identification of the rational numbers in the Lean files with central charges of worldsheet theories | —    | a modelling statement, never Tier A                                                               |

*Remark 5.9* (What a faithful formalization would need). To make Proposition 5.4 a theorem one would need, at the least: a definition of a (chiral) conformal field theory or vertex operator algebra with a stress tensor and its central charge; the construction of the  $bc$  system as such an object for each  $\lambda$ ; a proof that its central charge is Equation (5.14), which is a finite computation with formal power series once the definitions exist; and additivity of  $c$  under tensor products. None of these is available in Mathlib at the time of writing, to our knowledge. The path-integral steps of Sections 5.2 and 5.3 are formal manipulations even on paper and are further still from formalization. Chapter 38 returns to this.

#### Chapter summary

- Diffeomorphisms and Weyl transformations are gauge redundancies of the Polyakov path integral. Dividing by their volume is done by the Faddeev–Popov method: choose a fiducial metric  $\hat{g}$  and pay the Jacobian  $\Delta_{FP}[\hat{g}]$ .
- $\Delta_{FP}$  is the determinant of the operator taking a vector  $v$  to the traceless part of  $\nabla_\alpha v_\beta + \nabla_\beta v_\alpha$ . Written as a Grassmann integral it becomes the ghost action  $\frac{1}{2\pi} \int d^2z (b\bar{\partial}c + \bar{b}\partial\bar{c})$ .
- The ghosts form a conformal field theory with  $h_b = 2$ ,  $h_c = -1$  and  $c = -26$ . For general weight,  $c_{bc}(\lambda) = 1 - 3(2\lambda - 1)^2$ ; the commuting system has the opposite sign.
- The Weyl anomaly is proportional to the total central charge. Consistency requires  $c_{\text{matter}} = 26$  for the bosonic string and  $15$  for the superstring; with free fields,  $D = 26$  and  $D = 10$ . These are budgets for a conformal field theory, not only counts of dimensions; a K3 surface uses  $c = 6$ .
- Physical states correspond to matter primaries of weight  $(1, 1)$ .
- There are five superstring theories in ten dimensions. Type IIA and IIB are exchanged by T-duality, as are the two heterotic strings.
- Lean checks the arithmetic of the final step (values of two polynomials, three linear identities, one uniqueness statement). Everything connecting that arithmetic to the worldsheet is Tier L.

## Exercises

**Exercise 5.1** (★). Repeat Example 5.1 in  $\mathbb{R}^3$ , declaring the direction of  $\vec{x}$  to be pure gauge, so that the orbits are spheres and  $\text{Vol} = 4\pi$ . Compute  $Z = \frac{1}{4\pi} \int d^3x e^{-|\vec{x}|^2}$  directly and with the slice “positive  $z$ -axis”, and identify the Faddeev–Popov Jacobian. (Answer:  $\Delta = r^2$ ,  $Z = \sqrt{\pi}/4$ .)

**Exercise 5.2** (★). Verify Equation (5.8) for  $n = 2$ : expand  $\exp(-\sum_{ij} \bar{\theta}_i M_{ij} \theta_j)$  to second order and show that the coefficient of  $\theta_1 \bar{\theta}_1 \theta_2 \bar{\theta}_2$  is  $M_{11}M_{22} - M_{12}M_{21}$ .

**Exercise 5.3** (★). In  $n$  worldsheet dimensions, count the components of a symmetric traceless tensor and of a vector. Show that they agree only for  $n = 2$ . What does this say about the attempt to gauge away the whole metric of a membrane worldvolume ( $n = 3$ ) with diffeomorphisms and Weyl transformations?

**Exercise 5.4** (★★). For  $\lambda = 2$ , compute the  $(z - w)^{-2}$  term of  $T(z)T(w)$  from the single contractions and show that it equals  $2T(w)$ . You will need to Taylor expand fields at  $z$  around  $w$  and to keep track of one reordering sign per term.

**Exercise 5.5** (★★). Show directly from Equation (5.13) that exchanging the names  $b \leftrightarrow c$  together with  $\lambda \rightarrow 1 - \lambda$  maps  $T$  to itself (use  $:c\partial b: = -:\partial b c:$ ), and conclude that  $c_{bc}(\lambda) = c_{bc}(1 - \lambda)$ . Then state and prove in Lean:

```
theorem fermionicGhostCharge_symm (lam : ℚ) :
  fermionicGhostCharge lam = fermionicGhostCharge (1 - lam) := by
  sorry
```

(Hint: `unfold fermionicGhostCharge; ring`.)

**Exercise 5.6** (★★). The Lean file proves uniqueness of  $D = 26$  but not of  $D = 10$ . State and prove the missing theorem: for  $D : \mathbb{Q}$ , if  $D * 1 + D * (1 / 2) + \text{fermionicGhostCharge } 2 + \text{bosonicGhostCharge } (3 / 2) = 0$  then  $D = 10$ . Which two existing theorems should you rewrite with before calling `linarith`?

**Exercise 5.7** (★★). Prove in Lean that `fermionicGhostCharge (1 / 2) = 1`. Explain in two sentences why this supports the value  $c = \frac{1}{2}$  for one real fermion, which the second file records as the constant `centralChargeFermion`. Is your explanation part of what Lean checked?

**Exercise 5.8** (★★). (a) For the bosonic string on  $\mathbb{R}^{1,d-1}$  times an internal theory, give the internal central charge as a function of  $d$ . (b) Do the same for the superstring, and find the internal central charge for  $d = 4$  and  $d = 6$ . (c) A Calabi–Yau threefold has real dimension 6. Check that its free-field count of  $c$  matches (b) for  $d = 4$ , and that K3 matches  $d = 6$ .

**Exercise 5.9** (★★★). Starting from the anomaly equation  $\frac{1}{Z} \frac{\partial Z}{\partial \omega} = \frac{c}{24\pi} \sqrt{g} (R - 2\nabla^2 \omega)$  with  $\mu = 0$  and  $c = D - 26$  (Remark 5.6), integrate in  $\omega$  to show that

$$Z[e^{2\omega}g] = Z[g] \exp\left(\frac{c}{24\pi} \int d^2\sigma \sqrt{g} (g^{\alpha\beta} \partial_\alpha \omega \partial_\beta \omega + R\omega)\right).$$

For which sign of  $D - 26$  does the Liouville field have a kinetic term of the conventional sign? Compare with [Tong §5.3.2, ll. 6940–6985].

## Further reading

- [Tong §5.1, ll. 6300–6613]: the Faddeev–Popov procedure for the string, in the form followed here. [Tong §5.2, ll. 6614–6839]: the ghost conformal field theory and the full  $TT$  product for  $\lambda = 2$ .
- [Tong §5.3, ll. 6840–6987]: critical central charge, the superstring count and non-critical strings. [Tong §5.4, ll. 6988–7116]: vertex operators.
- [Tong §2.5, ll. 2948–3008] and [Tong §8.3.3, ll. 11828–11835]: the list of superstring theories and their behaviour under T-duality.
- [Tong §6.1.1, ll. 7329–7412]: the sum over worldsheet topologies and the string coupling, which complete the definition of the path integral begun here.
- [Asp §1, ll. 112–190]: why type II and heterotic strings on K3 are the natural setting for the second half of this book.
- BRST quantization and the general first-order system are treated in Polchinski’s textbook, to which Tong refers [Tong §5.4, ll. 6997–7001]; that book is not in this book’s library, and the Lean file cites its equations (2.7.20) and (10.4.10) for the two ghost-charge formulas without our having checked them against it.

## Chapter 6

# Strings in Background Fields and the Low-Energy Effective Action

So far the string has moved in flat, empty spacetime. Yet Chapter 3 found that the closed string itself contains a massless symmetric tensor, an antisymmetric tensor and a scalar: the graviton  $G_{\mu\nu}$ , the Kalb–Ramond field  $B_{\mu\nu}$  and the dilaton  $\Phi$ . This chapter answers the question: *what happens to a string that moves through a spacetime in which these three fields are switched on, and which such spacetimes are allowed?* The worldsheet theory becomes an interacting two-dimensional field theory whose coupling constants are the spacetime fields themselves. The Weyl symmetry, which Chapter 5 showed to be indispensable, survives quantization only if these couplings do not run. The conditions for that, the vanishing of three beta functions, are Einstein’s equations coupled to  $B$  and  $\Phi$ , and they follow from a spacetime action, the low-energy effective action. A student should care for three reasons. First, this is how general relativity comes out of string theory, rather than being put in. Second, the string coupling  $g_s$  is revealed to be the value of a field,  $g_s = e^{\Phi_0}$ . Third, and most important for this book, the field  $B$  has a gauge symmetry that makes it invisible locally when it is constant, while its integral over a two-cycle of a torus remains physical. That observation, and the Hamiltonian that goes with it, is the seed of the generalized metric of Chapter 15 and of the duality group of Chapter 11.

Conventions are those of the book:  $\ell_s = \sqrt{\alpha'}$ , and the spacetime signature is mostly plus. As in Chapters 4 and 5 the worldsheet is Euclidean, with metric  $g_{\alpha\beta}$  and coordinates  $\sigma^\alpha$ ,  $\alpha = 1, 2$ ; this is why a factor  $i$  accompanies the  $B$ -field coupling below. The spacetime dimension is  $D$ . We follow [Tong §7, ll. 9085–10380] throughout.

## 6.1 From gravitons to a curved background

### 6.1.1 The non-linear sigma model

The Polyakov action of Chapter 2 contains the flat metric of spacetime through the contraction  $\partial_\alpha X^\mu \partial_\beta X^\nu \eta_{\mu\nu}$ . The obvious way to describe a string in a curved spacetime is to replace the constant matrix by a function of the position of the string:

$$S = \frac{1}{4\pi\alpha'} \int d^2\sigma \sqrt{g} g^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu G_{\mu\nu}(X). \quad [\text{Tong (7.1), ll. 9088–9097}] \quad (6.1)$$

Here  $X^\mu(\sigma)$ ,  $\mu = 0, \dots, D-1$ , are the embedding coordinates,  $g_{\alpha\beta}$  is the worldsheet metric,  $g = \det g_{\alpha\beta}$ , and  $G_{\mu\nu}(X)$  is the spacetime metric evaluated at the point  $X(\sigma)$ . (In this chapter  $G_{\mu\nu}$  always denotes the metric, never the Einstein tensor.) A two-dimensional field theory whose fields are coordinates on a curved manifold, the *target space*, is called a *non-linear*

*sigma model*. The action (6.1) is still invariant under reparametrizations of the worldsheet, because every index  $\alpha$  is contracted, and under Weyl rescalings  $g_{\alpha\beta} \rightarrow e^{2\omega(\sigma)} g_{\alpha\beta}$ , because in two dimensions  $\sqrt{g} \rightarrow e^{2\omega} \sqrt{g}$  and  $g^{\alpha\beta} \rightarrow e^{-2\omega} g^{\alpha\beta}$ . It is also invariant under changes of coordinates in the target space,  $X^\mu \rightarrow X'^\mu(X)$ , provided  $G_{\mu\nu}$  transforms as a tensor: a change of spacetime coordinates is a *field redefinition* of the two-dimensional theory.

### 6.1.2 Why the background is made of gravitons

There is something to justify. The string already contains a graviton. A classical gravitational field should therefore be a state of many such gravitons, in the way a classical radio wave is a coherent state of photons; one is not free to postulate a second, independent metric. To see that (6.1) passes this test, write the metric as flat space plus a fluctuation,  $G_{\mu\nu}(X) = \delta_{\mu\nu} + h_{\mu\nu}(X)$  (Euclidean target space, for the path integral). The action splits as  $S = S_{\text{Poly}} + V$  with

$$V = \frac{1}{4\pi\alpha'} \int d^2\sigma \sqrt{g} g^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu h_{\mu\nu}(X), \quad [\text{Tong (7.2), ll. 9126–9136}] \quad (6.2)$$

and the partition function becomes

$$Z = \int \mathcal{D}X \mathcal{D}g e^{-S_{\text{Poly}} - V} = \int \mathcal{D}X \mathcal{D}g e^{-S_{\text{Poly}}} \left(1 - V + \frac{1}{2}V^2 - \dots\right). \quad (6.3)$$

For a plane wave,  $h_{\mu\nu}(X) = \zeta_{\mu\nu} e^{ip \cdot X}$  with  $\zeta_{\mu\nu}$  symmetric and traceless,  $V$  is exactly the integrated vertex operator of the graviton state met in Chapter 5: an operator that, inserted once in the path integral, creates or absorbs one graviton of momentum  $p$  and polarization  $\zeta$ . The term  $V^n/n!$  inserts  $n$  of them. The exponential  $e^{-V}$  is therefore a coherent superposition of states with any number of gravitons, and a general  $h_{\mu\nu}(X)$  is a superposition of plane waves. [Tong §7, ll. 9098–9150]

#### Physical picture: a laser beam of gravitons

A laser beam is a classical electromagnetic wave, and at the same time it is a coherent state  $\exp(\alpha a^\dagger)|0\rangle$  of photons. In the same way the curved metric in (6.1) is a coherent state of the closed string's own massless spin-two excitation. Nothing new has been added to the theory: a string in a curved background moves through a condensate of other strings, each in its graviton state. The same is true of the backgrounds  $B_{\mu\nu}$  and  $\Phi$ .

## 6.2 The $\alpha'$ expansion

In conformal gauge,  $g_{\alpha\beta} = \delta_{\alpha\beta}$ , the flat-space action is free. The action (6.1) is not:

$$S = \frac{1}{4\pi\alpha'} \int d^2\sigma G_{\mu\nu}(X) \partial_\alpha X^\mu \partial^\alpha X^\nu. \quad [\text{Tong (7.3), ll. 9152–9160}] \quad (6.4)$$

To exhibit the interactions, expand around a string that sits at a point  $\bar{x}^\mu$  and fluctuates a little,

$$X^\mu(\sigma) = \bar{x}^\mu + \sqrt{\alpha'} Y^\mu(\sigma). \quad (6.5)$$

The factor  $\sqrt{\alpha'}$  makes  $Y^\mu$  dimensionless, so that “ $Y$  is small” is a meaningful statement. Taylor-expanding the metric gives

$$\frac{1}{\alpha'} G_{\mu\nu}(X) \partial X^\mu \partial X^\nu = \left[ G_{\mu\nu}(\bar{x}) + \sqrt{\alpha'} G_{\mu\nu,\omega}(\bar{x}) Y^\omega + \frac{\alpha'}{2} G_{\mu\nu,\omega\rho}(\bar{x}) Y^\omega Y^\rho + \dots \right] \partial Y^\mu \partial Y^\nu, \quad (6.6)$$

where a comma denotes a partial derivative with respect to  $X$ . The first term is the free kinetic term. Every further Taylor coefficient is the coupling constant of an interaction vertex



with two derivatives: the cubic vertex has strength  $\sqrt{\alpha'} \partial G$ , the quartic one  $\alpha' \partial^2 G$ , and so on. *The sigma model has infinitely many coupling constants, packaged into the function  $G_{\mu\nu}(X)$ .* If the metric varies on a length scale  $r_c$ , the radius of curvature, then  $\partial G \sim 1/r_c$  and the dimensionless coupling is

$$\frac{\sqrt{\alpha'}}{r_c} = \frac{\ell_s}{r_c}. \quad [\text{Tong (7.4), ll. 9163–9205}] \quad (6.7)$$

Perturbation theory on the worldsheet is an expansion in powers of  $\ell_s/r_c$ . It is reliable when spacetime is nearly flat on the scale of the string. Since the curvature is built from two derivatives of the metric, each additional worldsheet loop costs a factor  $\alpha'/r_c^2$ .

**Common pitfall: two expansions, two meanings of “loop”**

String theory has two unrelated perturbation expansions. The  $\alpha'$  *expansion* counts loops of the two-dimensional field theory on a worldsheet of *fixed* topology; its parameter is  $\alpha'/r_c^2$  and it measures how far the string is from being a point. The  $g_s$  *expansion* counts handles of the worldsheet, which are loops in *spacetime*; its parameter is  $g_s^2$  (see Section 6.5). A “one-loop beta function” in this chapter is one loop on the worldsheet, computed on a sphere, and is a tree-level statement in spacetime. When  $r_c \sim \ell_s$  the first expansion fails, but the worldsheet path integral still defines the problem; when  $g_s \sim 1$  the second fails, and perturbation theory offers no definition at all. [Tong §7.1, ll. 9206–9220]

## 6.3 Weyl invariance at one loop gives Einstein's equations

### 6.3.1 Why the couplings must not run

Classically (6.4) is conformally invariant for *any* metric  $G_{\mu\nu}$ . Quantum mechanically, an interacting field theory needs an ultraviolet cut-off, and after renormalization its couplings generally depend on the energy scale  $\mu$  at which they are measured. A theory with a preferred scale is not conformally invariant. In most of physics that is a fact of life: the classical scale invariance of Yang–Mills theory is broken in exactly this way. For the string it would be fatal, because on the worldsheet conformal symmetry is what remains of the gauge symmetries of Chapter 5, and an anomalous gauge symmetry makes the theory inconsistent. The scale dependence of a coupling is measured by its beta function. Here there is one for each coupling, that is, a beta *functional* of the metric, schematically

$$\beta_{\mu\nu}(G) \sim \mu \frac{\partial G_{\mu\nu}(X; \mu)}{\partial \mu}, \quad \text{and consistency requires} \quad \beta_{\mu\nu}(G) = 0. \quad (6.8)$$

[Tong §7.1.1, ll. 9222–9241]

### 6.3.2 The one-loop computation

The computation is short once coordinates are chosen well. Around any point  $\bar{x}$  one can choose *Riemann normal coordinates*, in which the metric at  $\bar{x}$  is  $\delta_{\mu\nu}$ , its first derivatives vanish, and its second derivatives are given by the Riemann tensor:

$$G_{\mu\nu}(X) = \delta_{\mu\nu} - \frac{\alpha'}{3} R_{\mu\lambda\nu\kappa}(\bar{x}) Y^\lambda Y^\kappa + O(Y^3). \quad [\text{Tong §7.1.1, ll. 9246–9252}] \quad (6.9)$$

The cubic vertex of (6.6) is gone, and to quartic order

$$S = \frac{1}{4\pi} \int d^2\sigma \left[ \partial Y^\mu \partial Y^\nu \delta_{\mu\nu} - \frac{\alpha'}{3} R_{\mu\lambda\nu\kappa} Y^\lambda Y^\kappa \partial Y^\mu \partial Y^\nu \right]. \quad (6.10)$$

This is a theory of  $D$  free massless scalars with a quartic, two-derivative interaction whose strength is the curvature of spacetime. The one-loop divergence comes from closing two of the four legs of the vertex,  $Y^\lambda$  and  $Y^\kappa$ , into a loop. In position space that means replacing  $Y^\lambda(\sigma)Y^\kappa(\sigma)$  by its expectation value. The propagator of Chapter 4,

$$\langle Y^\lambda(\sigma) Y^\kappa(\sigma') \rangle = -\frac{1}{2} \delta^{\lambda\kappa} \ln |\sigma - \sigma'|^2, \quad (6.11)$$

diverges as  $\sigma' \rightarrow \sigma$ . In dimensional regularization, with worldsheet dimension  $d = 2 + \epsilon$ , the coincident limit is

$$\langle Y^\lambda(\sigma) Y^\kappa(\sigma) \rangle = 2\pi \delta^{\lambda\kappa} \int \frac{d^{2+\epsilon}k}{(2\pi)^{2+\epsilon}} \frac{1}{k^2} \rightarrow \frac{\delta^{\lambda\kappa}}{\epsilon}. \quad [\text{Tong §7.1.1, ll. 9274–9296}] \quad (6.12)$$

Contracting  $\delta^{\lambda\kappa}$  with the Riemann tensor in the vertex produces the Ricci tensor,  $R_{\mu\lambda\nu\kappa} \delta^{\lambda\kappa} = R_{\mu\nu}$ . The divergent part of the quartic term is therefore a term *of the same form as the kinetic term*, namely  $\epsilon^{-1} R_{\mu\nu} \partial Y^\mu \partial Y^\nu$  with a numerical coefficient. A divergence of the form of a term already present in the action is removed by renormalizing the corresponding coupling, here the metric. Part of the divergence is absorbed in a rescaling of the field,  $Y^\mu \rightarrow Y^\mu + (\alpha'/6\epsilon) R^\mu{}_\nu Y^\nu$ ; the rest requires

$$G_{\mu\nu} \rightarrow G_{\mu\nu} + \frac{\alpha'}{\epsilon} R_{\mu\nu}. \quad [\text{Tong (7.5), ll. 9297–9312}] \quad (6.13)$$

A pole  $1/\epsilon$  in dimensional regularization corresponds to a logarithm of the cut-off, hence of the scale  $\mu$ , and the coefficient of the pole is the beta function:

**Proposition 6.1** (One-loop metric beta function [L]). *To first order in  $\alpha'$  the sigma model (6.1) has*

$$\beta_{\mu\nu}(G) = \alpha' R_{\mu\nu}, \quad (6.14)$$

*so that it is conformally invariant at this order if and only if the target space is Ricci flat,  $R_{\mu\nu} = 0$ . [Tong (7.6), ll. 9313–9323]*

The vacuum Einstein equations have appeared as the condition for a two-dimensional quantum field theory to have no scale. They were not assumed at any point.

### 6.3.3 The same result from the Weyl anomaly

Chapters 4 and 5 diagnosed the breakdown of Weyl invariance by the trace of the stress tensor,  $T^\alpha{}_\alpha \neq 0$ . It is instructive to see (6.14) in that language. Take the worldsheet metric  $g_{\alpha\beta} = e^{2\phi} \delta_{\alpha\beta}$ . In exactly two dimensions  $\phi$  drops out of (6.1). In  $d = 2 + \epsilon$  dimensions it does not:  $\sqrt{g} = e^{(2+\epsilon)\phi}$  while  $g^{\alpha\beta} = e^{-2\phi} \delta^{\alpha\beta}$ , leaving a factor  $e^{\epsilon\phi} \approx 1 + \epsilon\phi$ . Inserting the renormalized metric (6.13), the product  $\epsilon\phi \times (\alpha'/\epsilon) R_{\mu\nu}$  is finite and survives the limit  $\epsilon \rightarrow 0$ :

$$S = \frac{1}{4\pi\alpha'} \int d^2\sigma \partial_\alpha X^\mu \partial^\alpha X^\nu [G_{\mu\nu}(X) + \alpha' \phi R_{\mu\nu}(X)]. \quad [\text{Tong §7.1.1, ll. 9328–9357}] \quad (6.15)$$

The action now depends on the conformal factor, which is the statement that Weyl invariance is broken. With  $T^\alpha{}_\alpha = -2\pi \partial S / \partial \phi$  in conformal gauge,

$$T^\alpha{}_\alpha = -\frac{1}{2} R_{\mu\nu} \partial X^\mu \partial X^\nu \equiv -\frac{1}{2\alpha'} \beta_{\mu\nu} \partial X^\mu \partial X^\nu, \quad (6.16)$$

and the second equality, which *defines*  $\beta_{\mu\nu}$  as the coefficient of  $\partial X \partial X$  in the trace, again gives  $\beta_{\mu\nu} = \alpha' R_{\mu\nu}$ . [Tong §7.1.1, ll. 9358–9378] The mechanism is typical of anomalies: a symmetry that holds only in  $d = 2$ , a regulator that works only away from  $d = 2$ , and a finite remainder  $\epsilon \times 1/\epsilon$ .

### 6.3.4 Worked example: Ricci flow of a sphere

When the beta function does not vanish it still has a meaning: it tells how the target-space metric changes with the scale,

$$\mu \frac{\partial G_{\mu\nu}}{\partial \mu} = \alpha' R_{\mu\nu}, \quad [\text{Tong (7.7), ll. 9380–9398}] \quad (6.17)$$

an equation known as *Ricci flow*.

**Example 6.2** (The two-sphere shrinks in the infrared). Let the target space be a round two-sphere of radius  $r$ ,  $G_{\mu\nu} = r^2 \hat{g}_{\mu\nu}$  with  $\hat{g}$  the metric of the unit sphere. A direct computation (the Ricci scalar of this metric is  $2/r^2$ , and in two dimensions  $R_{\mu\nu} = \frac{1}{2} R G_{\mu\nu}$ ) gives  $R_{\mu\nu} = \hat{g}_{\mu\nu}$ , independent of  $r$ . Because the sphere is a symmetric space, the flow can only change its radius, and (6.17) reduces to one ordinary differential equation,

$$\mu \frac{dr^2}{d\mu} = \alpha' \quad \implies \quad r^2(\mu) = r_0^2 + \alpha' \ln \frac{\mu}{\mu_0}. \quad (6.18)$$

(The source displays the right-hand side as  $\alpha'/2\pi$  [Tong §7.1.2, ll. 9399–9409]; we quote what (6.17) gives literally. The sign, and the conclusions below, do not depend on the constant.) The radius grows towards the ultraviolet and shrinks towards the infrared. Since the coupling is  $\ell_s/r$ , the model is weakly coupled at high energy, *asymptotically free*, and strongly coupled at low energy. For numbers, start with  $r_0^2 = 10\alpha'$  at the scale  $\mu_0$ , where the loop parameter is  $\alpha'/r_0^2 = 0.1$ . The one-loop formula predicts  $r^2 = \alpha'$ , coupling of order one, at  $\ln(\mu/\mu_0) = -9$ , that is at  $\mu = e^{-9}\mu_0 \approx 1.2 \times 10^{-4}\mu_0$ . Below that scale the one-loop result cannot be trusted; the model develops a mass gap there [Tong §7.1.2, ll. 9399–9415]. A sphere alone is not a consistent string background: it is not Ricci flat. Exercise 6.6 shows how  $H$ -flux rescues a three-sphere.

#### Historical note

According to Tong, the relation between the sigma-model beta function and Einstein’s equations was first shown by Friedan in his 1980 PhD thesis, and the beta functions of Section 6.6 are derived in Callan, Friedan, Martinec and Perry, “Strings in Background Fields” (1985) [Tong §7.2.3, footnote, ll. 9633–9639]. Ricci flow, in a version generalized by a scalar that string theorists recognize as the dilaton, was used by Perelman in the proof of the Poincaré conjecture [Tong §7.1.2, ll. 9416–9418].

## 6.4 The Kalb–Ramond field: the string as a charged object

### 6.4.1 The coupling

The vertex operator of the antisymmetric massless state has the same form as that of the graviton, with an antisymmetric polarization. Exponentiating it, in the same way as (6.3), adds a second term to the action:

$$S = \frac{1}{4\pi\alpha'} \int d^2\sigma \sqrt{g} \left[ G_{\mu\nu}(X) \partial_\alpha X^\mu \partial_\beta X^\nu g^{\alpha\beta} + i B_{\mu\nu}(X) \partial_\alpha X^\mu \partial_\beta X^\nu \epsilon^{\alpha\beta} \right], \quad (6.19)$$

where  $\epsilon^{\alpha\beta}$  is the antisymmetric tensor with  $\sqrt{g} \epsilon^{12} = +1$  [Tong (7.8), ll. 9451–9466]. The factor  $i$  appears because the worldsheet is Euclidean and the term contains exactly one “time” derivative. The new term does not contain the worldsheet metric at all (the  $\sqrt{g}$  cancels against the one in  $\epsilon^{\alpha\beta}$ ), so it is trivially Weyl invariant. The field  $B_{\mu\nu} = -B_{\nu\mu}$  is the *Kalb–Ramond field*, or simply the  $B$ -field.

To understand it, recall how a point particle of unit charge couples to the electromagnetic potential  $A_\mu$ : through  $\int d\tau A_\mu(X) \dot{X}^\mu$  [Tong (7.9), ll. 9477–9489]. Geometrically this is the integral along the worldline of the one-form  $A = A_\mu dX^\mu$ : a one-form is what can be integrated over a one-dimensional object. The worldsheet is two-dimensional, so the natural object to integrate over it is a two-form,  $B = \frac{1}{2}B_{\mu\nu} dX^\mu \wedge dX^\nu$ . Its pull-back to the worldsheet is

$$X^*B = \frac{1}{2}B_{\mu\nu}(\partial_1 X^\mu \partial_2 X^\nu - \partial_2 X^\mu \partial_1 X^\nu) d\sigma^1 \wedge d\sigma^2 = \frac{1}{2}B_{\mu\nu} \partial_\alpha X^\mu \partial_\beta X^\nu \epsilon^{\alpha\beta} \sqrt{g} d^2\sigma,$$

so the second term of (6.19) is simply

$$S_B = \frac{i}{2\pi\alpha'} \int_\Sigma X^*B. \quad (6.20)$$

The string is charged under  $B$  in the same sense in which a particle is charged under  $A$ , with charge density equal to the string tension  $1/2\pi\alpha'$ . [Tong §7.2.1, ll. 9490–9504]

### 6.4.2 Gauge symmetry and the field strength $H$

The coupling  $\int A$  is unchanged by  $A_\mu \rightarrow A_\mu + \partial_\mu \alpha$  up to endpoint terms, because the integrand changes by a total derivative. The analogue for the two-form is

$$B_{\mu\nu} \longrightarrow B_{\mu\nu} + \partial_\mu C_\nu - \partial_\nu C_\mu, \quad \text{i.e. } B \rightarrow B + dC, \quad [\text{Tong (7.11), ll. 9505–9511}] \quad (6.21)$$

with  $C = C_\mu dX^\mu$  an arbitrary one-form. Since pull-back commutes with the exterior derivative,  $S_B$  changes by  $\frac{i}{2\pi\alpha'} \int_\Sigma d(X^*C)$ , which vanishes by Stokes' theorem on a closed worldsheet. (For an open string a boundary term remains; it is cancelled by a shift of the gauge field that lives on the D-brane on which the string ends, see Chapter 13 and [Tong §7.6.1, ll. 10880–10905].) The gauge-invariant field strength is the three-form  $H = dB$ ,

$$H_{\mu\nu\rho} = \partial_\mu B_{\nu\rho} + \partial_\nu B_{\rho\mu} + \partial_\rho B_{\mu\nu}, \quad [\text{Tong §7.2.1, ll. 9512–9518}] \quad (6.22)$$

the analogue of  $F = dA$ . It is sometimes called torsion, because it enters the sigma model as an antisymmetric part of the connection.

### 6.4.3 Only $H$ appears in the equation of motion

The claim that  $H$  is “what the string feels locally” can be made precise by varying  $S_B$ . Take conformal gauge and  $X^\lambda \rightarrow X^\lambda + \delta X^\lambda$ :

$$\delta S_B = \frac{i}{4\pi\alpha'} \int d^2\sigma \epsilon^{\alpha\beta} \left[ \partial_\lambda B_{\mu\nu} \delta X^\lambda \partial_\alpha X^\mu \partial_\beta X^\nu + 2B_{\mu\nu} \partial_\alpha \delta X^\mu \partial_\beta X^\nu \right].$$

Integrate the second term by parts. The derivative  $\partial_\alpha$  falls either on  $\partial_\beta X^\nu$ , which gives zero because  $\epsilon^{\alpha\beta} \partial_\alpha \partial_\beta = 0$ , or on  $B_{\mu\nu}(X)$ , which gives  $-2\delta X^\mu \partial_\lambda B_{\mu\nu} \partial_\alpha X^\lambda \partial_\beta X^\nu$ . After renaming indices, the bracket multiplies  $\delta X^\lambda \epsilon^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu$ , which is antisymmetric in  $\mu\nu$ , by  $\partial_\lambda B_{\mu\nu} - 2\partial_\mu B_{\lambda\nu}$ . Antisymmetrizing the last term in  $\mu\nu$  turns it into  $\partial_\mu B_{\nu\lambda} + \partial_\nu B_{\lambda\mu}$ , and the three terms assemble into (6.22):

$$\delta S_B = \frac{i}{4\pi\alpha'} \int d^2\sigma H_{\lambda\mu\nu}(X) \delta X^\lambda \epsilon^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu. \quad (6.23)$$

For a flat metric the variation of the first term of (6.19) is  $-\frac{1}{2\pi\alpha'} \int \delta X^\lambda \partial^\alpha \partial_\alpha X_\lambda$ , so the equation of motion is

$$\partial^\alpha \partial_\alpha X_\lambda = \frac{i}{2} H_{\lambda\mu\nu} \epsilon^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu. \quad (6.24)$$

This is the string's Lorentz force law. Just as  $m\ddot{x}_\lambda = F_{\lambda\mu}\dot{x}^\mu$  contains  $F$  and not  $A$ , the string's equation of motion contains  $H$  and not  $B$ . A constant  $B$ -field has  $H = 0$  and exerts no force. Section 6.9 shows that it is nevertheless physical when the target space has non-contractible two-cycles, exactly as a constant  $A$  along a circle is physical (the Aharonov–Bohm effect) although  $F = 0$ . The derivation is ours; we confirmed (6.23) by a symbolic check of the Euler–Lagrange expression for a polynomial  $B$ -field in three target dimensions.

## 6.5 The dilaton and the string coupling

The third massless field is the scalar  $\Phi(X)$ , the *dilaton*. Its coupling to the worldsheet cannot be guessed from a vertex operator as easily, because the naive operator is not primary [Tong §7.2.2, ll. 9519–9524]. The result is

$$S = \frac{1}{4\pi\alpha'} \int d^2\sigma \sqrt{g} \left[ G_{\mu\nu} \partial_\alpha X^\mu \partial_\beta X^\nu g^{\alpha\beta} + i B_{\mu\nu} \partial_\alpha X^\mu \partial_\beta X^\nu \epsilon^{\alpha\beta} + \alpha' \Phi(X) R^{(2)} \right], \quad (6.25)$$

where  $R^{(2)}$  is the Ricci scalar of the *worldsheet* metric [Tong (7.12), ll. 9525–9538]. Two features are unusual. The dilaton term vanishes on a flat worldsheet, which is why it was invisible in conformal gauge on the plane. And it is *not* Weyl invariant: under  $g \rightarrow e^{2\omega}g$  one has  $\sqrt{g} R^{(2)} \rightarrow \sqrt{g} (R^{(2)} - 2\nabla^2\omega)$ , and  $\int \sqrt{g} \Phi(X) \nabla^2\omega$  does not vanish unless  $\Phi$  is constant. We return to this in Section 6.6; first, the constant case.

### 6.5.1 A constant dilaton counts handles

For  $\Phi(X) = \lambda$ , a constant, the last term of (6.25) is

$$S_{\text{dilaton}} = \lambda \cdot \frac{1}{4\pi} \int d^2\sigma \sqrt{g} R^{(2)} = \lambda \chi, \quad \chi = 2 - 2h, \quad (6.26)$$

where  $\chi$  is the Euler number of the worldsheet and  $h$  its number of handles; by the Gauss–Bonnet theorem  $\chi$  depends only on the topology. [Tong (6.4)–(6.5), ll. 7355–7386; §7.2.2, ll. 9552–9560] A worldsheet with  $h$  handles is therefore weighted in the path integral by

$$e^{-\lambda\chi} = (e^\lambda)^{2h-2} = g_s^{2h-2}, \quad g_s = e^\lambda. \quad (6.27)$$

Adding a handle, which in spacetime means adding a closed-string loop, costs  $g_s^2$ :  $g_s$  is the *string coupling constant* (Figure 6.1). The sphere is weighted by  $g_s^{-2}$ , the torus by  $g_s^0$ . [Tong §6.1.1, ll. 7389–7405]

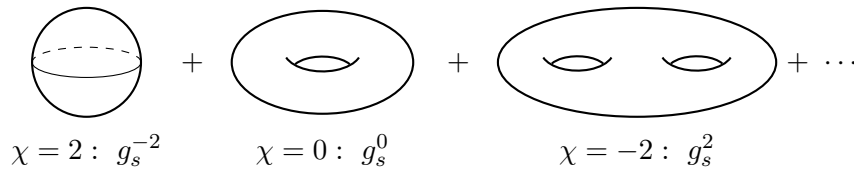


Figure 6.1: The sum over worldsheet topologies. A constant dilaton  $\Phi = \lambda$  contributes  $e^{-\lambda\chi}$  to the weight of each surface, so every handle costs a factor  $g_s^2 = e^{2\lambda}$ .

If the dilaton varies, the role of  $\lambda$  is played by its asymptotic value,

$$\Phi_0 = \lim_{X \rightarrow \infty} \Phi(X), \quad g_s = e^{\Phi_0}. \quad [\text{Tong (7.13)–(7.14), ll. 9561–9571}] \quad (6.28)$$

**Physical picture: a coupling constant that is a field**

In quantum electrodynamics the coupling  $e$  is a number that one measures and cannot explain. In string theory the corresponding number is the expectation value of a dynamical field. It can differ from place to place, and in principle it is determined by the equations of motion, like the metric. The price is a stronger condition for perturbation theory: not only  $g_s \ll 1$  asymptotically, but  $e^{\Phi(X)} \ll 1$  wherever the string goes [Tong §7.2.2, ll. 9572–9582].

## 6.6 The three beta functions

The dilaton term in (6.25) breaks Weyl invariance already classically. Why is that acceptable? The key is the explicit factor of  $\alpha'$  in front of it, present on dimensional grounds:  $\Phi$  and  $\sqrt{g} R^{(2)} d^2\sigma$  are dimensionless, while the other two terms carry two derivatives of  $X$ , of dimension  $\text{length}^2$ , to be compensated by  $1/\alpha'$ . But  $\alpha'$  is also the loop-counting parameter of the sigma model (Section 6.2). The *classical* Weyl variation of the dilaton term is therefore of the same order,  $\alpha'^1$ , as the *one-loop* Weyl anomaly (6.15) of the  $G$  and  $B$  terms, and the two can cancel. [Tong §7.2.3, ll. 9583–9596] This is what happens. The form of the classical contribution can be read off from the Weyl variation quoted below (6.25): integrating by parts moves  $\nabla^2$  onto  $\Phi(X(\sigma))$ , and the chain rule produces  $\partial_\mu \partial_\nu \Phi \partial_\alpha X^\mu \partial_\beta X^\nu$ , which has the structure of  $\beta_{\mu\nu}(G)$  in (6.16). Fixing its coefficient requires the careful treatment cited in the historical note; we quote the result.

With three background fields the trace of the stress tensor has three structures,

$$\langle T^\alpha{}_\alpha \rangle = -\frac{1}{2\alpha'} \beta_{\mu\nu}(G) g^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu - \frac{i}{2\alpha'} \beta_{\mu\nu}(B) \epsilon^{\alpha\beta} \partial_\alpha X^\mu \partial_\beta X^\nu - \frac{1}{2} \beta(\Phi) R^{(2)}, \quad (6.29)$$

[Tong (7.15), ll. 9597–9609] and the one-loop results are

$$\beta_{\mu\nu}(G) = \alpha' R_{\mu\nu} + 2\alpha' \nabla_\mu \nabla_\nu \Phi - \frac{\alpha'}{4} H_{\mu\lambda\kappa} H_\nu{}^{\lambda\kappa}, \quad (6.30)$$

$$\beta_{\mu\nu}(B) = -\frac{\alpha'}{2} \nabla^\lambda H_{\lambda\mu\nu} + \alpha' \nabla^\lambda \Phi H_{\lambda\mu\nu}, \quad (6.31)$$

$$\beta(\Phi) = -\frac{\alpha'}{2} \nabla^2 \Phi + \alpha' \nabla_\mu \Phi \nabla^\mu \Phi - \frac{\alpha'}{24} H_{\mu\nu\lambda} H^{\mu\nu\lambda}. \quad (6.32)$$

[Tong §7.2.3, ll. 9610–9628] Here  $\nabla_\mu$  is the covariant derivative of the spacetime metric  $G_{\mu\nu}$ , which also raises indices. A consistent background of the bosonic string at this order must satisfy [L]

$$\beta_{\mu\nu}(G) = \beta_{\mu\nu}(B) = \beta(\Phi) = 0. \quad (6.33)$$

Each equation is recognizable before they are combined. It helps to recognize each equation before combining them.

- $\beta(G) = 0$  is Einstein's equation with sources. For  $\Phi$  constant it reads  $R_{\mu\nu} = \frac{1}{4} H_{\mu\lambda\kappa} H_\nu{}^{\lambda\kappa}$ : the energy of the  $H$ -field curves spacetime, as the energy of  $F_{\mu\nu}$  does in Einstein–Maxwell theory.
- $\beta(B) = 0$  can be written  $\nabla^\lambda (e^{-2\Phi} H_{\lambda\mu\nu}) = 0$ , since  $\nabla^\lambda (e^{-2\Phi} H_{\lambda\mu\nu}) = e^{-2\Phi} (\nabla^\lambda H_{\lambda\mu\nu} - 2\nabla^\lambda \Phi H_{\lambda\mu\nu}) = -\frac{2}{\alpha'} e^{-2\Phi} \beta_{\mu\nu}(B)$ . This is the analogue of Maxwell's equation  $\nabla^\lambda F_{\lambda\mu} = 0$ , with a position-dependent coupling  $e^{-2\Phi}$ .
- $\beta(\Phi) = 0$  is a wave equation for the dilaton, sourced by  $H^2$ . In Section 6.8 we will see that for  $D \neq 26$  it acquires a constant term, which is the central-charge condition of Chapter 5.

All terms are of order  $\alpha'$ : they are the leading terms of an expansion in  $\alpha'/r_c^2$ . At the next order Einstein's equations are corrected; for  $H = 0$  and constant  $\Phi$  the two-loop result is  $\beta_{\mu\nu} = \alpha' R_{\mu\nu} + \frac{1}{2}\alpha'^2 R_{\mu\lambda\rho\sigma} R_{\nu}^{\lambda\rho\sigma} + \dots$  [Tong §7.3.2, ll. 9793–9812]. Such terms are negligible when  $r_c \gg \ell_s$  and are the first sign that the string is not a point.

## 6.7 The low-energy effective action

### 6.7.1 An action whose equations of motion are the beta functions

We now change the point of view. The conditions (6.33) are differential equations for fields on spacetime. Do they follow from an action principle in  $D = 26$  dimensions? They do [L]:

$$S = \frac{1}{2\kappa_0^2} \int d^{26}X \sqrt{-G} e^{-2\Phi} \left( R - \frac{1}{12} H_{\mu\nu\lambda} H^{\mu\nu\lambda} + 4 \partial_\mu \Phi \partial^\mu \Phi \right), \quad [\text{Tong (7.16), ll. 9640–9657}] \quad (6.34)$$

written in Lorentzian signature, with  $G = \det G_{\mu\nu}$  and  $R$  the spacetime Ricci scalar. This is the *low-energy effective action* of the bosonic string, in the *string frame*. The constant  $\kappa_0$  is not fixed by the equations of motion. Its dimension is: the action is dimensionless,  $R$  has dimension  $\text{length}^{-2}$  and the measure  $\text{length}^{26}$ , so  $\kappa_0^2 \sim \ell_s^{24}$ . The variation of (6.34) is

$$\begin{aligned} \delta S = \frac{1}{2\kappa_0^2 \alpha'} \int d^{26}X \sqrt{-G} e^{-2\Phi} & \left[ \delta G_{\mu\nu} \beta^{\mu\nu}(G) - \delta B_{\mu\nu} \beta^{\mu\nu}(B) \right. \\ & \left. - (2\delta\Phi + \frac{1}{2} G^{\mu\nu} \delta G_{\mu\nu}) (\beta^\lambda{}_\lambda(G) - 4\beta(\Phi)) \right], \end{aligned} \quad (6.35)$$

[Tong §7.3, ll. 9658–9668] so the equations of motion are equivalent to (6.33).

It is a good exercise in reading such a formula to verify the simplest part of it, the dilaton variation. Only  $\delta\Phi$  is switched on. From (6.34), writing  $\mathcal{L} = R - \frac{1}{12} H^2 + 4(\partial\Phi)^2$ ,

$$\begin{aligned} \delta S &= \frac{1}{2\kappa_0^2} \int \sqrt{-G} e^{-2\Phi} \left[ -2\mathcal{L} \delta\Phi + 8 \partial_\mu \Phi \partial^\mu \delta\Phi \right] \\ &= \frac{1}{2\kappa_0^2} \int \sqrt{-G} e^{-2\Phi} \delta\Phi \left[ -2\mathcal{L} - 8\nabla^2 \Phi + 16(\partial\Phi)^2 \right], \end{aligned}$$

where the last step integrates by parts and uses  $\nabla_\mu (e^{-2\Phi} \nabla^\mu \Phi) = e^{-2\Phi} (\nabla^2 \Phi - 2(\partial\Phi)^2)$ . The bracket equals  $-2[R - \frac{1}{12} H^2 + 4\nabla^2 \Phi - 4(\partial\Phi)^2]$ . On the other side, from (6.30) and (6.32),

$$\begin{aligned} \beta^\lambda{}_\lambda(G) - 4\beta(\Phi) &= \alpha' [R + 2\nabla^2 \Phi - \frac{1}{4} H^2] - 4\alpha' [-\frac{1}{2} \nabla^2 \Phi + (\partial\Phi)^2 - \frac{1}{24} H^2] \\ &= \alpha' [R + 4\nabla^2 \Phi - 4(\partial\Phi)^2 - \frac{1}{12} H^2], \end{aligned}$$

using  $-\frac{1}{4} + \frac{1}{6} = -\frac{1}{12}$ . The two agree, including the factor  $-2/\alpha'$  predicted by (6.35). Likewise the  $B$  equation of motion of (6.34) is  $\nabla^\lambda (e^{-2\Phi} H_{\lambda\mu\nu}) = 0$ , which we showed above to be  $\beta(B) = 0$ . The metric variation is longer and is left to the references. One caveat: the bosonic string also has a tachyon, ignored here; the results are of interest because they carry over to the superstring, where it is absent [Tong §7.4.5, ll. 10443–10452].

Something remarkable has happened. The starting point was one string in flat space; consistency of its motion in a condensate of its own gravitons now dictates how spacetime must curve. The same action is reached by another road: compute graviton scattering amplitudes from the string path integral and ask which field theory reproduces them at energies far below  $1/\ell_s$  [Tong §7.3, ll. 9669–9688].

### 6.7.2 String frame and Einstein frame

The action (6.34) differs from the Einstein–Hilbert action by the overall factor  $e^{-2\Phi}$ . Its origin is clear from (6.27): the action was derived on the sphere, which carries the weight  $g_s^{-2}$ ; with a varying dilaton this becomes  $e^{-2\Phi(X)}$ . The dilaton kinetic term also appears to have the wrong sign, but with a field-dependent factor in front of  $R$  the kinetic terms of  $G$  and  $\Phi$  are mixed and no conclusion can be drawn before they are disentangled. [Tong §7.3.1, ll. 9690–9701]

To disentangle them, split the dilaton into its constant and varying parts,  $\tilde{\Phi} = \Phi - \Phi_0$ , and define a new metric in  $D$  dimensions by a Weyl rescaling of *spacetime*,

$$\tilde{G}_{\mu\nu}(X) = e^{2\omega} G_{\mu\nu}(X), \quad \omega = -\frac{2\tilde{\Phi}}{D-2}. \quad [\text{Tong (7.17)–(7.18), ll. 9702–9715}] \quad (6.36)$$

This is a relabelling of the fields, not a symmetry. We need the behaviour of each ingredient of (6.34). The Ricci scalars of two conformally related metrics obey

$$\tilde{R} = e^{-2\omega} [R - 2(D-1)\nabla^2\omega - (D-2)(D-1)\partial_\mu\omega\partial^\mu\omega] \quad [\text{Tong §7.3.1, ll. 9722–9726}] \quad (6.37)$$

(we confirmed this identity by a symbolic check for conformally flat  $G$  in  $D = 2, 3, 4$ ). Exchanging the roles of the two metrics, i.e.  $\omega \rightarrow -\omega$ , gives  $R = e^{2\omega} [\tilde{R} + 2(D-1)\tilde{\nabla}^2\omega - (D-2)(D-1)(\tilde{\partial}\omega)^2]$ , where a tilde means that indices are contracted with  $\tilde{G}$ . Furthermore  $\sqrt{-G} = e^{-D\omega} \sqrt{-\tilde{G}}$ ; every inverse metric contributes  $G^{\mu\nu} = e^{2\omega} \tilde{G}^{\mu\nu}$ ; and, by the choice of  $\omega$ ,  $e^{-2\tilde{\Phi}} = e^{(D-2)\omega}$ . The prefactor of the curvature term is then

$$\sqrt{-G} e^{-2\tilde{\Phi}} e^{2\omega} = \sqrt{-\tilde{G}} e^{(-D+(D-2)+2)\omega} = \sqrt{-\tilde{G}},$$

which is the purpose of (6.36): the factor in front of  $\tilde{R}$  is now constant. The term  $\tilde{\nabla}^2\omega$  is a total derivative and drops out. The  $(\partial\omega)^2$  terms are  $-(D-2)(D-1)(\tilde{\partial}\omega)^2$  from the curvature and, from the dilaton kinetic term,  $4(\partial\Phi)^2 = 4e^{2\omega}(\tilde{\partial}\tilde{\Phi})^2 = e^{2\omega}(D-2)^2(\tilde{\partial}\omega)^2$ . Their sum is

$$[(D-2)^2 - (D-2)(D-1)](\tilde{\partial}\omega)^2 = -(D-2)(\tilde{\partial}\omega)^2 = -\frac{4}{D-2}(\tilde{\partial}\tilde{\Phi})^2.$$

Finally  $H^2$  contains three inverse metrics, so it acquires  $e^{(-D+(D-2)+6)\omega} = e^{4\omega} = e^{-8\tilde{\Phi}/(D-2)}$ . For  $D = 26$  the numbers are  $4/(D-2) = \frac{1}{6}$  and  $8/(D-2) = \frac{1}{3}$ , and the action becomes

$$S = \frac{1}{2\kappa^2} \int d^{26}X \sqrt{-\tilde{G}} \left( \tilde{R} - \frac{1}{12} e^{-\tilde{\Phi}/3} H_{\mu\nu\lambda} H^{\mu\nu\lambda} - \frac{1}{6} \partial_\mu \tilde{\Phi} \partial^\mu \tilde{\Phi} \right), \quad [\text{Tong (7.19), ll. 9727–9744}] \quad (6.38)$$

with

$$\kappa^2 = \kappa_0^2 e^{2\Phi_0} \sim \ell_s^{24} g_s^2. \quad [\text{Tong (7.20), ll. 9756–9760}] \quad (6.39)$$

This is the *Einstein frame*. The gravitational term has the standard form with  $8\pi G_N = \kappa^2$ , and the dilaton is an ordinary scalar with a kinetic term of the correct sign: the apparent wrong sign in the string frame was an artefact of the mixing. There is no potential for  $\tilde{\Phi}$ , so in the bosonic string nothing fixes  $g_s$  [Tong §7.3.1, ll. 9745–9752].

#### Common pitfall: which metric measures lengths?

$G_{\mu\nu}$  is the metric that the string feels: it is the one in (6.1), and lengths compared with  $\ell_s$  (the radius  $R$  of a circle in Chapter 7, the statement  $R \rightarrow \alpha'/R$  in Chapter 8) are string-frame lengths.  $\tilde{G}_{\mu\nu}$  is the metric in which Newton's constant is constant. The choice exists because there is a massless scalar that can be absorbed in the ruler. A massless dilaton



would also mediate a long-range force that violates the equivalence principle, which is one reason why any realistic model must give  $\Phi$  a mass [Tong §7.3.1, ll. 9773–9791]; this is part of the moduli problem of Chapter 26.

**Example 6.3** (String length versus Planck length). Define the Planck length in  $D$  dimensions by  $8\pi G_N = \kappa^2 = \ell_p^{D-2}$ . The dimensional analysis that gave (6.39) gives  $\kappa^2 \sim \ell_s^{D-2} g_s^2$  in any  $D$ , hence

$$\frac{\ell_p}{\ell_s} \sim g_s^{2/(D-2)}.$$

For the bosonic string,  $D = 26$ , the exponent is  $\frac{1}{12}$ : a coupling  $g_s = 0.1$  gives  $\ell_p/\ell_s \approx 0.83$ . For a superstring,  $D = 10$ , the exponent is  $\frac{1}{4}$  and  $g_s = 0.1$  gives  $\ell_p/\ell_s \approx 0.56$ . In both cases weak coupling means  $\ell_p < \ell_s$ : the string scale is reached *before* the Planck scale as one goes to short distances [Tong §7.3.1, ll. 9761–9772]. (Compactification on a space  $X$  divides  $\kappa^2$  by  $\text{Vol}(X)$  [Tong §7.4.1, ll. 10007–10020].)

**What carries over to the superstring.** Each of the superstring theories listed in Chapter 5 contains the same three massless fields. The part of its ten-dimensional effective action that involves only them has the form (6.34) with  $d^{26}X \rightarrow d^{10}X$ , where for the heterotic theories  $H$  contains an additional Chern–Simons term; the remaining bosonic fields (Ramond–Ramond forms, or the heterotic gauge field) enter through a second piece  $S_2$  [Tong §7.3.3, ll. 9825–9935]. [L] The sector  $(G, B, \Phi)$  is called the *NS–NS sector*. It is the sector on which T-duality acts geometrically, and it is the field content of double field theory: Chapter 17 rewrites (6.34) so that  $G$  and  $B$  enter only through the generalized metric and  $\sqrt{-G}e^{-2\Phi}$  is a single field,  $e^{-2d}$ . The Ramond–Ramond fluxes return in Chapters 25 and 26.

## 6.8 The critical dimension revisited: the linear dilaton

The function  $\beta(\Phi)$  multiplies  $R^{(2)}$  in (6.29), exactly like the central charge in the Weyl anomaly  $T^\alpha_\alpha = -\frac{c}{12}R^{(2)}$  of Chapters 4 and 5. The two are the same object:  $\beta(\Phi)$  is the total central charge seen as a function of the background, and the constant part that we have so far set to zero by taking  $D = 26$  is its leading term. For general  $D$ ,

$$\beta(\Phi) = \frac{D-26}{6} - \frac{\alpha'}{2}\nabla^2\Phi + \alpha'\nabla_\mu\Phi\nabla^\mu\Phi - \frac{\alpha'}{24}H_{\mu\nu\lambda}H^{\mu\nu\lambda} = 0, \quad [\text{Tong (7.28), ll. 10289–10302}] \quad (6.40)$$

and the string-frame action acquires the term  $-2(D-26)/3\alpha'$  inside the bracket of (6.34), a potential for the dilaton. The constant is of order  $\alpha'^0$  while the other terms are of order  $\alpha'^1$ : a tree-level term is being balanced against one-loop terms, and in general one should then not trust the truncation of the expansion. [Tong §7.4.4, ll. 10303–10336]

**Example 6.4** (The linear dilaton). There is one solution of (6.40) that can be trusted. Take flat space,  $H = 0$ , and a dilaton linear in one coordinate,  $\Phi = V_\mu X^\mu$ . Then  $\nabla_\mu\nabla_\nu\Phi = 0$ , so  $\beta(G) = \beta(B) = 0$ , and (6.40) becomes the algebraic condition

$$V_\mu V^\mu = \frac{26-D}{6\alpha'}. \quad (6.41)$$

With the mostly-plus signature  $V$  is spacelike for  $D < 26$  and timelike for  $D > 26$ :  $\Phi = \sqrt{(26-D)/6\alpha'} X^1$  or  $\Phi = \sqrt{(D-26)/6\alpha'} X^0$ . [Tong §7.4.4, ll. 10337–10365] For numbers: in  $D = 25$  the dilaton grows by one unit, and  $g_s^{\text{eff}} = e^\Phi$  by a factor  $e$ , over the distance  $\sqrt{6}\ell_s \approx 2.4\ell_s$ ; in  $D = 2$  one finds  $V^2 = 4/\alpha'$  and the same happens over  $\ell_s/2$ . The

background can be trusted because the sigma model is still free, with a modified stress tensor of central charge  $c = D + 6\alpha' V^2 = 26$  [Tong §7.4.4, ll. 10361–10441]. The lesson is that “ $D = 26$ ” really means “total central charge 26” (Chapter 5), and that the price of  $D \neq 26$  is a coupling that becomes strong somewhere in spacetime.

## 6.9 Constant backgrounds on a torus: what survives of $B$

Part II studies the simplest solutions of (6.33): a flat torus  $T^d$  with constant metric  $G_{ij}$ , constant  $B_{ij}$  and constant dilaton,  $i, j = 1, \dots, d$ . All three beta functions vanish trivially:  $R_{\mu\nu} = 0$ ,  $H = dB = 0$ ,  $\partial\Phi = 0$ . The constant  $G_{ij}$  obviously matters, since it fixes the sizes and angles of the torus. Does a constant  $B_{ij}$  matter? Locally it is pure gauge:  $B = dC$  with  $C_\nu = \frac{1}{2}B_{\mu\nu}X^\mu$ . In infinite flat space it could therefore be removed. On a torus it cannot, for two related reasons.

### 6.9.1 Periods of $B$

The one-form  $C_\nu = \frac{1}{2}B_{\mu\nu}X^\mu$  is not periodic, so it is not a legitimate gauge parameter on the torus, in the same way that a constant  $A_\theta$  on a circle cannot be gauged away by a single-valued function. What cannot be gauged away is the integral of  $B$  over a two-cycle. Take  $T^2$  with coordinates  $X^1 \sim X^1 + 2\pi R_1$ ,  $X^2 \sim X^2 + 2\pi R_2$ , and a worldsheet that is itself a torus,  $\sigma^{1,2} \in [0, 2\pi)$ , wrapped on the target by the linear map

$$X^1 = R_1(a\sigma^1 + b\sigma^2), \quad X^2 = R_2(c\sigma^1 + e\sigma^2), \quad a, b, c, e \in \mathbb{Z}.$$

The integers are winding numbers, as in Chapter 7, now around both cycles of the worldsheet (Figure 6.2). From (6.20),

$$S_B = \frac{i}{2\pi\alpha'} B_{12}R_1R_2 (ae - bc) (2\pi)^2 = 2\pi i b_{12} N, \quad b_{12} = \frac{B_{12}R_1R_2}{\alpha'} = \frac{1}{4\pi^2\alpha'} \int_{T^2} B, \quad (6.42)$$

with  $N = ae - bc$ . The integer  $N$  is the (signed) number of times the worldsheet covers the target torus. The path integral weight  $e^{-S_B} = e^{-2\pi i b_{12} N}$  depends on  $b_{12}$ : a constant  $B$ -field is observable, although it exerts no force. It is unchanged when  $b_{12} \rightarrow b_{12} + 1$ . In the source’s words,  $B$  plays the role of a theta parameter: the constant  $B$ -term is a total derivative and gives only topological contributions, and an integer shift changes the action by an integer multiple of  $2\pi i$  [GPR §2.4, ll. 1351–1371]. [L] So the dimensionless periods  $b_{ij}$  are *angles*:

$$b_{ij} \sim b_{ij} + \Theta_{ij}, \quad \Theta_{ij} = -\Theta_{ji} \in \mathbb{Z}. \quad (6.43)$$

These integer shifts are the first elements of the duality group  $O(d, d; \mathbb{Z})$  that we meet; they are taken up in Chapters 10 and 11. The normalization in (6.42) was derived here from Tong’s action; GPR use coordinates of period  $2\pi$  and a dimensionless  $B$ , in which the shift is by integers, as in (6.43).

### 6.9.2 $B$ shifts the momentum: the Hamiltonian of a wound string

The second way to see the effect of a constant  $B$  is canonical, and it leads directly to the central object of Part III. Pass to a Lorentzian worldsheet with coordinates  $(\tau, \sigma)$ ,  $\dot{X} = \partial_\tau X$ ,  $X' = \partial_\sigma X$ . For constant  $G_{ij}$  and  $B_{ij}$  the Lagrangian density of (6.19) is

$$\mathcal{L} = \frac{1}{4\pi\alpha'} \left[ G_{ij}(\dot{X}^i \dot{X}^j - X'^i X'^j) + 2B_{ij} \dot{X}^i X'^j \right],$$

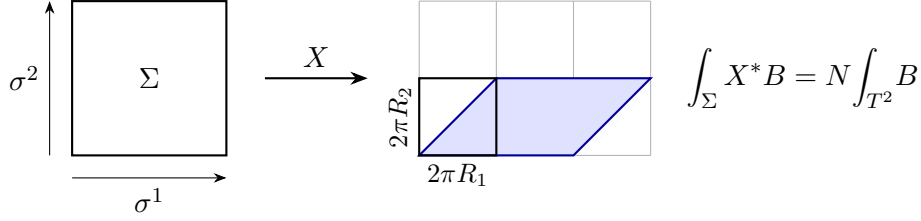


Figure 6.2: A toroidal worldsheet wrapped on a target two-torus, drawn in the covering plane (grid: copies of the unit cell). The image of the worldsheet for  $(a, b, c, e) = (2, 1, 0, 1)$  is the shaded parallelogram; its area is  $N = ae - bc = 2$  unit cells, so  $e^{-S_B} = e^{-4\pi i b_{12}}$ .

where the overall sign of the  $B$  term depends on the orientation convention for  $\epsilon^{\alpha\beta}$  and can be absorbed in  $B \rightarrow -B$ . The term  $2B_{ij}\dot{X}^i X'^j$  is a total derivative when  $B$  is constant, so it does not change the equations of motion, but it changes the canonical momentum:

$$2\pi\alpha' P_i = 2\pi\alpha' \frac{\partial \mathcal{L}}{\partial \dot{X}^i} = G_{ij}\dot{X}^j + B_{ij}X'^j. \quad (6.44)$$

This is the string analogue of  $p = m\dot{x} + qA$ , and it agrees with [GPR (2.4.7), ll. 1104–1106] in units  $\alpha' = 1$ . For a string that winds the torus,  $X'^j \neq 0$  on average, and the quantized momentum  $\oint P_i$  is no longer the kinetic momentum: *a constant B-field shifts the momentum of wound strings*, in proportion to their winding. This is how  $B$  enters the spectrum in Chapter 10.

The Hamiltonian density is  $\mathcal{H} = P_i \dot{X}^i - \mathcal{L}$ . The  $B$  term is linear in velocities and cancels, leaving  $\mathcal{H} = \frac{1}{4\pi\alpha'} G_{ij}(\dot{X}^i \dot{X}^j + X'^i X'^j)$ ; but  $\mathcal{H}$  must be expressed through  $P$ , not  $\dot{X}$ . Writing  $\Pi_i = 2\pi\alpha' P_i$  and solving (6.44),  $\dot{X} = G^{-1}(\Pi - BX')$ . Then, using  $B^\top = -B$ ,

$$\begin{aligned} \dot{X}^\top G \dot{X} &= (\Pi - BX')^\top G^{-1} (\Pi - BX') \\ &= \Pi^\top G^{-1} \Pi + X'^\top B G^{-1} \Pi - \Pi^\top G^{-1} B X' - X'^\top B G^{-1} B X', \end{aligned}$$

and therefore

$$\mathcal{H} = \frac{1}{4\pi\alpha'} \begin{pmatrix} X' \\ \Pi \end{pmatrix}^\top \begin{pmatrix} G - B G^{-1} B & B G^{-1} \\ -G^{-1} B & G^{-1} \end{pmatrix} \begin{pmatrix} X' \\ \Pi \end{pmatrix}. \quad (6.45)$$

The  $2d \times 2d$  matrix is the *generalized metric*  $\mathcal{H}(G, B)$ , in exactly the block form fixed in this book's conventions and used by GPR for the zero-mode spectrum [GPR (2.4.9), ll. 1113–1121; (2.4.17)–(2.4.18), ll. 1262–1275]. We confirmed (6.45), and the identity  $\mathcal{H}\eta\mathcal{H} = \eta$  with  $\eta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , by a symbolic check for  $d = 2$ . The energy of a string on a torus depends on  $G$  and  $B$  only through  $\mathcal{H}$ , and it treats the winding density  $X'$  and the momentum density  $\Pi$  on the same footing. Chapter 15 takes this matrix as its starting point, and Chapter 31 builds this programme's dual-scale principle on it.

## 6.10 What the machine has checked

Nothing in this chapter has been formalized in the strict sense. The statements above involve a two-dimensional quantum field theory, its renormalization, and Riemannian geometry of the target space. Mathlib has smooth manifolds, but at the time of writing, to our knowledge, no Levi-Civita connection or Ricci tensor in a form that would allow Proposition 6.1 even to be *stated*, and no rigorous definition of the sigma-model path integral exists on paper. All results of Sections 6.3, 6.6 and 6.7 are therefore Tier L.

The library does contain one older module whose subject is the effective action (6.34), as the target of the reduction of double field theory, the file `ActionCurvature.lean` of the library `DoubleFieldTheory`. We mention it as a pointer forward to Chapter 17, and because it is a clear example of what the book calls an *arithmetic shadow*. It models  $\sqrt{-G}$ ,  $e^{-2\Phi}$ ,  $R$ ,  $(\nabla\Phi)^2$  and  $H^2$  by *integers*.

#### In Lean: `DoubleFieldTheory/ActionCurvature.lean`

Two definitions and the three theorems closest to this chapter, in the namespace `DoubleFieldTheory.ActionCurvature`:

```
def DFTRicciComponents (R_geom kin_phi H_sq : Int) : Int :=
  R_geom + 4 * kin_phi - H_sq

theorem dft_ricci_expansion (R_geom kin_phi H_sq : Int) :
  DFTRicciComponents R_geom kin_phi H_sq + H_sq
    = R_geom + 4 * kin_phi := by ...

theorem dft_ricci_physical_reduction (R_geom : Int) :
  DFTRicciComponents R_geom 0 0 = R_geom := by ...

def ContractedEinstein (D R : Int) : Int :=
  (D - 2) * R

theorem contracted_einstein_dim2 (R : Int) :
  ContractedEinstein 2 R = 0 := by ...
```

**Plain reading.** For all integers  $r, k, h$  one has  $(r + 4k - h) + h = r + 4k$ , then  $r + 4 \cdot 0 - 0 = r$ , and finally  $(2 - 2) \cdot r = 0$ . The file also proves that the two integer products `DilatonMeasure` and `ActionLagrangian` are commutative (`dilaton_measure_symm`, `dft_action_nsns_equivalence`) and that  $t + \delta = t$  when  $\delta = 0$  (`connection_trace_conservation`).

**Proof idea.** Unfold the definition with `dsimp`; the linear-arithmetic decision procedure `omega` closes the goal, or `Int.mul_comm` is rewritten once.

**What this is a shadow of.** `DFTRicciComponents` mirrors the *shape* of the bracket in (6.34): the coefficient 4 of the dilaton kinetic term and the minus sign of  $H^2$  are there; the coefficient  $\frac{1}{12}$  is absorbed in the integer argument `H_sq`. The theorem `dft_ricci_physical_reduction` is the arithmetic shadow of the remark that for  $\Phi$  constant and  $H = 0$  the Lagrangian of (6.34) is the Einstein–Hilbert one. The theorem `contracted_einstein_dim2` is the shadow of a fact used silently in Section 6.5: in two dimensions  $R_{\alpha\beta} = \frac{1}{2}g_{\alpha\beta}R^{(2)}$ , the Einstein tensor vanishes, and  $\int \sqrt{g} R^{(2)}$  has no dynamics. (The trace of the Einstein tensor in dimension  $D$  is  $-\frac{1}{2}(D-2)R$ ; the Lean definition is the integer  $(D-2)R$ , proportional to it.) None of the theorems involves a metric, a derivative or an integral, and none asserts that the double-field-theory action reduces to (6.34). The docstrings of the file describe that reduction in words; docstrings are not checked by the kernel.

We ran `#print axioms` on a scratch copy of the file: the six theorems depend only on `propext` and, for those closed by `omega`, `Quot.sound`; both belong to the three standard axioms of Lean.

*Remark 6.5* (What a faithful formalization would need). Two targets are within reach of present tools. (i) The algebra of Section 6.9: for real matrices  $G$  (invertible, symmetric) and  $B$  (antisymmetric), the identity behind (6.45) and  $\mathcal{H}\eta\mathcal{H} = \eta$ . The newer library `DualScaleStream2` proves statements of this kind about  $\mathcal{H}(G, B)$ , namely `etaR_genMetric_sq`

and `genMetric_symm` (Chapter 15). (ii) Once (6.37) is granted, the Einstein-frame computation reduces to the identity  $(D-2)^2 - (D-2)(D-1) = -(D-2)$ , which a student can prove in Lean in one line (Exercise 6.4). The beta functions are out of reach until Riemannian curvature is available in Mathlib, and their derivation from a path integral is out of reach for deeper reasons. Chapter 38 returns to this.

| statement                                                                                                                           | tier | evidence                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------|------|----------------------------------------------------------------------------------------------------------------------------|
| $(r+4k-h)+h = r+4k$ , $r+4\cdot 0-0 = r$ , $(2-2)r = 0$<br>over $\mathbb{Z}$                                                        | [A]  | <code>dft_ricci_expansion</code> ,<br><code>dft_ricci_physical_reduction</code> ,<br><code>contracted_einstein_dim2</code> |
| $\beta_{\mu\nu}(G) = \alpha' R_{\mu\nu}$ at one loop; the beta functions<br>(6.30)–(6.32); the action (6.34)                        | [L]  | Tong §7.1, §7.2.3, §7.3                                                                                                    |
| Dilaton equation of (6.34) $\Leftrightarrow \beta^\lambda{}_\lambda - 4\beta(\Phi) = 0$ ;<br>Einstein frame (6.38); identity (6.37) | [L]  | Tong §7.3, §7.3.1; derivations here;<br>symbolic check                                                                     |
| $g_s = e^{\Phi_0}$ ; weight $g_s^{2h-2}$                                                                                            | [L]  | Tong §6.1.1, §7.2.2                                                                                                        |
| String equation of motion (6.24) contains only $H$                                                                                  | [L]  | derived here; symbolic check                                                                                               |
| $b_{ij} \sim b_{ij} + \Theta_{ij}$ , $\Theta \in \mathbb{Z}$                                                                        | [L]  | GPR §2.4; derivation here                                                                                                  |
| Hamiltonian (6.45) = generalized metric                                                                                             | [L]  | GPR (2.4.9), (2.4.17); symbolic check                                                                                      |
| Identification of the integers in the Lean file with<br>curvatures and measures                                                     | —    | a modelling statement, never Tier A                                                                                        |

### Chapter summary

- A string in a background is a two-dimensional sigma model (6.25); the background fields  $G_{\mu\nu}$ ,  $B_{\mu\nu}$ ,  $\Phi$  are its coupling functions and are coherent states of the string's own massless modes.
- Worldsheet perturbation theory is an expansion in  $\alpha'/r_c^2$ , distinct from the one in  $g_s$ . Weyl invariance requires the beta functions to vanish. At one loop  $\beta_{\mu\nu}(G) = \alpha' R_{\mu\nu} + \dots$ : Einstein's equations are a consistency condition.
- The string is charged under the two-form  $B$ . The gauge symmetry  $B \rightarrow B + dC$  makes only  $H = dB$  visible in the equation of motion.
- The dilaton couples to the worldsheet curvature. Its constant part counts handles,  $g_s = e^{\Phi_0}$ , and its classical Weyl variation cancels against one-loop effects.
- The beta-function equations follow from the string-frame action (6.34); a Weyl rescaling of spacetime brings it to the Einstein frame (6.38), with  $\kappa^2 \sim \ell_s^{24} g_s^2$ .
- On a torus a constant  $B$  is locally pure gauge but its periods are angles,  $b_{ij} \sim b_{ij} + 1$ . It shifts the canonical momentum of wound strings, and the Hamiltonian depends on  $(G, B)$  through the generalized metric  $\mathcal{H}(G, B)$ .
- In Lean, only integer identities mirroring the shape of the effective Lagrangian are checked (Tier A as arithmetic); every physical statement of the chapter is Tier L.

## Exercises

**Exercise 6.1** (★). (a) Verify directly from (6.22) that  $H$  is invariant under (6.21). (b) Show that a constant  $B_{\mu\nu}$  on  $\mathbb{R}^D$  equals  $\partial_\mu C_\nu - \partial_\nu C_\mu$  for  $C_\nu = \frac{1}{2} B_{\mu\nu} X^\mu$  (mind the sign). (c) Explain why this  $C$  is not an admissible gauge parameter on a torus, and compare with a constant gauge potential  $A_\theta$  on a circle.

**Exercise 6.2** (\*). A closed-string amplitude with  $h$  handles is weighted by  $g_s^{2h-2}$ . The Euler number of a surface with  $h$  handles and  $b$  boundaries is  $\chi = 2 - 2h - b$  [Tong §6.3, ll. 8129–8130]. Which power of  $g_s$  weights the disk, and why does this suggest that the tension of a D-brane scales as  $1/g_s$  (Chapter 13)?

**Exercise 6.3** (\*\*). Repeat the derivation of the Einstein frame for general  $D$  and show that

$$S = \frac{1}{2\kappa^2} \int d^D X \sqrt{-\tilde{G}} \left( \tilde{R} - \frac{1}{12} e^{-8\tilde{\Phi}/(D-2)} H^2 - \frac{4}{D-2} (\tilde{\partial}\tilde{\Phi})^2 \right).$$

Evaluate the two coefficients for  $D = 10$ . Why does the construction fail for  $D = 2$ ?

**Exercise 6.4** (\*\*, Lean). State and prove in Lean 4 with Mathlib the identity used in the Einstein-frame computation: for every rational (or real)  $D$ ,  $(D-2)^2 - (D-2)(D-1) = -(D-2)$ . Then state and prove: if  $D \neq 2$  then  $-(D-2) \cdot (2/(D-2))^2 = -4/(D-2)$ . Which tactic closes the first goal, and what extra step does the second need? Say precisely what your theorems do *not* prove about the effective action.

**Exercise 6.5** (\*\*). For  $d = 1$  there is no  $B$ -field and  $G_{11} = R^2/\alpha'$  in coordinates of period  $2\pi\sqrt{\alpha'}$ . Write (6.45) in this case and show that the exchange  $X' \leftrightarrow \Pi$  together with  $G_{11} \rightarrow 1/G_{11}$  leaves  $\mathcal{H}$  invariant. Which transformation of  $R$  is this (Chapter 8)? For  $d = 2$ , verify by hand that  $\det \mathcal{H}(G, B) = 1$ .

**Exercise 6.6** (\*\*\*). *A three-sphere supported by H-flux.* Let the target space contain a round  $S^3$  of radius  $r$  with  $H = h\varepsilon$ , where  $\varepsilon$  is the volume form of the sphere, and constant dilaton. (a) Using  $R_{\mu\nu} = (2/r^2)G_{\mu\nu}$  on  $S^3$  and  $\varepsilon_{\mu\lambda\kappa}\varepsilon_\nu{}^{\lambda\kappa} = 2G_{\mu\nu}$ , show that  $\beta(G) = \beta(B) = 0$  if and only if  $h = 2/r$ . (b) Show that then  $-\frac{\alpha'}{24}H^2 = -\alpha'/r^2$ , so that by (6.40) the sphere contributes  $3 - 6\alpha'/r^2$  instead of 3 to the central-charge budget. (c) Assume the flux is quantized as  $\frac{1}{4\pi^2\alpha'} \int_{S^3} H = k \in \mathbb{Z}$  (standard, not checked against a source in this book's library). With  $\text{Vol}(S^3) = 2\pi^2 r^3$  show that  $r^2 = k\alpha'$  and that the central charge is  $3 - 6/k$ . Compare with the first two terms of the expansion in  $1/k$  of  $3k/(k+2)$ , the exact central charge of the  $SU(2)$  Wess–Zumino–Witten model at level  $k$  (standard, not in this book's library). For which  $k$  is the one-loop result reliable?

**Exercise 6.7** (\*\*\*). Consider (6.34) in  $D$  dimensions with  $H = 0$  on the ansatz  $ds^2 = -dt^2 + \sum_i a_i(t)^2 (dx^i)^2$ ,  $\Phi = \Phi(t)$ , and define the shifted dilaton  $e^{-2d} = \sqrt{-G} e^{-2\Phi}$ , i.e.  $d = \Phi - \frac{1}{2} \sum_i \ln a_i$ . (a) Show that, up to a total derivative,  $\sqrt{-G} e^{-2\Phi} (R + 4\partial_\mu \Phi \partial^\mu \Phi)$  equals  $e^{-2d} [\sum_i (\dot{a}_i/a_i)^2 - 4\dot{d}^2]$  (we confirmed this by a symbolic check for three spatial dimensions). (b) Conclude that the action is invariant under  $a_i \rightarrow 1/a_i$  for any single  $i$  with  $d$  fixed. This *scale-factor duality* is the cosmological form of T-duality; it is used in Chapter 32. (c) Which combination of  $G$  and  $\Phi$  is *not* invariant? Relate your answer to the role of  $e^{-2d}$  in Chapter 17.

## Further reading

- [Tong §7.1, ll. 9151–9418]: the sigma model, the  $\alpha'$  expansion, the one-loop beta function in the counterterm and the Weyl-anomaly language, and Ricci flow. [Tong §7.2, ll. 9419–9628]: the couplings to  $B$  and  $\Phi$ , the string coupling and the three beta functions; the original references in its footnote are not in this book's library.
- [Tong §7.3, ll. 9629–9941]: the effective action, the two frames,  $\alpha'$  and  $g_s$  corrections, the superstring actions. [Tong §7.4, ll. 9942–10380]: simple solutions, among them compactifications and the linear dilaton.

- [GPR §2.4, ll. 1053–1121 and 1345–1371]: the toroidal sigma model with constant  $G$  and  $B$  and the integer shifts of  $B$  (Chapters 10 and 11). [Tong §7.6.1, ll. 10844–10905]: how the gauge symmetry of  $B$  is completed on open strings by the gauge field on a D-brane (Chapter 13).





## Part II

# Compactification and T-Duality



## Chapter 7

# Compactification on a Circle: Momentum and Winding

Suppose one of the 25 spatial directions of the bosonic string is not a line but a circle of radius  $R$ . What does an observer in the remaining 25 dimensions see? For a point particle the answer has been known since Kaluza and Klein: a tower of particles with masses  $|n|/R$ ,  $n \in \mathbb{Z}$ , charged under a gauge field that descends from the higher-dimensional metric. This chapter derives that answer and then asks the same question of a string. The string gives a second tower, with masses  $|w|R/\alpha'$ ,  $w \in \mathbb{Z}$ , coming from strings that wrap the circle. No point particle can do that. The result is the mass formula

$$M^2 = \frac{n^2}{R^2} + \frac{w^2 R^2}{\alpha'^2} + \frac{2}{\alpha'}(N + \tilde{N} - 2), \quad N - \tilde{N} = nw, \quad (7.1)$$

which we derive step by step from the mode expansion of Chapter 2 and the constraints of Chapter 3. A student should care because the rest of Part II and all of Part III start from this formula. T-duality (Chapter 8) is the observation that (7.1) does not change under  $R \rightarrow \alpha'/R$ ,  $n \leftrightarrow w$ . The Narain lattice (Chapter 10) is the set of pairs  $(p_L, p_R)$  introduced below. The doubled coordinates of double field theory (Chapter 14) are the variables conjugate to  $n$  and  $w$ . The “dual-scale” principle of Chapter 31 is a statement about the first two terms of (7.1). The chapter closes with the Lean declarations that encode the momentum part of (7.1), and with one theorem, `massForm_circle`, which proves that two independently written libraries compute the same number.

Conventions are those of the book:  $\hbar = c = 1$ , mostly-plus signature,  $\alpha'$  of dimension length<sup>2</sup>, string length  $\ell_s = \sqrt{\alpha'}$ , tension  $T = 1/(2\pi\alpha')$ , closed strings with  $\sigma \sim \sigma + 2\pi$  and  $\sigma^\pm = \tau \pm \sigma$ . The compact coordinate is  $X^{25}$ , written  $X$  when no confusion can arise; the non-compact coordinates are  $X^\mu$  with  $\mu = 0, \dots, 24$ . The integer  $n$  is the momentum number and  $w$  the winding number. Tong writes  $m$  for our  $w$ ; Giveon, Porrati and Rabinovici write  $m$  as well and use dimensionless momenta, as explained in Remark 7.2.

## 7.1 Kaluza–Klein reduction of a field

### 7.1.1 A scalar field on $\mathbb{R}^{1,D-2} \times S^1$

Before the string, take the simplest object that can live on a spacetime with one circular direction: a free massless real scalar field  $\Phi(x^\mu, y)$  in  $D$  dimensions, where  $x^\mu$  are coordinates on  $\mathbb{R}^{1,D-2}$  and  $y \sim y + 2\pi R$  is the coordinate on the circle. Its action is

$$S = -\frac{1}{2} \int d^{D-1}x \int_0^{2\pi R} dy \left( \partial_\mu \Phi \partial^\mu \Phi + (\partial_y \Phi)^2 \right). \quad (7.2)$$

The field must be single valued on the circle,  $\Phi(x, y + 2\pi R) = \Phi(x, y)$ , so it has a Fourier series

$$\Phi(x, y) = \sum_{n \in \mathbb{Z}} \Phi_n(x) e^{iny/R}, \quad \Phi_{-n} = \Phi_n^*, \quad (7.3)$$

where the second condition makes  $\Phi$  real [Tong §8.1.1, ll. 11283–11310]. Insert (7.3) into (7.2). The  $y$ -integral of a product of two exponentials is

$$\int_0^{2\pi R} dy e^{i(n+m)y/R} = 2\pi R \delta_{n+m,0},$$

so only pairs  $(n, -n)$  survive. For the  $y$ -derivative term,  $\partial_y \Phi = \sum_n (in/R) \Phi_n e^{iny/R}$ , and the pair  $(n, m = -n)$  contributes  $(in/R)(-in/R) \Phi_n \Phi_{-n} = (n^2/R^2) |\Phi_n|^2$ . The result is

$$S = -\frac{2\pi R}{2} \sum_{n \in \mathbb{Z}} \int d^{D-1}x \left( \partial_\mu \Phi_n \partial^\mu \Phi_{-n} + \frac{n^2}{R^2} |\Phi_n|^2 \right). \quad (7.4)$$

This is the action of infinitely many fields in  $D - 1$  dimensions. The field  $\Phi_0$  is real and massless. For each  $n \geq 1$  the pair  $\Phi_{\pm n}$  is one complex field whose mass can be read off from the term without derivatives:

$$M_n^2 = \frac{n^2}{R^2} \quad (7.5)$$

[L] [Tong §8.1.1, eq. (8.3), ll. 11311–11335].

There is a one-line way to remember (7.5). A massless particle in  $D$  dimensions has  $p_M p^M = 0$ . Split the momentum as  $p_M = (p_\mu, p_y)$ . An observer who cannot resolve the circle measures the  $(D - 1)$ -dimensional mass  $M^2 = -p_\mu p^\mu = p_y^2$ . The wavefunction  $e^{ip_y y}$  is single valued only if  $p_y = n/R$ . Momentum along a hidden direction looks like mass.

#### Physical picture: why a small circle is invisible

The modes with  $n \neq 0$  cost an energy of at least  $1/R$ . An experiment with energy  $E \ll 1/R$  cannot produce them, and sees only  $\Phi_0(x)$ : a field that does not depend on  $y$  at all. In this sense the extra dimension disappears at low energy; equivalently, a probe of wavelength  $\lambda \gg R$  cannot resolve the circle. In ordinary units the first Kaluza–Klein mass is  $M_1 c^2 = \hbar c / R$  with  $\hbar c \simeq 197.3 \text{ MeV fm}$ . A circle of radius  $R = 10^{-19} \text{ m} = 10^{-4} \text{ fm}$  would give  $M_1 c^2 \simeq 1.97 \text{ TeV}$ , and a smaller circle gives a heavier tower. For a field this is the whole story: the smaller the circle, the better it hides. For a string it is half of the story, as Section 7.2 shows.

### 7.1.2 The Kaluza–Klein gauge field and the charge of the tower

A field theory that contains gravity has more than scalars. The  $D$ -dimensional metric  $G_{MN}$  splits, from the  $(D - 1)$ -dimensional point of view, into a metric  $\tilde{G}_{\mu\nu}$ , a vector  $A_\mu$  and a scalar  $\sigma$ , packaged as

$$ds^2 = \tilde{G}_{\mu\nu} dx^\mu dx^\nu + e^{2\sigma} (dy + A_\mu dx^\mu)^2 \quad (7.6)$$

[Tong §8.1, eq. (8.2), ll. 11198–11212]. All three are taken independent of  $y$  when we work at energies far below  $1/R$ . Each has a clear meaning.

- $e^\sigma$  rescales the length of the circle: its proper circumference is  $2\pi R e^\sigma$ . A constant shift of  $\sigma$  is a change of  $R$ . Nothing in the reduced action fixes the value of  $\sigma$ ; there is no potential for it. A massless scalar of this kind is called a *modulus* [Tong §8.1, ll. 11257–11271]. In the language of the worldsheet,  $R$  labels a family of conformal field theories connected by an exactly marginal deformation [GPR §2.1, ll. 458–575].

- $A_\mu$  is a gauge field. A coordinate change  $y \rightarrow y + \Lambda(x)$  that depends on the position in the non-compact directions is a diffeomorphism of the  $D$ -dimensional space. It leaves (7.6) of the same form, because  $dy + A_\mu dx^\mu \rightarrow dy + (A_\mu + \partial_\mu \Lambda) dx^\mu$ : the shift of  $y$  can be absorbed by  $A_\mu \rightarrow A_\mu - \partial_\mu \Lambda$ . A position-dependent rotation of the circle is a  $U(1)$  gauge transformation. (Tong's sign of  $\Lambda$  is the opposite one; nothing depends on it.) Inserting the ansatz in the Einstein–Hilbert action gives gravity in  $D - 1$  dimensions coupled to a Maxwell field  $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$  and to  $\sigma$  [L] [Tong §8.1, ll. 11212–11256]. We do not repeat that computation; the statement we need is only which fields appear.

The Fourier modes of (7.3) are *charged* under this  $U(1)$ . Under  $y \rightarrow y + \Lambda(x)$  the mode  $\Phi_n$  picks up a phase,

$$\Phi_n(x) \rightarrow e^{in\Lambda(x)/R} \Phi_n(x), \quad (7.7)$$

which is the transformation law of a field of charge  $n/R$ . Rescaling the gauge field to  $A'_\mu = A_\mu/R$  makes the charges integers [Tong §8.1.1, ll. 11336–11349]. The reason is simple: the gauge symmetry is translation along the circle, and the conserved charge of a translation is momentum. *Kaluza–Klein charge is momentum in the hidden direction.*

The closed bosonic string has two more massless fields, the two-form  $B_{MN}$  and the dilaton  $\Phi$  (Chapters 3 and 6). The dilaton reduces to a scalar. The two-form reduces to a two-form  $B_{\mu\nu}$  and a second vector,

$$\tilde{A}_\mu = B_{\mu 25}. \quad (7.8)$$

So the low-energy theory in 25 dimensions contains a metric, a two-form, *two*  $U(1)$  gauge fields  $A_\mu$  and  $\tilde{A}_\mu$ , and two massless scalars  $\Phi$  and  $\sigma$  [Tong §8.1, ll. 11272–11283]. We have just seen which states carry the charge of  $A_\mu$ . No mode of any field in the low-energy action is charged under  $\tilde{A}_\mu$ . The string will supply the missing charged states.

## 7.2 The string on a circle: two integers

Now let a closed string move on  $\mathbb{R}^{1,24} \times S^1$  with

$$X^{25} \sim X^{25} + 2\pi R. \quad (7.9)$$

The identification changes the string in two ways [Tong §8.2, ll. 11351–11372].

### 7.2.1 Momentum is quantized

The centre of mass of the string behaves like a particle. Its wavefunction contains the factor  $e^{ipx}$ , where  $x$  is the centre-of-mass position on the circle and  $p$  the total momentum  $p^{25}$ . The points  $x$  and  $x + 2\pi R$  are the same point, so the wavefunction must agree there:  $e^{2\pi ipR} = 1$ , that is

$$p = \frac{n}{R}, \quad n \in \mathbb{Z}. \quad (7.10)$$

This is the same integer as in (7.5). Nothing here is special to strings.

### 7.2.2 Winding is possible

The second change is special to strings. A closed string is a map from the circle  $\sigma \in [0, 2\pi)$  into spacetime, and so far we required  $X^M(\tau, \sigma + 2\pi) = X^M(\tau, \sigma)$ . With the identification (7.9) this is too strong for  $M = 25$ . The string is closed if the two ends of the  $\sigma$ -interval are at the same *point of the circle*, and the coordinate values of one point differ by multiples of  $2\pi R$ :

$$X^{25}(\tau, \sigma + 2\pi) = X^{25}(\tau, \sigma) + 2\pi R w, \quad w \in \mathbb{Z}. \quad (7.11)$$

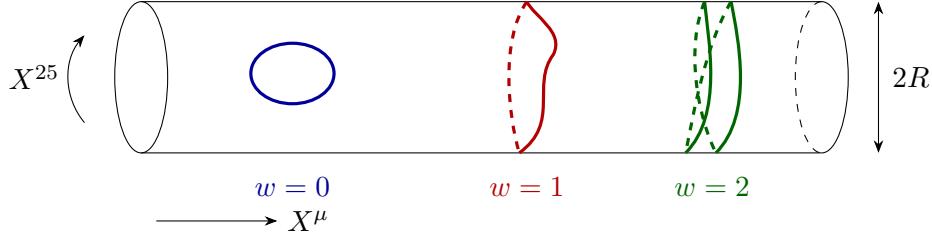


Figure 7.1: Closed strings on  $\mathbb{R} \times S^1$ , drawn as a cylinder; only one non-compact direction is shown. The string with  $w = 0$  can shrink to a point. The strings with  $w = 1$  and  $w = 2$  cannot: they wrap the circle, and their length is at least  $2\pi R|w|$ . Dashed parts of the curves lie on the far side of the cylinder.

The integer  $w$  is the *winding number*: the number of times the string wraps the circle, counted with orientation (Figure 7.1). Three facts about  $w$  are worth stating before any computation.

1.  $w$  is topological. A closed curve on a circle cannot be unwound by a continuous deformation; in mathematical terms  $w$  labels the class of the map in  $\pi_1(S^1) = \mathbb{Z}$ . It cannot change as the free string evolves in time, because  $X(\tau, \sigma + 2\pi) - X(\tau, \sigma)$  is continuous in  $\tau$  and takes values in the discrete set  $2\pi R\mathbb{Z}$ .
2. In interactions, winding is conserved additively: a string with  $w = 1$  and a string with  $w = -1$  can join into an unwound string. It is a charge, and Section 7.6 identifies the gauge field that couples to it.
3. A wound string cannot shrink to a point. Its length is at least  $2\pi R|w|$ , so its energy is at least length times tension,

$$M_{\text{winding}} = 2\pi R|w| \cdot T = 2\pi R|w| \cdot \frac{1}{2\pi\alpha'} = \frac{|w|R}{\alpha'}. \quad (7.12)$$

This estimate is classical. The quantum mass formula below contains exactly this term.

#### Common pitfall: “winding is just momentum in disguise”

Momentum  $n$  and winding  $w$  are independent integers with different origins. The integer  $n$  exists because the *wavefunction* must be single valued; it is a quantum effect, and a particle has it too. The integer  $w$  exists because the *classical configuration*  $X(\sigma)$  need only be single valued up to the identification; it is present before quantization, and a particle does not have it. The two look alike in the final formulas, and that is a result (Chapter 8), not an assumption.

## 7.3 Zero modes: $p_L$ and $p_R$

### 7.3.1 The mode expansion with winding

In conformal gauge every  $X^M$  obeys the wave equation  $\partial_+ \partial_- X^M = 0$ , and the general solution is a left-mover plus a right-mover,  $X = X_L(\sigma^+) + X_R(\sigma^-)$  (Section 2.5). The identification (7.9) does not change the local physics, so the wave equation, the oscillators and their commutators are exactly as before. Only the part of  $X^{25}$  that is linear in  $\tau$  and  $\sigma$  changes. Write the most general linear piece as

$$X^{25}(\tau, \sigma) = x + a\tau + b\sigma + \text{oscillators}.$$

The oscillator terms  $e^{-ik\sigma^\pm}$  are periodic in  $\sigma$ , so  $X(\tau, \sigma + 2\pi) - X(\tau, \sigma) = 2\pi b$ , and (7.11) fixes  $b = wR$ . For  $a$ , recall that the momentum density is  $\Pi = \dot{X}/(2\pi\alpha')$  (Equation (2.36)), so the total momentum is

$$p = \int_0^{2\pi} d\sigma \frac{\dot{X}^{25}}{2\pi\alpha'} = \frac{a}{\alpha'} \quad \implies \quad a = \alpha' p = \frac{\alpha' n}{R},$$

where the oscillators drop out of the integral and (7.10) was used in the last step. Therefore

$$X^{25}(\tau, \sigma) = x + \frac{\alpha' n}{R} \tau + wR \sigma + \text{oscillators} \quad (7.13)$$

[Tong §8.2, ll. 11373–11380]. Now split the linear part into functions of  $\sigma^\pm = \tau \pm \sigma$ . Using  $\tau = \frac{1}{2}(\sigma^+ + \sigma^-)$  and  $\sigma = \frac{1}{2}(\sigma^+ - \sigma^-)$ ,

$$\frac{\alpha' n}{R} \tau + wR \sigma = \frac{1}{2} \left( \frac{\alpha' n}{R} + wR \right) \sigma^+ + \frac{1}{2} \left( \frac{\alpha' n}{R} - wR \right) \sigma^-.$$

Comparing with the non-compact expansion Equation (2.35), where the coefficient of  $\sigma^\pm$  is  $\frac{1}{2}\alpha' p$ , we see that the left- and right-movers now carry *different* momenta:

$$\boxed{p_L = \frac{n}{R} + \frac{wR}{\alpha'}, \quad p_R = \frac{n}{R} - \frac{wR}{\alpha'}} \quad (7.14)$$

and the expansion of the compact coordinate is

$$\begin{aligned} X_L^{25}(\sigma^+) &= \frac{1}{2}x + \frac{1}{2}\alpha' p_L \sigma^+ + i\sqrt{\frac{\alpha'}{2}} \sum_{k \neq 0} \frac{1}{k} \tilde{\alpha}_k^{25} e^{-ik\sigma^+}, \\ X_R^{25}(\sigma^-) &= \frac{1}{2}x + \frac{1}{2}\alpha' p_R \sigma^- + i\sqrt{\frac{\alpha'}{2}} \sum_{k \neq 0} \frac{1}{k} \alpha_k^{25} e^{-ik\sigma^-} \end{aligned} \quad (7.15)$$

[L] [Tong §8.2, eq. (8.4), ll. 11381–11420]. We write  $k$  for the mode number, because  $n$  is now the momentum integer. The expansions of  $X^\mu$ ,  $\mu = 0, \dots, 24$ , are unchanged. Two checks of (7.14):

- *Dimensions.*  $n/R$  is an inverse length.  $wR/\alpha'$  is length over length<sup>2</sup>, also an inverse length.
- *Decompactification.* For  $w = 0$  one has  $p_L = p_R = p$  and (7.15) is Equation (2.35). A string that does not wind does not know, classically, that the direction is compact.

The inverse relations are

$$\frac{n}{R} = \frac{1}{2}(p_L + p_R), \quad \frac{wR}{\alpha'} = \frac{1}{2}(p_L - p_R). \quad (7.16)$$

The sum of the two chiral momenta is the ordinary momentum. The difference is the winding. In the language of Chapter 2 the zero-mode oscillators are

$$\tilde{\alpha}_0^{25} = \sqrt{\frac{\alpha'}{2}} p_L, \quad \alpha_0^{25} = \sqrt{\frac{\alpha'}{2}} p_R, \quad (7.17)$$

while in the non-compact directions  $\alpha_0^\mu = \tilde{\alpha}_0^\mu = \sqrt{\alpha'/2} p^\mu$  as before (Remark 2.2 anticipated this).

*Remark 7.1* (A first look at the dual coordinate). The splitting into  $X_L$  and  $X_R$  suggests a second combination,  $\tilde{X} \equiv X_L - X_R$ . Its zero-mode part follows from (7.15) and (7.14):

$$\tilde{X}^{25} = \frac{1}{2}\alpha'(p_L - p_R)\tau + \frac{1}{2}\alpha'(p_L + p_R)\sigma + \dots = wR\tau + \frac{\alpha'n}{R}\sigma + \dots$$

Compare with (7.13): in  $\tilde{X}$  the winding  $w$  plays the role of momentum and  $n$  the role of winding, and  $\tilde{X}$  is periodic with period  $2\pi\alpha'/R$  when  $X$  has period  $2\pi R$ . This is derived here only as algebra; its meaning is the subject of Chapters 8 and 14.

*Remark 7.2* (Other normalizations). Giveon, Porrati and Rabinovici use dimensionless momenta

$$p_{L,R}^{\text{GPR}} = \frac{1}{\sqrt{2}} \left( \frac{\sqrt{\alpha'}}{R} n \pm \frac{R}{\sqrt{\alpha'}} w \right) = \sqrt{\frac{\alpha'}{2}} p_{L,R}$$

[GPR §2.2, eq. (2.2.5), ll. 576–650]. Their  $p_{L,R}$  are our zero-mode oscillators (7.17). With this normalization  $L_0$  contains  $\frac{1}{2}p^2$  with no factor of  $\alpha'$ . The Lean file of Section 7.7 sets  $\alpha' = 1$  and uses (7.14). When comparing formulas between sources, first check which of the two normalizations is in use; a stray factor of 2 in a mass formula nearly always comes from here.

#### Physical picture: a two-component charge

A state of the string on a circle carries a pair of integers  $(n, w)$ . The pair  $(p_L, p_R)$  is the same information in a basis that depends on  $R$ . As  $R$  varies, the integers stay fixed and the point  $(p_L, p_R)$  moves in the plane. One combination does not move at all:

$$\frac{\alpha'}{4}(p_L^2 - p_R^2) = \frac{\alpha'}{4} \cdot 4 \frac{n}{R} \frac{wR}{\alpha'} = nw. \quad (7.18)$$

This indefinite quadratic form on the charges is the first appearance of the  $O(1, 1)$  structure that Chapters 10 and 11 generalize to  $O(d, d)$ . The positive-definite combination  $\frac{1}{2}(p_L^2 + p_R^2)$  does depend on  $R$ , and it is what enters the mass.

## 7.4 The mass formula and level matching

### 7.4.1 Derivation

The physical-state conditions are the same as in Chapter 3, because they come from the worldsheet gauge symmetry and not from the shape of spacetime:

$$(L_0 - 1)|\text{phys}\rangle = 0, \quad (\tilde{L}_0 - 1)|\text{phys}\rangle = 0, \quad (7.19)$$

with the normal-ordering constant  $a = 1$  of the critical dimension  $D = 26$ , and

$$L_0 = \frac{1}{2}\alpha_0 \cdot \alpha_0 + N, \quad \tilde{L}_0 = \frac{1}{2}\tilde{\alpha}_0 \cdot \tilde{\alpha}_0 + \tilde{N}, \quad (7.20)$$

where  $N$  and  $\tilde{N}$  are the right- and left-moving levels of Equation (3.23). The dot product runs over all 26 directions. The only new ingredient is that the 25th component of  $\alpha_0$  differs from that of  $\tilde{\alpha}_0$ . Using (7.17),

$$\begin{aligned} \frac{1}{2}\alpha_0 \cdot \alpha_0 &= \frac{1}{2} \cdot \frac{\alpha'}{2} (p_\mu p^\mu + p_R^2) = \frac{\alpha'}{4} (-M^2 + p_R^2), \\ \frac{1}{2}\tilde{\alpha}_0 \cdot \tilde{\alpha}_0 &= \frac{\alpha'}{4} (-M^2 + p_L^2). \end{aligned} \quad (7.21)$$



Here the mass is *defined* as the 25-dimensional observer defines it,

$$M^2 \equiv - \sum_{\mu=0}^{24} p_{\mu} p^{\mu}, \quad (7.22)$$

with the sum over non-compact directions only: momentum along the circle is counted as mass, as in Section 7.1. Inserting (7.21) in (7.19) gives two expressions for the same  $M^2$ :

$$M^2 = p_R^2 + \frac{4}{\alpha'}(N - 1), \quad M^2 = p_L^2 + \frac{4}{\alpha'}(\tilde{N} - 1) \quad (7.23)$$

[L] [Tong §8.2, ll. 11421–11455]. (In lightcone gauge one chooses the lightcone directions inside  $\mathbb{R}^{1,24}$ ; then  $N, \tilde{N}$  count the 24 transverse oscillators, of which one is  $\alpha_{-k}^{25}$ , and the same two equations follow.) For  $p_L = p_R = 0$  these reduce to Equation (3.31).

*Subtract* the two equations in (7.23):  $0 = p_L^2 - p_R^2 + \frac{4}{\alpha'}(\tilde{N} - N)$ . With (7.18),  $p_L^2 - p_R^2 = 4nw/\alpha'$ , so

$$\boxed{N - \tilde{N} = nw} \quad (7.24)$$

[L] [Tong §8.2, eq. (8.5), ll. 11456–11464]. *Add* the two equations and divide by two. From (7.14),

$$\frac{1}{2}(p_L^2 + p_R^2) = \frac{1}{2} \left[ \left( \frac{n}{R} + \frac{wR}{\alpha'} \right)^2 + \left( \frac{n}{R} - \frac{wR}{\alpha'} \right)^2 \right] = \frac{n^2}{R^2} + \frac{w^2 R^2}{\alpha'^2},$$

because the cross terms  $\pm 2nw/\alpha'$  cancel. Hence

$$\boxed{M^2 = \frac{n^2}{R^2} + \frac{w^2 R^2}{\alpha'^2} + \frac{2}{\alpha'}(N + \tilde{N} - 2)} \quad (7.25)$$

[L] [Tong §8.2, eq. (8.6), ll. 11465–11490]. This is the formula announced in (7.1), in the book's conventions. The algebra from (7.23) to (7.24) and (7.25) was also confirmed by a symbolic check (sympy).

### 7.4.2 Reading the three terms

- $n^2/R^2$  is the Kaluza–Klein term (7.5). It is large for a small circle.
- $w^2 R^2/\alpha'^2$  is the square of the classical energy (7.12) of a string stretched  $|w|$  times around the circle. It is large for a *large* circle. The two terms respond to  $R$  in opposite ways, and their sum cannot be made small by any choice of  $R$  once both  $n$  and  $w$  are non-zero: by the inequality of arithmetic and geometric means,

$$\frac{n^2}{R^2} + \frac{w^2 R^2}{\alpha'^2} \geq \frac{2|nw|}{\alpha'}, \quad \text{with equality iff } R^2 = \alpha' |n/w|. \quad (7.26)$$

- $\frac{2}{\alpha'}(N + \tilde{N} - 2)$  is the oscillator energy with the zero-point constant  $-2$ , exactly as in flat space.

The three terms are not independent, because (7.24) ties the levels to the charges. Eliminating  $\tilde{N} = N - nw$ ,

$$\alpha' M^2 = \left( \frac{n\sqrt{\alpha'}}{R} - \frac{wR}{\sqrt{\alpha'}} \right)^2 + 4(N - 1) = \left( \frac{n\sqrt{\alpha'}}{R} + \frac{wR}{\sqrt{\alpha'}} \right)^2 + 4(\tilde{N} - 1), \quad (7.27)$$

which is (7.23) again and is the quickest form for computing a table.

### 7.4.3 Why level matching changes

In flat space level matching  $N = \tilde{N}$  expresses the fact that no point on a closed string is special: the operator  $L_0 - \tilde{L}_0$  generates rigid shifts  $\sigma \rightarrow \sigma + \epsilon$ , and physical states must be invariant (Section 3.4.2). The same is true on the circle, but the generator has a new piece. From (7.20) and (7.21),

$$L_0 - \tilde{L}_0 = N - \tilde{N} + \frac{\alpha'}{4}(p_R^2 - p_L^2) = N - \tilde{N} - nw. \quad (7.28)$$

The term  $-nw$  has a direct meaning. For a wound string, shifting  $\sigma$  by  $\epsilon$  moves every point of the string along the circle by  $wR\epsilon$ , because of the term  $wR\sigma$  in (7.13). A translation in spacetime by a distance  $\delta$  acts on a state of momentum  $p = n/R$  as the phase  $e^{ip\delta} = e^{inw\epsilon}$ . So part of the  $\sigma$ -shift is a spacetime translation, and the phase it produces must be compensated by the oscillators. The condition  $N - \tilde{N} = nw$  says that the total phase vanishes.

#### Common pitfall: imposing $N = \tilde{N}$ on a circle

On a circle  $N = \tilde{N}$  is required only when  $nw = 0$ . States with both momentum and winding *must* have unequal levels. For instance  $(n, w) = (1, 1)$  with  $N = \tilde{N} = 0$  is not a physical state, and its “mass”  $\alpha'M^2 = \alpha'/R^2 + R^2/\alpha' - 4$  appears nowhere in the spectrum. The lightest state with  $(n, w) = (1, 1)$  has  $N = 1, \tilde{N} = 0$ .

## 7.5 The lightest states as a function of the radius

It is convenient to measure the radius in string units,

$$r \equiv \frac{R}{\sqrt{\alpha'}}, \quad \alpha'M^2 = \frac{n^2}{r^2} + w^2r^2 + 2(N + \tilde{N} - 2), \quad N - \tilde{N} = nw. \quad (7.29)$$

Table 7.1 lists the lightest states. Each line stands for the states with the given  $(n, w)$  and their images under  $(n, w) \rightarrow (-n, -w)$ , which have the same mass. Figure 7.2 plots the same functions.

Several features of the table deserve comment.

*The tachyon and its tower.* The state with all quantum numbers zero is the tachyon of Chapter 3,  $\alpha'M^2 = -4$ . Its momentum modes  $(n, 0)$  have  $\alpha'M^2 = n^2/r^2 - 4$ : they are the 26-dimensional tachyon moving along the circle, and they are tachyonic in 25 dimensions as long as  $|n| < 2r$ . Its winding modes  $(0, w)$  have  $\alpha'M^2 = w^2r^2 - 4$  and are tachyonic for  $|w| < 2/r$ . The bosonic string has this instability at every radius; it is a property of the bosonic vacuum and is absent in the superstring (Chapter 5). We keep these states in the table because they are useful labels of the towers, not because they are physical particles.

*The massless level.* The states with  $n = w = 0$  and  $N = \tilde{N} = 1$  are massless for every  $R$ . They are the subject of Section 7.6.

*Two limits.* As  $r \rightarrow \infty$  the momentum states  $(n, 0)$  at fixed level become degenerate and dense:  $n/R$  turns into the continuous momentum of a non-compact direction, as it must. The winding states become infinitely heavy and leave the spectrum. As  $r \rightarrow 0$  the roles are reversed: momentum states decouple, which is the field-theory expectation, but the *winding* states become dense. A continuum of states is the signature of a non-compact dimension. So the spectrum of a string on a very small circle looks like that of a string on a very large one. This observation, made precise, is T-duality (Chapter 8).

*The special radius.* At  $r = 1$  the table contains extra massless states. Setting  $M^2 = 0$  in (7.29) at  $r = 1$  and solving with level matching gives, besides the massless level itself, exactly the following [Tong §8.2.3, ll. 11561–11612]:

Table 7.1: The lightest closed-string states on a circle, with  $r = R/\sqrt{\alpha'}$ . The last three columns evaluate  $\alpha'M^2$  at  $r = \frac{1}{2}, 1, 2$ . All entries follow from (7.29); they were recomputed with exact rational arithmetic (symbolic check).

| $n$     | $w$     | $N$ | $\tilde{N}$ | $\alpha'M^2$     | $r = \frac{1}{2}$ | $r = 1$ | $r = 2$         |
|---------|---------|-----|-------------|------------------|-------------------|---------|-----------------|
| 0       | 0       | 0   | 0           | -4               | -4                | -4      | -4              |
| $\pm 1$ | 0       | 0   | 0           | $r^{-2} - 4$     | 0                 | -3      | $-\frac{15}{4}$ |
| 0       | $\pm 1$ | 0   | 0           | $r^2 - 4$        | $-\frac{15}{4}$   | -3      | 0               |
| $\pm 2$ | 0       | 0   | 0           | $4r^{-2} - 4$    | 12                | 0       | -3              |
| 0       | $\pm 2$ | 0   | 0           | $4r^2 - 4$       | -3                | 0       | 12              |
| 0       | 0       | 1   | 1           | 0                | 0                 | 0       | 0               |
| $\pm 1$ | $\pm 1$ | 1   | 0           | $(r - r^{-1})^2$ | $\frac{9}{4}$     | 0       | $\frac{9}{4}$   |
| $\pm 1$ | $\mp 1$ | 0   | 1           | $(r - r^{-1})^2$ | $\frac{9}{4}$     | 0       | $\frac{9}{4}$   |
| $\pm 1$ | 0       | 1   | 1           | $r^{-2}$         | 4                 | 1       | $\frac{1}{4}$   |
| 0       | $\pm 1$ | 1   | 1           | $r^2$            | $\frac{1}{4}$     | 1       | 4               |

- $N = \tilde{N} = 0$ :  $n^2 + w^2 = 4$  with  $nw = 0$ , that is  $(n, w) = (\pm 2, 0)$  or  $(0, \pm 2)$ . Four scalars.
- $N + \tilde{N} = 1$ :  $n^2 + w^2 = 2$  with  $nw = N - \tilde{N} = \pm 1$ . For  $nw = +1$ ,  $(n, w) = \pm(1, 1)$  with  $N = 1, \tilde{N} = 0$ : the states  $\alpha_{-1}^\mu |0; p; n, w\rangle$ . For  $nw = -1$ ,  $(n, w) = \pm(1, -1)$  with  $N = 0, \tilde{N} = 1$ : the states  $\tilde{\alpha}_{-1}^\mu |0; p; n, w\rangle$ . Four vectors. (Tong's text prints " $N = 1, \tilde{N} = 0$ " for both pairs; level matching (7.24) and the oscillator  $\tilde{\alpha}_{-1}$  that Tong writes for the second pair fix the assignment given here.)

Massless charged vectors signal a non-abelian gauge symmetry. The gauge group is enhanced from  $U(1) \times U(1)$  to  $SU(2) \times SU(2)$  [L]. We return to this point in Chapter 8, where  $r = 1$  is identified as the self-dual radius.

**Example 7.3** (Worked example: the spectrum at  $R = 2\sqrt{\alpha'}$ ). Take  $r = 2$ . We list all states with  $\alpha'M^2 \leq 1$ , using (7.27).

Level  $N = \tilde{N} = 0$ , so  $nw = 0$ . Momentum modes have  $\alpha'M^2 = n^2/4 - 4$ :

$$\alpha'M^2 = -4, -\frac{15}{4}, -3, -\frac{7}{4}, 0, \frac{9}{4}, \dots \quad \text{for} \quad |n| = 0, 1, 2, 3, 4, 5, \dots$$

The modes with  $|n| \leq 3$  are tachyonic, the modes  $n = \pm 4$  are massless, and  $|n| \geq 5$  is above our cut. Winding modes:  $\alpha'M^2 = 4w^2 - 4$ , giving 0 for  $|w| = 1$  and 12 for  $|w| = 2$ .

Level  $N = \tilde{N} = 1$ , again  $nw = 0$ :  $\alpha'M^2 = n^2/4$  or  $4w^2$ . This gives 0 for  $n = w = 0, \frac{1}{4}$  for  $n = \pm 1$  and 1 for  $n = \pm 2$ . The winding states start at 4.

Levels  $(N, \tilde{N}) = (1, 0)$  or  $(0, 1)$ , so  $nw = \pm 1$  and  $|n| = |w| = 1$ :  $\alpha'M^2 = (2 - \frac{1}{2})^2 = \frac{9}{4}$ . Too heavy. One may also try  $(N, \tilde{N}) = (2, 0)$  with  $nw = 2$ : the choice  $(n, w) = (2, 1)$  gives  $\alpha'M^2 = (1 - 2)^2 + 4 = 5$  and  $(1, 2)$  gives  $(\frac{1}{2} - 4)^2 + 4 = \frac{65}{4}$ .

The massless states at  $r = 2$  are therefore: the level  $N = \tilde{N} = 1$ ,  $n = w = 0$ ; two scalars with  $(n, w) = (\pm 4, 0)$ ; and two scalars with  $(n, w) = (0, \pm 1)$ . The last four are an instance of the accidental massless scalars that occur at  $R^2 = n^2\alpha'/4$  and  $R^2 = 4\alpha'/w^2$  [Tong §8.2.3, ll. 11561–11586]. In physical units, if  $\sqrt{\alpha'} = 10^{-34}$  m, the first massive Kaluza–Klein state of the graviton multiplet,  $\alpha'M^2 = \frac{1}{4}$ , has  $M = 1/(2\sqrt{\alpha'})$ , which is  $\hbar c/(2 \times 10^{-34} \text{ m}) \simeq 9.9 \times 10^{17} \text{ GeV}$ .

Repeat the exercise at  $r = \frac{1}{2}$  and the same list of masses appears, with the roles of  $n$  and  $w$  exchanged: the columns  $r = \frac{1}{2}$  and  $r = 2$  of Table 7.1 are permutations of each other.

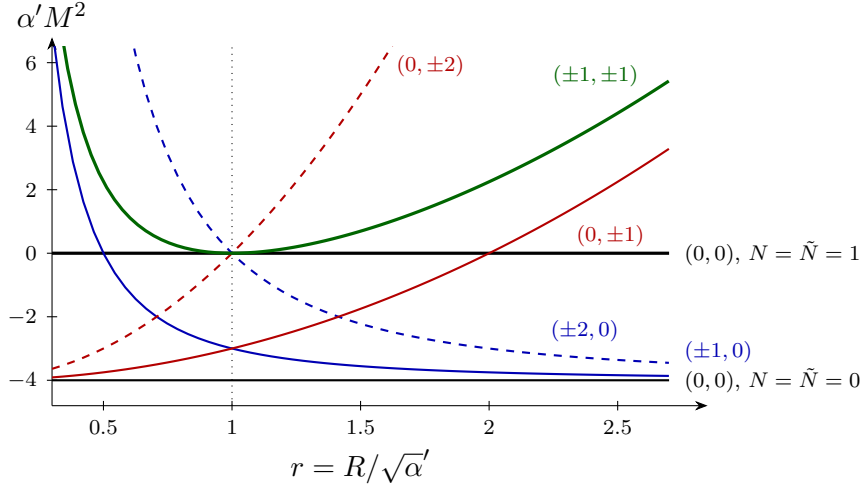


Figure 7.2:  $\alpha' M^2$  for the states of Table 7.1 with  $N + \tilde{N} \leq 1$ , labelled by  $(n, w)$ . Momentum modes (blue) become light at large  $r$ , winding modes (red) at small  $r$ . The curve  $(r - 1/r)^2$  (green) belongs to the vector states with  $|n| = |w| = 1$ ; it touches zero only at  $r = 1$ , where the curves  $(\pm 2, 0)$  and  $(0, \pm 2)$  also cross zero. The picture is symmetric under  $r \rightarrow 1/r$  with blue and red exchanged; Chapter 8 explains why.

## 7.6 The $U(1) \times U(1)$ gauge symmetry

### 7.6.1 Massless states at a generic radius

At a generic radius the only massless states are those with  $n = w = 0$  and  $N = \tilde{N} = 1$ ,

$$\alpha_{-1}^M \tilde{\alpha}_{-1}^N |0; p\rangle, \quad M, N \in \{\mu, 25\}.$$

In 26 non-compact dimensions these were the graviton, the  $B$ -field and the dilaton. A 25-dimensional observer sorts them by how many indices point along the circle [Tong §8.2.1, ll. 11491–11526]:

- $\alpha_{-1}^\mu \tilde{\alpha}_{-1}^\nu |0; p\rangle$ : a symmetric traceless tensor, an antisymmetric tensor and a trace. These are the 25-dimensional metric  $G_{\mu\nu}$ , two-form  $B_{\mu\nu}$  and dilaton  $\Phi$ .
- $\alpha_{-1}^\mu \tilde{\alpha}_{-1}^{25} |0; p\rangle$  and  $\alpha_{-1}^{25} \tilde{\alpha}_{-1}^\mu |0; p\rangle$ : two vectors. The symmetric combination comes from  $G_{\mu 25}$  and is the Kaluza–Klein gauge field  $A_\mu$  of (7.6). The antisymmetric combination comes from  $B_{\mu 25}$  and is  $\tilde{A}_\mu$  of (7.8).
- $\alpha_{-1}^{25} \tilde{\alpha}_{-1}^{25} |0; p\rangle$ : one scalar, the fluctuation of  $G_{25 25}$ , that is the radius modulus  $\sigma$ .

A count of states confirms that nothing is lost. In lightcone gauge the 26-dimensional massless level has  $24 \times 24 = 576$  states. In 25 dimensions the transverse index takes 23 values plus the circle direction, and

$$\underbrace{23 \times 23}_{G_{\mu\nu}, B_{\mu\nu}, \Phi} + \underbrace{23 + 23}_{A_\mu, \tilde{A}_\mu} + \underbrace{1}_{\sigma} = 529 + 46 + 1 = 576.$$

The string reproduces the massless content of the Kaluza–Klein reduction of Section 7.1.2. The existence of the massless scalar  $\sigma$  is the string’s way of saying that  $R$  is a free parameter. Its vertex operator is built from  $\partial X^{25} \bar{\partial} X^{25}$  (Chapter 4); adding it to the worldsheet action changes the coefficient of the kinetic term of  $X^{25}$ , which is a change of  $R$ . In the terminology of [GPR §2.1, ll. 458–575] it is a truly marginal operator, and  $R$  is a coordinate on the moduli space of the conformal field theory.

### 7.6.2 Charges: momentum and winding

Which states are charged under the two gauge fields? Tong computes the three-point amplitude of one photon and two tachyon modes with charges  $(n, w)$  and  $(-n, -w)$ , and finds it proportional to  $p_L \pm p_R$ , with the upper sign for  $A_\mu$  and the lower sign for  $\tilde{A}_\mu$  [L] [Tong §8.2.2, ll. 11527–11560]. By (7.16),

$$\text{charge under } A_\mu \propto p_L + p_R = \frac{2n}{R}, \quad \text{charge under } \tilde{A}_\mu \propto p_L - p_R = \frac{2wR}{\alpha'}. \quad (7.30)$$

The first statement is (7.7) again. The second can be understood without any amplitude, from the coupling of the string to a background  $B$ -field (Chapter 6). In Lorentzian conformal gauge that coupling is

$$S_B = -\frac{1}{4\pi\alpha'} \int d\tau d\sigma \epsilon^{\alpha\beta} B_{MN}(X) \partial_\alpha X^M \partial_\beta X^N = -\frac{1}{2\pi\alpha'} \int d\tau d\sigma B_{MN} \dot{X}^M X'^N, \quad (7.31)$$

with  $\epsilon^{\tau\sigma} = +1$ ; the overall sign is a convention for  $B$  [Tong §7.2.1, eq. (7.8), ll. 9425–9500]. Consider a string that winds the circle  $w$  times and is otherwise point-like:  $X^\mu = X^\mu(\tau)$  and  $X'^{25} = wR$ . Only the components  $B_{\mu 25} = -B_{25\mu}$  contribute, and  $X'^\mu = 0$  removes the term with  $N = \mu$ , so

$$S_B = -\frac{1}{2\pi\alpha'} \int d\tau B_{\mu 25} \dot{X}^\mu \int_0^{2\pi} d\sigma X'^{25} = -\frac{wR}{\alpha'} \int d\tau \tilde{A}_\mu \dot{X}^\mu. \quad (7.32)$$

This is the worldline action of a point particle of charge  $wR/\alpha'$  (up to the sign convention) in the gauge field  $\tilde{A}_\mu$ , compare Tong's eq. (7.9). The gauge transformation of  $\tilde{A}_\mu$  descends from the gauge symmetry  $B \rightarrow B + d\lambda$  of the two-form with  $\lambda = \lambda_{25}(x) dy$ .

#### Physical picture: two gauge fields, two charges

The metric gauges translations along the circle, and the conserved charge of translations is momentum  $n$ . The two-form couples to the string the way a Maxwell potential couples to a charged particle, and when the string is wrapped on the circle it *is* a charged particle in 25 dimensions, with charge  $w$ . A point-particle theory has  $A_\mu$  but no states charged under  $\tilde{A}_\mu$ . String theory fills both charge lattices. The integer vector  $(n, w) \in \mathbb{Z}^2$  is the charge vector of  $U(1) \times U(1)$ ; in the chiral basis  $U(1)_L \times U(1)_R$  the charges are  $(p_L, p_R)$ .

## 7.7 What the machine has checked

Three Lean files touch the content of this chapter. They belong to three different layers of the library (Chapter 35), and they illustrate three different levels of faithfulness to the physics. None of them contains a string, a Hilbert space or a Virasoro operator. What they contain is the algebra of the zero-mode part of (7.25).

### 7.7.1 Stream 1: the momenta as rational functions

The file `StringTheoryFormalization/UseCases/TDualityMassSpectrum.lean`, namespace `StringTheory.UseCases.TDuality`, begins with three definitions.

#### In Lean: $p_L, p_R$ and the momentum part of the mass

```
def leftMomentum (n w R : ℚ) : ℚ := n / R + w * R

def rightMomentum (n w R : ℚ) : ℚ := n / R - w * R
```

```
def momentumMassSq (n w R : ℚ) : ℚ :=
  (leftMomentum n w R) ^ 2 + (rightMomentum n w R) ^ 2
```

**Reading.** These are (7.14) in units  $\alpha' = 1$ , as functions  $\mathbb{Q}^3 \rightarrow \mathbb{Q}$ . Three things differ from the physics. (i) The radius is a rational number, not a positive real, and may be negative. (ii)  $n$  and  $w$  are rationals, not integers; integrality plays no role in the theorems below. (iii) `momentumMassSq` is  $p_L^2 + p_R^2$ , which by the computation above (7.25) equals

$$p_L^2 + p_R^2 = 2 \left( \frac{n^2}{R^2} + w^2 R^2 \right),$$

twice the momentum–winding part of  $\alpha' M^2$ . The oscillator term  $\frac{2}{\alpha'}(N + \tilde{N} - 2)$  and the level-matching condition are not modelled. The file’s header comment quotes the oscillator term of (7.23) with the coefficient  $2/\alpha'$  where Tong’s equation has  $4/\alpha'$ . Comments are not checked by Lean, and no declaration in the file involves that term, so no theorem is affected; a reader comparing the file with this chapter should use (7.23).

#### In Lean: how the momenta respond to $R \rightarrow 1/R$ , $n \leftrightarrow w$

```
theorem tduality_fixes_left_momentum (n w R : ℚ) (hR : R ≠ 0) :
  leftMomentum w n (1 / R) = leftMomentum n w R := by ...

theorem tduality_flips_right_momentum (n w R : ℚ) (hR : R ≠ 0) :
  rightMomentum w n (1 / R) = - rightMomentum n w R := by ...

theorem tduality_invariant_mass_squared (n w R : ℚ) (hR : R ≠ 0) :
  momentumMassSq w n (1 / R) = momentumMassSq n w R := by ...
```

**Reading.** For all rationals  $n, w$  and all rational  $R \neq 0$ : exchanging the first two arguments and replacing  $R$  by  $1/R$  leaves  $n/R + wR$  unchanged, changes the sign of  $n/R - wR$ , and therefore leaves the sum of squares unchanged. These are identities between rational functions. They are the algebraic core of T-duality and are discussed in Chapter 8; we quote them here because they are statements about the objects defined in this chapter. They say nothing about the oscillators, about level matching (which is invariant for the separate reason that  $nw = wn$ ), or about interactions. **Proof idea.** Unfold the definitions, clear denominators with `field_simp` using  $R \neq 0$ , and let `ring` verify the polynomial identity. The third theorem rewrites with the first two and uses  $(-x)^2 = x^2$ .

A companion file in the same directory, `NarainLattice.lean`, restates the same two momenta under the names `pL` and `pR` (its namespace is that of `Stream 1` with `NarainLattice` in place of `TDuality`). It proves the Lean form of (7.18):

```
def narainNorm (n w R : ℚ) : ℚ := (pL n w R ^ 2 - pR n w R ^ 2) / 2

theorem narain_norm_R_independent (n w R : ℚ) (hR : R ≠ 0) :
  narainNorm n w R = 2 * n * w := by ...
```

In units  $\alpha' = 1$  this is  $\frac{1}{2}(p_L^2 - p_R^2) = 2nw$ : the indefinite form does not depend on  $R$ . It is the identity behind level matching, although the file does not mention levels. Chapter 10 takes it up.

### 7.7.2 Stream 2: the circle as the $d = 1$ generalized metric

The newer library `DualScaleStream2` works with real matrices of arbitrary size  $d$  and with the generalized metric  $\mathcal{H}(G, B)$  of Chapter 15. For  $d = 1$  and  $B = 0$  the metric on the circle is the  $1 \times 1$  matrix  $G = (R^2)$  in units  $\alpha' = 1$ , and

$$\mathcal{H} = \begin{pmatrix} R^2 & 0 \\ 0 & R^{-2} \end{pmatrix}, \quad Z = \begin{pmatrix} w \\ n \end{pmatrix}, \quad Z^\top \mathcal{H} Z = R^2 w^2 + \frac{n^2}{R^2}.$$

The order  $Z = (w, n)$ , winding first, is the convention of this module; it follows Hull and Zwiebach, as the file header records. The last expression is the momentum–winding part of (7.29). The theorem `massForm_circle` in `DualScaleStream2/DFT/GeneralizedMetric.lean` states this, not for a formula written afresh, but for the *Stream 1 definition itself*.

#### In Lean: the bridge between the two libraries

```
noncomputable def massForm (H : Matrix (Charge d) (Charge d) ℝ)
  (Z : Charge d → ℝ) : ℝ :=
  Z ·v (H *v Z)

theorem massForm_circle (n w R : ℚ) (hR : R ≠ 0) :
  massForm (genMetric (d := 1) (![(R : ℝ) ^ 2]) 0)
    (Sum.elim ![(w : ℝ)] ![(n : ℝ)]) =
    ((StringTheory.UseCases.TDuality.momentumMassSq n w R : ℚ) : ℝ) / 2 := by ...
```

**Reading.** `Charge 1` is the index type  $\text{Fin } 1 \oplus \text{Fin } 1$  of a doubled charge vector; `Sum.elim ![(w)] ![(n)]` is the vector whose first block is  $w$  and whose second block is  $n$ . For rational  $n, w$  and rational  $R \neq 0$ , all cast to real numbers, the quadratic form  $Z^\top \mathcal{H}(R^2, 0)Z$  equals one half of the rational number `momentumMassSq n w R`, cast to  $\mathbb{R}$ . The factor  $\frac{1}{2}$  is the factor 2 noted in the previous box. **Why it matters.** The two libraries were written independently, with different number systems ( $\mathbb{Q}$  and  $\mathbb{R}$ ) and different data structures (three scalars; a block matrix acting on a sum type). A reader might worry that they formalize two different things under the same name. This theorem removes the worry for the circle: the kernel has checked that they define the same function of  $(n, w, R)$ . It does not show that either of them is the mass of a string; that identification is the content of Sections 7.3 and 7.4 and is Tier L. **Proof idea.** Compute the  $1 \times 1$  inverse  $(R^2)^{-1}$  from the adjugate formula; at  $B = 0$  the generalized metric is block diagonal, so the quadratic form splits into a winding term and a momentum term; then unfold the Stream 1 definitions, push the casts through with `push_cast`, and finish with `field_simp` and `ring`.

### 7.7.3 The foundation layer: an arithmetic shadow

The third file, `StringTheoryFoundation/Duality/T_Duality.lean`, has no imports beyond core Lean. Its namespace is `StringTheory.Foundation.Duality.T_Duality`.

#### In Lean: a state is a pair of integers

```
structure CircleState where
  n : Int
  w : Int
  deriving Repr, DecidableEq

def tDualState (s : CircleState) : CircleState :=
  { n := s.w, w := s.n }
```



```

theorem t_duality_involution (s : CircleState) :
  tDualState (tDualState s) = s := by ...

def selfDualMassFactor (n w : Int) : Int :=
  n * n + w * w

theorem self_dual_mass_symmetry (n w : Int) :
  selfDualMassFactor n w = selfDualMassFactor w n := by ...

theorem t_duality_spectrum_invariance (n w : Int) :
  n * n + w * w = w * w + n * n := by ...

```

**Reading.** A `CircleState` records the charge vector  $(n, w) \in \mathbb{Z}^2$  of Section 7.6 and nothing else: no level, no momentum  $p^\mu$ , no radius. The first theorem says that swapping the two components twice is the identity. The other two say that  $n^2 + w^2$  is symmetric in  $n$  and  $w$ ; the last is commutativity of integer addition. The name `selfDualMassFactor` refers to the fact that at  $r = 1$  the momentum–winding part of (7.29) is  $n^2 + w^2$ . These statements are *arithmetic shadows* of the physics: true, kernel-checked, and much weaker than their names suggest. There is no radius in the file, so “spectrum invariance” here cannot refer to  $R \rightarrow \alpha'/R$ . **Proof idea.** Definitional unfolding; the integer identities are closed by `omega`.

#### 7.7.4 Tier table and what a faithful formalization would need

| Statement                                                                                                                                                                          | Tier       | Where                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-----------------------------------------------------------------------------|
| KK tower $M_n^2 = n^2/R^2$ ; fields $G, B, \Phi, A, \tilde{A}, \sigma$ in 25 dimensions                                                                                            | [L]        | Tong §8.1                                                                   |
| Mode expansion with winding;<br>$p_{L,R} = n/R \pm wR/\alpha'$                                                                                                                     | [L]        | Tong §8.2                                                                   |
| $M^2 = p_R^2 + \frac{4}{\alpha'}(N - 1) = p_L^2 + \frac{4}{\alpha'}(\tilde{N} - 1)$ , hence (7.24), (7.25)                                                                         | [L]        | Tong §8.2                                                                   |
| Charges $p_L \pm p_R$ under $A_\mu, \tilde{A}_\mu$<br>$n/R + wR$ fixed, $n/R - wR$ negated, sum of squares invariant under $(n, w, R) \rightarrow (w, n, 1/R)$ , over $\mathbb{Q}$ | [L]<br>[A] | Tong §8.2.2<br><code>tduality_*</code> (three theorems)                     |
| $\frac{1}{2}(p_L^2 - p_R^2) = 2nw$ for all $R \neq 0$ , over $\mathbb{Q}$ , $\alpha' = 1$                                                                                          | [A]        | <code>narain_norm_R_independent</code>                                      |
| $Z^\top \mathcal{H}(R^2, 0) Z = \frac{1}{2} \text{momentumMassSq}$                                                                                                                 | [A]        | <code>massForm_circle</code>                                                |
| $(n, w) \mapsto (w, n)$ is an involution; $n^2 + w^2$ symmetric, over $\mathbb{Z}$                                                                                                 | [A]        | <code>t_duality_involution</code> ,<br><code>self_dual_mass_symmetry</code> |
| Table 7.1, Example 7.3, inequality (7.26)                                                                                                                                          | —          | symbolic check, not Tier A                                                  |
| “ <code>momentumMassSq</code> is the mass of a string state”                                                                                                                       | [L]        | identification, never Tier A                                                |

The Tier A theorems in this table depend only on the three standard axioms of Lean’s classical mathematics, namely `propext`, `Classical.choice` and `Quot.sound`; this was confirmed with `#print axioms for massForm_circle` and `tduality_invariant_mass_squared`.

A formalization of (7.25) that deserved the name would need, at the least: (a) a Fock space for the oscillators  $\alpha_k, \tilde{\alpha}_k$  with the level operators  $N, \tilde{N}$ ; (b) a zero-mode sector labelled by  $(n, w) \in \mathbb{Z}^2$  and  $p^\mu$ , with the operators  $L_0, \tilde{L}_0$  of (7.20) acting on the tensor product; (c) the physical-state conditions (7.19) as hypotheses, and (7.24), (7.25) as theorems about eigenvalues; (d) a real radius  $R > 0$  and an explicit  $\alpha'$ . Step (c) is, once (a) and (b) exist, precisely the algebra of Section 7.4. Steps (a) and (b) are the real work and are not in the library today (Chapter 38).



## Chapter summary

- A field on  $\mathbb{R}^{1,D-2} \times S^1$  is a tower of fields with masses  $|n|/R$ . The metric component  $G_{\mu 25}$  is a  $U(1)$  gauge field  $A_\mu$ , and the mode  $n$  has charge  $n$ . The radius is a modulus: a massless scalar without potential.
- A closed string on a circle carries two integers. Momentum  $n$  comes from single-valuedness of the wavefunction; winding  $w$  from the boundary condition  $X(\sigma + 2\pi) = X(\sigma) + 2\pi R w$ . Winding is topological, conserved, and costs energy  $|w|R/\alpha'$ .
- The left- and right-movers carry  $p_{L,R} = n/R \pm wR/\alpha'$ . The combination  $\frac{\alpha'}{4}(p_L^2 - p_R^2) = nw$  is independent of  $R$ .
- $M^2 = n^2/R^2 + w^2 R^2/\alpha'^2 + \frac{2}{\alpha'}(N + \tilde{N} - 2)$  with  $N - \tilde{N} = nw$ , where  $M$  is the mass seen in 25 dimensions.
- At generic  $R$  the massless states are  $G_{\mu\nu}, B_{\mu\nu}, \Phi$ , two gauge fields  $A_\mu, \tilde{A}_\mu$  and the modulus  $\sigma$ . Momentum is the charge of  $A_\mu$ ; winding is the charge of  $\tilde{A}_\mu = B_{\mu 25}$ . At  $R = \sqrt{\alpha'}$  further states become massless.
- In Lean: the rational functions  $n/R \pm wR$  and their behaviour under  $(n, w, R) \rightarrow (w, n, 1/R)$  [A]; the identity of the Stream 1 and Stream 2 circle mass forms, `massForm_circle` [A]. The oscillators, level matching and the identification with a string are not formalized.

## Exercises

**Exercise 7.1** (★ Massive field on a circle). Repeat the reduction of Section 7.1 for a scalar of mass  $m$  in  $D$  dimensions, adding  $-\frac{1}{2}m^2\Phi^2$  to (7.2). Show that  $M_n^2 = m^2 + n^2/R^2$ . For which  $m^2 < 0$  and  $R$  are all modes with  $n \neq 0$  non-tachyonic? Compare with the momentum modes of the string tachyon in Table 7.1.

**Exercise 7.2** (★ The two quadratic forms). From (7.14) verify  $\frac{\alpha'}{4}(p_L^2 - p_R^2) = nw$  and  $\frac{1}{2}(p_L^2 + p_R^2) = n^2/R^2 + w^2 R^2/\alpha'^2$ . Express both in terms of the dimensionless momenta of Remark 7.2.

**Exercise 7.3** (★ Accidental massless scalars). Show that a state with  $N = \tilde{N} = 0$  is massless exactly when  $r = |n|/2$  with  $w = 0$ , or  $r = 2/|w|$  with  $n = 0$ . List the radii in the interval  $\frac{1}{2} \leq r \leq 2$  at which this happens, and the states involved.

**Exercise 7.4** (★★ Twisted boundary conditions). A complex scalar may obey  $\Phi(x, y + 2\pi R) = e^{2\pi i \theta} \Phi(x, y)$  with a fixed  $\theta \in [0, 1)$ . Find the Kaluza–Klein masses. For  $\theta = \frac{1}{2}$  show that there is no massless mode and that the lightest mode has  $M = 1/(2R)$ . Explain why a *real* scalar allows only  $\theta \in \{0, \frac{1}{2}\}$ .

**Exercise 7.5** (★★ States with both charges are never tachyonic). Fix  $(n, w, N, \tilde{N})$  with  $nw \neq 0$  and regard (7.25) as a function of  $R$ . Show that it is minimized at  $R^2 = \alpha'|n/w|$ , and use level matching to show that the minimum is  $\alpha' M_{\min}^2 = 4(\max(N, \tilde{N}) - 1) \geq 0$ . Which states reach  $M^2 = 0$ , and at which radius for given  $(n, w)$ ?

**Exercise 7.6** (★★ Counting tachyons). Show that the number of tachyonic states at radius  $r$  is  $\#\{n \in \mathbb{Z} : |n| < 2r\} + \#\{w \in \mathbb{Z} : |w| < 2/r\} - 1$ . Evaluate it at  $r = 2$  and compare with Example 7.3. At which  $r$  is the number smallest, and what is it there?

**Exercise 7.7** (★★ Lean: the factor of two). State and prove in Lean, in the namespace of `momentumMassSq`:

```
theorem momentumMassSq_eq (n w R : ℚ) (hR : R ≠ 0) :
  momentumMassSq n w R = 2 * (n ^ 2 / R ^ 2 + w ^ 2 * R ^ 2)
```

Hint: `unfold`, then `field_simp`, then `ring`. Then prove the bound  $4nw \leq \text{momentumMassSq } n \ w \ R$ , the Lean form of (7.26). Hint: first establish  $p_L^2 - p_R^2 = 4nw$  as a `have`, then call `nlinarith` with the hint `sq_nonneg (n / R - w * R)`.

**Exercise 7.8** (★★ Lean: level matching as a predicate). In the setting of `T_Duality.lean` define

```
def levelMatched (s : CircleState) (N Nt : Int) : Prop := N - Nt = s.n * s.w
```

and prove that `levelMatched (tDualState s) N Nt ↔ levelMatched s N Nt`. Which lemma about integers does the proof use? State in one sentence what this theorem does and does not say about the string spectrum.

**Exercise 7.9** (★★★ The circle from the general theory). The file `GeneralizedMetric.lean` proves, for every  $d$ , the two theorems `massForm_covariant` and `tduality_inverts_metric`:

$$Z^T (g^T \mathcal{H} g) Z = (gZ)^T \mathcal{H} (gZ), \quad \eta \mathcal{H}(G, 0) \eta = \mathcal{H}(G^{-1}, 0).$$

Using only these two statements and  $\eta^T = \eta$ , derive on paper the invariance of  $R^2 w^2 + n^2 / R^2$  under  $R \rightarrow 1/R$ ,  $n \leftrightarrow w$ . Then write the corresponding Lean statement for  $d = 1$  and outline how `massForm_circle` would let you transport `tduality_invariant_mass_squared` from  $\mathbb{Q}$  to the matrix setting.

## Further reading

- Tong, *String Theory*, §8.1–8.2: the text this chapter follows most closely [Tong §8.1–8.2.3, ll. 11183–11625]. The coupling of the string to  $B_{\mu\nu}$  used in Section 7.6 is in [Tong §7.2.1, ll. 9425–9500].
- Giveon, Porrati and Rabinovici, *Target space duality in string theory*: the circle in canonical language, with dimensionless  $p_{L,R}$ , the conjugate momentum and the Hamiltonian  $H = L_0 + \tilde{L}_0$  [GPR §2.2, ll. 576–760]; moduli as couplings of truly marginal operators [GPR §2.1, ll. 458–575].
- Alvarez, Alvarez-Gaumé and Lozano, *An introduction to T-duality in string theory*: the introduction places the circle among the dualities of statistical mechanics and field theory, and states that the equivalence  $R \rightarrow \alpha'/R$  of the interacting theory requires a shift of the dilaton [AAL §1, ll. 95–117]. This is the starting point of Chapters 8 and 9.

## Chapter 8

# T-Duality and the Self-Dual Radius

In Chapter 7 we compactified one direction of space on a circle of radius  $R$  and found that a closed string carries two integers there: a momentum number  $n$ , as any quantum particle on a circle does, and a winding number  $w$ , which only an extended object can have. This chapter asks what happens when the circle is made small. For a point particle the answer is dull: the Kaluza–Klein momentum states become infinitely heavy, the circle disappears from low-energy physics, and that is all. For a string the answer is that *nothing singular happens at all*: the theory on a circle of radius  $R$  is the same theory as on a circle of radius  $\alpha'/R$ , with the roles of momentum and winding exchanged. This statement is called *T-duality*. It is the simplest example of a phenomenon with no counterpart in field theory, it is the seed from which the duality group  $O(d, d; \mathbb{Z})$ , the doubled coordinates and the generalized metric of Parts II and III grow, and it is the origin of the “two scales” in the title of this book.

We shall be careful about one thing that is often blurred. “T-duality is a symmetry” can mean four different statements of increasing strength: that the mass formula is invariant; that the two conformal field theories are the same; that string perturbation theory agrees at every loop order; and that the full theory is the same. The evidence, and the part that a machine has checked, differ from level to level, and a student should know at each moment which level is being discussed. Section 8.2 lays out this ladder; the rest of the chapter climbs it.

## 8.1 The symmetry of the mass formula

### 8.1.1 Data carried over from the circle

We use the conventions of Chapter 7:  $\alpha'$  has dimension length<sup>2</sup>, the string length is  $\ell_s = \sqrt{\alpha'}$ , the compact coordinate is  $X \equiv X^{25} \sim X + 2\pi R$ , the momentum along the circle is  $n/R$  with  $n \in \mathbb{Z}$ , and the string may close only up to a winding,  $X(\sigma + 2\pi) = X(\sigma) + 2\pi w R$  with  $w \in \mathbb{Z}$ . (Tong calls the winding number  $m$ ; we write  $w$  throughout.) The zero modes are packaged into the left- and right-moving momenta

$$p_L = \frac{n}{R} + \frac{wR}{\alpha'}, \quad p_R = \frac{n}{R} - \frac{wR}{\alpha'}, \quad (8.1)$$

which appear in the mode expansion  $X = X_L(\sigma^+) + X_R(\sigma^-)$ ,  $\sigma^\pm = \tau \pm \sigma$ ,

$$\begin{aligned} X_L(\sigma^+) &= \frac{1}{2}x + \frac{1}{2}\alpha' p_L \sigma^+ + i\sqrt{\frac{\alpha'}{2}} \sum_{k \neq 0} \frac{1}{k} \tilde{\alpha}_k e^{-ik\sigma^+}, \\ X_R(\sigma^-) &= \frac{1}{2}x + \frac{1}{2}\alpha' p_R \sigma^- + i\sqrt{\frac{\alpha'}{2}} \sum_{k \neq 0} \frac{1}{k} \alpha_k e^{-ik\sigma^-} \end{aligned} \quad (8.2)$$

[Tong §8.2, eq. (8.4), ll. 11351–11415]. As a check, adding the two lines gives  $X = x + \frac{1}{2}\alpha'(p_L + p_R)\tau + \frac{1}{2}\alpha'(p_L - p_R)\sigma + \dots = x + \alpha'\frac{n}{R}\tau + wR\sigma + \dots$ : the centre of mass moves with momentum  $n/R$  and the string winds  $w$  times. The  $L_0$  and  $\tilde{L}_0$  constraints give the mass seen by an observer in the 25 non-compact dimensions,

$$M^2 = p_L^2 + \frac{4}{\alpha'}(\tilde{N} - 1) = p_R^2 + \frac{4}{\alpha'}(N - 1), \quad (8.3)$$

where  $N$  and  $\tilde{N}$  are the oscillator levels of the  $\alpha_k$  and the  $\tilde{\alpha}_k$  [Tong §8.2, ll. 11430–11452]. The *difference* of the two expressions is the level-matching condition, and their *average* is the mass formula:

$$p_L^2 - p_R^2 = \frac{4nw}{\alpha'} \implies N - \tilde{N} = nw, \quad (8.4)$$

$$M^2 = \frac{1}{2}(p_L^2 + p_R^2) + \frac{2}{\alpha'}(N + \tilde{N} - 2) = \frac{n^2}{R^2} + \frac{w^2 R^2}{\alpha'^2} + \frac{2}{\alpha'}(N + \tilde{N} - 2) \quad (8.5)$$

[Tong §8.2, eqs. (8.5)–(8.6), ll. 11458–11480]. The two new terms have transparent meanings. A state with momentum  $n/R$  along a hidden direction looks, in the lower-dimensional world, like a particle of mass  $|n|/R$ . A string wound  $w$  times around a circle of circumference  $2\pi R$  has length  $2\pi|w|R$  and therefore rest energy  $2\pi|w|RT = |w|R/\alpha'$ , with  $T = 1/(2\pi\alpha')$  the tension [Tong §8.2, ll. 11482–11487].

### 8.1.2 The duality map

**Definition 8.1.** The *T-duality map* on the labels of closed-string states on a circle is

$$\mathbb{T}: \quad R \mapsto \tilde{R} = \frac{\alpha'}{R}, \quad (n, w) \mapsto (\tilde{n}, \tilde{w}) = (w, n), \quad (N, \tilde{N}) \mapsto (\tilde{N}, N). \quad (8.6)$$

**Proposition 8.2** (T-duality of the zero modes). *Under  $\mathbb{T}$ ,*

$$p_L \mapsto p_L, \quad p_R \mapsto -p_R. \quad (8.7)$$

*Consequently the mass formula (8.5) and the level-matching condition (8.4) are invariant: the states at radius  $R$  and the states at radius  $\alpha'/R$  are in one-to-one correspondence, with equal masses.*

*Proof.* Substitute (8.6) into (8.1):

$$\tilde{p}_L = \frac{\tilde{n}}{\tilde{R}} + \frac{\tilde{w}\tilde{R}}{\alpha'} = \frac{wR}{\alpha'} + \frac{n}{\alpha'} \cdot \frac{\alpha'}{R} = \frac{wR}{\alpha'} + \frac{n}{R} = p_L, \quad \tilde{p}_R = \frac{wR}{\alpha'} - \frac{n}{R} = -p_R.$$

The mass (8.3) contains only  $p_L^2$ ,  $p_R^2$ ,  $N$  and  $\tilde{N}$ , so it is unchanged. Level matching involves the product  $nw = wn$ , also unchanged. The map is a bijection of  $\mathbb{Z}^2$  because it is its own inverse.  $\square$

The rule (8.7) is what Tong and Giveon–Porrati–Rabinovici state [Tong §8.3, ll. 11627–11640, 11665–11670; GPR §2.2, eq. (2.2.11), ll. 731–743]. GPR use the dimensionless momenta  $\sqrt{\alpha'/2}p_{L,R}$  and the dimensionless radius  $R/\sqrt{\alpha'}$ ; in those variables the map reads  $R \mapsto 1/R$ . The proposition is a statement of elementary algebra, and it is the part of this chapter that has been checked in Lean (Section 8.7).

Two features of (8.7) deserve emphasis, because they are stronger than the invariance of  $M^2$ .

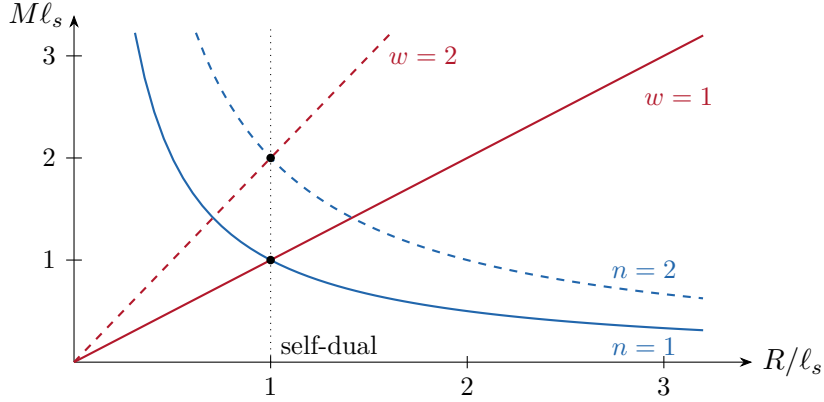


Figure 8.1: The zero-mode contributions  $|n|/R$  (momentum) and  $|w|R/\alpha'$  (winding) to the mass, in string units. The picture is symmetric under reflection  $R/\ell_s \mapsto \ell_s/R$  together with the exchange of the two families of curves. Light momentum states signal a large circle; light winding states signal a large *dual* circle. The full  $M^2$  of (8.5) adds the oscillator term, which does not depend on  $R$ .

- *The two chiral halves are treated differently.* The left-movers do not notice the duality at all, while the right-moving momentum is reflected. We shall see in Section 8.3 that the same is true of the whole right-moving field, oscillators included. T-duality is a parity transformation acting on one chirality of the worldsheet theory.
- *The two Kaluza–Klein gauge fields are exchanged.* In Chapter 7 we saw that the charge under the gauge field  $A_\mu$  coming from the metric is proportional to  $p_L + p_R = 2n/R$ , and the charge under  $\tilde{A}_\mu$  coming from the  $B$ -field is proportional to  $p_L - p_R = 2wR/\alpha'$  [Tong §8.2.2, ll. 11527–11560]. Since  $\mathbb{T}$  exchanges  $p_L + p_R \leftrightarrow p_L - p_R$ , it exchanges  $A_\mu \leftrightarrow \tilde{A}_\mu$ : what one observer calls “the photon of geometry” the dual observer calls “the photon of the  $B$ -field”.

#### Physical picture: what it means to move and what it means to wind

Take  $R \gg \ell_s$ . Winding states cost an energy  $\sim R/\alpha'$  and are out of reach; momentum states are spaced by  $1/R$  and, as  $R \rightarrow \infty$ , merge into a continuum. A continuum of momentum states is precisely how a low-energy observer recognises a large dimension. Now take  $R \ll \ell_s$ . The momentum states are heavy and drop out, but the winding states are now spaced by  $R/\alpha' = 1/\tilde{R}$  and form a continuum as  $R \rightarrow 0$ : the spectrum looks exactly as if a *new* large dimension of radius  $\tilde{R} = \alpha'/R$  had opened up [Tong §8.3, ll. 11643–11654]. A point particle has no winding states, which is why Kaluza–Klein field theory on a small circle simply loses a dimension, while string theory does not.

### 8.1.3 A worked example: $R = 2\ell_s$ against $R = \ell_s/2$

**Example 8.3.** Set  $\alpha' = 1$  and compare the circle of radius  $R = 2$  with its dual of radius  $\tilde{R} = \frac{1}{2}$ . At  $R = 2$ , (8.5) reads  $M^2 = \frac{1}{4}n^2 + 4w^2 + 2(N + \tilde{N} - 2)$  with  $N - \tilde{N} = nw$ ; at  $\tilde{R} = \frac{1}{2}$  it reads  $M^2 = 4\tilde{n}^2 + \frac{1}{4}\tilde{w}^2 + 2(N + \tilde{N} - 2)$ . Listing all label sets  $(n, w; N, \tilde{N})$  with  $M^2 \leq \frac{9}{4}$  gives Table 8.1.

Three lessons can be read off. First, the two lists of masses and multiplicities coincide, as Proposition 8.2 guarantees; the table was generated by enumerating integers, not by using the proposition, so it is an independent check. Second, a *given* label does not keep its mass:

Table 8.1: Lightest sectors of the bosonic string on a circle,  $\alpha' = 1$ . “Sectors” counts label sets  $(n, w; N, \tilde{N})$ , not states: a sector with  $N = 1$  contains one state for each polarization of the oscillator. The two columns are related by  $n \leftrightarrow w$ .

| $M^2$           | sectors | $(n, w; N, \tilde{N})$ at $R = 2$                                                     | $(\tilde{n}, \tilde{w}; N, \tilde{N})$ at $\tilde{R} = \frac{1}{2}$                 |
|-----------------|---------|---------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| -4              | 1       | (0, 0; 0, 0)                                                                          | (0, 0; 0, 0)                                                                        |
| $-\frac{15}{4}$ | 2       | ( $\pm 1, 0; 0, 0$ )                                                                  | (0, $\pm 1; 0, 0$ )                                                                 |
| -3              | 2       | ( $\pm 2, 0; 0, 0$ )                                                                  | (0, $\pm 2; 0, 0$ )                                                                 |
| $-\frac{7}{4}$  | 2       | ( $\pm 3, 0; 0, 0$ )                                                                  | (0, $\pm 3; 0, 0$ )                                                                 |
| 0               | 5       | ( $\pm 4, 0; 0, 0$ ), (0, $\pm 1; 0, 0$ ), (0, 0; 1, 1)                               | (0, $\pm 4; 0, 0$ ), ( $\pm 1, 0; 0, 0$ ), (0, 0; 1, 1)                             |
| $\frac{1}{4}$   | 2       | ( $\pm 1, 0; 1, 1$ )                                                                  | (0, $\pm 1; 1, 1$ )                                                                 |
| 1               | 2       | ( $\pm 2, 0; 1, 1$ )                                                                  | (0, $\pm 2; 1, 1$ )                                                                 |
| $\frac{9}{4}$   | 8       | ( $\pm 5, 0; 0, 0$ ), ( $\pm 3, 0; 1, 1$ ), $\pm(1, 1; 1, 0)$ ,<br>$\pm(1, -1; 0, 1)$ | (0, $\pm 5; 0, 0$ ), (0, $\pm 3; 1, 1$ ), $\pm(1, 1; 1, 0)$ ,<br>$\pm(-1, 1; 0, 1)$ |

the state  $(n, w) = (1, 0)$  with  $N = \tilde{N} = 0$  has  $M^2 = -\frac{15}{4}$  at  $R = 2$  but  $M^2 = 0$  at  $R = \frac{1}{2}$ . T-duality is not the statement that each state is insensitive to  $R$ ; it is the statement that the two *collections* of states agree after relabelling. Third, the bosonic string has tachyons ( $M^2 < 0$ ) in both descriptions. This disease of the bosonic string is unrelated to the circle and is cured in the superstring (Chapter 5); T-duality maps tachyons to tachyons.

## 8.2 Four meanings of “T-duality is a symmetry”

Proposition 8.2 is an identity between two lists of numbers. It is natural to ask how much more is true. We distinguish four levels.

**Level 1: the free spectrum.** The sets of masses, with multiplicities and charges, agree at radius  $R$  and  $\alpha'/R$ . Equivalently the one-loop (torus) partition function agrees, because it is a trace over the spectrum. This is Proposition 8.2, extended from the zero modes to all states in Section 8.3.

**Level 2: the conformal field theory.** There is a one-to-one map between the local operators of the two worldsheet theories under which *all* operator product expansions agree. Since tree-level string scattering amplitudes are correlation functions of vertex operators on the sphere (Chapter 5), level 2 implies that tree-level interactions agree. Two theories can have the same spectrum without being equivalent, in the same way that two drums of different shapes can sound the same notes, so level 2 is a genuinely stronger claim.

**Level 3: string perturbation theory.** The sum over worldsheets of all genera agrees. This requires a new ingredient: the string coupling must be transformed together with  $R$  (Section 8.4).

**Level 4: the exact theory.** The duality holds beyond perturbation theory. For the bosonic string this has no sharp meaning, since its perturbation series is already ill-defined because of the tachyon [GPR §2.3, ll. 1023–1029]. The best available argument, reviewed in Section 8.5, is that T-duality is a remnant of a *gauge* symmetry, and gauge symmetries cannot be broken by quantum corrections.

Levels 1–3 are results of the literature, [L]. Only the zero-mode algebra underlying level 1 is kernel-checked in this book’s library, [A]. No level is a statement about experiments: a symmetry of string theory on a circle is a mathematical property of a model.

## 8.3 The dual coordinate

### 8.3.1 Flipping the right-movers

The rule  $p_L \mapsto p_L$ ,  $p_R \mapsto -p_R$  suggests defining a new worldsheet field in which the whole right-moving half of  $X$  is reflected:

$$Y(\tau, \sigma) = X_L(\sigma^+) - X_R(\sigma^-) \quad (8.8)$$

[Tong §8.3, ll. 11665–11678]. Let us find out what kind of field  $Y$  is. Insert the expansions (8.2):

$$\begin{aligned} Y &= y + \frac{1}{2}\alpha'(p_L - p_R)\tau + \frac{1}{2}\alpha'(p_L + p_R)\sigma + \text{oscillators} \\ &= y + wR\tau + \alpha'\frac{n}{R}\sigma + \dots = y + \alpha'\frac{w}{\tilde{R}}\tau + n\tilde{R}\sigma + \dots, \quad \tilde{R} = \frac{\alpha'}{R}. \end{aligned} \quad (8.9)$$

Compare with  $X = x + \alpha'\frac{n}{R}\tau + wR\sigma + \dots$ . The field  $Y$  is a free boson that advances by  $2\pi n\tilde{R}$  when  $\sigma \rightarrow \sigma + 2\pi$ , and whose centre of mass carries momentum  $w/\tilde{R}$ . In other words,  $Y$  is a coordinate on a circle of radius  $\tilde{R} = \alpha'/R$ , on which the string has winding number  $n$  and momentum number  $w$ . The oscillators of  $Y$  are  $\tilde{\alpha}_k$  and  $-\alpha_k$ , which is the rule

$$\alpha_k \mapsto -\alpha_k, \quad \tilde{\alpha}_k \mapsto \tilde{\alpha}_k \quad (8.10)$$

of [GPR §2.2, eq. (2.2.12), ll. 742–750]. The level operators  $N = \sum_{k>0} \alpha_{-k}\alpha_k$  and  $\tilde{N}$  are quadratic in the oscillators and therefore invariant, which justifies the last entry of (8.6). So are the contributions  $\frac{\alpha'}{4}p_R^2 + N$  and  $\frac{\alpha'}{4}p_L^2 + \tilde{N}$  of the compact direction to  $L_0$  and  $\tilde{L}_0$ , and more generally all the Virasoro generators, which are quadratic in  $\partial_{\pm}X$ .

The relation between  $X$  and  $Y$  can be written without splitting into movers. From  $\partial_{\tau} = \partial_+ + \partial_-$  and  $\partial_{\sigma} = \partial_+ - \partial_-$ ,

$$\partial_{\tau}X = X'_L + X'_R = \partial_{\sigma}Y, \quad \partial_{\sigma}X = X'_L - X'_R = \partial_{\tau}Y,$$

or, covariantly,

$$\partial_{\alpha}X = \epsilon_{\alpha\beta}\partial^{\beta}Y, \quad \epsilon_{\tau\sigma} = -\epsilon_{\sigma\tau} = +1, \quad (8.11)$$

where the index is raised with the worldsheet Minkowski metric [Tong §8.3, eq. (8.9), ll. 11679–11690]. This is a two-dimensional version of electric–magnetic duality: the one-form  $dX$  is exchanged with its Hodge dual. It exchanges the equation of motion  $\partial_{\alpha}\partial^{\alpha}X = 0$  with the Bianchi identity  $\epsilon^{\alpha\beta}\partial_{\alpha}\partial_{\beta}Y = 0$ , and the Noether charge of translations (momentum,  $\oint d\sigma \partial_{\tau}X$ ) with the topological charge (winding,  $\oint d\sigma \partial_{\sigma}X$ ). Note that (8.11) is non-local:  $Y$  is obtained from  $X$  by integrating its derivatives, not by a pointwise change of variables. No redefinition of the target-space coordinate  $X \mapsto f(X)$  produces  $Y$ .

### 8.3.2 From the spectrum to the operator algebra

Pass to the Euclidean worldsheet and complex coordinates as in Chapter 4, where left-movers become holomorphic and right-movers antiholomorphic:  $X(z, \bar{z}) = X(z) + \bar{X}(\bar{z})$ . The basic operator products are

$$X(z)X(w) \sim -\frac{\alpha'}{2}\ln(z-w), \quad \bar{X}(\bar{z})\bar{X}(\bar{w}) \sim -\frac{\alpha'}{2}\ln(\bar{z}-\bar{w}), \quad X(z)\bar{X}(\bar{w}) \sim 0. \quad (8.12)$$

Each is quadratic in the field to which the reflection is applied, so the fields  $Y(z) = X(z)$  and  $\bar{Y}(\bar{z}) = -\bar{X}(\bar{z})$  obey the same operator products, and the stress tensors  $T = -\frac{1}{\alpha'}\partial X\partial X$ ,  $\bar{T} = -\frac{1}{\alpha'}\bar{\partial}\bar{X}\bar{\partial}\bar{X}$  are literally unchanged. Tong summarises: all OPEs of  $Y$  coincide with

those of  $X$ , and this is sufficient to ensure that all interactions defined in the conformal field theory are the same [Tong §8.3, ll. 11675–11678].

It is instructive to see how this works for the operators that carry momentum and winding. A state with labels  $(n, w)$  is created by the vertex operator

$$V_{n,w}(z, \bar{z}) = : \exp(ip_L X(z) + ip_R \bar{X}(\bar{z})) : = : \exp(ip_L Y(z) + i(-p_R) \bar{Y}(\bar{z})) : \quad (8.13)$$

[Tong §8.2.2, ll. 11536–11545]. The second form is the key point: the *same* operator, expressed through  $Y$ , is a vertex operator with momenta  $(p_L, -p_R)$ , that is, with labels  $(w, n)$  on the circle of radius  $\tilde{R}$ . Nothing has been transformed; an operator algebra has been given a second set of names. With the rule of Chapter 4 that  $e^{ikX(z)}$  has weight  $\alpha' k^2/4$ , the weights of  $V_{n,w}$  and the product of two such operators are

$$(h, \bar{h}) = \left( \frac{\alpha' p_L^2}{4}, \frac{\alpha' p_R^2}{4} \right), \quad h - \bar{h} = nw \in \mathbb{Z}, \quad (8.14)$$

$$V_{n,w}(z, \bar{z}) V_{n',w'}(0) \sim z^{\alpha' p_L p'_L/2} \bar{z}^{\alpha' p_R p'_R/2} V_{n+n', w+w'}(0) + \dots, \quad (8.15)$$

$$\frac{\alpha'}{2} (p_L p'_L - p_R p'_R) = nw' + wn'.$$

The exponents depend on the right-moving momenta only through the product  $p_R p'_R$ , which is invariant under the simultaneous reflection of both factors. So the operator product coefficients agree in the two descriptions: this is level 2. The last identity in (8.15), which follows by expanding (8.1), shows that the difference of the two exponents is an integer, so the product is single-valued when  $z$  circles the origin. That integer,  $nw' + wn'$ , is a bilinear form on the charge lattice  $\mathbb{Z}^2 \ni (n, w)$  which is independent of  $R$  and invariant under  $n \leftrightarrow w$ ; it will reappear in Chapter 10 as the form  $\eta$  of signature  $(1, 1)$  whose automorphisms make up the duality group  $O(1, 1; \mathbb{Z})$  (Chapter 11).

*Remark 8.4.* A complete treatment must also attach sign factors (“cocycles”) to the operators  $V_{n,w}$  so that they commute rather than anticommute when  $nw' + wn'$  is odd, and must check that the duality map can be chosen compatibly with them. This point is standard but is not discussed in the sources of this book’s library, and we do not pursue it.

#### Common pitfall: “the physics does not depend on $R$ ”

T-duality does *not* say that physics is independent of the radius. The spectrum at  $R = 2\ell_s$  differs from the spectrum at  $R = 3\ell_s$ ; an observer can measure the spacing of the light tower and distinguish them. What cannot be distinguished is  $R = 2\ell_s$  from  $R = \ell_s/2$ . The symmetry is a discrete identification of the parameter space, not the absence of a parameter.

## 8.4 The path integral, the string coupling and the dilaton

### 8.4.1 A derivation that does not split into movers

The argument of Section 8.3 used the special solvability of the free boson. There is a second derivation which works directly in the path integral, and which is the prototype of the Buscher procedure of Chapter 9, where it is applied to curved backgrounds. We follow [Tong §8.3.1, ll. 11716–11803]; the general version is [AAL §2.1, ll. 215–300 and 336–376].

Write  $X = R\varphi$  with  $\varphi \sim \varphi + 2\pi$ , so that the radius appears as a coupling constant. On a Euclidean worldsheet,

$$S[\varphi] = \frac{R^2}{4\pi\alpha'} \int d^2\sigma \partial_\alpha \varphi \partial^\alpha \varphi. \quad (8.16)$$



*Step 1: gauge the shift symmetry.* The action is invariant under  $\varphi \rightarrow \varphi + \lambda$ . Introduce a worldsheet gauge field  $A_\alpha \rightarrow A_\alpha - \partial_\alpha \lambda$  and the covariant derivative  $D_\alpha \varphi = \partial_\alpha \varphi + A_\alpha$ . By itself this would change the theory, so add a Lagrange multiplier  $\theta$  that forces the gauge field to be trivial:

$$S[\varphi, \theta, A] = \frac{R^2}{4\pi\alpha'} \int d^2\sigma D_\alpha \varphi D^\alpha \varphi + \frac{i}{2\pi} \int d^2\sigma \theta \epsilon^{\alpha\beta} \partial_\alpha A_\beta. \quad (8.17)$$

*Step 2: integrate out  $\theta$ .* This sets  $\epsilon^{\alpha\beta} \partial_\alpha A_\beta = 0$ . On a worldsheet with the topology of the plane,  $A$  is then pure gauge and can be set to zero, giving back (8.16). On a worldsheet with handles a flat gauge field can still have holonomies around the non-contractible cycles. If  $\theta$  is itself periodic with period  $2\pi$ , the sum over its winding sectors restricts these holonomies to multiples of  $2\pi$ , and those are removed by gauge transformations  $\lambda$  that wind around the cycle, which are allowed because  $\varphi$  is periodic. So (8.17), divided by the volume of the gauge group, defines the same theory as (8.16) on every worldsheet.

*Step 3: integrate in the other order.* Fix the gauge  $\varphi = 0$  instead and integrate the multiplier term by parts. The action becomes quadratic in  $A$  with no derivatives:

$$S = \frac{R^2}{4\pi\alpha'} \int d^2\sigma A_\alpha A^\alpha - \frac{i}{2\pi} \int d^2\sigma \epsilon^{\alpha\beta} (\partial_\alpha \theta) A_\beta.$$

The Gaussian integral is done by solving the equation of motion of  $A$ ,

$$\frac{R^2}{2\pi\alpha'} A^\beta = \frac{i}{2\pi} \epsilon^{\alpha\beta} \partial_\alpha \theta \implies A^\beta = i \frac{\alpha'}{R^2} \epsilon^{\alpha\beta} \partial_\alpha \theta,$$

and substituting back. Using  $\epsilon^{\alpha\beta} \epsilon_{\gamma\beta} = \delta_\gamma^\alpha$  in Euclidean signature, the first term gives  $-\frac{\alpha'}{4\pi R^2} (\partial\theta)^2$  and the second  $+\frac{\alpha'}{2\pi R^2} (\partial\theta)^2$ , so

$$S[\theta] = \frac{\alpha'}{4\pi R^2} \int d^2\sigma \partial_\alpha \theta \partial^\alpha \theta = \frac{\tilde{R}^2}{4\pi\alpha'} \int d^2\sigma \partial_\alpha \theta \partial^\alpha \theta, \quad \tilde{R} = \frac{\alpha'}{R}. \quad (8.18)$$

This is (8.16) for a  $2\pi$ -periodic boson on a circle of radius  $\tilde{R}$ . The equation of motion of  $A$ , at  $\varphi = 0$  where  $A_\alpha$  stands for  $D_\alpha \varphi$ , is the Euclidean form of (8.11) with  $Y = \tilde{R}\theta$ : the factor  $i$  is the usual consequence of the Wick rotation. The derivation never refers to the genus of the worldsheet, so it supports level 3 of Section 8.2. One thing was discarded on the way: the Gaussian integral over  $A$  produces a field-independent factor, proportional to  $\sqrt{\alpha'}/R$  for each mode, which Tong sets aside with the remark that a more careful treatment turns it into a shift of the dilaton [Tong §8.3.1, ll. 11795–11803]. We now obtain that shift in two independent ways.

### 8.4.2 The dilaton shift from the effective action

Recall from Chapter 6 that the string coupling is the expectation value of the dilaton,  $g_s = e^\Phi$ , and that a worldsheet of genus  $g$  is weighted by  $g_s^{2g-2}$ . The low-energy effective action in 26 dimensions is multiplied by  $1/g_s^2$ . Reducing on the circle, the integral over  $X^{25}$  gives a factor  $2\pi R$ , so the 25-dimensional gravitational action comes with the coefficient

$$\frac{2\pi R}{\ell_s^{24} g_s^2} \quad (8.19)$$

up to a number. An observer in 25 dimensions measures this combination, which is an inverse Newton constant, not  $R$  and  $g_s$  separately. If the two descriptions are to be indistinguishable we need  $R/g_s^2 = \tilde{R}/\tilde{g}_s^2$ , that is,

$$\tilde{g}_s = g_s \sqrt{\frac{\tilde{R}}{R}} = \frac{\sqrt{\alpha'}}{R} g_s, \quad \tilde{\Phi} = \Phi - \ln \frac{R}{\sqrt{\alpha'}} \quad (8.20)$$

[Tong §8.3, eq. (8.10), ll. 11691–11715]. This is a plausibility argument, as Tong says. It can be restated by saying that the 25-dimensional dilaton  $\Phi_{25} = \Phi - \frac{1}{2} \ln(R/\sqrt{\alpha'})$  is T-duality invariant: indeed  $\tilde{\Phi} - \frac{1}{2} \ln(\tilde{R}/\sqrt{\alpha'}) = \Phi - \ln(R/\sqrt{\alpha'}) + \frac{1}{2} \ln(R/\sqrt{\alpha'})$ . The same rule, in the form  $\tilde{\phi} = \phi - \frac{1}{2} \ln g_{00}$  with  $g_{00} = R^2/\alpha'$  the metric component along the circle, comes out of a regularized treatment of the Gaussian determinant [AAL §2.1, eqs. (2.1.6)–(2.1.7), ll. 296–312]; it is one of the Buscher rules of Chapter 9. Two consistency checks: at  $R = \sqrt{\alpha'}$  the coupling does not change, as it must at a fixed point; and applying (8.20) twice returns  $g_s$ , since  $(\sqrt{\alpha'}/\tilde{R})(\sqrt{\alpha'}/R) = 1$ .

### 8.4.3 The dilaton shift from the sum over genera

Giveon, Porrati and Rabinovici obtain the same result from the partition function. Let  $r = R/\sqrt{\alpha'}$  be the dimensionless radius. On a worldsheet of genus  $g$  the compact boson has  $2g$  integers, its windings around the  $a$ - and  $b$ -cycles. After a Poisson resummation over half of them, the sum over these sectors becomes a sum over  $g$  copies of the pairs  $(p_L, p_R)$  that is manifestly invariant under  $r \rightarrow 1/r$  with the integers exchanged, multiplied by an explicit power of the radius:

$$Z_g(r) = r^{1-g} F_g(r), \quad F_g(1/r) = F_g(r) \quad \implies \quad Z_g(1/r) = r^{2g-2} Z_g(r) \quad (8.21)$$

[GPR §2.3, eqs. (2.3.6)–(2.3.9), ll. 930–997]. One power of  $r$  comes from the volume of the zero mode of  $X$ , and  $r^{-g}$  from the resummation. At genus one the factor is absent:  $Z_1(1/r) = Z_1(r)$ . This is level 1, since the torus amplitude is a trace over the spectrum and the integers  $n, w$  are summation variables [GPR §2.2, ll. 778–786]. At genus  $g \neq 1$  the partition functions differ, but only by a power of  $r$  that depends on the genus exactly as the weight  $g_s^{2g-2}$  does. Hence

$$\sum_{g \geq 0} \tilde{g}_s^{2g-2} Z_g(1/r) = \sum_{g \geq 0} (\tilde{g}_s r)^{2g-2} Z_g(r) = \sum_{g \geq 0} g_s^{2g-2} Z_g(r) \quad \text{iff} \quad \tilde{g}_s = \frac{g_s}{r}, \quad (8.22)$$

which is (8.20) again. GPR write the weight as  $e^{(1-g)\Phi_{\text{GPR}}}$ , so that  $\Phi_{\text{GPR}} = -2\Phi$ , and state the result as  $\Phi_{\text{GPR}} \rightarrow \Phi_{\text{GPR}} + 2 \ln r$  [GPR §2.3, eqs. (2.3.10)–(2.3.11), ll. 998–1029]; converting,  $\tilde{\Phi} = \Phi - \ln r$ , in agreement with (8.20). Their conclusion is worth quoting in substance: a compactification of small radius is equivalent to one of large radius *provided* the dilaton expectation value is changed as well.

#### Common pitfall: forgetting the coupling

“ $R \rightarrow \alpha'/R$  is a symmetry of string theory” is false as it stands: by (8.21) only the torus amplitude is invariant under the inversion of  $R$  alone. The symmetry is  $(R, g_s) \rightarrow (\alpha'/R, g_s \sqrt{\alpha'}/R)$ . For  $R \ll \ell_s$  the dual coupling is *larger* than the original one: a weakly coupled string on a very small circle is dual to a string on a large circle whose ten- or twenty-six-dimensional coupling is stronger, although the lower-dimensional coupling (8.19) is the same.

## 8.5 The self-dual radius

### 8.5.1 The fixed point and the moduli space

The map  $R \mapsto \alpha'/R$  on the positive reals has exactly one fixed point,

$$R = \frac{\alpha'}{R} \iff R^2 = \alpha' \iff R = \sqrt{\alpha'} = \ell_s, \quad (8.23)$$

the *self-dual radius*. The radius is not a fixed parameter of the theory but the expectation value of a massless scalar field, the radion  $\sigma$  of Chapter 7; such a parameter is called a *modulus*, and the set of its inequivalent values is the moduli space. Naively this is the half-line  $R \in (0, \infty)$ . T-duality tells us that  $R$  and  $\alpha'/R$  label the same theory, so the true moduli space is the quotient

$$\mathcal{M}_{S^1} = (0, \infty) / (R \sim \alpha'/R) \cong [\sqrt{\alpha'}, \infty). \quad (8.24)$$

In the logarithmic coordinate  $x = \ln(R/\sqrt{\alpha'})$  the duality is the reflection  $x \mapsto -x$  of the real line, and  $\mathcal{M}_{S^1} = \mathbb{R}/\mathbb{Z}_2 = [0, \infty)$ : a half-line whose end point is the self-dual radius (Figure 8.2). Every circle compactification is equivalent to one with  $R \geq \ell_s$ . The coordinate  $x$  is also the natural one for measuring distances in moduli space, a fact that becomes important in Chapter 27.

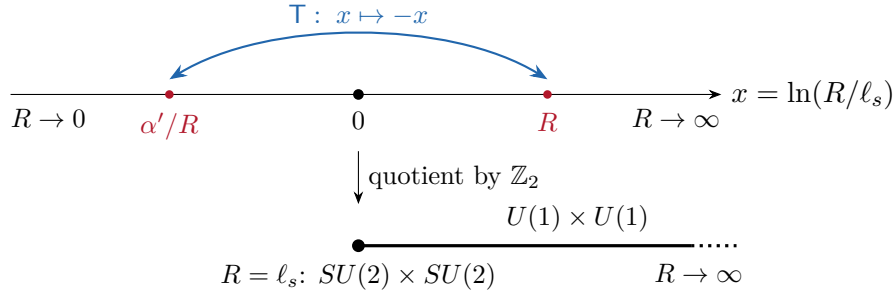


Figure 8.2: Top: T-duality acts on the logarithm of the radius as a reflection. Bottom: the moduli space of the circle compactification is the quotient, a half-line. At its end point the gauge symmetry is enhanced.

### 8.5.2 Extra massless states

End points of moduli spaces are where something happens. Set  $R = \sqrt{\alpha'}$  in (8.5):

$$\alpha' M^2 = n^2 + w^2 + 2(N + \tilde{N}) - 4, \quad N - \tilde{N} = nw. \quad (8.25)$$

We look for all solutions with  $M^2 = 0$ . Since  $N + \tilde{N} \geq |N - \tilde{N}| = |nw|$ , we need  $n^2 + w^2 + 2|nw| \leq 4$ , that is  $(|n| + |w|)^2 \leq 4$ . The possibilities are therefore:

- (i)  $n = w = 0$ ,  $N = \tilde{N} = 1$ : the states  $\alpha_{-1}^M \tilde{\alpha}_{-1}^{M'} |0; p\rangle$  that are massless at every radius. With both indices non-compact they give the graviton, the  $B$ -field and the dilaton in 25 dimensions; with one index along the circle, the two gauge fields  $A_\mu$ ,  $\tilde{A}_\mu$ ; with both along the circle, the radion [Tong §8.2.1, ll. 11491–11526].
- (ii)  $(n, w) = (\pm 2, 0)$  or  $(0, \pm 2)$ ,  $N = \tilde{N} = 0$ : four scalars, momentum and winding modes of the tachyon.
- (iii)  $(n, w) = \pm(1, 1)$ , so  $nw = 1$  and  $(N, \tilde{N}) = (1, 0)$ : the states  $\alpha_{-1}^M |n, w\rangle$ . For  $M = \mu$  these are two *vectors*; for  $M = 25$ , two scalars.
- (iv)  $(n, w) = \pm(1, -1)$ , so  $nw = -1$  and  $(N, \tilde{N}) = (0, 1)$ : the states  $\tilde{\alpha}_{-1}^M |n, w\rangle$ ; again two vectors and two scalars.

This agrees with the list of [Tong §8.2.3, ll. 11561–11612] (where the levels of the last item should read  $N = 0$ ,  $\tilde{N} = 1$ , as level matching requires) and with the direct enumeration

of Example 8.3 repeated at  $R = 1$ , which finds nine label sets at  $M^2 = 0$ . Away from the self-dual radius the new states are massive. For the vectors of item (iii),

$$M^2 = \frac{1}{R^2} + \frac{R^2}{\alpha'^2} - \frac{2}{\alpha'} = \left(\frac{1}{R} - \frac{R}{\alpha'}\right)^2 = p_R^2, \quad M \simeq \frac{2}{\alpha'} |R - \sqrt{\alpha'}| \quad \text{near } R = \sqrt{\alpha'}. \quad (8.26)$$

### 8.5.3 Enhanced gauge symmetry and T-duality as a gauge transformation

Massless vectors that carry charge under  $U(1) \times U(1)$  can only be consistent as the gauge bosons of a non-abelian group. The charges are  $(p_L + p_R, p_L - p_R) \propto (n, w)$ , equal to  $\pm(1, 1)$  and  $\pm(1, -1)$ ; together with the two neutral photons this is the content of

$$U(1) \times U(1) \longrightarrow SU(2) \times SU(2) \quad \text{at } R = \sqrt{\alpha'} \quad (8.27)$$

[Tong §8.2.3, ll. 11608–11621]. In terms of  $p_L, p_R$ : the vectors of item (iii) have  $p_R = 0$ ,  $p_L = \pm 2/\sqrt{\alpha'}$  and are charged only under the left-moving combination of the photons; those of item (iv) have  $p_L = 0$  and are charged only under the right-moving one. So one  $SU(2)$  is carried by left-movers and one by right-movers. The scalars organise accordingly. There are  $1 + 4 + 2 + 2 = 9$  of them, and they fill the representation  $(\mathbf{3}, \mathbf{3})$ ; GPR describe them as scalars of spin  $(1, 1)$  under  $SU(2)_L \times SU(2)_R$  [GPR §2.6.1, ll. 2416–2422]. Moving away from the self-dual radius is giving an expectation value to one of these scalars, the radion, and (8.26) is the Higgs mechanism: the charged vectors acquire a mass proportional to the expectation value, and  $SU(2) \times SU(2)$  is broken to its Cartan subgroup  $U(1) \times U(1)$ .

On the worldsheet the enhancement is visible as extra conserved currents. At any radius  $\partial X(z)$  is a holomorphic current of weight  $(1, 0)$ . By (8.14) the operator  $V_{n,w}$  has weights  $(1, 0)$  precisely when  $p_R = 0$  and  $\alpha' p_L^2 = 4$ , which at  $R = \sqrt{\alpha'}$  is solved by  $(n, w) = \pm(1, 1)$ . So exactly at the self-dual radius there are three holomorphic currents,

$$J^3(z) \propto i \partial X(z), \quad J^\pm(z) = : \exp(\pm 2iX(z)/\sqrt{\alpha'}) :, \quad (8.28)$$

and three antiholomorphic ones  $\bar{J}^3, \bar{J}^\pm$  built in the same way from  $\bar{X}$ ; they generate the two  $SU(2)$  current algebras [GPR §2.6.1, ll. 2474–2481].

We can now see what T-duality is at the self-dual point. It acts as  $\bar{X} \mapsto -\bar{X}$  and trivially on  $X(z)$ , hence

$$\bar{J}^3 \longmapsto -\bar{J}^3, \quad \bar{J}^\pm \longmapsto \bar{J}^\mp, \quad J^a \longmapsto J^a. \quad (8.29)$$

This is a rotation by  $\pi$  about the 1-axis of  $SU(2)_R$ : the non-trivial element of the Weyl group of  $SU(2)_R$ . It is therefore a *gauge transformation* of the 25-dimensional theory. A small deformation  $R = \sqrt{\alpha'}(1 + \delta)$  is a perturbation of the worldsheet action by the operator  $\delta \cdot J^3 \bar{J}^3$ , which the rotation maps to  $-\delta \cdot J^3 \bar{J}^3$ , i.e. to  $R = \sqrt{\alpha'}(1 - \delta) \approx \alpha'/R$ . The theories at  $R$  and at  $\alpha'/R$  are thus related by a gauge transformation of the enhanced symmetry [GPR §2.6.1, ll. 2436–2481]. This is the conceptual reason to believe in level 4 of Section 8.2: two configurations related by a gauge transformation are not merely similar but are the same physical state, and gauge redundancies are not broken by loop or non-perturbative effects.

*Remark 8.5.* The bosonic mechanism of enhancement uses momentum and winding modes of the tachyon and does not operate in the type II superstrings; it does operate in the heterotic string [Tong §8.2.3, ll. 11622–11625]. T-duality itself survives in all of them, with modifications discussed in Chapter 13.

## 8.6 Minimal length as physics

### 8.6.1 What an observer can measure

All three of our sources draw the same moral: T-duality is a mechanism by which string theory exhibits a minimal length. As the circle shrinks below  $R = \sqrt{\alpha'}$  “the theory acts as if the circle is growing again, with winding modes playing the role of momentum modes” [Tong §8.3, ll. 11659–11662]; a small radius “is completely equivalent to a very large one” [GPR §2.2, ll. 784–793]. To turn the slogan into a statement we must say how a length is measured. An observer in the non-compact dimensions has no ruler to lay along the circle. What she can do is find the lightest tower of equally spaced states and *define* the radius as the inverse spacing, because this is how a Kaluza–Klein circle shows up in experiments. For  $R > \ell_s$  the lightest tower is the momentum tower, with spacing  $1/R$ ; for  $R < \ell_s$  it is the winding tower, with spacing  $R/\alpha'$ . The operationally defined radius is therefore

$$R_{\text{obs}} = \max\left(R, \frac{\alpha'}{R}\right) \geq \sqrt{\alpha'}. \quad (8.30)$$

Álvarez, Álvarez-Gaumé and Lozano make the same point with correlation functions: distances extracted from the large-separation behaviour of correlators come out differently for pure momentum probes and for pure winding probes, the latter seeing a circle of radius  $\alpha'/R$ , and this leads “in a natural way” to restricting the outcome of distance measurements to the interval  $d \in (1, \infty)$  in string units [AAL §6, ll. 1289–1301]. The parameter  $R$  can be made as small as we like; the *measured* size cannot.

**Example 8.6.** Let  $R = 0.1 \ell_s$ . The momentum states have masses  $10|n|/\ell_s$ ; the winding states have masses  $0.1|w|/\ell_s$ . The light tower has spacing  $0.1/\ell_s$ , and the observer reports a circle of radius  $R_{\text{obs}} = 10 \ell_s$ . If she also measures the lower-dimensional Newton constant she finds the value appropriate to a circle of radius  $10 \ell_s$  and coupling  $\tilde{g}_s = 10 g_s$ , by (8.20). There is no measurement whose outcome is “ $0.1 \ell_s$ ”.

### 8.6.2 The dual-scale function

The function (8.30) is duality invariant but not smooth at the self-dual point. This programme uses instead the simplest smooth invariant built from the two radii, the *effective* or *dual scale*

$$R_{\text{eff}}(R) = R + \frac{\alpha'}{R} = 2\sqrt{\alpha'} \cosh x, \quad x = \ln \frac{R}{\sqrt{\alpha'}}. \quad (8.31)$$

It treats  $R$  and  $\tilde{R}$  on an equal footing, and:

**Proposition 8.7.** *For all  $R > 0$ : (a)  $R_{\text{eff}}(\alpha'/R) = R_{\text{eff}}(R)$ ; (b)  $R_{\text{eff}}(R) \geq 2\sqrt{\alpha'}$ , with equality if and only if  $R = \sqrt{\alpha'}$ ; (c)  $R_{\text{obs}} \leq R_{\text{eff}} \leq 2R_{\text{obs}}$ .*

*Proof.* (a) is the symmetry of a sum. (b) follows from

$$R + \frac{\alpha'}{R} - 2\sqrt{\alpha'} = \left(\sqrt{R} - \sqrt{\alpha'/R}\right)^2 = \frac{(R - \sqrt{\alpha'})^2}{R} \geq 0,$$

which vanishes only for  $R = \sqrt{\alpha'}$ ; this is the inequality between the arithmetic and the geometric mean of  $R$  and  $\alpha'/R$ , whose product is fixed. (c) A sum of two positive numbers lies between the larger of them and twice the larger.  $\square$

Part (c) says that  $R_{\text{eff}}$  agrees with the operational radius up to a factor of at most two and coincides with it asymptotically,  $R_{\text{eff}}/R_{\text{obs}} \rightarrow 1$  as  $R \rightarrow 0$  or  $\infty$ . A closely related invariant appears directly in the mass formula. With  $r = R/\sqrt{\alpha'}$  the zero-mode part of (8.5) is

$$\alpha' M_{\text{zero modes}}^2 = \frac{n^2}{r^2} + w^2 r^2 = \begin{pmatrix} w & n \end{pmatrix} \begin{pmatrix} r^2 & 0 \\ 0 & r^{-2} \end{pmatrix} \begin{pmatrix} w \\ n \end{pmatrix}, \quad (8.32)$$

a quadratic form in the doubled charge vector  $Z = (w, n)$  whose matrix,  $\mathcal{H} = \text{diag}(r^2, r^{-2})$ , is the  $d = 1$ ,  $B = 0$  generalized metric of Chapter 15. T-duality acts on it as  $\mathcal{H} \mapsto \eta \mathcal{H} \eta$  with  $\eta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , which exchanges the two diagonal entries. Its trace is invariant and equals

$$\text{tr } \mathcal{H} = r^2 + r^{-2} = \left(r + \frac{1}{r}\right)^2 - 2 = \frac{R_{\text{eff}}^2}{\alpha'} - 2 \geq 2. \quad (8.33)$$

The bound  $\text{tr } \mathcal{H} \geq 2$  is the  $d = 1$  case of the inequality  $\text{tr } G + \text{tr } G^{-1} \geq 2d$  for tori proved in Chapter 31.

The status of these statements must be kept apart. The inequality of Proposition 8.7(b) and the identity (8.33) are mathematics, **[A]** in the units  $\alpha' = 1$  (Section 8.7); the remaining parts of the proposition are proved above by hand. That T-duality implies a minimal *measurable* compactification radius for closed-string probes is the view of the literature quoted above, **[L]**. That  $R_{\text{eff}}$ , rather than  $R_{\text{obs}}$  or any other invariant, is “the” physical scale, and that it regulates cosmological singularities, is a proposal of this programme, **[C]**, examined in Chapters 31 and 32. One caveat from the literature belongs here already: the minimal length concerns closed strings as probes. Polchinski reports evidence that D-branes probe length scales shorter than the string length, “in conflict with the idea that the string length is a minimum distance” [Pol, ll. 3020–3025]. The minimal length of this section is a property of perturbative closed-string physics on a circle, not a theorem about quantum gravity.

## 8.7 What the machine has checked

Four Lean files touch the material of this chapter. They differ greatly in how much they say, and we go through them from the most faithful to the least. As everywhere in this book, a kernel-checked statement is a statement about the Lean objects that appear in it; that those objects model a string on a circle is our reading and is never **[A]**. According to `#print axioms`, the theorems quoted below use no axioms beyond the standard `propext`, `Classical.choice` and `Quot.sound`.

### 8.7.1 Zero-mode T-duality over the rationals

File `StringTheoryFormalization/UseCases/TDualityMassSpectrum.lean`, namespace `StringTheory.UseCases.TDuality`. Units  $\alpha' = 1$ ; argument order  $(n, w, R)$ .

#### In Lean: the momenta and their transformation

```
def leftMomentum (n w R : ℚ) : ℚ := n / R + w * R

def rightMomentum (n w R : ℚ) : ℚ := n / R - w * R

def momentumMassSq (n w R : ℚ) : ℚ :=
  (leftMomentum n w R) ^ 2 + (rightMomentum n w R) ^ 2

theorem tduality_fixes_left_momentum (n w R : ℚ) (hR : R ≠ 0) :
  leftMomentum w n (1 / R) = leftMomentum n w R := by ...
```

```
theorem tduality_flips_right_momentum (n w R : ℚ) (hR : R ≠ 0) :
  rightMomentum w n (1 / R) = - rightMomentum n w R := by ...
```

**Reading.** The definitions are (8.1) at  $\alpha' = 1$ , as functions  $\mathbb{Q}^3 \rightarrow \mathbb{Q}$ . Note that  $n, w$  are arbitrary rationals here, not integers: the theorems hold for all rational values, in particular the integral ones, and integrality is never used. The two theorems are exactly (8.7): swapping the first two arguments and replacing  $R$  by  $1/R$  fixes the first function and negates the second, for every  $R \neq 0$  (positivity of  $R$  is not needed). **Proof idea.** Unfold, clear denominators with `field_simp` using  $R \neq 0$ , and close the polynomial identity with `ring`.

#### In Lean: invariance of the zero-mode mass

```
theorem tduality_invariant_mass_squared (n w R : ℚ) (hR : R ≠ 0) :
  momentumMassSq w n (1 / R) = momentumMassSq n w R := by ...
```

**Reading.** The quantity  $p_L^2 + p_R^2 = 2(n^2/R^2 + w^2R^2)$  is invariant under the duality map. By (8.5) this is *twice* the zero-mode contribution to  $M^2$ ; the factor is harmless but should be remembered when comparing with other files. The oscillator term  $\frac{2}{\alpha'}(N + \tilde{N} - 2)$  and level matching are not modelled: the theorem is the zero-mode core of level 1 of Section 8.2, and says nothing about levels 2–4. **Proof idea.** Rewrite with the two previous theorems; then  $(-p_R)^2 = p_R^2$  by `ring`. The proof thus follows the logic of Proposition 8.2: the stronger statement about  $p_L, p_R$  first, the mass as a corollary.

#### In Lean: the self-dual radius

```
theorem self_dual_radius_unique (R : ℚ) (hR : 0 < R)
  (hFix : (1 : ℚ) / R = R) : R = 1 := by ...
```

**Reading.** A positive rational fixed by  $R \mapsto 1/R$  equals 1. This is the uniqueness half of (8.23) in units  $\alpha' = 1$ ; existence ( $1/1 = 1$ ) is trivial and not stated. **Proof idea.** `field_simp` turns the hypothesis into  $1 = R^2$  (up to orientation); `nlinarith`, given the hints  $(R - 1)^2 \geq 0$  and  $(R + 1)^2 \geq 0$  and  $R > 0$ , excludes the root  $R = -1$ .

#### Common pitfall: units hidden in a number field

Working over  $\mathbb{Q}$  with  $\alpha' = 1$  is not an innocent choice. For general  $\alpha'$  the fixed-point equation is  $R^2 = \alpha'$ , which has *no* rational solution when, say,  $\alpha' = 2$ . A version of `self_dual_radius_unique` with a parameter  $\alpha'$  must live over  $\mathbb{R}$  and conclude  $R = \sqrt{\alpha'}$  (Exercise 8.8). Setting  $\alpha' = 1$  is legitimate, since it is a choice of unit of length, but then “ $R = 1$ ” means “ $R = \ell_s$ ”.

The same momenta are consumed by a theorem of the newer library, `massForm_circle` in the namespace `DualScaleStream2.DFT`, which states that the quadratic form (8.32), evaluated with the generalized metric of  $G = R^2$ ,  $B = 0$  on the charge vector  $(w, n)$ , equals `momentumMassSq n w R` divided by two. It is discussed in Chapter 15; we mention it because it fixes the convention  $Z = (w, n)$  and confirms the factor of two noted above.

### 8.7.2 The dual-scale bound over the reals

File `DualScaleStream2/DualScale/TraceBound.lean`, with namespace `DualScaleStream2.DualScale`. Here `dualScale G` is defined as the trace of the generalized

metric at  $B = 0$ , for a real  $d \times d$  matrix  $G$ ; the file proves  $\text{dualScale } G = \text{tr } G + \text{tr } G^{-1}$  (`dualScale_eq`), its invariance under  $G \mapsto G^{-1}$  (`dualScale_inv`) and the bound  $2d$  for positive-definite  $G$  (`dualScale_ge`), which belong to Chapter 31. Two statements concern the circle.

#### In Lean: the circle case of the dual scale

```
theorem dualScale_circle (R : ℝ) (hR : R ≠ 0) :
  dualScale (!![R ^ 2] : Matrix (Fin 1) (Fin 1) ℝ)
    = (R + R⁻¹) ^ 2 - 2 := by ...

theorem circle_effective_scale_ge_two (R : ℝ) (hR : 0 < R) :
  2 ≤ R + R⁻¹ := by ...
```

**Reading.** The first theorem is (8.33): for the  $1 \times 1$  metric  $G = (R^2)$  the trace of the generalized metric is  $(R + 1/R)^2 - 2$ . The second is Proposition 8.7(b) at  $\alpha' = 1$ , over the real numbers:  $R + 1/R \geq 2$  for  $R > 0$ . The equality case and the invariance under  $R \mapsto 1/R$  are not part of these two statements (the invariance is the  $d = 1$  case of `dualScale_inv`). Nothing in either statement refers to strings, measurement or cosmology. **Proof idea.** For the first, compute the inverse of a  $1 \times 1$  matrix through the adjugate formula, then `field_simp` and `ring`. For the second, rewrite  $R + R^{-1} - 2 = (R - 1)^2/R$  and use that a square divided by a positive number is non-negative: the proof of Proposition 8.7(b), line by line.

### 8.7.3 Arithmetic shadows in the older libraries

Two further files carry T-duality in their names. They belong to the older layer of the project, in which physical quantities are replaced by integers or natural numbers, and their theorems are far weaker than their documentation strings suggest. We record them so that the reader who meets them in the repository knows what they are.

#### In Lean: DualScaleValidation.UseCase1

```
def alpha_prime : Nat := 1
def effective_dual_scale_numerator (R : Nat) : Nat := R * R + alpha_prime

theorem buscher_log_involution (x : Int) : -(-x) = x := by ...
theorem dual_scale_self_dual_value :
  effective_dual_scale_numerator 1 = 2 := by ...
theorem self_dual_is_global_minimum (R : Nat) (h : R ≥ 1) :
  effective_dual_scale_numerator R ≥ 2 := by ...
```

**Reading.** The first theorem is the fact that negation of integers is an involution; it is an arithmetic shadow of “ $x \mapsto -x$  squares to the identity” in Figure 8.2, with the real coordinate  $x$  replaced by an integer. The second is  $1 \cdot 1 + 1 = 2$ . The third says  $R^2 + 1 \geq 2$  for natural numbers  $R \geq 1$ . Since  $R^2 + \alpha' = R \cdot R_{\text{eff}}(R)$ , this is a statement about  $R \cdot R_{\text{eff}}$ , not about  $R_{\text{eff}}$ : the inequality equivalent to  $R_{\text{eff}} \geq 2$  would be  $R^2 + 1 \geq 2R$ , which is not what is stated, and on natural numbers  $R \geq 1$  the dual region  $R < 1$  is absent altogether. The file’s comments speak of minimal length and of the elimination of cosmological singularities; none of that is contained in these statements. A faithful version is `circle_effective_scale_ge_two` above. **Proof idea.** `omega` and `rfl`.



**In Lean: StringTheory.StringDynamics (file TDualityGysin.lean)**

```

structure TDualState where
  momentumNum : ℤ -- n (KK momentum)
  windingNum : ℤ -- w (winding number)

def tDualityAction (s : TDualState) : TDualState :=
  { momentumNum := s.windingNum
    windingNum := s.momentumNum }

theorem tduality_involution (s : TDualState) :
  tDualityAction (tDualityAction s) = s := by ...

```

**Reading.** Swapping the two components of a pair of integers twice gives back the pair: a correct, if minimal, formal statement of  $T^2 = 1$  on the charges, with no radius, mass or Hilbert space involved. The same file defines `tDualityRadius` as the quotient  $\alpha'/R$  of two reals (its positivity hypotheses are unused) and `gysinPushforward`, which evaluates an integer sequence at 0; despite its name the latter constructs no cohomology and no fibre integration, as the file's own header states. **Proof idea.** `simp` after unfolding.

## 8.7.4 Tier table and what is missing

Table 8.2: Epistemic status of the main claims of this chapter.

| Claim                                                                                                                   | Tier | Basis                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------------|
| $p_L \mapsto p_L, p_R \mapsto -p_R$ under $R \mapsto 1/R, n \leftrightarrow w$<br>( $\alpha' = 1$ , over $\mathbb{Q}$ ) | [A]  | <code>tduality_fixes_left_momentum</code> ,<br><code>tduality_flips_right_momentum</code> |
| $p_L^2 + p_R^2$ invariant                                                                                               | [A]  | <code>tduality_invariant_mass_squared</code>                                              |
| $R = 1$ is the only positive rational fixed point of<br>$R \mapsto 1/R$                                                 | [A]  | <code>self_dual_radius_unique</code>                                                      |
| $\text{tr } \mathcal{H} = (R + 1/R)^2 - 2; R + 1/R \geq 2$ on $\mathbb{R}_{>0}$                                         | [A]  | <code>dualScale_circle</code> ,<br><code>circle_effective_scale_ge_two</code>             |
| $(n, w) \mapsto (w, n)$ is an involution of $\mathbb{Z}^2$                                                              | [A]  | <code>tduality_involution</code>                                                          |
| Full spectrum, oscillators included, agrees (level 1)                                                                   | [L]  | Tong §8.3; GPR §2.2                                                                       |
| All OPEs agree; tree amplitudes agree (level 2)                                                                         | [L]  | Tong §8.3; AAL §2.1                                                                       |
| All genera agree with $\tilde{g}_s = g_s \sqrt{\alpha'}/R$ (level 3)                                                    | [L]  | GPR §2.3; Tong §8.3.1                                                                     |
| $SU(2) \times SU(2)$ at $R = \sqrt{\alpha'}$ ; T-duality is a Weyl reflection                                           | [L]  | Tong §8.2.3; GPR §2.6.1                                                                   |
| Minimal measurable radius for closed-string probes                                                                      | [L]  | Tong §8.3; GPR §2.2; AAL §6                                                               |
| Masses in Table 8.1; nine massless label sets at $R = \ell_s$                                                           | —    | enumeration by script; symbolic check, not Tier A                                         |
| $R_{\text{eff}}$ is the physical scale; it regulates singularities                                                      | [C]  | this programme                                                                            |

What would a faithful formalization of T-duality on a circle need? In increasing order of difficulty: (i) the integrality of  $n, w$  and the level-matching constraint, so that the *set* of states and its bijection can be stated, which is a modest extension of the present file; (ii) the Fock space of the oscillators and a unitary operator implementing (8.10), intertwining the two Hamiltonians, which needs infinite-dimensional Hilbert spaces but no new mathematics; (iii) the vertex operator algebra of the lattice  $\mathbb{Z}^2$  with the form  $nw' + wn'$ , and the statement that (8.6) induces an isomorphism of it, which is level 2 and would require a library of vertex

algebras that Mathlib does not currently provide; (iv) higher-genus partition functions and (8.21), which need theta functions on Riemann surfaces. None of (ii)–(iv) exists in this project.

### Chapter summary

- T-duality of the circle is the map  $R \mapsto \alpha'/R$ ,  $n \leftrightarrow w$ . On the zero modes it acts as  $p_L \mapsto p_L$ ,  $p_R \mapsto -p_R$ ; on the field as  $X = X_L + X_R \mapsto Y = X_L - X_R$ , that is  $\partial_\alpha X = \epsilon_{\alpha\beta} \partial^\beta Y$ ; on the oscillators as  $\alpha_k \mapsto -\alpha_k$ ,  $\tilde{\alpha}_k \mapsto \tilde{\alpha}_k$ .
- “Symmetry” has four levels: spectrum, conformal field theory, perturbation theory, exact theory. The mass formula shows the first; the invariance of all OPEs under a one-sided reflection shows the second; the path integral and the genus expansion show the third, provided  $g_s \mapsto g_s \sqrt{\alpha'}/R$ ; the interpretation as a gauge transformation supports the fourth.
- T-duality is a relabelling of one theory, not an invariance of each state, and not the statement that physics is independent of  $R$ .
- The self-dual radius  $R = \sqrt{\alpha'}$  is the unique fixed point. There, four extra massless vectors and eight extra massless scalars appear,  $U(1)^2$  is enhanced to  $SU(2) \times SU(2)$ , and T-duality is the Weyl reflection of one  $SU(2)$ . The moduli space is the half-line  $R \geq \sqrt{\alpha'}$ .
- Closed strings cannot measure a radius below  $\sqrt{\alpha'}$ :  $R_{\text{obs}} = \max(R, \alpha'/R)$ . The smooth invariant  $R_{\text{eff}} = R + \alpha'/R \geq 2\sqrt{\alpha'}$  is the basic object of the dual-scale programme; the inequality is [\[A\]](#), its physical role is [\[L\]](#) or [\[C\]](#) as specified in Section 8.6.
- Lean has checked the zero-mode algebra over  $\mathbb{Q}$  at  $\alpha' = 1$ , the uniqueness of the fixed point, and  $R + 1/R \geq 2$  over  $\mathbb{R}$ . It has not checked anything about oscillators, operator algebras, loops or the dilaton.

## Exercises

**Exercise 8.1** (★). Extend Table 8.1 to the circle of radius  $R = 3\ell_s$ : list all label sets with  $\alpha' M^2 \leq 1$ , and write down the corresponding list at  $\tilde{R} = \ell_s/3$  without further computation. Which states are massless, and is either radius a point of enhanced gauge symmetry?

**Exercise 8.2** (★). Show that the map  $(R, g_s) \mapsto (\alpha'/R, g_s \sqrt{\alpha'}/R)$  is an involution and that  $g_s^2/R$  and  $\Phi - \frac{1}{2} \ln(R/\sqrt{\alpha'})$  are invariant under it. For  $R = \ell_s/5$  and  $g_s = 0.01$ , find the dual radius and coupling, and compare the two values of the combination (8.19).

**Exercise 8.3** (★). Besides the self-dual radius there are radii at which only extra massless *scalars* appear. With  $N = \tilde{N} = 0$ , show that a pure winding state is massless when  $R^2 = 4\alpha'/w^2$  and a pure momentum state when  $R^2 = n^2\alpha'/4$  [Tong §8.2.3, ll. 11565–11590]. Check that these two families of special radii are exchanged by T-duality, and explain why no massless charged *vector* can occur at any  $R \neq \sqrt{\alpha'}$ . (Hint: for  $N = 1$ ,  $\tilde{N} = 0$  show that  $M^2 = p_R^2$ .)

**Exercise 8.4** (★★). Starting from  $Y = X_L - X_R$ , derive (8.11). Show that under  $X \rightarrow Y$  the equation of motion and the identity  $\partial_\tau \partial_\sigma X = \partial_\sigma \partial_\tau X$  are exchanged. Compute  $\oint d\sigma \partial_\sigma Y$  and  $\frac{1}{2\pi\alpha'} \oint d\sigma \partial_\tau Y$  from (8.9) and interpret the results as winding and momentum on the dual circle.

**Exercise 8.5** (★★). Prove the identity  $\frac{\alpha'}{2}(p_L p'_L - p_R p'_R) = nw' + wn'$  of (8.15). Deduce  $h - \bar{h} = nw$  for  $V_{n,w}$ , and show that the bilinear form  $\langle (n, w), (n', w') \rangle = nw' + wn'$  on  $\mathbb{Z}^2$  is even, has signature  $(1, 1)$ , and is preserved by exactly four integer matrices:  $\pm 1$  and  $\pm$  the swap. Which of these is T-duality, and what do the others do to  $X_L$  and  $X_R$ ?

**Exercise 8.6** (★★). Show that  $R_{\text{eff}} = 2\sqrt{\alpha'} \cosh x$  and  $\text{tr } \mathcal{H} = 2 \cosh 2x$ , with  $x = \ln(R/\sqrt{\alpha'})$ . Expand both to second order around the self-dual point. Show that any smooth duality-invariant function  $f(R)$  has  $df/dx = 0$  at  $x = 0$ , so that the self-dual radius is a critical point of every duality-invariant potential for the radion. Why does this not by itself show that the radius is stabilized there?

**Exercise 8.7** (★★, Lean). In the namespace `StringTheory.UseCases.TDuality`, state and prove that T-duality exchanges the two  $U(1)$  charges:

```
theorem tduality_swaps_charges (n w R : ℚ) (hR : R ≠ 0) :
  leftMomentum w n (1 / R) + rightMomentum w n (1 / R)
    = leftMomentum n w R - rightMomentum n w R := by
  sorry
```

Give a proof that uses the two existing theorems and `ring`, and a second one from the definitions with `field_simp`. Which is more robust against a change of the definitions?

**Exercise 8.8** (★★★, Lean). State and prove the self-dual radius with  $\alpha'$  restored: for real  $R > 0$  and  $\alpha' > 0$  with  $\alpha'/R = R$ , one has  $R = \sqrt{\alpha'}$  (`Real.sqrt`). A possible route: derive  $R^2 = \alpha'$  with `field_simp`, rewrite the goal with it, and use `Real.sqrt_sq`. Is the hypothesis  $\alpha' > 0$  needed? Then explain why the analogous statement over  $\mathbb{Q}$  cannot even be formulated with a conclusion of the form  $R = c$  for  $\alpha' = 2$ .

**Exercise 8.9** (★★★). (Genus one by hand.) Let  $\tau = \tau_1 + i\tau_2$ ,  $r = R/\sqrt{\alpha'}$ ,  $q = e^{2\pi i \tau}$ . Use the Poisson formula  $\sum_{m \in \mathbb{Z}} e^{-\pi a(m+b)^2} = a^{-1/2} \sum_{k \in \mathbb{Z}} e^{-\pi k^2/a + 2\pi i k b}$  to show that

$$\sum_{m,n \in \mathbb{Z}} \exp\left(-\frac{\pi r^2}{\tau_2} |m + n\tau|^2\right) = \frac{\sqrt{\tau_2}}{r} \sum_{k,n \in \mathbb{Z}} q^{\frac{1}{2}P_L^2} \bar{q}^{\frac{1}{2}P_R^2}, \quad P_{L,R} = \frac{1}{\sqrt{2}} \left(\frac{k}{r} \pm nr\right),$$

where  $P_{L,R} = \sqrt{\alpha'/2} p_{L,R}$  are the dimensionless momenta of GPR. Including the factor  $r$  from the zero mode, conclude that  $Z_1(1/r) = Z_1(r)$ , the case  $g = 1$  of (8.21). (The identity can be tested numerically, for instance at  $r = 1.3$ ,  $\tau = 0.23 + 0.9i$ , where both sides equal 1.01830...)

## Further reading

- The spectrum on a circle, enhanced symmetry, T-duality, the dual coordinate, the dilaton shift and the path-integral derivation: [Tong §8.2–8.3.1, ll. 11351–11803].
- Circle duality in the operator formalism, the one-loop and higher-genus partition functions and the dilaton: [GPR §2.2–2.3, ll. 576–1029]. Duality as a gauge symmetry and the Weyl reflection: [GPR §2.6.1, ll. 2377–2490].
- Abelian duality by gauging an isometry, global issues and the dilaton determinant: [AAL §2.1, ll. 215–376]; the operational definition of distance: [AAL §6, ll. 1289–1320].
- In this book: the general Buscher rules in Chapter 9; the lattice of  $(p_L, p_R)$  in Chapter 10; the group generated by dualities, shifts and basis changes in Chapter 11; open strings and D-branes under T-duality in Chapter 13; the dual-scale bound for tori in Chapter 31.



## Chapter 9

# Buscher Rules: T-Duality from the Worldsheet

In Chapter 8 T-duality appeared as a property of a spectrum: the mass formula of the string on a circle does not change under  $R \rightarrow \alpha'/R$ ,  $n \leftrightarrow w$ . That argument uses the exact solution of a free theory. It says nothing about a string moving in a curved background, where no mode expansion is available, and it does not explain *why* the two descriptions agree. This chapter answers a sharper question: given a sigma model with metric  $G$ , Kalb–Ramond field  $B$  and dilaton  $\Phi$ , and given only that the background does not depend on one coordinate  $\theta$ , what is the dual background  $(\tilde{G}, \tilde{B}, \tilde{\Phi})$ ? The answer is a set of algebraic formulas, the *Buscher rules*, and the derivation is a short manipulation of the worldsheet path integral: make the shift symmetry of  $\theta$  local, add a Lagrange multiplier that undoes the gauging, and integrate out the fields in two different orders. One order returns the original model; the other gives the dual one, with the Lagrange multiplier promoted to the dual coordinate.

A student should care for three reasons. First, the method is the prototype for many dualities in field theory: electric–magnetic duality in four dimensions and particle–vortex duality in three can be derived in the same way (a standard fact, not checked against a source in this book’s library). Second, the rules are used every time T-duality is applied to a background that is not a flat torus: D-brane solutions (Chapter 13), backgrounds with flux (Chapter 25), and the torus fibres of Chapter 24. Third, the rules look complicated in components but are one matrix conjugation of the generalized metric  $\mathcal{H}(G, B)$ ; seeing this here prepares the  $O(d, d; \mathbb{Z})$  action of Chapters 10 and 11 and the doubled formalism of Chapters 14 and 15. The chapter closes with an account of what the Lean file `DoubleFieldTheory/TDualityBuscher.lean` states. It is an integer model of the one-circle case, and we describe its scope precisely.

**Conventions.** The worldsheet is Euclidean with coordinates  $\sigma^a$ ,  $a = 1, 2$ , and metric  $h_{ab}$ ;  $\varepsilon^{ab}$  is the antisymmetric symbol with  $\varepsilon^{12} = +1$ . Target-space coordinates are  $X^i$ ,  $i = 0, \dots, d-1$ . As in the rest of the book  $\alpha'$  has dimension  $\text{length}^2$  and  $\ell_s = \sqrt{\alpha'}$ . In this chapter the coordinate along the isometry is called  $\theta = X^0$  and is given the period

$$\theta \sim \theta + 2\pi\ell_s, \tag{9.1}$$

so that the metric components are dimensionless. A circle of physical radius  $R$  is then described by  $G_{00} = R^2/\alpha'$ , because the circumference is  $\sqrt{G_{00}} \cdot 2\pi\ell_s = 2\pi R$ . This choice matters for the dilaton and for the global analysis; Section 9.5 explains what changes for other periods.

## 9.1 The sigma model with an isometry

The starting point is the non-linear sigma model of Chapter 6, written in Euclidean signature,

$$S = \frac{1}{4\pi\alpha'} \int d^2\sigma \left[ \sqrt{h} h^{ab} G_{ij}(X) \partial_a X^i \partial_b X^j + i \varepsilon^{ab} B_{ij}(X) \partial_a X^i \partial_b X^j + \alpha' \sqrt{h} R^{(2)} \Phi(X) \right], \quad (9.2)$$

where  $G_{ij}$  is the target-space metric,  $B_{ij} = -B_{ji}$  the Kalb–Ramond field (called “torsion” in part of the literature),  $\Phi$  the dilaton and  $R^{(2)}$  the scalar curvature of  $h_{ab}$  [AAL §2.1, ll. 217–235]. The factor  $i$  in the  $B$ -term is a consequence of the Wick rotation: a term with one time derivative, such as  $\varepsilon^{ab} \partial_a X^i \partial_b X^j$ , picks up one factor of  $i$  under  $\tau \rightarrow -i\tau$ , whereas the kinetic term, with two, stays real. The  $B$ -term does not involve  $h_{ab}$  at all: it is the integral of the two-form  $B$  pulled back to the worldsheet.

Assume that the background has an *abelian isometry*: there is a vector field  $k$  such that the action is invariant under  $\delta X^i = \epsilon k^i(X)$ . Invariance of the three terms of (9.2) requires  $k$  to be a Killing vector of  $G$ , the field strength  $H = dB$  to be invariant, and  $k^i \partial_i \Phi = 0$  [GPR §4.1, ll. 5935–5953]. Under these conditions one can choose *adapted coordinates*  $X^i = (\theta, x^\alpha)$ ,  $\alpha = 1, \dots, d-1$ , in which  $k = \partial_\theta$  and none of  $G$ ,  $B$ ,  $\Phi$  depends on  $\theta$ ; for  $B$  this may require a gauge transformation  $B \rightarrow B + d\lambda$  [GPR §4.1, ll. 5954–5958]. In adapted coordinates the isometry is the shift  $\theta \rightarrow \theta + \epsilon$  and the Lagrangian of (9.2) splits as

$$\begin{aligned} 4\pi\alpha' \mathcal{L} = & \sqrt{h} h^{ab} (G_{00} \partial_a \theta \partial_b \theta + 2G_{0\alpha} \partial_a \theta \partial_b x^\alpha + G_{\alpha\beta} \partial_a x^\alpha \partial_b x^\beta) \\ & + i \varepsilon^{ab} (2B_{0\alpha} \partial_a \theta \partial_b x^\alpha + B_{\alpha\beta} \partial_a x^\alpha \partial_b x^\beta) + \alpha' \sqrt{h} R^{(2)} \Phi. \end{aligned} \quad (9.3)$$

The factor 2 in the mixed  $B$ -term comes from the two orderings  $(i, j) = (0, \alpha)$  and  $(\alpha, 0)$ :  $B_{\alpha 0} \varepsilon^{ab} \partial_a x^\alpha \partial_b \theta = B_{0\alpha} \varepsilon^{ab} \partial_a \theta \partial_b x^\alpha$ , since both  $B$  and  $\varepsilon$  are antisymmetric. The field  $\theta$  enters (9.3) only through its derivative  $\partial_a \theta$ . The whole construction depends on this fact.

For the derivations below we take the worldsheet metric flat,  $h_{ab} = \delta_{ab}$ , and write  $A \cdot B = A_a B_a$ . Nothing is lost: in two dimensions one can always choose  $h_{ab} = e^{2\omega} \delta_{ab}$  locally, the kinetic term  $\sqrt{h} h^{ab}$  is independent of  $\omega$ , and the  $\varepsilon$ -terms never contained  $h_{ab}$ . The curvature term is the only place where  $h_{ab}$  matters, and it returns in Section 9.4.

## 9.2 Warm-up: the circle in three steps

Take first a single field  $\theta$  with constant  $G_{00} = R^2/\alpha'$  and nothing else:

$$S[\theta] = \frac{G_{00}}{4\pi\alpha'} \int d^2\sigma \partial_a \theta \partial_a \theta. \quad (9.4)$$

With  $\theta = \ell_s \varphi$  this is Tong’s action  $\frac{R^2}{4\pi\alpha'} \int (\partial\varphi)^2$  for a field  $\varphi$  of period  $2\pi$  [Tong §8.3.1, ll. 11718–11731]. We follow his argument and keep track of the normalizations.

**Step 1: gauge the shift symmetry.** The symmetry  $\theta \rightarrow \theta + \lambda$  with constant  $\lambda$  is made local by introducing a worldsheet gauge field  $A_a$  with  $A_a \rightarrow A_a - \partial_a \lambda$  and replacing  $\partial_a \theta$  by the covariant derivative  $D_a \theta = \partial_a \theta + A_a$  [Tong §8.3.1, ll. 11732–11735]. This changes the theory:  $A_a$  is a new integration variable.

**Step 2: add a Lagrange multiplier that undoes Step 1.** Introduce one more scalar  $\tilde{\theta}$  and the *master action*

$$S[\theta, A, \tilde{\theta}] = \frac{1}{4\pi\alpha'} \int d^2\sigma \left[ G_{00} D_a \theta D_a \theta + 2i \tilde{\theta} \varepsilon^{ab} \partial_a A_b \right]. \quad (9.5)$$

The new term is gauge invariant because  $\varepsilon^{ab}\partial_a\partial_b\lambda = 0$ . In terms of fields of period  $2\pi$ ,  $\tilde{\theta} = \ell_s\vartheta$  and  $A = \ell_s\mathcal{A}$ , it reads  $\frac{i}{2\pi}\int\vartheta\varepsilon^{ab}\partial_a\mathcal{A}_b$ , which is the normalization of Tong's (8.12) [Tong §8.3.1, ll. 11736–11746]. The functional integral over  $\tilde{\theta}$  of  $\exp(-S)$  is a functional delta function,

$$\int \mathcal{D}\tilde{\theta} \exp\left(-\frac{i}{2\pi\alpha'}\int d^2\sigma \tilde{\theta} F\right) \propto \delta[F], \quad F = \varepsilon^{ab}\partial_a A_b = \partial_1 A_2 - \partial_2 A_1.$$

This is why the multiplier term must be imaginary in Euclidean signature: a real exponent would give a Gaussian damping, not a constraint. On a worldsheet with the topology of the plane or the sphere,  $F = 0$  implies  $A_a = \partial_a\chi$  for some function  $\chi$ ; the gauge transformation  $\lambda = \chi$  then sets  $A_a = 0$  and (9.5) reduces to (9.4). On worldsheets with non-contractible cycles this conclusion needs an extra argument, given in Section 9.5.

**Step 3: integrate in the other order.** Alternatively, use the gauge freedom first to set  $\theta = 0$ , so that  $D_a\theta = A_a$ , and integrate the multiplier term by parts,

$$\int d^2\sigma \tilde{\theta} \varepsilon^{ab}\partial_a A_b = -\int d^2\sigma \varepsilon^{ab}\partial_a \tilde{\theta} A_b = \int d^2\sigma \varepsilon^{ab} A_a \partial_b \tilde{\theta}, \quad (9.6)$$

where the last step renames  $a \leftrightarrow b$  and uses  $\varepsilon^{ba} = -\varepsilon^{ab}$ . The action is now quadratic in  $A_a$  with no derivatives acting on it:

$$4\pi\alpha' \mathcal{L} = G_{00} A_a A_a + 2A_a j_a, \quad j_a = i\varepsilon^{ab}\partial_b \tilde{\theta}. \quad (9.7)$$

A Gaussian integral is evaluated by completing the square,  $G_{00}A \cdot A + 2A \cdot j = G_{00}(A + j/G_{00})^2 - j \cdot j/G_{00}$ . The shifted integral gives a field-independent factor (for constant  $G_{00}$ ), and what remains is the value of the action at the stationary point

$$A_a = -\frac{i}{G_{00}}\varepsilon^{ab}\partial_b \tilde{\theta}. \quad (9.8)$$

Since  $\varepsilon^{ab}\varepsilon^{ac} = \delta^{bc}$ , one has  $j \cdot j = -\partial_b \tilde{\theta} \partial_b \tilde{\theta}$ . The two factors of  $i$  turn the sign of  $-j \cdot j/G_{00}$  into a plus and produce a kinetic term of the correct sign:

$$\tilde{S}[\tilde{\theta}] = \frac{1}{4\pi\alpha'} \frac{1}{G_{00}} \int d^2\sigma \partial_a \tilde{\theta} \partial_a \tilde{\theta}. \quad (9.9)$$

This is again a circle sigma model, with  $\tilde{G}_{00} = 1/G_{00} = \alpha'/R^2$ . If  $\tilde{\theta}$  has the same period  $2\pi\ell_s$  as  $\theta$  (which Section 9.5 shows to be the case), the dual radius is  $\tilde{R} = \ell_s\sqrt{\tilde{G}_{00}} = \alpha'/R$  [Tong §8.3.1, ll. 11770–11799]. No mode expansion was used.

#### Momentum becomes winding

Combine the two halves of the argument. Without fixing the gauge, the stationary point (9.8) reads  $D_a\theta = -\frac{i}{G_{00}}\varepsilon^{ab}\partial_b \tilde{\theta}$ , because only the gauge-invariant combination  $D_a\theta = \partial_a\theta + A_a$  enters. The multiplier constraint allows the other gauge choice,  $A_a = 0$ , in which  $D_a\theta = \partial_a\theta$ . The two fields are therefore related by

$$G_{00}\partial_a\theta = -i\varepsilon^{ab}\partial_b \tilde{\theta}. \quad (9.10)$$

The left-hand side is the Noether current of the shift symmetry of  $\theta$ ; its charge is the momentum along the circle. The right-hand side is a current that is conserved identically,  $\partial_a(\varepsilon^{ab}\partial_b \tilde{\theta}) = 0$ , whatever the equations of motion; its charge counts how many times  $\tilde{\theta}$  winds. So the duality maps the momentum of  $\theta$  to the winding of  $\tilde{\theta}$ , which is the exchange  $n \leftrightarrow w$  of Chapter 8. After continuation to Lorentzian signature, where the

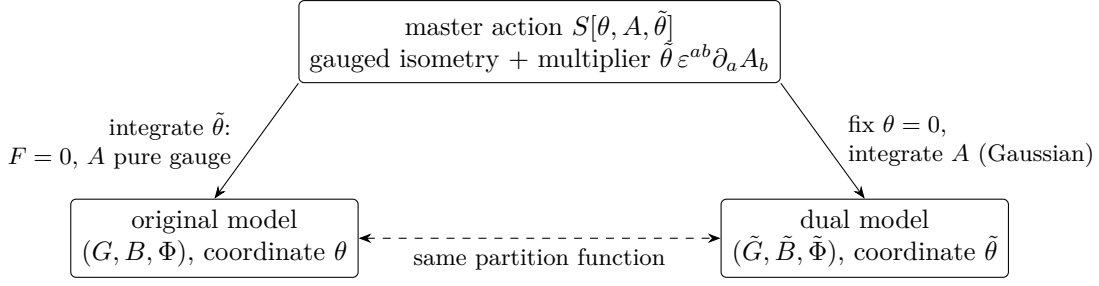


Figure 9.1: The logic of the Buscher procedure. Both sigma models are obtained from one path integral by performing the integrations in different orders; they are therefore two descriptions of the same quantum theory, provided the global conditions of Section 9.5 hold.

factor  $i$  disappears, (9.10) is Tong's relation  $\partial_\alpha X = \epsilon_{\alpha\beta} \partial^\beta Y$  between the coordinate  $X = X_L + X_R$  and the dual coordinate  $Y = X_L - X_R$  [Tong §8.3, ll. 11673–11689].

### 9.3 A general background: deriving the Buscher rules

Return to the general Lagrangian (9.3). Since  $\theta$  appears only through  $\partial_a \theta$ , Steps 1–3 can be merged: after gauge fixing  $\theta = 0$  the covariant derivative is just  $A_a$ , which we rename  $V_a$ . The master action is obtained from (9.3) by the substitution  $\partial_a \theta \rightarrow V_a$ , with  $V_a$  an independent field, plus the multiplier term [AAL §2.1, eq. (2.1.2), ll. 236–248]:

$$4\pi\alpha' \mathcal{L}_{\text{master}} = G_{00} V \cdot V + 2G_{0\alpha} V \cdot \partial x^\alpha + G_{\alpha\beta} \partial x^\alpha \cdot \partial x^\beta + i\epsilon^{ab} (2B_{0\alpha} V_a \partial_b x^\alpha + B_{\alpha\beta} \partial_a x^\alpha \partial_b x^\beta) + 2i\tilde{\theta} \epsilon^{ab} \partial_a V_b + \alpha' R^{(2)} \Phi. \quad (9.11)$$

Integrating over  $\tilde{\theta}$  imposes  $\epsilon^{ab} \partial_a V_b = 0$ , hence  $V_a = \partial_a \theta$  on a topologically trivial worldsheet, and one is back at (9.3) [AAL §2.1, ll. 249–250]. The version with an explicit gauge field and  $\theta$  kept, due to Roček and Verlinde, is equivalent and makes the geometric meaning clearer: the dual model is obtained by gauging the isometry with a flat connection [AAL §2.1, eq. (2.1.8), ll. 342–370].

Now integrate over  $V_a$  instead. Use (9.6) for the multiplier term and collect everything linear in  $V_a$ :

$$4\pi\alpha' \mathcal{L}_{\text{master}} = G_{00} V \cdot V + 2V \cdot J + (\text{terms without } V), \quad J_a = G_{0\alpha} \partial_a x^\alpha + i\epsilon^{ab} K_b, \quad (9.12)$$

with the abbreviation  $K_b = \partial_b \tilde{\theta} + B_{0\alpha} \partial_b x^\alpha$ . Completing the square as before, the stationary point is

$$V_a = -\frac{J_a}{G_{00}} = -\frac{1}{G_{00}} \left( G_{0\alpha} \partial_a x^\alpha + i\epsilon^{ab} (B_{0\alpha} \partial_b x^\alpha + \partial_b \tilde{\theta}) \right), \quad (9.13)$$

in agreement with [AAL §2.1, eq. (2.1.3), ll. 250–257], and the  $V$ -dependent terms are replaced by  $-J \cdot J / G_{00}$ . It remains to expand  $J \cdot J$ :

$$\begin{aligned} J \cdot J &= G_{0\alpha} G_{0\beta} \partial x^\alpha \cdot \partial x^\beta + 2i G_{0\alpha} \epsilon^{ab} \partial_a x^\alpha K_b - \epsilon^{ab} \epsilon^{ac} K_b K_c \\ &= G_{0\alpha} G_{0\beta} \partial x^\alpha \cdot \partial x^\beta - K \cdot K + 2i G_{0\alpha} \epsilon^{ab} \partial_a x^\alpha K_b, \\ K \cdot K &= \partial \tilde{\theta} \cdot \partial \tilde{\theta} + 2B_{0\alpha} \partial \tilde{\theta} \cdot \partial x^\alpha + B_{0\alpha} B_{0\beta} \partial x^\alpha \cdot \partial x^\beta. \end{aligned} \quad (9.14)$$

Insert this into  $-J \cdot J / G_{00}$  and sort the result by structure. The terms symmetric in the worldsheet indices build the dual metric, those with  $\epsilon^{ab}$  build the dual  $B$ -field:



- $\partial\tilde{\theta} \cdot \partial\tilde{\theta}$  has coefficient  $1/G_{00}$ ;
- $2\partial\tilde{\theta} \cdot \partial x^\alpha$  has coefficient  $B_{0\alpha}/G_{00}$ ;
- $\partial x^\alpha \cdot \partial x^\beta$  receives  $-(G_{0\alpha}G_{0\beta} - B_{0\alpha}B_{0\beta})/G_{00}$  in addition to  $G_{\alpha\beta}$ ;
- the term  $-\frac{2i}{G_{00}}G_{0\alpha}\varepsilon^{ab}\partial_a x^\alpha\partial_b\tilde{\theta} = +\frac{2i}{G_{00}}G_{0\alpha}\varepsilon^{ab}\partial_a\tilde{\theta}\partial_b x^\alpha$  has the form of the mixed  $B$ -term in (9.3) with  $B_{0\alpha}$  replaced by  $G_{0\alpha}/G_{00}$ ;
- the term  $-\frac{2i}{G_{00}}G_{0\alpha}B_{0\beta}\varepsilon^{ab}\partial_a x^\alpha\partial_b x^\beta$  is antisymmetrized by  $\varepsilon^{ab}$  and adds  $-(G_{0\alpha}B_{0\beta} - G_{0\beta}B_{0\alpha})/G_{00}$  to  $B_{\alpha\beta}$ .

The outcome is a sigma model of the form (9.3) for the coordinates  $(\tilde{\theta}, x^\alpha)$ .

**Theorem 9.1** (Buscher rules). *Let  $(G, B, \Phi)$  be independent of  $\theta = X^0$  and  $G_{00} \neq 0$ . Integrating out  $V_a$  in (9.11) gives the sigma model with background*

$$\begin{aligned}\tilde{G}_{00} &= \frac{1}{G_{00}}, & \tilde{G}_{0\alpha} &= \frac{B_{0\alpha}}{G_{00}}, & \tilde{G}_{\alpha\beta} &= G_{\alpha\beta} - \frac{G_{0\alpha}G_{0\beta} - B_{0\alpha}B_{0\beta}}{G_{00}}, \\ \tilde{B}_{0\alpha} &= \frac{G_{0\alpha}}{G_{00}}, & \tilde{B}_{\alpha\beta} &= B_{\alpha\beta} - \frac{G_{0\alpha}B_{0\beta} - G_{0\beta}B_{0\alpha}}{G_{00}},\end{aligned}\tag{9.15}$$

together with the dilaton shift of Section 9.4,  $\tilde{\Phi} = \Phi - \frac{1}{2} \log G_{00}$ . [L] [AAL §2.1, eqs. (2.1.4)–(2.1.6), ll. 258–303; GPR §4.1, eq. (4.1.9), ll. 6032–6065]

The extracted text of both sources garbles the layout of these equations. The assignment of right-hand sides above is fixed by the derivation just given, and it agrees with GPR's (4.1.9) once  $B_{a0} = -B_{0a}$  is used there.

Three features deserve comment.

*Metric and B-field mix.* The mixed components are exchanged,  $\tilde{G}_{0\alpha} \leftrightarrow B_{0\alpha}/G_{00}$  and  $\tilde{B}_{0\alpha} \leftrightarrow G_{0\alpha}/G_{00}$ . In Kaluza–Klein language (Chapter 7)  $G_{0\alpha}/G_{00}$  is the gauge field that couples to momentum and  $B_{0\alpha}$  is the gauge field that couples to winding. The exchange of the two gauge fields is the background-field version of  $n \leftrightarrow w$ .

*A compact form.* In terms of  $E_{ij} = G_{ij} + B_{ij}$  the five rules (9.15) become four:

$$\tilde{E}_{00} = \frac{1}{E_{00}}, \quad \tilde{E}_{0\alpha} = \frac{E_{0\alpha}}{E_{00}}, \quad \tilde{E}_{\alpha 0} = -\frac{E_{\alpha 0}}{E_{00}}, \quad \tilde{E}_{\alpha\beta} = E_{\alpha\beta} - \frac{E_{\alpha 0}E_{0\beta}}{E_{00}}.\tag{9.16}$$

To check, note  $E_{00} = G_{00}$ , and take symmetric and antisymmetric parts: for instance  $\tilde{E}_{0\alpha} + \tilde{E}_{\alpha 0} = (E_{0\alpha} - E_{\alpha 0})/E_{00} = 2B_{0\alpha}/G_{00} = 2\tilde{G}_{0\alpha}$ . The last rule is a Schur complement: it is what remains of a matrix after one row and column have been eliminated, as expected from a Gaussian integration.

*The rules are an involution.* The dual background is independent of  $\tilde{\theta}$ , so the procedure can be repeated. Applying (9.16) twice gives  $\tilde{\tilde{E}}_{00} = E_{00}$ ,  $\tilde{\tilde{E}}_{0\alpha} = (E_{0\alpha}/E_{00})E_{00} = E_{0\alpha}$ ,  $\tilde{\tilde{E}}_{\alpha 0} = E_{\alpha 0}$ , and  $\tilde{\tilde{E}}_{\alpha\beta} = E_{\alpha\beta} - E_{\alpha 0}E_{0\beta}/E_{00} - (-E_{\alpha 0}/E_{00})(E_{0\beta}/E_{00})E_{00} = E_{\alpha\beta}$ . Dualizing twice returns the original model [AAL §2.1, ll. 317–318].

#### Signs and the orientation of the worldsheet

The signs of the mixed components in (9.15)–(9.16) depend on conventions that differ between papers: the sign of the  $B$ -term in (9.2), the sign of  $\varepsilon^{12}$ , and the sign of the multiplier term. Flipping the orientation of the worldsheet,  $\sigma^1 \rightarrow -\sigma^1$ , is equivalent to  $B \rightarrow -B$  [GPR §2.4, ll. 1423–1424] and changes  $\tilde{G}_{0\alpha}$  and  $\tilde{B}_{0\alpha}$  by a sign, which can be absorbed by  $\tilde{\theta} \rightarrow -\tilde{\theta}$ . The components  $\tilde{G}_{00}$ ,  $\tilde{G}_{\alpha\beta}$  and  $\tilde{B}_{\alpha\beta}$  are quadratic in the mixed fields and are not affected. Always check a formula taken from the literature on one

example before using it.

## 9.4 The dilaton shift

The Gaussian integral over  $V_a$  yields more than the stationary value of the action. It also gives a determinant. At each worldsheet point the integral over the two components of  $V_a$  contributes a factor proportional to  $1/G_{00}(x(\sigma))$ , so that, formally,

$$\int \mathcal{D}V e^{-S_{\text{master}}} = e^{-\tilde{S}} \times \prod_{\sigma} \frac{1}{G_{00}(x(\sigma))}. \quad (9.17)$$

When  $G_{00}$  depends on  $x^\alpha$  this factor depends on the fields that remain and cannot be dropped. An infinite product over worldsheet points has no meaning without regularization. Buscher's result is that, with a regularization that respects conformal invariance, the product equals the exponential of a local term of exactly the form of the dilaton coupling in (9.2), and the net effect is

$$\tilde{\Phi} = \Phi - \frac{1}{2} \log G_{00} \quad (9.18)$$

[L] [AAL §2.1, eqs. (2.1.6)–(2.1.7), ll. 294–316]. We give three arguments for (9.18), in increasing order of rigour.

**(i) The low-energy observer.** For a circle of radius  $R$  the effective action in the non-compact dimensions is multiplied by  $2\pi R e^{-2\Phi} = 2\pi R/g_s^2$ : the circumference comes from the integral over the circle and  $e^{-2\Phi}$  multiplies the string-frame action of Chapter 6. An observer who cannot distinguish  $R$  from  $\tilde{R} = \alpha'/R$  must find the same lower-dimensional Newton constant in both descriptions,  $R/g_s^2 = \tilde{R}/\tilde{g}_s^2$ , hence

$$\tilde{g}_s = g_s \sqrt{\tilde{R}/R} = g_s \frac{\sqrt{\alpha'}}{R} \quad (9.19)$$

[Tong §8.3, ll. 11691–11714]. With  $g_s = e^\Phi$  and  $G_{00} = R^2/\alpha'$  this is  $\tilde{\Phi} = \Phi - \frac{1}{2} \log(R^2/\alpha')$ , i.e. (9.18).

**(ii) Counting on a lattice.** The next argument is a heuristic of our own, not taken from the sources, and is valid for constant  $G_{00}$ . It shows where the Euler number comes from. Replace the worldsheet by a cell decomposition with  $N_0$  vertices,  $N_1$  links and  $N_2$  faces, so that  $N_0 - N_1 + N_2 = \chi = 2 - 2g$ . The field  $\theta$  lives on vertices, the one-form  $V$  on links, and the multiplier  $\tilde{\theta}$ , which multiplies the curl of  $V$ , on faces. The reparametrization-invariant measure for  $\theta$  is  $\prod_{\text{vertices}} \sqrt{G_{00}} d\theta$ , and for the dual field it is  $\prod_{\text{faces}} \sqrt{\tilde{G}_{00}} d\tilde{\theta} = \prod_{\text{faces}} G_{00}^{-1/2} d\tilde{\theta}$ . The Gaussian integral over  $V$  gives  $G_{00}^{-1/2}$  per link. Starting from the original model with its invariant measure and passing through the master action, the partition functions are related by

$$Z_{\text{orig}} = G_{00}^{N_0/2} \cdot G_{00}^{-N_1/2} \cdot G_{00}^{N_2/2} Z_{\text{dual}} = G_{00}^{\chi/2} Z_{\text{dual}},$$

where the last factor converts  $\prod d\tilde{\theta}$  into the invariant dual measure. For constant  $\Phi$  the dilaton term in (9.2) equals  $\Phi\chi$ , since  $\frac{1}{4\pi} \int \sqrt{h} R^{(2)} = \chi$ , so each model carries a weight  $e^{-\Phi\chi}$ . Equality of  $e^{-\Phi\chi} Z_{\text{orig}}$  and  $e^{-\tilde{\Phi}\chi} Z_{\text{dual}}$  requires  $e^{-\tilde{\Phi}\chi} = e^{-\Phi\chi} G_{00}^{\chi/2}$ , which is (9.18). The local factors cancel between vertices, links and faces; only the topological remainder survives, and a term proportional to  $\chi$  is a dilaton.

(iii) **The regularized determinant.** On the sphere, and in the gauged formulation, one can write  $A = \partial\alpha$ ,  $\bar{A} = \bar{\partial}\beta$  in complex coordinates. The change of variables from  $(A, \bar{A})$  to  $(\alpha, \beta)$  has Jacobian  $\det \Delta$ , with  $\Delta = -\frac{1}{\sqrt{h}}\partial_a\sqrt{h}h^{ab}\partial_b$  the scalar Laplacian, and the Gaussian integral over  $\alpha, \beta$  produces  $(\det \Delta_{G_{00}})^{-1}$  with  $\Delta_{G_{00}} = -\frac{1}{\sqrt{h}}\partial_a\sqrt{h}h^{ab}G_{00}\partial_b$ . The measure factor is thus the ratio  $\det \Delta / \det \Delta_{G_{00}}$  [AAL §4, eqs. (4.0.6)–(4.0.9), ll. 1013–1040]. Writing  $G_{00} = 1 + \sigma$  and using the heat-kernel expansion to first order in  $\sigma$ , Alvarez, Alvarez-Gaumé and Lozano find

$$\frac{\det \Delta_{G_{00}}}{\det \Delta} = \exp\left(-\frac{1}{8\pi} \int d^2\sigma \sqrt{h} R^{(2)} \log G_{00}\right) \quad (9.20)$$

[L] [AAL §4, eqs. (4.0.10)–(4.0.16), ll. 1048–1111]. The inverse of (9.20) multiplies  $e^{-\tilde{S}}$ . Compared with the dilaton term  $\exp\left(-\frac{1}{4\pi} \int \sqrt{h} R^{(2)} \Phi\right)$  it amounts to replacing  $\Phi$  by  $\Phi - \frac{1}{2} \log G_{00}$ . The shift is needed for consistency and cannot be chosen otherwise: without it the dual model is in general not conformally invariant, that is, its beta functions (Chapter 6) do not vanish even if those of the original model do [AAL §4, ll. 945–953]. Buscher showed conformal invariance of the dual model with the shift to first order in  $\alpha'$ ; the gauging argument of Roček and Verlinde extends the equivalence to all orders [AAL §2.1, ll. 305–307 and 371–376].

**Proposition 9.2** (An invariant measure). *Under (9.15) and (9.18),  $\det \tilde{G} = \det G / G_{00}^2$  and therefore*

$$\sqrt{\det \tilde{G}} e^{-2\tilde{\Phi}} = \sqrt{\det G} e^{-2\Phi}. \quad (9.21)$$

*Proof.* Write  $G$  in Kaluza–Klein form. With  $a_\alpha = G_{0\alpha}/G_{00}$  one has  $G_{ij}dX^i dX^j = G_{00}(d\theta + a_\alpha dx^\alpha)^2 + g_{\alpha\beta} dx^\alpha dx^\beta$ , where  $g_{\alpha\beta} = G_{\alpha\beta} - G_{0\alpha}G_{0\beta}/G_{00}$ , and  $\det G = G_{00} \det g$  because the change of basis  $d\theta \rightarrow d\theta + a_\alpha dx^\alpha$  has unit determinant. For the dual metric the same decomposition gives  $\tilde{g}_{\alpha\beta} = \tilde{G}_{\alpha\beta} - \tilde{G}_{0\alpha}\tilde{G}_{0\beta}/\tilde{G}_{00} = \tilde{G}_{\alpha\beta} - B_{0\alpha}B_{0\beta}/G_{00} = g_{\alpha\beta}$ . Hence  $\det \tilde{G} = \tilde{G}_{00} \det g = \det G / G_{00}^2$ , and  $\sqrt{\det \tilde{G}} e^{-2\tilde{\Phi}} = (\sqrt{\det G} / G_{00}) e^{-2\Phi} G_{00}$ .  $\square$

The proof shows more than was claimed: the metric  $g_{\alpha\beta}$  seen by a lower-dimensional observer is duality invariant. The combination in (9.21) is the measure of the string-frame effective action; in double field theory it is written  $e^{-2d}$  and  $d$  is called the *duality-invariant dilaton* (Chapter 17). For one circle,  $\sqrt{\det \tilde{G}} = R/\ell_s = e^x$  with  $x = \log(R/\ell_s)$ , so that

$$2d = 2\Phi - x, \quad x \mapsto -x, \quad \Phi \mapsto \Phi - x. \quad (9.22)$$

These three linear expressions are what the Lean file of Section 9.8 encodes.

## 9.5 Global issues

### 9.5.1 The period of the Lagrange multiplier

On a worldsheet of genus  $g \geq 1$  the constraint  $\varepsilon^{ab}\partial_a V_b = 0$ , that is  $dV = 0$ , does not imply  $V = d\theta$  for a single-valued function  $\theta$ . A closed one-form can have non-zero *holonomies*  $\nu_\gamma = \oint_\gamma V$  around the non-contractible cycles  $\gamma$ . For the original model we need exactly the closed forms  $V = d\theta$  with  $\theta$  a map to the circle (9.1); these have  $\nu_\gamma \in 2\pi\ell_s\mathbb{Z}$ , the integer being the number of times the worldsheet cycle winds around the target circle. Holonomies that are not of this form would describe configurations that do not exist in the original model (Figure 9.2).

The multiplier enforces this if, and only if, it is itself periodic with the right period. Let  $\tilde{\theta} \sim \tilde{\theta} + 2\pi\tilde{L}$ . On the torus  $d\tilde{\theta}$  is the sum of an exact form and a harmonic form with periods  $\oint_a d\tilde{\theta} = 2\pi\tilde{L}m_a$ ,  $\oint_b d\tilde{\theta} = 2\pi\tilde{L}m_b$ , and the path integral over  $\tilde{\theta}$  includes a sum over the integers  $m_a, m_b$ . The multiplier term is defined by its integrated form (9.6),  $\frac{i}{2\pi\alpha'} \int V \wedge d\tilde{\theta}$ . The exact

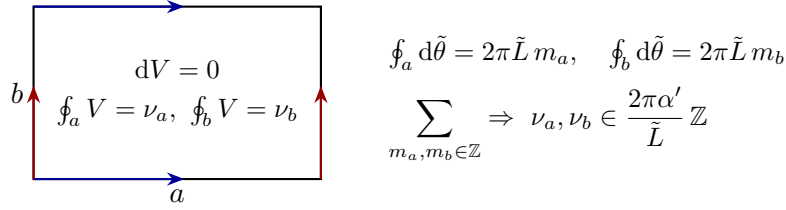


Figure 9.2: A torus worldsheet, drawn as a rectangle with opposite sides identified, and its two cycles  $a$  and  $b$ . The sum over the windings of the multiplier  $\tilde{\theta}$  restricts the holonomies of the flat gauge field  $V$  to a lattice.

part of  $d\tilde{\theta}$  imposes  $dV = 0$  as before. For two closed one-forms on the torus, Riemann's bilinear identity gives  $\int V \wedge d\tilde{\theta} = \oint_a V \oint_b d\tilde{\theta} - \oint_b V \oint_a d\tilde{\theta}$ , so the harmonic part contributes

$$\sum_{m_a, m_b \in \mathbb{Z}} \exp \left[ -\frac{i\tilde{L}}{\alpha'} (\nu_a m_b - \nu_b m_a) \right] \propto \sum_{k_a, k_b \in \mathbb{Z}} \delta \left( \nu_a - \frac{2\pi\alpha'}{\tilde{L}} k_a \right) \delta \left( \nu_b - \frac{2\pi\alpha'}{\tilde{L}} k_b \right), \quad (9.23)$$

by the Poisson formula  $\sum_m e^{-imy} = 2\pi \sum_k \delta(y - 2\pi k)$ . The holonomies are forced into  $2\pi(\alpha'/\tilde{L})\mathbb{Z}$ . If  $\theta$  has period  $2\pi L$  we need them in  $2\pi L\mathbb{Z}$ . The two periods are therefore tied together:

$$L\tilde{L} = \alpha'. \quad (9.24)$$

This is the content of the statements “the holonomies are gauge trivial if  $\theta$  has periodicity  $2\pi$ ” [Tong §8.3.1, ll. 11759–11769] and “this fixes the periodicity of  $\tilde{\theta}$  to be  $2\pi$ ” [GPR §4.1, ll. 6077–6126], both in normalizations in which  $L = \tilde{L}$ ; the same analysis with Riemann's bilinear identity is in [AAL §2.1, ll. 455–558]. With our choice  $L = \ell_s$  one gets  $\tilde{L} = \ell_s$ , as assumed in Section 9.2. Abelian isometries need only the torus analysis; higher genus adds more pairs of cycles, each treated in the same way.

#### Where did the radius go?

The rule  $\tilde{G}_{00} = 1/G_{00}$  is a statement about components in coordinates with  $L\tilde{L} = \alpha'$ . If you prefer  $X \sim X + 2\pi R$  with  $G_{XX} = 1$ , then  $L = R$ , the dual coordinate has period  $2\pi\alpha'/R$  and  $\tilde{G}_{XX} = 1$ : the dual radius  $\alpha'/R$  sits in the period, not in the metric. For the dilaton the formula  $\tilde{\Phi} = \Phi - \frac{1}{2} \log G_{00}$  is correct as written only when  $L = \tilde{L} = \ell_s$ ; for general  $L$  it reads  $\tilde{\Phi} = \Phi - \frac{1}{2} \log(L^2 G_{00}/\alpha')$ , as argument (i) of Section 9.4 shows when the circumference is written as  $2\pi L\sqrt{G_{00}}$ . A sign of trouble is a formula such as  $\log G_{00}$  with a dimensionful  $G_{00}$ .

For a non-compact isometry (a line instead of a circle) the periods of  $d\tilde{\theta}$  are arbitrary real numbers instead of multiples of  $2\pi\tilde{L}$  [AAL §2.1, ll. 509–510]. This is the limit  $L \rightarrow \infty$ ,  $\tilde{L} \rightarrow 0$  of (9.24). The sum in (9.23) becomes an integral, which sets  $\nu_a = \nu_b = 0$ : a field  $\theta$  valued in a line has no winding sectors, and the flat gauge field is pure gauge.

### 9.5.2 Fixed points of the isometry

The rules require  $G_{00} \neq 0$ . At a fixed point of the isometry the Killing vector vanishes,  $G_{00} = k^i k_i = 0$ , and the stationary point (9.13) is singular [AAL §2.1, ll. 331–332]. The simplest example is the plane in polar coordinates,  $ds^2 = dr^2 + r^2 d\vartheta^2$ , with the rotation  $\vartheta \rightarrow \vartheta + \epsilon$  and constant dilaton  $\Phi_0$ . In our conventions  $\theta = \ell_s \vartheta$  and  $G_{00} = r^2/\alpha'$ , and Theorem 9.1 gives

$$d\tilde{s}^2 = dr^2 + \frac{\alpha'}{r^2} d\tilde{\theta}^2, \quad \tilde{\Phi} = \Phi_0 - \log \frac{r}{\ell_s}. \quad (9.25)$$

A metric  $dr^2 + f(r)^2 d\tilde{\theta}^2$  has scalar curvature  $-2f''/f$ ; with  $f = \ell_s/r$  this is  $-4/r^2$ . The flat plane is mapped to a space whose curvature and string coupling  $e^{\tilde{\Phi}} = g_s \ell_s/r$  both diverge at  $r = 0$ . Nothing is wrong with the string on the plane. The dual sigma-model description breaks down where the orbit of the isometry shrinks to zero size. A common interpretation (standard; not checked against a source in this book's library) is that strings wound around the small orbit are light there, and a description in terms of  $\tilde{G}, \tilde{B}, \tilde{\Phi}$  alone does not contain them. In such cases it is safer to work with the master action, which is regular [AAL §2.1, ll. 331–332 and 574–576].

### 9.5.3 What the local derivation does not tell you

Alvarez, Alvarez-Gaumé and Lozano list further limitations of the derivation in adapted coordinates [AAL §2.1, ll. 319–334]. The choice of coordinates hides general covariance and with it the global topology of the dual manifold: locally the dual is  $(M/S^1) \times S^1$ , but whether the product is twisted is not visible; their example is the round  $S^3$ , whose dual with respect to a fixed-point-free  $U(1)$  is  $S^2 \times S^1$  [AAL §2.1, ll. 559–573]. Isometries that do not commute with the one being dualized are in general not symmetries of the dual sigma model. The covariant formulation in terms of the Killing vector  $k$  replaces  $G_{00} \rightarrow k^2$ ,  $G_{0\alpha} \rightarrow k_\alpha$ ,  $B_{0\alpha} \rightarrow v_\alpha$ , where  $\iota_k H = -dv$  [AAL §2.1, eq. (2.1.23), ll. 578–604].

## 9.6 Worked examples

**Example 9.3** (The circle, with numbers). Take  $R = 2\ell_s$  and  $g_s = 0.1$ . Then  $G_{00} = 4$ ,  $\tilde{G}_{00} = \frac{1}{4}$ , so  $\tilde{R} = \ell_s/2 = \alpha'/R$ . The dilaton shifts by  $-\frac{1}{2} \log 4 = -\log 2$ , so  $\tilde{g}_s = g_s/2 = 0.05$ , in agreement with (9.19). The invariant (9.21) is  $\sqrt{\tilde{G}_{00}}/g_s^2 = 2/0.01 = 200$  before and  $\frac{1}{2}/0.0025 = 200$  after. In the variables of (9.22),  $x = \log 2$  goes to  $-\log 2$  and  $2d = 2 \log 0.1 - \log 2 = \log 0.005$  is unchanged. At the self-dual radius  $R = \ell_s$  one has  $G_{00} = 1$ ,  $x = 0$ , and neither the metric nor the dilaton changes.

**Example 9.4** (A tilted torus with  $B$ -field). Let  $d = 2$  with constant background

$$G = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & \frac{3}{2} \\ -\frac{3}{2} & 0 \end{pmatrix}, \quad \det G = 5.$$

Dualizing along  $X^0$ :  $\tilde{G}_{00} = \frac{1}{2}$ ,  $\tilde{G}_{01} = B_{01}/G_{00} = \frac{3}{4}$ ,  $\tilde{B}_{01} = G_{01}/G_{00} = \frac{1}{2}$ , and  $\tilde{G}_{11} = 3 - (1 \cdot 1 - \frac{3}{2} \cdot \frac{3}{2})/2 = 3 + \frac{5}{8} = \frac{29}{8}$ . There is no  $\tilde{B}_{\alpha\beta}$  with  $\alpha, \beta \geq 1$  in two dimensions. Thus

$$\tilde{G} = \begin{pmatrix} \frac{1}{2} & \frac{3}{4} \\ \frac{3}{4} & \frac{29}{8} \end{pmatrix}, \quad \tilde{B} = \begin{pmatrix} 0 & \frac{1}{2} \\ -\frac{1}{2} & 0 \end{pmatrix}, \quad \det \tilde{G} = \frac{29}{16} - \frac{9}{16} = \frac{5}{4} = \frac{\det G}{G_{00}^2},$$

as Proposition 9.2 requires, and  $\tilde{\Phi} = \Phi - \frac{1}{2} \log 2$ . Dualizing again:  $1/\tilde{G}_{00} = 2$ ,  $\tilde{B}_{01}/\tilde{G}_{00} = 1$ ,  $\tilde{G}_{01}/\tilde{G}_{00} = \frac{3}{2}$  and  $\frac{29}{8} - (\frac{9}{16} - \frac{1}{4}) \cdot 2 = \frac{29}{8} - \frac{5}{8} = 3$ : the original background returns.

**Example 9.5** ( $B$ -field becomes tilt). Let  $G = \text{diag}(G_{00}, G_{11})$  be a rectangular two-torus with  $B_{01} = b$ . The rules give

$$\tilde{G} = \begin{pmatrix} \frac{1}{G_{00}} & \frac{b}{G_{00}} \\ \frac{b}{G_{00}} & G_{11} + \frac{b^2}{G_{00}} \end{pmatrix}, \quad \tilde{B} = 0, \quad \text{i.e.} \quad ds^2 = \frac{1}{G_{00}} (d\tilde{\theta} + b dx^1)^2 + G_{11} (dx^1)^2. \quad (9.26)$$

The dual torus carries no  $B$ -field, but it is no longer rectangular: the  $B$ -field has turned into the angle between the two cycles. This is the simplest instance of the exchange of the Kähler modulus  $\rho$  (which contains  $B$ ) and the complex-structure modulus  $\tau$  (which contains the angle) studied in Chapter 12. The shift  $b \rightarrow b + 1$ , which is a symmetry of the original model because it changes the action by  $2\pi i$  times an integer, becomes the geometric transformation  $\tilde{\theta} \rightarrow \tilde{\theta} - x^1$  of the dual torus.

## 9.7 Preview: one matrix instead of five rules

The rules (9.15) are non-linear in  $(G, B)$ . They become linear when  $G$  and  $B$  are packaged into the *generalized metric* of Chapter 15, the symmetric  $2d \times 2d$  matrix

$$\mathcal{H}(G, B) = \begin{pmatrix} G - BG^{-1}B & BG^{-1} \\ -G^{-1}B & G^{-1} \end{pmatrix}. \quad (9.27)$$

Let  $e = \text{diag}(1, 0, \dots, 0)$  be the projector on the dualized direction and define

$$T_\theta = \begin{pmatrix} 1 - e & -e \\ -e & 1 - e \end{pmatrix}, \quad T_\theta^\top \eta T_\theta = \eta, \quad T_\theta^2 = 1, \quad \eta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (9.28)$$

**Proposition 9.6.** *With  $(\tilde{G}, \tilde{B})$  given by (9.15),*

$$\mathcal{H}(\tilde{G}, \tilde{B}) = T_\theta^\top \mathcal{H}(G, B) T_\theta. \quad (9.29)$$

We have verified (9.29) for  $d = 2$  and  $d = 3$  with fully symbolic  $G$  and  $B$  using sympy; this is a symbolic check, not a Lean theorem. The proof for all  $d$  is a block-matrix computation (Exercise 9.9); the general  $O(d, d; \mathbb{Z})$  action is the subject of Chapter 11. For  $d = 1$  it is immediate:  $\mathcal{H} = \text{diag}(G_{00}, 1/G_{00})$ ,  $T_\theta = -\eta$ , and conjugation swaps the two entries. The matrix  $T_\theta$  has integer entries and preserves  $\eta$ , so it belongs to  $O(d, d; \mathbb{Z})$ . It is, up to signs, the *factorized duality*  $D_0$  of Giveon, Porrati and Rabinovici, whose off-diagonal blocks are  $+e$  [GPR §2.4, eq. (2.4.29), ll. 1397–1421]. The two matrices are related by  $B \rightarrow -B$ , the orientation convention of the pitfall box in Section 9.3: with  $\mathcal{H}(G, -B)$  on both sides, (9.29) holds with  $+e$  (also checked symbolically). GPR state that the sigma-model duality acts on the background “exactly as in the flat case” [GPR §4.1, ll. 6025–6031]. In the same way (9.16) is the fractional-linear map  $\tilde{E} = ((1 - e)E - e)(-eE + (1 - e))^{-1}$ .

Three consequences follow. Dualities along different circles are conjugations by commuting matrices; dualizing all  $d$  directions with  $B = 0$  gives  $\eta \mathcal{H}(G, 0) \eta = \mathcal{H}(G^{-1}, 0)$ , the matrix version of  $R \rightarrow \alpha'/R$ ; and the quantity (9.21) is the only place where the dilaton enters, which is why double field theory uses  $(\mathcal{H}, d)$  as its field variables (Chapter 17).

## 9.8 What the machine has checked

The file `DoubleFieldTheory/TDualityBuscher.lean` belongs to one of the older libraries of the project. It contains no sigma model, no path integral and no metric with more than one component. It models the one-circle case with integers and proves arithmetic identities. We quote every theorem of the file and say what each one states.

### 9.8.1 The matrix part

The imported module `DoubleFieldTheory.GeneralizedGeometry` defines a structure `Mat2` of  $2 \times 2$  integer matrices with fields `a b c d`, the product `MatMul`, the transpose `MatTranspose`,

the matrix  $\text{InversionGen} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  and the diagonal matrix  $\text{GenMetric } g \text{ ginv} = \text{diag}(g, g_{\text{inv}})$  with two *independent* integer arguments.

#### Congruence action and the self-dual point

```
def CongruenceAction (M H : Mat2) : Mat2 :=
  MatMul (MatTranspose M) (MatMul H M)

theorem t_duality_congruence (H : Mat2) (hH : H = GenMetric 1 1) :
  CongruenceAction InversionGen H = GenMetric 1 1 := by ...

theorem inversion_generator_properties :
  MatMul InversionGen InversionGen = Identity2 ∧ Det2 InversionGen = -1 := by
  decide
```

**Reading.**  $\text{CongruenceAction}$  is  $M^T H M$ , the  $d = 1$  form of (9.29). The first theorem says: if  $H$  is the  $2 \times 2$  identity matrix then  $\sigma^T H \sigma$  is the identity matrix, where  $\sigma = \eta$  for  $d = 1$ . It concerns one fixed matrix, the generalized metric at the self-dual radius  $G_{00} = 1$ ; it does not state  $\text{diag}(g, g^{-1}) \mapsto \text{diag}(g^{-1}, g)$  for other radii. The second theorem says  $\sigma^2 = 1$  and  $\det \sigma = -1$ . **Proof idea.** After substituting the hypothesis, both sides of the first statement reduce by computation to the same structure and `rfl` closes the goal; the second is a closed statement about integers settled by `decide`. **[A]**

### 9.8.2 The logarithmic radius and the dilaton

The remaining declarations use an integer  $x$  in the role of  $\log(R/\ell_s)$  and an integer `phi` in the role of  $\Phi$ , following (9.22).

#### Four integer identities

```
def BuscherLogMap (x : Int) : Int := -x
def BuscherDilatonMap (phi x : Int) : Int := phi - x
def TwoDilaton (phi x : Int) : Int := 2 * phi - x

theorem buscher_log_involution (x : Int) :
  BuscherLogMap (BuscherLogMap x) = x := by ...

theorem buscher_dilaton_involution (phi x : Int) :
  BuscherDilatonMap (BuscherDilatonMap phi x) (BuscherLogMap x) = phi :=
  by ...

theorem buscher_dilaton_measure_invariance (phi x : Int) :
  TwoDilaton (phi - x) (-x) = TwoDilaton phi x := by ...

theorem self_dual_radius_rigidity (x : Int) (hx : BuscherLogMap x = x) :
  x = 0 := by ...
```

**Reading.** For all integers:  $-(-x) = x$ ;  $(\varphi - x) - (-x) = \varphi$ ;  $2(\varphi - x) - (-x) = 2\varphi - x$ ; and  $-x = x$  implies  $x = 0$ . These are the *arithmetic shadows* of four statements of this chapter: the map  $R \rightarrow \alpha'/R$  is an involution; the dilaton shift (9.18) is undone by a second dualization (because the second shift is computed with the dual radius); the combination  $2d$  of (9.22) is invariant; and the self-dual radius is the only fixed point. **Proof idea.** Each proof unfolds the definitions with `dsimp` and calls `omega`, the decision procedure for linear integer arithmetic. **[A]** for the integer identities.

**Honest scope.** An integer cannot represent  $\log(R/\ell_s)$  for a general radius, so these theorems are not statements about the real numbers  $x$  and  $\Phi$ . The four identities are linear and hold verbatim over  $\mathbb{R}$  (a one-line `ring` or `linarith` proof), but that transfer is not in the file. The documentation strings in the file describe the physics in stronger terms than the statements justify; when the two differ, read the statement. The identification of  $x$  with a logarithmic radius and of  $\phi$  with the dilaton is an interpretation and is never **[A]**. None of the following is formalized in this file, nor, as far as a search of the catalogue of compiled declarations shows, elsewhere in the project: the master action, the Gaussian integration, the rules (9.15) for  $d > 1$ , the determinant (9.20), and the periodicity argument (9.24).

**What is proved elsewhere.** The newer library `DualScaleStream2` works with Mathlib matrices. Its file `TDuality/Factorized.lean` defines the factorized duality for any  $d$ , with GPR's signs, and proves that it is an involution and an element of  $O(d, d; \mathbb{Z})$ . Its file `DFT/GeneralizedMetric.lean` proves that the full duality inverts the metric at  $B = 0$ :

```
def factorized (k : Fin d) : Matrix (Charge d) (Charge d) ℤ :=
  fromBlocks (1 - proj k) (proj k) (proj k) (1 - proj k)

theorem factorized_mul_self (k : Fin d) :
  factorized k * factorized k = 1 := by ...

theorem factorized_isODD (k : Fin d) : IsODD (factorized k) := by ...

theorem tduality_inverts_metric (G : Matrix (Fin d) (Fin d) ℝ)
  (hG : IsUnit G.det) :
  etaR d * genMetric G 0 * etaR d = genMetric G-1 0 := by ...
```

These are statements about real and integer matrices for all  $d$ ; they are discussed in Chapters 11 and 15. A faithful formalization of Theorem 9.1 as an algebraic statement would define the map  $(G, B) \mapsto (\tilde{G}, \tilde{B})$  on real matrices with  $G_{00} \neq 0$  and prove Proposition 9.6 and the involution property. This is within reach of Mathlib's block-matrix and Schur-complement lemmas and is a natural project for a reader (see also Chapter 38). A formalization of the path-integral derivation is not, since Mathlib has no functional integral.

| Claim                                                                                                      | Tier       | Evidence                                                                                                   |
|------------------------------------------------------------------------------------------------------------|------------|------------------------------------------------------------------------------------------------------------|
| $\sigma^\top \mathbf{1} \sigma = \mathbf{1}$ , $\sigma^2 = 1$ , $\det \sigma = -1$ (integer $2 \times 2$ ) | <b>[A]</b> | <code>t_duality_congruence</code> ,<br><code>inversion_generator_properties</code>                         |
| $-(-x) = x$ ; $-x = x \Rightarrow x = 0$ over $\mathbb{Z}$                                                 | <b>[A]</b> | <code>buscher_log_involution</code> ,<br><code>self_dual_radius_rigidity</code>                            |
| $(\varphi - x) - (-x) = \varphi$ ; $2(\varphi - x) + x = 2\varphi - x$ over $\mathbb{Z}$                   | <b>[A]</b> | <code>buscher_dilaton_involution</code> ,<br><code>buscher_dilaton_measure_invariance</code>               |
| $D_k^2 = 1$ , $D_k \in O(d, d; \mathbb{Z})$ ;<br>$\eta \mathcal{H}(G, 0) \eta = \mathcal{H}(G^{-1}, 0)$    | <b>[A]</b> | <code>factorized_mul_self</code> , <code>factorized_isODD</code> ,<br><code>tduality_inverts_metric</code> |
| Buscher rules (9.15) from the master action                                                                | <b>[L]</b> | AAL §2.1; GPR §4.1; derived on the page                                                                    |
| Dilaton shift (9.18) from the regularized determinant                                                      | <b>[L]</b> | AAL §4                                                                                                     |
| Equivalence on all genera given $L\tilde{L} = \alpha'$                                                     | <b>[L]</b> | AAL §2.1; GPR §4.1                                                                                         |
| $\mathcal{H}(\tilde{G}, \tilde{B}) = T_\theta^\top \mathcal{H} T_\theta$ for $d = 2, 3$                    | —          | symbolic check (sympy)                                                                                     |

### Chapter summary

- If  $G$ ,  $B$ ,  $\Phi$  do not depend on a coordinate  $\theta$ , the sigma model can be rewritten with an auxiliary vector  $V_a$  in place of  $\partial_a \theta$  and a Lagrange multiplier  $\tilde{\theta}$  imposing  $\varepsilon^{ab} \partial_a V_b = 0$ .
- Integrating out  $\tilde{\theta}$  returns the original model. Integrating out  $V_a$ , a Gaussian integral, gives a sigma model for  $(\tilde{\theta}, x^\alpha)$  with the Buscher background (9.15):  $\tilde{G}_{00} = 1/G_{00}$ , and



the mixed components of  $G$  and  $B$  are exchanged.

- The determinant of the Gaussian integral shifts the dilaton,  $\tilde{\Phi} = \Phi - \frac{1}{2} \log G_{00}$ ; then  $\sqrt{\det G} e^{-2\Phi}$  is invariant.
- On worldsheets of higher genus the equivalence requires the periods of  $\theta$  and  $\tilde{\theta}$  to satisfy  $L\tilde{L} = \alpha'$ . The rules fail where the Killing vector vanishes.
- In terms of the generalized metric the rules are one conjugation by an element of  $O(d, d; \mathbb{Z})$ .
- In Lean, the file for this chapter proves integer identities that mirror the one-circle case ( $x \rightarrow -x$ ,  $\Phi \rightarrow \Phi - x$ ,  $2d$  invariant, unique fixed point  $x = 0$ ). The rules themselves are Tier L.

## Exercises

**Exercise 9.1** (★). Evaluate the ordinary integral  $\int d^2V \exp(-\frac{1}{4\pi\alpha'}(gV \cdot V + 2V \cdot j))$  for a constant  $g > 0$  and a complex vector  $j_a$ . Identify the stationary point, the exponent and the prefactor, and explain which of the three becomes the dilaton shift.

**Exercise 9.2** (★). Dualize the background of Example 9.4 along  $X^1$  instead of  $X^0$  (relabel the coordinates, apply (9.15), relabel back). Check  $\det \tilde{G} = \det G / G_{11}^2$ .

**Exercise 9.3** (★). Derive (9.16) from (9.15), and conversely.

**Exercise 9.4** (★★). Repeat the derivation of Section 9.3 in complex worldsheet coordinates, starting from GPR's action  $\frac{1}{2\pi} \int d^2z E_{ij} \partial X^i \bar{\partial} X^j$  with gauge fields  $A, \bar{A}$  and multiplier term  $\bar{\theta}(\partial \bar{A} - \bar{\partial} A)$  [GPR §4.1, ll. 5976–6000]. Show that the integration over  $A, \bar{A}$  leads directly to (9.16), up to the sign conventions discussed in the pitfall box.

**Exercise 9.5** (★★). GPR write the dilaton coupling as  $-\frac{1}{8\pi} \int d^2z \phi R^{(2)}$  and the shift as  $\phi' = \phi + \log G_{00}$  [GPR §4.1, ll. 5947–5953 and 6067–6071]. Find the relation between their  $\phi$  and the  $\Phi$  of this chapter that makes the two statements agree. (Hint: compare the coefficient of the curvature term and apply both shifts to the circle.)

**Exercise 9.6** (★★). The round three-sphere of radius  $r_0$  can be written as  $ds^2 = \frac{r_0^2}{4} [d\vartheta^2 + \sin^2\vartheta d\phi^2 + (d\psi + \cos\vartheta d\phi)^2]$ , with  $\psi \sim \psi + 4\pi$ . Set  $B = 0$  and dualize along  $\psi$  (first rescale  $\psi$  so that its period is  $2\pi\ell_s$ ). Show that the dual metric is the product of a round  $S^2$  and a circle, with  $\tilde{B} \propto \cos\vartheta d\tilde{\theta} \wedge d\phi$ , and that the dilaton is constant. Compare with [AAL §2.1, ll. 570–573].

**Exercise 9.7** (★★, Lean). State and prove in Lean the  $d = 1$  Buscher rule that is missing from the file:

```
-- exercise: not yet proved (not in the library)
theorem buscher_circle_shadow (g ginv : Int) :
  CongruenceAction InversionGen (GenMetric g ginv) = GenMetric ginv g
```

(Hint: `simp [CongruenceAction, MatMul, MatTranspose, InversionGen, GenMetric].`) Why does this statement not need the hypothesis  $g \cdot g_{\text{inv}} = 1$ ?

**Exercise 9.8** (★★, Lean). Using Mathlib, restate the last two theorems of the second box of Section 9.8 (the invariance of `TwoDilaton` and the uniqueness of the fixed point) for real `x` and `phi`, and prove them. Which tactic replaces `omega`, and why does `omega` not apply?

**Exercise 9.9** (★★★). Prove Proposition 9.6 for all  $d$ . (Hint: write  $\mathcal{H} = \begin{pmatrix} 1 & B \\ 0 & 1 \end{pmatrix} \begin{pmatrix} G & 0 \\ 0 & G^{-1} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -B & 1 \end{pmatrix}$ , and use the Kaluza–Klein decomposition of the proof of Proposition 9.2 to reduce the statement to  $d = 1$  blocks.)

## Further reading

- The path-integral argument for one circle, including the remark on holonomies: [Tong §8.3.1, ll. 11716–11803]; the dilaton from the effective action: [Tong §8.3, ll. 11691–11714].
- Buscher’s formulation, the gauging of Roček and Verlinde, the covariant form of the rules and the global analysis: [AAL §2.1, ll. 217–604]. The dilaton and the regularized determinant: [AAL §4, ll. 943–1111].
- Factorized duality in sigma models in complex coordinates and the periodicity of the multiplier: [GPR §4.1, ll. 5904–6127]; the matrix  $D_i$  in the flat case: [GPR §2.4, ll. 1397–1424].
- Buscher’s original papers are cited in AAL as reference [16] and in GPR as references [33]–[36]; they are not part of this book’s library and have not been checked directly.

## Chapter 10

# Toroidal Compactification and the Narain Lattice

A string on a circle carries two integers, the momentum number  $n$  and the winding number  $w$ , and Chapters 7 and 8 showed that the whole radius dependence of the spectrum sits in the way these two integers are split into a left-moving and a right-moving momentum. This chapter asks what happens when  $d$  dimensions are curled up at once, on a torus  $T^d$  with an arbitrary constant metric  $G_{ij}$  and two-form  $B_{ij}$ . The  $2d$  integers of momentum and winding form a lattice  $\Gamma^{d,d}$  with an inner product of signature  $(d, d)$ ; it is *even* and *self-dual*, and as an abstract lattice it is the same for every torus. All the dependence on the  $d^2$  parameters  $(G, B)$  is in the choice of a  $d$ -dimensional positive subspace of  $\mathbb{R}^{d,d}$ , and the set of such choices is the homogeneous space  $O(d, d; \mathbb{R})/(O(d) \times O(d))$ . A student should care for three reasons. First, the duality group  $O(d, d; \mathbb{Z})$  of Chapter 11, the generalized metric of Chapter 15 and the doubled coordinates of Chapter 14 are all read off from this lattice. Second, the same two words, even and self-dual, describe the cohomology lattice of a K3 surface (Chapters 18 and 20) and organise the moduli space of strings on K3 (Chapter 23). Third, the rank-two case is small enough to be checked by the Lean kernel; Section 10.8 states exactly what was checked.

## 10.1 The torus and its background fields

### 10.1.1 Coordinates of period $2\pi$

A flat torus  $T^d$  is the quotient of  $\mathbb{R}^d$  by a lattice. Let  $\hat{e}_1, \dots, \hat{e}_d$  be  $d$  linearly independent vectors in  $\mathbb{R}^d$ , with the dimension of a length, and identify

$$x \sim x + 2\pi \sum_{i=1}^d m^i \hat{e}_i, \quad m^i \in \mathbb{Z}. \quad (10.1)$$

For  $d = 1$  and  $\hat{e}_1 = R$  this is the circle  $X \sim X + 2\pi R$  of Chapter 7. Instead of Cartesian coordinates it is convenient to use coordinates *along the lattice vectors*: write  $x = \sum_i X^i \hat{e}_i$ . The numbers  $X^i$  are dimensionless, and the identification (10.1) becomes

$$X^i \sim X^i + 2\pi m^i, \quad m^i \in \mathbb{Z}, \quad (10.2)$$

the same for every torus [GPR §2.4, eq. (2.4.2), ll. 1055–1058]. The price is that the metric is no longer the unit matrix. The squared length of a displacement is  $ds^2 = \hat{G}_{ij} dX^i dX^j$  with  $\hat{G}_{ij} = \hat{e}_i \cdot \hat{e}_j$ , a constant, symmetric, positive definite  $d \times d$  matrix with the dimension of

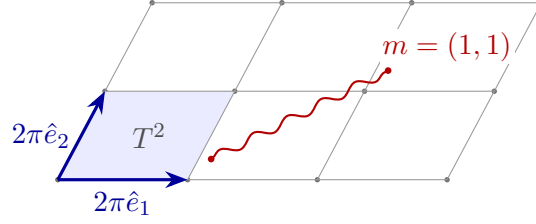


Figure 10.1: A two-torus is the plane modulo the lattice generated by  $2\pi\hat{e}_1$  and  $2\pi\hat{e}_2$  (shaded: one cell). In the coordinates  $X^i$  the cell is always  $[0, 2\pi)^2$ , and the geometry sits in  $G_{ij} = \hat{e}_i \cdot \hat{e}_j / \alpha'$ . Wavy line: a closed string with winding numbers  $m = (1, 1)$ .

length squared. Shape and size of the torus are encoded in  $\hat{G}_{ij}$ , and the coordinate ranges are fixed once and for all (Figure 10.1).

Because T-duality involves  $\alpha'$  explicitly, it pays to measure lengths in string units. We define the dimensionless metric

$$G_{ij} = \frac{\hat{G}_{ij}}{\alpha'} = \frac{\hat{e}_i \cdot \hat{e}_j}{\alpha'} \quad (10.3)$$

[GPR §2.2, ll. 805–821]. For the circle,  $G_{11} = R^2/\alpha'$ ; the self-dual radius  $R = \sqrt{\alpha'}$  of Chapter 8 is  $G_{11} = 1$ . Until Section 10.3.3 restores  $\alpha'$  in the mass formula, every quantity in this chapter is dimensionless.

### 10.1.2 The action and its $d^2$ parameters

Chapter 6 showed that a string couples to a background metric and to a background two-form, the Kalb–Ramond field. On a torus both may be taken constant. With a flat worldsheet metric, worldsheet coordinates  $(\tau, \sigma)$ ,  $\sigma \in [0, 2\pi)$ , and the notation  $\dot{X} = \partial_\tau X$ ,  $X' = \partial_\sigma X$ , the action for the  $d$  compact coordinates is

$$S = \frac{1}{4\pi} \int d\tau \int_0^{2\pi} d\sigma \left[ G_{ij} (\dot{X}^i \dot{X}^j - X'^i X'^j) + 2B_{ij} \dot{X}^i X'^j \right], \quad (10.4)$$

where  $B_{ij} = -B_{ji}$  is the two-form in units of  $\alpha'$  [GPR §2.4, eq. (2.4.1), ll. 1031–1053]. The overall sign of the  $B$  term is a convention; the one used here reproduces the canonical momentum of GPR, eq. (2.4.7), and the generalized metric fixed in the conventions of this book. For  $d = 1$  there is no two-form, and after the rescaling  $X \rightarrow X/R$  the first term is the circle action of Chapter 7.

Count the parameters. A symmetric matrix  $G$  has  $d(d+1)/2$  independent entries and an antisymmetric matrix  $B$  has  $d(d-1)/2$ , so there are

$$\frac{d(d+1)}{2} + \frac{d(d-1)}{2} = d^2 \quad (10.5)$$

real parameters [GPR §2.4, eqs. (2.4.3)–(2.4.4), ll. 1068–1089]. They are conveniently assembled into one matrix, the *background matrix*

$$E = G + B, \quad G = \frac{1}{2}(E + E^T), \quad B = \frac{1}{2}(E - E^T) \quad (10.6)$$

[GPR §2.4, eq. (2.4.5), ll. 1089–1096]. Each parameter multiplies an operator that can be added to the worldsheet action without spoiling conformal invariance, because the action stays quadratic (the model is “Gaussian”). Such parameters are called *moduli*: in the lower-dimensional effective theory each of them is a massless scalar field without potential. One may wonder whether the term with  $B$  matters at all, since for constant  $B$  the integrand  $B_{ij} \dot{X}^i X'^j = \frac{1}{2} B_{ij} \epsilon^{\alpha\beta} \partial_\alpha X^i \partial_\beta X^j$  is a total derivative. It does not change the equations of motion. It changes the quantum theory of strings that wind, as we now show.

## 10.2 Canonical quantization of the zero modes

### 10.2.1 Canonical momentum versus velocity

From (10.4) the momentum density conjugate to  $X^i(\sigma)$  is

$$P_i = \frac{\partial \mathcal{L}}{\partial \dot{X}^i} = \frac{1}{2\pi} (G_{ij} \dot{X}^j + B_{ij} X'^j) \quad (10.7)$$

[GPR §2.4, eq. (2.4.7), ll. 1107–1110]. The equations of motion are the free wave equation  $\ddot{X}^i = X''^i$ , and a closed string on the torus must close only up to the identification (10.2). The general solution is therefore

$$X^i(\tau, \sigma) = x^i + m^i \sigma + v^i \tau + \text{oscillators}, \quad X^i(\tau, \sigma + 2\pi) = X^i(\tau, \sigma) + 2\pi m^i. \quad (10.8)$$

The integers  $m^i$  are the *winding numbers*: the string wraps  $m^i$  times around the  $i$ -th cycle (Figure 10.1). The constant  $v^i$  is the centre-of-mass velocity. The total momentum is the integral of (10.7); the oscillators are periodic and drop out, so

$$p_i \equiv \int_0^{2\pi} d\sigma P_i = G_{ij} v^j + B_{ij} m^j. \quad (10.9)$$

Now comes the step that carries all the physics. The operator  $p_i$  is conjugate to the centre-of-mass position  $x^i$ ,  $[x^i, p_j] = i\delta_j^i$ , so it generates translations, and a wavefunction  $e^{ip_i x^i}$  must be single valued under  $x^i \rightarrow x^i + 2\pi$ . Hence

$$p_i = n_i \in \mathbb{Z} \quad (10.10)$$

[GPR §2.4, eq. (2.4.8), ll. 1110–1115]. It is the *canonical* momentum that is quantized, not the velocity. Solving (10.9) for the velocity gives

$$v^i = G^{ij} (n_j - B_{jk} m^k), \quad (10.11)$$

where  $G^{ij}$  is the inverse matrix of  $G_{ij}$  [GPR §2.4, eq. (2.4.30), ll. 1425–1438]. A string with winding but  $n = 0$  therefore still moves when  $B \neq 0$ .

#### Physical picture: $B$ is a theta angle

A charged particle on a ring threaded by a magnetic flux has canonical momentum  $p = m\dot{x} + qA$ . The vector potential is locally pure gauge and does not enter the classical motion, but  $p$  is quantized in integers, so the allowed *velocities* are shifted by the flux, and the energy levels depend on it. A flux of one quantum can be removed by relabelling the integer. The constant  $B$ -field does the same thing to a winding string: the winding number  $m^k$  plays the role of the charge,  $B_{jk}$  the role of the vector potential, and (10.11) is the shifted velocity. GPR compare with the theta term of four-dimensional gauge theory, in whose presence a monopole acquires electric charge [GPR §2.4, ll. 1199–1206]. Two consequences follow at once:  $B$  does matter quantum mechanically, and a shift of  $B_{ij}$  by an antisymmetric *integer* matrix can be absorbed into  $n_j \rightarrow n_j + \Theta_{jk} m^k$  and changes nothing [GPR §2.4, ll. 1351–1371]. The second fact is the first element of the duality group of Chapter 11.

### 10.2.2 The Hamiltonian

The Hamiltonian density is  $\mathcal{H} = P_i \dot{X}^i - \mathcal{L}$ . Inserting (10.7), the term with  $B$  cancels between  $P_i \dot{X}^i$  and  $\mathcal{L}$ , which is the Hamiltonian form of the statement that a total derivative carries no energy:

$$\mathcal{H} = \frac{1}{4\pi} (\dot{X}^\top G \dot{X} + X'^\top G X'). \quad (10.12)$$

A Hamiltonian must be written in terms of momenta. With  $\dot{X} = G^{-1}(2\pi P - BX')$  from (10.7) and  $B^\top = -B$ ,

$$\begin{aligned} \dot{X}^\top G \dot{X} &= (2\pi P - BX')^\top G^{-1} (2\pi P - BX') \\ &= (2\pi)^2 P^\top G^{-1} P - X'^\top B G^{-1} B X' + 4\pi X'^\top B G^{-1} P. \end{aligned} \quad (10.13)$$

In the second line we used  $(BX')^\top = -X'^\top B$  twice: once to turn  $(BX')^\top G^{-1} B X'$  into  $-X'^\top B G^{-1} B X'$ , and once to combine the two cross terms, which are equal because each is a number and equals its own transpose. Therefore

$$\mathcal{H} = \frac{1}{4\pi} \left[ (2\pi)^2 P^\top G^{-1} P + X'^\top (G - B G^{-1} B) X' + 4\pi X'^\top B G^{-1} P \right] \quad (10.14)$$

[GPR §2.4, eq. (2.4.9), ll. 1116–1140]. Notice that  $G - B G^{-1} B = G + B^\top G^{-1} B$  is positive definite, so the energy is bounded below, as it must be.

For the zero modes,  $P_i = n_i/2\pi$  and  $X'^i = m^i$  are constant along the string; integrating (10.14) over  $\sigma$  multiplies it by  $2\pi$  and gives

$$H_0 = \frac{1}{2} \left[ n^\top G^{-1} n + m^\top (G - B G^{-1} B) m + 2 m^\top B G^{-1} n \right] \quad (10.15)$$

[GPR §2.4, eq. (2.4.11), ll. 1183–1191]. This quadratic form in the  $2d$  integers is best written with a single  $2d \times 2d$  matrix. Collect winding and momentum numbers in the *doubled charge vector*  $Z = (m^i, n_i)$ , windings first, and define

$$\mathcal{H}(G, B) = \begin{pmatrix} G - B G^{-1} B & B G^{-1} \\ -G^{-1} B & G^{-1} \end{pmatrix}, \quad H_0 = \frac{1}{2} Z^\top \mathcal{H} Z \quad (10.16)$$

[GPR §2.4, eqs. (2.4.17)–(2.4.18), ll. 1267–1290, where the matrix is called  $M$ ]. The two off-diagonal blocks contribute  $m^\top B G^{-1} n - n^\top G^{-1} B m$ , and the second term is the transpose of the first with  $B^\top = -B$ , so both equal  $m^\top B G^{-1} n$ , in agreement with (10.15). The matrix  $\mathcal{H}$  is the *generalized metric*; it is symmetric and positive definite, and Chapter 15 is devoted to it. The Lean definition `genMetric` (in the file `DFT/GeneralizedMetric.lean` of `DualScaleStream2`) is this matrix, with the same ordering  $Z = (w, n)$ .

## 10.3 Left- and right-moving momenta

### 10.3.1 An orthonormal frame

On the circle the zero modes were packaged into  $p_L$  and  $p_R$ , the momenta of the left-moving and right-moving parts of  $X$ . The same is possible on  $T^d$ , with one new ingredient: to take the square of a momentum we need an orthonormal frame. Choose a real invertible  $d \times d$  matrix  $e$ , a *vielbein*, with

$$G = e^\top e, \quad \text{so that} \quad G^{-1} = e^{-1} e^{-\top}, \quad (10.17)$$

where  $e^{-\top}$  denotes the inverse transpose. Geometrically the columns of  $\sqrt{\alpha'} e$  are the Cartesian components of the lattice vectors  $\hat{e}_i$  of (10.1). The choice of  $e$  is not unique:  $e$  and  $Oe$  with  $O \in O(d)$  give the same  $G$ . A covector with coordinate components  $k_i$  has frame components  $e^{-\top} k$ , and its squared length is  $k^\top G^{-1} k$ .

**Definition 10.1** (Left and right momenta). For a state with momentum numbers  $n \in \mathbb{Z}^d$  and winding numbers  $m \in \mathbb{Z}^d$  on the torus with background matrix  $E = G + B$ ,

$$p_L = \frac{1}{\sqrt{2}} e^{-\top} (n + E^\top m), \quad p_R = \frac{1}{\sqrt{2}} e^{-\top} (n - E m), \quad (10.18)$$

both vectors in  $\mathbb{R}^d$  with the Euclidean scalar product. Explicitly  $E^\top = G - B$ , so  $p_L$  contains  $n + (G - B)m$  and  $p_R$  contains  $n - (G + B)m$ .

This is GPR's eq. (2.4.12) [GPR §2.4, ll. 1192–1198] rewritten with column vectors; GPR normalise their frame by  $\sum_a e_i^a e_j^a = 2G_{ij}$ , which hides the factor  $1/\sqrt{2}$  that we keep explicit. For  $d = 1$ ,  $e = R/\sqrt{\alpha'}$  and

$$p_{L,R} = \frac{1}{\sqrt{2}} \left( \frac{\sqrt{\alpha'}}{R} n \pm \frac{R}{\sqrt{\alpha'}} m \right) = \sqrt{\frac{\alpha'}{2}} \left( \frac{n}{R} \pm \frac{mR}{\alpha'} \right) \quad (10.19)$$

[GPR §2.2, eq. (2.2.5), ll. 620–649]. The bracket on the right is the dimensionful  $p_{L,R}$  of Chapter 7; the dimensionless momenta of this chapter are those multiplied by  $\sqrt{\alpha'}/2$ , which are exactly the oscillator zero modes  $\tilde{\alpha}_0, \alpha_0$  of that chapter.

Where do (10.18) come from? Split the zero-mode part of (10.8) into a function of  $\tau + \sigma$  and a function of  $\tau - \sigma$ :  $m\sigma + v\tau = \frac{1}{2}(v + m)(\tau + \sigma) + \frac{1}{2}(v - m)(\tau - \sigma)$ . With (10.11),

$$G(v + m) = n - Bm + Gm = n + E^\top m, \quad G(v - m) = n - Bm - Gm = n - Em.$$

So  $p_L$  and  $p_R$  are, up to normalisation and the change to the orthonormal frame, the left-moving and right-moving velocities with the index lowered. This is why they, and not  $n$  and  $m$  separately, appear in vertex operators and in the Virasoro generators.

### 10.3.2 The two quadratic identities

**Proposition 10.2** (Sum and difference of squares). *For all  $n, m \in \mathbb{R}^d$ , every symmetric invertible  $G$  and every antisymmetric  $B$ ,*

$$p_L^2 + p_R^2 = Z^\top \mathcal{H}(G, B) Z, \quad (10.20)$$

$$p_L^2 - p_R^2 = 2 n \cdot m = Z^\top \eta Z, \quad \eta = \begin{pmatrix} 0 & \mathbf{1}_d \\ \mathbf{1}_d & 0 \end{pmatrix}, \quad (10.21)$$

where  $Z = (m, n)$  and  $n \cdot m = \sum_i n_i m^i$ . In particular the difference does not depend on  $G$  or  $B$ .

*Proof.* By (10.17),  $p_L^2 = \frac{1}{2}(n + E^\top m)^\top G^{-1}(n + E^\top m)$  and similarly for  $p_R^2$  with  $E^\top \rightarrow -E$ . Expand both:

$$\begin{aligned} 2p_L^2 &= n^\top G^{-1} n + 2 n^\top G^{-1} E^\top m + m^\top E G^{-1} E^\top m, \\ 2p_R^2 &= n^\top G^{-1} n - 2 n^\top G^{-1} E m + m^\top E^\top G^{-1} E m. \end{aligned}$$

The two quadratic terms in  $m$  are equal. Indeed  $E G^{-1} E^\top = (G + B) G^{-1} (G - B) = G - B + B - B G^{-1} B = G - B G^{-1} B$ , and  $E^\top G^{-1} E = (G - B) G^{-1} (G + B)$  gives the same result. In

the difference they cancel and the cross terms add up to  $2n^\top G^{-1}(E^\top + E)m = 4n^\top G^{-1}Gm = 4n \cdot m$ ; dividing by 2 gives (10.21). In the sum the cross terms give  $2n^\top G^{-1}(E^\top - E)m = -4n^\top G^{-1}Bm = 4m^\top BG^{-1}n$ , so  $p_L^2 + p_R^2 = n^\top G^{-1}n + m^\top (G - BG^{-1}B)m + 2m^\top BG^{-1}n$ , which is  $2H_0$  of (10.15).  $\square$

Both identities, the relation  $\mathcal{H}\eta\mathcal{H} = \eta$  used below, and  $\det \mathcal{H} = 1$  were also confirmed for  $d = 2$  with symbolic  $G$  and  $B$  by computer algebra (a symbolic check, not a Lean proof). The content of the proposition is that the  $2d$  integers carry two quadratic forms. One,  $\mathcal{H}$ , is positive definite and depends on the background: it is the energy. The other,  $\eta$ , is indefinite and knows nothing about the background: it will turn out to be the spin of the state.

### 10.3.3 Mass formula and level matching on $T^d$

The oscillators are not affected by the compactification except through their inner product: in the coordinates  $X^i$  the modes obey  $[\alpha_k^i, \alpha_l^j] = [\tilde{\alpha}_k^i, \tilde{\alpha}_l^j] = k G^{ij} \delta_{k+l,0}$ , the number operators are  $N = \sum_{k>0} \alpha_{-k}^i G_{ij} \alpha_k^j$  and the same with tildes, and the complete Hamiltonian of the compact sector is

$$H = \frac{1}{2} Z^\top \mathcal{H} Z + N + \tilde{N} \quad (10.22)$$

[GPR §2.4, eqs. (2.4.32)–(2.4.34), ll. 1448–1489]. For the bosonic string on  $\mathbb{R}^{1,25-d} \times T^d$  the two mass-shell conditions of Chapter 7 hold unchanged, with  $N$  and  $\tilde{N}$  now counting the oscillators of all 24 transverse directions. In the dimensionless momenta of this chapter they read

$$\frac{\alpha'}{2} M^2 = p_R^2 + 2(N - 1), \quad \frac{\alpha'}{2} M^2 = p_L^2 + 2(\tilde{N} - 1). \quad (10.23)$$

(Multiply the circle formulas  $M^2 = p_R^2 + \frac{4}{\alpha'}(N - 1)$  by  $\alpha'/2$  and use (10.19).) Half the sum and the difference of the two conditions give, with Proposition 10.2,

$$M^2 = \frac{1}{\alpha'} Z^\top \mathcal{H}(G, B) Z + \frac{2}{\alpha'} (N + \tilde{N} - 2), \quad (10.24)$$

$$N - \tilde{N} = \frac{1}{2} (p_L^2 - p_R^2) = n \cdot m. \quad (10.25)$$

These are the  $T^d$  versions of the mass formula and of the level-matching condition  $N - \tilde{N} = nw$  of Chapter 7. Level matching is the statement that the state is invariant under rigid shifts of  $\sigma$ ; it is a constraint on the integers alone, with no  $G$  or  $B$  in it. That is the first hint that  $n \cdot m$  is more fundamental than the energy.

It is a common mistake to read (10.24) as “momentum  $n$ , energy  $n^\top G^{-1}n$ ” and to conclude that  $B$  is invisible. That is true only for  $m = 0$ . Completing the square in (10.15),

$$H_0 = \frac{1}{2} (n - Bm)^\top G^{-1} (n - Bm) + \frac{1}{2} m^\top G m, \quad (10.26)$$

the kinetic energy of the shifted momentum plus the stretching energy of the winding string: as  $B$  varies, a state with fixed labels  $(n, m)$  changes its energy continuously.

## 10.4 The Narain lattice

### 10.4.1 Lattices with an indefinite inner product

We need a few definitions from lattice theory; Chapter 18 has the general theory.

**Definition 10.3.** A *lattice*  $\Gamma$  of rank  $r$  is the set of integer combinations  $\sum_a k^a f_a$  of  $r$  linearly independent vectors  $f_a$  in a real vector space  $V$  of dimension  $r$  equipped with a nondegenerate symmetric bilinear form  $\langle \cdot, \cdot \rangle$ . Its *Gram matrix* in the basis  $f_a$  is  $\mathcal{G}_{ab} = \langle f_a, f_b \rangle$ . The lattice is



*integral* if all  $\langle v, v' \rangle$ ,  $v, v' \in \Gamma$ , are integers, and *even* if all  $\langle v, v \rangle$  are even integers. The *dual lattice* is

$$\Gamma^* = \{u \in V : \langle u, v \rangle \in \mathbb{Z} \text{ for all } v \in \Gamma\},$$

and  $\Gamma$  is *self-dual*, or *unimodular*, if  $\Gamma^* = \Gamma$ . The *signature*  $(n_+, n_-)$  counts the positive and negative eigenvalues of the form on  $V$ .

An even lattice is integral, because  $\langle v, v' \rangle = \frac{1}{2}(\langle v+v', v+v' \rangle - \langle v, v \rangle - \langle v', v' \rangle)$ . An integral lattice is contained in its dual. Self-duality is decided by one number:

**Lemma 10.4.** *An integral lattice is self-dual if and only if its Gram matrix has determinant  $\pm 1$ .*

*Proof.* Let  $f^{*a}$  be the basis of  $V$  dual to  $f_a$ ,  $\langle f^{*a}, f_b \rangle = \delta_b^a$ . A vector  $u = \sum u_a f^{*a}$  pairs integrally with all of  $\Gamma$  exactly when all  $u_a$  are integers, so  $\Gamma^*$  is the lattice generated by the  $f^{*a}$ . Expanding  $f_a = \sum_b \mathcal{G}_{ab} f^{*b}$  shows that  $\Gamma \subseteq \Gamma^*$  is the image of  $\mathbb{Z}^r$  under the integer matrix  $\mathcal{G}$ . This image is all of  $\mathbb{Z}^r$  if and only if  $\mathcal{G}^{-1}$  is again an integer matrix, that is if and only if  $\det \mathcal{G} = \pm 1$  (if  $\mathcal{G}^{-1}$  is integral then  $\det \mathcal{G} \cdot \det \mathcal{G}^{-1} = 1$  in  $\mathbb{Z}$ ; conversely Cramer's rule gives  $\mathcal{G}^{-1} = \text{adj } \mathcal{G} / \det \mathcal{G}$ ).  $\square$

The argument in brackets, “an integer inverse forces  $\det = \pm 1$ ”, is precisely the certificate lemma `isUnimodular_of_mul_eq_one` of Section 10.8.

### 10.4.2 Definition and first properties

Put the left and right momenta of a state side by side,  $(p_L, p_R) \in \mathbb{R}^d \oplus \mathbb{R}^d$ , and give this  $2d$ -dimensional space the *Lorentzian* inner product

$$\langle (p_L, p_R), (p'_L, p'_R) \rangle = p_L \cdot p'_L - p_R \cdot p'_R, \quad (10.27)$$

which has signature  $(d, d)$ ; we write  $\mathbb{R}^{d,d}$  for the space with this form.

**Definition 10.5** (Narain lattice). The *Narain lattice* of the background  $E = G + B$  is

$$\Gamma^{d,d}(E) = \{(p_L(n, m), p_R(n, m)) : n, m \in \mathbb{Z}^d\} \subset \mathbb{R}^{d,d},$$

with  $p_{L,R}$  given by (10.18).

**Theorem 10.6** ([L] in general; [A] for the statements listed in Section 10.8). *For every background,  $\Gamma^{d,d}(E)$  is an even self-dual lattice of signature  $(d, d)$ . In the basis obtained by setting one of the  $2d$  integers  $(m, n)$  to one and the others to zero, its Gram matrix is  $\eta$  of (10.21), for every  $G$  and  $B$ .*

*Proof.* The map  $(m, n) \mapsto (p_L, p_R)$  is linear and injective (from  $p_L - p_R = \sqrt{2} e^{-\text{T}} G m$  and then  $p_L + p_R$  one recovers  $m$  and  $n$ ), so the images of the  $2d$  unit vectors are a basis of a lattice of rank  $2d$ . Polarizing (10.21),

$$\langle (p_L, p_R), (p'_L, p'_R) \rangle = n \cdot m' + n' \cdot m = Z^{\text{T}} \eta Z', \quad (10.28)$$

so the Gram matrix is  $\eta$ . The norm of a lattice vector is  $2n \cdot m \in 2\mathbb{Z}$ : the lattice is even. Finally  $\eta^2 = \mathbf{1}$ , so  $\eta$  is its own integer inverse and  $\det \eta = (-1)^d$ ; by Lemma 10.4 the lattice is self-dual. The signature is that of  $\mathbb{R}^{d,d}$ .  $\square$

The statement is GPR's [GPR §2.4, eq. (2.4.16), ll. 1249–1258], where it is attributed to Narain. For  $d = 1$  the Gram matrix is

$$U = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad (10.29)$$

the *hyperbolic plane*: two null vectors, pure momentum  $(n, m) = (1, 0)$  and pure winding  $(0, 1)$ , with mutual pairing one. For general  $d$  the pairing (10.28) only couples  $n_i$  with  $m^i$  for the same  $i$ , so

$$\Gamma^{d,d} \cong \underbrace{U \oplus U \oplus \cdots \oplus U}_d, \quad (10.30)$$

one hyperbolic plane per circle. This is no accident of our construction. Even self-dual lattices of signature  $(n_+, n_-)$  exist only for  $n_+ - n_-$  divisible by 8, and when both  $n_+$  and  $n_-$  are positive the lattice is unique up to isometry [Asp §2.3, ll. 559–566]. For signature  $(d, d)$  it is (10.30); for signature  $(3, 19)$  it is the K3 lattice  $3U \oplus 2E_8(-1)$  of Chapter 20, and the three copies of  $U$  in it are the same matrix (10.29).

#### Common pitfall: “the lattice depends on the radius”

As an abstract lattice, a free abelian group with an integer bilinear form,  $\Gamma^{d,d}$  is the same for all  $G$  and  $B$ : it is  $\mathbb{Z}^{2d}$  with the form  $\eta$ . What depends on the background is the *embedding* in  $\mathbb{R}^{d,d} = \mathbb{R}_L^d \oplus \mathbb{R}_R^d$ , that is, how each lattice vector is split into a left and a right part. Both statements are usually abbreviated as “the Narain lattice”, and one must know which is meant. The spectrum sees the embedding; the duality group of Chapter 11 is the symmetry group of the abstract lattice.

### 10.4.3 The circle: changing $R$ is a Lorentz boost

For  $d = 1$  the geometry can be drawn. From (10.19), with  $r = R/\sqrt{\alpha'}$ ,

$$p_L + p_R = \sqrt{2} \frac{n}{r}, \quad p_L - p_R = \sqrt{2} m r. \quad (10.31)$$

The combinations  $p_L \pm p_R$  are light-cone coordinates of the plane  $\mathbb{R}^{1,1}$ , and changing the radius rescales one by  $r^{-1}$  and the other by  $r$ . That is a Lorentz boost of rapidity  $\lambda = \ln r$ :

$$\begin{pmatrix} p_L \\ p_R \end{pmatrix}_r = \begin{pmatrix} \cosh \lambda & -\sinh \lambda \\ -\sinh \lambda & \cosh \lambda \end{pmatrix} \begin{pmatrix} p_L \\ p_R \end{pmatrix}_{r=1}, \quad \lambda = \ln r. \quad (10.32)$$

(Check: at  $r = 1$ ,  $p_{L,R} = (n \pm m)/\sqrt{2}$ , and the first row gives  $\frac{1}{\sqrt{2}}[(\cosh \lambda - \sinh \lambda)n + (\cosh \lambda + \sinh \lambda)m] = \frac{1}{\sqrt{2}}(n/r + mr)$ .) A boost preserves  $p_L^2 - p_R^2$ : each lattice point slides along its hyperbola  $p_L^2 - p_R^2 = 2nm$  (Figure 10.2). It does not preserve  $p_L^2 + p_R^2$ , which is why the masses change. The moduli space of the circle is the group of boosts,  $O(1, 1; \mathbb{R})$  modulo reflections, a half-line parametrised by  $r > 0$ . T-duality is  $\lambda \rightarrow -\lambda$ : the lattice at  $1/r$  is the mirror image under  $p_R \rightarrow -p_R$  of the lattice at  $r$ , with the labels  $n \leftrightarrow m$  exchanged.

## 10.5 Why modular invariance forces an even self-dual lattice

So far evenness and self-duality were *observed*. The deeper statement is that they are *required*: a set of momenta  $(p_L, p_R)$  defines a consistent closed-string theory only if it is an even self-dual lattice [Asp §3.5, ll. 1985–1988]. The derivation below is standard; its steps are carried out on the page and have not been checked line by line against a source in this book's library. The starting point, the one-loop partition function as a trace over the oscillators and a sum over  $(p_L, p_R)$ , is [GPR §2.2, eq. (2.2.13), ll. 766–785].

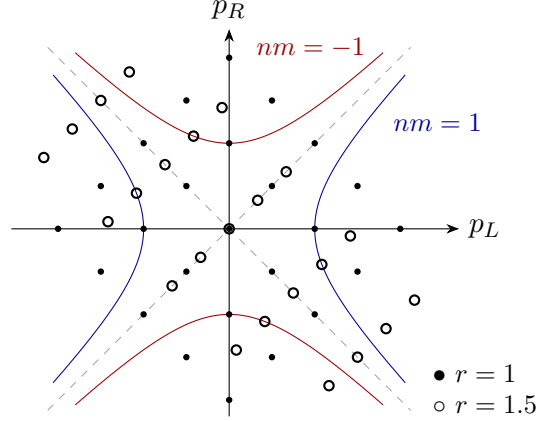


Figure 10.2: The Narain lattice  $\Gamma^{1,1}$  in the  $(p_L, p_R)$  plane at the self-dual radius  $r = 1$  (filled dots) and at  $r = 1.5$  (open circles). The dashed lines are the light cone  $p_L^2 = p_R^2$ , on which the pure-momentum and pure-winding states lie. Changing  $r$  is a Lorentz boost: every point moves along its hyperbola  $p_L^2 - p_R^2 = 2nm$ . At  $r = 1$  four points,  $(n, m) = \pm(1, 1)$  with  $p_R = 0$  and  $\pm(1, -1)$  with  $p_L = 0$ , sit at distance  $\sqrt{2}$  on the axes; these are the extra massless states of the self-dual radius (Chapter 8).

### 10.5.1 The torus partition function

The one-loop vacuum amplitude of a closed string is a path integral over worldsheets with the topology of a torus (not the target torus  $T^d$ ). A flat worldsheet torus is specified, up to rescaling, by one complex number  $\tau = \tau_1 + i\tau_2$ ,  $\tau_2 > 0$ : let a closed string of length  $2\pi$  propagate for Euclidean time  $2\pi\tau_2$ , rotate it by an angle  $2\pi\tau_1$ , and glue the ends. Propagation is generated by  $H = L_0 + \tilde{L}_0 - \frac{c}{12}$  and the rotation by  $L_0 - \tilde{L}_0$ , so that, with  $q = e^{2\pi i\tau}$ ,

$$Z(\tau, \bar{\tau}) = \text{tr } q^{\tilde{L}_0 - c/24} \bar{q}^{L_0 - c/24}. \quad (10.33)$$

Here  $\tilde{L}_0 = \frac{1}{2}p_L^2 + \tilde{N}$  and  $L_0 = \frac{1}{2}p_R^2 + N$  for the  $d$  compact bosons, and  $c = d$  (Chapter 4). The trace factorises into oscillators and zero modes. Each oscillator  $\tilde{\alpha}_{-k}$  contributes a geometric series  $1 + q^k + q^{2k} + \dots = (1 - q^k)^{-1}$ , so the  $d$  left-moving towers give  $q^{-d/24} \prod_{k \geq 1} (1 - q^k)^{-d} = \eta(\tau)^{-d}$ , where  $\eta(\tau) = q^{1/24} \prod_{k \geq 1} (1 - q^k)$  is the Dedekind eta function (no relation to the matrix  $\eta$ ). Therefore

$$Z_\Gamma(\tau, \bar{\tau}) = \frac{\Theta_\Gamma(\tau, \bar{\tau})}{|\eta(\tau)|^{2d}}, \quad \Theta_\Gamma(\tau, \bar{\tau}) = \sum_{(p_L, p_R) \in \Gamma} q^{\frac{1}{2}p_L^2} \bar{q}^{\frac{1}{2}p_R^2} = \sum_{\Gamma} e^{i\pi\tau p_L^2 - i\pi\bar{\tau} p_R^2}. \quad (10.34)$$

Different values of  $\tau$  can describe the same torus. The parallelogram spanned by 1 and  $\tau$  generates the same lattice in  $\mathbb{C}$  as the one spanned by 1 and  $\tau + 1$ , and, after a rotation and rescaling, as the one spanned by 1 and  $-1/\tau$ . These two maps,

$$T : \tau \mapsto \tau + 1, \quad S : \tau \mapsto -\frac{1}{\tau}, \quad (10.35)$$

generate the *modular group*  $SL(2, \mathbb{Z})$  (acting modulo  $\pm 1$ ). The path integral does not know which parallelogram we used, so a consistent theory must have  $Z(\tau + 1) = Z(\tau)$  and  $Z(-1/\tau) = Z(\tau)$ . This is *modular invariance*. If it failed, the integral over  $\tau$  that defines the one-loop amplitude would depend on an arbitrary choice of fundamental region.

### 10.5.2 $T$ requires an even lattice

Under  $\tau \rightarrow \tau + 1$ ,  $|\eta(\tau)|$  is unchanged, since  $\eta(\tau + 1) = e^{i\pi/12}\eta(\tau)$  is a pure phase. Each term of  $\Theta_\Gamma$  is multiplied by

$$e^{i\pi(p_L^2 - p_R^2)}. \quad (10.36)$$

Terms with different  $(p_L^2, p_R^2)$  are linearly independent functions of  $\tau$ , so the sum is invariant if and only if every phase is 1:

$$p_L^2 - p_R^2 \in 2\mathbb{Z} \quad \text{for all } (p_L, p_R) \in \Gamma. \quad (10.37)$$

This is evenness with respect to the Lorentzian form (10.27). Physically,  $\tau \rightarrow \tau + 1$  is a rotation of the closed string by a full turn, generated by  $L_0 - \tilde{L}_0$ , the worldsheet spin. A full turn must act trivially, so the spin  $N - \tilde{N} + \frac{1}{2}(p_R^2 - p_L^2)$  of every state, physical or not, must be an integer. For the toroidal lattice  $\frac{1}{2}(p_L^2 - p_R^2) = n \cdot m$ : evenness holds because momentum and winding numbers are integers.

### 10.5.3 $S$ requires a self-dual lattice

For  $S$  we need the Poisson summation formula: for a lattice  $\Gamma \subset V$  and a rapidly decreasing function  $f$ ,

$$\sum_{v \in \Gamma} f(v) = \frac{1}{\text{vol}(\Gamma)} \sum_{u \in \Gamma^*} \hat{f}(u), \quad \hat{f}(u) = \int_V dx f(x) e^{-2\pi i \langle u, x \rangle}, \quad (10.38)$$

where  $\text{vol}(\Gamma) = \sqrt{|\det \mathcal{G}|}$  is the volume of a unit cell. (The formula is usually stated with a Euclidean pairing; the Lorentzian one differs by  $x_R \rightarrow -x_R$ , which changes neither the measure nor the Gaussian below.) Take  $f(x) = e^{i\pi\tau x_L^2 - i\pi\bar{\tau} x_R^2}$ . The Fourier transform is a product of  $2d$  one-dimensional Gaussian integrals  $\int dx e^{i\pi\tau x^2 - 2\pi i u x} = (-i\tau)^{-1/2} e^{-i\pi u^2/\tau}$ , and the complex conjugate formula for the right-movers, so

$$\hat{f}(u) = (-i\tau)^{-d/2} (i\bar{\tau})^{-d/2} e^{i\pi(-1/\tau)u_L^2 - i\pi(-1/\bar{\tau})u_R^2}, \quad (-i\tau)(i\bar{\tau}) = |\tau|^2.$$

Inserting this in (10.38) and renaming  $\tau \rightarrow -1/\tau$  gives

$$\Theta_\Gamma(-1/\tau) = \frac{|\tau|^d}{\text{vol}(\Gamma)} \Theta_{\Gamma^*}(\tau). \quad (10.39)$$

The eta function obeys  $\eta(-1/\tau) = (-i\tau)^{1/2}\eta(\tau)$ , hence  $|\eta(-1/\tau)|^{2d} = |\tau|^d |\eta(\tau)|^{2d}$ , and the factors of  $|\tau|$  cancel:

$$Z_\Gamma(-1/\tau) = \frac{1}{\text{vol}(\Gamma)} Z_{\Gamma^*}(\tau). \quad (10.40)$$

If  $\Gamma = \Gamma^*$  then  $\text{vol}(\Gamma) = 1$  by Lemma 10.4 and  $Z$  is invariant. Conversely, assume invariance under both  $T$  and  $S$ . By (10.37) the lattice is even, hence integral, hence  $\Gamma \subseteq \Gamma^*$ . Compare the constant terms,  $q^0 \bar{q}^0$ , of  $\Theta_\Gamma$  and  $\Theta_{\Gamma^*}/\text{vol}(\Gamma)$ . Only the zero vector has  $p_L = p_R = 0$ , so the constant terms are 1 and  $1/\text{vol}(\Gamma)$ . Thus  $\text{vol}(\Gamma) = 1$ , the Gram determinant is  $\pm 1$ , and  $\Gamma = \Gamma^*$ .

**Theorem 10.7 ([L]).** *Let  $\Gamma \subset \mathbb{R}^{d,d}$  be a lattice of rank  $2d$ . The partition function (10.34) is invariant under  $SL(2, \mathbb{Z})$  if and only if  $\Gamma$  is even and self-dual.*

**Physical picture: what the two conditions mean**

Evenness says that every operator of the theory has integer spin, so that correlation functions are single valued on the worldsheet. Self-duality says that the set of charges is as large as it can be: the dual lattice is the set of all charges whose vertex operators are mutually local with those of  $\Gamma$ , and  $\Gamma^* = \Gamma$  means that none of them has been left out. A proper sublattice of  $\Gamma^{d,d}$ , for instance “momentum states only”, passes the  $T$  test and fails the  $S$  test. A point particle on a torus has only momentum states. Modular invariance, a purely stringy requirement, forces the winding states to be there. This is the sharpest form of the statement of Chapter 7 that a closed string cannot be compactified without winding sectors.

## 10.6 The moduli space

### 10.6.1 The generalized metric is an element of $O(d, d; \mathbb{R})$

The group  $O(d, d; \mathbb{R})$  consists of the real  $2d \times 2d$  matrices preserving the form  $\eta$ ,

$$g^\top \eta g = \eta, \quad g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \iff a^\top c + c^\top a = 0, \quad b^\top d + d^\top b = 0, \quad a^\top d + c^\top b = \mathbf{1} \quad (10.41)$$

[GPR §2.4, eqs. (2.4.13)–(2.4.15), ll. 1211–1242]. (In the block matrix,  $d$  is a  $d \times d$  block; the clash of letters is traditional.) Its dimension is found by counting:  $g$  has  $4d^2$  entries, and  $g^\top \eta g = \eta$  is an equation between symmetric matrices, hence  $d(2d + 1)$  conditions, leaving

$$\dim O(d, d; \mathbb{R}) = 4d^2 - d(2d + 1) = d(2d - 1). \quad (10.42)$$

The generalized metric (10.16) is itself such a matrix. The quickest way to see it is to factorise. With the vielbein of (10.17) define

$$g_E = \begin{pmatrix} \mathbf{1} & B \\ 0 & \mathbf{1} \end{pmatrix} \begin{pmatrix} e^\top & 0 \\ 0 & e^{-1} \end{pmatrix} = \begin{pmatrix} e^\top & B e^{-1} \\ 0 & e^{-1} \end{pmatrix}. \quad (10.43)$$

Both factors satisfy (10.41): for the first,  $a = d = \mathbf{1}$ ,  $c = 0$  and  $b^\top d + d^\top b = B^\top + B = 0$ ; for the second,  $b = c = 0$  and  $a^\top d = e e^{-1} = \mathbf{1}$ . Multiplying out,

$$g_E g_E^\top = \begin{pmatrix} e^\top e + B e^{-1} e^{-\top} B^\top & B e^{-1} e^{-\top} \\ e^{-1} e^{-\top} B^\top & e^{-1} e^{-\top} \end{pmatrix} = \begin{pmatrix} G - B G^{-1} B & B G^{-1} \\ -G^{-1} B & G^{-1} \end{pmatrix} = \mathcal{H}(G, B) \quad (10.44)$$

[GPR §2.4, eqs. (2.4.20)–(2.4.23), ll. 1294–1325; GPR write  $G = e e^\top$ , which exchanges  $e$  and  $e^\top$ ]. Since  $O(d, d; \mathbb{R})$  is closed under transposition [GPR §2.4, ll. 1243–1248],  $\mathcal{H} = g_E g_E^\top$  belongs to it:

$$\mathcal{H} \eta \mathcal{H} = \eta, \quad \text{equivalently} \quad (\eta \mathcal{H})^2 = \mathbf{1}. \quad (10.45)$$

The factorisation (10.43) has a plain reading: every toroidal background is obtained from the square torus at the self-dual radius with no  $B$ -field,  $E = \mathbf{1}$ , by a linear change of frame  $e \in GL(d, \mathbb{R})$  followed by a shift of  $B$ . The counting agrees:  $d^2$  parameters in  $e$ , plus  $d(d-1)/2$  in  $B$ , minus  $d(d-1)/2$  for the rotations  $e \rightarrow Oe$  that do not change  $G$ , is  $d^2$ .

### 10.6.2 Planes in $\mathbb{R}^{d,d}$ and the coset

Because of (10.45) the matrices

$$P_{\pm} = \frac{1}{2}(\mathbf{1} \pm \eta\mathcal{H}) \quad (10.46)$$

are complementary projectors,  $P_{\pm}^2 = P_{\pm}$ ,  $P_+P_- = 0$ ,  $P_+ + P_- = \mathbf{1}$ . Proposition 10.2 says what they project on:

$$p_L^2 = \frac{1}{2}Z^T(\mathcal{H} + \eta)Z = Z^T\eta P_+Z, \quad p_R^2 = \frac{1}{2}Z^T(\mathcal{H} - \eta)Z = -Z^T\eta P_-Z. \quad (10.47)$$

So  $P_+$  keeps the left-moving part of a charge vector and  $P_-$  the right-moving part. The image  $\Pi$  of  $P_+$  is a  $d$ -dimensional subspace of  $\mathbb{R}^{d,d}$  on which the form  $\eta$  is positive definite, the image of  $P_-$  is its orthogonal complement  $\Pi^{\perp}$ , on which  $\eta$  is negative definite, and  $\mathcal{H} = \eta(P_+ - P_-)$  is recovered from  $\Pi$ . We arrive at the cleanest description of a toroidal compactification [Asp §3.5, ll. 1989–2006]:

*Fix the lattice  $\Gamma^{d,d} = \mathbb{Z}^{2d}$  with the form  $\eta$  once and for all. A background is a choice of a positive definite  $d$ -plane  $\Pi \subset \mathbb{R}^{d,d}$ . The left and right momenta of a charge vector are its orthogonal projections on  $\Pi$  and  $\Pi^{\perp}$ .*

The group  $O(d, d; \mathbb{R})$  acts on the set of positive  $d$ -planes, and the action is transitive (any orthonormal basis of a positive plane, completed by an orthonormal basis of its complement, is mapped to any other such basis by an element of the group). The subgroup that maps a given  $\Pi$  to itself consists of the independent rotations of  $\Pi$  and of  $\Pi^{\perp}$ , that is, of  $p_L$  and of  $p_R$ , which change neither  $p_L^2$  nor  $p_R^2$ . Hence

**Theorem 10.8 ([L]).** *The moduli space of toroidal backgrounds is locally*

$$\mathcal{M}_d = \frac{O(d, d; \mathbb{R})}{O(d; \mathbb{R}) \times O(d; \mathbb{R})}, \quad \dim \mathcal{M}_d = d(2d - 1) - 2 \cdot \frac{d(d - 1)}{2} = d^2, \quad (10.48)$$

and  $E = G + B \mapsto \mathcal{H}(G, B)$  is a parametrisation of it.

[GPR §2.4, ll. 1207–1210 and 1259–1266]. The dimension agrees with (10.5), as it must. A third way to get  $d^2$ : a plane near  $\Pi$  is the graph of a linear map  $\Pi \rightarrow \Pi^{\perp}$ , a  $d \times d$  matrix. This is how the moduli appear in the spectrum. The states  $\alpha_{-1}^i \tilde{\alpha}_{-1}^j |0\rangle$  with both oscillators along the torus have  $N = \tilde{N} = 1$ ,  $n = m = 0$ , hence  $M^2 = 0$  by (10.24); they are  $d^2$  massless scalars in the uncompactified dimensions, and their vertex operators are the  $d^2$  marginal operators  $\partial X^i \bar{\partial} X^j$  that change  $E_{ij}$ . For  $d = 1$  the coset is the half-line of Figure 10.2; for  $d = 2$  it has four dimensions and factorises into two copies of the upper half-plane, the subject of Chapter 12.

#### Common pitfall: “locally”

Two different planes can give the same set of pairs  $(p_L^2, p_R^2)$ , and then the same physics. If  $g$  is an *integer* matrix in  $O(d, d; \mathbb{R})$ , it maps the lattice  $\mathbb{Z}^{2d}$  to itself, and the backgrounds  $\mathcal{H}$  and  $g^T \mathcal{H} g$  differ only by the relabelling  $Z \rightarrow gZ$  of the charges:  $Z^T(g^T \mathcal{H} g)Z = (gZ)^T \mathcal{H}(gZ)$ . The true moduli space is the quotient  $O(d, d; \mathbb{Z}) \backslash O(d, d; \mathbb{R}) / (O(d) \times O(d))$  [Asp §3.5, eq. (77), ll. 1983–1986]. The relabelling statement, with the bijection made explicit, is the Lean theorem `spectrum_equivalence` quoted below; the group  $O(d, d; \mathbb{Z})$  is the subject of Chapter 11. GPR write the action as  $\mathcal{H} \rightarrow g \mathcal{H} g^T$  [GPR §2.4, eq. (2.4.19), ll. 1288–1293]; the two conventions differ by  $g \leftrightarrow g^T$ , which is harmless because the group is closed under transposition.

**Historical note**

GPR attribute the even self-dual momentum lattice to Narain (their ref. [236], 1986), the coset (10.48) to Narain and to Narain, Sarmadi and Witten (ref. [238], 1987), and the matrix (10.16) to Giveon, Rabinovici and Veneziano (ref. [146], 1989) [GPR §2.4, ll. 1207–1210, 1249–1258, 1267–1274; refs. ll. 10712, 10822–10824].

**10.7 Worked examples**

**Example 10.9** (The  $SU(3) \times SU(3)$  point of  $T^2$ ). Take  $d = 2$  and the background matrix

$$E = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad G = \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & \frac{1}{2} \\ -\frac{1}{2} & 0 \end{pmatrix} \quad (10.49)$$

[GPR §2.4.2, eq. (2.4.64), ll. 2026–2036]. Geometrically, the two lattice vectors have length  $\sqrt{\alpha'}$  and make an angle of  $60^\circ$  ( $\cos \theta = G_{12}/\sqrt{G_{11}G_{22}} = \frac{1}{2}$ ), and half a unit of  $B$ -flux threads the torus. We compute

$$G^{-1} = \frac{1}{3} \begin{pmatrix} 4 & -2 \\ -2 & 4 \end{pmatrix}, \quad BG^{-1} = \frac{1}{3} \begin{pmatrix} -1 & 2 \\ -2 & 1 \end{pmatrix}, \quad G - BG^{-1}B = \frac{1}{3} \begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix},$$

so that

$$\mathcal{H} = \frac{1}{3} \begin{pmatrix} 4 & 2 & -1 & 2 \\ 2 & 4 & -2 & 1 \\ -1 & -2 & 4 & -2 \\ 2 & 1 & -2 & 4 \end{pmatrix}, \quad \det \mathcal{H} = 1. \quad (10.50)$$

Consider the state  $(n_1, n_2; m^1, m^2) = (1, 0; 1, 0)$ . Then  $E^\top m = (1, 1)$ ,  $Em = (1, 0)$ , so  $n + E^\top m = (2, 1)$  and  $n - Em = (0, 0)$ . Hence  $p_R = 0$  and

$$p_L^2 = \frac{1}{2} (2, 1) G^{-1} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \frac{1}{2} \cdot \frac{1}{3} (16 - 8 + 4) = 2,$$

consistent with  $p_L^2 - p_R^2 = 2n \cdot m = 2$ . By (10.25) the state needs  $N - \tilde{N} = 1$ ; with  $N = 1$ ,  $\tilde{N} = 0$  the mass formula (10.23) gives  $\frac{\alpha'}{2} M^2 = p_R^2 + 2(N - 1) = 0$ . If the oscillator  $\alpha_{-1}^\mu$  points along the uncompactified directions this is a *massless vector* carrying winding and momentum.

Enumerating all integers with  $|n_i|, |m^i| \leq 3$  (a finite computer search; larger integers have larger  $p_L^2 + p_R^2$  because  $\mathcal{H}$  is positive definite) gives Table 10.1. There are exactly six lattice vectors with  $(p_L^2, p_R^2) = (2, 0)$ . In the orthonormal frame  $e = \begin{pmatrix} 1 & 1/2 \\ 0 & \sqrt{3}/2 \end{pmatrix}$  their left momenta are

$$p_L = \pm\sqrt{2}(1, 0), \quad \pm\sqrt{2}\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right), \quad \pm\sqrt{2}\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right),$$

the vertices of a regular hexagon of radius  $\sqrt{2}$ : the root system of  $SU(3)$ . Together with the two vectors  $\alpha_{-1}^\mu \tilde{\alpha}_{-1}^i |0\rangle$  that exist at every point of the moduli space, they fill out the  $6 + 2 = 8$  gauge bosons of  $SU(3)$ , and the six vectors with  $(p_L^2, p_R^2) = (0, 2)$  give a second  $SU(3)$ . This generalises the  $SU(2) \times SU(2)$  of the self-dual radius (Chapter 8). For comparison, at  $E = \mathbf{1}$  the same search finds four vectors of each kind,  $\pm(1, 0; 1, 0)$ ,  $\pm(0, 1; 0, 1)$  and  $\pm(1, 0; -1, 0)$ ,  $\pm(0, 1; 0, -1)$ : the gauge group is  $SU(2)^2 \times SU(2)^2$  [GPR §2.4.2, ll. 2019–2026]. That the extra vectors are non-abelian gauge bosons is a statement about interactions ([L]); the counting supports it but does not prove it.



Table 10.1: The shortest vectors of  $\Gamma^{2,2}$  at the point (10.49) (exhaustive search over  $|n_i|, |m^i| \leq 3$ ). Last column: lightest level-matched state, from  $\frac{\alpha'}{2} M^2 = \frac{1}{2}(p_L^2 + p_R^2) + N + \tilde{N} - 2$ .

| $(p_L^2, p_R^2)$             | number | $n \cdot m$ | examples $(n_1, n_2; m^1, m^2)$     | lightest $\frac{\alpha'}{2} M^2$ |
|------------------------------|--------|-------------|-------------------------------------|----------------------------------|
| $(0, 0)$                     | 1      | 0           | $(0, 0; 0, 0)$                      | $-2$                             |
| $(\frac{2}{3}, \frac{2}{3})$ | 18     | 0           | $(1, 0; 0, 0),$<br>$(0, 1; -1, 0)$  | $-\frac{4}{3}$                   |
| $(2, 0)$                     | 6      | 1           | $(1, 0; 1, 0),$<br>$(0, 1; -1, 1)$  | $0 \quad (N = 1, \tilde{N} = 0)$ |
| $(0, 2)$                     | 6      | $-1$        | $(1, 0; -1, 1),$<br>$(0, 1; 0, -1)$ | $0 \quad (N = 0, \tilde{N} = 1)$ |

## 10.8 What the machine has checked

Four Lean files bear on this chapter, written in two independent “streams” of the project that meet here. None of them defines a string, a worldsheet or a partition function: they contain the arithmetic core of Theorem 10.6, statements about the integers  $(n, w)$  and about explicit integer matrices. All depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`.

### 10.8.1 Stream 1: the rank-two lattice from the momenta

The file `StringTheoryFormalization/UseCases/NarainLattice.lean` works over  $\mathbb{Q}$  with  $\alpha' = 1$  and uses the unnormalised momenta of Chapter 7 (no factor  $1/\sqrt{2}$ ), which it compensates by a factor  $\frac{1}{2}$  in the norm:

```
def pL (n w R : ℚ) : ℚ := n / R + w * R
def pR (n w R : ℚ) : ℚ := n / R - w * R
def narainNorm (n w R : ℚ) : ℚ := (pL n w R ^ 2 - pR n w R ^ 2) / 2

theorem narain_norm_R_independent (n w R : ℚ) (hR : R ≠ 0) :
  narainNorm n w R = 2 * n * w := by ...
```

#### In Lean: the norm forgets the radius

**Reading.** For all rationals  $n, w$  and every nonzero rational  $R$ ,  $\frac{1}{2}[(n/R + wR)^2 - (n/R - wR)^2] = 2nw$ . This is (10.21) for  $d = 1, B = 0$ , with rational data; here  $n$  and  $w$  are not even required to be integers. **Proof idea.** Unfold the definitions, clear the denominator  $R$  with `field_simp` (this is where  $R \neq 0$  is used), and close the polynomial identity with `ring`. **What it is not.** Nothing is said about  $d > 1, B \neq 0$  or irrational radii; Proposition 10.2 in that generality is proved on the page, not in Lean.

The file then *forgets* the momenta and keeps the integer quadratic form:

```
def Q (n w : ℤ) : ℤ := 2 * n * w
theorem narain_form_even (n w : ℤ) : Even (Q n w) := ⟨n * w, by unfold Q; ring⟩

def B (n w n' w' : ℤ) : ℤ := n * w' + n' * w
theorem B_is_polarization_of_Q (n w n' w' : ℤ) :
  2 * B n w n' w' = Q (n + n') (w + w') - Q n w - Q n' w' := by ...

def gram : Matrix (Fin 2) (Fin 2) ℤ :=
  !![B 1 0 1 0, B 1 0 0 1; B 0 1 1 0, B 0 1 0 1]
```



```
theorem gram_eq_hyperbolic : gram = !![0, 1; 1, 0] := by ...
theorem narain_gram_unimodular : gram.det = -1 := by ...
```

#### In Lean: even, and Gram determinant -1

**Reading.** (i) For all integers  $n, w$  the integer  $2nw$  is even; Mathlib’s **Even a** means  $\exists r, a = r + r$ , and the witness supplied is  $r = nw$ . (ii) The bilinear form  $B$  (no relation to the Kalb–Ramond field; the name is local to this file) is the polarization of  $Q$ ; this is (10.28) for  $d = 1$ . (iii) The Gram matrix of  $B$  in the basis  $(1, 0), (0, 1)$  is the matrix  $U$  of (10.29), and the determinant of this one matrix is  $-1$ . **Proof idea.** `ring` for (i) and (ii); `norm_num` evaluates the four entries; `Matrix.det_fin_two` is the  $2 \times 2$  determinant formula. **What it is not.** `narainNorm` and  $Q$  are linked only by the shared expression  $2nw$ ; no Lean theorem says “the set of pairs  $(p_L, p_R)$  is a lattice isometric to  $U$ ”, and the step from  $\det = -1$  to  $\Gamma = \Gamma^*$  is Lemma 10.4, proved on the page.

### 10.8.2 Stream 2: Gram matrices, a certificate lemma and the hyperbolic plane

`DualScaleStream2/Lattice/Basic.lean` represents a lattice of rank  $n$  by its Gram matrix alone and defines the two properties of this chapter as predicates on it:

```
abbrev Gram (n : ℕ) := Matrix (Fin n) (Fin n) ℤ
def IsEvenDiag {n : ℕ} (G : Gram n) : Prop := ∀ i, Even (G i i)
def IsUnimodular {n : ℕ} (G : Gram n) : Prop := G.det = 1 ∨ G.det = -1

theorem isUnimodular_of_mul_eq_one {n : ℕ} (G H : Gram n) (h : H * G = 1) :
  IsUnimodular G := by ...

theorem even_quadratic_form_of_even_diag {n : ℕ} (G : Gram n)
  (hsymm : GT = G) (heven : IsEvenDiag G) (x : Fin n → ℤ) :
  Even (x ·v (G *v x)) := by ...
```

#### In Lean: two general lemmas about integer Gram matrices

**Reading.** Both hold for every rank  $n$ . The first says that an integer matrix with an integer left inverse has determinant  $\pm 1$ ; it is one half of Lemma 10.4, and Stream 2 discharges every unimodularity claim this way, by exhibiting an inverse. The second says that a symmetric integer matrix with even diagonal gives an even number  $x^T G x$  for *every* integer vector  $x$ : evenness need only be checked on a basis. **Proof idea.** For the first,  $\det H \cdot \det G = \det \mathbf{1} = 1$  in  $\mathbb{Z}$ , and the only ways to write 1 as a product of two integers are  $1 \cdot 1$  and  $(-1)(-1)$ . For the second, split  $G = D + T + T^T$  into its diagonal and strictly upper triangular parts; then  $x^T G x = \sum_i G_{ii} x_i^2 + 2 x^T T x$  and both terms are even. **What it is not.** `IsUnimodular` is a statement about a determinant. The equivalence with  $\Gamma^* = \Gamma$  is not formalised, because the dual lattice is not defined in these files.

`DualScaleStream2/Lattice/Hyperbolic.lean` then posits the hyperbolic plane as a literal matrix and checks its properties entry by entry:

```
def hyperbolicU : Gram 2 := !![0, 1; 1, 0]

theorem hyperbolicU_symm : hyperbolicUT = hyperbolicU := by ...
theorem hyperbolicU_evenDiag : IsEvenDiag hyperbolicU := by ...
theorem hyperbolicU_mul_self : hyperbolicU * hyperbolicU = 1 := by ...
theorem hyperbolicU_unimodular : IsUnimodular hyperbolicU :=
  isUnimodular_of_mul_eq_one _ _ hyperbolicU_mul_self
```

```

theorem hyperbolicU_congruence :
  (!! [1, 1; 1, -1] : Gram 2)T * hyperbolicU * !! [1, 1; 1, -1]
  = !! [2, 0; 0, -2] := by ...

theorem hyperbolicU_eq_narain_gram :
  hyperbolicU = StringTheory.UseCases.NarainLattice.gram := by ...

```

**In Lean:**  $U$  is even and unimodular, and the two streams agree

**Reading.** The matrix  $U$  is symmetric, has even diagonal, squares to the identity and is therefore unimodular by the certificate lemma; with `even_quadratic_form_of_even_diag`,  $x^T U x$  is even for all  $x \in \mathbb{Z}^2$ . In the congruence  $P^T U P = \text{diag}(2, -2)$  the columns of  $P$  are the charge vectors  $(1, 1)$  and  $(1, -1)$  of norm  $\pm 2$ , the special points of Figure 10.2; concluding “signature  $(1, 1)$ ” from it uses Sylvester’s law of inertia, which is [L] here. The last theorem says that the matrix written down in Stream 2 and the matrix *computed* in Stream 1 by polarizing  $Q(n, w) = 2nw$  are equal. **Proof idea.** There are only four entries; `fin_cases` enumerates them and each case closes by `rfl`, `simp` or `norm_num`. **Why the bridge matters.** The  $U$  that enters the K3 lattice  $3U \oplus 2E_8(-1)$  in Chapter 20 is, by a kernel-checked equality, the momentum-winding Gram matrix of one circle, not merely a matrix that looks the same in two files.

### 10.8.3 Stream 2: general $d$

For arbitrary  $d$  the relevant declarations are in `DualScaleStream2/TDuality/Factorized.lean` and `Spectrum.lean`. The index type of a doubled charge vector is `Charge d`, defined as `Fin d  $\oplus$  Fin d`, and `eta d` is the block matrix from `Blocks 0 1 1 0`, the  $\eta$  of (10.21).

```

def chargeNorm (Z : Charge d  $\rightarrow$   $\mathbb{Z}$ ) :  $\mathbb{Z}$  := Z  $\cdot_v$  (eta d  $\cdot_v$  Z)

theorem chargeNorm_sumElim (n w : Fin d  $\rightarrow$   $\mathbb{Z}$ ) :
  chargeNorm (Sum.elim n w) = 2 * (n  $\cdot_v$  w) := by ...
theorem chargeNorm_even (Z : Charge d  $\rightarrow$   $\mathbb{Z}$ ) : Even (chargeNorm Z) := by ...
theorem chargeNorm_invariant (g : Matrix (Charge d) (Charge d)  $\mathbb{Z}$ )
  (hg : IsODD g) (Z : Charge d  $\rightarrow$   $\mathbb{Z}$ ) :
  chargeNorm (g  $\cdot_v$  Z) = chargeNorm Z := by ...

theorem spectrum_equivalence (g : Matrix (Charge d) (Charge d)  $\mathbb{Z}$ )
  (hg : IsODD g) (H : Matrix (Charge d) (Charge d)  $\mathbb{R}$ ) :
   $\exists e : (\text{Charge } d \rightarrow \mathbb{Z}) \simeq (\text{Charge } d \rightarrow \mathbb{Z}), \forall Z : \text{Charge } d \rightarrow \mathbb{Z},$ 
  massForm ((toReal g)T * H * toReal g) (fun i => (Z i :  $\mathbb{R}$ )) =
  massForm H (fun i => (e Z i :  $\mathbb{R}$ ))  $\wedge$ 
  chargeNorm (e Z) = chargeNorm Z := by ...

```

**In Lean:**  $\Gamma^{d,d}$  is even for every  $d$ ; dual backgrounds match charge by charge

**Reading.** `chargeNorm Z` is  $Z^T \eta Z$ . For every  $d$  and every integer vector  $Z$ , split into its two blocks, it equals  $2n \cdot w$ , hence is even: the evenness half of Theorem 10.6 for all  $d$ . (Since  $\eta$  is symmetric it does not matter which block is called momentum; the docstrings say  $Z = (n, w)$ , while `massForm` is used with windings first, as in (10.16).) `IsODD g` is the equation  $g^T \eta g = \eta$  for an integer matrix. `spectrum_equivalence` says: for every such  $g$  and *every* real matrix  $H$ , not only a generalized metric, there is a bijection  $e$  of  $\mathbb{Z}^{2d}$ , namely  $Z \mapsto gZ$ , with  $Z^T (g^T H g) Z = (eZ)^T H (eZ)$  and equal charge norms: the content of the pitfall box “locally” in Section 10.6. **Proof idea.** The inverse of  $g$  is the integer matrix  $\eta g^T \eta$  (`isODD_left_inv`, `isODD_right_inv`), which gives the bijection; the mass identity

Table 10.2: Epistemic status of the claims of this chapter.

| Claim                                                                                                | Tier | Evidence                                                                                                                                             |
|------------------------------------------------------------------------------------------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\frac{1}{2}(p_L^2 - p_R^2) = 2nw$ for $d = 1$ , $B = 0$ , rational $n, w, R$ , unnormalised momenta | [A]  | <code>narain_norm_R_independent</code>                                                                                                               |
| $2nw$ is even; its polarization; Gram matrix $= U$ ; $\det U = -1$                                   | [A]  | <code>narain_form_even</code> ,<br><code>B_is_polarization_of_Q</code> ,<br><code>gram_eq_hyperbolic</code> ,<br><code>narain_gram_unimodular</code> |
| $U$ symmetric, even diagonal, $U^2 = \mathbf{1}$ , unimodular, congruent to $\text{diag}(2, -2)$     | [A]  | <code>hyperbolicU_symm</code> , <code>_evenDiag</code> ,<br><code>_mul_self</code> , <code>_unimodular</code> ,<br><code>_congruence</code>          |
| Stream 2's $U$ equals Stream 1's Gram matrix                                                         | [A]  | <code>hyperbolicU_eq_narain_gram</code>                                                                                                              |
| Integer inverse $\Rightarrow \det = \pm 1$ ; even diagonal $\Rightarrow$ even form (any rank)        | [A]  | <code>isUnimodular_of_mul_eq_one</code> ,<br><code>even_quadratic_form_of_even_diag</code>                                                           |
| $Z^T \eta Z = 2n \cdot w$ is even and $O(d, d; \mathbb{Z})$ -invariant, all $d$                      | [A]  | <code>chargeNorm_sumElim</code> ,<br><code>chargeNorm_even</code> ,<br><code>chargeNorm_invariant</code>                                             |
| $H$ and $g^T H g$ have the same zero-mode spectrum, all $d$                                          | [A]  | <code>spectrum_equivalence</code>                                                                                                                    |
| Proposition 10.2 (general $d, G, B$ ); Lemma 10.4                                                    | —    | proved on the page; the former also checked symbolically for $d = 2$                                                                                 |
| Signature of $U$ is $(1, 1)$ (Sylvester)                                                             | [L]  | quoted in <code>Hyperbolic.lean</code>                                                                                                               |
| Modular invariance $\Leftrightarrow$ even self-dual (Theorem 10.7)                                   | [L]  | Asp §3.5; derivation on the page                                                                                                                     |
| Moduli space $O(d, d; \mathbb{R})/O(d) \times O(d)$ ; uniqueness of $\Gamma^{d,d}$                   | [L]  | GPR §2.4; Asp §2.3                                                                                                                                   |
| Identification of $(\mathbb{Z}^2, U)$ with the charge lattice of a physical string                   | —    | modelling step; never Tier A                                                                                                                         |

is associativity of matrix multiplication after casting  $\mathbb{Z} \rightarrow \mathbb{R}$  (`massForm_covariant`); the norm identity is  $g^T \eta g = \eta$ . **What it is not.** Only the zero-mode quadratic form is concerned: no oscillators, no partition function. Unimodularity of  $\eta$  for general  $d$  would follow from `eta_mul_self` ( $\eta^2 = \mathbf{1}$ ) by the certificate argument, but is not stated as an `IsUnimodular` theorem, because `Gram n` is indexed by `Fin n` and `eta d` by `Fin d ⊕ Fin d` (Exercise 10.7).

#### 10.8.4 Tier table and what a faithful formalization would need

Table 10.2 collects the status of every claim of this chapter.

A faithful formalization of this chapter would need, in increasing order of difficulty: (i) Proposition 10.2 over  $\mathbb{R}$  with a positive definite  $G$  and its inverse, within reach of Mathlib's matrix library; (ii) lattices as  $\mathbb{Z}$ -submodules of a real quadratic space with their duals, so that “self-dual” means  $\Gamma^* = \Gamma$  rather than  $\det = \pm 1$ ; (iii) the theta series (10.34) and Poisson summation in indefinite signature, for Theorem 10.7; (iv) the coset (10.48) as a manifold. Items (iii) and (iv) are research-level projects (Chapter 38).

##### Chapter summary

- A torus  $T^d$  with constant  $G$  and  $B$  has  $d^2$  parameters  $E = G + B$ . The canonical momentum is quantized,  $p_i = n_i \in \mathbb{Z}$ , the velocity is  $G^{-1}(n - Bm)$ :  $B$  acts on winding strings like a theta angle, and integer shifts of  $B$  are invisible.

- $p_L = \frac{1}{\sqrt{2}}e^{-\top}(n + E^\top m)$ ,  $p_R = \frac{1}{\sqrt{2}}e^{-\top}(n - Em)$  with  $G = e^\top e$ . Two identities hold:  $p_L^2 + p_R^2 = Z^\top \mathcal{H} Z$  (energy, depends on the background) and  $p_L^2 - p_R^2 = 2n \cdot m = Z^\top \eta Z$  (spin, does not).
- $M^2 = \frac{1}{\alpha'} Z^\top \mathcal{H} Z + \frac{2}{\alpha'}(N + \tilde{N} - 2)$  and  $N - \tilde{N} = n \cdot m$ .
- The vectors  $(p_L, p_R)$  form the Narain lattice  $\Gamma^{d,d} \cong U^{\oplus d}$ : even, self-dual, signature  $(d, d)$ , Gram matrix  $\eta$  for every background. Only its embedding in  $\mathbb{R}_L^d \oplus \mathbb{R}_R^d$  varies; for  $d = 1$  varying  $R$  is a Lorentz boost.
- Modular invariance requires exactly this:  $T$  forces evenness (integer spin),  $S$  forces self-duality (Poisson resummation). A lattice of momentum states alone fails.
- A background is a positive  $d$ -plane in  $\mathbb{R}^{d,d}$ ; the moduli space is locally  $O(d, d; \mathbb{R})/O(d) \times O(d)$ , of dimension  $d^2$ , and globally a quotient by  $O(d, d; \mathbb{Z})$ .
- Lean has checked the rank-two lattice in two independent ways and that they agree, two general lemmas on integer Gram matrices, and evenness, invariance and the spectrum bijection for all  $d$ ; partition functions and the coset are not formalised.

## Exercises

**Exercise 10.1** (★). Show that  $\dim O(d, d; \mathbb{R}) = d(2d - 1)$  by linearising  $g = \mathbf{1} + \epsilon X$  in (10.41): the condition becomes  $(\eta X)^\top = -\eta X$ . Deduce (10.48), and list  $\dim \mathcal{M}_d$  for  $d = 1, 2, 4, 6$ . Which of these numbers is the number of massless scalars  $\alpha_{-1}^i \tilde{\alpha}_{-1}^j |0\rangle$ ?

**Exercise 10.2** (★). Derive (10.12) from (10.4) and (10.7), and then the completed square (10.26). Use it to show that  $B \rightarrow B + \Theta$ ,  $n \rightarrow n + \Theta m$  leaves  $H_0$  and  $n \cdot m$  invariant for every antisymmetric integer matrix  $\Theta$ . For  $d = 2$ ,  $G = \mathbf{1}$ ,  $B_{12} = b$ , follow the states  $(n; m) = (0, 1; 1, 0)$  and  $(0, 0; 1, 0)$  from  $b = 0$  to  $b = -1$  and show that they exchange their energies 1 and  $\frac{1}{2}$ . Write the  $4 \times 4$  matrix  $g$  acting on  $Z = (m, n)$  and check (10.41).

**Exercise 10.3** (★★). Prove  $\mathcal{H}\eta\mathcal{H} = \eta$  by block multiplication, without using the factorisation (10.43). Then show that  $P_\pm$  of (10.46) are projectors of rank  $d$  (hint:  $\text{tr}(\eta\mathcal{H}) = \text{tr}(BG^{-1}) - \text{tr}(G^{-1}B) = 0$ ) and that their images are orthogonal with respect to  $\eta$ .

**Exercise 10.4** (★★). At  $E = \mathbf{1}$  for  $d = 2$ , find all  $(n; m)$  with  $(p_L^2, p_R^2) = (2, 0)$  and all with  $(1, 1)$ , and compute the mass of the lightest level-matched state in each class. Compare with Table 10.1. Then move to  $G = \mathbf{1}$ ,  $B_{12} = \frac{1}{2}$ : do any of the  $(2, 0)$  vectors survive?

**Exercise 10.5** (★★). For an integer  $k \geq 2$  let  $\Gamma_k = \{(n, m) : m \in k\mathbb{Z}\} \subset \Gamma^{1,1}$  (only windings divisible by  $k$ ). Compute its Gram matrix,  $\text{vol}(\Gamma_k)$  and  $\Gamma_k^*$ . Which condition of Theorem 10.7 fails, what is  $Z_{\Gamma_k}(-1/\tau)$  according to (10.40), and which vectors of  $\Gamma_k^*$  violate evenness?

**Exercise 10.6** (★★, Lean). In the namespace of `NarainLattice.lean` state and prove: (a)  $\mathbb{Q} \mathbf{n} \mathbf{w} = \mathbb{Q} \mathbf{w} \mathbf{n}$ , the invariance of the norm under  $n \leftrightarrow w$ ; (b) the symmetry  $\mathbb{B} \mathbf{n} \mathbf{w} \mathbf{n}' \mathbf{w}' = \mathbb{B} \mathbf{n}' \mathbf{w}' \mathbf{n} \mathbf{w}$ ; (c)  $\mathbb{Q} \mathbf{n} \mathbf{w} = \mathbb{B} \mathbf{n} \mathbf{w} \mathbf{n} \mathbf{w}$ . Each proof is one line: `unfold` followed by `ring`.

**Exercise 10.7** (★★★, Lean). Define the Gram matrix of  $U \oplus U$  as a literal `Gram 4` in the ordering  $(m^1, m^2, n_1, n_2)$ , so that it equals  $\eta$  for  $d = 2$ . Prove that it is unimodular using `isUnimodular_of_mul_eq_one` with the matrix itself as the inverse, and that  $x^\top G x$  is even for all  $x$  using `even_quadratic_form_of_even_diag`. Then explain what would be needed to state the same result for `eta d` with  $d$  arbitrary (hint: Mathlib's `Matrix.reindex` along `finSumFinEquiv`, an equivalence  $\text{Fin } d \oplus \text{Fin } d \simeq \text{Fin } (d + d)$ ).

## Further reading

- Toroidal backgrounds,  $p_{L,R}$ , the matrix  $M = \mathcal{H}$  and the coset: [GPR §2.4, ll. 1031–1345]; the circle, its partition function and the dimensionless metric: [GPR §2.2, ll. 576–821]; oscillators under duality: [GPR §2.4, ll. 1423–1520]; enhanced symmetry: [GPR §2.4.1–2.4.2, ll. 1836–1866, 2019–2036].
- The Grassmannian picture and modular invariance: [Asp §3.5, ll. 1965–2011]; even self-dual lattices: [Asp §2.3, ll. 541–566]. Toroidal duality among abelian and non-abelian dualities, and the caveat that duals of momentum operators are winding operators only for flat compact isometries: [AAL §1, ll. 105–135; §2.1, ll. 605–612].
- In this book: the lattice theory is developed in Chapter 18; the discrete group  $O(d, d; \mathbb{Z})$  and its generators in Chapter 11; the case  $d = 2$  in Chapter 12; the generalized metric as a field in Chapter 15; the same coset with signature  $(4, 20)$  in Chapter 23.



# Chapter 11

## The Duality Group $O(d, d; \mathbb{Z})$

On a circle, the closed string cannot tell the radius  $R$  from  $\alpha'/R$  (Chapter 8). On a  $d$ -dimensional torus the background is a metric  $G$  and a two-form  $B$ , together  $d^2$  real numbers (Chapter 10), and the corresponding question is: *which different pairs  $(G, B)$  describe the same string physics?* The answer is a group, the *duality group*  $O(d, d; \mathbb{Z})$  of integer  $2d \times 2d$  matrices preserving the pairing between momentum and winding. This chapter defines the group, shows why its defining condition is forced by level matching, describes three families of elements with a direct physical meaning, and proves that two backgrounds related by a group element have the same spectrum, state by state. That proof is short, it is complete, and it has been checked in Lean; we give it on the page first and then compare with the formal statement. The student should care for two reasons. First, the moduli space of toroidal compactifications is the quotient of the space of pairs  $(G, B)$  by this group, so every statement about “the space of string vacua on a torus” depends on it. Second, the whole of double field theory (Chapters 14, 15 and 17) is an attempt to make the continuous version of this group manifest in a field theory.

### 11.1 What has to be explained

#### 11.1.1 The data: charges, the generalized metric and the pairing

We use the conventions of Chapter 10. The torus is  $T^d = \mathbb{R}^d/2\pi\mathbb{Z}^d$  with coordinates  $X^i \sim X^i + 2\pi$ ; the metric  $G_{ij}$  (symmetric, positive definite) and the Kalb–Ramond field  $B_{ij}$  (antisymmetric) are constant and dimensionless, lengths being measured in units of  $\sqrt{\alpha'}$ ; and  $E = G + B$  is the *background matrix*. A closed-string state carries  $d$  integer winding numbers  $m^i$  and  $d$  integer momentum numbers  $n_i$ . We collect them in the doubled charge vector

$$Z = \begin{pmatrix} m \\ n \end{pmatrix} \in \mathbb{Z}^{2d}, \quad \text{windings first,} \quad (11.1)$$

which is the ordering of Giveon, Porrati and Rabinovici [GPR §2.4, eq. (2.4.18), ll. 1278–1289] and of the Lean module `DualScaleStream2/DFT/GeneralizedMetric.lean`. Two matrices act on such vectors:

$$\mathcal{H}(G, B) = \begin{pmatrix} G - BG^{-1}B & BG^{-1} \\ -G^{-1}B & G^{-1} \end{pmatrix}, \quad \eta = \begin{pmatrix} 0 & \mathbf{1}_d \\ \mathbf{1}_d & 0 \end{pmatrix}, \quad (11.2)$$

the generalized metric [GPR §2.4, eq. (2.4.17), ll. 1266–1274] and the constant matrix that pairs momentum with winding. In terms of them the mass formula and the level-matching condition

of Chapter 10 read

$$\alpha' M^2 = Z^\top \mathcal{H}(G, B) Z + 2(N + \tilde{N} - 2), \quad N - \tilde{N} = \frac{1}{2} Z^\top \eta Z = n \cdot m, \quad (11.3)$$

where  $N, \tilde{N} \in \{0, 1, 2, \dots\}$  are the oscillator levels of the two chiralities [GPR §2.4, eqs. (2.4.11), (2.4.16), (2.4.33), ll. 1183–1191, 1249–1256, 1459–1463]. The division of labour in (11.3) is the key to the whole chapter: *the background enters only through  $\mathcal{H}$ , and the constraint involves only  $\eta$ , which knows nothing about the background.*

It is useful to have the quadratic form written as a sum of squares. Because  $B^\top = -B$ ,

$$\begin{aligned} (n - Bm)^\top G^{-1} (n - Bm) &= n^\top G^{-1} n - 2n^\top G^{-1} Bm + m^\top B^\top G^{-1} Bm \\ &= n^\top G^{-1} n - 2n^\top G^{-1} Bm - m^\top B G^{-1} Bm, \end{aligned}$$

and the two off-diagonal blocks of  $\mathcal{H}$  contribute  $m^\top B G^{-1} n - n^\top G^{-1} Bm = -2n^\top G^{-1} Bm$  (the first term is a number, equal to its transpose  $n^\top G^{-1} B^\top m$ ). Hence

$$Z^\top \mathcal{H}(G, B) Z = (n - Bm)^\top G^{-1} (n - Bm) + m^\top G m. \quad (11.4)$$

The first term is the kinetic energy of the centre of mass, with the momentum shifted by the  $B$ -field exactly as a vector potential shifts the momentum of a charged particle; the second is the energy stored in the stretched string. Both are positive, so  $\mathcal{H}(G, B)$  is positive definite.

### 11.1.2 What a symmetry must do

Suppose we relabel the charges by an integer matrix  $g$ , that is  $Z \mapsto gZ$ . For the relabelling to lose no state and create none,  $g$  must be a bijection of  $\mathbb{Z}^{2d}$ , so  $g^{-1}$  must also have integer entries. The relabelled state has mass form

$$(gZ)^\top \mathcal{H} (gZ) = Z^\top (g^\top \mathcal{H} g) Z. \quad (11.5)$$

Read from right to left, this says: the numbers  $Z^\top \mathcal{H}' Z$  computed with the matrix  $\mathcal{H}' = g^\top \mathcal{H} g$  are the numbers  $Z^\top \mathcal{H} Z$  in a different order. So the zero-mode energies of the backgrounds  $\mathcal{H}$  and  $\mathcal{H}'$  agree as lists, with multiplicity, for *every* invertible integer  $g$ . This is not yet a symmetry of the string, because the physical states are those that satisfy the level-matching constraint in (11.3), and the constraint must hold for  $gZ$  if and only if it holds for  $Z$  with the same  $N - \tilde{N}$ . We therefore need

$$(gZ)^\top \eta (gZ) = Z^\top \eta Z \quad \text{for all } Z \in \mathbb{Z}^{2d}. \quad (11.6)$$

The matrix  $g^\top \eta g - \eta$  is symmetric, and a symmetric matrix  $S$  with  $Z^\top S Z = 0$  for all integer  $Z$  vanishes: taking  $Z$  a basis vector gives  $S_{II} = 0$ , and taking  $Z$  the sum of two basis vectors gives  $2S_{IJ} = 0$ . Condition (11.6) is therefore equivalent to the matrix equation  $g^\top \eta g = \eta$ . This is the definition of the next section, and we have found it by asking for nothing more than compatibility with level matching.

#### Physical picture: two kinds of information

The pairing  $\eta$  is kinematics. It records that momentum and winding are conjugate charges, and  $\frac{1}{2} Z^\top \eta Z = n \cdot m$  is the amount by which the left- and right-moving oscillator levels must differ. It is the same for all tori. The generalized metric  $\mathcal{H}$  is dynamics: it says how much energy a given charge costs on a given torus. A duality is a relabelling of charges that respects the kinematics. It changes the dynamics from  $\mathcal{H}$  to  $g^\top \mathcal{H} g$ , and the theorem of this chapter is that the change is invisible in the spectrum.



## 11.2 The groups $O(d, d; \mathbb{R})$ and $O(d, d; \mathbb{Z})$

**Definition 11.1.** Let  $\eta$  be the matrix in (11.2). The group  $O(d, d; \mathbb{R})$  is the set of real  $2d \times 2d$  matrices  $g$  with

$$g^\top \eta g = \eta, \quad (11.7)$$

and  $O(d, d; \mathbb{Z})$  is the subset of those with integer entries [GPR §2.4, eqs. (2.4.13)–(2.4.15), ll. 1213–1242, and ll. 1366–1367; AMN §3.1, eq. (3.4), ll. 569–601].

The name comes from the signature of  $\eta$ . For any  $v \in \mathbb{R}^d$  the vectors  $(v, v)$  and  $(v, -v)$  satisfy  $\eta(v, \pm v) = \pm(v, \pm v)$ , so  $\eta$  has  $d$  eigenvalues  $+1$  and  $d$  eigenvalues  $-1$ . In the basis of these eigenvectors  $\eta$  becomes  $\text{diag}(\mathbf{1}_d, -\mathbf{1}_d)$ , and (11.7) is the definition of the pseudo-orthogonal group of signature  $(d, d)$ , the analogue of the Lorentz group  $O(1, 3)$  with  $d$  time-like and  $d$  space-like directions. In the language of Chapter 10 the two eigenspaces are the left-moving and right-moving momenta at the point  $G = \mathbf{1}$ ,  $B = 0$ .

Writing  $g$  in  $d \times d$  blocks and multiplying out,

$$g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} : \quad g^\top \eta g = \begin{pmatrix} a^\top c + c^\top a & a^\top d + c^\top b \\ b^\top c + d^\top a & b^\top d + d^\top b \end{pmatrix}, \quad (11.8)$$

so that (11.7) is the set of three conditions

$$a^\top c + c^\top a = 0, \quad b^\top d + d^\top b = 0, \quad a^\top d + c^\top b = \mathbf{1} \quad (11.9)$$

[GPR §2.4, eq. (2.4.15), ll. 1236–1242]; the fourth block is the transpose of the third. (The letter  $d$  denotes both the dimension and the lower-right block. This is the notation of the literature, and the context always decides.)

**Proposition 11.2** (Group law and inverses). *Let  $g, h$  satisfy (11.7). Then*

- (i)  $\eta^2 = \mathbf{1}$  and  $\eta$  itself satisfies (11.7);
- (ii)  $gh$  satisfies (11.7);
- (iii)  $g$  is invertible, with  $g^{-1} = \eta g^\top \eta$ ; if  $g$  has integer entries, so has  $g^{-1}$ ;
- (iv)  $g^{-1}$  and  $g^\top$  satisfy (11.7); equivalently  $g \eta g^\top = \eta$ ;
- (v)  $\det g = \pm 1$ .

Consequently  $O(d, d; \mathbb{R})$  and  $O(d, d; \mathbb{Z})$  are groups under matrix multiplication.

*Proof.* (i)  $\eta^2 = \mathbf{1}$  is immediate from the block form, and then  $\eta^\top \eta \eta = \eta \eta \eta = \eta$  because  $\eta$  is symmetric. (ii)  $(gh)^\top \eta (gh) = h^\top (g^\top \eta g) h = h^\top \eta h = \eta$ . (iii) Multiply (11.7) on the left by  $\eta$ :  $(\eta g^\top \eta) g = \eta^2 = \mathbf{1}$ . A left inverse of a square matrix is a two-sided inverse, so  $g^{-1} = \eta g^\top \eta$ , whose entries are those of  $g$  rearranged:

$$g^{-1} = \eta g^\top \eta = \begin{pmatrix} d^\top & b^\top \\ c^\top & a^\top \end{pmatrix}. \quad (11.10)$$

(iv) From (iii),  $g(\eta g^\top \eta) = \mathbf{1}$ ; multiplying on the right by  $\eta$  gives  $g \eta g^\top = \eta$ , which is (11.7) for  $g^\top$  [GPR §2.4, ll. 1244–1248]. For the inverse,  $(g^{-1})^\top \eta g^{-1} = (\eta g \eta) \eta (\eta g^\top \eta) = \eta (g \eta g^\top) \eta = \eta^3 = \eta$ . (v) Take the determinant of (11.7):  $(\det g)^2 \det \eta = \det \eta$  and  $\det \eta = (-1)^d \neq 0$ .  $\square$

No separate argument was needed for the integrality of  $g^{-1}$ . For a general integer matrix the inverse involves a division by the determinant; here (11.10) shows the inverse explicitly. This is the step that makes  $Z \mapsto gZ$  a bijection of the charge lattice and not only an injection.

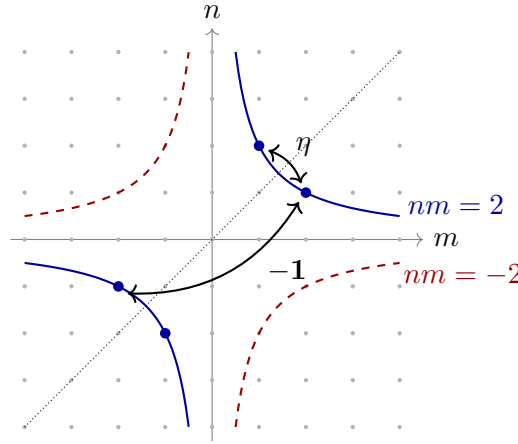


Figure 11.1: The charge lattice of the circle,  $d = 1$ . The level sets of  $\frac{1}{2}Z^\top \eta Z = nm$  are hyperbolas; level matching places a state with  $N - \tilde{N} = 2$  on the solid one. The group  $O(1, 1; \mathbb{Z})$  has four elements and maps each hyperbola to itself:  $\eta$  is the reflection in the dotted diagonal and  $-1$  the reflection in the origin. The marked points are the orbit of  $(m, n) = (2, 1)$ .

**How large is the group?** The real group is a Lie group. Writing  $g = \mathbf{1} + \epsilon X$  in (11.7) and keeping first order in  $\epsilon$  gives  $X^\top \eta + \eta X = 0$ , that is,  $\eta X$  is antisymmetric. An antisymmetric  $2d \times 2d$  matrix has  $d(2d - 1)$  independent entries, so

$$\dim O(d, d; \mathbb{R}) = d(2d - 1) : \quad 1, 6, 15, \dots \quad \text{for } d = 1, 2, 3, \dots \quad (11.11)$$

In blocks,  $X = \begin{pmatrix} \alpha & \beta \\ \gamma & -\alpha^\top \end{pmatrix}$  with  $\alpha$  arbitrary ( $d^2$  parameters) and  $\beta, \gamma$  antisymmetric ( $\frac{1}{2}d(d - 1)$  each). We shall meet the three kinds of block again as the three kinds of finite transformation. The integer group  $O(d, d; \mathbb{Z})$  is discrete; it is finite for  $d = 1$  and infinite for  $d \geq 2$ .

**Example 11.3** (The case  $d = 1$ , completely). For  $d = 1$  the blocks are numbers and (11.9) reads  $2ac = 0$ ,  $2bd = 0$ ,  $ad + bc = 1$ . If  $c = 0$  then  $ad = 1$  and hence  $b = 0$ ; if  $a = 0$  then  $bc = 1$  and hence  $d = 0$ . So

$$O(1, 1; \mathbb{R}) = \left\{ \begin{pmatrix} \lambda & 0 \\ 0 & \lambda^{-1} \end{pmatrix}, \begin{pmatrix} 0 & \lambda \\ \lambda^{-1} & 0 \end{pmatrix} : \lambda \in \mathbb{R} \setminus \{0\} \right\}, \quad (11.12)$$

a one-parameter group with four connected components, in agreement with (11.11). With  $\lambda = e^\theta$  the first family is the Lorentz boost of Chapter 10 in light-cone coordinates. Integer entries force  $\lambda = \pm 1$ :

$$O(1, 1; \mathbb{Z}) = \{\mathbf{1}, -\mathbf{1}, \eta, -\eta\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2. \quad (11.13)$$

Here  $\eta : (m, n) \mapsto (n, m)$  is the exchange of winding and momentum of Chapter 8, and  $-\mathbf{1} : (m, n) \mapsto (-m, -n)$  is the reflection  $X \rightarrow -X$  of the circle. Figure 11.1 shows the four-element orbit of a charge. The duality group of the circle is small; the interesting structure starts at  $d = 2$ .

**Common pitfall:**  $O(d, d)$  is not  $O(2d)$

The condition is  $g^\top \eta g = \eta$ , not  $g^\top g = \mathbf{1}$ . An element of  $O(d, d; \mathbb{R})$  in general changes the Euclidean length  $Z^\top Z$  of a charge vector, and it changes  $Z^\top \mathcal{H} Z$  at fixed  $\mathcal{H}$ ; this is why a generic element of the *real* group maps a background to a physically different

one [GPR §2.4, ll. 1259–1262]. The elements of  $O(d, d; \mathbb{R})$  that are also orthogonal in the ordinary sense form the compact subgroup  $O(d) \times O(d)$  of rotations of left- and right-moving momenta separately, which appeared in Chapter 10 as the denominator of the moduli space.

### 11.3 Dual backgrounds have the same spectrum

We can now state and prove the main theorem. It is formulated for an arbitrary real matrix  $\mathcal{H}$ ; that  $g^\top \mathcal{H} g$  is again the generalized metric of some torus is a separate fact, proved in Section 11.4.

**Theorem 11.4** (Spectrum equivalence). *Let  $g \in O(d, d; \mathbb{Z})$  and let  $\mathcal{H}$  be a real  $2d \times 2d$  matrix. Put  $\mathcal{H}' = g^\top \mathcal{H} g$ . Then the map*

$$e : \mathbb{Z}^{2d} \rightarrow \mathbb{Z}^{2d}, \quad e(Z) = gZ,$$

*is a bijection, and for every  $Z \in \mathbb{Z}^{2d}$*

$$Z^\top \mathcal{H}' Z = e(Z)^\top \mathcal{H} e(Z), \quad e(Z)^\top \eta e(Z) = Z^\top \eta Z. \quad (11.14)$$

*Proof. Bijection.* By Proposition 11.2(iii) the matrix  $g^{-1} = \eta g^\top \eta$  has integer entries, so  $Z \mapsto g^{-1}Z$  maps  $\mathbb{Z}^{2d}$  to itself and is a two-sided inverse of  $e$ . *Mass form.*  $e(Z)^\top \mathcal{H} e(Z) = (gZ)^\top \mathcal{H} (gZ) = Z^\top g^\top \mathcal{H} g Z = Z^\top \mathcal{H}' Z$ , which is (11.5). This step uses only the rule  $(gZ)^\top = Z^\top g^\top$  and would hold for any matrix  $g$ . *Charge norm.*  $e(Z)^\top \eta e(Z) = Z^\top (g^\top \eta g) Z = Z^\top \eta Z$  by (11.7).  $\square$

The proof is three lines long because the work was done in choosing the variables. In terms of  $(G, B)$  the map  $\mathcal{H} \mapsto g^\top \mathcal{H} g$  is a complicated non-linear transformation, the Buscher rules of Chapter 9 being a special case, and a direct check that the zero-mode Hamiltonian of Chapter 10 is invariant would be laborious. In terms of  $\mathcal{H}$  and  $Z$  it is linear algebra.

**Corollary 11.5** (Full perturbative mass spectrum). *Let  $\mathcal{H} = \mathcal{H}(G, B)$  and suppose  $\mathcal{H}' = g^\top \mathcal{H} g$  equals  $\mathcal{H}(G', B')$  for some background  $(G', B')$ . Then the closed-string spectra (11.3) on the two backgrounds coincide, including multiplicities.*

*Proof.* A state of the primed theory is labelled by a charge  $Z$  and an oscillator state at levels  $(N, \tilde{N})$  with  $N - \tilde{N} = \frac{1}{2} Z^\top \eta Z$ . Assign to it the charge  $e(Z)$  and the same oscillator occupation numbers in the unprimed theory. By (11.14) the constraint is again satisfied and  $\alpha' M^2$  is the same. The assignment is a bijection because  $e$  is, and because the number of oscillator states at given levels  $(N, \tilde{N})$  is a count of partitions that does not depend on the background (Chapter 10; the oscillators obey  $[\alpha_k^i, \alpha_l^j] = k G^{ij} \delta_{k+l,0}$  for every  $G$  [GPR §2.4, eq. (2.4.32), ll. 1446–1456], and a change of  $G$  is a change of basis in the space they span).  $\square$

#### Common pitfall: equal spectra are necessary, not sufficient

Theorem 11.4 and its corollary compare lists of masses. Two quantum theories with the same spectrum need not be the same theory; one must also match all correlation functions. Giveon, Porrati and Rabinovici are explicit that the spectrum condition is only “a necessary condition”, and that the equivalence of the theories is established separately, by exhibiting the dualities as residual gauge symmetries [GPR §2.4, ll. 1336–1348]. For the factorized dualities the path-integral derivation of Chapter 9 provides such an argument on every worldsheet. In this chapter, and in the Lean development, “same spectrum”

means exactly the statement (11.14).

## 11.4 How the group acts on backgrounds

Theorem 11.4 compares  $\mathcal{H}$  with  $g^\top \mathcal{H} g$ . To read it as a statement about tori we must know that  $g^\top \mathcal{H}(G, B) g$  is again of the form  $\mathcal{H}(G', B')$ , and we want a formula for  $(G', B')$ . Both follow from a characterization of generalized metrics that does not mention  $G$  and  $B$ .

**Lemma 11.6** (Which matrices are generalized metrics). *A real  $2d \times 2d$  matrix  $\mathcal{H}$  is of the form  $\mathcal{H}(G, B)$ , with  $G$  symmetric positive definite and  $B$  antisymmetric, if and only if  $\mathcal{H}$  is symmetric, positive definite and satisfies*

$$\mathcal{H} \eta \mathcal{H} = \eta. \quad (11.15)$$

The pair  $(G, B)$  is then unique.

*Proof.* ( $\Rightarrow$ ) Symmetry of  $\mathcal{H}(G, B)$  is visible in (11.2), using  $(BG^{-1})^\top = -G^{-1}B$ , and positivity is (11.4). For (11.15) write  $\mathcal{H} = \begin{pmatrix} P & Q \\ Q^\top & S \end{pmatrix}$  with  $P = G - BG^{-1}B$ ,  $Q = BG^{-1}$ ,  $S = G^{-1}$ . Then

$$\mathcal{H} \eta \mathcal{H} = \begin{pmatrix} PQ^\top + QP & PS + Q^2 \\ Q^\top Q^\top + SP & Q^\top S + SQ \end{pmatrix}, \quad (11.16)$$

and one computes  $Q^\top S + SQ = -G^{-1}BG^{-1} + G^{-1}BG^{-1} = 0$ ,  $PS + Q^2 = \mathbf{1} - BG^{-1}BG^{-1} + BG^{-1}BG^{-1} = \mathbf{1}$ , and  $PQ^\top + QP = -(G - BG^{-1}B)G^{-1}B + BG^{-1}(G - BG^{-1}B) = 0$ .

( $\Leftarrow$ ) Let  $\mathcal{H}$  be symmetric and positive definite with blocks  $P, Q, S$  as above. The diagonal block  $S$  of a positive definite matrix is positive definite, so we may set  $G = S^{-1}$  and  $B = QS^{-1}$ ; this is the only possible choice, which proves uniqueness. The lower-right block of (11.15),  $Q^\top S + SQ = 0$ , becomes  $S(B^\top + B)S = 0$  because  $Q = BS$  and  $Q^\top = S^\top B^\top$ ; hence  $B$  is antisymmetric. The upper-right block,  $PS + Q^2 = \mathbf{1}$ , gives  $P = (\mathbf{1} - BSBS)S^{-1} = G - BG^{-1}B$ . So  $\mathcal{H} = \mathcal{H}(G, B)$ .  $\square$

Equation (11.15) says, in view of Definition 11.1 and the symmetry of  $\mathcal{H}$ , that a generalized metric is itself an element of  $O(d, d; \mathbb{R})$ . Equivalently  $(\eta \mathcal{H})^2 = \mathbf{1}$ : the matrix  $\eta \mathcal{H}$  is an involution, and its two eigenspaces carry the information about the background.

**Lemma 11.7** (The background as a graph). *For  $\mathcal{H} = \mathcal{H}(G, B)$  and  $E = G + B$  the eigenspaces of  $\eta \mathcal{H}$  are the  $d$ -dimensional subspaces*

$$\begin{aligned} W_+ &= \{(m, Em) : m \in \mathbb{R}^d\} & (\text{eigenvalue } +1), \\ W_- &= \{(m, -E^\top m) : m \in \mathbb{R}^d\} & (\text{eigenvalue } -1). \end{aligned} \quad (11.17)$$

*Proof.*  $\eta \mathcal{H} Z = Z$  is equivalent to  $\mathcal{H} Z = \eta Z$ . For  $Z = (m, Em)$  the lower block of  $\mathcal{H} Z$  is  $-G^{-1}Bm + G^{-1}(G + B)m = m$  and the upper block is  $(G - BG^{-1}B)m + BG^{-1}(G + B)m = Gm + Bm = Em$ , so  $\mathcal{H} Z = (Em, m) = \eta Z$ . For  $Z = (m, -E^\top m)$ , with  $E^\top = G - B$ , the lower block is  $-G^{-1}Bm - G^{-1}(G - B)m = -m$  and the upper block is  $(G - BG^{-1}B)m - BG^{-1}(G - B)m = (G - B)m$ , so  $\mathcal{H} Z = -\eta Z$ . The two subspaces have dimension  $d$  and intersect only in 0, so they are the full eigenspaces.  $\square$

In the language of Chapter 10, the right-moving momentum is proportional to  $n - Em$  and the left-moving one to  $n + E^\top m$  [GPR §2.4, eq. (2.4.12), ll. 1193–1198]. Thus  $W_+$  is the space of (real) charges that are purely left-moving and  $W_-$  the space of those that are purely right-moving. Choosing a background is the same as choosing how to split  $\mathbb{R}^{2d}$  into left and right.

**Theorem 11.8** (Action on the background matrix). *Let  $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in O(d, d; \mathbb{R})$  and let  $(G, B)$  be a background with  $E = G + B$ . Then  $g^\top \mathcal{H}(G, B) g = \mathcal{H}(G', B')$ , where  $G'$  and  $B'$  are the symmetric and antisymmetric parts of*

$$E' = (a^\top E + c^\top)(b^\top E + d^\top)^{-1}, \quad (11.18)$$

and the inverse on the right-hand side exists.

*Proof.* Put  $\mathcal{H}' = g^\top \mathcal{H} g$ . It is symmetric, and positive definite because  $g$  is invertible. Using Proposition 11.2(iv) in the form  $g \eta g^\top = \eta$ ,

$$\mathcal{H}' \eta \mathcal{H}' = g^\top \mathcal{H} (g \eta g^\top) \mathcal{H} g = g^\top (\mathcal{H} \eta \mathcal{H}) g = g^\top \eta g = \eta,$$

so by Lemma 11.6  $\mathcal{H}' = \mathcal{H}(G', B')$  for a unique background. To find it, note that (11.7) gives  $\eta g^\top = g^{-1} \eta$ , hence  $\eta \mathcal{H}' = g^{-1} (\eta \mathcal{H}) g$ : the involutions of the two backgrounds are conjugate, and their  $+1$  eigenspaces are related by  $W'_+ = g^{-1} W_+$ . With (11.10) and Lemma 11.7,

$$g^{-1} \begin{pmatrix} m \\ Em \end{pmatrix} = \begin{pmatrix} (d^\top + b^\top E) m \\ (c^\top + a^\top E) m \end{pmatrix} \stackrel{!}{=} \begin{pmatrix} m' \\ E' m' \end{pmatrix}.$$

The map  $m \mapsto m' = (b^\top E + d^\top) m$  is the composition of the bijection  $g^{-1} : W_+ \rightarrow W'_+$  with the projection of the graph  $W'_+$  onto its first  $d$  coordinates, which is also a bijection. Hence  $b^\top E + d^\top$  is invertible, and  $E'(b^\top E + d^\top) = a^\top E + c^\top$ .  $\square$

We have checked (11.18) against  $g^\top \mathcal{H} g$  with exact rational arithmetic (sympy) for random backgrounds and random products of the generators below in  $d = 1, 2, 3$ ; this is a symbolic check and not a Lean theorem. Formula (11.18) is a matrix version of the fractional linear action  $\tau \mapsto (a\tau + b)/(c\tau + d)$  of  $SL(2, \mathbb{Z})$  on the upper half plane, and for  $d = 2$  it contains two copies of that action (Chapter 12).

**Common pitfall:**  $g^\top \mathcal{H} g$  or  $g \mathcal{H} g^\top$ ?

Giveon, Porrati and Rabinovici let the group act by  $\mathcal{H} \mapsto g \mathcal{H} g^\top$  and obtain  $E' = (aE + b)(cE + d)^{-1}$  [GPR §2.4, eqs. (2.4.19)–(2.4.24), ll. 1290–1334]. We use  $\mathcal{H} \mapsto g^\top \mathcal{H} g$ , the form in which the Lean theorem `spectrum_equivalence` is stated, and obtain (11.18). The two conventions are exchanged by  $g \leftrightarrow g^\top$ , and since the transpose of a group element is a group element (Proposition 11.2(iv)) both describe the same set of transformations of backgrounds. The price is that a matrix that shifts  $B$  in one convention is upper triangular, and in the other lower triangular. When reading a paper or a Lean file, first find out which of the two actions is meant and in which order the charges are listed.

## 11.5 Three families of elements

Two families of symmetries of the toroidal spectrum are evident from the worldsheet action, and a third is the generalization of  $R \rightarrow \alpha'/R$  [GPR §2.4, ll. 1349–1421; AMN §3.1, eq. (3.10), ll. 660–725]. For each we give the matrix in our convention (background  $\mathcal{H} \mapsto g^\top \mathcal{H} g$ , charges  $Z \mapsto gZ$ ), verify (11.7), find the action on  $E$  from (11.18) and on the charges, and state the physical reason why it is a symmetry.

### 11.5.1 Integer shifts of the $B$ -field

Let  $\Theta$  be a  $d \times d$  matrix, and define

$$S_\Theta = \begin{pmatrix} \mathbf{1} & 0 \\ -\Theta & \mathbf{1} \end{pmatrix} = g_\Theta^\top, \quad g_\Theta = \begin{pmatrix} \mathbf{1} & \Theta \\ 0 & \mathbf{1} \end{pmatrix}, \quad (11.19)$$

where  $g_\Theta$  is the matrix of [GPR §2.4, eq. (2.4.25), ll. 1351–1367] and of the Lean definition `thetaShift`. *Membership:* with  $a = d = \mathbf{1}$ ,  $b = 0$ ,  $c = -\Theta$  the conditions (11.9) read  $-\Theta - \Theta^\top = 0$ ,  $0 = 0$  and  $\mathbf{1} = \mathbf{1}$ . So  $S_\Theta \in O(d, d; \mathbb{R})$  exactly when  $\Theta$  is antisymmetric, and  $S_\Theta \in O(d, d; \mathbb{Z})$  when in addition its entries are integers. *Action on the background:* (11.18) gives  $E' = (E + c^\top)(\mathbf{1})^{-1} = E - \Theta^\top = E + \Theta$ , that is

$$G' = G, \quad B' = B + \Theta. \quad (11.20)$$

*Action on the charges:*  $S_\Theta(m, n) = (m, n - \Theta m)$ . This agrees with (11.4): replacing  $B$  by  $B + \Theta$  replaces  $n - Bm$  by  $(n - \Theta m) - Bm$ , and nothing else changes.

*Physical meaning.* The  $B$ -field term of the worldsheet action is the integral of the pull-back of a closed two-form, so it is a total derivative and only feels the topology of the map from the worldsheet to the torus. With the normalization of Chapters 6 and 10, shifting  $B_{ij}$  by an integer changes the action of every closed worldsheet by an integer multiple of  $2\pi$ , and the weight  $e^{iS}$  in the path integral does not change [GPR §2.4, ll. 1368–1371]. Such a shift is a *large gauge transformation* of the two-form: it cannot be undone by  $B \rightarrow B + d\Lambda$  with a single-valued  $\Lambda$ , yet it relates physically identical configurations. The situation is that of the theta angle of a gauge theory, or of a particle on a ring threaded by magnetic flux: the spectrum is periodic in the flux, and one period of flux shifts the integer that labels the momentum eigenstates. Here one period of  $B$  shifts the momentum numbers by the winding numbers,  $n \mapsto n - \Theta m$ . The components of  $B$  are therefore periodic variables, and there are  $\frac{1}{2}d(d-1)$  independent shifts, none for the circle and one for  $T^2$ .

### 11.5.2 Changes of basis of the torus lattice

Let  $A \in GL(d, \mathbb{Z})$ , an integer matrix with  $\det A = \pm 1$ , so that  $A^{-1}$  is also an integer matrix. Define

$$L_A = \begin{pmatrix} A^\top & 0 \\ 0 & A^{-1} \end{pmatrix} = g_A^\top, \quad g_A = \begin{pmatrix} A & 0 \\ 0 & (A^\top)^{-1} \end{pmatrix} \quad (11.21)$$

[GPR §2.4, eq. (2.4.26), ll. 1372–1381]. *Membership:*  $b = c = 0$ , and the third condition in (11.9) is  $a^\top d = AA^{-1} = \mathbf{1}$ . Conversely a block-diagonal matrix  $\text{diag}(a, d)$  satisfies (11.7) if and only if  $d = (a^\top)^{-1}$ : once the action on windings is chosen, the action on momenta is forced. *Action on the background:*  $E' = AE(A^{-\top})^{-1} = AEA^\top$ , that is

$$G' = AGA^\top, \quad B' = ABA^\top. \quad (11.22)$$

*Action on the charges:*  $L_A(m, n) = (A^\top m, A^{-1}n)$  [GPR §2.4, ll. 1383–1396].

*Physical meaning.* The torus is  $\mathbb{R}^d$  divided by a lattice, and the coordinates  $X^i \sim X^i + 2\pi$  are adapted to a basis of that lattice. The lattice has many bases (Figure 11.2). New coordinates  $X'$  with  $X = A^\top X'$  have the same periodicities if and only if  $A$  and  $A^{-1}$  are integer matrices, and then  $dX^\top G dX = dX'^\top (AGA^\top) dX'$ , which is (11.22). Nothing has been done to the torus;  $G$  and  $AGA^\top$  are two descriptions of one Riemannian manifold. A string that winds  $m'$  times around the new cycles winds  $m = A^\top m'$  times around the old ones, and the phase  $n^\top X = n^\top A^\top X'$  of a momentum eigenstate shows  $n' = An$ , that is  $n = A^{-1}n'$ . These are the two blocks of  $L_A$ . Permutations of the circles and the reflection  $X^k \rightarrow -X^k$

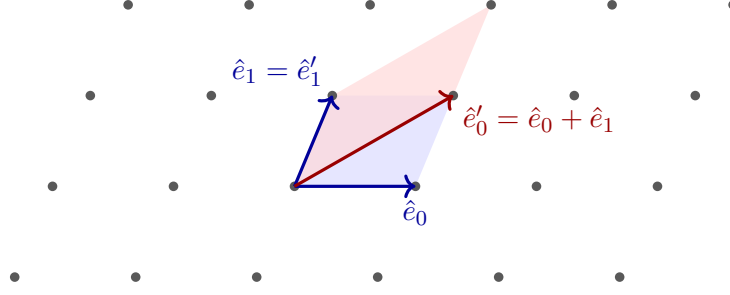


Figure 11.2: One lattice, two bases. The basis  $(\hat{e}'_0, \hat{e}'_1)$  is obtained from  $(\hat{e}_0, \hat{e}_1)$  by  $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \in GL(2, \mathbb{Z})$ . The two shaded cells have the same area and define the same torus. The Gram matrices of the two bases are  $G$  and  $G' = AGA^T$ ; the element  $L_A$  of  $O(2, 2; \mathbb{Z})$  records how winding and momentum numbers are relabelled.

of one of them are special cases. The subgroup  $\{L_A\}$  is the group of large diffeomorphisms of the torus; string theory inherits it from geometry, and a point particle on the torus has it too.

### 11.5.3 Factorized dualities

Let  $e_k$  be the  $d \times d$  matrix whose only non-zero entry is a 1 in position  $(k, k)$ . The *factorized duality* along the  $k$ -th circle is

$$D_k = \begin{pmatrix} \mathbf{1} - e_k & e_k \\ e_k & \mathbf{1} - e_k \end{pmatrix} \quad (11.23)$$

[GPR §2.4, eq. (2.4.29), ll. 1399–1414]. It is symmetric, so here the two conventions of the pitfall box agree. *Membership:* the matrix  $e_k$  is a symmetric projector,  $e_k^2 = e_k$ , hence  $e_k(\mathbf{1} - e_k) = 0$  and  $(\mathbf{1} - e_k)^2 = \mathbf{1} - e_k$ . The conditions (11.9) become  $2(\mathbf{1} - e_k)e_k = 0$  (twice) and  $(\mathbf{1} - e_k)^2 + e_k^2 = \mathbf{1}$ . The same identities give

$$D_k^2 = \begin{pmatrix} (\mathbf{1} - e_k)^2 + e_k^2 & 2(\mathbf{1} - e_k)e_k \\ 2e_k(\mathbf{1} - e_k) & e_k^2 + (\mathbf{1} - e_k)^2 \end{pmatrix} = \mathbf{1}, \quad D_k D_l = D_l D_k, \quad (11.24)$$

the second because diagonal matrices commute. *Action on the charges:*  $D_k$  exchanges  $m^k$  with  $n_k$  and leaves the other  $2d - 2$  charges alone. *Action on the background:* take  $k = 0$  and split the indices as  $i = (0, \alpha)$ . Then in (11.18)

$$a^\top E + c^\top = (\mathbf{1} - e_0)E + e_0 = \begin{pmatrix} 1 & 0 \\ E_{\alpha 0} & E_{\alpha \beta} \end{pmatrix}, \quad b^\top E + d^\top = e_0 E + (\mathbf{1} - e_0) = \begin{pmatrix} E_{00} & E_{0\beta} \\ 0 & \mathbf{1} \end{pmatrix}.$$

The second matrix is upper triangular with inverse  $\begin{pmatrix} 1/E_{00} & -E_{0\beta}/E_{00} \\ 0 & \mathbf{1} \end{pmatrix}$ , and  $E_{00} = G_{00} > 0$ . Multiplying,

$$E' = \begin{pmatrix} \frac{1}{E_{00}} & -\frac{E_{0\beta}}{E_{00}} \\ \frac{E_{\alpha 0}}{E_{00}} & E_{\alpha \beta} - \frac{E_{\alpha 0} E_{0\beta}}{E_{00}} \end{pmatrix}. \quad (11.25)$$

These are the Buscher rules of Chapter 9 in their compact form, up to the sign of the two mixed entries. For a product background,  $E_{0\beta} = E_{\alpha 0} = 0$  and  $E_{00} = R^2/\alpha'$ , (11.25) is  $R^2/\alpha' \mapsto \alpha'/R^2$  with the other directions untouched:  $D_k$  is the circle duality of Chapter 8 performed on one circle [GPR §2.4, ll. 1419–1421].

|               | matrix $g$                                                                             | background             | $(m, n) \mapsto$          | origin                                 |
|---------------|----------------------------------------------------------------------------------------|------------------------|---------------------------|----------------------------------------|
| $B$ -shift    | $S_\Theta = \begin{pmatrix} \mathbf{1} & 0 \\ -\Theta & \mathbf{1} \end{pmatrix}$      | $B \mapsto B + \Theta$ | $(m, n - \Theta m)$       | large gauge transf.                    |
| basis change  | $L_A = \begin{pmatrix} A^\top & 0 \\ 0 & A^{-1} \end{pmatrix}$                         | $E \mapsto AEA^\top$   | $(A^\top m, A^{-1}n)$     | large diffeomorphism                   |
| fact. duality | $D_k = \begin{pmatrix} \mathbf{1} - e_k & e_k \\ e_k & \mathbf{1} - e_k \end{pmatrix}$ | rules (11.25)          | $m^k \leftrightarrow n_k$ | stringy, $R_k \rightarrow \alpha'/R_k$ |

Table 11.1: The three families of elements of  $O(d, d; \mathbb{Z})$ , in the convention  $\mathcal{H} \mapsto g^\top \mathcal{H} g$ ,  $Z = (m, n) \mapsto gZ$ . Here  $\Theta$  is an antisymmetric integer matrix and  $A \in GL(d, \mathbb{Z})$ .

*Remark 11.9* (The sign of the mixed entries). Chapter 9 uses the matrix  $T_\theta$ , which is  $D_0$  with  $e_0$  replaced by  $-e_0$  in the off-diagonal blocks, and finds (11.25) with the opposite sign in the first row and the first column. The two are related by a basis change: for the reflection  $r_0 = \mathbf{1} - 2e_0 \in GL(d, \mathbb{Z})$  of the dualized coordinate one has  $(\mathbf{1} - e_0)r_0 = \mathbf{1} - e_0$  and  $e_0 r_0 = -e_0$ , hence

$$T_\theta = D_0 L_{r_0}. \quad (11.26)$$

By (11.22),  $L_{r_0}$  flips the sign of exactly the entries  $E_{0\beta}$  and  $E_{\alpha 0}$ . So the two forms of the rules differ by the orientation of the dual circle, which is a matter of convention, and both  $D_0$  and  $T_\theta$  belong to  $O(d, d; \mathbb{Z})$ . Since (11.25) was derived for every  $d$  from Theorem 11.8, this also proves for every  $d$  the statement of Chapter 9 that the Buscher rules are the conjugation  $\mathcal{H} \mapsto T_\theta^\top \mathcal{H} T_\theta$ .

*Physical meaning.* Unlike the first two families,  $D_k$  is not a symmetry of the classical sigma-model action under a change of variables that a point particle could follow. It exchanges a Kaluza–Klein momentum with a winding number, and only an extended object has winding. The argument that it is a symmetry is either the spectrum (Theorem 11.4), or the worldsheet path integral of Chapter 9, or the enhanced gauge symmetry at the self-dual radius (Chapter 8).

**Proposition 11.10** (All circles at once).  $D_0 D_1 \cdots D_{d-1} = \eta$ . Consequently  $\eta^\top \mathcal{H}(G, 0) \eta = \mathcal{H}(G^{-1}, 0)$ : dualizing every circle of a torus without  $B$ -field inverts the metric.

*Proof.* Let  $\hat{e}_k = \text{diag}(e_k, e_k)$ . These are commuting projectors with  $\hat{e}_k \hat{e}_l = 0$  for  $k \neq l$  and  $\sum_k \hat{e}_k = \mathbf{1}$ , they commute with  $\eta$ , and  $D_k = \mathbf{1} + \hat{e}_k(\eta - \mathbf{1})$ . In the product  $\prod_k (\mathbf{1} + \hat{e}_k(\eta - \mathbf{1}))$  every term containing two different projectors vanishes, leaving  $\mathbf{1} + \sum_k \hat{e}_k(\eta - \mathbf{1}) = \eta$ . For the second statement put  $g = \eta$  in (11.18):  $a = d = 0$ ,  $b = c = \mathbf{1}$  and  $E' = E^{-1}$ , which for  $B = 0$  is  $G' = G^{-1}$ .  $\square$

For general  $B$  the full duality is  $E \mapsto E^{-1}$ , the direct generalization of  $R^2/\alpha' \mapsto \alpha'/R^2$ ; the symmetric part of  $E^{-1}$  is  $(G - BG^{-1}B)^{-1}$ , not  $G^{-1}$  (Exercise 11.3).

Table 11.1 collects the three families.

## 11.6 Worked example: $d = 2$ with explicit $4 \times 4$ matrices

For  $T^2$  the charge vector is  $Z = (m^0, m^1; n_0, n_1)$  and  $\frac{1}{2}Z^\top \eta Z = m^0 n_0 + m^1 n_1$ . There is one independent  $B$ -shift, generated by  $\Theta = \varepsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ , and we take  $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  of Figure 11.2 as



a sample basis change. From (11.19), (11.21) and (11.23):

$$S_\varepsilon = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}, \quad L_A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (11.27)$$

$$D_0 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad D_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \quad D_0 D_1 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} = \eta.$$

The reader should check one of them against (11.7) by hand; for  $S_\varepsilon$ , for instance,  $\eta S_\varepsilon$  has rows  $(0, -1, 1, 0)$ ,  $(1, 0, 0, 1)$ ,  $(1, 0, 0, 0)$ ,  $(0, 1, 0, 0)$ , and multiplying by  $S_\varepsilon^\top$  from the left returns  $\eta$ . On the charges,  $S_\varepsilon Z = (m^0, m^1; n_0 - m^1, n_1 + m^0)$  and  $L_A Z = (m^0, m^0 + m^1; n_0 - n_1, n_1)$ , and one verifies directly that both preserve  $m^0 n_0 + m^1 n_1$ : for  $S_\varepsilon$  the two new terms are  $-m^0 m^1 + m^1 m^0 = 0$ .

**A background with numbers.** Take a rectangular torus with radii  $R_0 = 2\sqrt{\alpha'}$  and  $R_1 = \sqrt{\alpha'}$  and half a unit of  $B$ -field:

$$G = \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix}, \quad B = \frac{1}{2}\varepsilon, \quad E = \begin{pmatrix} 4 & \frac{1}{2} \\ -\frac{1}{2} & 1 \end{pmatrix}, \quad \mathcal{H} = \begin{pmatrix} \frac{17}{4} & 0 & 0 & \frac{1}{2} \\ 0 & \frac{17}{16} & -\frac{1}{8} & 0 \\ 0 & -\frac{1}{8} & \frac{1}{4} & 0 \\ \frac{1}{2} & 0 & 0 & 1 \end{pmatrix}. \quad (11.28)$$

The blocks of  $\mathcal{H}$  follow from  $BG^{-1}B = -\text{diag}(\frac{1}{4}, \frac{1}{16})$ , so that  $G - BG^{-1}B = \text{diag}(\frac{17}{4}, \frac{17}{16})$ , and from  $BG^{-1} = \begin{pmatrix} 0 & 1/2 \\ -1/8 & 0 \end{pmatrix}$ . As a check, (11.4) gives the same quadratic form:  $\frac{1}{4}(n_0 - \frac{1}{2}m^1)^2 + (n_1 + \frac{1}{2}m^0)^2 + 4(m^0)^2 + (m^1)^2$ .

Now dualize the circle of radius  $R_0$ . From (11.25) with  $E_{00} = 4$ ,  $E_{01} = \frac{1}{2}$ ,  $E_{10} = -\frac{1}{2}$ ,  $E_{11} = 1$ :

$$E' = \begin{pmatrix} \frac{1}{4} & -\frac{1}{8} \\ -\frac{1}{8} & 1 + \frac{1}{16} \end{pmatrix} = G', \quad B' = 0. \quad (11.29)$$

The dual torus has no  $B$ -field at all. Its first circle has radius  $\sqrt{\alpha'}/2 = \alpha'/R_0$ , as expected, and its two cycles are no longer orthogonal: the angle  $\vartheta$  between them satisfies  $\cos \vartheta = G'_{01}/\sqrt{G'_{00}G'_{11}} = -\frac{1}{8}/\sqrt{\frac{17}{64}} = -1/\sqrt{17}$ , so  $\vartheta \approx 104^\circ$ . *The  $B$ -field of one description is the tilt of the torus in the other.* The generalized metric of the dual background can be computed in two ways, from (11.29) and (11.2), or as  $D_0 \mathcal{H} D_0$ , which exchanges the first and third rows and columns of (11.28); both give

$$\mathcal{H}' = \begin{pmatrix} \frac{1}{4} & -\frac{1}{8} & 0 & 0 \\ -\frac{1}{8} & \frac{17}{16} & 0 & 0 \\ 0 & 0 & \frac{17}{4} & \frac{1}{2} \\ 0 & 0 & \frac{1}{2} & 1 \end{pmatrix}, \quad G'^{-1} = \begin{pmatrix} \frac{17}{4} & \frac{1}{2} \\ \frac{1}{2} & 1 \end{pmatrix}, \quad \det G' = \frac{1}{4}.$$

Table 11.2 lists a few states. Each row is one instance of (11.14).

**The elements do not commute.** Conjugating the  $B$ -shift by the duality gives

$$D_0 S_\varepsilon D_0 = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} = L_{A'}, \quad A' = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}, \quad (11.30)$$

| charge $Z$ on the dual torus | $Z^\top \mathcal{H}' Z$ | $D_0 Z$ on the original torus | $(D_0 Z)^\top \mathcal{H} (D_0 Z)$ | $N - \tilde{N}$ |
|------------------------------|-------------------------|-------------------------------|------------------------------------|-----------------|
| $(1, 0; 0, 0)$               | $\frac{1}{4}$           | $(0, 0; 1, 0)$                | $\frac{1}{4}$                      | 0               |
| $(0, 0; 1, 0)$               | $\frac{17}{4}$          | $(1, 0; 0, 0)$                | $\frac{17}{4}$                     | 0               |
| $(0, 1; 1, 0)$               | $\frac{85}{16}$         | $(1, 1; 0, 0)$                | $\frac{85}{16}$                    | 0               |
| $(1, 0; 1, 0)$               | $\frac{9}{2}$           | $(1, 0; 1, 0)$                | $\frac{9}{2}$                      | 1               |
| $(1, 1; 0, 1)$               | $\frac{33}{16}$         | $(0, 1; 1, 1)$                | $\frac{33}{16}$                    | 1               |

Table 11.2: States of the background (11.28) and of its dual (11.29), with  $Z = (m^0, m^1; n_0, n_1)$ . A string wound once around the small circle of the dual torus (first row) is a string with one unit of momentum on the large circle of the original torus. With the oscillators,  $\alpha' M^2$  is the second column plus  $2(N + \tilde{N} - 2)$ ; for the fourth row the lightest state has  $N = 1$ ,  $\tilde{N} = 0$  and  $\alpha' M^2 = \frac{5}{2}$  in both descriptions.

a basis change. In words: the stringy periodicity  $B_{01} \rightarrow B_{01} + 1$  of the original torus is, in the dual description, the geometric fact that the lattice of the dual torus can be described in a sheared basis. This is visible in the example: shifting  $B = \frac{1}{2}\varepsilon$  to  $\frac{3}{2}\varepsilon$  changes the off-diagonal entry of  $G'$  from  $-\frac{1}{8}$  to  $-\frac{3}{8}$  and  $G'_{11}$  from  $\frac{17}{16}$  to  $\frac{25}{16}$ , which is the change of basis  $\hat{e}'_1 \rightarrow \hat{e}'_1 - \hat{e}'_0$  of the dual lattice ( $G'_{01} - G'_{00} = -\frac{3}{8}$ ,  $G'_{11} - 2G'_{01} + G'_{00} = \frac{25}{16}$ ). Since  $D_0 S_\varepsilon D_0 \neq S_\varepsilon$ , the group is not abelian, and the element  $D_0 S_\varepsilon$  has infinite order. Relation (11.30) is the matrix form of the exchange of the Kähler and complex-structure moduli of  $T^2$ , the simplest example of mirror symmetry (Chapter 12).

## 11.7 The whole group, and what lies outside it

### 11.7.1 Relations and generation

Within each family the composition law is simple. From the block forms one finds

$$S_\Theta S_{\Theta'} = S_{\Theta + \Theta'}, \quad L_A L_{A'} = L_{A'A}, \quad L_A^{-1} S_\Theta L_A = S_{A\Theta A^\top}, \quad L_\pi D_k L_\pi^{-1} = D_{\pi(k)}, \quad (11.31)$$

where in the last relation  $\pi$  is a permutation matrix (the labelling of  $\pi(k)$  depends on the convention for permutation matrices). So the  $B$ -shifts form an abelian group isomorphic to  $\mathbb{Z}^{d(d-1)/2}$ , the basis changes form a copy of  $GL(d, \mathbb{Z})$ , the basis changes normalize the shifts, and all factorized dualities are conjugate to  $D_0$ . Elements of the form  $S_\Theta L_A$  make up the *geometric subgroup*: they map a torus with a  $B$ -field to the same torus in other coordinates and another gauge (Exercise 11.4). Everything that is specific to strings comes from a single additional element,  $D_0$ .

**Theorem 11.11** (Generation). *The integer  $B$ -shifts  $S_\Theta$ , the basis changes  $L_A$  and the factorized dualities  $D_k$  generate  $O(d, d; \mathbb{Z})$ . [L] [GPR §2.4, ll. 1422–1423]*

Aldazabal, Marqués and Núñez state the corresponding decomposition for the continuous group, where  $A$  ranges over  $GL(d, \mathbb{R})$  and  $\Theta$  over real antisymmetric matrices [AMN §3.1, eq. (3.10), ll. 660–717]. Neither source proves the statement on the page, and it is not formalized in the Lean development; we use it only as Tier L background. Nothing proved in this chapter depends on it: Theorem 11.4 holds for every element of the group, whether or not it has been written as a product of generators. For  $d = 1$  the theorem is a consequence of Example 11.3: there are no shifts,  $GL(1, \mathbb{Z}) = \{\pm 1\}$  gives  $L_{-1} = -\mathbf{1}$ ,  $D_0 = \eta$ , and these two elements generate the four elements of (11.13). For  $d = 2$  the group is described in Chapter 12.

A word on the list of families. With the action  $\mathcal{H} \mapsto g^\top \mathcal{H} g$  the matrix  $g_\Theta$  of (11.19), with  $\Theta$  in the upper right block, is also in the group, but it does not shift  $B$ . From (11.18),  $E' = E(\Theta^\top E + \mathbf{1})^{-1}$ , that is

$$E'^{-1} = E^{-1} - \Theta, \quad g_\Theta = \eta S_{-\Theta} \eta. \quad (11.32)$$

It is a  $B$ -shift conjugated by the full duality: it shifts the antisymmetric part of  $E^{-1}$ . In double field theory such elements are called  $\beta$ -transformations [AMN §3.1, ll. 721–723]. They are not a fourth family of generators, since  $\eta = D_0 \cdots D_{d-1}$ .

### 11.7.2 What is not in the group

*Worldsheet parity.* Reversing the orientation of the string,  $\sigma \rightarrow -\sigma$ , maps  $B \rightarrow -B$ . On the doubled charges this is  $\Omega = \text{diag}(\mathbf{1}, -\mathbf{1})$ , and indeed  $\Omega^\top \mathcal{H}(G, B) \Omega = \mathcal{H}(G, -B)$ . But  $\Omega^\top \eta \Omega = -\eta$ : parity exchanges left- and right-movers and flips the sign of  $n \cdot m$ , so it is not an element of  $O(d, d; \mathbb{Z})$  [GPR §2.4, ll. 1423–1427]. It is a symmetry of the bosonic string, of a different kind, exchanging  $N$  with  $\tilde{N}$ .

*Continuous transformations.* A generic element of  $O(d, d; \mathbb{R})$  is not a symmetry. It maps a background to a physically different one, and this is how the real group sweeps out the space of all backgrounds: by Theorem 11.8 and Chapter 10,

$$\mathcal{M}_d = O(d, d; \mathbb{Z}) \setminus O(d, d; \mathbb{R}) / (O(d) \times O(d)), \quad \dim \mathcal{M}_d = d(2d - 1) - 2 \cdot \frac{1}{2} d(d - 1) = d^2, \quad (11.33)$$

the  $d^2$  being the components of  $E = G + B$  [GPR §2.4, ll. 1263–1266]. The real group acts on the space of backgrounds; the integer subgroup tells us which points of that space are the same theory. Double field theory, a classical field theory, works with the continuous group: Aldazabal, Marqués and Núñez drop the  $\mathbb{Z}$  as “irrelevant for our purposes of introducing DFT at the classical level” [AMN §3.1, ll. 569–572]. Integrality is a property of the quantum spectrum of charged states, which knows that  $Z$  is an integer vector.

*Fixed points.* An element  $g \neq \pm \mathbf{1}$  may fix a particular background,  $g^\top \mathcal{H} g = \mathcal{H}$ . Then  $e(Z) = gZ$  is a symmetry acting *within* one theory, and states are permuted among themselves. The self-dual circle  $G = 1$ , fixed by  $\eta$ , is the basic example (Chapter 8); such points are the orbifold points of  $\mathcal{M}_d$ , and they are where extra massless states appear.

## 11.8 What the machine has checked

The three files `DualScaleStream2/TDuality/ODD.lean`, `Factorized.lean` and `Spectrum.lean` formalize the algebra of this chapter for arbitrary  $d$ , over Mathlib’s block matrices. The index type of a charge vector is the disjoint union `Fin d ⊕ Fin d`, so a charge vector is a function `Charge d → ℤ` and `fromBlocks a b c d` is the block matrix (11.8). The doc-comments of `ODD.lean` call the two halves “momenta” and “windings” in that order, whereas `GeneralizedMetric.lean` lists windings first, as we do. For the theorems of `ODD.lean` and `Factorized.lean` the order does not matter, because  $\eta$  and  $D_k$  are unchanged when the two halves are exchanged; it matters only for the interpretation of `thetaShift`, discussed below. All theorems quoted here depend only on the standard axioms `propext`, `Classical.choice` and `Quot.sound`.

### The group as a predicate; closure under product and inverse

```
abbrev Charge (d : ℕ) := Fin d ⊕ Fin d
def eta (d : ℕ) : Matrix (Charge d) (Charge d) ℤ := fromBlocks 0 1 1 0
def IsODD (g : Matrix (Charge d) (Charge d) ℤ) : Prop :=
```

```

gT * eta d * g = eta d

theorem eta_mul_self : eta d * eta d = 1 := by ...
theorem eta_isODD : IsODD (eta d) := by ...
theorem isODD_mul (g h : Matrix (Charge d) (Charge d) ℤ)
  (hg : IsODD g) (hh : IsODD h) : IsODD (g * h) := by ...
theorem isODD_left_inv (g : Matrix (Charge d) (Charge d) ℤ) (hg : IsODD g) :
  (eta d * gT * eta d) * g = 1 := by ...
theorem isODD_right_inv (g : Matrix (Charge d) (Charge d) ℤ) (hg : IsODD g) :
  g * (eta d * gT * eta d) = 1 := by ...
theorem isODD_inv_isODD (g : Matrix (Charge d) (Charge d) ℤ) (hg : IsODD g) :
  IsODD (eta d * gT * eta d) := by ...

```

**Reading.** `IsODD g` is (11.7) for an integer matrix. The six theorems are items (i)–(iv) of Proposition 11.2:  $\eta^2 = 1$ ;  $\eta$  is in the group; products stay in the group;  $\eta g^T \eta$  is a left and a right inverse of  $g$  with integer entries by construction; and that inverse is in the group.

**What is not there.** The group is a predicate on matrices. No `Group` or `Subgroup` structure is declared, the identity matrix is not shown to satisfy `IsODD` by a named theorem, and  $\det g = \pm 1$  and the closure under transposition are not stated (Exercise 11.7).

**Proof idea.** `isODD_mul` is the calc chain of our proof of (ii). `isODD_left_inv` regroups  $(\eta g^T \eta)g = \eta(g^T \eta g)$  and rewrites with the hypothesis and `eta_mul_self`; the right inverse follows from Mathlib’s `mul_eq_one_comm` for square matrices over a commutative ring.

[A]

### The three families are in the group

```

def thetaShift (Θ : Matrix (Fin d) (Fin d) ℤ) :
  Matrix (Charge d) (Charge d) ℤ := fromBlocks 1 Θ 0 1
def basisChange (A B : Matrix (Fin d) (Fin d) ℤ) :
  Matrix (Charge d) (Charge d) ℤ := fromBlocks A 0 0 B
def proj (k : Fin d) : Matrix (Fin d) (Fin d) ℤ := single k k 1
def factorized (k : Fin d) : Matrix (Charge d) (Charge d) ℤ :=
  fromBlocks (1 - proj k) (proj k) (proj k) (1 - proj k)

theorem thetaShift_isODD (Θ : Matrix (Fin d) (Fin d) ℤ) (hΘ : ΘT = -Θ) :
  IsODD (thetaShift Θ) := by ...
theorem basisChange_isODD (A B : Matrix (Fin d) (Fin d) ℤ)
  (h : AT * B = 1) : IsODD (basisChange A B) := by ...
theorem factorized_isODD (k : Fin d) : IsODD (factorized k) := by ...
theorem factorized_mul_self (k : Fin d) :
  factorized k * factorized k = 1 := by ...
theorem factorized_comm (k l : Fin d) :
  factorized k * factorized l = factorized l * factorized k := by ...
theorem factorized_two : factorized (0 : Fin 2) * factorized 1 = eta 2 := by ...

```

**Reading.** `thetaShift Θ` is  $g_\Theta$  of (11.19), and  $S_\Theta$  is its transpose; `basisChange A B` is  $\text{diag}(A, B)$ , and  $L_A = \text{basisChange } A^T A^{-1}$ ; `factorized k` is  $D_k$ . The first three theorems are one direction each: if  $\Theta$  is antisymmetric then  $g_\Theta$  is in the group, if  $A^T B = 1$  then  $\text{diag}(A, B)$  is; the converses shown in Section 11.5 are not formalized. The hypothesis  $A^T B = 1$  replaces the matrix inverse, so  $GL(d, \mathbb{Z})$  never has to be mentioned. The next two are (11.24). The last is Proposition 11.10 for  $d = 2$  only; the statement for general  $d$  is not formalized (Exercise 11.9). The physical readings attached to these matrices (large gauge transformation, change of coordinates,  $R \rightarrow \alpha'/R$ ) are in the doc-comments, not in the statements, and are not Tier A. In particular no Lean theorem in these three files says that  $g_\Theta$  shifts  $B$ ; that is `genMetric_bshift` in `DFT/BShift.lean`, which states

$g_\Theta \mathcal{H}(G, B) g_\Theta^\top = \mathcal{H}(G, B + \Theta)$  over  $\mathbb{R}$ , the convention of GPR, equivalent to our (11.20) because  $S_\Theta = g_\Theta^\top$  (Chapter 15). **Proof idea.** Unfold to `fromBlocks`, multiply blockwise with `fromBlocks_multiply`, and compare the four blocks. For  $D_k$  the only input is  $e_k^2 = e_k$  (`proj_mul_self`, from Mathlib's algebra of matrix units), after which `abel` closes each block. `factorized_two` is a comparison of two explicit  $4 \times 4$  integer matrices, the last matrix in (11.27), entry by entry (`fin_cases`, `rfl`). [A]

### The charge norm: even and invariant

```
def chargeNorm (Z : Charge d → ℤ) : ℤ := Z ·v (eta d *v Z)

theorem chargeNorm_sumElim (n w : Fin d → ℤ) :
  chargeNorm (Sum.elim n w) = 2 * (n ·v w) := by ...
theorem chargeNorm_even (Z : Charge d → ℤ) : Even (chargeNorm Z) := by ...
theorem chargeNorm_invariant (g : Matrix (Charge d) (Charge d) ℤ)
  (hg : IsODD g) (Z : Charge d → ℤ) :
  chargeNorm (g *v Z) = chargeNorm Z := by ...
```

**Reading.** `chargeNorm Z` is  $Z^\top \eta Z$ . If the two halves of  $Z$  are  $n$  and  $w$ , it equals  $2n \cdot w$ ; it is even for every integer  $Z$ , which is the evenness of the Narain lattice of Chapter 10 for all  $d$ ; and it does not change under  $Z \mapsto gZ$  with  $g$  in the group. The last statement is the second equation in (11.14). The converse found in Section 11.1, that invariance for all  $Z$  forces  $g^\top \eta g = \eta$ , is not formalized. **Proof idea.** For invariance, move  $g$  across the dot product with Mathlib's `dotProduct_mulVec` and `vecMul_mulVec`, which produces  $g^\top \eta g$ , and rewrite with the hypothesis. [A]

### Spectrum equivalence

```
noncomputable def massForm (H : Matrix (Charge d) (Charge d) ℝ)
  (Z : Charge d → ℝ) : ℝ := Z ·v (H *v Z)
theorem massForm_covariant (g H : Matrix (Charge d) (Charge d) ℝ)
  (Z : Charge d → ℝ) :
  massForm (g⊤ * H * g) Z = massForm H (g *v Z) := by ...

noncomputable def toReal (g : Matrix (Charge d) (Charge d) ℤ) :
  Matrix (Charge d) (Charge d) ℝ := g.map (Int.cast : ℤ → ℝ)

theorem spectrum_equivalence (g : Matrix (Charge d) (Charge d) ℤ)
  (hg : IsODD g) (H : Matrix (Charge d) (Charge d) ℝ) :
  ∃ e : (Charge d → ℤ) ≃ (Charge d → ℤ), ∀ Z : Charge d → ℤ,
    massForm ((toReal g)⊤ * H * toReal g) (fun i => (Z i : ℝ)) =
    massForm H (fun i => (e Z i : ℝ)) ∧
    chargeNorm (e Z) = chargeNorm Z := by ...
```

**Reading.** For every  $d$ , every integer  $g$  with  $g^\top \eta g = \eta$  and every real  $2d \times 2d$  matrix  $H$ , there exists a bijection  $e$  of the set of integer charge vectors such that, for all  $Z$ , the number  $Z^\top (g^\top H g) Z$  equals  $e(Z)^\top H e(Z)$  and  $e(Z)$  has the same charge norm as  $Z$ . This is Theorem 11.4 word for word, including the generality in  $H$ : the matrix need not be symmetric, positive or of the form  $\mathcal{H}(G, B)$ . The symbol  $\simeq$  is Lean's type of bijections given together with their inverse, and `toReal` and the casts  $(Z i : \mathbb{R})$  are needed because the group and the charges live over  $\mathbb{Z}$  while the background lives over  $\mathbb{R}$ . The statement is existential; the proof supplies  $e(Z) = gZ$  with inverse  $Z \mapsto (\eta g^\top \eta)Z$ . **What it does not say.** It does not mention oscillators,  $\alpha'$ , the mass formula (11.3) as a whole, the partition function or correlation functions, and it does

not say that  $g^\top \mathcal{H}(G, B) g$  is again a generalized metric. Corollary 11.5, Lemma 11.6, Lemma 11.7 and Theorem 11.8 are proved on the page only. (The “only if” half of Lemma 11.6 has a partial Lean counterpart, `etaR_genMetric_sq` and `genMetric_symm` in `DFT/GeneralizedMetric.lean`; see Chapter 15.) **Proof idea.** The bijection is assembled from `isODD_left_inv` and `isODD_right_inv`. For the mass form one first shows that casting to  $\mathbb{R}$  commutes with the matrix–vector product (`push_cast`), then applies `massForm_covariant`, which is (11.5) and needs no hypothesis on  $g$ . The charge-norm half is `chargeNorm_invariant`. **[A]**

Two further files of the same directory, treated in Chapter 12, contain statements we have used: `thetaShift_mul` and `basisChange_mul` are the first two relations in (11.31) (for  $g_\Theta$  and  $\text{diag}(A, B)$ ). The theorem `mirror_conjugates_tauShift` is the relation (11.30), stated as  $D_0 \text{diag}(A, B) D_0 = g_\Theta^\top$  for one explicit pair of  $2 \times 2$  matrices. Finally `eta_one_reindex` states that for  $d = 1$  the matrix  $\eta$ , reindexed from `Fin 1`  $\oplus$  `Fin 1` to `Fin 2`, is equal to the matrix `hyperbolicU` of the hyperbolic plane  $U$  of Chapter 18.

| Claim                                                                                                                                              | Tier       | Evidence                                                                                                                                       |
|----------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| $\eta^2 = \mathbf{1}$ ; closure of $O(d, d; \mathbb{Z})$ under product and inverse; $g^{-1} = \eta g^\top \eta$                                    | <b>[A]</b> | <code>eta_mul_self</code> , <code>isODD_mul</code> , <code>isODD_left_inv</code> , <code>isODD_right_inv</code> , <code>isODD_inv_isODD</code> |
| $g_\Theta$ , $\text{diag}(A, B)$ with $A^\top B = \mathbf{1}$ , $D_k$ , $\eta$ lie in $O(d, d; \mathbb{Z})$                                        | <b>[A]</b> | <code>thetaShift_isODD</code> , <code>basisChange_isODD</code> , <code>factorized_isODD</code> , <code>eta_isODD</code>                        |
| $D_k^2 = \mathbf{1}$ , $D_k D_l = D_l D_k$ ; $D_0 D_1 = \eta$ for $d = 2$                                                                          | <b>[A]</b> | <code>factorized_mul_self</code> , <code>factorized_comm</code> , <code>factorized_two</code>                                                  |
| $Z^\top \eta Z = 2n \cdot w$ is even and $O(d, d; \mathbb{Z})$ -invariant                                                                          | <b>[A]</b> | <code>chargeNorm_sumElim</code> , <code>chargeNorm_even</code> , <code>chargeNorm_invariant</code>                                             |
| Theorem 11.4: bijection of charges matching mass form and charge norm                                                                              | <b>[A]</b> | <code>spectrum_equivalence</code>                                                                                                              |
| $\det g = \pm 1$ ; $g^\top \in O(d, d; \mathbb{Z})$ ; $D_0 \cdots D_{d-1} = \eta$ for all $d$ ; Lemmas 11.6 and 11.7; Theorem 11.8; Corollary 11.5 | —          | proved on the page; (11.18) also checked symbolically (sympy)                                                                                  |
| The three families generate $O(d, d; \mathbb{Z})$                                                                                                  | <b>[L]</b> | GPR §2.4, ll. 1422–1423                                                                                                                        |
| Equivalence of the full conformal field theories, not only of the spectra                                                                          | <b>[L]</b> | GPR §2.4, ll. 1336–1348; Chapter 9                                                                                                             |
| Identification of <code>massForm</code> and <code>chargeNorm</code> with the mass and level matching of a physical string                          | <b>[L]</b> | Chapter 10; never Tier A                                                                                                                       |

### Chapter summary

- With  $Z = (m, n)$ , the background enters the closed-string spectrum only through  $Z^\top \mathcal{H}(G, B) Z$  and level matching only through  $\frac{1}{2} Z^\top \eta Z = n \cdot m$ . A relabelling  $Z \mapsto gZ$  respects level matching for all charges if and only if  $g^\top \eta g = \eta$ .
- $O(d, d; \mathbb{Z})$  is the group of integer matrices with  $g^\top \eta g = \eta$ . The inverse is  $\eta g^\top \eta$ , automatically an integer matrix; the transpose of an element is an element;  $\det g = \pm 1$ . The real group has dimension  $d(2d - 1)$ , and  $O(1, 1; \mathbb{Z}) = \{\pm \mathbf{1}, \pm \eta\}$ .
- Spectrum equivalence: for  $g \in O(d, d; \mathbb{Z})$  the map  $Z \mapsto gZ$  is a bijection of the charge lattice with  $Z^\top (g^\top \mathcal{H} g) Z = (gZ)^\top \mathcal{H} (gZ)$  and  $(gZ)^\top \eta (gZ) = Z^\top \eta Z$ . The proof is three lines, and it is the Lean theorem `spectrum_equivalence`.
- A matrix is a generalized metric if and only if it is symmetric, positive definite and satisfies  $\mathcal{H} \eta \mathcal{H} = \eta$ . Hence  $g^\top \mathcal{H}(E) g = \mathcal{H}(E')$  with  $E' = (a^\top E + c^\top)(b^\top E + d^\top)^{-1}$ .

- Three families: integer  $B$ -shifts (large gauge transformations,  $B \mapsto B + \Theta$ ,  $n \mapsto n - \Theta m$ ); basis changes (large diffeomorphisms,  $E \mapsto AEA^\top$ ); factorized dualities ( $m^k \leftrightarrow n_k$ , Buscher rules,  $R_k \rightarrow \alpha'/R_k$ ). Their product over all circles is  $\eta$ , which acts as  $E \mapsto E^{-1}$ .
- That these families generate the whole group is Tier L and is not formalized. World-sheet parity  $B \rightarrow -B$  is not in the group, and a generic real element maps a background to an inequivalent one.

## Exercises

**Exercise 11.1** ( $\star$  Relations). Verify the four relations (11.31) by block multiplication. For the last one take  $d = 3$  and the cyclic permutation, write  $L_\pi$  and  $D_0$  as  $6 \times 6$  matrices, and find which  $D_j$  results.

**Exercise 11.2** ( $\star$  The table). For the background (11.28), apply  $D_1$  instead of  $D_0$ . Compute  $E'$  from (11.25) (with the roles of the indices 0 and 1 exchanged), show that  $B' = 0$  again, and find the radii and the angle of the dual torus. Reproduce two rows of the analogue of Table 11.2.

**Exercise 11.3** ( $\star\star$  The full duality). Show that  $\eta$  acts as  $E \mapsto E^{-1}$ , and that the symmetric and antisymmetric parts of  $E^{-1} = (G+B)^{-1}$  are  $G' = (G - BG^{-1}B)^{-1}$  and  $B' = -G^{-1}B(G - BG^{-1}B)^{-1}$ . (Hint: compare the blocks of  $\eta \mathcal{H}(G, B) \eta$  with those of  $\mathcal{H}(G', B')$ .) Evaluate both for the background (11.28) and check the result against  $D_1$  applied to (11.29).

**Exercise 11.4** ( $\star\star$  The geometric subgroup). Show that the elements of  $O(d, d; \mathbb{Z})$  with vanishing upper right block,  $b = 0$ , are exactly the products  $S_\Theta L_A$  with  $A \in GL(d, \mathbb{Z})$  and  $\Theta$  antisymmetric and integer, and that they form a subgroup. Show that  $S_\Theta L_A$  maps  $(G, B)$  to  $(AGA^\top, A(B + \Theta)A^\top)$  (the shift acts first; why?), and conclude that a point particle on the torus, whose spectrum is  $n^\top G^{-1}n$ , cannot distinguish the two backgrounds when  $B = 0 = \Theta$ .

**Exercise 11.5** ( $\star\star$  Which elements act trivially?). Show that  $g^\top \mathcal{H}g = \mathcal{H}$  for *all* generalized metrics  $\mathcal{H}$  if and only if  $g = \pm \mathbf{1}$ . (Hint: use  $\mathcal{H}(G, 0)$  with diagonal  $G$  first, then a non-diagonal  $G$ , then  $B \neq 0$ .) Conclude that the group acting faithfully on backgrounds is  $O(d, d; \mathbb{Z})/\{\pm \mathbf{1}\}$ , and describe it for  $d = 1$ .

**Exercise 11.6** ( $\star\star$  Infinitesimal transformations). Let  $X = \begin{pmatrix} \alpha & \beta \\ \gamma & -\alpha^\top \end{pmatrix}$  with  $\beta, \gamma$  antisymmetric. (a) Verify  $X^\top \eta + \eta X = 0$ . (b) Compute  $\exp X$  when only  $\gamma \neq 0$  and when only  $\alpha \neq 0$ , and compare with  $S_\Theta$  and  $L_A$ . (c) Using (11.18) to first order, show that  $\delta E = \alpha^\top E + E\alpha + \gamma^\top - E\beta^\top E$ . Which term is non-linear in the background, and why does it carry the name  $\beta$ -transformation?

**Exercise 11.7** ( $\star\star$  Lean: the transpose). State and prove in Lean: `theorem isODD_transpose (g : Matrix (Charge d) (Charge d) ℤ) (hg : IsODD g) : IsODD g⊤`. Follow the proof of Proposition 11.2(iv): start from `isODD_right_inv g hg`, multiply on the right by `eta d`, and simplify with `Matrix.mul_assoc` and `eta_mul_self`. Then deduce that  $S_\Theta$  satisfies `IsODD` from `thetaShift_isODD`.

**Exercise 11.8** ( $\star\star$  Lean: small elements). State and prove `IsODD (1 : Matrix (Charge d) (Charge d) ℤ)` and `IsODD (-1 : Matrix (Charge d) (Charge d) ℤ)`. Then obtain the second statement in a different way, from `basisChange_isODD` with  $A = B = -\mathbf{1}$ ; you will need a lemma saying that `basisChange (-1) (-1) = -1`.



**Exercise 11.9** (★★★ Lean: all circles at once). (a) Prove `factorized (0 : Fin 3) * factorized 1 * factorized 2 = eta 3` in the style of `factorized_two`. (b) Formulate Proposition 11.10 for general  $d$  in Lean. One possibility uses `Finset.noncommProd` together with `factorized_comm`; another avoids products altogether by proving  $D_k = \mathbf{1} + \hat{e}_k(\eta - \mathbf{1})$  and an induction over a list of distinct indices. Explain which hypotheses of the on-page proof each formulation needs.

**Exercise 11.10** (★★★ Symmetries of the most symmetric torus). Let  $\mathcal{H} = \mathbf{1}$ , that is  $G = \mathbf{1}$ ,  $B = 0$ : a product of  $d$  self-dual circles. (a) Show that an integer matrix  $g$  with  $g^\top g = \mathbf{1}$  is a signed permutation matrix, so that there are  $2^{2d}(2d)!$  of them. (b) Show that such a  $g$  lies in  $O(d, d; \mathbb{Z})$  if and only if it commutes with  $\eta$ , and that there are  $2^d \cdot 2^d \cdot d!$  such elements: 4, 32, 384 for  $d = 1, 2, 3$ . Identify the three factors with reflections of circles, factorized dualities and permutations of circles. (c) For  $d = 2$  the permutation matrix  $g$  that exchanges  $m^1$  with  $n_0$  satisfies  $g^\top \mathcal{H} g = \mathcal{H}$  but is not in  $O(2, 2; \mathbb{Z})$ . Which part of the proof of Theorem 11.4 fails for it? Exhibit a charge  $Z$  that is allowed at oscillator levels  $N = \tilde{N}$  while  $gZ$  is not. This shows that preserving the form of  $\mathcal{H}$  is not enough; level matching is what singles out  $O(d, d; \mathbb{Z})$ .

## Further reading

- The definition of  $O(d, d)$  in block form, the action on  $E$  by fractional linear transformations, the three families with their physical interpretation, the generation statement and worldsheet parity: [GPR §2.4, ll. 1213–1427]. The transformation of the oscillators, which completes the map between the two Hilbert spaces: [GPR §2.4, ll. 1428–1497].
- The same group from the point of view of double field theory, with the generalized momentum, the decomposition into diffeomorphisms, shifts and factorized T-dualities, the Buscher rules and  $\beta$ -transformations: [AMN §3.1, ll. 567–779].
- The Lean sources are `ODD.lean`, `Factorized.lean` and `Spectrum.lean` in the directory `DualScaleStream2/TDuality/`; their headers list what is and what is not proved. For the real generalized metric see `DualScaleStream2/DFT/GeneralizedMetric.lean` and Chapter 15; for  $d = 2$  see Chapter 12.



## Chapter 12

# The Two-Torus: $\tau$ , $\rho$ and Mirror Symmetry

A string on a circle has one modulus, the radius, and one duality,  $R \rightarrow \alpha'/R$  (Chapter 8). On a  $d$ -torus there are  $d^2$  moduli and the duality group is  $O(d, d; \mathbb{Z})$  (Chapters 10 and 11). The first case in which this group is more than a reflection is  $d = 2$ , and it is the case a student should know by heart, for three reasons. First, the four real moduli of  $T^2$  combine into two complex numbers,  $\tau$  and  $\rho$ , each living in the upper half-plane, and the abstract group  $O(2, 2; \mathbb{Z})$  becomes two copies of the modular group  $SL(2, \mathbb{Z})$  acting by fractional linear maps, which can be drawn. Second, the circle duality applied to one side of the torus *exchanges*  $\tau$  and  $\rho$ : the shape of one torus is the size of the other. This is the simplest example of mirror symmetry, and the only one in this book where everything can be computed by hand. Third, the torus  $T^2$  is a factor of the compactification space  $K3 \times T^2$  of Chapter 24, and the hyperbolic geometry of its moduli space is the model for the distance statements of Chapter 27. The question of this chapter is therefore: *what are  $\tau$  and  $\rho$ , how does each generator of  $O(2, 2; \mathbb{Z})$  act on them, and which of these statements have been checked by the Lean kernel?*

**Conventions.** As in Chapter 10, the torus coordinates have period  $2\pi$ ,  $X^i \sim X^i + 2\pi$ , the metric  $G_{ij} = \hat{e}_i \cdot \hat{e}_j / \alpha'$  and the antisymmetric field  $B_{ij}$  are dimensionless (lengths are measured in units of  $\sqrt{\alpha'}$ ), the doubled charge vector is  $Z = (m^1, m^2, n_1, n_2)$  with windings first, and the generalized metric is

$$\mathcal{H}(G, B) = \begin{pmatrix} G - BG^{-1}B & BG^{-1} \\ -G^{-1}B & G^{-1} \end{pmatrix}, \quad \frac{1}{2}(p_L^2 + p_R^2) = \frac{1}{2}Z^T \mathcal{H} Z. \quad (12.1)$$

A duality  $g \in O(2, 2; \mathbb{Z})$  acts on backgrounds by

$$\mathcal{H} \longmapsto g \mathcal{H} g^T, \quad (12.2)$$

which is the convention of our main source [GPR §2.4, eq. (2.4.19), ll. 1288–1293] and of the Lean theorem `genMetric_bshift`. The transposed convention  $\mathcal{H} \mapsto g^T \mathcal{H} g$  of Chapter 10 describes the same group; the difference matters only when one asks which matrix does what, and we return to it in Section 12.7.

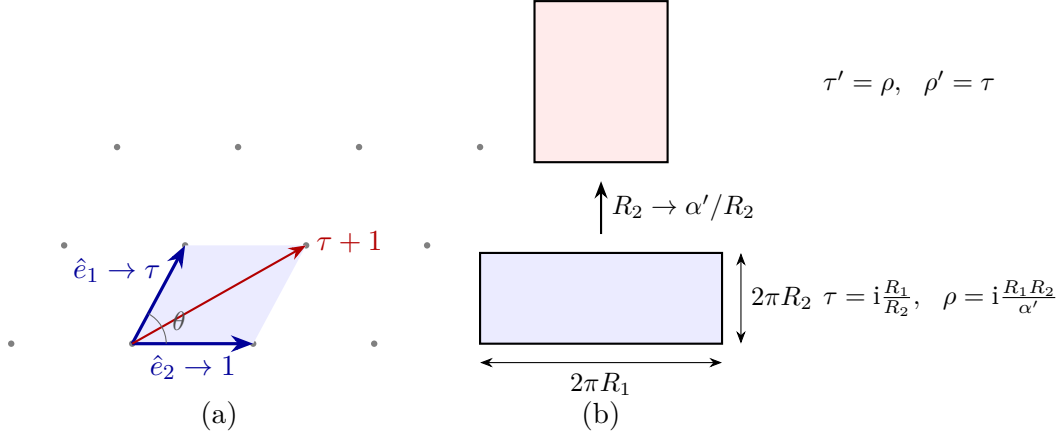


Figure 12.1: (a) The lattice of a two-torus, rotated and rescaled so that  $\hat{e}_2 = 1$ ; then  $\hat{e}_1 = \tau$ . Replacing  $\hat{e}_1$  by  $\hat{e}_1 + \hat{e}_2$  (red) gives the same lattice, hence the same torus, with  $\tau + 1$  in place of  $\tau$ . (b) A rectangular torus with  $R_1/R_2 = 8/3$  and  $R_1 R_2 = \frac{3}{2}\alpha'$ , and its image under T-duality on the second circle: the ratio of the sides and the area in string units have been exchanged.

## 12.1 Four real numbers, two complex numbers

### 12.1.1 The shape of the torus: $\tau$

A flat two-torus is the plane divided by the lattice generated by two vectors  $2\pi\hat{e}_1$ ,  $2\pi\hat{e}_2$ . Identify the plane with  $\mathbb{C}$  and regard  $\hat{e}_1, \hat{e}_2$  as complex numbers. Rotating the plane and rescaling all lengths does not change the *shape* of the torus, that is, its structure as a Riemann surface; both operations multiply  $\hat{e}_1$  and  $\hat{e}_2$  by the same non-zero complex number. The only invariant of the pair under such a multiplication is the ratio

$$\tau = \frac{\hat{e}_1}{\hat{e}_2} \in \mathbb{C}, \quad (12.3)$$

and we order the two vectors so that  $\text{Im } \tau > 0$ . This number is the *complex structure modulus* of the torus. To express it through the metric, multiply numerator and denominator by  $\bar{\hat{e}}_2$ :  $\tau = \hat{e}_1 \bar{\hat{e}}_2 / |\hat{e}_2|^2$ . The real part of  $\hat{e}_1 \bar{\hat{e}}_2$  is the scalar product  $\hat{e}_1 \cdot \hat{e}_2 = \alpha' G_{12}$ , and its imaginary part is, up to the sign fixed by  $\text{Im } \tau > 0$ , the area of the parallelogram spanned by the two vectors,  $\alpha' \sqrt{\det G}$ . Hence

$$\tau = \tau_1 + i\tau_2 = \frac{G_{12}}{G_{22}} + i \frac{\sqrt{\det G}}{G_{22}}, \quad \det G = G_{11}G_{22} - G_{12}^2 > 0 \quad (12.4)$$

[GPR §2.4.2, eq. (2.4.55), ll. 1883–1892]. After a rotation and rescaling that bring  $\hat{e}_2$  to the number 1, the torus is the parallelogram with corners  $0, 1, \tau, 1 + \tau$  (Figure 12.1). The angle  $\theta$  between the two sides satisfies  $\cos \theta = \tau_1/|\tau|$ , and the ratio of their lengths is  $|\tau|$ ; a rectangular torus has  $\tau_1 = 0$ , a square one  $\tau = i$ .

#### Common pitfall: which ratio is $\tau$ ?

Many texts define  $\tau = \hat{e}_2/\hat{e}_1$ , the inverse of (12.3) up to orientation. We follow GPR, where  $G_{22}$  stands in the denominator of (12.4). With the other convention  $\tau$  is replaced by  $-1/\tau$  composed with a reflection, every statement of this chapter survives, but the matrices that represent “ $\tau \rightarrow \tau + 1$ ” and “ $\tau \rightarrow \tau/(\tau + 1)$ ” exchange their roles. Always check a formula on the rectangular torus before using it.

### 12.1.2 The size and the $B$ -field: $\rho$

The metric has three independent components and  $\tau$  uses two combinations of them. The third is the overall scale, which we measure by the area. With coordinates of period  $2\pi$  the area of the torus is  $(2\pi)^2 \alpha' \sqrt{\det G}$ . In  $d = 2$  an antisymmetric matrix has one component,  $B_{12} = -B_{21} = b$ . The *complexified Kähler modulus* combines the two:

$$\rho = \rho_1 + i\rho_2 = B_{12} + i\sqrt{\det G} \quad (12.5)$$

[GPR §2.4.2, eq. (2.4.55), ll. 1883–1892]. The name comes from complex geometry: on a complex manifold with a compatible metric, the Kähler form is the two-form built from the metric and the complex structure; on  $T^2$  it is the area form, and its integral is the area. One half-plane of the  $T^2$  moduli space “represents the Kähler form and  $B$ -field on the torus” [Asp §5.1, ll. 2750–2762]. Why the area and  $B$  belong together in one complex number, rather than being two unrelated real parameters, is not obvious at this point; the mass formula of Section 12.2 will show that  $\rho$  enters the spectrum exactly as  $\tau$  does.

The map  $(G_{11}, G_{12}, G_{22}, B_{12}) \mapsto (\tau, \rho)$  is invertible. Solving (12.4) and (12.5) for the metric gives  $G_{22} = \rho_2/\tau_2$ ,  $G_{12} = \tau_1 G_{22}$  and  $G_{11} = (\det G + G_{12}^2)/G_{22} = \rho_2(\tau_2^2 + \tau_1^2)/\tau_2$ , that is

$$E = G + B = \frac{\rho_2}{\tau_2} \begin{pmatrix} |\tau|^2 & \tau_1 \\ \tau_1 & 1 \end{pmatrix} + \rho_1 \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \quad (12.6)$$

[GPR §2.4.2, eq. (2.4.56), ll. 1893–1917]. One checks  $\det G = (\rho_2/\tau_2)^2(|\tau|^2 - \tau_1^2) = \rho_2^2$ , as it must be. The moduli space of backgrounds is therefore, before any duality is taken into account,

$$\{(\tau, \rho)\} = \mathbb{H} \times \mathbb{H}, \quad \mathbb{H} = \{z \in \mathbb{C} : \text{Im } z > 0\}, \quad (12.7)$$

two copies of the upper half-plane (the symbol  $\mathbb{H}$  is not to be confused with the generalized metric  $\mathcal{H}$ ). This is the  $d = 2$  case of the coset  $O(2, 2; \mathbb{R})/(O(2) \times O(2))$  of Chapter 10, and each factor is  $SL(2, \mathbb{R})/U(1)$  [GPR §2.4.2, ll. 1876–1882; Asp §5.1, eq. (100), ll. 2750–2762].

**Example 12.1** (Rectangular torus). Two orthogonal circles of radii  $R_1, R_2$  and  $B = 0$  have  $G = \text{diag}(R_1^2, R_2^2)/\alpha'$ , so

$$\tau = i \frac{R_1}{R_2}, \quad \rho = i \frac{R_1 R_2}{\alpha'}. \quad (12.8)$$

The first is the ratio of the sides, the second the area in units of  $(2\pi)^2 \alpha'$ . T-duality on the second circle,  $R_2 \rightarrow \alpha'/R_2$ , gives  $\tau' = iR_1 R_2/\alpha' = \rho$  and  $\rho' = iR_1/R_2 = \tau$ . For  $R_1 = 2\sqrt{\alpha'}$ ,  $R_2 = \frac{3}{4}\sqrt{\alpha'}$  one has  $\tau = \frac{8}{3}i$ ,  $\rho = \frac{3}{2}i$  (Figure 12.1b): a long thin torus of moderate area is dual to a moderately elongated torus of large area. This exchange is the subject of Section 12.5.

## 12.2 The spectrum in terms of $\tau$ and $\rho$

The zero-mode spectrum of the string on  $T^2$  is governed by the left and right momenta  $p_L, p_R$  of Chapter 10, whose squared lengths are

$$p_L^2 = \frac{1}{2} v_L^T G^{-1} v_L, \quad v_L = n + (G - B)m; \quad p_R^2 = \frac{1}{2} v_R^T G^{-1} v_R, \quad v_R = n - (G + B)m. \quad (12.9)$$

Here  $n = (n_1, n_2)$  are the momentum numbers and  $m = (m^1, m^2)$  the winding numbers; below we write  $m_i$  for  $m^i$  to keep the formulas light. We now express (12.9) through  $\tau$  and  $\rho$ . The computation has three steps.

**Step 1: the length of a covector.** Inverting the metric in (12.6),

$$G^{-1} = \frac{1}{\rho_2 \tau_2} \begin{pmatrix} 1 & -\tau_1 \\ -\tau_1 & |\tau|^2 \end{pmatrix}, \quad v^\top G^{-1} v = \frac{v_1^2 - 2\tau_1 v_1 v_2 + |\tau|^2 v_2^2}{\rho_2 \tau_2} = \frac{|v_1 - \tau v_2|^2}{\rho_2 \tau_2}$$

for any real  $v = (v_1, v_2)$ . The complex number  $v_1 - \tau v_2$  is, up to a phase and the factor  $(\rho_2 \tau_2)^{-1/2}$ , the covector  $v$  written in an orthonormal frame.

**Step 2: the map  $v \mapsto v_1 - \tau v_2$  applied to  $Gm$  and  $Bm$ .** From (12.6),

$$\begin{aligned} (Gm)_1 - \tau(Gm)_2 &= \frac{\rho_2}{\tau_2} \left[ |\tau|^2 m_1 + \tau_1 m_2 - \tau(\tau_1 m_1 + m_2) \right] = \frac{\rho_2}{\tau_2} \left[ \tau(\bar{\tau} - \tau_1) m_1 + (\tau_1 - \tau) m_2 \right] \\ &= \frac{\rho_2}{\tau_2} (-i\tau_2) (\tau m_1 + m_2) = -i\rho_2 (m_2 + \tau m_1), \end{aligned}$$

where we used  $|\tau|^2 = \tau\bar{\tau}$  and  $\bar{\tau} - \tau_1 = \tau_1 - \tau = -i\tau_2$ . For the  $B$ -field,  $Bm = \rho_1(m_2, -m_1)$  and therefore  $(Bm)_1 - \tau(Bm)_2 = \rho_1(m_2 + \tau m_1)$ . The same complex combination  $m_2 + \tau m_1$  of the winding numbers appears in both.

**Step 3: assemble.** For  $v_L = n + Gm - Bm$  the coefficient of  $(m_2 + \tau m_1)$  is  $-i\rho_2 - \rho_1 = -\rho$ ; for  $v_R = n - Gm - Bm$  it is  $+i\rho_2 - \rho_1 = -\bar{\rho}$ . Hence:

**Proposition 12.2** (Zero-mode spectrum on  $T^2$ ). *With  $\tau, \rho$  defined by (12.4) and (12.5),*

$$p_L^2 = \frac{|(n_1 - \tau n_2) - \rho(m_2 + \tau m_1)|^2}{2\rho_2 \tau_2}, \quad p_R^2 = \frac{|(n_1 - \tau n_2) - \bar{\rho}(m_2 + \tau m_1)|^2}{2\rho_2 \tau_2}. \quad (12.10)$$

This is [GPR §2.4.2, eq. (2.4.58), ll. 1953–1967]; we have also confirmed by a symbolic check (sympy) that (12.10) equals (12.9) identically in the eight variables, and that  $p_L^2 - p_R^2 = 2(n_1 m_1 + n_2 m_2)$ . The last identity can be seen directly: with  $a = n_1 - \tau n_2$  and  $c = m_2 + \tau m_1$ ,  $|a - \rho c|^2 - |a - \bar{\rho} c|^2 = -2 \operatorname{Re}[(\rho - \bar{\rho}) \bar{a} c] = 4\rho_2 \operatorname{Im}(\bar{a} c)$  and  $\operatorname{Im}(\bar{a} c) = \tau_2(n_1 m_1 + n_2 m_2)$ , so the moduli cancel, as the level-matching condition requires.

The mass formula of Chapter 10 becomes

$$\frac{\alpha'}{2} M^2 = \frac{p_L^2 + p_R^2}{2} + N + \tilde{N} - 2, \quad N - \tilde{N} = n_1 m_1 + n_2 m_2. \quad (12.11)$$

#### Physical picture: what $\tau$ and $\rho$ do to the spectrum

Set the windings to zero. Then  $p_L^2 = p_R^2 = |n_1 - \tau n_2|^2 / (2\rho_2 \tau_2)$ : the Kaluza–Klein masses are the squared lengths of the vectors of the *dual* lattice, whose shape is governed by  $\tau$  and whose scale is  $1/\rho_2$  — a large torus has light momentum modes. Now set the momenta to zero:  $p_L^2 = p_R^2 = |\rho|^2 |m_2 + \tau m_1|^2 / (2\rho_2 \tau_2)$ . For  $B = 0$  this is  $\rho_2 |m_2 + \tau m_1|^2 / (2\tau_2)$ , the squared length of the wound string, which grows with the area. The  $B$ -field does not affect a state with momentum only, but it shifts the effective momentum of a wound string: for  $m \neq 0$  the numerator contains  $n_1 - \rho_1 m_2$  and  $n_2 + \rho_1 m_1$ . A shift of  $\rho_1$  by an integer can thus be absorbed in a relabelling of the integers  $n_i$ .

The value of (12.10) is that symmetries can be read off. Any transformation of  $(\tau, \rho)$  that can be compensated by an invertible integer relabelling of  $(n_1, n_2, m_1, m_2)$ , leaving the set of pairs  $(p_L^2, p_R^2)$  unchanged, maps a background to a physically equivalent one. For example, exchanging  $\tau \leftrightarrow \rho$  in the numerator of  $p_L^2$  gives  $n_1 - \rho n_2 - \tau m_2 - \tau \rho m_1$ , which is the original expression  $n_1 - \tau n_2 - \rho m_2 - \rho \tau m_1$  with  $n_2$  and  $m_2$  exchanged: a momentum number has been traded for a winding number in the second direction, which is what T-duality on the second circle does. We now go through the generators systematically.

### 12.3 The duality group of the two-torus

We use the three families of generators of  $O(d, d; \mathbb{Z})$  from Chapter 11 [GPR §2.4, eqs. (2.4.25), (2.4.26), (2.4.29), ll. 1355–1414], in block form with respect to  $Z = (m, n)$ :

$$g_\Theta = \begin{pmatrix} \mathbf{1} & \Theta \\ 0 & \mathbf{1} \end{pmatrix}, \quad g_A = \begin{pmatrix} A & 0 \\ 0 & (A^\top)^{-1} \end{pmatrix}, \quad g_{D_i} = \begin{pmatrix} \mathbf{1} - e_i & e_i \\ e_i & \mathbf{1} - e_i \end{pmatrix}, \quad (12.12)$$

with  $\Theta$  an antisymmetric integer matrix,  $A \in GL(2, \mathbb{Z})$ , and  $e_i$  the matrix with a single 1 at position  $(i, i)$ . In Lean these are `thetaShift`, `basisChange` and `factorized`. Inserting them in (12.2) and comparing blocks with (12.1) gives their action on  $E = G + B$ :

$$g_\Theta : B \rightarrow B + \Theta, \quad g_A : G \rightarrow AGA^\top, \quad B \rightarrow ABA^\top. \quad (12.13)$$

For the first, this is the content of `genMetric_bshift` [A]. For the second,  $g_A \mathcal{H} g_A^\top$  has lower right block  $A^{-\top} G^{-1} A^{-1} = (AGA^\top)^{-1}$  and upper right block  $ABG^{-1} A^{-1} = (ABA^\top)(AGA^\top)^{-1}$ .

#### 12.3.1 The modular group of the shape: $SL(2, \mathbb{Z})_\tau$

Let  $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$  have integer entries and  $\det A = 1$ . The transformation  $G \rightarrow AGA^\top$  is what the metric  $G_{ij} = \hat{e}_i \cdot \hat{e}_j / \alpha'$  undergoes when the lattice basis is changed to

$$\hat{e}'_1 = a \hat{e}_1 + b \hat{e}_2, \quad \hat{e}'_2 = c \hat{e}_1 + d \hat{e}_2. \quad (12.14)$$

Since  $\det A = 1$  and the inverse matrix is again integral, the new vectors generate the same lattice: the torus has not changed, only its description. Dividing the two equations and using (12.3),

$$\tau' = \frac{\hat{e}'_1}{\hat{e}'_2} = \frac{a\tau + b}{c\tau + d}. \quad (12.15)$$

What happens to  $\rho$ ? The area of the cell is unchanged because  $\det(AGA^\top) = (\det A)^2 \det G$ . For the  $B$ -field we need a small lemma, which is special to  $2 \times 2$  matrices: with  $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ ,

$$A J A^\top = (\det A) J \quad \text{for every } 2 \times 2 \text{ matrix } A. \quad (12.16)$$

Indeed the  $(1, 2)$  entry of the left side is  $ad - bc$ , the diagonal entries are  $ab - ba = 0$  and  $cd - dc = 0$ , and the result is antisymmetric. This is the Lean theorem `mul_jMat_mul_transpose` [A]. Since  $B = \rho_1 J$ , a basis change with  $\det A = 1$  leaves  $B$ , hence  $\rho$ , untouched:

$$SL(2, \mathbb{Z})_\tau : \quad \tau \rightarrow \frac{a\tau + b}{c\tau + d}, \quad \rho \rightarrow \rho. \quad (12.17)$$

The group is generated by  $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ ,  $\tau \rightarrow \tau + 1$ , and  $S = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ ,  $\tau \rightarrow -1/\tau$  (the fact that  $S$  and  $T$  generate  $SL(2, \mathbb{Z})$  is standard; not checked against a source in this book's library). Geometrically  $T$  is the change of basis drawn in Figure 12.1a and  $S$  exchanges the two sides of the cell, with a sign to preserve the orientation. These symmetries “reflect the fact that the target space is a 2-d torus, and thus unmodified by modular transformations of its complex structure” [GPR §2.4.2, ll. 1948–1949]: they are present for a point particle on  $T^2$  as well and are not stringy.

A basis change with  $\det A = -1$  reverses the orientation. For the reflection  $R_1 = \text{diag}(-1, 1)$ , that is  $X^1 \rightarrow -X^1$ , one finds  $G_{12} \rightarrow -G_{12}$  and, by (12.16),  $B \rightarrow -B$ :

$$R : \quad (\tau, \rho) \rightarrow (-\bar{\tau}, -\bar{\rho}) \quad (12.18)$$

[GPR §2.4.2, eq. (2.4.60), ll. 1980–1986].

### 12.3.2 The modular group of the size: $SL(2, \mathbb{Z})_\rho$

In  $d = 2$  the antisymmetric integer matrices are  $\Theta = tJ$ ,  $t \in \mathbb{Z}$ , and by (12.13)

$$T'^t = g_{tJ} : \quad \rho \rightarrow \rho + t, \quad \tau \rightarrow \tau. \quad (12.19)$$

The reason this is a symmetry was given in Chapter 11: the  $B$ -term of the worldsheet action is a total derivative, it only measures how many times the worldsheet wraps the torus, and an integer shift of  $B_{12}$  changes the action by a multiple of  $2\pi$  [GPR §2.4, ll. 1368–1371]. It “expresses the periodicity in the antisymmetric torsion  $B_{12}$ ” [GPR §2.4.2, l. 1950]. The  $B$ -field is thus an angle, like the  $\theta$ -parameter of a gauge theory, and  $\rho_1$  and  $\tau_1$  are both periodic variables.

The analogy between (12.19) and  $\tau \rightarrow \tau + 1$  suggests that there is also a symmetry

$$S' : \quad \rho \rightarrow -\frac{1}{\rho}, \quad \tau \rightarrow \tau, \quad (12.20)$$

and (12.10) confirms it. Write  $a = n_1 - \tau n_2$  and  $c = m_2 + \tau m_1$ , so that  $p_L^2 = |a - \rho c|^2 / (2\rho_2 \tau_2)$ . Under  $\rho \rightarrow -1/\rho$  the imaginary part becomes  $\rho_2/|\rho|^2$ , and a state with new charges  $a', c'$  has

$$p_L'^2 = \frac{|\rho|^2 |a' + c'/\rho|^2}{2\rho_2 \tau_2} = \frac{|\rho a' + c'|^2}{2\rho_2 \tau_2},$$

which equals the old  $p_L^2$  for  $a' = c$ ,  $c' = -a$ ; the same substitution works for  $p_R^2$  with  $\bar{\rho}$ . In terms of the integers,  $a' = c$  means  $(n'_1, n'_2) = (m_2, -m_1)$  and  $c' = -a$  means  $(m'_1, m'_2) = (n_2, -n_1)$ : momenta and windings are exchanged, with a rotation by  $90^\circ$  in the lattice. For  $B = 0$  it sends the area  $\rho_2$  to  $1/\rho_2$  at fixed shape. This is “a stringy symmetry that for  $B = 0$  takes the volume in target space into its inverse” [GPR §2.4.2, ll. 1950–1952], the two-dimensional version of  $R \rightarrow \alpha'/R$ . Together,

$$SL(2, \mathbb{Z})_\rho : \quad \rho \rightarrow \frac{a'\rho + b'}{c'\rho + d'}, \quad \tau \rightarrow \tau. \quad (12.21)$$

The two groups act on different variables, so they commute as transformations of  $\mathbb{H} \times \mathbb{H}$ . At the level of the  $4 \times 4$  integer matrices this is a statement that needs proof. For the translations it follows from (12.16):

$$g_A g_{tJ} = \begin{pmatrix} A & tAJ \\ 0 & A^{-\top} \end{pmatrix}, \quad g_{tJ} g_A = \begin{pmatrix} A & tJA^{-\top} \\ 0 & A^{-\top} \end{pmatrix}, \quad (12.22)$$

and  $AJ = JA^{-\top}$  is equivalent to  $AJA^\top = J$ , that is, to  $\det A = 1$ . This is the Lean theorem `basisChange_comm_thetaShift` [A] (Section 12.7).

### 12.3.3 The exchanges: factorized duality and the full group

The factorized dualities  $D_1, D_2$  of (12.12) are T-duality on the first and on the second circle. Their action on the moduli, obtained by inserting (12.6) in (12.2) and reading off the new  $G$  and  $B$  (a symbolic check; the case  $D_2$  is [GPR §2.4.2, eq. (2.4.59), ll. 1969–1978]), is

$$D_2 : (\tau, \rho) \rightarrow (\rho, \tau), \quad D_1 : (\tau, \rho) \rightarrow \left(-\frac{1}{\rho}, -\frac{1}{\tau}\right), \quad D_1 D_2 = \eta : (\tau, \rho) \rightarrow \left(-\frac{1}{\tau}, -\frac{1}{\rho}\right). \quad (12.23)$$

The last entry is the inversion  $E \rightarrow E^{-1}$  of Chapter 11; on  $T^2$  it is the product  $SS'$  of the two modular inversions. The matrix identity  $D_1 D_2 = \eta$  is `factorized_two` [A]. The asymmetry

between  $D_1$  and  $D_2$  is a consequence of the convention (12.3), in which  $\hat{e}_2$  is the reference side. As integer matrices  $D_1 = -g_S S' D_2$  (symbolic check), so both dualities exchange the complex structure with the Kähler modulus, and they differ by an element of  $SL(2, \mathbb{Z})_\tau \times SL(2, \mathbb{Z})_\rho$ .

Finally, worldsheet parity  $\sigma \rightarrow -\sigma$  reverses the sign of  $B$  and nothing else:

$$W : (\tau, \rho) \rightarrow (\tau, -\bar{\rho}). \quad (12.24)$$

It exchanges  $p_L$  and  $p_R$ , changes the sign of  $p_L^2 - p_R^2$ , and is therefore *not* an element of  $O(2, 2; \mathbb{Z})$ , although it is a symmetry of the bosonic string [GPR §2.4.2, eq. (2.4.61), ll. 1986–1996].

**Theorem 12.3** (Duality group of  $T^2$  [L]). *The elements  $S, T$  (acting on  $\tau$ ),  $S', T'$  (acting on  $\rho$ ), the exchange  $D_2$  and the reflection  $R$  generate  $O(2, 2; \mathbb{Z})$ . The subgroup generated by  $S, T, S', T'$  is normal, and the quotient is  $\mathbb{Z}_2 \times \mathbb{Z}_2$ , generated by the classes of  $D_2$  and  $R$ :*

$$O(2, 2; \mathbb{Z}) \cong [SL(2, \mathbb{Z})_\tau \times SL(2, \mathbb{Z})_\rho] \rtimes (\mathbb{Z}_2 \times \mathbb{Z}_2), \quad (12.25)$$

where the product in brackets is understood with  $(-\mathbf{1}, -\mathbf{1})$  identified with the identity (Remark 12.4). Including  $W$ , a minimal set of generators of the full symmetry group is  $S, T, D_2, W$ , with  $S' = D_2 S D_2$ ,  $T' = D_2 T D_2$  and  $R = D_2 W D_2 W$ .

[GPR §2.4.2, ll. 1876–1882 and 1997–2000; generation of  $O(d, d; \mathbb{Z})$  by (12.12): GPR §2.4, ll. 1422–1423]. GPR write the right-hand side as “ $SL(2, \mathbb{Z}) \times SL(2, \mathbb{Z}) \otimes_S [\mathbb{Z}_2 \times \mathbb{Z}_2]$ ” and stress that it is *not* the naive factorization into two modular groups that the local form  $O(2, 2; \mathbb{R}) \sim SL(2, \mathbb{R}) \times SL(2, \mathbb{R})$  might suggest. The relations  $T' = D_2 T D_2$  and  $S' = D_2 S D_2$  are immediate from (12.23), since  $D_2$  simply renames the variable on which  $T$  or  $S$  acts; as integer matrices we have checked  $D_2 g_T D_2 = g_J$  symbolically, and that  $D_2$  and  $R$  commute, so that they generate a  $\mathbb{Z}_2 \times \mathbb{Z}_2$ .

*Remark 12.4* (The element  $-\mathbf{1}$ ). The matrix  $-\mathbf{1}_2 \in SL(2, \mathbb{Z})_\tau$  acts trivially on  $\tau$  but not on the charges:  $g_{-\mathbf{1}} = -\mathbf{1}_4$  sends  $Z \rightarrow -Z$ . The same is true of  $S'^2$ : from  $S' = D_2 S D_2$  and  $g_S^2 = -\mathbf{1}_4$  one gets  $S'^2 = -\mathbf{1}_4$  (symbolic check). Thus the two centres are mapped to the *same* element of  $O(2, 2; \mathbb{Z})$ , and the subgroup generated by  $S, T, S', T'$  is  $(SL(2, \mathbb{Z}) \times SL(2, \mathbb{Z}))/\mathbb{Z}_2$ . On the moduli space only  $PSL(2, \mathbb{Z}) = SL(2, \mathbb{Z})/\{\pm \mathbf{1}\}$  acts effectively in each factor. The state  $Z$  and the state  $-Z$  are a particle and its antiparticle;  $-\mathbf{1}_4$  is charge conjugation.

#### Common pitfall: “ $O(2, 2)$ is $SL(2) \times SL(2)$ ”

Over the real numbers and for the connected component this is true, and it is why the moduli space is a product of two half-planes. Over the integers, the two factors are exchanged by an element of the group itself. Consequently the moduli space of the string on  $T^2$  is *not* the product of two modular curves, but its quotient by the exchange  $\tau \leftrightarrow \rho$  and by  $(\tau, \rho) \rightarrow (-\bar{\tau}, -\bar{\rho})$ : a torus of shape  $\tau$  and size  $\rho$  is the same string background as one of shape  $\rho$  and size  $\tau$ .

## 12.4 The upper half-plane

Both moduli live in  $\mathbb{H}$  and both are acted on by  $SL(2, \mathbb{Z})$  through fractional linear maps. This section collects the geometry of that action. Nothing in it is specific to strings; the same half-plane appears as the moduli space of the worldsheet torus in Chapter 10 and as that of the type IIB axio-dilaton in Chapter 26. Except where a source is given, the statements of this section are standard and derived on the page; they have not been checked against a source in this book’s library.

### 12.4.1 Fractional linear maps

For a real matrix  $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$  with  $ad - bc = 1$  and  $z \in \mathbb{H}$  put  $\gamma z = (az + b)/(cz + d)$ . Multiplying numerator and denominator by  $c\bar{z} + d$ ,

$$\gamma z = \frac{ac|z|^2 + bd + adz + bc\bar{z}}{|cz + d|^2} \implies \operatorname{Im}(\gamma z) = \frac{(ad - bc) \operatorname{Im} z}{|cz + d|^2} = \frac{\operatorname{Im} z}{|cz + d|^2} > 0, \quad (12.26)$$

so  $\gamma$  maps  $\mathbb{H}$  to itself; the denominator cannot vanish because  $z$  is not real. Two facts make this an action of the group: the identity matrix acts trivially, and

$$\gamma(\gamma' z) = (\gamma\gamma')z. \quad (12.27)$$

The second is a short computation with fractions, and is kernel-checked for complex matrices as `mobius_compose` [A] (Section 12.7). Since  $\gamma$  and  $-\gamma$  give the same map, the group acting effectively is  $PSL(2, \mathbb{R})$ . The action is transitive:  $z = x + iy$  is the image of  $i$  under  $\begin{pmatrix} \sqrt{y} & x/\sqrt{y} \\ 0 & 1/\sqrt{y} \end{pmatrix}$ . The matrices fixing  $i$  satisfy  $ai + b = i(ci + d)$ , that is  $a = d$ ,  $b = -c$ ,  $a^2 + b^2 = 1$ : they form the rotation group  $SO(2) = U(1)$ . This proves  $\mathbb{H} = SL(2, \mathbb{R})/U(1)$ , the statement used in (12.7).

### 12.4.2 The fundamental domain

Two points of  $\mathbb{H}$  related by an element of the modular group  $SL(2, \mathbb{Z})$  describe the same torus. A *fundamental domain* is a region containing exactly one representative of each orbit, up to boundary identifications. The standard choice is

$$\mathcal{F} = \{z \in \mathbb{H} : |\operatorname{Re} z| \leq \frac{1}{2}, |z| \geq 1\}, \quad (12.28)$$

drawn in Figure 12.2. That every  $z$  can be brought into  $\mathcal{F}$  is shown by an algorithm which is also the practical way to reduce a given modulus:

1. apply a power of  $T$  to achieve  $|\operatorname{Re} z| \leq \frac{1}{2}$ ;
2. if  $|z| < 1$ , apply  $S$ : by (12.26) this *increases* the imaginary part,  $\operatorname{Im}(-1/z) = \operatorname{Im} z / |z|^2$ ; return to step 1.

The algorithm terminates. Indeed, all values taken by the imaginary part are of the form  $\operatorname{Im} z / |cz + d|^2$  with integers  $(c, d)$ , and only finitely many pairs of integers satisfy  $|cz + d| < 1$ , because the numbers  $cz + d$  form a lattice in  $\mathbb{C}$ . A strictly increasing sequence in a set with finitely many values above its starting point must stop. In Lean’s mathematical library this is the theorem `ModularGroup.exists_smul_mem_fd` of Mathlib (file `Mathlib/NumberTheory/Modular.lean`); it is part of the library this project builds on, not a result of this project. That no two interior points of  $\mathcal{F}$  are equivalent is the companion statement `ModularGroup.eq_smul_self_of_mem_fdo_mem_fdo`.

Three points of  $\mathcal{F}$  are special because they are fixed by a non-trivial element of  $PSL(2, \mathbb{Z})$ :  $z = i$  is fixed by  $S$  (order 2),  $z = e^{i\pi/3}$  by  $TS : z \rightarrow 1 - 1/z$  (order 3), and its  $T$ -image  $e^{2\pi i/3}$  by  $ST$ . At these points the quotient  $SL(2, \mathbb{Z}) \backslash \mathbb{H}$  is not smooth but has a conical (“orbifold”) singularity. For  $\tau$ , these are the square and the hexagonal lattice, whose rotation symmetry is  $\mathbb{Z}_4$  and  $\mathbb{Z}_6$  instead of the generic  $\mathbb{Z}_2$ . In the string moduli space the two points

$$(\tau, \rho) = (i, i) \quad \text{and} \quad (\tau, \rho) = (e^{i\pi/3}, e^{i\pi/3}) \quad (12.29)$$

are the backgrounds with gauge symmetry  $(SU(2) \times SU(2))_L \times (SU(2) \times SU(2))_R$  and  $SU(3)_L \times SU(3)_R$  respectively [GPR §2.4.2, eqs. (2.4.63)–(2.4.65), ll. 2019–2052]; the second was studied in Chapter 10. Orbifold points of the moduli space and enhanced gauge symmetry go together: the extra massless vectors exist because a finite duality group maps the background to itself.



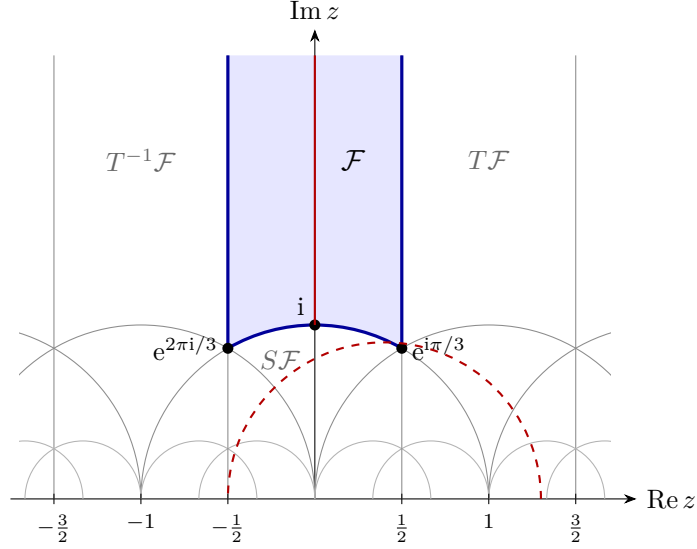


Figure 12.2: The fundamental domain  $\mathcal{F}$  of  $SL(2, \mathbb{Z})$  in the upper half-plane (shaded) and some of its images. The vertical sides are identified by  $T$ , the two halves of the arc by  $S$ . The marked points have non-trivial stabilizers. Red: two geodesics of the Poincaré metric, a vertical line (solid) and a semicircle centred on the real axis (dashed).

### 12.4.3 The Poincaré metric

Is there a natural notion of distance between two tori? A metric on  $\mathbb{H}$  that does not distinguish between equivalent descriptions must be invariant under  $SL(2, \mathbb{Z})$ ; we show that there is a metric invariant under all of  $SL(2, \mathbb{R})$  and that it is unique up to a constant. Differentiating  $\gamma z$  gives  $d(\gamma z) = [a(cz + d) - c(az + b)]dz/(cz + d)^2 = dz/(cz + d)^2$ . Combined with (12.26),

$$\frac{|d(\gamma z)|^2}{(\text{Im } \gamma z)^2} = \frac{|dz|^2}{|cz + d|^4} \cdot \frac{|cz + d|^4}{(\text{Im } z)^2} \implies ds^2 = \frac{dx^2 + dy^2}{y^2} \text{ is invariant.} \quad (12.30)$$

This is the *Poincaré metric*. Uniqueness: an invariant metric is determined by its value at  $i$  because the action is transitive, and at  $i$  it must be invariant under the rotations that fix  $i$ , hence proportional to  $dx^2 + dy^2$ . Its Gaussian curvature follows from the formula  $K = -e^{-2u}(\partial_x^2 + \partial_y^2)u$  for a metric  $e^{2u}(dx^2 + dy^2)$ : with  $u = -\ln y$  one has  $(\partial_x^2 + \partial_y^2)u = 1/y^2$  and

$$K = -y^2 \cdot \frac{1}{y^2} = -1. \quad (12.31)$$

The half-plane is the hyperbolic plane, of constant negative curvature. The fundamental domain, although it extends to  $y = \infty$ , has finite area:

$$\text{Area}(\mathcal{F}) = \int_{-1/2}^{1/2} dx \int_{\sqrt{1-x^2}}^{\infty} \frac{dy}{y^2} = \int_{-1/2}^{1/2} \frac{dx}{\sqrt{1-x^2}} = 2 \arcsin \frac{1}{2} = \frac{\pi}{3}. \quad (12.32)$$

On the string moduli space  $\mathbb{H} \times \mathbb{H}$  the corresponding metric is

$$ds^2 = \frac{|d\tau|^2}{\tau_2^2} + \frac{|d\rho|^2}{\rho_2^2} = -\frac{1}{4} \text{tr} (d\mathcal{H} d\mathcal{H}^{-1}), \quad (12.33)$$

where the second equality is a symbolic check (sympy) using (12.6). The right side is manifestly invariant under  $\mathcal{H} \rightarrow g\mathcal{H}g^T$  for constant  $g$ , and it treats  $\tau$  and  $\rho$  symmetrically, as mirror symmetry demands. The expression is not arbitrary: the action of double field theory

contains the term  $\frac{1}{8}\mathcal{H}^{MN}\partial_M\mathcal{H}^{KL}\partial_N\mathcal{H}_{KL}$  [HHZ §1, eq. (1.15), ll. 334–346], which for moduli depending only on the non-compact coordinates  $x^\mu$  is proportional to the scalar kinetic term  $g_{ij}\partial_\mu\phi^i\partial^\mu\phi^j$ , with  $\phi^i = (\tau_1, \tau_2, \rho_1, \rho_2)$  and  $g_{ij}$  the metric (12.33) (Chapter 17). The reduction itself is not carried out in this book; the normalization of (12.33) is a convention. For the circle,  $\mathcal{H} = \text{diag}(R^2/\alpha', \alpha'/R^2)$  gives  $ds^2 = 2(dR/R)^2$ , which is invariant under  $R \rightarrow \alpha'/R$ .

#### 12.4.4 Geodesics and distances

The geodesics of (12.30) are the paths  $(x(s), y(s))$  extremizing  $\int L ds$  with  $L = \frac{1}{2}(\dot{x}^2 + \dot{y}^2)/y^2$ . Rather than writing the Euler–Lagrange equations we use two conserved quantities.  $L$  does not depend on  $x$ , so the momentum conjugate to  $x$  is conserved;  $L$  does not depend on  $s$ , so the “energy”, here equal to  $L$  itself, is conserved:

$$p = \frac{\dot{x}}{y^2} = \text{const}, \quad \frac{\dot{x}^2 + \dot{y}^2}{y^2} = 1, \quad (12.34)$$

where we fixed the constant by choosing  $s$  to be the arc length. The energy is a sum of two non-negative terms; that  $(\dot{y}/y)^2 \geq 0$  is the content of the (elementary) Lean lemma `poincare_geodesic_kinetic_energy_nonneg`.

*Case  $p = 0$ .* Then  $x$  is constant and  $\dot{y}/y = \pm 1$ : the geodesic is the vertical line  $y = y_0 e^{\pm s}$ . The distance between  $iy_0$  and  $iy_1$  is  $|\ln(y_1/y_0)|$ .

*Case  $p \neq 0$ .* Substituting  $\dot{x} = py^2$  in the energy gives  $\dot{y}^2 = y^2(1 - p^2y^2)$ , so  $dx/dy = \dot{x}/\dot{y} = \pm py/\sqrt{1 - p^2y^2}$ , which integrates to  $x - x_0 = \mp \frac{1}{p}\sqrt{1 - p^2y^2}$ , that is

$$(x - x_0)^2 + y^2 = \frac{1}{p^2} : \quad (12.35)$$

a semicircle centred on the real axis, of radius  $1/|p|$ . An arc-length parametrization is  $x = x_0 + r \tanh s$ ,  $y = r/\cosh s$  (symbolic check: it solves the Euler–Lagrange equations  $\ddot{x} = 2\dot{x}\dot{y}/y$ ,  $\ddot{y} = (\dot{y}^2 - \dot{x}^2)/y$  with unit speed). The arc of the unit circle bounding  $\mathcal{F}$  is a geodesic, and so are its vertical sides.

**Proposition 12.5** (Distance formula). *The geodesic distance between  $z, w \in \mathbb{H}$  is given by*

$$\cosh d(z, w) = 1 + \frac{|z - w|^2}{2 \operatorname{Im} z \operatorname{Im} w}. \quad (12.36)$$

*Proof.* The right side is invariant under  $SL(2, \mathbb{R})$ : from  $\gamma z - \gamma w = (z - w)/((cz + d)(cw + d))$  and (12.26) the factors  $|cz + d|^2|cw + d|^2$  cancel. The left side is invariant because the metric is. By transitivity we may take  $z = i$ ; a rotation about  $i$  then turns the geodesic through  $i$  and  $w$  into the vertical one, so that  $w = ie^d$ . There the right side is  $1 + (e^d - 1)^2/(2e^d) = \frac{1}{2}(e^d + e^{-d})$ .  $\square$

#### Physical picture: infinite distance and light towers

Take a rectangular torus without  $B$ -field and let its area grow at fixed shape,  $\rho = i\rho_2 \rightarrow i\infty$ . By the case  $p = 0$  this is a geodesic, and the distance travelled is  $\Delta = \ln(\rho_2^{\text{final}}/\rho_2^{\text{initial}})$ : the decompactification limit is at *infinite* distance, although the fundamental domain has finite area. Along this path the momentum states of (12.10) have  $M^2 \propto |n_1 - \tau n_2|^2/(\rho_2 \tau_2)$ , so a whole tower of states becomes light as

$$M \propto \rho_2^{-1/2} = e^{-\Delta/2} M_{\text{initial}},$$

exponentially in the distance (the coefficient  $\frac{1}{2}$  depends on the normalization chosen in

(12.33)). In the opposite direction  $\rho_2 \rightarrow 0$  the winding states do the same, as they must by  $S'$ . The cusp  $\tau \rightarrow i\infty$  is the limit of a long thin torus, with one light momentum tower and one light winding tower. That infinite distances in moduli space come with exponentially light towers is the pattern elevated to a conjecture about quantum gravity in Chapter 27; on  $T^2$  it is a computation. For the circle, the combination  $R + \alpha'/R$  of Chapter 31 equals  $2\sqrt{\alpha'} \cosh u$  with  $u = \ln(R/\sqrt{\alpha'})$  proportional to the distance from the self-dual point: it is minimal there and grows exponentially in both directions.

## 12.5 Mirror symmetry as a factorized duality

For Calabi–Yau manifolds, mirror symmetry is a duality that “relates the complex-structure deformations of one CY background to the Kähler-class deformations of its mirror partner” [GPR §1, ll. 334–340]; see also [Asp §3.4, ll. 1840–1842]. The torus  $T^2$  is the Calabi–Yau manifold of complex dimension one, and for it the statement is a theorem we have already met:

$$D_2 : \quad (\tau, \rho) \longleftrightarrow (\rho, \tau) \quad \text{[L]} \quad (12.37)$$

“For a complex torus, mirror symmetry is identical to what is called ‘factorized duality’” [GPR §1, ll. 344–345], and “an  $R \leftrightarrow 1/R$  transformation on one of the circles in the  $T^2$  is a mirror map in the sense that the notions of deformation of complex structure and complexified Kähler form are interchanged” [Asp §5.1, ll. 2826–2829]. The rectangular torus of Example 12.1 shows the mechanism in one line. Three consequences deserve emphasis.

**The mirror manifold is a different manifold.** The torus with moduli  $(\tau, \rho)$  and the torus with moduli  $(\rho, \tau)$  are different Riemannian manifolds: in Figure 12.1b one is long and thin, the other large and nearly square. A point particle distinguishes them. The string does not, because it has winding states in addition to momentum states, and the duality exchanges  $n_2 \leftrightarrow m_2$ .

**Non-geometric symmetries are mirrors of geometric ones.** By Theorem 12.3,  $T' = D_2 T D_2$  and  $S' = D_2 S D_2$ . The periodicity of the  $B$ -field is the mirror image of the freedom to change the lattice basis by  $\hat{e}_1 \rightarrow \hat{e}_1 + \hat{e}_2$ , and the inversion of the area is the mirror image of the exchange of the two sides of the cell. In general, conjugation by  $D_2$  is an isomorphism  $SL(2, \mathbb{Z})_\tau \rightarrow SL(2, \mathbb{Z})_\rho$ . This is how mirror symmetry is used in practice, also for Calabi–Yau threefolds: a quantity that is hard to compute on the Kähler side, where the string corrections live, is computed on the complex-structure side of the mirror, where classical geometry suffices [GPR §1, ll. 338–340].

**The other circle.** Had we dualized the first circle instead, (12.23) gives  $(\tau, \rho) \rightarrow (-1/\rho, -1/\tau)$ : still an exchange of shape and size, composed with the two inversions. Conjugating the shift  $T$  by  $D_1$  we find, following a point  $(\tau, \rho)$  through the three maps of  $g = D_1 g T D_1$  in the action (12.2) (first  $D_1$ , then  $T$  on the first entry, then  $D_1$ ),

$$(\tau, \rho) \rightarrow \left(-\frac{1}{\rho}, -\frac{1}{\tau}\right) \rightarrow \left(1 - \frac{1}{\rho}, -\frac{1}{\tau}\right) \rightarrow \left(\tau, \frac{-1}{1 - 1/\rho}\right) = \left(\tau, \frac{\rho}{1 - \rho}\right). \quad (12.38)$$

This is the element  $\begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix} = S' T' S'^{-1}$  of  $SL(2, \mathbb{Z})_\rho$ : a shift by one unit not of  $\rho$  but of  $-1/\rho$ . As an integer matrix it is the *lower* triangular  $\begin{pmatrix} 1 & 0 \\ j & 1 \end{pmatrix} = g_{-j}^T$ . This precise identity is what has been formalized, as `mirror_conjugates_tauShift`; see Section 12.7 for its reading in both conventions for the action.

*Remark 12.6* (Type II strings). For the superstring, T-duality on one circle exchanges the type IIA and type IIB theories [Asp §5.1, Proposition 5, ll. 2821–2825]. Hence type IIA on a torus with moduli  $(\tau, \rho)$  is type IIB on the torus with moduli  $(\rho, \tau)$ , and the modular group that is geometric in one description acts on the Kähler modulus in the other [Asp §5.1, ll. 2826–2831]. This is the form in which the statement is used for  $K3 \times T^2$  in Chapter 24.

*Remark 12.7* (A third half-plane). The one-loop partition function depends on  $\tau$ ,  $\rho$  and the modulus  $\sigma$  of the worldsheet torus, and it is symmetric under the exchange of the target-space complex structure with the worldsheet one,  $Z(\tau, \rho; \sigma) = Z(\sigma, \rho; \tau)$ ; together with  $\tau \leftrightarrow \rho$  this is a triality [GPR §2.4.2, eq. (2.4.62), ll. 2005–2016] [L]. We do not use it in this book.

## 12.6 Worked examples

**Example 12.8** (A square torus in disguise). A background is given by

$$G = \begin{pmatrix} 13 & 8 \\ 8 & 5 \end{pmatrix}, \quad B_{12} = \frac{7}{3}.$$

*Moduli.*  $\det G = 65 - 64 = 1$ , so by (12.4) and (12.5)  $\tau = \frac{8}{5} + \frac{1}{5}i$  and  $\rho = \frac{7}{3} + i$ . The cell has the area of the self-dual square torus, but it looks very elongated:  $|\tau|^2 = \frac{65}{25}$ , and the angle between the sides has  $\cos \theta = \tau_1/|\tau| = 8/\sqrt{65} \approx 0.992$ ,  $\theta \approx 7^\circ$ .

*Reduction of  $\tau$ .* Step 1:  $T^{-2}$  gives  $\tau - 2 = -\frac{2}{5} + \frac{1}{5}i$ . Step 2:  $|\tau - 2|^2 = \frac{4}{25} + \frac{1}{25} = \frac{1}{5} < 1$ , so apply  $S$ :  $-1/z = -\bar{z}/|z|^2 = 5(\frac{2}{5} + \frac{1}{5}i) = 2 + i$ . Step 1 again:  $T^{-2}$  gives  $i$ . Hence  $\tau \sim i$ : the lattice is the square lattice in a skew basis. Indeed  $G = AA^T$  with  $A = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}$ ,  $\det A = 1$ , and (12.15) gives  $(2i + 3)/(i + 2) = \frac{1}{5}(3 + 2i)(2 - i) = \frac{1}{5}(8 + i)$ .

*Reduction of  $\rho$ .*  $T'^{-2}$  gives  $\rho - 2 = \frac{1}{3} + i$ , which lies in  $\mathcal{F}$ . The background is equivalent to the square torus of unit area with  $B_{12} = \frac{1}{3}$ . It is *not* equivalent to the  $SU(2)^2$  point  $(i, i)$ . The orbit of  $(i, i)$  under the whole group consists of pairs  $(\gamma i, \gamma' i)$ , because  $D_2$  and  $R$  map  $(i, i)$  to itself; and  $\frac{1}{3} + i$  is an interior point of  $\mathcal{F}$ , hence not equivalent to the point  $i$  of  $\mathcal{F}$ .

*Distance to the enhanced point.* By (12.36),  $\cosh d = 1 + \frac{(1/3)^2}{2} = \frac{19}{18}$ ,  $d \approx 0.332$ .

**Example 12.9** (The two enhanced symmetry points). GPR give the  $SU(3)$  point as  $E = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  [GPR §2.4.2, eq. (2.4.64), ll. 2026–2036]. Its symmetric and antisymmetric parts are  $G = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1 \end{pmatrix}$  and  $B_{12} = \frac{1}{2}$ , so  $\det G = \frac{3}{4}$  and

$$\tau = \frac{1/2}{1} + i \frac{\sqrt{3}/2}{1} = e^{i\pi/3}, \quad \rho = \frac{1}{2} + i \frac{\sqrt{3}}{2} = e^{i\pi/3},$$

in agreement with (12.29). The point is fixed by  $TS$  acting on  $\tau$ , by  $T'S'$  acting on  $\rho$ , and by the mirror map  $D_2$ , because  $\tau = \rho$ . As a check on (12.10), take  $(n_1, n_2; m_1, m_2) = (1, 0; 1, 0)$ :  $a = 1$ ,  $c = \tau$ , and with  $\rho = \tau = \omega := e^{i\pi/3}$ ,  $2\rho_2\tau_2 = \frac{3}{2}$ ,

$$p_L^2 = \frac{2}{3} |1 - \omega^2|^2 = \frac{2}{3} \cdot 3 = 2, \quad p_R^2 = \frac{2}{3} |1 - \bar{\omega}\omega|^2 = 0,$$

using  $\omega^2 = e^{2\pi i/3}$  and  $|1 - e^{2\pi i/3}|^2 = 3$ . With  $N = 1$ ,  $\tilde{N} = 0$ , (12.11) gives  $M^2 = 0$ : one of the six charged massless vectors of  $SU(3)_L$  found in Chapter 10. A finite search over  $|n_i|, |m_i| \leq 2$  (symbolic check) finds six vectors with  $(p_L^2, p_R^2) = (2, 0)$  at this point, four at  $(i, i)$  and none at  $(\tau, \rho) = (i, 2i)$ .

How far apart are the two enhanced points? In each half-plane, (12.36) with  $z = i$ ,  $w = \omega$  gives  $|z - w|^2 = \frac{1}{4} + (1 - \frac{\sqrt{3}}{2})^2 = 2 - \sqrt{3}$  and

$$\cosh d = 1 + \frac{2 - \sqrt{3}}{\sqrt{3}} = \frac{2}{\sqrt{3}}, \quad d = \ln \frac{2 + 1}{\sqrt{3}} = \frac{1}{2} \ln 3 \approx 0.549,$$

This is also the distance in the quotient  $SL(2, \mathbb{Z}) \backslash \mathbb{H}$ : the images of  $\omega$  are the points  $\omega + k$ ,  $k \in \mathbb{Z}$ , of which  $\omega$  and  $\omega - 1$  are nearest to  $i$ , and points with  $\text{Im} \leq \frac{\sqrt{3}}{2}/3$  (because  $|c\omega + d|^2 = c^2 + cd + d^2$  is 1 or at least 3), which are at distance at least  $\ln(2\sqrt{3}) \approx 1.24$  from  $i$ . Since both moduli move, the distance between the two enhanced points in the metric (12.33) is  $\sqrt{2}d \approx 0.777$ .

## 12.7 What the machine has checked

Two files in the directory `DualScaleStream2/TDuality` treat  $d = 2$ , namely `Mirror.lean` and `SL2Product.lean`. They work entirely with  $4 \times 4$  integer matrices, built from three definitions valid for every  $d$  (Chapter 11): `thetaShift`, which maps  $\Theta$  to  $\begin{pmatrix} 1 & \Theta \\ 0 & 1 \end{pmatrix}$ ; `basisChange`, which maps  $A, B$  to  $\text{diag}(A, B)$ ; and `factorized`, with `factorized k` equal to  $D_{k+1}$  (Lean counts from 0, so `factorized 0` is  $D_1$ ). Neither file mentions  $\tau$  or  $\rho$ : the complex moduli, the half-plane and Proposition 12.2 are not formalized. What is proved are matrix identities in the charge lattice, whose translation into statements about moduli is the (unformalized, symbolic-check) dictionary of Section 12.3.

### In Lean: the mirror conjugate of the $\tau$ -shift (`Mirror.lean`)

```
def tauShift : Matrix (Fin 2) (Fin 2) ℤ := !![1, 1; 0, 1]
def tauShiftDual : Matrix (Fin 2) (Fin 2) ℤ := !![1, 0; -1, 1]
theorem tauShift_dual_spec : tauShiftT * tauShiftDual = 1 := by ...
theorem tauShift_isODD : IsODD (basisChange tauShift tauShiftDual) := ...
def mirrorTheta : Matrix (Fin 2) (Fin 2) ℤ := !![0, -1; 1, 0]
theorem mirrorTheta_antisymm : mirrorThetaT = -mirrorTheta := by ...
theorem mirror_conjugates_tauShift :
  factorized (0 : Fin 2) * basisChange tauShift tauShiftDual * factorized 0 =
    (thetaShift mirrorTheta)T := by ...
```

**Reading.** With  $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  and  $\Theta = -J$ : the pair  $(T, (T^T)^{-1})$  satisfies the compatibility condition, so  $g_T \in O(2, 2; \mathbb{Z})$ ;  $\Theta$  is antisymmetric; and  $D_1 g_T D_1 = g_{-J}^T$ , the lower triangular matrix of (12.38). This is one identity between two fixed  $4 \times 4$  integer matrices.

**Proof idea.** `ext` reduces the matrix equation to its 16 entries, `fin_cases` enumerates them, and each is closed by `rfl`: the kernel evaluates both sides. **What it does not say.** Nothing about  $\tau$ ,  $\rho$  or the generalized metric, and nothing about  $D_2$ ; the companion  $D_2 g_T D_2 = g_J$ , which is GPR's  $T' = D_2 T D_2$ , is a symbolic check in this book (Exercise 12.9).

### Common pitfall: one identity, two readings

How `mirror_conjugates_tauShift` translates to moduli depends on the convention for the action. With  $\mathcal{H} \rightarrow g\mathcal{H}g^T$  (this chapter, GPR, `genMetric_bshift`),  $g_T$  is  $\tau \rightarrow \tau + 1$  and the right side is  $\rho \rightarrow \rho/(1 - \rho)$ , as in (12.38). With  $\mathcal{H} \rightarrow g^T\mathcal{H}g$  (Chapter 10, `massForm_covariant`), the right side is the plain  $B$ -shift  $\rho \rightarrow \rho - 1$ , and  $g_T$  acts as  $G \rightarrow T^T G T$ , which is  $\tau \rightarrow \tau/(\tau + 1)$ . The docstring of the Lean file describes the left side in the first reading and the right side in the second; it also states explicitly that no sign or orientation of the  $\rho$ -shift is claimed. In either convention the content is the same:  $D_1$  conjugates a parabolic generator of  $SL(2, \mathbb{Z})_\tau$  into a parabolic generator of  $SL(2, \mathbb{Z})_\rho$ . Both readings have been confirmed symbolically.

**In Lean:  $SL(2, \mathbb{Z})_\tau$  commutes with the  $\rho$ -translations (SL2Product.lean)**

```

def jMat : Matrix (Fin 2) (Fin 2) ℤ := !![0, 1; -1, 0]
theorem mul_jMat_mul_transpose (A : Matrix (Fin 2) (Fin 2) ℤ) :
  A * jMat * A⊤ = A.det • jMat := by ...
theorem basisChange_mul {d : ℕ} (A B A' B' : Matrix (Fin d) (Fin d) ℤ) :
  basisChange A B * basisChange A' B' = basisChange (A * A') (B * B') := by ...
theorem thetaShift_mul {d : ℕ} (Θ Θ' : Matrix (Fin d) (Fin d) ℤ) :
  thetaShift Θ * thetaShift Θ' = thetaShift (Θ + Θ') := by ...
theorem basisChange_comm_thetaShift (A B : Matrix (Fin 2) (Fin 2) ℤ)
  (hAB : A⊤ * B = 1) (hdet : A.det = 1) (t : ℤ) :
  basisChange A B * thetaShift (t • jMat) =
    thetaShift (t • jMat) * basisChange A B := by ...

```

**Reading.** The first theorem is (12.16), for all integer  $2 \times 2$  matrices. The next two say that basis changes multiply blockwise and that  $\Theta$ -shifts add,  $g_\Theta g_{\Theta'} = g_{\Theta + \Theta'}$ , for every  $d$ . The last is (12.22): *if  $A^\top B = 1$  and  $\det A = 1$ , then  $g_A$  commutes with  $T'^t = g_t J$  for every integer  $t$ .* **Proof idea.** From  $\det A = 1$  and the first theorem,  $A J A^\top = J$ ; multiplying  $AJ$  by  $1 = A^\top B$  on the right and regrouping gives  $AJ = JB$ ; then both products are expanded blockwise and compared entry by entry. **What it does not say.** Only one direction is formalized. The converse — for  $t \neq 0$  the two matrices commute *only if*  $\det A = 1$  — follows on paper from  $AJ - JA^{-\top} = (1 - 1/\det A)AJ$  and has been confirmed symbolically, but is not in Lean. The inversion  $S'$  and hence the full  $SL(2, \mathbb{Z})_\rho$  do not appear, nor does Theorem 12.3.

The half-plane geometry of Section 12.4 is touched by two files of an older library of this project (StringTheoryFormalization), written for other purposes.

**In Lean: composition of Möbius maps (Frontier/SL2CSymmetry.lean)**

```

structure MöbiusTransform where
  (a b c d : ℂ)
  det_one : a * d - b * c = 1
  noncomputable def MöbiusTransform.act (M : MöbiusTransform) (z : ℂ) : ℂ :=
    (M.a * z + M.b) / (M.c * z + M.d)
  theorem möbius_compose (M N : MöbiusTransform) (z : ℂ)
    (h : N.c * z + N.d ≠ 0)
    (h' : M.c * (N.act z) + M.d ≠ 0) :
    (MöbiusTransform.mk
      (M.a * N.a + M.b * N.c)
      (M.a * N.b + M.b * N.d)
      (M.c * N.a + M.d * N.c)
      (M.c * N.b + M.d * N.d)
    (by
      have key : (M.a * N.a + M.b * N.c) * (M.c * N.b + M.d * N.d)
        - (M.a * N.b + M.b * N.d) * (M.c * N.a + M.d * N.c)
        = (M.a * M.d - M.b * M.c) * (N.a * N.d - N.b * N.c) := by ring
      rw [key, M.det_one, N.det_one, one_mul])).act z =
      M.act (N.act z) := by ...
  theorem möbius_id_act (z : ℂ) : möbiusId.act z = z := by ...

```

**Reading.** For complex matrices of determinant one, the matrix product (whose determinant is shown to be one inside the statement) acts as the composition of the two maps, at every  $z$  where the two denominators do not vanish; the identity matrix acts trivially. This is (12.27) over  $\mathbb{C}$ ; it contains the real and the integer case. **Proof idea.** Unfold, clear denominators with `field_simp`, finish with `ring`. **What it does not say.** That

$\mathbb{H}$  is preserved (12.26), anything about  $SL(2, \mathbb{Z})$  or a fundamental domain (for these see Mathlib’s `ModularGroup`), or about the metric. The file was written for the conformal group of the sphere (Chapter 4); its theorems named `ward_identity_translation` and `ward_identity_dilatation` have the conclusion `True` and prove nothing.

**In Lean: what exists about geodesics** (`Frontier/ModuliGeodesics.lean`)

```
theorem poincare_geodesic_kinetic_energy_nonneg (y y' : ℝ) (hy : 0 < y) :
  let energy := (y')^2 / y^2
  0 ≤ energy := by ...
```

**Reading.** For real  $y > 0$  and any real  $y'$ , the number  $y'^2/y^2$  is non-negative: the  $y$ -part of the conserved energy in (12.34) is a square. It is a statement about two real numbers, proved by `positivity`; no metric, geodesic equation or conservation law is formalized. The other declarations of this file must be read with the same care, as its own documentation says: `poincare_geodesic_atlas_curvature` checks by `rfl` that a structure field whose default value is  $-1$  has the value  $-1$  — it is not the computation (12.31); `geodesic_equation_kummer_locus` has a conclusion of the form “ $\dots \vee \text{True}$ ”; and `wp_geodesic_completeness` is proved by extending a path by itself. None of them is evidence for a statement of this chapter.

| Claim                                                                                                               | Tier | Evidence                                                                                                  |
|---------------------------------------------------------------------------------------------------------------------|------|-----------------------------------------------------------------------------------------------------------|
| $AJA^T = (\det A)J$ for integer $2 \times 2$ $A$                                                                    | [A]  | <code>mul_jMat_mul_transpose</code>                                                                       |
| $\det A = 1 \Rightarrow g_A$ commutes with $g_{tJ}$                                                                 | [A]  | <code>basisChange_comm_thetaShift</code>                                                                  |
| Converse of the previous line                                                                                       | —    | on the page; symbolic check                                                                               |
| $g_{AgA'} = g_{AA'}$ , $g_{\Theta}g_{\Theta'} = g_{\Theta+\Theta'}$ (all $d$ )                                      | [A]  | <code>basisChange_mul</code> , <code>thetaShift_mul</code>                                                |
| $g_T \in O(2, 2; \mathbb{Z})$ ; $D_1 g_T D_1 = g_{-J}^T$ ; $D_1 D_2 = \eta$                                         | [A]  | <code>tauShift_isODD</code> ,<br><code>mirror_conjugates_tauShift</code> ,<br><code>factorized_two</code> |
| $g_{\Theta} \mathcal{H}(G, B) g_{\Theta}^T = \mathcal{H}(G, B + \Theta)$ (real matrices)                            | [A]  | <code>genMetric_bshift</code>                                                                             |
| Möbius maps compose like matrices (over $\mathbb{C}$ )                                                              | [A]  | <code>mobius_compose</code>                                                                               |
| $p_{L,R}^2$ in terms of $\tau, \rho$ , (12.10)                                                                      | [L]  | GPR (2.4.58); derivation and symbolic check                                                               |
| Action of $S, T, S', T', D_1, D_2, R, W$ on $(\tau, \rho)$                                                          | [L]  | GPR (2.4.57)–(2.4.61); symbolic check ( $D_1$ : this book)                                                |
| $O(2, 2; \mathbb{Z}) \cong [SL(2, \mathbb{Z}) \times SL(2, \mathbb{Z})] \rtimes (\mathbb{Z}_2 \times \mathbb{Z}_2)$ | [L]  | GPR §2.4.2                                                                                                |
| Mirror symmetry of $T^2 =$ factorized duality                                                                       | [L]  | GPR §1; Asp §5.1                                                                                          |
| Metric (12.33), curvature, geodesics, distance formula                                                              | —    | derived on the page; symbolic checks                                                                      |

Table 12.1: Epistemic status of the claims of this chapter. The identification of the integer matrices with dualities of a physical string background is never Tier A. All Tier A entries depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`.

**What a faithful formalization would need.** (i) A definition of  $\tau$  and  $\rho$  as points of Mathlib’s `UpperHalfPlane` from a positive definite real  $G$  and an antisymmetric  $B$ , with the bijection (12.6). (ii) The theorem that  $g_A \mathcal{H} g_A^T$  has moduli  $(A \cdot \tau, \rho)$  for  $A \in SL(2, \mathbb{Z})$ , where Mathlib’s action of  $SL(2, \mathbb{Z})$  on the half-plane can be reused, and the analogous statements for  $g_{tJ}$ ,  $D_1$ ,  $D_2$ . (iii) Proposition 12.2 as an identity of real polynomials divided by  $\rho_2 \tau_2$  — within reach of `field_simp` and `ring`. (iv) The structure theorem (12.25), which needs



$O(2, 2; \mathbb{Z})$  as a group and a generation argument; this is the hard part.

### Chapter summary

- The moduli of  $T^2$  are  $\tau = (G_{12} + i\sqrt{\det G})/G_{22}$ , the shape, and  $\rho = B_{12} + i\sqrt{\det G}$ , the area in string units with the  $B$ -field; the moduli space is locally  $\mathbb{H} \times \mathbb{H}$ .
- $p_L^2$  and  $p_R^2$  are  $|(n_1 - \tau n_2) - \rho(m_2 + \tau m_1)|^2 / (2\rho_2 \tau_2)$  with  $\rho$ , respectively  $\bar{\rho}$ ; dualities are the changes of  $(\tau, \rho)$  that can be undone by relabelling the integers.
- $SL(2, \mathbb{Z})_\tau$  (basis changes) is geometric;  $SL(2, \mathbb{Z})_\rho$  is generated by the integer  $B$ -shift  $T'$  and the area inversion  $S'$ . The two commute; together with the exchange  $D_2$  and the reflection  $R$  they generate  $O(2, 2; \mathbb{Z})$ .
- Mirror symmetry of  $T^2$  is T-duality on one circle,  $\tau \leftrightarrow \rho$ ; it maps the geometric modular group to the stringy one,  $T' = D_2 T D_2$ ,  $S' = D_2 S D_2$ .
- Each half-plane carries the Poincaré metric, of curvature  $-1$ ; geodesics are vertical lines and semicircles;  $\cosh d = 1 + |z - w|^2 / (2 \operatorname{Im} z \operatorname{Im} w)$ . Decompactification is at infinite distance, with a tower of mass  $\propto e^{-\Delta/2}$ .
- Kernel-checked:  $AJA^\top = (\det A)J$ , the commutation of  $g_A$  ( $\det A = 1$ ) with  $g_{tJ}$  (one direction),  $D_1 g_T D_1 = g_{-J}^\top$ ,  $D_1 D_2 = \eta$ , composition of Möbius maps. Not formalized:  $\tau$ ,  $\rho$ , their transformation laws, the group structure, the hyperbolic geometry.

## Exercises

**Exercise 12.1** (★). For  $G = \begin{pmatrix} 5 & 3 \\ 3 & 2 \end{pmatrix}$  and  $B_{12} = \frac{5}{2}$  compute  $\tau$  and  $\rho$ , and reduce both to  $\mathcal{F}$ . Why is the reduced  $\rho$  not unique? (Answer:  $\tau \sim i$ ;  $\rho \sim \pm \frac{1}{2} + i$ , two boundary points identified by  $T'$ .)

**Exercise 12.2** (★). Verify (12.26) and (12.27) by direct computation, and show that  $\gamma z - \gamma w = (z - w) / ((cz + d)(cw + d))$ .

**Exercise 12.3** (★★). Using (12.10), show that  $W : (\tau, \rho) \rightarrow (\tau, -\bar{\rho})$  exchanges  $p_L^2$  and  $p_R^2$  when combined with  $m_i \rightarrow -m_i$ . Conclude from  $p_L^2 - p_R^2 = 2n \cdot m$  that  $W$  cannot be represented by an element of  $O(2, 2; \mathbb{Z})$  acting on  $Z$ .

**Exercise 12.4** (★★). Derive the action of  $D_1$  in (12.23) from (12.10): show that the background  $(-1/\rho, -1/\tau)$  with  $n_1$  and  $m_1$  exchanged has the same  $p_L^2$  and  $p_R^2$  as  $(\tau, \rho)$ . (Hint: multiply the numerator by  $|\tau\rho|^{-1}$ .)

**Exercise 12.5** (★★). Show that the backgrounds  $E(g, b) = \begin{pmatrix} g & b \\ -b & g \end{pmatrix}$  with  $g^2 + b^2 = 1$  have  $\tau = i$  and  $|\rho| = 1$ , and that  $E^{-1} = E(g, -b)$ . Which element of the full symmetry group fixes every point of this circle? [GPR §2.4.2, eqs. (2.4.66)–(2.4.67), ll. 2053–2075]

**Exercise 12.6** (★★). Find the geodesic through  $z = i$  and  $w = 1 + 2i$  (centre and radius of the semicircle) and compute  $d(z, w)$  from (12.36). (Answer: centre  $x_0 = 2$ , radius  $\sqrt{5}$ ;  $\cosh d = \frac{3}{2}$ .)

**Exercise 12.7** (★★★). The domain  $\mathcal{F}$  is a hyperbolic triangle with angles  $\frac{\pi}{3}, \frac{\pi}{3}, 0$ . The Gauss–Bonnet theorem gives the area of a geodesic triangle of curvature  $-1$  as  $\pi$  minus the sum of its angles. Compare with (12.32), and compute the area of the fundamental domain of the subgroup generated by  $T^2$  and  $S$ .

**Exercise 12.8** (★, Lean). State and prove in Lean that  $J^2 = -1$  (for `jMat`) and that the square of `tauShift` is `!![1, 2; 0, 1]`. The proof pattern of `mirrorTheta_antisymm` suffices: `ext`, then `fin_cases`, then `rfl` or `decide`.



**Exercise 12.9** ( $\star\star$ , Lean). State and prove the  $D_2$  companion of `mirror_conjugates_tauShift`:

```
example : factorized (1 : Fin 2) * basisChange tauShift tauShiftDual * factorized 1
  = thetaShift jMat := by sorry
```

This is GPR's  $T' = D_2 T D_2$ . Then use `factorized_mul_self` to deduce  $g_T = D_2 g_J D_2$ .

**Exercise 12.10** ( $\star\star\star$ , Lean). Formulate the converse of `basisChange_comm_thetaShift`: if  $A^\top B = 1$ ,  $t \neq 0$  and the two matrices commute, then  $\det A = 1$ . (Hint: compare the upper right blocks to get  $tAJ = tJB$ , multiply by  $A^\top$  on the right, and use `mul_jMat_mul_transpose`; you will need that  $J \neq 0$  and that  $\mathbb{Z}$  has no zero divisors.)

## Further reading

- The  $d = 2$  example, with the generators  $S, T, S', T', D_2, R, W$ , the fixed points and the triality with the worldsheet modulus: [GPR §2.4.2, ll. 1874–2075]. The generators of  $O(d, d; \mathbb{Z})$  for general  $d$ : [GPR §2.4, ll. 1340–1423].
- Mirror symmetry as factorized duality, and its place among target-space dualities: [GPR §1, ll. 334–348].
- The moduli space of conformal field theories on  $T^2$  as a product of two copies of  $SL(2)/U(1)$ , and T-duality on one circle as the mirror map between type IIA and type IIB: [Asp §5.1, ll. 2750–2762 and 2819–2831]. Mirror symmetry for K3, where the distinction between complex structure and Kähler form is more subtle: [Asp §3.4, ll. 1836–1888], taken up in Chapter 23.
- The generalized-metric form of the kinetic term: [HHZ §1, ll. 334–346]; Chapters 15 and 17.
- Lean sources: the files `Mirror.lean`, `SL2Product.lean`, `ODD.lean` and `Factorized.lean`, all in the directory `DualScaleStream2/TDuality`. For the modular group and its fundamental domain see the Mathlib file `NumberTheory/Modular.lean`.



## Chapter 13

# Open Strings, D-Branes and T-Duality

So far T-duality has been a statement about closed strings: a closed string can wind around a circle, winding is exchanged with momentum, and the theories at radius  $R$  and  $\alpha'/R$  are the same (Chapters 7 and 8). An open string has two free ends and cannot wind: it can always be unwound and shrunk to a point. What, then, does T-duality do to a theory that contains open strings? This chapter answers the question, and the answer changed the subject. The dual of an open string with freely moving ends is an open string whose ends are *stuck* on a hypersurface. The hypersurface is not part of the background that we put in by hand; it fluctuates, it has a tension, and it carries a gauge field. It is a *D-brane*. A student should care for two reasons. First, the argument is a clean example of how a duality forces new objects into a theory. Second, D-branes and their non-dynamical cousins, the orientifold planes, are the sources whose charges must add up to zero on a compact space. That bookkeeping is the *tadpole condition* that governs the compactifications on  $K3 \times T^2$  studied in Chapters 24 and 25.

Everything physical in this chapter is Tier [\[L\]](#): it is taken from the lectures of Polchinski (key Pol) and Tong, and none of it is formalized in this book's Lean library. Section 13.7 says precisely what is and is not checked, and Section 13.8 turns the central mechanism into two small Lean statements that a student can prove.

### 13.1 Open strings and their boundary conditions

#### 13.1.1 The boundary term

Take the Polyakov action in conformal gauge (Chapter 2) for a worldsheet that is a strip,  $-\infty < \tau < \infty$  and  $0 \leq \sigma \leq \pi$ :

$$S = -\frac{1}{4\pi\alpha'} \int d\tau \int_0^\pi d\sigma \partial_\alpha X^\mu \partial^\alpha X_\mu, \quad (13.1)$$

with the index  $\alpha \in \{\tau, \sigma\}$  raised by the worldsheet Minkowski metric  $\text{diag}(-1, +1)$ . Vary  $X^\mu \rightarrow X^\mu + \delta X^\mu$  and integrate by parts:

$$\delta S = \frac{1}{2\pi\alpha'} \int d^2\sigma (\partial^\alpha \partial_\alpha X_\mu) \delta X^\mu - \frac{1}{2\pi\alpha'} \int d\tau \left[ \partial_\sigma X_\mu \delta X^\mu \right]_{\sigma=0}^{\sigma=\pi} + (\text{terms at } \tau = \pm\infty) \quad (13.2)$$

[Tong §3, ll. 3040–3073; Pol §1.1, ll. 65–90]. The terms at initial and final time vanish because  $\delta X^\mu$  is held fixed there, as always in the principle of least action. The first term gives the

wave equation  $\partial_+ \partial_- X^\mu = 0$ , as for the closed string. The second term is new. For a closed string there is no boundary and it is absent; for the open string it must vanish separately at each end, because variations near  $\sigma = 0$  and near  $\sigma = \pi$  are independent:

$$\partial_\sigma X_\mu \delta X^\mu = 0 \quad \text{at } \sigma = 0 \text{ and at } \sigma = \pi. \quad (13.3)$$

For each direction  $\mu$  and each end there are two ways to satisfy (13.3) [Tong §3, ll. 3074–3104].

**Definition 13.1** (Neumann and Dirichlet conditions). The *Neumann* (N) condition in the direction  $\mu$  is  $\partial_\sigma X^\mu = 0$  at the end of the string;  $\delta X^\mu$  is unrestricted. The *Dirichlet* (D) condition is  $\delta X^\mu = 0$ , that is,  $X^\mu = c^\mu$  is a constant at the end of the string; then  $\partial_\tau X^\mu = 0$  there.

The two conditions say opposite things about the flow of momentum. The momentum current along the string is  $P_\mu^\sigma = -\frac{1}{2\pi\alpha'} \partial_\sigma X_\mu$  (Chapter 2), so the Neumann condition states that no momentum flows off the end of the string. With Neumann conditions in all directions the end is free, and one can show that it moves at the speed of light [Tong §3, ll. 3078–3094]. The Dirichlet condition instead fixes the position and lets momentum flow out: something must absorb it. The Neumann condition is the only one compatible with Poincaré invariance in all  $D$  dimensions [Pol §1.1, ll. 87–90], because a constant vector  $c^\mu$  singles out a place.

**Definition 13.2** ( $Dp$ -brane). Impose Neumann conditions in the directions  $a = 0, 1, \dots, p$  and Dirichlet conditions  $X^I = c^I$  in the directions  $I = p+1, \dots, D-1$ , at both ends. The ends then lie on the  $(p+1)$ -dimensional hypersurface  $\{X^I = c^I\}$  of spacetime, called a  *$Dp$ -brane*;  $p$  is the number of its spatial dimensions.

The brane breaks the Lorentz group as  $SO(1, D-1) \rightarrow SO(1, p) \times SO(D-p-1)$  [Tong §3, ll. 3118–3139]. A D0-brane is a particle, a D1-brane a string-like object, a D2-brane a membrane. Neumann conditions in all directions are the case  $p = D-1$ : a space-filling brane, so that “the end can be anywhere” [Tong §3, ll. 3150–3153]. Dirichlet conditions in all directions, time included, describe the D-instanton [Tong §3, ll. 3154–3159; Pol §1.1, ll. 109–111]. Figure 13.1 shows the two kinds of end.

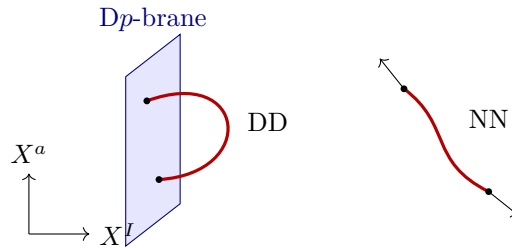


Figure 13.1: Left: in the transverse directions  $X^I$  the ends obey Dirichlet conditions and sit on the brane; along the brane they obey Neumann conditions and slide freely. Right: with Neumann conditions in every direction both ends are free.

### 13.1.2 Mode expansions

We solve the wave equation as  $X^\mu = X_L^\mu(\sigma^+) + X_R^\mu(\sigma^-)$  with  $\sigma^\pm = \tau \pm \sigma$ . We keep the conventions of Chapter 7, where the oscillators  $\tilde{\alpha}_k$  sit in  $X_L$  and  $\alpha_k$  in  $X_R$ , and we write  $k$

for the mode number:

$$\begin{aligned} X_L^\mu(\sigma^+) &= \frac{1}{2}x^\mu + \alpha' p^\mu \sigma^+ + i\sqrt{\frac{\alpha'}{2}} \sum_{k \neq 0} \frac{1}{k} \tilde{\alpha}_k^\mu e^{-ik\sigma^+}, \\ X_R^\mu(\sigma^-) &= \frac{1}{2}x^\mu + \alpha' p^\mu \sigma^- + i\sqrt{\frac{\alpha'}{2}} \sum_{k \neq 0} \frac{1}{k} \alpha_k^\mu e^{-ik\sigma^-} \end{aligned} \quad (13.4)$$

[Tong §3, eq. (3.4), ll. 3160–3180]. The coefficient of  $p^\mu$  is  $\alpha'$  and not  $\frac{1}{2}\alpha'$  as for the closed string; we shall see in a moment that this makes  $p^\mu$  the spacetime momentum when  $\sigma$  runs over  $[0, \pi]$  only.

**Proposition 13.3** (Boundary conditions on the modes). *For the expansion (13.4) on the strip  $0 \leq \sigma \leq \pi$ :*

(i) *Neumann conditions at both ends hold if and only if  $\alpha_k = \tilde{\alpha}_k$  for all  $k \neq 0$ . Then*

$$X(\tau, \sigma) = x + 2\alpha' p \tau + i\sqrt{2\alpha'} \sum_{k \neq 0} \frac{\alpha_k}{k} e^{-ik\tau} \cos k\sigma. \quad (13.5)$$

(ii) *Dirichlet conditions  $X(\tau, 0) = c$ ,  $X(\tau, \pi) = d$  hold if and only if  $\alpha_k = -\tilde{\alpha}_k$  for all  $k \neq 0$ , the term linear in  $\tau$  is absent, and the zero modes are replaced by a term linear in  $\sigma$ :*

$$X(\tau, \sigma) = c + \frac{(d-c)}{\pi} \sigma - \sqrt{2\alpha'} \sum_{k \neq 0} \frac{\alpha_k}{k} e^{-ik\tau} \sin k\sigma. \quad (13.6)$$

*Proof.* (i) Since  $\partial_\sigma = \partial_+ - \partial_-$ , one has  $\partial_\sigma X = \sqrt{\alpha'/2} \sum_k (\tilde{\alpha}_k e^{-ik\sigma^+} - \alpha_k e^{-ik\sigma^-})$ ; the zero-mode terms cancel. At  $\sigma = 0$  this is  $\sqrt{\alpha'/2} \sum_k (\tilde{\alpha}_k - \alpha_k) e^{-ik\tau}$ , which vanishes for all  $\tau$  if and only if  $\tilde{\alpha}_k = \alpha_k$ . With this,  $\partial_\sigma X \propto \sum_k \alpha_k e^{-ik\tau} \sin k\sigma$ , which vanishes at  $\sigma = \pi$  precisely because  $k$  is an integer. Adding  $X_L$  and  $X_R$  and using  $e^{-ik\sigma^+} + e^{-ik\sigma^-} = 2e^{-ik\tau} \cos k\sigma$  gives (13.5). (ii) Now require  $\partial_\tau X = 0$  at the ends. Since  $\partial_\tau = \partial_+ + \partial_-$ , the same computation with a relative sign gives  $\tilde{\alpha}_k = -\alpha_k$  and no term  $2\alpha' p \tau$ . The general solution of the wave equation may still contain a term linear in  $\sigma$ , obtained by giving  $X_L$  and  $X_R$  opposite linear terms  $\pm \frac{d-c}{2\pi} \sigma^\pm$ ; it is fixed by the positions of the two ends. Finally  $-e^{-ik\sigma^+} + e^{-ik\sigma^-} = 2i e^{-ik\tau} \sin k\sigma$ , and  $i \cdot i\sqrt{\alpha'/2} \cdot 2 = -\sqrt{2\alpha'}$ , which gives (13.6).  $\square$

The mode conditions are those of [Tong §3, eq. (3.5) and below, ll. 3181–3202]; the two expansions were checked symbolically with sympy against (13.4). In both cases only one set of oscillators survives: the boundary reflects left-movers into right-movers. An open string has, roughly, half the degrees of freedom of a closed string. For mixed conditions, N at one end and D at the other, see Exercise 13.3.

**Momentum.** For the Neumann expansion the total momentum is the integral of the momentum density  $P^\tau = \frac{1}{2\pi\alpha'} \partial_\tau X$ :

$$P = \frac{1}{2\pi\alpha'} \int_0^\pi d\sigma \partial_\tau X = \frac{1}{2\pi\alpha'} (2\alpha' p) \pi = p,$$

the oscillators dropping out because  $\int_0^\pi \cos k\sigma d\sigma = 0$  for integer  $k \neq 0$  [Tong §3, ll. 3206–3230]. This confirms the normalization of (13.4). In a Dirichlet direction (13.6) has no momentum zero mode at all:  $x^I$  is replaced by the number  $c^I$  and  $p^I$  by zero. States of the open string therefore have wavefunctions that depend only on the coordinates *along* the brane [Tong §3.1, ll. 3231–3238]. The fields that the open string gives rise to live on the brane.

**Common pitfall**

“Momentum is not conserved in the Dirichlet directions, so the theory is inconsistent.” Translation invariance transverse to the brane is broken by the brane, exactly as it is broken by any heavy object placed at a point. The momentum that flows off the end of the string is absorbed by the brane. This is possible only if the brane can recoil, that is, only if it is a dynamical object; Section 13.5 shows that it is.

## 13.2 The open-string spectrum

Quantization proceeds as for the closed string (Chapter 3), but with one set of oscillators  $[\alpha_k^\mu, \alpha_l^\nu] = k \delta_{k+l,0} \eta^{\mu\nu}$ . In light-cone gauge, with the light-cone directions chosen along the brane, the mass of a state measured by an observer on the brane is

$$M^2 = \frac{1}{\alpha'} (N - 1), \quad N = \sum_i \sum_{k>0} \alpha_{-k}^i \alpha_k^i, \quad (13.7)$$

where  $i$  runs over the  $D - 2 = 24$  transverse directions, both those along the brane and those perpendicular to it. Lorentz invariance of the quantum theory again requires  $D = 26$  and the normal-ordering constant  $a = 1$ , the same values as for the closed string [Tong §3.1, ll. 3248–3290]. Compared with the closed-string formula  $M^2 = \frac{2}{\alpha'} (N + \tilde{N} - 2)$  of Chapter 3, there is one oscillator sum instead of two and an overall factor  $\frac{1}{4}$  at equal level; the factor comes from  $\alpha'p$  versus  $\frac{1}{2}\alpha'p$  in the mode expansions. The lowest states are as follows [Tong §3.1.1–3.1.2, ll. 3306–3368].

- $N = 0$ : a tachyon with  $M^2 = -1/\alpha'$ , confined to the brane. It signals that the bosonic D-brane is unstable and decays into closed strings. In the type II superstring the tachyon is absent for Dp-branes with  $p$  even (IIA) or odd (IIB), which are stable because they carry Ramond–Ramond charge [Tong §3.1.4, ll. 3397–3406].
- $N = 1$ , oscillator along the brane,  $\alpha_{-1}^a |0; p\rangle$ : a massless vector of  $SO(1, p)$ , a photon  $A_a$  on the brane.
- $N = 1$ , oscillator transverse to the brane,  $\alpha_{-1}^I |0; p\rangle$ :  $D - p - 1$  massless scalars  $\phi^I$  on the brane, which form a vector of the transverse rotation group  $SO(D - p - 1)$ .

**Stretched strings.** Take two parallel Dp-branes at transverse positions  $c^I$  and  $d^I$  and a string that goes from the first to the second. Its expansion is (13.6). The term linear in  $\sigma$  contributes to the Virasoro constraint like a momentum:  $\partial_\sigma X^I = (d^I - c^I)/\pi$  plays the role that  $2\alpha'p$  plays in (13.5). The mass formula becomes

$$M^2 = \frac{|d - c|^2}{(2\pi\alpha')^2} + \frac{1}{\alpha'} (N - 1) \quad (13.8)$$

[Tong §3.3, ll. 3489–3520]. The first term has a classical meaning, which is also an independent check: a static string of length  $L = |d - c|$  and tension  $T = 1/(2\pi\alpha')$  has energy  $TL$ , and  $(TL)^2$  is the first term. The ground state of the stretched string is tachyonic only if  $L < 2\pi\sqrt{\alpha'}$ , that is, when the branes approach each other to within a distance of the order of the string length.

### 13.3 T-duality of open strings

#### 13.3.1 A paradox at small radius

Compactify  $X^{25} \sim X^{25} + 2\pi R$  and consider open strings with Neumann conditions in all directions. The momentum along the circle is quantized,  $p^{25} = n/R$ , and an observer in the 25 non-compact dimensions sees, from (13.7),

$$M^2 = \frac{n^2}{R^2} + \frac{1}{\alpha'}(N - 1). \quad (13.9)$$

There is no winding term: an open string has no winding number. As  $R \rightarrow 0$  all states with  $n \neq 0$  become infinitely heavy and no new light states appear. The open strings behave like fields: the compact dimension disappears and they live in  $D - 1$  dimensions. But a theory of open strings also contains closed strings (two ends can join), and we saw in Chapter 8 that for closed strings the compact dimension *reappears* as  $R \rightarrow 0$ , because winding states become light. So the closed strings live in  $D$  dimensions and the open strings in  $D - 1$  [Pol §2.2, ll. 679–693].

#### Physical picture

The interior of an open string is made of the same material as a closed string, so it must still vibrate in all  $D$  dimensions of the dual spacetime. What distinguishes an open string is its ends. The only consistent reading of the paradox is that the *ends* are confined to a  $(D - 1)$ -dimensional hyperplane of the dual spacetime while the rest of the string is not [Pol §2.2, ll. 694–697].

#### 13.3.2 Neumann becomes Dirichlet

The dual coordinate of Chapter 8 is  $Y = X_L - X_R$  for the compact direction, and it obeys

$$\partial_\tau X = \partial_\sigma Y, \quad \partial_\sigma X = \partial_\tau Y. \quad (13.10)$$

These relations are local on the worldsheet and hold equally on the strip. Read at the boundary, the second one says

$$\partial_\sigma X = 0 \text{ (Neumann for } X) \iff \partial_\tau Y = 0 \text{ (Dirichlet for } Y), \quad (13.11)$$

and the first one says the converse with  $X$  and  $Y$  exchanged. *T-duality exchanges Neumann and Dirichlet conditions* [Tong §8.3.2, ll. 11804–11825; Pol §2.2, ll. 697–700]. In the language of Proposition 13.3 the statement is one line. The oscillators of  $Y$  are  $\tilde{\alpha}_k^Y = \tilde{\alpha}_k$  and  $\alpha_k^Y = -\alpha_k$ . Hence

$$\alpha_k = \tilde{\alpha}_k \text{ (N for } X) \iff \alpha_k^Y = -\tilde{\alpha}_k^Y \text{ (D for } Y).$$

We can also see it in the expansions. Subtracting instead of adding the two lines of (13.4) with  $\alpha_k = \tilde{\alpha}_k$  gives

$$Y(\tau, \sigma) = y_0 + 2\alpha' p \sigma + \sqrt{2\alpha'} \sum_{k \neq 0} \frac{\alpha_k}{k} e^{-ik\tau} \sin k\sigma, \quad (13.12)$$

which is of the Dirichlet form (13.6) (with  $\alpha_k \rightarrow -\alpha_k$ ), where  $y_0$  is an undetermined constant and the separation of the ends is  $d - c = 2\pi\alpha' p$ .

### 13.3.3 Where the ends are

Equation (13.11) says that each end has a fixed value of  $Y$ . Are the two values related? Integrate along the string and use (13.10):

$$Y(\tau, \pi) - Y(\tau, 0) = \int_0^\pi d\sigma \partial_\sigma Y = \int_0^\pi d\sigma \partial_\tau X = 2\pi\alpha' p^{25} = \frac{2\pi\alpha' n}{R} = 2\pi n \tilde{R}, \quad \tilde{R} = \frac{\alpha'}{R}, \quad (13.13)$$

[Pol §2.2, eq. (31), ll. 701–720]. The dual coordinate lives on a circle of radius  $\tilde{R}$ , so  $2\pi n \tilde{R}$  is a whole number of turns: the two ends are at the *same* point of the dual circle, and the string wraps the circle  $n$  times on the way. The momentum number of the original description has become a winding number, as for closed strings. The winding of an open string is meaningful here precisely because its ends are no longer free to unwind it. The same argument applied to a path on a worldsheet that connects the ends of two different strings shows that all ends lie on one and the same hyperplane [Pol §2.2, ll. 720–727]. Figure 13.2 below shows strings with  $n = 0$  and  $n = 1$  in the more general setting of several branes.

The masses agree, as they must. In the dual picture the string has length  $L = 2\pi n \tilde{R}$  and (13.8) gives  $M^2 = (2\pi n \tilde{R})^2 / (2\pi\alpha')^2 + \frac{1}{\alpha'}(N - 1) = n^2 \tilde{R}^2 / \alpha'^2 + \dots = n^2 / R^2 + \dots$ , which is (13.9). The paradox is resolved. For  $R \ll \sqrt{\alpha'}$  the natural variables are the dual ones: spacetime has a large circle of radius  $\tilde{R}$ , closed strings move on all of it, and open strings end on a D24-brane at one point of it.

**Proposition 13.4** (T-duality and brane dimension). *T-duality along a circle tangent to a  $Dp$ -brane gives a  $D(p - 1)$ -brane localized at a point of the dual circle. T-duality along a circle transverse to a  $Dp$ -brane gives a  $D(p + 1)$ -brane wrapped on the dual circle.* [L] [Pol §2.4, ll. 879–881; Tong §8.3.2, ll. 11820–11825]

*Proof.* A tangent direction carries the Neumann condition, which becomes Dirichlet by (13.11); the number of Neumann directions drops by one. A transverse direction carries the Dirichlet condition, and the first relation of (13.10) turns it into Neumann.  $\square$

Dualizing  $k$  circles of the theory of freely moving open strings, which is a theory of space-filling D25-branes, gives  $D(25 - k)$ -branes. In the superstring, where T-duality exchanges the type IIA and IIB theories, the rule  $p \mapsto p \pm 1$  is consistent with the fact that IIA has stable branes of even  $p$  and IIB of odd  $p$  [Tong §8.3.3, ll. 11828–11833].

#### Historical note

Dirichlet conditions were first studied as a variant of string theory in its own right. Their association with a hyperplane in spacetime came later, from compactifications of open strings; it was then observed that the hyperplane is dynamical and can be produced by T-duality [Pol §1.1, ll. 103–108]. Tong remarks that Dirichlet conditions were rarely considered until the mid-1990s [Tong §3, ll. 3105–3111; §8.3.2, ll. 11826–11827].

## 13.4 Chan–Paton factors and Wilson lines

### 13.4.1 Labels on the ends

It is consistent with Poincaré and conformal invariance to let each end of an open string carry a label  $i \in \{1, \dots, N\}$  that has no dynamics of its own: its Hamiltonian vanishes, so an end prepared in the state  $i$  stays in it [Pol §1.2, ll. 227–240]. A state of the string is then  $|k; ij\rangle$ , and a general state is a combination  $\sum_{i,j} \lambda_{ij} |k; ij\rangle$  with an  $N \times N$  matrix  $\lambda$ , the *Chan–Paton factor*. In an amplitude the right end of one string must be in the same state as the left



end of the next, so every open-string amplitude contains a trace  $\text{tr}(\lambda^1 \lambda^2 \dots)$ , invariant under  $\lambda \mapsto U \lambda U^{-1}$  with  $U \in U(N)$ . The  $N^2$  massless vectors  $\lambda_{ij} \alpha_{-1}^\mu |k; ij\rangle$  transform in the adjoint representation, and the worldsheet symmetry  $U(N)$  becomes a gauge symmetry in spacetime [Pol §1.2, ll. 241–271]. In the brane language of Tong, the label says on which of  $N$  coincident branes the end sits [Tong §3.3, ll. 3522–3540].

### 13.4.2 A Wilson line is a set of positions

On a circle a gauge field can have a constant component that cannot be removed by a periodic gauge transformation:

$$A_{25} = \frac{1}{2\pi R} \text{diag}(\theta_1, \theta_2, \dots, \theta_N), \quad W = \exp\left(i \oint A_{25} dX^{25}\right) = \text{diag}(e^{i\theta_1}, \dots, e^{i\theta_N}). \quad (13.14)$$

The field strength vanishes, but the holonomy  $W$ , the *Wilson line*, is gauge invariant, and each  $\theta_i$  is an angle defined modulo  $2\pi$ . For generic angles it breaks  $U(N) \rightarrow U(1)^N$ . Locally  $A_{25}$  is pure gauge,  $A_{25} = -i\Lambda^{-1} \partial_{25} \Lambda$  with  $\Lambda = \text{diag}(e^{i\theta_i X^{25}/2\pi R})$ . One may remove  $A_{25}$  with  $\Lambda$ , but  $\Lambda$  is not periodic, and the price is that charged fields are no longer periodic: a string in the state  $|ij\rangle$ , whose two ends have opposite charges under the  $i$ -th and  $j$ -th  $U(1)$ , picks up the phase  $e^{i(\theta_j - \theta_i)}$  under  $X^{25} \rightarrow X^{25} + 2\pi R$  [Pol §2.3, ll. 732–774]. A wavefunction  $e^{ipX^{25}}$  with this periodicity has

$$p_{ij}^{25} = \frac{2\pi n + \theta_j - \theta_i}{2\pi R}, \quad n \in \mathbb{Z}. \quad (13.15)$$

This is the familiar statement that a constant vector potential shifts the momentum of a charged particle,  $p \rightarrow p - qA$ , here with charges  $\mp 1$  at the two ends.

Now repeat the computation (13.13) with the fractional momentum:

$$Y(\pi) - Y(0) = 2\pi\alpha' p_{ij}^{25} = (2\pi n + \theta_j - \theta_i) \tilde{R} \quad (13.16)$$

[Pol §2.3, eq. (34), ll. 774–777]. The ends of an  $ij$  string are no longer at the same point. Up to a common additive constant, an end in the state  $i$  sits at

$$Y_i = \theta_i \tilde{R} = 2\pi\alpha' (A_{25})_{ii} \quad (13.17)$$

[Pol §2.3, eq. (35), ll. 779–783]. There are  $N$  hyperplanes, one for each value of the Chan–Paton label, at positions on the dual circle given by the eigenvalues of the Wilson line (Figure 13.2). The angle  $\theta_i \in [0, 2\pi)$  becomes the point  $Y_i \in [0, 2\pi\tilde{R})$ : a gauge-theory modulus has become a geometric modulus.

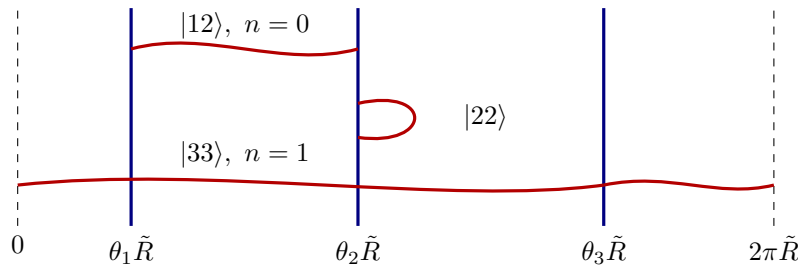


Figure 13.2: Three D-branes on the dual circle (the dashed lines are identified), at the positions given by a Wilson line with angles  $\theta_1, \theta_2, \theta_3$ . A  $|12\rangle$  string stretches between two branes; a  $|22\rangle$  string with  $n = 0$  can shrink to zero length; a  $|33\rangle$  string with  $n = 1$  winds once around the circle. After [Pol §2.3–2.4, Figs. 5–6, ll. 765–800].

The spectrum confirms the picture. With (13.15) in (13.9),

$$M_{ij}^2 = \left( \frac{2\pi n + \theta_j - \theta_i}{2\pi R} \right)^2 + \frac{1}{\alpha'}(N-1) = \left( \frac{(2\pi n + \theta_j - \theta_i) \tilde{R}}{2\pi\alpha'} \right)^2 + \frac{1}{\alpha'}(N-1) \quad (13.18)$$

[Pol §2.4, eq. (36), ll. 810–832]. The second form is (13.8) with  $L = |2\pi n + \theta_j - \theta_i| \tilde{R}$ , the length of a string that goes from brane  $i$  to brane  $j$  and  $n$  times around the circle. The two forms are equal because  $\tilde{R}/\alpha' = 1/R$ . What is Higgsing by a Wilson line in one description is the tension of a stretched string in the other.

**Example 13.5** (Two branes, with numbers). Work in units  $\alpha' = 1$ . Let  $R = \frac{1}{4}$ , so  $\tilde{R} = 4$  and the dual circle has circumference  $8\pi \approx 25.1$ . Take  $N = 2$  with  $\theta_1 = 0$  and  $\theta_2 = \pi$ : the two branes sit at  $Y_1 = 0$  and  $Y_2 = 4\pi$ , at diametrically opposite points. Table 13.1 lists the lightest states at level  $N = 1$  (the vectors and scalars). There are exactly two massless

Table 13.1: States at oscillator level 1 for  $\alpha' = 1$ ,  $R = \frac{1}{4}$ ,  $\theta = (0, \pi)$ . The column  $L$  is the length of the stretched string in the dual picture; the last two columns are computed independently from the two forms of (13.18).

| sector                   | $n$     | $L =  2\pi n + \theta_j - \theta_i  \tilde{R}$ | $(L/2\pi\alpha')^2$ | $(p_{ij}^{25})^2$ |
|--------------------------|---------|------------------------------------------------|---------------------|-------------------|
| $ 11\rangle,  22\rangle$ | 0       | 0                                              | 0                   | 0                 |
| $ 12\rangle$             | 0       | $4\pi$                                         | 4                   | 4                 |
| $ 21\rangle$             | 0       | $4\pi$                                         | 4                   | 4                 |
| $ 11\rangle,  22\rangle$ | $\pm 1$ | $8\pi$                                         | 16                  | 16                |
| $ 12\rangle$             | 1       | $12\pi$                                        | 36                  | 36                |

vectors,  $|11\rangle$  and  $|22\rangle$  at  $n = 0$ : the gauge group is  $U(1)^2$ . The off-diagonal vectors, the W-bosons of the breaking  $U(2) \rightarrow U(1)^2$ , have mass  $M = 2$ , which is  $(\theta_2 - \theta_1)/2\pi R = \pi/(2\pi \cdot \frac{1}{4})$  in the original picture and  $TL = 4\pi/2\pi$  in the dual one. The ground state of the  $|12\rangle$  string has  $M^2 = 4 - 1 = 3 > 0$  and is not tachyonic, in agreement with  $L = 4\pi > 2\pi\sqrt{\alpha'}$ . If instead  $\theta_2 \rightarrow \theta_1$ , the two branes coincide, the  $|12\rangle$  and  $|21\rangle$  vectors become massless, and  $U(2)$  is restored.

## 13.5 D-branes are dynamical

### 13.5.1 The massless fields on the brane

Consider the massless states of Section 13.2 in the T-dual picture of one dualized circle, for a string with both ends on the brane  $i$  [Pol §2.4, ll. 833–854]:

$$\alpha_{-1}^\mu |k; ii\rangle, \quad V = \partial_t X^\mu; \quad \alpha_{-1}^{25} |k; ii\rangle, \quad V = \partial_t X^{25} = \partial_n Y, \quad (13.19)$$

where  $\partial_t$  and  $\partial_n$  are the derivatives tangent and normal to the boundary of the worldsheet, and the equality is (13.10) at the boundary. The first set is a gauge field along the brane. The second was, before dualizing, the component  $A_{25}$  of the gauge field along the circle. Equation (13.17) tells us what a constant value of this field means in the dual picture: it is the *position* of the brane. A value that varies slowly along the brane,  $A_{25}(\xi)$ , is a brane whose transverse position  $Y(\xi) = 2\pi\alpha' A_{25}(\xi)$  varies from place to place, that is, a brane that is bent, and the quanta of the field are small fluctuations of its shape. This matches Tong’s identification  $X^I(\xi) = 2\pi\alpha' \phi^I(\xi)$  of the transverse scalars  $\phi^I$  [Tong §3.2, ll. 3446–3456].

### The same logic as for gravity

In Chapters 3 and 6 we started with strings in flat spacetime, found a massless closed-string state, and recognized it as a fluctuation of the geometry: the background is dynamical. Here we start with a flat hyperplane, find a massless open-string state, and recognize it as a fluctuation of the shape of the hyperplane. In a theory with gravity there could be no perfectly rigid object anyway [Pol §2.4, ll. 805–809, 848–857].

**Coincident branes.** When no two of the  $N$  branes coincide, the massless vectors are the  $N$  states  $|ii\rangle$  and the gauge group is  $U(1)^N$ . If  $m$  branes coincide,  $\theta_1 = \dots = \theta_m$ , the strings stretched between them have zero length, there are  $m^2$  massless vectors, and the gauge group is enhanced to  $U(m)$ ; in the original description the Wilson line leaves  $U(m)$  unbroken. At the same time  $m^2$  massless scalars appear: the  $m$  positions are promoted to an  $m \times m$  matrix [Pol §2.4, ll. 860–871; Tong §3.3, ll. 3527–3549]. In Chapter 8 the closed-string gauge symmetry was enhanced at a special radius; here the open-string gauge symmetry is enhanced at special brane positions.

## 13.5.2 The action and the tension

A dynamical object needs an action. Lorentz and reparametrization invariance fix the leading term to be the volume of the worldvolume, as for the Nambu–Goto string of Chapter 2: the Dirac action  $S = -T_p \int d^{p+1}\xi \sqrt{-\det \gamma_{ab}}$ , with  $\gamma_{ab} = \partial_a X^\mu \partial_b X^\nu \eta_{\mu\nu}$  the induced metric and  $T_p$  the tension [Tong §3.2, eq. (3.6), ll. 3417–3440]. Including the gauge field on the brane and the closed-string background fields  $G$ ,  $B$  and the dilaton  $\Phi$  of Chapter 6, the low-energy action is

$$S_p = -T_p \int d^{p+1}\xi e^{-\Phi} \sqrt{-\det (G_{ab} + B_{ab} + 2\pi\alpha' F_{ab})} \quad (13.20)$$

[L] [Pol §2.5, eq. (38), ll. 886–898], the Dirac–Born–Infeld action;  $G_{ab}$  and  $B_{ab}$  are pulled back to the brane. (Polchinski writes  $\det^{1/2}$  without the sign; the minus sign under the root is needed in Lorentzian signature.) Three features can be understood with the tools of this chapter.

- The factor  $e^{-\Phi} = 1/g_s$  appears because the action arises from the disk, the tree level of open strings. A D-brane is therefore heavy at weak coupling.
- The dependence on  $F$  follows from T-duality. Let the brane extend along  $X^1, X^2$  with constant  $F_{12}$ , in the gauge  $A_2 = X^1 F_{12}$ . Dualize  $X^2$ . By (13.17) the dual brane sits at  $Y^2 = 2\pi\alpha' X^1 F_{12}$ : it is a tilted brane of one dimension less. Its length element is  $dX^1 \sqrt{1 + (\partial_1 Y^2)^2} = dX^1 \sqrt{1 + (2\pi\alpha' F_{12})^2}$ , which is  $\sqrt{\det(\delta_{ab} + 2\pi\alpha' F_{ab})}$  for a  $2 \times 2$  antisymmetric  $F$  [Pol §2.5, eqs. (39)–(40), ll. 899–915]. A magnetic field on a brane is T-dual to a tilt.
- Only the combination  $B + 2\pi\alpha' F$  can appear. The worldsheet couplings are  $\frac{1}{2\pi\alpha'} \int_M B + \int_{\partial M} A$ ; the transformation  $\delta B = d\zeta$  produces a boundary term which is cancelled by  $\delta A = -\zeta/2\pi\alpha'$ , and  $B + 2\pi\alpha' F$  is the invariant [Pol §2.5, eq. (41), ll. 915–933].

T-duality also relates the tensions of branes of different dimension. Wrap a  $Dp$ -brane on a torus with radii  $R_1, \dots, R_p$ . Its mass is  $T_p e^{-\Phi} \prod_{i=1}^p 2\pi R_i$ . Dualize the circle  $R_p$ . By Proposition 13.4 the same object is a  $D(p-1)$ -brane wrapped on the remaining circles, with mass  $T_{p-1} e^{-\tilde{\Phi}} \prod_{i=1}^{p-1} 2\pi R_i$ . The couplings are related by the dilaton shift of Chapter 8,  $e^{\tilde{\Phi}} = e^{\Phi} \sqrt{\alpha'}/R_p$ . Equating the two masses,

$$T_p e^{-\tilde{\Phi}} \frac{\sqrt{\alpha'}}{R_p} 2\pi R_p = T_{p-1} e^{-\tilde{\Phi}} \implies$$

$$T_{p-1} = 2\pi\sqrt{\alpha'} T_p \quad (13.21)$$

[L] [Pol §2.5, eqs. (42)–(44), ll. 934–973; dilaton: §2.1, eq. (30), ll. 662–668]. Dimensional analysis agrees:  $T_p$  is a mass per unit  $p$ -volume, so  $T_{p-1}/T_p$  is a length, and the only length available is  $\sqrt{\alpha'}$ . The radius  $R_p$  has dropped out, as it must. The absolute normalization requires a one-loop computation, the exchange of closed strings between two branes [Pol §2.6, ll. 990–1011], which we do not reproduce.

## 13.6 Orientifolds and tadpoles: a look ahead

The constructions of Chapters 24 and 25 need one more object, and it arises in the same way. An *unoriented* string theory is obtained by keeping only states invariant under worldsheet parity  $\Omega$ , which exchanges left- and right-movers,  $X_L \leftrightarrow X_R$ . What does  $\Omega$  do to the dual coordinate? Since  $Y = X_L - X_R$ ,

$$\Omega : \quad Y(\tau, \sigma) \longmapsto -Y(\tau, \pi - \sigma). \quad (13.22)$$

In the dual description  $\Omega$  is worldsheet parity *combined with* the spacetime reflection  $Y \rightarrow -Y$ . Projecting onto invariant states therefore does not remove half of the states at each point. It relates the state of a string at  $Y$  to that of the orientation-reversed string at  $-Y$ . The dual spacetime is the circle modulo the reflection, the interval  $0 \leq Y \leq \pi\tilde{R}$ , and the two fixed points  $Y = 0$  and  $Y = \pi\tilde{R}$  are the locations of *orientifold planes*. With  $k$  dualized circles the dual space is  $T^k/\mathbb{Z}_2$ , which has  $2^k$  fixed planes [Pol §2.7, ll. 1211–1250]. For  $k = 4$  this quotient is the orbifold limit of a K3 surface (Chapter 21), and for  $k = 2$  it is the base of the  $K3 \times T^2$  orientifolds discussed in Chapter 24.

Three facts about orientifold planes matter later [Pol §2.7, ll. 1250–1283].

- They are not dynamical. No string states are tied to them, so they have no fluctuations of shape, unlike D-branes.
- Away from the fixed planes the local physics is that of the oriented theory. Each D-brane at  $Y_i = \theta_i \tilde{R}$  has an image at  $-Y_i$ ; a Wilson line of  $SO(N)$  has eigenvalue angles  $\{\theta_1, -\theta_1, \dots, \theta_{N/2}, -\theta_{N/2}\}$ . There are  $\frac{1}{2}N$  independent branes for an  $N$ -valued Chan–Paton label.
- When  $m$  branes coincide away from a fixed plane the gauge group is  $U(m)$ ; when they sit on a fixed plane, strings to the images also become massless and the group is  $SO(2m)$ .

Like D-branes, orientifold planes are sources for the closed-string fields: they have a tension and, in the superstring, a Ramond–Ramond charge. For the type II orientifold with  $9 - p$  dualized circles the charge of one plane is  $\mp 2^{p-5}$  in units of the  $Dp$ -brane charge  $\mu_p$ , and there are  $2^{9-p}$  planes, so the total is

$$2^{9-p} \cdot (\mp 2^{p-5}) \mu_p = \mp 16 \mu_p, \quad (13.23)$$

independent of  $p$  [L] [Pol Lecture 3, eq. (93), ll. 1968–1984]. Now the key point. On a compact space the lines of flux that leave a charge have nowhere to go; Gauss’s law forces the total charge to vanish. A net tension would merely make the background evolve; a net Ramond–Ramond charge makes the field equations inconsistent. With the negative sign the charge of the planes is cancelled by exactly 16 D-branes (plus their 16 images), which is the T-dual description of the type I string with gauge group  $SO(32)$  [Pol Lecture 3, ll. 1985–1990]. The corresponding count for the bosonic string is  $2^{25-p} \cdot 2^{p-13} = 2^{12}$  branes [Pol §2.7, eq. (66) and below, ll. 1424–1443].

**Why this is called a tadpole**

A source with nothing to absorb its flux shows up in perturbation theory as a diagram in which a massless closed string is emitted into the vacuum: a one-point function, or “tadpole”. Its cancellation is the requirement that such one-point functions add up to zero. In Chapter 25 the same Gauss law appears for D3-brane charge, where the curvature of the compact space itself contributes  $\chi/24$  and fluxes and branes make up the balance. The logic is the one met here: *on a compact space, count all sources of a given charge, with signs, and demand zero.*

**Common pitfall**

Do not count branes and Chan–Paton indices as the same thing in an orientifold. There are 16 dynamical D-branes on the interval but a 32-valued label, because each brane has an image. Sources, and the Lean files of this project’s older libraries, normalize charges per brane, per brane–image pair or per fixed plane; before comparing two tadpole sums, check the unit (Chapter 25).

## 13.7 What the machine has checked

Nothing in this chapter has been formalized. The project’s Lean libraries contain no definition of an open string, a boundary condition, a Chan–Paton factor, a Wilson line or a D-brane as a boundary condition, and the catalogue of compiled declarations has no module on these topics. Every physical statement above is [L], with the source indicated. We say this plainly because the neighbouring chapters do contain kernel-checked statements, and the open-string story does not inherit their status. What the library does contain is the following.

**Adjacent results in the library**

The closed-string ancestor of (13.11) is the sign flip of the right-moving zero mode, proved in `StringTheoryFormalization/UseCases/TDualityMassSpectrum.lean` (namespace `StringTheory.UseCases.TDuality`) and discussed in Chapter 8:

```
def rightMomentum (n w R : ℚ) : ℚ := n / R - w * R

theorem tduality_flips_right_momentum (n w R : ℚ) (hR : R ≠ 0) :
  rightMomentum w n (1 / R) = - rightMomentum n w R := by ...
```

**Reading.** For rational numbers  $n, w, R$  with  $R \neq 0$ , exchanging the first two arguments and replacing  $R$  by  $1/R$  negates the function  $n/R - wR$ . This is the zero-mode part of “T-duality reflects the right-movers” at  $\alpha' = 1$ . It says nothing about oscillators, about a worldsheet with boundary, or about open strings.

On the other side of this chapter, `DualScaleStream2.Flux.tadpole_budget` (file `DualScaleStream2/Flux/Tadpole.lean`, discussed in Chapter 25) states that for natural numbers with  $flux + n = 24$  one has  $n \leq 24$ , and  $n = 24$  if  $flux = 0$ . It is an arithmetic shadow of a charge-cancellation condition and models neither branes nor Gauss’s law.

## 13.8 A formalization exercise

What would a first honest formalization of this chapter look like? The central mechanism, (13.11) in the form given below it, is a statement about *mode data*: two families of amplitudes, an involution that negates one of them, and two predicates exchanged by the involution. None

Table 13.2: Status of the main statements of this chapter.

| Statement                                                                      | Tier     | Where                 |
|--------------------------------------------------------------------------------|----------|-----------------------|
| N/D conditions on modes; spectrum; stretched strings                           | [L]      | Tong §3               |
| T-duality exchanges N and D; $p \mapsto p \pm 1$                               | [L]      | Pol §2.2, Tong §8.3.2 |
| Wilson line = brane positions (13.17)                                          | [L]      | Pol §2.3              |
| Branes are dynamical; DBI action; $T_{p-1} = 2\pi\sqrt{\alpha'} T_p$           | [L]      | Pol §2.4–2.5          |
| Orientifold planes; 16 branes cancel the charge (13.23)                        | [L]      | Pol §2.7, §3          |
| Closed strings, over $\mathbb{Q}$ : <code>tduality_flips_right_momentum</code> | [A]      | Chapter 8             |
| Statements of Section 13.8                                                     | exercise | not in the library    |

of this needs analysis, Hilbert spaces or a worldsheet. It is therefore a realistic target for a student who has read Chapter 34. We propose two statements. Both were elaborated by the author against the project’s toolchain and Mathlib in a scratch file. They are *not* part of the compiled library or its catalogue: in this book’s bookkeeping they are exercises, not Tier A results.

### Statement 1: the boundary-condition involution.

```

/-- Oscillator amplitudes of one free boson on the strip, by mode number k. -/
structure ModeData where
  left  :  $\mathbb{Z} \rightarrow \mathbb{C}$       -- left-movers (alpha-tilde_k in the text)
  right :  $\mathbb{Z} \rightarrow \mathbb{C}$       -- right-movers (alpha_k in the text)

/-- T-duality: reflect the right-movers. -/
def tdual (m : ModeData) : ModeData := ⟨m.left, fun k => - m.right k⟩

def IsNeumann (m : ModeData) : Prop :=  $\forall k, m.\text{right } k = m.\text{left } k$ 

def IsDirichlet (m : ModeData) : Prop :=  $\forall k, m.\text{right } k = - m.\text{left } k$ 

theorem tdual_involutive (m : ModeData) : tdual (tdual m) = m := by sorry

theorem neumann_iff_dirichlet_tdual (m : ModeData) :
  IsNeumann m  $\leftrightarrow$  IsDirichlet (tdual m) := by sorry

```

In words: reflecting the right-movers twice is the identity, and a configuration satisfies the Neumann relation of Proposition 13.3 if and only if its reflection satisfies the Dirichlet relation. The first proof is a case split on the structure followed by `simp`; the second unfolds the definitions and uses that negation is injective. The statement is deliberately modest. It identifies `ModeData` with nothing physical; the identification of  $\alpha_k, \tilde{\alpha}_k$  with the modes of a string, and the derivation of the two relations from  $\partial_\sigma X = 0$  and  $\partial_\tau X = 0$ , remain [L].

**Statement 2: Wilson lines and positions.** The second target is the arithmetic core of (13.16) and (13.18); the real parameter  $a$  stands for  $\alpha'$ .

```

noncomputable def wilsonMomentum (n :  $\mathbb{Z}$ ) (θi θj R :  $\mathbb{R}$ ) :  $\mathbb{R}$  :=
  (2 * Real.pi * n + θj - θi) / (2 * Real.pi * R)

theorem endpoint_separation (n :  $\mathbb{Z}$ ) (θi θj R a :  $\mathbb{R}$ ) (hR : R ≠ 0) :
  2 * Real.pi * a * wilsonMomentum n θi θj R
  = (2 * Real.pi * n + θj - θi) * (a / R) := by sorry

theorem stretched_mass_eq_momentum (n :  $\mathbb{Z}$ ) (θi θj R a :  $\mathbb{R}$ )

```

```

(hR : R ≠ 0) (hα : α ≠ 0) :
((2 * Real.pi * n + θj - θi) * (α / R) / (2 * Real.pi * α)) ^ 2
= (wilsonMomentum n θi θj R) ^ 2 := by sorry

```

In words:  $2\pi\alpha'$  times the fractional momentum equals the angle difference times the dual radius  $\alpha'/R$ ; and the squared energy of a classical string of that length and tension  $1/2\pi\alpha'$  equals the squared momentum. Both close with `unfold`, the fact `Real.pi_ne_zero` and `field_simp`. Note which hypotheses are needed:  $R \neq 0$  in both,  $\alpha' \neq 0$  only in the second, and positivity nowhere.

**What a faithful formalization would need.** In increasing order of difficulty: (i) a type of boundary conditions for  $D$  directions, a function  $\text{Fin } D \rightarrow \text{Bool}$ , with T-duality along direction  $i$  flipping the  $i$ -th entry, and the theorem that the number of Neumann directions changes by exactly one (Proposition 13.4; see Exercise 13.7); (ii) the solution space of the wave equation on the strip, with the theorem that the Neumann and Dirichlet conditions on smooth solutions are *equivalent* to the mode relations, which is Proposition 13.3 and needs Fourier series on an interval; (iii) the Fock space of the oscillators and the mass operator, so that (13.18) becomes a statement about a spectrum and not only about its zero-mode part; (iv) the identification of a Wilson line as a flat connection on a circle, with holonomy in a maximal torus of  $U(N)$ , and the map from holonomies to configurations of  $N$  points on the dual circle modulo permutations. Step (i) is an afternoon; (ii) and (iii) are projects; (iv) needs moduli of flat connections. The count (13.23) is elementary arithmetic, but a faithful formalization would have to derive the charge of an orientifold plane from an amplitude, which is far out of reach.

#### Chapter summary

- The boundary term of the open-string action vanishes for Neumann conditions  $\partial_\sigma X = 0$  (free end, no momentum flow) or Dirichlet conditions  $X = c$  (fixed end). On the modes:  $\alpha_k = \tilde{\alpha}_k$ , respectively  $\alpha_k = -\tilde{\alpha}_k$  with no momentum zero mode.
- Ends with Dirichlet conditions in  $D - p - 1$  directions lie on a  $Dp$ -brane. The spectrum  $M^2 = (N - 1)/\alpha'$  contains a photon on the brane and one massless scalar per transverse direction. A string stretched over a distance  $L$  has the extra term  $(L/2\pi\alpha')^2$ .
- T-duality,  $X = X_L + X_R \mapsto Y = X_L - X_R$ , exchanges  $\partial_\sigma$  and  $\partial_\tau$  and therefore Neumann and Dirichlet conditions. It maps  $Dp$  to  $D(p \mp 1)$  for a circle tangent or transverse to the brane. Open-string momentum  $n/R$  becomes winding between the ends,  $\Delta Y = 2\pi n \tilde{R}$ .
- A Wilson line  $\text{diag}(\theta_i)/2\pi R$  becomes brane positions  $Y_i = \theta_i \tilde{R}$ ; symmetry breaking is the mass of stretched strings, and coincident branes give  $U(m)$ .
- The scalar that is T-dual to  $A_{25}$  moves the brane: D-branes are dynamical, with action (13.20) and tensions related by  $T_{p-1} = 2\pi\sqrt{\alpha'} T_p$ .
- For unoriented strings T-duality produces orientifold planes at the fixed points of  $T^k/\mathbb{Z}_2$ . They are not dynamical but carry charge. On a compact space the total charge must vanish, which fixes the number of branes (16 plus images in type I). This Gauss-law counting is the prototype of the tadpole conditions of Chapter 25.
- All of this is [L]. Nothing about open strings is kernel-checked in this book's library; Section 13.8 proposes first targets.

## Exercises

**Exercise 13.1** (\*). Derive (13.2) from (13.1), keeping track of every sign. Why is “Neumann at one end, Dirichlet at the other” an allowed choice?

**Exercise 13.2** (\*). In units  $\alpha' = 1$  take  $R = \frac{1}{2}$  and three branes with Wilson-line angles  $\theta = (0, \frac{2\pi}{3}, \frac{4\pi}{3})$ . Find the positions of the branes on the dual circle, and the masses of all level-one states with  $M^2 \leq 4$ , once from each form of (13.18). What is the unbroken gauge group? Is any ground state tachyonic apart from those of the  $|ii\rangle$  strings with  $n = 0$ ?

**Exercise 13.3** (\*\*). Consider a direction with a Neumann condition at  $\sigma = 0$  and a Dirichlet condition  $X = c$  at  $\sigma = \pi$ . Show that the mode relation at  $\sigma = 0$  is still  $\alpha_k = \tilde{\alpha}_k$ , but that the condition at  $\sigma = \pi$  now forces the mode numbers to be half-integers,  $k \in \mathbb{Z} + \frac{1}{2}$ , and that there is neither a momentum nor a stretching term. Write the analogue of (13.5). What does T-duality along this direction do to the pair of conditions?

**Exercise 13.4** (\*\*). A D1-brane is wrapped on the diagonal of a square two-torus with radii  $R_1 = R_2 = R$ , along  $X^2 = X^1$ . Dualize the direction  $X^2$ . Using (13.17) backwards, show that the result is a D2-brane wrapped on the dual torus carrying a constant field strength, and find  $2\pi\alpha' F_{12}$ . Check that the flux  $\frac{1}{2\pi} \int F$  through the dual torus is an integer. What is this integer for a brane along  $X^2 = qX^1$  with  $q \in \mathbb{Z}$ ?

**Exercise 13.5** (\*\*). Iterate (13.21) to express  $T_p$  in terms of  $T_0$ . A Dp-brane is wrapped on a  $p$ -torus with all radii equal to  $R$ . Show that at fixed  $T_0$  and fixed coupling its mass is  $T_0 e^{-\Phi} (R/\sqrt{\alpha'})^p$ , and interpret the result at the self-dual radius. Then verify that the mass is unchanged if *all*  $p$  circles are dualized, provided the dilaton is shifted  $p$  times.

**Exercise 13.6** (\*\*, Lean). Prove `tdual_involutive` and `neumann_iff_dirichlet_tdual` of Section 13.8. Then define the level of a finitely supported configuration as a finite sum of products `m.right (-k) * m.right k` and show that it is invariant under `tdual`. Which property of the integrand do you use, and where did it appear in Chapter 8?

**Exercise 13.7** (\*\*, Lean). Let `bc : Fin D → Bool` record the boundary conditions, with `true` for Neumann. Define `flip i bc := Function.update bc i (!bc i)` and `numNeumann bc := (Finset.univ.filter fun j => bc j = true).card`. State and prove: if `bc i = true` then `numNeumann (flip i bc) + 1 = numNeumann bc`, and the analogous statement for `bc i = false`. Show also that `flip i` is an involution. This is Proposition 13.4 as combinatorics.

**Exercise 13.8** (\*\*\*). In the unoriented theory on the interval  $0 \leq Y \leq \pi\tilde{R}$ , place  $m$  branes at a generic point  $Y_0$  (with their images at  $-Y_0$ ) and move them to the fixed plane  $Y = 0$ . Using (13.8), list the strings that become massless in the limit and count the massless vectors, given that the projection  $\Omega$  keeps the antisymmetric combinations of brane–image strings in the  $SO$  case. Show that the count goes from  $m^2 = \dim U(m)$  to  $m(2m - 1) = \dim SO(2m)$ . Then explain why  $m + \frac{1}{2}$  branes can sit at a fixed plane [Pol §2.7, ll. 1262–1283].

## Further reading

- Open strings, boundary conditions, the spectrum on a brane, the Dirac action and multiple branes: [Tong §3, ll. 3040–3549]. T-duality for open strings in one page: [Tong §8.3.2–8.3.3, ll. 11804–11833].



- The route followed in this chapter, from Neumann conditions through T-duality to D-branes, Chan–Paton factors, Wilson lines, brane dynamics and the Born–Infeld action: [Pol §1.1–1.2, ll. 60–271; §2.1–2.5, ll. 482–989].
- Orientifolds and the cancellation of the total charge: [Pol §2.7, ll. 1211–1300, 1424–1443; Lecture 3, ll. 1968–2007].
- In this book: the closed-string dual coordinate in Chapter 8; the worldsheet derivation of T-duality in Chapter 9; the orbifold  $T^4/\mathbb{Z}_2$  in Chapter 21; branes, fluxes and tadpoles on  $K3 \times T^2$  in Chapters 24 and 25; how to write the Lean exercises in Chapter 34.



**Part III**

**Double Field Theory**



## Chapter 14

# Doubled Coordinates and the Strong Constraint

A point particle on a circle has one integer quantum number, its momentum  $n$ , and one coordinate  $x$  conjugate to it: the wave function of the state  $|n\rangle$  is  $e^{inx}$ . A closed string on a circle has *two* integers, momentum  $n$  and winding  $w$  (Chapter 7), and T-duality exchanges them (Chapter 8). If we want a field theory in which T-duality acts as simply as a rotation acts on ordinary fields, the winding number should be treated exactly like the momentum, and then it needs a coordinate of its own. This chapter answers three questions. What is the coordinate  $\tilde{x}$  conjugate to winding, and where does it come from? If fields depend on both  $x$  and  $\tilde{x}$ , what becomes of the level-matching condition of the closed string? And how does one get back from  $2d$  coordinates to the  $d$  coordinates of the space we observe, without breaking the duality symmetry that motivated the doubling? The answers are the *doubled torus*, the *weak constraint*  $\partial_a \tilde{\partial}^a \phi = 0$ , and the *strong constraint* or *section condition*, whose solutions are the maximal totally null subspaces of the doubled space. The duality group  $O(d, d; \mathbb{Z})$  of Chapter 11 permutes these solutions. This is the kinematical basis of double field theory; the fields that live on the doubled space and their action are the subject of Chapters 15 to 17.

Throughout,  $d$  is the number of compact directions, indices  $a, b = 1, \dots, d$  label them, and  $x^\mu$  are the coordinates of the non-compact Minkowski factor. Repeated indices are summed.

### 14.1 Why double: a coordinate for the winding number

#### 14.1.1 The dual coordinate on the worldsheet

Take one compact direction,  $X \sim X + 2\pi R$ . In the conventions of this book the closed-string solution splits into a left-moving and a right-moving part,  $X(\tau, \sigma) = X_L(\sigma^+) + X_R(\sigma^-)$  with  $\sigma^\pm = \tau \pm \sigma$ , and the zero-mode parts are

$$X_L = x_L + \frac{\alpha'}{2} p_L \sigma^+ + \dots, \quad X_R = x_R + \frac{\alpha'}{2} p_R \sigma^- + \dots, \quad p_{L,R} = \frac{n}{R} \pm \frac{wR}{\alpha'}, \quad (14.1)$$

where the dots are oscillators. Adding the two lines gives  $X = x + \alpha' \frac{n}{R} \tau + wR \sigma + \dots$  with  $x = x_L + x_R$ : the centre of mass moves with momentum  $p = n/R$  and the string closes only up to  $w$  turns,  $X(\sigma + 2\pi) = X(\sigma) + 2\pi R w$ . The *difference* of the two halves is an equally good free field,

$$\tilde{X} \equiv X_L - X_R = \tilde{x} + wR \tau + \alpha' \frac{n}{R} \sigma + \dots, \quad \tilde{x} = x_L - x_R, \quad (14.2)$$

which Tong calls “the new direction in spacetime” and which satisfies  $\partial_\tau X = \partial_\sigma \tilde{X}$ ,  $\partial_\sigma X = \partial_\tau \tilde{X}$  [Tong §8.3, ll. 11664–11683]. Compare (14.2) with the expansion of  $X$ . In  $\tilde{X}$  the coefficient

of  $\tau$  is  $\alpha' \cdot (wR/\alpha')$ , so  $\tilde{X}$  has centre-of-mass momentum  $wR/\alpha' = w/\tilde{R}$  with  $\tilde{R} = \alpha'/R$ , and the coefficient of  $\sigma$  is  $n\tilde{R}$ , so it winds  $n$  times around a circle of radius  $\tilde{R}$ . The field  $\tilde{X}$  is the coordinate of the T-dual circle, and its zero mode  $\tilde{x} \sim \tilde{x} + 2\pi\alpha'/R$  is conjugate to the winding number in the same way as  $x$  is conjugate to the momentum.

The combination  $\tilde{x}$  is invisible in the sigma-model action, which contains only  $X$ . It becomes visible as soon as one writes the operator that creates a state with both charges. The zero-mode part of the vertex operator of a state with momenta  $(p_L, p_R)$  is  $\exp(ip_L X_L + ip_R X_R)$  (Chapters 7 and 10), and the elementary identity

$$p_L X_L + p_R X_R = \frac{1}{2}(p_L + p_R)(X_L + X_R) + \frac{1}{2}(p_L - p_R)(X_L - X_R) = \frac{n}{R} X + \frac{wR}{\alpha'} \tilde{X} \quad (14.3)$$

shows that a state with  $w \neq 0$  has a wave function that depends on  $\tilde{x}$ . The doubling is therefore not an extra assumption: the zero modes  $x_L$  and  $x_R$  of a compact boson are independent operators, and  $(x, \tilde{x})$  is a basis for them adapted to momentum and winding. Hull and Zwiebach stress that “the toroidal zero modes are doubled; no new oscillators are added” [HZ §1, ll. 286–288].

### 14.1.2 From the string field to fields on a doubled torus

Let us now take  $d$  compact directions and use, as Hull and Zwiebach do, dimensionless coordinates of period  $2\pi$ :  $x^a \sim x^a + 2\pi$  and  $\tilde{x}_a \sim \tilde{x}_a + 2\pi$ . The radii and, more generally, the constant metric and  $B$ -field of the torus then sit in the background matrix  $E = G + B$  and not in the periodicities [HZ §2.2, ll. 573–650]. The momentum and winding operators have integer eigenvalues and

$$[x^a, p_b] = i\delta_b^a, \quad [\tilde{x}_a, w^b] = i\delta_a^b, \quad p_a = -i\frac{\partial}{\partial x^a} \equiv -i\partial_a, \quad w^a = -i\frac{\partial}{\partial \tilde{x}_a} \equiv -i\tilde{\partial}^a \quad (14.4)$$

[HZ §2.2, eq. (2.20) and ll. 700–765]. A general state of the first-quantized closed string is a superposition over the oscillator content  $I$  and over all charges,

$$|\Psi\rangle = \sum_I \int dk \sum_{n, w \in \mathbb{Z}^d} \hat{\phi}_I(k_\mu, n_a, w^a) \mathcal{O}_I |k_\mu, n_a, w^a\rangle, \quad (14.5)$$

where  $\mathcal{O}_I$  is a product of oscillators [HZ §1, eq. (1.1), ll. 205–216]. In string field theory the coefficient functions  $\hat{\phi}_I$  are the dynamical variables. A Fourier transformation in  $k_\mu$  and  $n_a$  turns them into functions of  $x^\mu$  and  $x^a$ , as in any Kaluza–Klein reduction. Nothing prevents us from transforming in  $w^a$  as well:

$$\phi_I(x^\mu, x^a, \tilde{x}_a) = \sum_{n, w \in \mathbb{Z}^d} \hat{\phi}_I(x^\mu; n, w) e^{i(n_a x^a + w^a \tilde{x}_a)}. \quad (14.6)$$

The component fields of the closed string on  $\mathbb{R}^{1, D-d-1} \times T^d$  are thus fields on  $\mathbb{R}^{1, D-d-1} \times T^{2d}$ , where the *doubled torus*  $T^{2d}$  has the periodic coordinates  $(x^a, \tilde{x}_a)$  [HZ §1, eq. (1.2), ll. 217–230]. The tensor indices of the fields are *not* doubled: the massless level  $N = \tilde{N} = 1$  still gives a symmetric tensor  $h_{ij}$ , an antisymmetric tensor  $b_{ij}$  and a scalar  $d$  with  $i, j = 0, \dots, D-1$ , each now a function of  $(x^\mu, x^a, \tilde{x}_a)$  [HZ §1, eq. (1.10), ll. 333–343]. Double field theory is the theory of these three fields; it keeps the states that would be massless in the decompactification limit but keeps *all* their momentum and winding excitations [AMN §3, ll. 473–492].

We collect coordinates, derivatives and charges in  $2d$ -component objects. In this chapter we follow the ordering of the Lean library, momentum first:

$$\mathbb{X} = \begin{pmatrix} x^a \\ \tilde{x}_a \end{pmatrix}, \quad \partial_M = \begin{pmatrix} \partial_a \\ \tilde{\partial}^a \end{pmatrix}, \quad Z = \begin{pmatrix} n_a \\ w^a \end{pmatrix} \in \mathbb{Z}^{2d}, \quad e^{iZ \cdot \mathbb{X}} = e^{i(n_a x^a + w^a \tilde{x}_a)}, \quad (14.7)$$

where  $Z \cdot \mathbb{X}$  is the ordinary dot product. Hull and Zwiebach and the review of Aldazabal, Marqués and Núñez put the winding entry first,  $v = (m, n)$  and  $X^M = (\tilde{x}_i, x^i)$  [HZ §5.1, eq. (5.4), ll. 3686–3700; AMN §3.3, eq. (3.25), ll. 970–978]. For everything in this chapter the ordering is immaterial, because the only structure we use is the pairing

$$Z \circ Z' \equiv Z^\top \eta Z' = n \cdot w' + w \cdot n', \quad \eta = \begin{pmatrix} 0 & \mathbf{1}_d \\ \mathbf{1}_d & 0 \end{pmatrix}, \quad (14.8)$$

which is symmetric under the exchange of the two blocks. (The ordering does matter for the block form of the generalized metric; see Chapter 15.)

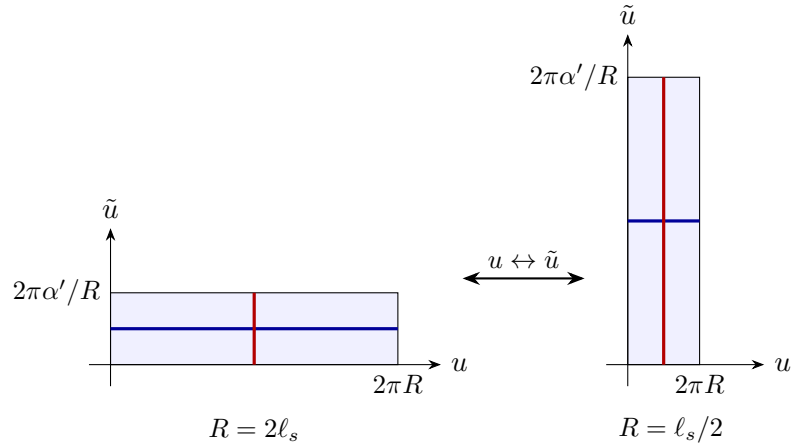


Figure 14.1: The doubled torus for  $d = 1$  in coordinates of physical length,  $u = Rx$  and  $\tilde{u} = (\alpha'/R)\tilde{x}$ . Opposite sides of each rectangle are identified. The horizontal circle (blue) is the spacetime circle, the vertical circle (red) is its T-dual. The area  $4\pi^2\alpha'$  does not depend on  $R$ ; T-duality exchanges the two axes and maps the left rectangle to the right one.

**Example 14.1** (The doubled torus of a circle). For  $d = 1$  and a circle of radius  $R$ , introduce coordinates of physical length  $u = Rx \sim u + 2\pi R$  and  $\tilde{u} = (\alpha'/R)\tilde{x} \sim \tilde{u} + 2\pi\alpha'/R$  [HZ §2.2, eq. (2.36), ll. 1012–1024]. The mode (14.7) becomes  $\exp(i\frac{n}{R}u + i\frac{wR}{\alpha'}\tilde{u})$ , in agreement with (14.3). Take  $R = 2\ell_s$  with  $\ell_s = \sqrt{\alpha'}$ . The doubled torus is a rectangle with sides  $4\pi\ell_s$  and  $\pi\ell_s$  (Figure 14.1, left). The mode  $(n, w) = (1, 0)$  has wave number  $1/(2\ell_s)$  along  $u$ ; the mode  $(0, 1)$  has wave number  $2/\ell_s$  along  $\tilde{u}$ , four times larger. These wave numbers are the contributions  $n/R$  and  $wR/\alpha'$  to the mass formula of Chapter 7. For any  $R$  the area is  $2\pi R \cdot 2\pi\alpha'/R = 4\pi^2\alpha'$ : shrinking the spacetime circle stretches the dual circle, and the doubled torus cannot be made small in both directions at once.

#### Physical picture: who sees which coordinate

A wave packet built from momentum modes is localized in  $x$  and spread out in  $\tilde{x}$ ; a packet built from winding modes is localized in  $\tilde{x}$ . When  $R \gg \ell_s$  the winding modes cost an energy of order  $R/\alpha'$  each, low-energy observers excite only momentum modes, and all fields they can prepare are independent of  $\tilde{x}$ : they see a circle of radius  $R$ . When  $R \ll \ell_s$  the roles are exchanged and the light fields depend only on  $\tilde{x}$ : the observers see a large circle of radius  $\alpha'/R$  [Tong §8.3, ll. 11643–11652]. Hull and Zwiebach put it this way: the doubled torus contains the original torus together with all the tori related to it by T-duality, and T-duality changes “which  $T^d$  subspace of the doubled torus is to be regarded as part of the spacetime” [HZ §1, ll. 123–128].

## 14.2 Level matching becomes a differential constraint

### 14.2.1 The operator $\Delta$

A closed string has no preferred origin of  $\sigma$ , and physical states must be invariant under rigid shifts of  $\sigma$  (“there are no special points in a closed string”, [AMN §3.1, eq. (3.2), ll. 524–529]). The generator of these shifts is  $L_0 - \bar{L}_0$ , so every physical state obeys the level-matching condition. On a torus it reads

$$L_0 - \bar{L}_0 = N - \bar{N} - p_a w^a = 0 \quad (14.9)$$

[HZ §1, eq. (1.3), ll. 230–234], which for  $d = 1$  is the condition  $N - \bar{N} = nw$  of Chapter 7; we write  $\bar{N}$  or  $\tilde{N}$  interchangeably for the second number operator. The derivation is one line: the zero modes contribute  $\frac{\alpha'}{4}(p_L^2 - p_R^2)$  to the difference of the two Virasoro generators, and with (14.1)

$$\frac{\alpha'}{4}(p_L^2 - p_R^2) = \frac{\alpha'}{4} \cdot 4 \frac{n}{R} \cdot \frac{wR}{\alpha'} = nw.$$

The radius has dropped out. This remains true for a general torus:  $p_a w^a$  contains no metric and no  $B$ -field.

In string field theory the two physical-state conditions play different roles. The mass-shell condition  $L_0 + \bar{L}_0 - 2 = 0$  is the linearized *equation of motion* of the component fields. Level matching is imposed *off shell*, as a constraint on the string field and on the gauge parameters,  $(L_0 - \bar{L}_0)|\Psi\rangle = 0$  [HZ §1, eqs. (1.6)–(1.7), ll. 289–298; §2.2, ll. 936–941]. Let us translate it into a statement about the component fields (14.6). Acting on the mode  $e^{iZ \cdot \mathbb{X}}$ , the operators (14.4) give  $p_a w^a \rightarrow (-i)^2 \partial_a \tilde{\partial}^a = -\partial_a \tilde{\partial}^a$ . Hence a component field at oscillator level  $(N_I, \bar{N}_I)$  satisfies

$$(N_I - \bar{N}_I) \phi_I = \frac{\alpha'}{2} \Delta \phi_I, \quad \Delta \equiv -\frac{2}{\alpha'} \frac{\partial}{\partial x^a} \frac{\partial}{\partial \tilde{x}_a} = -\frac{1}{\alpha'} \partial^\top \eta \partial \quad (14.10)$$

[HZ §1, eq. (1.9), ll. 310–323; §2.2, eqs. (2.32)–(2.33), ll. 907–935]. As a check,  $-\partial_a \tilde{\partial}^a e^{i(n \cdot x + w \cdot \tilde{x})} = -(in_a)(iw^a) e^{i(\dots)} = n \cdot w e^{i(\dots)}$ , which reproduces (14.9).

**Definition 14.2** (Weak constraint). A field  $\phi(x^\mu, x^a, \tilde{x}_a)$  at level  $N = \bar{N}$  satisfies the *weak constraint* if

$$\Delta \phi = 0 \quad \Longleftrightarrow \quad \partial_a \tilde{\partial}^a \phi = 0 \quad \Longleftrightarrow \quad \eta^{MN} \partial_M \partial_N \phi = 0. \quad (14.11)$$

All fields of double field theory,  $h_{ij}$ ,  $b_{ij}$  and  $d$ , have  $N = \bar{N} = 1$  and obey (14.11), and so do the gauge parameters [HZ §1, ll. 338–350]. In Fourier space the constraint is algebraic. Since  $\eta^{MN} \partial_M \partial_N e^{iZ \cdot \mathbb{X}} = -(Z \circ Z) e^{iZ \cdot \mathbb{X}}$  and  $Z \circ Z = 2n \cdot w$  by (14.8),

$$\Delta \phi = 0 \quad \Longleftrightarrow \quad \hat{\phi}(Z) = 0 \text{ unless } Z \circ Z = 0 \quad \Longleftrightarrow \quad \hat{\phi}(n, w) = 0 \text{ unless } n \cdot w = 0. \quad (14.12)$$

The allowed charges are the integer points of the null cone of  $\eta$  in  $\mathbb{R}^{2d}$ . For  $N \neq \bar{N}$  the cone is replaced by the hyperboloid  $\frac{1}{2}Z \circ Z = N - \bar{N}$  [HZ §5.1, eqs. (5.3)–(5.6), ll. 3676–3717; §5.3, eqs. (5.19)–(5.20), ll. 3922–3935]. Figure 14.2 shows both for  $d = 1$ .

### 14.2.2 Two operators, two metrics

It is instructive to set the mass-shell operator next to  $\Delta$ . With the generalized metric  $\mathcal{H}(E)$  of the torus, to be studied in Chapter 15 (its block form depends on the ordering of  $\partial_M$ ; we do not need it here), the zero-mode part of  $L_0 + \bar{L}_0$  is the Laplacian  $\square = \frac{1}{\alpha'} \partial^\top \mathcal{H}(E) \partial$ , while  $\Delta = -\frac{1}{\alpha'} \partial^\top \eta \partial$ . In the words of Hull and Zwiebach,  $\square$  is a Laplacian for the metric  $\mathcal{H}(E)$  and  $\Delta$  is one for the  $O(d, d)$ -invariant metric  $\eta$ ; “no background fields are required” in  $\Delta$



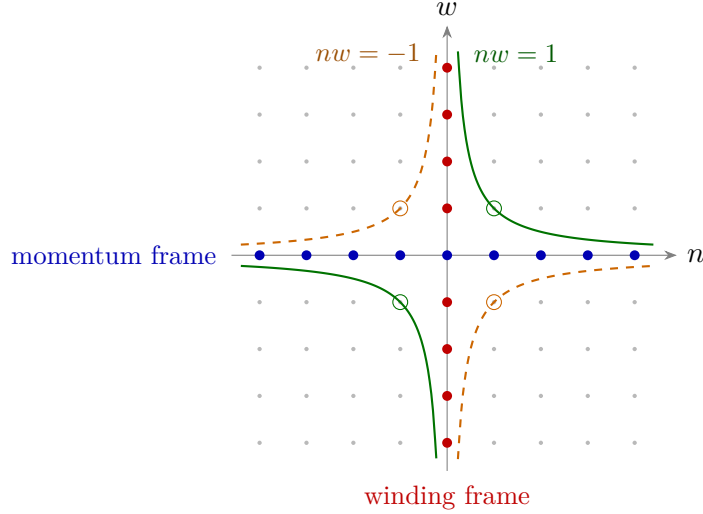


Figure 14.2: The charge lattice  $\mathbb{Z}^2$  of a circle. The null cone  $nw = 0$  consists of the two axes; its integer points (large dots) are the charges allowed at  $N = \bar{N}$ . States with  $N - \bar{N} = \pm 1$  have charges on the hyperbolas  $nw = \pm 1$ , which contain only the four circled lattice points. The inversion  $\eta$  reflects the picture in the diagonal  $n = w$ .

[HZ §2.2, eqs. (2.30)–(2.33), ll. 893–935]. For the circle of Example 14.1 we have  $\partial_x = R \partial_u$  and  $\partial_{\tilde{x}} = (\alpha'/R) \partial_{\tilde{u}}$ , so

$$\square|_{T^2} = \frac{\partial^2}{\partial u^2} + \frac{\partial^2}{\partial \tilde{u}^2}, \quad \Delta = -\frac{2}{\alpha'} R \cdot \frac{\alpha'}{R} \frac{\partial}{\partial u} \frac{\partial}{\partial \tilde{u}} = -2 \frac{\partial}{\partial u} \frac{\partial}{\partial \tilde{u}}. \quad (14.13)$$

On the mode  $\exp(i \frac{n}{R} u + i \frac{wR}{\alpha'} \tilde{u})$  the first operator gives  $-(n^2/R^2 + w^2 R^2/\alpha'^2)$ , the zero-mode part of the mass formula, which depends on  $R$ ; the second gives  $2nw$ , which does not. The doubled space thus carries two metrics: a positive-definite one,  $\mathcal{H}$ , which encodes the moduli and measures energies, and an indefinite one of signature  $(d, d)$ ,  $\eta$ , which is fixed once and for all and encodes level matching. The rest of this chapter uses only  $\eta$ .

### 14.2.3 What the weak constraint allows

For  $d = 1$  the condition  $nw = 0$  forces  $n = 0$  or  $w = 0$ . Every Fourier mode depends on  $x$  alone or on  $\tilde{x}$  alone, and the general solution of the weak constraint is  $\phi = f(x) + g(\tilde{x})$  (Exercise 14.1). For  $d \geq 2$  there are genuinely mixed solutions, such as

$$\phi = e^{i(x^1 + \tilde{x}_2)}, \quad n = (1, 0), \quad w = (0, 1), \quad n \cdot w = 0. \quad (14.14)$$

This is the content of the remark that string field theory “is a theory of constrained fields, but the constraint still allows fields with non-trivial dependence on both  $x^a$  and  $\tilde{x}_a$  if  $d > 1$ ” [HZ §1, ll. 320–323].

**Example 14.3** (Counting allowed charges on  $T^2$ ). Let  $d = 2$  and consider the  $3^4 = 81$  charges  $Z = (n_1, n_2; w^1, w^2)$  with all entries in  $\{-1, 0, 1\}$ . The weak constraint is  $n_1 w^1 + n_2 w^2 = 0$ . If  $w = 0$  all 9 choices of  $n$  are allowed; if  $n = 0$  and  $w \neq 0$  there are 8 more. If both are non-zero we need  $n \perp w$ . A vector with one vanishing entry, such as  $n = (\pm 1, 0)$ , is orthogonal to the two vectors  $w = (0, \pm 1)$ ; there are 4 such  $n$ , giving  $4 \cdot 2 = 8$  pairs. A vector with two non-vanishing entries, such as  $n = (1, 1)$ , is orthogonal to  $w = \pm(1, -1)$ ; again  $4 \cdot 2 = 8$  pairs. In total  $9 + 8 + 8 + 8 = 33$  of the 81 charges are null, and 16 of them are mixed. (A direct enumeration by script confirms 33, and gives 129 of 625 for entries in  $\{-2, \dots, 2\}$ ; this is a symbolic check.) A field satisfying the weak constraint can contain all 33 modes at once.

### 14.3 Products of fields and the strong constraint

#### 14.3.1 The weak constraint is not preserved by multiplication

An interacting field theory multiplies fields. Let  $\phi = e^{iZ \cdot \mathbb{X}}$  and  $\phi' = e^{iZ' \cdot \mathbb{X}}$  with  $Z$  and  $Z'$  null. The product has charge  $Z + Z'$  and, by bilinearity and symmetry of the pairing,

$$(Z + Z') \circ (Z + Z') = Z \circ Z + 2Z \circ Z' + Z' \circ Z' = 2Z \circ Z', \quad (14.15)$$

so that  $\Delta(\phi\phi') = 0$  if and only if  $Z \circ Z' = 0$ : the two charges must be *mutually* orthogonal, not only null [HZ §5.1, eq. (5.7), ll. 3722–3735]. In position space the same fact is the Leibniz rule for a second-order operator,

$$\eta^{MN} \partial_M \partial_N (AB) = (\eta^{MN} \partial_M \partial_N A) B + A (\eta^{MN} \partial_M \partial_N B) + 2\eta^{MN} \partial_M A \partial_N B, \quad (14.16)$$

with the cross term  $\eta^{MN} \partial_M A \partial_N B = \partial_a A \tilde{\partial}^a B + \tilde{\partial}^a A \partial_a B$ .

**Example 14.4** (The simplest failure). For  $d = 1$  take  $A = e^{ix}$  (one unit of momentum) and  $B = e^{i\tilde{x}}$  (one unit of winding). Both satisfy the weak constraint. Their product  $e^{i(x+\tilde{x})}$  has  $(n, w) = (1, 1)$  and  $\partial_x \partial_{\tilde{x}}(AB) = -AB \neq 0$ . Indeed  $(1, 1)$  lies on the hyperbola  $nw = 1$  of Figure 14.2: it is the charge of a state with  $N - \bar{N} = 1$ , not of a field with  $N = \bar{N}$ . For  $d = 2$  the two mixed modes with  $Z = (1, 0; 0, 1)$  and  $Z' = (0, 1; 1, 0)$  are both null, but  $Z \circ Z' = n \cdot w' + w \cdot n' = 1 + 1 = 2$ .

String field theory deals with this by building a projector into the string product. For a function  $A = \sum_Z \hat{A}(Z) e^{iZ \cdot \mathbb{X}}$  on the doubled torus one defines

$$[[A]] \equiv \sum_{Z \in \mathbb{Z}^{2d}} \delta_{Z \circ Z, 0} \hat{A}(Z) e^{iZ \cdot \mathbb{X}}, \quad \Delta[[A]] = 0, \quad [[[[A]]]] = [[A]], \quad (14.17)$$

and replaces products  $AB$  in the gauge transformations by  $[[AB]]$  [HZ §5.1, eqs. (5.8)–(5.13), ll. 3736–3822]. The price is high: the projected product is not associative (Exercise 14.5), the gauge algebra does not satisfy the Jacobi identity, and sign factors (cocycles) appear in the vertices [HZ §5.1, ll. 3646–3670; §5.3, ll. 3953–3958]. Hull and Zwiebach constructed the weakly constrained theory to cubic order in the fields [HZ §1, ll. 137–141].

#### 14.3.2 The strong constraint

The alternative is to demand more of the fields, so that products are automatically allowed.

**Definition 14.5** (Strong constraint, section condition). A collection  $\mathcal{F}$  of fields and gauge parameters on the doubled space satisfies the *strong constraint* (or *section condition*) if

$$\eta^{MN} \partial_M \partial_N A = 0 \quad \text{and} \quad \eta^{MN} \partial_M A \partial_N B = \partial_a A \tilde{\partial}^a B + \tilde{\partial}^a A \partial_a B = 0 \quad \text{for all } A, B \in \mathcal{F}. \quad (14.18)$$

By (14.16) this is equivalent to requiring  $\eta^{MN} \partial_M \partial_N$  to annihilate all fields, all gauge parameters *and all their products*, which is how the review of Aldazabal, Marqués and Núñez states it [AMN §3.3, eqs. (3.29)–(3.32), ll. 1021–1050]. In Fourier space, let  $S \subset \mathbb{Z}^{2d}$  be the set of all charges that occur in any member of  $\mathcal{F}$ . The first condition in (14.18) says  $Z \circ Z = 0$  for  $Z \in S$ ; the second, applied to two single modes, gives  $-(Z \circ Z') e^{i(Z+Z') \cdot \mathbb{X}} = 0$ . Therefore

$$\mathcal{F} \text{ satisfies the strong constraint} \iff Z \circ Z' = 0 \text{ for all } Z, Z' \in S. \quad (14.19)$$

A set with this property is called *totally null* or *totally isotropic*; Hull and Zwiebach call the null cone the “large space” and a totally null set of charges a “small space” [HZ §5.3, ll. 3922–3965]. Property (14.19) is exactly what the Lean predicate `IsSection` expresses (Section 14.6).

**Common pitfall: null is not totally null**

The null cone is not a linear space. In Figure 14.2 the charges  $(1, 0)$  and  $(0, 1)$  are both null but their sum  $(1, 1)$  is not. A set of charges in which every element is null can fail to be a section; what is needed is that every *pair* is orthogonal. The two notions agree only for sets closed under addition, by the polarization identity (14.15) (Exercise 14.8). Correspondingly, the weak constraint is a linear condition on one field, while the strong constraint is a condition on a whole collection of fields and is not linear: the sum of two solutions belonging to different sections, such as  $f(x) + g(\tilde{x})$ , solves the weak but not the strong constraint.

## 14.4 Solutions: maximal isotropic subspaces

### 14.4.1 At most half of the coordinates

**Proposition 14.6.** *Let  $V \subset \mathbb{R}^{2d}$  be a linear subspace that is totally null for  $\eta$ . Then  $\dim V \leq d$ , and the bound is attained.*

*Proof.* Let  $V^\perp = \{u \in \mathbb{R}^{2d} : u \circ v = 0 \text{ for all } v \in V\}$ . The map  $\mathbb{R}^{2d} \rightarrow V^*$ ,  $u \mapsto (u \circ \cdot)|_V$ , is surjective:  $u \mapsto u \circ \cdot$  is an isomorphism onto the dual of  $\mathbb{R}^{2d}$  because  $\eta$  is non-degenerate ( $\eta^2 = \mathbf{1}$ ), and every linear functional on  $V$  extends to  $\mathbb{R}^{2d}$ . Its kernel is  $V^\perp$ ; hence  $\dim V^\perp = 2d - \dim V$ . Total nullity means  $V \subseteq V^\perp$ , so  $\dim V \leq 2d - \dim V$ . The subspace  $\{(n, 0)\}$  has dimension  $d$  and is totally null by (14.8).  $\square$

A totally null subspace of the maximal dimension  $d$  is called *maximal isotropic* or, in this context, a *section*. Now let  $S$  be as in (14.19) and let  $V$  be its real span. By bilinearity  $V$  is totally null, so  $r = \dim V \leq d$ . Choose a basis  $Z_1, \dots, Z_r$  of  $V$  and put  $y^k = Z_k \cdot \mathbb{X}$ . Every mode with  $Z = \sum_k c_k Z_k \in V$  is a function of the  $y^k$  alone,  $e^{iZ \cdot \mathbb{X}} = e^{ic_k y^k}$ . We have shown:

**Corollary 14.7.** *Fields that satisfy the strong constraint depend on at most  $d$  linear combinations  $y^k = Z_k \cdot \mathbb{X}$  of the  $2d$  doubled coordinates, and the charges  $Z_k$  are null and mutually orthogonal. In particular, the fields are constant along the directions  $\eta V$ , because  $(\eta Z') \cdot Z = Z' \circ Z = 0$  for  $Z, Z' \in V$ .*

For  $V = \{(n, 0)\}$  the constant directions are  $\eta V = \{(0, n)\}$ , the  $\tilde{x}$  directions:  $\tilde{\partial}^a(\dots) = 0$ . This is the *supergravity frame*, in which  $h_{ij}$ ,  $b_{ij}$  and  $d$  reduce to the ordinary metric fluctuation, Kalb–Ramond field and dilaton on the undoubled space [HZ §1, ll. 338–343; AMN §3.3, ll. 1050–1058]. Hull and Zwiebach note that in such a frame, with null coordinates  $y^a$  and complementary coordinates  $\tilde{y}_a$ , one has  $\Delta = -\frac{2}{\alpha'} \partial_{y^a} \partial_{\tilde{y}_a}$ ; for fields independent of  $\tilde{y}$  the constraint holds identically, all products satisfy it, and neither projectors nor cocycles are needed [HZ §5.3, eq. (5.21), ll. 3946–3960].

### 14.4.2 A gallery of sections

- (i) *Momentum frame*  $\{(n, 0)\}$ : fields depend on  $x^a$  only. This is the description natural at large radii.
- (ii) *Winding frame*  $\{(0, w)\}$ : fields depend on  $\tilde{x}_a$  only, the “dual versions of these fields” [HZ §1, ll. 340–342].
- (iii) *Mixed coordinate frames*. For a subset  $K \subseteq \{1, \dots, d\}$  keep  $w^a$  for  $a \in K$  and  $n_a$  for  $a \notin K$ . For  $d = 2$ ,  $K = \{2\}$  this is  $\{(n_1, 0; 0, w^2)\}$ , and  $Z \circ Z' = n_1 \cdot 0 + 0 \cdot w'^2 +$  (same with primes exchanged)  $= 0$ . The mode (14.14) lives in this frame. There are

$2^d$  coordinate frames; they describe the torus with the circles in  $K$  replaced by their T-duals.

- (iv) *Tilted frames.* Let  $\Theta$  be an antisymmetric  $d \times d$  matrix. The graph  $\{(n, \Theta n)\}$  is totally null because  $n \cdot \Theta n' + \Theta n \cdot n' = n^\top (\Theta + \Theta^\top) n' = 0$ , and so is  $\{(\Theta w, w)\}$ . Conversely, a  $d$ -dimensional totally null subspace that projects isomorphically onto the momentum frame is the graph of a linear map  $\Theta$ , and total nullity forces  $\Theta^\top = -\Theta$ . The sections near the momentum frame are therefore parametrized by the  $d(d-1)/2$  entries of an antisymmetric matrix. For  $d = 1$  there is no such parameter, and Figure 14.2 shows why: the only null lines are the two axes.

Is every solution of the strong constraint obtained from the supergravity frame by a duality rotation? Aldazabal, Marqués and Núñez assert that a possible solution is  $\tilde{\partial}^i(\dots) = 0$  “or any  $O(D, D)$  rotation of this. Actually, it can be proven that this is the only solution” [AMN §3.3, ll. 1050–1056]. [L] Over the real numbers this is an instance of Witt’s theorem, which says that the orthogonal group of a non-degenerate quadratic form acts transitively on isotropic subspaces of a given dimension (standard; not checked against a source in this book’s library). Item (iv) and Exercise 14.6 prove the statement on the page for sections that are graphs over the momentum frame or over the winding frame. Nothing of this classification has been formalized.

#### Common pitfall: the strong constraint is not level matching

The weak constraint is the level-matching condition of the states with  $N = \bar{N}$  on a torus, as derived above. The strong constraint is *stronger than anything string theory requires* of those states: the 16 mixed charges of Example 14.3 are perfectly good string states, but no single section contains all of them. The review of Aldazabal, Marqués and Núñez warns that, in a background-independent formulation, “the level matching condition should not be identified with the strong constraint (or any weaker version)” [AMN §3.3, ll. 1059–1070]. The strong constraint is a sufficient condition for the gauge algebra of double field theory to close (Chapter 16; [AMN §3.3, ll. 1078–1083]). With it, the theory restricted to one section is a conventional local field theory written in a duality-covariant way [HZ §5.3, ll. 3956–3964]; the genuinely stringy, weakly constrained theory is known to cubic order only.

## 14.5 $O(d, d)$ rotates one frame into another

Recall from Chapter 11 that  $g \in O(d, d; \mathbb{Z})$  means that  $g$  is an integer  $2d \times 2d$  matrix with  $g^\top \eta g = \eta$ . The group acts on charges by  $Z \mapsto gZ$  and on the doubled coordinates linearly, preserving their periodicities because  $g$  and  $g^{-1}$  are integer matrices [HZ §1, ll. 276–283; §4.1, eq. (4.1), ll. 2803–2812; AMN §3.3, eq. (3.27), ll. 996–1005]. Two facts make the constraints duality covariant.

**Lemma 14.8.** *If  $g^\top \eta g = \eta$  then  $g^{-1} = \eta g^\top \eta$  and  $g \eta g^\top = \eta$ .*

*Proof.* Multiply  $g^\top \eta g = \eta$  from the left by  $\eta$  and use  $\eta^2 = \mathbf{1}$ :  $(\eta g^\top \eta) g = \mathbf{1}$ , so  $g$  is invertible with the stated inverse. Then  $g(\eta g^\top \eta) = \mathbf{1}$ , and multiplying from the right by  $\eta$  gives  $g \eta g^\top = \eta$ .  $\square$

**Proposition 14.9** (Covariance of the constraints). *Let  $g^\top \eta g = \eta$ . (a) For all  $Z, Z'$ :  $(gZ) \circ (gZ') = Z \circ Z'$ . In particular  $g$  maps null charges to null charges and totally null sets to totally null sets. (b) Under the change of coordinates  $\mathbb{X}' = g\mathbb{X}$  the operator  $\eta^{MN} \partial_M \partial_N$  keeps its form.*

*Proof.* (a)  $(gZ)^\top \eta (gZ') = Z^\top (g^\top \eta g) Z' = Z^\top \eta Z'$ . (b) By the chain rule  $\partial = g^\top \partial'$ , so  $\partial^\top \eta \partial = \partial'^\top (g \eta g^\top) \partial' = \partial'^\top \eta \partial'$  by Lemma 14.8.  $\square$

So “if a given configuration solves the strong constraint, any T-duality transformation of it will also do” [AMN §3.3, ll. 1044–1047], and any  $T^d$  obtained from the  $x$ -torus or the  $\tilde{x}$ -torus by acting with  $O(d, d; \mathbb{Z})$  is again totally null [HZ §5.3, ll. 3940–3946]. Let us see what the three families of generators of Chapter 11 do to the gallery above.

- The full inversion  $g = \eta$  sends  $(n, w) \mapsto (w, n)$ . It maps the momentum frame onto the winding frame: the theory written in the coordinates  $x$  becomes the same theory written in  $\tilde{x}$ . For a rectangular torus this is  $R_a \mapsto \alpha'/R_a$  in every direction [HZ §1, ll. 276–283].
- A factorized duality in the direction  $k$  exchanges  $n_k \leftrightarrow w^k$  and leaves the other entries alone. Starting from the momentum frame, the dualities in the directions  $k \in K$  produce the mixed coordinate frame labelled by  $K$ . For  $d = 2$  and  $k = 2$  the matrix is the permutation  $(n_1, n_2; w^1, w^2) \mapsto (n_1, w^2; w^1, n_2)$ , which sends  $(n_1, n_2; 0, 0)$  to  $(n_1, 0; 0, n_2)$ .
- A  $\Theta$ -shift  $g_\Theta = \begin{pmatrix} 1 & \Theta \\ 0 & 1 \end{pmatrix}$ , with  $\Theta$  an antisymmetric integer matrix, sends  $(n, w) \mapsto (n + \Theta w, w)$ . It fixes every charge of the momentum frame, which is why an integer shift of the  $B$ -field is invisible in supergravity, and it tilts the winding frame into  $\{(\Theta w, w)\}$ . For  $d = 2$  and  $\Theta = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$  the image of  $(0, 0; w^1, w^2)$  is  $(w^2, -w^1; w^1, w^2)$ , and  $Z \circ Z' = (w^2 w'^1 - w^1 w'^2) + (w'^2 w^1 - w'^1 w^2) = 0$ , as it must be. A basis change  $\text{diag}(A, (A^\top)^{-1})$  maps each of the two standard frames to itself.

For the circle the whole story fits in Figure 14.2. The equations  $g^\top \eta g = \eta$  for  $g = \begin{pmatrix} a & b \\ c & e \end{pmatrix}$  read  $ac = 0$ ,  $be = 0$ ,  $ae + bc = 1$ , whose integer solutions are  $g = \pm \mathbf{1}$  and  $g = \pm \eta$ . The group  $O(1, 1; \mathbb{Z})$  has four elements;  $\pm \mathbf{1}$  preserve each axis, and  $\pm \eta$  exchange the two axes, which are the only two sections.

#### Physical picture: a choice of polarization

The following analogy, which is an aid to intuition and not a derivation, may help. The situation resembles the choice of a polarization in quantum mechanics. Phase space has  $2d$  coordinates  $(q, p)$  and a fixed antisymmetric form; wave functions depend on half of them, say on  $q$ , and a linear canonical transformation trades the  $q$ -representation for the  $p$ -representation without changing the physics. Here the doubled space has  $2d$  coordinates and a fixed *symmetric* form  $\eta$ ; fields depend on a maximal null half, and a duality transformation trades one half for another. In neither case is there a preferred half. What selects the  $x$ -frame in practice is dynamics, that is the second metric  $\mathcal{H}$ : at large radius the modes that depend on  $x$  are the light ones.

#### Historical note

Hull and Zwiebach place their construction in a line of earlier work that they cite: the closed string field theory on tori of Kugo and Zwiebach, which already showed how T-duality is realised as a symmetry of the string field theory; Tseytlin’s first-quantized approach with separate left- and right-moving coordinates; Siegel’s duality-covariant field theory for the massless sector under the assumption that all momenta are mutually orthogonal, which is the strongly constrained theory in today’s language; and Hull’s doubled sigma models, in which T-duality changes which  $T^d \subset T^{2d}$  is regarded as spacetime [HZ §1, ll. 107–148; §5.3, ll. 3960–3967; refs. [2], [6]–[8], ll. 4079–4092].

## 14.6 What the machine has checked

The Lean library contains the *charge-lattice shadow* of this chapter: everything that can be said about the strong constraint in Fourier space, on the integer lattice  $\mathbb{Z}^d \oplus \mathbb{Z}^d$ , for every  $d$ . It contains no differential operators, no functions on a torus and no Fourier series; the dictionary (14.12), (14.19) between fields and sets of charges is our reading and is not [A]. Two files are involved:

- `DualScaleStream2/TDuality/ODD.lean`, namespace `DualScaleStream2.TDuality`, discussed in detail in Chapter 11;
- `DualScaleStream2/DFT/SectionCondition.lean`, namespace `DualScaleStream2.DFT`.

Running `#print axioms` on the six theorems of the second file reports the three standard axioms `propext`, `Classical.choice`, `Quot.sound` and nothing else.

### In Lean: the ambient data (from `ODD.lean` and `Factorized.lean`)

```
abbrev Charge (d : ℕ) := Fin d ⊕ Fin d

def eta (d : ℕ) : Matrix (Charge d) (Charge d) ℤ := fromBlocks 0 1 1 0

def IsODD (g : Matrix (Charge d) (Charge d) ℤ) : Prop := gT * eta d * g = eta d

def chargeNorm (Z : Charge d → ℤ) : ℤ := Z ·v (eta d *v Z)

theorem chargeNorm_sumElim (n w : Fin d → ℤ) :
  chargeNorm (Sum.elim n w) = 2 * (n ·v w) := by ...
```

**Reading.** A charge is a function from the index type  $\text{Fin } d \oplus \text{Fin } d$  to  $\mathbb{Z}$ ; `Sum.elim n w` is the charge with momentum part  $n$  (left summand) and winding part  $w$  (right summand), which fixes the convention  $Z = (n, w)$  of (14.7). The matrix `eta d` is  $\eta$  of (14.8), `IsODD g` is the equation  $g^T \eta g = \eta$  as a predicate on integer matrices, and `chargeNorm` is  $Z \circ Z$ , shown to equal  $2n \cdot w$ . The same file proves `eta_mul_self` ( $\eta^2 = 1$ ), `eta_isODD`, `isODD_mul` (closure under products) and `thetaShift_isODD`, `basisChange_isODD` for the generators used in Section 14.5. There is no group structure and no statement that these matrices generate  $O(d, d; \mathbb{Z})$ .

### In Lean: the pairing, sections, and level matching

```
def etaPair (Z Z' : Charge d → ℤ) : ℤ := Z ·v (eta d *v Z')

def IsSection (S : Set (Charge d → ℤ)) : Prop :=
  ∀ Z ∈ S, ∀ Z' ∈ S, etaPair Z Z' = 0

theorem etaPair_self (Z : Charge d → ℤ) : etaPair Z Z = chargeNorm Z := by ...

theorem levelMatching_iff (n w : Fin d → ℤ) :
  chargeNorm (Sum.elim n w) = 0 ↔ n ·v w = 0 := by ...
```

**Reading.** `etaPair` is  $Z \circ Z'$ . `IsSection S` says that the set  $S$  is totally null, which is the right-hand side of (14.19). Note what the definition does *not* ask:  $S$  need not be a subgroup, need not have rank  $d$ , and need not be maximal. The empty set and  $\{0\}$  are sections in this sense; the name records the intended use. `levelMatching_iff` says that over the integers  $2(n \cdot w) = 0$  if and only if  $n \cdot w = 0$ , i.e. that a charge is  $\eta$ -null exactly

when (14.12) holds. The number operators  $N, \bar{N}$  do not occur; the theorem is the  $N = \bar{N}$  case of (14.9) once one accepts that  $\frac{1}{2}Z \circ Z$  represents  $p_a w^a$ . **Proof idea.** `etaPair_self` holds by unfolding (`rfl`): the two definitions are the same expression. For the equivalence, rewrite with `chargeNorm_sumElim` and let the linear-arithmetic procedure `omega` decide  $2k = 0 \leftrightarrow k = 0$ .

#### In Lean: the two standard frames

```
def momentumFrame (d : ℕ) : Set (Charge d → ℤ) :=
  {Z | ∃ n : Fin d → ℤ, Z = Sum.elim n (0 : Fin d → ℤ)}

def windingFrame (d : ℕ) : Set (Charge d → ℤ) :=
  {Z | ∃ w : Fin d → ℤ, Z = Sum.elim (0 : Fin d → ℤ) w}

theorem eta_momentum_to_winding (n : Fin d → ℤ) :
  eta d *v Sum.elim n (0 : Fin d → ℤ) = Sum.elim (0 : Fin d → ℤ) n := by ...

theorem momentumFrame_isSection : IsSection (momentumFrame d) := by ...

theorem windingFrame_isSection : IsSection (windingFrame d) := by ...
```

**Reading.** The frames are items (i) and (ii) of the gallery, as sets of integer charges. The first theorem is  $\eta(n, 0) = (0, n)$ : the full inversion sends each pure-momentum charge to the pure-winding charge with the same integers. The other two say that  $(n, 0) \circ (n', 0) = 0$  and  $(0, w) \circ (0, w') = 0$  for all integer vectors, for every  $d$ . That these frames have the maximal rank  $d$  (Proposition 14.6) is not stated in Lean. **Proof idea.** Destructure the membership hypotheses to obtain  $n, n'$ ; compute  $\eta Z'$  blockwise with Mathlib's `fromBlocks_mulVec`; the dot product of `Sum.elim n 0` with `Sum.elim 0 n'` splits into  $n \cdot 0 + 0 \cdot n'$ , which `simp` evaluates to 0.

#### In Lean: dualities map sections to sections

```
theorem isSection_image (g : Matrix (Charge d) (Charge d) ℤ) (hg : IsODD g)
  (S : Set (Charge d → ℤ)) (hS : IsSection S) :
  IsSection ((fun Z => g *v Z) '' S) := by ...
```

**Reading.** This is part (a) of Proposition 14.9 for sets: if  $S$  is totally null and  $g^\top \eta g = \eta$ , then the image  $\{gZ : Z \in S\}$  is totally null. It is a one-directional statement (the converse also holds, by Lemma 14.8, but is not in the library), it concerns integer matrices only, and it does not say that every section arises as an image of `momentumFrame`. Combined with `eta_isODD`, `thetaShift_isODD` and `momentumFrame_isSection` it yields, without further work, that the winding frame and all tilted frames  $g_\Theta(\text{winding frame})$  are sections. **Proof idea.** The proof is the computation in the text: a lemma `key` rewrites  $(gZ) \cdot \eta(gZ')$  as  $Z \cdot (g^\top \eta g) Z'$  using associativity of `mulVec` and `dotProduct_mulVec`; the hypothesis `hg` replaces the bracket by  $\eta$ ; what remains is an instance of `hS`.

What would a faithful formalization need? First, the *linear algebra* of this chapter: Proposition 14.6, maximality, and the transitivity of the orthogonal group on maximal isotropic subspaces. Mathlib has bilinear and quadratic forms and orthogonal complements, so the dimension bound is within reach (Exercise 14.9). Second, the *analysis*: smooth functions on  $T^{2d}$ , their Fourier coefficients, and the statement that  $\eta^{MN} \partial_M \partial_N$  acts on the coefficient  $\hat{\phi}(Z)$  by multiplication with  $-Z \circ Z$ ; this would turn (14.12) and (14.19) into theorems and connect `IsSection` to differential equations. Third, the step from the operator  $L_0 - \bar{L}_0$  of



Table 14.1: Epistemic status of the main claims of this chapter.

| Claim                                                                                          | Tier | Basis                                                  |
|------------------------------------------------------------------------------------------------|------|--------------------------------------------------------|
| $Z \circ Z = 2n \cdot w$ ; null $\Leftrightarrow n \cdot w = 0$ (over $\mathbb{Z}$ , all $d$ ) | [A]  | chargeNorm_sumElim,<br>levelMatching_iff, etaPair_self |
| Momentum and winding frames are totally null                                                   | [A]  | momentumFrame_isSection,<br>windingFrame_isSection     |
| $\eta$ maps $(n, 0)$ to $(0, n)$                                                               | [A]  | eta_momentum_to_winding                                |
| $O(d, d; \mathbb{Z})$ maps totally null sets to totally null sets                              | [A]  | isSection_image                                        |
| Component fields live on $T^{2d}$ ; $L_0 - \bar{L}_0 = 0$ becomes $\Delta\phi = 0$             | [L]  | HZ §1, §2.2                                            |
| Products need $Z \circ Z' = 0$ ; projector $[[\cdot]]$ ; small and large spaces                | [L]  | HZ §5.1, §5.3                                          |
| Strong constraint and its duality covariance                                                   | [L]  | AMN §3.3                                               |
| Every solution is a rotation of the supergravity frame                                         | [L]  | AMN §3.3 (asserted there)                              |
| $\dim V \leq d$ for totally null $V$ ; tilted frames $\leftrightarrow$ antisymmetric $\Theta$  | —    | proved on the page; not formalized                     |
| Counts 33/81 and 129/625 in Example 14.3                                                       | —    | enumeration by script; symbolic check                  |
| Identification of <b>IsSection</b> with the strong constraint on fields                        | —    | reading; never Tier A                                  |

a conformal field theory to  $\Delta$ , which requires a formalized Fock space of the compact boson and is far from the present library.

#### Chapter summary

- A compact boson has two independent zero modes. In the basis  $x = x_L + x_R$ ,  $\tilde{x} = x_L - x_R$  the first is conjugate to momentum and the second to winding; a state with charges  $(n, w)$  has the wave function  $e^{i(n \cdot x + w \cdot \tilde{x})}$ . The component fields of closed string field theory on  $T^d$  are functions on the doubled torus  $T^{2d}$ .
- Level matching at  $N = \bar{N}$  becomes the weak constraint  $\partial_a \tilde{\partial}^a \phi = 0$ , i.e. all Fourier charges are null,  $Z \circ Z = 2n \cdot w = 0$ . The operator involves only  $\eta$ , never the radii or the  $B$ -field.
- Products of weakly constrained fields violate the constraint unless the charges are mutually orthogonal. The strong constraint demands  $Z \circ Z' = 0$  for all charges in all fields: the set of charges is totally null.
- A totally null subspace of  $(\mathbb{R}^{2d}, \eta)$  has dimension at most  $d$ . Strongly constrained fields depend on at most  $d$  coordinates: the momentum frame (supergravity), the winding frame,  $2^d$  mixed frames, and frames tilted by an antisymmetric  $\Theta$ .
- $O(d, d; \mathbb{Z})$  preserves  $\circ$ , hence both constraints, and maps sections to sections: T-duality changes which half of the doubled torus is called spacetime.
- Kernel-checked (Tier A): nullity  $\Leftrightarrow n \cdot w = 0$ , the two standard frames are sections,  $\eta$  maps one to the other, and  $O(d, d; \mathbb{Z})$ -images of sections are sections, all on the integer charge lattice. Not formalized: fields, derivatives, the dimension bound, and the classification of sections.



## Exercises

**Exercise 14.1** (★). Let  $d = 1$ . Using the Fourier expansion (14.6), show that the general solution of  $\partial_x \partial_{\tilde{x}} \phi = 0$  on the doubled torus is  $\phi = f(x) + g(\tilde{x})$  with  $f, g$  periodic. Which of these solutions satisfy the strong constraint when  $\mathcal{F} = \{\phi\}$ ?

**Exercise 14.2** (★). For a rectangular torus with radii  $R_1, \dots, R_d$ , use the coordinates of Example 14.1 in each direction to show that the volume of the doubled torus is  $(4\pi^2 \alpha')^d$ . For  $d = 1$  and  $R = 3\ell_s$ , give the two periods, and the wave numbers of the modes  $(n, w) = (2, 0)$  and  $(0, 2)$  along  $u$  and  $\tilde{u}$ . At which radius is the doubled torus a square?

**Exercise 14.3** (★). Verify the zero-mode expansion (14.2) from (14.1) and check  $\partial_\tau X = \partial_\sigma \tilde{X}$  and  $\partial_\sigma X = \partial_\tau \tilde{X}$  on the zero modes. Then verify the identity (14.3).

**Exercise 14.4** (★★). Repeat the count of Example 14.3 for  $d = 2$  and entries in  $\{-2, \dots, 2\}$  by hand or by a short program, and confirm 129 null charges out of 625. How many of them lie in the momentum frame, in the winding frame, and in the mixed frame  $\{(n_1, 0; 0, w^2)\}$ ? Exhibit two null charges that lie in no common section.

**Exercise 14.5** (★★). Let  $d = 1$ ,  $A = e^{ix}$ ,  $B = e^{i\tilde{x}}$ ,  $C = e^{-i\tilde{x}}$ . With the projector (14.17), compute  $[[[AB]]C]]$  and  $[[A[[BC]]]]$  and conclude that the projected product is not associative. Explain why no such failure can occur if  $A, B, C$  belong to a collection that satisfies the strong constraint.

**Exercise 14.6** (★★). Let  $V \subset \mathbb{R}^{2d}$  be a  $d$ -dimensional totally null subspace whose projection to the winding part is an isomorphism. (a) Show that  $V = \{(\Theta w, w)\}$  for a unique matrix  $\Theta$  and that  $\Theta^\top = -\Theta$ . (b) Show that  $g_\Theta = \begin{pmatrix} 1 & \Theta \\ 0 & 1 \end{pmatrix}$  satisfies  $g_\Theta^\top \eta g_\Theta = \eta$  and maps the winding frame onto  $V$ . (c) Deduce that  $V$  is the image of the momentum frame under  $g_\Theta \eta$ . For which  $\Theta$  does this rotation preserve the integer lattice?

**Exercise 14.7** (★★, Lean). State and prove in Lean, importing `DualScaleStream2.DFT`. **SectionCondition**: (a) a subset of a section is a section; (b) every element  $Z$  of a section satisfies `chargeNorm Z = 0`. Part (a) is a one-line term; for (b) rewrite with `etaPair_self` from right to left. Then use `isSection_image`, `eta_isODD` and `momentumFrame_isSection` to show that the image of `momentumFrame d` under `eta d` is a section, without unfolding any definition.

**Exercise 14.8** (★★★, Lean). Formalize the polarization argument of (14.15): if a set  $S$  of charges is closed under addition and every  $Z \in S$  satisfies `etaPair Z Z = 0`, then `IsSection S`. You will need the symmetry of `etaPair` (prove it from  $\eta^\top = \eta$ ) and its additivity in each argument (Mathlib: `mulVec_add`, `add_dotProduct`, `dotProduct_add`); the final step  $2k = 0 \Rightarrow k = 0$  is done by `omega`.

**Exercise 14.9** (★★★, Lean). Formulate Proposition 14.6 in Lean for the real form  $\eta_{\mathbb{R}}$  on  $\mathbb{R}^{2d}$ : for a submodule  $V$  on which the bilinear form vanishes identically, `Module.finrank ℝ V ≤ d`. Locate in Mathlib the statement that the orthogonal complement of  $V$  with respect to a non-degenerate bilinear form has dimension  $2d - \dim V$ , and reduce the bound to it. (This statement is not in the book's library; treat it as a small project.)

## Further reading

- Doubled torus, string field components depending on  $x$  and  $\tilde{x}$ , the constraint  $\Delta = 0$  and the massless fields: [HZ §1, ll. 100–350]. Mode expansions, zero modes, the operators  $\square$  and  $\Delta$ , rectangular tori: [HZ §2.2, ll. 573–1030].

- Null charges, the projector  $[[\cdot]]$ , failure of the Jacobi identity, large and small spaces, totally null subspaces: [HZ §5.1, ll. 3644–3858; §5.3, ll. 3920–3967].
- T-duality basics, the generalized momentum and level matching in  $O(D, D)$  form: [AMN §3.1, ll. 494–660]. Double space, the strong constraint, the supergravity frame and the caveat about level matching: [AMN §3.3, ll. 919–1084].
- The dual coordinate  $X_L - X_R$  on the worldsheet: [Tong §8.3, ll. 11626–11683].
- In this book: momentum and winding in Chapter 7; the group  $O(d, d; \mathbb{Z})$  and its generators in Chapter 11; the metric  $\mathcal{H}$  on the doubled space in Chapter 15; the gauge algebra whose closure requires the strong constraint in Chapter 16; the action in Chapter 17; the architecture of the Lean library in Chapter 35.

# Chapter 15

## The Generalized Metric

A closed string on a torus  $T^d$  sees a metric  $G$  and a two-form  $B$ , and the duality group of Chapter 11 acts on the pair  $(G, B)$  by a complicated non-linear rule, of which the Buscher rules of Chapter 9 are a sample. This chapter answers a practical question: *is there a single object, built from  $G$  and  $B$ , on which every duality acts linearly?* There is. It is the  $2d \times 2d$  matrix

$$\mathcal{H}(G, B) = \begin{pmatrix} G - BG^{-1}B & BG^{-1} \\ -G^{-1}B & G^{-1} \end{pmatrix},$$

the *generalized metric*. It met us briefly in Chapter 10 as the matrix of the zero-mode Hamiltonian. Here we take it apart. We derive it again, by a route that produces a factorized form from which every property can be read off; we prove by hand that  $\mathcal{H}$  is symmetric, positive definite and an element of  $O(d, d; \mathbb{R})$ ; we show how a  $B$ -shift, a change of lattice basis, and the inversion dualities act on it; we prove that the linear action on  $\mathcal{H}$  is the fractional linear action  $E \mapsto (aE + b)(cE + d)^{-1}$  on the background matrix  $E = G + B$ ; and we work one example with numbers from beginning to end. A student should care because the rest of Part III is written in this language: double field theory (Chapters 14, 16 and 17) is what one obtains by letting  $\mathcal{H}$  depend on the doubled coordinates and asking for an action principle. Every algebraic identity proved in this chapter has a kernel-checked counterpart in Lean, valid for every  $d$  over the real numbers, and Section 15.8 states exactly what those theorems say.

### 15.1 From the zero-mode Hamiltonian to $\mathcal{H}$

#### 15.1.1 Setting and conventions

We use the conventions of Chapter 10. The torus coordinates  $X^i$ ,  $i = 1, \dots, d$ , are dimensionless with period  $2\pi$ ; the constant background consists of a symmetric positive definite matrix  $G_{ij}$ , measured in units of  $\alpha'$  (for a circle of radius  $R$ ,  $G_{11} = R^2/\alpha'$ ), and an antisymmetric matrix  $B_{ij}$ ; the worldsheet action is

$$S = \frac{1}{4\pi} \int d\tau \int_0^{2\pi} d\sigma \left( G_{ij} (\dot{X}^i \dot{X}^j - X'^i X'^j) + 2B_{ij} \dot{X}^i X'^j \right), \quad (15.1)$$

which is the conformal-gauge form of [GPR §2.4, eq. (2.4.1), ll. 1031–1053] for a flat worldsheet; the overall sign of the  $B$ -term is a convention, and we fix it so that the canonical momentum (15.4) below agrees with GPR. The combination

$$E = G + B, \quad G = \frac{1}{2}(E + E^T), \quad B = \frac{1}{2}(E - E^T), \quad (15.2)$$

is the *background matrix* [GPR §2.4, eq. (2.4.5), ll. 1090–1097]: a general real  $d \times d$  matrix whose symmetric part is positive definite, hence  $d^2$  real parameters.

### 15.1.2 The zero modes as a particle in a magnetic field

The full canonical analysis was carried out in Chapter 10. For the present purpose it is enough, and more transparent, to keep only the zero modes. Insert  $X^i(\tau, \sigma) = x^i(\tau) + m^i \sigma$  into (15.1), where the integers  $m^i$  are the winding numbers. The  $\sigma$ -integral gives a factor  $2\pi$ , and we are left with the Lagrangian of a particle on the torus,

$$L_0 = \frac{1}{2} \dot{x}^\top G \dot{x} - \frac{1}{2} m^\top G m + \dot{x}^\top B m. \quad (15.3)$$

The first term is kinetic energy. The second comes from the stretching: a string wound  $m$  times has length  $2\pi\sqrt{\alpha' m^\top G m}$  and rest mass tension times length,  $\sqrt{m^\top G m/\alpha'}$ , and  $H_0$  measures the mass *squared* in units of  $1/\alpha'$  (see (15.15)). The third term has the form  $\dot{x} \cdot A$  with a constant “vector potential”  $A_i = B_{ij}m^j$ : a winding string feels  $B$  the way a charge on a ring feels an enclosed flux. The canonical momentum is

$$p = \frac{\partial L_0}{\partial \dot{x}} = G \dot{x} + B m \quad \Longleftrightarrow \quad \dot{x} = G^{-1}(p - B m), \quad (15.4)$$

which is the zero mode of [GPR §2.4, eq. (2.4.7), ll. 1105–1110]. Because  $p_i$  generates translations of a coordinate of period  $2\pi$ , single-valuedness of the wave function  $e^{ip_i x^i}$  forces  $p_i = n_i \in \mathbb{Z}$  [GPR §2.4, eq. (2.4.8), ll. 1110–1115]. It is the canonical momentum that is quantized, not the velocity. The Hamiltonian is the Legendre transform,  $H_0 = p^\top \dot{x} - L_0 = \frac{1}{2} \dot{x}^\top G \dot{x} + \frac{1}{2} m^\top G m$ ; the  $B$ -term cancels, because a term linear in velocities carries no energy. Expressed through the momentum with (15.4),

$$\boxed{H_0 = \frac{1}{2} (n - B m)^\top G^{-1} (n - B m) + \frac{1}{2} m^\top G m} \quad (15.5)$$

This is the most useful form of the zero-mode energy, and we shall refer to it as the *completed square*. It shows at a glance that  $H_0 \geq 0$ , that  $B$  enters only through the shifted momentum  $n - B m$ , and that for  $B = 0$  momentum and winding decouple.

Expanding the square, and using  $B^\top = -B$  in the form  $(B m)^\top G^{-1} (B m) = -m^\top B G^{-1} B m$  and  $-n^\top G^{-1} B m = +m^\top B G^{-1} n$  (a number equals its transpose), we recover

$$H_0 = \frac{1}{2} [n^\top G^{-1} n + m^\top (G - B G^{-1} B) m + 2 m^\top B G^{-1} n] \quad (15.6)$$

[GPR §2.4, eq. (2.4.11), ll. 1183–1191].

**Definition 15.1** (Generalized metric). Let  $G$  be an invertible real  $d \times d$  matrix and  $B$  a real  $d \times d$  matrix. The *generalized metric* is the real  $2d \times 2d$  block matrix

$$\mathcal{H}(G, B) = \begin{pmatrix} G - B G^{-1} B & B G^{-1} \\ -G^{-1} B & G^{-1} \end{pmatrix}. \quad (15.7)$$

A *background* is a pair  $(G, B)$  with  $G$  symmetric positive definite and  $B$  antisymmetric. With the *doubled charge vector*  $Z = (m, n) \in \mathbb{Z}^{2d}$ , windings first, (15.6) reads  $H_0 = \frac{1}{2} Z^\top \mathcal{H}(G, B) Z$ .

This is the matrix  $M(E)$  of [GPR §2.4, eqs. (2.4.17)–(2.4.18), ll. 1267–1290], the matrix  $\mathcal{H}(E)$  of [HZ §2.2, eq. (2.17), ll. 655–665], where it appears as the Hamiltonian density in the variables  $(X', 2\pi P)$ , and the matrix called  $\mathcal{H}^{MN}$  in [HHZ §2, eq. (2.7), ll. 452–457]. The definition deliberately asks nothing of  $B$  and only invertibility of  $G$ : this is how the Lean definition `genMetric` is set up, and we shall keep track of which property needs which hypothesis.

### 15.1.3 The factorized form

The completed square (15.5) is a matrix identity in disguise. The map  $(m, n) \mapsto (m, n - Bm)$  is linear, with matrix  $\begin{pmatrix} \mathbf{1} & 0 \\ -B & \mathbf{1} \end{pmatrix}$ , and the quadratic form  $m^\top G m + n^\top G^{-1} n$  has matrix  $\text{diag}(G, G^{-1})$ . So:

**Proposition 15.2** (Factorization). *For every invertible  $G$  and every  $B$ ,*

$$\mathcal{H}(G, B) = \underbrace{\begin{pmatrix} \mathbf{1} & B \\ 0 & \mathbf{1} \end{pmatrix}}_{U_B} \begin{pmatrix} G & 0 \\ 0 & G^{-1} \end{pmatrix} \underbrace{\begin{pmatrix} \mathbf{1} & 0 \\ -B & \mathbf{1} \end{pmatrix}}_{L_B}. \quad (15.8)$$

If  $B$  is antisymmetric, then  $L_B = U_B^\top$  and  $\mathcal{H}(G, B) = U_B \mathcal{H}(G, 0) U_B^\top$ .

*Proof.* Multiply from the right:  $\text{diag}(G, G^{-1}) L_B = \begin{pmatrix} G & 0 \\ -G^{-1}B & G^{-1} \end{pmatrix}$ . Then

$$U_B \begin{pmatrix} G & 0 \\ -G^{-1}B & G^{-1} \end{pmatrix} = \begin{pmatrix} G + B(-G^{-1}B) & BG^{-1} \\ -G^{-1}B & G^{-1} \end{pmatrix} = \mathcal{H}(G, B).$$

No property of  $B$  was used. Finally  $U_B^\top = \begin{pmatrix} \mathbf{1} & 0 \\ B^\top & \mathbf{1} \end{pmatrix}$ , which equals  $L_B$  exactly when  $B^\top = -B$ .  $\square$

The three-factor form is [HHZ §2, eq. (2.10), ll. 474–487]. It says that every background is obtained from a background without  $B$ -field by “switching on”  $B$  with the unipotent matrix  $U_B$ . We symbolically checked (15.8) with sympy for  $d = 2$  and a generic (not antisymmetric)  $B$ .

#### Common pitfall: four conventions for one matrix

Two independent choices produce four versions of (15.7) in the literature. (i) *Order of the doubled vector.* We, GPR and HZ list windings first,  $Z = (m, n)$ , equivalently  $(X', 2\pi P)$ , so that  $G$  sits in the upper left block. HHZ and AMN order the doubled coordinates as  $X^M = (\tilde{x}_i, x^i)$  and the momenta as  $P^M = (\tilde{p}^i, p_i)$  and write the metric with lower indices,

$$\mathcal{H}_{MN} = \begin{pmatrix} g^{ij} & -g^{ik}b_{kj} \\ b_{ik}g^{kj} & g_{ij} - b_{ik}g^{kl}b_{lj} \end{pmatrix}$$

[HHZ §1, eq. (1.9), ll. 256–262 and footnote 3, ll. 314–316; AMN §3.1, eq. (3.8), ll. 640–651]. This is  $\eta \mathcal{H}(G, B) \eta$  in our notation: rows and columns are swapped in blocks. By Theorem 15.4 below it is also  $\mathcal{H}(G, B)^{-1}$ , which is why HHZ call (15.7) the metric “with upper indices”  $\mathcal{H}^{MN}$  and (15.7)’s inverse the generalized metric proper [HHZ §2, eqs. (2.22)–(2.25), ll. 620–660]. (ii) *Sign of  $B$ .* Reversing the orientation of the worldsheet,  $\sigma \rightarrow -\sigma$ , sends  $B \rightarrow -B$  [GPR §2.4, ll. 1423–1425] and reverses the sign of both off-diagonal blocks. Before comparing a formula with another text or with a Lean file, find the order of the charge vector and the block that contains  $G^{-1}$ . The Lean module described in Section 15.8 uses (15.7) verbatim, with the winding block first.

## 15.2 The algebraic properties of $\mathcal{H}$

Throughout,  $\eta = \begin{pmatrix} 0 & \mathbf{1} \\ \mathbf{1} & 0 \end{pmatrix}$  is the  $O(d, d)$  form of Chapters 10 and 11,  $\eta^2 = \mathbf{1}$ , and  $O(d, d; \mathbb{R})$  is the group of real matrices  $g$  with  $g^\top \eta g = \eta$ . On charge vectors  $\frac{1}{2} Z^\top \eta Z = m \cdot n$ , the quantity fixed by level matching.

### 15.2.1 Symmetry

**Proposition 15.3.** *If  $G$  is symmetric and invertible and  $B$  is antisymmetric, then  $\mathcal{H}(G, B)^\top = \mathcal{H}(G, B)$ .*

*Proof.* The transpose of a block matrix transposes each block and exchanges the off-diagonal ones. First  $(G^{-1})^\top = (G^\top)^{-1} = G^{-1}$ . Then, block by block,

$$\begin{aligned} (G - BG^{-1}B)^\top &= G^\top - B^\top(G^{-1})^\top B^\top = G - (-B)G^{-1}(-B) = G - BG^{-1}B, \\ (-G^{-1}B)^\top &= -B^\top(G^{-1})^\top = BG^{-1}, \quad (BG^{-1})^\top = G^{-1}B^\top = -G^{-1}B, \end{aligned}$$

and the lower right block is  $G^{-1}$ , already shown symmetric.  $\square$

Both hypotheses are needed: for a generic  $B$  the two off-diagonal blocks are not transposes of each other (we confirmed with sympy that the identity fails for generic  $B$  at  $d = 2$ ). Physically, symmetry costs nothing, since only the symmetric part of a matrix contributes to a quadratic form  $Z^\top \mathcal{H} Z$ ; the content of the proposition is that the particular representative (15.7) is already symmetric.

### 15.2.2 $\mathcal{H}$ is an element of $O(d, d; \mathbb{R})$

**Theorem 15.4** (The  $O(d, d)$  constraint [A]). *For every invertible  $G$  and every  $B$ ,*

$$\mathcal{H} \eta \mathcal{H} = \eta, \quad \text{equivalently} \quad (\eta \mathcal{H})^2 = \mathbf{1}, \quad \text{equivalently} \quad \mathcal{H}^{-1} = \eta \mathcal{H} \eta. \quad (15.9)$$

*If moreover  $\mathcal{H}$  is symmetric (Proposition 15.3), then  $\mathcal{H}^\top \eta \mathcal{H} = \eta$ , that is,  $\mathcal{H} \in O(d, d; \mathbb{R})$ .*

The literature states this for backgrounds: “the matrix  $\mathcal{H}(E)$  satisfies the constraint  $\mathcal{H}^{-1} = \eta \mathcal{H} \eta$ ” [HZ §2.2, l. 665];  $\mathcal{H} \eta \mathcal{H} = \eta^{-1}$  [HHZ §2, eqs. (2.8)–(2.9), ll. 458–466]; “this is an  $O(D, D)$  element” [AMN §3.3, eq. (3.23), ll. 944–950]. That antisymmetry of  $B$  is not needed is an observation made while formalizing; it is the statement `etaR_genMetric_sq`. We give two proofs, since each teaches something.

*First proof: from the factorization.* The three equivalent forms follow from one another by multiplying with  $\eta = \eta^{-1}$ . We prove the first. Two one-line computations, valid for any  $B$  and any invertible  $G$ :

$$L_B \eta U_B = \begin{pmatrix} 0 & \mathbf{1} \\ \mathbf{1} & -B \end{pmatrix} \begin{pmatrix} \mathbf{1} & B \\ 0 & \mathbf{1} \end{pmatrix} = \begin{pmatrix} 0 & \mathbf{1} \\ \mathbf{1} & B - B \end{pmatrix} = \eta, \quad U_B \eta L_B = \eta, \quad (15.10)$$

$$\begin{pmatrix} G & 0 \\ 0 & G^{-1} \end{pmatrix} \eta \begin{pmatrix} G & 0 \\ 0 & G^{-1} \end{pmatrix} = \begin{pmatrix} 0 & GG^{-1} \\ G^{-1}G & 0 \end{pmatrix} = \eta. \quad (15.11)$$

(The second identity in (15.10) is checked in the same way:  $U_B \eta = \begin{pmatrix} B & \mathbf{1} \\ \mathbf{1} & 0 \end{pmatrix}$ , and multiplying by  $L_B$  replaces the upper left block by  $B - B = 0$ .) With  $D = \text{diag}(G, G^{-1})$  and (15.8),

$$\mathcal{H} \eta \mathcal{H} = U_B D (L_B \eta U_B) D L_B = U_B (D \eta D) L_B = U_B \eta L_B = \eta.$$

For the last claim,  $\mathcal{H}^\top \eta \mathcal{H} = \mathcal{H} \eta \mathcal{H} = \eta$  when  $\mathcal{H}^\top = \mathcal{H}$ .  $\square$

*Second proof: four blocks.* This is the computation the Lean proof performs, and a useful exercise in handling non-commuting blocks. First

$$\eta \mathcal{H} = \begin{pmatrix} -G^{-1}B & G^{-1} \\ G - BG^{-1}B & BG^{-1} \end{pmatrix} \equiv \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix},$$

since multiplying by  $\eta$  from the left exchanges the two block rows. The square of a block matrix is  $\begin{pmatrix} \alpha^2 + \beta\gamma & \alpha\beta + \beta\delta \\ \gamma\alpha + \delta\gamma & \gamma\beta + \delta^2 \end{pmatrix}$ , and

$$\begin{aligned} \alpha^2 + \beta\gamma &= G^{-1}BG^{-1}B + G^{-1}(G - BG^{-1}B) = \mathbf{1}, \\ \alpha\beta + \beta\delta &= -G^{-1}BG^{-1} + G^{-1}BG^{-1} = 0, \\ \gamma\alpha + \delta\gamma &= -(G - BG^{-1}B)G^{-1}B + BG^{-1}(G - BG^{-1}B) \\ &= -B + BG^{-1}BG^{-1}B + B - BG^{-1}BG^{-1}B = 0, \\ \gamma\beta + \delta^2 &= (G - BG^{-1}B)G^{-1} + BG^{-1}BG^{-1} = \mathbf{1}. \end{aligned}$$

The only facts used are  $G^{-1}G = GG^{-1} = \mathbf{1}$  and associativity.  $\square$

**Corollary 15.5.** *Let  $(G, B)$  be a background. Then (i)  $\det \mathcal{H} = 1$ ; (ii) if  $\lambda$  is an eigenvalue of  $\mathcal{H}$  then so is  $1/\lambda$ , with the same multiplicity; (iii)  $\text{tr } \mathcal{H} = \text{tr } G + \text{tr } G^{-1} + \text{tr } (B^\top G^{-1}B) \geq 2d$ , with equality if and only if  $G = \mathbf{1}$  and  $B = 0$ .*

*Proof.* (i) From (15.8):  $\det U_B = \det L_B = 1$  and  $\det \text{diag}(G, G^{-1}) = \det G \det G^{-1} = 1$ . (ii)  $\mathcal{H}^{-1} = \eta \mathcal{H} \eta = \eta \mathcal{H} \eta^{-1}$  is similar to  $\mathcal{H}$ , so  $\mathcal{H}$  and  $\mathcal{H}^{-1}$  have the same characteristic polynomial. (iii) The trace of (15.7) is  $\text{tr } G - \text{tr}(BG^{-1}B) + \text{tr } G^{-1}$  and  $-B = B^\top$ . If  $g_1, \dots, g_d > 0$  are the eigenvalues of  $G$ , then  $\text{tr } G + \text{tr } G^{-1} = \sum_a (g_a + 1/g_a) \geq 2d$  because  $x + 1/x - 2 = (\sqrt{x} - 1/\sqrt{x})^2 \geq 0$ , with equality only for all  $g_a = 1$ ; and  $\text{tr}(B^\top G^{-1}B) = \sum_j (Be_j)^\top G^{-1}(Be_j) \geq 0$  since  $G^{-1}$  is positive definite, with equality only if every column  $Be_j$  vanishes.  $\square$

Part (iii) with  $B = 0$  is the trace form of the dual-scale inequality  $R + \alpha'/R \geq 2\sqrt{\alpha'}$  of Chapter 31; in Lean it is `dualScale_ge` [A]. The extension to  $B \neq 0$  just given is proved on this page and is not formalized.

### 15.2.3 Positive definiteness

**Proposition 15.6.** *For a background  $(G, B)$  the matrix  $\mathcal{H}(G, B)$  is positive definite: for every real  $Z = (m, n) \neq 0$ ,*

$$Z^\top \mathcal{H}(G, B) Z = m^\top G m + (n - Bm)^\top G^{-1} (n - Bm) > 0. \quad (15.12)$$

*Proof.* By Proposition 15.2,  $Z^\top \mathcal{H} Z = (L_B Z)^\top \text{diag}(G, G^{-1})(L_B Z)$  with  $L_B Z = (m, n - Bm)$ ; this is (15.12), the completed square (15.5) read backwards. Both terms are non-negative since  $G$  and  $G^{-1}$  are positive definite. If the sum vanishes then  $m = 0$ , and then  $n - Bm = n = 0$ .  $\square$

“The metric  $\mathcal{H}_{MN} \dots$  is positive definite if  $g_{ij}$  is” [HHZ §1, ll. 284–285]. The doubled space thus carries two different metrics: the constant  $\eta$  of signature  $(d, d)$ , which knows nothing about the background and measures the level-matching quantity  $m \cdot n$ , and the positive definite  $\mathcal{H}$ , which contains the background and measures the energy. The constraint (15.9) ties them together.

**Common pitfall: “ $\mathcal{H}$  is orthogonal, so it cannot be positive definite”**

In the compact group  $O(2d)$  the only symmetric positive definite element is the identity.  $O(d, d; \mathbb{R})$  is not compact, and it contains a  $d^2$ -dimensional family of symmetric positive definite elements, namely the matrices  $\mathcal{H}(G, B)$ . For  $d = 1$  they are  $\text{diag}(R^2/\alpha', \alpha'/R^2)$ : a *boost* in the light-cone directions of  $\eta$ , with rapidity  $\log(R^2/\alpha')$ . Note also that `genMetric_symm` and `etaR_genMetric_sq` do not include positivity; in the Lean develop-

ment positive definiteness of  $G$  enters only in `dualScale_ge`.

### 15.2.4 The involution $\mathcal{H}\eta$ and the left–right split

By (15.9) the matrix

$$\mathcal{S} = \mathcal{H}\eta, \quad \mathcal{S}^2 = \mathcal{H}\eta\mathcal{H}\eta = \eta\eta = \mathbf{1}, \quad (15.13)$$

is an involution [HHZ §1, eqs. (1.12)–(1.14), ll. 296–308; §2, eq. (2.6)]. An involution is determined by its two eigenspaces, so the background must be encoded in them.

**Lemma 15.7** (Eigenspaces). *Let  $(G, B)$  be a background and  $E = G + B$ . The eigenspaces of  $\mathcal{S} = \mathcal{H}\eta$  are*

$$V_+ = \left\{ \begin{pmatrix} Eu \\ u \end{pmatrix} : u \in \mathbb{R}^d \right\} \text{ (eigenvalue } +1), \quad V_- = \left\{ \begin{pmatrix} -E^\top u \\ u \end{pmatrix} : u \in \mathbb{R}^d \right\} \text{ (eigenvalue } -1). \quad (15.14)$$

The form  $\eta$  is positive definite on  $V_+$ , negative definite on  $V_-$ , and  $V_-$  is the  $\eta$ -orthogonal complement of  $V_+$ .

*Proof.*  $\mathcal{S} \begin{pmatrix} Eu \\ u \end{pmatrix} = \mathcal{H} \begin{pmatrix} u \\ Eu \end{pmatrix}$ . The upper block is  $(G - BG^{-1}B)u + BG^{-1}(G + B)u = Gu + Bu = Eu$  and the lower block is  $-G^{-1}Bu + G^{-1}(G + B)u = u$ . For  $V_-$ , with  $E^\top = G - B$ :  $\mathcal{H} \begin{pmatrix} u \\ -E^\top u \end{pmatrix}$  has upper block  $(G - BG^{-1}B)u - BG^{-1}(G - B)u = (G - B)u = E^\top u$  and lower block  $-G^{-1}Bu - G^{-1}(G - B)u = -u$ , which is  $-\begin{pmatrix} -E^\top u \\ u \end{pmatrix}$ . Both spaces have dimension  $d$ , and they meet only in 0 (different eigenvalues), so they fill  $\mathbb{R}^{2d}$ . For  $v = \begin{pmatrix} Eu \\ u \end{pmatrix}$  we find  $v^\top \eta v = 2u^\top Eu = 2u^\top Gu > 0$  because the antisymmetric part drops out of a quadratic form; similarly  $v^\top \eta v = -2u^\top Gu < 0$  on  $V_-$ . Orthogonality:  $\begin{pmatrix} Eu \\ u \end{pmatrix}^\top \eta \begin{pmatrix} -E^\top u' \\ u' \end{pmatrix} = u^\top E^\top u' - u^\top E^\top u' = 0$ .  $\square$

In Chapter 10 the left- and right-moving momenta were found to be proportional to  $n + E^\top m$  and  $n - Em$  [GPR §2.4, eq. (2.4.12), ll. 1193–1198]. A charge vector  $Z = (m, n)$  has  $\eta Z = (n, m) \in V_-$  exactly when  $n = -E^\top m$ , that is, when its left-moving momentum vanishes; likewise  $\eta Z \in V_+$  means  $p_R = 0$ . (That  $\eta Z$  appears, and not  $Z$ , is index bookkeeping:  $\mathcal{S} = \mathcal{H}\eta$  first lowers the index of its argument with  $\eta$ . The matrix  $\eta\mathcal{H}$  used in Chapter 11 has the eigenspaces  $\eta V_\pm$ .) *Choosing a background is choosing how to split the doubled space into left-movers and right-movers*, and  $\mathcal{H}$  is the bookkeeping device for that split:  $\mathcal{H} = \mathcal{S}\eta$ , where  $\mathcal{S}$  is  $+1$  on  $V_+$  and  $-1$  on  $V_-$ .

## 15.3 The mass form and its covariance

### 15.3.1 The spectrum in terms of $\mathcal{H}$

Restoring the oscillators and the non-compact momenta (Chapter 10), the mass of a closed-string state in the uncompactified dimensions is

$$M^2 = \frac{1}{\alpha'} Z^\top \mathcal{H}(G, B) Z + \frac{2}{\alpha'} (N + \tilde{N} - 2), \quad N - \tilde{N} = \frac{1}{2} Z^\top \eta Z = m \cdot n, \quad (15.15)$$

where  $N, \tilde{N}$  are the oscillator levels. The zero-mode part is [GPR §2.4, eq. (2.4.18), ll. 1278–1290]. AMN write the same structure,  $M^2 = (N + \tilde{N} - 2) + P^\top \mathcal{H} P$  with  $N - \tilde{N} = \frac{1}{2} P^\top \eta P$ , in a review-level normalization that absorbs our relative factor 2 and with the momentum block first [AMN §3.1, eqs. (3.6)–(3.9), ll. 620–656]; we keep the convention of Chapter 10 and check it against the circle below. All dependence on the background sits in the quadratic form

$$\mu_{\mathcal{H}}(Z) = Z^\top \mathcal{H} Z, \quad (15.16)$$



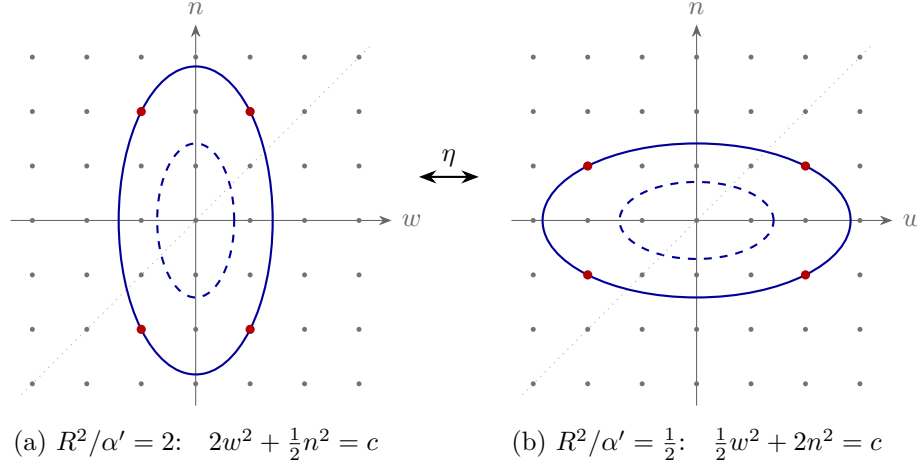


Figure 15.1: Level sets  $\mu_{\mathcal{H}}(Z) = c$  of the mass form for  $d = 1$  (solid:  $c = 4$ , dashed:  $c = 1$ ) in the plane of winding  $w$  and momentum  $n$ . The lattice  $\mathbb{Z}^2$  of charges is the same in both panels; only the ellipse depends on the background. The full duality  $\eta$  reflects the plane in the dotted diagonal; it maps the lattice to itself and the ellipses of  $R^2/\alpha' = 2$  to those of  $R^2/\alpha' = \frac{1}{2}$ . The four marked states with  $\mu = 4$  in (a) are mapped to the four marked states with  $\mu = 4$  in (b). Area is preserved because  $\det \mathcal{H} = 1$ .

which we call the *mass form*; this is the Lean definition `massForm`. The two metrics of the doubled space have now each found their job:  $\mathcal{H}$  gives the mass,  $\eta$  gives the level-matching constraint.

As a check on normalizations take  $d = 1$ ,  $G = R^2/\alpha'$ ,  $B = 0$ , and  $Z = (w, n)$ . Then  $\mathcal{H} = \text{diag}(R^2/\alpha', \alpha'/R^2)$  and

$$\frac{1}{\alpha'} \mu_{\mathcal{H}}(Z) = \frac{1}{\alpha'} \left( \frac{R^2}{\alpha'} w^2 + \frac{\alpha'}{R^2} n^2 \right) = \frac{n^2}{R^2} + \frac{w^2 R^2}{\alpha'^2}, \quad (15.17)$$

the circle formula of Chapter 7. In terms of the left- and right-moving momenta  $p_{L,R} = n/R \pm wR/\alpha'$  of Chapter 8 this is  $\frac{1}{2}(p_L^2 + p_R^2)$ , because the cross terms  $\pm 2nw/\alpha'$  cancel in the sum. Figure 15.1 shows the level sets of  $\mu_{\mathcal{H}}$ .

### 15.3.2 Covariance

**Proposition 15.8** (Covariance of the mass form **[A]**). *For all real  $2d \times 2d$  matrices  $g$ ,  $\mathcal{H}$  and all  $Z \in \mathbb{R}^{2d}$ ,*

$$\mu_{g^{\top} \mathcal{H} g}(Z) = \mu_{\mathcal{H}}(gZ). \quad (15.18)$$

*Proof.*  $Z^{\top} (g^{\top} \mathcal{H} g) Z = (gZ)^{\top} \mathcal{H} (gZ)$ , since  $(gZ)^{\top} = Z^{\top} g^{\top}$ . □

This is `massForm_covariant`. No hypothesis on  $g$  is needed, and none appears in the Lean statement. The group enters when we ask for more: if  $g \in O(d, d; \mathbb{Z})$  then  $Z \mapsto gZ$  is a bijection of the charge lattice that preserves  $Z^{\top} \eta Z$ , hence level matching, and the complete spectra (15.15) of the backgrounds  $\mathcal{H}$  and  $g^{\top} \mathcal{H} g$  coincide state by state. That argument is the subject of Chapter 11; what remains for this chapter is to show that the transformed matrix is again the generalized metric of a background and to identify that background.

**Common pitfall:  $g^\top \mathcal{H} g$  versus  $g \mathcal{H} g^\top$** 

Chapter 11 and `massForm_covariant` transform the background by  $\mathcal{H} \mapsto g^\top \mathcal{H} g$  and the charges by  $Z \mapsto gZ$ . GPR, HHZ and the Lean file `BShift.lean` use  $\mathcal{H} \mapsto g \mathcal{H} g^\top$  [GPR §2.4, eq. (2.4.19), ll. 1290–1293; HHZ §2, eq. (2.16), ll. 557–561]; the charges must then transform with the inverse transpose,

$$\mu_{g \mathcal{H} g^\top}(Z') = \mu_{\mathcal{H}}(Z) \quad \text{for} \quad Z' = (g^\top)^{-1} Z = \eta g \eta Z,$$

where the last form holds for  $g \in O(d, d; \mathbb{R})$ . The two conventions are exchanged by  $g \leftrightarrow g^\top$ , and the transpose of a group element is a group element [GPR §2.4, ll. 1243–1248], so the set of dual backgrounds is the same. In the remainder of this chapter we use  $\mathcal{H} \mapsto g \mathcal{H} g^\top$ , because in this convention the matrix that shifts  $B$  is upper triangular like  $U_B$ , and the action on  $E$  takes its customary form.

## 15.4 How dualities act on $\mathcal{H}$

We now conjugate  $\mathcal{H}(G, B)$  with the elements of  $O(d, d; \mathbb{R})$  described in Chapter 11 and read off the new background. The factorization (15.8) reduces each computation to the multiplication of triangular or diagonal block matrices.

### 15.4.1 The $B$ -shift acts by conjugation

**Theorem 15.9** ( $B$ -shift [A]). *Let  $G$  be invertible,  $B$  arbitrary and  $\Theta$  antisymmetric. With  $g_\Theta = \begin{pmatrix} 1 & \Theta \\ 0 & 1 \end{pmatrix}$ ,*

$$g_\Theta \mathcal{H}(G, B) g_\Theta^\top = \mathcal{H}(G, B + \Theta). \quad (15.19)$$

*Proof.* Matrices of the form  $U_X$  multiply by adding their arguments:  $g_\Theta U_B = U_\Theta U_B = U_{B+\Theta}$ . On the other side,

$$L_B g_\Theta^\top = \begin{pmatrix} 1 & 0 \\ -B & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ \Theta^\top & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -B + \Theta^\top & 1 \end{pmatrix} = L_{B+\Theta} \quad \text{because } \Theta^\top = -\Theta.$$

Hence  $g_\Theta \mathcal{H}(G, B) g_\Theta^\top = U_{B+\Theta} \text{diag}(G, G^{-1}) L_{B+\Theta} = \mathcal{H}(G, B + \Theta)$  by Proposition 15.2.  $\square$

The proof shows where the hypothesis is used: only antisymmetry of  $\Theta$  matters. The Lean theorem `genMetric_bshift`, to which the badge [A] refers, is stated for a background ( $G$  symmetric with invertible determinant,  $B$  antisymmetric); the slightly more general statement above is proved on this page only. For a symmetric  $\Theta \neq 0$  the identity fails (sympy,  $d = 2$ ), and so do the three other candidates  $g^\top \mathcal{H} g$  with  $\pm \Theta$  and  $g \mathcal{H} g^\top$  with  $-\Theta$ ; the convention is forced. The matrix  $g_\Theta$  belongs to  $O(d, d; \mathbb{R})$  (the condition  $g^\top \eta g = \eta$  reduces to  $\Theta + \Theta^\top = 0$ ; in Lean, `thetaShiftR_preserves_eta`), and by the preceding pitfall box the charges transform as

$$Z' = (g_\Theta^\top)^{-1} Z = \begin{pmatrix} 1 & 0 \\ \Theta & 1 \end{pmatrix} \begin{pmatrix} m \\ n \end{pmatrix} = \begin{pmatrix} m \\ n + \Theta m \end{pmatrix}, \quad n' - (B + \Theta)m = n - Bm. \quad (15.20)$$

The shifted momentum  $n - Bm$  of the completed square is unchanged, and so is the energy.

**Physical picture: when is a  $B$ -shift a symmetry?**

Equation (15.19) holds for every real antisymmetric  $\Theta$ : it is a statement about how  $O(d, d; \mathbb{R})$  moves through the space of backgrounds, and a generic real shift produces a

*different* theory with a different spectrum. The shift is a symmetry only if (15.20) maps the charge lattice to itself, which requires  $\Theta m \in \mathbb{Z}^d$  for all integer  $m$ , that is, integer entries. Then  $g_\Theta \in O(d, d; \mathbb{Z})$ . The worldsheet reason is that the constant  $B$ -term is a total derivative, and an integer shift changes the action by a multiple of  $2\pi$  [GPR §2.4, eq. (2.4.25) and following, ll. 1351–1371]. So  $B_{ij}$  is a periodic variable, an angle, exactly like the flux through a ring in the Aharonov–Bohm effect. That the  $B$ -shift is a symmetry of the path integral is [L]; what is [A] is the matrix identity (15.19).

### 15.4.2 Change of lattice basis

For  $A \in GL(d, \mathbb{R})$  put  $g_A = \text{diag}(A, (A^\top)^{-1})$  [GPR §2.4, eq. (2.4.26), ll. 1373–1381]. It preserves  $\eta$ , because  $A^\top(A^\top)^{-1} = \mathbf{1}$ . One checks  $g_A U_B = \begin{pmatrix} A & AB \\ 0 & A^{-\top} \end{pmatrix} = U_{ABA^\top} g_A$ , and transposing (for antisymmetric  $B$ ),  $L_B g_A^\top = g_A^\top L_{ABA^\top}$ . Therefore

$$g_A \mathcal{H}(G, B) g_A^\top = U_{ABA^\top} g_A \begin{pmatrix} G & 0 \\ 0 & G^{-1} \end{pmatrix} g_A^\top L_{ABA^\top} = \mathcal{H}(AGA^\top, ABA^\top), \quad (15.21)$$

since  $g_A \text{diag}(G, G^{-1}) g_A^\top = \text{diag}(AGA^\top, (AGA^\top)^{-1})$ . Both  $G$  and  $B$  transform as tensors,  $E' = AEA^\top$ . For  $A \in GL(d, \mathbb{Z})$  this is a relabelling of the cycles of the same torus. This identity has no Lean counterpart in the modules described in this chapter (Exercise 15.9).

### 15.4.3 Inversion: T-duality proper

**Theorem 15.10** (Full duality at  $B = 0$  [A]). *For every invertible  $G$ ,*

$$\eta \mathcal{H}(G, 0) \eta = \mathcal{H}(G^{-1}, 0). \quad (15.22)$$

*Proof.*  $\mathcal{H}(G, 0) = \text{diag}(G, G^{-1})$ . Left multiplication by  $\eta$  exchanges the block rows, right multiplication the block columns, so  $\eta \text{diag}(G, G^{-1}) \eta = \text{diag}(G^{-1}, G)$ , and this is  $\mathcal{H}(G^{-1}, 0)$  because  $(G^{-1})^{-1} = G$ .  $\square$

For the circle this is  $R^2/\alpha' \rightarrow \alpha'/R^2$ , that is  $R \rightarrow \alpha'/R$ , together with  $Z = (w, n) \mapsto \eta Z = (n, w)$ : the T-duality of Chapter 8. For a rectangular torus  $G = \text{diag}(R_1^2, \dots, R_d^2)/\alpha'$  all radii are inverted at once. AMN call  $\eta$  the “inversion metric” for this reason [AMN §3.1, ll. 763–767]. One can also invert a single radius. The *factorized duality*

$$D_k = \begin{pmatrix} \mathbf{1} - e_k & e_k \\ e_k & \mathbf{1} - e_k \end{pmatrix}, \quad (e_k)_{ij} = \delta_{ik} \delta_{jk}, \quad (15.23)$$

[GPR §2.4, eq. (2.4.29), ll. 1399–1421] exchanges  $m^k \leftrightarrow n_k$  and leaves the other charges alone. For diagonal  $G$  and  $B = 0$  the matrix  $\mathcal{H}$  is diagonal, and conjugation by the permutation matrix  $D_k$  exchanges its  $k$ -th and  $(d+k)$ -th diagonal entries:  $R_k^2/\alpha' \leftrightarrow \alpha'/R_k^2$ , the other radii unchanged. Since  $D_k^2 = \mathbf{1}$ ,  $D_k = D_k^\top$  and  $D_1 D_2 \cdots D_d = \eta$ , the full duality is the product of the  $d$  factorized ones.

With  $B \neq 0$ , or  $G$  not diagonal,  $\eta \mathcal{H}(G, B) \eta$  and  $D_k \mathcal{H}(G, B) D_k$  are still generalized metrics, but reading off the new  $G$  and  $B$  block by block becomes awkward. The background matrix  $E$  does it in one line.

## 15.5 The background matrix and fractional linear transformations

**Theorem 15.11** (Action on  $E$  [L]). *Let  $(G, B)$  be a background,  $E = G + B$ , and  $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in O(d, d; \mathbb{R})$ . Then  $cE + d$  is invertible, the symmetric part  $G'$  of*

$$E' = g(E) := (aE + b)(cE + d)^{-1} \quad (15.24)$$

*is positive definite, and with  $B'$  the antisymmetric part of  $E'$ ,*

$$g \mathcal{H}(G, B) g^\top = \mathcal{H}(G', B'). \quad (15.25)$$

This is [GPR §2.4, eqs. (2.4.19)–(2.4.24), ll. 1290–1334] and [HHZ §1, eq. (1.3), ll. 143–149; §2, eqs. (2.11)–(2.16), ll. 495–561]. Those texts argue through a vielbein (see Section 15.5.1); we give a proof through the involution  $\mathcal{S}$  of (15.13), which needs no choice of frame.

*Proof. Step 1:  $\mathcal{S}$  transforms by conjugation.* From  $g^\top \eta g = \eta$  we get  $g^\top \eta = \eta g^{-1}$ . Put  $\mathcal{H}' = g \mathcal{H} g^\top$  and  $\mathcal{S}' = \mathcal{H}' \eta$ . Then  $\mathcal{S}' = g \mathcal{H} g^\top \eta = g \mathcal{H} \eta g^{-1} = g \mathcal{S} g^{-1}$ . Hence  $\mathcal{S}'$  is an involution with eigenspaces  $V'_\pm = g V_\pm$ .

*Step 2:  $gV_+$  is a graph.* By Lemma 15.7,

$$g \begin{pmatrix} Eu \\ u \end{pmatrix} = \begin{pmatrix} (aE + b)u \\ (cE + d)u \end{pmatrix}. \quad (15.26)$$

Suppose  $(cE + d)u = 0$ . Then  $v' = g \begin{pmatrix} Eu \\ u \end{pmatrix}$  has vanishing lower block, so  $v'^\top \eta v' = 0$ . But  $g$  preserves  $\eta$ , so  $v'^\top \eta v' = \begin{pmatrix} Eu \\ u \end{pmatrix}^\top \eta \begin{pmatrix} Eu \\ u \end{pmatrix} = 2u^\top G u$ , which vanishes only for  $u = 0$ . Thus  $cE + d$  is invertible. Writing  $u' = (cE + d)u$ , (15.26) becomes  $\begin{pmatrix} E'u' \\ u' \end{pmatrix}$  with  $E'$  as in (15.24):  $V'_+ = \{\begin{pmatrix} E'u' \\ u' \end{pmatrix}\}$ . The same computation gives  $2u'^\top G' u' = v'^\top \eta v' = 2u^\top G u > 0$  for  $u' \neq 0$ , so  $G'$  is positive definite and  $(G', B')$  is a background.

*Step 3: the two involutions agree.* Let  $\mathcal{S}'' = \mathcal{H}(G', B')\eta$ . By Lemma 15.7 its  $+1$  eigenspace is  $\{\begin{pmatrix} E'u' \\ u' \end{pmatrix}\} = V'_+$ , and its  $-1$  eigenspace is the  $\eta$ -orthogonal complement of  $V'_+$ . The  $-1$  eigenspace of  $\mathcal{S}'$  is  $gV_- = g(V_+^\perp) = (gV_+)^{\perp} = V_+'^{\perp}$ , because  $g$  preserves  $\eta$ . Two involutions with the same eigenspaces are equal, so  $\mathcal{H}' = \mathcal{S}' \eta = \mathcal{S}'' \eta = \mathcal{H}(G', B')$ .  $\square$

Since the proof realizes  $g(E)$  through the linear action of  $g$  on subspaces, the group law  $g_1(g_2(E)) = (g_1 g_2)(E)$  is automatic. As a symbolic check we verified (15.25) with sympy for  $d = 2$ , symbolic  $G$  and  $B$ , and  $g$  a product of seven generators. The theorem is Tier L: the Lean modules of this chapter prove the special cases  $g = g_\Theta$  and  $g = \eta$  at  $B = 0$ , not the general statement.

### Physical picture: homogeneous coordinates

The modular group acts on the upper half plane by  $\tau \mapsto (a\tau + b)/(c\tau + d)$  because it acts *linearly* on the column  $\begin{pmatrix} \tau \\ 1 \end{pmatrix}$ , and columns that differ by a factor describe the same point. Equation (15.26) is the same mechanism with matrices: the  $2d \times d$  matrix  $\begin{pmatrix} E \\ 1 \end{pmatrix}$  spans the subspace  $V_+$ , the group acts linearly on it, and right multiplication by  $(cE + d)^{-1}$  restores the normalization of the lower block. The condition that replaces  $\text{Im } \tau > 0$  is positivity of the symmetric part of  $E$ . For  $d = 2$  this analogy becomes an identity:  $O(2, 2)$  contains two commuting copies of  $SL(2)$  acting on the complex structure  $\tau$  and on  $\rho = B_{12} + i\sqrt{\det G}$  (Chapter 12).

**The three families again.** With (15.24) the results of Section 15.4 take one line each:

$$\begin{aligned} g_\Theta : \quad E' &= (E + \Theta) \mathbf{1}^{-1} = E + \Theta, & G' &= G, \quad B' = B + \Theta; \\ g_A : \quad E' &= AE((A^\top)^{-1})^{-1} = AEA^\top; \\ \eta : \quad E' &= (0 \cdot E + \mathbf{1})(\mathbf{1} \cdot E + 0)^{-1} = E^{-1}. \end{aligned} \quad (15.27)$$

The last line is the general form of Theorem 15.10. To split  $E^{-1}$  into its parts, write  $E^{-1} \pm E^{-\top} = E^{-1}(E^\top \pm E)E^{-\top}$ :

$$G' = E^{-1}GE^{-\top} = (G - BG^{-1}B)^{-1}, \quad B' = -E^{-1}BE^{-\top}, \quad (15.28)$$

where the second form of  $G'$  uses  $E^\top G^{-1}E = (G - B)G^{-1}(G + B) = G - BG^{-1}B$ . So the upper left block of  $\mathcal{H}$  has a meaning: *it is the inverse of the  $T$ -dual metric*. With  $B \neq 0$  the dual metric is not  $G^{-1}$ : for an invertible  $B$  much larger than  $G$  it is approximately  $-B^{-1}GB^{-1} = (B^{-1})^\top GB^{-1}$ , which tends to zero as  $B$  grows, whatever the size of  $G$ .

For the factorized duality (15.23),  $E' = ((\mathbf{1} - e_k)E + e_k)(e_k E + \mathbf{1} - e_k)^{-1}$ . For  $d = 2$ ,  $k = 1$  the matrix to invert is  $\begin{pmatrix} E_{11} & E_{12} \\ 0 & 1 \end{pmatrix}$ , and one finds

$$E' = \frac{1}{E_{11}} \begin{pmatrix} 1 & -E_{12} \\ E_{21} & E_{11}E_{22} - E_{21}E_{12} \end{pmatrix}, \quad (15.29)$$

which, separated into symmetric and antisymmetric parts, are the Buscher rules of Chapter 9:  $G'_{11} = 1/G_{11}$ ,  $G'_{12} = -B_{12}/G_{11}$ ,  $B'_{12} = -G_{12}/G_{11}$ ,  $G'_{22} = G_{22} - (G_{12}^2 - B_{12}^2)/G_{11}$ . Metric and  $B$ -field components with one leg along the dualized circle are exchanged. The signs of the mixed components are opposite to those written in Chapter 9; the two forms differ by the reflection  $A = \text{diag}(-1, 1)$  of the dual coordinate, a basis change (15.21), and describe the same physics. We checked (15.29) against  $D_1\mathcal{H}D_1$  with sympy.

### 15.5.1 Vielbeins and the coset

Choose a frame  $e$  with  $G = ee^\top$  (for instance the Cholesky factor). Then  $\text{diag}(G, G^{-1}) = \text{diag}(e, e^{-\top}) \text{diag}(e, e^{-\top})^\top$  and Proposition 15.2 becomes

$$\mathcal{H}(G, B) = g_E g_E^\top, \quad g_E = U_B \begin{pmatrix} e & 0 \\ 0 & e^{-\top} \end{pmatrix} = \begin{pmatrix} e & B e^{-\top} \\ 0 & e^{-\top} \end{pmatrix} \in O(d, d; \mathbb{R}), \quad (15.30)$$

[GPR §2.4, eqs. (2.4.20)–(2.4.23), ll. 1294–1325; HHZ §2, eqs. (2.11)–(2.15), ll. 495–534]. The matrix  $g_E$  maps the reference background  $E = \mathbf{1}$  to  $E$ :  $g_E(\mathbf{1}) = (e + B e^{-\top})e^\top = G + B$ . It is a generalized vielbein:  $\mathcal{H} = g_E \mathbf{1} g_E^\top$  is to  $g_E$  what  $G = e \mathbf{1} e^\top$  is to  $e$ . The frame is not unique;  $g_E \rightarrow g_E k$  with  $k k^\top = \mathbf{1}$  and  $k \in O(d, d; \mathbb{R})$  gives the same  $\mathcal{H}$ . Such  $k$  form the maximal compact subgroup  $O(d) \times O(d)$ , the independent rotations of  $p_L$  and  $p_R$ , and we recover the statement of Chapter 10 that  $\mathcal{H}$  parametrizes the coset  $O(d, d; \mathbb{R})/(O(d) \times O(d))$  of dimension  $d(2d - 1) - d(d - 1) = d^2$  [GPR §2.4, ll. 1259–1266; AMN §3.6, ll. 1306–1312; HHZ §2, ll. 563–570].

## 15.6 Worked example: a slanted two-torus with a $B$ -field

Take  $d = 2$ ,

$$G = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \quad B = b\varepsilon, \quad \varepsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad b = \frac{1}{2}, \quad E = \begin{pmatrix} 2 & \frac{3}{2} \\ \frac{1}{2} & 1 \end{pmatrix}.$$

In string units the two cycles have lengths  $2\pi\sqrt{2}$  and  $2\pi$  and meet at  $45^\circ$ , since  $\cos\theta = G_{12}/\sqrt{G_{11}G_{22}} = 1/\sqrt{2}$ ;  $\det G = 1$ , so the area is that of the self-dual square torus.

*Step 1: the blocks.*  $G^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$  and  $BG^{-1} = \frac{1}{2}\varepsilon G^{-1} = \begin{pmatrix} -1/2 & 1 \\ -1/2 & 1/2 \end{pmatrix}$ . For any symmetric  $2 \times 2$  matrix,  $\varepsilon G^{-1} \varepsilon = -G/\det G$ , so

$$G - BG^{-1}B = \left(1 + \frac{b^2}{\det G}\right)G = \frac{|\rho|^2}{\rho_2^2}G, \quad \rho = b + i\sqrt{\det G} = \frac{1}{2} + i, \quad (15.31)$$

here  $\frac{5}{4}G$ . Therefore

$$\mathcal{H} = \begin{pmatrix} \frac{5}{2} & \frac{5}{4} & -\frac{1}{2} & 1 \\ \frac{5}{4} & \frac{5}{4} & -\frac{1}{2} & \frac{1}{2} \\ -\frac{1}{2} & -\frac{1}{2} & 1 & -1 \\ 1 & \frac{1}{2} & -1 & 2 \end{pmatrix}. \quad (15.32)$$

*Step 2: checks.* The matrix is symmetric (Proposition 15.3). Its trace is  $\frac{27}{4}$ , in agreement with Corollary 15.5(iii):  $\text{tr } G + \text{tr } G^{-1} + b^2 \text{tr } G/\det G = 3 + 3 + \frac{3}{4}$ , and  $\frac{27}{4} \geq 4$ . Its characteristic polynomial is

$$\lambda^4 - \frac{27}{4}\lambda^3 + \frac{193}{16}\lambda^2 - \frac{27}{4}\lambda + 1,$$

which is palindromic, as (i) and (ii) of the corollary demand: the constant term is  $\det \mathcal{H} = 1$  and the roots come in pairs,  $\lambda \approx 0.2329, 4.2946$  and  $\lambda \approx 0.6266, 1.5960$ , with  $0.2329 \times 4.2946 \approx 0.6266 \times 1.5960 \approx 1.000$ .

*Step 3: some masses.* Write  $Z = (m^1, m^2; n_1, n_2)$ . The diagonal entries of (15.32) are the values of  $\mu_{\mathcal{H}}$  on the four unit charges: a string wound once around the first cycle has  $\mu = \frac{5}{2}$ , not  $G_{11} = 2$ ; the  $B$ -field has made it heavier by the factor  $|\rho|^2/\rho_2^2$ . For  $Z = (1, 0; 0, 1)$  the completed square (15.12) is quickest:  $n - Bm = (0, 1) - \frac{1}{2}(0, -1) = (0, \frac{3}{2})$ , so

$$\mu_{\mathcal{H}}(Z) = m^T G m + (n - Bm)^T G^{-1} (n - Bm) = 2 + \frac{9}{4} \cdot 2 = \frac{13}{2},$$

which agrees with  $\mathcal{H}_{11} + 2\mathcal{H}_{14} + \mathcal{H}_{44} = \frac{5}{2} + 2 + 2$ . Here  $m \cdot n = 0$ , so  $N = \tilde{N}$ , and the lightest such state,  $N = \tilde{N} = 1$ , has  $\alpha' M^2 = \frac{13}{2}$  by (15.15). For  $Z = (1, 0; 1, 0)$  we get  $\mu = \frac{5}{2} - 1 + 1 = \frac{5}{2}$  and  $m \cdot n = 1$ , so  $N = \tilde{N} + 1$ ; with  $N = 1, \tilde{N} = 0$  the mass is  $\alpha' M^2 = \frac{5}{2} + 2(1 - 2) = \frac{1}{2}$ .

*Step 4: an integer B-shift.* Take  $\Theta = -\varepsilon$ , so  $b \rightarrow -\frac{1}{2}$ . By (15.20) the state  $Z = (1, 0; 0, 1)$  is mapped to  $Z' = (m, n + \Theta m) = (1, 0; 0, 2)$ , and indeed in the new background  $n' - B'm = (0, 2) + \frac{1}{2}(0, -1) = (0, \frac{3}{2})$  as before:  $\mu = \frac{13}{2}$  again. The backgrounds  $b = \frac{1}{2}$  and  $b = -\frac{1}{2}$  have the same spectrum although no state keeps its label.

*Step 5: inversion.* With  $\det E = \frac{5}{4}$ ,

$$E^{-1} = \frac{4}{5} \begin{pmatrix} 1 & -\frac{3}{2} \\ -\frac{1}{2} & 2 \end{pmatrix}, \quad G' = \frac{4}{5} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} = \frac{G^{-1}}{|\rho|^2}, \quad b' = -\frac{2}{5} = -\frac{b}{|\rho|^2}.$$

One checks  $G' = (G - BG^{-1}B)^{-1} = (\frac{5}{4}G)^{-1}$ , as (15.28) predicts. In terms of  $\rho' = b' + i\sqrt{\det G'} = -\frac{2}{5} + \frac{4}{5}i$  this is  $\rho' = -1/\rho$ : the full duality acts on  $\rho$  as the modular inversion (Chapter 12). The dual torus has area  $\frac{4}{5}$  of the self-dual one, and the state  $Z = (1, 0; 0, 1)$  reappears as  $\eta Z = (0, 1; 1, 0)$  with the same  $\mu = \frac{13}{2}$ .

## 15.7 From moduli to fields

So far  $G$  and  $B$  were constants, and  $\mathcal{H}$  was one point in the moduli space of the torus. Double field theory takes the step suggested by the notation: it lets the generalized metric depend on the doubled coordinates of Chapter 14,  $\mathcal{H} = \mathcal{H}(x, \tilde{x})$ , with the same algebraic form as

(15.7) at each point, “the same form ... but here the fields are non-constant” [AMN §3.3, eqs. (3.22)–(3.26), ll. 929–985]. Three features of the constant case survive and organize the field theory [HHZ §1, ll. 250–291]:

- $\mathcal{H}$  is a symmetric  $O(D, D)$  tensor with  $\mathcal{H}^{-1} = \eta\mathcal{H}\eta$ ; indices are raised and lowered with the constant  $\eta$ , never with  $\mathcal{H}$ ;
- a global  $h \in O(D, D)$  acts linearly,  $\mathcal{H}'(X') = h\mathcal{H}(X)h^\top$  with  $X' = hX$ , while on  $E = g + b$  the same transformation is the non-linear (15.24) [HHZ §2, eq. (2.16), ll. 557–561];
- the dilaton joins in the  $O(D, D)$  scalar  $e^{-2d} = \sqrt{g}e^{-2\phi}$  [AMN §3.3, eq. (3.24), ll. 959–965].

The gain is practical. Gauge transformations that are non-linear in  $E$  become linear in  $\mathcal{H}$ , and the action of Chapter 17 and the proof of its gauge invariance simplify considerably [HHZ §6, ll. 2412–2418]. The price is the constraint (15.9): the  $2D(2D+1)/2$  components of a symmetric matrix are not independent fields, only  $D^2$  of them are, and variations of  $\mathcal{H}$  must respect  $\mathcal{H}\eta\mathcal{H} = \eta$ . None of this field-theoretic structure is formalized in the Lean modules of this chapter, which treat a single constant background.

#### Historical note: whose “generalized metric”?

HHZ recall that the matrix (15.7) “appeared in the early T-duality literature”, where it “defines the first-quantized Hamiltonian for closed strings in a toroidal background” [HHZ §1, ll. 279–283]. In generalized geometry (Chapter 16) the coordinates are not doubled,  $\mathcal{H}$  is a metric on the bundle  $T \oplus T^*$ , and Gualtieri gave the name generalized metric to the involution  $\mathcal{S}$ ; it was later proposed for  $\mathcal{H}$  instead. On the doubled space, HHZ remark, the name “is a misnomer”:  $\mathcal{H}$  “is better regarded as a conventional metric on the doubled space” obeying a constraint [HHZ §1, ll. 310–333].

## 15.8 What the machine has checked

The results of Sections 15.2 to 15.4 have been formalized. The library is `DualScaleStream2`, and the two short modules are

`DFT/GeneralizedMetric.lean`      and      `DFT/BShift.lean`,

both in the namespace `DualScaleStream2.DFT`. Everything is stated over  $\mathbb{R}$ , for an arbitrary natural number  $d$ , with Mathlib’s matrices. We go through every theorem of the two files. A third, older file of a different library formalizes only the integer matrix  $\eta$  and is described at the end. All theorems quoted here depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`. We confirmed this with the command `#print axioms` for three of them: `etaR_genMetric_sq`, `genMetric_bshift` and `massForm_circle`.

### 15.8.1 Definitions

#### In Lean: $\eta$ and $\mathcal{H}$ over $\mathbb{R}$

```
abbrev Charge (d : ℕ) := Fin d ⊕ Fin d          -- in TDuality/ODD.lean

noncomputable def etaR (d : ℕ) : Matrix (Charge d) (Charge d) ℝ :=
  fromBlocks 0 1 1 0

noncomputable def genMetric (G B : Matrix (Fin d) (Fin d) ℝ) :
  Matrix (Charge d) (Charge d) ℝ :=
  fromBlocks (G - B * G⁻¹ * B) (B * G⁻¹) (-(G⁻¹ * B)) G⁻¹

theorem etaR_mul_self : etaR d * etaR d = 1 := by ...
```

**Reading.** Rows and columns of a doubled matrix are indexed by the disjoint union  $\text{Fin } d \oplus \text{Fin } d$ , not by  $\text{Fin } (2*d)$ . This is what makes block matrices painless: `fromBlocks A B C D` is literally the matrix  $\begin{pmatrix} A & B \\ C & D \end{pmatrix}$ , and no index arithmetic is ever needed. `genMetric` is (15.7) character by character. It is defined for *all* real matrices  $G$ ,  $B$ : no symmetry, no positivity, not even invertibility is built in. `etaR_mul_self` says  $\eta^2 = 1$ .

#### Common pitfall: the inverse of a non-invertible matrix

In Mathlib  $G^{-1}$  is a total function: it is the adjugate divided by the determinant when  $\det G$  is a unit, and a junk value (the zero matrix, over a field) otherwise. So `genMetric G B` always “exists”, but  $G^{-1}G = 1$  may be used only under the hypothesis `hG : IsUnit G.det`, which over  $\mathbb{R}$  means  $\det G \neq 0$ . Every theorem below that needs the inverse carries this hypothesis explicitly. The two facts used are the Mathlib lemmas `Matrix.nonsing_inv_mul` and `Matrix.mul_nonsing_inv`.

### 15.8.2 The theorems of `GeneralizedMetric.lean`

#### In Lean: the constraint and the symmetry

```
theorem etaR_genMetric_sq (G B : Matrix (Fin d) (Fin d) ℝ)
  (hG : IsUnit G.det) :
  (etaR d * genMetric G B) * (etaR d * genMetric G B) = 1 := by ...

theorem genMetric_symm (G B : Matrix (Fin d) (Fin d) ℝ) (hG : IsUnit G.det)
  (hGs : GT = G) (hB : BT = -B) :
  (genMetric G B)T = genMetric G B := by ...
```

**Reading.** The first is  $(\eta\mathcal{H})^2 = 1$ , Theorem 15.4, for every  $G$  with invertible determinant and every  $B$ ; there is no hypothesis on  $B$ . It is equivalent to  $\mathcal{H}^{-1} = \eta\mathcal{H}\eta$  and to  $\mathcal{H}\eta\mathcal{H} = \eta$ , but those two forms are not stated in the file (Exercise 15.8). The second is Proposition 15.3. Together they give  $\mathcal{H}^T\eta\mathcal{H} = \eta$ , membership in  $O(d, d; \mathbb{R})$ ; this combination is also not stated as a theorem. **Proof idea.** The second proof of Theorem 15.4: compute  $\eta\mathcal{H}$  with `Matrix.fromBlocks_multiply`, square it with the same lemma, rewrite 1 as `fromBlocks 1 0 0 1`, and split the equation into four block equations with `Matrix.fromBlocks_inj`. Each block is closed by `simp only` with distributivity, associativity and the two inverse lemmas, followed by `abel`. For the symmetry, `Matrix.fromBlocks_transpose` and `Matrix.transpose_nonsing_inv` do the work. Positive definiteness (Proposition 15.6) is not formalized.

A remark on tactics. Matrices do not commute, so the tactic `ring` does not apply. `abel` normalizes in the additive group only, which suffices once `simp only` has expanded all products and cancelled  $G^{-1}G$ . This division of labour is the pattern of every proof in the two files.

#### In Lean: T-duality is metric inversion

```
theorem tduality_inverts_metric (G : Matrix (Fin d) (Fin d) ℝ)
  (hG : IsUnit G.det) :
  etaR d * genMetric G 0 * etaR d = genMetric G-1 0 := by ...
```

**Reading.** Theorem 15.10:  $\eta\mathcal{H}(G, 0)\eta = \mathcal{H}(G^{-1}, 0)$ , for the full duality  $\eta$  and vanishing  $B$  only. Neither a factorized duality  $D_k$  nor the inversion  $E \mapsto E^{-1}$  with  $B \neq 0$  is covered. **Proof idea.** Rewrite both sides as block-diagonal matrices (this needs  $(G^{-1})^{-1} = G$ , Mathlib’s `Matrix.nonsing_inv_nonsing_inv`, hence `hG`); two applications



of `Matrix.fromBlocks_multiply` swap the diagonal blocks.

### In Lean: the mass form, covariance, and the bridge to the circle

```

noncomputable def massForm (H : Matrix (Charge d) (Charge d) ℝ)
  (Z : Charge d → ℝ) : ℝ :=
  Z ·v (H *v Z)

theorem massForm_covariant (g H : Matrix (Charge d) (Charge d) ℝ)
  (Z : Charge d → ℝ) :
  massForm (gT * H * g) Z = massForm H (g *v Z) := by ...

theorem massForm_circle (n w R : ℚ) (hR : R ≠ 0) :
  massForm (genMetric (d := 1) (![(R : ℝ) ^ 2]) 0)
    (Sum.elim ![(w : ℝ)] ![(n : ℝ)]) =
    ((StringTheory.UseCases.TDuality.momentumMassSq n w R : ℚ) : ℝ) / 2
  := by ...

```

**Reading.** `massForm H Z` is  $Z^T \mathcal{H} Z$  for a real vector  $Z$  and *any* real matrix. `massForm_covariant` is Proposition 15.8, with no hypothesis on  $g$ : it is not a statement about  $O(d, d; \mathbb{R})$ . `massForm_circle` is (15.17) in units  $\alpha' = 1$ : for rational  $n, w, R$  with  $R \neq 0$ , the mass form of the  $1 \times 1$  background  $G = (R^2)$ ,  $B = 0$  on the vector with first (winding) component  $w$  and second (momentum) component  $n$  equals one half of `momentumMassSq n w R`, which an older library defines as  $p_L^2 + p_R^2$  with  $p_{L,R} = n/R \pm wR$  (`leftMomentum, rightMomentum`; Chapter 8). The theorem ties two independently written developments together and fixes the convention  $Z = (w, n)$  of `genMetric`. The charges are rational here because the older library works over  $\mathbb{Q}$ ; that they are integers in physics is not part of the statement. **Proof idea.** Covariance is  $(gZ) \cdot \mathcal{H}(gZ) = Z \cdot g^T \mathcal{H} g Z$  via `Matrix.dotProduct_transpose_mulVec`. For the circle, the inverse of a  $1 \times 1$  matrix is computed from the adjugate, the block form is expanded, and the remaining identity  $R^2 w^2 + n^2 / R^2 = \frac{1}{2} [(n/R + wR)^2 + (n/R - wR)^2]$  is closed by `field_simp` and `ring` (real numbers do commute).

### 15.8.3 The theorems of `BShift.lean`

#### In Lean: the $B$ -shift

```

noncomputable def thetaShiftR (Θ : Matrix (Fin d) (Fin d) ℝ) :
  Matrix (Charge d) (Charge d) ℝ :=
  fromBlocks 1 Θ 0 1

theorem genMetric_bshift (G B Θ : Matrix (Fin d) (Fin d) ℝ)
  (hG : IsUnit G.det) (hGs : GT = G) (hB : BT = -B) (hΘ : ΘT = -Θ) :
  thetaShiftR Θ * genMetric G B * (thetaShiftR Θ)T
  = genMetric G (B + Θ) := by ...

theorem thetaShiftR_preserves_eta (Θ : Matrix (Fin d) (Fin d) ℝ)
  (hΘ : ΘT = -Θ) :
  (thetaShiftR Θ)T * etaR d * thetaShiftR Θ = etaR d := by ...

theorem genMetric_bshift_cancel (G B Θ : Matrix (Fin d) (Fin d) ℝ) :
  genMetric G (B + Θ + -Θ) = genMetric G B := by ...

```

**Reading.** `genMetric_bshift` is Theorem 15.9 for a background:  $g_\Theta \mathcal{H}(G, B) g_\Theta^T = \mathcal{H}(G, B + \Theta)$  for every *real* antisymmetric  $\Theta$ . Integrality plays no role and is not

mentioned; the statement is about the continuous group. Of the four hypotheses, the proof script uses only  $\mathbf{h}\Theta$ , in agreement with our proof by hand; the other three record the intended physical setting.  $\mathbf{thetaShiftR\_preserves\_eta}$  is  $g_\Theta \in O(d, d; \mathbb{R})$ , one direction only (if  $\Theta$  is antisymmetric then  $\eta$  is preserved).  $\mathbf{genMetric\_bshift\_cancel}$  says that  $\mathcal{H}(G, B + \Theta - \Theta) = \mathcal{H}(G, B)$ ; it holds because  $B + \Theta + (-\Theta) = B$  as matrices, needs no hypothesis, and has no physical content beyond that. **Proof idea.** Unfold, transpose and multiply the blocks ( $\mathbf{Matrix.fromBlocks\_transpose}$ ,  $\mathbf{Matrix.fromBlocks\_multiply}$  twice), substitute  $\Theta^\top = -\Theta$ , split into four blocks; the upper left block,  $G - BG^{-1}B - \Theta G^{-1}B - (B + \Theta)G^{-1}\Theta = G - (B + \Theta)G^{-1}(B + \Theta)$ , needs  $\mathbf{abel}$ , the other three are closed by  $\mathbf{simp}$  only. The module comment records that the convention  $g\mathcal{H}g^\top$  with  $+\Theta$  in the upper right block was selected by a sympy experiment before the formal proof was attempted, the three other candidates being false.

#### 15.8.4 An older formalization of $\eta$

An older library, `StringTheoryFormalization`, contains a module `ODDMetric.lean` (in its directory `StringDynamics`), which predates the two files above:

**In Lean:**  $\eta$  as an integer matrix on  $\mathbf{Fin} (2 * D)$

```
def oddMetric : Matrix (Fin (2 * D)) (Fin (2 * D)) ℤ :=
  Matrix.of (fun i j =>
    if i.val < D ∧ j.val = i.val + D then 1
    else if j.val < D ∧ i.val = j.val + D then 1
    else 0)

theorem odd_metric_symm : (oddMetric D)ᵀ = oddMetric D := by ...

theorem odd_metric_d1 : oddMetric 1 = !![0, 1; 1, 0] := by ...
```

**Reading.** `oddMetric D` is the integer  $2D \times 2D$  matrix with entries 1 at the positions  $(i, i + D)$  and  $(i + D, i)$ , that is,  $\eta$  written without block structure. The two theorems say that it is symmetric for every  $D$  and that for  $D = 1$  it is the matrix  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ . Nothing about the generalized metric, the signature of  $\eta$  or a group action is proved in this file, although its header comment announces the duality group. **Proof idea.** Entrywise: after  $\mathbf{simp}$  the symmetry is a statement about natural numbers closed by  $\mathbf{omega}$ ; the case  $D = 1$  is a check of four entries by  $\mathbf{fin\_cases}$ . Comparing the two encodings of  $\eta$  is instructive: with `Fin (2 * D)` even symmetry requires index arithmetic, while with `Fin d ⊕ Fin d` and `fromBlocks` it is one  $\mathbf{simp}$  (Exercise 15.7).

## 15.8.5 Tier table

| Statement                                                                                                              | Where                                                     | Tier |
|------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|------|
| $\eta^2 = \mathbf{1}$ over $\mathbb{R}$                                                                                | <code>etaR_mul_self</code>                                | [A]  |
| $(\eta\mathcal{H})^2 = \mathbf{1}$ , $G$ invertible, any $B$                                                           | <code>etaR_genMetric_sq</code>                            | [A]  |
| $\mathcal{H}^\top = \mathcal{H}$ for a background                                                                      | <code>genMetric_symm</code>                               | [A]  |
| $\eta\mathcal{H}(G, 0)\eta = \mathcal{H}(G^{-1}, 0)$                                                                   | <code>tduality_inverts_metric</code>                      | [A]  |
| $\mu_{g^\top\mathcal{H}g}(Z) = \mu_{\mathcal{H}}(gZ)$                                                                  | <code>massForm_covariant</code>                           | [A]  |
| $d = 1$ mass form $= \frac{1}{2}(p_L^2 + p_R^2)$ over $\mathbb{Q}$                                                     | <code>massForm_circle</code>                              | [A]  |
| $g_\Theta\mathcal{H}(G, B)g_\Theta^\top = \mathcal{H}(G, B + \Theta)$                                                  | <code>genMetric_bshift</code>                             | [A]  |
| $g_\Theta^\top\eta g_\Theta = \eta$ over $\mathbb{R}$                                                                  | <code>thetaShiftR_preserves_eta</code>                    | [A]  |
| integer $\eta$ symmetric; explicit for $D = 1$                                                                         | <code>odd_metric_symm</code> , <code>odd_metric_d1</code> | [A]  |
| $\text{tr } \mathcal{H}(G, 0) \geq 2d$ for positive definite $G$                                                       | <code>dualScale_ge</code> (Chapter 31)                    | [A]  |
| Factorization, $\det \mathcal{H} = 1$ , positivity, eigenspaces                                                        | this chapter, by hand                                     | —    |
| $g\mathcal{H}(E)g^\top = \mathcal{H}(g(E))$ for all $g \in O(d, d; \mathbb{R})$                                        | GPR (2.4.24), HHZ (2.16)                                  | [L]  |
| $\mathcal{H}$ parametrizes $O(d, d; \mathbb{R})/O(d) \times O(d)$                                                      | GPR §2.4, AMN §3.6                                        | [L]  |
| <code>massForm</code> is the closed-string mass; integer $\Theta$ -shift and $\eta$ are exact symmetries of the string | GPR §2.4                                                  | [L]  |

The rows marked “—” are proved on the page and not in Lean; they are elementary, and formalizing them is a reasonable first project (Exercise 15.9). The last row is the identification of a Lean object with a physical system and would not become Tier A even if every matrix identity were formalized: a faithful formalization would need the worldsheet theory, its Hilbert space and its Hamiltonian, none of which exists in these libraries.

## Chapter summary

- The zero-mode energy of a closed string on  $T^d$  is the completed square  $\frac{1}{2}m^\top Gm + \frac{1}{2}(n - Bm)^\top G^{-1}(n - Bm)$ . Its matrix in the basis  $Z = (m, n)$  is the generalized metric  $\mathcal{H}(G, B)$ , and the completed square is the factorization  $\mathcal{H} = U_B \text{diag}(G, G^{-1}) U_B^\top$ .
- $\mathcal{H}$  is symmetric, positive definite, has  $\det \mathcal{H} = 1$  and satisfies  $\mathcal{H}\eta\mathcal{H} = \eta$ : it is a symmetric element of  $O(d, d; \mathbb{R})$ . The constraint alone needs neither symmetry of  $G$  nor antisymmetry of  $B$ .
- $\mathcal{S} = \mathcal{H}\eta$  is an involution whose eigenspaces, the graphs of  $E$  and  $-E^\top$ , are the purely left- and right-moving directions.
- Dualities act linearly,  $\mathcal{H} \mapsto g\mathcal{H}g^\top$ , and on  $E = G + B$  by  $E \mapsto (aE + b)(cE + d)^{-1}$ . Special cases:  $B \rightarrow B + \Theta$ ;  $E \rightarrow AE A^\top$ ;  $E \rightarrow E^{-1}$ , which at  $B = 0$  is  $G \rightarrow G^{-1}$ ; Buscher’s rules for  $D_k$ . The upper left block  $G - BG^{-1}B$  of  $\mathcal{H}$  is the inverse of the T-dual metric.
- The mass is  $\alpha' M^2 = Z^\top \mathcal{H} Z + 2(N + \tilde{N} - 2)$  with  $N - \tilde{N} = m \cdot n$ ;  $\mathcal{H}$  measures energy,  $\eta$  measures level matching.
- Lean (Tier A, all  $d$ , over  $\mathbb{R}$ ): the constraint, the symmetry, inversion at  $B = 0$ , the  $B$ -shift by conjugation, covariance of the mass form, and the agreement with the circle formula. The general fractional linear action, positivity and the coset structure are Tier L or proved by hand only.

## Exercises

**Exercise 15.1** ( $\star$ ). For  $d = 1$  write  $r = R^2/\alpha'$ . (a) Write  $\mathcal{H}$ , verify (15.9) directly, and find the eigenvalues and eigenvectors of  $\mathcal{S} = \mathcal{H}\eta$ ; compare with Lemma 15.7. (b) Show that  $\text{tr } \mathcal{H} \geq 2$  with equality exactly at the self-dual radius. (c) Why is there no  $B$ -field for  $d = 1$ ?

**Exercise 15.2** (★). Show that  $U_B U_{B'} = U_{B+B'}$  and  $U_B^{-1} = U_{-B}$ . Use (15.8) to write  $\mathcal{H}(G, B)^{-1}$  as a product of three matrices and verify, without expanding into blocks, that it equals  $\eta \mathcal{H}(G, B) \eta$ . (Hint:  $\eta U_B \eta = L_{-B}$ .)

**Exercise 15.3** (★★). Let  $(G, B)$  be a background. Show that  $\frac{1}{2}(\mathcal{H} + \eta)$  and  $\frac{1}{2}(\mathcal{H} - \eta)$  are positive semidefinite of rank  $d$ , and that their kernels are  $\eta V_-$  and  $\eta V_+$  respectively. Conclude that  $\mu_{\mathcal{H}}(Z) \geq |Z^\top \eta Z| = 2|m \cdot n|$  for every charge, and interpret the inequality in terms of  $p_L^2$  and  $p_R^2$ .

**Exercise 15.4** (★★). For  $d = 2$  prove  $\varepsilon G^{-1} \varepsilon = -G / \det G$  for symmetric  $G$  and derive (15.31). With the parametrization  $G = \frac{\rho_2}{\tau_2} \begin{pmatrix} 1 & \tau_1 \\ \tau_1 & |\tau|^2 \end{pmatrix}$ ,  $B = \rho_1 \varepsilon$ , show that

$$\text{tr } \mathcal{H} = \frac{(1 + |\tau|^2)(1 + |\rho|^2)}{\tau_2 \rho_2},$$

and find the backgrounds with  $\text{tr } \mathcal{H} = 4$ . The formula is symmetric under  $\tau \leftrightarrow \rho$ ; which duality of Chapter 12 does this reflect?

**Exercise 15.5** (★★). For the background of Section 15.6 compute the factorized dual  $E' = D_1(E)$  from (15.29). (Answer:  $G' = \begin{pmatrix} 1/2 & -1/4 \\ -1/4 & 5/8 \end{pmatrix}$ ,  $b' = -\frac{1}{2}$ .) Find the image of the state  $Z = (1, 0; 0, 1)$  and check that its mass form is again  $\frac{13}{2}$ .

**Exercise 15.6** (★★). Show that for  $r, s \in O(d)$  the matrix  $k = \frac{1}{2} \begin{pmatrix} r+s & r-s \\ r-s & r+s \end{pmatrix}$  satisfies  $k^\top \eta k = \eta$  and  $k k^\top = \mathbf{1}$ , and that every element of  $O(d, d; \mathbb{R})$  with  $k k^\top = \mathbf{1}$  is of this form. (Hint: such a  $k$  commutes with  $\eta$ ; diagonalize  $\eta$ .) Which background does this subgroup leave fixed under  $\mathcal{H} \mapsto k \mathcal{H} k^\top$ , and what is  $k(\mathbf{1})$  in (15.24)?

**Exercise 15.7** (★★, Lean). Create a new file that imports `DualScaleStream2.DFT.BShift`. State and prove that the matrix `etaR d` is symmetric. Three tactics suffice: unfold the definition, rewrite with the lemma `Matrix.fromBlocks_transpose`, and call `simp`. Compare the length of your proof with that of `odd_metric_symm`, and explain the difference.

**Exercise 15.8** (★★, Lean). State and prove in Lean the form  $\mathcal{H} \eta \mathcal{H} = \eta$  of the constraint: `genMetric G B * etaR d * genMetric G B = etaR d` under `hG : IsUnit G.det`. Hint: multiply the theorem `etaR_genMetric_sq` from the left by  $\eta$ . Apart from it you need only the lemmas `mul_assoc`, `etaR_mul_self`, `one_mul` and `mul_one`.

**Exercise 15.9** (★★★, Lean). Formalize (15.21): for  $A$  with invertible determinant, symmetric invertible  $G$  and any  $B$ ,  $g_A \mathcal{H}(G, B) g_A^\top = \mathcal{H}(A G A^\top, A B A^\top)$  with  $g_A = \text{fromBlocks } A \ 0 \ 0 \ (A^{-1})^\top$ . Begin with the lemma  $(A G A^\top)^{-1} = (A^{-1})^\top G^{-1} A^{-1}$ ; then follow the pattern of `genMetric_bshift`. Which of the hypotheses did you really need?

**Exercise 15.10** (★★★). Prove the converse of Section 15.2: a real symmetric positive definite  $2d \times 2d$  matrix  $\mathcal{H}$  with  $\mathcal{H} \eta \mathcal{H} = \eta$  is  $\mathcal{H}(G, B)$  for a unique background. (Hint: the lower right block  $S$  is positive definite; set  $G = S^{-1}$ , read  $B$  off the upper right block, and use two of the block equations of the constraint.) Deduce (15.25) without using eigenspaces, and compare with the argument of Chapter 11.

## Further reading

- The generalized metric as the matrix of the toroidal Hamiltonian, the embedding  $E \mapsto g_E$ , the fractional linear action and the three families of generators: [GPR §2.4, ll. 1031–1425]; transformation of the oscillators, [GPR §2.4, ll. 1489–1520].

- The Hamiltonian density in the doubled variables  $(X', 2\pi P)$  and the constraint  $\mathcal{H}^{-1} = \eta\mathcal{H}\eta$ : [HZ §2.2, eqs. (2.14)–(2.18), ll. 600–680].
- $\mathcal{H}$  as an  $O(D, D)$  tensor field, the involution  $S$ , the three-factor form, the vielbein construction and index conventions: [HHZ §1, ll. 250–333; §2, ll. 403–660].
- A review-level summary of the mass formula, the inversion metric and the generalized fields of DFT: [AMN §3.1, ll. 596–665 and 745–770; §3.3, ll. 919–1000; §3.6, ll. 1260–1312].
- Lean sources: the three modules named in Section 15.8. The trace bound of Chapter 31 is proved in the file `DualScale/TraceBound.lean` of `DualScaleStream2`.



## Chapter 16

# Generalized Geometry: Courant and Dorfman Brackets

The massless fields of the closed string are a metric  $g_{ij}$ , a two-form  $b_{ij}$  and a dilaton (Chapter 6). The metric comes with a gauge symmetry, diffeomorphisms, whose parameter is a vector field. The two-form comes with a second gauge symmetry,  $b \mapsto b + d\tilde{\lambda}$ , whose parameter is a one-form. In Chapter 15 the two fields were packaged into one object, the generalized metric  $\mathcal{H}$ . This chapter answers the obvious next question: *can the two gauge symmetries be packaged as well, and what algebra do they then form?*

The answer is yes, and it is instructive in an unexpected way. The combined parameter is a *generalized vector*  $V = v + \xi$ , a vector field plus a one-form. The combined transformation is a *generalized Lie derivative*. Acting on another generalized vector it defines a product, the *Dorfman bracket*, and the antisymmetric part of that product is the *Courant bracket*. Neither of the two is a Lie bracket: the first is not antisymmetric, the second violates the Jacobi identity. A student who has learned that “symmetries form Lie algebras” should see at least once, in full detail, how a perfectly consistent gauge symmetry escapes that slogan, and why nothing goes wrong. In double field theory the same brackets reappear in  $O(D, D)$  covariant form as the D- and C-brackets, and there the closure of the gauge algebra is no longer automatic: it holds if the strong constraint of Chapter 14 is imposed. At the end we read, line by line, what the two Lean modules attached to this chapter actually prove; they are integer toy models, and part of the exercise is to say precisely which features of the brackets they capture and which they do not.

Conventions.  $D$  is the number of spacetime dimensions and  $i, j, k = 1, \dots, D$ . We order doubled objects as in Chapter 14, vector (momentum) entry first:  $X^M = (x^i, \tilde{x}_i)$ ,  $\partial_M = (\partial_i, \tilde{\partial}^i)$ ,  $M = 1, \dots, 2D$ . The review of Aldazabal, Marqués and Núñez and the papers of Hohm, Hull and Zwiebach put the tilde entry first [AMN §3.4, eq. (3.33), ll. 1096–1099; HHZ §1, footnote, ll. 313–315]; since every formula below is written with  $\eta$ -contracted indices, the ordering does not affect them.

## 16.1 Two gauge symmetries and their algebra

### 16.1.1 Diffeomorphisms and $b$ -field gauge transformations

An infinitesimal diffeomorphism with parameter a vector field  $\lambda = \lambda^i \partial_i$  acts on every tensor by the Lie derivative  $L_\lambda$ . On a scalar  $\phi$ , a vector  $V$  and a one-form  $\omega$ ,

$$L_\lambda \phi = \lambda^j \partial_j \phi, \quad (L_\lambda V)^i = \lambda^j \partial_j V^i - V^j \partial_j \lambda^i = [\lambda, V]^i, \quad (L_\lambda \omega)_i = \lambda^j \partial_j \omega_i + \partial_i \lambda^j \omega_j, \quad (16.1)$$

[AMN §3.1, eqs. (3.14)–(3.15), ll. 840–846]. The middle expression is the Lie bracket of vector fields. It is antisymmetric and satisfies the Jacobi identity  $[[\lambda_1, \lambda_2], \lambda_3] + \text{cyclic} = 0$ ; consequently  $[L_{\lambda_1}, L_{\lambda_2}] = L_{[\lambda_1, \lambda_2]}$  and diffeomorphisms close into a Lie algebra. The second symmetry acts only on the two-form:

$$\delta_{\tilde{\lambda}} b_{ij} = \partial_i \tilde{\lambda}_j - \partial_j \tilde{\lambda}_i = (d\tilde{\lambda})_{ij}, \quad \delta_{\tilde{\lambda}} g_{ij} = 0, \quad (16.2)$$

with  $\tilde{\lambda} = \tilde{\lambda}_i dx^i$  a one-form [AMN §3.1, eq. (3.16), ll. 855–859]. The field strength  $H = db$  is invariant because  $d^2 = 0$ . For the same reason the parameter has its own redundancy:  $\tilde{\lambda} = d\chi$  does nothing at all. This “gauge symmetry of the gauge symmetry” will be responsible for most of the unusual features below.

We shall use three operations on differential forms: the exterior derivative  $d$ , the contraction  $\iota_v$  with a vector field,  $(\iota_v \omega)_i = v^j \omega_{ji}$  on a one-form and  $(\iota_v B)_j = v^i B_{ij}$  on a two-form, and the Lie derivative. They obey

$$L_v = d \iota_v + \iota_v d, \quad [L_v, d] = 0, \quad [L_v, \iota_u] = \iota_{[v, u]}, \quad [L_v, L_u] = L_{[v, u]}. \quad (16.3)$$

The first identity is Cartan’s formula; the student should check it on a one-form using (16.1):  $(\iota_v d\omega)_i = v^j (\partial_j \omega_i - \partial_i \omega_j)$  and  $(d\iota_v \omega)_i = \partial_i (v^j \omega_j)$  add up to the last expression in (16.1). (These identities are standard; they are not taken from a source in this book’s library.)

### 16.1.2 The algebra of the combined transformations

Let  $\delta_{(\lambda, \tilde{\lambda})} b = L_{\lambda} b + d\tilde{\lambda}$  be the combined variation. Apply two of them in succession. A variation acts on the field only, not on the parameters, so

$$\delta_1 \delta_2 b = \delta_1 (L_{\lambda_2} b + d\tilde{\lambda}_2) = L_{\lambda_2} (L_{\lambda_1} b + d\tilde{\lambda}_1),$$

and therefore, using  $[L_{\lambda_2}, L_{\lambda_1}] = L_{[\lambda_2, \lambda_1]}$  and  $L d = dL$ ,

$$[\delta_1, \delta_2] b = L_{[\lambda_2, \lambda_1]} b + d(L_{\lambda_2} \tilde{\lambda}_1 - L_{\lambda_1} \tilde{\lambda}_2). \quad (16.4)$$

The commutator is again a combined transformation, with parameters  $\lambda_{12} = [\lambda_2, \lambda_1]$  and  $\tilde{\lambda}_{12} = L_{\lambda_2} \tilde{\lambda}_1 - L_{\lambda_1} \tilde{\lambda}_2$ : diffeomorphisms act on the one-form parameters, and one-form transformations commute among themselves. But  $\tilde{\lambda}_{12}$  is *not unique*. Because only  $d\tilde{\lambda}_{12}$  enters, we may add any exact form to it; for instance

$$\tilde{\lambda}'_{12} = L_{\lambda_2} \tilde{\lambda}_1 - L_{\lambda_1} \tilde{\lambda}_2 - \kappa d(\iota_{\lambda_2} \tilde{\lambda}_1 - \iota_{\lambda_1} \tilde{\lambda}_2) \quad (16.5)$$

is equally good for every real number  $\kappa$ . The bracket on pairs  $(\lambda, \tilde{\lambda})$  is thus determined by the physics only up to exact one-forms. The choice  $\kappa = \frac{1}{2}$  will turn out to be the Courant bracket, singled out by a symmetry that is invisible here: it treats  $v$  and  $\xi$  on the same footing under  $O(D, D)$ .

#### Physical picture: a symmetry with a redundant parameter

Electromagnetism has a gauge parameter  $\chi$  that is a function, and nothing acts trivially except constants. The Kalb–Ramond field  $b$  is one step higher: its parameter is a one-form, and a whole function’s worth of parameters,  $\tilde{\lambda} = d\chi$ , acts trivially. Whenever this happens, “the” algebra of gauge parameters is ambiguous, and one should expect identities such as antisymmetry or Jacobi to hold only *up to trivial parameters*. The transformations of the fields themselves still compose perfectly well: (16.4) is an honest identity between operators acting on  $b$ .



## 16.2 Generalized vectors and the natural pairing

**Definition 16.1** (Generalized tangent bundle). Let  $M$  be a  $D$ -dimensional manifold with tangent bundle  $T$  and cotangent bundle  $T^*$ . A *generalized vector* is a section  $V = v + \xi$  of  $E = T \oplus T^*$ , the formal sum of a vector field  $v$  and a one-form  $\xi$ . In components  $V^M = (v^i, \xi_i)$ . The *natural pairing* of  $X = v + \xi$  and  $Y = u + \zeta$  is

$$\langle X, Y \rangle = \iota_v \zeta + \iota_u \xi = v^i \zeta_i + u^i \xi_i = X^M \eta_{MN} Y^N, \quad \eta = \begin{pmatrix} 0 & \mathbf{1}_D \\ \mathbf{1}_D & 0 \end{pmatrix}. \quad (16.6)$$

This is the convention of Hohm, Hull and Zwiebach, without a factor  $\frac{1}{2}$  [HHZ §1, ll. 306–322]; part of the mathematical literature includes the  $\frac{1}{2}$ . The pairing needs no metric on  $M$ : a one-form eats a vector, and that is all that is used. It is symmetric and indefinite. Indeed  $\langle X, X \rangle = 2\iota_v \xi$ , so every pure vector and every pure one-form is null, while  $v \pm \xi$  with  $\xi_i = v^i$  have norm  $\pm 2|v|^2$ . Diagonalizing  $\eta$  gives  $D$  eigenvalues  $+1$  and  $D$  eigenvalues  $-1$ : the signature is  $(D, D)$ , and the group of fibrewise linear maps preserving the pairing is  $O(D, D; \mathbb{R})$ , the group of Chapter 11 now acting on  $E$  rather than on the charge lattice. The index  $M$  is raised and lowered with  $\eta$ :  $V_M = \eta_{MN} V^N = (\xi_i, v^i)$ ,  $\partial^M = \eta^{MN} \partial_N = (\tilde{\partial}^i, \partial_i)$ .

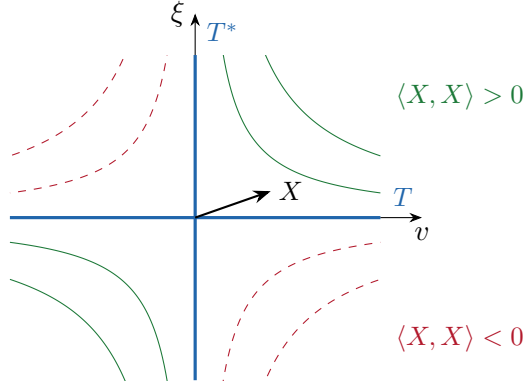


Figure 16.1: The fibre of  $T \oplus T^*$  for  $D = 1$ . The quadratic form is  $\langle X, X \rangle = 2v\xi$ ; its level sets are hyperbolas and its null cone consists of the two axes, which are the fibres of  $T$  and of  $T^*$ . Both are maximal isotropic subspaces, the linear-algebra counterpart of the “sections” of Chapter 14.

**Three kinds of  $O(D, D)$  transformations.** Write an element of  $O(D, D; \mathbb{R})$  in  $D \times D$  blocks acting on the column  $(v, \xi)^T$ . Three families of elements will be used (cf. Chapter 11); a fourth, discrete element is the matrix  $\eta$  itself, which exchanges  $v$  and  $\xi$ :

$$h_A = \begin{pmatrix} A & 0 \\ 0 & (A^T)^{-1} \end{pmatrix}, \quad e^B = \begin{pmatrix} \mathbf{1} & 0 \\ B & \mathbf{1} \end{pmatrix}, \quad e^\beta = \begin{pmatrix} \mathbf{1} & \beta \\ 0 & \mathbf{1} \end{pmatrix}, \quad B^T = -B, \quad \beta^T = -\beta. \quad (16.7)$$

The  $B$ -transform maps  $v + \xi \mapsto v + \xi + Bv$ , where  $(Bv)_i = B_{ij}v^j$ . In the language of forms this is  $\xi \mapsto \xi - \iota_v B$  with our convention  $(\iota_v B)_j = v^i B_{ij}$ ; since  $-B$  is as good a two-form as  $B$ , nothing depends on this sign. It shears  $T$  into the graph of  $B$  and leaves  $T^*$  pointwise fixed. Let us verify that it preserves the pairing, and see where antisymmetry is used:

$$\langle e^B X, e^B Y \rangle = v^i (\zeta_i + B_{ij}u^j) + u^i (\xi_i + B_{ij}v^j) = \langle X, Y \rangle + v^i u^j (B_{ij} + B_{ji}) = \langle X, Y \rangle.$$

In matrix form  $(e^B)^T \eta e^B = \eta + \text{diag}(B + B^T, 0)$ , which equals  $\eta$  exactly when  $B$  is antisymmetric.

**Common pitfall: there is no  $B$ -shear in one dimension**

For  $D = 1$  an antisymmetric  $1 \times 1$  matrix is zero, so  $O(1, 1)$  contains no non-trivial  $B$ -transform. A  $2 \times 2$  matrix  $\begin{pmatrix} 1 & 0 \\ b & 1 \end{pmatrix}$  with a number  $b \neq 0$  in the corner is *not* in  $O(1, 1)$ :  $(e^B)^\top \eta e^B = \begin{pmatrix} 2b & 1 \\ 1 & 0 \end{pmatrix}$ . The off-diagonal entries agree with  $\eta$ ; the upper-left entry does not. We return to this in Section 16.7, because the Lean module contains exactly this matrix.

## 16.3 The generalized Lie derivative

### 16.3.1 Definition and first properties

We now work on the doubled space of Chapter 14, where fields may depend on  $x^i$  and on  $\tilde{x}_i$ , and specialize later. The gauge parameter is a generalized vector  $\xi^M = (\lambda^i, \tilde{\lambda}_i)$ .

**Definition 16.2** (Generalized Lie derivative). On a generalized scalar  $S$ , a generalized vector  $V^M$  of weight zero and a tensor  $A_{MN}$ ,

$$\mathcal{L}_\xi S = \xi^P \partial_P S, \quad (16.8)$$

$$\mathcal{L}_\xi V^M = \xi^P \partial_P V^M + (\partial^M \xi_P - \partial_P \xi^M) V^P, \quad (16.9)$$

$$\mathcal{L}_\xi A_{MN} = \xi^P \partial_P A_{MN} + (\partial_M \xi^P - \partial^P \xi_M) A_{PN} + (\partial_N \xi^P - \partial^P \xi_N) A_{MP}, \quad (16.10)$$

and similarly for more indices: each index contributes two terms. A *density* of weight  $\omega$  receives in addition the term  $\omega \partial_P \xi^P$  times itself; the dilaton combination  $e^{-2d}$  has  $\omega = 1$ , so that  $\mathcal{L}_\xi e^{-2d} = \partial_M (\xi^M e^{-2d})$  is a total derivative and  $e^{-2d}$  can serve as integration measure (Chapter 17).

[HHZ §3.2, eqs. (3.11), (3.20)–(3.22), ll. 781–878; AMN §3.4, eqs. (3.34), (3.39), ll. 1101–1140]. Compare (16.9) with the ordinary Lie derivative (16.1) in  $2D$  dimensions,  $\xi^P \partial_P V^M - \partial_P \xi^M V^P$ . The new term  $\partial^M \xi_P V^P$  involves  $\eta$  twice, once to raise the derivative index and once to lower the parameter index. Three consequences follow in one line each.

(a) *The pairing is invariant.* Apply (16.10) to the constant tensor  $\eta_{MN}$ . The transport term vanishes, and

$$\mathcal{L}_\xi \eta_{MN} = (\partial_M \xi^P - \partial^P \xi_M) \eta_{PN} + (\partial_N \xi^P - \partial^P \xi_N) \eta_{MP} = \partial_M \xi_N - \partial_N \xi_M + \partial_N \xi_M - \partial_M \xi_N = 0.$$

An ordinary Lie derivative of a constant tensor does not vanish for an arbitrary vector field; here the extra term is designed to cancel it [HHZ §3.2, eqs. (3.16)–(3.17), ll. 822–834; AMN §3.4, eq. (3.38), ll. 1126–1129]. Since also  $\mathcal{L}_\xi$  obeys the Leibniz rule on products, the condition  $\mathcal{H}\eta\mathcal{H} = \eta^{-1}$  that makes  $\mathcal{H}$  an  $O(D, D)$  matrix (Chapter 15) is preserved by gauge transformations  $\delta_\xi \mathcal{H} = \mathcal{L}_\xi \mathcal{H}$  [HHZ §3.2, eqs. (3.18)–(3.19), ll. 838–850].

(b) *The antisymmetric combination is an infinitesimal  $O(D, D)$  rotation.* Write (16.9) as  $\mathcal{L}_\xi V^M = \xi^P \partial_P V^M + K^M_P V^P$  with  $K_{MP} = \partial_M \xi_P - \partial_P \xi_M$ . Because  $K_{MP} = -K_{PM}$ , the matrix  $K^M_P$  lies in the Lie algebra  $\mathfrak{so}(D, D)$ . A generalized diffeomorphism transports a generalized vector along  $\xi$  and rotates it, point by point, by an element of the duality Lie algebra.

(c) *Trivial parameters.* If  $\xi^M = \partial^M \chi$  then  $K_{MP} = \partial_M \partial_P \chi - \partial_P \partial_M \chi = 0$  and the transport term is  $\partial^P \chi \partial_P (\dots)$ , which vanishes by the strong constraint. Hence  $\mathcal{L}_{\xi+\partial\chi} = \mathcal{L}_\xi$  on all fields [HHZ §3.2, eq. (3.15), ll. 815–818]. This is the doubled form of “ $\tilde{\lambda} = d\chi$  does nothing”.

### 16.3.2 Reduction to the supergravity frame

Impose the strong constraint in the supergravity frame,  $\tilde{\partial}^i = 0$  on everything. Then  $\partial^M = (\tilde{\partial}^i, \partial_i) = (0, \partial_i)$ : a raised derivative index has only a lower, form-type component. Take  $\xi = \lambda + \tilde{\lambda}$  and  $V = u + \zeta$  in (16.9).

For the vector component  $M = {}^i$  the term  $\partial^M \xi_P$  is  $\tilde{\partial}^i \xi_P = 0$ , and  $\xi^P \partial_P = \lambda^j \partial_j$ , so

$$(\mathcal{L}_\xi V)^i = \lambda^j \partial_j u^i - \partial_j \lambda^i u^j = [\lambda, u]^i.$$

For the form component  $M = {}_i$  we need  $\xi_P V^P = \tilde{\lambda}_j u^j + \lambda^j \zeta_j$  with the derivative acting on  $\xi$  only, and  $\partial_P \xi_i V^P = \partial_j \tilde{\lambda}_i u^j$ :

$$\begin{aligned} (\mathcal{L}_\xi V)_i &= \lambda^j \partial_j \zeta_i + \partial_i \tilde{\lambda}_j u^j + \partial_i \lambda^j \zeta_j - \partial_j \tilde{\lambda}_i u^j \\ &= (L_\lambda \zeta)_i - u^j (\partial_j \tilde{\lambda}_i - \partial_i \tilde{\lambda}_j) = (L_\lambda \zeta)_i - (\iota_u d\tilde{\lambda})_i. \end{aligned}$$

Altogether

$$\boxed{\mathcal{L}_{\lambda+\tilde{\lambda}}(u+\zeta) = [\lambda, u] + L_\lambda \zeta - \iota_u d\tilde{\lambda}} \quad (\tilde{\partial}^i = 0). \quad (16.11)$$

Read the right-hand side physically. The first two terms are an ordinary diffeomorphism acting on the vector and on the one-form. The third is an infinitesimal  $B$ -transform (16.7) with the *exact* two-form  $d\tilde{\lambda}$ : precisely the two-form that (16.2) adds to  $b$ . So the generalized Lie derivative knows both symmetries, and it knows that  $\tilde{\lambda}$  enters only through  $d\tilde{\lambda}$ . Acting on the generalized metric parameterized by  $(g, b)$  it gives, in the supergravity frame,

$$\mathcal{L}_\xi g_{ij} = L_\lambda g_{ij}, \quad \mathcal{L}_\xi b_{ij} = L_\lambda b_{ij} + \partial_i \tilde{\lambda}_j - \partial_j \tilde{\lambda}_i, \quad (16.12)$$

[AMN §3.4, eqs. (3.35)–(3.37), ll. 1101–1123] [\[L\]](#). (AMN write the last term as  $2\partial_{[i}\tilde{\lambda}_{j]}$  with unit-weight antisymmetrization.) Without the restriction  $\tilde{\partial}^i = 0$  the transformations of  $g$  and  $b$  are non-linear in the fields and mix  $\partial$  with  $\tilde{\partial}$ ; remarkably they are still *linear* in  $\mathcal{H}$  [HHZ §3.1, eqs. (3.1)–(3.10), ll. 680–776].

## 16.4 The Dorfman bracket

For ordinary vector fields the Lie bracket *is* the Lie derivative,  $[X, Y] = L_X Y$ . Hohm, Hull and Zwiebach copy this [HHZ §3.3, eq. (3.29), ll. 946–953]:

**Definition 16.3** (D-bracket, Dorfman bracket). The  $D$ -bracket of two generalized vectors on the doubled space is  $[A, B]_D = \mathcal{L}_A B$ . When nothing depends on  $\tilde{x}$  it becomes, by (16.11), the *Dorfman bracket* on  $T \oplus T^*$ , usually written as a product:

$$X \circ Y = (v + \xi) \circ (u + \zeta) = [v, u] + L_v \zeta - \iota_u d\xi. \quad (16.13)$$

**Proposition 16.4** (Properties of the Dorfman bracket). *For generalized vectors  $X = v + \xi$ ,  $Y = u + \zeta$ ,  $Z$  on  $M$  and a function  $\phi$ :*

- (i)  $X \circ Y + Y \circ X = d\langle X, Y \rangle$ ;
- (ii)  $d\phi \circ Y = 0$  and  $X \circ d\phi = d(L_v \phi) = d\langle X, d\phi \rangle$ ;
- (iii) (*Leibniz identity*)  $X \circ (Y \circ Z) = (X \circ Y) \circ Z + Y \circ (X \circ Z)$ ;
- (iv)  $L_v \langle Y, Z \rangle = \langle X \circ Y, Z \rangle + \langle Y, X \circ Z \rangle$ .

*Proof.* (i) The vector parts cancel,  $[v, u] + [u, v] = 0$ . The form part is  $L_v \zeta - \iota_v d\zeta + L_u \xi - \iota_u d\xi = d\iota_v \zeta + d\iota_u \xi$  by Cartan's formula (16.3). (ii) With  $X = d\phi$  there is no vector part and  $\iota_u dd\phi = 0$ . With  $Y = d\phi$ ,  $X \circ d\phi = L_v d\phi = dL_v \phi$ . (iii) Let  $Z = w + \rho$ . The vector parts give the Jacobi identity of the Lie bracket in the form  $[v, [u, w]] = [[v, u], w] + [u, [v, w]]$ . The form parts are

$$\begin{aligned} X \circ (Y \circ Z) : & \quad L_v (L_u \rho - \iota_w d\zeta) - \iota_{[u, w]} d\xi, \\ (X \circ Y) \circ Z : & \quad L_{[v, u]} \rho - \iota_w d(L_v \zeta - \iota_u d\xi), \\ Y \circ (X \circ Z) : & \quad L_u (L_v \rho - \iota_w d\xi) - \iota_{[v, w]} d\zeta. \end{aligned}$$

Subtract the last two lines from the first and collect by the form involved. The  $\rho$  terms are  $([L_v, L_u] - L_{[v,u]})\rho = 0$ . The  $\zeta$  terms are  $-L_v\iota_w d\zeta + \iota_w L_v d\zeta + \iota_{[v,w]} d\zeta = (-[L_v, \iota_w] + \iota_{[v,w]})d\zeta = 0$ , where  $dL_v = L_v d$  was used. For the  $\xi$  terms note  $d\iota_u d\xi = L_u d\xi$  (Cartan's formula and  $d^2 = 0$ ), so they are  $-\iota_{[u,w]} d\xi - \iota_w L_u d\xi + L_u \iota_w d\xi = ([L_u, \iota_w] - \iota_{[u,w]})d\xi = 0$ . (iv) This is  $\mathcal{L}_X \eta = 0$  together with the Leibniz rule for  $\mathcal{L}_X$  acting on the scalar  $\eta_{MN} Y^M Z^N$ .  $\square$

Property (iii) is what Hohm, Hull and Zwiebach call a “Jacobi-like identity” [HHZ §3.3, eqs. (3.31)–(3.32), ll. 977–1003]. It says that  $X \circ$  is a derivation of the product  $\circ$ , or equivalently

$$[\mathcal{L}_X, \mathcal{L}_Y] Z = \mathcal{L}_{X \circ Y} Z, \quad (16.14)$$

which is the closure of the gauge algebra. For an antisymmetric bracket the Leibniz identity *is* the Jacobi identity; the Dorfman bracket satisfies it without being antisymmetric. An algebra with such a product is called a *Leibniz algebra*. Property (i) measures the failure of antisymmetry, and (ii) shows that this failure is harmless: the symmetric part is an exact one-form, and exact one-forms generate no transformation.

## 16.5 The Courant bracket and its Jacobiator

**Definition 16.5** (C-bracket, Courant bracket). The *C-bracket* is the antisymmetric part of the D-bracket,

$$\begin{aligned} [\xi_1, \xi_2]_C^M &= \frac{1}{2} (\mathcal{L}_{\xi_1} \xi_2 - \mathcal{L}_{\xi_2} \xi_1)^M \\ &= \xi_1^N \partial_N \xi_2^M - \xi_2^N \partial_N \xi_1^M - \frac{1}{2} (\xi_1^P \partial^M \xi_{2P} - \xi_2^P \partial^M \xi_{1P}), \end{aligned} \quad (16.15)$$

[HHZ §1, eq. (1.19), ll. 383–400; AMN §3.5, eq. (3.47), ll. 1206–1212]. When nothing depends on  $\tilde{x}$  it becomes the *Courant bracket* on  $T \oplus T^*$ ,

$$[X, Y]_C = [v, u] + L_v \zeta - L_u \xi - \frac{1}{2} d(\iota_v \zeta - \iota_u \xi). \quad (16.16)$$

The first two terms on the right of (16.15) are the Lie bracket in  $2D$  dimensions; the last is the  $\eta$ -dependent correction. To get (16.16) from (16.13), antisymmetrize and use Cartan's formula in the form  $\iota_u d\xi = L_u \xi - d\iota_u \xi$ :

$\frac{1}{2}(X \circ Y - Y \circ X) = [v, u] + \frac{1}{2}(L_v \zeta - \iota_u d\xi) - \frac{1}{2}(L_u \xi - \iota_v d\zeta) = [v, u] + L_v \zeta - L_u \xi + \frac{1}{2} d\iota_u \xi - \frac{1}{2} d\iota_v \zeta$ , which is (16.16). Comparing with (16.5), the Courant bracket is the choice  $\kappa = \frac{1}{2}$  (with the opposite overall sign, because composing field variations reverses the order, as in (16.4)). By Proposition 16.4(i) the two brackets differ by an exact form,

$$X \circ Y = [X, Y]_C + \frac{1}{2} d\langle X, Y \rangle, \quad [A, B]_D^M = [A, B]_C^M + \frac{1}{2} \partial^M (B^N A_N), \quad (16.17)$$

the second form being the doubled statement [HHZ §3.3, eq. (3.30), ll. 955–962]. It was shown by Hull and Zwiebach, in a paper not in this book's library, that the C-bracket reduces to the Courant bracket on any maximal isotropic section, not only on  $\tilde{\partial} = 0$ ; we quote the statement from [HHZ §3.3, ll. 963–976] [L].

### 16.5.1 Failure of the Jacobi identity

**Theorem 16.6** (Jacobiator of the Courant bracket). *For generalized vectors  $X, Y, Z$  on  $M$  define the Jacobiator  $\text{Jac}(X, Y, Z) = [[X, Y]_C, Z]_C + [[Y, Z]_C, X]_C + [[Z, X]_C, Y]_C$  and the function*

$$\text{Nij}(X, Y, Z) = \frac{1}{6} (\langle [X, Y]_C, Z \rangle + \langle [Y, Z]_C, X \rangle + \langle [Z, X]_C, Y \rangle). \quad (16.18)$$

*Then  $\text{Jac}(X, Y, Z) = d\text{Nij}(X, Y, Z)$ . In particular the vector part of the Jacobiator vanishes and its one-form part is exact.*

*Proof.* Abbreviate  $[\cdot, \cdot] = [\cdot, \cdot]_C$  and let  $\sum$  denote the sum over the three cyclic permutations of  $(X, Y, Z)$ . By (16.17) and Proposition 16.4(ii),

$$[[X, Y], Z] = [X, Y] \circ Z - \frac{1}{2} d\langle [X, Y], Z \rangle = (X \circ Y) \circ Z - \underbrace{\frac{1}{2} d\langle X, Y \rangle \circ Z}_{=0} - \frac{1}{2} d\langle [X, Y], Z \rangle.$$

Summing,  $\text{Jac} = S - 3 \text{dNij}$  with  $S = \sum (X \circ Y) \circ Z = \sum [X, Y] \circ Z$ . Now use the Leibniz identity,  $(X \circ Y) \circ Z = X \circ (Y \circ Z) - Y \circ (X \circ Z)$ . Under the cyclic relabelling  $(X, Y, Z) \rightarrow (Z, X, Y)$  the second term becomes  $X \circ (Z \circ Y)$ , so

$$S = \sum X \circ (Y \circ Z - Z \circ Y) = 2 \sum X \circ [Y, Z] =: 2T.$$

On the other hand  $S = \sum [Y, Z] \circ X$  after relabelling, so by Proposition 16.4(i)  $S + T = \sum ([Y, Z] \circ X + X \circ [Y, Z]) = \sum d\langle [Y, Z], X \rangle = 6 \text{dNij}$ . The two relations give  $T = 2 \text{dNij}$ ,  $S = 4 \text{dNij}$  and  $\text{Jac} = (4 - 3) \text{dNij}$ .  $\square$

The doubled version reads  $J(\xi_1, \xi_2, \xi_3)^M = \frac{1}{6} \partial^M (\langle [\xi_1, \xi_2]_C, \xi_3 \rangle + \text{cyclic})$  when the strong constraint holds; AMN write it as one third of the symmetrized D-bracket, which is the same thing because the symmetrized D-bracket is  $\frac{1}{2} \partial^M \langle \cdot, \cdot \rangle$  [AMN §3.5, eqs. (3.49)–(3.51), ll. 1228–1248]. The Jacobiator is a trivial parameter in the sense of Section 16.3(c):  $\mathcal{L}_{\text{Jac}} = 0$ . This is the resolution of the apparent paradox. Operators always satisfy the Jacobi identity,  $[[\mathcal{L}_1, \mathcal{L}_2], \mathcal{L}_3] + \text{cyclic} = 0$ ; with closure in the form  $[\mathcal{L}_1, \mathcal{L}_2] = \mathcal{L}_{[1,2]_C}$  (which follows from (16.14) because the symmetric part acts trivially) this implies  $\mathcal{L}_{\text{Jac}(\xi_1, \xi_2, \xi_3)} = 0$ , which does *not* force  $\text{Jac} = 0$  when the map  $\xi \mapsto \mathcal{L}_\xi$  has a kernel.

#### Where the brackets come from

In the mathematical literature the pairing, the generalized metric and the Courant bracket are the basic structures of generalized geometry, in which coordinates are not doubled but  $T$  and  $T^*$  are combined into  $E = T \oplus T^*$  [HHZ §1, ll. 310–322]. The C-bracket was introduced by Siegel and was recognized by Hull and Zwiebach as the  $O(D, D)$  covariant extension of the Courant bracket to doubled fields [HHZ §1, ll. 399–401]. Independently of this, the cubic double field theory derived from closed string field theory already showed a bracket of gauge parameters that fails the Jacobi identity; Hull and Zwiebach attribute this to the homotopy Lie algebra structure of closed string field theory [HZ §1, ll. 400–406; §3.1, eqs. (3.7)–(3.9), ll. 1713–1763].

### 16.5.2 A worked example with numbers

**Example 16.7** (A non-zero Jacobiator on the plane). On  $M = \mathbb{R}^2$  with coordinates  $(x^1, x^2)$  take

$$X = \partial_1, \quad Y = \frac{1}{2}(x^1)^2 dx^2, \quad Z = \partial_2.$$

For a vector  $v$  and a one-form  $\zeta$ , (16.16) gives  $[v, \zeta]_C = L_v \zeta - \frac{1}{2} d\iota_v \zeta$  and  $[\zeta, v]_C = -[v, \zeta]_C$ .

*First brackets.*  $L_{\partial_1} Y = x^1 dx^2$  and  $\iota_{\partial_1} Y = 0$ , so  $[X, Y]_C = x^1 dx^2$ . Next  $L_{\partial_2} Y = 0$  and  $\iota_{\partial_2} Y = \frac{1}{2}(x^1)^2$ , so  $[Y, Z]_C = -(0 - \frac{1}{2} d\frac{1}{2}(x^1)^2) = \frac{1}{2} x^1 dx^1$ . Finally  $[Z, X]_C = [\partial_2, \partial_1] = 0$ .

*Second brackets.*  $[[X, Y]_C, Z]_C = -(L_{\partial_2}(x^1 dx^2) - \frac{1}{2} d(x^1)) = \frac{1}{2} dx^1$ , and  $[[Y, Z]_C, X]_C = -(L_{\partial_1}(\frac{1}{2} x^1 dx^1) - \frac{1}{2} d(\frac{1}{2} x^1)) = -\frac{1}{2} dx^1 + \frac{1}{4} dx^1 = -\frac{1}{4} dx^1$ . Hence

$$\text{Jac}(X, Y, Z) = \frac{1}{2} dx^1 - \frac{1}{4} dx^1 + 0 = \frac{1}{4} dx^1 \neq 0.$$

*Check against Theorem 16.6.* The three pairings are

$$\langle [X, Y]_C, Z \rangle = \iota_{\partial_2}(x^1 dx^2) = x^1, \quad \langle [Y, Z]_C, X \rangle = \frac{1}{2} x^1, \quad \langle [Z, X]_C, Y \rangle = 0,$$

so  $N_{ij} = \frac{1}{6} \cdot \frac{3}{2} x^1 = \frac{1}{4} x^1$  and  $dN_{ij} = \frac{1}{4} dx^1$ , in agreement. At the point  $x^1 = 2$  one has  $N_{ij} = \frac{1}{2}$ . The Dorfman products of the same three objects are  $X \circ Y = x^1 dx^2$ ,  $Y \circ X = -x^1 dx^2$ ,  $Y \circ Z = x^1 dx^1$ ,  $Z \circ Y = 0$ : the pair  $(Y, Z)$  shows the failure of antisymmetry,  $Y \circ Z + Z \circ Y = d(\frac{1}{2}(x^1)^2) = d\langle Y, Z \rangle$ , as Proposition 16.4(i) requires. (All values in this example were confirmed by a symbolic check in sympy.)

## 16.6 Closure on the doubled space needs the strong constraint

On  $T \oplus T^*$  closure, (16.14), was a consequence of Cartan calculus. On the doubled space it is not automatic. A direct computation with (16.9) and (16.15) gives, for arbitrary dependence on  $(x, \tilde{x})$ ,

$$([\mathcal{L}_{\xi_1}, \mathcal{L}_{\xi_2}] - \mathcal{L}_{[\xi_1, \xi_2]_C}) V^M = F^M, \quad (16.19)$$

$$F^M = \frac{1}{2} (\xi_1^N \partial^Q \xi_{2N} - \xi_2^N \partial^Q \xi_{1N}) \partial_Q V^M - (\partial^Q \xi_1^M \partial_Q \xi_{2P} - \partial^Q \xi_2^M \partial_Q \xi_{1P}) V^P. \quad (16.20)$$

Every term of  $F^M$  contains a contraction  $\partial^Q A \partial_Q B = \eta^{MN} \partial_M A \partial_N B$  of derivatives of two *different* objects. This is exactly the expression that the strong constraint sets to zero (Chapter 14); the weak constraint  $\partial^Q \partial_Q A = 0$  on each object separately is of no help. Hence:

**Proposition 16.8** (Closure). *If all fields and gauge parameters satisfy the strong constraint, then  $[\mathcal{L}_{\xi_1}, \mathcal{L}_{\xi_2}] = \mathcal{L}_{[\xi_1, \xi_2]_C}$  on every generalized tensor.* [L]

[HHZ §3.2, eqs. (3.23)–(3.28), ll. 883–942; AMN §3.5, eqs. (3.44)–(3.48), ll. 1174–1226]. The extension from vectors to arbitrary tensors uses only the Leibniz rule and  $\mathcal{L}_\xi \eta = 0$ . Because  $\mathcal{L}_{\partial_X} = 0$ , one may equally write  $\mathcal{L}_{[\xi_1, \xi_2]_D}$  on the right. We checked (16.19)–(16.20) symbolically for  $D = 1$  and  $D = 2$  with arbitrary functions as components (a symbolic check, not Tier A).

*Remark 16.9* (A sign in the sources). In the extracted text of HHZ the closure relation is printed with a minus sign,  $[\hat{\mathcal{L}}_{\xi_1}, \hat{\mathcal{L}}_{\xi_2}] = -\hat{\mathcal{L}}_{[\xi_1, \xi_2]_C}$ , together with  $[\delta_{\xi_1}, \delta_{\xi_2}] = \delta_{\xi_{12}}$ ,  $\xi_{12} = -[\xi_1, \xi_2]_C$ , and the violation term (3.24) has the opposite overall sign to (16.20) [HHZ §3.2, ll. 883–942]. AMN print the plus sign, eq. (3.48). With the definitions (16.9), (16.15) and composition of *operators*, the symbolic check fixes the plus sign. Composition of *field variations* reverses the order, as we saw in (16.4); reading the HHZ commutator in that order accounts for both signs at once. The structure of the result is the same in both sources.

**Example 16.10** (Closure fails without the strong constraint). Take  $D = 1$ , coordinates  $(x, \tilde{x})$ , and components ordered as  $(v, \xi)$ :

$$\xi_1 = (0, x), \quad \xi_2 = (\tilde{x}, 0), \quad V = (1, 0).$$

Each component is annihilated by  $\partial^Q \partial_Q = 2\partial_x \tilde{\partial}$ , so the weak constraint holds. The strong constraint fails for the pair: contracting the derivatives of the component  $x$  of  $\xi_1$  with those of the component  $\tilde{x}$  of  $\xi_2$  gives  $\partial^Q(x) \partial_Q(\tilde{x}) = \partial_x(x) \tilde{\partial}(\tilde{x}) = 1$ . With  $\xi_{1M} = (x, 0)$  and  $\xi_{2M} = (0, \tilde{x})$  we evaluate (16.20). The first term of (16.20) contains  $\partial_Q V^M$  and vanishes because  $V$  is constant. In the second term only  $P = 1$  contributes ( $V^P = \delta_1^P$ ):  $\partial^Q \xi_1^M \partial_Q \xi_{21} = 0$  because  $\xi_{21} = 0$ , while  $\partial^Q \xi_2^M \partial_Q \xi_{11}$  is non-zero only for  $M = 1$ , where it equals  $\partial \tilde{x} \cdot \partial_x x = 1$ . Hence

$$F^M = -(0 - 1) \delta_1^M = (1, 0) \neq 0.$$

A direct evaluation confirms it:  $\mathcal{L}_{\xi_1} \xi_2 = (x, 0)$ ,  $\mathcal{L}_{\xi_2} \xi_1 = (0, \tilde{x})$ ,  $[\xi_1, \xi_2]_C = \frac{1}{2}(x, -\tilde{x})$ , and the two sides of the would-be closure relation differ by the constant generalized vector  $(1, 0)$ . The commutator of two gauge transformations is then *not* a gauge transformation, and an action invariant under each  $\mathcal{L}_\xi$  would have no reason to exist.

**Physical picture: why closure is non-negotiable**

A gauge symmetry removes unphysical polarizations. If  $\delta_1$  and  $\delta_2$  are symmetries of an action, so is  $[\delta_1, \delta_2]$ ; if that commutator is not of the form  $\delta_{12}$ , the theory has more invariances than parameters, or the “symmetries” are not symmetries of any action. Supergravity is safe, because diffeomorphisms and two-form gauge transformations do form a group. Double field theory inherits this safety on every section, and the strong constraint is the  $O(D, D)$  invariant way of saying “on some section”. AMN stress that the closure condition itself,  $F^M = 0$ , is weaker than the strong constraint and leaves room for configurations that violate the strong constraint, which they use later in dimensional reductions of double field theory [AMN §3.3, ll. 1070–1082; §3.5, ll. 1250–1262] [L].

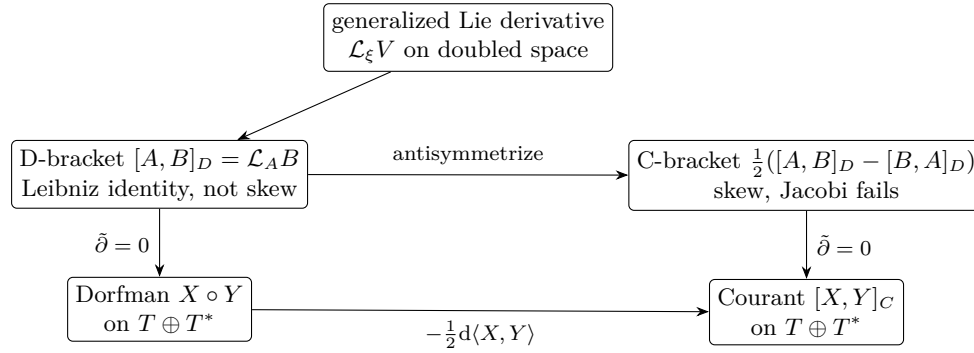


Figure 16.2: The four brackets of this chapter. Vertical arrows impose the strong constraint in the supergravity frame; horizontal arrows remove the symmetric part, which is a trivial gauge parameter.

### 16.6.1 $B$ -transforms as symmetries of the bracket

Which fibrewise  $O(D, D)$  rotations of (16.7) are compatible with the bracket? Take a two-form  $B(x)$ , not necessarily closed, acting by  $e^B(v + \xi) = v + \xi + \iota_v B$  with  $(\iota_v B)_j = v^i B_{ij}$  (the matrix of (16.7) with lower-left block  $B^\top$ ). Then

$$e^B X \circ e^B Y = e^B (X \circ Y) + \iota_u \iota_v dB, \quad (\iota_u \iota_v dB)_k = v^i u^j (dB)_{ijk}. \quad (16.21)$$

The proof is a short application of (16.3): the new terms on the left are  $L_v \iota_u B - \iota_u d\iota_v B$ , the new term on the right is  $\iota_{[v, u]} B$ , and  $L_v \iota_u - \iota_{[v, u]} = \iota_u L_v = \iota_u (d\iota_v + \iota_v d)$ . (Standard; confirmed by a symbolic check for  $D = 3$ , not taken from a source in this book’s library.) Thus a *closed*  $B$  is a symmetry of the Dorfman bracket, hence of the Courant bracket. Diffeomorphisms and closed  $B$ -transforms are exactly the symmetries we started from in Section 16.1, since an exact  $B = d\tilde{\lambda}$  is closed: the bracket is an algebraic structure on  $T \oplus T^*$  whose automorphisms contain the gauge group of the  $(g, b)$  system. (That there are no further automorphisms preserving the pairing is a standard result of generalized geometry; it is not checked against a source in this book’s library.) If  $dB = H \neq 0$ , (16.21) defines the  $H$ -twisted bracket  $X \circ_H Y = X \circ Y + \iota_u \iota_v H$ , appropriate when  $b$  is defined only patchwise and  $H$  is a non-trivial flux (Chapter 25 meets such fluxes in a different setting).

## 16.7 What the machine has checked

Two modules of the older `DoubleFieldTheory` library belong to this chapter:



DoubleFieldTheory/GeneralizedGeometry.lean,  
DoubleFieldTheory/CourantAlgebroid.lean.

Both model a generalized vector by a *pair of integers*. There are no manifolds, no functions and no derivatives in these files; in the terminology of this book they are *arithmetic shadows*. All theorems quoted below are proved by unfolding definitions, an occasional rewrite with commutativity, and `omega` or `rfl`; the three we inspected with `#print axioms` depend on `propext` and `Quot.sound` only.

### 16.7.1 The pairing and $\eta$

#### In Lean: generalized vectors and the natural pairing

```
structure GenVector where
  v : Int
  xi : Int
  deriving Repr, DecidableEq

def CourantPairing (X Y : GenVector) : Int :=
  X.xi * Y.v + Y.xi * X.v

theorem courant_pairing_symm (X Y : GenVector) :
  CourantPairing X Y = CourantPairing Y X := by ...

def ODD_Eta : Mat2 := { a := 0, b := 1, c := 1, d := 0 }

theorem odd_metric_eval (X Y : GenVector) :
  BilinearForm ODD_Eta X Y = CourantPairing X Y := by ...
```

**Reading.** For  $D = 1$  and integer components, (16.6) is symmetric, and it equals  $X^T \eta Y$  with  $\eta$  the  $2 \times 2$  matrix of (16.6). `Mat2` is a structure with four integer entries and `BilinearForm` spells out  $X^T M Y$ . **Proof idea.** Unfold the definitions; what remains is commutativity of integer multiplication, which `omega` handles after one explicit `Int.mul_comm`. This is a faithful formalization of the pointwise,  $D = 1$  linear algebra of Definition 16.1. **[A]**

#### In Lean: the $O(1,1)$ condition, inversion and the “ $B$ -twist”

```
def IsODD (M : Mat2) : Prop :=
  MatMul (MatTranspose M) (MatMul ODD_Eta M) = ODD_Eta

theorem odd_inversion_generator : IsODD InversionGen := by
  rfl

def BTwist (b : Int) : Mat2 :=
  { a := 1, b := 0, c := b, d := 1 }

theorem btwist_preserves_eta (b : Int) :
  (MatMul (MatTranspose (BTwist b)) (MatMul ODD_Eta (BTwist b))).b
  = ODD_Eta.b ∧
  (MatMul (MatTranspose (BTwist b)) (MatMul ODD_Eta (BTwist b))).c
  = ODD_Eta.c := by ...
```

**Reading.** `IsODD` is  $M^T \eta M = \eta$  for integer  $2 \times 2$  matrices. `InversionGen` is the matrix  $\eta$  itself, which exchanges  $v$  and  $\xi$  (the T-duality of Chapter 8), and it satisfies the condition. For `BTwist` the theorem asserts *only the two off-diagonal entries* of  $M^T \eta M$ . That is all that is true: by the pitfall box in Section 16.2 the upper-left entry is  $2b$ , so `IsODD (BTwist`



b) holds only for  $b = 0$ . The theorem is correct as stated, but it must not be read as “the  $B$ -shift is an  $O(D, D)$  element”: for that one needs  $D \geq 2$  and an antisymmetric block, as in (16.7). [A] for the literal statement. The same file also contains a diagonal generalized metric, an “energy”  $|v| + |\xi|$  and an arithmetic identity standing in for chiral projectors; the generalized metric proper is treated, over  $\mathbb{R}$  and with matrices, by the newer module described in Chapter 15.

## 16.7.2 The bracket models

### In Lean: CourantSection, CBracket and its antisymmetry

```
structure CourantSection where
  v : Int
  alpha : Int
  deriving Repr, DecidableEq

def CBracket (X Y : CourantSection) : CourantSection :=
  { v := X.v * Y.v - Y.v * X.v,
    alpha := X.v * Y.alpha - Y.v * X.alpha }

theorem cbracket_antisymm (X Y : CourantSection) :
  (CBracket X Y).alpha = - ((CBracket Y X).alpha) := by ...

theorem cbracket_v_zero (X Y : CourantSection) :
  (CBracket X Y).v = 0 := by ...

theorem jacobiator_vector_vanishes (X Y Z : CourantSection) :
  JacVector X Y Z = 0 := by ...
```

**Reading.** A section is a pair of integers  $(v, a)$ . The “Lie bracket” slot is  $vu - uv$ , which is zero for integers, and the form slot is  $vb - ua$ . The first theorem says the form slot is antisymmetric; the second that the vector slot is always 0; the third that the sum of the vector slots of the three double brackets is 0, which follows at once from the second.

**Proof idea.** Unfold, rewrite with commutativity, **omega**. [A]

Is there a geometric situation that this integer model describes exactly? Yes, a small one. On the real line take the two-parameter family

$$X_{(v,a)} = v \partial_x + a e^{2x} dx, \quad v, a \in \mathbb{Z}.$$

For  $D = 1$  the form part of (16.16) is  $(v\beta)' - (u\alpha)' - \frac{1}{2}(v\beta - u\alpha)' = \frac{1}{2}(v\beta - u\alpha)'$  for constant  $v, u$ ; with  $\alpha = ae^{2x}$ ,  $\beta = be^{2x}$  this is  $(vb - ua)e^{2x}$ , and the vector part  $[v\partial_x, u\partial_x]$  vanishes. So  $[X_{(v,a)}, X_{(u,b)}]_C = X_{(0, vb-ua)}$ , which is **CBracket**. On this family the Courant bracket is a Lie bracket, because  $N_{ij}$  vanishes identically (Exercise 16.5), so the model cannot exhibit Theorem 16.6: in Lean the vector part of the Jacobiator vanishes because *every* vector part vanishes, not because of the mechanism in our proof. The identification of **CBracket** with this family is our reading, not something the Lean file states.

### In Lean: the integer “Dorfman bracket”

```
def DorfmanBracket (X Y : CourantSection) : CourantSection :=
  { v := X.v * Y.v - Y.v * X.v,
    alpha := 2 * (X.v * Y.alpha) - (Y.v * X.alpha) }

theorem dorfman_cbracket_diff (X Y : CourantSection) :
  (DorfmanBracket X Y).alpha - (CBracket X Y).alpha
```

```

    = X.v * Y.alpha := by ...

theorem dorfman_symmetric_exact (X Y : CourantSection) :
  (DorfmanBracket X Y).alpha + (DorfmanBracket Y X).alpha =
  (X.v * Y.alpha + Y.v * X.alpha) := by ...

```

**Reading.** Two ring identities:  $(2vb - ua) - (vb - ua) = vb$  and  $(2vb - ua) + (2ua - vb) = vb + ua$ . The right-hand side of the second has the form of the pairing (16.6), so it is the integer shadow of Proposition 16.4(i) with  $d$  replaced by the identity. **[A]** as arithmetic. **What it is not.** In the geometric theory (16.17) forces the difference  $X \circ Y - [X, Y]_C$  to be *half* the symmetric part, i.e. symmetric in  $X, Y$ ; here it is  $vb$ , which is not symmetric. Equivalently, the antisymmetric part of this `DorfmanBracket` is  $3(vb - ua)$ , three times `CBracket` rather than twice. So the pair  $(\text{DorfmanBracket}, \text{CBracket})$  does not satisfy Definition 16.5. On the family  $X_{(v,a)}$  above, the true Dorfman product is  $X_{(v,a)} \circ X_{(u,b)} = X_{(0, 2vb)}$ , because  $d\alpha = 0$  on a line; the model with `alpha := 2 * (X.v * Y.alpha)` would satisfy both relations (Exercise 16.7).

#### In Lean: section condition and “closure”

```

structure FieldDeriv where
  dx : Int
  dtx : Int
  deriving Repr, DecidableEq

def SectionContract (Phi Psi : FieldDeriv) : Int :=
  Phi.dx * Psi.dtx + Phi.dtx * Psi.dx

theorem strong_section_condition (Phi Psi : FieldDeriv)
  (hPhi : Phi.dtx = 0) (hPsi : Psi.dtx = 0) :
  SectionContract Phi Psi = 0 := by ...

def LieCommutator (Lx Ly : Int → Int) (f : Int) : Int :=
  Lx (Ly f) - Ly (Lx f)

theorem gen_lie_closure (a b : Int) (f : Int) :
  LieCommutator (fun x => a * x) (fun x => b * x) f = 0 := by ...

```

**Reading.** `FieldDeriv` records two integers, standing for the values of  $\partial_x \Phi$  and  $\tilde{\partial} \Phi$  at a point for  $D = 1$ . The first theorem says: if both tilde-derivatives are 0 then  $\partial_x \Phi \tilde{\partial} \Psi + \tilde{\partial} \Phi \partial_x \Psi = 0$ . This is the easy direction “supergravity frame  $\Rightarrow$  strong constraint”, pointwise; the converse classification of solutions is the subject of Chapter 14 and its own Lean module. The second theorem says that multiplication by  $a$  and multiplication by  $b$  commute on  $\mathbb{Z}$ . Despite its name it contains no generalized Lie derivative and no bracket; it is not a formalization of Proposition 16.8. **[A]** for the literal statements.

| Claim                                                                           | Where                                   | Tier       |
|---------------------------------------------------------------------------------|-----------------------------------------|------------|
| Pairing symmetric, equals $X^\top \eta Y$ ( $D = 1, \mathbb{Z}$ )               | <code>courant_pairing_symm</code> ,     | <b>[A]</b> |
| $\eta \in O(1, 1; \mathbb{Z})$                                                  | <code>odd_metric_eval</code>            |            |
| Off-diagonal entries of $(e^B)^\top \eta e^B$                                   | <code>odd_inversion_generator</code>    | <b>[A]</b> |
| Integer toy bracket: skew form slot, zero vector slot                           | <code>btwist_preserves_eta</code>       | <b>[A]</b> |
|                                                                                 | <code>cbracket_antisymm</code> ,        | <b>[A]</b> |
|                                                                                 | <code>cbracket_v_zero</code> ,          |            |
|                                                                                 | <code>jacobiator_vector_vanishes</code> |            |
| $\tilde{\partial} = 0 \Rightarrow$ contraction vanishes ( $D = 1$ , point-wise) | <code>strong_section_condition</code>   | <b>[A]</b> |
| $\mathcal{L}_\xi \eta = 0$ ; reduction (16.11), (16.12)                         | HHZ §3.2, AMN §3.4                      | <b>[L]</b> |
| Leibniz identity; $\text{Jac} = \text{dNij}$                                    | proved above; HHZ §3.3, AMN §3.5        | <b>[L]</b> |
| Closure under the strong constraint                                             | HHZ §3.2, AMN §3.5                      | <b>[L]</b> |
| Formula (16.20), identity (16.21)                                               | symbolic check (sympy)                  | —          |

**What a faithful formalization would need.** The brackets are first-order bidifferential operators. A Lean statement of Proposition 16.4 needs either (a) smooth vector fields and differential forms on a manifold with  $\text{d}$ ,  $\iota$ ,  $L$  and the identities (16.3), or (b) an algebraic substitute: a commutative ring  $R$  with commuting derivations  $\partial_1, \dots, \partial_D$  (polynomials suffice for Example 16.7), generalized vectors as elements of  $R^{2D}$ , and (16.9) as a definition. Route (b) is within reach of `ring`-style automation; Theorem 16.6 then becomes a finite polynomial identity in the components and their first and second derivatives. Neither route is carried out in the present library **[C]** (a proposal, recorded again in Chapter 38).

#### Chapter summary

- A generalized vector  $V = v + \xi$  is a section of  $T \oplus T^*$ . The natural pairing  $\langle X, Y \rangle = \iota_v \zeta + \iota_\xi \eta$  has signature  $(D, D)$  and structure group  $O(D, D)$ .
- The generalized Lie derivative (16.9) is transport plus an  $\mathfrak{so}(D, D)$  rotation. It preserves  $\eta$ , and for  $\tilde{\partial} = 0$  it unifies diffeomorphisms with  $b \mapsto b + \text{d}\tilde{\lambda}$ .
- The D-bracket (Dorfman bracket)  $\mathcal{L}_X Y$  obeys the Leibniz identity but is not skew; its symmetric part is  $\text{d}\langle X, Y \rangle$ .
- The C-bracket (Courant bracket) is its skew part. Its Jacobiator is  $\text{dNij}$ , a trivial gauge parameter, so the gauge *transformations* still compose consistently.
- On the doubled space, closure  $[\mathcal{L}_1, \mathcal{L}_2] = \mathcal{L}_{[1,2]_C}$  fails by terms  $\partial^Q A \partial_Q B$  and holds under the strong constraint.
- Closed  $B$ -transforms and diffeomorphisms are the symmetries of the bracket; a non-closed  $B$  twists the bracket by  $H = \text{d}B$ .
- Lean: the  $D = 1$  integer pairing and  $\eta$  are formalized faithfully. The bracket theorems are arithmetic statements about pairs of integers; they capture skewness of one slot and nothing of the differential structure. The integer `DorfmanBracket` is not the symmetrization partner of `CBracket`.

## Exercises

**Exercise 16.1** ( $\star$ ). Diagonalize  $\eta$  for  $D = 1$  and exhibit two generalized vectors of norm  $\langle X, X \rangle = +2$  and  $-2$ . For general  $D$ , show that  $T$  and  $T^*$  are maximal isotropic, and that the graph  $\{v + Bv\}$  of a two-form  $B$  is isotropic if and only if  $B$  is antisymmetric.

**Exercise 16.2** (\*). Verify Cartan’s formula  $L_v = d\iota_v + \iota_v d$  on a one-form in components, and deduce  $d\iota_u d\xi = L_u d\xi$ , the step used in the proof of Proposition 16.4(iii).

**Exercise 16.3** (\*). Starting from (16.13), show that the Dorfman bracket differs from (16.5) in that the exact term is added with coefficient 1 for one of the two parameters and 0 for the other. Which is which, and why is this asymmetric choice the one that makes (16.14) hold exactly rather than up to trivial parameters?

**Exercise 16.4** (\*\*). Derive the second form of (16.17) directly from (16.9) and (16.15), without setting  $\hat{\partial} = 0$ . Conclude that  $[A, B]_D + [B, A]_D = \partial^M (A^N B_N)$  holds with no constraint at all.

**Exercise 16.5** (\*\*). For the family  $X_{(v,a)} = v\partial_x + ae^{2x}dx$  on the line, compute  $N_{ij}$  of three members  $X_{(v,a)}$ ,  $X_{(u,b)}$ ,  $X_{(w,c)}$  from (16.18) and show that it vanishes identically. Then replace  $e^{2x}$  by  $e^{kx}$ : for which  $k$  does the family close under the Courant bracket, and what replaces  $vb - ua$ ?

**Exercise 16.6** (\*\*). Repeat Example 16.10 with  $\xi_2 = (x, 0)$  in place of  $(\tilde{x}, 0)$ . Show that the strong constraint now holds for all pairs among  $\xi_1, \xi_2, V$  and that  $F^M = 0$ . Then apply the exchange  $v \leftrightarrow \xi$  (the matrix  $\eta$ ) to all three objects and to the coordinates, and check that closure still holds: the constraint and the bracket are  $O(D, D)$  covariant.

**Exercise 16.7** (\*\*, Lean). (a) State and prove in Lean that the antisymmetric part of the library’s integer Dorfman bracket is three times the C-bracket:

```
theorem dorfman_antisymm_three (X Y : CourantSection) :
  (DorfmanBracket X Y).alpha - (DorfmanBracket Y X).alpha
    = 3 * (CBracket X Y).alpha := by
  sorry
```

(b) Define `DorfmanExp X Y` with form slot  $2 * (X.v * Y.alpha)$  and vector slot 0, and prove that its antisymmetric part is  $2 * (CBracket X Y).alpha$  and its symmetric part is twice the pairing  $vb + ua$ . Hint: `dsimp` with the definitions, then `omega`. (We have checked that both statements compile against the library.)

**Exercise 16.8** (\*\*, Lean). Prove  $\neg \text{IsODD (BTwist 1)}$  and  $\text{IsODD (BTwist 0)}$ . Hint: `unfold IsODD` and decide for the first, `rfl` for the second. Then prove that the upper-left entry of  $(e^B)^\top \eta e^B$  is  $2 * b$  for all  $b : \text{Int}$ .

**Exercise 16.9** (\*\*\*). Prove (16.21) in full, and use it with Proposition 16.4 to show that the  $H$ -twisted Dorfman bracket  $X \circ_H Y = X \circ Y + \iota_u \iota_v H$  satisfies the Leibniz identity if and only if  $dH = 0$ . This is the Bianchi identity of the three-form flux, recovered from the consistency of a bracket.

**Exercise 16.10** (\*\*\*, Lean). Carry out route (b) of Section 16.7 in the smallest case. Over polynomials in one variable, define generalized vectors as pairs of polynomials, the Dorfman product by  $(v, \alpha) \circ (u, \beta) = (vu' - uv', (v\beta)')$  (why is the  $\iota_u d\alpha$  term absent for  $D = 1$ ?), and prove the Leibniz identity. Which Mathlib structure provides the derivative and its product rule?

## Further reading

- Generalized Lie derivative,  $\mathcal{L}_\xi \eta = 0$ , closure and the violation term: [HHZ §3.1–3.2, ll. 670–942]. D-bracket, its symmetric part and Jacobi-like identity, relation to the Dorfman bracket: [HHZ §3.3, ll. 946–1003].

- A pedagogical account with weights, the tensor  $Y^{MN}{}_{PQ} = \eta^{MN}\eta_{PQ}$ , the closure constraint and the Jacobiator: [AMN §3.4–3.5, ll. 1087–1262]; the supergravity symmetries being unified: [AMN §3.1, ll. 840–862].
- The string field theory origin of a gauge algebra that fails the Jacobi identity: [HZ §3.1, ll. 1705–1763].
- The mathematical literature on Courant algebroids and generalized complex geometry (Hitchin; Gualtieri; Courant; Roytenberg) is cited by [HHZ refs. [15]–[17], ll. 2505–2522; AMN ref. [84], ll. 5803–5804] but is not in this book’s library; statements attributed to it above are derived on the page.
- Next: the gauge-invariant action built from  $\mathcal{H}$  and  $d$ , Chapter 17.



# Chapter 17

## The Double Field Theory Action

The last three chapters assembled a kinematics: doubled coordinates subject to the strong constraint (Chapter 14), a generalized metric  $\mathcal{H}$  that packages the metric  $g$  and the Kalb–Ramond field  $b$  into one  $O(D, D)$  tensor (Chapter 15), and a gauge algebra governed by the C-bracket (Chapter 16). None of this is dynamics yet. The question of this chapter is: *is there an action, written only with  $\mathcal{H}$ , the derivatives  $\partial_M$  and one scalar, whose every index is contracted with an  $O(D, D)$ -invariant tensor, and which becomes the familiar low-energy action of the string when nothing depends on the winding coordinates?* The answer is yes, and the action has four terms. A student should care for two reasons. First, the construction explains a fact that looks accidental in Chapter 6: the relative coefficients  $1 : 4 : -\frac{1}{12}$  of  $R$ ,  $(\partial\phi)^2$  and  $H^2$  in the NS–NS action are forced by duality. Second, the chapter is a clean case study of what the word “formalized” can and cannot mean today: the Lean files attached to this chapter prove statements about integers, and we shall say precisely how far those are from the action itself.

Throughout,  $D$  is the number of spacetime dimensions, indices  $i, j, \dots = 1, \dots, D$  label ordinary coordinates,  $M, N, \dots = 1, \dots, 2D$  label doubled ones, and we drop the overall constant  $1/2\kappa_D^2$  in front of every action, as our sources do. The signature is mostly plus and  $\sqrt{g}$  is shorthand for  $\sqrt{|\det g_{ij}|}$ .

### 17.1 A dilaton that T-duality leaves alone

The low-energy action for the massless NS–NS fields of the closed string is

$$S_{\text{NS}} = \int d^D x \sqrt{g} e^{-2\phi} \left[ R + 4(\partial\phi)^2 - \frac{1}{12} H_{ijk} H^{ijk} \right], \quad H_{ijk} = \partial_i b_{jk} + \partial_j b_{ki} + \partial_k b_{ij}, \quad (17.1)$$

where  $R$  is the Ricci scalar of  $g_{ij}$ ,  $\phi$  is the dilaton and  $H = db$  is the field strength of the Kalb–Ramond field [AMN §3.2, eqs. (3.17)–(3.18), ll. 868–884; BW, eq. (7), ll. 343–361]. Its equations of motion are the conditions for Weyl invariance of the worldsheet theory at one loop (Chapter 6).

Look at the measure  $\sqrt{g} e^{-2\phi}$ . Neither factor is invariant under T-duality. Take one direction to be a circle of radius  $R$  and use a coordinate  $y \sim y + 2\pi\sqrt{\alpha'}$ , so that  $g_{yy} = r^2$  with  $r = R/\sqrt{\alpha'}$  dimensionless. T-duality sends  $r \mapsto 1/r$ , so  $\sqrt{g} \mapsto \sqrt{g}/r^2$ . But the Buscher rules of Chapter 9 also shift the dilaton,  $\phi \mapsto \phi - \ln r$ , because the string coupling measured by a lower-dimensional observer must not change. Hence  $e^{-2\phi} \mapsto r^2 e^{-2\phi}$  and the *product* is invariant:

$$\sqrt{g} e^{-2\phi} \mapsto \frac{\sqrt{g}}{r^2} \cdot r^2 e^{-2\phi} = \sqrt{g} e^{-2\phi}. \quad (17.2)$$

This suggests taking the product as the fundamental variable.

**Definition 17.1** (duality-invariant dilaton). The field  $d$  defined by

$$e^{-2d} = \sqrt{g} e^{-2\phi} \quad \Longleftrightarrow \quad d = \phi - \frac{1}{4} \ln |\det g| \quad (17.3)$$

is called the  $O(D, D)$  dilaton, or shifted dilaton. [AMN §3.3, eq. (3.24), ll. 956–965]

Under a global  $O(D, D)$  transformation  $X' = hX$  the field  $d$  is a scalar,  $d'(X') = d(X)$ , while  $g$  and  $b$  transform by the fractional linear rule of Chapter 11 [HHZ §1, eq. (1.3), ll. 141–149]. The price is that  $d$  is not a scalar under diffeomorphisms:  $e^{-2d}$  is a *density*. Under an infinitesimal diffeomorphism generated by a vector field  $\xi^i$  one has  $\delta\phi = \xi^i \partial_i \phi$  and  $\delta\sqrt{g} = \partial_i(\xi^i \sqrt{g})$ , the latter because  $\delta g_{ij} = \nabla_i \xi_j + \nabla_j \xi_i$  gives  $\delta\sqrt{g} = \frac{1}{2} \sqrt{g} g^{ij} \delta g_{ij} = \sqrt{g} \nabla_i \xi^i$ . Therefore

$$\delta(e^{-2d}) = \partial_i(\xi^i e^{-2d}), \quad \text{equivalently} \quad \delta d = \xi^i \partial_i d - \frac{1}{2} \partial_i \xi^i. \quad (17.4)$$

The doubled version simply replaces  $i$  by  $M$ : with a gauge parameter  $\xi^M = (\tilde{\xi}_i, \xi^i)$  that unifies the one-form parameter of  $b$ -field gauge transformations with the vector field of diffeomorphisms,

$$\delta_\xi(e^{-2d}) = \partial_M(\xi^M e^{-2d}), \quad \delta_\xi d = \xi^M \partial_M d - \frac{1}{2} \partial_M \xi^M \quad (17.5)$$

[HHZ §1, eq. (1.8), ll. 216–230; §4.3, eq. (4.29), ll. 1348–1353; AMN §3.4, eq. (3.34), ll. 1100–1112]. A density is exactly what an integration measure must be: if a Lagrangian is  $e^{-2d}$  times a gauge scalar  $\mathcal{S}$  (meaning  $\delta_\xi \mathcal{S} = \xi^M \partial_M \mathcal{S}$ ), then  $\delta_\xi(e^{-2d} \mathcal{S}) = \partial_M(\xi^M e^{-2d} \mathcal{S})$  is a total derivative and the action is invariant. This one-line argument is the backbone of the proof of gauge invariance in Section 17.5.

#### Physical picture: what $d$ measures

Compactify on a torus of volume  $V$ . The  $D$ -dimensional action (17.1) becomes a lower-dimensional one with prefactor  $V e^{-2\phi}$ : the effective gravitational coupling of the observer who cannot resolve the torus. T-duality changes the torus and the ten-dimensional coupling  $g_s = e^\phi$  separately, but it is an exact symmetry of string perturbation theory, so it cannot change what the lower-dimensional observer measures. The field  $d$  is that invariant coupling, promoted to a local field. String gas cosmology (Chapter 32) uses the same variable under the name “shifted dilaton”,  $\varphi = 2\phi - \ln V = 2d$  [BW, eq. (9), ll. 395–404].

## 17.2 Ingredients and index conventions

In this chapter we follow the ordering of HHZ and AMN, winding coordinate first:

$$X^M = \begin{pmatrix} \tilde{x}_i \\ x^i \end{pmatrix}, \quad \partial_M = \begin{pmatrix} \tilde{\partial}_i \\ \partial_i \end{pmatrix}, \quad \eta_{MN} = \begin{pmatrix} 0 & \mathbf{1} \\ \mathbf{1} & 0 \end{pmatrix} = \eta^{MN} \quad (17.6)$$

[HHZ §1, eqs. (1.4)–(1.6), ll. 150–188]. Indices are raised and lowered with  $\eta$  only, never with  $\mathcal{H}$ . With this ordering the generalized metric with *upper* indices is the matrix that Chapter 15 calls  $\mathcal{H}(G, B)$ ,

$$\mathcal{H}^{MN} = \begin{pmatrix} g_{ij} - b_{ik} g^{kl} b_{lj} & b_{ik} g^{kj} \\ -g^{ik} b_{kj} & g^{ij} \end{pmatrix}, \quad \mathcal{H}_{MN} = \eta_{MP} \mathcal{H}^{PQ} \eta_{QN} = \begin{pmatrix} g^{ij} & -g^{ik} b_{kj} \\ b_{ik} g^{kj} & g_{ij} - b_{ik} g^{kl} b_{lj} \end{pmatrix}, \quad (17.7)$$

and the second is the inverse of the first,  $\mathcal{H}^{MP} \mathcal{H}_{PN} = \delta^M_N$  [HHZ §1, eqs. (1.9)–(1.11), ll. 256–291]. The only property of the block form that we need repeatedly is this: *when  $\tilde{\partial}^i = 0$ , a derivative index  $M$  is forced into the lower block, and the lower-right block of  $\mathcal{H}^{MN}$  is  $g^{ij}$ .* For instance

$$\mathcal{H}^{MN} \partial_M \partial_N f = (g - b g^{-1} b)_{ij} \tilde{\partial}^i \tilde{\partial}^j f + 2 (b g^{-1})^j_i \tilde{\partial}^i \partial_j f + g^{ij} \partial_i \partial_j f \xrightarrow{\tilde{\partial}=0} g^{ij} \partial_i \partial_j f. \quad (17.8)$$



In the opposite frame,  $\partial_i = 0$ , the surviving block is  $g - bg^{-1}b$ ; for  $b = 0$  this is  $g_{ij}$ , which is the *inverse* of the dual metric  $g' = g^{-1}$  and so plays on  $\tilde{x}$ -space the role that  $g^{ij}$  plays on  $x$ -space. Finally, all fields and gauge parameters obey the strong constraint of Chapter 14,

$$\eta^{MN} \partial_M A \partial_N B = 0, \quad \eta^{MN} \partial_M \partial_N A = 0 \quad \text{for all fields and parameters } A, B, \quad (17.9)$$

so that, up to an  $O(D, D)$  rotation, nothing depends on  $\tilde{x}$  [HHZ §1, ll. 228–236; AMN §3.3, eqs. (3.29)–(3.32), ll. 1026–1058].

#### Common pitfall: reading the wrong block

The statement “ $\mathcal{H}^{MN} \partial_M \partial_N$  reduces to  $g^{ij} \partial_i \partial_j$ ” depends on the ordering (17.6). With the momentum-first ordering of Chapter 14 the same operator is written with the blocks of  $\mathcal{H}$  swapped, and a careless reader finds  $g_{ij} \partial_i \partial_j$ , an expression with its indices in impossible positions. Index position is the safeguard:  $\partial_i$  carries a lower index and can only be contracted with  $g^{ij}$ .

## 17.3 The action

### 17.3.1 Which terms can appear?

We look for a Lagrangian with two derivatives (as in (17.1)), multiplied by the density  $e^{-2d}$ , built from  $\mathcal{H}$ ,  $d$  and  $\partial_M$  with all indices contracted. Two observations cut the list of candidates down to four.

*The  $\mathbb{Z}_2$  of  $b \mapsto -b$ .* The NS–NS action is invariant under  $b \mapsto -b$  (it contains  $b$  only through  $H^2$ ). In the doubled language this reflection acts as  $\tilde{\partial}^i \mapsto -\tilde{\partial}^i$  and flips the sign of the off-diagonal blocks of  $\mathcal{H}$ :  $\partial \mapsto Z\partial$ ,  $\mathcal{H} \mapsto Z\mathcal{H}Z$  with  $Z = \text{diag}(-1, 1)$ . This  $Z$  is *not* an element of  $O(D, D)$ , because  $Z\eta Z = -\eta \neq \eta$ . A term in which every index is contracted directly between  $\partial_M$ ,  $\mathcal{H}^{MN}$  and  $\mathcal{H}_{MN}$  is invariant, since each contraction produces  $ZZ = 1$ ; a term that needs an explicit  $\eta$  is not [HHZ §4.1, eqs. (4.2)–(4.6), ll. 1040–1085].

*Integration by parts.* Terms with two derivatives on one field, such as  $\mathcal{H}^{MN} \partial_M \partial_N d$  and  $\partial_M \partial_N \mathcal{H}^{MN}$ , can be traded for products of first derivatives, because the density  $e^{-2d}$  produces a factor  $-2\partial_M d$  when the derivative is moved onto it.

What remains is

$$\mathcal{H}^{MN} \partial_M \mathcal{H}^{KL} \partial_N \mathcal{H}_{KL}, \quad \mathcal{H}^{MN} \partial_N \mathcal{H}^{KL} \partial_L \mathcal{H}_{MK}, \quad \partial_M d \partial_N \mathcal{H}^{MN}, \quad \mathcal{H}^{MN} \partial_M d \partial_N d \quad (17.10)$$

[HHZ §4.1, eqs. (4.7)–(4.9), ll. 1086–1106]. Note that there is no candidate with only two factors of  $\mathcal{H}$  and no explicit  $\eta$ : a term like  $\partial_M \mathcal{H}^{KL} \partial^M \mathcal{H}_{KL}$  needs  $\eta^{MN}$  to raise the index, and it vanishes anyway by (17.9).

### 17.3.2 The result

**Theorem 17.2** (Hohm–Hull–Zwiebach action). *The action*

$$S_{DFT} = \int d^D x d^D \tilde{x} e^{-2d} \left[ \frac{1}{8} \mathcal{H}^{MN} \partial_M \mathcal{H}^{KL} \partial_N \mathcal{H}_{KL} - \frac{1}{2} \mathcal{H}^{MN} \partial_N \mathcal{H}^{KL} \partial_L \mathcal{H}_{MK} - 2 \partial_M d \partial_N \mathcal{H}^{MN} + 4 \mathcal{H}^{MN} \partial_M d \partial_N d \right] \quad (17.11)$$

is invariant under global  $O(D, D)$  transformations and, when the strong constraint (17.9) holds, under the gauge transformations  $\delta_\xi \mathcal{H}^{MN} = \hat{\mathcal{L}}_\xi \mathcal{H}^{MN}$  (generalized Lie derivative, Chapter 16) and (17.5). [L][HHZ §1, eq. (1.15), ll. 336–352; §4.1, eqs. (4.10)–(4.11), ll. 1107–1126]

For reference, the generalized Lie derivative of  $\mathcal{H}$  is

$$\hat{\mathcal{L}}_\xi \mathcal{H}^{MN} = \xi^P \partial_P \mathcal{H}^{MN} + (\partial^M \xi_P - \partial_P \xi^M) \mathcal{H}^{PN} + (\partial^N \xi_P - \partial_P \xi^N) \mathcal{H}^{MP} \quad (17.12)$$

[HHZ §1, eqs. (1.16)–(1.17), ll. 362–372]: a transport term, and for each index the difference of a “covariant” and a “contravariant” rotation, so that  $\eta^{MN}$  itself is invariant.

The global invariance is immediate, and this is the point of the generalized-metric formulation. Under  $X' = hX$  with  $h^\top \eta h = \eta$  the fields transform linearly,  $\mathcal{H}'(X') = h \mathcal{H}(X) h^\top$  for the upper-index matrix,  $\partial' = (h^{-1})^\top \partial$  and  $d'(X') = d(X)$  [HHZ §1, eq. (1.10), ll. 272–278]. Every index in (17.11) is contracted between an upper and a lower position, so all factors of  $h$  cancel in pairs, and the measure is invariant because  $|\det h| = 1$ . If some directions are compact, the periodicities of  $(x, \tilde{x})$  must also be preserved and the group is reduced to the discrete subgroup  $O(d, d; \mathbb{Z})$  of the  $d$  toroidal directions [HHZ §1, ll. 191–192]. Compare this with the earlier formulation in terms of  $E_{ij} = g_{ij} + b_{ij}$ , where the same symmetry acts by fractional linear transformations and had to be verified term by term [HHZ §1, eqs. (1.1)–(1.3), ll. 100–149 and ll. 245–252]. Gauge invariance is less obvious; we return to it in Section 17.5.

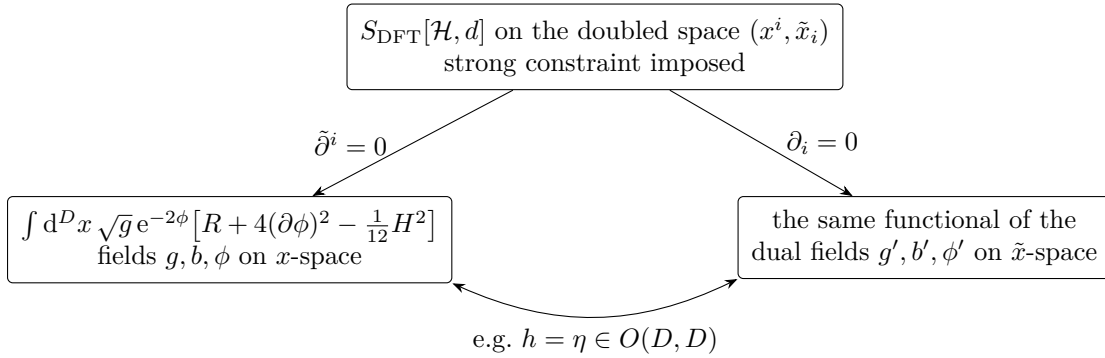


Figure 17.1: One action, several supergravity frames. Every solution of the strong constraint is an  $O(D, D)$  rotation of the frame  $\tilde{\partial}^i = 0$  [AMN §3.3, ll. 1050–1056]; because  $S_{\text{DFT}}$  is  $O(D, D)$  invariant, each frame yields the NS–NS action for the correspondingly rotated fields.

## 17.4 Reduction to the NS–NS action

We now set  $\tilde{\partial}^i = 0$  (the supergravity frame) in (17.11) and work out each term. The  $\tilde{x}$ -integral then gives a constant volume factor that we absorb in the normalization. Write  $\mathcal{L} = e^{-2d}(T_1 + T_2 + T_3 + T_4)$  for the four terms in the order displayed.

**The dilaton terms.** By the rule stated below (17.7), the derivative indices select the block  $g^{ij}$ :

$$T_4 = 4 g^{ij} \partial_i d \partial_j d, \quad T_3 = -2 \partial_i d \partial_j g^{ij}. \quad (17.13)$$

**The first metric term.** In  $T_1$  both derivative indices are lower-block, so

$$T_1 = \frac{1}{8} g^{ij} \partial_i \mathcal{H}^{KL} \partial_j \mathcal{H}_{KL}.$$

The contraction over  $K, L$  is a sum over the four blocks of (17.7), each block of  $\mathcal{H}^{KL}$  being multiplied entry by entry with the block of  $\mathcal{H}_{KL}$  in the same position. Set  $b = 0$  first. The two diagonal blocks give  $\partial_i g_{kl} \partial_j g^{kl}$  and  $\partial_i g^{kl} \partial_j g_{kl}$ , which are equal after contraction with the symmetric  $g^{ij}$ , hence

$$T_1|_{b=0} = \frac{1}{4} g^{ij} \partial_i g^{kl} \partial_j g_{kl}. \quad (17.14)$$

Now restore  $b$  and collect the terms with two derivatives of  $b$  and no undifferentiated  $b$ . The upper-right blocks contribute

$$\partial_i (bg^{-1})_k{}^l \partial_j (-g^{-1}b)^{kl} \longrightarrow -(\partial_i b_{km}) g^{ml} g^{kn} (\partial_j b_{nl}) = -\partial_i b_{km} \partial_j b^{km},$$

where the arrow means that we keep only the part in which both derivatives act on  $b$ , and where  $b^{km} = g^{kn} g^{ml} b_{nl}$ . The lower-left blocks give the same, so

$$T_1|_{(\partial b)^2} = \frac{1}{8} g^{ij} \cdot 2 \cdot (-\partial_i b_{kl} \partial_j b^{kl}) = -\frac{1}{4} \partial_i b_{jk} \partial^i b^{jk}. \quad (17.15)$$

The remaining  $b$ -dependent pieces of  $T_1$  contain undifferentiated  $b$ ; they come from  $\partial(bg^{-1}b)$  in the upper-left block and from  $\partial g^{-1}$  hidden in  $bg^{-1}$ , and they cancel among themselves. This must happen: a constant shift of  $b$  is an  $O(D, D)$  transformation (the  $B$ -shift of Chapter 11) that preserves the frame  $\tilde{\partial} = 0$ , so undifferentiated  $b$  cannot survive.

**The second metric term.** In  $T_2 = -\frac{1}{2} \mathcal{H}^{MN} \partial_N \mathcal{H}^{KL} \partial_L \mathcal{H}_{MK}$  the indices  $N$  and  $L$  are lower-block. For  $b = 0$  the matrices are block diagonal, which forces  $M$  and then  $K$  into the lower block as well, and

$$T_2|_{b=0} = -\frac{1}{2} g^{ij} \partial_j g^{kl} \partial_l g_{ik}. \quad (17.16)$$

The terms quadratic in  $\partial b$  are found in the same way as for  $T_1$  (Exercise 17.3):

$$T_2|_{(\partial b)^2} = -\frac{1}{2} \partial_i b_{jk} \partial^j b^{ki}. \quad (17.17)$$

**Assembling  $H^2$ .** Expand the square of the three-form in (17.1). There are  $3 \times 3 = 9$  products; three of them pair a term with itself and six pair different terms. Relabelling indices,

$$H_{ijk} H^{ijk} = 3 \partial_i b_{jk} \partial^i b^{jk} + 6 \partial_i b_{jk} \partial^j b^{ki}, \quad \text{so} \quad -\frac{1}{12} H^2 = -\frac{1}{4} \partial_i b_{jk} \partial^i b^{jk} - \frac{1}{2} \partial_i b_{jk} \partial^j b^{ki}. \quad (17.18)$$

The two pieces are exactly (17.15) and (17.17). The coefficients  $\frac{1}{8}$  and  $-\frac{1}{2}$  of the  $O(D, D)$  action therefore *produce* the  $\frac{1}{12}$  of supergravity, and they also fix the gauge-invariant combination  $H = db$ : neither  $T_1$  nor  $T_2$  is invariant under  $b \mapsto b + d\tilde{\lambda}$  alone.

**Proposition 17.3** (reduction, first form). *For  $\tilde{\partial}^i = 0$  the Lagrangian of (17.11) equals, with no integration by parts,*

$$\mathcal{L}^{(0)} = e^{-2d} \left[ \frac{1}{4} g^{ij} \partial_i g_{kl} \partial_j g^{kl} - \frac{1}{2} g^{ij} \partial_j g^{kl} \partial_l g_{ik} - 2 \partial_i d \partial_j g^{ij} + 4 g^{ij} \partial_i d \partial_j d - \frac{1}{12} H^2 \right]. \quad (17.19)$$

[L][HHZ §4.1, eqs. (4.12)–(4.13), ll. 1127–1170]

We derived every term above except the cancellation of undifferentiated  $b$ . As an independent symbolic check we evaluated both sides with sympy in  $D = 3$  for a non-diagonal polynomial metric, a generic polynomial  $b$  and a non-constant  $d$ , building  $\mathcal{H}$  from (17.7); the difference is identically zero, and the separate identities (17.15), (17.17) hold for the full  $b$ -dependence of  $T_1$  and  $T_2$ , not only at quadratic order.

**From first derivatives to the Ricci scalar.** Equation (17.19) contains no second derivatives, whereas  $R$  does. The two differ by a total derivative, just as the Einstein–Hilbert Lagrangian  $\sqrt{g} R$  differs by a total derivative from a Lagrangian quadratic in Christoffel symbols. The shortest way to see the match uses the scalar  $\mathcal{R}$  of the next section, so we state the result here and complete the argument there.

**Theorem 17.4** (reduction to supergravity). *With  $\tilde{\partial}^i = 0$ ,  $\mathcal{H}$  parametrized by (17.7) and  $d$  by (17.3),*

$$\mathcal{L}_{DFT}\Big|_{\tilde{\partial}=0} = \sqrt{g} e^{-2\phi} \left[ R + 4(\partial\phi)^2 - \frac{1}{12} H^2 \right] + \partial_i (e^{-2d} V^i), \quad V^i = \partial_j g^{ij} + g^{ij} \partial_j \ln |\det g|. \quad (17.20)$$

[L][HHZ §4.1, ll. 1127–1185; AMN §3.6, eq. (3.71) and ll. 1458–1477]

## 17.5 The generalized scalar curvature

### 17.5.1 Integrating by parts into the form $e^{-2d}\mathcal{R}$

The Einstein–Hilbert Lagrangian is “density times scalar”, and gauge invariance follows from that structure alone. The Lagrangian in (17.11) is not of this form: its four terms are not separately gauge scalars. We now bring it to that form. Take the two dilaton terms and use  $\partial_M e^{-2d} = -2e^{-2d} \partial_M d$ :

$$-2e^{-2d} \partial_M d \partial_N \mathcal{H}^{MN} = \partial_M (e^{-2d}) \partial_N \mathcal{H}^{MN} = \partial_M (e^{-2d} \partial_N \mathcal{H}^{MN}) - e^{-2d} \partial_M \partial_N \mathcal{H}^{MN}, \quad (17.21)$$

$$\begin{aligned} 4e^{-2d} \mathcal{H}^{MN} \partial_M d \partial_N d &= -4e^{-2d} \mathcal{H}^{MN} \partial_M d \partial_N d + 8e^{-2d} \mathcal{H}^{MN} \partial_M d \partial_N d \\ &= -4e^{-2d} \mathcal{H}^{MN} \partial_M d \partial_N d - 4\partial_M (e^{-2d}) \mathcal{H}^{MN} \partial_N d \\ &= -4e^{-2d} \mathcal{H}^{MN} \partial_M d \partial_N d - 4\partial_M (e^{-2d} \mathcal{H}^{MN} \partial_N d) + 4e^{-2d} \partial_M (\mathcal{H}^{MN} \partial_N d). \end{aligned} \quad (17.22)$$

Adding the two lines and expanding the last bracket gives

$$\mathcal{L}_{DFT} = e^{-2d} \mathcal{R} + \partial_M (e^{-2d} [\partial_N \mathcal{H}^{MN} - 4\mathcal{H}^{MN} \partial_N d]), \quad (17.23)$$

with

$$\begin{aligned} \mathcal{R} &= 4\mathcal{H}^{MN} \partial_M \partial_N d - \partial_M \partial_N \mathcal{H}^{MN} - 4\mathcal{H}^{MN} \partial_M d \partial_N d + 4\partial_M \mathcal{H}^{MN} \partial_N d \\ &\quad + \frac{1}{8} \mathcal{H}^{MN} \partial_M \mathcal{H}^{KL} \partial_N \mathcal{H}_{KL} - \frac{1}{2} \mathcal{H}^{MN} \partial_M \mathcal{H}^{KL} \partial_K \mathcal{H}_{NL} \end{aligned} \quad (17.24)$$

[HHZ §4.2, eqs. (4.24)–(4.27), ll. 1272–1338]. (The last term is the second term of (17.11) with its dummy indices renamed,  $M \leftrightarrow N$  and then  $K \leftrightarrow L$ , using the symmetry of  $\mathcal{H}$ .)

**Theorem 17.5** (gauge scalar). *If the strong constraint holds,  $\delta_\xi \mathcal{R} = \xi^M \partial_M \mathcal{R}$ . Consequently  $S_{DFT} = \int e^{-2d} \mathcal{R}$  (up to a boundary term) is gauge invariant. [L][HHZ §4.3, eqs. (4.28)–(4.47), ll. 1340–1528; AMN §3.6, eqs. (3.66)–(3.67), ll. 1405–1421]*

*Idea of the proof.* Every object  $W$  built from  $\mathcal{H}$ ,  $d$  and partial derivatives transforms as  $\delta_\xi W = \hat{\mathcal{L}}_\xi W + \Delta_\xi W$ , where the first term is what a generalized tensor with the index structure of  $W$  would do and  $\Delta_\xi W$  is the failure. Partial derivatives are not covariant, so for example  $\Delta_\xi(\partial_M d) = -\frac{1}{2} \partial_M (\partial \cdot \xi)$ : differentiating (17.5) gives  $\xi^P \partial_P \partial_M d + \partial_M \xi^P \partial_P d - \frac{1}{2} \partial_M \partial_P \xi^P$ ; the first two terms are the generalized Lie derivative of a covector, because the term  $-\partial^P \xi_M \partial_P d$  that would complete it vanishes by the strong constraint, and the last term is the failure. The map  $\Delta_\xi$  obeys the Leibniz rule, so one computes it on each of the six terms of (17.24) and discards everything of the form  $\partial^P(\cdot) \partial_P(\cdot)$  by the strong constraint. The four terms containing  $d$  fail together by

$$\Delta_\xi [4\mathcal{H}^{MN} \partial_M \partial_N d - \partial_M \partial_N \mathcal{H}^{MN} + 4\partial_M \mathcal{H}^{MN} \partial_N d - 4\mathcal{H}^{MN} \partial_M d \partial_N d] = \partial_M \partial_N \xi^P \partial_P \mathcal{H}^{MN}, \quad (17.25)$$

all terms with three derivatives of  $\xi$  having cancelled inside the bracket, and the two pure- $\mathcal{H}$  terms fail by exactly the opposite amount [HHZ §4.3, eqs. (4.30)–(4.47), ll. 1354–1528]. We do not reproduce the two pages of algebra. The point to retain is where the strong constraint enters: not in writing the action, but in its gauge invariance.

### 17.5.2 $\mathcal{R}$ is the dilaton equation of motion

Vary  $d$  in (17.11). The variation hits the prefactor, giving  $-2\delta d\mathcal{L}$ , and the two explicit dilaton terms:

$$\delta_d\mathcal{L} = -2\delta d\mathcal{L} + e^{-2d}[-2\partial_M\delta d\partial_N\mathcal{H}^{MN} + 8\mathcal{H}^{MN}\partial_M d\partial_N\delta d].$$

Integrate the bracket by parts, again using  $\partial_M e^{-2d} = -2e^{-2d}\partial_M d$ . Writing  $T_{\mathcal{H}}$  for the two pure- $\mathcal{H}$  terms of (17.11), the coefficient of  $e^{-2d}\delta d$  is

$$-2[T_{\mathcal{H}} - 2\partial d\partial\mathcal{H} + 4\mathcal{H}\partial d\partial d] + [2\partial\partial\mathcal{H} - 4\partial d\partial\mathcal{H}] + [16\mathcal{H}\partial d\partial d - 8\partial\mathcal{H}\partial d - 8\mathcal{H}\partial\partial d]$$

in an obvious shorthand, and collecting terms,

$$\delta_d S = -2 \int e^{-2d} \mathcal{R} \delta d. \quad (17.26)$$

So the dilaton equation of motion is  $\mathcal{R} = 0$  [HHZ §4.2, ll. 1272–1294; AMN §3.7, eq. (3.79), l. 1522]. In particular the Lagrangian in the form  $e^{-2d}\mathcal{R}$  vanishes on every solution, a property it shares with the string-frame NS–NS action.

### 17.5.3 Reduction of $\mathcal{R}$ and the end of the reduction proof

Setting  $\tilde{\partial} = 0$  in (17.24) must give the dilaton equation of motion of supergravity, and it does:

$$\mathcal{R}|_{\tilde{\partial}=0} = R + 4\nabla^i\nabla_i\phi - 4(\partial\phi)^2 - \frac{1}{12}H^2, \quad (17.27)$$

to be compared with the supergravity field equation [AMN §3.2, eq. (3.21), ll. 899–907]. This is an exact identity, without total derivatives. It is the one formula of this chapter that we did not derive by hand in general: the second derivatives of  $g$  hidden in  $-\partial_M\partial_N\mathcal{H}^{MN} \rightarrow -\partial_i\partial_j g^{ij}$  and in  $4g^{ij}\partial_i\partial_j d$  (recall  $d = \phi - \frac{1}{4}\ln|\det g|$ ) assemble into  $R$ . We verified it as a symbolic check with sympy in  $D = 3$  with the same non-diagonal  $g$ , generic  $b$  and  $\phi$  as before, and in Section 17.6 we verify it by hand for a family of cosmological backgrounds.

Given (17.27), the theorem follows in two lines. Since  $\nabla_i(e^{-2\phi}\nabla^i\phi) = e^{-2\phi}[\nabla^2\phi - 2(\partial\phi)^2]$ ,

$$\sqrt{g}e^{-2\phi}[4\nabla^2\phi - 4(\partial\phi)^2] = 4\partial_i(\sqrt{g}e^{-2\phi}g^{ij}\partial_j\phi) + 4\sqrt{g}e^{-2\phi}(\partial\phi)^2, \quad (17.28)$$

which turns the combination  $4\nabla^2\phi - 4(\partial\phi)^2$  of the field equation into the  $+4(\partial\phi)^2$  of the action. Inserting (17.27) and (17.28) into (17.23) gives (17.20) with  $V^i = \partial_j g^{ij} - 4g^{ij}\partial_j d + 4g^{ij}\partial_j\phi$ , and  $-4\partial_j d + 4\partial_j\phi = \partial_j \ln|\det g|$  by (17.3). The vector  $V^i$  does not contain the dilaton: it is the familiar boundary term that relates  $\sqrt{g}R$  to the first-order “ $\Gamma\Gamma$ ” Lagrangian of general relativity. We checked (17.20), including the sign of each term of  $V^i$ , symbolically in  $D = 2$ .

#### Common pitfall: $4(\partial\phi)^2$ versus $-4(\partial\phi)^2$

The action (17.1) has  $+4(\partial\phi)^2$ ; the scalar (17.27) has  $-4(\partial\phi)^2 + 4\nabla^2\phi$ . Both are correct and they differ by the total derivative (17.28). The “wrong-sign” kinetic term of the dilaton in the string frame is not a ghost: after the Weyl rescaling to the Einstein frame (Chapter 6) the kinetic term has the healthy sign. A related trap:  $\mathcal{R}$  is *not* the Ricci scalar of the  $2D$ -dimensional metric  $\mathcal{H}_{MN}$ . Its name records only that it is a gauge scalar with two derivatives that reduces to  $R + \dots$ ; HHZ state that they “do not have an understanding of these as curvatures” [HHZ §6, ll. 2446–2451].

### 17.5.4 The generalized Ricci tensor

Varying  $\mathcal{H}$  is more delicate than varying  $d$ , because  $\mathcal{H}$  is constrained:  $\mathcal{H}\eta\mathcal{H} = \eta^{-1}$ . Write the unconstrained variation as  $\delta S = \int e^{-2d} \delta \mathcal{H}^{MN} \mathcal{K}_{MN}$ . With  $S^M_N = \mathcal{H}^{MP} \eta_{PN}$ , which satisfies  $S^2 = 1$ , an allowed variation obeys  $\delta \mathcal{H} = -S \delta \mathcal{H} S^T$ , and the projectors  $\frac{1}{2}(1 \pm S)$  select the part of  $\mathcal{K}$  that such variations can see. The field equation is

$$\mathcal{R}_{MN} \equiv \frac{1}{4}(1 - S^T)\mathcal{K}(1 + S) + \frac{1}{4}(1 + S^T)\mathcal{K}(1 - S) = 0, \quad (17.29)$$

which combines the  $g$  and  $b$  equations of supergravity in one  $O(D, D)$  tensor [HHZ §4.4, eqs. (4.48)–(4.58), ll. 1536–1650]. The counting is instructive:  $\delta \mathcal{H}$  maps the  $+1$  eigenspace of  $S$  to the  $-1$  eigenspace and back, so it has  $D \times D$  independent components, the  $\frac{1}{2}D(D+1)$  of  $\delta g$  plus the  $\frac{1}{2}D(D-1)$  of  $\delta b$ .

## 17.6 Worked example: cosmological backgrounds and scale-factor duality

Let  $D = n + 1$ , let all fields depend on time  $t$  only, put  $b = 0$  and

$$ds^2 = -dt^2 + \sum_{i=1}^n a_i(t)^2 (dx^i)^2, \quad h_i \equiv \frac{\dot{a}_i}{a_i}, \quad e^{-2d} = \left( \prod_i a_i \right) e^{-2\phi} \quad (17.30)$$

[BW, eq. (8), ll. 373–393]. We write  $h_i$  for the Hubble rates to avoid a clash with the three-form  $H$ .

**The DFT side.** Only  $\partial_t$  is non-zero and  $\mathcal{H}^{tt} = g^{tt} = -1$ . In (17.19), the second and third terms contain  $\partial_t g^{tt}$  or  $\partial_t g_{tt}$  and vanish. The first term needs

$$\partial_t g_{kl} \partial_t g^{kl} = \sum_i (2a_i \dot{a}_i) (-2\dot{a}_i / a_i^3) = -4 \sum_i h_i^2,$$

hence  $\frac{1}{4}g^{tt}(-4 \sum h_i^2) = + \sum h_i^2$ ; the fourth is  $4g^{tt}\dot{d}^2 = -4\dot{d}^2$ . Therefore

$$\mathcal{L}_{\text{DFT}} = e^{-2d} \left[ \sum_{i=1}^n h_i^2 - 4\dot{d}^2 \right]. \quad (17.31)$$

This took three lines. Scale-factor duality,  $a_i \mapsto 1/a_i$  for any subset of directions with  $d$  fixed, flips  $h_i \mapsto -h_i$  and is manifest [BW, eq. (10), ll. 405–431]. In DFT language it is the  $O(D, D)$  element that exchanges  $x^i \leftrightarrow \tilde{x}_i$  in those directions; it acts on  $\mathcal{H} = \text{diag}(g, g^{-1})$  by swapping  $a_i^2 \leftrightarrow a_i^{-2}$  and leaves (17.11) invariant term by term.

**The supergravity side.** For (17.30) the Ricci scalar is  $R = 2 \sum_i \dot{h}_i + \sum_i h_i^2 + (\sum_i h_i)^2$  (standard; for  $n = 3$  and equal  $a_i$  it is the familiar  $6\dot{h} + 12h^2$ ). From (17.30),  $\dot{\phi} = \dot{d} + \frac{1}{2} \sum_i h_i$ , so  $4(\partial\phi)^2 = -4\dot{\phi}^2 = -4\dot{d}^2 - 4\dot{d} \sum h_i - (\sum h_i)^2$ . The  $(\sum h_i)^2$  terms cancel and

$$\begin{aligned} \sqrt{g} e^{-2\phi} [R + 4(\partial\phi)^2] &= e^{-2d} \left[ 2 \sum_i \dot{h}_i + \sum_i h_i^2 - 4\dot{d}^2 - 4\dot{d} \sum_i h_i \right] \\ &= e^{-2d} \left[ \sum_i h_i^2 - 4\dot{d}^2 \right] + \frac{d}{dt} \left( 2e^{-2d} \sum_i h_i \right), \end{aligned}$$

because  $\frac{d}{dt}(2e^{-2d} \sum h_i) = e^{-2d} [2 \sum \dot{h}_i - 4\dot{d} \sum h_i]$ . This is Theorem 17.4 for this family, by hand; one checks that  $-e^{-2d} V^t$  with  $V^t = g^{tt} \partial_t \ln |\det g| = -2 \sum h_i$  is the same boundary term. On the supergravity side the duality is hidden:  $R$  and  $\phi$  each change under  $a_i \mapsto 1/a_i$ , and the invariance appears only after the cancellation of  $(\sum h_i)^2$ .

**Solutions, with numbers.** There are three equations. Varying  $\ln a_i$  in (17.31) gives  $\frac{d}{dt}(e^{-2d}h_i) = 0$ . Varying  $d$  gives  $\mathcal{R} = 0$ , and evaluating (17.24) on (17.30) term by term (only the first, third, and fifth terms survive) gives  $\mathcal{R} = -4\ddot{d} + 4\dot{d}^2 + \sum h_i^2$ . The third is the equation of  $g_{tt}$ , which we lost by fixing  $g_{tt} = -1$ : restoring  $g_{tt} = -N(t)^2$  one finds  $\mathcal{L} = e^{-2\hat{d}}N^{-1}[\sum h_i^2 - 4\dot{\hat{d}}^2]$  with  $e^{-2\hat{d}} = \prod a_i e^{-2\phi}$  (symbolic check), and varying  $N$  at  $N = 1$  gives the constraint  $\sum h_i^2 = 4\dot{\hat{d}}^2$ . Try the power-law ansatz  $a_i = t^{p_i}$ ,  $e^{-2d} = t$ , for which  $h_i = p_i/t$ ,  $\dot{d} = -1/2t$  and  $\ddot{d} = 1/2t^2$ . The first equation holds because  $e^{-2d}h_i = p_i$  is constant. The constraint reads  $\sum p_i^2/t^2 = 1/t^2$  and the dilaton equation reads  $(-2 + 1 + \sum p_i^2)/t^2 = 0$ ; both hold if and only if

$$\sum_{i=1}^n p_i^2 = 1. \quad (17.32)$$

For  $n = 3$  isotropic directions,  $p = 1/\sqrt{3} \approx 0.577$  and the physical dilaton is  $\phi = d + \frac{1}{2} \sum p_i \ln t = \frac{1}{2}(\sqrt{3} - 1) \ln t \approx 0.366 \ln t$ : a decelerated expansion with growing coupling. Its scale-factor dual has  $p = -1/\sqrt{3}$  and  $\phi = -\frac{1}{2}(\sqrt{3} + 1) \ln t \approx -1.366 \ln t$ : contraction with a coupling that decreases. The two are one point of the constraint sphere (17.32) and its antipode; in the DFT variables  $(\mathcal{H}, d)$  they are related by a constant  $O(D, D)$  matrix and share the same  $d(t)$ .

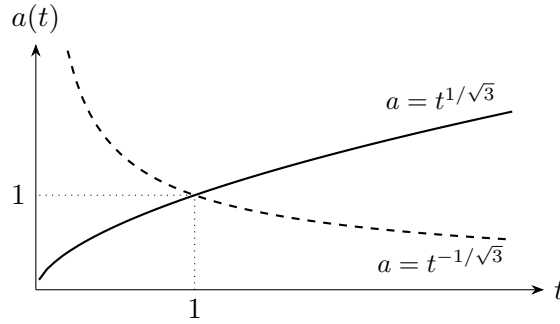


Figure 17.2: The isotropic  $n = 3$  solution of (17.32) and its scale-factor dual. Both have the same  $e^{-2d} = t$ ; they cross at the self-dual value  $a = 1$ , that is  $R = \sqrt{\alpha'}$  in the units of Section 17.1.

## 17.7 Beyond the strong constraint: what is and is not known

It is tempting to read (17.11) as a theory of fields on a  $2D$ -dimensional space. With the strong constraint it is not: the constraint implies that, after an  $O(D, D)$  rotation, no field depends on  $\tilde{x}$ , and the action is then the NS–NS action in a particular notation [HHZ §1, ll. 228–236; AMN §3.3, ll. 1050–1058]. What has been gained is (i) a formulation in which  $O(D, D)$  acts linearly and the duality invariance of the dimensionally reduced theory is visible before reduction, (ii) a geometric unification of diffeomorphisms and  $b$ -field gauge transformations, and (iii) a starting point for relaxing the constraint. The status of (iii), according to the sources of this book, is as follows.

- *Weak constraint only.* String field theory requires only  $\partial^M \partial_M A = 0$  on each field, not on products. A doubled action with this weaker condition was constructed to cubic order in fluctuations around a flat background; extending it to higher order needs new structures, including a projector onto the kernel of  $\partial^M \partial_M$ , and HHZ call this “the most important outstanding question”. [L][HHZ §6, ll. 2454–2467]

- *Closure constraints.* What gauge invariance really uses is closure of the gauge algebra and the vanishing of a small number of obstruction tensors. The strong constraint solves these conditions, but it is not the only solution: the flux formulation of the action keeps the terms  $\Delta_{(SC)}\mathcal{R}$  that the strong constraint would set to zero, and generalized Scherk–Schwarz reductions provide configurations that depend on both  $x$  and  $\tilde{x}$  in a controlled way. [L][AMN §3.5, eqs. (3.44)–(3.46), ll. 1176–1259; §3.6, ll. 1443–1472; §3.3, ll. 1071–1083; §5.6, ll. 3703–3720]
- *Link to the worldsheet.* Under the strong constraint the generalized Ricci-flatness equation (17.29) is the condition for conformal invariance of a doubled sigma model; whether this persists beyond the strong constraint is left open in AMN. [L][AMN §7, ll. 5046–5058]

Whether a consistent, fully doubled field theory exists to all orders is therefore [C] in the classification of this book. Nothing in later chapters depends on it: the uses we make of  $\mathcal{H}$  and  $d$  for  $K3 \times T^2$  (Chapter 24) and for the dual-scale principle (Chapter 31) are within the strongly constrained theory or concern constant backgrounds.

## 17.8 What the machine has checked

The honest summary is short: *none of the theorems of this chapter is formalized*. There is no manifold, no derivative, no matrix-valued field and no integral in the two Lean files attached to it. What they contain is an *arithmetic shadow*: each physical quantity is replaced by one integer, and the theorems are identities of integer arithmetic that the corresponding physical statement would imply. We quote them because a student who opens the repository will find them under names that promise more, and should learn to read the statement rather than the name. The files are `DoubleFieldTheory/ActionCurvature.lean` (namespace `DoubleFieldTheory.ActionCurvature`, no imports) and `DoubleFieldTheory/PhysicsDSL.lean` (namespace `DoubleFieldTheory.PhysicsDSL`). We ran `#print axioms` on every theorem quoted below: each depends on a subset of `propext` and `Quot.sound`, and `dsl_self_dual_minimum` on no axiom at all.

### In Lean: the measure and the Lagrangian as integer products

```
def DilatonMeasure (sqrt_g e_minus_2phi : Int) : Int :=
  sqrt_g * e_minus_2phi

theorem dilaton_measure_symm (s e : Int) :
  DilatonMeasure s e = DilatonMeasure e s := by ...

def ActionLagrangian (density ricci : Int) : Int :=
  density * ricci

theorem dft_action_nsns_equivalence (density R_geom : Int) :
  ActionLagrangian density R_geom = ActionLagrangian R_geom density := by ...
```

**Reading.** `DilatonMeasure` is the product of two integers, named after the two factors of (17.3). The first theorem says that this product does not change when its two arguments are exchanged, i.e.  $s \cdot e = e \cdot s$  in  $\mathbb{Z}$ ; the second says the same for `ActionLagrangian`. Both proofs unfold the definition and rewrite with `Int.mul_comm`. Neither statement involves T-duality: the invariance (17.2) would be  $(s/r^2) \cdot (r^2 e) = s \cdot e$ , which needs a field, not  $\mathbb{Z}$  (Exercise 17.8). And despite its name the second theorem does not relate two actions; the equivalence of Theorem 17.4 is [L].



## In Lean: the three-term decomposition

```

def DFTRicciComponents (R_geom kin_phi H_sq : Int) : Int :=
  R_geom + 4 * kin_phi - H_sq

theorem dft_ricci_expansion (R_geom kin_phi H_sq : Int) :
  DFTRicciComponents R_geom kin_phi H_sq + H_sq = R_geom + 4 * kin_phi := by ...

theorem dft_ricci_physical_reduction (R_geom : Int) :
  DFTRicciComponents R_geom 0 0 = R_geom := by ...

theorem connection_trace_conservation (trace_H_initial delta_trace : Int)
  (hd : delta_trace = 0) :
  trace_H_initial + delta_trace = trace_H_initial := by ...

```

**Reading.** `DFTRicciComponents` is the integer  $R + 4k - h$ . To match the bracket of (17.1) one must read  $k$  as  $(\partial\phi)^2$  and  $h$  as  $\frac{1}{12}H^2$ , not as  $H^2$ : the coefficient  $\frac{1}{12}$ , which Section 17.4 showed to be the non-trivial output of the reduction, is absent from the file. The two theorems say  $(R + 4k - h) + h = R + 4k$  and  $R + 4 \cdot 0 - 0 = R$ ; both are closed by the linear-arithmetic tactic `omega` after unfolding. For example `DFTRicciComponents 5 3 2 = 15`. The third theorem says  $a + \delta = a$  when  $\delta = 0$ ; its docstring relates it to compatibility of a connection with  $\mathcal{H}$ , but no connection occurs in the statement. What would be a faithful version of the second theorem? It is the statement that (17.27) equals  $R$  when  $\phi$  is constant and  $H = 0$ , a theorem about fields.

## In Lean: the trace of the Einstein tensor, as arithmetic

```

def ContractedEinstein (D R : Int) : Int :=
  (D - 2) * R

theorem contracted_einstein_dim2 (R : Int) :
  ContractedEinstein 2 R = 0 := by ...

```

**Reading.**  $(2 - 2) \cdot R = 0$  for every integer  $R$ . The physics behind the name: the Einstein tensor  $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R$  has trace  $g^{\mu\nu}G_{\mu\nu} = R - \frac{D}{2}R = -\frac{1}{2}(D - 2)R$ , which vanishes identically on a two-dimensional worldsheet (Chapter 2). The Lean definition drops the factor  $-\frac{1}{2}$ , which is harmless for the vanishing at  $D = 2$  and wrong for any other use; e.g. `ContractedEinstein 4 7 = 14` whereas the trace is  $-7$ .

## In Lean: the notation layer and the one-component section condition

```

structure FieldDeriv where
  dx : Int
  dtx : Int

def SectionContract (Phi Psi : FieldDeriv) : Int :=
  Phi.dx * Psi.dtx + Phi.dtx * Psi.dx

notation:65 "∂_M " Φ " ∂^M " Ψ => SectionContract Φ Ψ

theorem dsl_strong_section_condition (Φ Ψ : FieldDeriv)
  (hΦ : Φ.dtx = 0) (hΨ : Ψ.dtx = 0) :
  (∂_M Φ ∂^M Ψ) = 0 := by ...

def Reff (R : Nat) : Nat :=
  R * R + 1

```

```
theorem dsl_effective_radius_strictly_super_planckian (R : Nat) (h : R ≥ 1) :
  R_eff(R) ≥ 2 := by ...
```

**Reading.** The first two declarations live in `DoubleFieldTheory/CourantAlgebroid.lean`; `PhysicsDSL.lean` adds *notation* so that Lean source resembles the formulas of a physics paper, and restates earlier theorems in that notation (`dsl_courant_pairing_symm`, `dsl_cbracket_antisymm`, `dsl_cbracket_self_vanishes`, `dsl_dorfman_cbracket_diff`; see Chapter 16). A `FieldDeriv` is a pair of integers standing for the values of  $\partial_x A$  and  $\tilde{\partial}^x A$  of one field at one point for  $D = 1$ , and `SectionContract` is  $\eta^{MN} \partial_M A \partial_N B$  for such pairs. The theorem says: if the second entries of both pairs vanish, the contraction vanishes. This is one direction only (supergravity frame  $\Rightarrow$  strong constraint), for  $D = 1$ , at one point. The all- $d$  statement on the charge lattice, including  $O(d, d; \mathbb{Z})$  covariance, is `IsSection` and `isSection_image` of Chapter 14. Finally `R_eff R` is  $R^2 + 1$ , the numerator of  $R + 1/R$  in units  $\alpha' = 1$ , and the last theorem says  $R^2 + 1 \geq 2$  for natural numbers  $R \geq 1$ . This is weaker than the dual-scale inequality  $R + 1/R \geq 2$ , which would read  $R^2 + 1 \geq 2R$ ; the faithful real-number statement is `circle_effective_scale_ge_two` in `DualScaleStream2/DualScale/TraceBound.lean` (Chapter 31). The docstring’s conclusion about singularities is an interpretation, **[C]** at best, and not part of what is proved.

#### Common pitfall: trusting docstrings

The Lean kernel checks statements and proofs. It does not read comments. The docstrings of `ActionCurvature.lean` contain correct physics (they quote (17.11) and (17.20)), equation numbers of papers that we have not checked against the papers and that do not match the numbering of HHZ used in this chapter, and claims of verification placed next to one-line integer identities. When you cite a formal result, cite the statement after the colon, never the prose above it.

Table 17.1: Epistemic status of the main claims of this chapter.

| Claim                                                                                                                  | Tier       | Basis                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------|------------|--------------------------------------------------------------------------------------------------------------------|
| $s \cdot e = e \cdot s$ , $(R + 4k - h) + h = R + 4k$ , $(2 - 2)R = 0$ in $\mathbb{Z}$                                 | <b>[A]</b> | <code>dilaton_measure_symm</code> ,<br><code>dft_ricci_expansion</code> ,<br><code>contracted_einstein_dim2</code> |
| $\tilde{\partial} A = \tilde{\partial} B = 0 \Rightarrow \eta^{MN} \partial_M A \partial_N B = 0$ (integers, $D = 1$ ) | <b>[A]</b> | <code>dsl_strong_section_condition</code>                                                                          |
| $e^{-2d} = \sqrt{g} e^{-2\phi}$ is an $O(D, D)$ scalar and a gauge density                                             | <b>[L]</b> | AMN §3.3–3.4; HHZ §1, §4.3                                                                                         |
| Action (17.11): $O(D, D)$ and gauge invariance                                                                         | <b>[L]</b> | HHZ §1, §4                                                                                                         |
| Reduction (17.19), (17.20), (17.27)                                                                                    | <b>[L]</b> | HHZ §4.1–4.2, AMN §3.6; symbolic checks in $D = 2, 3$                                                              |
| Pieces (17.13)–(17.18); cosmological example                                                                           | —          | derived on the page; not formalized                                                                                |
| Identification of any Lean object above with $\mathcal{R}$ , $d$ or $S_{\text{DFT}}$                                   | —          | a reading; never <b>[A]</b>                                                                                        |
| A consistent DFT with only the weak constraint, to all orders                                                          | <b>[C]</b> | open (HHZ §6)                                                                                                      |

### 17.8.1 What a faithful formalization would need

It is useful to separate three levels, in increasing order of difficulty.

*Level 1: algebra at a point.* Everything in Section 17.4 is polynomial algebra once the first derivatives are treated as independent variables. Fix a commutative ring, an invertible symmetric matrix  $g$ , and for each  $i$  a symmetric matrix  $\gamma_i$  (standing for  $\partial_i g$ ) and an anti-symmetric matrix  $\beta_i$  (standing for  $\partial_i b$ ). Add scalars  $\delta_i$  for  $\partial_i d$ , define  $\partial_i g^{-1} := -g^{-1} \gamma_i g^{-1}$  and the derivative of each block of (17.7) by the Leibniz rule. Then Proposition 17.3 becomes an identity between matrix polynomials, of the same kind as the theorems about `genMetric` proved in `DualScaleStream2/DFT/GeneralizedMetric.lean` (Chapter 15), for instance `tduality_inverts_metric`. It needs only Mathlib’s `Matrix` library, block matrices and traces. This would be a genuine [A] statement about the four coefficients  $\frac{1}{8}, -\frac{1}{2}, -2, 4$ , and it is the natural next step (Exercise 17.9 is its  $D = 1$  case).

*Level 2: fields on  $\mathbb{R}^{2D}$ .* To state Theorem 17.5 one needs smooth maps  $\mathbb{R}^{2D} \rightarrow$  matrices, partial derivatives and their commutation, the strong constraint as a hypothesis on a family of functions, and the generalized Lie derivative. Mathlib’s calculus (`fderiv`, iterated derivatives, symmetry of second derivatives) suffices in principle; the obstacle is the volume of index bookkeeping, since Mathlib has no tensor-index notation and each contraction must be written as a finite sum. Integration by parts as in (17.23) would be stated as an identity of integrands (“equal up to a divergence”), avoiding measure theory.

*Level 3: the geometric statement.* Theorem 17.4 mentions the Ricci scalar of a pseudo-Riemannian metric. A formal version on a manifold needs metrics of indefinite signature, the Levi-Civita connection, curvature, densities and differential forms with `db`. To our knowledge these were not all available in Mathlib when the libraries of this book were compiled; we have not audited Mathlib for this chapter, so treat this sentence as a pointer, not a fact. On  $\mathbb{R}^D$  with global coordinates one can define  $R$  by the coordinate formula and bypass manifolds, which reduces Level 3 to Level 2.

#### Chapter summary

- The combination  $e^{-2d} = \sqrt{g} e^{-2\phi}$  is invariant under T-duality;  $d$  is an  $O(D, D)$  scalar and  $e^{-2d}$  a density under generalized diffeomorphisms.
- The DFT action (17.11) has four terms built from  $\mathcal{H}$ ,  $d$  and  $\partial_M$  without explicit  $\eta$ ;  $O(D, D)$  invariance is manifest, gauge invariance requires the strong constraint.
- With  $\tilde{\partial} = 0$  it reduces to (17.19) exactly and to the NS–NS action up to a dilaton-independent boundary term; the coefficients  $\frac{1}{8}$  and  $-\frac{1}{2}$  generate the  $-\frac{1}{12}H^2$ .
- Up to a total derivative  $\mathcal{L} = e^{-2d}\mathcal{R}$  with  $\mathcal{R}$  a gauge scalar, equal to the dilaton equation of motion; the  $\mathcal{H}$  equation defines a generalized Ricci tensor. Neither is known to be a curvature in the Riemannian sense.
- For cosmological backgrounds the action is  $e^{-2d}[\sum h_i^2 - 4\dot{d}^2]$ , and scale-factor duality is manifest.
- With the strong constraint DFT is a reformulation of supergravity; a fully doubled theory beyond cubic order is open.
- Lean: only integer-arithmetic shadows are kernel-checked. A matrix-polynomial version of the reduction is within reach of Mathlib and would be the first faithful [A] result here.

## Exercises

**Exercise 17.1** (★). Show that the two forms of (17.5) are equivalent. Then set  $\tilde{\partial} = 0$  and  $\xi^M = (\tilde{\xi}_i, 0)$ , a pure  $b$ -field gauge transformation, and conclude that  $d$ , hence  $\phi$ , is inert, as it must be.

**Exercise 17.2** (★). Verify that  $Z = \text{diag}(-1, 1)$  satisfies  $Z\eta Z = -\eta$ , and that  $\mathcal{H}^{MN} \mapsto (Z\mathcal{H}Z)^{MN}$  is the map  $b \mapsto -b$  in (17.7). Which of the following are even under this  $\mathbb{Z}_2$ :  $\mathcal{H}^{MN}\partial_M d\partial_N d$ ;  $\eta^{MN}\partial_M d\partial_N d$ ;  $\partial_M \mathcal{H}^{MN}\partial_N d$ ?

**Exercise 17.3** (★★). Derive (17.17). Hint: at quadratic order in  $\partial b$  and zeroth order in undifferentiated  $b$  you may take the undifferentiated factor  $\mathcal{H}^{MN}$  block diagonal, so that  $M$  and  $N$  are both lower-block; both  $\partial_N \mathcal{H}^{KL}$  and  $\partial_L \mathcal{H}_{MK}$  must then be taken from off-diagonal blocks, with  $K$  in the upper block. Then prove (17.18) by expanding the nine products.

**Exercise 17.4** (★★). Fill in the algebra leading to (17.26): list the seven terms, and check that the coefficient of each of the six structures of (17.24) is  $-2$  times the one displayed there.

**Exercise 17.5** (★★). *Constant  $H$ -flux on a flat three-torus.* Let  $D = 3$ ,  $g_{ij} = \delta_{ij}$ ,  $\phi = 0$  and  $b_{12} = -b_{21} = h x^3$  with  $h$  constant. Compute  $H_{123}$ ,  $H_{ijk}H^{ijk}$  and  $\mathcal{R}$  from (17.27). (Answer:  $\mathcal{R} = -h^2/2$ .) Is this background a solution? Compute  $\mathcal{H}^{MN}$  and observe that it depends on  $x^3$  only through a  $B$ -shift matrix; relate this to the monodromy  $x^3 \rightarrow x^3 + 2\pi$ .

**Exercise 17.6** (★★). *Linear dilaton.* Let  $g_{ij} = \eta_{ij}$  be flat Minkowski space,  $b = 0$  and  $\phi = v_i x^i$  with constant  $v$ . Show that  $d = \phi$ , that  $\mathcal{R} = -4v \cdot v$  directly from (17.24), and that the  $\mathcal{H}$  equation is satisfied. Conclude that the background solves the DFT equations if and only if  $v$  is null. What replaces this condition in the noncritical string, where (17.1) contains an extra constant  $c$  [BW, eq. (7), ll. 343–364]?

**Exercise 17.7** (★★). Show that the trace of the Einstein tensor in  $D$  dimensions is  $-\frac{1}{2}(D-2)R$ . State precisely which integer identity `contracted_einstein_dim2` proves, and write a corrected definition over  $\mathbb{Q}$  whose value is the trace for every  $D$ . Does your corrected version still make the  $D = 2$  theorem provable by `simp`?

**Exercise 17.8** (★, Lean). State and prove in Lean 4 with Mathlib the invariance (17.2): for real  $r \neq 0$  and any real  $w$ ,  $r^{-1} \cdot (r^2 w) = r w$ . (Here  $r = \sqrt{g}$  for the circle and  $w = e^{-2\phi}$ ; the left side is  $\sqrt{g'} e^{-2\phi'}$ .) One tactic suffices: `field_simp`. Explain why the statement cannot even be formulated for the integer-valued `DilatonMeasure`.

**Exercise 17.9** (★★★, Lean). *Level 1 in  $D = 1$ .* Let  $g \neq 0$  and  $\gamma$  be real numbers,  $\gamma$  standing for  $\partial_x g$ , so that  $\partial_x g^{-1} = -\gamma/g^2$ . With  $\mathcal{H}^{MN} = \text{diag}(g, g^{-1})$  show on paper that the first term of (17.11) is  $\frac{1}{8}g^{-1} \cdot 2\gamma(-\gamma/g^2) = -\frac{1}{4}g^{-1}(\gamma/g)^2$  and that the second vanishes. Prove the displayed identity in Lean (`field_simp` followed by `ring`). Then formulate the general statement for  $d \times d$  real matrices using `Matrix.fromBlocks` and `Matrix.trace`, following the style of `genMetric`; proving it is a small research project.

## Further reading

- The action in terms of  $\mathcal{H}$  and  $d$ , its relation to the earlier  $E_{ij}$  formulation and the overview of results: [HHZ §1, ll. 66–400]. Construction of the action, the  $\mathbb{Z}_2$  argument and the reduction: [HHZ §4.1, ll. 1007–1266]. Generalized scalar curvature and proof of gauge invariance: [HHZ §4.2–4.3, ll. 1268–1528]. Generalized Ricci tensor: [HHZ §4.4, ll. 1530–1650]. Open problems: [HHZ §6, ll. 2411–2470].

- The NS–NS action and its equations of motion, the fields  $\mathcal{H}$  and  $d$ , the strong constraint: [AMN §3.2–3.3, ll. 850–1084]. Generalized Lie derivative and the density  $e^{-2d}$ : [AMN §3.4, ll. 1085–1125]. Closure constraints: [AMN §3.5, ll. 1173–1259]. Frame (flux) formulation of the action and its reduction to the generalized-metric form: [AMN §3.6, ll. 1260–1477].
- Scale-factor duality and the shifted dilaton in cosmology: [BW, ll. 340–445].
- In this book: the NS–NS action from worldsheet Weyl invariance in Chapter 6; the dilaton shift in Chapter 9; the strong constraint in Chapter 14;  $\mathcal{H}$  and its Lean formalization in Chapter 15; generalized Lie derivatives and the C-bracket in Chapter 16; string gas cosmology in Chapter 32; a catalogue of which Lean libraries are faithful and which are arithmetic shadows in Chapter 35.



## Part IV

# K3 Surfaces and Lattices





# Chapter 18

## Lattices: Even, Unimodular, $E_8$ and $U$

Every charge that has appeared in this book so far has been an integer: a momentum number  $n$ , a winding number  $w$ , a vector  $Z \in \mathbb{Z}^{2d}$  of both. Integers that can be added form a *lattice*, and the physics equips that lattice with an integer-valued inner product: for the circle, the level-matching pairing  $2nw$  of Chapter 7; for the torus, the form  $\eta$  of Chapters 10 and 11. In Part IV the same structure returns in geometry. The second cohomology of a K3 surface is a lattice of rank 22 whose inner product counts intersections of surfaces, and almost everything we shall say about K3 surfaces (their moduli, their symmetries, the strings that propagate on them) is phrased in terms of that lattice. The question of this chapter is therefore a purely algebraic one: *what is an integral lattice, which invariants distinguish one lattice from another, and what are the two building blocks,  $U$  and  $E_8$ , from which the K3 lattice is assembled?*

A student should care for two reasons. First, the classification is surprisingly rigid: an indefinite lattice that is even and unimodular is determined by two integers, its signature. This is why one can write down  $H^2(K3, \mathbb{Z})$  without ever seeing a K3 surface (Chapter 20). Secondly, the chapter is a good place to learn how finite, explicit mathematics is handed to a proof assistant. Every statement about  $E_8$  below is a statement about one  $8 \times 8$  integer matrix, and the Lean proofs are *certificates*: an inverse matrix, a triangular factorization, a sum of squares, each of which the kernel checks by arithmetic.

Conventions. A lattice is written  $\Lambda$ , its pairing  $(x \cdot y)$ , and  $x^2 := (x \cdot x)$ . Signatures are written  $(n_+, n_-)$ . For a lattice  $\Lambda$  and an integer  $m$ ,  $\Lambda(m)$  is the same group with the pairing multiplied by  $m$ . Direct sums are orthogonal. These are the conventions of our main source [Huy §14.0.1–0.2, ll. 12645–12705].

### 18.1 Integral lattices and Gram matrices

**Definition 18.1** (Lattice). A *lattice* is a free  $\mathbb{Z}$ -module  $\Lambda$  of finite rank  $n$  together with a symmetric bilinear form  $(\cdot): \Lambda \times \Lambda \rightarrow \mathbb{Z}$  which is non-degenerate: if  $(x \cdot y) = 0$  for all  $y$ , then  $x = 0$ . [Huy §14.0.1, ll. 12645–12649]

“Free of rank  $n$ ” means that there is a basis  $e_1, \dots, e_n$  such that every element is uniquely  $x = \sum_i x_i e_i$  with  $x_i \in \mathbb{Z}$ . Once a basis is chosen, the pairing is encoded in the *Gram matrix*

$$G_{ij} := (e_i \cdot e_j), \quad (x \cdot y) = \sum_{i,j} x_i G_{ij} y_j = x^\top G y, \quad (18.1)$$

a symmetric integer matrix; non-degeneracy is the statement  $\det G \neq 0$ . Note what is *not* assumed: the form need not be positive. The word “lattice” may suggest a grid of points in

Euclidean space, but the lattices of string theory are mostly *indefinite*: they contain nonzero vectors with  $x^2 = 0$  and vectors with  $x^2 < 0$ .

**Change of basis.** A second basis  $e'_j = \sum_i P_{ij}e_i$  is related to the first by an integer matrix  $P$  whose inverse is also an integer matrix, that is  $P \in GL(n, \mathbb{Z})$ , and then  $\det P = \pm 1$  (because  $\det P \cdot \det P^{-1} = 1$  in  $\mathbb{Z}$ ). The Gram matrix changes by a *congruence*,

$$G' = P^T G P, \quad P \in GL(n, \mathbb{Z}). \quad (18.2)$$

A lattice is thus an equivalence class of symmetric integer matrices under (18.2), and an *invariant* of the lattice is any quantity that does not change under it. Two lattices are *isomorphic*,  $\Lambda \cong \Lambda'$ , if there is a group isomorphism preserving the pairing; an isomorphism of  $\Lambda$  with itself is an *isometry*, and the isometries form the *orthogonal group*

$$O(\Lambda) = \{ g \in GL(n, \mathbb{Z}) : g^T G g = G \}. \quad (18.3)$$

For  $G = \eta$  this is the T-duality group  $O(d, d; \mathbb{Z})$  of Chapter 11. Deciding whether two given Gram matrices are congruent is hard in general, which is why invariants matter.

**Definition 18.2** (Even and odd). A lattice is *even* if  $x^2 \in 2\mathbb{Z}$  for every  $x \in \Lambda$ , and *odd* otherwise. [Huy §14.0.1, ll. 12649–12651]

Evenness is a condition on infinitely many vectors, but it can be read off from  $n$  numbers.

**Lemma 18.3** (Evenness from the diagonal). *Let  $G$  be a symmetric integer matrix. Then  $x^T G x$  is even for every  $x \in \mathbb{Z}^n$  if and only if every diagonal entry  $G_{ii}$  is even.*

*Proof.* Separate the diagonal terms from the off-diagonal ones and use  $G_{ij} = G_{ji}$ :

$$x^T G x = \sum_i G_{ii} x_i^2 + \sum_{i \neq j} G_{ij} x_i x_j = \sum_i G_{ii} x_i^2 + 2 \sum_{i < j} G_{ij} x_i x_j. \quad (18.4)$$

The second sum is even whatever the  $G_{ij}$  are. If all  $G_{ii}$  are even, the first sum is even too. Conversely,  $x = e_i$  gives  $x^T G x = G_{ii}$ .  $\square$

The “if” direction is the Lean theorem `even_quadratic_form_of_even_diag` of Section 18.8; its proof is exactly the splitting (18.4), with the matrix written as diagonal plus strictly upper triangular plus the transpose of the latter. Symmetry is essential: for the non-symmetric matrix  $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$  the diagonal is even but  $x^T G x = x_1 x_2$  is odd at  $x = (1, 1)$ .

**Example 18.4** (Three lattices of rank at most two). (i)  $\langle m \rangle$  denotes the rank-one lattice  $\mathbb{Z}e$  with  $e^2 = m$ . It is even exactly when  $m$  is even. (ii)  $\langle 1 \rangle \oplus \langle -1 \rangle$  has  $G = \text{diag}(1, -1)$  and  $x^2 = x_1^2 - x_2^2$ ; it is odd. (iii) The lattice with  $G = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  has  $x^2 = 2x_1 x_2$  and is even. Lattices (ii) and (iii) have the same rank and, as we shall see, the same determinant and the same signature. They are nevertheless not isomorphic, because one is even and the other is not, and parity is an invariant.

#### Why string theory wants even lattices

For a closed string on a circle the level-matching condition of Chapter 7 reads  $N - \tilde{N} = nw$ . The right-hand side is one half of the norm of the charge vector  $(n, w)$  in the lattice (iii) above. Level matching can be satisfied by integer oscillator levels only if that half-norm is an integer, that is, if the lattice is even. For  $d$  compact directions the statement is that the momenta  $(p_L, p_R)$  form an even lattice with  $p_L^2 - p_R^2 = 2m_i n^i \in 2\mathbb{Z}$ , and that this lattice

is self-dual (Section 18.2) [GPR §2.4, ll. 1249–1255]. The two conditions together are what modular invariance of the worldsheet theory on a torus requires [Asp §3.5, ll. 1987–1988]. The usual refinement is that evenness gives invariance of the partition function under  $\tau \rightarrow \tau + 1$  and self-duality gives invariance under  $\tau \rightarrow -1/\tau$ , by Poisson resummation over the lattice; this is standard, is not spelled out in the sources of this book’s library, and is the subject of Chapter 10.

## 18.2 The dual lattice, the discriminant and unimodularity

### 18.2.1 The discriminant

Under the congruence (18.2),  $\det G' = (\det P)^2 \det G = \det G$ . The determinant of a Gram matrix is therefore an invariant of the lattice, the *discriminant*  $\text{disc } \Lambda := \det G$  [Huy §14.0.1, ll. 12651–12652]. Its sign and its absolute value both carry information, and both have a concrete meaning which we now derive.

### 18.2.2 The dual lattice

Extend the pairing  $\mathbb{Q}$ -linearly to  $\Lambda_{\mathbb{Q}} := \Lambda \otimes \mathbb{Q}$ , the rational vector space with the same basis. The *dual lattice* is

$$\Lambda^* := \{ x \in \Lambda_{\mathbb{Q}} : (x \cdot y) \in \mathbb{Z} \text{ for all } y \in \Lambda \}, \quad (18.5)$$

which is a concrete model of  $\text{Hom}_{\mathbb{Z}}(\Lambda, \mathbb{Z})$  [Huy §14.0.1, ll. 12659–12666]. Because the pairing is integral on  $\Lambda$ , we have  $\Lambda \subseteq \Lambda^*$ . The dual lattice has the *dual basis*  $e_1^*, \dots, e_n^*$  defined by  $(e_i^* \cdot e_j) = \delta_{ij}$ . To find it, write  $e_i^* = \sum_k C_{ik} e_k$  with rational  $C$ ; the defining condition is  $\sum_k C_{ik} G_{kj} = \delta_{ij}$ , so

$$e_i^* = \sum_k (G^{-1})_{ik} e_k, \quad (e_i^* \cdot e_j^*) = (G^{-1} G G^{-1})_{ij} = (G^{-1})_{ij}. \quad (18.6)$$

In words: *the Gram matrix of the dual lattice is the inverse Gram matrix*. In the other direction,  $e_j = \sum_i G_{ji} e_i^*$ , since both sides have the same pairings with all  $e_k$ . In coordinates with respect to the dual basis,  $\Lambda^* = \mathbb{Z}^n$  and  $\Lambda = G \mathbb{Z}^n$ . The index of the subgroup  $G \mathbb{Z}^n \subseteq \mathbb{Z}^n$  is  $|\det G|$  (bring  $G$  to diagonal form  $\text{diag}(d_1, \dots, d_n)$  by integer row and column operations, which change neither the index nor  $|\det G|$ ; for a diagonal matrix the quotient is  $\prod_i \mathbb{Z}/d_i \mathbb{Z}$ ). Hence:

**Proposition 18.5.** *The discriminant group  $A_{\Lambda} := \Lambda^*/\Lambda$  is a finite abelian group of order  $|\text{disc } \Lambda|$ . [Huy §14.0.1, ll. 12667–12672]*

**Example 18.6** (The lattice  $A_2$ ). Let  $G = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$ , the Gram matrix of two vectors of length  $\sqrt{2}$  at an angle of  $120^\circ$  (the hexagonal lattice). Then  $\det G = 3$  and  $G^{-1} = \frac{1}{3} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ . The dual basis vector  $e_1^* = \frac{1}{3}(2e_1 + e_2)$  is not in  $\Lambda$ , but  $3e_1^*$  is, and  $e_2^* = 2e_1^* - e_1$  differs from  $2e_1^*$  by a lattice vector. So  $A_{\Lambda} \cong \mathbb{Z}/3\mathbb{Z}$ , generated by the class of  $e_1^*$ , of order  $3 = \det G$  as Proposition 18.5 demands. The norm  $(e_1^*)^2 = \frac{2}{3}$  is well defined modulo 2 on  $A_{\Lambda}$ ; this is the *discriminant form*  $q_{\Lambda}: A_{\Lambda} \rightarrow \mathbb{Q}/2\mathbb{Z}$  of an even lattice [Huy §14.0.1, ll. 12682–12690], which will be needed for sublattices of the K3 lattice in Chapter 20.

### 18.2.3 Unimodular lattices

**Definition 18.7.** A lattice is *unimodular* (or *self-dual*) if  $\Lambda^* = \Lambda$ ; equivalently if  $A_{\Lambda}$  is trivial; equivalently if  $\text{disc } \Lambda = \pm 1$ . [Huy §14.0.1, ll. 12672–12673]

The equivalences are Proposition 18.5. A fourth equivalent form is the one that a computer can check most cheaply.

**Lemma 18.8** (Inverse certificate). *A symmetric integer matrix  $G$  satisfies  $\det G = \pm 1$  if and only if there is an integer matrix  $H$  with  $HG = \mathbf{1}$ .*

*Proof.* If  $HG = \mathbf{1}$  with  $H$  integral, then  $\det H \cdot \det G = 1$  is a factorization of 1 in  $\mathbb{Z}$ , so  $\det G = \pm 1$ . Conversely, Cramer's rule gives  $G^{-1} = \text{adj}(G)/\det G$ , where the adjugate  $\text{adj}(G)$  is a matrix of minors of  $G$  and hence integral; if  $\det G = \pm 1$ , then  $H = \pm \text{adj}(G)$  is integral.  $\square$

By (18.6) the lemma says:  $\Lambda$  is unimodular exactly when the dual basis vectors already lie in  $\Lambda$ . The first half of the proof is the Lean lemma `isUnimodular_of_mul_eq_one`. It is the workhorse of the formalization, because it replaces the computation of an  $8 \times 8$  determinant (a sum of  $8! = 40320$  terms in the Leibniz expansion) by the verification of 64 integer identities.

**Unimodular does not mean “ $\det = +1$ ”, and says nothing about the sign of the form**

The three lattices  $\langle 1 \rangle \oplus \langle 1 \rangle$ ,  $\langle 1 \rangle \oplus \langle -1 \rangle$  and  $U = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  are all unimodular, with discriminants  $+1, -1, -1$ . The rank-8 lattices  $E_8$  and  $U^{\oplus 4}$  are both even and unimodular with discriminant  $+1$ , yet the first is positive definite and the second has signature  $(4, 4)$ . To distinguish them one needs the signature, which is the subject of the next section; for  $E_8$  it requires a separate proof (Section 18.6.4).

### 18.2.4 Sums, twists and sublattices

For an orthogonal direct sum the Gram matrix is block diagonal, so  $\text{disc}(\Lambda_1 \oplus \Lambda_2) = \text{disc} \Lambda_1 \cdot \text{disc} \Lambda_2$ , and a sum of even (unimodular) lattices is even (unimodular). For the twist,  $\text{disc} \Lambda(m) = m^{\text{rk} \Lambda} \text{disc} \Lambda$  [Huy §14.0.3, ll. 12819–12823]. If  $\Lambda_1 \subseteq \Lambda$  is a sublattice of the same rank, a basis of  $\Lambda_1$  is  $e' = eP$  with an integer matrix  $P$  of  $|\det P| = (\Lambda : \Lambda_1)$ , the index. Then  $G_1 = P^T G P$  gives

$$\text{disc} \Lambda_1 = \text{disc} \Lambda \cdot (\Lambda : \Lambda_1)^2 \quad [\text{Huy eq. (0.1), ll. 12708–12710}]. \quad (18.7)$$

This formula is a quick way to prove unimodularity: if a lattice of discriminant 4 has an integral overlattice of index 2, the overlattice is unimodular. We use it for  $E_8$  in Example 18.14.

## 18.3 Signature

Over the real numbers every non-degenerate symmetric bilinear form can be diagonalized with entries  $\pm 1$ : there is a real invertible  $P$  with  $P^T G P = \text{diag}(1, \dots, 1, -1, \dots, -1)$ . The numbers  $n_+$  and  $n_-$  of positive and negative entries form the *signature*  $(n_+, n_-)$ , and  $\tau := n_+ - n_-$  is the *index*. A lattice is *definite* if  $n_+ = 0$  or  $n_- = 0$  and *indefinite* otherwise [Huy §14.0.1, ll. 12653–12658]. The definition makes sense because of

**Theorem 18.9** (Sylvester's law of inertia). *The numbers  $n_{\pm}$  do not depend on the diagonalizing matrix  $P$ .*

*Proof.* Two diagonalizations give two splittings  $\mathbb{R}^n = V_+ \oplus V_- = V'_+ \oplus V'_-$ , where the form is positive definite on  $V_+$ ,  $V'_+$  and negative definite on  $V_-$ ,  $V'_-$ , and  $n_+ = \dim V_+$ ,  $n'_+ = \dim V'_+$ . A vector  $x \in V_+ \cap V'_-$  has  $x^2 > 0$  unless  $x = 0$ , and also  $x^2 \leq 0$ ; so  $V_+ \cap V'_- = 0$ , hence  $\dim V_+ + \dim V'_- \leq n = \dim V'_+ + \dim V'_-$ , that is  $n_+ \leq n'_+$ . By symmetry  $n_+ = n'_+$ .  $\square$

Two remarks make the signature computable. First, the sign of the discriminant is  $(-1)^{n_-}$ , because  $\det(P^\top GP) = (\det P)^2 \det G$ . Secondly,  $P$  need not be integral: *any* real or rational congruence to a diagonal matrix reveals the signature, and one reads off the signs of the diagonal entries. The standard way to produce such a congruence is Gaussian elimination without row exchanges, which factorizes

$$G = LDL^\top, \quad L \text{ unit lower triangular}, \quad D = \text{diag}(d_1, \dots, d_n), \quad (18.8)$$

whenever no zero pivot is met. Then  $x^\top Gx = \sum_k d_k y_k^2$  with  $y = L^\top x$ , which is “completing the square” one variable at a time. Since the upper-left  $k \times k$  block of (18.8) is the same factorization of the upper-left block  $G_{(k)}$  of  $G$ , and  $\det L_{(k)} = 1$ , the pivots are ratios of leading principal minors:

$$d_1 d_2 \cdots d_k = \det G_{(k)}, \quad d_k = \frac{\det G_{(k)}}{\det G_{(k-1)}}. \quad (18.9)$$

In particular  $G$  is positive definite if and only if all pivots are positive, if and only if all leading principal minors are positive (Sylvester’s criterion).

Signatures add under direct sums, and the twist by  $-1$  exchanges  $n_+$  and  $n_-$ . In physics language,  $n_+$  and  $n_-$  count left- and right-moving compact bosons (Chapter 10); in geometry, for the lattice  $H^2(X, \mathbb{Z})$  of a four-manifold, they count self-dual and anti-self-dual harmonic two-forms (Chapter 19).

## 18.4 The hyperbolic plane $U$

**Definition 18.10.** The *hyperbolic plane* is the lattice  $U = \mathbb{Z}e \oplus \mathbb{Z}f$  with  $e^2 = f^2 = 0$  and  $(e \cdot f) = 1$ , that is, with Gram matrix  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ . [Huy §14.0.3, ll. 12781–12787]

For  $x = ae + bf$  we have  $x^2 = 2ab$ . The lattice is even (the diagonal of  $G$  is zero; Lemma 18.3), and unimodular with  $\text{disc } U = -1$ ; indeed  $G^2 = \mathbf{1}$ , so  $G$  is its own integer inverse, and the dual basis is  $e^* = f$ ,  $f^* = e$ . The change of variables  $u = e + f$ ,  $v = e - f$  gives  $u^2 = 2$ ,  $v^2 = -2$ ,  $(u \cdot v) = 0$ , or in matrix form

$$P^\top \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} P = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}, \quad P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad (18.10)$$

so by Theorem 18.9 the signature is  $(1, 1)$ . Note that  $\det P = -2$ : the vectors  $u, v$  span a sublattice  $\langle 2 \rangle \oplus \langle -2 \rangle$  of index 2 and discriminant  $-4 = (-1) \cdot 2^2$ , in agreement with (18.7). Over  $\mathbb{Z}$  the lattice  $U$  *cannot* be diagonalized, since a diagonal unimodular lattice of rank two is odd (Example 18.4). Finally  $U(-1) \cong U$ , by  $e \mapsto e$ ,  $f \mapsto -f$  [Huy §14.0.3, l. 12831].

**Isometries.** A vector is *isotropic* if its norm is zero and *primitive* if it is not an integer multiple  $kx'$ ,  $|k| \geq 2$ , of another lattice vector. An isometry  $g$  of  $U$  must send  $e$  to a primitive isotropic vector, and  $2ab = 0$  with  $ae + bf$  primitive leaves only  $\pm e, \pm f$ . The image of  $f$  is then fixed by  $g(f)^2 = 0$  and  $(g(e) \cdot g(f)) = 1$ . Hence

$$O(U) = \{\pm \mathbf{1}, \pm \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\} \cong (\mathbb{Z}/2\mathbb{Z})^2 \quad [\text{Huy Ex. 14.2.1, ll. 13159–13164}], \quad (18.11)$$

which is the group  $O(1, 1; \mathbb{Z})$  found in Chapter 11. So  $U$  is an indefinite lattice with a *finite* orthogonal group. This is exceptional and is tied to the existence of isotropic vectors in rank two [Huy §14.2, ll. 13147–13157]; for  $U^{\oplus d}$  with  $d \geq 2$  the group  $O(d, d; \mathbb{Z})$  is infinite.

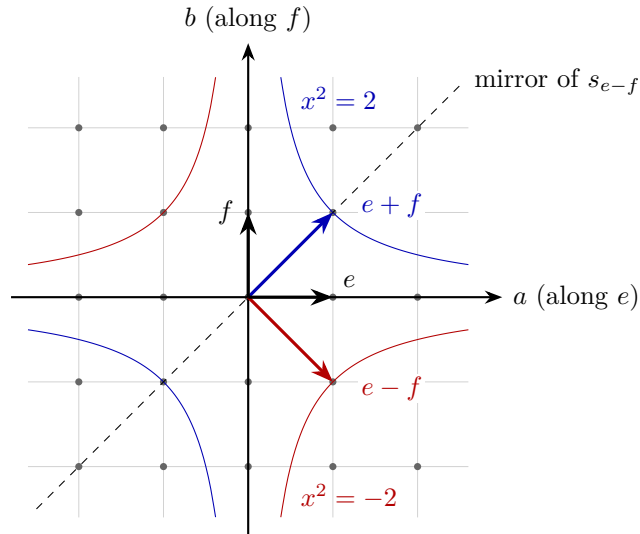


Figure 18.1: The hyperbolic plane  $U$  in the coordinates  $x = ae + bf$ , where  $x^2 = 2ab$ . The two axes are the isotropic (null) directions. The hyperbolas  $x^2 = \pm 2$  each contain exactly two lattice points,  $\pm(e + f)$  and  $\pm(e - f)$ . The reflection  $s_{e-f}$  of Section 18.7 is the mirror image in the dashed line: it exchanges  $e$  and  $f$ .

#### $U$ is the charge lattice of a circle

Identify  $a$  with the momentum number  $n$  and  $b$  with the winding number  $w$  of a closed string on a circle (Chapter 7). Then  $\frac{1}{2}x^2 = nw$  is the right-hand side of the level-matching condition, the two null directions are the pure-momentum and pure-winding states, and the nontrivial isometry  $e \leftrightarrow f$  is T-duality,  $n \leftrightarrow w$  (Chapter 8). The radius  $R$  does not appear: the lattice  $U$  with its pairing is the same for all radii. What depends on  $R$  is how  $U \otimes \mathbb{R} = \mathbb{R}^{1,1}$  is split into a positive line (left-movers) and a negative line (right-movers), and the mass formula measures the position of a charge vector relative to that splitting. The moduli space of the circle is the space of such splittings modulo  $O(U)$ . This picture generalizes verbatim to tori ( $U^{\oplus d}$ , Chapter 10) and to K3 surfaces (Chapter 23). In Lean, the Gram matrix `hyperbolicU` is proved equal to the Narain Gram matrix of the older library by `hyperbolicU_eq_narain_gram`; the identification of either matrix with a physical string is a modelling statement and is not Tier A.

**$U$  always splits off.** If a lattice  $\Lambda$  contains two vectors  $e, f$  with  $e^2 = f^2 = 0$  and  $(e \cdot f) = 1$ , then  $\Lambda = U \oplus U^\perp$ . Indeed, for  $\alpha \in \Lambda$  put  $\alpha' := \alpha - (e \cdot \alpha)f - (f \cdot \alpha)e$ . Then  $(\alpha' \cdot e) = (\alpha \cdot e) - (e \cdot \alpha) \cdot 1 - 0 = 0$  and likewise  $(\alpha' \cdot f) = 0$ , so  $\alpha' \in U^\perp$ , and  $\alpha$  is the sum of an element of  $U$  and one of  $U^\perp$  [Huy Ex. 14.0.3, ll. 12789–12794]. The argument uses only that the Gram matrix of  $e, f$  is invertible over  $\mathbb{Z}$ ; it works for any unimodular sublattice. For a non-unimodular sublattice  $\Lambda_1$  it fails, and  $\Lambda_1 \oplus \Lambda_1^\perp$  is in general only of finite index in  $\Lambda$  [Huy eq. (0.2), ll. 12717–12729].

## 18.5 The classification of indefinite even unimodular lattices

We have met three invariants of an even unimodular lattice: the rank, the discriminant  $\pm 1$  and the signature. The discriminant is  $(-1)^{n_-}$  and the rank is  $n_+ + n_-$ , so only the signature is independent. The remarkable fact is that for indefinite lattices there is nothing else.

**Theorem 18.11** (Classification; Tier L). *Let  $(n_+, n_-)$  be given. An even unimodular lattice of signature  $(n_+, n_-)$  exists if and only if*

$$n_+ - n_- \equiv 0 \pmod{8}. \quad (18.12)$$

*If  $n_+ > 0$  and  $n_- > 0$ , the lattice is unique up to isomorphism; for  $\tau = n_+ - n_- \geq 0$  it is*

$$II_{n_+, n_-} \cong E_8^{\oplus \tau/8} \oplus U^{\oplus n_-}, \quad \text{and for } \tau \leq 0: \quad II_{n_+, n_-} \cong E_8(-1)^{\oplus |\tau|/8} \oplus U^{\oplus n_+}. \quad (18.13)$$

[L] [Huy Thm. 14.1.1 and Cor. 14.1.3, ll. 12906–12932; Asp §2.3, ll. 559–565]

Huybrechts calls the theorem a classical result of Milnor and refers for the proof to Milnor–Husemoller, *Symmetric bilinear forms*, Ch. II, and Serre, *A course in arithmetic*, Ch. V [Huy ll. 12906–12907; bibliography ll. 19517, 19808]. Neither is in this book’s library and the theorem is not formalized in the Lean development; everywhere in this book it is Tier L. We use it as a black box and we now unpack what it says.

**Existence.** The “if” part is constructive once  $E_8$  and  $U$  are known to be even and unimodular of signatures  $(8, 0)$  and  $(1, 1)$ : the right-hand sides of (18.13) are even, unimodular, and have signature  $(8 \cdot \tau/8 + n_-, n_-) = (n_+, n_-)$ . This part *is* within reach of the elementary methods of this chapter, and Sections 18.6 and 18.6.4 supply it. The “only if” part, the congruence (18.12), is deep; we do not prove it. It is the algebraic counterpart of the fact that a modular-invariant lattice partition function needs  $n_L - n_R \in 8\mathbb{Z}$  [Asp §3.5, ll. 1987–1988].

**Uniqueness.** This is the part that is used most. Here are the instances that occur in this book.

| signature | $\tau$ | lattice                                  | where it appears                                 |
|-----------|--------|------------------------------------------|--------------------------------------------------|
| $(1, 1)$  | 0      | $U$                                      | charge lattice of a circle (Chapter 7)           |
| $(d, d)$  | 0      | $U^{\oplus d}$                           | Narain lattice of $T^d$ (Chapter 10)             |
| $(1, 9)$  | –8     | $E_8(-1) \oplus U$                       | Enriques lattice [Huy ll. 12886–12893]           |
| $(3, 19)$ | –16    | $E_8(-1)^{\oplus 2} \oplus U^{\oplus 3}$ | $H^2(K3, \mathbb{Z})$ (Chapter 20)               |
| $(4, 20)$ | –16    | $E_8(-1)^{\oplus 2} \oplus U^{\oplus 4}$ | Mukai lattice $H^*(K3, \mathbb{Z})$ (Chapter 22) |
| $(1, 25)$ | –24    | $E_8(-1)^{\oplus 3} \oplus U$            | Niemeier lattices (Chapter 29)                   |

The K3 row shows how the theorem is used in practice. Topology gives three facts about  $H^2(X, \mathbb{Z})$  of a K3 surface  $X$ : it is unimodular (Poincaré duality), even (the manifold is spin), and of signature  $(3, 19)$  (the signature theorem). The theorem then hands us the Gram matrix without any further geometry [Asp §2.3, ll. 523–636; Huy Prop. 1.3.5, ll. 600–609]. This argument is carried out in Chapters 19 and 20.

#### Definite lattices are not classified by their signature

The uniqueness statement needs  $n_{\pm} > 0$ . For definite lattices it fails badly: in rank 24 there are 24 different even unimodular negative definite lattices, the Niemeier lattices, of which  $E_8(-1)^{\oplus 3}$  is the simplest [Huy §14.4.1, ll. 13741–13760]. They all become isomorphic after adding one copy of  $U$ :  $N \oplus U \cong II_{1,25}$  [Huy Cor. 14.4.1, ll. 13744–13746]. Conversely, inside the single lattice  $II_{1,25}$  there are 24 inequivalent ways of choosing a primitive null vector [Huy ll. 13754–13757]. Adding a hyperbolic plane erases the distinction between definite lattices of the same rank.

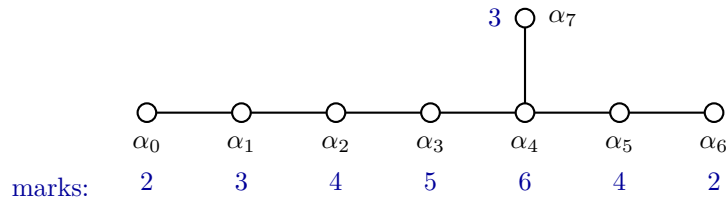


Figure 18.2: The Dynkin diagram of  $E_8$  in the labelling of the Lean file `E8.lean`: a chain  $\alpha_0, \dots, \alpha_6$  with  $\alpha_7$  attached to  $\alpha_4$ , so that the arms at the branch node have 4, 2 and 1 further nodes. The numbers in the second row are the first row of the inverse Cartan matrix (18.17); they are the coefficients of the highest root.

## 18.6 The lattice $E_8$

### 18.6.1 From a graph to a Gram matrix

To any graph  $\Gamma$  with simple edges one associates a lattice  $\Lambda(\Gamma)$  with one basis vector  $\alpha_i$  per vertex and

$$(\alpha_i \cdot \alpha_j) = \begin{cases} 2 & i = j, \\ -1 & i, j \text{ joined by an edge,} \\ 0 & \text{otherwise} \end{cases} \quad [\text{Huy §14.0.3 v), ll. 12835–12838}]. \quad (18.14)$$

Such a lattice is even by Lemma 18.3. The connected graphs for which it is positive definite are exactly the Dynkin diagrams of type  $A_n, D_n, E_6, E_7, E_8$  [Huy ll. 12839–12841]; the basis vectors are then called *simple roots* and the Gram matrix is the *Cartan matrix* of the corresponding simply-laced Lie algebra. Among all of them only  $E_8$  is unimodular: the discriminant groups are  $\mathbb{Z}/(n+1)$  for  $A_n$ , of order 4 for  $D_n$ , and  $\mathbb{Z}/3, \mathbb{Z}/2, 0$  for  $E_6, E_7, E_8$  [Huy ll. 12868–12872].

For the graph of Figure 18.2 the recipe (18.14) gives

$$E_8 := \begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 2 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 0 & -1 \\ 0 & 0 & 0 & 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 2 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 2 \end{pmatrix}, \quad (18.15)$$

which is the Lean matrix `cartanE8`. The only entries off the tridiagonal band are the two  $-1$ 's in positions (4, 7) and (7, 4), which record the extra edge.

#### Every book labels the nodes differently

The matrix printed by Huybrechts [Huy §14.0.3 iii), ll. 12796–12818] has its branch at the third node of the chain and is not equal to (18.15); Bourbaki's labelling is different again. The matrices are related by a permutation matrix  $P$  as in (18.2), so they define the same lattice. When comparing a formula with Lean code, first match the labelling: the inverse matrix, the pivots and the highest root all change their appearance under  $P$ .

### 18.6.2 The determinant from the diagram

The determinant of a graph lattice can be computed by removing leaves.



**Lemma 18.12** (Leaf recursion). *Let  $v$  be a vertex of  $\Gamma$  joined to exactly one other vertex  $u$ . Then*

$$\det \Lambda(\Gamma) = 2 \det \Lambda(\Gamma \setminus v) - \det \Lambda(\Gamma \setminus \{u, v\}).$$

*Proof.* Order the basis so that  $u$  and  $v$  come last. The Gram matrix is

$$G = \begin{pmatrix} G'' & b & 0 \\ b^\top & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix},$$

where  $G''$  is the Gram matrix of  $\Gamma \setminus \{u, v\}$ . Expand along the last row. The entry 2 contributes  $2 \det \Lambda(\Gamma \setminus v)$ . The entry  $-1$  in position  $(n, n-1)$  contributes  $(-1) \cdot (-1)^{2n-1} \det M = \det M$ , where  $M = \begin{pmatrix} G'' & 0 \\ b^\top & -1 \end{pmatrix}$  is obtained by deleting the last row and the column of  $u$ ; and  $\det M = -\det G''$ .  $\square$

For the chain  $A_n$  the lemma gives  $a_n = 2a_{n-1} - a_{n-2}$  with  $a_0 = 1$ ,  $a_1 = 2$ , hence  $\det A_n = n + 1$ . For  $E_8$  take  $v = \alpha_7$  and  $u = \alpha_4$ . Removing  $\alpha_7$  leaves the chain  $A_7$ ; removing also  $\alpha_4$  leaves  $A_4 \sqcup A_2$  (the nodes 0–3 and 5–6). Therefore

$$\det E_8 = 2 \cdot 8 - 5 \cdot 3 = 1. \quad (18.16)$$

The same computation for the diagram with a chain of  $n - 1$  nodes and the extra node attached to the third node from one end gives  $\det E_n = 2n - 3(n - 3) = 9 - n$ : the values 3, 2, 1 for  $n = 6, 7, 8$  match the orders of the discriminant groups quoted above,  $\det E_9 = 0$  says that the next diagram is degenerate, and  $\det E_{10} = -1$  says that  $E_{10}$  is again even and unimodular, now indefinite (a numerical check finds signature  $(9, 1)$ ), hence isomorphic to  $E_8 \oplus U$  by Theorem 18.11. The number 8 is singled out as the rank at which the determinant hits 1. Equation (18.16) is a proof on the page; Lean obtains  $\det E_8 = 1$  by another route (`cartanE8_det`, below).

### 18.6.3 The explicit inverse

By Lemma 18.8,  $\det E_8 = \pm 1$  is equivalent to the existence of an integer inverse. It is

$$E_8^{-1} = \begin{pmatrix} 2 & 3 & 4 & 5 & 6 & 4 & 2 & 3 \\ 3 & 6 & 8 & 10 & 12 & 8 & 4 & 6 \\ 4 & 8 & 12 & 15 & 18 & 12 & 6 & 9 \\ 5 & 10 & 15 & 20 & 24 & 16 & 8 & 12 \\ 6 & 12 & 18 & 24 & 30 & 20 & 10 & 15 \\ 4 & 8 & 12 & 16 & 20 & 14 & 7 & 10 \\ 2 & 4 & 6 & 8 & 10 & 7 & 4 & 5 \\ 3 & 6 & 9 & 12 & 15 & 10 & 5 & 8 \end{pmatrix}, \quad (18.17)$$

the Lean matrix `cartanE8Inv`. The reader should check a few entries of  $E_8^{-1}E_8 = \mathbf{1}$  by hand. For the first row: the  $(0, 0)$  entry is  $2 \cdot 2 + 3 \cdot (-1) = 1$ , the  $(0, 1)$  entry is  $2 \cdot (-1) + 3 \cdot 2 + 4 \cdot (-1) = 0$ , and the  $(0, 4)$  entry, which feels the branch, is  $-5 + 2 \cdot 6 - 4 - 3 = 0$ . In general the first row satisfies  $2m_i = \sum_{j \sim i} m_j$  at every node except  $\alpha_0$ , where  $2m_0 - m_1 = 1$ : each mark is half the sum of its neighbours. This is a discrete harmonic condition, and it makes the row easy to reconstruct from the diagram alone.

The inverse has a meaning. By (18.6) its rows are the coordinates of the dual basis vectors  $\omega_i = \sum_j (E_8^{-1})_{ij} \alpha_j$  (the *fundamental weights*), with  $(\omega_i \cdot \alpha_j) = \delta_{ij}$  and  $(\omega_i \cdot \omega_j) = (E_8^{-1})_{ij}$ . For  $A_2$  the fundamental weights were not in the lattice (Example 18.6); for  $E_8$  they are: *the weight*

*lattice of  $E_8$  equals its root lattice.* The diagonal entries 2, 6, 12, 20, 30, 14, 4, 8 are the norms  $\omega_i^2$ . Only  $\omega_0$  has norm 2, so it is itself a root,  $\omega_0 = 2\alpha_0 + 3\alpha_1 + 4\alpha_2 + 5\alpha_3 + 6\alpha_4 + 4\alpha_5 + 2\alpha_6 + 3\alpha_7$ . A computer enumeration (a symbolic check, not Tier A) confirms that this is the highest of the 120 positive roots; its coefficients sum to 29.

#### 18.6.4 Positive definiteness

Evenness and  $\det = 1$  do not yet distinguish  $E_8$  from  $U^{\oplus 4}$ . We need the signature, and we get it from the factorization (18.8). The leading principal minors of (18.15) are the determinants of  $A_1, \dots, A_7$  and of  $E_8$  itself, that is 2, 3, 4, 5, 6, 7, 8, 1. By (18.9) the pivots are

$$D = \text{diag}\left(2, \frac{3}{2}, \frac{4}{3}, \frac{5}{4}, \frac{6}{5}, \frac{7}{6}, \frac{8}{7}, \frac{1}{8}\right), \quad \prod_k d_k = 1. \quad (18.18)$$

All are positive, so  $E_8$  is positive definite, of signature (8, 0); the last pivot is small but not zero (for the diagram  $E_9$  of Section 18.6.2, with one more node on the long arm, the last pivot is  $0/9 = 0$ ). To make the argument checkable line by line, let us carry out the elimination. Write  $q(x) := x^\top E_8 x = 2 \sum_i x_i^2 - 2 \sum_{i \sim j} x_i x_j$ , where the second sum runs over the seven edges. Completing the square in  $x_0$ ,

$$2x_0^2 - 2x_0x_1 = 2\left(x_0 - \frac{1}{2}x_1\right)^2 - \frac{1}{2}x_1^2,$$

leaves  $x_1$  with the coefficient  $2 - \frac{1}{2} = \frac{3}{2}$ . Completing the square in  $x_1$ ,  $\frac{3}{2}x_1^2 - 2x_1x_2 = \frac{3}{2}\left(x_1 - \frac{2}{3}x_2\right)^2 - \frac{2}{3}x_2^2$ , leaves  $x_2$  with  $2 - \frac{2}{3} = \frac{4}{3}$ . Along the chain the pivot recursion is  $d_{k+1} = 2 - 1/d_k$ , solved by  $d_k = (k+2)/(k+1)$ . At the branch node  $x_4$  is coupled to both  $x_5$  and  $x_7$ , so the square contains both, and a cross term  $x_5x_7$  is generated which the next step removes. The result is

$$q(x) = 2y_0^2 + \frac{3}{2}y_1^2 + \frac{4}{3}y_2^2 + \frac{5}{4}y_3^2 + \frac{6}{5}y_4^2 + \frac{7}{6}y_5^2 + \frac{8}{7}y_6^2 + \frac{1}{8}y_7^2, \quad (18.19)$$

$$\begin{aligned} y_0 &= x_0 - \frac{1}{2}x_1, & y_1 &= x_1 - \frac{2}{3}x_2, & y_2 &= x_2 - \frac{3}{4}x_3, & y_3 &= x_3 - \frac{4}{5}x_4, \\ y_4 &= x_4 - \frac{5}{6}x_5 - \frac{5}{6}x_7, & y_5 &= x_5 - \frac{6}{7}x_6 - \frac{5}{7}x_7, & y_6 &= x_6 - \frac{5}{8}x_7, & y_7 &= x_7. \end{aligned} \quad (18.20)$$

This is the identity  $E_8 = LDL^\top$  with  $y = L^\top x$ ; the coefficients in (18.20) are the entries of the Lean matrix `e8L`. Both the matrix identity and the polynomial identity (18.19) were checked with `sympy` before being given to Lean. Positive definiteness now follows in two steps. (a)  $q(x) \geq 0$ , as a sum of squares with positive weights. (b) If  $q(x) = 0$ , then every  $y_k = 0$ ; reading (18.20) from the end,  $x_7 = 0$ , then  $x_6 = 0$ ,  $x_5 = 0$ ,  $\dots$ ,  $x_0 = 0$ . The substitution  $x \mapsto y$  is invertible precisely because it is unit triangular.

**Example 18.13** (The sum of the simple roots). Take  $x = (1, 1, 1, 1, 1, 1, 1, 1)$ , that is  $\beta = \alpha_0 + \dots + \alpha_7$ . Directly,  $q = 2 \cdot 8 - 2 \cdot 7 = 2$ : eight vertices, seven edges. Through (18.20),

$$y = \left(\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, -\frac{2}{3}, -\frac{4}{7}, \frac{3}{8}, 1\right), \quad q = \frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \frac{8}{15} + \frac{8}{21} + \frac{9}{56} + \frac{1}{8} = 2,$$

as one confirms with the common denominator 840:  $420 + 140 + 70 + 42 + 448 + 320 + 135 + 105 = 1680$ . So  $\beta$  is a vector of norm 2, a root. The same count shows that for any tree the sum of all basis vectors has norm  $2(\#\text{vertices} - \#\text{edges}) = 2$ .

**Example 18.14** ( $E_8$  inside  $\mathbb{R}^8$ , and the 240 roots). Positive definiteness means that  $E_8$  can be drawn as a lattice of points in Euclidean  $\mathbb{R}^8$ . A model which is standard (it is not taken from a source in this book's library; we verify what we use) is

$$\Gamma_8 := \{x \in \mathbb{Z}^8 \cup (\mathbb{Z} + \frac{1}{2})^8 : \sum_i x_i \in 2\mathbb{Z}\}$$

with the Euclidean inner product. In terms of the standard unit vectors  $\varepsilon_1, \dots, \varepsilon_8$  of  $\mathbb{R}^8$  put

$$\alpha_k = \varepsilon_{7-k} - \varepsilon_{6-k} \quad (0 \leq k \leq 5), \quad \alpha_6 = \frac{1}{2}(\varepsilon_1 + \varepsilon_8) - \frac{1}{2}(\varepsilon_2 + \dots + \varepsilon_7), \quad \alpha_7 = \varepsilon_1 + \varepsilon_2.$$

These eight vectors lie in  $\Gamma_8$  and their Gram matrix is exactly (18.15) (symbolic check; for instance  $(\alpha_5 \cdot \alpha_6) = (\varepsilon_2 - \varepsilon_1) \cdot \alpha_6 = -\frac{1}{2} - \frac{1}{2} = -1$  and  $(\alpha_6 \cdot \alpha_7) = \frac{1}{2} - \frac{1}{2} = 0$ ). Unimodularity is visible without any determinant: the integer vectors with even coordinate sum form the lattice  $D_8$  of index 2 in  $\mathbb{Z}^8$ , so  $\text{disc } D_8 = 1 \cdot 2^2 = 4$  by (18.7), and  $\Gamma_8 \supset D_8$  has index 2, so  $\text{disc } \Gamma_8 = 4/2^2 = 1$ . Since the  $\alpha_k$  span a sublattice of  $\Gamma_8$  of discriminant 1, the same formula shows that the index is 1: they span all of  $\Gamma_8$ .

The roots can now be counted. Integer vectors of norm 2 are  $\pm\varepsilon_i \pm \varepsilon_j$  with  $i < j$ :  $4\binom{8}{2} = 112$  of them, all of even sum. Half-integer vectors of norm 2 have all coordinates  $\pm\frac{1}{2}$  with an even number of minus signs:  $2^7 = 128$ . Together  $112 + 128 = 240$  roots, in agreement with [Huy Ex. 14.2.3, ll. 13197–13201]. A computer search that starts from the simple roots of (18.15) and closes under reflections also stops at 240 vectors (symbolic check).

### 18.6.5 $E_8(-1)$

In geometry the form is negative on the relevant classes: the exceptional curves  $C_i \cong \mathbb{P}^1$  of the minimal resolution of an  $E_8$  surface singularity  $x^2 + y^3 + z^5 = 0$  have  $C_i^2 = -2$  and meet each other according to the  $E_8$  Dynkin diagram [Huy §14.0.3v, ll. 12842–12860]. The lattice that occurs in  $H^2(K3, \mathbb{Z})$  is therefore the twist  $E_8(-1)$  with Gram matrix  $-E_8$ , which is even, unimodular (its discriminant is  $(-1)^8 \cdot 1 = 1$ ) and negative definite [Huy ll. 12829–12831]. In Lean it is `e8Neg`. The Lie algebra  $E_8$  and the gauge group of the heterotic string play no role in this chapter; only the lattice does.

## 18.7 Reflections and the Weyl group

### 18.7.1 Reflections in $(-2)$ -vectors

In a real vector space with a non-degenerate symmetric form, the reflection in the hyperplane orthogonal to a vector  $\delta$  with  $\delta^2 \neq 0$  is

$$s_\delta(x) = x - 2 \frac{(x \cdot \delta)}{\delta^2} \delta \quad [\text{Huy §7.5.4, ll. 6031–6034}]. \quad (18.21)$$

It fixes  $\delta^\perp$  and sends  $\delta \mapsto -\delta$ . On a lattice this map is in general not defined over  $\mathbb{Z}$ , because of the denominator  $\delta^2$ . It is integral when  $\delta^2 = \pm 1$  or  $\pm 2$ . In an even lattice the first case cannot occur, and we are left with  $\delta^2 = \pm 2$ . A vector with  $\delta^2 = -2$  is called a *root* or  *$(-2)$ -class*, and for it (18.21) becomes

$$s_\delta(x) = x + (x \cdot \delta) \delta, \quad \text{as a matrix} \quad S_\delta = \mathbf{1} + \delta \delta^\top G \quad [\text{Huy §14.2.1, ll. 13166–13176}], \quad (18.22)$$

where  $\delta$  is a column vector and we used  $(x \cdot \delta) = \delta^\top G x$ . (For a positive definite lattice such as  $E_8$  one uses vectors of norm  $+2$  and  $s_\delta(x) = x - (x \cdot \delta) \delta$ ; passing to  $E_8(-1)$  changes both signs and gives the same linear map.)

**Proposition 18.15.** *Let  $G$  be a symmetric integer matrix,  $\delta \in \mathbb{Z}^n$ ,  $N := \delta^\top G \delta$  and  $S := \mathbf{1} + \delta \delta^\top G$ . Then*

$$S^\top G S - G = (2 + N) G \delta \delta^\top G, \quad S^2 - \mathbf{1} = (2 + N) \delta \delta^\top G. \quad (18.23)$$

*In particular, if  $N = -2$  then  $S$  is an isometry of  $G$  and an involution.*

*Proof.* The key is that  $\delta^\top G \delta$  is a number, so  $(\delta \delta^\top G)(\delta \delta^\top G) = \delta(\delta^\top G \delta) \delta^\top G = N \delta \delta^\top G$ . Hence  $S^2 = \mathbf{1} + 2\delta \delta^\top G + N \delta \delta^\top G$ . Similarly, with  $S^\top = \mathbf{1} + G \delta \delta^\top$  (here symmetry of  $G$  is used),

$$S^\top G S = (\mathbf{1} + G \delta \delta^\top) G (\mathbf{1} + \delta \delta^\top G) = G + 2G \delta \delta^\top G + N G \delta \delta^\top G. \quad \square$$

The factor  $(2 + N)$  explains why the norm  $-2$  is special for the formula (18.22). The Lean theorems `reflection_isometry` and `reflection_involution` are the two consequences at  $N = -2$ , for arbitrary rank; the general identities (18.23) are proved above on the page and were also checked with `sympy`, but are not themselves Lean theorems.

**Example 18.16** (T-duality is a reflection). In  $U$  the vector  $\delta = e - f$  has  $\delta^2 = -2$ , and  $s_\delta(e) = e + (e \cdot (e - f))(e - f) = e - (e - f) = f$ . So the reflection  $s_{e-f}$  exchanges  $e$  and  $f$  (Figure 18.1): with the identification of Section 18.4, the T-duality  $n \leftrightarrow w$  of a circle is the reflection in the  $(-2)$ -vector  $e - f$ . The only roots of  $U$  are  $\pm(e - f)$ , since  $2ab = -2$  forces  $(a, b) = \pm(1, -1)$ . Hence the Weyl group of  $U$  is  $\mathbb{Z}/2\mathbb{Z}$ , a proper subgroup of  $O(U) \cong (\mathbb{Z}/2\mathbb{Z})^2$  [Huy ll. 13181–13182]; the element  $-\mathbf{1}$  is not a product of reflections in roots.

**Example 18.17** (A simple reflection of  $E_8(-1)$ ). Take  $G = -E_8$  and  $\delta = \alpha_4$ , the branch node. Then  $(x \cdot \alpha_4) = -(E_8 x)_4 = x_3 - 2x_4 + x_5 + x_7$ , so  $s_{\alpha_4}$  changes only the fourth coordinate:

$$s_{\alpha_4}(x) = (x_0, x_1, x_2, x_3, x_3 - x_4 + x_5 + x_7, x_5, x_6, x_7).$$

In particular  $s_{\alpha_4}(\alpha_3) = \alpha_3 + \alpha_4$ , again a root, and  $s_{\alpha_4}(\alpha_4) = -\alpha_4$ . Applying it twice returns  $x_3 - (x_3 - x_4 + x_5 + x_7) + x_5 + x_7 = x_4$ , as it must.

### 18.7.2 The Weyl group

The *Weyl group*  $W(\Lambda) \subseteq O(\Lambda)$  is the subgroup generated by all reflections  $s_\delta$  in roots [Huy §14.2.1, ll. 13177–13180]. The following facts are Tier L and are not formalized.

- For  $E_8(-1)$ , which has 240 roots, the Weyl group has order  $2^{14} \cdot 3^5 \cdot 5^2 \cdot 7 = 4! \cdot 6! \cdot 8! = 696\,729\,600$  and is the whole orthogonal group  $O(E_8(-1))$  [Huy Ex. 14.2.3, ll. 13197–13201]. That it is generated by the eight *simple* reflections  $s_{\alpha_0}, \dots, s_{\alpha_7}$  is standard Lie theory (not checked against a source in this book’s library).
- For the K3 lattice, every isometry of trivial spinor norm (equivalently, every isometry preserving the orientation of the three positive directions) is a product of reflections in  $(-2)$ -classes [Huy Thm. 14.2.2, ll. 13184–13195; §7.5.4, ll. 6028–6043]. This is the lattice-theoretic input to the global structure of the K3 moduli space in Chapters 20 and 23.
- In general  $W$  can have infinite index in  $O(\Lambda)$  [Huy ll. 13205–13210].

#### Reflections as monodromies

Reflections in  $(-2)$ -classes are what a K3 surface does to its own cohomology when it is taken around a singular point of its moduli space. If a family of K3 surfaces degenerates by shrinking a sphere  $C$  with  $C^2 = -2$  to a point (an  $A_1$  singularity), then transporting  $H^2(X, \mathbb{Z})$  once around the degeneration acts as the reflection  $s_\delta$  in the class  $\delta = [C]$  [Huy §6, ll. 5398–5401]. This is the Picard–Lefschetz formula. In type IIA string theory on a K3 surface, a shrinking sphere means that a brane wrapped on it becomes massless, and a non-abelian gauge symmetry appears whose roots are the  $(-2)$ -classes of the shrunk spheres; the roots all have the same length, so the gauge algebra is of ADE type [\[L\]](#) [Asp §4.3, ll. 2464–2485, 2551–2565]; see Chapter 23. The same source notes that the simplest instance is the circle, where at a special radius a pair  $\pm\alpha$  of charge vectors enhances  $U(1)$

to  $SU(2)$  [Asp ll. 2472–2477]. Compare Example 18.16: for the circle the reflection  $s_{e-f}$  is T-duality, whose fixed point is the self-dual radius of Chapter 8.

## 18.8 What the machine has checked

The five files `Basic.lean`, `E8.lean`, `E8PosDef.lean`, `Hyperbolic.lean` and `Reflection.lean` of the directory `DualScaleStream2/Lattice/` formalize the finite, explicit part of this chapter. A design decision should be stated first: *a lattice is represented by its Gram matrix in a fixed basis*. There is no type of abstract  $\mathbb{Z}$ -modules with a form, no notion of lattice isomorphism, and therefore no statement in Lean that could express Theorem 18.11. What Lean proves are properties of particular integer matrices, together with a few general lemmas about all integer matrices of a given size. All theorems quoted below depend only on the standard axioms `propext`, `Classical.choice` and `Quot.sound`, according to the project’s axiom audit [docs/VERIFIED\_FOUNDATION.md, ll. 7–16].

### Definitions and the two general lemmas (`Basic.lean`)

```
abbrev Gram (n : ℕ) := Matrix (Fin n) (Fin n) ℤ
def IsEvenDiag {n : ℕ} (G : Gram n) : Prop := ∀ i, Even (G i i)
def IsUnimodular {n : ℕ} (G : Gram n) : Prop := G.det = 1 ∨ G.det = -1

theorem isUnimodular_of_mul_eq_one {n : ℕ} (G H : Gram n) (h : H * G = 1) :
  IsUnimodular G := by ...

theorem even_quadratic_form_of_even_diag {n : ℕ} (G : Gram n)
  (hsymm : GT = G) (heven : IsEvenDiag G) (x : Fin n → ℤ) :
  Even (x ·v (G ·v x)) := by ...
```

**Reading.** `Gram n` is just the type of  $n \times n$  integer matrices; symmetry and non-degeneracy are not part of the type and appear as hypotheses where needed. `IsEvenDiag` is a property of a *basis* (even diagonal), not the definition of an even lattice; the second theorem upgrades it to the statement that  $x^T G x$  is even for all integer  $x$ , which is one direction of Lemma 18.3. The first theorem is one direction of Lemma 18.8: an integer left inverse implies  $\det G = \pm 1$ . The converse directions are true and elementary but are not stated in Lean.

**Proof idea.** For the certificate:  $\det(HG) = \det H \det G = 1$  in  $\mathbb{Z}$ , and the library lemma `Int.eq_one_or_neg_one_of_mul_eq_one` finishes. For evenness: write  $G = D + U + U^T$  with  $D$  diagonal and  $U$  strictly upper triangular, show  $x^T U^T x = x^T U x$ , and conclude  $x^T G x = x^T D x + 2x^T U x$ , which is (18.4).

### $E_8$ and $U$ : symmetric, even diagonal, unimodular (`E8.lean`, `Hyperbolic.lean`)

```
theorem cartanE8Inv_mul : cartanE8Inv * cartanE8 = 1 := by ...
theorem cartanE8_symm : cartanE8T = cartanE8 := by ...
theorem cartanE8_evenDiag : IsEvenDiag cartanE8 := by ...
theorem cartanE8_unimodular : IsUnimodular cartanE8 :=
  isUnimodular_of_mul_eq_one _ _ cartanE8Inv_mul

def e8Neg : Gram 8 := -cartanE8
theorem e8Neg_unimodular : IsUnimodular e8Neg := by ...

def hyperbolicU : Gram 2 := !![0, 1; 1, 0]
theorem hyperbolicU_mul_self : hyperbolicU * hyperbolicU = 1 := by ...
```

```

theorem hyperbolicU_unimodular : IsUnimodular hyperbolicU :=
  isUnimodular_of_mul_eq_one _ _ hyperbolicU_mul_self
theorem hyperbolicU_congruence :
  (!! [1, 1; 1, -1] : Gram 2)T * hyperbolicU * (!! [1, 1; 1, -1]
    = !! [2, 0; 0, -2] := by ...

```

**Reading.** `cartanE8` and `cartanE8Inv` are the literal matrices (18.15) and (18.17). The inverse was computed outside Lean and is not trusted: if one entry were wrong, `cartanE8Inv_mul` would not compile. Unimodularity of  $E_8$ , of  $E_8(-1)$  and of  $U$  are then one-line applications of the certificate lemma. The companions `e8Neg_symm`, `e8Neg_evenDiag`, `hyperbolicU_symm` and `hyperbolicU_evenDiag` complete the list. `hyperbolicU_congruence` is (18.10); it exhibits one positive and one negative direction, but the step from there to “the signature of  $U$  is  $(1, 1)$ ” is Theorem 18.9, which is proved on the page and not in Lean. That the matrix `cartanE8` is the root lattice of the Lie algebra  $E_8$  is a matter of definition (Equation (18.14)), not a theorem.

**Proof idea.** Each matrix identity is proved by enumerating the index pairs (`fin_cases i <;> fin_cases j`, 64 cases for  $E_8$ ) and evaluating both sides. This is a decision procedure for literal matrices of small integers, not an argument that would extend to a family of Cartan matrices.

#### det $E_8 = 1$ and positive definiteness (`E8PosDef.lean`)

```

noncomputable def e8D : Matrix (Fin 8) (Fin 8) ℝ :=
  Matrix.diagonal ![2, 3/2, 4/3, 5/4, 6/5, 7/6, 8/7, 1/8]
noncomputable def cartanE8R : Matrix (Fin 8) (Fin 8) ℝ :=
  cartanE8.map (Int.cast : ℤ → ℝ)

theorem e8_LDL : e8L * e8D * e8LT = cartanE8R := by ...
theorem cartanE8_det : cartanE8R.det = 1 := by ...
theorem cartanE8_posDef : cartanE8R.PosDef := by ...

```

**Reading.** `e8L` is the unit lower triangular real matrix whose transpose produces the variables (18.20), and `e8D` holds the pivots (18.18). `e8_LDL` is the factorization (18.8) for  $E_8$  over  $\mathbb{R}$ . `cartanE8_det` sharpens  $\det = \pm 1$  to the integer statement  $\det E_8 = +1$ . `cartanE8_posDef` says that the real matrix is positive definite in Mathlib’s sense: symmetric, and  $x^T E_8 x > 0$  for every nonzero  $x \in \mathbb{R}^8$ . These are theorems about one fixed matrix. They do not say that  $E_8$  is the *unique* positive definite even unimodular lattice of rank 8 (true, Tier L, not in our sources in this form), and the Lean development does not contain a notion of signature of a matrix: the pair `sigE8Neg` =  $(0, 8)$  used in Chapter 20 is a definition, whose justification by `cartanE8_posDef` is made on paper.

**Proof idea.** For the determinant:  $\det L = \det L^T = 1$  because the matrices are unit triangular,  $\det D = 2 \cdot \frac{3}{2} \cdots \frac{8}{7} \cdot \frac{1}{8} = 1$  telescopes, multiplicativity gives  $\det E_8 = 1$  over  $\mathbb{R}$ , and `Int.cast_det` transports the result to  $\mathbb{Z}$ . For positive definiteness: unfold  $x^T E_8 x$  into an explicit polynomial in  $x_0, \dots, x_7$ ; introduce  $y_0, \dots, y_7$  by (18.20); prove (18.19) by the tactic `ring`; then argue by contradiction exactly as in steps (a), (b) of Section 18.6.4: a sum of non-negative terms that is  $\leq 0$  forces every  $y_k^2 = 0$  (eight calls of `linarith`), and back-substitution forces  $x = 0$ .

**Why the obvious route failed.** It is instructive that `cartanE8_posDef` does *not* use `e8_LDL`. The natural plan is: from  $E_8 = LDL^T$  deduce  $x^T E_8 x = (L^T x)^T D (L^T x)$ , then use that  $D$  is positive and  $L^T$  is invertible. On paper this is two lines. According to the project’s notes, the automated attempts along this route went through the literal 64-entry product

$LDL^T$  of real matrices and failed; the notes attribute this to the cost of unfolding large definitions (we have not re-run the failed attempts). The theorem was closed only after the *statement to be checked* was changed: the scalar polynomial identity (18.19) was derived and verified with sympy outside Lean, and inside Lean it is a single call to `ring`, which normalizes polynomials efficiently and never unfolds a matrix product [LL.md §S2.7, ll. 69–73; docs/STREAM2\_WORKFLOW.md, ll. 168–170]. The lesson recorded there is “pick the proof route before picking the tier”: when a goal times out, look for a formulation that avoids large definitional unfolding before using a stronger prover (Chapter 37). A related incident concerns `e8_LDL` itself: one automated attempt abandoned it because single-file compilation ran with Lean’s default limit of 200 000 heartbeats instead of the project’s 1 000 000; a stricter limit can reject a valid proof but can never accept an invalid one [LL.md §S2.4, ll. 47–54].

#### Reflections (Reflection.lean)

```
def reflection (L : Gram n) (v : Fin n → ℤ) : Gram n :=
  1 + vecMulVec v v * L
def latticeNorm (L : Gram n) (v : Fin n → ℤ) : ℤ := v ·v (L *v v)

theorem vecMulVec_mul_self (L : Gram n) (v : Fin n → ℤ) :
  vecMulVec v v * L * vecMulVec v v
    = (latticeNorm L v) • vecMulVec v v := by ...

theorem reflection_isometry (L : Gram n) (hL : LT = L) (v : Fin n → ℤ)
  (hv : latticeNorm L v = -2) :
  (reflection L v)T * L * reflection L v = L := by ...

theorem reflection_involution (L : Gram n) (v : Fin n → ℤ)
  (hv : latticeNorm L v = -2) :
  reflection L v * reflection L v = 1 := by ...

theorem e8Neg_simpleRoot_norm (i : Fin 8) :
  latticeNorm e8Neg (Pi.single i 1) = -2 := by ...
theorem e8Neg_weyl_isometry (i : Fin 8) :
  (reflection e8Neg (Pi.single i 1))T * e8Neg
    * reflection e8Neg (Pi.single i 1) = e8Neg := ...
```

**Reading.** `vecMulVec v v` is the matrix  $vv^T$ , so `reflection L v` is  $S = \mathbf{1} + vv^T L$  of (18.22). The two main theorems hold for every rank  $n$  and every integer matrix  $L$  (symmetric, for the isometry): if  $v^T L v = -2$  then  $S^T L S = L$  and  $S^2 = \mathbf{1}$ . These are the  $N = -2$  cases of Proposition 18.15. The last two theorems specialize to the eight simple roots of  $E_8(-1)$  (`Pi.single i 1` is the  $i$ -th standard basis vector). They do not cover the other 232 roots one by one, although `reflection_isometry` applies to any of them once its norm is computed; and nothing is proved about the group generated by the reflections, its order, or its relation to  $O(E_8(-1))$ .

**Proof idea.** `vecMulVec_mul_self` is the observation that  $v^T L v$  is a scalar. With it,  $(vv^T L)^2 = -2vv^T L$ ; expanding  $(\mathbf{1} + Lvv^T)L(\mathbf{1} + vv^T L)$  with the tactic `noncomm_ring` and substituting leaves  $L$ . No case analysis is involved, which is why the result holds in every rank.

| Claim                                                                                                                                                            | Tier | Evidence                                                                                                       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----------------------------------------------------------------------------------------------------------------|
| Integer left inverse $\Rightarrow \det G = \pm 1$ ; even diagonal and symmetric $\Rightarrow x^\top G x$ even                                                    | [A]  | isUnimodular_of_mul_eq_one, even_quadratic_form_of_even_diag                                                   |
| $E_8, E_8(-1), U$ : symmetric, even diagonal, unimodular                                                                                                         | [A]  | cartanE8_symm, cartanE8_evenDiag, cartanE8_unimodular, e8Neg_unimodular, hyperbolicU_unimodular and companions |
| $E_8^{-1}$ is (18.17); $E_8 = LDL^\top$ ; $\det E_8 = 1$ ; $E_8$ positive definite over $\mathbb{R}$                                                             | [A]  | cartanE8Inv_mul, e8_LDL, cartanE8_det, cartanE8_posDef                                                         |
| $P^\top U P = \text{diag}(2, -2)$ ; $U$ equals the Narain Gram matrix of the older library                                                                       | [A]  | hyperbolicU_congruence, hyperbolicU_eq_narain_gram                                                             |
| Reflection in a $(-2)$ -vector is an isometry and an involution (any rank); the eight simple reflections of $E_8(-1)$                                            | [A]  | reflection_isometry, reflection_involution, e8Neg_weyl_isometry                                                |
| Converses of the two general lemmas; $ A_\Lambda  =  \text{disc } \Lambda $ ; Sylvester's law; leaf recursion and $\det E_n = 9 - n$ ; Proposition 18.15; $O(U)$ | —    | proved on the page                                                                                             |
| $\Gamma_8$ model, 240 roots, highest root, signature of $E_{10}$                                                                                                 | —    | symbolic checks (sympy), count of roots also [L] via Huy Ex. 14.2.3                                            |
| Classification of indefinite even unimodular lattices; $n_+ - n_- \equiv 0 \pmod{8}$                                                                             | [L]  | Huy Thm. 14.1.1, Cor. 14.1.3; Asp §2.3                                                                         |
| $ W(E_8)  = 696\,729\,600 =  O(E_8) $ ; generation of $O^+$ of the K3 lattice by reflections; 24 Niemeier lattices                                               | [L]  | Huy Ex. 14.2.3, Thm. 14.2.2, §14.4.1                                                                           |
| Charge lattices of consistent string backgrounds are even and self-dual                                                                                          | [L]  | GPR §2.4; Asp §3.5; identification of a Lean matrix with a physical lattice is never Tier A                    |

### Chapter summary

- A lattice is a free  $\mathbb{Z}$ -module with a non-degenerate symmetric integer pairing; in a basis it is a Gram matrix  $G$ , defined up to  $G \mapsto P^\top G P$  with  $P \in GL(n, \mathbb{Z})$ . Invariants: rank, parity (even or odd), discriminant  $\det G$ , signature  $(n_+, n_-)$ .
- A symmetric  $G$  gives an even lattice if and only if its diagonal is even. The dual lattice has Gram matrix  $G^{-1}$ , and  $|\Lambda^* / \Lambda| = |\det G|$ . Unimodular means  $\Lambda^* = \Lambda$ , or  $\det G = \pm 1$ , or:  $G$  has an integer inverse. The last form is a certificate that a proof assistant can check entry by entry.
- The signature is read off from any real diagonalization (Sylvester). Gaussian elimination  $G = LDL^\top$  gives pivots equal to ratios of leading minors; all pivots positive means positive definite.
- $U$  has Gram matrix  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ : even, unimodular, signature  $(1, 1)$ ,  $O(U) \cong (\mathbb{Z}/2)^2$ . It is the momentum–winding lattice of a circle, and it splits off orthogonally from any lattice that contains it.
- $E_8$  is the graph lattice of the Dynkin diagram of Figure 18.2: even,  $\det E_8 = 2 \cdot 8 - 5 \cdot 3 = 1$ , positive definite with pivots  $2, \frac{3}{2}, \dots, \frac{8}{7}, \frac{1}{8}$ , with 240 roots.  $E_8(-1)$  is its negative.
- Classification (Tier L): even unimodular lattices exist only for  $n_+ - n_- \equiv 0 \pmod{8}$ , and an indefinite one is determined by its signature:  $E_8(\pm 1)^{\oplus |\tau|/8} \oplus U^{\oplus \min(n_+, n_-)}$ . Hence  $II_{3,19} = E_8(-1)^{\oplus 2} \oplus U^{\oplus 3}$ . Definite lattices are not determined by their rank.
- For  $\delta^2 = -2$  the reflection  $s_\delta(x) = x + (x \cdot \delta)\delta$  is an integral isometry of order two; in general  $S^\top G S - G = (2 + \delta^2)G\delta\delta^\top G$ . The reflections generate the Weyl group. T-duality of the circle is the reflection  $s_{e-f}$  of  $U$ .



- Lean proves all the finite statements about the matrices  $E_8$ ,  $-E_8$ ,  $U$  and the general reflection lemma. The classification theorem, the Weyl group and every physical identification are outside the formalization.

## Exercises

**Exercise 18.1** (★ Parity is an invariant). Show directly from (18.2) that if  $G$  has even diagonal and  $P \in GL(n, \mathbb{Z})$ , then  $P^T G P$  has even diagonal. Deduce that  $\langle 1 \rangle \oplus \langle -1 \rangle \not\cong U$ , and find all vectors of norm 0 and of norm 1 in each of the two lattices.

**Exercise 18.2** (★ Discriminant groups). Compute  $G^{-1}$  and the discriminant group  $A_\Lambda$  for (a)  $A_3$  (a chain of three nodes), (b)  $U(2)$ , (c)  $\langle 2 \rangle \oplus \langle -2 \rangle$ . Lattices (b) and (c) have the same rank, signature and discriminant. Are they isomorphic? (Hint: look for vectors of norm 2.)

**Exercise 18.3** (★ Marks from the diagram). Using only the rule “each mark is half the sum of its neighbours, except at  $\alpha_0$  where  $2m_0 - m_1 = 1$ ”, reconstruct the first row of (18.17) starting from the three ends of the diagram. Then do the same for the row of  $\omega_7$ , where the inhomogeneous condition sits at  $\alpha_7$ .

**Exercise 18.4** (★★ The family  $E_n$ ). Prove  $\det E_n = 9 - n$  from Lemma 18.12. Compute the pivots of  $E_7$  and  $E_6$  in a labelling analogous to Figure 18.2 and confirm that both are positive definite. Which pivot of  $E_9$  vanishes, and what is the null vector? (It is the vector of marks of the affine diagram.)

**Exercise 18.5** (★★  $U$  splits off, with numbers). In  $\Lambda = E_8(-1) \oplus U$  with basis  $\alpha_0, \dots, \alpha_7, e, f$ , let  $e' = e$  and  $f' = f + \alpha_0 + e$ . Check that  $e'^2 = f'^2 = 0$  and  $(e' \cdot f') = 1$ . Compute the projection  $\alpha \mapsto \alpha'$  of Section 18.4 for  $\alpha = \alpha_0$  and  $\alpha = \alpha_1$ , and verify that the images of  $\alpha_0, \dots, \alpha_7$  span a copy of  $E_8(-1)$  orthogonal to  $e', f'$ . Conclude that the decomposition  $II_{1,9} = E_8(-1) \oplus U$  is far from unique.

**Exercise 18.6** (★★ Reflections of norm +2 and  $\pm 1$ ). Repeat Proposition 18.15 for  $S = 1 - c\delta\delta^T G$  with a rational constant  $c$ , and show that  $S$  is an isometry if and only if  $c = 0$  or  $c = 2/\delta^2$ . For which values of  $\delta^2$  is  $S$  integral for every integral  $G$  and  $\delta$ ? Show that in  $U$  the reflection in  $e + f$  is  $-s_{e-f}$ , and identify it with an element of (18.11).

**Exercise 18.7** (★★ Lean:  $A_2$  is not unimodular). Define in Lean the Gram matrix of  $A_2$  and state that its determinant is 3:

```
-- exercise: not yet proved (not in the library)
def cartanA2 : Gram 2 := !![2, -1; -1, 2]
theorem cartanA2_det : cartanA2.det = 3 := by
  simp [cartanA2, Matrix.det_fin_two]; norm_num
```

Repair the proof if necessary, and deduce that `IsUnimodular cartanA2` is false. Then prove that it has even diagonal, imitating the proof of `hyperbolicU_evenDiag`.

**Exercise 18.8** (★★ Lean: the missing converse). State and prove in Lean the converse of `even_quadratic_form_of_even_diag`: if `Even (x ·v (G *v x))` holds for every  $x$ , then `IsEvenDiag G`. Hint: specialize to the basis vector `Pi.single i 1`. The proof of the theorem `e8Neg_simpleRoot_norm` shows which `simp` lemmas evaluate the quadratic form on a basis vector. Does your proof need symmetry of  $G$ ?

**Exercise 18.9** (\*\*\* Lean: a non-simple root). Let  $\theta$  be the highest root, in Lean the vector `![2, 3, 4, 5, 6, 4, 2, 3]` of type `Fin 8  $\rightarrow$   $\mathbb{Z}$` . Prove that `latticeNorm e8Neg` takes the value  $-2$  on it, and use `reflection_isometry` to conclude that the corresponding reflection is an isometry of `e8Neg`. Explain why no new general theorem is needed, and estimate what a Lean proof that there are exactly 240 integer vectors of norm  $-2$  would require (the search space must first be bounded; use (18.19)).

**Exercise 18.10** (\*\*\* Positive definiteness of all  $A_n$ ). Show that the pivots of the chain  $A_n$  are  $d_k = (k+1)/k$ ,  $k = 1, \dots, n$ , and write  $x^\top A_n x$  as an explicit sum of squares. Formulate the statement “ $A_n$  is positive definite for every  $n$ ” in Lean. Why can the proof technique of `cartanE8_posDef` (a fixed list of eight variables and one call of `ring`) not be reused, and what would replace it? (An alternative identity:  $x^\top A_n x = x_1^2 + \sum_{k=1}^{n-1} (x_k - x_{k+1})^2 + x_n^2$ .)

## Further reading

- [Huy ch. 14, ll. 12627–13010, 13141–13215, 13734–13760] is the reference for everything in this chapter and much more (discriminant forms, primitive embeddings, Nikulin’s theorems). §14.0 (ll. 12645–12895) contains the definitions and the list of standard lattices, §14.1 (ll. 12899–12932) the classification, §14.2 (ll. 13141–13210) the orthogonal group and reflections, §14.4 (ll. 13734–13760) Niemeier lattices. The proofs of the classification are in the books of Serre and of Milnor–Husemoller cited in Section 18.5; they are not in this book’s library.
- [Asp §2.3, ll. 520–636] derives the lattice structure of  $H_2(K3, \mathbb{Z})$  (signature, self-duality, evenness) in the language of a physicist and states the classification; [Asp §4.3, ll. 2464–2485] explains how vectors of norm  $-2$  become roots of a gauge group.
- [GPR §2.4, ll. 1249–1258] states the even self-dual property of the Narain lattice and its relation to  $O(d, d; \mathbb{R})$ ; see Chapters 10 and 11.
- The Lean files of this chapter carry long doc-comments stating what is and is not proved; `E8PosDef.lean` is the one to read for the certificate technique. Chapter 20 assembles  $E_8(-1)^{\oplus 2} \oplus U^{\oplus 3}$  and Chapter 22 adds the fourth copy of  $U$ .

## Chapter 19

# K3 Surfaces: Definition, Examples and Topology

Up to this point the compact spaces of this book were tori. A torus is flat, its string spectrum can be written in closed form, and its duality group  $O(d, d; \mathbb{Z})$  acts on an explicit lattice of momenta and windings (Chapters 10 and 11). The price is that a flat space has trivial holonomy and therefore breaks no supersymmetry at all. The next simplest compact space on which a string can propagate consistently is curved, and in real dimension four there is essentially one choice: the K3 surface. This chapter answers three questions. What is a K3 surface, and how does one write one down? What are its topological invariants, in particular the numbers 24, 22, 20 and  $(3, 19)$  that will reappear in every later chapter of Part IV? And why do physicists care: what is special about the holonomy of a K3 surface, and what does it do to supersymmetry? The reader needs no algebraic geometry beyond what is recalled here; the two tools we use repeatedly are Chern classes, treated as formal polynomials, and the Riemann–Roch formula, treated as a black box with a precise statement. The lattice that the invariants determine,  $3U \oplus 2E_8(-1)$ , was constructed in Chapter 18 and is studied as the cohomology of a K3 surface in Chapter 20.

### 19.1 The definition and what each condition means

Let  $X$  be a compact connected complex manifold of complex dimension two, a *complex surface*. It carries the sheaf  $\mathcal{O}_X$  of holomorphic functions, the rank-two bundle  $\Omega_X$  of holomorphic one-forms  $a_1 dz_1 + a_2 dz_2$ , and its top exterior power, the *canonical bundle*  $\omega_X = \Omega_X^2 = \Lambda^2 \Omega_X$ , whose local sections are the holomorphic two-forms  $a dz_1 \wedge dz_2$ . Its first Chern class is  $c_1(\omega_X) = -c_1(T_X)$ , where  $T_X$  is the holomorphic tangent bundle; the class  $K = -c_1(T_X)$  is called the canonical class.

**Definition 19.1** (K3 surface). A *complex K3 surface* is a compact connected complex manifold  $X$  of complex dimension two such that

$$\Omega_X^2 \cong \mathcal{O}_X \quad \text{and} \quad H^1(X, \mathcal{O}_X) = 0. \quad (19.1)$$

[Huy ch. 1, Def. 3.1, ll. 464–465; algebraic version over any field: Def. 1.1, ll. 158–161]

Both conditions have a direct meaning.

**Trivial canonical bundle.** The isomorphism  $\Omega_X^2 \cong \mathcal{O}_X$  says that there is a holomorphic two-form  $\Omega$  on  $X$  that vanishes nowhere. It is unique up to a constant: if  $\Omega'$  is a second one,

the ratio  $\Omega'/\Omega$  is a holomorphic function on a compact connected manifold, hence constant by the maximum principle. So

$$h^{2,0}(X) := \dim H^0(X, \Omega_X^2) = 1. \quad (19.2)$$

[Asp §2.1, ll. 260–265] In particular  $c_1(T_X) = 0$ . The form  $\Omega$  also gives a pairing  $\Omega_X \times \Omega_X \rightarrow \omega_X \cong \mathcal{O}_X$ ,  $(\alpha, \beta) \mapsto \alpha \wedge \beta / \Omega$ , which is alternating and non-degenerate at every point: a *holomorphic symplectic structure*. It identifies the tangent bundle with the cotangent bundle,  $T_X \cong \Omega_X$  [Huy ch. 1 §1, ll. 165–170]; we use this identification in Section 19.4.

**No holomorphic one-forms, no first cohomology.** The second condition excludes the other compact surfaces with a nowhere-vanishing holomorphic two-form, the complex tori  $\mathbb{C}^2/\Gamma$ : on a torus  $dz_1 \wedge dz_2$  trivializes the canonical bundle, but  $H^1(\mathcal{O})$  is two-dimensional. We will see in Section 19.4 that  $H^1(X, \mathcal{O}_X) = 0$  forces  $H^1(X, \mathbb{Z}) = 0$ , so the first Betti number of a K3 surface vanishes; in fact a K3 surface is simply connected (Theorem 19.3 below).

*Remark 19.2* (Two definitions in use). Physics texts often define a K3 surface as a compact complex *Kähler* surface with  $h^{1,0} = 0$  and  $K = 0$  [Asp §2.1, ll. 229–233]. The Kähler condition need not be imposed: every complex K3 surface in the sense of Definition 19.1 is Kähler, a fact that is “of great importance, but not easy to prove” [Huy ch. 1 §3.1, ll. 495–497]. [L] On a Kähler manifold  $h^{1,0} = h^{0,1}$ , so the two definitions agree. Not every complex K3 surface is projective, however; the algebraic ones form a proper subset [Huy ch. 1 §3.1, ll. 486–493].

#### Why “K3”?

The name was coined shortly after the first ascent, in 1954, of the mountain K2 in the Karakoram; it is explained as honouring Kummer, Kähler and Kodaira, with an allusion to that mountain. It does not mean “Kummer’s third surface”, a reading that circulates in part of the physics literature. [Asp §2, ll. 198–211 and footnote 1, ll. 237–241]

The fact that makes the subject tractable is that the definition, which constrains only holomorphic data, pins down the underlying smooth manifold completely.

**Theorem 19.3** (One diffeomorphism type). *Any two complex K3 surfaces are diffeomorphic. In particular every K3 surface is diffeomorphic to a smooth quartic in  $\mathbb{P}^3$ , and is simply connected.* [L] [Asp §2.1, Theorem 1, l. 236; Huy ch. 1, Remark 3.6(i), ll. 626–632]

Consequently *all topological invariants of a K3 surface can be computed on one example*. This is the strategy of Aspinwall’s lectures, followed in Section 19.2. Huybrechts instead derives the same numbers from the definition alone by Riemann–Roch; we do this in Section 19.4. The agreement of the two routes checks every number in this chapter.

#### “The” K3 surface

Physicists say “the K3 surface” because of Theorem 19.3: there is one smooth four-manifold. As *complex* manifolds K3 surfaces come in a continuous family with twenty complex parameters, and the choice of a Ricci-flat metric adds further parameters (Chapters 20 and 23). Topological statements ( $\chi = 24$ , signature  $(3, 19)$ ) hold for all of them; statements about curves, Picard groups or symmetries depend on which member of the family is meant.

## 19.2 The first example: a quartic surface

Let  $f(x_0, x_1, x_2, x_3)$  be a homogeneous polynomial of degree  $n$  and

$$S_n = \{[x_0 : x_1 : x_2 : x_3] \in \mathbb{P}^3 : f(x) = 0\}. \quad (19.3)$$

We assume  $S_n$  smooth: the four partial derivatives  $\partial f/\partial x_i$  have no common zero on  $S_n$ . The standard example is the Fermat surface  $f = x_0^n + x_1^n + x_2^n + x_3^n$  [Asp §2.1, eq. (4), ll. 246–250]. We ask for which  $n$  the surface  $S_n$  is a K3 surface, and answer the question twice.

### 19.2.1 Writing down the holomorphic two-form

In the patch  $x_0 \neq 0$  use affine coordinates  $y_i = x_i/x_0$  ( $i = 1, 2, 3$ ) and write  $F(y) = f(1, y_1, y_2, y_3)$ , with partial derivatives  $F_i = \partial F/\partial y_i$ . On the surface  $F = 0$  we have  $dF = F_1 dy_1 + F_2 dy_2 + F_3 dy_3 = 0$ . Wedge this relation with  $dy_2$ :  $F_1 dy_1 \wedge dy_2 = -F_3 dy_3 \wedge dy_2 = F_3 dy_2 \wedge dy_3$ . Doing the same with  $dy_1$  one finds that the three expressions

$$\Omega = \frac{dy_1 \wedge dy_2}{F_3} = \frac{dy_2 \wedge dy_3}{F_1} = \frac{dy_3 \wedge dy_1}{F_2} \quad (19.4)$$

agree on  $S_n$  wherever they are defined [Asp §2.1, eq. (6), ll. 266–293]. Smoothness says that at each point at least one  $F_i$  is nonzero. Where  $F_3 \neq 0$  the implicit function theorem makes  $(y_1, y_2)$  local coordinates on  $S_n$ , and the first expression is a holomorphic two-form without zeros; similarly for the other two. Hence  $\Omega$  is holomorphic and nowhere zero on the whole patch  $x_0 \neq 0$ . This is the residue of the meromorphic three-form  $dy_1 \wedge dy_2 \wedge dy_3/F$  along  $F = 0$  [Huy ch. 1, Example 1.3(i), eqs. (1.1)–(1.2), ll. 194–206].

The question is what happens on the “hyperplane at infinity”  $x_0 = 0$ . Aspinwall calls the check “straightforward but rather tedious”; it takes six lines. In the patch  $x_1 \neq 0$  use  $u_0 = x_0/x_1$ ,  $u_2 = x_2/x_1$ ,  $u_3 = x_3/x_1$  and  $G(u) = f(u_0, 1, u_2, u_3)$ . On the overlap

$$y_1 = \frac{1}{u_0}, \quad y_2 = \frac{u_2}{u_0}, \quad y_3 = \frac{u_3}{u_0}, \quad F(y) = u_0^{-n} G(u), \quad (19.5)$$

the last equality because  $f$  is homogeneous of degree  $n$ . Then

$$dy_1 \wedge dy_2 = \left( -\frac{du_0}{u_0^2} \right) \wedge \left( \frac{du_2}{u_0} - \frac{u_2 du_0}{u_0^2} \right) = -\frac{du_0 \wedge du_2}{u_0^3}, \quad (19.6)$$

$$F_3 = u_0^{-n} \frac{\partial G}{\partial u_3} \frac{\partial u_3}{\partial y_3} = u_0^{1-n} G_3, \quad (19.7)$$

where we used  $u_3 = y_3/y_1$ , so  $\partial u_3/\partial y_3 = 1/y_1 = u_0$ , while  $u_0$  and  $u_2$  do not depend on  $y_3$ . Dividing,

$$\Omega = -u_0^{n-4} \frac{du_0 \wedge du_2}{G_3}. \quad (19.8)$$

The form  $du_0 \wedge du_2/G_3$  is the analogue of (19.4) in the second patch, holomorphic and nowhere zero there. So along the curve  $u_0 = 0$  the form  $\Omega$  has a zero of order  $n - 4$  if  $n > 4$ , a pole of order  $4 - n$  if  $n < 4$ , and extends to a nowhere-vanishing holomorphic two-form exactly when

$$n = 4. \quad (19.9)$$

For  $n \neq 4$  equation (19.8) is still informative: it says that the canonical bundle of  $S_n$  is  $\mathcal{O}(n - 4)$  restricted to  $S_n$ , the adjunction formula  $\omega_S \cong (\omega_{\mathbb{P}^3} \otimes \mathcal{O}(n))|_S$  with  $\omega_{\mathbb{P}^3} = \mathcal{O}(-4)$  [Huy ch. 1, Example 1.3(i), ll. 190–194].

The second condition of Definition 19.1 holds for every  $n$ : from the exact sequence  $0 \rightarrow \mathcal{O}(-n) \rightarrow \mathcal{O} \rightarrow \mathcal{O}_S \rightarrow 0$  on  $\mathbb{P}^3$  and the vanishing  $H^1(\mathbb{P}^3, \mathcal{O}) = H^2(\mathbb{P}^3, \mathcal{O}(-n)) = 0$  one reads off  $H^1(S, \mathcal{O}_S) = 0$  [Huy ch. 1, Example 1.3(i), ll. 185–189]. Topologically this is the Lefschetz hyperplane theorem, which also gives  $\pi_1(S) = 0$  [Asp §2.1, ll. 253–254 and eq. (13), ll. 364–366]. Therefore:

**Proposition 19.4.** *A smooth quartic surface in  $\mathbb{P}^3$  is a K3 surface. [L]*

### 19.2.2 Chern classes by adjunction

The same conclusion, and the Euler characteristic as a bonus, follows from a formal computation. Let  $x \in H^2(\mathbb{P}^3, \mathbb{Z})$  be the class of a hyperplane;  $x^3$  integrates to 1 over  $\mathbb{P}^3$  and  $x^4 = 0$ . The total Chern class of projective space is  $c(T_{\mathbb{P}^k}) = (1+x)^{k+1}$ , and the normal bundle  $N_S$  of a degree- $n$  hypersurface is  $\mathcal{O}(n)|_S$  with  $c(N_S) = 1+nx$ . Since  $T_{\mathbb{P}^3}|_S = T_S \oplus N_S$  as smooth bundles and the total Chern class is multiplicative,

$$c(T_S) = \frac{(1+x)^4}{1+nx} = (1+4x+6x^2)(1-nx+n^2x^2) + O(x^3) = 1 + (4-n)x + (n^2-4n+6)x^2. \quad (19.10)$$

[Asp §2.1, eqs. (7)–(11), ll. 296–337] So  $c_1(T_S) = (4-n)x$ , which vanishes exactly for  $n = 4$ , in agreement with (19.8). The top Chern class integrates to the Euler characteristic. To integrate over  $S$  we integrate over  $\mathbb{P}^3$  after multiplying with the class dual to  $S$ , which is  $nx$ :

$$\chi(S_n) = \int_{S_n} c_2(T_S) = \int_{\mathbb{P}^3} nx \cdot (n^2-4n+6)x^2 = n(n^2-4n+6), \quad \chi(S_4) = 4 \cdot 6 = 24. \quad (19.11)$$

[Asp §2.1, eq. (12), ll. 337–363]

**Example 19.5** (The surfaces of degree 1 to 5). Formula (19.10) can be tested on surfaces the reader already knows. Besides  $\chi$  we compute two further characteristic numbers, whose meaning is explained in Sections 19.4 and 19.5: the holomorphic Euler characteristic  $\chi(\mathcal{O}) = (c_1^2 + c_2)/12$  and the signature  $\sigma = (c_1^2 - 2c_2)/3$ . With  $c_1^2[S_n] = n(4-n)^2$  and  $c_2[S_n] = n(n^2-4n+6)$ ,

$$\chi(\mathcal{O}_{S_n}) = \frac{n(n^2-6n+11)}{6}, \quad \sigma(S_n) = -\frac{n(n-2)(n+2)}{3}. \quad (19.12)$$

All  $S_n$  are simply connected, so  $b_1 = b_3 = 0$  and  $b_2 = \chi - 2$ ; the numbers  $b_{\pm}$  of positive and negative eigenvalues of the intersection form follow from  $b_+ + b_- = b_2$ ,  $b_+ - b_- = \sigma$ .

| $n$ | $c_1$ | $\chi$ | $\chi(\mathcal{O})$ | $\sigma$ | $b_2$ | $(b_+, b_-)$ | surface                                        |
|-----|-------|--------|---------------------|----------|-------|--------------|------------------------------------------------|
| 1   | $3x$  | 3      | 1                   | 1        | 1     | (1, 0)       | the plane $\mathbb{P}^2$                       |
| 2   | $2x$  | 4      | 1                   | 0        | 2     | (1, 1)       | the quadric $\mathbb{P}^1 \times \mathbb{P}^1$ |
| 3   | $x$   | 9      | 1                   | -5       | 7     | (1, 6)       | cubic: $\mathbb{P}^2$ blown up in 6 points     |
| 4   | 0     | 24     | 2                   | -16      | 22    | (3, 19)      | <b>K3</b>                                      |
| 5   | $-x$  | 55     | 5                   | -35      | 53    | (9, 44)      | quintic surface (general type)                 |

The first three rows reproduce known answers:  $\chi(\mathbb{P}^2) = 3$ ; the quadric is a product of two spheres with  $\chi = 2 \cdot 2$  and hyperbolic intersection form  $U$ ; blowing up a point adds one class of self-intersection  $-1$ , so six blow-ups of  $\mathbb{P}^2$  give  $\chi = 3 + 6$  and  $(b_+, b_-) = (1, 6)$ . The row  $n = 4$  contains every number of this chapter. It is the only row with  $c_1 = 0$ : positive  $c_1$  ( $n < 4$ ) gives rational surfaces, negative  $c_1$  ( $n > 4$ ) surfaces of general type, and K3 sits on the boundary, as a Ricci-flat space should. (The table is a symbolic check with sympy; the identification of the low-degree surfaces is standard and not checked against a source in this book's library.)

## 19.3 More examples

The quartic is one member of a short list of classical constructions [Huy ch. 1, Example 1.3, ll. 185–270]. Each gives K3 surfaces depending on parameters, and by Theorem 19.3 each must have  $\chi = 24$ , which we verify as a check of the method.

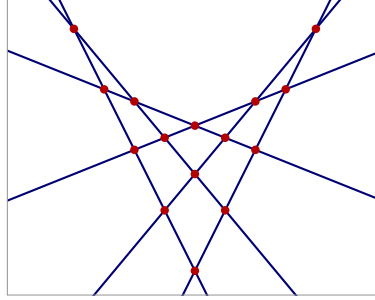


Figure 19.1: A degenerate branch sextic: six lines in general position in the plane meet in  $\binom{6}{2} = 15$  points (dots). The double cover of the plane branched along the six lines has an ordinary double point over each dot; resolving the 15 points gives a smooth K3 surface.

### 19.3.1 Complete intersections

Let  $S \subset \mathbb{P}^{m+2}$  be a smooth surface cut out by  $m$  homogeneous equations of degrees  $d_1, \dots, d_m$ . The normal bundle is  $\bigoplus_i \mathcal{O}(d_i)|_S$  and (19.10) becomes

$$c(T_S) = \frac{(1+x)^{m+3}}{\prod_i (1+d_i x)} \implies c_1(T_S) = \left(m+3 - \sum_i d_i\right)x. \quad (19.13)$$

So  $S$  is a K3 surface if and only if  $\sum_i d_i = m+3$ . A linear equation ( $d_i = 1$ ) only lowers the dimension of the ambient projective space, so assume all  $d_i \geq 2$ ; then  $2m \leq m+3$  leaves exactly three cases [Huy ch. 1, Example 1.3(ii), ll. 212–217]:

| ambient space  | degrees $(d_i)$ | $c_2(T_S)$ | degree $\prod d_i$ | $\chi = \prod d_i \cdot c_2$ |
|----------------|-----------------|------------|--------------------|------------------------------|
| $\mathbb{P}^3$ | (4)             | $6x^2$     | 4                  | 24                           |
| $\mathbb{P}^4$ | (2, 3)          | $4x^2$     | 6                  | 24                           |
| $\mathbb{P}^5$ | (2, 2, 2)       | $3x^2$     | 8                  | 24                           |

For instance, for (2, 3):  $(1+x)^5 = 1+5x+10x^2$ ,  $1/[(1+2x)(1+3x)] = 1-5x+19x^2$ , and the  $x^2$  coefficient of the product is  $10-25+19=4$ . The class dual to  $S$  is  $\prod d_i x^m$ , which supplies the factor 6. The three Euler characteristics agree, as they must. (The  $c_2$  column and  $\chi$  are derived here and checked with sympy; the source states only the list of degrees.)

### 19.3.2 Double planes

Let  $C \subset \mathbb{P}^2$  be a smooth curve of degree six and  $\pi : X \rightarrow \mathbb{P}^2$  the double cover branched along  $C$ : in an affine chart,  $X = \{w^2 = g(y_1, y_2)\}$  with  $g$  the sextic equation. Then  $\pi_* \mathcal{O}_X \cong \mathcal{O} \oplus \mathcal{O}(-3)$ , which gives  $H^1(X, \mathcal{O}_X) = 0$ , and the canonical bundle formula for branched covers gives  $\omega_X \cong \pi^*(\omega_{\mathbb{P}^2} \otimes \mathcal{O}(3)) \cong \mathcal{O}_X$  because  $\omega_{\mathbb{P}^2} = \mathcal{O}(-3)$ . So  $X$  is a K3 surface [L] [Huy ch. 1, Example 1.3(iv), ll. 260–266]. The degree six is forced: a branch curve of degree  $2k$  gives  $\omega_X = \pi^* \mathcal{O}(k-3)$ . For the Euler characteristic, remove the branch curve, where the cover is two-to-one, and add it back once:

$$\chi(X) = 2(\chi(\mathbb{P}^2) - \chi(C)) + \chi(C) = 2 \cdot 3 - \chi(C) = 6 - (2 - 2g) = 6 + 18 = 24, \quad (19.14)$$

where the genus of a smooth plane curve of degree  $d$  is  $g = (d-1)(d-2)/2 = 10$  for  $d = 6$ .

The branch curve may degenerate. If  $C$  is the union of six lines in general position (Figure 19.1), the double cover acquires  $\binom{6}{2} = 15$  rational double points, one over each intersection point, and blowing them up produces a smooth K3 surface; the canonical bundle does not change under this resolution [Huy ch. 1, Example 1.3(iv), ll. 267–270]. The Euler characteristic is again 24 (Exercise 19.1).

### 19.3.3 Kummer surfaces

Let  $A = \mathbb{C}^2/\Gamma$  be a complex two-torus and  $\iota : x \mapsto -x$ . The fixed points of  $\iota$  are the  $2^4 = 16$  points of order two. The quotient  $A/\iota$  has 16 singular points, each locally  $\mathbb{C}^2/\{\pm 1\}$ , and its minimal resolution  $X$  is a K3 surface, the Kummer surface of  $A$ . [L] [Huy ch. 1, Example 1.3(iii), ll. 218–259, and Example 3.3, ll. 499–507] The two conditions are inherited from the torus:  $dz_1 \wedge dz_2$  is invariant under  $\iota$  and descends to a nowhere-vanishing two-form, while  $\iota$  acts as  $-1$  on  $H^1(A, \mathcal{O}_A)$  (spanned by the classes of  $d\bar{z}_1, d\bar{z}_2$ ), so nothing invariant survives:  $H^1(X, \mathcal{O}_X) = 0$  although  $H^1(A, \mathcal{O}_A) \neq 0$ . Topologically, by the same cut-and-paste as above,

$$\chi(X) = \frac{\chi(A) - 16}{2} + 16 \cdot \chi(S^2) = \frac{0 - 16}{2} + 32 = 24. \quad (19.15)$$

This is the example closest to string theory, since  $A/\iota$  is the orbifold  $T^4/\mathbb{Z}_2$  on which strings can be quantized exactly; Chapter 21 is devoted to it. It is also the only one of our constructions that produces non-projective K3 surfaces, namely whenever the torus  $A$  is not projective [Huy ch. 1, Example 3.3, ll. 508–513].

### 19.3.4 Elliptic K3 surfaces

Many K3 surfaces admit a map  $\pi : X \rightarrow \mathbb{P}^1$  whose generic fibre is an elliptic curve  $y^2 = 4x^3 - g_2x - g_3$ , where  $g_2$  and  $g_3$  are homogeneous polynomials of degree 8 and 12 on the base [Huy ch. 1 §4.4, ll. 721–728]. The fibre degenerates where the discriminant  $\Delta = g_2^3 - 27g_3^2$ , of degree 24, vanishes: generically there are 24 singular fibres, each a sphere with two points identified ( $\chi = 1$ ), while smooth fibres have  $\chi(T^2) = 0$ . Hence  $\chi(X) = 24$  once more. (The fibre count is standard; only the degrees of  $g_2, g_3$  are taken from the source.) In F-theory the 24 points are the positions of 24 seven-branes; the same number controls the tadpole condition of Chapter 25.

## 19.4 Hodge numbers from Riemann–Roch

We now forget the examples and derive the invariants from Definition 19.1 alone. The *Hodge numbers* of a compact complex manifold are

$$h^{p,q}(X) = \dim H^q(X, \Omega_X^p), \quad (19.16)$$

the dimension of the space of  $\bar{\partial}$ -closed  $(p, q)$ -forms modulo  $\bar{\partial}$ -exact ones. [Huy ch. 1 §2.4, ll. 406–410; Asp §2.1, eq. (1), ll. 217–222] For a surface  $0 \leq p, q \leq 2$ , so there are nine of them. Two general tools constrain them.

**Serre duality.** On a compact complex surface,  $H^q(X, E) \cong H^{2-q}(X, E^* \otimes \omega_X)^*$  for every holomorphic vector bundle  $E$ . With  $\omega_X \cong \mathcal{O}_X$  and  $(\Omega_X^p)^* \cong \Omega_X^{2-p}$  this becomes  $h^{p,q} = h^{2-p, 2-q}$ : the Hodge diamond is symmetric under the rotation by  $180^\circ$ .

**Hirzebruch–Riemann–Roch.** For a holomorphic vector bundle  $E$  of rank  $r$  on a compact complex surface, the holomorphic Euler characteristic  $\chi(X, E) = \sum_q (-1)^q \dim H^q(X, E)$  is the integral of a characteristic class,

$$\begin{aligned} \chi(X, E) &= \int_X \text{ch}(E) \text{td}(X), \\ \text{ch}(E) &= r + c_1(E) + \frac{1}{2}(c_1(E)^2 - 2c_2(E)), \quad \text{td}(X) = 1 + \frac{1}{2}c_1 + \frac{1}{12}(c_1^2 + c_2), \end{aligned} \quad (19.17)$$



where  $c_i = c_i(T_X)$ . The left side is analytic (it counts solutions of differential equations); the right side is topological. [Huy ch. 1 §2.4, eq. (2.6), ll. 416–419; the expansions of  $\text{ch}$  and  $\text{td}$  to degree two are standard]

#### 19.4.1 Step 1: $\chi(\mathcal{O}_X) = 2$ and Noether’s formula give $c_2 = 24$

By definition  $h^{0,0} = 1$  (holomorphic functions are constant) and  $h^{0,1} = 0$ . Serre duality gives  $h^{0,2} = \dim H^2(X, \mathcal{O}_X) = \dim H^0(X, \omega_X) = h^{2,0} = 1$ . Hence

$$\chi(X, \mathcal{O}_X) = h^{0,0} - h^{0,1} + h^{0,2} = 1 - 0 + 1 = 2. \quad (19.18)$$

[Huy ch. 1 §2.3, ll. 348–351] Now apply (19.17) to  $E = \mathcal{O}_X$ , for which  $\text{ch} = 1$ . Only the degree-four part of the Todd class contributes to the integral, and we obtain *Noether’s formula*

$$\chi(X, \mathcal{O}_X) = \frac{c_1^2(X) + c_2(X)}{12}. \quad (19.19)$$

For a K3 surface  $c_1 = 0$ , so  $2 = c_2(X)/12$ , that is,

$$\chi_{\text{top}}(X) = \int_X c_2(X) = 24. \quad (19.20)$$

[Huy ch. 1 §2.4, ll. 420–426] This agrees with the direct count (19.11) on the quartic, and with the cut-and-paste counts of Section 19.3. Note what was used: only  $c_1 = 0$ ,  $h^{0,1} = 0$  and the uniqueness of  $\Omega$ .

#### 19.4.2 Step 2: no holomorphic one-forms

For a complex K3 surface the exponential sequence  $0 \rightarrow \mathbb{Z} \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_X^* \rightarrow 0$  (the second map is  $f \mapsto e^{2\pi i f}$ ) gives a long exact sequence that begins

$$0 \rightarrow H^1(X, \mathbb{Z}) \rightarrow H^1(X, \mathcal{O}_X) \rightarrow H^1(X, \mathcal{O}_X^*) \rightarrow H^2(X, \mathbb{Z}) \rightarrow H^2(X, \mathcal{O}_X) \rightarrow \cdots \quad (19.21)$$

The injectivity on the left holds because  $H^0(X, \mathcal{O}_X) = \mathbb{C} \rightarrow H^0(X, \mathcal{O}_X^*) = \mathbb{C}^*$  is surjective. Since  $H^1(X, \mathcal{O}_X) = 0$  we get  $H^1(X, \mathbb{Z}) = 0$ , hence  $b_1 = 0$  and, by Poincaré duality,  $b_3 = 0$ . On a compact complex surface  $H^1(X, \mathbb{C}) \cong H^1(X, \mathcal{O}_X) \oplus H^0(X, \Omega_X)$ , so  $b_1 = 0$  forces

$$h^{1,0} = \dim H^0(X, \Omega_X) = 0, \quad (19.22)$$

and by Serre duality  $h^{1,2} = h^{1,0} = 0$ ,  $h^{2,1} = h^{0,1} = 0$ . [Huy ch. 1 §3.2, ll. 519–528, and §3.3, ll. 561–566] Because  $T_X \cong \Omega_X$ , the same statement reads  $H^0(X, T_X) = 0$ : a K3 surface has no holomorphic vector fields, hence no continuous group of holomorphic symmetries.

#### 19.4.3 Step 3: $h^{1,1} = 20$

Apply (19.17) to  $E = \Omega_X$ , of rank two. Its Chern classes are  $c_1(\Omega_X) = -c_1(T_X) = 0$  and  $c_2(\Omega_X) = c_2(T_X)$ , with  $\int c_2 = 24$ . Therefore  $\text{ch}(\Omega_X) = 2 - c_2$  and

$$\chi(X, \Omega_X) = \int_X (2 - c_2) \left(1 + \frac{c_2}{12}\right) = 2 \cdot \frac{24}{12} - 24 = -20. \quad (19.23)$$

On the other hand  $\chi(X, \Omega_X) = h^{1,0} - h^{1,1} + h^{1,2} = -h^{1,1}$  by Step 2. Hence

$$h^{1,1}(X) = 20. \quad (19.24)$$

[Huy ch. 1 §2.4, ll. 427–432 (there written as  $2h^0(\Omega_X) - h^1(\Omega_X) = -20$ )]

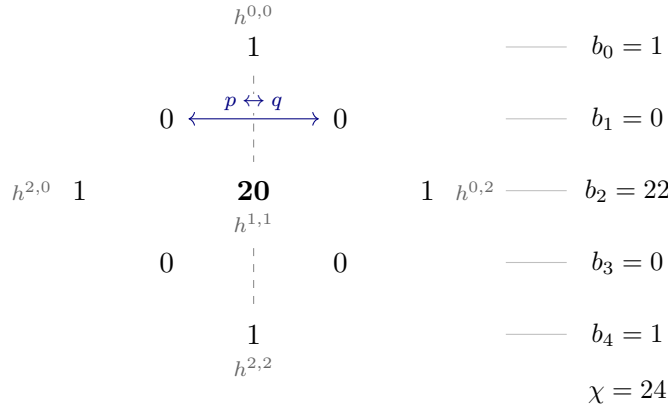


Figure 19.2: The Hodge diamond of a K3 surface. Row  $k$  (from the top,  $k = 0, \dots, 4$ ) lists  $h^{p,q}$  with  $p + q = k$  and sums to the Betti number  $b_k$ . Reflection in the dashed vertical axis is complex conjugation,  $h^{p,q} = h^{q,p}$ ; rotation by  $180^\circ$  is Serre duality,  $h^{p,q} = h^{2-p,2-q}$ .

#### 19.4.4 The Hodge diamond and the Betti numbers

The nine numbers are customarily arranged in a diamond, with  $h^{p,q}$  in row  $p + q$ ; see Figure 19.2. [Huy ch. 1, eq. (2.7), ll. 437–447; Asp §2.1, eq. (14), ll. 377–404] The diamond is the same for every K3 surface, projective or not, and over any field. Since a K3 surface is Kähler, the Hodge decomposition  $H^k(X, \mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(X)$  holds and the rows of the diamond sum to the Betti numbers (Figure 19.2):

$$b_0 = 1, \quad b_1 = 0, \quad b_2 = h^{2,0} + h^{1,1} + h^{0,2} = 22, \quad b_3 = 0, \quad b_4 = 1. \quad (19.25)$$

There are two checks. First,  $\sum_k (-1)^k b_k = 1 + 22 + 1 = 24$  reproduces (19.20); read the other way,  $\chi = 24$  and  $b_1 = b_3 = 0$  give  $b_2 = 22$  without Hodge theory [Huy ch. 1 §3.3, ll. 576–585]. Second, for the quartic the row  $n = 4$  of the table in Example 19.5 gave the same  $b_2 = 22$  from a Chern class computation on  $\mathbb{P}^3$ . (This is Aspinwall’s route to  $h^{1,1} = 22 - 2 = 20$  [Asp §2.1, ll. 370–379].)

Moreover  $H^2(X, \mathbb{Z})$  has no torsion, so

$$H^2(X, \mathbb{Z}) \cong \mathbb{Z}^{22}. \quad (19.26)$$

[Huy ch. 1 §3.2, ll. 535–540, and §3.3, l. 586] [\[L\]](#)

##### What the Hodge numbers count in physics

When a ten-dimensional field theory is compactified on  $X$ , massless fields in the lower-dimensional theory come from harmonic forms on  $X$ , and there are  $b_k$  independent harmonic  $k$ -forms. For K3,  $b_1 = 0$  means that the metric or a vector potential with one leg along  $X$  yields no massless vectors, in line with the absence of continuous isometries;  $b_2 = 22$  means that a two-form field such as  $B$  gives 22 massless scalars, one for each two-cycle. The split  $22 = 1 + 20 + 1$  also has a meaning: deformations of the complex structure are counted by  $H^1(X, T_X) \cong H^1(X, \Omega_X)$ , of dimension  $h^{1,1} = 20$ , again because  $T_X \cong \Omega_X$ . The resulting moduli space is the subject of Chapter 23.

#### 19.4.5 Line bundles and curves: a first look at the Picard lattice

For a line bundle  $L$  formula (19.17) with  $c_1(X) = 0$  reads

$$\chi(X, L) = \frac{(L)^2}{2} + 2, \quad (19.27)$$

where  $(L)^2 = \int_X c_1(L)^2$  is the self-intersection number [Huy ch. 1 §2.3, eq. (2.5), ll. 359–361]. Since the left side is an integer,  $(L)^2$  is even for every line bundle. If  $C \subset X$  is a smooth curve of genus  $g$ , the adjunction formula  $2g - 2 = (C)^2 + K \cdot C$  with  $K = 0$  gives

$$(C)^2 = 2g - 2 : \quad (19.28)$$

a rational curve ( $g = 0$ ) on a K3 surface has self-intersection  $-2$ , as have the 16 exceptional spheres of a Kummer surface. The exact sequence (19.21) also identifies the group of line bundles  $\text{Pic}(X) = H^1(X, \mathcal{O}_X^*)$  with the kernel of  $H^2(X, \mathbb{Z}) \rightarrow H^2(X, \mathcal{O}_X)$ , the integral classes of type  $(1, 1)$ :

$$\text{Pic}(X) \cong H^2(X, \mathbb{Z}) \cap H^{1,1}(X), \quad \rho(X) := \text{rk Pic}(X) \leq h^{1,1} = 20. \quad (19.29)$$

[Huy ch. 1 §3.3, eqs. (3.2)–(3.3), ll. 568–575] The *Picard number*  $\rho$  depends on the complex structure and takes every value from 0 to 20. [L] It returns in Chapter 20.

## 19.5 The intersection form: unimodular, even, signature (3, 19)

On a compact oriented four-manifold the cup product defines a symmetric bilinear form on  $H^2(X, \mathbb{Z})$  modulo torsion,

$$(\alpha, \beta) = \int_X \alpha \wedge \beta \in \mathbb{Z}, \quad (19.30)$$

equal to the number of intersection points, counted with orientation, of two surfaces representing the dual homology classes [Asp §2.3, eq. (19), ll. 514–522]. For a K3 surface this makes  $\mathbb{Z}^{22}$  into a lattice in the sense of Chapter 18, the *K3 lattice*. It has three properties, which together determine it.

### 19.5.1 Unimodular

Poincaré duality states that for a basis  $\{e_i\}$  of  $H_2(X, \mathbb{Z})$  there is a dual basis  $\{e_j^*\}$  of  $H_2(X, \mathbb{Z})$  with  $(e_i, e_j^*) = \delta_{ij}$ . Hence the Gram matrix of the intersection form is invertible over  $\mathbb{Z}$  and has determinant  $\pm 1$ : the lattice is unimodular (self-dual). [Asp §2.3, ll. 537–542; Huy ch. 1 §3.3, ll. 586–588]

### 19.5.2 Signature by the Hirzebruch signature theorem

Let  $b_+$  and  $b_-$  be the numbers of positive and negative eigenvalues of the form on  $H^2(X, \mathbb{R})$ , and  $\sigma = b_+ - b_-$  its signature (the *index*). The signature theorem expresses  $\sigma$  by the first Pontryagin class,  $\sigma = \frac{1}{3} \int_X p_1$ , and for a complex surface  $p_1 = c_1^2 - 2c_2$ . Therefore

$$\sigma(X) = \frac{c_1^2 - 2c_2}{3} = \frac{0 - 2 \cdot 24}{3} = -16. \quad (19.31)$$

[Huy ch. 1, proof of Prop. 3.5, ll. 614–619; Asp §2.3, eq. (20), ll. 523–536, in the form  $\sigma = -\frac{2}{3}\chi$ ] Together with  $b_+ + b_- = 22$ ,

$$b_+ = \frac{22 - 16}{2} = 3, \quad b_- = \frac{22 + 16}{2} = 19. \quad (19.32)$$

### 19.5.3 Signature again, from Hodge theory

The number  $b_+ = 3$  can be understood from form by form. Let  $J$  be a Kähler form on  $X$  and  $\Omega$  the holomorphic two-form. Work at a point in flat coordinates  $z_1 = x_1 + ix_2$ ,  $z_2 = x_3 + ix_4$ , with  $e_i = dx_i$  and volume form  $\text{vol} = e_1 \wedge e_2 \wedge e_3 \wedge e_4$ , in which  $J = \frac{i}{2}(dz_1 \wedge d\bar{z}_1 + dz_2 \wedge d\bar{z}_2)$  and  $\Omega = dz_1 \wedge dz_2$ . Expanding,

$$\begin{aligned} J &= e_1 \wedge e_2 + e_3 \wedge e_4, \\ \text{Re } \Omega &= e_1 \wedge e_3 - e_2 \wedge e_4 = e_1 \wedge e_3 + e_4 \wedge e_2, \\ \text{Im } \Omega &= e_1 \wedge e_4 + e_2 \wedge e_3. \end{aligned} \tag{19.33}$$

These are exactly the three self-dual two-forms of  $\mathbb{R}^4$  (those with  $*\alpha = \alpha$ ) [Asp §2.4, eq. (37), ll. 824–837]. Each squares to  $\alpha \wedge \alpha = 2 \text{vol} > 0$ , and they are mutually orthogonal, e.g.  $J \wedge \text{Re } \Omega = 0$  because no term of the product contains all four  $e_i$ . Globally,

$$\int_X J \wedge J = 2 \text{Vol}(X) > 0, \quad \int_X \Omega \wedge \bar{\Omega} > 0, \quad \int_X \Omega \wedge \Omega = 0, \quad \int_X J \wedge \Omega = 0, \tag{19.34}$$

the last two for reasons of type: a  $(4, 0)$ - or  $(3, 1)$ -form on a surface vanishes. So the classes  $[J]$ ,  $[\text{Re } \Omega]$ ,  $[\text{Im } \Omega]$  span a positive definite three-plane in  $H^2(X, \mathbb{R})$  [Asp §2.4, ll. 835–857]. The Hodge index theorem says that there is nothing more: on the orthogonal complement of  $[J]$  inside  $H^{1,1}$  the form is negative definite (in the flat model, the  $(1, 1)$ -form  $e_1 \wedge e_2 - e_3 \wedge e_4$  is orthogonal to  $J$ , anti-self-dual, and squares to  $-2 \text{vol}$ ). For any compact Kähler surface this gives

$$b_+ = 2h^{2,0} + 1, \quad b_- = h^{1,1} - 1, \tag{19.35}$$

which for K3 is  $(b_+, b_-) = (3, 19)$  and  $\sigma = 2 + 2 - 20 = -16$ , in agreement with (19.31). (Formula (19.35) is standard; our sources state the index theorem for  $H^{1,1}$  [Huy ch. 1, Remark 3.4, ll. 547–558] and the dimensions 3 and 19 [Asp §2.4, eqs. (35)–(36), ll. 807–816].) The row  $n = 5$  of Example 19.5 offers an independent test of (19.35): there  $h^{2,0} = \chi(\mathcal{O}) - 1 = 4$  and indeed  $b_+ = 9 = 2 \cdot 4 + 1$ .

### 19.5.4 Even

Finally,  $(\alpha, \alpha) \in 2\mathbb{Z}$  for all  $\alpha \in H^2(X, \mathbb{Z})$ : the lattice is even. For classes of line bundles we saw this in (19.27), but line bundles span at most a sublattice of rank 20. The general statement is topological: by Wu’s formula the intersection form of a closed oriented smooth four-manifold is even if and only if the second Stiefel–Whitney class  $w_2$  vanishes, and for a complex surface  $w_2 \equiv c_1 \pmod{2}$ . A K3 surface has  $c_1 = 0$ , so its form is even. [L] [Huy ch. 1, proof of Prop. 3.5, ll. 610–613; Asp §2.3, ll. 543–551] The condition  $w_2 = 0$  is also the condition for spinors to exist on  $X$ , which we need in Section 19.6. The quartic is the only even row in the table of Example 19.5 besides the quadric:  $c_1 = (4 - n)x$  is even exactly for even  $n$ .

### 19.5.5 The result

Indefinite even unimodular lattices are classified by their signature (Chapter 18):  $b_+ - b_-$  must be divisible by 8 and the lattice is then unique. For  $(3, 19)$  this gives:

**Theorem 19.6** (The K3 lattice). *For a complex K3 surface  $X$ , the group  $H^2(X, \mathbb{Z})$  with its intersection form is an even unimodular lattice of signature  $(3, 19)$ , and hence*

$$H^2(X, \mathbb{Z}) \cong U \oplus U \oplus U \oplus E_8(-1) \oplus E_8(-1). \tag{19.36}$$

[L] [Huy ch. 1, Prop. 3.5, ll. 600–619]

The bookkeeping is: rank  $3 \cdot 2 + 2 \cdot 8 = 22$ ; each hyperbolic plane  $U$  has signature  $(1, 1)$  and each  $E_8(-1)$  has signature  $(0, 8)$ , giving  $(3, 3 + 16) = (3, 19)$  and  $\sigma = -16 = 2 \cdot (-8)$ . The isomorphism (19.36) is abstract; exhibiting 22 explicit cycles with this Gram matrix is best done on a Kummer surface (Chapter 21).

#### Signs and orientations

Which of 3 and 19 counts the positive directions depends on the orientation. We use the orientation given by the complex structure, for which  $\int J \wedge J > 0$ ; with it the self-dual harmonic forms are the three positive directions and  $E_8$  enters with its *negative* definite form, written  $E_8(-1)$  or  $-E_8$ . Reversing the orientation turns  $(3, 19)$  into  $(19, 3)$  and  $\sigma$  into  $+16$ . Also do not confuse  $\chi_{\text{top}} = 24$  with  $\chi(\mathcal{O}_X) = 2$ ; Noether's formula (19.19) relates them,  $24 = 12 \cdot 2$ .

## 19.6 Holonomy, hyperkähler structure and supersymmetry

### 19.6.1 Ricci-flat metrics and $SU(2)$ holonomy

So far no metric was used. Physics needs one: a string background must, to lowest order in  $\alpha'$ , solve the vacuum Einstein equations  $R_{\mu\nu} = 0$  (Chapter 6). For a Kähler metric the Ricci tensor defines a closed  $(1, 1)$ -form representing  $2\pi c_1(T_X)$  (a standard fact), so  $c_1 = 0$  is necessary for Ricci-flatness. Yau's theorem states that it is sufficient: on a compact Kähler manifold with  $K = 0$  and a fixed complex structure, every Kähler class contains a unique Ricci-flat Kähler metric. [L] [Asp §2.4, ll. 841–843] No such metric on a K3 surface is known in closed form [Asp §2.4, ll. 860–862]; what follows uses only its existence.

The *holonomy group* of a Riemannian manifold  $M$  of dimension  $d$  is the group of all linear maps of a tangent space  $T_p M$  obtained by parallel transport, with the Levi-Civita connection, around closed loops based at  $p$ . It is a subgroup of  $O(d)$ , of  $SO(d)$  if  $M$  is orientable, and special geometries correspond to smaller subgroups: the holonomy lies in  $U(d/2)$  if and only if the metric is Kähler, in  $SU(d/2)$  if and only if it is Ricci-flat Kähler (Calabi–Yau), and in  $Sp(d/4)$  if and only if it is hyperkähler [Asp §2.2, ll. 410–434]. The logic is that a tensor is covariantly constant if and only if it is invariant under the holonomy group:  $U(d/2)$  is the subgroup of  $SO(d)$  preserving a complex structure,  $SU(d/2)$  preserves in addition a holomorphic volume form, and  $Sp(d/4)$  preserves a quaternionic structure. For  $d = 4$  the last two coincide,  $SU(2) \cong Sp(1)$ , so:

**Proposition 19.7.** *A K3 surface with its Ricci-flat Kähler metric has holonomy  $SU(2)$  and is hyperkähler: the metric is Kähler with respect to a whole two-sphere of complex structures  $aI + bJ + cK$ ,  $a^2 + b^2 + c^2 = 1$ , where  $I, J, K$  satisfy the quaternion relations. [L] [Asp §2.2, ll. 472–482; §2.4, ll. 870–874]*

The sphere is visible in (19.33). The rotation group of  $\mathbb{R}^4$  is locally a product,  $SO(4) \cong (SU(2)_+ \times SU(2)_-)/\mathbb{Z}_2$ . A vector transforms as  $(\mathbf{2}, \mathbf{2})$  and the six-dimensional space of two-forms splits as

$$\Lambda^2 \mathbb{R}^4 = \Lambda^+ \oplus \Lambda^- = (\mathbf{3}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{3}), \quad (19.37)$$

self-dual plus anti-self-dual. If the holonomy group is  $SU(2)_-$ , it acts trivially on  $\Lambda^+$ : the bundle of self-dual two-forms is flat, and because  $X$  is simply connected it has three globally defined covariantly constant sections,  $J$ ,  $\text{Re } \Omega$  and  $\text{Im } \Omega$  [Asp §2.4, ll. 819–823]. Covariantly constant forms are harmonic, and we recover  $b_+ = 3$ : *the three positive directions of the K3 lattice are the three parallel two-forms of the hyperkähler metric*. An  $SO(3)$  rotation of the triple does not change the metric but changes which combination is called the Kähler form

and which the holomorphic two-form; these rotations sweep out the two-sphere of complex structures. The dimension count  $\dim SU(2) = 3 < 6 = \dim SO(4)$ , with a complement of dimension 3, is the only part of this paragraph that has a counterpart in Lean (Section 19.7).

### 19.6.2 Why K3 preserves half of the supersymmetry

A supersymmetry of the compactified theory requires a covariantly constant spinor on  $X$ : the supersymmetry parameter must be invariant under parallel transport, that is, under the holonomy group. A Dirac spinor in four Euclidean dimensions has four components, two of each chirality, transforming as

$$\mathbf{4} = (\mathbf{2}, \mathbf{1}) \oplus (\mathbf{1}, \mathbf{2}) \quad \text{under} \quad SU(2)_+ \times SU(2)_-. \quad (19.38)$$

On a generic four-manifold the holonomy is all of  $SO(4)$ , neither summand contains an invariant, and no supersymmetry survives. On a flat torus the holonomy is trivial and all four components are constant. On K3 the holonomy  $SU(2)_-$  leaves the two components of  $(\mathbf{2}, \mathbf{1})$  invariant and mixes the other two. In ten dimensions, the 16-component Majorana–Weyl supercharge decomposes under  $so(1, 9) \supset so(1, 5) \oplus su(2)_+ \oplus su(2)_-$  as

$$\mathbf{16} \rightarrow (\mathbf{4}, \mathbf{2}, \mathbf{1}) \oplus (\mathbf{4}', \mathbf{1}, \mathbf{2}), \quad (19.39)$$

and only the first summand, with  $4 \cdot 2 = 8$  of the 16 components, is invariant under the holonomy. [Asp §4.1, eq. (89), ll. 2273–2295] The rule is:

*Compactification on a smooth K3 surface preserves half as many supersymmetries as compactification on a torus  $T^4$ .*

The heterotic and type I strings, with 16 supercharges in ten dimensions, keep 8 on K3, which is the minimal supersymmetry in six dimensions [Asp §4.1, ll. 2288–2293]; the type II strings, with 32, keep 16.

#### Why physicists love K3

Half of the supersymmetry is the useful amount. With all of it (tori) everything is fixed by symmetry; with a quarter or less (Calabi–Yau threefolds) quantum corrections are hard to control. With half, the moduli space of the compactified theory is still determined exactly by symmetry and by the lattice  $H^2(X, \mathbb{Z})$ , yet the internal space is curved, has no isometries and has a rich topology. K3 is therefore the simplest laboratory for exact statements about strings on curved spaces, such as string dualities (Chapter 23). It is also the only one of its kind: a compact Ricci-flat Kähler surface whose holonomy is exactly  $SU(2)$  is a K3 surface (standard; not checked against a source in this book’s library).

*Remark 19.8* (The Dirac index). The two parallel spinors can be counted topologically. On a spin four-manifold the index of the Dirac operator is  $-\frac{1}{24} \int p_1 = -\sigma/8$ , which for K3 equals  $16/8 = 2$ . For a Kähler surface with  $c_1 = 0$  this is the Todd genus  $\chi(\mathcal{O}_X) = c_2/12 = 2$  of (19.18): positive-chirality spinors are sections of  $\Lambda^{0,0} \oplus \Lambda^{0,2}$ , with zero modes 1 and  $\bar{\Omega}$ . (Standard index theory; not checked against a source in this book’s library. The arithmetic  $16/8 = 2$  appears in Lean, Section 19.7.)

## 19.7 What the machine has checked

It is important to be precise here, because the distance between the mathematics of this chapter and its formal counterpart is larger than anywhere in Parts II and III. *No complex manifold, sheaf, cohomology group, Chern class or metric is defined in any of the Lean libraries*

of this project. A faithful formalization of  $\chi(K3) = 24$  would need compact complex surfaces, coherent sheaf cohomology with Serre duality, Chern classes and the Hirzebruch–Riemann–Roch theorem, which the project does not supply. The four files attached to this chapter contain the *arithmetic shadow* of Sections 19.4 to 19.6: the numbers 1, 0, 20, 22, 24, 3, 19,  $-16$  are entered as data, and the kernel checks that they are consistent with each other under the formulas of this chapter. Every statement above about actual K3 surfaces is therefore Tier L. All declarations below depend on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`.

### 19.7.1 The Hodge table and its checksums

The file `Frontier/HodgeNumbers.lean` of the library `StringTheoryFormalization` (namespace `StringTheory.Frontier`) stores the diamond of Figure 19.2 as a function on  $\{0, 1, 2\}^2$ :

```
def k3HodgeNumber : Fin 3 → Fin 3 → ℕ
| ⟨0, _⟩, ⟨0, _⟩ => 1   -- h^{0,0} = 1
| ⟨0, _⟩, ⟨1, _⟩ => 0   -- h^{0,1} = 0
| ⟨0, _⟩, ⟨2, _⟩ => 1   -- h^{0,2} = 1
| ⟨1, _⟩, ⟨0, _⟩ => 0   -- h^{1,0} = 0
| ⟨1, _⟩, ⟨1, _⟩ => 20  -- h^{1,1} = 20
| ⟨1, _⟩, ⟨2, _⟩ => 0   -- h^{1,2} = 0
| ⟨2, _⟩, ⟨0, _⟩ => 1   -- h^{2,0} = 1
| ⟨2, _⟩, ⟨1, _⟩ => 0   -- h^{2,1} = 0
| ⟨2, _⟩, ⟨2, _⟩ => 1   -- h^{2,2} = 1

theorem k3_euler_characteristic :
  (∑ p : Fin 3, ∑ q : Fin 3,
    (if (p.val + q.val) % 2 = 0 then 1 else -1 : ℤ) *
    (k3HodgeNumber p q : ℤ)) = 24 := by ...

theorem k3_hodge_symmetry (p q : Fin 3) :
  k3HodgeNumber p q = k3HodgeNumber q p := by ...

theorem k3_serre_duality (p q : Fin 3) :
  k3HodgeNumber p q =
    k3HodgeNumber ⟨2 - p.val, by omega⟩ ⟨2 - q.val, by omega⟩ := by ...

theorem k3_b2 :
  k3HodgeNumber ⟨0, by norm_num⟩ ⟨2, by norm_num⟩ +
  k3HodgeNumber ⟨1, by norm_num⟩ ⟨1, by norm_num⟩ +
  k3HodgeNumber ⟨2, by norm_num⟩ ⟨0, by norm_num⟩ = 22 := by ...
```

#### In Lean: a nine-entry table sums to 24 and to 22

**Reading.** `k3HodgeNumber` is a definition by cases, a lookup table; the number 20 in it is typed in, not derived. `k3_euler_characteristic` says that  $\sum_{p,q} (-1)^{p+q} h^{p,q} = 24$  for this table, the Hodge-theoretic form of  $\chi = \sum (-1)^k b_k$ . `k3_hodge_symmetry` and `k3_serre_duality` say that the table is invariant under  $(p, q) \mapsto (q, p)$  and  $(p, q) \mapsto (2 - p, 2 - q)$ ; `k3_b2` says that its middle row sums to 22. **Proof idea.** The sums are expanded with `Fin.sum_univ_three` and evaluated by `simp`; the two symmetries are checked on all nine pairs by `fin_cases p <=> fin_cases q <=> rfl`. **What it is not.** These are checksums on the table: a typing error in one entry would very likely break one of them. They are not proofs of Noether’s formula, of Hodge symmetry or of Serre duality, which are theorems about all compact Kähler manifolds and need the objects listed above. The file also contains `hodge11_from_kummer`, which despite its name checks only that the entry 20 equals  $|\text{Fin } 16| + 4$ , that is,  $20 = 16 + 4$ .



### 19.7.2 Signature from the table

The file `UseCases/K3SignatureTheorem.lean` of the same library (namespace `StringTheory.UseCases.K3Signature`) takes the Hodge-index route (19.35), not the Hirzebruch route (19.31):

```
def bPlus : ℕ := 2 * k3HodgeNumber ⟨2, by norm_num⟩ ⟨0, by norm_num⟩ + 1
def bMinus : ℕ := k3HodgeNumber ⟨1, by norm_num⟩ ⟨1, by norm_num⟩ - 1

theorem bPlus_eq_three : bPlus = 3 := by ...
theorem bMinus_eq_nineteen : bMinus = 19 := by ...
theorem betti_sum_eq_b2 : bPlus + bMinus = 22 := by ...
theorem k3_signature_eq_neg_sixteen :
  (bPlus : ℤ) - (bMinus : ℤ) = -16 := by ...
```

**In Lean:**  $2 \cdot 1 + 1 = 3$ ,  $20 - 1 = 19$ ,  $3 - 19 = -16$

**Reading.** `bPlus` and `bMinus` are *defined* by the right-hand sides of (19.35), evaluated on the table. The theorems say that these natural numbers are 3 and 19, that they add up to 22, and that their difference, computed in  $\mathbb{Z}$ , is  $-16$ . **Proof idea.** Unfold the definitions and let `decide` evaluate; the sum is rewritten into the three table entries of `k3_b2` and closed by that theorem, so  $b_+ + b_- = b_2$  is a genuine dependency and not a second copy of the number 22; the last step is `norm_num`. **What it is not.** The Hodge index theorem is the content of the *definition*, which the kernel cannot check: had we defined `bPlus` as  $2h^{2,0} + 2$ , Lean would prove  $\sigma = -14$  without complaint. Neither an intersection form nor the signature theorem appears. The cast to  $\mathbb{Z}$  matters: in  $\mathbb{N}$ , truncated subtraction gives  $3 - 19 = 0$ .

### 19.7.3 Two older libraries: the same numbers as bare integers

Two files of the older libraries described in Chapter 35 define small arithmetic functions and evaluate them at the K3 values. The first is `DoubleFieldTheory/K3Topology.lean`, with namespace `DoubleFieldTheory.K3Topology`:

```
def K3BettiSum (b0 b1 b2 b3 b4 : Int) : Int :=
  b0 - b1 + b2 - b3 + b4
theorem k3_euler_characteristic :
  K3BettiSum 1 0 22 0 1 = 24 := by decide

def LatticeRank (n_e8 n_u : Int) : Int := n_e8 * 8 + n_u * 2
def LatticeSig (n_e8 n_u : Int) : Int := n_e8 * (-8) + n_u * 0
theorem k3_intersection_lattice_rank_sig :
  LatticeRank 2 3 = 22 ∧ LatticeSig 2 3 = -16 := by decide

def DiracIndex (sig : Int) : Int := (-sig) / 8
theorem k3_atiyah_singer_dirac_index :
  DiracIndex (-16) = 2 := by decide
```

Further evaluations of the same kind, in this file and in `StringTheoryFoundation/K3/K3Surfaces.lean`, which stores the diamond as a record with nine fields:

| declaration                            | file                    | content                                                    |
|----------------------------------------|-------------------------|------------------------------------------------------------|
| <code>k3_hirzebruch_signature</code>   | <code>K3Topology</code> | $3 - 19 = -16$                                             |
| <code>k3_su2_holonomy_reduction</code> | <code>K3Topology</code> | $3 < 6$ and $6 - 3 = 3$ (constants for $su(2)$ , $so(4)$ ) |
| <code>k3_lattice_rank_valid</code>     | <code>K3Surfaces</code> | $3 \cdot 2 + 2 \cdot 8 = 22$                               |
| <code>k3_signature_is_minus_16</code>  | <code>K3Surfaces</code> | $3 - 19 = -16$ in $\mathbb{Z}$                             |



**In Lean: evaluations of integer formulas**

**Reading.** Each theorem evaluates an explicit integer expression:  $1 - 0 + 22 - 0 + 1 = 24$ ;  $2 \cdot 8 + 3 \cdot 2 = 22$  and  $2 \cdot (-8) + 3 \cdot 0 = -16$ , the rank and signature bookkeeping below Theorem 19.6;  $16/8 = 2$ , the arithmetic of Remark 19.8. **Proof idea.** `decide` or `rfl`: the kernel computes both sides. **What it is not.** The names `k3_hirzebruch_signature` and `k3_atiyah_singer_dirac_index` refer to the theorems that *motivate* the formulas; the formulas themselves are definitions, and neither theorem is stated. `LatticeRank` counts summands and does not mention a lattice; the lattices  $U$  and  $E_8$  with their Gram matrices live in a different library (Chapter 18). One detail deserves a warning: `DiracIndex` uses integer division, which rounds, so `DiracIndex (-15)` also evaluates to an integer although no spin four-manifold has signature  $-15$ . A companion statement, `k3_parallel_chiral_spinor_index`, checks  $2 - 0 = 2$ .

**19.7.4 The atlas finding: one fact, several proofs**

The project’s theorem atlas (Chapter 36) compares all statements across libraries by textual similarity and by shared dependencies. The fact  $\chi(K3) = 24$  is its clearest example of duplication. It is proved independently five times:

| declaration                                                                         | starting point                                         |
|-------------------------------------------------------------------------------------|--------------------------------------------------------|
| <code>DoubleFieldTheory.K3Topology.k3_euler_characteristic</code>                   | Betti numbers as arguments                             |
| <code>StringTheory.Frontier.k3_euler_characteristic</code>                          | the Hodge table                                        |
| <code>SocrateAI.Moonshine.k3_euler_eq_24</code>                                     | constants <code>k3_b0</code> , ..., <code>k3_b4</code> |
| <code>Lean5Corpus.Problems.KummerModularity.k3_euler_characteristic_identity</code> | $1 + 22 + 1$ equals a constant set to 24               |
| <code>StringTheory.Foundation.Atlas.atlas_k3_euler_characteristic</code>            | fields of a record <code>defaultK3</code>              |

The first and third have the highest cross-library statement similarity in the whole atlas (cosine 0.631), while the first two share almost no dependencies (dependency overlap 0.009). [atlas.md, section “Most similar statements across libraries”] Agreement between such proofs is weak evidence: each starts from its own hand-entered copy of the Betti or Hodge numbers, so the five proofs confirm five times that  $1 + 22 + 1 = 24$ , not that  $b_2(K3) = 22$ .

The newer library does it differently. `k3HodgeNumber` is one of the atlas hubs, used by twelve theorems, and the statements that need K3 data import that single table instead of retyping it:

```
-- DualScaleStream2/Flux/Tadpole.lean
theorem euler_K3 :
  eulerFromHodge StringTheory.Frontier.k3HodgeNumber = 24 := by ...

-- DualScaleStream2/Lattice/K3T2Signature.lean
theorem sigK3_matches_hodge :
  sigK3.pos = StringTheory.UseCases.K3Signature.bPlus ∧
  sigK3.neg = StringTheory.UseCases.K3Signature.bMinus := by ...
```

The second statement is the more interesting one. Its left side is the signature  $(3, 19)$  computed from the decomposition  $3U \oplus 2E_8(-1)$ , its right side is (19.35) evaluated on the Hodge table. It is a machine-checked consistency check between two independently tabulated literature inputs, the formal counterpart of our remark that (19.31) and (19.35) agree.

### 19.7.5 Tier table

| Statement                                                                                       | Tier | Evidence                                                                                                                   |
|-------------------------------------------------------------------------------------------------|------|----------------------------------------------------------------------------------------------------------------------------|
| The examples of Section 19.3 are K3; all K3 are diffeomorphic                                   | [L]  | Huy ch. 1, Ex. 1.3, Rem. 3.6; Asp Thm. 1                                                                                   |
| $\chi = 24$ , Hodge diamond, $b_2 = 22$ ; $H^2(X, \mathbb{Z}) \cong 3U \oplus 2E_8(-1)$         | [L]  | Huy ch. 1 §2.4, §3.3, Prop. 3.5; Asp §2.1                                                                                  |
| Ricci-flat metric, holonomy $SU(2)$ , half of the supersymmetry                                 | [L]  | Asp §2.2, §2.4, §4.1                                                                                                       |
| The table <code>k3HodgeNumber</code> has alternating sum 24, middle row 22, and both symmetries | [A]  | <code>k3_euler_characteristic</code> , <code>k3_b2</code> , <code>k3_hodge_symmetry</code> , <code>k3_serre_duality</code> |
| $2h^{2,0} + 1 = 3$ , $h^{1,1} - 1 = 19$ , difference $-16$ , on that table                      | [A]  | <code>bPlus_eq_three</code> , <code>bMinus_eq_nineteen</code> , <code>k3_signature_eq_neg_sixteen</code>                   |
| Lattice-side and Hodge-side $(b_+, b_-)$ agree                                                  | [A]  | <code>sigK3_matches_hodge</code>                                                                                           |
| $1-0+22-0+1 = 24$ , $3-19 = -16$ , $2\cdot 8+3\cdot 2 = 22$ , $16/8 = 2$                        | [A]  | <code>DoubleFieldTheory.K3Topology</code> , <code>K3Surfaces</code>                                                        |
| The table <i>is</i> the Hodge diamond of a K3 surface                                           | —    | identification; by the rules of this book never Tier A                                                                     |

#### Chapter summary

- A K3 surface is a compact complex surface with trivial canonical bundle and  $H^1(X, \mathcal{O}_X) = 0$ . It carries a nowhere-vanishing holomorphic two-form  $\Omega$ , unique up to scale, and  $T_X \cong \Omega_X$ . Examples: quartics in  $\mathbb{P}^3$ , complete intersections  $(2, 3)$  and  $(2, 2, 2)$ , double planes branched over a sextic, Kummer surfaces, elliptic fibrations with 24 singular fibres. All are diffeomorphic.
- $\chi(\mathcal{O}_X) = 2$  and Noether's formula give  $\chi = c_2 = 24$ ; Riemann–Roch for  $\Omega_X$  gives  $h^{1,1} = 20$ ; the Hodge diamond is  $1/0\ 0/1\ 20\ 1/0\ 0/1$  and  $b_2 = 22$ .
- The intersection form on  $H^2(X, \mathbb{Z}) \cong \mathbb{Z}^{22}$  is unimodular (Poincaré duality), even (Wu's formula,  $c_1 = 0$ ), and has signature  $(3, 19)$ ,  $\sigma = -16$  (Hirzebruch, or Hodge index  $b_+ = 2h^{2,0} + 1$ ). Hence it is  $3U \oplus 2E_8(-1)$ .
- The three positive directions are spanned by  $J$ ,  $\text{Re } \Omega$ ,  $\text{Im } \Omega$ , the parallel self-dual forms of the Ricci-flat metric. Holonomy  $SU(2) = Sp(1)$ : K3 is hyperkähler and preserves half of the supersymmetry.
- In Lean, the numbers are data and the kernel checks their mutual consistency (Tier A arithmetic); the geometry is Tier L. The fact  $\chi = 24$  is proved in five places.

## Exercises

**Exercise 19.1** (★). (a) Expand  $(1+x)^6/(1+2x)^3$  to second order, confirm  $c_1 = 0$ ,  $c_2 = 3x^2$  for the complete intersection  $(2, 2, 2)$ , and explain the factor 8 in  $\chi = 24$ . (b) For the six-line double plane of Figure 19.1: show that the union  $C$  of the six lines has  $\chi(C) = 6 \cdot 2 - 15 = -3$ , that the singular double cover has  $\chi = 2(3+3) - 3 = 9$ , and that replacing each of the 15 double points by a sphere gives  $\chi = 24$ .

**Exercise 19.2** (★). Show that the Fermat quartic  $x_0^4 + x_1^4 + x_2^4 + x_3^4 = 0$  is smooth over  $\mathbb{C}$ . Then count: the space of quartic polynomials in four variables has dimension  $\binom{7}{3} = 35$ , the group  $GL(4, \mathbb{C})$  of coordinate changes has dimension 16. How many parameters do quartic K3 surfaces have, and how does this compare with  $h^{1,1} = 20$ ? (The missing deformation leads to K3 surfaces that are not quartics, see Chapter 20.)

**Exercise 19.3** (★★). A smooth hypersurface  $X \subset \mathbb{P}^2 \times \mathbb{P}^1$  of bidegree  $(3, 2)$  is a K3 surface [Huy ch. 1 §4.1, ll. 680–684]. Let  $a, b$  be the hyperplane classes of the two factors, with  $a^3 = 0$ ,  $b^2 = 0$ ,  $\int a^2 b = 1$ . From  $c(T_X) = (1+a)^3(1+b)^2/(1+3a+2b)$  show that  $c_1 = 0$ ,  $c_2 = 3a^2 + 6ab$  (using  $b^2 = 0$ ), and  $\chi = \int (3a + 2b) c_2 = 24$ .

**Exercise 19.4** (★★). Use (19.27) and Serre duality,  $h^2(L) = h^0(L^*)$ , to show: if  $L$  is a line bundle on a K3 surface with  $(L)^2 \geq -2$ , then  $L$  or  $L^*$  has a nonzero section. Conclude that every class  $\delta \in \text{Pic}(X)$  with  $(\delta)^2 = -2$  is, up to sign, the class of an effective curve. Why does the argument fail for  $(L)^2 = -4$ ?

**Exercise 19.5** (★★). For the degree- $n$  surface  $S_n \subset \mathbb{P}^3$  one has  $h^{2,0} = \binom{n-1}{3}$  (the holomorphic two-forms are  $P\Omega$  with  $P$  a polynomial of degree  $n-4$ , by (19.8)). Using (19.12) and  $b_2 = \chi - 2$ , prove the polynomial identity  $b_+(S_n) = 2\binom{n-1}{3} + 1$  for all  $n$ , confirming (19.35). Deduce  $h^{1,1}(S_n)$  and check it against  $n = 1, \dots, 4$ .

**Exercise 19.6** (★★). Let  $\omega_1 = J$ ,  $\omega_2 = \text{Re } \Omega$ ,  $\omega_3 = \text{Im } \Omega$  be the forms (19.33) on  $\mathbb{R}^4$  with the Euclidean metric, and define  $4 \times 4$  matrices by  $(I_a)_{ij} = (\omega_a)_{ij}$ , the components of  $\omega_a = \frac{1}{2}(\omega_a)_{ij} e_i \wedge e_j$ . Show that  $I_a^2 = -1$ , that  $I_1 I_2 = \pm I_3$  and cyclic (the sign depends on conventions), and that  $aI_1 + bI_2 + cI_3$  squares to  $-1$  if and only if  $a^2 + b^2 + c^2 = 1$ : this is the two-sphere of complex structures of Proposition 19.7.

**Exercise 19.7** (★, Lean). Using `k3HodgeNumber`, state and prove in Lean that the five rows of the Hodge diamond sum to  $1, 0, 22, 0, 1$ . For instance the second row is `k3HodgeNumber 1 0 + k3HodgeNumber 0 1 = 0`. Which of `rfl`, `decide`, `simp [k3HodgeNumber]` close the goals? Then derive `k3_b2` from your middle-row statement.

**Exercise 19.8** (★ ★ ★, Lean). Formalize the uniqueness statement hidden in (19.11): define `def chiHypersurface (n : ℤ) : ℤ := n * (n^2 - 4*n + 6)` and prove `theorem chi_eq_24_iff (n : ℤ) : chiHypersurface n = 24 ↔ n = 4`. Hint:  $n^3 - 4n^2 + 6n - 24 = (n-4)(n^2 + 6)$ ; obtain the factorized equation with `linear_combination`, split it with `mul_eq_zero`, and exclude  $n^2 + 6 = 0$  with `positivity` or `nlinarith [sq_nonneg n]`. Explain in one sentence why this theorem, unlike the five theorems of Section 19.7 that evaluate to 24, quantifies over something, and what it still does not say about surfaces.

## Further reading

- D. Huybrechts, *Lectures on K3 Surfaces*, ch. 1: definitions and examples [Huy ch. 1 §1, ll. 155–270], classical invariants by Riemann–Roch [Huy ch. 1 §2, ll. 272–447], complex K3 surfaces and the lattice  $H^2(X, \mathbb{Z})$  [Huy ch. 1 §3, ll. 449–665], further constructions including elliptic K3 surfaces and the classical Kummer quartic with 16 nodes [Huy ch. 1 §4, ll. 672–746].
- P. Aspinwall, *K3 Surfaces and String Duality*: definition, the quartic and the Hodge diamond [Asp §2.1, ll. 213–404]; holonomy and hyperkähler structures [Asp §2.2, ll. 402–505]; self-dual forms and Yau’s theorem [Asp §2.4, ll. 775–874]; counting supersymmetries under compactification [Asp §4.1, ll. 2258–2295].
- In this book: even unimodular lattices, Chapter 18; periods and the Torelli theorem, Chapter 20; the Kummer construction and  $T^4/\mathbb{Z}_2$ , Chapter 21; the role of  $\chi = 24$  in tadpole cancellation, Chapter 25, and in the elliptic genus, Chapter 28.



## Chapter 20

# The K3 Lattice, Periods and the Torelli Theorem

Chapter 19 showed that all K3 surfaces are one and the same smooth four-manifold, with  $b_2 = 22$  and signature  $-16$ , while as complex manifolds they form a family with twenty complex parameters. This chapter answers the question that these two facts raise: *how does one tell two K3 surfaces apart, and what does the space of all of them look like?* The answer is a piece of linear algebra. The second cohomology  $H^2(X, \mathbb{Z})$  with its intersection pairing is always the same lattice,  $3U \oplus 2E_8(-1)$  of Chapter 18. A complex structure is recorded by the position of one complex line, spanned by the class of the holomorphic two-form, relative to that lattice; a Ricci-flat metric is recorded by the position of a positive three-plane. The Torelli theorem says that nothing else is needed.

A student of string theory should care because this is the model for every moduli space that follows. For a torus the moduli space was  $O(d, d; \mathbb{Z}) \backslash O(d, d; \mathbb{R}) / (O(d) \times O(d))$ : the positions of a positive  $d$ -plane relative to the Narain lattice, up to lattice automorphisms (Chapters 10 and 11). For K3 we shall find  $O(\Gamma^{3,19}) \backslash O(3, 19) / (O(3) \times O(19)) \times \mathbb{R}_+$ , a space of the same kind, and in Chapter 23 the  $B$ -field will enlarge  $(3, 19)$  to  $(4, 20)$ . The points of the moduli space at which the three-plane becomes orthogonal to a lattice vector of square  $-2$  are the K3 analogue of the self-dual radius of Chapter 8: there the surface develops an ADE singularity and the type IIA string acquires a non-abelian gauge symmetry.

Conventions.  $X$  is a complex K3 surface (Chapter 19),  $\Lambda := 3U \oplus 2E_8(-1)$  is the K3 lattice, also written  $\Gamma^{3,19}$ ;  $(x \cdot y)$  is the pairing,  $x^2 := (x \cdot x)$ , and signatures are written  $(n_+, n_-)$ . The holomorphic two-form and its cohomology class are both written  $\Omega$  (Huybrechts writes  $\sigma$ ). Vectors of positive square are called space-like, following Aspinwall.

## 20.1 The intersection form on $H^2(X, \mathbb{Z})$

### 20.1.1 From cycles to a lattice

On a compact oriented four-manifold two closed surfaces in general position meet in finitely many points, and counting these points with orientation signs gives an integer

$$\alpha_1 \cdot \alpha_2 = \#(\alpha_1 \cap \alpha_2), \quad \alpha_1, \alpha_2 \in H_2(X, \mathbb{Z}) \quad [\text{Asp } \S 2.3, \text{ ll. 514--518}]. \quad (20.1)$$

In terms of differential forms the same pairing is  $(a \cdot b) = \int_X a \wedge b$  for closed two-forms  $a, b$ ; because two-forms commute under the wedge product the pairing is symmetric. Since a K3 surface is simply connected,  $H_2(X, \mathbb{Z})$  has no torsion and is isomorphic to  $\mathbb{Z}^{22}$  as a group [Asp §2.3, ll. 513–514 and footnote 4]. A free abelian group with a symmetric integer pairing is a

lattice in the sense of Chapter 18, provided the pairing is non-degenerate. We now determine its three invariants: discriminant, parity and signature.

**Unimodular, from Poincaré duality.** Poincaré duality states that for a basis  $\{e_i\}$  of  $H_2(X, \mathbb{Z})$  there is a second basis  $\{e_j^*\}$  with

$$e_i \cdot e_j^* = \delta_{ij} \quad [\text{Asp §2.3, ll. 536–545}]. \quad (20.2)$$

Let  $G_{ij} = e_i \cdot e_j$  be the Gram matrix and write  $e_j^* = \sum_k P_{kj} e_k$  with an integer matrix  $P$ . Then (20.2) reads  $GP = \mathbf{1}$ : the Gram matrix has an integer inverse. By Chapter 18 this is one of the equivalent forms of unimodularity:  $\det G \cdot \det P = 1$  in  $\mathbb{Z}$  forces  $\det G = \pm 1$ , and the dual lattice  $\Lambda^*$  equals  $\Lambda$ . Duality also identifies the lattice of integral homology  $H_2(X, \mathbb{Z})$  with that of integral cohomology  $H^2(X, \mathbb{Z})$ , and from now on we do not distinguish them. Huybrechts quotes the same fact in the form: the intersection form of a compact oriented four-manifold, modulo torsion, is unimodular [Huy ch. 1 §3.3, ll. 586–589].

**Even, from Wu's formula.** The intersection form of a compact smooth four-manifold  $M$  is even if and only if the second Stiefel–Whitney class  $w_2(M)$  vanishes, and for any almost complex structure  $w_2(M) \equiv c_1 \pmod{2}$  [Huy ch. 1, proof of Prop. 3.5, ll. 610–613]. The mechanism behind the first statement is the congruence  $x^2 \equiv (x \cdot w_2) \pmod{2}$  for all  $x \in H^2(M, \mathbb{Z})$  (this is the usual form of Wu's formula; it is standard, and our sources quote only its consequence). For a K3 surface  $c_1 = 0$ , so

$$x^2 \in 2\mathbb{Z} \quad \text{for all } x \in H^2(X, \mathbb{Z}) \quad [\text{Asp §2.3, ll. 546–552}]. \quad (20.3)$$

The condition  $w_2 = 0$  is the condition for  $M$  to admit spinors. The evenness of the K3 lattice is therefore the lattice-theoretic trace of the fact that fermions can be defined on a K3 surface.

A student can test (20.3) on holomorphic curves without knowing Wu's formula. For a smooth curve  $C$  of genus  $g$  in a surface with trivial canonical bundle the adjunction formula gives

$$C^2 = 2g - 2 \quad [\text{Asp §2.6, eq. (53), ll. 1178–1184}], \quad (20.4)$$

because the degrees of the normal bundle and of the tangent bundle of  $C$ , which are  $C^2$  and  $2 - 2g$ , must add up to  $c_1(X)|_C = 0$ . On a quartic  $X \subset \mathbb{P}^3$  a plane section is a smooth plane quartic curve, of genus  $(4 - 1)(4 - 2)/2 = 3$ , so  $C^2 = 4$ ; and indeed two planes meet in a line, which meets the quartic in four points [Asp §2.5, ll. 964–967]. A sphere ( $g = 0$ ) always has  $C^2 = -2$ . Such curves will be the main actors of Section 20.4. This test is not a proof of evenness, because for a fixed complex structure the classes of curves do not span  $H^2(X, \mathbb{Z})$  [Huy ch. 1, footnote 11, ll. 641–643].

**Signature (3, 19), by two routes.** The signature theorem gives  $b_+ - b_- = \frac{1}{3}(c_1^2 - 2c_2) = -\frac{2}{3}\chi(X) = -16$  [Asp §2.3, eq. (20), ll. 527–535; Huy ll. 614–618], as computed in Chapter 19. With  $b_+ + b_- = b_2 = 22$  one finds  $(b_+, b_-) = (3, 19)$ . The second route uses Hodge theory and tells us *which* three directions are positive. Decompose

$$H^2(X, \mathbb{R}) = (H^{2,0} \oplus H^{0,2})_{\mathbb{R}} \oplus H_{\mathbb{R}}^{1,1}, \quad \dim = 2h^{2,0} + h^{1,1} = 2 + 20. \quad (20.5)$$

On the first summand, spanned by  $\text{Re } \Omega$  and  $\text{Im } \Omega$ , the form is positive definite (we prove this in Section 20.2). On  $H_{\mathbb{R}}^{1,1}$  the Hodge index theorem allows exactly one positive direction, that of a Kähler class [Huy ch. 1, Remark 3.4, ll. 553–557]. Hence

$$b_+ = 2h^{2,0} + 1 = 3, \quad b_- = h^{1,1} - 1 = 19. \quad (20.6)$$

The two routes use different inputs (characteristic classes; Hodge numbers) and agree. The second one is what the Lean theorem `sigK3_matches_hodge` of Section 20.7 compares with the lattice decomposition.

### 20.1.2 The theorem and what it does not say

An indefinite even unimodular lattice is determined up to isomorphism by its signature, and exists only if  $n_+ - n_- \equiv 0 \pmod{8}$  (Chapter 18; [Asp §2.3, eq. (23), ll. 557–566]). For  $(3, 19)$  we have  $3 - 19 = -16$ , and the lattice with that signature built from the two blocks of Chapter 18 is  $U^{\oplus 3} \oplus E_8(-1)^{\oplus 2}$ , with  $3 \cdot (1, 1) + 2 \cdot (0, 8) = (3, 19)$ .

**Theorem 20.1** (The K3 lattice; Tier L). *For every complex K3 surface  $X$  there is an isometry of lattices*

$$H^2(X, \mathbb{Z}) \cong \Lambda := E_8(-1) \oplus E_8(-1) \oplus U \oplus U \oplus U. \quad (20.7)$$

[L] [Huy ch. 1, Prop. 3.5, eq. (3.4), ll. 600–618; Asp §2.3, eq. (24), ll. 567–631]

**Example 20.2** (The Gram matrix, checked by machine arithmetic). In a basis adapted to (20.7) the Gram matrix is the block-diagonal  $22 \times 22$  integer matrix  $G_\Lambda = \text{diag}(-E_8, -E_8, U, U, U)$ , where  $E_8$  is the Cartan matrix of Chapter 18 and  $U = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ . Its invariants follow block by block: the diagonal entries are  $-2$  and  $0$ , hence even;  $\det G_\Lambda = (\det(-E_8))^2 (\det U)^3 = (+1)^2 (-1)^3 = -1$ ; each  $U$  has eigenvalues  $\pm 1$  and each  $-E_8$  is negative definite, giving 3 positive and  $16 + 3 = 19$  negative eigenvalues. A symbolic check (sympy, exact determinant; numerical eigenvalues) confirms  $\det = -1$  and the inertia  $(3, 19)$ . In Lean the blocks are certified separately (Chapter 18) and the signature arithmetic is the theorem `sigK3_eq`.

#### “Abstractly isomorphic” is not “equal”

Theorem 20.1 does not hand us 22 surfaces inside  $X$  that intersect according to  $G_\Lambda$ . It says that such a basis of *classes* exists, and the proof finds it by classification, not by geometry. Huybrechts remarks that it would be interesting to see the decomposition directly on some K3 surface by writing down cycles, and proposes the Kummer surface as a candidate [Huy ll. 620–623]; see Chapter 21. Moreover, the isometry in (20.7) is far from unique: it can be composed with any element of the infinite group  $O(\Lambda)$ . Choosing one is extra data, called a marking (Definition 20.7), and keeping track of this choice is the main bookkeeping task of the chapter.

A second caveat concerns the converse. A simply connected complex surface whose intersection form is (20.7) need not be diffeomorphic to a K3 surface, although by Freedman’s results it is homeomorphic to one [Huy ch. 1, Remark 3.6(ii), ll. 633–639]. The lattice determines the topological type of the manifold, but not its smooth structure.

## 20.2 Periods

### 20.2.1 The two relations satisfied by $\Omega$

The lattice is the same for all K3 surfaces, so it cannot distinguish them. What varies is the position of the Hodge decomposition (20.5) relative to the lattice. To measure it, choose a basis  $\{e_i\}$  of two-cycles and integrate the holomorphic two-form:

$$\varpi_i = \int_{e_i} \Omega, \quad i = 1, \dots, 22 \quad [\text{Asp §2.3, eq. (26), ll. 651–655}]. \quad (20.8)$$

These 22 complex numbers are the *periods* of  $X$ . They are the coordinates of the class  $\Omega \in H^2(X, \mathbb{C}) = \Lambda \otimes \mathbb{C}$  in the basis dual to  $\{e_i\}$ . Since  $\Omega$  is defined only up to a nonzero complex factor (Chapter 19), the meaningful object is the point  $[\Omega] = [\varpi_1 : \cdots : \varpi_{22}]$  of the projective space  $\mathbb{P}(\Lambda_{\mathbb{C}}) \cong \mathbb{P}^{21}$ .

Not every point of  $\mathbb{P}^{21}$  can occur. Write  $\Omega = x + iy$  with  $x, y \in H^2(X, \mathbb{R})$ . In local coordinates  $\Omega = f dz_1 \wedge dz_2$ , so  $\Omega \wedge \Omega$  is a  $(4, 0)$ -form and vanishes on a complex surface, where only  $(2, 2)$ -forms can be integrated. Hence

$$0 = \int_X \Omega \wedge \Omega = (x + iy) \cdot (x + iy) = (x^2 - y^2) + 2i(x \cdot y) \quad [\text{Asp } \S 2.3, \text{ eq. (29), ll. 672–682}], \quad (20.9)$$

so  $x^2 = y^2$  and  $x \cdot y = 0$ . On the other hand  $\Omega \wedge \bar{\Omega} = |f|^2 dz_1 \wedge dz_2 \wedge d\bar{z}_1 \wedge d\bar{z}_2$  is a positive multiple of the volume form (in real coordinates  $z_k = u_k + iv_k$  it equals  $4|f|^2 du_1 \wedge dv_1 \wedge du_2 \wedge dv_2$ ), so

$$0 < \int_X \Omega \wedge \bar{\Omega} = (x + iy) \cdot (x - iy) = x^2 + y^2 \quad [\text{Asp } \S 2.3, \text{ eq. (30), ll. 683–694}]. \quad (20.10)$$

Together:  $x^2 = y^2 > 0$  and  $x \cdot y = 0$ . The real and imaginary parts of  $\Omega$  are orthogonal vectors of equal positive length. They span a plane in  $H^2(X, \mathbb{R}) \cong \mathbb{R}^{3,19}$  on which the form is positive definite, a *space-like two-plane*, and the ordered pair  $(x, y)$  orients it. Rescaling  $\Omega \mapsto \lambda \Omega$  with  $\lambda = re^{i\theta}$  rotates  $(x, y)$  by the angle  $\theta$  and stretches it by  $r$ : plane and orientation are unchanged. Complex conjugation  $(x, y) \mapsto (x, -y)$  reverses the orientation [Asp §2.3, ll. 697–702]. This proves the claim used for (20.6), and it shows that the data carried by the periods is exactly an oriented positive two-plane.

## 20.2.2 The period domain

**Definition 20.3** (Period domain). For a lattice  $\Lambda$  of signature  $(n_+, n_-)$  with  $n_+ \geq 2$ ,

$$D := \{[x] \in \mathbb{P}(\Lambda_{\mathbb{C}}) \mid x^2 = 0, (x \cdot \bar{x}) > 0\} \quad [\text{Huy ch. 6 } \S 1.1, \text{ ll. 4281–4290}]. \quad (20.11)$$

The second condition is well posed because  $(\lambda x \cdot \overline{\lambda x}) = |\lambda|^2(x \cdot \bar{x})$ . For the K3 lattice,  $D$  is an open subset (in the classical topology) of a smooth quadric hypersurface in  $\mathbb{P}^{21}$ , hence a complex manifold of dimension  $21 - 1 = 20$ . This is the number  $h^{1,1} = 20$  of complex-structure deformations found in Chapter 19, and it is the first sign that periods are good coordinates.

**Proposition 20.4** (Three descriptions of  $D$ ; Tier L). *The map  $[x] \mapsto \mathbb{R} \operatorname{Re} x \oplus \mathbb{R} \operatorname{Im} x$  identifies  $D$  with the Grassmannian  $\operatorname{Gr}^{po}(2, \Lambda_{\mathbb{R}})$  of oriented positive two-planes, and*

$$D \cong \operatorname{Gr}^{po}(2, \Lambda_{\mathbb{R}}) \cong O(n_+, n_-) / (SO(2) \times O(n_+ - 2, n_-)). \quad (20.12)$$

[L] [Huy ch. 6, Prop. 1.5, ll. 4347–4363; Asp §2.3, eq. (31), ll. 726–747]

*Idea.* We have already seen that  $x^2 = 0, (x \cdot \bar{x}) > 0$  make  $\operatorname{Re} x, \operatorname{Im} x$  an orthogonal basis of equal positive length of a plane, and that  $\lambda x$  gives the same oriented plane. Conversely an oriented positive plane with oriented orthonormal basis  $e_1, e_2$  is the image of  $e_1 + ie_2$ . For the second isomorphism,  $O(n_+, n_-)$  acts transitively on oriented positive planes, and the stabilizer of one of them consists of rotations inside the plane,  $SO(2)$ , times arbitrary isometries of its orthogonal complement, which has signature  $(n_+ - 2, n_-)$ .  $\square$



As a check, count real dimensions with  $\dim O(p, q) = \frac{1}{2}(p + q)(p + q - 1)$ :

$$\dim_{\mathbb{R}} D = \dim O(3, 19) - \dim SO(2) - \dim O(1, 19) = 231 - 1 - 190 = 40 = 2 \cdot 20. \quad (20.13)$$

The count can also be done by hand: a two-plane in  $\mathbb{R}^{22}$  is given by two vectors (44 numbers) up to  $GL(2, \mathbb{R})$  (4 numbers), and positivity is an open condition.

A point of  $D$  determines the whole Hodge decomposition, not only  $H^{2,0}$ : set  $H^{2,0} = \mathbb{C}x$ ,  $H^{0,2} = \mathbb{C}\bar{x}$ , and let  $H^{1,1}$  be the orthogonal complement of both. This is the unique Hodge structure on  $\Lambda$  with  $H^{2,0} = \mathbb{C}x$  for which  $H^{1,1} \perp H^{2,0}$  [Huy ch. 6, Prop. 1.2, ll. 4304–4318]. The orthogonality is geometric: for a  $(1, 1)$ -form  $a$  the product  $a \wedge \Omega$  is of type  $(3, 1)$  and vanishes. In particular the period determines which integral classes are of type  $(1, 1)$ :

$$\mathrm{NS}(X) = \mathrm{Pic}(X) = H^2(X, \mathbb{Z}) \cap H^{1,1}(X) = \{v \in H^2(X, \mathbb{Z}) \mid (v \cdot \Omega) = 0\}, \quad (20.14)$$

the *Picard* or Néron–Severi lattice, of rank  $\rho(X) \leq 20$  [Huy ch. 1, eqs. (3.2)–(3.3), ll. 570–575; Asp §2.5, eq. (41), ll. 921–930]. (The second equality in (20.14) holds because an integral class is real, so  $(v \cdot \Omega) = 0$  implies  $(v \cdot \bar{\Omega}) = 0$ .) The condition  $(v \cdot \Omega) = 0$  is one complex linear equation on the period for each  $v$ . A generic period satisfies none of them, and the corresponding K3 surface contains no curves at all:  $\rho = 0$  [Asp §2.5, ll. 931–935]. If  $X$  is algebraic,  $\mathrm{NS}(X)$  has signature  $(1, \rho - 1)$ , because it is orthogonal to the positive plane of  $\Omega$  and contains the class of a hyperplane section [Asp §2.5, ll. 1018–1022]. Its orthogonal complement in  $H^2(X, \mathbb{Z})$  is the *transcendental lattice*  $T(X)$ , of signature  $(2, 20 - \rho)$ ; it is rarely unimodular [Asp §2.5, ll. 1027–1029].

**Example 20.5** (A period point with  $\rho = 20$ ). Let  $e_k, f_k$  ( $k = 1, 2, 3$ ) be the standard bases of the three copies of  $U$ :  $e_k^2 = f_k^2 = 0$ ,  $e_k \cdot f_k = 1$ . Put

$$x = e_1 + f_1, \quad y = e_2 + f_2, \quad \Omega = x + iy.$$

Then  $x^2 = 2(e_1 \cdot f_1) = 2$ ,  $y^2 = 2$  and  $x \cdot y = 0$  because the two copies of  $U$  are orthogonal. Therefore  $\Omega^2 = x^2 - y^2 + 2ix \cdot y = 0$  and  $\Omega \cdot \bar{\Omega} = x^2 + y^2 = 4 > 0$ : the point  $[\Omega]$  lies in  $D$ . By (20.14) the Picard lattice consists of the integral vectors orthogonal to  $x$  and  $y$ . A vector  $ae_1 + bf_1$  has  $(ae_1 + bf_1) \cdot x = a + b$ , so the orthogonal complement of  $x$  in the first  $U$  is spanned by  $e_1 - f_1$ , of square  $-2$ . Hence

$$\mathrm{NS} = \mathbb{Z}(e_1 - f_1) \oplus \mathbb{Z}(e_2 - f_2) \oplus U \oplus E_8(-1)^{\oplus 2}, \quad \rho = 1 + 1 + 2 + 16 = 20,$$

of signature  $(1, 19)$  as predicted, and  $T = \mathbb{Z}x \oplus \mathbb{Z}y$  with Gram matrix  $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$ , of determinant  $4 \neq \pm 1$ . The period has the largest possible Picard number. That a K3 surface with this period exists is guaranteed by Theorem 20.9 below; we did not have to write down an equation. We return to this surface in Example 20.16. (Symbolic check of all pairings: sympy.)

*Remark 20.6* (The case  $n_+ = 2$ : upper half-planes). For algebraic K3 surfaces the hyperplane class  $\ell$  stays of type  $(1, 1)$ , the period is confined to  $\ell^\perp$  of signature  $(2, 19)$ , and the quartic surfaces, for example, have the moduli space  $O(\Lambda_C) \backslash O(2, 19) / (O(2) \times O(19))$  of complex dimension  $\frac{1}{2}(210 - 1 - 171) = 19$ . This matches the  $35 - 16 = 19$  coefficients of a quartic polynomial modulo  $GL(4, \mathbb{C})$ ; one of the 20 deformations is lost [Asp §2.5, eq. (44), ll. 968–983]. For  $n_+ = 2$  the period domain has a familiar shape. Split  $\Lambda_{\mathbb{R}} = U_{\mathbb{R}} \oplus W$  and send  $\zeta \in W_{\mathbb{C}}$  to  $x = e - \frac{1}{2}\zeta^2 f + \zeta$ . Then  $x^2 = 2 \cdot (-\frac{1}{2}\zeta^2) + \zeta^2 = 0$  identically, and with  $\zeta = a + ib$  one finds  $(x \cdot \bar{x}) = -\mathrm{Re} \zeta^2 + \zeta \cdot \bar{\zeta} = -(a^2 - b^2) + (a^2 + b^2) = 2b^2$ . So  $D$  is the tube domain  $\{\zeta \mid (\mathrm{Im} \zeta)^2 > 0\}$  (this is [Huy ch. 6, Prop. 1.7, ll. 4384–4396] with  $\zeta = \sqrt{2}z$ ). For  $W$  of signature  $(1, 0)$  this is the union of the upper and the lower half-plane: the parameter  $\tau$  of Chapter 12 is a period in exactly this sense, and  $D$  for signature  $(2, n_-)$  is its higher-dimensional analogue. For  $n_+ = 2$ ,  $D$  has two components exchanged by complex conjugation; for the full K3 lattice,  $n_+ = 3$ , it is connected [Huy ch. 6, Remark 1.6, ll. 4365–4373].

## 20.3 Markings, the period map and the Torelli theorems

### 20.3.1 Marked K3 surfaces

To compare the periods of two different surfaces they must be written in the same coordinates, that is, in the fixed lattice  $\Lambda$ .

**Definition 20.7** (Marking). A *marking* of a K3 surface  $X$  is an isometry  $\varphi: H^2(X, \mathbb{Z}) \rightarrow \Lambda$ . Two marked surfaces  $(X, \varphi)$ ,  $(X', \varphi')$  are equivalent if there is a biholomorphic map  $g: X \rightarrow X'$  with  $\varphi \circ g^* = \varphi'$ . The set of equivalence classes is denoted  $N$ . [Huy ch. 7 §2.1, ll. 5600–5606; Asp §2.3, ll. 656–660]

Concretely, a marking is a choice of which 22 cycles are used in (20.8). The *period map* is

$$\mathcal{P}: N \rightarrow D \subset \mathbb{P}(\Lambda_{\mathbb{C}}), \quad (X, \varphi) \mapsto [\varphi_{\mathbb{C}}(\Omega_X)], \quad (20.15)$$

and by Section 20.2 it does take values in  $D$ . Three theorems describe it. All three are Tier L; none is formalized in this book's Lean libraries, and none could be stated there at present, because the libraries contain no complex manifolds.

**Theorem 20.8** (Local Torelli; Tier L). *A K3 surface  $X_0$  has a smooth universal deformation space  $\text{Def}(X_0)$  of complex dimension 20, and the period map  $\mathcal{P}: \text{Def}(X_0) \rightarrow D$  is a local isomorphism.* [L] [Huy ch. 6, Cor. 2.7 and Prop. 2.8, ll. 4646–4670]

The proof explains why it is true. First-order deformations of the complex structure are classified by  $H^1(X_0, T_{X_0})$ , where  $T$  is the holomorphic tangent bundle. The tangent space of  $D$  at the line  $\ell = H^{2,0}$  is  $\text{Hom}(\ell, \ell^\perp/\ell) \cong \text{Hom}(H^{2,0}, H^{1,1})$  [Huy ch. 6, Remark 1.1, ll. 4296–4302], and the differential of  $\mathcal{P}$  sends a deformation  $v$  to the map  $\Omega \mapsto v \lrcorner \Omega$ . Because  $\Omega$  vanishes nowhere, contraction with it is an isomorphism of bundles  $T_X \rightarrow \Omega_X^1$ , hence an isomorphism  $H^1(T_X) \rightarrow H^1(\Omega_X^1) = H^{1,1}$  of 20-dimensional spaces. In words: to first order a change of complex structure is the same thing as a change of the class of  $\Omega$  in the (1,1) directions. Gluing the local deformation spaces makes  $N$  a 20-dimensional complex manifold, which however is not Hausdorff [Huy ch. 7 §2.1, ll. 5607–5616].

**Theorem 20.9** (Surjectivity of the period map; Tier L). *On each connected component  $N^o$  of  $N$  the period map  $\mathcal{P}: N^o \rightarrow D$  is surjective: every point of the period domain is the period of a marked K3 surface.* [L] [Huy ch. 7, Thm. 4.1, ll. 5892–5895; first proved by Todorov, ll. 5884–5885]

**Theorem 20.10** (Global Torelli; Tier L). *Two complex K3 surfaces  $X, X'$  are isomorphic if and only if there is a Hodge isometry  $\psi: H^2(X, \mathbb{Z}) \rightarrow H^2(X', \mathbb{Z})$ , that is, an isometry of lattices whose complexification maps  $H^{2,0}(X)$  to  $H^{2,0}(X')$ . Moreover, if  $\psi$  maps at least one Kähler class of  $X$  to a Kähler class of  $X'$ , then  $\psi = f^*$  for a unique isomorphism  $f: X' \rightarrow X$ .* [L] [Huy ch. 7, Thm. 5.3, ll. 5969–5972]

In the language of markings: if  $\mathcal{P}(X, \varphi) = \mathcal{P}(X', \varphi')$  then  $\psi = \varphi'^{-1} \circ \varphi$  is a Hodge isometry, so  $X \cong X'$ . The period is a complete invariant of the complex structure. Aspinwall states the combination of Theorems 20.9 and 20.10 as one sentence: the moduli space of complex structures on a marked K3 surface, including orbifold points, is the space of possible periods [Asp §2.3, Theorem 2, ll. 721–725].

#### A Hodge isometry need not come from a map of surfaces

The first half of Theorem 20.10 says that  $X$  and  $X'$  are isomorphic; it does not say that the given  $\psi$  is induced by an isomorphism. The isometry  $-\text{id}$  is a Hodge isometry of every

K3 surface and acts trivially on  $D$ , but it is never of the form  $f^*$ , because it maps Kähler classes to their negatives [Huy ch. 7 §5.2, ll. 5982–5984]. The reflections of Section 20.4 in classes of curves are further examples. This is the reason for the Kähler-cone condition in the second half, and the reason why  $N$  fails to be Hausdorff: several marked surfaces, differing by such isometries, sit over the same period point and cannot be separated.

### 20.3.2 Forgetting the marking

To obtain the moduli space of unmarked surfaces one divides by the changes of marking. Which elements of  $O(\Lambda)$  should be regarded as trivial? Those induced by diffeomorphisms of the underlying four-manifold. Let  $O^+(\Lambda) \subset O(\Lambda)$  be the subgroup of index two that preserves the orientation of the positive three-planes; equivalently, the isometries of spinor norm  $+1$  [Huy ch. 7 §5.4, ll. 6031–6041]. Every element of  $O^+(\Lambda)$  is induced by a diffeomorphism and no element outside it is [Asp §2.3, ll. 757–761]; Huybrechts proves the closely related statement that the monodromy group of K3 surfaces is exactly  $O^+(H^2(X, \mathbb{Z}))$  [Huy ch. 7, Prop. 5.5, ll. 6049–6051]. Therefore the moduli space of complex structures on a K3 surface, including orbifold points, is

$$\mathcal{M}_c \cong O^+(\Gamma^{3,19}) \backslash O^+(3, 19) / (O(2) \times O(1, 19))^+ \quad \text{[L] [Asp §2.3, eq. (32), ll. 762–766],} \quad (20.16)$$

where the superscripts  $+$  restrict to the index-two subgroups preserving the orientation of the space-like directions. The structure is that of Chapter 11: a Teichmüller space  $D$  (all periods), divided by a modular group  $O^+(\Gamma^{3,19})$  (relabellings of the cycles). But there is a difference of substance. The action of  $O^+(\Gamma^{3,19})$  on  $D$  is ergodic, and  $\mathcal{M}_c$  is not Hausdorff [Asp §2.3, ll. 767–769; Huy ch. 6 §1.3, ll. 4409–4412]. Underlying this is the fact that the stabilizer  $SO(2) \times O(1, 19)$  of a point is not compact, so that the discrete group need not act properly discontinuously. A geometric symptom: a K3 surface may have an infinite automorphism group, which then fixes its period point [Huy ch. 6, Example 1.9, ll. 4414–4420]. Aspinwall notes that  $\mathcal{M}_c$  does not seem to make a natural appearance in string theory, whereas the moduli spaces which do appear are Hausdorff [Asp §2.3, ll. 770–774]. The next two sections construct the first of them.

## 20.4 $(-2)$ -curves, reflections and the Kähler cone

A complex structure fixes the plane of  $\Omega$ . A Kähler metric adds a Kähler class  $J \in H_{\mathbb{R}}^{1,1}$  with  $J^2 = \int_X J \wedge J > 0$ . Which classes are Kähler? By (20.6) the form on  $H_{\mathbb{R}}^{1,1} \cong \mathbb{R}^{1,19}$  is Lorentzian, so the set  $\{\alpha \mid \alpha^2 > 0\}$  is the interior of a light-cone with two components. The *positive cone*  $\mathcal{C}_X$  is the component containing a Kähler class, and the Kähler cone satisfies  $\mathcal{K}_X \subset \mathcal{C}_X$  [Huy ch. 7 §3.2, ll. 5812–5818]. It is cut out of  $\mathcal{C}_X$  by the spheres contained in  $X$ .

**Definition 20.11** (Roots). A *root* of a lattice is a vector  $\delta$  with  $\delta^2 = -2$ . A  $(-2)$ -*curve* on  $X$  is a smooth rational curve  $C \cong \mathbb{P}^1 \subset X$ ; by (20.4) its class is a root of  $\text{NS}(X)$ .

There is a converse: if  $\delta \in \text{NS}(X)$  is a root, then  $\delta$  or  $-\delta$  is the class of an effective curve. The reason is the Riemann–Roch theorem: the line bundle  $L$  with  $c_1(L) = \delta$  has  $\chi(L) = 2 + \frac{1}{2}\delta^2 = 1 > 0$ , so  $L$  or its dual has a section [Huy ch. 8 §2.2, footnote 3, ll. 6486–6488; Asp §2.6, ll. 1203–1206]. If  $\text{NS}(X) = \mathbb{Z}\delta$ , that curve is a single smooth  $\mathbb{P}^1$  [Huy ch. 7, proof of Prop. 5.5, ll. 6055–6059].

**Theorem 20.12** (Kähler cone; Tier L). *The Kähler cone of a K3 surface  $X$  is*

$$\mathcal{K}_X = \{\alpha \in \mathcal{C}_X \mid (\alpha \cdot C) > 0 \text{ for all smooth rational curves } C \subset X\}. \quad (20.17)$$

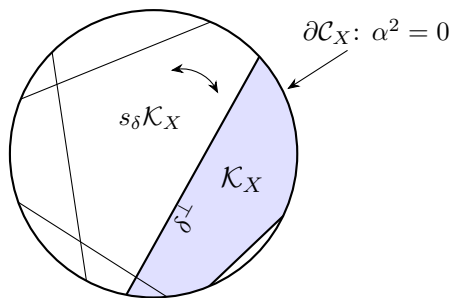


Figure 20.1: A section  $\alpha^2 = 1$  of the positive cone  $\mathcal{C}_X \subset \mathbb{R}^{1,19}$ , drawn as a disc (a hyperbolic space in the projective model, where hyperplanes are chords). The walls  $\delta^\perp$  of roots cut it into chambers; the shaded chamber is the Kähler cone, and the reflection  $s_\delta$  maps it to the adjacent chamber. Schematic: for  $\rho = 20$  there are in general infinitely many walls, accumulating only at the boundary.

In particular, if  $\text{Pic}(X) = 0$  then  $\mathcal{K}_X = \mathcal{C}_X$ . [L] [Huy ch. 8, Thm. 5.2 and Cor. 5.3, ll. 7209–7218]

The condition is necessary for an obvious reason:  $(J \cdot C) = \int_C J$  is the area of  $C$  in the Kähler metric, and areas of holomorphic curves are positive. The content of the theorem is that it is sufficient.

### 20.4.1 Walls, chambers and the Weyl group

Each root  $\delta \in \text{NS}(X)$  defines a wall  $\delta^\perp \cap \mathcal{C}_X$ . The walls are locally finite in  $\mathcal{C}_X$ , and the connected components of their complement are called *chambers* [Huy ch. 8 §2.2, ll. 6494–6519]. By Theorem 20.12 the Kähler cone is one of the chambers. The reflection

$$s_\delta(x) = x + (x \cdot \delta)\delta, \quad \delta^2 = -2 \quad [\text{Huy ch. 8 §2.2, ll. 6483–6493}], \quad (20.18)$$

is an integral isometry and an involution (Chapter 18, where this is proved on the page and in Lean). It fixes the wall  $\delta^\perp$  pointwise, sends  $\delta \mapsto -\delta$ , and preserves the positive cone, because  $(s_\delta(x) \cdot x) = x^2 + (x \cdot \delta)^2 > 0$  for  $x \in \mathcal{C}_X$ , so that  $x$  and  $s_\delta(x)$  lie in the same component. Since  $\delta$  is of type  $(1, 1)$ ,  $s_\delta$  fixes  $\Omega$ : it is a Hodge isometry. The group  $W$  generated by all  $s_\delta$  is the *Weyl group* of  $X$ .

**Proposition 20.13** (Tier L).  *$W$  acts simply transitively on the set of chambers, and it is generated by the reflections in the walls of any one chamber.* [L] [Huy ch. 8, Remark 2.5 and Prop. 2.6, ll. 6542–6563]

The source proves this for a lattice of signature  $(1, n)$ , such as  $\text{NS}(X)$  of an algebraic K3 surface, and treats the Kähler cone of a general K3 surface by the same method in its §8.5.

Figure 20.1 shows the picture. It completes the Torelli theorem: given a Hodge isometry  $\psi$  between  $X$  and  $X'$ , first replace  $\psi$  by  $-\psi$  if necessary so that positive cones correspond; then  $\psi(\mathcal{K}_X)$  is one of the chambers of  $X'$ , and by Proposition 20.13 exactly one  $w \in W$  moves it onto  $\mathcal{K}_{X'}$ . The corrected isometry  $w \circ \psi$  satisfies the hypothesis of the second half of Theorem 20.10 and is induced by a unique isomorphism. So the group of Hodge isometries of  $X$  is  $\{\pm 1\} \times W$  extended by the automorphisms of  $X$ .

#### Compare the circle

In Chapter 18 the T-duality  $n \leftrightarrow w$  of a circle was identified with the reflection  $s_{e-f}$  of  $U$  in the root  $e - f$ . The half-line of radii  $R > 0$  is cut by one “wall”, the self-dual

radius  $R = \sqrt{\alpha'}$  fixed by the reflection, into two chambers exchanged by it, and either chamber is a fundamental domain. For K3 surfaces the Kähler cone plays the role of the fundamental domain  $R \geq \sqrt{\alpha'}$ , the walls are the loci where a sphere shrinks to zero area, and on the wall something becomes massless in both cases (Section 20.6). The analogy is structural; it is not a statement that  $s_\delta$  for a geometric  $(-2)$ -curve is a T-duality.

**Reflections as monodromies and as generators.** Reflections also arise from families. If a family of K3 surfaces over a disc acquires an  $A_1$  singularity at the centre, transporting  $H^2$  once around the centre acts as  $s_\delta$ , where  $\delta$  is the class of the exceptional  $(-2)$ -curve [Huy ch. 6 §5, ll. 5398–5401]. This is how  $O^+(\Lambda)$  entered (20.16): by a theorem essentially due to Wall, every element of  $O^+(\Lambda)$  is a product of reflections  $s_{\delta_i}$  in roots [Huy ch. 7 §5.4, ll. 6041–6044], and each reflection is a monodromy, hence induced by a diffeomorphism. Note the two different uses of roots. For the monodromy statement *all* roots of the rank-22 lattice  $\Lambda$  are used. For the Weyl group of a given surface only the roots in  $\text{NS}(X)$ , which depend on the period, are used.

## 20.5 The moduli space of Einstein metrics

### 20.5.1 From a two-plane to a three-plane

A Kähler metric on a K3 surface whose Ricci tensor is proportional to the metric is in fact Ricci-flat, so “Einstein” and “Ricci-flat” can be used interchangeably [Asp §2.4, ll. 787–789]. By Yau’s theorem, for a fixed complex structure and a fixed Kähler class there is exactly one Ricci-flat Kähler metric [Asp §2.4, ll. 841–844]. The data of such a metric are therefore the two-plane of  $\Omega$  together with a vector  $J$  that is

- orthogonal to the plane, because  $J$  is of type  $(1, 1)$  and  $J \wedge \Omega$  is of type  $(3, 1)$ ;
- space-like, because  $J^2 = \int_X J \wedge J$  is, up to normalization, the volume of  $X$  [Asp §2.4, eq. (38), ll. 849–856].

Thus a Ricci-flat Kähler metric determines a positive *three*-plane

$$\Sigma = \langle \text{Re } \Omega, \text{Im } \Omega, J \rangle \subset H^2(X, \mathbb{R}) \cong \mathbb{R}^{3,19}, \quad (20.19)$$

and a number, the volume. Since  $b_+ = 3$ , the plane  $\Sigma$  is a *maximal* positive subspace. It has an intrinsic meaning. The Hodge star  $*$  of a Riemannian metric on a four-manifold squares to  $+1$  on two-forms, and harmonic two-forms split into self-dual and anti-self-dual ones:  $H^2(X, \mathbb{R}) = H^+ \oplus H^-$ . From  $\alpha \wedge * \alpha = |\alpha|^2 \text{vol}$  one reads off that  $\alpha^2 > 0$  for  $\alpha = * \alpha$  and  $\alpha^2 < 0$  for  $\alpha = -* \alpha$ , so  $\dim H^+ = 3$  and  $\dim H^- = 19$  [Asp §2.4, eqs. (33)–(36), ll. 790–817]. In a local orthonormal coframe  $e^1, \dots, e^4$  with  $dz_1 = e^1 + ie^2$ ,  $dz_2 = e^3 + ie^4$  the self-dual forms are spanned by

$$s_1 = e^{12} + e^{34}, \quad s_2 = e^{13} + e^{42}, \quad s_3 = e^{14} + e^{23}, \quad e^{ij} := e^i \wedge e^j, \quad (20.20)$$

and one checks directly that  $\frac{1}{2}(dz_1 \wedge d\bar{z}_1 + dz_2 \wedge d\bar{z}_2) = s_1$  is the Kähler form, while  $dz_1 \wedge dz_2 = (e^{13} - e^{24}) + i(e^{14} + e^{23}) = s_2 + is_3$  is the holomorphic two-form. Hence

$$\Sigma = H^+ \quad [\text{Asp §2.4, eq. (37) and ll. 825–840}]. \quad (20.21)$$

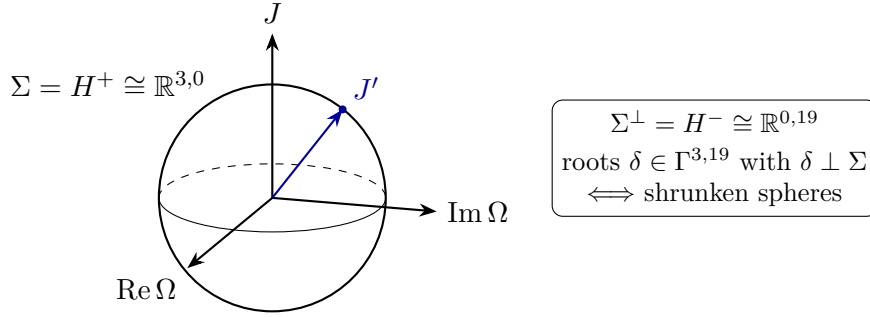


Figure 20.2: The positive three-plane  $\Sigma \subset \mathbb{R}^{3,19}$  of a Ricci-flat metric. A point  $J'$  of the unit sphere in  $\Sigma$  selects a Kähler direction, and the plane orthogonal to it inside  $\Sigma$  is the period plane of the corresponding complex structure. All points of the sphere give the same metric.

### One metric, a two-sphere of complex structures

The metric determines  $\Sigma = H^+$  but *not* its splitting into  $\Omega$  and  $J$ . Any unit vector  $\hat{j} \in \Sigma$  may be declared to be the direction of the Kähler form; the orthogonal plane in  $\Sigma$  then plays the role of  $\Omega$ . A K3 surface regarded as a Riemannian manifold therefore carries a whole two-sphere  $S^2$  of complex structures, all with the same Ricci-flat metric [Asp §2.4, ll. 864–874]. This is the hyperkähler structure: the holonomy  $SU(2) \cong Sp(1)$  of Chapter 19 commutes with three complex structures  $I_1, I_2, I_3$  obeying the quaternion relations, and the combinations  $aI_1 + bI_2 + cI_3$  with  $a^2 + b^2 + c^2 = 1$  form the  $S^2$ . For the string this means that the splitting into “complex structure moduli” and “Kähler moduli”, which organizes Calabi–Yau threefolds, has no invariant meaning on K3. Figure 20.2 illustrates the geometry.

## 20.5.2 The moduli space

**Theorem 20.14** (Moduli of Einstein metrics; Tier L). *The moduli space of Einstein metrics on a K3 surface, including orbifold points, is the Grassmannian of oriented positive three-planes in  $\mathbb{R}^{3,19}$ , times the volume, modulo the diffeomorphisms acting on the lattice:*

$$\begin{aligned} \mathcal{M}_E &\cong O^+(\Gamma^{3,19}) \backslash O^+(3, 19) / (SO(3) \times O(19)) \times \mathbb{R}_+ \\ &\cong O(\Gamma^{3,19}) \backslash O(3, 19) / (O(3) \times O(19)) \times \mathbb{R}_+. \end{aligned} \quad (20.22)$$

[L] [Asp §2.4, Theorem 3 and eqs. (39)–(40), ll. 877–903]

The second form follows from the first because the extra generator  $-\mathbf{1}$  is central in  $O(3, 19)$ , so adjoining it on the left and on the right changes nothing [Asp §2.4, ll. 900–903]. Let us count dimensions:

$$\dim \mathcal{M}_E = \dim O(3, 19) - \dim O(3) - \dim O(19) + 1 = 231 - 3 - 171 + 1 = 58. \quad (20.23)$$

Two independent checks. (i) A positive three-plane near a given  $\Sigma$  is the graph of a linear map  $\Sigma \rightarrow \Sigma^\perp$ , which has  $3 \cdot 19 = 57$  entries; add 1 for the volume. (ii) Complex structures have 40 real parameters (20.13) and Kähler classes 20; the pairs  $(\Omega, J)$  fibre over the metrics with fibre  $S^2$ , and  $40 + 20 - 2 = 58$ . Both agree with (20.23).

In contrast to (20.16), the stabilizer  $O(3) \times O(19)$  is now compact, the discrete group acts properly discontinuously, and  $\mathcal{M}_E$  is Hausdorff [Asp §2.4, ll. 903–906]. Adding the Kähler class has repaired the defect of  $\mathcal{M}_c$ . Comparing with Chapter 10,  $\mathcal{M}_E$  without the volume factor has exactly the form of a Narain moduli space with  $(d, d)$  replaced by  $(3, 19)$ : a maximal

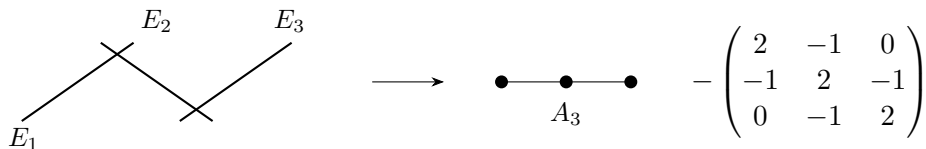


Figure 20.3: The resolution of  $\mathbb{C}^2/\mathbb{Z}_4$ : three  $(-2)$ -curves (drawn as lines), adjacent ones meeting transversally in one point; the dual graph, which is the Dynkin diagram  $A_3$ ; and the intersection matrix  $E_i \cdot E_j$ , minus the Cartan matrix of  $SU(4)$ .

positive subspace moving relative to an even self-dual lattice. In the torus case the positive subspace was the space of left-moving momenta; here it is the space of self-dual harmonic forms. Chapter 23 adds the  $B$ -field and the volume to  $\Sigma$ , obtaining a positive four-plane in  $\mathbb{R}^{4,20}$  and a moduli space of real dimension  $4 \cdot 20 = 80$  that is literally of Narain type.

## 20.6 ADE singularities and enhanced gauge symmetry

### 20.6.1 Where the metric degenerates

The theorems above included “orbifold points”. We can now say where they are. Let  $\delta \in \Gamma^{3,19}$  be a root with  $\delta \perp \Sigma$ . Then  $\delta \perp \Omega$ , so  $\delta \in \text{NS}(X)$  by (20.14), and  $\pm\delta$  is the class of a rational curve  $C$ . But also  $\delta \perp J$ , so the area  $(J \cdot C)$  vanishes: by Theorem 20.12  $J$  is not a Kähler class but lies on a wall of the Kähler cone, and the sphere  $C$  has shrunk to a point.

**Theorem 20.15** (Orbifold points; Tier L). *A point of  $\mathcal{M}_E$  corresponds to an orbifold if and only if the three-plane  $\Sigma$  is orthogonal to a root of  $\Gamma^{3,19}$ . [L] [Asp §2.6, Theorem 4 and its proof, ll. 1203–1217]*

Curves of genus  $g \geq 1$  cannot shrink in this way: their classes have  $C^2 = 2g - 2 \geq 0$  and cannot be orthogonal to  $\Sigma$ , whose orthogonal complement is negative definite [Asp §2.6, ll. 1214–1216]. The simplest local model is the blow-down of one  $(-2)$ -curve, which produces the singularity  $\mathbb{C}^2/\mathbb{Z}_2$ , that is  $x_0x_1 = x_2^2$  in  $\mathbb{C}^3$  with  $x_0 = \xi^2, x_1 = \eta^2, x_2 = \xi\eta$  [Asp §2.6, eq. (50), ll. 1082–1090]. The process is continuous in the moduli space: keep the complex structure fixed and move  $J$  until  $J \cdot C = 0$  [Asp §2.6, ll. 1196–1202].

In general, let  $R_\Sigma := \{\delta \in \Gamma^{3,19} \mid \delta^2 = -2, \delta \perp \Sigma\}$ . These roots lie in the negative definite space  $\Sigma^\perp$ , so they form a finite root system, and the Gram matrix of a set of simple roots has  $-2$  on the diagonal and is negative definite. This is, up to sign, the defining property of the Cartan matrix of a simply-laced Lie algebra, and the classification is the same: connected components are of type  $A_n, D_n, E_6, E_7, E_8$  [Asp §2.6, ll. 1218–1228]. Geometrically, the simple roots of one component are spheres which intersect according to the Dynkin diagram and which are all contracted to one point. For  $A_{n-1}$  the singularity is  $\mathbb{C}^2/\mathbb{Z}_n$ , that is  $x_0x_1 = x_2^n$ , and its resolution is a chain of  $n - 1$  spheres [Asp §2.6, ll. 1229–1330]; see Figure 20.3.

**Example 20.16** (A maximally singular K3 surface). Continue Example 20.5:  $\Omega = (e_1 + f_1) + i(e_2 + f_2)$ , and choose  $J = e_3 + f_3$ . Then  $J^2 = 2 > 0$  and  $J \perp \Omega$ , so  $\Sigma = \langle e_1 + f_1, e_2 + f_2, e_3 + f_3 \rangle$  is a positive three-plane with Gram matrix  $2 \cdot \mathbf{1}_3$ , and

$$\Sigma^\perp \cap \Gamma^{3,19} = \mathbb{Z}(e_1 - f_1) \oplus \mathbb{Z}(e_2 - f_2) \oplus \mathbb{Z}(e_3 - f_3) \oplus E_8(-1)^{\oplus 2} = A_1(-1)^{\oplus 3} \oplus E_8(-1)^{\oplus 2},$$

a negative definite lattice of rank 19 and discriminant  $(-2)^3 = -8$ . In an orthogonal sum of negative definite even lattices a vector of square  $-2$  has all its components but one equal to zero (the squares of the components are even, non-positive, and add up to  $-2$ ). So  $R_\Sigma$  is the



union of the roots of the summands:  $2 + 2 + 2 + 240 + 240 = 486$  roots, a root system of type  $3A_1 + 2E_8$  and rank 19, the largest rank that  $\Sigma^\perp \cong \mathbb{R}^{0,19}$  allows. The corresponding point of  $\mathcal{M}_E$  is a K3 orbifold with two  $E_8$  singularities and three  $\mathbb{C}^2/\mathbb{Z}_2$  singularities, in which 19 independent spheres have been shrunk. Moving  $J$  to  $J_t = e_3 + tf_3$  with  $t \neq 1$ ,  $t > 0$ , gives  $J_t \cdot (e_3 - f_3) = t - 1 \neq 0$  and resolves one of the  $A_1$  points, the sphere acquiring area  $|t - 1|$ . (Symbolic check of the rank, the discriminant and the orthogonality relations: sympy. The count 240 is Tier L, Chapter 18.)

## 20.6.2 What the string sees

Compactify the type IIA string on a K3 surface (Chapter 23). The Ramond–Ramond three-form, reduced on the 22 two-cycles, gives 22 abelian gauge fields; the one-form and the dual of the three-form give two more, so the gauge group is generically  $U(1)^{24}$ , and perturbative string theory cannot produce charged states for it [Asp §4.3, ll. 2433–2442]. The heterotic string on  $T^4$  has the same moduli space and also  $U(1)^{24}$ , but there non-abelian gauge groups appear by the mechanism of Chapter 8: a lattice vector  $\alpha$  with  $\alpha^2 = -2$  becomes orthogonal to the positive plane, the state has  $p_L = 0$ ,  $p_R^2 = -2$  in the conventions of the source, and it is a massless vector boson [Asp §4.3, ll. 2450–2483]. If the two theories are dual, the type IIA string must do the same.

**Proposition 20.17** (Aspinwall’s Proposition 2; Tier L, conditional). *If a K3 surface has an orbifold singularity, a type IIA string compactified on it can exhibit a non-abelian gauge group, and the ADE type of the singularity is the ADE type of the gauge algebra; for instance  $\mathbb{C}^2/\mathbb{Z}_n$  gives  $SU(n)$ .* [L] [Asp §4.3, Proposition 2, ll. 2525–2531]

Two qualifications belong to the statement. First, it rests on the duality between the type IIA string on K3 and the heterotic string on  $T^4$ , which the source presents as a proposition that cannot at present be proved, together with the assumptions made in constructing the moduli spaces [Asp §4.3, ll. 2527–2528]. Its tier is that of a well-tested conjecture of the literature. Secondly, the singularity is necessary but not sufficient: the component of the  $B$ -field along the vanishing cycle must also vanish, while the solvable orbifold conformal field theories have  $B = \frac{1}{2}$  there. This is why string theory on an orbifold is perfectly regular and yet shows no enhanced symmetry [Asp §4.3, ll. 2530–2546]. In Example 20.16 the enhanced gauge algebra at  $B = 0$  would be  $E_8 \oplus E_8 \oplus SU(2)^3$ , of rank 19, inside the total rank 24.

### Three meanings of “ $E_8$ ” in this chapter

(i) The summand  $E_8(-1)$  of the abstract lattice  $\Lambda$ , present for every K3 surface, including smooth ones with no curves at all. (ii) An  $E_8$  singularity, present only at special points of  $\mathcal{M}_E$ , where eight spheres with the  $E_8$  intersection matrix have zero area. (iii) An  $E_8$  gauge symmetry of the type IIA string, present only if in addition the  $B$ -field vanishes on these spheres. The implication is (iii)  $\Rightarrow$  (ii)  $\Rightarrow$  (i) and none of the converses holds. The Lean results below concern (i) only.

## 20.7 What the machine has checked

Almost everything in this chapter is geometry and analysis on complex manifolds, and none of that is formalized: the Lean libraries of this book contain no manifolds, no cohomology and no Hodge structures. What is formalized is the finite arithmetic which the geometry rests on. Three files of the directory `DualScaleStream2/Lattice/` are relevant: `K3T2Signature.lean`, `E8.lean` and `Reflection.lean`. The last two were presented in Chapter 18; here we describe



what they contribute to the K3 lattice. All theorems quoted depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`.

#### Signature bookkeeping (K3T2Signature.lean)

```
structure Signature where
  pos : ℕ
  neg : ℕ
  deriving DecidableEq, Repr

namespace Signature
instance : Add Signature := ⟨fun a b => ⟨a.pos + b.pos, a.neg + b.neg⟩⟩
def rank (s : Signature) : ℕ := s.pos + s.neg
def index (s : Signature) : ℤ := (s.pos : ℤ) - s.neg
end Signature

def sigU : Signature := ⟨1, 1⟩
def sigE8Neg : Signature := ⟨0, 8⟩
def sigK3 : Signature := sigE8Neg + sigE8Neg + sigU + sigU + sigU
def sigMukai : Signature := sigK3 + sigU

theorem sigK3_eq : sigK3 = ⟨3, 19⟩ := by rfl
theorem rank_K3 : sigK3.rank = 22 := by ...
theorem index_mod_eight :
  sigK3.index % 8 = 0 ∧ sigMukai.index % 8 = 0 ∧ sigK3T2.index % 8 = 0 := by ...
```

**Reading.** A `Signature` is a bare pair of natural numbers. Nothing in the structure ties it to a quadratic form. `sigU` and `sigE8Neg` are *definitions*: the numbers (1,1) and (0,8) are entered by hand. They are justified on paper by Sylvester’s law together with the Lean theorems `hyperbolicU_congruence` and `cartanE8_posDef` of Chapter 18, but no Lean statement derives the pairs from those theorems, since the library has no definition of the signature of a matrix. `sigK3_eq` then says: adding the pair (0,8) twice and the pair (1,1) three times gives (3,19). `rank_K3` says  $3 + 19 = 22$ . `index_mod_eight` says that  $3 - 19 = -16$ ,  $4 - 20 = -16$  and  $6 - 22 = -16$  are divisible by 8 (for integers Lean’s `%` returns a non-negative remainder, and  $-16 = (-2) \cdot 8 + 0$ ); `sigK3T2`, of value (6,22), belongs to Chapter 24. This verifies the necessary condition of the classification theorem at three points. It is not the theorem that every even unimodular lattice satisfies  $n_+ \equiv n_- \pmod{8}$ , which remains Tier L.

**Proof idea.** Each statement reduces by unfolding definitions to a closed equation between numerals, which the kernel evaluates: `rfl` for `sigK3_eq` and for the three conjuncts of `index_mod_eight`, `simp` with the definitions followed by `rfl` for `rank_K3`.

#### Two routes to (3,19) compared (K3T2Signature.lean)

```
theorem sigK3_matches_hodge :
  sigK3.pos = StringTheory.UseCases.K3Signature.bPlus ∧
  sigK3.neg = StringTheory.UseCases.K3Signature.bMinus := by ...
```

**Reading.** In an older library of the project the numbers `bPlus` and `bMinus` are defined from a table of the Hodge numbers of K3 by the formulas (20.6),  $b_+ = 2h^{2,0} + 1$  and  $b_- = h^{1,1} - 1$ , and the theorems `bPlus_eq_three`, `bMinus_eq_nineteen` and `k3_signature_eq_neg_sixteen` evaluate them. The present theorem says that the lattice-decomposition route and the Hodge route produce the same two natural numbers. This is the formal counterpart of the paragraph “Signature (3,19), by two routes” in Section 20.1. Both inputs are tabulated Tier L facts. Their agreement is a consistency

check of the bookkeeping; it is not a derivation of the signature of a K3 surface, and the formulas (20.6) themselves (the Hodge index theorem) are not proved in Lean.

**Proof idea.** Both sides unfold to the numerals 3 and 19; `rfl` closes each half.

### The summand $E_8(-1)$ (E8.lean)

```
def e8Neg : Gram 8 := -cartanE8

theorem e8Neg_symm : e8NegT = e8Neg := by ...
theorem e8Neg_evenDiag : IsEvenDiag e8Neg := by ...
theorem e8Neg_unimodular : IsUnimodular e8Neg := by ...
```

**Reading.** `Gram 8` is the type of  $8 \times 8$  integer matrices and `cartanE8` is the Cartan matrix of Chapter 18. The three theorems say that the matrix  $-E_8$  is symmetric, has even diagonal entries (all equal to  $-2$ ), and has determinant  $\pm 1$ . Combined with the general lemma `even_quadratic_form_of_even_diag` of Chapter 18 the first two give that  $x^T(-E_8)x$  is even for every integer vector  $x$ . These are exactly the three properties that Section 20.1 derived for  $H^2(X, \mathbb{Z})$  from Poincaré duality and Wu’s formula, verified here for one block of the right-hand side of (20.7). The same properties of  $U$  are `hyperbolicU_symm`, `hyperbolicU_evenDiag` and `hyperbolicU_unimodular`. The  $22 \times 22$  matrix  $G_\Lambda$  of Example 20.2 is not assembled in Lean, and no Lean statement asserts that a direct sum of even unimodular Gram matrices is even unimodular, although on paper this is immediate: the diagonal is the union of the diagonals and the determinant is the product.

**Proof idea.** Symmetry and the diagonal are decided entry by entry (`fin_cases`). Unimodularity uses a certificate: the integer matrix `cartanE8Inv` satisfies `cartanE8Inv * cartanE8 = 1` (theorem `cartanE8Inv_mul`, 64 entries checked by the kernel), hence  $(-E_8^{-1})(-E_8) = 1$ , and `isUnimodular_of_mul_eq_one` converts an integer inverse into  $\det = \pm 1$ . Note that this is the argument used for (20.2): Poincaré duality provides an integer inverse of the Gram matrix.

### Reflections in $(-2)$ -vectors (Reflection.lean)

```
def reflection (L : Gram n) (v : Fin n → ℤ) : Gram n :=
  1 + vecMulVec v v * L
def latticeNorm (L : Gram n) (v : Fin n → ℤ) : ℤ := v ·v (L *v v)

theorem reflection_isometry (L : Gram n) (hL : LT = L) (v : Fin n → ℤ)
  (hv : latticeNorm L v = -2) :
  (reflection L v)T * L * reflection L v = L := by ...

theorem e8Neg_simpleRoot_norm (i : Fin 8) :
  latticeNorm e8Neg (Pi.single i 1) = -2 := by ...

theorem e8Neg_weyl_isometry (i : Fin 8) :
  (reflection e8Neg (Pi.single i 1))T * e8Neg
  * reflection e8Neg (Pi.single i 1) = e8Neg := ...
```

**Reading.** `reflection L v` is the matrix  $1 + vv^T L$  of the map (20.18). `reflection_isometry` holds for every rank  $n$  and every symmetric integer matrix  $L$ , so in particular for the  $22 \times 22$  matrix  $G_\Lambda$  and any root  $\delta$  of the K3 lattice, even though  $G_\Lambda$  is not defined in Lean: the reflection in a vector of square  $-2$  preserves the form. Its companion `reflection_involution` gives  $s_\delta^2 = 1$ . `Pi.single i 1` is the

$i$ -th standard basis vector, that is, the  $i$ -th simple root  $\alpha_i$ . `e8Neg_simpleRoot_norm` says  $\alpha_i^2 = -2$ , and `e8Neg_weyl_isometry` says that each of the eight simple reflections preserves  $-E_8$ . The theorem covers eight reflections. It does not define the Weyl group, does not show that these eight generate it, and says nothing about the other 232 roots, though `reflection_isometry` would apply to each of them once its norm is computed. No Lean statement connects these matrices with curves on a surface, with monodromy, or with gauge bosons.

**Proof idea.** The key identity is  $(vv^T L)(vv^T L) = (v^T L v) vv^T L$  (`vecMulVec_mul_self`); at norm  $-2$  the quadratic term in the expansion of  $(\mathbf{1} + Lvv^T)L(\mathbf{1} + vv^T L)$  cancels the two linear terms. The  $E_8$  statements are instances: the norm is computed by `fin_cases` and `simp`, and `e8Neg_weyl_isometry` is `reflection_isometry` applied to `e8Neg_symm` and `e8Neg_simpleRoot_norm`.

**What a faithful formalization would need.** To state even Theorem 20.1 in Lean one needs singular cohomology of a manifold with its cup product, Poincaré duality, characteristic classes for Wu’s formula and the signature theorem, and the classification of indefinite unimodular forms. To state the Torelli theorem one needs in addition complex manifolds, Hodge decomposition and deformation theory. None of this was available in the Mathlib version used by the project, and the programme did not attempt it. A realistic intermediate goal, purely algebraic, would be (compare Chapter 38): define lattices as  $\mathbb{Z}$ -modules with a form, define isometries and direct sums, prove that  $G_\Lambda$  is even and unimodular, and prove that  $s_\delta \in O(\Lambda)$  generate a group acting on the set of roots. Exercises 20.6 and 20.7 are first steps.

| Statement                                                                                                          | Tier | Where                                                                                                          |
|--------------------------------------------------------------------------------------------------------------------|------|----------------------------------------------------------------------------------------------------------------|
| $(0, 8) + (0, 8) + 3(1, 1) = (3, 19)$ ; rank 22; $-16 \equiv 0 \pmod{8}$                                           | [A]  | <code>sigK3_eq</code> , <code>rank_K3</code> , <code>index_mod_eight</code>                                    |
| Lattice route and Hodge route give the same pair                                                                   | [A]  | <code>sigK3_matches_hodge</code>                                                                               |
| $-E_8$ symmetric, even diagonal, $\det = \pm 1$                                                                    | [A]  | <code>e8Neg_symm</code> , <code>e8Neg_evenDiag</code> ,<br><code>e8Neg_unimodular</code>                       |
| $s_v$ is an isometry and an involution if $v^T L v = -2$ ; the eight simple reflections of $-E_8$                  | [A]  | <code>reflection_isometry</code> ,<br><code>reflection_involution</code> ,<br><code>e8Neg_weyl_isometry</code> |
| $G_\Lambda$ : $\det = -1$ , inertia $(3, 19)$ ; Examples 20.5 and 20.16                                            | —    | symbolic checks (sympy) and derivations on the page                                                            |
| $H^2(X, \mathbb{Z})$ is even, unimodular, of signature $(3, 19)$ , hence $\cong \Lambda$                           | [L]  | Huy ch. 1 Prop. 3.5; Asp §2.3                                                                                  |
| Local Torelli, surjectivity, global Torelli; Kähler cone; $W$ simply transitive on chambers; $\text{Mon}(X) = O^+$ | [L]  | Huy ch. 6–8                                                                                                    |
| $\mathcal{M}_c, \mathcal{M}_E$ as double cosets; orbifold points $\Leftrightarrow$ root $\perp \Sigma$ ; ADE       | [L]  | Asp §2.3–2.6                                                                                                   |
| ADE singularity with $B = 0 \Rightarrow$ non-abelian gauge symmetry of type IIA                                    | [L]  | Asp §4.3, conditional on IIA/heterotic duality                                                                 |
| A Lean <b>Signature</b> or Gram matrix “is” the cohomology of a K3 surface                                         | —    | an identification, never Tier A                                                                                |

### Chapter summary

- $H^2(X, \mathbb{Z})$  of a K3 surface with the intersection pairing is a lattice of rank 22. It is unimodular by Poincaré duality (the dual basis is an integer inverse of the Gram

matrix), even by Wu's formula and  $c_1 = 0$ , and of signature  $(3, 19)$  by the signature theorem or by  $b_+ = 2h^{2,0} + 1$ . Classification then gives  $H^2(X, \mathbb{Z}) \cong 3U \oplus 2E_8(-1)$ , abstractly.

- The class  $\Omega = x + iy$  satisfies  $\Omega^2 = 0$ ,  $\Omega \cdot \bar{\Omega} > 0$ :  $x \perp y$ ,  $x^2 = y^2 > 0$ . The period is the point  $[\Omega]$  of the 20-dimensional period domain  $D \cong O(3, 19)/(SO(2) \times O(1, 19))$ , equivalently an oriented positive two-plane. It determines the Hodge decomposition and the Picard lattice  $\Omega^\perp \cap H^2(X, \mathbb{Z})$ .
- Torelli theorems (Tier L): the period map is a local isomorphism, surjective, and a Hodge isometry implies isomorphism. A Hodge isometry comes from a unique isomorphism if it respects Kähler cones. The moduli space  $\mathcal{M}_c$  of complex structures is a double coset by  $O^+(\Gamma^{3,19})$  and is not Hausdorff.
- Roots  $\delta^2 = -2$  of  $\text{NS}(X)$  are, up to sign, classes of curves;  $(-2)$ -curves bound the Kähler cone, which is one chamber of the positive cone; the Weyl group generated by  $s_\delta(x) = x + (x \cdot \delta)\delta$  permutes the chambers simply transitively.
- A Ricci-flat metric is a positive three-plane  $\Sigma = H^+ = \langle \Omega, J \rangle$  and a volume:  $\mathcal{M}_E \cong O(\Gamma^{3,19}) \backslash O(3, 19)/(O(3) \times O(19)) \times \mathbb{R}_+$ , of dimension  $57 + 1 = 58$ , Hausdorff, of Narain type. One metric carries an  $S^2$  of complex structures.
- $\Sigma \perp \text{root} \Leftrightarrow$  orbifold point; the roots orthogonal to  $\Sigma$  form an ADE root system of rank  $\leq 19$ ; with vanishing  $B$ -field the type IIA string has the corresponding non-abelian gauge symmetry (conditional on string duality).
- Lean checks the signature arithmetic, the evenness and unimodularity of the block  $-E_8$ , and that reflections in  $(-2)$ -vectors are involutive isometries. It does not contain K3 surfaces, periods or the Torelli theorem.

## Exercises

**Exercise 20.1** ( $\star$ ). Solve  $b_+ + b_- = 22$ ,  $b_+ - b_- = -16$ . Then repeat the computation for the four-torus  $T^4$ , which has  $\chi = 0$ ,  $c_1 = 0$  and  $b_2 = 6$ : find its signature, and use the classification of Chapter 18 to show that  $H^2(T^4, \mathbb{Z}) \cong 3U$ , given that it is even. Compare with the K3 lattice: which summand is “new”?

**Exercise 20.2** ( $\star$ ). A smooth quartic surface may contain a line  $\ell \cong \mathbb{P}^1$ . Check that the Fermat quartic  $x_0^4 + x_1^4 + x_2^4 + x_3^4 = 0$  contains the line  $x_0 = \zeta x_1$ ,  $x_2 = \zeta x_3$  for  $\zeta^4 = -1$ . Use (20.4) to find  $\ell^2$ , and compute the Gram matrix of the sublattice of  $\text{NS}(X)$  spanned by the hyperplane class  $h$  and  $\ell$ , using  $h \cdot \ell = 1$ . Verify that its signature is  $(1, 1)$ , in accordance with the signature  $(1, \rho - 1)$  of the Picard lattice.

**Exercise 20.3** ( $\star\star$ ). Let  $\Omega = x + iy$  with  $x = e_1 + 2f_1$  and  $y = \sqrt{2}(e_2 + f_2)$  in the notation of Example 20.5. (a) Verify that  $[\Omega] \in D$ . (b) Show that a lattice vector  $v = ae_1 + bf_1 + ce_2 + df_2 + \dots$  is orthogonal to  $\Omega$  if and only if  $2a + b = 0$  and  $c + d = 0$ , and conclude that  $\rho = 20$  again. (c) Replace  $\sqrt{2}$  by a transcendental number  $t$  and  $x$  by  $e_1 + t'f_1$  with  $t'$  chosen such that  $x^2 = y^2$ . What is  $\rho$  now? The Picard number is not constant on  $D$ : it jumps at special periods.

**Exercise 20.4** ( $\star\star$ ). Let  $\delta, \delta'$  be roots of a lattice and  $g$  an isometry. (a) Prove  $g \circ s_\delta = s_{g(\delta)} \circ g$ . (b) If  $\delta \cdot \delta' = 0$ , show that  $s_\delta$  and  $s_{\delta'}$  commute. (c) If  $\delta \cdot \delta' = 1$ , show that  $s_\delta(\delta') = \delta + \delta'$  is a root and that  $(s_\delta s_{\delta'})^3 = 1$ . These are the relations of the Weyl group of  $A_2$ , the symmetric group  $S_3$ . Which singularity and which gauge group correspond to this configuration?

**Exercise 20.5** (★★). In Example 20.16 replace  $J$  by  $J' = e_3 + f_3 + \epsilon w$ , where  $w$  is a vector in the first  $E_8(-1) \otimes \mathbb{R}$  with  $(w \cdot \alpha_i) \neq 0$  for all simple roots except  $\alpha_0$  and  $\alpha_1$ , and  $\epsilon$  is small. Check that  $J'$  is still orthogonal to  $\Omega$  and space-like. Which roots of the first  $E_8(-1)$  remain orthogonal to  $\Sigma$ ? Describe the singularities of the resulting orbifold and the gauge algebra at  $B = 0$ . (Hint: a root orthogonal to  $w$  must be a combination of simple roots orthogonal to  $w$ , if  $w$  is chosen in the closure of the fundamental chamber.)

**Exercise 20.6** (★★, Lean). State and prove in Lean, in the style of `sigK3_eq`, that the signature pair of  $3U$  is  $(3, 3)$ , that its rank is 6 and that its index is divisible by 8. Then state (without proving anything about lattices) the pair for  $E_8(-1)^{\oplus 3} \oplus U^{\oplus k}$  and find, by `decide`, for which  $k \leq 4$  the rank equals 26. Which part of Exercise 20.1 is invisible to this arithmetic?

**Exercise 20.7** (★★★, Lean). Define in Lean the vector  $v : \text{Fin } 8 \rightarrow \mathbb{Z}$  of the root  $\alpha_3 + \alpha_4$  of  $E_8(-1)$  and prove `latticeNorm e8Neg v = -2`. Deduce from `reflection_isometry` that the reflection in this non-simple root is an isometry of `e8Neg`. Then prove the matrix identity  $s_{\alpha_4} s_{\alpha_3} s_{\alpha_4} = s_{\alpha_3 + \alpha_4}$  for the corresponding `reflection` matrices (a finite computation: `ext i j, fin_cases, simp`). This is the identity of part (a) of the exercise on isometries above with  $g = s_{\alpha_4}$ .

**Exercise 20.8** (★★★). (A period domain that is an upper half-plane.) Let  $\Lambda' = U \oplus \langle 2 \rangle$ , with basis  $e, f, w$  and  $w^2 = 2$ ; its signature is  $(2, 1)$ . Using Remark 20.6, show that  $D$  is the union of the upper and lower half-planes with coordinate  $\zeta = \tau w$ , and that the reflection in the root  $e - f$  acts on the period  $[1 : -\tau^2 : \tau]$  (coordinates along  $e, f, w$ ) by  $\tau \mapsto -1/\tau$ . Compare with the T-duality and S-transformations of Chapter 12.

## Further reading

- The K3 lattice: [Huy ch. 1 §3.3, Prop. 3.5 and Remarks 3.6–3.7, ll. 561–660]; the physicist’s derivation with Poincaré duality and the period relations, [Asp §2.3, ll. 502–774].
- Period domains, their Grassmannian and tube-domain descriptions, and the action of  $O(\Lambda)$ : [Huy ch. 6 §1, ll. 4272–4420]; local Torelli, [Huy ch. 6 §2, Prop. 2.8, ll. 4654–4670].
- Moduli of marked surfaces, surjectivity of the period map and the global Torelli theorem via monodromy: [Huy ch. 7 §§2–5, ll. 5594–6060].
- Chambers, walls, the Weyl group and the Kähler cone: [Huy ch. 8 §2, ll. 6423–6563, and §5, ll. 7195–7222].
- Einstein metrics, the three-plane  $\Sigma$  and hyperkähler rotation: [Asp §2.4, ll. 777–908]; algebraic K3 surfaces and the transcendental lattice, [Asp §2.5, ll. 910–1029]; orbifolds, blow-ups and the ADE classification, [Asp §2.6, ll. 1033–1330]; enhanced gauge symmetry, [Asp §4.3, ll. 2428–2560].
- The three Lean files of Section 20.7 are short and carry detailed module documentation. The lattice theory behind them is Chapter 18; the explicit cycles realizing part of  $\Lambda$  on a Kummer surface are in Chapter 21; the extension to  $\Gamma^{4,20}$  is in Chapters 22 and 23.



# Chapter 21

## Kummer Surfaces and the Orbifold $T^4/\mathbb{Z}_2$

The K3 surfaces of Chapter 19 were defined by equations, and their lattice  $3U \oplus 2E_8(-1)$  was obtained in Chapter 20 from general theorems. Neither description lets one *see* the 22 two-cycles, and neither is a space on which a string can be quantized exactly. This chapter studies the one K3 surface for which both are possible. Take a four-torus, identify  $x$  with  $-x$ , and repair the 16 singular points that result. The outcome is a K3 surface, the *Kummer surface*, and every number attached to K3 can be counted on it by hand:  $22 = 6 + 16$ ,  $24 = 8 + 16$ ,  $20 = 4 + 16$ ,  $(3, 19) = (3, 3) + (0, 16)$ , and, for the string,  $80 = 16 + 64$ . The questions we answer are these. Why are there exactly 16 singular points and what do they look like? What does “blowing up” do, in coordinates, and why does the new sphere have self-intersection  $-2$ ? Why do the  $6 + 16$  obvious classes *not* generate  $H^2(X, \mathbb{Z})$ , and what is the rank-16 *Kummer lattice* that they span after saturation? Finally, how does the string know about the 16 spheres when they have zero size? The answer to the last question, twisted sectors, is the reason why string theory on an orbifold is consistent although the space is singular. The torus side of the story uses Chapter 10; the lattice vocabulary (discriminant, index,  $U$ , even lattices) is that of Chapter 18.

### 21.1 The orbifold $T^4/\mathbb{Z}_2$

#### 21.1.1 Sixteen fixed points

Let  $\Lambda \subset \mathbb{R}^4$  be a lattice of rank four and  $A = \mathbb{R}^4/\Lambda$  the four-torus. The map  $\iota: x \mapsto -x$  preserves  $\Lambda$  and therefore acts on  $A$ ; since  $\iota^2 = 1$  it generates a group  $\mathbb{Z}_2$ . We write  $S_0 = A/\iota$  for the quotient, in which the points  $x$  and  $-x$  are identified, and follow the notation of [Asp §2.6, ll. 1064–1072]. A point of  $A$  is fixed by  $\iota$  when  $x \equiv -x$  modulo  $\Lambda$ , that is when  $2x \in \Lambda$ :

$$\text{Fix}(\iota) = \frac{1}{2}\Lambda/\Lambda \cong (\mathbb{Z}/2\mathbb{Z})^4, \quad |\text{Fix}(\iota)| = 2^4 = 16. \quad (21.1)$$

If  $\lambda_1, \dots, \lambda_4$  is a basis of  $\Lambda$ , the fixed points are  $p_\epsilon = \frac{1}{2}(\epsilon_1\lambda_1 + \dots + \epsilon_4\lambda_4)$  with  $\epsilon = (\epsilon_1, \dots, \epsilon_4) \in \mathbb{F}_2^4$ , where  $\mathbb{F}_2 = \{0, 1\}$  is the field with two elements. These are the points of order two of the group  $A$  [Huy ch. 1, Example 1.3(iii), ll. 218–222]. The labelling by the vector space  $\mathbb{F}_2^4$  looks like bookkeeping now; in Section 21.4 the affine geometry of  $\mathbb{F}_2^4$  will decide which combinations of cycles are integral.

**Example 21.1** (The square torus). Take  $A = \mathbb{C}^2/(\mathbb{Z} + i\mathbb{Z})^2$  with coordinates  $(z_1, z_2)$ , as in [Asp eq. (49), ll. 1066–1072]. In each factor the fixed points of  $z \mapsto -z$  are  $0, \frac{1}{2}, \frac{i}{2}, \frac{1}{2} + \frac{i}{2}$ , so the 16 fixed points are the pairs  $(z_1, z_2)$  with both entries in this list. One factor alone gives

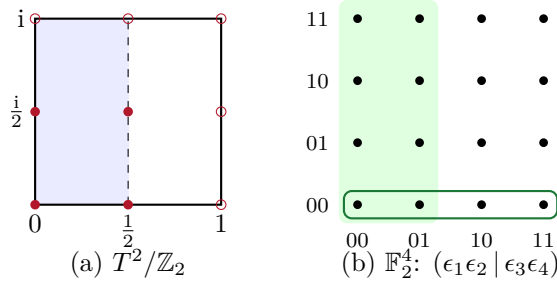


Figure 21.1: (a) A fundamental domain of  $z \mapsto -z$  on the square torus is the shaded half; the four filled dots are the fixed points (open dots are their translates). (b) The 16 fixed points of  $T^4/\mathbb{Z}_2$  as the points of  $\mathbb{F}_2^4$ ; horizontal position is  $(\epsilon_1\epsilon_2)$ , vertical position is  $(\epsilon_3\epsilon_4)$ . The framed row is an affine plane (four points: the fixed points on one invariant two-torus); the shaded block  $\epsilon_1 = 0$  is an affine hyperplane (eight points).

the “pillow”  $T^2/\mathbb{Z}_2$  of Figure 21.1: fold the fundamental square along its middle line and sew the edges; the result is a sphere with four corners.

### 21.1.2 What the quotient looks like near a fixed point

Away from the 16 points the action of  $\mathbb{Z}_2$  is free and  $S_0$  is a smooth flat manifold. Near a fixed point  $p$  put  $y = x - p$ ; then  $\iota$  acts as  $y \mapsto -y$ , and a neighbourhood of the image of  $p$  in  $S_0$  is a neighbourhood of the origin in

$$\mathbb{R}^4/\{\pm 1\} = \mathbb{C}^2/\mathbb{Z}_2, \quad (\xi, \eta) \sim (-\xi, -\eta). \quad (21.2)$$

This is a cone: the sphere  $|y| = r$  is mapped to  $S^3/\{\pm 1\} = \mathbb{RP}^3$ , so  $\mathbb{C}^2/\mathbb{Z}_2$  is the cone over  $\mathbb{RP}^3$  with its vertex at the fixed point. It is not a manifold there, because a small punctured neighbourhood of a point in a four-manifold is simply connected, while  $\pi_1(\mathbb{RP}^3) = \mathbb{Z}_2$ . The space  $S_0$  is an *orbifold*: it is covered by patches of the form  $\mathbb{R}^n/G_i$  with  $G_i$  finite [Asp §2.6, ll. 1042–1049]. Such a singular point, locally  $\mathbb{C}^2/\mathbb{Z}_2$ , is called an  $A_1$  singularity or ordinary double point; the name will be justified in Section 21.2.

### 21.1.3 Why $S_0$ deserves to be called a K3 surface

A K3 surface was defined in Chapter 19 by a trivial canonical bundle and the absence of holomorphic one-forms, and was shown to be simply connected. We check the three properties for  $S_0$ . Choose a complex structure on  $\mathbb{R}^4$  with coordinates  $(z_1, z_2)$ , so that  $A$  is a complex torus.

1. The holomorphic two-form  $dz_1 \wedge dz_2$  is invariant, since both factors change sign. It descends to  $S_0$  and has no zeros. The Kähler form  $\frac{i}{2}(dz_1 \wedge d\bar{z}_1 + dz_2 \wedge d\bar{z}_2)$  is invariant for the same reason [Asp ll. 1073–1077].
2. The one-forms  $dz_1, dz_2$  are odd and do not descend:  $S_0$  has no holomorphic one-forms, although  $A$  has two.
3.  $S_0$  is simply connected [Asp ll. 1074–1076]. Indeed, let  $\gamma$  be the closed loop  $s \mapsto s\lambda_1$ ,  $s \in [0, 1]$ , which generates one factor of  $\pi_1(A) = \mathbb{Z}^4$ . Its second half, from the fixed point  $\frac{1}{2}\lambda_1$  to  $\lambda_1 \equiv 0$ , is the image under  $\iota$  of the first half run backwards. In  $S_0$  the loop is therefore of the form  $\delta \cdot \delta^{-1}$  and can be contracted. The new closed loops of  $S_0$ , images of paths from  $y$  to  $-y$ , are of the same form once the path is slid through a fixed point.



**Holonomy  $\mathbb{Z}_2 \subset SU(2)$ : a flat space that breaks half the supersymmetry**

The torus is flat, so parallel transport around a contractible loop is trivial. In  $S_0$  there are new closed loops: the image of a path in  $A$  from  $y$  to  $-y$ . A vector transported along it comes back multiplied by  $-1$ , because the identification  $y \sim -y$  acts on tangent vectors too. The holonomy group of  $S_0$  is thus  $\mathbb{Z}_2 = \{\pm \mathbf{1}_2\}$ , and  $-\mathbf{1}_2 = \text{diag}(-1, -1)$  has determinant  $+1$ : it lies in  $SU(2)$  [Asp ll. 1331–1335]. A spinor on a four-dimensional space transforms under  $SU(2)_+ \times SU(2)_-$ ; if the holonomy lies in one factor, the spinors of the other chirality are untouched (Chapter 19). Hence  $S_0$  preserves exactly the same half of the supersymmetry as a smooth K3 surface with its full  $SU(2)$  holonomy, while being flat everywhere except at 16 points. All the curvature of K3, in particular the whole of  $\chi = 24$ , has been pushed into these points.

 **$T^2/\mathbb{Z}_2$  is not a “two-dimensional K3”**

For one complex dimension,  $z \mapsto -z$  has determinant  $-1$  on  $dz$ : the holomorphic one-form is projected out, and the holonomy  $-1 \in U(1)$  is not in  $SU(1) = \{1\}$ . The pillow of Figure 21.1(a) is a sphere with four conical points of deficit angle  $\pi$ ; the total deficit  $4\pi = 2\pi\chi(S^2)$  is the Gauss–Bonnet theorem. It is positively curved and preserves no supersymmetry by itself. The rule is that  $\mathbb{C}^n/G$  is a local model of a Calabi–Yau orbifold only when  $G \subset SU(n)$ ; for  $n = 2$  this is what makes  $dz_1 \wedge dz_2$  invariant [Asp ll. 1331–1333].

## 21.2 The local model: resolving $\mathbb{C}^2/\mathbb{Z}_2$

### 21.2.1 The singularity as a quadric cone

Functions on  $\mathbb{C}^2/\mathbb{Z}_2$  are functions of  $(\xi, \eta)$  that are even under  $(\xi, \eta) \mapsto (-\xi, -\eta)$ . Every even polynomial is a polynomial in the three quadratic monomials

$$x_0 = \xi^2, \quad x_1 = \eta^2, \quad x_2 = \xi\eta, \quad \text{which satisfy} \quad f := x_0x_1 - x_2^2 = 0. \quad (21.3)$$

Conversely a point of the surface  $\{f = 0\} \subset \mathbb{C}^3$  determines  $(\xi, \eta)$  up to the common sign, so  $\mathbb{C}^2/\mathbb{Z}_2$  is the hypersurface  $f = 0$  [Asp eq. (50), ll. 1082–1091]. The gradient  $(\partial_0 f, \partial_1 f, \partial_2 f) = (x_1, x_0, -2x_2)$  vanishes only at the origin, which is the single singular point. With  $x_0 = u + iv$ ,  $x_1 = u - iv$ ,  $x_2 = iw$  the equation becomes  $u^2 + v^2 + w^2 = 0$ , the case  $n = 2$  of the  $A_{n-1}$  singularity  $x^2 + y^2 + z^n = 0$  in the  $A$ – $D$ – $E$  list of surface singularities [Asp ll. 1226–1230 and eq. (58), ll. 1523–1527].

### 21.2.2 Blowing up in coordinates

The idea of a blow-up is to remember not only *that* a path reaches the singular point but *from which direction*. In  $\mathbb{C}^3$  a direction is a point  $[s_0 : s_1 : s_2] \in \mathbb{P}^2$ , and the blow-up of  $\mathbb{C}^3$  at the origin is the set of pairs (point, line through the origin containing it),  $\{x_i s_j = x_j s_i\} \subset \mathbb{C}^3 \times \mathbb{P}^2$  [Asp eq. (51), ll. 1094–1106]. A path in the cone  $f = 0$  approaching the origin along  $(x_0 t, x_1 t, x_2 t)$  lands on a direction with  $s_0 s_1 - s_2^2 = 0$ . The closure of the cone minus its vertex inside the blow-up, its *proper transform*  $\widetilde{\mathbb{C}^2/\mathbb{Z}_2}$ , therefore replaces the vertex by the conic

$$E = \{s_0 s_1 = s_2^2\} \subset \mathbb{P}^2, \quad [\xi : \eta] \mapsto [\xi^2 : \eta^2 : \xi\eta], \quad (21.4)$$

which the displayed parametrization identifies with  $\mathbb{P}^1$  [Asp ll. 1123–1131]. In words:  $E$  is the set of complex lines  $\ell \subset \mathbb{C}^2$  through the origin (the points  $\pm(\xi, \eta)$  lie on the same line), and the resolved space is

$$\widetilde{\mathbb{C}^2/\mathbb{Z}_2} = \{(\ell, q) : \ell \in \mathbb{P}^1, q \in \ell/\{\pm 1\}\}. \quad (21.5)$$

It is fibred over  $E = \mathbb{P}^1$  with fibre  $\ell/\{\pm 1\} \cong \mathbb{C}$  (the quotient of a complex line by  $\pm 1$  is again a complex line, with coordinate the square). We now make this explicit. Cover  $\mathbb{P}^1$  by the charts  $\xi \neq 0$  and  $\eta \neq 0$  and use

$$\text{chart 1: } t = \frac{\eta}{\xi} = \frac{x_2}{x_0}, \quad a = \xi^2 = x_0; \quad \text{chart 2: } t' = \frac{\xi}{\eta} = \frac{x_2}{x_1}, \quad a' = \eta^2 = x_1. \quad (21.6)$$

In chart 1 the cone is parametrized by  $(x_0, x_1, x_2) = (a, at^2, at)$ , which indeed solves  $x_0x_1 = x_2^2$ , and  $(a, t) \in \mathbb{C}^2$  are unconstrained coordinates: the space is smooth. The exceptional curve is  $E = \{a = 0\}$ , with  $t$  as its coordinate. On the overlap

$$t' = \frac{1}{t}, \quad a' = t^2 a. \quad (21.7)$$

Compare with the tautological line bundle  $\mathcal{O}(-1)$  over  $\mathbb{P}^1$ , whose fibre over  $\ell$  is the line  $\ell$  itself: its fibre coordinates are  $\xi$  and  $\eta = t\xi$ , so the transition function is  $t$ . The transition function of our bundle is  $t^2$ , hence

$$\widetilde{\mathbb{C}^2/\mathbb{Z}_2} = \text{total space of } \mathcal{O}_{\mathbb{P}^1}(-2), \quad (21.8)$$

with  $E$  as the zero section [Asp II. 1134–1137].

**Proposition 21.2** (The exceptional curve is a  $(-2)$ -curve).  $E \cdot E = -2$ .

*Two derivations.* (i) The self-intersection of a curve is the degree of its normal bundle, computed as the number of zeros minus the number of poles of any meromorphic section [Asp II. 1139–1173]. Take the section  $a = 1$  in chart 1. In chart 2 it reads  $a' = t^2 = 1/t'^2$ : no zeros and a double pole at  $t' = 0$ . The degree is  $0 - 2 = -2$ . (ii) For a smooth curve  $C$  of genus  $g$  in a surface with trivial canonical bundle, the adjunction formula gives  $C \cdot C = 2g - 2$  [Asp eq. (53), II. 1178–1182]. For  $g = 0$  this is  $-2$ . Route (ii) assumes the canonical bundle of the resolved space to be trivial; this is Proposition 21.3.  $\square$

A negative self-intersection means that  $E$  is rigid: it cannot be moved to a nearby holomorphic curve, since two distinct holomorphic curves intersect non-negatively [Asp II. 1173–1177].

**Proposition 21.3** (The resolution is crepant). *The two-form  $d\xi \wedge d\eta$  on  $\mathbb{C}^2/\mathbb{Z}_2$  extends to a holomorphic two-form without zeros on  $\widetilde{\mathbb{C}^2/\mathbb{Z}_2}$ . Explicitly  $d\xi \wedge d\eta = \frac{1}{2} da \wedge dt = -\frac{1}{2} da' \wedge dt'$ .*

*Proof.* In chart 1,  $\xi = a^{1/2}$  and  $\eta = ta^{1/2}$ , so  $d\xi = \frac{1}{2}a^{-1/2}da$  and  $d\eta = a^{1/2}dt + \frac{1}{2}ta^{-1/2}da$ . The wedge product is  $\frac{1}{2}a^{-1/2}a^{1/2}da \wedge dt = \frac{1}{2}da \wedge dt$ : the square roots cancel, and nothing special happens at  $a = 0$ . From (21.7),  $da' \wedge dt' = (t^2da + 2ta dt) \wedge (-t^{-2}dt) = -da \wedge dt$ . (Both Jacobians were confirmed by a symbolic check.)  $\square$

#### Blowing up a smooth point is different

If one blows up the origin of  $\mathbb{C}^2$  itself, without the quotient, the coordinates are  $(\xi, t)$  with  $\eta = t\xi$ , the exceptional curve has normal bundle  $\mathcal{O}(-1)$  and self-intersection  $-1$ , and  $d\xi \wedge d\eta = \xi d\xi \wedge dt$  *vanishes* along  $E = \{\xi = 0\}$ . The canonical bundle changes,  $\omega_{\tilde{A}} \cong \mathcal{O}(\sum E_i)$  for the blow-up  $\tilde{A}$  of the torus in 16 points [Huy ch. 1, II. 244–246]. Blowing up a smooth point of a K3 surface destroys the K3 property; blowing up an  $A_1$  point preserves it.

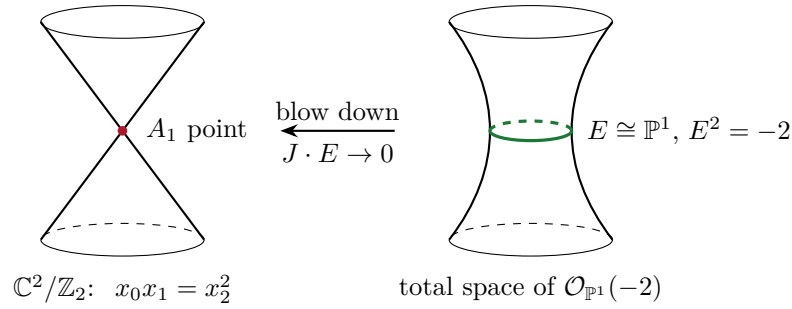


Figure 21.2: The  $A_1$  singularity and its resolution (real two-dimensional cartoon of a complex surface). The vertex of the cone is replaced by a sphere  $E$  whose area is  $J \cdot E$ , the integral of the Kähler form. Letting  $J \cdot E \rightarrow 0$  with the complex structure fixed recovers the orbifold.

### 21.2.3 The global construction, in two orders

Performing the local surgery at all 16 points of  $S_0$  gives a smooth compact complex surface  $X = \text{Km}(A)$  with a nowhere vanishing holomorphic two-form (Proposition 21.3 glued to  $dz_1 \wedge dz_2$ ), with no holomorphic one-forms and simply connected: a K3 surface [L] [Asp ll. 1188–1194; Huy ch. 1, Example 1.3(iii), ll. 218–250]. One can also resolve in the opposite order [Huy ll. 225–240 and §3.2.5, ll. 2015–2031]: first blow up the 16 fixed points of the smooth torus, obtaining  $\sigma: \tilde{A} \rightarrow A$  with sixteen  $(-1)$ -curves  $E_i$ ; the involution lifts to  $\tilde{\iota}$  on  $\tilde{A}$ , acting in the coordinates of the pitfall box as  $(\xi, t) \mapsto (-\xi, t)$ ; its fixed locus is the union of the curves  $E_i = \{\xi = 0\}$ , and the quotient has the smooth coordinates  $(a, t) = (\xi^2, t)$  of (21.6):

$$\begin{array}{ccc} \tilde{A} & \xrightarrow{\sigma} & A \\ \pi \downarrow & & \downarrow \\ X & \longrightarrow & A/\iota = S_0 \end{array} \quad \begin{array}{l} \pi: \tilde{A} \rightarrow X \text{ is a double cover branched along } \bigcup E_i, \\ \bar{E}_i := \pi(E_i) \subset X, \quad \pi^* \mathcal{O}(\bar{E}_i) \cong \mathcal{O}(2E_i). \end{array} \quad (21.9)$$

The second order is the one used for computations in cohomology, because everything is pulled back from or pushed forward to the smooth surface  $\tilde{A}$ .

#### The size of $E$ is a modulus

Nothing in the algebraic construction fixes the *size* of the new sphere. With a Kähler form  $J$  on  $X$  the area of  $\bar{E}_i$  is  $J \cdot \bar{E}_i = \int_{\bar{E}_i} J$ . Keeping the complex structure fixed, one may move  $J$  until it becomes orthogonal to  $\bar{E}_i$ ; the sphere then has zero area and the surface is the orbifold again [Asp ll. 1195–1202]. The orbifold  $S_0$  is thus not a different space from the Kummer surface but a boundary point of its family of metrics: in the language of Chapter 20, a Ricci-flat metric is a positive three-plane  $\Sigma \subset H^2(X, \mathbb{R})$ , and the metric is an orbifold metric exactly when  $\Sigma$  is orthogonal to a root, a class  $\alpha$  with  $\alpha^2 = -2$  [L] [Asp Theorem 4, ll. 1208–1216]. At  $S_0$  the plane  $\Sigma$  is orthogonal to all sixteen  $\bar{E}_i$  at once. Near each resolved point the Ricci-flat metric is approximated by an explicitly known asymptotically flat metric on  $\mathcal{O}_{\mathbb{P}^1}(-2)$  [Asp ll. 1503–1507]. Because  $\Sigma$  is spanned by  $J$  and the real and imaginary parts of the holomorphic two-form  $\Omega$ , one can also make  $\bar{E}_i$  disappear by rotating  $\Omega$  instead of  $J$ : the singularity is then *deformed*,  $x_0x_1 - x_2^2 = \varepsilon$ , rather than blown up, and for K3 the two procedures are related by a rotation of  $\Sigma$  [Asp ll. 1513–1530]. Each  $A_1$  point therefore carries three real metric parameters, the components of  $\Sigma$  along  $\bar{E}_i$ .

## 21.3 Counting: $b_2 = 6 + 16$ and $\chi = 24$

### 21.3.1 The cohomology of the orbifold

With rational coefficients the cohomology of a quotient by a finite group is the invariant part of the cohomology of the covering space (average a closed form over the group). For the torus,  $H^k(A, \mathbb{Q}) = \Lambda^k H^1(A, \mathbb{Q})$  is spanned by the products  $dx^{m_1} \wedge \cdots \wedge dx^{m_k}$ , and  $\iota^* dx^m = -dx^m$ . Therefore  $\iota^* = (-1)^k$  on  $H^k$  [Huy §3.2.5, ll. 2037–2038], and

$$b_k(S_0) = \binom{4}{k} \text{ for } k \text{ even, } 0 \text{ for } k \text{ odd: } (b_0, \dots, b_4) = (1, 0, 6, 0, 1), \quad \chi(S_0) = 8. \quad (21.10)$$

The six surviving two-forms  $dx^m \wedge dx^n$  are dual to the six coordinate two-tori  $T_{mn} \subset A$ . The Hodge decomposition of the invariant forms is just as simple. On the torus  $h^{p,q}(A) = \binom{2}{p} \binom{2}{q}$ , spanned by  $dz_I \wedge d\bar{z}_J$ , and the form has sign  $(-1)^{p+q}$  under  $\iota$ :

$$h^{2,0} = h^{0,2} = 1 \text{ (} dz_1 \wedge dz_2 \text{ and its conjugate), } h^{1,1} = 4 \text{ (} dz_i \wedge d\bar{z}_j \text{), } h^{1,0} = h^{0,1} = 0. \quad (21.11)$$

### 21.3.2 Adding the sixteen spheres

Let  $N_i$  be a small conical neighbourhood of the  $i$ -th singular point and  $\tilde{N}_i$  the neighbourhood of  $\bar{E}_i$  that replaces it.  $N_i$  is contractible;  $\tilde{N}_i$  is a disc bundle over  $S^2$  and contracts onto  $\bar{E}_i$ . Their common boundary is  $\mathbb{RP}^3$ , which has the rational cohomology of  $S^3$ . Writing  $X = (X \setminus \bigcup \tilde{N}_i) \cup \bigcup \tilde{N}_i$ , the Mayer–Vietoris sequence shows that each replacement adds exactly one class to  $H^2(\cdot, \mathbb{Q})$ , the class of  $\bar{E}_i$ , and changes nothing in degrees 1 and 3. Hence

$$b_2(X) = 6 + 16 = 22, \quad b_1(X) = b_3(X) = 0, \quad \chi(X) = 1 + 22 + 1 = 24. \quad (21.12)$$

Since the curves  $\bar{E}_i$  are holomorphic, their classes are of type  $(1, 1)$ , and (21.11) becomes  $h^{1,1}(X) = 4 + 16 = 20$ ,  $h^{2,0}(X) = 1$ : the Hodge diamond of Chapter 19. The same grouping, 16 classes from the singular points, the  $(2, 0)$  and  $(0, 2)$  forms and 4 more  $(1, 1)$  forms from the torus, is the one used in the physics literature [DRS §2, ll. 513–519]. For the signature, the wedge product on the invariant classes  $\Lambda^2 \mathbb{R}^4$  has signature  $(3, 3)$  (self-dual and anti-self-dual forms), and the sixteen classes  $\bar{E}_i$  are mutually orthogonal with square  $-2$ :

$$(3, 3) + (0, 16) = (3, 19), \quad (21.13)$$

in agreement with the signature found from the Hirzebruch theorem in Chapter 19.

**Example 21.4** (The number 24 in three ways). (a) *Betti numbers*: (21.12). (b) *Cut and paste*: the Euler characteristic is additive under decompositions and multiplicative under unbranched covers. Remove the 16 fixed points from  $A$ :  $\chi = 0 - 16$ . The quotient of the rest is free:  $\chi = -8$ . Glue in 16 spheres:  $\chi(X) = -8 + 16 \cdot 2 = 24$ . Gluing in 16 points instead gives  $\chi(S_0) = -8 + 16 = 8$ , as in (21.10). (c) *Elliptic fibration*: let  $A = E_1 \times E_2$  be a product of elliptic curves. The projection to  $E_1/\pm \cong \mathbb{P}^1$  makes  $X$  an elliptic surface whose smooth fibres are copies of  $E_2$  ( $\chi = 0$ ). Over each of the four branch points the fibre of  $S_0$  is  $E_2/\pm \cong \mathbb{P}^1$  and passes through four singular points; in  $X$  it becomes a configuration of five spheres, the central one meeting the four  $\bar{E}_i$  once each: a fibre of Kodaira type  $I_0^*$  [Huy ch. 11, Example 1.11(i), ll. 9500–9507]. Five spheres glued at four points have  $\chi = 5 \cdot 2 - 4 = 6$ , and  $\chi(X) = 4 \cdot 6 = 24$ ; equivalently, the discriminant of an elliptic K3 surface has degree 24 and vanishes to order 6 at an  $I_0^*$  fibre [Huy §11.2.4, ll. 9725–9735]. All three counts agree with  $c_2(X) = 24$  from Noether’s formula in Chapter 19.

## 21.4 The intersection form and the Kummer lattice

The count  $22 = 6 + 16$  is a statement about vector spaces. The integral statement is subtler and is the mathematical core of this chapter: the  $6 + 16$  obvious classes span a sublattice of index  $2^{11}$  in  $H^2(X, \mathbb{Z})$ .

### 21.4.1 The torus classes are rescaled by two

Let  $\alpha, \alpha' \in H^2(\tilde{A}, \mathbb{Z})$  be  $\tilde{\iota}$ -invariant classes and  $\pi_*\alpha, \pi_*\alpha'$  their push-forwards to  $X$ . Since  $\pi$  has degree two and  $\alpha$  is invariant,  $\pi^*\pi_*\alpha = 2\alpha$ . On the other hand  $(\pi^*\beta \cdot \pi^*\beta')_{\tilde{A}} = 2(\beta \cdot \beta')_X$  for all classes on  $X$ , again because  $\pi$  has degree two. Combining the two,  $4(\alpha \cdot \alpha')_{\tilde{A}} = 2(\pi_*\alpha \cdot \pi_*\alpha')_X$ , that is

$$(\pi_*\alpha \cdot \pi_*\alpha')_X = 2(\alpha \cdot \alpha')_{\tilde{A}} \quad [\text{Huy eq. (2.5), ll. 2037–2044}]. \quad (21.14)$$

Two consequences follow. First,  $\pi_*[E_i] = [\bar{E}_i]$  and  $E_i^2 = -1$  give  $\bar{E}_i^2 = -2$ : a third derivation of Proposition 21.2. Second,  $H^2(A, \mathbb{Z})$  with the basis  $dx^m \wedge dx^n$  is  $U^{\oplus 3}$ , the three hyperbolic pairs being  $(T_{12}, T_{34})$ ,  $(T_{13}, T_{42})$ ,  $(T_{14}, T_{23})$  with  $T_{12} \cdot T_{34} = 1$  and so on. Its image in  $H^2(X, \mathbb{Z})$  carries twice this form:

$$\pi_*H^2(A, \mathbb{Z}) \cong U(2)^{\oplus 3}, \quad \text{disc } U(2)^{\oplus 3} = \det \begin{pmatrix} 0 & 2 \\ 2 & 0 \end{pmatrix}^3 = -2^6. \quad (21.15)$$

The factor two has a transparent meaning. A two-torus  $T_{12}$  in generic position and its mirror image  $\iota(T_{12})$  are different in  $A$  and become one torus in  $X$ . The images of  $T_{12}$  and  $T_{34}$  meet where  $(T_{12} \cup \iota T_{12}) \cap (T_{34} \cup \iota T_{34})$ , four points of  $A$ , is mapped to: two points of  $X$ .

### 21.4.2 Half-integral classes from invariant tori

**Lemma 21.5.** *Let  $T \subset A$  be a two-torus through a fixed point, invariant under  $\iota$ , for instance  $T_{12}$  placed at  $(x^3, x^4) = (0, 0)$ . It contains four fixed points, forming an affine plane  $P \subset \mathbb{F}_2^4$ . The proper transform in  $X$  of the image of  $T$  is a sphere  $C$  with*

$$[C] = \frac{1}{2}(\pi_*[T] - \sum_{i \in P} [\bar{E}_i]) \in H^2(X, \mathbb{Z}), \quad C^2 = -2, \quad C \cdot \bar{E}_i = 1 \quad (i \in P). \quad (21.16)$$

*Proof.* The statement is topological, so we may choose the complex structure on  $A$  for which  $T$  is a holomorphic curve ( $z_1 = x^1 + ix^2$  for  $T_{12}$ ).  $T/\iota$  is the pillow of Figure 21.1(a), a sphere through four singular points. In  $\tilde{A}$  the proper transform  $\tilde{T}$  of  $T$  has class  $\sigma^*[T] - \sum_{i \in P} [E_i]$ , since  $T$  passes once through each blown-up point. The map  $\tilde{T} \rightarrow C$  is two-to-one, so  $\pi_*[\tilde{T}] = 2[C]$ ; this is the first formula. Then, by  $\pi_*[T]^2 = 2T^2 = 0$  and the orthogonality of  $\pi_*[T]$  to the  $\bar{E}_i$ ,  $C^2 = \frac{1}{4}(0 + 4 \cdot (-2)) = -2$  and  $C \cdot \bar{E}_i = \frac{1}{2} \cdot (-1) \cdot (-2) = 1$ . The value  $C^2 = -2$  is what *every* smooth sphere in a K3 surface must have, which checks the formula.  $\square$

So  $\frac{1}{2}(\pi_*[T] - \sum_P \bar{E}_i)$  is an integral class although neither half is: the sublattice  $\pi_*H^2(A, \mathbb{Z}) \oplus \mathbb{Z}\bar{E}_i$  is not all of  $H^2(X, \mathbb{Z})$ . The class  $F = 2C + \sum_{i \in P} \bar{E}_i = \pi_*[T]$  is the  $I_0^*$  fibre of Example 21.4(c); its five components form the affine Dynkin diagram  $\tilde{D}_4$  of Figure 21.3, and  $F^2 = 0$  as it must be for a fibre.

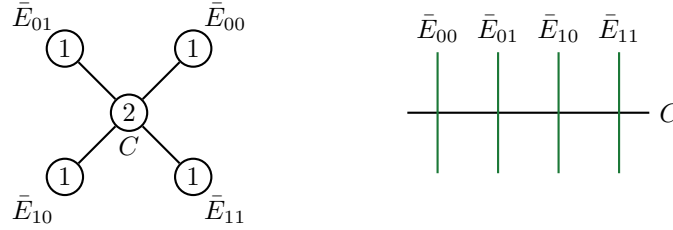


Figure 21.3: The image of an invariant two-torus. Left: dual graph  $\tilde{D}_4$ ; nodes are  $(-2)$ -spheres, edges are transverse intersection points, the numbers are the multiplicities in the fibre class  $F = 2C + \sum \bar{E}_i$ . Right: the curves themselves. The labels are the  $(\epsilon_1\epsilon_2)$  of Figure 21.1(b).

### 21.4.3 From planes to hyperplanes: the Kummer lattice

Take two *parallel* invariant tori  $T, T'$  (the same  $T_{12}$  placed at two different fixed values of  $(x^3, x^4)$ ), with planes  $P, P'$ . They are homologous, so subtracting the two classes (21.16) eliminates the torus part:

$$[C'] - [C] = \frac{1}{2} \sum_{i \in P} \bar{E}_i - \frac{1}{2} \sum_{i \in P'} \bar{E}_i \equiv \frac{1}{2} \sum_{i \in W} \bar{E}_i \pmod{\bigoplus \mathbb{Z} \bar{E}_i}, \quad W = P \cup P'. \quad (21.17)$$

The union of two parallel affine planes in  $\mathbb{F}_2^4$  is an affine hyperplane, a set of 8 points defined by one equation  $a \cdot \epsilon = c$  (in Figure 21.1(b): two rows). Every affine hyperplane arises in this way: choose a basis of  $\Lambda$  adapted to  $W$  and take two parallel invariant tori spanned by two of the basis vectors. Hence, for each of the 30 affine hyperplanes  $W \subset \mathbb{F}_2^4$ , the class  $\frac{1}{2} \sum_{i \in W} \bar{E}_i$  lies in  $H^2(X, \mathbb{Z})$ ; adding two complementary hyperplanes, so does  $\frac{1}{2} \sum_{\text{all } 16} \bar{E}_i$ , in agreement with the square root  $L$  of  $\mathcal{O}(\sum \bar{E}_i)$  that defines the double cover  $\pi$  [Huy ch. 1, ll. 249–251].

**Definition 21.6** (Kummer lattice). The Kummer lattice  $K$  is the saturation of  $\bigoplus_i \mathbb{Z} \bar{E}_i$  in  $H^2(X, \mathbb{Z})$ , i.e. the smallest primitive sublattice containing all  $\bar{E}_i$ , equivalently  $K = (\bigoplus \mathbb{Z} \bar{E}_i)^{\perp\perp}$  [Huy Def. 14.3.13, ll. 13469–13476]. (A sublattice  $M \subseteq \Lambda$  is *primitive* if  $\Lambda/M$  has no torsion:  $nv \in M$  with  $v \in \Lambda$  implies  $v \in M$ .)

**Theorem 21.7** (Structure of the Kummer lattice; [L]).  $K$  is generated by the  $\bar{E}_i$  together with the  $\frac{1}{2} \sum_{i \in W} \bar{E}_i$ ,  $W$  an affine hyperplane of  $\mathbb{F}_2^4$  [Huy §14.3.3, ll. 13482–13487]. Moreover [Huy Prop. 14.3.14, ll. 13497–13503]:

- (i)  $K^\perp = \pi_* H^2(A, \mathbb{Z}) \cong U(2)^{\oplus 3}$ ;
- (ii) the index of  $\bigoplus \mathbb{Z} \bar{E}_i \cong \langle -2 \rangle^{\oplus 16}$  in  $K$  is  $2^5$ ;
- (iii)  $K$  is even, negative definite, of rank 16 and  $\text{disc } K = 2^6$ ;
- (iv) the discriminant group is  $A_K \cong (\mathbb{Z}/2\mathbb{Z})^6$ .

$K$  does not depend on the torus  $A$ , and its roots (vectors of square  $-2$ ) are exactly the  $\pm \bar{E}_i$  [Huy ll. 13494–13496].

We have shown on the page that the listed generators *are* integral; that there are no further integral classes in  $K \otimes \mathbb{Q}$  is the part we take from the literature. Granting it, the numbers (ii)–(iii) can be checked by hand, which is a useful exercise in the lattice techniques of Chapter 18.

**Example 21.8** (The index  $2^5$  and the discriminant  $2^6$ ). A vector of  $K$  is  $x = \frac{1}{2} \sum_i n_i \bar{E}_i$  with  $n_i \in \mathbb{Z}$ , and the condition is on the parities:  $(n_i \bmod 2) \in \mathcal{C}$ , where  $\mathcal{C} \subseteq \mathbb{F}_2^{16}$  is the linear span of the indicator functions of the hyperplanes. The indicator function of  $\{a \cdot \epsilon = c\}$  is the affine function  $\epsilon \mapsto a \cdot \epsilon + c + 1$  on  $\mathbb{F}_2^4$ . The span is the space of all affine functions  $c_0 + c_1 \epsilon_1 + \cdots + c_4 \epsilon_4$ , of dimension 5:  $|\mathcal{C}| = 2^5 = 32$  words, namely 0, the all-ones word, and the 30 hyperplanes (weight 8). The index of  $\bigoplus \mathbb{Z} \bar{E}_i$  in  $K$  is  $|\mathcal{C}| = 2^5$ . The index formula of Chapter 18,  $\text{disc}(\text{sublattice}) = \text{disc}(\text{lattice}) \cdot \text{index}^2$ , gives

$$\text{disc } K = \frac{\text{disc} \langle -2 \rangle^{\oplus 16}}{(2^5)^2} = \frac{2^{16}}{2^{10}} = 2^6.$$

*Integrality*: two hyperplanes meet in 0, 4 or 8 points, so  $\frac{1}{2} \sum_W \bar{E}_i \cdot \frac{1}{2} \sum_{W'} \bar{E}_i = -\frac{1}{2} |W \cap W'| \in \{0, -2, -4\}$ . *Evenness*:  $(\frac{1}{2} \sum_W \bar{E}_i)^2 = \frac{1}{4} \cdot 8 \cdot (-2) = -4$ . *Roots*: a vector with some half-integral  $n_i/2$  has at least 8 of them, hence square  $\leq -4$ ; the vectors of square  $-2$  are the  $\pm \bar{E}_i$  only. *Global check*:  $H^2(X, \mathbb{Z})$  is unimodular and  $K, K^\perp$  are primitive, so  $|\text{disc } K| = |\text{disc } K^\perp|$  [Huy II. 13512–13515]: indeed  $2^6 = |-2^6|$  by (21.15). The index of  $K^\perp \oplus K$  in  $H^2(X, \mathbb{Z})$  is  $\sqrt{2^6} \cdot 2^6 = 2^6$ , and that of the naive lattice  $U(2)^{\oplus 3} \oplus \langle -2 \rangle^{\oplus 16}$  is  $2^6 \cdot 2^5 = 2^{11} = \sqrt{2^6} \cdot 2^{16}$ . (An explicit basis of  $K$  was built and its Gram determinant  $2^6$ , integrality and evenness were confirmed by a symbolic check; the code  $\mathcal{C}$  was enumerated by machine: 32 words of weights 0, 8, 16.) The remaining factor  $2^6$  is made of “mixed” classes such as (21.16), which glue  $K^\perp$  to  $K$ .

#### 22 = 6 + 16 is not a lattice decomposition

It is tempting to write  $H^2(\text{Km}(A), \mathbb{Z}) = U^{\oplus 3} \oplus \langle -2 \rangle^{\oplus 16}$ . The right-hand side has discriminant  $-2^{16}$  and is not unimodular; with  $U(2)$  in place of  $U$  it is  $-2^{22}$ . The true lattice  $3U \oplus 2E_8(-1)$  contains  $U(2)^{\oplus 3} \oplus \langle -2 \rangle^{\oplus 16}$  with index  $2^{11}$ . In particular  $\sum_i \bar{E}_i^2 = -32$  is the trace of a Gram matrix of a sublattice in one basis, not an invariant of  $H^2(X, \mathbb{Z})$ .

### 21.4.4 Which K3 surfaces are Kummer?

The Picard lattice of  $X = \text{Km}(A)$  contains the  $\bar{E}_i$  and the images of the algebraic classes of  $A$ , and the transcendental lattices are related by the rescaling (21.14):

$$\rho(X) = \rho(A) + 16, \quad T(X) \cong T(A)(2), \quad 16 \leq \rho(X) \leq 20 \quad (\text{over } \mathbb{C}) \quad (21.18)$$

[Huy §3.2.5, II. 2084–2095; Asp II. 1192–1194]. A Kummer surface need not be algebraic, because a complex torus need not be [Asp II. 1193–1194]; for a generic torus  $\rho(A) = 0$  (standard). At the other extreme, the square torus of Example 21.1 is  $E_i \times E_i$  with  $E_i$  the elliptic curve with complex multiplication by  $i$ ; it has  $\rho(A) = 4$  (the classes of the two factors, the diagonal and the graph of multiplication by  $i$ ; a standard fact, not checked against this book’s library) and its Kummer surface has the maximal Picard number 20.

**Theorem 21.9** (Nikulin; [L]). *For a complex K3 surface  $X$  the following are equivalent: (i)  $X$  is a Kummer surface; (ii) there is a primitive embedding  $K \hookrightarrow NS(X)$ . Both imply (iii): there is a primitive embedding  $T(X) \hookrightarrow U(2)^{\oplus 3}$ ; and (iii) implies (i) if  $X$  is projective [Huy Thm. 14.3.17, II. 13545–13552]. Moreover  $X$  is Kummer if and only if it contains 16 disjoint smooth rational curves [Huy Rem. 14.3.19, II. 13595–13597].*

The proof combines the global Torelli theorem of Chapter 20 with the surjectivity of the period map for tori, applied to  $K^\perp \cong U(2)^{\oplus 3}$  [Huy II. 13578–13586]. As an application, the Fermat quartic  $x_0^4 + x_1^4 + x_2^4 + x_3^4 = 0$  of Chapter 19 has  $T(X) \cong \mathbb{Z}(8) \oplus \mathbb{Z}(8)$ , which embeds in  $U(2)^{\oplus 3}$ ; hence it is a Kummer surface, although no torus is visible in its equation [Huy Example 14.3.18, II. 13588–13593].

**Historical note**

The term “orbifold” is due to W. Thurston; the objects first appeared in the mathematical literature as “V-manifolds” [Asp II. 1036–1039]. Theorem 21.9 was first stated by Nikulin [Huy I. 13544], and the global Torelli theorem was first proved for Kummer surfaces [Huy II. 13579–13581].

## 21.5 Strings on the orbifold: twisted sectors

A point particle on  $S_0$  sees a singular space: its wave equation needs boundary conditions at the 16 cone points that the geometry does not supply. The string does better. String theory on  $A/\iota$  is a perfectly regular conformal field theory, built from the free theory on the torus by a three-step prescription [GPR §3.1, II. 4118–4148]:

1. define the action of the group on all worldsheet fields, here  $g: X^m \mapsto -X^m$  and, by worldsheet supersymmetry,  $\psi^m \mapsto -\psi^m$  for the four torus directions  $m = 1, \dots, 4$ ;
2. keep only the invariant states, with the projector  $P = \frac{1}{2}(1 + g)$ ;
3. add the strings that close only up to the group action, the *twisted sector*,

$$X^m(\sigma + 2\pi, \tau) = -X^m(\sigma, \tau), \quad (21.19)$$

and project these onto invariant states as well.

Step 3 has no analogue for particles. A string obeying (21.19) is an open curve on the torus whose end-points are exchanged by  $\iota$ , hence a *closed* curve on  $S_0$ ; leaving it out would be as arbitrary as leaving out winding strings on a circle (Chapter 7).

### 21.5.1 The untwisted sector: 16 moduli and no gauge fields

The untwisted states are the states of the torus compactification of Chapter 10 that are even under  $g$ . A state with momentum and winding  $(n_m, w^m)$  is sent to  $(-n_m, -w^m)$ , so only the combinations  $|n, w\rangle + |-n, -w\rangle$  (times an even oscillator polynomial) and  $|n, w\rangle - |-n, -w\rangle$  (times an odd one) survive. For the massless NS–NS fields the rule is: count internal indices.

- The scalars  $G_{mn}$  and  $B_{mn}$ , with vertex operators  $\partial X^m \bar{\partial} X^n$ , have two internal indices and are even. All  $10 + 6 = 16$  of them survive. They parametrize the Narain moduli space  $O(4, 4)/(O(4) \times O(4))$  of the torus, of dimension  $4 \cdot 4 = 16$ .
- The  $4 + 4$  Kaluza–Klein and winding gauge fields  $G_{\mu m}$ ,  $B_{\mu m}$  have one internal index and are odd. They are projected out. This is the string’s way of saying  $b_1(X) = 0$ : a K3 surface has no continuous isometries and no one-cycles for  $B$  to wrap.

The T-duality group  $O(4, 4; \mathbb{Z})$  of the torus (Chapter 11) commutes with  $g = -\mathbf{1}$  and acts on the untwisted sector; it will reappear as a subgroup of the duality group  $O(\Gamma_{4,20})$  of strings on K3 in Chapter 23.

### 21.5.2 The twisted sector: $16 \times 4$ moduli

The boundary condition (21.19) has three consequences.

(i) *Localization*. Writing  $X^m = x^m + \text{oscillators}$ , the constant part must satisfy  $x \equiv -x$  modulo  $\Lambda$ : the centre of mass sits at one of the 16 fixed points (21.1). There is one copy of the twisted Hilbert space for each fixed point [GPR II. 4362–4369], and a twisted string cannot



carry momentum or winding on the torus [GPR ll. 4379–4380]. Twisted strings are stuck at the singularities; they are the degrees of freedom of the sixteen spheres of zero size.

(ii) *Moding*. An antiperiodic field has half-integer modes,  $X^m \sim \sum_{r \in \mathbb{Z}+1/2} \alpha_r^m e^{-ir(\tau-\sigma)}/r + \dots$ . By worldsheet supersymmetry  $\psi^m$  picks up the same extra sign: in the twisted NS sector the four  $\psi^m$  are *integer* moded, in the twisted R sector half-integer moded.

(iii) *Zero-point energy*. Use the mnemonic of [Pol ll. 165–196]: a periodic boson contributes  $-\frac{1}{24}$ , an antiperiodic boson  $+\frac{1}{48}$ , and fermions contribute with the opposite sign. In light-cone gauge there are 8 transverse directions, 4 along the uncompactified space and 4 along the torus. For one side (left- or right-moving) of the NS sector

$$E_0^{\text{untw}} = 8\left(-\frac{1}{24} - \frac{1}{48}\right) = -\frac{1}{2}, \quad E_0^{\text{tw}} = 4\left(-\frac{1}{24} - \frac{1}{48}\right) + 4\left(\frac{1}{48} + \frac{1}{24}\right) = -\frac{1}{4} + \frac{1}{4} = 0. \quad (21.20)$$

This is the same arithmetic as for an open string with  $\nu$  mixed Neumann–Dirichlet directions in Chapter 13,  $E_0 = -\frac{1}{2} + \frac{\nu}{8}$  [Pol eq. (98), ll. 2160–2180], at  $\nu = 4$ . In the untwisted sector the ground state is tachyonic and is removed by the GSO projection, and the massless states need one oscillator  $\psi_{-1/2}$ . In the twisted sector the *ground state itself* is massless. It is degenerate, because the four integer-moded  $\psi^m$  have zero modes  $\psi_0^m$  obeying a Clifford algebra:  $2^{4/2} = 4$  states, a spinor of the internal  $SO(4)$ , of which the GSO projection keeps one chirality, 2 states. Combining left- and right-movers,

$$2 \times 2 = 4 \text{ massless NS–NS scalars at each fixed point} = \mathbf{3} \oplus \mathbf{1}. \quad (21.21)$$

(The zero-point energies are from the cited source. The counting of the GSO-projected zero-mode states and their survival under  $P$  is the standard result for this orbifold; it is not checked against a source in this book’s library.) The decomposition  $\mathbf{3} \oplus \mathbf{1}$  refers to the  $SU(2)$  rotating the three-plane  $\Sigma$ . The triplet is the three metric parameters of the physics box in Section 21.2: giving it an expectation value blows up or deforms the  $A_1$  point. The singlet is the flux of the  $B$ -field through the sphere,  $\int_{\bar{E}_i} B$ .

**Example 21.10** ( $80 = 16 + 64$ , and two refinements). The untwisted sector has 16 moduli and the twisted sectors  $16 \times 4 = 64$ , together

$$16 + 64 = 80 = 4 \cdot 20 = \dim \frac{O(4, 20)}{O(4) \times O(20)},$$

the dimension of the moduli space of conformal field theories on K3 [Asp ll. 1670 and 1948]. The count can be split according to geometric origin [Asp ll. 1650–1670]:

$$\underbrace{10}_{\text{flat metrics on } T^4} + \underbrace{16 \times 3}_{\text{triplets}} = 58 \text{ (Einstein metrics on K3)}, \quad \underbrace{6}_{B_{mn}} + \underbrace{16 \times 1}_{\text{singlets}} = 22 = b_2 \text{ (} B\text{-field)}.$$

The number 58 is  $3 \cdot 19 + 1$ , the dimension of the space of positive three-planes in  $\mathbb{R}^{3,19}$  plus the volume (Chapter 20); the number  $22 = 6 + 16$  is (21.12) once more. The massless spectrum of the singular orbifold thus knows the Betti number of the smooth K3 surface. In M-theory language the 16 twisted two-forms come with 16 of the 22 gauge fields obtained by reducing the three-form on  $H^2(X)$  [DRS ll. 511–519].

#### The solvable orbifold is not the point of enhanced gauge symmetry

When all sixteen spheres shrink, M-theory on  $T^4/\mathbb{Z}_2$  has the non-abelian gauge group  $SU(2)^{16} \times U(1)^6$ , one  $SU(2)$  per  $A_1$  point [DRS ll. 515–519], and for the type IIA string an  $A$ – $D$ – $E$  singularity likewise gives the corresponding gauge group [Asp Prop. 2, ll. 2517–2529].

In the perturbative type IIA string, however, no new massless vectors can appear [Asp II. 2441–2443], and the orbifold conformal field theory is perfectly regular. There is no contradiction: a vanishing sphere is necessary but not sufficient. The gauge symmetry is enhanced when, in addition,  $B = 0$  along the class of the sphere, whereas the conformal field theory orbifold has  $B = \frac{1}{2}$  there. The free-field orbifold and the point with gauge group  $SU(2)^{16}$  are different points of the 80-dimensional moduli space, with the same singular metric [L] [Asp II. 2530–2547]. The singlet in (21.21) is the modulus that interpolates between them.

A classical observer sees sixteen cone points of infinite curvature. The string sees, at each of them, four massless scalars whose expectation values are the size, the shape and the  $B$ -flux of a sphere that is not yet there. Turning them on moves the theory into the smooth Kummer surfaces, which are dense in the moduli space of K3 surfaces [Huy II. 6200–6202]: the solvable model is a point *of* the K3 moduli space, not a caricature of it (compare Chapter 31 for the theme that at sizes of order  $\sqrt{\alpha'}$  “singular” and “smooth” are not geometric notions).

## 21.6 What the machine has checked

Three files of the repository contain declarations about the Kummer construction:

```
KummerBlowup.lean          in StringTheoryFormalization/StringDynamics/
KummerTadpole.lean         in DualScaleM24Formalization/Moonshine/
Problem9_KummerModularity.lean in Lean5Corpus/Problems/
```

All of them belong to the older part of the corpus that records *arithmetic shadows*: integers and one literal matrix. None of them defines a torus, an involution, a blow-up, a cohomology group or a lattice embedding, so nothing in the preceding sections is Tier A as a statement about surfaces. What is kernel-checked is the arithmetic that those sections reduce to. The theorems below depend only on the standard axioms `propext`, `Classical.choice` and `Quot.sound`.

### The $16 \times 16$ Gram matrix (KummerBlowup.lean, namespace StringTheory.StringDynamics)

```
def kummerExceptionalDivisors : Fin 16 → String :=
  fun i => s!"E_{i.val}"

def kummerIntersectionForm (i j : Fin 16) : ℤ :=
  if i = j then -2 else 0

theorem kummer_lattice_contribution :
  (∑ i : Fin 16, kummerIntersectionForm i i) = -32 := by ...

theorem exceptional_self_intersection (i : Fin 16) :
  kummerIntersectionForm i i = -2 := by ...
```

**Reading.** `kummerIntersectionForm` is the matrix  $-2 \cdot \mathbf{1}_{16}$ , the Gram matrix of  $\langle -2 \rangle^{\oplus 16}$  in the basis  $\bar{E}_i$ . The first definition only produces the names "E\_0", ..., "E\_15". The theorem `exceptional_self_intersection` says that every diagonal entry of this matrix is  $-2$ ; `kummer_lattice_contribution` says that its trace is  $-32$ . They are statements about a matrix that was *defined* to have these entries. The mathematical content of Proposition 21.2, that the curve produced by the blow-up has normal bundle  $\mathcal{O}(-2)$ , is an input, not an output. Note also that the matrix describes  $\bigoplus \mathbb{Z}\bar{E}_i$ , not the Kummer lattice  $K$  of Definition 21.6, which is larger by index  $2^5$ ; and by the pitfall box of Section 21.4 the number  $-32$  is a trace in one basis, not a lattice invariant. The companion

`kummer_sublattice_rank` in `MukaiLattice.lean` states that the index type `Fin 16` has 16 elements.

**Proof idea.** `simp` unfolds the definition; on the diagonal the condition `i = i` is true and the entry is  $-2$ . For the trace, `Finset.sum_const` turns the sum of a constant over `Fin 16` into  $16 \cdot (-2)$ .

### Counterparts in the other libraries

```
-- DualScaleM24Formalization/Moonshine/KummerTadpole.lean
-- (namespace SocrateAI.Moonshine)
def num_kummer_singularities : Nat := 16
def exceptional_self_intersection : Int := -2
theorem k3_euler_eq_24 : k3_euler = 24 := by rfl
theorem kummer_exceptional_intersection (i j : Fin 16) :
  (if i = j then exceptional_self_intersection else 0) =
  (if i = j then -2 else 0) := by rfl
theorem picard_rank_kummer_maximal : k3_b2 - transcendental_rank = 20 := by rfl
theorem hodge_h11_eq_20 : k3_b2 - 2 * hodge_h20 = 20 := by rfl

-- Lean5Corpus/Problems/Problem9_KummerModularity.lean
theorem k3_euler_characteristic_identity :
  k3_betti_0 + k3_betti_2 + k3_betti_4 = k3_euler_char := by rfl
theorem kummer_cycle_balance :
  kummer_exceptional_divisors + kummer_torus_cycles = 20 := by rfl
theorem singular_k3_picard_number_equals_twenty :
  picard_number singular_kummer_shioda_tate = 20 := by rfl

-- StringTheoryFoundation/ModularForms/FermatModularBridge.lean
theorem kummer_fixed_points_dim4 :
  (2 : Nat) ^ 4 = kummer_fixed_points_count := by rfl
```

**Reading.** Here `k3_euler` is defined as  $1 - 0 + 22 - 0 + 1$  from the constants `k3_b0`, ..., `k3_b4`, so `k3_euler_eq_24` is the last equality of (21.12); `k3_euler_characteristic_identity` is the same sum  $1 + 22 + 1 = 24$  with natural numbers. `kummer_cycle_balance` is  $16 + 4 = 20$ , the count  $h^{1,1} = 4 + 16$  below (21.12); `hodge_h11_eq_20` is  $22 - 2 \cdot 1 = 20$ ; `kummer_fixed_points_dim4` is  $2^4 = 16$ , the cardinality in (21.1). `kummer_exceptional_intersection` compares two if-expressions that become identical once the constant  $-2$  is unfolded. `HodgeNumbers.lean` has one more statement of  $20 = 16 + 4$ , `hodge11_from_kummer`. The second half of `KummerTadpole.lean` (`rr_tadpole_cancellation`:  $16 \cdot 4 + 4 \cdot (-16) = 0$ ) concerns seven-brane charges and is discussed in Chapter 25.

**Two names that claim more than the statement.** `picard_rank_kummer_maximal` is  $22 - 2 = 20$ . By (21.18) the Picard number of a Kummer surface is  $16 + \rho(A)$  and takes every value from 16 to 20; the value 20 belongs to special tori such as that of Example 21.1, not to Kummer surfaces in general. In `singular_kummer_shioda_tate` the Shioda–Tate formula  $\rho = 2 + \text{rk } MW + \sum_v (m_v - 1)$  is instantiated with the triple  $(2, 0, 18)$ . For the natural fibration of  $\text{Km}(E_1 \times E_2)$  of Example 21.4(c) the four  $I_0^*$  fibres contribute  $4 \cdot 4 = 16$ , so that  $\rho = 18 + \text{rk } MW$  [Huy Example 11.3.5, ll. 9881–9885], and  $\rho = 20$  is reached with the triple  $(2, 2, 16)$ . Lean checks that  $2 + 0 + 18 = 20$ ; it does not know which fibration, if any, has these data.

**Proof idea.** All of these are closed by `rfl` (or `decide`): both sides reduce to the same numeral by evaluation.

| Claim                                                                                                                       | Tier | Where                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------|
| $2^4 = 16$ ; trace of $-2 \cdot \mathbf{1}_{16}$ is $-32$ ; $1 + 22 + 1 = 24$ ;<br>$16 + 4 = 20$ ; $22 - 2 = 20$            | [A]  | declarations quoted above                                               |
| These integers are the fixed-point number,<br>$\sum \bar{E}_i^2$ , $\chi$ , $h^{1,1}$ and $\rho_{\max}$ of a Kummer surface | [L]  | identification; never Tier A                                            |
| $\text{Km}(A)$ is a K3 surface; $\bar{E}_i^2 = -2$ ; $b_2 = 6 + 16$ ;<br>Nikulin's criterion                                | [L]  | Asp §2.6; Huy ch. 1, §3.2.5, Thm. 14.3.17                               |
| $K$ has index $2^5$ over $\bigoplus \mathbb{Z}\bar{E}_i$ , $\text{disc } K = 2^6$ , $K^\perp \cong U(2)^{\oplus 3}$         | [L]  | Huy Prop. 14.3.14; numbers re-derived in Example 21.8 (symbolic check)  |
| Twisted sectors: $E_0 = 0$ , 4 scalars per fixed point, $16 + 64 = 80$                                                      | [L]  | Pol eq. (98); GPR §3.1; Asp l. 1670; GSO count standard, unsourced here |
| $B = \frac{1}{2}$ at the conformal field theory orbifold                                                                    | [L]  | Asp ll. 2530–2547                                                       |

**What a faithful formalization would need.** The lattice part of this chapter is within reach of the Gram-matrix approach of Chapter 18: take the rational basis matrix  $P$  of Example 21.8 (eleven  $\bar{E}_i$  and five half-sums) and let the kernel check that  $P^\top(-2 \cdot \mathbf{1})P$  has integer entries, even diagonal and determinant  $2^6$ ; adding  $U(2)^{\oplus 3}$  and six glue vectors of the type (21.16), one could check unimodularity of the full rank-22 Gram matrix. The geometric part (that the blow-up of  $A/\iota$  is a K3 surface) needs complex surfaces, blow-ups and sheaf cohomology, which the libraries used in this book do not provide; see Chapter 38.

#### Chapter summary

- $x \mapsto -x$  on  $T^4 = \mathbb{R}^4/\Lambda$  has the  $2^4 = 16$  fixed points  $\frac{1}{2}\Lambda/\Lambda \cong \mathbb{F}_2^4$ . The quotient is flat with holonomy  $\mathbb{Z}_2 \subset SU(2)$ , simply connected, with an invariant  $dz_1 \wedge dz_2$ : a K3 orbifold.
- Locally  $\mathbb{C}^2/\mathbb{Z}_2 = \{x_0x_1 = x_2^2\}$ . Its blow-up is the total space of  $\mathcal{O}_{\mathbb{P}^1}(-2)$ , with coordinates  $(a, t)$ ,  $a' = t^2a$ ; the exceptional sphere has  $E^2 = -2$  and  $d\xi \wedge d\eta = \frac{1}{2}da \wedge dt$  has no zeros. The area  $J \cdot E$  is a modulus, and  $J \cdot E \rightarrow 0$  returns the orbifold.
- $b_2 = 6 + 16 = 22$ ,  $h^{1,1} = 4 + 16 = 20$ , signature  $(3, 3) + (0, 16) = (3, 19)$ , and  $\chi = 24$  in three ways:  $1 + 22 + 1$ ;  $\frac{0-16}{2} + 32$ ;  $4 \times \chi(I_0^*) = 4 \times 6$ .
- Integrally,  $\pi_*H^2(A, \mathbb{Z}) \cong U(2)^{\oplus 3}$  and the  $\bar{E}_i$  span  $\langle -2 \rangle^{\oplus 16}$ , but  $H^2(X, \mathbb{Z})$  is larger by index  $2^{11}$ . The Kummer lattice  $K \supset \bigoplus \mathbb{Z}\bar{E}_i$  (index  $2^5$ , discriminant  $2^6$ ) is generated by half-sums over affine hyperplanes of  $\mathbb{F}_2^4$ , which come from pairs of parallel invariant tori. A K3 surface is Kummer iff  $K$  embeds primitively in  $NS(X)$  (Nikulin).
- The string on  $T^4/\mathbb{Z}_2$  has 16 untwisted moduli and, in each of 16 twisted sectors with zero-point energy 0, four massless scalars  $\mathbf{3} \oplus \mathbf{1}$  (blow-up and  $B$ -flux):  $16 + 64 = 80 = \dim O(4, 20)/(O(4) \times O(20))$ . The solvable orbifold has  $B = \frac{1}{2}$  through each vanishing sphere and no enhanced gauge symmetry.
- Lean checks the arithmetic ( $2^4 = 16$ , trace  $-32$ ,  $1 + 22 + 1 = 24$ ,  $16 + 4 = 20$ ); the geometry and the Kummer lattice are Tier L.

## Exercises

**Exercise 21.1** ( $\star$  Fixed points in other dimensions). Show that  $x \mapsto -x$  on  $T^d$  has  $2^d$  fixed points. For  $d = 2$  verify the Gauss–Bonnet theorem for the pillow. For  $d = 4$  explain why  $\chi(T^4/\mathbb{Z}_2) = 8$  is not  $\frac{1}{2}\chi(T^4)$ .

**Exercise 21.2** (★ The quadric cone). (a) Show that every polynomial in  $(\xi, \eta)$  that is even under  $(\xi, \eta) \mapsto (-\xi, -\eta)$  is a polynomial in  $x_0, x_1, x_2$  of (21.3). (b) Show that  $f = x_0x_1 - x_2^2$  is singular only at the origin. (c) Find the linear change of variables that brings  $f$  to  $u^2 + v^2 + w^2$ , and show that the deformed surface  $u^2 + v^2 + w^2 = \varepsilon$ ,  $\varepsilon > 0$ , contains the real two-sphere of radius  $\sqrt{\varepsilon}$ , which shrinks to the singular point as  $\varepsilon \rightarrow 0$ .

**Exercise 21.3** (★★ The  $I_0^*$  fibre). With  $F = 2C + \sum_{i \in P} \bar{E}_i$  as in Lemma 21.5, verify  $F^2 = 0$ ,  $F \cdot C = 0$  and  $F \cdot \bar{E}_i = 0$  from the intersection numbers (21.16). Write the  $5 \times 5$  Gram matrix of  $(C, \bar{E}_i)$  and show that it is negative semi-definite with kernel spanned by  $F$ . Deduce the contribution 4 of each fibre to the Shioda–Tate formula.

**Exercise 21.4** (★★ Discriminants and indices). Compute the discriminants of  $U^{\oplus 3} \oplus \langle -2 \rangle^{\oplus 16}$ ,  $U(2)^{\oplus 3} \oplus \langle -2 \rangle^{\oplus 16}$  and  $U(2)^{\oplus 3} \oplus K$ . Using the index formula of Chapter 18, find the index of the latter two in the unimodular lattice  $H^2(X, \mathbb{Z})$  and check  $2^{11} = 2^5 \cdot 2^6$ . Why can the first lattice not be a finite-index sublattice of  $H^2(X, \mathbb{Z})$  in which the  $\bar{E}_i$  are the exceptional curves?

**Exercise 21.5** (★★★ Planes are not enough). Let  $P \subset \mathbb{F}_2^4$  be an affine plane (four points). (a) Show that  $v = \frac{1}{2} \sum_{i \in P} \bar{E}_i$  has even square and integral pairing with every element of  $K$ , so that integrality alone does not exclude  $v$  from  $H^2(X, \mathbb{Z})$ . (b) Show that nevertheless  $v \notin H^2(X, \mathbb{Z})$  when  $P$  is the plane of an invariant torus  $T$ : otherwise Lemma 21.5 would give  $\frac{1}{2}\pi_*[T] \in H^2(X, \mathbb{Z})$ , contradicting the primitivity of  $\pi_*H^2(A, \mathbb{Z})$  stated in Theorem 21.7(i). (c) Conclude that  $v \in K^* \setminus K$  and find its order in  $A_K$ .

**Exercise 21.6** (★★ Zero-point energies). Generalize (21.20) to a  $\mathbb{Z}_2$  reflection of  $\nu$  of the eight transverse directions and recover  $E_0 = -\frac{1}{2} + \frac{\nu}{8}$ . For which  $\nu$  is the twisted NS ground state massless? Show that the twisted R sector has  $E_0 = 0$  for every  $\nu$ .

**Exercise 21.7** (★ Lean: the off-diagonal entries). State and prove in Lean that distinct exceptional curves are disjoint in the toy Gram matrix:

```
theorem exceptional_disjoint (i j : Fin 16) (h : i ≠ j) :
  kummerIntersectionForm i j = 0 := by
  sorry -- hint: simp [kummerIntersectionForm, h]
```

Then state and prove that the matrix is symmetric. (Hint: `eq_comm` inside the `if`.)

**Exercise 21.8** (★★★ Lean: the code behind the Kummer lattice). Model  $\mathbb{F}_2^4$  as `Fin 4`  $\rightarrow$  `Bool` and an affine function by its five coefficients. Prove by `decide` that every non-constant affine function takes the value `true` exactly 8 times, and that two of them which are neither equal nor complementary are both `true` on exactly 4 points. Relate this to Example 21.8.

## Further reading

- [Asp §2.6, ll. 1031–1230 and 1503–1530]: orbifolds, the blow-up of  $\mathbb{C}^2/\mathbb{Z}_2$  as used here,  $(-2)$ -curves, Theorem 4, resolution versus deformation; [Asp §4.3, ll. 2440–2480 and 2500–2547]: enhanced gauge symmetry and the  $B$ -field at orbifold points (Chapter 23).
- [Huy ch. 1, ll. 218–257]: the algebraic construction and  $\omega_X \cong \mathcal{O}_X$ ; [Huy §3.2.5, ll. 2015–2096]: cohomology, the factor 2,  $\rho(X) = \rho(A) + 16$ ; [Huy §14.3.3, ll. 13460–13600]: the Kummer lattice and Nikulin’s theorem; [Huy ch. 11, ll. 9500–9507, 9725–9735, 9881–9885]: the  $I_0^*$  fibration.
- [GPR §3.1, ll. 4118–4148, 4320–4332, 4362–4380]: the orbifold prescription and twisted sectors (for a  $\mathbb{Z}_3$  heterotic orbifold); [Pol ll. 165–196 and eq. (98), ll. 2160–2180]: zero-point energies.

- [DRS §2, ll. 511–541]:  $T^4/\mathbb{Z}_2$  in M-theory, the grouping  $16 + 4 + 2$  of the two-forms, and the type IIB orientifold used in Chapter 25.

## Chapter 22

# The Mukai Lattice and Fourier–Mukai Transforms

In Chapter 20 the second cohomology  $H^2(X, \mathbb{Z})$  of a K3 surface  $X$ , with its intersection form, turned out to be the even unimodular lattice  $3U \oplus 2E_8(-1)$  of signature  $(3, 19)$ , and the position of the holomorphic two-form inside it determined the surface. String theory on a K3 surface, however, is governed by a lattice of signature  $(4, 20)$ , not  $(3, 19)$  (Chapter 23). Where do the extra two directions come from? The answer of this chapter is: from  $H^0(X, \mathbb{Z})$  and  $H^4(X, \mathbb{Z})$ , the two one-dimensional pieces of cohomology that an algebraic geometer usually ignores because they carry “no information”. Put together with  $H^2$  and given a pairing with one carefully chosen sign, they form the *Mukai lattice*

$$\tilde{H}(X, \mathbb{Z}) = H^0(X, \mathbb{Z}) \oplus H^2(X, \mathbb{Z}) \oplus H^4(X, \mathbb{Z}) \cong 4U \oplus 2E_8(-1),$$

even, unimodular, of signature  $(4, 20)$ . The question we answer is: *why is this the natural lattice, and what acts on it?* Two facts make it natural. First, every coherent sheaf  $E$  on  $X$  (a vector bundle, a point, a curve with a line bundle on it) has a *Mukai vector*  $v(E) \in \tilde{H}(X, \mathbb{Z})$ , and the Riemann–Roch theorem becomes the one-line statement  $\chi(E, F) = -\langle v(E), v(F) \rangle$ . Second, symmetries of the *derived category* of sheaves on  $X$ , the Fourier–Mukai transforms, act on  $\tilde{H}(X, \mathbb{Z})$  by lattice isometries that mix  $H^0$ ,  $H^2$  and  $H^4$ . For a physicist the first fact says that  $\tilde{H}(X, \mathbb{Z})$  is the charge lattice of branes wrapped on  $X$ , and the second says that there are dualities that exchange a point-like brane with a brane filling the whole surface — the K3 analogue of exchanging momentum and winding on a circle (Chapter 8).

The chapter is organized as follows. Section 22.1 derives the Mukai vector and the Mukai pairing from the Hirzebruch–Riemann–Roch formula, step by step. Section 22.2 studies the resulting lattice. Section 22.3 introduces derived categories and Fourier–Mukai transforms at the level needed to understand how they act on cohomology, and works out the action in a rank-three example. Section 22.4 gives the physical reading. Section 22.5 states precisely what has been machine-checked: the linear algebra of the pairing is checked for every integral Gram matrix; the sheaf theory and the derived categories are not formalized at all.

## 22.1 From Riemann–Roch to the Mukai pairing

### 22.1.1 The Euler pairing

Let  $X$  be a K3 surface (Chapter 19): a compact complex surface with trivial canonical bundle  $\omega_X \cong \mathcal{O}_X$  and  $H^1(X, \mathcal{O}_X) = 0$ . For two coherent sheaves  $E, F$  on  $X$  the groups  $\text{Ext}^i(E, F)$  are finite-dimensional complex vector spaces:  $\text{Ext}^0 = \text{Hom}$  is the space of sheaf

homomorphisms,  $\text{Ext}^1(E, F)$  classifies extensions  $0 \rightarrow F \rightarrow G \rightarrow E \rightarrow 0$ , and  $\text{Ext}^1(E, E)$  is the space of first-order deformations of  $E$ . For sheaves on a surface,  $\text{Ext}^i$  vanishes unless  $0 \leq i \leq 2$ . The *Euler pairing* is the alternating sum

$$\chi(E, F) := \sum_{i=0}^2 (-1)^i \dim \text{Ext}^i(E, F). \quad (22.1)$$

Two properties make it the right object to study. It is additive in short exact sequences in each argument, so it only depends on coarse, topological data of  $E$  and  $F$ . And on a K3 surface it is *symmetric*: Serre duality on a surface with trivial canonical bundle reads

$$\text{Ext}^i(E, F) \cong \text{Ext}^{2-i}(F, E)^*, \quad (22.2)$$

so the three terms of  $\chi(E, F)$  are those of  $\chi(F, E)$  in reverse order, with the same signs because  $(-1)^i = (-1)^{2-i}$  [Huy ch. 9 §1.2, ll. 7393–7411 and 7444–7447]. A symmetric integer-valued bilinear form is exactly the structure of a lattice (Chapter 18). The task is to find the lattice.

For  $E = \mathcal{O}_X$  one has  $\text{Ext}^i(\mathcal{O}_X, F) = H^i(X, F)$ , hence  $\chi(\mathcal{O}_X, F) = \chi(F)$ , the usual holomorphic Euler characteristic. More generally, for a vector bundle  $E$  with dual  $E^*$ ,  $\chi(E, F) = \chi(E^* \otimes F)$  [Huy ch. 9 §1.2, ll. 7447–7449].

### 22.1.2 Hirzebruch–Riemann–Roch on a K3 surface

The Hirzebruch–Riemann–Roch theorem computes  $\chi(F)$  from characteristic classes:  $\chi(F) = \int_X \text{ch}(F) \text{td}(X)$ . We need three ingredients.

- The Chern character of a sheaf of rank  $r$  with Chern classes  $c_1, c_2$  is  $\text{ch}(F) = (r, c_1, \text{ch}_2)$  with  $\text{ch}_2 = \frac{1}{2}c_1^2 - c_2$ , the three entries lying in  $H^0, H^2, H^4$ . We identify  $H^0(X, \mathbb{Z}) = \mathbb{Z}$  (generated by 1) and  $H^4(X, \mathbb{Z}) = \mathbb{Z}$  (generated by the class of a point, so that  $\int_X$  reads off the  $H^4$  entry).
- The Todd class of a surface is  $\text{td} = 1 + \frac{1}{2}c_1(X) + \frac{1}{12}(c_1(X)^2 + c_2(X))$ . For a K3 surface  $c_1(X) = 0$  and  $\int_X c_2(X) = \chi_{\text{top}}(X) = 24$  (Chapter 19), so

$$\text{td}(X) = (1, 0, 2), \quad \sqrt{\text{td}(X)} = (1, 0, 1), \quad (22.3)$$

the square root being taken in the cohomology ring:  $(1, 0, 1)^2 = (1, 0, 2)$  because the  $H^4$  entry squares to zero.

- The dual Chern character  $\text{ch}^*$ , defined by  $\text{ch}_i^* = (-1)^i \text{ch}_i$ , i.e.  $\text{ch}^*(E) = (r, -c_1, \text{ch}_2)$ ; for a vector bundle it equals  $\text{ch}(E^*)$ .

With these,  $\chi(F) = \text{ch}_2(F) + 2 \text{rk}(F)$  and, replacing  $F$  by “ $E^* \otimes F$ ”,

$$\chi(E, F) = \int_X \text{ch}^*(E) \text{ch}(F) \text{td}(X) = \int_X \left( \text{ch}^*(E) \sqrt{\text{td}(X)} \right) \left( \text{ch}(F) \sqrt{\text{td}(X)} \right) \quad (22.4)$$

[Huy ch. 9 §1.2, eqs. (1.2)–(1.3), ll. 7440–7459]. The second form is the key idea: *split the Todd class symmetrically between the two sheaves*. Each factor then depends on one sheaf only.

**Definition 22.1** (Mukai vector). The Mukai vector of a coherent sheaf (or a bounded complex of sheaves)  $E$  on a K3 surface is

$$v(E) := \text{ch}(E) \sqrt{\text{td}(X)} = (\text{rk } E, c_1(E), \text{ch}_2(E) + \text{rk } E) = (\text{rk } E, c_1(E), \chi(E) - \text{rk } E). \quad (22.5)$$

[Huy ch. 9, Def. 1.2, ll. 7461–7464]



The second equality is the product  $(r, c_1, \text{ch}_2) \cdot (1, 0, 1) = (r, c_1, \text{ch}_2 + r)$ ; the third uses  $\chi(E) = \text{ch}_2 + 2r$ . We write a general Mukai vector as  $v = (r, c, s)$  with  $r, s \in \mathbb{Z}$  and  $c \in H^2(X, \mathbb{Z})$ . That  $s$  is an integer even though  $\text{ch}_2 = \frac{1}{2}c_1^2 - c_2$  contains a  $\frac{1}{2}$  follows from the evenness of the K3 lattice, or directly from  $s = \chi(E) - \text{rk } E$  [Huy ch. 16 §3.1, ll. 15616–15624].

### 22.1.3 Deriving the pairing

Now compute (22.4) in components. Write  $v(E) = (r, c, s)$  and  $v(F) = (r', c', s')$ . The factor belonging to  $E$  is  $\text{ch}^*(E)\sqrt{\text{td}} = (r, -c, s)$ : the dual only flips the sign of the  $H^2$  entry. The product of  $(r, -c, s)$  and  $(r', c', s')$  in the cohomology ring has  $H^4$  entry  $rs' + sr' - (c.c')$ , where  $(c.c')$  is the intersection form on  $H^2$ . Hence

$$\chi(E, F) = rs' + sr' - (c.c'). \quad (22.6)$$

The right-hand side is symmetric and bilinear in the two vectors, as it must be. One would like to call it “the” pairing on  $H^0 \oplus H^2 \oplus H^4$ , but it restricts to *minus* the intersection form on  $H^2$ . To keep the intersection form on  $H^2$  — the form used throughout Chapter 20 — one defines the pairing with the opposite overall sign.

**Definition 22.2** (Mukai pairing). For  $\alpha = (\alpha_0, \alpha_2, \alpha_4)$  and  $\beta = (\beta_0, \beta_2, \beta_4)$  in  $H^*(X, \mathbb{Z})$ ,

$$\langle \alpha, \beta \rangle := (\alpha_2.\beta_2) - \alpha_0\beta_4 - \alpha_4\beta_0. \quad (22.7)$$

[Huy ch. 9, Def. 1.4, ll. 7475–7480]

In words: the Mukai pairing differs from the ordinary intersection pairing on  $H^*(X, \mathbb{Z})$  (which pairs  $H^0$  with  $H^4$  by  $+\alpha_0\beta_4$ ) only by a sign on  $H^0 \oplus H^4$ . Equation (22.6) becomes

**Theorem 22.3** (Riemann–Roch in Mukai form [L]). *For coherent sheaves (or bounded complexes)  $E, F$  on a K3 surface,*

$$\chi(E, F) = -\langle v(E), v(F) \rangle. \quad (22.8)$$

[Huy ch. 9, eq. (1.4), ll. 7481–7482]

All Todd-class corrections have disappeared into the definition of  $v$ . In particular the self-pairing of a Mukai vector is

$$v^2 := \langle v, v \rangle = (c.c) - 2rs. \quad (22.9)$$

#### Common pitfall: three signs

- (1) The minus sign in (22.8) is a convention forced by wanting  $\langle \cdot, \cdot \rangle$  to restrict to  $+(\cdot, \cdot)$  on  $H^2$ ; a sheaf with many deformations has *large positive*  $v^2$ .
- (2) The Mukai vector is  $\text{ch}\sqrt{\text{td}}$ , not  $\text{ch}$ : forgetting the factor  $(1, 0, 1)$  gives  $(1, 0, 0)$  for the structure sheaf, whose square is 0 instead of  $-2$ .
- (3) The third entry of  $v$  is  $\text{ch}_2 + r$ , not the second Chern class  $c_2$ ; for a rank-two bundle with  $c_1 = 0$  and  $c_2 = 24$  it equals  $-22$ .

### 22.1.4 Worked examples

**Example 22.4** (Structure sheaf).  $\mathcal{O}_X$  has rank 1,  $c_1 = 0$ ,  $\text{ch}_2 = 0$ , so  $v(\mathcal{O}_X) = (1, 0, 1)$  and  $v^2 = 0 - 2 \cdot 1 \cdot 1 = -2$ . Theorem 22.3 gives  $\chi(\mathcal{O}_X, \mathcal{O}_X) = 2$ . Check it directly:  $\text{Ext}^i(\mathcal{O}_X, \mathcal{O}_X) = H^i(X, \mathcal{O}_X)$  has dimensions 1, 0, 1 for a K3 surface, so  $\chi = 1 - 0 + 1 = 2$ . The arithmetic step  $\langle (1, 0, 1), (1, 0, 1) \rangle = -2$  is the Lean theorem `structureSheaf_mukai_sq` of Section 22.5.

**Example 22.5** (A point). The skyscraper sheaf  $k(x)$  of a point  $x \in X$  has rank 0,  $c_1 = 0$  and  $\chi(k(x)) = 1$ , so  $v(k(x)) = (0, 0, 1)$  and  $v^2 = 0$ . Directly:  $\text{Ext}^i(k(x), k(x)) \cong \Lambda^i T_x X$  has dimensions 1, 2, 1, so  $\chi = 1 - 2 + 1 = 0$ . The two-dimensional  $\text{Ext}^1$  is the tangent space to  $X$  at  $x$ : a point deforms by moving on the surface. (The identification of  $\text{Ext}^i(k(x), k(x))$  with  $\Lambda^i T_x X$  is standard; not checked against a source in this book’s library.)

**Example 22.6** (Line bundles). For  $L \in \text{Pic}(X)$  with  $\ell = c_1(L)$ ,  $\text{ch}(L) = e^\ell = (1, \ell, \ell^2/2)$  and  $v(L) = (1, \ell, \ell^2/2 + 1)$  [Huy ch. 9, Ex. 1.3, ll. 7469–7471]. Then  $v(L)^2 = \ell^2 - 2(\ell^2/2 + 1) = -2$  for every  $L$ . For two line bundles with classes  $\ell, m$ ,

$$-\langle v(L), v(M) \rangle = -\left[(\ell \cdot m) - \frac{m^2}{2} - 1 - \frac{\ell^2}{2} - 1\right] = \frac{1}{2}(m - \ell)^2 + 2,$$

which is the classical Riemann–Roch formula  $\chi(N) = \frac{1}{2}c_1(N)^2 + 2$  for the line bundle  $N = L^* \otimes M$ , as it should be since  $\chi(L, M) = \chi(L^* \otimes M)$ . Numerically: on a smooth quartic  $X \subset \mathbb{P}^3$  with hyperplane class  $h$ ,  $h^2 = 4$ , the bundle  $\mathcal{O}_X(1)$  has  $v = (1, h, 3)$  and  $\chi(\mathcal{O}_X, \mathcal{O}_X(1)) = -\langle (1, 0, 1), (1, h, 3) \rangle = 3 + 1 = 4$ , the four linear forms on  $\mathbb{P}^3$ .

**Example 22.7** (A rational curve). If  $C \cong \mathbb{P}^1$  is a smooth rational curve in  $X$ , then  $\mathcal{O}_C(k)$ , viewed as a sheaf on  $X$ , has  $v = (0, [C], k + 1)$  and  $v^2 = [C]^2 = -2$ , the self-intersection of a  $(-2)$ -curve (Chapter 21) [Huy ch. 16, Ex. 2.7(ii), ll. 15509–15513].

**Example 22.8** (The tangent bundle).  $T_X$  has rank 2,  $c_1 = 0$ ,  $c_2 = 24$ , so  $\text{ch}_2 = -24$  and  $v(T_X) = (2, 0, -22)$ . Then  $v^2 = 0 - 2 \cdot 2 \cdot (-22) = 88$  [Huy ch. 9 §1.3, ll. 7523–7526]. We use this number below.

### 22.1.5 What $v^2$ measures

A sheaf  $E$  is *simple* if its only endomorphisms are scalars,  $\text{Hom}(E, E) = \mathbb{C}$ . By Serre duality (22.2) then also  $\text{Ext}^2(E, E) \cong \mathbb{C}$ , and Theorem 22.3 gives

$$\dim \text{Ext}^1(E, E) = 2 - \chi(E, E) = v(E)^2 + 2. \quad (22.10)$$

Since a dimension is non-negative, every simple sheaf has  $v^2 \geq -2$  [Huy ch. 9 §1.3, ll. 7485–7491]. The extreme case  $v^2 = -2$  means  $\text{Ext}^1(E, E) = 0$ : the sheaf is *rigid*, it has no deformations. Line bundles and  $\mathcal{O}_C(k)$  are rigid; a point has  $v^2 = 0$  and a two-dimensional deformation space; the tangent bundle, which is simple for a complex K3 surface, has  $\dim \text{Ext}^1(T_X, T_X) = 88 + 2 = 90$  [Huy ch. 9 §1.3, ll. 7523–7526]. More generally, for a primitive Mukai vector  $v$  and a generic polarization  $H$ , the moduli space  $M_H(v)$  of stable sheaves with Mukai vector  $v$  is, if non-empty, smooth and projective of dimension  $v^2 + 2$  [Huy ch. 10, Prop. 2.5, ll. 8816–8818]. Moreover the Serre duality pairing on  $\text{Ext}^1(E, E)$  is non-degenerate and alternating [Huy ch. 9, Prop. 1.1, ll. 7408–7417], which is why (22.10) is always even and why these moduli spaces carry a holomorphic symplectic form.

#### Physical picture: $v^2 + 2$ counts moduli of a brane

Think of  $E$  as a brane wrapped on  $X$  and of  $v(E)$  as its charge vector. Then  $\text{Ext}^1(E, E)$  is the space of massless deformations of the brane, and (22.10) says that the number of such moduli is fixed by the charge alone:  $v^2 + 2$ . Charges with  $v^2 = -2$  belong to rigid objects, which have no moduli; these are the charges that will generate reflections of the lattice in Section 22.3. Charges with  $v^2 = 0$ , such as the point  $(0, 0, 1)$ , have a two-dimensional moduli space, and this moduli space is (for primitive  $v$  and when non-empty) again a K3 surface — possibly a different one. This is the geometric origin of the dualities of this

chapter.

## 22.2 The Mukai lattice

### 22.2.1 Structure: $H^2 \oplus U(-1)$

In the basis  $\{1 \in H^0, [\text{pt}] \in H^4\}$  the restriction of (22.7) to  $H^0 \oplus H^4$  has Gram matrix

$$U(-1) = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}, \quad (22.11)$$

the hyperbolic plane  $U$  of Chapter 18 with its form negated. It is even (zero diagonal), unimodular ( $\det = -1$ ; in fact  $U(-1)^2 = 1$ , so  $U(-1)$  is its own integral inverse) and of signature  $(1, 1)$ : the vectors  $(1, -1)$  and  $(1, 1)$  have squares  $+2$  and  $-2$ . Moreover  $U(-1) \cong U$  as lattices: the change of basis  $[\text{pt}] \mapsto -[\text{pt}]$  turns one Gram matrix into the other. Therefore

$$\tilde{H}(X, \mathbb{Z}) := (H^*(X, \mathbb{Z}), \langle \cdot, \cdot \rangle) = H^2(X, \mathbb{Z}) \oplus U(-1) \cong (3U \oplus 2E_8(-1)) \oplus U = 4U \oplus 2E_8(-1). \quad (22.12)$$

This is the *extended K3 lattice* or *Mukai lattice*, written  $\tilde{\Lambda}$  or  $II_{4,20}$ ; it is even, unimodular, of signature  $(4, 20)$  and rank 24 [Huy ch. 14 §0.3(vi), ll. 12873–12880; ch. 16 §3.1, ll. 15601–15606]. Each property is inherited summand by summand: a direct sum of even lattices is even, determinants multiply, and signatures add,  $(3, 19) + (1, 1) = (4, 20)$ . Figure 22.1 summarizes the structure.

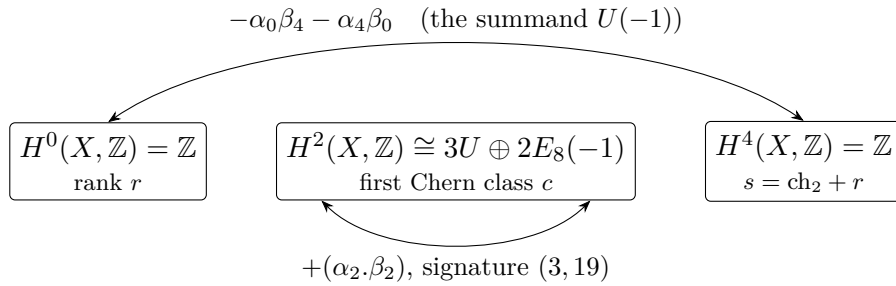


Figure 22.1: The Mukai lattice.  $H^2$  is paired with itself by the intersection form;  $H^0$  is paired with  $H^4$  with a minus sign. Total signature  $(3, 19) + (1, 1) = (4, 20)$ .

#### Common pitfall: two pairings on the same group

Aspinwall describes  $\Gamma^{4,20}$  as the lattice of total cohomology with the *unflipped* rule “the product of two cycles is zero unless their dimensions add up to 4” [Asp §3.4, ll. 1952–1957], i.e.  $H^2 \oplus U$ . Huybrechts uses  $H^2 \oplus U(-1)$ . Both are the same abstract lattice  $II_{4,20}$  because  $U \cong U(-1)$ , but formulas written in coordinates differ by  $s \mapsto -s$ . Riemann–Roch in the form (22.8), and all Lean statements of this chapter, use the Mukai sign (22.7).

### 22.2.2 The ring structure and $B$ -field shifts

$H^*(X, \mathbb{Z})$  is not only a lattice but a ring:  $(r, c, s) \cdot (r', c', s') = (rr', rc' + r'c, rs' + r's + (c \cdot c'))$ . For  $B \in H^2(X, \mathbb{Z})$  put

$$\exp(B) := (1, B, \tfrac{1}{2}(B \cdot B)), \quad \exp(B) \cdot (r, c, s) = (r, c + rB, s + (B \cdot c) + \tfrac{r}{2}(B \cdot B)). \quad (22.13)$$

Evenness of  $H^2$  makes  $\frac{1}{2}(B \cdot B)$  an integer. Multiplication by  $\exp(B)$  is called a  *$B$ -field shift*.

**Lemma 22.9.** *Multiplication by  $\exp(B)$  is an isometry of the Mukai lattice, with inverse  $\exp(-B)$ . [Huy ch. 14, Lemma 2.8, ll. 13265–13281 (stated there for  $\Lambda \oplus U$ )]*

*Proof.* Let  $v = (r, c, s)$ ,  $w = (r', c', s')$  and denote the shifted vectors by  $\hat{v}, \hat{w}$ . Then

$$\begin{aligned} (\hat{c}.\hat{c}') &= (c.c') + r'(B.c) + r(B.c') + rr'(B.B), \\ -\hat{r}\hat{s}' &= -rs' - r(B.c') - \frac{1}{2}rr'(B.B), \quad -\hat{s}\hat{r}' = -sr' - r'(B.c) - \frac{1}{2}rr'(B.B). \end{aligned}$$

Adding the three lines, all terms containing  $B$  cancel and  $\langle \hat{v}, \hat{w} \rangle = \langle v, w \rangle$ . The inverse follows from  $\exp(B)\exp(-B) = (1, 0, \frac{1}{2}B^2 + \frac{1}{2}B^2 - B^2) = (1, 0, 0)$ . (We also confirmed the cancellation symbolically with sympy for a general symmetric Gram matrix; this is a symbolic check, not a Lean theorem.)  $\square$

If  $L$  is a line bundle with  $c_1(L) = \ell$ , then  $\text{ch}(E \otimes L) = \text{ch}(E) e^\ell$ , so  $v(E \otimes L) = \exp(\ell) \cdot v(E)$ : tensoring with a line bundle acts on Mukai vectors by a  $B$ -field shift [Huy ch. 16 §2.4, ll. 15554–15556]. For instance  $\exp(\ell) \cdot (1, 0, 1) = (1, \ell, \ell^2/2 + 1) = v(L)$ , reproducing Example 22.6, and Lemma 22.9 explains at once why  $v(L)^2 = v(\mathcal{O}_X)^2 = -2$ .

### 22.2.3 The Hodge structure

For a complex K3 surface the Mukai lattice carries a weight-two Hodge structure extending the one on  $H^2$  (Chapter 20):

$$\tilde{H}^{2,0}(X) := H^{2,0}(X), \quad \tilde{H}^{1,1}(X) := H^{1,1}(X) \oplus (H^0 \oplus H^4)(X, \mathbb{C}), \quad (22.14)$$

so  $H^0$  and  $H^4$  are declared to be of type  $(1, 1)$  [Huy ch. 16 §3.1, ll. 15601–15612]. The integral  $(1, 1)$  classes form the *extended Néron–Severi lattice*

$$N(X) = \mathbb{Z} \oplus \text{NS}(X) \oplus \mathbb{Z} \cong \text{NS}(X) \oplus U, \quad (22.15)$$

which is exactly where Mukai vectors of sheaves live, because  $c_1(E)$  is always an algebraic class [Huy ch. 16 §2.4 and §3.1, ll. 15533–15545 and 15612–15618]. If the Picard number of  $X$  is  $\rho$ , then  $N(X)$  has rank  $\rho + 2$  and signature  $(2, \rho)$  for projective  $X$  (signature  $(1, \rho - 1)$  of  $\text{NS}(X)$  plus  $(1, 1)$ ).

## 22.3 Derived categories and Fourier–Mukai transforms

### 22.3.1 Why complexes

The Euler pairing and the Mukai vector are additive, so they extend from sheaves to formal differences of sheaves, and negative ranks appear:  $-v(E)$  is not the Mukai vector of any sheaf if  $\text{rk } E > 0$ . The natural objects carrying *all* vectors of  $N(X)$  are bounded complexes of coherent sheaves  $E^\bullet = (\cdots \rightarrow E^{i-1} \rightarrow E^i \rightarrow E^{i+1} \rightarrow \cdots)$ , with  $v(E^\bullet) = \sum_i (-1)^i v(E^i)$ . The *bounded derived category*  $D^b(X)$  has these complexes as objects; its morphisms are obtained from maps of complexes by first identifying homotopic maps and then formally inverting quasi-isomorphisms (maps inducing isomorphisms on all cohomology sheaves) [Huy ch. 16 §1.1, ll. 15130–15185]. Three features are all we need:

- the *shift*  $E \mapsto E[1]$ ,  $E[1]^i = E^{i+1}$ , which moves a complex one step to the left. Since it changes the parity of every position,  $v(E[1]) = -v(E)$  [Huy ch. 16 §1.1–1.2, ll. 15185–15188 and 15221–15224];

- sheaves sit inside  $D^b(X)$  as complexes concentrated in degree 0, and one has

$$\mathrm{Ext}^i(E, F) = \mathrm{Hom}_{D^b(X)}(E, F[i]),$$

so (22.1) makes sense for complexes [Huy ch. 16 §1.1, ll. 15204–15211];

- Serre duality holds in  $D^b(X)$ ; for a K3 surface the Serre functor is the double shift  $E \mapsto E[2]$  [Huy ch. 16 §1.3, ll. 15244–15253].

#### Physical picture: branes and anti-branes

In the brane interpretation the shift  $[1]$  turns a brane into its anti-brane: all charges change sign,  $v \mapsto -v$ , while the mass is unchanged. A complex is a bound state of branes (even positions) and anti-branes (odd positions), the maps between consecutive terms playing the role of the tachyon fields that let them partially annihilate. This dictionary is standard in the D-brane literature but is not checked against a source in this book’s library; we use it only as intuition.

### 22.3.2 Fourier–Mukai transforms

**Definition 22.10.** Let  $X, Y$  be smooth projective varieties with projections  $q : X \times Y \rightarrow X$  and  $p : X \times Y \rightarrow Y$ , and let  $\mathcal{P} \in D^b(X \times Y)$ . The Fourier–Mukai transform with kernel  $\mathcal{P}$  is the functor

$$\Phi_{\mathcal{P}} : D^b(X) \rightarrow D^b(Y), \quad \Phi_{\mathcal{P}}(E) = p_*(q^*E \otimes \mathcal{P}), \quad (22.16)$$

all functors being derived. [Huy ch. 16, Def. 1.1, ll. 15259–15271]

The analogy with the classical Fourier transform  $\hat{f}(y) = \int f(x) e^{ixy} dx$  is exact in structure: pull the “function”  $E$  back to the product, multiply by a kernel, integrate along the fibres of  $p$ . The original example is the Poincaré line bundle on  $A \times \hat{A}$  for an abelian variety  $A$  and its dual, which is the algebro-geometric form of T-duality on a torus (Chapter 13; this remark is standard, not checked against a source in this book’s library). By a theorem of Orlov every exact equivalence  $D^b(X) \simeq D^b(Y)$  is of the form (22.16) [Huy ch. 16 §1.3, ll. 15300–15303], so nothing is lost by studying kernels. Elementary examples [Huy ch. 16 §2.3, ll. 15466–15478]: the structure sheaf of the diagonal  $\mathcal{O}_{\Delta}$  gives the identity;  $\Delta_*L$  gives  $E \mapsto E \otimes L$ ;  $\mathcal{O}_{\Delta}[1]$  gives the shift.

The same formula makes sense in cohomology. The *cohomological Fourier–Mukai transform* is

$$\Phi_{\mathcal{P}}^H : H^*(X, \mathbb{Q}) \rightarrow H^*(Y, \mathbb{Q}), \quad \alpha \mapsto p_*(q^*\alpha \cdot v(\mathcal{P})), \quad v(\mathcal{P}) = \mathrm{ch}(\mathcal{P})\sqrt{\mathrm{td}(X \times Y)}, \quad (22.17)$$

and the Grothendieck–Riemann–Roch theorem shows that it is compatible with the functor:  $\Phi_{\mathcal{P}}^H(v(E)) = v(\Phi_{\mathcal{P}}(E))$  [Huy ch. 16, Rem. 3.3, ll. 15659–15666]. The square root of the Todd class in Definition 22.1 is what makes this compatibility hold without correction factors.

**Theorem 22.11** (Mukai; Orlov [L]). *Let  $\Phi_{\mathcal{P}} : D^b(X) \xrightarrow{\sim} D^b(Y)$  be a derived equivalence between K3 surfaces. Then  $\Phi_{\mathcal{P}}^H$  maps integral classes to integral classes and is a Hodge isometry  $\tilde{H}(X, \mathbb{Z}) \xrightarrow{\sim} \tilde{H}(Y, \mathbb{Z})$ : it preserves the Mukai pairing and the Hodge structure (22.14). Conversely, two complex projective K3 surfaces are derived equivalent if and only if such a Hodge isometry exists, if and only if their transcendental lattices are Hodge isometric. [Huy ch. 16, Prop. 3.2, Prop. 3.5, Cor. 3.7, ll. 15628–15632, 15690–15692, 15725–15727]*

The reason the pairing is preserved is simple on the sublattice  $N(X)$  spanned by Mukai vectors: an equivalence of categories preserves all Ext groups, hence  $\chi(E, F) = \chi(\Phi(E), \Phi(F))$ , and by Theorem 22.3 this is the statement  $\langle v(E), v(F) \rangle = \langle v(\Phi E), v(\Phi F) \rangle$ . For transcendental classes one needs the adjoint kernel and the projection formula [Huy ch. 16, proof of Prop. 3.2, ll. 15634–15656]. What  $\Phi^H$  does *not* preserve is the grading:  $H^0, H^2, H^4$  get mixed. The second half of the theorem is the derived analogue of the Torelli theorem of Chapter 20: the ordinary Torelli theorem says that  $X \cong Y$  iff the Hodge structures on  $H^2$  are isometric; the derived version replaces  $H^2$  by  $\tilde{H}$  and “isomorphic” by “derived equivalent”. A consequence is that derived-equivalent K3 surfaces have the same Picard number, although their Néron–Severi lattices need not be isometric [Huy ch. 16, Cor. 2.8 and Rem. 2.10, ll. 15558–15561 and 15579–15582].

### 22.3.3 Three families of isometries

The group of autoequivalences of  $D^b(X)$  acts on  $\tilde{H}(X, \mathbb{Z})$  through the following building blocks.

- (i) *Shift.*  $E \mapsto E[1]$  acts by  $-\text{id}$ .
- (ii) *Line bundle twists.*  $E \mapsto E \otimes L$  acts by the  $B$ -field shift  $\exp(c_1(L))$  of (22.13) [Huy ch. 16, Ex. 3.4(ii), ll. 15678–15679].
- (iii) *Spherical twists.* An object  $E \in D^b(X)$  is *spherical* if  $\text{Ext}^i(E, E)$  is  $\mathbb{C}$  for  $i = 0, 2$  and zero otherwise — the Ext-groups have the dimensions of the cohomology of a two-sphere. By (22.10) a spherical object has  $v(E)^2 = -2$ . Line bundles and the sheaves  $\mathcal{O}_C(k)$  of Example 22.7 are spherical. To each spherical  $E$  is associated an autoequivalence  $T_E$ , the spherical twist, which acts on cohomology as the reflection

$$T_E^H = s_\delta : \alpha \mapsto \alpha + \langle \alpha, \delta \rangle \delta, \quad \delta = v(E), \quad \delta^2 = -2 \quad (22.18)$$

[Huy ch. 16, Defs. 2.5–2.6, eq. (2.3), Ex. 3.4(i), ll. 15481–15497, 15550–15552, 15670–15674].

That (22.18) is an isometry is a two-line computation valid in any lattice:  $\langle s_\delta \alpha, s_\delta \beta \rangle = \langle \alpha, \beta \rangle + 2\langle \alpha, \delta \rangle \langle \beta, \delta \rangle + \langle \alpha, \delta \rangle \langle \beta, \delta \rangle \delta^2$ , and the last two terms cancel precisely when  $\delta^2 = -2$ . Also  $s_\delta(\delta) = \delta - 2\delta = -\delta$  and  $s_\delta^2 = \text{id}$ . (At the level of categories  $T_E(E) \cong E[-1]$ , consistent with  $v \mapsto -v$ , but  $T_E^2$  is *not* the identity functor; it lies in the kernel of the representation on cohomology [Huy ch. 16 §2.3 and §3.4, ll. 15499–15501 and 15849–15856].)

**Example 22.12** (The twist by  $\mathcal{O}_X$  exchanges  $H^0$  and  $H^4$ ). Take  $\delta = v(\mathcal{O}_X) = (1, 0, 1)$ . For  $\alpha = (r, c, s)$  one finds  $\langle \alpha, \delta \rangle = -r - s$ , hence

$$T_{\mathcal{O}}^H(r, c, s) = (r, c, s) - (r + s)(1, 0, 1) = (-s, c, -r). \quad (22.19)$$

It is the identity on  $H^2$  and sends  $(r, 0, s) \mapsto (-s, 0, -r)$  [Huy ch. 16, Ex. 3.4(i), ll. 15672–15674]. A point,  $(0, 0, 1)$ , is sent to  $(-1, 0, 0)$ : the class of a rank-one object with all lower charges switched off, shifted by  $[1]$ . Figure 22.2 shows this reflection in the  $(r, s)$ -plane.

### 22.3.4 Worked example: a quartic surface of Picard number one

Let  $X \subset \mathbb{P}^3$  be a quartic with  $\text{NS}(X) = \mathbb{Z}h$ ,  $h^2 = 4$ . Then  $N(X) = \mathbb{Z}^3$  with coordinates  $(r, x, s)$  standing for  $(r, xh, s)$ , and (22.9) reads  $v^2 = 4x^2 - 2rs$ . The Gram matrix and the two basic isometries are

$$G = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 4 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad \exp(h) = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & 4 & 1 \end{pmatrix}, \quad T_{\mathcal{O}}^H = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix},$$

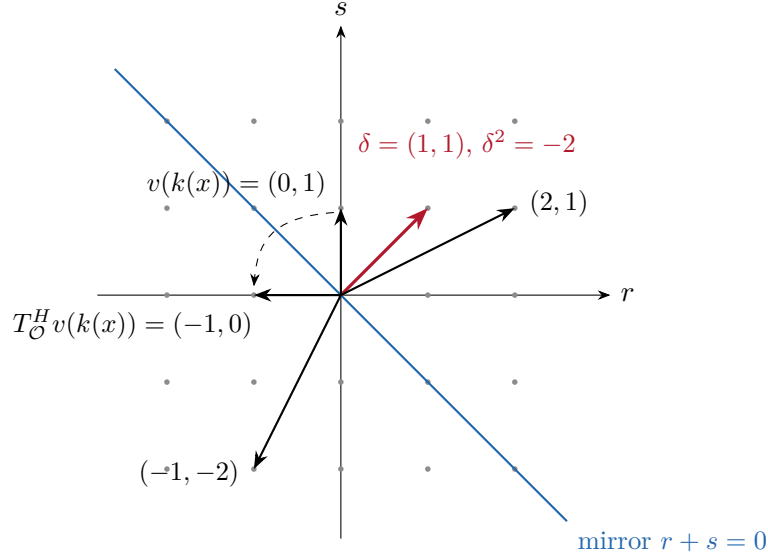


Figure 22.2: The  $H^0 \oplus H^4$  plane with coordinates  $(r, s)$  and quadratic form  $-2rs$ . The spherical twist by  $\mathcal{O}_X$  acts as the reflection  $(r, s) \mapsto (-s, -r)$  in the line  $r + s = 0$  orthogonal to  $\delta = (1, 1)$ . The null vector  $(0, 1)$  of a point goes to the null vector  $(-1, 0)$ , and  $(2, 1)$  goes to  $(-1, -2)$ ; both have the same square  $-2rs$ .

acting on column vectors. The second matrix is (22.13) with  $B = h$ :  $x \mapsto x + r$  and  $s \mapsto s + 4x + 2r$ . One checks  $M^T G M = G$  for both (we did, with sympy);  $\det G = -4$  and the signature is  $(2, 1)$ , in agreement with (22.15) for  $\rho = 1$ .

Now some numbers. (a)  $v(\mathcal{O}_X(1)) = \exp(h)(1, 0, 1)^T = (1, 1, 3)$ , as in Example 22.6, with  $v^2 = 4 - 6 = -2$ . (b) The composite  $T_{\mathcal{O}}^H \exp(h)$  sends the point class  $(0, 0, 1)$  to  $(-1, 0, 0)$  and the class  $(1, 0, 1)$  of  $\mathcal{O}_X$  to  $T_{\mathcal{O}}^H(1, 1, 3) = (-3, 1, -1)$ , again of square  $4 - 6 = -2$ . (c) The vector  $v = (2, 1, 1)$  is primitive with  $v^2 = 4 - 4 = 0$ , and  $v' = (-1, 0, 0)$  satisfies  $\langle v, v' \rangle = 1$ . By (22.10) stable sheaves with this Mukai vector (rank two,  $c_1 = h$ ) move in a two-dimensional family. The proof of the derived Torelli theorem shows that in this situation the moduli space  $M = M_H(v)$  is a K3 surface, that the existence of  $v'$  makes it a *fine* moduli space with a universal sheaf  $\mathcal{E}$  on  $M \times X$ , and that  $\Phi_{\mathcal{E}} : D^b(M) \rightarrow D^b(X)$  is an equivalence with  $\Phi_{\mathcal{E}}^H(0, 0, 1) = v$ : a point of  $M$  is a rank-two bundle on  $X$  [Huy ch. 16, Prop. 2.3, Ex. 3.4(iii) and proof of Prop. 3.5 step ii), ll. 15435–15438, 15682–15685, 15702–15711]. Rank has been exchanged for point-like charge by an isometry that is not in  $O(H^2)$ .

## 22.4 Physical reading: brane charges and the stringy moduli space

### 22.4.1 The lattice of Ramond–Ramond charges

Compactify type II string theory on a K3 surface. Besides the metric and the  $B$ -field, the theory has Ramond–Ramond gauge potentials, and their zero modes on  $X$  are counted by the Betti numbers  $b_0 + b_2 + b_4 = 1 + 22 + 1 = 24$ : they fill out  $H^0 \oplus H^2 \oplus H^4 = H^*(X)$  [Asp §4.4 (R-R moduli of type IIB), ll. 2587–2629]. D-branes (Chapter 13) are the objects charged under these potentials. In type IIA, a D4-brane wrapping  $X$   $r$  times, D2-branes wrapping a two-cycle Poincaré dual to  $c \in H^2(X, \mathbb{Z})$ , and D0-branes at points together carry a charge vector  $(r, c, s) \in H^*(X, \mathbb{Z})$ , and the correct integral charge of a brane carrying a gauge bundle  $E$  is not  $\text{ch}(E)$  but the Mukai vector  $\text{ch}(E)\sqrt{\text{td}(X)}$ : the curvature of  $X$  itself induces lower-brane

charge, which is the shift  $s = \text{ch}_2 + r$ . This identification of the Mukai vector with D-brane charge is standard in the D-brane literature [L], but it is *not* contained in the sources of this book’s library (Aspinwall’s lectures deliberately avoid D-branes, [Asp §1, ll. 156–165]); we record it as “standard; not checked against a source in this book’s library”. What *is* in our sources is the lattice and its automorphism group, to which we now turn.

### 22.4.2 The moduli space $O(\Gamma^{4,20}) \backslash O(4, 20) / (O(4) \times O(20))$

A non-linear sigma model on a K3 surface is specified by a Ricci-flat metric (58 real parameters) and a  $B$ -field class in  $H^2(X, \mathbb{R})$  ( $b_2 = 22$  parameters), 80 in all [Asp §3.2, ll. 1640–1654 and 1670]. The local structure of the moduli space is forced by the  $N = (4, 4)$  superconformal symmetry to be

$$\mathcal{T}_\sigma = \frac{O(4, 20)}{O(4) \times O(20)}, \quad (22.20)$$

the Grassmannian of positive-definite four-planes  $\Pi \subset \mathbb{R}^{4,20}$  [Asp §3.2–3.3, Theorem 5, eq. (64), ll. 1697–1713 and 1733]. Its dimension is  $4 \times 20 = 80$  — a useful check, since the dimension of  $O(p, q) / (O(p) \times O(q))$  is  $pq$  (Chapter 10). Compare the moduli space of Einstein metrics of unit volume,  $O(3, 19) / (O(3) \times O(19))$ , of dimension  $3 \times 19 = 57$ : the decomposition

$$\frac{O(4, 20)}{O(4) \times O(20)} \cong \frac{O(3, 19)}{O(3) \times O(19)} \times \mathbb{R}^{22} \times \mathbb{R}_+, \quad 80 = 57 + 22 + 1, \quad (22.21)$$

exhibits metric,  $B$ -field and volume [Asp §3.3, eq. (67), ll. 1770–1784]. This is the same counting as for the Narain moduli space of a torus,  $d^2 = \frac{1}{2}d(d+1) + \frac{1}{2}d(d-1)$  for  $G$  and  $B$  (Chapter 10), with the lattice  $\Gamma^{d,d}$  replaced by  $\Gamma^{4,20}$ .

The decomposition (22.21) is not canonical. It requires the choice of a primitive null vector  $w \in \Gamma^{4,20}$ ,  $w^2 = 0$ ; then  $w^\perp / w \cong \Gamma^{3,19}$  is identified with  $H^2(X, \mathbb{Z})$ , and a second null vector  $w^*$  with  $w \cdot w^* = 1$  completes the hyperbolic plane [Asp §3.3, ll. 1741–1753]. In Mukai’s language  $w$  is the generator of  $H^4$  — the class  $(0, 0, 1)$  of a point — and  $w^*$  spans  $H^0$  (up to the sign convention of the pitfall box above). Aspinwall’s conclusion is worth quoting in full: a point in the moduli space determines the conformal field theory uniquely, but one must choose  $w$  before one can speak of a K3 geometry with a metric and a  $B$ -field; “this choice is arbitrary and different choices will lead to potentially very different looking K3 surfaces which give the same conformal field theory. This may be viewed as a kind of T-duality” [Asp §3.4, ll. 1958–1963].

The discrete identifications are then the full isometry group of the lattice :

$$\mathcal{M}_\sigma = O(\Gamma^{4,20}) \backslash O(4, 20) / (O(4) \times O(20)) \quad (22.22)$$

[Asp §3.4, Theorem 6, eq. (76), ll. 1944–1948] [L], under the assumptions stated there (completeness and Hausdorffness of the moduli space). The part of  $O(\Gamma^{4,20})$  visible in the sigma-model action is the subgroup fixing  $w$ : classical diffeomorphisms, giving  $O(\Gamma^{3,19})$ , and integral shifts  $B \mapsto B + e$  with  $e \in H^2(X, \mathbb{Z})$ , which change the action by  $2\pi i \mathbb{Z}$  [Asp §3.3, ll. 1802–1828, eq. (69)]. Together they form  $O(\Gamma^{3,19}) \ltimes \Gamma^{3,19}$ . One further element, supplied by mirror symmetry, is needed to generate everything [Asp §3.4, ll. 1935–1943].

#### Physical picture: a dictionary between two literatures

The three kinds of isometries of Section 22.3 line up with the generators of the stringy duality group. The integral  $B$ -shift  $B \mapsto B + e$  is multiplication by  $\exp(e)$ , i.e. tensoring all branes with a line bundle of first Chern class  $e$ ; on the lattice it is the translation part of  $O(\Gamma^{3,19}) \ltimes \Gamma^{3,19}$ . Isometries of  $H^2$  are the classical symmetries of Chapter 20. The



isometries that move  $w = (0, 0, 1)$ , such as  $T_O^H$  in (22.19) or  $\Phi_{\mathcal{E}}^H$  of Section 22.3.4, are the genuinely stringy ones: they change which charges are called “point-like”, exactly as  $R \mapsto \alpha'/R$  exchanges momentum and winding. The purely lattice-theoretic counterpart is Wall’s theorem: for  $\Lambda$  even unimodular of signature  $(n_+, n_-)$  with  $n_{\pm} \geq 2$ ,  $O(\Lambda \oplus U)$  is generated by  $O(\Lambda)$ ,  $O(U)$  and the  $B$ -shifts  $\exp(B)$  [Huy ch. 14, Prop. 2.9, ll. 13282–13285]. Matching the *physical* duality group with the image of  $\text{Aut}(D^b(X))$  element by element is beyond our sources; treat it as a guide [C], not a theorem.

The relation to the rest of this book: on  $K3 \times T^2$  the charge lattice becomes  $\Gamma^{4,20} \oplus \Gamma^{2,2}$  of signature  $(6, 22)$  (Chapter 24); the number 24, the rank of the Mukai lattice, equals  $\chi_{\text{top}}(K3)$  and is the number that reappears in the tadpole (Chapter 25) and in Mathieu moonshine (Chapter 30), where symmetries of K3 sigma models are studied through their action on  $\Gamma^{4,20}$ .

## 22.5 What the machine has checked

Four Lean files concern this chapter. They are of very different strength, and the student should learn to tell them apart by reading the *statements*.

### 22.5.1 The Mukai pairing over an arbitrary Gram matrix

The module `DualScaleStream2/Lattice/Mukai.lean` formalizes Definition 22.2 as linear algebra over  $\mathbb{Z}$ . The lattice  $H^2$  is represented by an integral Gram matrix  $L : \text{Gram } n$  (an  $n \times n$  integer matrix, Chapter 18), and a Mukai vector is bare data:

```
structure MukaiVec (n : ℕ) where
  (r : ℤ)
  (c : Fin n → ℤ)
  (s : ℤ)

def mukaiPair {n : ℕ} (L : Gram n) (v w : MukaiVec n) : ℤ :=
  v.c ·v (L *v w.c) - v.r * w.s - v.s * w.r
```

Here the first term (a dot product with a matrix–vector product) is  $c_v^T L c_w$ , the intersection number  $(c_v, c_w)$  in coordinates. Nothing in `MukaiVec` refers to a sheaf: it is a triple  $(r, c, s)$ .

#### In Lean: symmetry, the norm formula, evenness

```
theorem mukaiPair_symm {n : ℕ} (L : Gram n) (hL : LT = L)
  (v w : MukaiVec n) : mukaiPair L v w = mukaiPair L w v := by ...

theorem mukaiPair_self {n : ℕ} (L : Gram n) (v : MukaiVec n) :
  mukaiPair L v v = v.c ·v (L *v v.c) - 2 * v.r * v.s := by ...

theorem mukaiPair_even {n : ℕ} (L : Gram n) (hL : LT = L)
  (heven : IsEvenDiag L) (v : MukaiVec n) :
  Even (mukaiPair L v v) := by ...
```

**Reading.** For every  $n$  and every integer matrix  $L$ : if  $L$  is symmetric then the Mukai pairing built from  $L$  is symmetric; the self-pairing is  $c^T L c - 2rs$ , which is (22.9); and if  $L$  is symmetric with even diagonal then every Mukai vector has even square, i.e.  $H^2 \oplus U(-1)$  is an even lattice whenever  $H^2$  is. **Proof idea.** Symmetry uses Mathlib’s `Matrix.dotProduct_transpose_mulVec` to swap the two  $H^2$  vectors and `ring` for the rest. Evenness rewrites with `mukaiPair_self`, applies the general lemma `even_quadratic_form_of_even_diag` of Chapter 18 to  $c^T L c$ , and notes that  $2rs$  is visi-

bly even. **Tier [A]**, as statements about integer matrices.

#### In Lean: the summand $U(-1)$ and the structure sheaf

```
def hyperbolicUNeg : Gram 2 := !![0, -1; -1, 0]

theorem hyperbolicUNeg_mul_self : hyperbolicUNeg * hyperbolicUNeg = 1 := by ...

theorem hyperbolicUNeg_unimodular : IsUnimodular hyperbolicUNeg :=
  isUnimodular_of_mul_eq_one _ _ hyperbolicUNeg_mul_self

theorem structureSheaf_mukai_sq {n : ℕ} (L : Gram n) :
  mukaiPair L ⟨1, 0, 1⟩ ⟨1, 0, 1⟩ = -2 := by ...
```

**Reading.** The matrix (22.11) squares to the identity, hence has an integral inverse, hence  $\det = \pm 1$  (`IsUnimodular` is literally “ $\det = 1 \vee \det = -1$ ”). And the vector  $(1, 0, 1)$  has Mukai square  $-2$  for *every* Gram matrix  $L$  — because its  $H^2$  component is zero,  $L$  never enters. **Proof idea.** A case split over the four matrix entries (`fin_cases`) for the first; the certificate lemma `isUnimodular_of_mul_eq_one` for the second; `simp` and `norm_num` for the third. **Tier [A]** for the arithmetic. That  $(1, 0, 1)$  is  $v(\mathcal{O}_X)$ , and that  $-\langle v, v \rangle$  is  $\chi(\mathcal{O}_X, \mathcal{O}_X) = 2$  (Example 22.4), is **[L]**: Lean knows nothing of sheaves here.

### 22.5.2 The signature $(4, 20)$

In `DualScaleStream2/Lattice/K3T2Signature.lean` a `Signature` is a pair of natural numbers with componentwise addition. The inputs `sigU`, defined as the pair  $(1, 1)$ , and `sigE8Neg`, defined as  $(0, 8)$ , are asserted, and then

```
def sigMukai : Signature := sigK3 + sigU

theorem sigMukai_eq : sigMukai = ⟨4, 20⟩ := by
  rfl
```

**Reading.**  $(0, 8) + (0, 8) + 3 \cdot (1, 1) + (1, 1) = (4, 20)$ . This is the bookkeeping step “signatures add” of (22.12), **[A]** as arithmetic of pairs. It is not a proof that some  $24 \times 24$  Gram matrix has four positive and twenty negative eigenvalues; the signatures of  $U$  and  $E_8(-1)$  enter as **[L]** inputs (see Chapter 18 for what is proved about  $E_8$ ). The same file checks  $4 - 20 \equiv 0 \pmod{8}$  in `index_mod_eight`, the necessary condition for an even unimodular lattice, at these specific numbers only.

### 22.5.3 Arithmetic shadows in the older libraries

#### In Lean: a record of four numbers

```
structure MukaiLattice where
  rank : ℕ := 24
  posSignature : ℕ := 4
  negSignature : ℕ := 20
  unimodular : Bool := true

def canonicalMukaiLattice : MukaiLattice := {}

theorem mukai_rank :
  canonicalMukaiLattice.rank = canonicalMukaiLattice.posSignature +
  canonicalMukaiLattice.negSignature := by rfl
```

**Reading** (StringTheoryFormalization/StringDynamics/MukaiLattice.lean).

`MukaiLattice` is not a lattice: it has no underlying group and no bilinear form. It is a record of three numbers and a Boolean flag that may be set to `true` or `false` at will. `mukai_rank` says  $24 = 4 + 20$  for the default values. The companion `kummer_sublattice_rank` says that the type `Fin 16` has 16 elements and mentions no lattice at all. These are *arithmetic shadows* in the sense of Chapter 35: [A] as numerals, silent about the mathematics their names evoke.

#### In Lean: Fourier–Mukai as bookkeeping — read the statement, not the name

```
structure DerivedCategory (X : Type*) where
  name : String

theorem fm_isEquivalence_of_mk (K3 K3hat : Type*)
  (src : DerivedCategory K3) (tgt : DerivedCategory K3hat)
  (ker : String) :
  (FourierMukaiTransform.mk src tgt ker true).isEquivalence = true := rfl

theorem fm_squared_is_shift :
  True := trivial
```

**Reading** (`StringTheoryFormalization/StringDynamics/FourierMukai.lean`). `DerivedCategory` wraps a character string; `FourierMukaiTransform` is a record with two such tags, a kernel `name` and a Boolean field `isEquivalence`. The first theorem says: if you build a record with the flag set to `true` and read the flag back, you get `true`. The second theorem is the proposition `True`. Neither says anything about functors, shifts or Theorem 22.11. The file’s own scope note records that an earlier declaration named `fm_lattice_isometry` was *false as stated* (the flag can be set to `false`) and was removed when the file was first compiled. Theorem 22.11 remains [L]; formalizing it would require derived categories of coherent sheaves, the Grothendieck–Riemann–Roch theorem and the cohomology of K3 surfaces, none of which is available in this corpus.

#### In Lean: the swap $(r, s) \mapsto (s, r)$ on three integers

```
structure MukaiVector where
  r : Int -- D0-brane rank / momentum
  c2 : Int -- c^2 = c · c in H^2(K3, Z)
  s : Int -- D4-brane charge / winding
  deriving Repr, DecidableEq

def mukai_norm (v : MukaiVector) : Int :=
  v.c2 - 2 * v.r * v.s

def buscher_reflection (v : MukaiVector) : MukaiVector :=
  { r := v.s, c2 := v.c2, s := v.r }

theorem mukai_norm_buscher_invariant (v : MukaiVector) :
  mukai_norm (buscher_reflection v) = mukai_norm v := by ...
```

**Reading** (`Lean5Corpus/Problems/Problem4_MukaiMonodromy.lean`). Here a “Mukai vector” is three integers  $(r, c_2, s)$ , where the single integer  $c_2$  stands for the *number* ( $c \cdot c$ ), not for a class  $c \in H^2$ . The theorem says  $c_2 - 2sr = c_2 - 2rs$ : commutativity of integer multiplication. Its companions `buscher_reflection_involution` (swapping twice is the identity) and `mukai_rank_and_signature` ( $24 = 24$  and  $4 - 20 = -16$ ) are equally elementary. [A] as integer arithmetic. The words “Buscher”, “monodromy” and “non-perturbative” in that file are interpretation [C]; note also that its comments label  $r$  as

D0 and  $s$  as D4 charge, the opposite of the convention of Section 22.4, which is harmless only because the statement is symmetric under  $r \leftrightarrow s$ .

*Remark 22.13* (How the swap relates to a genuine autoequivalence). The swap  $(r, c, s) \mapsto (s, c, r)$  is an isometry of the Mukai lattice; by (22.19) it equals  $-T_{\mathcal{O}}^H$  composed with  $c \mapsto -c$  on  $H^2$ . The isometry actually induced by the spherical twist is  $(r, c, s) \mapsto (-s, c, -r)$ . A record that remembers only the number  $(c.c)$  cannot distinguish  $c$  from  $-c$ , nor express the pairing of two different vectors, so the Lean statement cannot see these differences. A faithful version would be stated with `MukaiVec` and `mukaiPair`; see Exercises 22.7 and 22.9.

| Claim                                                                                                                      | Where                                                 | Tier         |
|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|--------------|
| Mukai pairing symmetric; $v^2 = c^T Lc - 2rs$ ; even if $L$ even                                                           | <code>mukaiPair_symm, _self, _even</code>             | [A]          |
| $U(-1)^2 = 1$ , $U(-1)$ unimodular                                                                                         | <code>hyperbolicUNeg_mul_self, _unimodular</code>     | [A]          |
| $\langle (1, 0, 1), (1, 0, 1) \rangle = -2$ for all $L$                                                                    | <code>structureSheaf_mukai_sq</code>                  | [A]          |
| $(3, 19) + (1, 1) = (4, 20)$                                                                                               | <code>sigMukai_eq</code>                              | [A]          |
| $24 = 4 + 20$ ; swap preserves $c_2 - 2rs$                                                                                 | <code>mukai_rank, mukai_norm_buscher_invariant</code> | [A] (shadow) |
| $v(\mathcal{O}_X) = (1, 0, 1)$ ; $\chi(E, F) = -\langle v(E), v(F) \rangle$                                                | Huy ch. 9 §1                                          | [L]          |
| $\tilde{H}(X, \mathbb{Z}) \cong 4U \oplus 2E_8(-1)$ , signature $(4, 20)$                                                  | Huy ch. 14                                            | [L]          |
| Derived equivalences act by Hodge isometries; derived Torelli                                                              | Huy ch. 16 §3                                         | [L]          |
| $\mathcal{M}_\sigma = O(\Gamma^{4,20}) \backslash O(4, 20) / (O(4) \times O(20))$                                          | Asp §3.4, Thm. 6                                      | [L]          |
| Mukai vector = D-brane charge                                                                                              | standard, not in library                              | [L]          |
| “Buscher reflection” reading of $(r, s) \mapsto (s, r)$ ;<br>match of $\text{Aut } D^b(X)$ with the physical duality group | this programme                                        | [C]          |

Table 22.1: Tier table for this chapter. All [A] entries depend only on the axioms `propext`, `Classical.choice`, `Quot.sound`.

### Chapter summary

- The Mukai vector  $v(E) = \text{ch}(E)\sqrt{\text{td } X} = (r, c_1, c_2 + r)$  and the Mukai pairing  $\langle \alpha, \beta \rangle = (\alpha_2, \beta_2) - \alpha_0 \beta_4 - \alpha_4 \beta_0$  turn Riemann–Roch on a K3 surface into  $\chi(E, F) = -\langle v(E), v(F) \rangle$ .
- $v^2 = (c.c) - 2rs$ ; a simple sheaf has  $v^2 + 2$  deformation parameters, so  $v^2 \geq -2$ .  $v(\mathcal{O}_X) = (1, 0, 1)$  has  $v^2 = -2$ ; a point has  $v = (0, 0, 1)$ ,  $v^2 = 0$ ;  $v(T_X)^2 = 88$ .
- $\tilde{H}(X, \mathbb{Z}) = H^2 \oplus U(-1) \cong 4U \oplus 2E_8(-1)$  is even, unimodular, of signature  $(4, 20)$  and rank 24.
- Derived equivalences of K3 surfaces act on  $\tilde{H}$  by Hodge isometries that mix  $H^0, H^2, H^4$ : shifts  $(-1)$ , line-bundle twists  $(\exp B)$ , spherical twists (reflections in  $(-2)$ -vectors). Two projective K3 surfaces are derived equivalent iff their Mukai lattices are Hodge isometric.
- Physically  $\tilde{H}(X, \mathbb{Z}) = \Gamma^{4,20}$  is the charge lattice, the sigma-model moduli space is  $O(\Gamma^{4,20}) \backslash O(4, 20) / (O(4) \times O(20))$  of dimension  $80 = 58 + 22$ , and the choice of the null vector  $w = (0, 0, 1)$  is the choice of a geometric interpretation.

- Lean checks the integer linear algebra of the pairing for all Gram matrices and the signature arithmetic. It does not contain sheaves, derived categories or Fourier–Mukai functors; the declarations with those names are bookkeeping records.

## Exercises

**Exercise 22.1** (★). Verify  $\sqrt{\text{td}(X)} = (1, 0, 1)$  from  $\text{td}(X) = (1, 0, 2)$ , and compute  $v(E)$  and  $v(E)^2$  for  $E = \mathcal{O}_X \oplus \mathcal{O}_X$ . Why does the result  $v^2 = -8 < -2$  not contradict the bound  $v^2 \geq -2$ ?

**Exercise 22.2** (★). Let  $I_Z$  be the ideal sheaf of  $n$  distinct points. Use additivity of  $v$  in  $0 \rightarrow I_Z \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_Z \rightarrow 0$  to show  $v(I_Z) = (1, 0, 1 - n)$  and  $v^2 = 2n - 2$ . Interpret  $v^2 + 2 = 2n$  geometrically.

**Exercise 22.3** (★). Show that  $U(-1) \cong U$  by exhibiting  $P \in GL(2, \mathbb{Z})$  with  $P^\top U(-1)P = U$ , and show that, in contrast,  $E_8(-1) \not\cong E_8$ .

**Exercise 22.4** (★★). Expand  $v(E)^2 \geq -2$  for a simple sheaf of rank  $r$  and Chern classes  $c_1, c_2$  to obtain  $\Delta(E) := 2rc_2 - (r-1)c_1^2 \geq 2(r^2 - 1)$  [Huy ch. 9, Rem. 1.5, ll. 7493–7499]. Check it on  $T_X$ .

**Exercise 22.5** (★★). In the quartic example of Section 22.3.4, compute the matrix  $M = T_{\mathcal{O}}^H \exp(h)$ , verify  $M^\top G M = G$ , and find the images of  $(0, 0, 1)$ ,  $(1, 0, 1)$  and  $(2, 1, 1)$ . Which of the images can be Mukai vectors of sheaves, and which only of complexes?

**Exercise 22.6** (★★). Show that  $\dim O(p, q)/(O(p) \times O(q)) = pq$  by counting dimensions of orthogonal groups, and verify  $80 = 57 + 22 + 1$  in (22.21). What is the analogous count for the  $(5, 21)$  Grassmannian that appears when the R-R moduli and the dilaton of type IIB are included [Asp §4.4, ll. 2587–2621]?

**Exercise 22.7** (★★, Lean). With `MukaiVec` and `mukaiPair` as in Section 22.5, define a function `swap` sending  $(r, c, s)$  to  $(s, c, r)$  and prove `mukaiPair L (swap v) (swap w) = mukaiPair L v w`. No hypothesis on `L` is needed. Explain why this statement is strictly stronger than the theorem `mukai_norm_buscher_invariant` quoted above.

**Exercise 22.8** (★★, Lean). State and prove in Lean: for a symmetric `L` and an integer vector `c` with  $c^\top L c = 2k$  (written with `mukaiPair`-style dot products), the Mukai vector  $(1, c, k+1)$  has Mukai square  $-2$ . This is  $v(L)^2 = -2$  of Example 22.6; which hypothesis replaces “ $\ell^2/2$  is an integer”?

**Exercise 22.9** (★★★, Lean). Define the reflection  $s_\delta(v) = v + \langle v, \delta \rangle \delta$  on `MukaiVec n` (componentwise) and prove that for symmetric `L` and a vector `d` with `mukaiPair L d d = -2` it preserves `mukaiPair`. Then specialize to  $\delta = (1, 0, 1)$  and recover (22.19). Hint: first prove bilinearity lemmas for `mukaiPair` in each argument.

**Exercise 22.10** (★★★). Prove Lemma 22.9 in Lean-ready form: write  $\exp(B)$  as a  $(n+2) \times (n+2)$  integer matrix  $M_B$  acting on  $(r, c, s)$ , write the Mukai Gram matrix  $\tilde{G}$  in block form, and show  $M_B^\top \tilde{G} M_B = \tilde{G}$  assuming  $B^\top L B$  is even. Compare  $M_B$  with the  $B$ -shift element of  $O(d, d; \mathbb{Z})$  in Chapter 11: what plays the role of the antisymmetric matrix there?

## Further reading

- Mukai vector, Mukai pairing and Riemann–Roch: [Huy ch. 9 §1, ll. 7369–7526].
- The extended K3 lattice,  $B$ -field shifts and Wall’s generators of  $O(\Lambda \oplus U)$ : [Huy ch. 14, ll. 12873–12880 and 13265–13285].
- Derived categories, Fourier–Mukai kernels, spherical twists: [Huy ch. 16 §§1–2, ll. 15107–15582]; the action on cohomology and the derived Torelli theorem: [Huy ch. 16 §3, ll. 15591–15740].
- Moduli of stable sheaves with a given Mukai vector: [Huy ch. 10 §2, ll. 8810–8822].
- The  $(4, 20)$  lattice in string theory, the role of the null vector  $w$  and the moduli space of sigma models: [Asp §3.2–3.4, ll. 1640–1963]; R–R moduli and the  $(5, 21)$  extension: [Asp §4.4, ll. 2587–2632]. These are developed in Chapter 23.

## Chapter 23

# Strings on K3: Moduli Space and Dualities

Two strands of this book meet in this chapter. In Parts II and III we studied strings on tori: the momentum and winding numbers form an even self-dual lattice  $\Gamma^{d,d}$ , the background fields  $G$  and  $B$  select a positive-definite  $d$ -plane in  $\mathbb{R}^{d,d}$ , and the duality group  $O(d, d; \mathbb{Z})$  is the isometry group of the lattice (Chapters 10 and 11). In Part IV we studied a purely geometric object, the K3 surface, and found that its cohomology carries the even unimodular lattice  $\Gamma^{3,19} = 3U \oplus 2E_8(-1)$  and, after adding  $H^0 \oplus H^4$ , the Mukai lattice  $\Gamma^{4,20}$  (Chapters 20 and 22). The question of this chapter is: *what happens when a string propagates on a K3 surface?*

The answer has the same shape as for a torus. The string sees, besides the Ricci-flat metric, a  $B$ -field; together they are 80 real parameters, and they assemble into the choice of a positive-definite *four*-plane  $\Pi \subset \mathbb{R}^{4,20}$ . The moduli space is

$$\mathcal{M}_\sigma = O(\Gamma^{4,20}) \setminus O(4, 20) / (O(4) \times O(20)), \quad (23.1)$$

which is literally a Narain moduli space — the one of the *heterotic* string on a four-torus. This coincidence is the first evidence for a non-perturbative duality between two very different string theories, type IIA on K3 and heterotic on  $T^4$ , and it predicts that a type IIA string acquires non-abelian gauge symmetry when a two-sphere inside the K3 surface shrinks to zero size. A student should care for two reasons: this is the simplest example in which a curved compactification is controlled exactly by a lattice, and it is the six-dimensional parent of the  $K3 \times T^2$  compactification that the rest of the book is about (Chapter 24).

Almost everything in this chapter is Tier [\[L\]](#): it is taken from Aspinwall’s lectures, with the derivations filled in. The machine-checked content is small and precisely delimited: signatures, ranks and parity of the lattices involved (Section 23.7). Our source makes its own assumptions explicit, and we repeat them where they enter.

## 23.1 The sigma model with K3 target

### 23.1.1 Action and conformal invariance

Let  $X$  be the target manifold and  $x : \Sigma_{\text{ws}} \rightarrow X$  the embedding of the worldsheet  $\Sigma_{\text{ws}}$  (a Riemann surface; we write  $\Sigma_{\text{ws}}$  because  $\Sigma$  will denote a three-plane below). In conformal gauge the bosonic part of the action is, in Aspinwall’s normalization,

$$S = \frac{i}{8\pi\alpha'} \int_{\Sigma_{\text{ws}}} (g_{ij} - B_{ij}) \partial x^i \bar{\partial} x^j d^2z - \frac{1}{2\pi} \int_{\Sigma_{\text{ws}}} \Phi R^{(2)} d^2z + \dots \quad (23.2)$$

[Asp §3.1, eq. (60), ll. 1559–1576]. Here  $g_{ij}$  is a Riemannian metric on  $X$ ,  $B_{ij}$  the components of a real two-form  $B$ ,  $\Phi$  the dilaton and  $R^{(2)}$  the scalar curvature of the worldsheet; the dots are fermionic terms. This is the same sigma model as in Chapter 6, written in complex worldsheet coordinates; the overall numerical factors differ from Tong’s normalization and play no role below, so we do not convert them (the coefficient of the dilaton term is garbled in our copy of the source; only its structure,  $\Phi$  times the Euler density, is used).

For the two-dimensional theory to define a string background it must be conformally invariant. As in Chapter 6, this is a condition order by order in  $\alpha'/R^2$ , where  $R$  is a typical radius of  $X$ . The simplest solution at leading order is:  $\Phi$  constant,  $dB = 0$ , and  $g_{ij}$  Ricci-flat [Asp §3.1, ll. 1583–1592]. The two compact four-dimensional examples used in this book are the flat torus  $T^4$  and the K3 surface with its Yau metric (Chapter 19).

### 23.1.2 Worldsheet supersymmetry and hyperkähler geometry

Add the worldsheet fermions  $\psi^i$ , which transform as sections of the (pulled-back) cotangent bundle of  $X$ . A supersymmetry variation has the form  $\delta_\epsilon x^i = \bar{\epsilon} l^i_j \psi^j$ . With one supersymmetry,  $l$  can be set to the identity. A second supersymmetry requires a second tensor  $l$ , which turns out to be an almost complex structure making  $X$  Kähler;  $N = 4$  requires three complex structures, that is, a hyperkähler manifold, and conversely [Asp §3.1, ll. 1601–1611]. A K3 surface with a Ricci-flat metric has holonomy  $SU(2) \cong Sp(1)$  and carries a two-sphere of complex structures  $aI + bJ + cK$ ,  $a^2 + b^2 + c^2 = 1$  (Chapter 19). Because a conformal field theory splits into left- and right-movers, each sector carries its own  $N = 4$  algebra: *the sigma model on a K3 surface is an  $N = (4, 4)$  superconformal field theory*. This has a consequence that distinguishes K3 from Calabi–Yau threefolds: for  $N = 4$  sigma models there are no perturbative corrections to the Ricci-flat metric beyond leading order, and there are arguments that this persists non-perturbatively, so the Ricci-flat metric is an exact solution [Asp §3.1, ll. 1618–1625] [L]. The central charge is  $c = 6$  on each side (four bosons and four fermions,  $4 + 4 \cdot \frac{1}{2}$ ; Chapter 4), and the  $N = 4$  algebra contains an affine  $\mathfrak{su}(2)$ . We will meet its characters again in Chapter 28.

#### Physical picture: why the torus analogy is not an accident

For a torus the metric is flat and the zero modes of the string are labelled by a lattice. For K3 the metric is curved and unknown in closed form, yet  $N = (4, 4)$  supersymmetry is strong enough to protect the theory from corrections. The price is that we cannot solve the model at a generic point; what we *can* determine is the space of all such models. The logic is that of Chapter 20: there, Yau’s theorem allowed us to replace “a Ricci-flat metric” by “a positive three-plane in  $\mathbb{R}^{3,19}$ ” without ever writing the metric down. Here the same trick is played one dimension higher.

## 23.2 Counting moduli: $58 + 22 = 80$

### 23.2.1 Three sets of parameters

The action (23.2) contains three kinds of data. We ask, for each, how many deformations change the conformal field theory.

**The metric.** It must be Ricci-flat. The moduli space of Einstein metrics on a K3 surface (orbifold points included) is

$$\mathcal{M}_E \cong O(\Gamma^{3,19}) \backslash O(3, 19) / (O(3) \times O(19)) \times \mathbb{R}_+, \quad (23.3)$$



the Grassmannian of positive three-planes  $\Sigma \subset \mathbb{R}^{3,19} = H^2(X, \mathbb{R})$ , modulo lattice isometries, times the volume [Asp §2.4, Theorem 3, eqs. (39)–(40), ll. 875–893] (Chapter 20). The plane  $\Sigma$  is spanned by the real and imaginary parts of the holomorphic two-form and by the Kähler form  $J$ . The dimension of  $\mathcal{M}_E$  is  $3 \times 19 + 1 = 58$ . Assuming every inequivalent Ricci-flat metric gives an inequivalent conformal field theory, the metric contributes 58 parameters [Asp §3.2, ll. 1639–1642].

**The  $B$ -field.** It enters through  $\int_{x(\Sigma_{\text{ws}})} B$ , the integral of  $B$  over the image of the worldsheet, a closed two-cycle. By Stokes' theorem an exact piece  $B = d\Lambda$  integrates to zero, so only the cohomology class  $[B] \in H^2(X, \mathbb{R})$  matters. Since  $b_2(K3) = 22$ , the  $B$ -field contributes 22 parameters [Asp §3.2, ll. 1643–1654].

**The dilaton.** For constant  $\Phi$  the last term of (23.2) is topological: by Gauss–Bonnet it contributes  $2\Phi(g - 1) = -\Phi \chi(\Sigma_{\text{ws}})$  on a worldsheet of genus  $g$ . A genus- $g$  amplitude is therefore weighted by  $\lambda^{2g-2}$  with the string coupling  $\lambda = e^\Phi$ , and nothing changes in the conformal field theory on a fixed worldsheet [Asp §3.2, ll. 1655–1667]. It is *not* a modulus of the conformal field theory. It returns in Section 23.5 as a modulus of the string theory.

$$\dim_{\mathbb{R}} \mathcal{M}_\sigma = 58 + 22 = 80 = 4 \times 20. \quad (23.4)$$

[Asp §3.2, l. 1670]. The factorization  $80 = 4 \cdot 20$  is the first hint of the answer.

### 23.2.2 The local form of the moduli space

To find the space itself, Aspinwall uses a holonomy argument, which we retrace.

*Step 1.* The symmetric space  $O(3, 19)/(O(3) \times O(19))$  in (23.3) has holonomy algebra  $\mathfrak{so}(3) \oplus \mathfrak{so}(19)$ . The  $\mathfrak{so}(3) \cong \mathfrak{su}(2)$  rotates the three-plane  $\Sigma$  into itself, i.e. it rotates the sphere of complex structures.

*Step 2.* In a conformal field theory left- and right-movers are independent, so each side has its own  $\mathfrak{su}(2)$ ; indeed each  $N = 4$  algebra contains an affine  $\mathfrak{su}(2)$ . Therefore the  $\mathfrak{so}(3)$  is promoted to  $\mathfrak{su}(2) \oplus \mathfrak{su}(2) \cong \mathfrak{so}(4)$ , which acts on the marginal operators, the tangent vectors of the moduli space [Asp §3.2, ll. 1674–1690].

*Step 3.* We need an 80-dimensional Riemannian manifold whose holonomy contains  $SO(4)$  acting non-trivially on every tangent direction, and which is non-compact (a K3 surface can be made arbitrarily large). The classification of holonomy groups by Berger and Simons then leaves one possibility [Asp §3.2, Theorem 5, ll. 1691–1713]:

**Theorem 23.1** (Local form; Aspinwall's Theorem 5, [L]). *Assuming that the deformations of  $g$  and  $B$  counted above are effective, every smooth neighbourhood in the moduli space of  $N = (4, 4)$  sigma models with K3 target is isomorphic to an open subset of*

$$\mathcal{T}_\sigma = \frac{O(4, 20)}{O(4) \times O(20)}. \quad (23.5)$$

The dimension can be checked directly. The group  $O(p, q)$  has the same dimension as  $O(p + q)$ , namely  $\frac{1}{2}(p + q)(p + q - 1)$ , so

$$\dim \frac{O(p, q)}{O(p) \times O(q)} = \frac{(p + q)(p + q - 1)}{2} - \frac{p(p - 1)}{2} - \frac{q(q - 1)}{2} = pq, \quad (23.6)$$

and for  $(p, q) = (4, 20)$  this is  $276 - 6 - 190 = 80$ . A more geometric way to see  $pq$ : the coset is the Grassmannian of positive-definite  $p$ -planes  $\Pi \subset \mathbb{R}^{p, q}$ ; a nearby plane is the graph of a

linear map  $\Pi \rightarrow \Pi^\perp$ , and such maps form a vector space of dimension  $pq$ . This is exactly how  $G + B$  appeared as a map in Chapter 10.

**Common pitfall: 80 is not “58 plus 22 Ramond–Ramond fields”**

The 22 extra moduli are periods of the Neveu–Schwarz  $B$ -field, present in every closed string theory. They are not Ramond–Ramond fields: type IIA has R–R potentials of odd degree only (a one-form and a three-form), a K3 surface has no odd cycles, and so the R–R sector contributes *no* scalars on K3 [Asp §4.2, ll. 2348–2353]. Nor is the dilaton among the 80: it is an 81st modulus, see (23.20). We stress this because the informal comment attached to the Lean declaration `witten_moduli_dim_is_80` (Section 23.7) gets precisely this point wrong; the theorem itself,  $4 \times 20 = 80$ , is unaffected.

## 23.3 Geometry of the four-plane

A point of  $\mathcal{T}_\sigma$  is a positive-definite four-plane  $\Pi \subset \mathbb{R}^{4,20}$ . How does one read off a metric, a  $B$ -field and a volume from  $\Pi$ ? The answer requires a choice, and the choice is the origin of the stringy dualities.

### 23.3.1 Choosing a null vector

Let  $\Gamma^{4,20} \subset \mathbb{R}^{4,20}$  be the even unimodular lattice of signature  $(4, 20)$ ; by the classification of Chapter 18 it is unique,  $\Gamma^{4,20} \cong 4U \oplus 2E_8(-1) \cong \Gamma^{3,19} \oplus U$ . Fix a primitive vector  $w \in \Gamma^{4,20}$  with  $w \cdot w = 0$ , and a second null vector  $w^*$  with  $w \cdot w^* = 1$ . Then  $w$  and  $w^*$  span a copy of  $U$  and

$$\frac{w^\perp}{w} \cap \Gamma^{4,20} \cong \Gamma^{3,19}, \quad \mathbb{R}^{4,20} = \mathbb{R}w \oplus \mathbb{R}w^* \oplus \mathbb{R}^{3,19} \quad (23.7)$$

[Asp §3.3, eq. (65), ll. 1741–1753]. Here  $w^\perp$  is the 23-dimensional space of vectors orthogonal to  $w$ ; it contains  $w$  itself because  $w$  is null, and dividing by  $w$  leaves 22 dimensions. In the language of Chapter 22,  $\mathbb{R}^{3,19} = H^2(X, \mathbb{R})$ ,  $w$  generates  $H^4(X, \mathbb{Z})$  and  $w^*$  generates  $H^0(X, \mathbb{Z})$  (up to the sign convention for the Mukai pairing discussed there).

### 23.3.2 From $\Pi$ to $(\Sigma, B, V)$ and back

Given  $\Pi$ , set  $\Sigma' = \Pi \cap w^\perp$ . Since  $\Pi$  is four-dimensional and  $w^\perp$  has codimension one,  $\Sigma'$  is a three-plane (it cannot be all of  $\Pi$ : otherwise  $w$  would be orthogonal to a maximal positive-definite subspace, hence lie in the negative-definite  $\Pi^\perp$ , contradicting  $w^2 = 0$ ,  $w \neq 0$ ). Project  $\Sigma'$  to  $w^\perp/w = \mathbb{R}^{3,19}$  to obtain a positive three-plane  $\Sigma$ : this is the Einstein metric of fixed volume. Let  $B' \in \Pi$  be the vector orthogonal to  $\Sigma'$  with  $B' \cdot w = 1$ , and decompose it as

$$B' = \alpha w + w^* + B, \quad \alpha \in \mathbb{R}, \quad B \in \mathbb{R}^{3,19} = H^2(X, \mathbb{R}) \quad (23.8)$$

[Asp §3.3, eq. (66), ll. 1755–1768]. The component  $B$  is the  $B$ -field. Let us make the inverse map explicit, which the source leaves to the reader.

**Proposition 23.2** (The four-plane in coordinates). *Let  $\sigma_1, \sigma_2, \sigma_3$  be an orthonormal basis of a positive three-plane  $\Sigma \subset \mathbb{R}^{3,19}$ , let  $B \in \mathbb{R}^{3,19}$  and  $\alpha \in \mathbb{R}$  with  $2\alpha + B \cdot B > 0$ . Then*

$$\xi_i = \sigma_i - (\sigma_i \cdot B) w \quad (i = 1, 2, 3), \quad \xi_4 = B' = \alpha w + w^* + B \quad (23.9)$$

*are mutually orthogonal, with  $\xi_i \cdot \xi_j = \delta_{ij}$  and*

$$B' \cdot B' = 2\alpha + B \cdot B. \quad (23.10)$$

*They span a positive-definite four-plane  $\Pi$  with  $\Pi \cap w^\perp = \text{span}(\xi_1, \xi_2, \xi_3)$ .*

*Proof.* Use  $w^2 = (w^*)^2 = 0$ ,  $w \cdot w^* = 1$  and  $w \cdot \sigma = w^* \cdot \sigma = 0$  for  $\sigma \in \mathbb{R}^{3,19}$ . Then  $\xi_i \cdot \xi_j = \sigma_i \cdot \sigma_j = \delta_{ij}$ , because every term containing  $w$  pairs to zero. Next  $\xi_i \cdot B' = \sigma_i \cdot B - (\sigma_i \cdot B)(w \cdot w^*) = 0$ : the correction  $-(\sigma_i \cdot B)w$  is exactly what is needed to make  $\xi_i$  orthogonal to  $B'$ . Finally  $B' \cdot B' = 2\alpha(w \cdot w^*) + B \cdot B$ . Since  $\xi_i \cdot w = 0$  and  $B' \cdot w = 1$ , the intersection with  $w^\perp$  is spanned by the  $\xi_i$ .  $\square$

Aspinwall identifies the norm (23.10) with the volume of the K3 surface,

$$V = \text{Vol}(X) = \int_X J \wedge J = 2\alpha + B \cdot B \quad (23.11)$$

(in units of  $\alpha'^2$ ), by matching the natural metrics on both sides of (23.12) below [Asp §3.3, eq. (68), ll. 1786–1797 and footnote 9] [L]. Proposition 23.2 makes this plausible: the condition that  $\Pi$  be positive-definite is precisely  $V > 0$ . Counting parameters,  $\Sigma$  has  $3 \times 19 = 57$ ,  $B$  has 22 and  $V$  has one:

$$\frac{O(4, 20)}{O(4) \times O(20)} \cong \frac{O(3, 19)}{O(3) \times O(19)} \times \mathbb{R}^{22} \times \mathbb{R}_+, \quad 80 = 57 + 22 + 1 \quad (23.12)$$

[Asp §3.3, eq. (67), ll. 1770–1784].

### 23.3.3 The discrete identifications

Which isometries  $g$  of the lattice are dualities, in the sense that  $\Pi$  and  $g\Pi$  define the same theory? Those visible in the action (23.2) fix  $w$ :

- Diffeomorphisms of  $X$  act on  $H^2(X, \mathbb{Z})$  by  $O^+(\Gamma^{3,19})$  (Chapter 20); worldsheet complex conjugation, which reverses the sign of  $B$ , supplies the remaining element of  $O(\Gamma^{3,19})$  [Asp §3.3, ll. 1802–1804 and 1823–1825].
- Integral shifts  $B \mapsto B + e$ ,  $e \in H^2(X, \mathbb{Z})$ : the action changes by  $2\pi i \times \text{integer}$ , so the path integral is unchanged [Asp §3.3, ll. 1804–1810].

It is instructive to see that the  $B$ -shift really is an isometry of  $\Gamma^{4,20}$ .

**Proposition 23.3** (The  $B$ -shift as a lattice isometry). *For  $e \in \Gamma^{3,19}$  define a linear map  $T_e$  of  $\mathbb{R}^{4,20}$  by*

$$T_e(w) = w, \quad T_e(x) = x - (x \cdot e)w \quad (x \in \mathbb{R}^{3,19}), \quad T_e(w^*) = w^* + e - \frac{1}{2}(e \cdot e)w. \quad (23.13)$$

*Then  $T_e$  preserves the inner product, maps  $\Gamma^{4,20}$  to itself, and sends the plane with data  $(\Sigma, B, \alpha)$  to the plane with data  $(\Sigma, B + e, \alpha - B \cdot e - \frac{1}{2}e \cdot e)$ . In particular the volume (23.11) is unchanged.*

*Proof.* Isometry:  $T_e(x) \cdot T_e(y) = x \cdot y$  since  $w$  is null and orthogonal to  $\mathbb{R}^{3,19}$ ;  $T_e(w^*) \cdot T_e(w) = 1$ ;  $T_e(w^*) \cdot T_e(x) = e \cdot x - (x \cdot e) = 0$ ; and  $T_e(w^*)^2 = e \cdot e - 2 \cdot \frac{1}{2}e \cdot e = 0$ . Integrality: the only non-obvious coefficient is  $\frac{1}{2}e \cdot e$ , which is an integer because  $\Gamma^{3,19}$  is even. Action on the plane:  $T_e(B') = \alpha w + w^* + e - \frac{1}{2}(e \cdot e)w + B - (B \cdot e)w$ , which is of the form (23.8) with the stated  $B$  and  $\alpha$ ; and  $T_e(\xi_i) = \sigma_i - (\sigma_i \cdot (B + e))w$ . Finally  $2\alpha' + (B + e)^2 = 2\alpha - 2B \cdot e - e^2 + B^2 + 2B \cdot e + e^2 = 2\alpha + B^2$ .  $\square$

The evenness of the lattice is therefore not a technicality: it is what makes the integral  $B$ -shift a symmetry, exactly as for the torus, where  $B_{ij} \mapsto B_{ij} + N_{ij}$  with  $N$  integer antisymmetric lies in  $O(d, d; \mathbb{Z})$  (Chapter 11). The maps  $T_e$  together with  $O(\Gamma^{3,19})$  form the “space group”  $O(\Gamma^{3,19}) \ltimes \Gamma^{3,19}$  [Asp §3.3, eq. (69), ll. 1826–1830]. These all fix  $w$ . One more element, moving  $w$ ,

is supplied by mirror symmetry: for an algebraic K3 surface with transcendental lattice  $\Lambda$  and “quantum Picard lattice”  $\Upsilon = \text{Pic}(X) \oplus U = \Lambda^\perp$ , split  $\Pi$  into the two-planes  $\Omega = \Pi \cap (\Lambda \otimes \mathbb{R})$  (complex structure) and  $\mathcal{U} = \Pi \cap (\Upsilon \otimes \mathbb{R})$  (Kähler form and  $B$ -field); the mirror map exchanges  $(\Lambda, \Omega) \leftrightarrow (\Upsilon, \mathcal{U})$ ; in a Greene–Plesser example with  $\Lambda \cong \Upsilon \cong \Gamma^{2,10}$  it is realized by an element of  $O(\Gamma^{4,20})$  exchanging two orthogonal  $\Gamma^{2,10}$  sublattices, and this element together with the space group generates all of  $O(\Gamma^{4,20})$  [Asp §3.4, eqs. (70)–(75), ll. 1860–1943].

**Theorem 23.4** (Global form; Aspinwall’s Theorem 6, [L]). *Given Theorem 23.1, completeness, and the assumption that the moduli space is Hausdorff, the moduli space of conformally invariant sigma models on a K3 surface is  $\mathcal{M}_\sigma = O(\Gamma^{4,20}) \backslash O(4, 20) / (O(4) \times O(20))$  [Asp §3.4, Theorem 6, eq. (76), ll. 1944–1950].*

The lattice  $\Gamma^{4,20}$  is the total integral cohomology  $H^*(X, \mathbb{Z})$  with the pairing that is non-zero only between degrees adding up to four — the Mukai lattice of Chapter 22 [Asp §3.4, ll. 1951–1957]. A point of  $\mathcal{M}_\sigma$  fixes the conformal field theory; only after choosing which null direction is called  $H^0$  (equivalently  $w = H^4$ ) can one speak of “the” K3 geometry, and different choices can give very different-looking K3 surfaces with the same conformal field theory. Aspinwall calls this “a kind of T-duality” [Asp §3.4, ll. 1958–1963].

## 23.4 The same coset from a torus: Narain revisited

The space (23.1) has the form of a Narain moduli space. To exploit this we need the torus story in the “plane in a lattice” language, allowing for different numbers  $n_L, n_R$  of left- and right-moving compact bosons.

### 23.4.1 Planes and momenta

Let  $\Gamma^{n_L, n_R} \subset \mathbb{R}^{n_L, n_R}$  be an even self-dual lattice (which forces  $n_L - n_R \in 8\mathbb{Z}$ ) and  $\Pi$  a positive-definite  $n_L$ -plane. Every vector decomposes as  $p = p_L + p_R$  with  $p_L \in \Pi$ ,  $p_R \in \Pi^\perp$ , so  $p_L \cdot p_L \geq 0$  and  $p_R \cdot p_R \leq 0$ . The lattice points are the winding and momentum states, with left and right conformal weights

$$h = \frac{1}{2} p_L \cdot p_L, \quad \bar{h} = -\frac{1}{2} p_R \cdot p_R, \quad (23.14)$$

and the moduli space of such theories is  $O(\Gamma^{n_L, n_R}) \backslash O(n_L, n_R) / (O(n_L) \times O(n_R))$ : the position of  $\Pi$  fixes the split of every charge into  $p_L$  and  $p_R$ , and the lattice isometries merely relabel the states [Asp §3.5, eqs. (77)–(78), ll. 1981–2001]. This is Chapter 10 in coordinate-free form. Note the convention: with Aspinwall we take the form positive on left-movers, and we write  $|p_R|^2 = -p_R \cdot p_R \geq 0$ . The identity  $p \cdot p = p_L^2 - |p_R|^2$  is the level-matching combination  $2n \cdot m$  of Chapter 10.

To recover  $G, B$  and Wilson lines, choose a null  $n_L$ -plane  $W$  spanned by lattice vectors, its dual null plane  $W^*$ , and the remaining negative-definite space  $V \cong \mathbb{R}^{n_R - n_L}$ , so  $\mathbb{R}^{n_L, n_R} = W \oplus W^* \oplus V$ . Then

$$\Pi = \{x + \psi(x) + A(x) : x \in W\}, \quad \psi : W \rightarrow W^*, \quad A : W \rightarrow V, \quad (23.15)$$

and  $\psi$ , viewed as a bilinear form on  $W$ , splits into its symmetric part  $G$  and antisymmetric part  $B$ . The torus is  $W/(\Gamma \cap W)$  with metric  $G$ ;  $A$  are the Wilson lines of the gauge group  $U(1)^{n_R - n_L} = V/(\Gamma \cap V)$  [Asp §3.5, eqs. (79)–(84), ll. 2003–2037]. The parameter count is  $\dim \text{Hom}(W, W^* \oplus V) = n_L \cdot n_R$ , as it must be by (23.6).

### 23.4.2 Worked example: the circle as a line in $\Gamma^{1,1}$

Take  $n_L = n_R = 1$ ,  $\Gamma^{1,1} = U$  with basis  $e, e^*$ ,  $e^2 = (e^*)^2 = 0$ ,  $e \cdot e^* = 1$ . A positive line is  $\Pi = \mathbb{R}(e + te^*)$  with  $t > 0$ , since  $(e + te^*)^2 = 2t$ . The unit vector along  $\Pi$  is  $u = (e + te^*)/\sqrt{2t}$  and the unit (norm  $-1$ ) vector along  $\Pi^\perp$  is  $u_\perp = (e - te^*)/\sqrt{2t}$ . For the state  $p = ne + me^*$ :

$$p_L^2 = (p \cdot u)^2 = \frac{(nt + m)^2}{2t} = \frac{1}{2}n^2t + nm + \frac{m^2}{2t}, \quad p_R \cdot p_R = -(p \cdot u_\perp)^2 = -\frac{(nt - m)^2}{2t}, \quad (23.16)$$

which reproduces [Asp §3.5, eq. (88), ll. 2160–2165] with  $t = \tan \theta$ . Compare with the book's convention (Chapter 7): the dimensionless left momentum is  $\sqrt{\alpha'/2}(n_{\text{mom}}/R + w_{\text{wind}}R/\alpha')$ , whose square is  $(n_{\text{mom}} + w_{\text{wind}}t)^2/(2t)$  with

$$t = R^2/\alpha'. \quad (23.17)$$

So Aspinwall's  $m$  is our momentum number, his  $n$  is our winding number, and the lattice isometry  $e \leftrightarrow e^*$  maps  $t \mapsto 1/t$ , i.e.  $R \mapsto \alpha'/R$ : T-duality (Chapter 8). *Numbers*: for  $R = 2\sqrt{\alpha'}$ ,  $t = 4$ , the state  $(n, m) = (1, -1)$  has  $p_L^2 = 9/8$ ,  $|p_R|^2 = 25/8$  and  $p^2 = p_L^2 - |p_R|^2 = -2 = 2nm$ . At the self-dual radius  $t = 1$  the same state has  $p_L = 0$  and  $|p_R|^2 = 2$ : a *root* of norm  $-2$  has become orthogonal to  $\Pi$ . This is the mechanism of Section 23.6 in its smallest example (Figure 23.1).

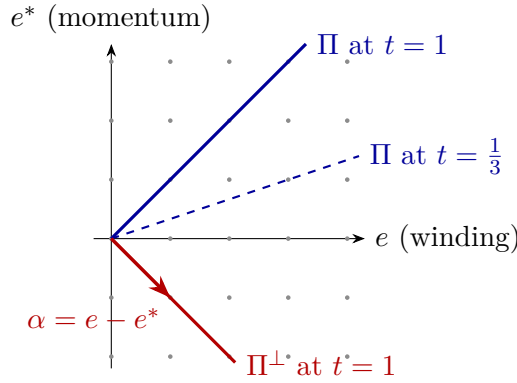


Figure 23.1: The Narain lattice  $\Gamma^{1,1}$  of a circle. The modulus  $t = R^2/\alpha'$  is the slope of the positive line  $\Pi$ . At  $t = 1$  the root  $\alpha = e - e^*$ ,  $\alpha^2 = -2$ , lies in  $\Pi^\perp$ : it has  $p_L = 0$ ,  $|p_R|^2 = 2$ , and gives a massless charged vector. Orthogonality is with respect to the hyperbolic form, for which the axes  $e, e^*$  are null.

### 23.4.3 The heterotic string on $T^4$

The heterotic string (Chapter 5) is a superstring on one side and a bosonic string on the other; the bosonic side has  $26 - 10 = 16$  extra compact bosons. Following Aspinwall we let the *left*-movers be the supersymmetric side — the opposite of the usual convention, forced by the comparison with K3 [Asp §3.5, footnote 11, ll. 2007–2011]. On  $T^4$  we then have  $n_L = 4$ ,  $n_R = 4 + 16 = 20$ , and the moduli space is (23.1) on the nose. In coordinates (23.15):

$$\underbrace{\frac{1}{2} \cdot 4 \cdot 5}_{G: 10} + \underbrace{\frac{1}{2} \cdot 4 \cdot 3}_{B: 6} + \underbrace{4 \times 16}_{\text{Wilson lines } A: 64} = 80. \quad (23.18)$$

The same 80 that was  $58 + 22$  on the K3 side is  $10 + 6 + 64$  here: the two descriptions cut the same space into entirely different pieces.

## 23.5 Type IIA on K3 and the duality with the heterotic string

### 23.5.1 Supersymmetry in six dimensions

We now pass from the worldsheet to spacetime: type IIA string theory on  $\mathbb{R}^{1,5} \times X$ . How much supersymmetry survives? Under  $\mathfrak{so}(1, 9) \supset \mathfrak{so}(1, 5) \oplus \mathfrak{so}(4)$  and  $\mathfrak{so}(4) \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$  the ten-dimensional spinor decomposes as

$$\mathbf{16} \rightarrow (\mathbf{4}, \mathbf{2}, \mathbf{1}) \oplus (\mathbf{4}', \mathbf{1}, \mathbf{2}). \quad (23.19)$$

The holonomy  $SU(2)$  of K3 is one of the two  $\mathfrak{su}(2)$  factors, say the second. A supercharge survives if it is invariant under holonomy (it must be covariantly constant), so only  $(\mathbf{4}, \mathbf{2}, \mathbf{1})$  survives: 8 of the 16 real components, one minimal spinor in six dimensions. *K3 preserves half of the supersymmetry that a torus would* [Asp §4.1, eq. (89), ll. 2268–2289]. Type IIA has two ten-dimensional supercharges of opposite chirality, 32 real components; on K3 we keep 16, and the opposite chiralities give  $N = (1, 1)$  in six dimensions [Asp §4.2, ll. 2333–2336]. The heterotic string has  $N = (1, 0)$  in ten dimensions: 16 supercharges, all preserved by the flat  $T^4$ , again  $N = (1, 1)$  in six dimensions [Asp §4.2, ll. 2374–2376].

### 23.5.2 The moduli space of the string theory

The holonomy of the moduli space of an  $N = (1, 1)$  theory in six dimensions is  $\mathfrak{sp}(1) \oplus \mathfrak{sp}(1) \cong \mathfrak{so}(4)$ . Scalars sit in two kinds of multiplets: the gravity multiplet contains one real scalar, the dilaton; each matter (vector) multiplet contains four scalars in the  $\mathbf{4}$  of  $\mathfrak{so}(4)$ . Hence the Teichmüller space is  $O(4, m)/(O(4) \times O(m)) \times \mathbb{R}$  with  $m$  the number of matter multiplets [Asp §4.2, eq. (91), ll. 2337–2346]. Since the R–R fields give no scalars (pitfall box above), comparison with Theorem 23.4 gives  $m = 20$  and

$$\mathcal{M}_{\text{IIA}} \cong \mathcal{M}_\sigma \times \mathbb{R}, \quad \dim = 80 + 1 = 81, \quad (23.20)$$

the conformal field theories sitting at the boundary  $\Phi \rightarrow -\infty$  of weak coupling [Asp §4.2, eq. (92), ll. 2354–2362]. The form  $O(4, m)$  is obtained here from spacetime supersymmetry alone, independently of Section 23.2; only  $m = 20$  is borrowed. Now 4 is the number of scalars in a multiplet and 20 the number of multiplets.

The gauge fields come from the R–R sector: the one-form gives one  $U(1)$ ; the three-form  $C_3$  reduced on the 22 two-cycles gives 22; and the six-dimensional dual of  $C_3$  ( $dC_3^* = *dC_3$ , a one-form in six dimensions) gives one more:

$$1 + 22 + 1 = 24 = b_0 + b_2 + b_4 = \text{rk } H^*(X, \mathbb{Z}) = \text{rk } \Gamma^{4,20} \quad (23.21)$$

[Asp §4.3, eq. (94), ll. 2434–2440]. In the standard multiplet structure 4 of the 24 vectors belong to the gravity multiplet and 20 to the matter multiplets (standard; not checked against a source in this book’s library). For the heterotic string on  $T^4$  the generic gauge group is also  $U(1)^{24}$ : 16 from the extra right-moving bosons, 4 Kaluza–Klein vectors from the metric, 4 from the  $B$ -field ( $b_1(T^4) = 4$ ) [Asp §4.3, ll. 2444–2449]. The charge lattice is  $\Gamma^{4,20}$  in both cases: the Narain lattice on one side,  $H^*(X, \mathbb{Z})$  on the other (Chapter 22).

### 23.5.3 The duality statement

**Conjecture 23.5** (String–string duality; Aspinwall’s Proposition 1, [L]). *The type IIA string compactified on a K3 surface is equivalent to the heterotic string compactified on  $T^4$ . The*



moduli spaces  $\mathcal{M}_\sigma \times \mathbb{R}$  are identified in the obvious way on  $\mathcal{M}_\sigma$ , while strong coupling of one theory is mapped to weak coupling of the other:

$$\Phi_{\text{Het}} = -\Phi_{\text{IIA}}, \quad g_{6,\text{Het}} = e^{2\Phi_{\text{Het}}} g_{6,\text{IIA}}, \quad (23.22)$$

where  $g_6$  is the six-dimensional spacetime metric and  $\Phi$  the dilaton [Asp §4.2, Proposition 1 and eq. (93), ll. 2377–2403].

We call it a conjecture because our source is explicit that it cannot be proved with the present definition of string theory; it is supported by the agreement of the moduli spaces, of the six-dimensional effective actions, and of the soliton spectra, and “nothing has been discovered . . . to disprove” it [Asp §4.2, ll. 2391–2413]. It is a statement in the literature (Tier [L] as a *conjecture with strong evidence*), not a theorem. Notice what is *not* claimed: the two worldsheet theories are not equivalent conformal field theories. The 80-dimensional spaces agree, but the map must reverse the sign of the dilaton, so that the perturbative expansion of one side is invisible in the other [Asp §4.2, ll. 2377–2388].

The rescaling in (23.22) has a simple consequence. A tension is a mass per length, i.e. it scales as (length)<sup>−2</sup>. If  $g_{\text{Het}} = e^{2\Phi_{\text{Het}}} g_{\text{IIA}}$ , lengths measured in the heterotic metric are  $e^{\Phi_{\text{Het}}}$  times IIA lengths, so an object of tension  $T$  in IIA units has tension

$$T^{(\text{Het units})} = e^{-2\Phi_{\text{Het}}} T = \frac{T}{\lambda_{\text{Het}}^2}. \quad (23.23)$$

The fundamental IIA string,  $T = 1/(2\pi\alpha')$ , is therefore a heavy object of tension  $\propto 1/\lambda_{\text{Het}}^2$  in the weakly coupled heterotic theory — the characteristic coupling dependence of a soliton — and vice versa. Each string is a soliton of the other; see the references in [Asp §4.2, ll. 2404–2409].

#### Historical note

The local form of the moduli space of  $N = (4, 4)$  theories is due to Seiberg (1988); the global form with the lattice  $\Gamma^{4,20}$  was obtained by Aspinwall and Morrison. The toroidal moduli space is due to Narain (1986) and to Narain and collaborators (1987). The duality between type IIA on K3 and the heterotic string on  $T^4$  was suggested by Hull and Townsend and by Witten (1995), and the appearance of enhanced gauge symmetry at orbifold points of K3 was first shown by Witten in the same paper [Asp, reference list [50], [52], [61], [62], [70], [71], ll. 5355–5398; §3.2, ll. 1671–1673; §3.3, ll. 1731–1732; §4.3, l. 2524].

## 23.6 Enhanced gauge symmetry

### 23.6.1 Heterotic side: roots orthogonal to $\Pi$

At special points of a Narain moduli space extra massless vectors appear (Chapter 8:  $SU(2) \times SU(2)$  at the self-dual radius). In the heterotic string a spacetime vector is created by a vertex operator whose supersymmetric (left) side carries the vector index,  $\partial X^\mu$  or its superpartner, and whose bosonic (right) side is any operator of weight  $\bar{h} = 1$ . Taking the right-moving operator to be a pure exponential of charge  $p$ , (23.14) requires  $-\frac{1}{2}p_R \cdot p_R = 1$ , and masslessness requires that no weight comes from the left momentum,  $p_L = 0$ . Hence

$$\mathcal{A} = \{\alpha \in \Gamma^{4,20} \cap \Pi^\perp : \alpha \cdot \alpha = -2\} \quad (23.24)$$

is the set of extra massless vectors [Asp §4.3, eq. (95), ll. 2465–2489]. Their charges under  $U(1)^{24}$  are the coordinates of  $\alpha$ , so  $\mathcal{A}$  is the root system of the non-abelian part of the gauge group,

with Cartan subalgebra inside  $\mathfrak{u}(1)^{24}$ . Three consequences follow at once: the rank stays 24; all roots have the same length, so the group is simply laced, of type  $A-D-E$ ; and since  $\Pi^\perp$  is negative-definite of dimension 20, the non-abelian part has rank at most 20.

Why only norm  $-2$  and not  $+2$ ? On the supersymmetric side the GSO projection prevents new left-moving states from becoming massless [Asp §4.3, ll. 2456–2461]. In the bosonic string on a circle both  $\alpha = e - e^*$  (norm  $-2$ ,  $p_L = 0$ ) and  $\beta = e + e^*$  (norm  $+2$ ,  $p_R = 0$ ) give massless vectors at  $t = 1$ , hence  $SU(2) \times SU(2)$ ; the heterotic string keeps only the first.

**Mass of the  $W$ -bosons near the special point.** Level matching fixes the mass of these states away from the enhanced point. We use the standard heterotic mass-shell conditions,  $\frac{\alpha'}{4}M^2 = \frac{1}{2}p_L^2 + N_L - \frac{1}{2} = \frac{1}{2}|p_R|^2 + N_R - 1$ , with the normal-ordering constants  $-\frac{1}{2}$  (NS sector) and  $-1$  (bosonic) of Chapters 3 and 5 (standard; not checked against a source in this book’s library). For a vector,  $N_L = \frac{1}{2}$  and  $N_R = 0$ . Both expressions agree precisely when  $p_L^2 - |p_R|^2 = -2$ , i.e.  $\alpha^2 = -2$ , and then

$$M^2 = \frac{2}{\alpha'} p_L(\alpha)^2, \quad (23.25)$$

where  $p_L(\alpha)$  is the orthogonal projection of  $\alpha$  to  $\Pi$ . The mass is the “distance” of the root from  $\Pi^\perp$ : the Higgs mechanism, drawn in a lattice.

**Example 23.6** ( $E_8 \times E_8$  without Wilson lines). Split  $\Gamma^{4,20} \cong \Gamma^{4,4} \oplus E_8(-1) \oplus E_8(-1)$  and take  $\Pi \subset \Gamma^{4,4} \otimes \mathbb{R}$ , i.e.  $A = 0$  in (23.15). Every vector of  $E_8(-1) \oplus E_8(-1)$  is orthogonal to  $\Pi$ , in particular all  $240 + 240$  roots of norm  $-2$ . The gauge group contains  $E_8 \times E_8$ , of dimension  $248 + 248$ , and the generic  $\Gamma^{4,4}$  part adds  $U(1)^8$ : total rank  $8 + 8 + 8 = 24$ . The largest possible enhancement has non-abelian rank 20, for instance  $E_8 \times E_8 \times SU(2)^4 \times U(1)^4$  [Asp §4.3, ll. 2490–2499 and 2551–2554].

### 23.6.2 Type IIA side: shrinking two-spheres

In type IIA perturbation theory the gauge fields are R–R fields, under which no perturbative string state is charged; the conformal field theory can only ever see  $U(1)^{24}$  [Asp §4.3, ll. 2440–2443]. If Conjecture 23.5 holds, the non-abelian gauge symmetry must nevertheless be there. Where?

Choose  $w$  as in Section 23.3; from now on  $\alpha$  denotes a root, and the coefficient called  $\alpha$  in (23.8) is written  $\alpha_{\text{vol}}$ . Suppose all roots  $\alpha \in \mathcal{A}$  lie in  $w^\perp/w = H^2(X, \mathbb{Z})$ . Use the basis (23.9) of  $\Pi$ . Since  $\alpha \cdot w = \alpha \cdot w^* = 0$ ,

$$\alpha \cdot \xi_i = \alpha \cdot \sigma_i, \quad \alpha \cdot B' = \alpha \cdot B, \quad (23.26)$$

so  $\alpha \perp \Pi$  is equivalent to *two* conditions:

1.  $\alpha \perp \Sigma$ . A class with  $\alpha^2 = -2$  orthogonal to the holomorphic two-form lies in the Picard lattice and  $\pm\alpha$  is the class of a rational curve  $C \cong S^2$ ; if moreover  $J \cdot \alpha = 0$  this curve has zero area, and  $X$  has an orbifold singularity there [Asp §2.6, Theorem 4, ll. 1196–1211] (Chapter 21). The roots orthogonal to  $\Sigma$  form the root system of the  $A-D-E$  singularity:  $\mathbb{C}^2/\mathbb{Z}_n$  corresponds to  $A_{n-1}$ , i.e.  $SU(n)$ .
2.  $\alpha \cdot B = 0$ : the  $B$ -field has no component along the vanishing cycle.

**Proposition 23.7** (Aspinwall’s Proposition 2 (Witten), [L], conditional on Conjecture 23.5). *A type IIA string on a K3 surface with an orbifold singularity of type  $A-D-E$ , with vanishing  $B$ -field through the collapsed cycles, has non-abelian gauge symmetry with the simply-laced Lie algebra of the same  $A-D-E$  type [Asp §4.3, Proposition 2, ll. 2523–2534].*



The volume  $V$  does not enter (23.26): the enhancement happens for a K3 of any size. The physical mechanism, outside perturbation theory, is that R–R charged solitons — D2-branes in the language of Chapter 13 — wrap the two-sphere; their mass is proportional to its area and they become massless when it shrinks [Asp §4.3, ll. 2555–2563]. They are the  $W$ -bosons.

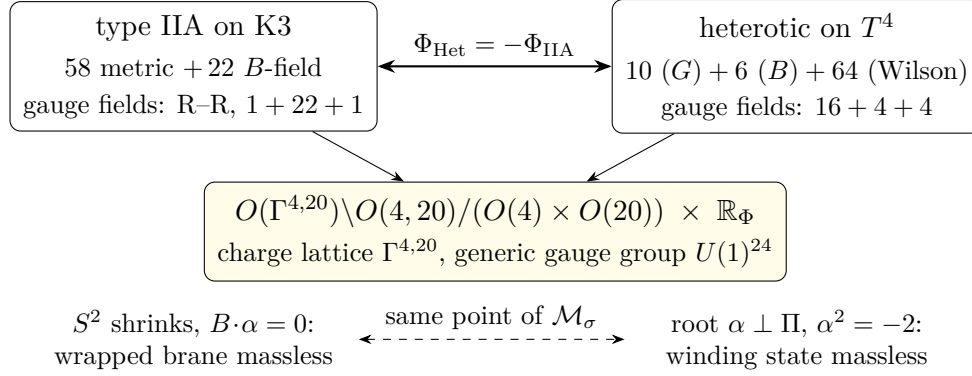


Figure 23.2: The two descriptions of the six-dimensional  $N = (1, 1)$  theory. The 80 moduli are counted differently on the two sides; the enhanced gauge symmetry is perturbative on the heterotic side and non-perturbative on the type IIA side.

### 23.6.3 Worked example: an $A_1$ singularity and the role of $B$

Let  $\alpha \in H^2(X, \mathbb{Z})$  be the class of a rational curve,  $\alpha^2 = -2$ . The basis (23.9) is orthogonal with  $\xi_i^2 = 1$  and  $\xi_4^2 = V$ , so by (23.26) the projection of  $\alpha$  to  $\Pi$  has length

$$p_L(\alpha)^2 = \sum_{i=1}^3 (\alpha \cdot \sigma_i)^2 + \frac{(\alpha \cdot B)^2}{V}, \quad V = 2\alpha_{\text{vol}} + B \cdot B. \quad (23.27)$$

The first term is the squared “size” of the curve measured by the three normalized hyperkähler forms; the second is the  $B$ -field flux through it.

(a) *Small smooth sphere.* Take  $(\alpha \cdot \sigma_i) = (0.06, 0, 0.08)$  and  $\alpha \cdot B = 0$ . Then  $p_L^2 = 0.0036 + 0.0064 = 0.01$  and by (23.25)  $M^2 = 0.02/\alpha'$ ,  $M \approx 0.14/\sqrt{\alpha'}$ : a light charged vector pair  $\pm\alpha$ . At  $\alpha \cdot \sigma_i = 0$  the pair is massless and, with the  $U(1)$  under which it is charged, fills out  $SU(2)$ ; the gauge group is  $SU(2) \times U(1)^{23}$ , of dimension  $3 + 23 = 26$ .

(b) *The orbifold conformal field theory.* An orbifold such as  $T^4/\mathbb{Z}_2$  (Chapter 21) is a perfectly regular conformal field theory, although its 16 two-spheres have zero size. There is no contradiction: the orbifold conformal field theory has  $B$ -field flux  $\frac{1}{2}$  through each collapsed sphere [Asp §4.3, ll. 2535–2547], that is  $\alpha \cdot B = \frac{1}{2}$  in our notation. Then (23.27) gives  $p_L^2 = 1/(4V)$  and

$$M^2 = \frac{1}{2V\alpha'} \neq 0. \quad (23.28)$$

For  $V = 100$  (a large K3 in units of  $\alpha'^2$ ) this is  $M \approx 0.07/\sqrt{\alpha'}$ : light, but not massless. The singular conformal field theory sits at  $\alpha \cdot B = 0$ , half a period away. Because  $B \sim B + e$ , the flux  $\alpha \cdot B$  is periodic with period one, and  $\frac{1}{2}$  is the point farthest from the enhancement locus.

(c) *A stringy enhancement at small volume.* Roots need not lie in  $H^2$ . Take  $\alpha = w^* - w$ :  $\alpha^2 = -2w \cdot w^* = -2$ . Now  $\alpha \cdot \xi_i = -(w^* - w) \cdot w(\sigma_i \cdot B) = -\sigma_i \cdot B$  and  $\alpha \cdot B' = \alpha_{\text{vol}} - 1$ . So  $\alpha \perp \Pi$  requires  $B \perp \Sigma$  and  $\alpha_{\text{vol}} = 1$ ; with  $B = 0$  this is  $V = 2$ . A *smooth* K3 surface of volume of order one in string units has enhanced  $SU(2)$  symmetry, whatever its shape. This is the K3 counterpart of the self-dual radius of Section 23.4.2, where the root  $e - e^*$

also pairs winding with momentum; Aspinwall states the general conclusion, that if the roots are not all orthogonal to  $w$  the surface may be smooth but its volume is of order one [Asp §4.3, ll. 2541–2550]. (The specific value  $V = 2$  depends on the normalization (23.11); the computation here is ours.)

**Common pitfall: “singular K3  $\Rightarrow$  non-abelian gauge symmetry”**

An orbifold singularity is necessary (for roots in  $H^2$ ) but not sufficient: the  $B$ -field through the vanishing cycle must also vanish. Conversely a smooth, small K3 can have enhanced symmetry, example (c). The locus of enhanced symmetry has real codimension *four* in  $\mathcal{M}_\sigma$  — three conditions on the metric and one on  $B$ , matching the four scalars of the vector multiplet that becomes part of the non-abelian theory.

*Remark 23.8* (Type IIB on K3). For type IIB the six-dimensional theory is chiral,  $N = (2, 0)$ , the holonomy is  $\mathfrak{sp}(2) \cong \mathfrak{so}(5)$ , and the R–R potentials of even degree *do* give scalars:  $b_0 + b_2 + b_4 = 24$ . With the dilaton the count is  $58 + 22 + 1 + 1 + 22 + 1 = 105 = 5 \times 21$  and the moduli space is  $O(\Gamma^{5,21}) \backslash O(5, 21) / (O(5) \times O(21))$ , in which the dilaton is no longer a separate factor [Asp §4.4, eq. (97) and table, ll. 2573–2621]. The pattern  $(3, 19) \rightarrow (4, 20) \rightarrow (5, 21)$  continues on  $K3 \times T^2$  with the lattice of signature  $(6, 22)$  (Chapter 24).

## 23.7 What the machine has checked

No part of this chapter’s physics — sigma models, moduli spaces as quotients, string duality — is formalized. What is formalized are the lattice-theoretic facts on which the argument rests, in the library `DualScaleStream2.Lattice`, and a handful of numerical identities in the older library `StringTheoryFoundation`. We list them with their exact scope.

### 23.7.1 Signatures and ranks

The file `DualScaleStream2/Lattice/K3T2Signature.lean` defines a signature as a pair of natural numbers with componentwise addition, takes  $\text{sgn } U = (1, 1)$  and  $\text{sgn } E_8(-1) = (0, 8)$  as *inputs*, and computes.

**In Lean:  $(3, 19) + (1, 1) = (4, 20)$  and the mod-8 condition**

```
def sigK3 : Signature := sigE8Neg + sigE8Neg + sigU + sigU + sigU
def sigMukai : Signature := sigK3 + sigU

theorem sigMukai_eq : sigMukai = ⟨4, 20⟩ := by ...

theorem index_mod_eight :
  sigK3.index % 8 = 0 ∧ sigMukai.index % 8 = 0 ∧
  sigK3T2.index % 8 = 0 := by ...
```

**Reading.** Adding the pairs  $(0, 8) + (0, 8) + 3 \cdot (1, 1) + (1, 1)$  gives  $(4, 20)$ , and  $4 - 20 = -16$  is divisible by 8 (as are  $3 - 19$  and  $6 - 22$ ). **Proof idea.** Both sides reduce to the same numerals by unfolding definitions (`rf1`). **Scope.** This is bookkeeping on pairs of numbers. That  $U$  and  $E_8(-1)$  *have* these signatures, that  $H^*(K3, \mathbb{Z})$  is  $4U \oplus 2E_8(-1)$ , and the theorem that even unimodular lattices exist only for  $n_+ - n_- \equiv 0 \pmod{8}$  — which is what forces  $n_L - n_R \in 8\mathbb{Z}$  in Section 23.4 — are [L] inputs. The companion theorems `sigK3_eq`, `rank_K3` (22) and `sigK3_matches_hodge` were discussed in Chapter 20.

### 23.7.2 Evenness of the Mukai lattice and the $B$ -shift

In Proposition 23.3 integrality of  $T_e$  rested on  $e \cdot e \in 2\mathbb{Z}$ . The statement that adjoining  $H^0 \oplus H^4$  preserves evenness is kernel-checked for an arbitrary even Gram matrix (Chapter 22):

**In Lean:** the extended lattice is even if  $H^2$  is

```
def mukaiPair {n : ℕ} (L : Gram n) (v w : MukaiVec n) : ℤ :=
  v.c ·v (L *v w.c) - v.r * w.s - v.s * w.r

theorem mukaiPair_even {n : ℕ} (L : Gram n) (hL : L⊤ = L) (heven : IsEvenDiag L)
  (v : MukaiVec n) : Even (mukaiPair L v v) := by ...

theorem hyperbolicUNeg_unimodular : IsUnimodular hyperbolicUNeg := ...
```

**Reading.** For a symmetric integer matrix  $L$  with even diagonal and any integer vector  $v = (r, c, s)$ , the number  $c^\top Lc - 2rs$  is even; and the  $2 \times 2$  Gram matrix  $\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$  of  $H^0 \oplus H^4$  has determinant  $\pm 1$ . **Proof idea.**  $c^\top Lc = \sum_i L_{ii}c_i^2 + 2\sum_{i<j} L_{ij}c_ic_j$  is even by the general lemma `even_quadratic_form_of_even_diag`, and  $2rs$  is even. **Scope.** Note the sign: the Mukai pairing uses  $U(-1)$  on  $H^0 \oplus H^4$ , Aspinwall's  $w, w^*$  span  $U$ . Since  $U(-1) \cong U$  (send  $w^* \mapsto -w^*$ ) this is a convention, not a discrepancy, but the isomorphism is not a Lean theorem. The map  $T_e$  of (23.13) itself is *not* formalized; see Exercise 23.8.

### 23.7.3 Roots and reflections

The enhanced gauge symmetry of Section 23.6 is governed by vectors of norm  $-2$ . For each such vector the reflection  $s_\alpha(x) = x + (x \cdot \alpha)\alpha$  is an isometry of the lattice; these reflections generate the Weyl group of the enhanced gauge algebra, and on the K3 side they are the Picard–Lefschetz monodromies around the singular surface (Chapter 20). The algebraic core is Tier [\[A\]](#):

**In Lean:** reflections in  $(-2)$ -vectors are integral isometries

```
def reflection (L : Gram n) (v : Fin n → ℤ) : Gram n :=
  1 + vecMulVec v v * L

theorem reflection_isometry (L : Gram n) (hL : L⊤ = L) (v : Fin n → ℤ)
  (hv : latticeNorm L v = -2) :
  (reflection L v)⊤ * L * reflection L v = L := by ...

theorem e8Neg_simpleRoot_norm (i : Fin 8) :
  latticeNorm e8Neg (Pi.single i 1) = -2 := by ...
```

**Reading.** For any symmetric integer Gram matrix  $L$  and any integer vector  $v$  with  $v^\top Lv = -2$ , the integer matrix  $S = 1 + vv^\top L$  satisfies  $S^\top LS = L$ . The eight basis vectors of  $E_8(-1)$  have norm  $-2$ . **Proof idea.** Expand  $S^\top LS = L + 2Lv v^\top L + (v^\top Lv) L v v^\top L$  and insert  $-2$ . **Scope.** This is the statement used in Example 23.6 only for the eight simple roots of one  $E_8(-1)$ ; the count of 240 roots, and the identification of  $\mathcal{A}$  with a gauge algebra, are [\[L\]](#). (Declarations in `DualScaleStream2/Lattice/Reflection.lean`.)

### 23.7.4 Arithmetic shadows of the duality

The older file `StringTheoryFoundation/StringTheory/WittenDuality.lean` is named after the duality but contains no string theory. Its four theorems are:

**In Lean: four numerical identities — read the statements, not the names**

```

def wittenModuliDimension (p q : Nat) : Nat := p * q
theorem witten_moduli_dim_is_80 :
  wittenModuliDimension 4 20 = 80 := by decide

theorem witten_duality_rank_match :
  defaultDualityLattice.b2_K3 + 2
  = defaultDualityLattice.t4_momentum + defaultDualityLattice.t4_gauge := by
  decide

theorem witten_6d_supercharges :
  defaultSixDSusy.numSupercharges = 16 := by decide

def exhibitsGaugeEnhancement (c : TwoCycle) : Prop :=
  c.intersectionSelf = -2 ∧ c.area = 0
theorem witten_gauge_enhancement_at_singularity :
  exhibitsGaugeEnhancement ⟨-2, 0⟩ := by ...

```

**Reading.** (i)  $4 \cdot 20 = 80$ . (ii) For a record with default fields 22, 4, 20:  $22 + 2 = 4 + 20$ . (iii) A record field defined as  $32/2$  equals 16. (iv) The pair  $(-2, 0)$  satisfies the predicate “first component is  $-2$  and second component is 0”. **Scope.** These are arithmetic shadows of (23.6), (23.21) and of Section 23.6. In (iv) the word “gauge enhancement” is the *name* of a predicate on two numbers; nothing about gauge fields is proved, and the  $B$ -field condition of Proposition 23.7 is absent from the predicate. The informal comments in that file also describe the 22 extra moduli as Ramond–Ramond periods and include the dilaton in the 80, which contradicts our source (pitfall box in Section 23.2); comments are not checked by the kernel. A faithful formalization would need at least the Grassmannian of positive four-planes in a real quadratic space of signature  $(4, 20)$  with  $O(4U \oplus 2E_8(-1))$  acting on it; Propositions 23.2 and 23.3 are elementary linear algebra and realistic first targets (Chapter 38).

**Chapter summary**

- A string on a K3 surface is described by an  $N = (4, 4)$  superconformal sigma model; the Ricci-flat metric is an exact solution. Its parameters are the metric (58) and the class of the  $B$ -field (22):  $80 = 4 \times 20$  in all. The dilaton is not among them.
- Locally the moduli space is the Grassmannian  $O(4, 20)/(O(4) \times O(20))$  of positive four-planes  $\Pi \subset \mathbb{R}^{4,20}$ ; globally one divides by  $O(\Gamma^{4,20})$ , the isometry group of the Mukai lattice  $H^*(X, \mathbb{Z})$  (under completeness and Hausdorff assumptions).
- After choosing a null vector  $w$ ,  $\Pi$  is spanned by  $\sigma_i - (\sigma_i \cdot B)w$  and  $B' = \alpha w + w^* + B$ , with  $B' \cdot B' = 2\alpha + B^2 = V$ . Integral  $B$ -shifts are lattice isometries because the lattice is even. Changing  $w$  is a T-duality-like relabelling.
- The same coset is the Narain moduli space of the heterotic string on  $T^4$  ( $10 + 6 + 64 = 80$ ); both theories have  $N = (1, 1)$  supersymmetry in six dimensions and generic gauge group  $U(1)^{24}$ . This supports the conjectural duality  $\text{IIA}/\text{K3} = \text{Het}/T^4$  with  $\Phi \mapsto -\Phi$ .
- Extra massless vectors are roots  $\alpha \in \Gamma^{4,20} \cap \Pi^\perp$ ,  $\alpha^2 = -2$ ; the gauge group is simply laced of rank 24. On the K3 side this means an  $A$ – $D$ – $E$  orbifold singularity *and* vanishing  $B$ -flux through the collapsed spheres; the orbifold conformal field theory has flux  $\frac{1}{2}$  and is regular.

| Claim                                                                                                        | Where                                                                                                                     | Tier                     |
|--------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------|
| $(3, 19) + (1, 1) = (4, 20)$ ; $\text{index} \equiv 0 \pmod{8}$ at $(3, 19), (4, 20), (6, 22)$               | <code>sigMukai_eq</code> , <code>index_mod_eight</code>                                                                   | <b>[A]</b>               |
| $H^2$ even $\Rightarrow H^0 \oplus H^2 \oplus H^4$ even; $U(-1)$ unimodular                                  | <code>mukaiPair_even</code> ,<br><code>hyperbolicUNeg_unimodular</code>                                                   | <b>[A]</b>               |
| Reflection in a $(-2)$ -vector is an integral isometry; simple roots of $E_8(-1)$ have norm $-2$             | <code>reflection_isometry</code> ,<br><code>e8Neg_simpleRoot_norm</code>                                                  | <b>[A]</b>               |
| $4 \cdot 20 = 80$ ; $22 + 2 = 4 + 20$ ; $32/2 = 16$                                                          | <code>witten_moduli_dim_is_80</code> ,<br><code>witten_duality_rank_match</code> ,<br><code>witten_6d_supercharges</code> | <b>[A]</b> (shadow)      |
| Propositions 23.2 and 23.3; formulas (23.25), (23.27); example (c)                                           | derived here; sympy check                                                                                                 | —                        |
| K3 sigma model is $N = (4, 4)$ , metric exact                                                                | Asp §3.1                                                                                                                  | <b>[L]</b>               |
| $\mathcal{M}_\sigma = O(\Gamma^{4,20}) \backslash O(4, 20) / (O(4) \times O(20))$ (under stated assumptions) | Asp §3.2–3.4, Thms. 5, 6                                                                                                  | <b>[L]</b>               |
| $V = 2\alpha + B^2$                                                                                          | Asp §3.3, eq. (68)                                                                                                        | <b>[L]</b>               |
| IIA on K3 $\leftrightarrow$ heterotic on $T^4$ , $\Phi \mapsto -\Phi$                                        | Asp §4.2, Prop. 1                                                                                                         | <b>[L]</b> (conjecture)  |
| $A$ – $D$ – $E$ enhanced gauge symmetry at orbifold points with $B \cdot \alpha = 0$                         | Asp §4.3, Prop. 2                                                                                                         | <b>[L]</b> (conditional) |
| Identification of any Lean object above with a physical system                                               | —                                                                                                                         | never <b>[A]</b>         |

Table 23.1: Tier table for this chapter. The **[A]** entries depend only on the axioms `propext`, `Classical.choice`, `Quot.sound`. Rows marked “derived here” are ordinary pen-and-paper mathematics of this chapter, checked symbolically but not formalized.

- Machine-checked: signature and parity bookkeeping, reflections in  $(-2)$ -vectors, numerical identities. Not machine-checked: anything about sigma models, moduli spaces, duality.

## Exercises

**Exercise 23.1** (★). Verify (23.6) and evaluate it for  $(3, 19)$ ,  $(4, 20)$ ,  $(5, 21)$  and  $(6, 22)$ . Match the first three numbers with a count of physical fields ((23.3) without the volume, (23.4), Remark 23.8).

**Exercise 23.2** (★). For the circle of Section 23.4.2 with  $t = R^2/\alpha' = 9$ , compute  $p_L^2$  and  $|p_R|^2$  for  $(n, m) = (1, -1)$ ,  $(1, 1)$  and  $(0, 1)$ , and check  $p_L^2 - |p_R|^2 = 2nm$  in each case. Which of the three states is a root? Find all  $t$  at which some root of  $\Gamma^{1,1}$  is orthogonal to  $\Pi$ .

**Exercise 23.3** (★). List the 24 abelian gauge fields of type IIA on K3 and of the heterotic string on  $T^4$  by their ten-dimensional origin. Which Lean theorem of Section 23.7 records the equality of the two counts, and what exactly does it state?

**Exercise 23.4** (★★). Show that the maps  $T_e$  of Proposition 23.3 satisfy  $T_e \circ T_f = T_{e+f}$ , and that for  $g \in O(\Gamma^{3,19})$  (extended by the identity on  $w, w^*$ ) one has  $g T_e g^{-1} = T_{g(e)}$ . Conclude that the  $T_e$  and  $O(\Gamma^{3,19})$  generate the semidirect product  $O(\Gamma^{3,19}) \ltimes \Gamma^{3,19}$ .

**Exercise 23.5** (★★). Let  $\alpha_1, \alpha_2 \in H^2(X, \mathbb{Z})$  with  $\alpha_1^2 = \alpha_2^2 = -2$ ,  $\alpha_1 \cdot \alpha_2 = 1$ , both orthogonal to  $\Pi$ . Show that  $\mathcal{A}$  contains exactly the six vectors  $\pm\alpha_1, \pm\alpha_2, \pm(\alpha_1 + \alpha_2)$  among integer combinations of  $\alpha_1, \alpha_2$ , identify the gauge group, give its dimension, and name the orbifold singularity of  $X$  [Asp §2.6, ll. 1227–1240].

**Exercise 23.6** (★★, Lean). In the style of `sigMukai_eq`, define `sigIIB : Signature := sigMukai + sigU` and prove `sigIIB = ⟨5, 21⟩`, `sigIIB.rank = 26` and `sigIIB.index % 8 = 0`. State in one sentence what these theorems do and do not say about type IIB string theory.

**Exercise 23.7** (★★, Lean). Using `mukaiPair` with an arbitrary `L : Gram n`, prove that the vector  $(r, c, s) = (1, 0, 1)$  has self-pairing  $-2$  (this is `structureSheaf_mukai_sq`) and that  $(1, 0, -1)$  has self-pairing  $+2$ . Relate the two vectors to the roots  $w^* \mp w$  of example (c) in Section 23.6.3, taking the sign convention  $U(-1)$  versus  $U$  into account.

**Exercise 23.8** (★★★, Lean). Formalize Proposition 23.3 for a general symmetric even Gram matrix `L : Gram n`: write the  $(n + 2) \times (n + 2)$  block Gram matrix of  $U \oplus L$ , the integer matrix of  $T_e$  (you will need  $(e \cdot_v (L *_v e)) / 2$  and the hypothesis `IsEvenDiag L`), and prove  $T_e^T (U \oplus L) T_e = U \oplus L$ . Which lemma of Section 23.7 supplies the divisibility by two?

**Exercise 23.9** (★★★). Work out (23.15) for the heterotic string on a circle with one Wilson line component switched on:  $\Gamma^{1,17} = U \oplus E_8(-1) \oplus E_8(-1)$ ,  $\Pi = \mathbb{R}(e + te^* + a)$  with  $a \in E_8(-1) \otimes \mathbb{R}$  in the first factor. Show that  $\Pi$  is positive iff  $2t + a \cdot a > 0$ , and that a root  $\beta$  of  $E_8(-1)$  remains orthogonal to  $\Pi$  iff  $\beta \cdot a = 0$ . For  $a$  proportional to a fundamental coweight, which subgroup of  $E_8$  survives? (Compare  $2t + a \cdot a$  with (23.10): the Wilson line plays the role of the  $B$ -field.)

## Further reading

- The sigma model, the count  $58 + 22$ , and Theorem 5: [Asp §3.1–3.2, ll. 1559–1725]. The null vector  $w$ , the decomposition into metric,  $B$ -field and volume, and the space group: [Asp §3.3, ll. 1727–1832]. Mirror symmetry for K3 and Theorem 6: [Asp §3.4, ll. 1836–1963].
- Narain moduli spaces as Grassmannians, including the heterotic case and the circle: [Asp §3.5, ll. 1967–2170]; compare Chapters 10 and 11.
- Supersymmetry counting, the IIA moduli space and the duality conjecture: [Asp §4.1–4.2, ll. 2189–2413]. Enhanced gauge symmetry: [Asp §4.3, ll. 2428–2566]. Type IIB and  $O(5, 21)$ : [Asp §4.4, ll. 2571–2640].
- $A$ – $D$ – $E$  singularities and  $(-2)$ -curves: [Asp §2.6, ll. 1172–1224]; Chapter 21. The Mukai lattice as  $H^*(X, \mathbb{Z})$  and its isometries from derived equivalences: Chapter 22. What becomes of  $\Gamma^{4,20}$  on  $K3 \times T^2$ : Chapter 24. Symmetries of K3 sigma models acting on  $\Gamma^{4,20}$  and their relation to  $M_{24}$ : Chapter 30.

## Part V

# **$K3 \times T^2$ and Fluxes**





## Chapter 24

# Compactification on $K3 \times T^2$

The ten-dimensional superstring has six dimensions too many. Chapter 23 removed four of them by wrapping the string on a K3 surface and found a six-dimensional theory whose moduli space is governed by the lattice  $\Gamma^{4,20}$ . This chapter removes the last two on a two-torus and asks the question a student should ask of any compactification: *what four-dimensional theory comes out?* Concretely: what is the topology of the six-dimensional internal space  $K3 \times T^2$ ; how much supersymmetry survives and why; how many massless scalars and gauge fields appear; on which space do the scalars take their values; and which discrete group identifies apparently different compactifications. The answers are unusually clean. The product  $K3 \times T^2$  is, up to free quotients, the *only* compact six-dimensional space of holonomy  $SU(2)$ , the four-dimensional theory has  $N = 4$  supersymmetry, and its 134 scalars and 28 gauge fields are organised by a single even unimodular lattice of signature  $(6, 22)$ . That lattice is obtained from the K3 lattice of Chapter 20 by adding hyperbolic planes, and the arithmetic of this addition is one of the things the Lean library checks. At the end of the chapter we lower the supersymmetry to  $N = 2$  and  $N = 1$  with orientifolds and fluxes; this is the setting of the tadpole condition of Chapter 25.

Two pieces of input are used throughout. From Chapter 19: a K3 surface  $X$  is a compact simply connected complex surface with trivial canonical bundle; its Betti numbers are  $(1, 0, 22, 0, 1)$ , its Hodge numbers are  $h^{2,0} = h^{0,2} = 1$ ,  $h^{1,1} = 20$ , and its Euler number is  $\chi(X) = 24$ . From Chapter 12: the flat two-torus with metric  $G$  and  $B$ -field  $B_{12}$  is described by two points  $\tau, \rho$  of the upper half-plane.

### 24.1 The topology of the product

#### 24.1.1 The Künneth formula

Let  $M$  and  $N$  be compact manifolds. A closed  $p$ -form on  $M$  and a closed  $q$ -form on  $N$  can be pulled back to  $M \times N$  and wedged together to give a closed  $(p + q)$ -form. The *Künneth formula* states that, with real coefficients, nothing else is needed:

$$H^k(M \times N; \mathbb{R}) \cong \bigoplus_{p+q=k} H^p(M; \mathbb{R}) \otimes H^q(N; \mathbb{R}), \quad \text{so} \quad b_k(M \times N) = \sum_{p+q=k} b_p(M) b_q(N). \quad (24.1)$$

(This is a standard theorem of algebraic topology; it is quoted, not proved, in this book and is not checked against a source in the book's library.) A convenient way to organise the convolution in (24.1) is the *Poincaré polynomial*  $P_M(t) = \sum_k b_k(M) t^k$ , for which (24.1) reads  $P_{M \times N} = P_M P_N$ . Because the Euler number is  $\chi(M) = \sum_k (-1)^k b_k(M) = P_M(-1)$ ,

multiplicativity follows at once:

$$\chi(M \times N) = P_M(-1) P_N(-1) = \chi(M) \chi(N). \quad (24.2)$$

**Example 24.1** (Betti numbers of  $K3 \times T^2$ ). The torus  $T^2 = S^1 \times S^1$  has  $P_{S^1} = 1 + t$ , hence  $P_{T^2} = (1 + t)^2 = 1 + 2t + t^2$ : one point class, two independent one-cycles (the  $a$ - and  $b$ -cycles), one volume class. For a K3 surface  $P_X = 1 + 22t^2 + t^4$ . Multiplying,

$$\begin{aligned} P_{K3 \times T^2}(t) &= (1 + 22t^2 + t^4)(1 + 2t + t^2) \\ &= 1 + 2t + (22 + 1)t^2 + (2 \cdot 22)t^3 + (1 + 22)t^4 + 2t^5 + t^6, \end{aligned} \quad (24.3)$$

so that

$$(b_0, \dots, b_6) = (1, 2, 23, 44, 23, 2, 1), \quad \chi(K3 \times T^2) = 1 - 2 + 23 - 44 + 23 - 2 + 1 = 0. \quad (24.4)$$

Each number has a face. The two one-cycles are those of the torus, because a K3 surface has none. The 23 two-cycles are the 22 two-cycles of K3 together with the torus itself. The 44 three-cycles are products of a two-cycle of K3 with a one-cycle of the torus; there is no other way to make degree three, since  $b_1(X) = b_3(X) = 0$ . Poincaré duality,  $b_k = b_{6-k}$ , is visible in (24.4). The Euler number vanishes because  $\chi(T^2) = 1 - 2 + 1 = 0$ ; by (24.2) *any* product with a torus has  $\chi = 0$ .

### 24.1.2 The Hodge diamond

Both factors are Kähler, and the Künneth decomposition respects the type of a form:  $h^{p,q}(M \times N) = \sum h^{a,b}(M) h^{p-a, q-b}(N)$ . In terms of the Hodge polynomial  $h_M(x, y) = \sum h^{p,q} x^p y^q$  this is again a product. With  $h_X = 1 + x^2 + y^2 + 20xy + x^2 y^2$  and  $h_{T^2} = (1 + x)(1 + y)$  one finds, for example,

$$\begin{aligned} h^{1,1}(K3 \times T^2) &= h^{1,1}(X) h^{0,0}(T^2) + h^{0,0}(X) h^{1,1}(T^2) = 20 + 1 = 21, \\ h^{2,1}(K3 \times T^2) &= h^{2,0}(X) h^{0,1}(T^2) + h^{1,1}(X) h^{1,0}(T^2) = 1 + 20 = 21, \\ h^{3,0}(K3 \times T^2) &= h^{2,0}(X) h^{1,0}(T^2) = 1, \quad h^{1,0}(K3 \times T^2) = h^{1,0}(T^2) = 1. \end{aligned} \quad (24.5)$$

The full diamond is shown in Figure 24.1. One checks  $b_2 = 2h^{2,0} + h^{1,1} = 2 + 21 = 23$  and  $b_3 = 2(h^{3,0} + h^{2,1}) = 2(1 + 21) = 44$ , in agreement with (24.4). (The product of the two Hodge polynomials was expanded with sympy as an independent symbolic check of Figure 24.1.)

#### Common pitfall: “ $K3 \times T^2$ is a Calabi–Yau threefold”

The product has a nowhere-vanishing holomorphic three-form,  $\Omega_X \wedge dz$ , so  $h^{3,0} = 1$  and  $c_1 = 0$ . In the loose sense (“compact Kähler with  $c_1 = 0$ ”) it is a Calabi–Yau threefold. In the strict sense used in most of string phenomenology—holonomy *exactly*  $SU(3)$ —it is not: a manifold of holonomy exactly  $SU(3)$  has  $h^{1,0} = h^{2,0} = 0$ , whereas here  $h^{1,0} = h^{2,0} = 1$ . The extra holomorphic forms are the fingerprints of the smaller holonomy group  $SU(2)$ , and smaller holonomy means *more* supersymmetry ( $N = 4$  instead of  $N = 2$  for type II strings). Formulas derived for strict Calabi–Yau threefolds, such as “type IIA gives  $h^{1,1}$  vector multiplets and  $h^{2,1} + 1$  hypermultiplets”, must not be applied blindly to  $K3 \times T^2$ .

## 24.2 Holonomy $SU(2)$ and the amount of supersymmetry

### 24.2.1 Counting unbroken supercharges

A supersymmetry of the ten-dimensional theory survives compactification on a manifold  $Y$  if its spinor parameter can be chosen covariantly constant on  $Y$ . Parallel transport around

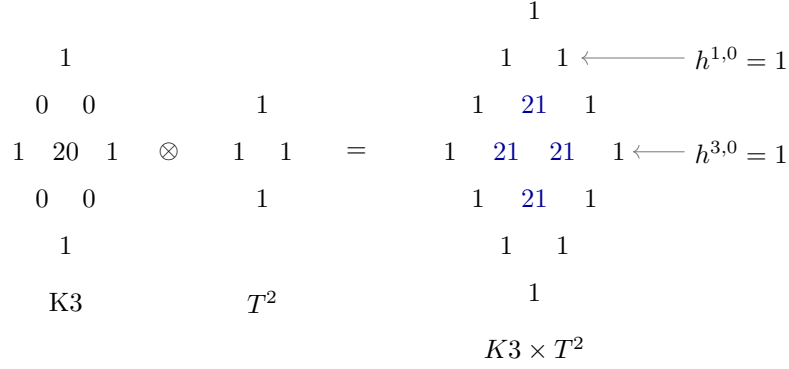


Figure 24.1: Hodge diamonds. Row  $k$  from the top lists  $h^{p,q}$  with  $p + q = k$ , with  $h^{k,0}$  at the right end. The diamond of the product is the “convolution” of the two factors; row sums give the Betti numbers 1, 2, 23, 44, 23, 2, 1.

a closed loop rotates a spinor by an element of the holonomy group of  $Y$ , so a covariantly constant spinor exists exactly for each component of the spinor on which the holonomy group acts trivially. The rule is therefore: decompose the ten-dimensional spinor under  $so(1, d - 1) \oplus so(10 - d)$ , restrict  $so(10 - d)$  to the holonomy algebra, and keep the singlets [Asp §4.1, ll. 2258–2265].

For a flat torus the holonomy is trivial and everything survives. For a K3 surface the holonomy is  $SU(2)$ . Aspinwall carries out the decomposition for one ten-dimensional Majorana–Weyl spinor, which has 16 real components:

$$so(1, 9) \supset so(1, 5) \oplus su(2) \oplus su(2), \quad \mathbf{16} \rightarrow (\mathbf{4}, \mathbf{2}, \mathbf{1}) \oplus (\mathbf{4}, \mathbf{1}, \mathbf{2}). \quad (24.6)$$

If the second  $su(2)$  is the holonomy algebra, the first summand consists of singlets and survives; the second does not. Eight of the sixteen components are kept: “compactification on a smooth K3 surface will preserve half as many supersymmetries as compactifying on a torus” [Asp §4.1, ll. 2273–2295]. Reducing further on  $T^2$  costs nothing. The count for the theories of interest is then as follows; recall that in four dimensions the minimal spinor has four real components, so  $N$  supersymmetries means  $4N$  real supercharges [Asp Table 2, ll. 2207–2254].

| ten-dimensional theory                       | internal space               | real supercharges | four-dimensional $N$ |
|----------------------------------------------|------------------------------|-------------------|----------------------|
| type IIA or IIB (32)                         | $T^6$                        | 32                | 8                    |
| type IIA or IIB (32)                         | $K3 \times T^2$              | 16                | 4                    |
| heterotic (16)                               | $T^6$                        | 16                | 4                    |
| heterotic (16)                               | $K3 \times T^2$              | 8                 | 2                    |
| type IIB orientifold ( $32 \rightarrow 16$ ) | $K3 \times T^2/\mathbb{Z}_2$ | 8                 | 2                    |
| the same, with generic flux                  | $K3 \times T^2/\mathbb{Z}_2$ | 4                 | 1                    |

The first four rows follow from (24.6); the second and third are the two sides of the duality of Section 24.4, and the fourth is the starting point of the  $N = 2$  heterotic/type II dual pairs [Asp §5.3, ll. 3074–3078]. The last two rows are discussed in Section 24.5 and are taken from [DRS §1, ll. 130–136; §4, ll. 1262–1266].

**Physical picture: why a curved space costs supersymmetry**

A supersymmetry transformation needs a spinor parameter  $\epsilon(y)$  defined over the whole internal space. On flat  $T^6$  we may take  $\epsilon$  constant, in any of its components. On a curved space, a “constant” spinor must mean covariantly constant, and curvature obstructs this: carrying  $\epsilon$  around a small loop rotates it by the curvature. A K3 surface is Ricci-flat but not flat; its curvature lies in one  $su(2)$  factor of  $so(4) = su(2) \oplus su(2)$  (this is what “holonomy  $SU(2)$ ” means). Spinors charged under that factor are rotated and cannot be constant; spinors charged only under the other factor do not feel the curvature at all. That is the content of (24.6). The torus factor of  $K3 \times T^2$  is flat and is invisible to this argument.

**24.2.2 Why  $K3 \times T^2$  and nothing else**

One may turn the argument around and ask for all compact six-manifolds on which a type II string yields  $N = 4$  in four dimensions, that is, all complex threefolds of holonomy  $SU(2)$ . Aspinwall’s argument runs as follows [Asp §5.2, ll. 2881–2894]. A covariantly constant holomorphic two-form forces  $h^{2,0} = 1$ , and an index argument then gives  $h^{1,0} = 1$ , so the manifold is not simply connected. By the Cheeger–Gromoll theorem the universal cover of a compact Ricci-flat manifold is isometric to  $M \times \mathbb{R}^n$  with  $M$  compact and simply connected; here it must be  $K3 \times \mathbb{R}^2$ . Hence *any complex threefold with holonomy  $SU(2)$  is isometric to  $K3 \times T^2$  or a quotient of it* [L]. The quotient group must preserve the holomorphic two-form of the K3 surface; such an action always has fixed points on K3, so to act freely it must be accompanied by a translation of  $T^2$ , and the group is therefore abelian. These free quotients give further  $N = 4$  theories with fewer gauge fields (Section 24.5). Compare Figure 24.1: the values  $h^{2,0} = h^{1,0} = 1$  are exactly what the product produces.

**24.3 The massless fields of the type IIA string on  $K3 \times T^2$** **24.3.1 Scalars:  $80 + 4 + 2 + 44 + 2 + 2 = 134$** 

Massless scalars in four dimensions arise from zero modes of the ten-dimensional fields on the internal space. For a  $p$ -form gauge field the rule is that each independent  $p$ -cycle of the internal space gives one scalar, namely the integral of the field over the cycle [Asp §4.2, ll. 2350–2355]. We go through the fields of the type IIA string.

*Metric and B-field.* Deformations of the Ricci-flat metric together with the periods of  $B$  are the deformations of the worldsheet conformal field theory. For the K3 factor there are  $58 + 22 = 80$  of them (Chapter 23); for the torus there are  $3 + 1 = 4$ , the three components of the flat metric and  $B_{12}$ . Are there mixed deformations, with one leg on K3 and one on  $T^2$ ? A mixed  $B$ -field zero mode would be a class in  $H^1(X) \otimes H^1(T^2)$ , and a mixed metric zero mode would require a harmonic one-form on K3 as well. Both are absent because  $b_1(X) = 0$ . This is the Künneth formula at work:  $b_2(K3 \times T^2) = 23 = 22 + 1$  has no mixed term, and the space of conformal field theories on  $K3 \times T^2$  is locally the product of the two factors,

$$\frac{O(4, 20)}{O(4) \times O(20)} \times \frac{O(2, 2)}{O(2) \times O(2)}, \quad \dim = 4 \cdot 20 + 2 \cdot 2 = 84. \quad (24.7)$$

Here we used that the Grassmannian  $O(p, q)/(O(p) \times O(q))$  of positive  $p$ -planes in  $\mathbb{R}^{p,q}$  has dimension  $pq$  (Chapter 10): a nearby plane is the graph of a linear map from the  $p$ -plane to its  $q$ -dimensional orthogonal complement.

*Ramond–Ramond fields.* Type IIA has a one-form  $C_1$  and a three-form  $C_3$ . The one-form integrated over the  $b_1(K3 \times T^2) = 2$  one-cycles gives 2 scalars; the three-form integrated over

the  $b_3(K3 \times T^2) = 44$  three-cycles gives 44 scalars. In addition,  $C_3$  with one leg along each of the  $b_1 = 2$  one-cycles is a two-form in four dimensions, and a two-form in four dimensions is dual to a scalar ( $db = \star da$ ): 2 more scalars. (The printed text of the source says “1-form” twice in this sentence; the second instance, which is paired with  $b_3 = 44$ , is the three-form.)

*Dilaton and axion.* The dilaton  $\Phi$  is one scalar, and the four-dimensional components  $B_{\mu\nu}$  of the  $B$ -field are dual to a second, the axion: 2 scalars.

Adding up, Aspinwall obtains [Asp §5.1, ll. 2743–2749]

$$\underbrace{80}_{K3 \text{ CFT}} + \underbrace{4}_{T^2 \text{ CFT}} + \underbrace{2}_{C_1} + \underbrace{44}_{C_3} + \underbrace{2}_{C_3 \rightarrow b_{\mu\nu}} + \underbrace{2}_{\Phi, B_{\mu\nu}} = 134. \quad (24.8)$$

### 24.3.2 Vectors: $2 + 2 + 1 + 23 = 28$

The same rules give the gauge fields; this count is not spelled out in the source and we derive it here. A  $p$ -form with one leg in spacetime and  $p - 1$  legs on an internal  $(p - 1)$ -cycle is a four-dimensional vector.

- The metric components  $g_{\mu m}$ ,  $m$  a torus direction, are the two Kaluza–Klein gauge fields of the two isometries of  $T^2$ .  $K3$  has no continuous isometries and contributes none.
- $B_{\mu m}$  gives  $b_1(K3 \times T^2) = 2$  vectors; these couple to string winding on the torus.
- $C_1$  gives  $b_0 = 1$  vector,  $C_\mu$  itself.
- $C_3$  gives  $b_2(K3 \times T^2) = 23$  vectors.

In total  $2 + 2 + 1 + 23 = 28$ . Aspinwall states the result in the form “the gauge group is always of rank 28” [Asp §5.1, ll. 2856–2858]. It is no accident that  $28 = 6 + 22$  is also the rank of the lattice that will appear in (24.15): the 28 electric charges of a state under the 28 gauge fields form a vector in a lattice of rank 28.

### 24.3.3 Assembling $N = 4$ multiplets

$N = 4$  supersymmetry in four dimensions has only two multiplets that contain scalars [Asp §5.1, ll. 2715–2723]: the supergravity multiplet, which contains one complex scalar (the dilaton–axion), and matter multiplets, which contain six real scalars each, transforming as a **6** of  $so(6)$ . (That the supergravity multiplet also contains six vectors, the graviphotons, and each matter multiplet one vector, is standard and is not checked against a source in this book’s library.) If  $m$  is the number of matter multiplets,

$$\text{scalars: } 2 + 6m = 134 \implies m = 22, \quad \text{vectors: } 6 + m = 28 \implies m = 22. \quad (24.9)$$

The two counts are independent and agree. The reader should appreciate that this is a non-trivial consistency check on Example 24.1: had we made an error in  $b_3$ , the number  $134 - 2$  would not have been divisible by six.

**Example 24.2** (The same numbers from the heterotic string on  $T^6$ ). The heterotic string of Chapter 10 compactified on  $T^6$  has the following massless scalars: the  $6 \cdot 7/2 = 21$  components of the torus metric, the  $6 \cdot 5/2 = 15$  components of the  $B$ -field, and  $16 \cdot 6 = 96$  Wilson lines of the sixteen Cartan gauge fields along the six circles. That is  $21 + 15 + 96 = 132 = 6 \cdot 22$ , the dimension of the Narain moduli space  $O(6, 22)/(O(6) \times O(22))$ . Together with the dilaton and the axion: 134. The vectors are  $g_{\mu m}$  (6),  $B_{\mu m}$  (6) and the sixteen Cartan gauge fields: 28. Both numbers agree with (24.8) and the vector count above, although on the type IIA side they were assembled from entirely different ingredients, the Betti numbers 2, 23, 44 of  $K3 \times T^2$ .

## 24.4 The moduli space and the duality group

### 24.4.1 The local form

Supersymmetry fixes the local geometry of the scalar manifold. The argument is the same as in six dimensions (Chapter 23): the holonomy algebra of the  $N = 4$  theory is  $u(4) \cong u(1) \oplus so(6)$ ; the dilaton–axion is charged only under  $u(1)$  and the matter scalars only under  $so(6)$ , so the moduli space factorises, and the Teichmüller space is [Asp §5.1 eq. (99), ll. 2717–2737]

$$\frac{O(6, m)}{O(6) \times O(m)} \times \frac{SL(2)}{U(1)}, \quad m = 22, \quad \dim = 6 \cdot 22 + 2 = 134. \quad (24.10)$$

For the heterotic string on  $T^6$  the interpretation is immediate (Example 24.2): the first factor is the Narain moduli space and the second, the upper half-plane  $SL(2)/U(1)$ , is parametrised by the dilaton–axion. For the type IIA string on  $K3 \times T^2$  it is less obvious which two of the 134 scalars split off.

### 24.4.2 The torus moduli $\tau$ and $\rho$

Recall from Chapter 12 that the four moduli of the conformal field theory on  $T^2$  form two complex numbers in the upper half-plane,

$$\tau = \frac{G_{12}}{G_{22}} + i \frac{\sqrt{\det G}}{G_{22}}, \quad \rho = B_{12} + i\sqrt{\det G}, \quad (24.11)$$

( $G$  measured in units of  $\alpha'$ ):  $\tau$  is the complex structure (shape) and  $\rho$  the complexified Kähler modulus (area and  $B$ -field). Correspondingly [Asp §5.1 eq. (100), ll. 2751–2764]

$$\frac{O(2, 2)}{O(2) \times O(2)} \cong \frac{SL(2)}{U(1)} \Big|_{\tau} \times \frac{SL(2)}{U(1)} \Big|_{\rho} \quad (24.12)$$

up to  $\mathbb{Z}_2$  identifications.

Which factor of (24.10) is the separate  $SL(2)/U(1)$ ? Aspinwall answers with a short computation [Asp §5.1 eqs. (101)–(102), ll. 2766–2806]. Start from the six-dimensional duality of Chapter 23, type IIA on  $K3$  = heterotic on  $T^4$ , under which  $\Phi_{\text{Het}} = -\Phi_{\text{IIA}}$  and the six-dimensional metrics are related by  $g_{6, \text{Het}} = e^{2\Phi_{\text{Het}}} g_{6, \text{IIA}}$  [Asp eq. (93), ll. 2405–2413]. Compactify both sides on a two-torus. The heterotic action is

$$S = \int d^6 X \sqrt{g_6} e^{-2\Phi_{6, \text{Het}}} (R + \dots) = \int d^4 X \sqrt{g_4} A_{\text{Het}} e^{-2\Phi_{6, \text{Het}}} (R + \dots), \quad (24.13)$$

where  $A_{\text{Het}}$  is the area of the torus in the heterotic metric; the coefficient of  $R$  is by definition  $e^{-2\Phi_{4, \text{Het}}}$ . Now express the area in the type IIA metric. An area scales with one power of the metric, so  $A_{\text{Het}} = e^{2\Phi_{6, \text{Het}}} A_{\text{IIA}}$ , and therefore

$$e^{-2\Phi_{4, \text{Het}}} = A_{\text{Het}} e^{-2\Phi_{6, \text{Het}}} = A_{\text{IIA}}. \quad (24.14)$$

The four-dimensional heterotic coupling is the inverse area of the type IIA torus. Since the heterotic dilaton–axion spans the separate factor in (24.10), in type IIA language *that factor is  $\rho$ , the area and  $B$ -field of the torus*. The other 132 scalars—the 80 moduli of the  $K3$  sigma model, the shape  $\tau$  of the torus, all 48 Ramond–Ramond scalars and the type IIA dilaton–axion—fill out the  $O(6, 22)$  coset:  $80 + 2 + 48 + 2 = 132$ .

### 24.4.3 The global form

The discrete identifications come from two sources. The conformal field theory on  $T^2$  is invariant under  $SL(2, \mathbb{Z})$  acting on  $\rho$  (integer shifts of  $B_{12}$  and the inversion of Chapter 12). The heterotic string on  $T^6$  is invariant under the isometry group  $O(\Gamma^{6,22})$  of its Narain lattice, the even unimodular lattice of signature  $(6, 22)$  (Chapters 10 and 11). Combining the two:

**Theorem 24.3** (Aspinwall’s Proposition 4, [L]). *Assume the six-dimensional duality between the type IIA string on  $K3$  and the heterotic string on  $T^4$  (Chapter 23). Then the type IIA string on  $K3 \times T^2$  is equivalent to the heterotic string on  $T^6$ , with moduli space*

$$\mathcal{M}_{N=4} = \left( O(\Gamma^{6,22}) \backslash O(6, 22) / (O(6) \times O(22)) \right) \times \left( SL(2, \mathbb{Z}) \backslash SL(2) / U(1) \right). \quad (24.15)$$

[Asp §5.1, Prop. 4 and eq. (103), ll. 2807–2816]

The word “theorem” is used in the sense of the source: a statement derived from stated assumptions, which include the six-dimensional duality and the completeness assumptions of Chapter 23; it is Tier L and conditional, not a mathematical theorem about a defined object.

#### Physical picture: three roles of one modular group

The  $SL(2, \mathbb{Z})$  in (24.15) acts on a complex scalar that has three different names [Asp §5.1, ll. 2840–2844]: for the heterotic string it is the dilaton–axion, so  $SL(2, \mathbb{Z})$  is a strong–weak coupling duality (S-duality) of the four-dimensional theory; for the type IIA string it is  $\rho$ , so the same  $SL(2, \mathbb{Z})$  is an ordinary T-duality of the torus, visible in perturbation theory; for the type IIB string, obtained from IIA by  $R \rightarrow \alpha'/R$  on one circle, which exchanges  $\rho$  and  $\tau$  (Example 24.4), it is the shape of the torus, and  $SL(2, \mathbb{Z})$  is a plain change of basis of the lattice defining  $T^2$ . A non-perturbative duality of one description is a geometric triviality of another. Conversely, the S-duality of the ten-dimensional type IIB string ends up inside  $O(\Gamma^{6,22})$  [Asp §5.1, ll. 2852–2855].

### 24.4.4 The lattice $\Gamma^{6,22}$ and its sublattices

By the classification of Chapter 18, an indefinite even unimodular lattice is determined by its signature  $(n_+, n_-)$ , which must satisfy  $n_+ - n_- \equiv 0 \pmod{8}$ . Hence

$$\underbrace{3U \oplus 2E_8(-1)}_{\Gamma^{3,19}=H^2(X, \mathbb{Z})} \subset \underbrace{\Gamma^{3,19} \oplus U}_{\Gamma^{4,20}=H^*(X, \mathbb{Z})} \subset \underbrace{\Gamma^{4,20} \oplus U \oplus U}_{\Gamma^{6,22}}, \quad (24.16)$$

with signatures obtained by adding  $(1, 1)$  for each hyperbolic plane  $U$  and  $(0, 8)$  for each  $E_8(-1)$ :

$$2(0, 8) + 3(1, 1) = (3, 19), \quad (3, 19) + (1, 1) = (4, 20), \quad (4, 20) + 2(1, 1) = (6, 22). \quad (24.17)$$

The indices are  $-16, -16, -16$ , all divisible by eight, and the ranks are 22, 24 and 28. The decomposition  $\Gamma^{6,22} \cong \Gamma^{4,20} \oplus \Gamma^{2,2}$  with  $\Gamma^{2,2} = U \oplus U$  is used by Aspinwall to describe the action of geometric symmetries of  $K3 \times T^2$  on the charge lattice [Asp §5.2, ll. 2898–2906]. It is summarised in Figure 24.2.

An isometry of a summand, extended by the identity on the orthogonal complement, is an isometry of the sum. Therefore

$$O(\Gamma^{4,20}) \times O(\Gamma^{2,2}) \subset O(\Gamma^{6,22}), \quad (24.18)$$

and both factors have been met before:  $O(\Gamma^{4,20})$  is the duality group of the K3 sigma model (Chapters 22 and 23) and  $O(\Gamma^{2,2}) = O(2, 2; \mathbb{Z})$  is the T-duality group of a two-torus, which contains the modular transformations of  $\tau$ , those of  $\rho$ , and the mirror exchange  $\tau \leftrightarrow \rho$  (Chapter 12). The inclusion (24.18) is proper by a wide margin:  $O(\Gamma^{6,22})$  also contains isometries that mix the K3 block with the torus block, and these have no description as symmetries of the conformal field theory on either factor.

#### Common pitfall: which $\Gamma^{2,2}$ ?

It is tempting to read (24.18) as “the perturbative T-duality group of the type IIA string on  $K3 \times T^2$ ,  $O(\Gamma^{4,20}) \times O(2, 2; \mathbb{Z})$ , sits inside  $O(\Gamma^{6,22})$ ”. This is not quite right. We have just seen that the Kähler modulus  $\rho$  of the type IIA torus is *not* among the 132 coordinates of the  $O(6, 22)$  coset; its modular group is the separate  $SL(2, \mathbb{Z})$  factor of (24.15). What is true is that the perturbative group sits inside the full modular group  $O(\Gamma^{6,22}) \times SL(2, \mathbb{Z})$ , with  $O(\Gamma^{4,20})$  and  $SL(2, \mathbb{Z})_\tau$  in the first factor and  $SL(2, \mathbb{Z})_\rho$  being the second. In the heterotic description the summand  $\Gamma^{2,2}$  is literally the Narain lattice of the last two circles of  $T^6 = T^4 \times T^2$ , containing  $\tau_{\text{Het}}$  and  $\rho_{\text{Het}}$ ; the duality map then sends the heterotic dilaton–axion to  $\rho_{\text{IIA}}$  by (24.14). (That it sends  $\rho_{\text{Het}}$  to the type IIA dilaton–axion is the natural completion of this statement; it is standard but is not spelled out in the source lines read for this chapter.)

**Example 24.4** (Rectangular torus: numbers). Take  $T^2$  rectangular with radii  $R_1 = 3\sqrt{\alpha'}$ ,  $R_2 = \frac{1}{2}\sqrt{\alpha'}$  and  $B_{12} = 0$ . In the conventions of Chapter 12,  $\tau = iR_1/R_2 = 6i$  and  $\rho = iR_1R_2/\alpha' = \frac{3}{2}i$ . T-duality on the second circle,  $R_2 \rightarrow \alpha'/R_2 = 2\sqrt{\alpha'}$ , gives  $\tau' = iR_1R_2/\alpha' = \frac{3}{2}i = \rho$  and  $\rho' = iR_1/R_2 = 6i = \tau$ : shape and size are exchanged, which is the statement that one T-duality is a mirror map and turns type IIA into type IIB [Asp §5.1, Prop. 5 and ll. 2826–2831]. Next apply the inversion of the separate  $SL(2, \mathbb{Z})$  in type IIA,  $\rho \rightarrow -1/\rho = \frac{2}{3}i$ , at fixed  $\tau$ : from  $R_1/R_2 = 6$  and  $R_1R_2 = \frac{2}{3}\alpha'$  we get  $R_1 = 2\sqrt{\alpha'}$ ,  $R_2 = \frac{1}{3}\sqrt{\alpha'}$ , that is, both radii inverted,  $R_i \rightarrow \alpha'/R_i$ , and the circles exchanged. By (24.14) the heterotic dual of this torus has  $e^{-2\Phi_{4,\text{Het}}} = A_{\text{IIA}} \propto \rho_2$ , so  $\rho_2 = \frac{3}{2} \rightarrow \frac{2}{3}$  is the exchange of a moderately weakly coupled heterotic string with a moderately strongly coupled one. In the language of Chapter 31, each circle contributes an effective scale  $R_i + \alpha'/R_i \geq 2\sqrt{\alpha'}$  that is unchanged by the inversion:  $3 + \frac{1}{3} = \frac{10}{3}$  and  $\frac{1}{2} + 2 = \frac{5}{2}$  in units of  $\sqrt{\alpha'}$  before, and  $2 + \frac{1}{2}, \frac{1}{3} + 3$  after.

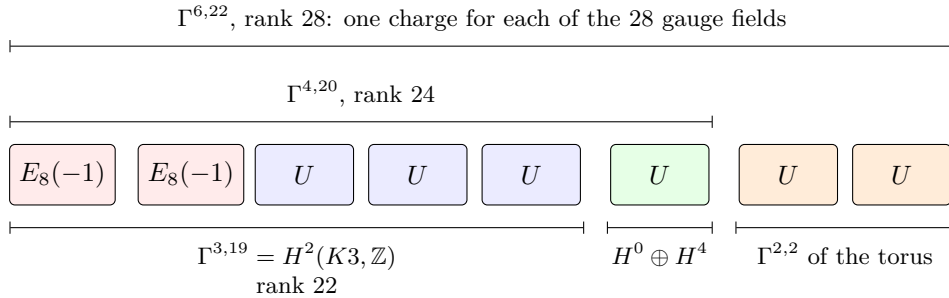


Figure 24.2: The charge lattice of the  $N = 4$  theory as an orthogonal sum. Each block  $U$  adds  $(1, 1)$  to the signature and each  $E_8(-1)$  adds  $(0, 8)$ .



## 24.5 Less supersymmetry: quotients, bundles, orientifolds and flux

$N = 4$  supersymmetry is too rigid for particle physics; the interest of  $K3 \times T^2$  is that it is the simplest starting point from which the supersymmetry can be lowered in a controlled way.

### 24.5.1 Free quotients: still $N = 4$ , smaller rank

A finite abelian group  $G$  acting on  $K3$  preserving the holomorphic two-form, combined with a translation of  $T^2$  (Section 24.2), acts freely on  $K3 \times T^2$  and preserves the  $SU(2)$  holonomy. Its action on  $\Gamma^{6,22} \cong \Gamma^{4,20} \oplus \Gamma^{2,2}$  is a lattice rotation on the first summand and a null shift on the second. In the dual heterotic description this is an asymmetric orbifold of  $T^6$ , and only the  $G$ -invariant sublattice of  $\Gamma^{6,22}$  survives; the rank of the gauge group drops from 28 to  $28 - M$ , where  $M$  is the rank of the sublattice of  $H^2(X, \mathbb{Z})$  on which  $G$  acts non-trivially ( $M = 8$  for  $G = \mathbb{Z}_2$ ) [Asp §5.2, ll. 2888–2906 and 2985–3000; Table 3, l. 2945]. Finite groups of symmetries of  $K3$  that preserve the holomorphic two-form are taken up again, from the point of view of the Mathieu group, in Chapter 30.

### 24.5.2 The heterotic string on $K3 \times T^2$ : $N = 2$

By the table of Section 24.2 the heterotic string on  $K3 \times T^2$  keeps 8 supercharges,  $N = 2$ . A heterotic compactification requires, besides the manifold, a gauge bundle  $E$  subject to the anomaly condition  $c_2(E) = c_2(TX)$ ; on  $K3$  this means  $c_2(E) = 24$ , integrated over the surface, so the bundle cannot be flat [Asp §5.3, ll. 3108–3120]. The number 24 is  $\chi(K3)$ : for a complex surface the integral of  $c_2$  of the tangent bundle is the Euler number. The many choices of bundle are what allows these models to be dual to type II strings on the many Calabi–Yau threefolds; the moduli space is a product  $\mathcal{M}_V \times \mathcal{M}_H$  of a Kähler manifold (vector multiplets) and a quaternionic Kähler manifold (hypermultiplets) [Asp §5.3 eq. (108), ll. 3125–3148].

### 24.5.3 The type IIB orientifold and flux: $N = 2$ and $N = 1$

Dasgupta, Rajesh and Sethi consider type IIB on  $K3 \times T^2$  divided by  $\Omega(-1)^{F_L} \mathbb{Z}_2$ , where  $\Omega$  is worldsheet parity,  $F_L$  the left-moving spacetime fermion number and  $\mathbb{Z}_2 : z \mapsto -z$  the reflection of the torus coordinate [DRS §4.1, ll. 1288–1292]. The reflection has four fixed points on  $T^2$ , namely  $z = 0, \frac{1}{2}, \frac{\tau}{2}, \frac{1+\tau}{2}$  in units of the periods, because  $z \equiv -z$  modulo the lattice means  $2z \in \mathbb{Z} + \tau\mathbb{Z}$ . At each fixed point sits an orientifold seven-plane (O7) wrapping  $K3$  and filling spacetime; its Ramond–Ramond charge is cancelled locally by D7-branes, and with a trivial gauge bundle the gauge group is  $SO(8)^4$  [DRS §4.1, ll. 1310–1314]. The quotient  $T^2/\mathbb{Z}_2$  is a sphere with four conical points (Figure 24.3). The orientifold projection halves the 16 supercharges of type IIB on  $K3 \times T^2$ :  $N = 2$ .

This orientifold is a special point in the moduli space of F-theory on the fourfold  $K3 \times K3$  [DRS §4.1, ll. 1292–1293], and with that comes a constraint. For M- and F-theory on a Calabi–Yau fourfold  $\mathcal{M}$  there is a tadpole proportional to  $\chi(\mathcal{M})/24$  which must be cancelled by  $n$  spacetime-filling branes and by flux [DRS §1 eq. (1.2), ll. 50–80]:

$$\frac{\chi(\mathcal{M})}{24} = \frac{1}{8\pi^2} \int_{\mathcal{M}} G \wedge G + n \quad \implies \quad \frac{1}{2} \int \frac{G}{2\pi} \wedge \frac{G}{2\pi} + n = 24 \quad \text{for } \mathcal{M} = K3 \times K3, \quad (24.19)$$

since  $\chi(K3 \times K3) = 24^2 = 576$  by (24.2) and  $576/24 = 24$  [DRS §3.1, ll. 637–641; §4.2, ll. 1362–1370]. The two forms of the flux term agree because  $\frac{1}{2}(2\pi)^{-2} = 1/(8\pi^2)$ . Without flux one

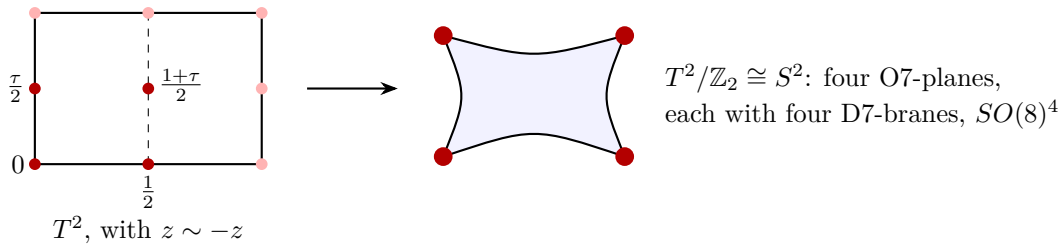


Figure 24.3: The reflection  $z \mapsto -z$  of the torus has four fixed points (dark dots; the light dots are their lattice translates). Half of the cell, to the left of the dashed line, is a fundamental domain; folding it gives a “pillow” with four corners. The seven-planes sit at the corners and wrap the  $K3$  surface.

needs  $n = 24$  D3-branes; after two T-dualities along the torus this becomes the type I string on  $K3 \times T^2$  with 24 D5-branes wrapping  $T^2$  [DRS §4, ll. 1267–1276]. With flux

$$\frac{G}{2\pi} = \alpha \wedge dz^1 d\bar{z}^2 + \beta \wedge d\bar{z}^1 dz^2 + \text{c.c.}, \quad \alpha \in H^{1,1}(K3), \quad \beta \in H^{2,0}(K3), \quad (24.20)$$

fewer branes are needed, and the supersymmetry depends on the flux: “if  $\beta = 0$ , the model has  $N = 2$  supersymmetry”, otherwise  $N = 1$  [DRS §4.2 eq. (4.4), ll. 1372–1382]. (The placement of the bars in (24.20) is reconstructed: the text extraction of the source drops them. We have fixed them by requiring that  $G$  be real and that every term be of type  $(2, 2)$  on  $K3 \times T^4$ , which supersymmetry demands [DRS §2.2, ll. 315–319]; for instance  $\beta \wedge dz^1 dz^2$  would be of type  $(4, 0)$ .) The flux also gives masses to some of the 134 moduli; this is the subject of Chapter 26, and the integrality and positivity properties of the flux term are treated in Chapter 25.

**Common pitfall: “ $\chi(K3 \times T^2) = 0$ , so there is no tadpole”**

The Euler number that enters (24.19) is that of the elliptically fibred *fourfold*, here  $K3 \times K3$  with  $\chi = 576$ , not that of the six-dimensional space on which the type IIB string is compactified. The threefold  $K3 \times T^2$  has  $\chi = 0$  and, without an orientifold, type II strings on it have no tadpole at all: that is the  $N = 4$  theory of Section 24.4. The 24 units of D3-brane charge of the orientifold come from the curvature of the  $K3$  surface wrapped by the seven-branes, and  $24 = \chi(K3)$ . Both statements,  $\chi(K3 \times T^2) = 0$  and  $\chi(K3 \times K3)/24 = 24$ , are checked in Lean below, and they refer to different manifolds.

## 24.6 What the machine has checked

Three Lean files touch the material of this chapter. None of them defines a manifold, a cohomology group or a lattice with a bilinear form. What they contain is the *arithmetic* of the chapter—signatures added as pairs of natural numbers, Euler numbers computed from tables—and the honest description is “arithmetic shadow” (Chapter 35). The inputs to that arithmetic (the Hodge numbers of  $K3$ , the signatures of  $U$  and  $E_8(-1)$ , the Künneth formula) are Tier L.

**In Lean: the signature tower (24.17)**

File `DualScaleStream2/Lattice/K3T2Signature.lean`, namespace `DualScaleStream2.Lattice`. A signature is a bare pair of naturals with componentwise addition:

```

structure Signature where
  pos : ℕ
  neg : ℕ

def sigU : Signature := ⟨1, 1⟩
def sigE8Neg : Signature := ⟨0, 8⟩
def sigK3 : Signature := sigE8Neg + sigE8Neg + sigU + sigU + sigU
def sigMukai : Signature := sigK3 + sigU
def sigT2 : Signature := sigU + sigU
def sigK3T2 : Signature := sigMukai + sigT2

theorem sigK3T2_eq : sigK3T2 = ⟨6, 22⟩ := by rfl
theorem rank_K3T2 : sigK3T2.rank = 28 := by ...
theorem index_mod_eight :
  sigK3.index % 8 = 0 ∧ sigMukai.index % 8 = 0 ∧ sigK3T2.index % 8 = 0 := by ...

```

**Reading.** `sigK3T2_eq` says that the pair obtained by adding  $(0, 8)$  twice and  $(1, 1)$  six times is  $(6, 22)$ ; `rank_K3T2` says  $6 + 22 = 28$ ; `index_mod_eight` says that  $3 - 19$ ,  $4 - 20$  and  $6 - 22$  are divisible by 8. Companion theorems `sigK3_eq`, `sigMukai_eq` and `rank_K3` give  $(3, 19)$ ,  $(4, 20)$  and 22, and `sigK3_matches_hodge` checks that  $(3, 19)$  agrees with the pair  $(2h^{2,0} + 1, h^{1,1} - 1)$  computed from the Hodge table of another library. **What is not said.** The structure `Signature` is not attached to any quadratic form: that  $U$  has signature  $(1, 1)$  and  $E_8(-1)$  has  $(0, 8)$  is entered by hand in `sigU` and `sigE8Neg`. The mod-8 theorem for even unimodular lattices is not proved; it is verified at three numerical instances. **Proof idea.** All terms are closed numerals, so both sides reduce to the same normal form and `rfl` (or `simp` followed by `rfl`) closes the goal.

### In Lean: Euler numbers from Hodge tables

File `DualScaleStream2/Flux/Tadpole.lean`, namespace `DualScaleStream2.Flux`:

```

def t2HodgeNumber : Fin 2 → Fin 2 → ℕ := fun _ _ => 1

def eulerFromHodge {m : ℕ} (h : Fin m → Fin m → ℕ) : ℤ :=
  ∑ p : Fin m, ∑ q : Fin m,
    (if (p.val + q.val) % 2 = 0 then 1 else -1 : ℤ) * (h p q : ℤ)

theorem euler_T2 : eulerFromHodge t2HodgeNumber = 0 := by ...

theorem k3t2_euler_zero :
  eulerFromHodge StringTheory.Frontier.k3HodgeNumber
    * eulerFromHodge t2HodgeNumber = 0 := by ...

theorem k3k3_anomaly : chiK3K3 % 24 = 0 ∧ chiK3K3 / 24 = 24 := by ...

theorem tadpole_budget (flux n : ℕ) (h : flux + n = 24) :
  n ≤ 24 ∧ (flux = 0 → n = 24) := by ...

```

**Reading.** `eulerFromHodge` is the alternating sum  $\sum_{p,q} (-1)^{p+q} h^{p,q}$  of an  $m \times m$  table. `euler_T2` evaluates it on the all-ones  $2 \times 2$  table of the torus:  $1 - 1 - 1 + 1 = 0$ . `k3t2_euler_zero` states that the *product of two integers*, the alternating sum of the K3 table (which a companion theorem `euler_K3` shows to be 24) and that of the torus table, is zero. It is the right-hand side of (24.2); the Künneth formula, which makes this product the Euler number of  $K3 \times T^2$ , is not formalised. `k3k3_anomaly` is  $576 \bmod 24 = 0$  and  $576/24 = 24$ , where `chiK3K3` is defined as the square of the K3 alternating sum. `tadpole_budget` is a statement about two natural numbers whose sum is 24; it says nothing about whether a flux with a given value exists. **Proof idea.** The finite sums

are unfolded (`Fin.sum_univ_succ`) and evaluated; the last two theorems are closed by `norm_num` and `omega`.

#### In Lean: dimension and supercharge bookkeeping

File `StringTheoryFoundation/StringTheory/K3xT2.lean`, namespace `StringTheory.Foundation.StringTheory.K3xT2`:

```
def dimRealK3xT2 : Nat := 4 + 2
theorem dim_real_k3xt2_is_6 : dimRealK3xT2 = 6 := by rfl

def targetSpacetimeDim : Nat := 10 - dimRealK3xT2
theorem target_spacetime_is_4d : targetSpacetimeDim = 4 := by rfl

def typeII_4D_supercharges : Nat := 32 / 2
def typeII_4D_N : Nat := typeII_4D_supercharges / 4
theorem typeII_k3xt2_susy_N4 : typeII_4D_N = 4 := by rfl

def heterotic_T6_supercharges : Nat := 16
theorem string_duality_susy_match :
  typeII_4D_supercharges = heterotic_T6_supercharges := by rfl

theorem euler_char_vanishes :
  eulerChar4D bettiK3 * eulerChar2D bettiT2 = 0 := by ...
```

**Reading.** These are the numerals of the table in Section 24.2:  $4 + 2 = 6$ ,  $10 - 6 = 4$ ,  $(32/2)/4 = 4$  and  $32/2 = 16$ . The division by two that represents “K3 breaks half of the supersymmetry” is written into the definition; the spinor decomposition (24.6) that justifies it is not formalised. The name `string_duality_susy_match` should be read with care: the theorem is  $16 = 16$  in  $\mathbb{N}$ , a necessary condition for the duality of Theorem 24.3, not evidence for it. `euler_char_vanishes` is a second, independent version of  $24 \cdot 0 = 0$ , this time from Betti-number records `bettiK3` = (1, 0, 22, 0, 1) and `bettiT2` = (1, 2, 1) rather than Hodge tables.

Two further declarations elsewhere in the repository are worth knowing. In the namespace `SocrateAI.Moonshine`, the Betti numbers (24.4) of the product are entered as seven definitions (`k3t2_b0` = 1, ..., `k3t2_b3` = 44, ...) and `k3t2_euler_char_eq_zero` verifies by `decide` that their alternating sum is zero; this is the left-hand side of (24.2), where `k3t2_euler_zero` is the right-hand side. Neither file derives the seven numbers from those of the factors (see Exercise 24.8). Second, the theorem `SocrateAI.FrontierTriad.total_k3t2_moduli_dim_eq` proves  $(3 \cdot 19 + 1) + 22 = 80$ . Despite its name and docstring, the number  $80 = 58 + 22$  is the dimension of the moduli space of the sigma model on  $K3$  *alone* (metric plus  $B$ -field, Chapter 23); the  $K3 \times T^2$  compactification has 84 such moduli by (24.7) and 134 scalars in all. The Lean statement is a correct sum of numerals; its name is a mislabel, and it should not be quoted as a fact about  $K3 \times T^2$ .

| Claim                                                                                             | Tier | Evidence                                        |
|---------------------------------------------------------------------------------------------------|------|-------------------------------------------------|
| $2(0, 8) + 6(1, 1) = (6, 22)$ ; rank 28; index $\equiv 0 \pmod 8$                                 | [A]  | sigK3T2_eq, rank_K3T2, index_mod_eight          |
| Alternating sums: torus table 0; K3 table times torus table 0                                     | [A]  | euler_T2, k3t2_euler_zero, euler_char_vanishes  |
| $1 - 2 + 23 - 44 + 23 - 2 + 1 = 0$                                                                | [A]  | k3t2_euler_char_eq_zero                         |
| $576/24 = 24$ ; $flux + n = 24 \Rightarrow n \leq 24$                                             | [A]  | k3k3_anomaly, tadpole_budget                    |
| Künneth formula; Betti and Hodge numbers of $K3 \times T^2$                                       | [L]  | standard; sympy check of (24.3), (24.5)         |
| Holonomy $SU(2) \Rightarrow K3 \times T^2$ or a quotient                                          | [L]  | Asp §5.2                                        |
| 134 scalars, $m = 22$ , moduli space (24.15)                                                      | [L]  | Asp §5.1 (conditional on IIA/heterotic duality) |
| Tadpole $\frac{1}{2} \int \frac{G}{2\pi} \wedge \frac{G}{2\pi} + n = 24$ ; $N = 2$ or 1 with flux | [L]  | DRS §1, §4                                      |
| “Signature $(6, 22)$ is the charge lattice of a physical theory”                                  | —    | an identification, never Tier A                 |

All Tier A entries depend only on the standard axioms `propext`, `Classical.choice` and `Quot.sound`.

#### Chapter summary

- Künneth: Poincaré polynomials multiply. For  $K3 \times T^2$  the Betti numbers are  $(1, 2, 23, 44, 23, 2, 1)$ , the Hodge numbers  $h^{1,1} = h^{2,1} = 21$ ,  $h^{1,0} = h^{2,0} = h^{3,0} = 1$ , and  $\chi = 24 \cdot 0 = 0$ .
- Holonomy  $SU(2)$  preserves half of the supercharges: type II on  $K3 \times T^2$  gives  $N = 4$  in four dimensions, the heterotic string gives  $N = 2$ . Every complex threefold of holonomy  $SU(2)$  is  $K3 \times T^2$  or a free quotient.
- Type IIA on  $K3 \times T^2$  has  $80 + 4 + 2 + 44 + 2 + 2 = 134$  massless scalars and  $2 + 2 + 1 + 23 = 28$  vectors: the supergravity multiplet plus 22 matter multiplets.
- The moduli space is  $O(\Gamma^{6,22}) \backslash O(6, 22) / (O(6) \times O(22)) \times SL(2, \mathbb{Z}) \backslash SL(2) / U(1)$ ; it is the same as for the heterotic string on  $T^6$ . The  $SL(2, \mathbb{Z})$  factor is heterotic S-duality, type IIA T-duality of  $\rho$ , and the type IIB torus shape.
- $\Gamma^{6,22} \cong \Gamma^{4,20} \oplus \Gamma^{2,2}$ , so  $O(\Gamma^{4,20}) \times O(2, 2; \mathbb{Z}) \subset O(\Gamma^{6,22})$ ; the signature arithmetic  $(3, 19) \rightarrow (4, 20) \rightarrow (6, 22)$  is kernel-checked, the lattices themselves are not formalised in these files.
- Orientifolding by  $\Omega(-1)^{F_L} \mathbb{Z}_2$  gives  $N = 2$ , four O7-planes, a D3 tadpole of  $24 = \chi(K3 \times K3)/24$ ; flux with a  $(2, 0)$  component lowers this to  $N = 1$ .

## Exercises

**Exercise 24.1** (\*). Compute the Betti numbers and the Euler number of  $T^4$ , of  $K3 \times K3$  and of  $K3 \times T^4$  from Poincaré polynomials. Check Poincaré duality in each case, and confirm  $\chi(K3 \times K3) = 576$ .

**Exercise 24.2** (\*). Using the Hodge polynomials of Section 24.1, compute the full Hodge diamond of  $K3 \times K3$ . Verify  $h^{4,0} = 1$ ,  $h^{2,0} = 2$ ,  $h^{1,1} = 40$ ,  $h^{3,1} = 40$ ,  $h^{2,2} = 404$  and recover  $\chi = 576$  from the alternating sum.

**Exercise 24.3** (\*). Repeat the scalar count of Section 24.3 for the type IIA string on  $T^6$  and show that the number of scalars is 70. (Use  $b_k(T^6) = \binom{6}{k}$ ; the conformal field theory on  $T^6$  has  $6 \cdot 6 = 36$  moduli. Do not forget the scalar dual to  $C_3$  with one internal leg, which here comes in  $b_1 = 6$  copies.) The  $N = 8$  theory has the coset  $E_{7(7)}/SU(8)$  of dimension  $133 - 63$  as its scalar manifold; this fact is not used elsewhere in the book.

**Exercise 24.4** (\*\*). Type IIB has Ramond–Ramond potentials  $C_0$ ,  $C_2$  and  $C_4$  with self-dual field strength. Count the massless scalars of type IIB on  $K3 \times T^2$  and show that the total is again 134. (Hints:  $C_2$  contributes  $b_2$  scalars and one more from its four-dimensional two-form component;  $C_4$  contributes  $b_4$  scalars, and its components with two legs in spacetime are related to these by self-duality and must not be counted again; the NS–NS sector is the same as for IIA.)

**Exercise 24.5** (\*\*). Show that  $O(p, q)/(O(p) \times O(q))$  has dimension  $pq$  by counting generators:  $\dim O(n) = n(n-1)/2$  and  $\dim O(p, q) = \dim O(p+q)$ . Deduce that the  $N = 4$  moduli space with  $m$  matter multiplets has dimension  $6m + 2$ , and that the rank reduction of Section 24.5 for  $G = \mathbb{Z}_2$  ( $M = 8$ ) leaves a moduli space of dimension 86.

**Exercise 24.6** (\*\*). A point  $z$  of the torus  $\mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$  is fixed by  $z \mapsto -z$  iff  $2z \in \mathbb{Z} + \tau\mathbb{Z}$ . (a) Show that there are exactly four fixed points. (b) The Euler number of a quotient by a  $\mathbb{Z}_2$  with  $k$  isolated fixed points is  $\chi(M/\mathbb{Z}_2) = (\chi(M) + k)/2$ . Conclude  $\chi(T^2/\mathbb{Z}_2) = 2$ , consistent with Figure 24.3. (c) The same formula applied to  $T^4/\mathbb{Z}_2$  with its 16 fixed points gives 8; blowing up each point replaces it by a sphere and adds 1 to  $\chi$ . Recover  $\chi(K3) = 24$  (Chapter 21).

**Exercise 24.7** (\*\*, Lean). In a file importing `DualScaleStream2.Lattice.K3T2Signature`, replace the three `sorry`s:

```
open DualScaleStream2.Lattice

example : sigT2 = ⟨2, 2⟩ := by sorry
example : sigK3T2.index = -16 := by sorry
example (a b : Signature) : (a + b).rank = a.rank + b.rank := by sorry
```

The first two should fall to `rfl` or `decide`; for the third unfold `Signature.rank`, use the simp lemmas `Signature.add_pos` and `Signature.add_neg`, and finish with `omega`. Explain why the third, unlike the first two, is a statement about infinitely many cases, and why the mod-8 property is *not* of this kind:  $(a + b).index \bmod 8$  can be anything even when  $a.index \bmod 8 = 0$ .

**Exercise 24.8** (\*\*\*, Lean). The repository checks  $\chi(K3 \times T^2) = 0$  twice, once as a product of two alternating sums and once as the alternating sum of seven hand-entered numbers, but never connects the two. Close the gap at the level of lists. Define the convolution of two lists of integers (the coefficient list of a product of polynomials) and the alternating sum of a list. (a) Prove by `decide` that the convolution of  $[1, 0, 22, 0, 1]$  and  $[1, 2, 1]$  is  $[1, 2, 23, 44, 23, 2, 1]$ . (b) Prove in general that the alternating sum of a convolution is the product of the alternating sums. (Hint: induct on the first list; you will need a lemma about the alternating sum of a list shifted by one place.) Which part of the Künneth theorem remains unformalised after (b)?

**Exercise 24.9** (\*\*\*). Let  $\Lambda = \Lambda_1 \oplus \Lambda_2$  be an orthogonal sum of lattices. (a) Show that  $O(\Lambda_1) \times O(\Lambda_2) \subset O(\Lambda)$ . (b) For  $\Lambda_1 = \Lambda_2 = U$  with bases  $(e_i, f_i)$ ,  $e_i^2 = f_i^2 = 0$ ,  $e_i \cdot f_i = 1$ , exhibit an isometry of  $U \oplus U$  which is not in  $O(U) \times O(U)$ , not even after exchanging the two summands. (Try  $e_1 \mapsto e_1$ ,  $e_2 \mapsto e_2 + e_1$ , and find the images of  $f_1, f_2$ .) (c) Interpret your isometry, under  $U \oplus U = \Gamma^{2,2}$ , as one of the generators of  $O(2, 2; \mathbb{Z})$  of Chapter 11.

## Further reading

- $N = 4$  theories in four dimensions, the count 134, the three roles of  $SL(2, \mathbb{Z})$  and Proposition 4: [Asp §5.1, ll. 2713–2858]. Threefolds of holonomy  $SU(2)$  and the free quotients: [Asp §5.2, ll. 2862–3000]. The passage to  $N = 2$ : [Asp §5.3, ll. 3063–3150].
- Supersymmetry counting by holonomy and the table of maximal supersymmetries: [Asp §4.1, ll. 2188–2295].
- The tadpole  $\chi/24$ , flux and the orientifold of  $K3 \times T^2$ : [DRS §1, ll. 40–130; §2.2, ll. 267–319; §3.1, ll. 636–660; §4–4.2, ll. 1262–1382].
- The lattices  $U$ ,  $E_8$  and the mod-8 condition: Chapter 18; the K3 lattice and the Mukai lattice: Chapters 20 and 22; the two-torus and  $O(2, 2; \mathbb{Z})$ : Chapters 11 and 12; how the tadpole is used: Chapters 25 and 26.





# Chapter 25

## Fluxes, Tadpoles and $\chi/24$

On a compact space a charge cannot be alone. The field lines that leave it must end somewhere, and if the space has no boundary and no infinity they can only end on other charges of the opposite sign. This elementary remark about Gauss's law becomes, in string theory, one of the sharpest consistency conditions a compactification has to pass: the *tadpole condition*. It is sharp because it is an equation between integers. For M-theory or F-theory on a Calabi–Yau fourfold  $X_8$  it reads

$$\frac{\chi(X_8)}{24} = n + \frac{1}{2} \int_{X_8} \frac{G}{2\pi} \wedge \frac{G}{2\pi}, \quad (25.1)$$

where  $\chi$  is the Euler number,  $n$  counts space-filling branes and  $G$  is the four-form flux. This chapter answers four questions about (25.1). Why is there such an equation at all? Why does the curvature of the internal space enter through the number  $\chi/24$ , and when is that number a positive integer? Why is the flux term an integer? And what does the equation look like in the perturbative type IIB description of our main example, the orientifold  $K3 \times T^2/\mathbb{Z}_2$  of Chapter 24, where the branes are D7- and D3-branes and the negative charges sit on orientifold planes? For the product of two K3 surfaces every number can be computed by hand:  $\chi(K3 \times K3) = 576$ , the budget is  $576/24 = 24$ , and the flux term is an integer because the K3 lattice of Chapter 20 is even. These last statements are arithmetic and lattice theory, and they are what the Lean library checks.

We use the following input. From Chapter 19: a K3 surface has Betti numbers  $(1, 0, 22, 0, 1)$ , Hodge numbers  $h^{0,0} = h^{2,0} = h^{0,2} = h^{2,2} = 1$ ,  $h^{1,1} = 20$ , and Euler number 24. From Chapters 18 and 20:  $H^2(K3; \mathbb{Z})$  with the cup product is the even unimodular lattice  $3U \oplus 2E_8(-1)$  of signature  $(3, 19)$ . From Chapter 13: a Dp-brane is a source of charge  $\mu_p$  for the Ramond–Ramond potential  $C_{p+1}$ .

### 25.1 Gauss's law on a compact space

#### 25.1.1 Electrodynamics first

Let  $M$  be a compact manifold without boundary, of dimension  $d$ , and let a gauge field with field strength  $F = dA$  be sourced by a charge density  $\rho$ . Gauss's law is the statement

$$d\star F = \rho \text{vol}_M, \quad (25.2)$$

where  $\star$  is the Hodge star of  $M$  and  $\text{vol}_M$  its volume form. Integrate both sides over  $M$ . The left-hand side is the integral of an exact form over a manifold without boundary, which

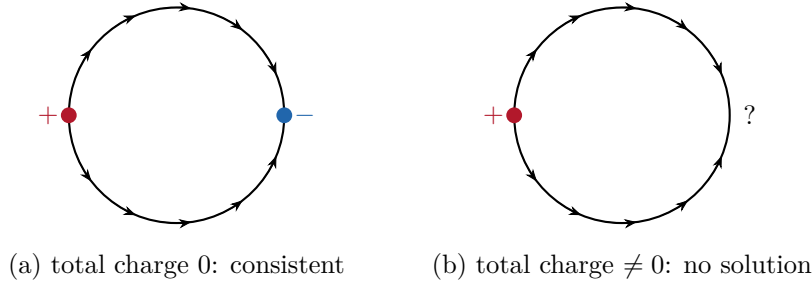


Figure 25.1: Gauss's law on a compact space, drawn for a circle. The flux leaving a positive charge in both directions must be absorbed by a negative charge (a). With a single charge (b) the two streams of flux meet head-on at the antipodal point and (25.2) has no solution.

vanishes by Stokes's theorem. Hence

$$Q_{\text{total}} = \int_M \rho \text{vol}_M = 0. \quad (25.3)$$

On  $\mathbb{R}^3$  the same integration over a ball gives the flux of the electric field through the boundary sphere, which can be anything; the difference is entirely the absence of a boundary. Two features of this one-line argument will survive every generalization below. First, the step uses only that  $\star F$  is a *globally defined* form: if it were defined only patch by patch the integral of its differential need not vanish. Second, the conclusion is insensitive to the metric; (25.3) is a topological constraint on the sources.

#### Physical picture: flux lines have no place to go

Polchinski's phrase for (25.3) is that with a nonzero total Ramond–Ramond source “the field equations are inconsistent, because R–R flux lines have no place to go in the compact space” [Pol Lecture 3, ll. 1987–1989]. He contrasts this with a nonzero total *tension*, which is only a source for the graviton and dilaton and makes those fields time dependent: an uncancelled tension changes the solution, an uncancelled charge means there is no solution [Pol Lecture 3, ll. 1985–1989].

### 25.1.2 Why the word “tadpole”

In perturbation theory, a term in the effective action that is *linear* in a field  $\phi$ ,  $\delta S = \int J \phi$ , produces a one-point function: a Feynman diagram with a single external leg ending in a blob, which looks like a tadpole. A linear term means that  $\phi = 0$  does not solve the equations of motion. If  $\phi$  has a potential, the vacuum merely shifts. For a gauge potential  $C$  whose constant mode has no potential at all, the equation of motion obtained by varying that constant mode is simply  $\int J = 0$ , and if the total source  $\int J$  does not vanish there is no solution. This is (25.3) in the language of diagrams. Sethi, Vafa and Witten introduce the subject in exactly these terms: a one-loop interaction “results in a one-point function for the  $B$  field of the effective two-dimensional theory, destabilizing the vacuum” [SVW §1, ll. 45–64].

### 25.1.3 A first example: orientifold planes and sixteen D-branes

The oldest stringy instance is the type I string and its T-duals (Chapter 13). T-dualizing type I on  $9 - p$  circles produces a torus  $T^{9-p}$  divided by the reflection of all its coordinates, combined with worldsheet orientation reversal. The reflection has  $2^{9-p}$  fixed points on the

| $p$ | compact space      | number of $Op$ -planes | charge of each (units of $\mu_p$ ) | total |
|-----|--------------------|------------------------|------------------------------------|-------|
| 9   | —                  | 1                      | −16                                | −16   |
| 7   | $T^2/\mathbb{Z}_2$ | 4                      | −4                                 | −16   |
| 5   | $T^4/\mathbb{Z}_2$ | 16                     | −1                                 | −16   |
| 3   | $T^6/\mathbb{Z}_2$ | 64                     | $-\frac{1}{4}$                     | −16   |

Table 25.1: Orientifold-plane charges from (25.4). In each row sixteen  $Dp$ -branes cancel the total. The last row agrees with the statement that  $T^6/\mathbb{Z}_2$  is a compact solution “with 16 D3-branes and 64 O3-planes”, so that the O3 tension is  $-\frac{1}{4}T_3$  [GKP §2.1, ll. 419–422].

torus; at each sits an *orientifold plane* ( $Op$ -plane), a  $p$ -dimensional object which is not dynamical but carries tension and Ramond–Ramond charge. In terms of the charge  $\mu_p$  of a  $Dp$ -brane the result is [Pol Lecture 3, eq. (93), ll. 1968–1984]

$$\mu'_p = -2^{p-5}\mu_p \quad \text{per } Op\text{-plane}, \quad 2^{9-p} \times (-2^{p-5})\mu_p = -16\mu_p \quad \text{in total} \quad (25.4)$$

(for the orthogonal projection; the symplectic projection has the opposite sign). The total is independent of  $p$ , as it must be, since T-duality cannot change whether the theory is consistent. Gauss’s law (25.3) on the compact torus then demands exactly 16  $Dp$ -branes, “giving the T-dual of  $SO(32)$ ” [Pol Lecture 3, ll. 1989–1990]. Table 25.1 lists the cases used in this chapter.

#### Common pitfall: counting branes and their images

In the covering torus every D-brane away from a fixed point has a mirror image under the reflection. Some authors count the brane and its image separately (“32 D-branes”), assign each half the charge, and double the orientifold charge accordingly; others rescale all charges by a common factor. Only the *ratio* of plane charge to brane charge,  $-2^{p-5}$ , and the vanishing of the total are convention independent. The Lean libraries use two bookkeepings of this cancellation (Section 25.6): the units of Table 25.1, and the same with all charges multiplied by 4. Both have the ratio  $-4$  at  $p = 7$ . A third bookkeeping, with sixteen “O7-planes” at the fixed points of  $T^4/\mathbb{Z}_2$  and ratio  $-2$ , matched no row of the table; it was corrected on 2026-09-18.

## 25.2 Curvature as a charge: the origin of $\chi/24$

### 25.2.1 The interaction $C \wedge X_8$

So far the sources were branes and planes. The new feature of compactifications to two, three and four dimensions is that the *curvature of the internal space itself* is a source. In type IIA string theory there is a one-loop interaction

$$\delta S = - \int B \wedge X_8(R), \quad (25.5)$$

where  $B$  is the Neveu–Schwarz two-form, the field that couples to the string, and  $X_8(R)$  is an eight-form built as a quartic polynomial in the curvature [SVW §1, eq. (1.1), ll. 45–54]. The same term appears in M-theory with  $B$  replaced by the three-form  $C$  that couples to the membrane, and in F-theory with the four-form that couples to the D3-brane [SVW §1, ll. 64–71]. In eleven-dimensional language it is an eight-derivative correction to supergravity, suppressed by six powers of the Planck mass, of the form  $-\int C \wedge X_8(R)$  [DRS §2.1, eq. (2.1), ll. 167–180].

Compactify on an eight-manifold  $N$ , so that spacetime is  $\mathbb{R}^n \times N$  with  $n = 2, 3, 4$  for type IIA, M-theory and F-theory respectively. If  $C$  is constant along  $N$  and has its legs along  $\mathbb{R}^n$ , (25.5) becomes  $-(\int_N X_8) \int_{\mathbb{R}^n} C$ : a term linear in  $C$ , a tadpole, with coefficient

$$I = - \int_N X_8 = \frac{\chi(N)}{24} \quad [\text{SVW §1, eq. (1.2) and ll. 93–107}]. \quad (25.6)$$

The first equality is the definition of  $I$ . The second holds for a Calabi–Yau fourfold; SVW obtain it by observing that the one-loop computation reduces to the worldsheet index  $\text{tr}(-1)^F$  of the sigma model on  $N$ , which is the Euler number, with a proportionality constant fixed in the original calculation. Becker and Becker reach the same conclusion from the differential-form side: for an eight-manifold admitting a nowhere-vanishing spinor the Pontryagin numbers obey  $p_1^2 - 4p_2 + 8\chi = 0$ , the polynomial  $X_8$  is proportional to  $P_1^2 - 4P_2$ , and therefore  $\int X_8$  is a fixed multiple of  $\chi$  [BB §2, eqs. (2.19)–(2.22), ll. 292–339]. [L]

#### Common pitfall: normalizations of $X_8$ and of $G$

The three papers normalize differently. BB write  $\int_{K^8} X_8 = -\chi/(4!(2\pi)^4)$  and a flux constraint  $\int F \wedge F + \chi/12 = 0$  for their field strength  $F$  [BB eqs. (2.22), (2.58), ll. 328–339, 706–713]; SVW use  $I = -\int X_8 = \chi/24$ ; DRS absorb a sign and write the source as  $+4\pi^2 X_8$  in (25.8) below. This book follows DRS and GVW: the integrally quantized object is  $G/2\pi$ , and the tadpole condition is (25.1). When you read another paper, fix the dictionary by the one convention-independent statement: with no flux, the number of branes is  $\chi/24$ .

### 25.2.2 Cancelling the tadpole with branes

The membrane couples to  $C$  by  $\int_{\mathbb{R}^3 \times \{y\}} C$  with unit charge. Placing  $n$  membranes at points  $y_1, \dots, y_n$  of  $N$ , stretched along  $\mathbb{R}^3$ , adds  $n$  to the coefficient of the term linear in  $C$ , with the sign such that branes (not anti-branes) compensate a positive  $I$ . So the simplest cure for the tadpole is to “include in the vacuum  $I$  strings, membranes, and three-branes in Type IIA, M-theory, and F-theory, respectively” [SVW §1, ll. 74–78]. Two obstructions are immediately visible [SVW §1, ll. 79–92]:

1.  $I$  may be *fractional*. Branes carry integer charge, so nothing can be done with branes alone unless  $\chi/24 \in \mathbb{Z}$ .
2.  $I$  may be a *negative* integer. Then anti-branes are needed. The vacuum is free of tadpoles, but supersymmetry is completely broken: the fourfold preserves supersymmetries of one eight-dimensional chirality, a brane preserves those of the same chirality by definition, and an anti-brane preserves only the opposite ones.

Hence, with branes alone, a supersymmetric tadpole-free vacuum exists if and only if  $\chi/24$  is a non-negative integer. [L]

### 25.2.3 Cancelling the tadpole with flux

There is a second contribution. Eleven-dimensional supergravity contains the Chern–Simons interaction  $\int C \wedge G \wedge G$  with  $G = dC$  [DRS §1, eq. (1.1), ll. 59–68]. If  $G$  has an expectation value along  $N$ , this term too is linear in the components of  $C$  along  $\mathbb{R}^3$ : a flux on the internal space carries membrane charge. The cleanest way to see all three contributions at once is the equation for the warp factor. With flux the metric is no longer a product but

$$ds^2 = e^{-\phi(y)} \eta_{\mu\nu} dx^\mu dx^\nu + e^{\frac{1}{2}\phi(y)} g_{\bar{a}\bar{b}} dy^{\bar{a}} dy^{\bar{b}}, \quad (25.7)$$

with  $g$  a Ricci-flat Kähler metric on the fourfold, and the warp factor obeys [DRS §2.1, eqs. (2.2), (2.5), ll. 180–228]

$$\Delta(e^{\frac{3}{2}\phi}) = \star \left\{ 4\pi^2 X_8 - \frac{1}{2} G \wedge G - 4\pi^2 \sum_{i=1}^n \delta^8(y - y_i) \right\}. \quad (25.8)$$

Here  $\Delta$  and  $\star$  are the Laplacian and Hodge star of  $g$ , and the  $\delta^8$  are eight-forms of unit integral. Equation (25.8) is Gauss’s law (25.2) for the electric component  $G_{\mu\nu\rho a} = \epsilon_{\mu\nu\rho} \partial_a e^{-\frac{3}{2}\phi}$  of the four-form: the warp factor plays the role of the electrostatic potential. Now integrate over the compact fourfold. The left-hand side is the integral of a Laplacian of a globally defined function and vanishes. Therefore

$$0 = 4\pi^2 \int_N X_8 - \frac{1}{2} \int_N G \wedge G - 4\pi^2 n \implies \int_N X_8 = n + \frac{1}{8\pi^2} \int_N G \wedge G. \quad (25.9)$$

Setting  $G = 0$  and comparing with the brane-only result  $n = \chi/24$  of (25.6) fixes  $\int_N X_8 = \chi/24$  in this normalization. Since  $8\pi^2 = 2(2\pi)^2$ , we obtain

$$\frac{\chi}{24} = \frac{1}{8\pi^2} \int_N G \wedge G + n = \frac{1}{2} \int_N \frac{G}{2\pi} \wedge \frac{G}{2\pi} + n, \quad (25.10)$$

which is (25.1) [DRS §1, eq. (1.2), ll. 70–80; GVW §2.1, eq. (2.1), ll. 94–110]. GVW allow anti-membranes as well and let  $n$  be the difference of the numbers of membranes and anti-membranes, “a positive or negative integer”. [L]

#### 25.2.4 The flux term is non-negative

Supersymmetry restricts the internal flux to be a  $(2, 2)$ -form that is *primitive*,  $J \wedge G = 0$  with  $J$  the Kähler form, and these conditions imply that  $G$  is self-dual,  $G = \star G$  [DRS §2.1, eqs. (2.3), (2.8)–(2.10), ll. 193–263]. For any real form,  $G \wedge \star G = |G|^2 \text{vol}$  with  $|G|^2 \geq 0$  pointwise. Hence

$$\int_N G \wedge G = \int_N G \wedge \star G = \int_N |G|^2 \text{vol} \geq 0, \quad \text{with equality only for } G = 0. \quad (25.11)$$

Combined with  $n \geq 0$  (no anti-branes, for supersymmetry), (25.10) becomes a *budget*: two non-negative quantities add up to the fixed number  $\chi/24$ . In particular

$$0 \leq n \leq \frac{\chi}{24}, \quad 0 \leq \frac{1}{2} \int_N \frac{G}{2\pi} \wedge \frac{G}{2\pi} \leq \frac{\chi}{24}, \quad G = 0 \iff n = \frac{\chi}{24}. \quad (25.12)$$

DRS state the middle inequality for  $K3 \times K3$  as  $\frac{1}{2} \int \frac{G}{2\pi} \wedge \frac{G}{2\pi} \leq 24$  [DRS §3.1, ll. 640–649]. The elementary step from (25.10) to (25.12), for natural numbers adding up to 24, is the Lean theorem `tadpole_budget` (Section 25.6).

#### Physical picture: an electrostatics problem with three kinds of charge

Read (25.8) as electrostatics on the fourfold. The curvature is a smooth background charge density of total charge  $+\chi/24$  (in units where a membrane has charge 1 and we have moved everything to one side). Membranes are point charges; self-dual flux is a smooth charge density of the same sign as the membranes. The warp factor is the potential they generate together, and it exists only if the total vanishes. The only situation in which one may forget all this is the non-compact one: on  $\mathbb{R}^8$ , or on a local model of a singularity, “flux can escape to infinity, and hence [the tadpole condition] no longer holds”; the flux measured at infinity appears as an extra term [GVW §2.2, ll. 380–399].

### 25.3 How large is $\chi/24$ ?

The left-hand side of (25.1) is pure topology. We compute it in three ways for  $K3 \times K3$ , and then ask what can be said in general.

#### 25.3.1 Three routes to $\chi(K3 \times K3) = 576$

*Route 1: Künneth.* The cohomology of a product is the tensor product of the cohomologies, so Poincaré polynomials multiply. With  $P_{K3}(t) = 1 + 22t^2 + t^4$ ,

$$P_{K3 \times K3}(t) = (1 + 22t^2 + t^4)^2 = 1 + 44t^2 + 486t^4 + 44t^6 + t^8, \quad (25.13)$$

so the Betti numbers are  $(1, 0, 44, 0, 486, 0, 44, 0, 1)$  and  $\chi = P(-1) = 1 + 44 + 486 + 44 + 1 = 576 = 24^2$ . The Euler number is multiplicative because  $P_{X \times Y}(-1) = P_X(-1)P_Y(-1)$ . The middle Betti number decomposes as

$$b_4 = 486 = \underbrace{1 \cdot 1}_{H^0 \otimes H^4} + \underbrace{22 \cdot 22}_{H^2 \otimes H^2} + \underbrace{1 \cdot 1}_{H^4 \otimes H^0}, \quad (25.14)$$

a decomposition we shall need in Section 25.4:  $G$ -flux lives in  $H^4$ .

*Route 2: Hodge numbers.* The Hodge numbers of a product are given by the same rule,  $h^{p,q}(X \times Y) = \sum_{r,s} h^{r,s}(X) h^{p-r,q-s}(Y)$ . From the K3 diamond one finds the  $5 \times 5$  table

$$(h^{p,q}(K3 \times K3)) = \begin{pmatrix} 1 & 0 & 2 & 0 & 1 \\ 0 & 40 & 0 & 40 & 0 \\ 2 & 0 & 404 & 0 & 2 \\ 0 & 40 & 0 & 40 & 0 \\ 1 & 0 & 2 & 0 & 1 \end{pmatrix}, \quad \chi = \sum_{p,q} (-1)^{p+q} h^{p,q} = 576. \quad (25.15)$$

For instance  $h^{2,2} = 1 \cdot 1 + 20 \cdot 20 + 1 \cdot 1 + 2 \cdot (1 \cdot 1) = 404$ : the last term counts  $h^{2,0}h^{0,2} + h^{0,2}h^{2,0}$ . Every nonzero entry has  $p + q$  even, so all signs are  $+$  and the entries add up to 576. Note  $h^{2,0} = 2$ : the product has two holomorphic two-forms, one from each factor. Its holonomy is  $SU(2) \times SU(2)$ , smaller than  $SU(4)$ , and M-theory on it has  $N = 4$  rather than  $N = 2$  supersymmetry in three dimensions [DRS §3.1, ll. 638–640].

*Route 3: Chern classes.* For a complex fourfold  $\chi = \int c_4$ . The tangent bundle of a product is the sum of the pull-backs, so total Chern classes multiply:  $c(K3 \times K3) = (1 + c_2^{(1)})(1 + c_2^{(2)})$ , where  $c_2^{(i)}$  is the second Chern class of the  $i$ -th factor, with  $\int_{K3} c_2 = 24$  and  $c_1 = 0$ . Hence

$$c_2 = c_2^{(1)} + c_2^{(2)}, \quad c_4 = c_2^{(1)}c_2^{(2)}, \quad \int c_4 = 24 \cdot 24 = 576, \quad \int c_2^2 = 2 \cdot 24 \cdot 24 = 1152. \quad (25.16)$$

All three routes give  $\chi/24 = 24$ , the value quoted by SVW and DRS [SVW §2.2, ll. 252–255; §2.4, l. 535; DRS §3.1, l. 640]. The arithmetic  $576 = 24 \cdot 24$  and  $576/24 = 24$  with remainder 0 is the Lean theorem `k3k3_anomaly`. [A] for the arithmetic, [L] for the Künneth formula.

#### 25.3.2 The general fourfold: $\chi$ is divisible by six

Is  $\chi/24$  always an integer? No, and the argument of SVW shows exactly how far one can go. Let  $N$  be a Calabi–Yau fourfold and  $\chi_p = \sum_{q=0}^4 (-1)^q h^{p,q}$  the index of the Dolbeault operator on  $(p, 0)$ -forms. The index theorem expresses  $\chi_p$  through Chern numbers; since  $c_1 = 0$  only  $\int c_2^2$  and  $\int c_4 = \chi$  can appear [SVW §2.1, eq. (2.1), ll. 125–155]:

$$\chi_0 = \frac{1}{720} \int (3c_2^2 - c_4), \quad \chi_1 = \frac{1}{180} \int (3c_2^2 - 31c_4), \quad \chi_2 = \frac{1}{120} \int (3c_2^2 + 79c_4). \quad (25.17)$$

Eliminate  $\int c_2^2$  with the first equation,  $\int c_2^2 = 240\chi_0 + \chi/3$ :

$$\chi_1 = \frac{720\chi_0 + \chi - 31\chi}{180} = 4\chi_0 - \frac{\chi}{6}, \quad \chi_2 = \frac{720\chi_0 + \chi + 79\chi}{120} = 6\chi_0 + \frac{2\chi}{3}. \quad (25.18)$$

For holonomy exactly  $SU(4)$  one has  $h^{0,q} = (1, 0, 0, 0, 1)$ , so  $\chi_0 = 2$ , and (25.18) becomes the formulas of SVW,  $\int c_2^2 = 480 + \chi/3$ ,  $\chi_1 = 8 - \chi/6$  and  $\chi_2 = 12 + 2\chi/3$  [SVW §2.1, ll. 158–204]. Since  $\chi_1$  and  $\chi_0$  are integers,  $\chi$  is divisible by 6; the possible fractional parts of  $\chi/24$  are  $0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}$  and “the obstruction takes values in  $\mathbb{Z}_4$ ” [SVW §2.1, ll. 121–124]. [L]

**Example 25.1** (Checking (25.18) on  $K3 \times K3$ ). Here  $\chi_0 = 1 - 0 + 2 - 0 + 1 = 4$  from the first row of (25.15), not 2, because the holonomy is smaller than  $SU(4)$ . The second row gives  $\chi_1 = -40 - 40 = -80$  and the third  $\chi_2 = 2 + 404 + 2 = 408$ . The formulas predict  $4 \cdot 4 - 576/6 = 16 - 96 = -80$  and  $6 \cdot 4 + 2 \cdot 576/3 = 24 + 384 = 408$ . The Chern numbers (25.16) give  $\chi_0 = (3 \cdot 1152 - 576)/720 = 2880/720 = 4$ . All agree (symbolic check with sympy).

**Example 25.2** (A fractional tadpole: the sextic fourfold). Let  $N \subset \mathbb{P}^5$  be a smooth hypersurface of degree six, a Calabi–Yau fourfold. By adjunction its total Chern class is  $c(N) = (1 + H)^6/(1 + 6H)$  with  $H$  the hyperplane class. Expanding,

$$\frac{(1 + H)^6}{1 + 6H} = 1 + 0 \cdot H + 15H^2 - 70H^3 + 435H^4 + \dots$$

The vanishing coefficient of  $H$  confirms  $c_1 = 0$ . Since  $\int_N H^4 = 6$  (a degree-six hypersurface meets a line in six points),  $\chi = 6 \cdot 435 = 2610$ . This is divisible by 6 ( $2610 = 6 \cdot 435$ ) but  $2610 = 24 \cdot 108 + 18$ , so

$$\frac{\chi}{24} = 108\frac{3}{4},$$

as stated in [SVW §2.1, ll. 205–213]. No number of membranes cancels this tadpole. DRS point out the way out: “for cases where  $\chi/24$  is not integral,  $G$ -flux is actually required” [DRS §1, ll. 70–72]. Such a flux cannot be an integral class  $\xi$ : then  $\frac{1}{2} \int \xi \wedge \xi \in \frac{1}{2}\mathbb{Z}$  could never supply the fractional part  $\frac{3}{4}$ . It lies in the shifted lattice described in Section 25.4.

### 25.3.3 Elliptic fibrations, and a negative example

For F-theory one needs  $N$  elliptically fibred over a threefold  $B$ , with a section and a smooth Weierstrass model. Then  $\chi$  can be written in terms of the Chern classes  $c_i$  of the base [SVW §2.2, eqs. (2.3), (2.10)–(2.12), ll. 258–386]:

$$\chi(N) = 12 \int_B c_1(c_2 + 30c_1^2), \quad \frac{\chi}{24} = \int_B \left(15c_1^3 + \frac{1}{2}c_1c_2\right). \quad (25.19)$$

Two checks. For  $B = K3 \times \mathbb{P}^1$  the fourfold is  $K3 \times K3$  with one factor elliptic;  $c_1(B)$  is twice the point class of  $\mathbb{P}^1$ , so  $c_1^3 = 0$  and  $\int c_1c_2 = 2 \cdot 24 = 48$ , giving  $\chi/24 = 24$ . For  $B = \mathbb{P}^3$ ,  $c_1 = 4H$  and  $c_2 = 6H^2$ , so  $\int c_1c_2 = 24$ ,  $\int c_1^3 = 64$  and  $\chi/24 = 15 \cdot 64 + 12 = 972$ , the value quoted by SVW. When  $N$  has full  $SU(4)$  holonomy the arithmetic genus of  $B$  is  $\int_B c_1c_2/24 = 1$ , and (25.19) becomes  $\chi/24 = 12 + 15 \int_B c_1^3$ , a positive integer: in this class supersymmetric tadpole cancellation by branes is always possible [SVW §2.2, eq. (2.4), ll. 270–291]. [L]

The sign obstruction is rarer. SVW find it in a conformal field theory rather than a smooth manifold: the orbifold  $T^8/(\mathbb{Z}_2 \times \mathbb{Z}_2)$  is a singular point of  $K3 \times K3$ , but *with discrete*



*torsion* its Hodge diamond becomes

$$\begin{pmatrix} 1 & 0 & 2 & 0 & 1 \\ 0 & 8 & 64 & 8 & 0 \\ 2 & 64 & 20 & 64 & 2 \\ 0 & 8 & 64 & 8 & 0 \\ 1 & 0 & 2 & 0 & 1 \end{pmatrix}, \quad \chi = 4 - 48 - 104 - 48 + 4 = -192 = -8 \cdot 24, \quad (25.20)$$

where we summed the rows with signs  $(-1)^{p+q}$ . Thus  $\chi/24 = -8$ : eight anti-branes are needed and supersymmetry is broken [SVW §2.3, ll. 429–515]. The large odd entries  $h^{2,1} = h^{1,2} = 64$  are what drives  $\chi$  negative.

## 25.4 Flux quantization and why $\frac{1}{2} \int G \wedge G$ is an integer

### 25.4.1 Dirac quantization of $G$

The flux term in (25.1) must be an integer whenever  $\chi/24$  is, for otherwise no integer  $n$  solves the equation. Where does this come from? The membrane couples to  $C$  as a charged particle couples to a gauge potential, and the usual Dirac argument (the phase  $\exp(i \int C)$  must be well defined when the membrane worldvolume is deformed across a four-cycle) requires the periods of  $G/2\pi$  over four-cycles to be integers:

$$\xi := \left[ \frac{G}{2\pi} \right] \in H^4(N; \mathbb{Z}) \quad (\text{first approximation}). \quad (25.21)$$

The precise statement carries a shift. The periods of  $G/2\pi$  coincide with those of  $c_2(N)/2$  modulo integers [GVW §5, ll. 1993–1996], so that  $\xi$  lives in a translate of the lattice  $H^4(N; \mathbb{Z})$ ; this shift “arises precisely when the intersection form on  $H^4(Y; \mathbb{Z})$  is not even”, and with the shifted condition the right-hand side of the tadpole formula is always integral [GVW §2.1, ll. 112–125]. DRS phrase the same fact as:  $[G/\pi]$  is always integral, and “if  $\chi/24 \in \mathbb{Z}$  then  $[G/2\pi]$  is an integer cohomology class” [DRS §2.1, ll. 230–234]. [\[L\]](#)

For  $K3 \times K3$  there is no shift, and we can see this directly. By (25.16),  $c_2/2 = \frac{1}{2}c_2^{(1)} + \frac{1}{2}c_2^{(2)}$ . The class  $c_2^{(1)}$  is 24 times the generator of  $H^4(K3_1; \mathbb{Z})$ , so  $c_2/2$  is 12 times a generator plus 12 times another: an integral class. Its periods are integers, and (25.21) holds as written. What remains is to understand why *half* the self-intersection of an integral class is an integer. That is a property of the lattice.

### 25.4.2 The lattice $H^4(K3 \times K3; \mathbb{Z})$

Let  $\Lambda = H^2(K3; \mathbb{Z}) \cong 3U \oplus 2E_8(-1)$  with Gram matrix  $L$  in a basis  $e_1, \dots, e_{22}$ , so  $L_{ik} = \int_{K3} e_i \wedge e_k$ . The Künneth decomposition (25.14) holds over  $\mathbb{Z}$  because the cohomology of  $K3$  has no torsion:

$$H^4(K3_1 \times K3_2; \mathbb{Z}) = \underbrace{\mathbb{Z}[K3_1 \times \text{pt}]^\vee \oplus \mathbb{Z}[\text{pt} \times K3_2]^\vee}_{\text{rank } 2} \oplus \underbrace{\Lambda_1 \otimes \Lambda_2}_{\text{rank } 484}. \quad (25.22)$$

The two classes in the first summand are the volume forms of the two factors; each squares to zero (a volume form of a four-manifold wedged with itself vanishes) and their product integrates to 1, so they span a hyperbolic plane  $U$ . On the second summand take the basis  $e_i \otimes f_j$ , represented by  $\pi_1^* e_i \wedge \pi_2^* f_j$ . Two-forms commute, so

$$\int_{K3_1 \times K3_2} (e_i \wedge f_j) \wedge (e_k \wedge f_l) = \int_{K3_1} e_i \wedge e_k \int_{K3_2} f_j \wedge f_l = L_{ik} L_{jl}. \quad (25.23)$$



The Gram matrix of  $\Lambda_1 \otimes \Lambda_2$  is therefore the *Kronecker product*  $L \otimes L$ , a  $484 \times 484$  integer matrix with entries  $(L \otimes L)_{(i,j),(k,l)} = L_{ik}L_{jl}$ . A flux in this summand is  $\xi = \sum_{i,j} x_{ij} e_i \otimes f_j$  with integers  $x_{ij}$ , and

$$\int \xi \wedge \xi = Q(x) := \sum_{i,j,k,l} x_{ij} L_{ik} L_{jl} x_{kl}. \quad (25.24)$$

For a decomposable flux  $\xi = a \otimes b$ , i.e.  $x_{ij} = a_i b_j$ , this factorizes:  $Q = (a \cdot a)(b \cdot b)$ , the product of the two self-intersections on the two K3 surfaces.

**Proposition 25.3** (Evenness of the flux lattice). *Let  $L_1, L_2$  be symmetric integer matrices and suppose every diagonal entry of  $L_1$  is even. Then  $x^\top (L_1 \otimes L_2) x$  is even for every integer vector  $x$ . Consequently  $\frac{1}{2} \int \xi \wedge \xi \in \mathbb{Z}$  for every  $\xi \in \Lambda_1 \otimes \Lambda_2$ .*

*Proof.* Write  $M = L_1 \otimes L_2$  and label its rows by pairs  $p = (i, j)$ .  $M$  is symmetric because  $M_{(i,j),(k,l)} = (L_1)_{ik}(L_2)_{jl} = (L_1)_{ki}(L_2)_{lj} = M_{(k,l),(i,j)}$ . Its diagonal is  $M_{pp} = (L_1)_{ii}(L_2)_{jj}$ , a product with an even factor, hence even. For any symmetric integer matrix,

$$x^\top M x = \sum_p M_{pp} x_p^2 + 2 \sum_{p < q} M_{pq} x_p x_q,$$

after choosing any total order on the labels. The second sum is visibly even and the first is even because each  $M_{pp}$  is.  $\square$

Two remarks. First, *nothing is assumed about  $L_2$  beyond symmetry*: one even factor suffices, and unimodularity plays no role. Second, the statement is about all of  $\Lambda_1 \otimes \Lambda_2$ , not only about decomposable elements, for which it would be trivial. The proposition is the Lean theorem `kronForm_even`, and its corollary is `flux_half_selfIntersection_integral`. **[A]**

Adding the hyperbolic plane of (25.22), which is also even, the whole lattice  $H^4(K3 \times K3; \mathbb{Z}) \cong U \oplus (\Lambda \otimes \Lambda)$  is even, consistently with the absence of a shift in the quantization condition. Its signature follows from that of  $\Lambda$ , (3, 19): a tensor product of a positive with a positive or a negative with a negative direction is positive, so

$$\begin{aligned} (n_+, n_-) &= (3 \cdot 3 + 19 \cdot 19 + 1, 2 \cdot 3 \cdot 19 + 1) = (371, 115), \\ 371 + 115 &= 486, \quad 371 - 115 = 256 = (-16)^2, \end{aligned} \quad (25.25)$$

the last equality being the multiplicativity of the signature,  $\sigma(K3) = -16$ . A supersymmetric flux is self-dual and so lies in the 371-dimensional positive part, in agreement with (25.11).

**Common pitfall: “the form is indefinite, so the flux term can be negative”**

The lattice  $H^4$  is indefinite, and an arbitrary integral class  $\xi$  can have  $\int \xi \wedge \xi < 0$ . Such a class is not a solution of the equations of motion with unbroken supersymmetry: it fails self-duality. The positivity (25.11) is a statement about the solutions, the evenness of Proposition 25.3 is a statement about the whole lattice. The tadpole budget uses both.

### 25.4.3 Worked example: three ways to spend a budget of 24

We follow [DRS §3.1, eqs. (3.1)–(3.5), ll. 660–728]. Take for each K3 a smooth hypersurface of degree (2, 2, 2) in  $\mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1$ . It has three natural divisor classes  $C_1, C_2, C_3$ , obtained by fixing a point in one of the three  $\mathbb{P}^1$  factors. Two distinct  $C_i$  meet where two coordinates are fixed, which leaves a quadratic equation on the third  $\mathbb{P}^1$ : two points. A  $C_i$  can be moved off itself, so  $C_i \cdot C_i = 0$ . The Gram matrix of the sublattice they span is

$$L_{\text{DRS}} = \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 0 \end{pmatrix}, \quad \det L_{\text{DRS}} = 16. \quad (25.26)$$

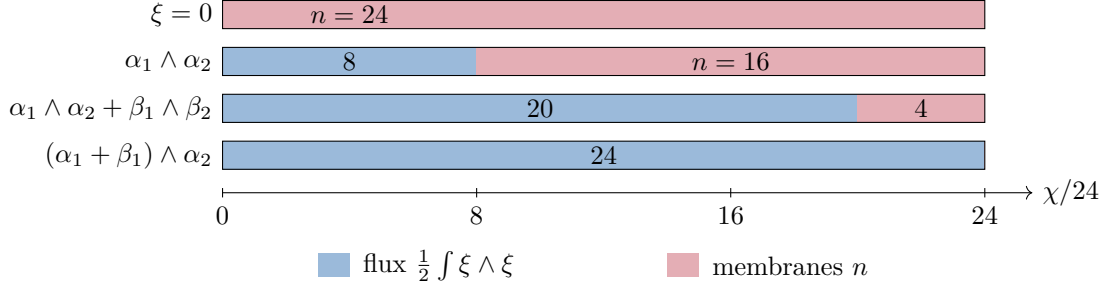


Figure 25.2: The tadpole budget of  $K3 \times K3$  for the four fluxes of Section 25.4.3. Flux and branes are both non-negative and add up to  $\chi/24 = 24$ .

It is symmetric with even (zero) diagonal, so Proposition 25.3 applies to it although it is far from unimodular. Take  $J = C_1 + C_2 + C_3$  as Kähler class, and

$$\alpha = C_1 - C_2, \quad \beta = C_1 - C_3.$$

With  $L_{\text{DRS}}$  one computes  $\alpha \cdot \alpha = 0 + 0 - 2 \cdot 2 = -4$ , likewise  $\beta \cdot \beta = -4$ ,  $\alpha \cdot \beta = C_1^2 - C_1 C_3 - C_2 C_1 + C_2 C_3 = 0 - 2 - 2 + 2 = -2$ , and  $J \cdot \alpha = (2 + 2) - (2 + 2) = 0 = J \cdot \beta$ . So  $\alpha, \beta$  are primitive  $(1, 1)$  classes. On a Kähler surface a primitive  $(1, 1)$ -form is anti-self-dual (the two-form computation in [DRS §2.1, eq. (2.7), ll. 240–241]), and on a product of two four-manifolds  $\star(a \wedge b) = \star_1 a \wedge \star_2 b$  for two-forms  $a, b$  pulled back from the factors, so the wedge product of two anti-self-dual two-forms is self-dual:  $(-a) \wedge (-b) = a \wedge b$ . With subscripts for the two K3 factors, three flux choices are:

| flux $\xi = G/2\pi$                    | $\int \xi \wedge \xi$ | $\frac{1}{2} \int \xi \wedge \xi$ | membranes $n$ |
|----------------------------------------|-----------------------|-----------------------------------|---------------|
| 0                                      | 0                     | 0                                 | 24            |
| $\alpha_1 \wedge \alpha_2$             | $(-4)(-4) = 16$       | 8                                 | 16            |
| $(\alpha_1 + \beta_1) \wedge \alpha_2$ | $(-12)(-4) = 48$      | 24                                | 0             |

using  $(\alpha + \beta)^2 = -4 - 4 + 2(-2) = -12$ . The second and third rows are DRS’s equations (3.4) and (3.5): “we can place 16 membranes” or “cancel the anomaly completely without branes”. A non-decomposable flux is just as easy with (25.24): for  $\xi = \alpha_1 \wedge \alpha_2 + \beta_1 \wedge \beta_2$ ,

$$\int \xi \wedge \xi = (\alpha \cdot \alpha)^2 + (\beta \cdot \beta)^2 + 2(\alpha \cdot \beta)^2 = 16 + 16 + 8 = 40, \quad n = 24 - 20 = 4.$$

All self-intersections are even, as Proposition 25.3 guarantees, and all four rows satisfy (25.12). Figure 25.2 displays the result.

#### Physical picture: flux freezes moduli

The second and third rows are possible only on special K3 surfaces. A generic K3 has no integral  $(1, 1)$  classes at all (its Picard lattice is trivial, Chapter 20), and DRS begin their construction with exactly this remark [DRS §3.1, ll. 673–676]. Once an integral class  $\xi$  is chosen, the requirement that it be of type  $(2, 2)$  and primitive is a condition on the complex structure and on the Kähler class: for non-zero  $G$  “we must adjust the complex structure of  $X$  so that  $G^{0,4} = G^{1,3} = 0$ ”, and the Kähler class so that  $G$  is primitive [GVW §2.4, eqs. (2.36)–(2.37), ll. 898–911]. Turning this around, a chosen flux restricts the moduli to the locus where it is supersymmetric. This is the mechanism of moduli stabilization, the subject of Chapter 26. Because both terms of the budget are non-negative, GVW note that there are only finitely many possible choices of  $n$  and  $G$  at a fixed value of the

total [GVW §2.2, ll. 401–405].

## 25.5 The type IIB picture: $K3 \times T^2/\mathbb{Z}_2$ with O7-planes and D7-branes

### 25.5.1 From M-theory on $K3 \times K3$ to the orientifold

F-theory is reached from M-theory by taking one K3 to be elliptically fibred and shrinking the fibre; M-theory on the fourfold goes over to type IIB on the base  $B$ , with the complex structure  $\tau$  of the fibre as the varying type IIB coupling [DRS §1, ll. 45–52]. Take the elliptic K3 at its orbifold point  $T^4/\mathbb{Z}_2 = (T_{\text{base}}^2 \times T_{\text{fibre}}^2)/\mathbb{Z}_2$ . Then the base of the fibration is  $T^2/\mathbb{Z}_2$  and the fourfold  $K3 \times T^4/\mathbb{Z}_2$  describes type IIB on

$$\frac{K3 \times T^2}{\Omega(-1)^{F_L} \mathbb{Z}_2}, \quad \mathbb{Z}_2 : z^1 \mapsto -z^1 \text{ on } T^2, \quad (25.27)$$

where  $\Omega$  is worldsheet parity and  $(-1)^{F_L}$  the left-moving fermion number [DRS §4.1, ll. 1289–1293; §4.2, ll. 1350–1352]. This orientifold “is a special point in the moduli space of F theory on  $K3 \times K3$ ”. The dictionary between the two pictures is explicit in DRS’s account of the sixteen fixed points [DRS §2.4, ll. 511–541]:

- M-theory on  $T^4/\mathbb{Z}_2$  has 22 gauge fields from reducing  $C$  on  $H^2(K3; \mathbb{Z})$ : sixteen from the twisted sectors (the  $A_1$  singularities of Chapter 21) and six untwisted.
- In the F-theory limit the two untwisted forms dual to the fibre and its Hodge dual drop out; the element  $\Omega(-1)^{F_L}$  acts on the doublet of two-forms  $(B, B')$  as  $-1 \in SL(2, \mathbb{Z})$ , so only components with one leg on  $T^2$  survive.
- “The 16 fixed points coalesce into 4 groups of coincident  $A_1$  singularities”, one group at each of the four fixed points of  $z^1 \mapsto -z^1$  on  $T^2$ . The gauge group is enhanced  $SU(2)^4 \rightarrow SO(8)$  at each, which in the orientifold arises “from placing 4 D7-branes at the location of each O7-plane”. The total gauge group is  $SO(8)^4$  [DRS §4.1, ll. 1311–1315].

### 25.5.2 Two tadpoles, not one

The orientifold (25.27) has *two* Gauss laws, and it is important to keep them apart.

*The seven-brane tadpole.* O7-planes and D7-branes fill spacetime and wrap K3; they are points on  $T^2/\mathbb{Z}_2$ . They are charged under the Ramond–Ramond eight-form  $C_8$ , whose field strength has flux lines spreading in the two transverse directions, the compact pillow  $T^2/\mathbb{Z}_2$ . Gauss’s law (25.3) on the pillow, with the  $p = 7$  row of Table 25.1, reads

$$\underbrace{4 \times (-4)}_{\text{O7-planes}} + \underbrace{16 \times (+1)}_{\text{D7-branes}} = 0 \quad (\text{units of } \mu_7). \quad (25.28)$$

With four D7-branes on top of each O7-plane the cancellation is *local*:  $-4 + 4 = 0$  at each fixed point, no  $C_8$  flux lines run across the pillow, the coupling  $\tau$  is constant, and perturbation theory applies everywhere. This is why the  $SO(8)^4$  point is the orientifold limit of F-theory (Figure 25.3). No choice of three-form flux enters (25.28).

*The three-brane tadpole.* D3-branes fill spacetime and are points in the full six-dimensional internal space. They are charged under the four-form  $D^+$  of type IIB, the field into which the membrane’s  $C_{\mu\nu\rho}$  lifts [DRS §2.4, eq. (2.27), ll. 479–486]. A  $G$ -flux with one leg along the

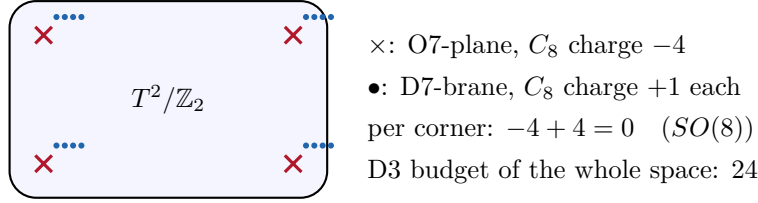


Figure 25.3: Charge bookkeeping on the pillow  $T^2/\mathbb{Z}_2$ , in units of the D7 charge  $\mu_7$ . Each seven-brane also wraps the K3 (not drawn). The seven-brane tadpole cancels corner by corner; the three-brane tadpole 24 must still be cancelled by D3-branes and three-form flux. The geometry of the pillow is described in Chapter 24.

elliptic fibre,  $G/2\pi = dz \wedge \omega - d\bar{z} \wedge \star \omega$  with  $\omega \in H^{1,2}(B)$ , lifts to the Neveu–Schwarz and Ramond–Ramond three-form field strengths

$$H = \omega - \star \omega, \quad H' = \omega \tau - \star \omega \bar{\tau}, \quad (25.29)$$

and the tadpole condition (25.1) becomes

$$\frac{\chi}{24} = n - \int_B H \wedge H', \quad n = \text{number of D3-branes}, \quad (25.30)$$

in the normalization of [DRS §2.4, eqs. (2.30)–(2.32), ll. 549–578]; the combination  $-\int H \wedge H'$  is the image of the non-negative M-theory flux term. It comes from the type IIB Chern–Simons interaction  $\int D^+ \wedge H \wedge H'$  [DRS §1, eq. (1.3), ll. 92–100], the descendant of  $\int C \wedge G \wedge G$ . In the conventions of GKP the same statement is the integrated Bianchi identity for the five-form,  $(2\kappa_{10}^2 T_3)^{-1} \int H_{(3)} \wedge F_{(3)} + Q_3^{\text{loc}} = 0$ , with  $Q_3^{\text{loc}}$  the D3 charge of all localized sources [GKP §2.1, eqs. (2.23)–(2.24), ll. 483–502]. For our orientifold the number on the left of (25.30) is  $\chi(K3 \times K3)/24 = 24$ ,

$$-\int H \wedge H' + n = 24, \quad (25.31)$$

which DRS write in its M-theory form  $\frac{1}{2} \int \frac{G}{2\pi} \wedge \frac{G}{2\pi} + n = 24$  on  $K3 \times T^4/\mathbb{Z}_2$  before lifting the flux [DRS §4.2, eqs. (4.4)–(4.5), ll. 1358–1390].

*Remark 25.4* (Where the 24 sits in the type IIB picture). In M-theory the number 24 is smeared over the curvature of  $K3 \times K3$ . In type IIB the internal space  $K3 \times T^2$  has  $\chi = 0$  (Chapter 24) and cannot be the source. The source is the curvature of the *seven-brane worldvolumes*: through a Chern–Simons coupling of the form  $\int C_{(4)} \wedge p_1(R)$ , “a D7-brane wrapped on K3 has  $-1$  unit of D3 charge”, and in F-theory the total background charge of all seven-branes is  $Q_3^{\text{D7}} = -\chi(X)/24$  [GKP §2.1, eqs. (2.20), (2.22), ll. 428–474]; SVW make the same observation, the induced charge of a worldvolume  $K$  being  $c_2(K)/24$  [SVW §2.2, ll. 387–399]. Note  $\chi(K3)/24 = 1$ . Sixteen D7-branes account for  $-16$  of the  $-24$ . The remaining  $-8$  is carried by the four O7-planes,  $-2$  each: TT state that “2 units of D3 brane charge are induced on the world volume of each O7-plane and one units of D3 brane charge in each D7-brane, giving rise to a total of 24 units”, cancelled as  $\frac{1}{2}N_{\text{flux}} + N_{\text{D3}} = 24$  [TT §2.2, eq. (2.3), ll. 164–171] (their sign convention is opposite to GKP’s). The two routes to 24,  $4 \cdot 2 + 16 \cdot 1$  and  $\chi(K3 \times K3)/24$ , agree; the arithmetic is `induced_d3_charge_matches_target`.

### 25.5.3 A chain of dualities as a consistency check

With no flux, (25.31) requires 24 D3-branes. Two T-dualities along the  $T^2$  invert both radii,  $R_i \mapsto \alpha'/R_i$  (Chapter 8), map  $\Omega(-1)^{F_L}\mathbb{Z}_2$  to  $\Omega$ , the O7/D7 system to the O9/D9 system of

type I, and each D3-brane to a D5-brane wrapped on the dual torus: “the resulting theory is type I on  $K3 \times T^2$  with 24 D5-branes wrapping the  $T^2$ ” [DRS §4, ll. 1270–1276; §4.1, ll. 1298–1311]. An S-duality turns this into the  $SO(32)$  heterotic string on  $K3 \times T^2$ . SVW use exactly this chain, in the opposite direction, as “a check on the proportionality constant in the formula  $I = \chi/24$ ”: F-theory on  $K3 \times K3$  is dual to the heterotic string on  $K3 \times T^2$ , where, if no gauge bundle is excited on the K3, the worldsheet anomaly is cancelled by 24 five-branes wrapped on  $T^2$ ; the eight-dimensional duality turns them into 24 three-branes [SVW §2.4, ll. 526–535]. [L]

#### Physical picture: what flux buys, and what it costs

Replacing some of the 24 D3-branes by three-form flux changes the four-dimensional physics. DRS’s flux (25.29) on  $K3 \times T^2/\mathbb{Z}_2$  is built from a class  $\alpha \in H^{1,1}(K3)$  and a class  $\beta \in H^{2,0}(K3)$ ; “if  $\beta = 0$ , the model has  $N = 2$  supersymmetry” and otherwise  $N = 1$  [DRS §4.2, eq. (4.4), ll. 1374–1383]. The metric is no longer a product but is warped, conformal to  $K3 \times T^2$  [DRS §4, ll. 1268–1270], and after the two T-dualities the type I and heterotic duals live on a space that is complex but not Kähler, with non-zero three-form flux (torsion) [DRS §4, ll. 1276–1286]. T-duality is here the tool that turns a flux compactification one understands into one that would be hard to find directly.

## 25.6 What the machine has checked

The physics of this chapter (the interaction  $C \wedge X_8$ , the identification  $\int X_8 = \chi/24$ , supersymmetry conditions, dualities) is Tier L throughout. What Lean checks is the integer arithmetic of the budget and the lattice-theoretic Proposition 25.3. The relevant files are `DualScaleStream2/Flux/Tadpole.lean` and `DualScaleStream2/Flux/Integrality.lean`, and four files in older libraries that model the orientifold charges by integer literals. All audited theorems depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`.

#### In Lean: Euler numbers and the anomaly of $K3 \times K3$

```
def eulerFromHodge {m : ℕ} (h : Fin m → Fin m → ℕ) : ℤ :=
  ∑ p : Fin m, ∑ q : Fin m,
    (if (p.val + q.val) % 2 = 0 then 1 else -1 : ℤ) * (h p q : ℤ)

theorem euler_K3 :
  eulerFromHodge StringTheory.Frontier.k3HodgeNumber = 24 := by ...

def chiK3K3 : ℤ := eulerFromHodge StringTheory.Frontier.k3HodgeNumber *
  eulerFromHodge StringTheory.Frontier.k3HodgeNumber

theorem k3k3_anomaly : chiK3K3 % 24 = 0 ∧ chiK3K3 / 24 = 24 := by ...
```

*Reading.* `eulerFromHodge` is the alternating sum  $\sum (-1)^{p+q} h^{p,q}$  of an arbitrary square table of natural numbers. `k3HodgeNumber` is the  $3 \times 3$  table  $(1, 0, 1; 0, 20, 0; 1, 0, 1)$ , entered by hand in another library; `euler_K3` says its alternating sum is 24. The Euler number of  $K3 \times K3$  is *defined* as the product  $24 \cdot 24$ : multiplicativity (Künneth) is an input, not a theorem. `k3k3_anomaly` then states that this integer is divisible by 24 with quotient 24. It does not know what a K3 surface is, and it says nothing about  $X_8$ . *Proof idea.* Unfold the double sum over `Fin 3`, evaluate, and finish with `norm_num` after rewriting with `euler_K3`. The companions `euler_T2` and `k3t2_euler_zero` are discussed in Chapter 24.

**In Lean: the budget (25.12)**

```
theorem tadpole_budget (flux n : ℕ) (h : flux + n = 24) :
  n ≤ 24 ∧ (flux = 0 → n = 24) := by ...
```

*Reading.* For natural numbers `flux` and `n` adding up to 24,  $n \leq 24$ , and  $n = 24$  if the flux term vanishes. The two physical inputs of (25.12) are encoded in the *types*: `flux : ℕ` is the positivity (25.11) together with the integrality of Proposition 25.3, and `n : ℕ` is the absence of anti-branes. The hypothesis `h` is the tadpole condition itself, assumed. The theorem constructs no flux and proves the existence of no vacuum. *Proof idea.* Linear arithmetic over  $\mathbb{N}$ , closed by `omega`.

**In Lean: Proposition 25.3**

```
def kronForm (L1 : Gram n) (L2 : Gram m) :
  Matrix (Fin n × Fin m) (Fin n × Fin m) ℤ :=
  kroneckerMap (· * ·) L1 L2

theorem kronForm_even (L1 : Gram n) (L2 : Gram m)
  (h1s : L1T = L1) (h2s : L2T = L2)
  (h1 : IsEvenDiag L1) (x : Fin n × Fin m → ℤ) :
  Even (x ·v (kronForm L1 L2 *v x)) := by ...

theorem flux_half_selfIntersection_integral (L1 : Gram n) (L2 : Gram m)
  (h1s : L1T = L1) (h2s : L2T = L2) (h1 : IsEvenDiag L1)
  (x : Fin n × Fin m → ℤ) :
  ∃ k : ℤ, x ·v (kronForm L1 L2 *v x) = 2 * k := by ...
```

*Reading.* `Gram n` is the type of  $n \times n$  integer matrices and `IsEvenDiag L` says every diagonal entry is even. `kronForm` is the matrix (25.23) indexed by pairs. The theorems say: for symmetric  $L_1, L_2$  with  $L_1$  even on the diagonal, and *every* integer vector  $x$  indexed by pairs,  $x^T(L_1 \otimes L_2)x$  is even, i.e. equals  $2k$  for an integer  $k$ . This is a theorem about all pairs of integer matrices of all sizes; it is genuinely general, and the example (25.26) is an instance. It is *not* a statement about the  $22 \times 22$  K3 matrix specifically: to apply it to  $3U \oplus 2E_8(-1)$  one needs that Gram matrix to be symmetric with even diagonal, for which the library provides the blocks `hyperbolicU_evenDiag` and `cartanE8_evenDiag` (Chapter 18). Nor does it cover the rank-two summand of (25.22), or say that fluxes are elements of this lattice; those are Tier L. *Proof idea.* The pair index is flattened by the bijection `finProdFinEquiv` between `Fin n × Fin m` and `Fin (n * m)`; symmetry and even diagonal survive the relabelling (`kronForm_symm`, `kronForm_evenDiag`); then the single-index lemma `even_quadratic_form_of_even_diag`, which is the computation in the proof of Proposition 25.3, is applied and the result transported back.

**In Lean: orientifold charges in the older libraries (arithmetic shadows)**

```
-- StringTheory.StringDynamics (TadpoleConstraint.lean)
def totalTadpole (braneStacks : Fin 4 → BraneStack) (H3 F3 : ℤ) : ℤ :=
  (∑ i, (braneStacks i).charge) + fluxTadpole H3 F3 - 24
theorem tadpole_cancellation (braneStacks : Fin 4 → BraneStack) (H3 F3 : ℤ)
  (h : totalTadpole braneStacks H3 F3 = 0) :
  (∑ i, (braneStacks i).charge) + fluxTadpole H3 F3 = 24 := by ...

-- StringTheory.Foundation.StringTheory.TadpoleCancellation (corrected 2026-09-18)
theorem d7_tadpole_cancellation : totalD7Charge + totalO7Charge = 0 := by rfl
theorem plane_brane_ratio : chargeO7Minus = -(2 ^ (7 - 5) : Int) * chargeD7Brane := by rfl
```



```

theorem oplane_total_independent :
  ∀ k < 7, (planes (k + 3) : Int) * planeCharge (k + 3) + 16 * 4 = 0 := by decide
theorem induced_d3_charge_matches_target :
  (numO7Planes : Int) * d3ChargePerO7 + (numD7Branes : Int) * d3ChargePerD7
    = d3TadpoleTarget := by rfl

-- SocrateAI.Moonshine (KummerTadpole.lean)
theorem rr_tadpole_cancellation : net_tadpole_charge = 0 := by rfl
-- DualScaleValidation.UseCase3, with its own copies of the definitions
theorem rr_tadpole_cancellation :
  total_d7_charge + total_o7_charge = 0 := rfl

```

*Reading.* These are statements about integer literals. `fluxTadpole` is the product of two integers; `tadpole_cancellation` says that if  $\sum_i q_i + H_3 F_3 - 24 = 0$  then  $\sum_i q_i + H_3 F_3 = 24$ , a rearrangement of its own hypothesis (the `euler` field of `BraneStack` is never used). `d7_tadpole_cancellation` unfolds to  $16 \cdot 1 + 4 \cdot (-4) = 0$ , exactly (25.28), and both `rr_tadpole_cancellation` to  $16 \cdot 4 + 4 \cdot (-16) = 0$ , the same row with all charges multiplied by 4 (`o7_d7_ratio` checks that the ratio is still  $-4$ ); all are proved by `rfl`. `oplane_total_independent` is the whole of Table 25.1 in units of  $\mu_p/4$ , for  $p = 3, \dots, 9$ , checked by `decide`. `induced_d3_charge_matches_target` is the arithmetic of Remark 25.4:  $4 \cdot 2 + 16 \cdot 1 = 24 \cdot 24/24$ .

*A correction.* Until 2026-09-18 the Foundation file placed sixteen O7<sup>-</sup>-planes at the fixed points of  $T^4/\mathbb{Z}_2$ , with 32 D7-branes of charge +2, and stated the D3 target as  $\chi(K3)/24 = 1$ . Sixteen planes belong to  $p = 5$  in Table 25.1, the ratio  $-2$  matched no row, and the orientifold of  $K3 \times T^2/\mathbb{Z}_2$  acts on the  $T^2$ , with four fixed points [TT §2.2, ll. 160–163; Sen l. 242]; the sixteen fixed points of  $T^4/\mathbb{Z}_2$  are the Kummer limit of the K3 factor (Chapter 21). The D3 target is  $\chi(K3 \times K3)/24 = 24$ ;  $\chi(K3)/24 = 1$  is the charge of a single wrapped D7-brane. A literature check by a sibling project flagged both points; they were verified against the pinned sources before the file was changed. The first edition of this chapter already noted that “the label ‘O7’ in that file does not match its numbers”. Finally,

```

theorem flux_tadpole_quantization_exact
  (s : StringTheory.Foundation.StringTheory.VafaSwampland.GVWFluxState) :
  StringTheory.Foundation.StringTheory.TadpoleCancellation.totalD7Charge +
  StringTheory.Foundation.StringTheory.TadpoleCancellation.totalO7Charge
    = 0 := ...

```

(in `StringTheoryFormalization/Frontier/FTermPotential.lean`) never uses its argument `s` and is proved by quoting `d7_tadpole_cancellation`. Despite its name it contains no flux and no quantization. A faithful formalization of (25.28) would need at least the fixed-point set of the involution on a torus as a finite type of cardinality  $2^{9-p}$  and the charge ratio  $-2^{p-5}$  as a function of  $p$ , so that the independence of the total from  $p$  becomes a theorem; see Exercise 25.8.

**Unification candidates.** The atlas of Chapter 36 compares statements across libraries by text similarity and by overlap of their dependencies. Under the heading “similar statements, essentially disjoint proofs” it lists two pairs from this chapter [`papers/book/generated/atlas.md`, ll. 254–262]:

- `rr_tadpole_cancellation` (in `UseCase3_FrontierTriad`) and `tadpole_cancellation` (in `StringDynamics`): similarity 0.514, dependency overlap 0.021;
- `k3t2_euler_zero` (in `Flux`) and `k3t2_euler_char_eq_zero` (in `Moonshine`): similarity 0.557, dependency overlap 0.009.

The reading above explains the finding: the tadpole of this chapter has been written down four times with four sets of private definitions. A single definition of the budget, with the Euler number supplied by `eulerFromHodge` and the flux term by `kronForm`, would let these statements share content.

| Claim                                                                                                                                      | Tier | Evidence                                                                                                                                                                     |
|--------------------------------------------------------------------------------------------------------------------------------------------|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alternating sum of the K3 Hodge table is 24;<br>$24^2 \bmod 24 = 0$ , $24^2/24 = 24$                                                       | [A]  | <code>euler_K3</code> , <code>k3k3_anomaly</code>                                                                                                                            |
| $\text{flux} + n = 24$ in $\mathbb{N} \Rightarrow n \leq 24$ , and $n = 24$ if<br>$\text{flux} = 0$                                        | [A]  | <code>tadpole_budget</code>                                                                                                                                                  |
| $L_1$ even-diagonal, $L_1, L_2$ symmetric $\Rightarrow$<br>$x^T(L_1 \otimes L_2)x$ even for all $x$                                        | [A]  | <code>kronForm_even</code> ,<br><code>flux_half_selfIntersection_integral</code>                                                                                             |
| $16 - 16 = 0$ and $64 - 64 = 0$ (ratio $-4$ );<br>Polchinski's table for $3 \leq p \leq 9$ ; $4 \cdot 2 + 16 \cdot 1 = 24$                 | [A]  | <code>d7_tadpole_cancellation</code> ,<br><code>rr_tadpole_cancellation</code> ,<br><code>oplane_total_independent</code> ,<br><code>induced_d3_charge_matches_target</code> |
| Betti and Hodge numbers of $K3 \times K3$ ,<br>(25.18), 2610, 972, $-192$ , signature (371, 115),<br>table of Section 25.4.3               | —    | symbolic checks (sympy), not Tier A                                                                                                                                          |
| $I = -\int X_8 = \chi/24$ ; tadpole condition (25.1);<br>obstructions                                                                      | [L]  | SVW §1; BB §2; DRS §1–2; GVW §2.1                                                                                                                                            |
| $6 \mid \chi$ ; (25.19); shifted quantization of $G$ ; su-<br>persymmetric $G$ primitive and self-dual                                     | [L]  | SVW §2.1–2.2; GVW §2.1, §5; DRS §2.1                                                                                                                                         |
| $Op$ -plane charge $-2^{p-5}\mu_p$ ; $SO(8)^4$ ; $24 \text{ D3} \leftrightarrow$<br>$24 \text{ D5} \leftrightarrow 24 \text{ five-branes}$ | [L]  | Pol Lecture 3; DRS §2.4, §4; SVW §2.4                                                                                                                                        |
| “ <code>kronForm L L</code> is the flux lattice of M-<br>theory on $K3 \times K3$ ”                                                        | —    | an identification, never Tier A                                                                                                                                              |

### Chapter summary

- On a compact space Gauss's law forces the total charge of every gauge field to vanish. A violation appears in perturbation theory as a one-point function, a tadpole, and means that the field equations have no solution.
- In compactifications on a Calabi–Yau fourfold the curvature carries charge  $\chi/24$  through the interaction  $C \wedge X_8$ . It must be balanced by  $n$  space-filling branes and by flux:  $\chi/24 = n + \frac{1}{2} \int \frac{G}{2\pi} \wedge \frac{G}{2\pi}$ .
- Supersymmetry requires  $n \geq 0$  and a primitive, hence self-dual,  $G$ , so both terms are non-negative: a budget, equal to  $576/24 = 24$  for  $K3 \times K3$ . Branes alone suffice only if  $\chi/24$  is a non-negative integer;  $\chi$  is always divisible by 6 but not always by 24 (sextic:  $108\frac{3}{4}$ ), and can be negative in orbifolds with discrete torsion ( $-8$ ).
- Flux is quantized,  $[G/2\pi] \in H^4(N; \mathbb{Z})$  up to a shift by  $c_2/2$  that is absent when the intersection form is even. On  $H^2(K3) \otimes H^2(K3)$  the form is the Kronecker product  $L \otimes L$ , which is even because  $L$  is; hence  $\frac{1}{2} \int \xi \wedge \xi \in \mathbb{Z}$ .
- In type IIB on  $K3 \times T^2/\mathbb{Z}_2$  there are two tadpoles. The seven-brane tadpole is cancelled locally,  $4 \times (-4) + 16 = 0$ , giving  $SO(8)^4$ . The three-brane tadpole is 24 and is shared between D3-branes and  $H \wedge H'$  flux; without flux it is T-dual to type I with 24 D5-branes.
- Lean checks the arithmetic ( $576/24 = 24$ , the budget inequality,  $64 - 64 = 0$ ) and the general evenness of Kronecker forms. The physical identifications are Tier L.



## Exercises

**Exercise 25.1** (★). Two point charges  $q_1, q_2$  sit on a circle of circumference  $\ell$  and the electric field obeys  $\partial_x E = \sum_i q_i \delta(x - x_i)$ . Show directly, by solving for the piecewise constant  $E(x)$  and demanding periodicity, that a solution exists if and only if  $q_1 + q_2 = 0$ . Which step fails on the real line?

**Exercise 25.2** (★). Compute the Betti numbers and the Euler number of  $K3 \times T^4$  and of  $T^8$  from Poincaré polynomials. What does the tadpole condition say about M-theory on these spaces with unbroken supersymmetry? (Use (25.12).)

**Exercise 25.3** (★★). A Calabi–Yau fourfold with holonomy exactly  $SU(4)$  has independent Hodge numbers  $h^{1,1}, h^{2,1}, h^{3,1}$  and  $h^{2,2}$ . (a) Write  $\chi_1$  and  $\chi$  in terms of them. (b) Use  $\chi_1 = 8 - \chi/6$  to derive  $h^{2,2} = 2(22 + 2h^{1,1} + 2h^{3,1} - h^{2,1})$  and  $\chi = 6(8 + h^{1,1} + h^{3,1} - h^{2,1})$ . (c) The sextic has  $h^{1,1} = 1$ ,  $h^{2,1} = 0$ ,  $h^{3,1} = 426$ . Recover  $\chi = 2610$ .

**Exercise 25.4** (★★). For the lattice (25.26), compute  $\frac{1}{2} \int \xi \wedge \xi$  for the nine decomposable fluxes  $\xi = a_1 \wedge a_2$  with  $a_1, a_2 \in \{\alpha, \beta, \alpha + \beta\}$ . Which of them fit in the budget, and with how many membranes? Show that every integral class  $a = xC_1 + yC_2 + zC_3$  orthogonal to  $J$  has  $a \cdot a = -2(x^2 + y^2 + z^2)$ , and conclude that only finitely many decomposable primitive fluxes satisfy (25.12).

**Exercise 25.5** (★★). Show that the rank-two summand in (25.22) has Gram matrix  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  and is orthogonal to  $\Lambda_1 \otimes \Lambda_2$ . Deduce the signature (25.25) and check it against  $\sigma(X \times Y) = \sigma(X)\sigma(Y)$ . Is  $U \oplus (\Lambda \otimes \Lambda)$  unimodular? (Hint:  $\det(A \otimes B) = (\det A)^m (\det B)^n$ .)

**Exercise 25.6** (★★, Lean). State and prove in Lean the strengthening of `tadpole_budget` that is symmetric in its two arguments: for `flux n : ℕ` with `flux + n = 24`, also `flux ≤ 24` and `n = 0 ↔ flux = 24`. Which tactic closes it? Then explain why the statement would be false for `flux n : ℤ` and what physical assumption the type `ℕ` encodes.

**Exercise 25.7** (★★★, Lean). Instantiate `flux_half_selfIntersection_integral` with  $L_1 = L_2 = L_{\text{DRS}}$  of (25.26), written as `!![0,2,2;2,0,2;2,2,0]`: prove the two symmetry hypotheses and `IsEvenDiag` (by `decide` or `fin_cases`). Then state, for `a b : Fin 3 → ℤ` and `x := fun p => a p.1 * b p.2`, the factorization  $x^\top (L_1 \otimes L_2) x = (a^\top L_1 a)(b^\top L_2 b)$  and prove it for general `Gram n` and `Gram m`. (Hint: use `Matrix.kroneckerMap_apply` and `Finset.sum_mul_sum`.)

**Exercise 25.8** (★★★, Lean). The library proves `oplane_total_independent` by `decide`, one `p` at a time. Prove instead, for all `p` with  $3 \leq p$  and  $p \leq 9$  and without case analysis, that `(planes p : ℤ) * planeCharge p = -64`, using `planes p := 2 ^ (9 - p)`, `planeCharge p := -(2 ^ (p - 3) : ℤ)` and `pow_add`. Which of the two bookkeepings of Section 25.6 does the  $p = 7$  case reproduce, and which row would the withdrawn “sixteen O7-planes” bookkeeping have needed?

## Further reading

- The tadpole  $\chi/24$ , its two obstructions, divisibility by six, the elliptic formula and the  $K3 \times K3$  duality check: [SVW §1–2, ll. 34–546]. Its Section 3 notes that blowing up a point of the base removes 120 branes [SVW §3, ll. 571–588].
- The supergravity solution with warp factor and the flux constraint from the Bianchi identity: [BB §2, eqs. (2.6)–(2.7), (2.56)–(2.59), ll. 117–146, 668–735].

- $G$ -flux on  $K3 \times K3$ , the lift to F-theory, the orientifold of  $K3 \times T^2$  and its type I and heterotic duals: [DRS §1–4, ll. 53–148, 156–266, 457–578, 636–808, 1262–1400].
- Flux quantization, its shift, vacua as states of one model, and the F-theory analogue of  $G$ : [GVW §2.1, ll. 84–135; §2.5, ll. 1068–1111]. The superpotential generated by  $G$  is the subject of Chapter 26.
- Orientifold-plane charges and the sixteen D-branes: [Pol Lecture 3, ll. 1968–2000]. D3 charge induced on wrapped seven-branes and the integrated Bianchi identity in type IIB: [GKP §2.1, ll. 419–502].

## Chapter 26

# Flux Superpotential and Moduli Stabilization

A compactification comes with parameters: the size of the internal space, its shape, the value of the string coupling. In Chapter 24 these parameters appeared as coordinates on a moduli space, and nothing in the equations of motion preferred one point of that space to another. In four dimensions each such parameter is a massless scalar field, and a massless scalar that couples to matter is a long-range force that nobody has seen. This is the *moduli problem*. The present chapter asks how a potential for the moduli can arise, what its general shape is, and which moduli it can fix. The answer of the last twenty-five years is: *fluxes*. The quantized field strengths that Chapter 25 was forced to introduce in order to cancel the tadpole do not only carry charge; they also cost energy, and the energy depends on the shape of the space through which they thread. We shall see that this energy is summarized in one holomorphic function, the Gukov–Vafa–Witten superpotential  $W = \int G \wedge \Omega$ ; that the resulting potential is non-negative because of a cancellation called *no-scale structure*; that its minima are the points where the covariant derivative  $DW$  vanishes; and that one modulus, the overall volume, always escapes. At the end we put next to this established picture the toy potential of this programme,  $V(R) \propto (R/\sqrt{\alpha'} - \sqrt{\alpha'}/R)^2$ , whose minimum is the self-dual radius of Chapter 8. As calculus it is elementary and machine-checked; as physics it is a proposal, and we shall be explicit about which is which.

Throughout, four-dimensional formulas are written in units where the reduced Planck mass is one ( $\kappa_4 = 1$ ), indices  $I, J$  run over all chiral superfields and  $i, j$  over all of them except the volume modulus  $\rho$ . A bar denotes complex conjugation.

### 26.1 The moduli problem

Suppose that a compactification has a continuous family of solutions labelled by a parameter  $\phi$ . Let  $\phi$  vary slowly over the four non-compact dimensions. Because every constant value of  $\phi$  solves the equations of motion, the four-dimensional action for  $\phi(x)$  contains no potential: it starts with the kinetic term  $-\frac{1}{2}g(\phi)(\partial\phi)^2$ . Hence  $\phi$  is a massless scalar field, a *modulus*. The radius  $R$  of the circle of Chapter 7 is the simplest example; the  $G_{ij}$  and  $B_{ij}$  of a torus (Chapter 10), the complex structure and Kähler parameters of K3 (Chapter 23) and the dilaton are others.

Why is this a problem and not a feature? The argument is given by Graña in three sentences which we paraphrase. [Gra §6, ll. 3914–3923]

1. A massless scalar mediates a force of infinite range, as the photon and the graviton do.

2. The coupling of a modulus to matter is not universal: the mass of the electron and the mass of the proton depend on the size of the internal space in different ways. Two test bodies of different composition would therefore fall with different accelerations, in conflict with tests of the equivalence principle, which hold to one part in  $10^{13}$ .
3. If the expectation value of the volume modulus “can be anything”, the theory predicts no couplings at all, since four-dimensional couplings are functions of the moduli.

The conclusion is that in a realistic vacuum “all ‘moduli’ should be massive”. A mass is the second derivative of a potential at its minimum, so we need a potential  $V(\phi)$  with an isolated minimum. Two warnings are in order before we look for one. First, a potential generated by a classical effect typically falls to zero at infinite volume or zero coupling, where the effect switches off; a generic potential therefore has a *runaway* direction and no minimum at finite distance. Graña notes that “for the GVW superpotential, the resulting potential has flat or runaway directions”. [Gra §6, ll. 3928–3930] Second, the amount of supersymmetry matters. With sixteen unbroken supercharges, as for type II strings on  $K3 \times T^2$  (Chapter 24), the scalar manifold is a fixed symmetric space and supersymmetry allows no potential on it (standard; not checked against a source in this book’s library). A potential requires breaking to at most four supercharges,  $N = 1$  in four dimensions, which is what orientifold planes and fluxes achieve in the type IIB model  $K3 \times T^2/\mathbb{Z}_2$  of Chapter 25.

## 26.2 The Gukov–Vafa–Witten superpotential

### 26.2.1 Flux restricts the geometry

Take M-theory on a compact Calabi–Yau fourfold  $Y$  with four-form flux  $G$ , the setting of Chapter 25. For  $G = 0$  the complex structure parameters  $t_i$  (there are  $h^{3,1}$  of them) and the Kähler parameters  $k_j$  (there are  $h^{1,1}$ ) are arbitrary. For  $G \neq 0$  this is no longer so. Supersymmetry requires [GVW §2.4, ll. 895–903]

$$G^{0,4} = G^{1,3} = 0, \quad G \wedge J = 0, \quad (26.1)$$

where  $G^{p,q}$  is the component of  $G$  with  $p$  holomorphic and  $q$  antiholomorphic indices and  $J$  is the Kähler form (GVW write  $K$  for it; we keep  $K$  for the Kähler potential). The logic of (26.1) deserves a moment of thought. The flux  $G$  is an *integral* cohomology class, fixed once and for all by quantization (Chapter 25); it does not know about the complex structure. The decomposition of a four-form into types  $(p, q)$ , on the other hand, depends on the complex structure: when the  $t_i$  change, the same class  $G$  acquires different components. Hence (26.1) is not a condition on  $G$  but a set of *equations for the moduli*, with  $G$  as a parameter. This is the mechanism of flux stabilization in one sentence.

### 26.2.2 A superpotential that reproduces the conditions

In a theory with four supercharges the potential for chiral fields is governed by a holomorphic function, the superpotential  $W$ , and in flat space the vacua preserving supersymmetry are the solutions of  $W = dW = 0$ . Gukov, Vafa and Witten looked for a  $W$  whose supersymmetric vacua are (26.1) and proposed [GVW §2.4, eq. (2.38), ll. 943–951]

$$W(t_i) = \frac{1}{2\pi} \int_Y \Omega \wedge G, \quad (26.2)$$

with  $\Omega$  the holomorphic four-form of  $Y$ . Let us check the claim. The form  $\Omega$  is of type  $(4, 0)$ , and the wedge product of a  $(4, 0)$  form with a  $(p, q)$  form can be integrated over  $Y$  only if

$(p, q) = (0, 4)$ . Therefore

$$W = 0 \iff \int_Y \Omega \wedge G^{0,4} = 0 \iff G^{0,4} = 0, \quad (26.3)$$

the last step because  $H^{0,4}(Y)$  is one-dimensional and spanned by  $\bar{\Omega}$ , with  $\int \Omega \wedge \bar{\Omega} \neq 0$ . Next, vary the complex structure. The derivative  $\partial\Omega/\partial t_i$  is no longer of pure type: it has a  $(4, 0)$  part proportional to  $\Omega$  and a  $(3, 1)$  part  $\chi_i$ , and the  $\chi_i$  span  $H^{3,1}(Y)$ . Given  $W = 0$ , the condition  $\partial_i W = 0$  says that  $\int \chi_i \wedge G = 0$  for all  $i$ , that is,  $G^{1,3} = 0$ . “So we have found the supersymmetric interaction that accounts for” the first two equations in (26.1). [GVW §2.4, ll. 961–965] The third one,  $G \wedge J = 0$ , follows in the same way from a second function  $\tilde{W} = \int_Y \mathcal{K} \wedge \mathcal{K} \wedge G$  of the complexified Kähler class  $\mathcal{K}$ . [GVW §2.4, eq. (2.41), ll. 979–988]

#### Why $W$ must be this integral: domain walls

GVW give a second, more physical argument. A five-brane wrapped on a four-cycle  $S \subset Y$  looks in the non-compact directions like a wall. On its two sides the flux differs,  $G_2 - G_1 = 2\pi[S]$ : the brane is the source of  $G$ , so crossing it changes the flux by one unit through the dual cycle. The tension of the wall is the volume of  $S$ , which for a supersymmetric cycle is  $|\int_S \Omega|$ . In a theory with four supercharges the tension of a supersymmetric domain wall equals the absolute value of the jump of  $W$  across it. Writing  $\int_S \Omega = \frac{1}{2\pi} \int_Y \Omega \wedge (G_2 - G_1)$  one reads off  $\Delta W$ , “a statement that is clearly compatible with” (26.2). [GVW §2.4, eq. (2.40), ll. 966–977] The superpotential is thus fixed by an object whose tension can be computed independently.

### 26.2.3 Type IIB: three-form flux and the axio-dilaton

Our main example lives in type IIB string theory, where the relevant fluxes are the Ramond–Ramond three-form  $F_3$  and the Neveu–Schwarz three-form  $H_3$  on a Calabi–Yau threefold (or orientifold)  $M$ . They enter through one complex combination, [GKP §2.1, ll. 168–169]

$$G_3 = F_3 - \tau H_3, \quad \tau = C_0 + i e^{-\phi}, \quad (26.4)$$

where  $\tau$  is the axio-dilaton:  $C_0$  is the Ramond–Ramond scalar and  $e^\phi = g_s$  the string coupling, so that weak coupling means large  $\text{Im } \tau$ . The superpotential is [GKP §2.3, eq. (2.40), ll. 799–808; Gra §5.4, eq. (5.43), ll. 3500–3509]

$$W = \int_M \Omega \wedge G_3 = \int_M \Omega \wedge F_3 - \tau \int_M \Omega \wedge H_3, \quad (26.5)$$

with  $\Omega$  now the holomorphic three-form. It is the F-theory limit of (26.2): on an elliptically fibred fourfold a four-form flux with one leg on the fibre decomposes as  $G = \sum_i H_i \wedge \theta^i$  with two three-forms  $H_i$  on the base and  $\theta^i$  the two one-forms of the fibre, [GVW §2.5, eq. (2.42), ll. 1085–1096] and the pair  $(H_1, H_2)$  corresponds to  $(F_3, H_3)$ ; GKP give the precise map,  $G_4 = -G_3 \wedge d\bar{w}/(\tau - \bar{\tau}) + \text{h.c.}$  with  $w$  the fibre coordinate. [GKP §2.3, eqs. (2.41)–(2.42), ll. 809–840] Three features of (26.5) control everything that follows.

- $W$  depends on the complex structure moduli  $z^\alpha$  through the periods of  $\Omega$ .
- $W$  depends on  $\tau$ , linearly. The dilaton couples differently to the two fluxes, and for this reason fluxes “generically” stabilize it. [GKP §1, ll. 97–100]
- $W$  does not depend on the Kähler moduli at all: “the Kähler moduli [...] do not appear in the superpotential”. [Gra §5.4, ll. 3506–3509]

If the three-cycles of  $M$  are  $A_K, B^K$  with periods  $Z^K = \int_{A^K} \Omega$ ,  $\mathcal{F}_K = \int_{B^K} \Omega$ , and the fluxes through them are integers (in units of  $(2\pi)^2 \alpha'$ ), then (26.5) is a finite sum: integers times periods. For the deformed conifold with  $M$  units of  $F_3$  through the  $A$ -cycle and  $-K$  units of  $H_3$  through the  $B$ -cycle, GKP obtain  $W = (2\pi)^2 \alpha' (M \mathcal{G}(z) - K \tau z)$  with  $z = \int_A \Omega$  and  $\mathcal{G}(z) = \int_B \Omega = \frac{z}{2\pi i} \ln z + \text{holomorphic}$ . [GKP §3, eqs. (3.4)–(3.10), ll. 1007–1090] We shall use a simpler model of the same type in Section 26.5.

*Remark 26.1* ( $W$  is a section, not a function). The form  $\Omega$  is defined only up to multiplication by a nowhere-vanishing holomorphic function of the moduli,  $\Omega \mapsto e^{-f(z)} \Omega$ . Then  $W \mapsto e^{-f} W$ : strictly speaking  $W$  is “a section of a line bundle over the moduli space”. [GVW §2.4, ll. 952–961] An observable cannot depend on this choice, and we shall verify below that the potential does not.

## 26.3 The $N = 1$ F-term potential

### 26.3.1 Ingredients

A four-dimensional  $N = 1$  supergravity theory with chiral superfields  $\phi^I$  is specified, as far as the scalar fields are concerned, by two functions: a real Kähler potential  $K(\phi, \bar{\phi})$  and a holomorphic superpotential  $W(\phi)$ . The kinetic terms are  $-K_{I\bar{J}} \partial_\mu \phi^I \partial^\mu \bar{\phi}^{\bar{J}}$  with the Kähler metric  $K_{I\bar{J}} = \partial_I \partial_{\bar{J}} K$ , which must be positive definite, and  $K^{I\bar{J}}$  denotes its inverse. The potential is, in the absence of D-terms, [Gra §5.2, eqs. (5.21)–(5.22), ll. 2990–3000; GKP §2.3, eq. (2.43), ll. 855–869]

$$V = e^K \left( K^{I\bar{J}} D_I W \overline{D_{\bar{J}} W} - 3 |W|^2 \right), \quad D_I W = \partial_I W + (\partial_I K) W. \quad (26.6)$$

This is the *F-term potential*;  $D_I W$  is the Kähler covariant derivative, and  $F_I = e^{K/2} D_I W$  is the order parameter of supersymmetry breaking. GKP carry an overall factor  $1/2\kappa_{10}^2$  from the ten-dimensional normalization; it plays no role for the location of minima.

Two structural facts should be checked once by every student.

**Kähler invariance.** Under  $K \mapsto K + f + \bar{f}$ ,  $W \mapsto e^{-f} W$  with  $f$  holomorphic, the kinetic terms are unchanged because  $\partial_I \partial_{\bar{J}} (f + \bar{f}) = 0$ . For the covariant derivative,

$$D_I (e^{-f} W) = e^{-f} (\partial_I W - f_I W) + (K_I + f_I) e^{-f} W = e^{-f} D_I W,$$

so  $D_I W$  transforms like  $W$ , which the plain derivative  $\partial_I W$  does not. Finally  $e^K \mapsto e^K |e^f|^2$  compensates  $|e^{-f}|^2$  from the bracket, and  $V$  is invariant. For the flux superpotential this is exactly the freedom of Remark 26.1, because the Kähler potential of the complex structure moduli is  $-\ln(-i \int_M \Omega \wedge \bar{\Omega})$ , see (26.7) below.

**The sign structure.** The first term in (26.6) is a positive definite quadratic form in the  $D_I W$ ; the second is negative. Global supersymmetry knows only the first. The term  $-3|W|^2$  is gravitational: it is the reason why supergravity, unlike global supersymmetry, admits supersymmetric vacua of negative energy.

**Proposition 26.2** (Supersymmetric points are critical points). *If  $D_I W = 0$  for all  $I$  at a point  $\phi_0$ , then  $\partial_I V(\phi_0) = 0$  for all  $I$ , and  $V(\phi_0) = -3e^K |W|^2 \leq 0$ .*

*Proof.* Write  $V = e^K Q - 3e^K W \bar{W}$  with  $Q = K^{J\bar{L}} D_J W \overline{D_{\bar{L}} W}$ . By the product rule every term of  $\partial_I (e^K Q)$  contains a factor  $D_J W$  or  $\overline{D_{\bar{L}} W}$  and vanishes at  $\phi_0$ . For the second piece,  $\bar{W}$  is antiholomorphic, so  $\partial_I (e^K W \bar{W}) = e^K \bar{W} (K_I W + \partial_I W) = e^K \bar{W} D_I W$ , which vanishes as well. The value of  $V$  is read off from (26.6).  $\square$

A supersymmetric vacuum is therefore Minkowski if  $W = 0$  there and anti-de Sitter if  $W \neq 0$ ; it is never de Sitter. The proposition does not say that a supersymmetric point is a *minimum*; this needs further information, which the no-scale structure will supply.

### 26.3.2 The Kähler potential of a type IIB flux compactification

For one Kähler modulus  $\rho$  (the overall volume;  $\text{Im } \rho \sim R^4$  in string units), the dilaton and the complex structure moduli, dimensional reduction of the ten-dimensional action gives [GKP §2.3, eqs. (2.37)–(2.38), ll. 773–789]

$$K = -3 \ln[-i(\rho - \bar{\rho})] - \ln[-i(\tau - \bar{\tau})] - \ln\left[-i \int_M \Omega \wedge \bar{\Omega}\right]. \quad (26.7)$$

Note that  $-i(\rho - \bar{\rho}) = 2 \text{Im } \rho > 0$ , so the logarithms are real. Graña uses  $T = -i\rho$ , in terms of which the first term reads  $-3 \ln(T + \bar{T})$ . [Gra §5.2, eq. (5.27), ll. 3075–3083] The coefficient 3 is the important number of this chapter.

## 26.4 No-scale structure

### 26.4.1 The cancellation

Because  $W$  in (26.5) does not depend on  $\rho$ , the covariant derivative in the  $\rho$  direction is purely the connection term,  $D_\rho W = K_\rho W$ . From  $K^{(\rho)} = -3 \ln[-i(\rho - \bar{\rho})]$ ,

$$K_\rho = -\frac{3}{\rho - \bar{\rho}}, \quad K_{\bar{\rho}} = +\frac{3}{\rho - \bar{\rho}}, \quad K_{\rho\bar{\rho}} = -\frac{3}{(\rho - \bar{\rho})^2} = \frac{3}{4(\text{Im } \rho)^2} > 0, \quad (26.8)$$

and therefore

$$K^{\rho\bar{\rho}} K_\rho K_{\bar{\rho}} = \left(-\frac{(\rho - \bar{\rho})^2}{3}\right) \left(-\frac{9}{(\rho - \bar{\rho})^2}\right) = 3. \quad (26.9)$$

This is the *no-scale identity*. The  $\rho$  contribution to the first term of (26.6) is  $K^{\rho\bar{\rho}} |K_\rho|^2 |W|^2 = 3|W|^2$ , and it cancels the negative term exactly:

$$V = e^K K^{i\bar{j}} D_i W \overline{D_j W} \geq 0, \quad (26.10)$$

where  $i, j$  run over  $\tau$  and the  $z^\alpha$  only (the metric is block diagonal because  $K$  is a sum). “In no-scale models the  $|D_\rho W|^2$  term cancels the negative term, leaving a nonnegative potential.” [GKP §2.3, ll. 866–869] [L] The coefficient matters: for  $K^{(\rho)} = -p \ln[-i(\rho - \bar{\rho})]$  the same computation gives  $K^{\rho\bar{\rho}} K_\rho K_{\bar{\rho}} = p$  and a residual term  $(p - 3)|W|^2$  of either sign (symbolic check with sympy for general  $p$ ). The value  $p = 3$  is tied to the fact that  $e^{-K^{(\rho)}}$  is proportional to the square of the six-dimensional volume in four-dimensional Planck units.

With several Kähler moduli the statement is  $K^{I\bar{J}} \partial_I K \partial_{\bar{J}} K = 3$ , [Gra §5.2, eq. (5.28), ll. 3086–3096] and it has an equally short proof.

**Lemma 26.3** (Homogeneity implies no-scale). *Let  $K = -\ln Y(x)$  where  $Y$  is a positive function of the real variables  $x^I = T^I + \bar{T}^I$ , homogeneous of degree  $p$ , and assume  $K_{IJ}$  invertible. Then  $K^{I\bar{J}} K_I K_{\bar{J}} = p$ .*

*Proof.* Since  $K$  depends on  $T^I$  only through  $x^I$ ,  $\partial_{T^I} \partial_{\bar{T}^J} K = K_{IJ}$ , the ordinary Hessian. Euler’s relation  $x^I Y_I = pY$  gives  $x^I K_I = -p$ . Differentiating with respect to  $x^J$ :  $K_J + x^I K_{IJ} = 0$ , so  $x^I = -K^{I\bar{J}} K_{\bar{J}}$ . Inserting this back,  $-p = x^I K_I = -K^{I\bar{J}} K_{\bar{J}} K_I$ .  $\square$

For a Calabi–Yau orientifold  $Y$  is the square of the volume, which is homogeneous of degree  $3/2$  in the four-cycle volumes  $\text{Re } T^I$ ; hence  $p = 3$ . We have checked the lemma symbolically for  $Y = x^1 x^2 x^3$  and  $Y = x^1 (x^2)^2 + (x^3)^3$ , and numerically for  $Y = ((x^1)^{3/2} - (x^2)^{3/2} - (x^3)^{3/2})^2$ .

| component of $G_3$   | detected by                                     | ISD? | effect                               |
|----------------------|-------------------------------------------------|------|--------------------------------------|
| $(2, 1)$ , primitive | —                                               | yes  | allowed, supersymmetric              |
| $(0, 3)$             | $W = \int \Omega \wedge G_3$                    | yes  | $V = 0$ , breaks SUSY via $D_\rho W$ |
| $(3, 0)$             | $D_\tau W \propto \int \Omega \wedge \bar{G}_3$ | no   | $V > 0$                              |
| $(1, 2)$             | $D_\alpha W = \int \chi_\alpha \wedge G_3$      | no   | $V > 0$                              |

Table 26.1: Hodge types of the three-form flux and the four-dimensional quantities that detect them; ISD means imaginary self-dual,  $*_6 G_3 = iG_3$ . [GKP §2.2–2.3, ll. 629–632, 706–709, 874–896; Gra §5.4, eqs. (5.44)–(5.47), ll. 3520–3552]

### 26.4.2 Consequences

Equation (26.10) has four consequences, all stated by GKP. [GKP §2.3, ll. 866–872]

1.  $V \geq 0$ , and  $V = 0$  exactly where  $D_i W = 0$  for all  $i \neq \rho$ . These points are global minima: *the minima of the flux potential are at  $D_i W = 0$ .*
2. These are  $n$  complex equations for the  $n$  fields  $\tau, z^\alpha$ . Generically the solutions are isolated, and all these fields are massive.
3. The conditions do not involve  $\rho$ , which “represents  $n$  equations on  $n$  moduli and leaves  $\rho$  undetermined”: the volume is a flat direction with  $V = 0$  along it.
4. “Generically at these solutions  $W \neq 0$ , so  $D_\rho W = -3W/(\rho - \bar{\rho})$  is nonzero and supersymmetry is broken.”

The last two items explain the name. Supersymmetry is broken, yet the vacuum energy is zero, and the scale of the breaking, measured by the gravitino mass  $m_{3/2} = e^{K/2}|W| \propto (\text{Im } \rho)^{-3/2}$  (standard; not checked against a source in this book’s library), slides with the undetermined field  $\rho$ : the theory has *no scale*.

### 26.4.3 What the conditions mean in ten dimensions

With (26.7) one has  $K_\tau = -1/(\tau - \bar{\tau})$  and  $\partial_\tau W = -\int \Omega \wedge H_3$ . Therefore

$$D_\tau W = -\int \Omega \wedge H_3 - \frac{1}{\tau - \bar{\tau}} \int \Omega \wedge (F_3 - \tau H_3) = \frac{1}{\bar{\tau} - \tau} \int_M \Omega \wedge \bar{G}_3, \quad \bar{G}_3 = F_3 - \bar{\tau} H_3, \quad (26.11)$$

since  $-(\tau - \bar{\tau})H_3 - (F_3 - \tau H_3) = -(F_3 - \bar{\tau} H_3)$ . This is GKP’s eq. (2.44) (the bar over  $G_{(3)}$  is lost in the text extraction; Graña’s eq. (5.44) writes it as  $\frac{i}{2}e^\phi \int \bar{G}_3 \wedge \Omega$ , which agrees because  $1/(\bar{\tau} - \tau) = ie^\phi/2$ ). For the complex structure moduli  $D_\alpha W = \int \chi_\alpha \wedge G_3$  with  $\chi_\alpha$  a basis of  $(2, 1)$  forms. [GKP §2.3, eq. (2.44), ll. 874–896] Reading off Hodge types as in Section 26.2:  $D_\tau W = 0$  removes the  $(3, 0)$  part of  $G_3$ ,  $D_\alpha W = 0$  removes the  $(1, 2)$  part, and what remains,  $(2, 1) \oplus (0, 3)$ , is precisely the flux that is *imaginary self-dual*,  $*_6 G_3 = iG_3$ , the condition GKP derive from the ten-dimensional equations of motion. [GKP §2.2, eq. (2.31), ll. 629–632, 706–709] The  $(0, 3)$  part is measured by  $W$  itself and is the part that breaks supersymmetry. Table 26.1 summarizes the dictionary. The agreement of the four-dimensional conditions with the ten-dimensional ones is the main evidence that (26.5) is the correct superpotential.

#### “No-scale” is a classical statement

The flat direction and the vanishing vacuum energy hold in the classical supergravity approximation. GKP warn that “there is no known reason that it should survive quantum corrections” and that already  $\alpha'$  corrections “presumably spoil the no-scale structure”.



[GKP §2.3, ll. 756–758] GVW make the same point for fourfolds. [GVW §2.4, ll. 1000–1008 and footnote 11, ll. 1036–1039] One should therefore not read (26.10) as a solution of the cosmological constant problem, and not read the flat direction as a prediction. It is the leading term of an expansion.

## 26.5 Worked example: fixing the dilaton with two periods

We now solve a model completely. Freeze the complex structure, so that the periods of  $\Omega$  are fixed complex numbers, and keep two of them,  $\Pi_1 = 1$  and  $\Pi_2 = i$ . With integer fluxes  $(f_1, f_2)$  of  $F_3$  and  $(h_1, h_2)$  of  $H_3$  through the two cycles, (26.5) becomes

$$W(\tau) = A - \tau B, \quad A = f_1 + if_2, \quad B = h_1 + ih_2. \quad (26.12)$$

This is a caricature: on a real Calabi–Yau the periods move with  $z^\alpha$  and the fluxes are constrained by the tadpole. It is, however, the two-period version of exactly the toy superpotential that the Lean file of Section 26.8 defines, and it shows every feature of the general discussion. The fields are  $\tau$  and  $\rho$ , and  $K = -\ln(2 \operatorname{Im} \tau) - 3 \ln(2 \operatorname{Im} \rho)$ .

**Step 1: the F-term.** By (26.11),  $D_\tau W = -(A - \bar{\tau}B)/(\tau - \bar{\tau})$ . It vanishes at

$$\bar{\tau}_0 = \frac{A}{B} \iff \tau_0 = \frac{(f_1 - if_2)(h_1 + ih_2)}{h_1^2 + h_2^2}, \quad \operatorname{Im} \tau_0 = \frac{f_1 h_2 - f_2 h_1}{h_1^2 + h_2^2}. \quad (26.13)$$

GKP’s conifold model has the same structure,  $\bar{\tau} = M\mathcal{G}(0)/K'z'(0)$ . [GKP §3, eqs. (3.15)–(3.17), ll. 1139–1160] The string coupling must be positive, so we need

$$N_{\text{flux}} := f_1 h_2 - f_2 h_1 > 0. \quad (26.14)$$

This integer is the intersection pairing of the two flux vectors: it is the toy version of  $\int H_3 \wedge F_3$ , the D3-brane charge carried by the flux, which entered the tadpole condition of Chapter 25 with a definite sign. [GKP §3, eq. (3.5), ll. 1019–1029] The same sign that makes the flux contribute positively to the tadpole makes the stabilized coupling positive.

**Step 2: the potential.** With  $K^{\tau\bar{\tau}} = -(\tau - \bar{\tau})^2$  the factors of  $\tau - \bar{\tau}$  cancel in  $K^{\tau\bar{\tau}}|D_\tau W|^2 = |A - \bar{\tau}B|^2$ , the no-scale identity removes  $-3|W|^2$ , and  $e^K = 1/(16 \operatorname{Im} \tau (\operatorname{Im} \rho)^3)$ :

$$V(\tau, \rho) = \frac{|A - \bar{\tau}B|^2}{16 \operatorname{Im} \tau (\operatorname{Im} \rho)^3} = \frac{|B|^2 |\tau - \tau_0|^2}{16 \operatorname{Im} \tau (\operatorname{Im} \rho)^3} \geq 0. \quad (26.15)$$

(Symbolic check of the cancellation with sympy.) The potential vanishes only at  $\tau = \tau_0$ , for every value of  $\rho$ ; away from  $\tau_0$  it is positive and decreases like  $(\operatorname{Im} \rho)^{-3}$ , a runaway towards infinite volume.

**Step 3: numbers.** Take  $(f_1, f_2) = (2, 0)$  and  $(h_1, h_2) = (0, 1)$ . Then  $A = 2$ ,  $B = i$ ,  $N_{\text{flux}} = 2$  and  $\tau_0 = 2i$ : the axion is fixed at  $C_0 = 0$  and the string coupling at  $g_s = 1/\operatorname{Im} \tau_0 = \frac{1}{2}$ . Along the line  $\tau = it$  one finds  $A - \bar{\tau}B = 2 - t$  and  $V \propto (2 - t)^2/t$ , shown in Figure 26.1. The value of the superpotential at the minimum is

$$W_0 = A - \tau_0 B = -\frac{2iN_{\text{flux}}}{\bar{B}} = 4 \neq 0, \quad (26.16)$$

so by item 4 above supersymmetry is broken, with  $D_\rho W = -3W_0/(\rho - \bar{\rho})$ , while  $V = 0$ . In this rigid toy  $W_0 \neq 0$  is unavoidable, since (26.16) vanishes only for  $N_{\text{flux}} = 0$ ; in a model

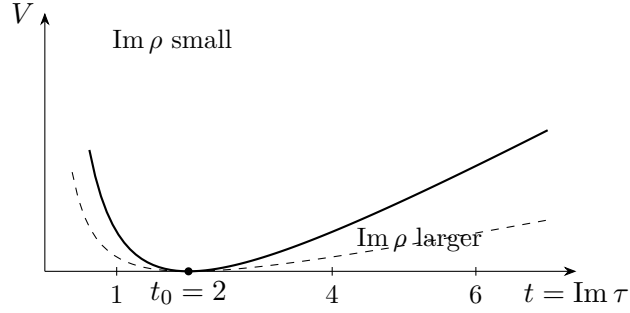


Figure 26.1: The toy flux potential (26.15) for  $A = 2$ ,  $B = i$  along  $\tau = it$ :  $V \propto (2-t)^2/t$ . The minimum at  $g_s = 1/t_0 = \frac{1}{2}$  has  $V = 0$  for every value of the volume modulus; increasing  $\text{Im } \rho$  lowers the whole curve (dashed) without moving the minimum. The volume is not stabilized.

with complex structure moduli one may try to tune  $W_0 = 0$  or  $|W_0| \ll 1$ . A second choice,  $(f_1, f_2) = (1, 0)$  and  $(h_1, h_2) = (1, 1)$ , gives  $N_{\text{flux}} = 1$ ,  $\tau_0 = \frac{1}{2}(1 + i)$  and  $W_0 = 1 - i$ ; this is strong coupling,  $g_s = 2$ , and one should use the  $SL(2, \mathbb{Z})$  duality of type IIB to map it to the fundamental domain of Chapter 12 (Exercise 26.3).

#### One period is not enough

With a single period  $\Pi$ ,  $W = (f - \tau h)\Pi$  and  $D_\tau W = 0$  gives  $\bar{\tau} = f/h$ , a *real* number: there is no vacuum in the upper half plane. GKP meet this obstruction in the conifold: near  $z = 0$  their superpotential is effectively independent of  $\tau$  and “the  $D_\tau$  equation cannot be satisfied”; the cure is “additional fluxes” on a second pair of cycles. [GKP §3, ll. 1118–1139] Stabilizing the dilaton needs at least two periods whose ratio is not real. The Lean structure `GVWSuperpotential` of Section 26.8 has one period, so its toy model has no supersymmetric vacuum with  $h_3 \neq 0$ ; this is a limitation of the toy, not a statement about string theory.

## 26.6 Beyond no-scale: fixing the volume

Since  $W$  receives no perturbative corrections, a dependence on the Kähler modulus can enter the superpotential only non-perturbatively, from Euclidean D3-branes or gaugino condensation on D7-branes, in the form  $W_{\text{np}} = B e^{-2\pi T}$  with  $T = -i\rho$ . [Gra §7.1–7.2, ll. 4738–4745, 4818–4826] Kachru, Kallosh, Linde and Trivedi (KKLT) assume that  $\tau$  and  $z^\alpha$  have been fixed by fluxes as above, leaving a constant  $W_0$ , and study

$$W = W_0 + B e^{-2\pi T}, \quad K = -3 \ln(T + \bar{T}). \quad (26.17)$$

[Gra §7.3, eq. (7.7), ll. 4855–4871] Set the axion to zero,  $T = \sigma$  real, and take  $W_0, B$  real. Then  $D_T W = -2\pi B e^{-2\pi\sigma} - \frac{3}{2\sigma}(W_0 + B e^{-2\pi\sigma})$ , and  $D_T W = 0$  gives

$$W_0 = -B e^{-2\pi\sigma_{\text{cr}}} \left(1 + \frac{4\pi}{3}\sigma_{\text{cr}}\right), \quad (26.18)$$

which is Graña’s eq. (7.8). At this point  $W = -\frac{4\pi}{3}\sigma_{\text{cr}} B e^{-2\pi\sigma_{\text{cr}}}$ , and by Proposition 26.2

$$V_{\text{min}} = -3 e^K |W|^2 = -\frac{3}{8\sigma_{\text{cr}}^3} \cdot \frac{16\pi^2 \sigma_{\text{cr}}^2 B^2 e^{-4\pi\sigma_{\text{cr}}}}{9} = -\frac{2\pi^2 B^2 e^{-4\pi\sigma_{\text{cr}}}}{3\sigma_{\text{cr}}}, \quad (26.19)$$

in agreement with eq. (7.9) of the same source. [Gra §7.3, eqs. (7.8)–(7.9), ll. 4873–4898] The volume is now fixed, the vacuum is supersymmetric and anti-de Sitter, and  $W_0 \neq 0$  is *nec-*

*essary*: the  $(0, 3)$  flux that broke supersymmetry in the no-scale model is what balances the exponential.

**Example 26.4** (How small must  $W_0$  be?). For  $B = 1$  and  $W_0 = -10^{-4}$ , solving (26.18) numerically gives  $\sigma_{\text{cr}} \approx 1.81$  and  $V_{\text{min}} \approx -5.0 \times 10^{-10}$ . This is not a large volume: the supergravity approximation needs  $\sigma \gg 1$ . Because  $\sigma_{\text{cr}}$  grows only like  $\frac{1}{2\pi} \ln(1/|W_0|)$ , a large volume needs an exponentially small  $|W_0|$ , or a smaller exponent. KKLT's illustrative figure replaces  $2\pi$  by  $a = 0.1$  (as for gaugino condensation in a gauge group of large rank); [Gra §7.4, ll. 5027–5029] then  $W_0 = -Be^{-a\sigma}(1 + \frac{2}{3}a\sigma)$ , and the same input gives  $\sigma_{\text{cr}} \approx 113.6$  and  $V_{\text{min}} = -a^2 B^2 e^{-2a\sigma_{\text{cr}}} / (6\sigma_{\text{cr}}) \approx -2.0 \times 10^{-15}$  (numerical root finding with `mpmath`).

Graña stresses that this construction “should be taken only as a toy model of complete moduli stabilization” and reviews the critiques. [Gra §7.3, ll. 4919–4926] Whether the further step to a de Sitter vacuum is consistent is the subject of the de Sitter conjecture of Chapter 27.

## 26.7 The dual-scale toy potential

### 26.7.1 Definition and elementary properties

None of the potentials above knows about T-duality: the flux potential runs away at large volume, and at small volume the supergravity approximation fails. This programme studies, as a model of what a T-duality invariant potential for a radius could look like, the function

$$V_{\text{ds}}(R) = V_0 \left( \frac{R}{\sqrt{\alpha'}} - \frac{\sqrt{\alpha'}}{R} \right)^2 = \frac{V_0}{\alpha'} \left[ \left( R + \frac{\alpha'}{R} \right)^2 - 4\alpha' \right], \quad V_0 > 0, \quad R > 0. \quad (26.20)$$

The second form shows that  $V_{\text{ds}}$  is a function of the dual-scale variable  $R_{\text{eff}} = R + \alpha'/R$  of Chapter 31 alone, namely  $V_{\text{ds}} = V_0 (R_{\text{eff}}^2 - R_{\text{eff,min}}^2) / \alpha'$  with  $R_{\text{eff,min}} = 2\sqrt{\alpha'}$ . Its properties are calculus:

1. *Duality invariance.*  $V_{\text{ds}}(\alpha'/R) = V_{\text{ds}}(R)$ , because  $R \mapsto \alpha'/R$  changes the sign of the bracket in the first form.
2. *Non-negativity and unique minimum.*  $V_{\text{ds}} \geq 0$  with equality if and only if  $R^2 = \alpha'$ , that is, at the self-dual radius  $R = \sqrt{\alpha'}$ . Equivalently  $R + \alpha'/R \geq 2\sqrt{\alpha'}$ , since  $R + \alpha'/R - 2\sqrt{\alpha'} = (R - \sqrt{\alpha'})^2 / R$ .
3. *Logarithmic variable.* With  $x = \ln(R/\sqrt{\alpha'})$ , T-duality is  $x \mapsto -x$  and  $V_{\text{ds}} = 4V_0 \sinh^2 x$ , an even function, equal to  $4V_0 x^2 (1 + O(x^2))$  near the minimum. The variable  $x$  has a canonical kinetic term up to a constant, since the kinetic term of a radius is proportional to  $(\partial R)^2 / R^2 = (\partial x)^2$  (Chapter 6), so the mass squared of the radion is proportional to  $V''(0) = 8V_0$  and is positive.
4. *No runaway.*  $V_{\text{ds}} \rightarrow \infty$  both for  $R \rightarrow \infty$  and for  $R \rightarrow 0$ ; this is what T-duality enforces and what the flux potential of Figure 26.1 lacks.

Figure 26.2 shows the potential. For a  $d$ -torus with metric  $G$  (in units  $\alpha' = 1$ ) and  $B = 0$  the natural generalization is  $V_{\text{ds}}(G) = V_0 (\text{tr } G + \text{tr } G^{-1} - 2d) = V_0 \text{tr}[(G - 1)G^{-1}(G - 1)]$ , which is invariant under  $G \mapsto G^{-1}$ , non-negative for positive definite  $G$ , and zero at  $G = 1$ ; for  $d = 1$ ,  $G = R^2$  it reduces to (26.20). The quantity  $\text{tr } G + \text{tr } G^{-1}$  is the trace of the generalized metric  $\mathcal{H}(G, 0)$  of Chapter 15.

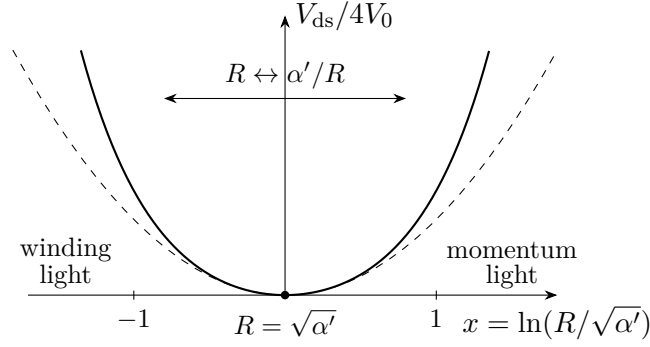


Figure 26.2: The dual-scale toy potential  $V_{\text{ds}} = 4V_0 \sinh^2 x$  (solid) and its quadratic approximation  $4V_0 x^2$  (dashed). T-duality is the reflection  $x \mapsto -x$ ; the unique minimum is its fixed point, the self-dual radius.

### 26.7.2 Where could such a potential come from?

There is a simple observation behind (26.20). In the conventions of Chapter 7 a state with momentum number  $n$  and winding number  $w$  contributes  $n^2/R^2 + w^2 R^2/\alpha'^2$  to  $M^2$ . For  $n^2 = w^2 = 1$ ,

$$\alpha' \left( \frac{1}{R^2} + \frac{R^2}{\alpha'^2} \right) - 2 = \left( \frac{R}{\sqrt{\alpha'}} - \frac{\sqrt{\alpha'}}{R} \right)^2 = \frac{V_{\text{ds}}(R)}{V_0}. \quad (26.21)$$

A population with equal numbers of momentum and winding quanta has a mass-squared budget that is minimized at the self-dual radius: momentum modes are cheap at large  $R$ , winding modes at small  $R$ , and the compromise is  $R = \sqrt{\alpha'}$ . This is the mechanism of string gas cosmology: the negative pressure of winding modes drives the radius to smaller values, the positive pressure of momentum modes to larger values, and “for an equal number of winding and momentum modes [...] the evolution is driven to the critical radius, the so-called self-dual radius [...] where the total pressure vanishes and T-duality is restored”. [BW §III.C, ll. 1109–1122] [L] The same authors point out what the mechanism does *not* do: it operates in the string frame and needs a rolling dilaton, which is “problematic” at late times, and the dilaton “still remains a serious challenge”. [BW §III.D, ll. 1183–1190; §V, ll. 1957–1958] Chapter 32 returns to this.

We therefore classify as follows. The statements 1–4 above are mathematics, [A] where a Lean declaration is named in Section 26.8 and elementary otherwise. That a gas of momentum and winding modes favours the self-dual radius is [L]. That (26.20) is the effective potential of a radius in a compactification on  $K3 \times T^2$ , in the Einstein frame, with the dilaton fixed, is [C]: it has not been derived from (26.6) with any  $K$  and  $W$ , and three obstacles are known. (i) With sixteen supercharges no potential is allowed (Section 26.1); the programme’s own Lean documentation records that the scenario is “in tension with  $N = 4$  non-renormalization”. (ii) At the self-dual radius additional states become massless (Chapter 8), so an effective description in terms of  $R$  alone breaks down exactly at the minimum; BW stress that these states “should therefore be considered in the low energy action”. [BW §III.C, ll. 1135–1138] (iii) The minimum has  $V_{\text{ds}} = 0$ , a Minkowski vacuum: by itself it gives no dark energy (Chapter 27).

## 26.8 What the machine has checked

Four Lean modules touch this chapter. They are very different in strength, and the docstrings of the files themselves say so; we follow them. Proof assistants are introduced in Chapter 34.

### 26.8.1 The frontier file on the F-term potential

The module `StringTheoryFormalization/Frontier/FTermPotential.lean` defines a structure `DilatonAxion` (a complex  $\tau$  with  $0 < \text{Im } \tau$ ), a real function `kahlerPotential` written as  $-\log(-\text{Im}(\tau - \bar{\tau}))$ , and the following toy superpotential. (A sign slip is worth noting:  $\text{Im}(\tau - \bar{\tau}) = 2 \text{Im } \tau$ , so the argument of the logarithm is negative. Mathlib’s `Real.log` is defined through the absolute value, so the value is nevertheless  $-\ln(2 \text{Im } \tau)$ , which is the  $\tau$  term of (26.7); the intended expression is  $-\ln[-i(\tau - \bar{\tau})]$  with  $-i(\tau - \bar{\tau}) = 2 \text{Im } \tau$  real and positive.)

```
structure GVWSuperpotential where
  f3 : ℤ
  h3 : ℤ
  period : ℂ
  complexMod : ℂ

noncomputable def GVWSuperpotential.eval (W : GVWSuperpotential) (τ : ℂ) : ℂ :=
  (W.f3 : ℂ) * W.period - τ * (W.h3 : ℂ) * W.period

noncomputable def fTermCondition (W : GVWSuperpotential) (τ : ℂ) : ℂ :=
  -(W.h3 : ℂ) * W.period
```

Thus `eval` is  $W(\tau) = (f_3 - \tau h_3)\Pi$ , the one-period case of (26.12), and `fTermCondition` is  $\partial_\tau W = -h_3\Pi$ : the plain derivative, *without* the term  $(\partial_\tau K)W$  of (26.6). It does not depend on  $\tau$ , and `kahlerPotential` is not used by any theorem. The three theorems named in the brief of this chapter are:

```
theorem fterm_potential_nonneg (W : GVWSuperpotential) (τ : ℂ) (hτ : 0 < τ.im) :
  (0 : ℝ) ≤ ||fTermCondition W τ|| ^ 2 := by ...

theorem no_scale_identity :
  (3 : ℚ) = 3 := rfl

theorem fterm_potential_minimum_susy
  (W : GVWSuperpotential) (τ : ℂ)
  (hW : W.eval τ = 0) (hDW : fTermCondition W τ = 0) :
  ||fTermCondition W τ|| ^ 2 = 0 := by ...
```

#### Plain-language reading

**fterm\_potential\_nonneg**: the squared norm of the complex number  $-h_3\Pi$  is non-negative. This is true of every complex number; the proof is the tactic `positivity`, and the hypothesis on  $\text{Im } \tau$  is not used. No  $e^K$ , no inverse Kähler metric and no potential  $V$  occur in the statement.

**no\_scale\_identity**: the rational number 3 equals 3, by `rfl`. The file’s own docstring calls it “vacuous”: the identity (26.9) appears only in a comment. Nothing about a Kähler metric is proved.

**fterm\_potential\_minimum\_susy**: *if*  $-h_3\Pi = 0$ , then its squared norm is 0. The proof rewrites with the hypothesis `hDW` and calls `simp`; the hypothesis `hW` is not used. It is one direction only and it does not mention a minimum.

A fourth theorem in the file, `flux_tadpole_quantization_exact`, re-exports the D7/O7 charge cancellation of Chapter 25 and says nothing about fluxes.

These theorems are kernel-checked, so they are **[A]** *for what they literally state*; what they state is far weaker than Proposition 26.2 or (26.10). The mathematics of Sections 26.3 and 26.4 is **[L]** and derived on the page; it is not formalized. A faithful formalization would need: the Kähler potential as a real function on the product of upper half planes; Wirtinger derivatives

$\partial_\rho, \partial_{\bar{\rho}}$  (or, more simply, real partial derivatives in  $\text{Re } \rho, \text{Im } \rho$ ); the statement  $K^{\rho\bar{\rho}} K_\rho K_{\bar{\rho}} = 3$  as a theorem about `Real.log`; the definition of  $V$  from  $K$  and  $W$ ; and then  $V \geq 0$  as a corollary. A realistic first step is the algebraic core, which needs no analysis at all: with  $u = \rho - \bar{\rho} \neq 0$ ,  $(-u^2/3) \cdot (-3/u) \cdot (3/u) = 3$  (Exercise 26.7). Chapter 38 lists this among the open formalization tasks.

### 26.8.2 The validation use case: integer shadows

The module `DualScaleValidation/UseCase1_ModuliStabilization.lean` works with natural numbers and integers, in units  $\alpha' = 1$  (`alpha_prime : Nat := 1`).

```
def effective_dual_scale_numerator (R : Nat) : Nat :=
  R * R + alpha_prime

theorem self_dual_is_global_minimum (R : Nat) (h : R ≥ 1) :
  effective_dual_scale_numerator R ≥ 2 := by ...

def moduli_potential (phi phi_0 : Int) : Int :=
  (phi - phi_0) * (phi - phi_0)

theorem moduli_potential_nonneg (phi phi_0 : Int) :
  moduli_potential phi phi_0 ≥ 0 := by ...

theorem moduli_potential_zero_iff (phi phi_0 : Int) :
  moduli_potential phi phi_0 = 0 ↔ phi = phi_0 := by ...
```

#### Plain-language reading

`self_dual_is_global_minimum`: for every natural number  $R \geq 1$ ,  $R^2 + 1 \geq 2$ . The proof obtains  $R \cdot R \geq 1$  from `Nat.mul_pos` and finishes with `omega`. The quantity  $R^2 + 1$  is  $R \cdot R_{\text{eff}}(R)$ , the *numerator* of  $R + 1/R$ , not  $R_{\text{eff}}$  itself. The inequality  $R + 1/R \geq 2$  is equivalent to  $R^2 + 1 \geq 2R$ , which is a different (and for  $R \geq 2$  stronger) statement. The theorem is therefore an arithmetic shadow: it records the value 2 at  $R = 1$  and that the numerator does not drop below it on the positive integers, where no radius  $R < 1$  exists and T-duality cannot be expressed.

`moduli_potential_nonneg` and `moduli_potential_zero_iff`: for integers,  $(\varphi - \varphi_0)^2 \geq 0$ , with equality if and only if  $\varphi = \varphi_0$ . This is the integer shadow of a quadratic potential with a unique minimum, such as the leading term  $4V_0 x^2$  of  $V_{\text{ds}}$ . The file documents that an earlier version over `Nat`, with truncated subtraction, had a whole ray of zeros; the present version over `Int` proves uniqueness. The module also contains `buscher_log_involution` ( $-(-x) = x$  for integers) and `moduli_vacuum_stability` (the “if” direction of the equivalence above).

### 26.8.3 The real statement: the trace bound

The module `DualScaleStream2/DualScale/TraceBound.lean`, discussed at length in Chapter 31, proves over  $\mathbb{R}$  the faithful version of property 2 of Section 26.7:

```
theorem circle_effective_scale_ge_two (R : ℝ) (hR : 0 < R) :
  2 ≤ R + R-1 := by ...

theorem dualScale_ge (G : Matrix (Fin d) (Fin d) ℝ) (hG : G.PosDef) :
  (2 * d : ℝ) ≤ dualScale G := by ...

theorem dualScale_inv (G : Matrix (Fin d) (Fin d) ℝ) (hG : IsUnit G.det) :
  dualScale G-1 = dualScale G := by ...
```

Here `dualScale`  $G$  is defined as the trace of the generalized metric at  $B = 0$  and `dualScale_eq` identifies it with  $\text{tr } G + \text{tr } G^{-1}$ . In words: for every positive real  $R$ ,  $R + 1/R \geq 2$ ; for every positive definite real  $d \times d$  matrix,  $\text{tr } G + \text{tr } G^{-1} \geq 2d$ ; and the dual scale is invariant under  $G \mapsto G^{-1}$ . The proof of the circle case uses the identity  $R + R^{-1} - 2 = (R - 1)^2/R$ ; the matrix case expands  $\text{tr}[(G - 1)G^{-1}(G - 1)] \geq 0$  and needs no eigenvalues. Together with `dualScale_one` ( $\mathcal{D}(1) = 2d$ ) these give:  $V_{\text{ds}}(G) \geq 0$ ,  $V_{\text{ds}}(G^{-1}) = V_{\text{ds}}(G)$  and  $V_{\text{ds}}(1) = 0$ . That  $G = 1$  is the *only* zero is elementary (Exercise 26.6) but is not among the compiled theorems.

## 26.8.4 The observables module

`DualScaleValidation/Observables.lean` imports the use case above and records four numerical relations as exact arithmetic, for instance

```
theorem dark_energy_cosmological_constant :
  dark_energy_w0 = -1 ∧ dark_energy_wa = 0 := by ...
```

where `dark_energy_w0 : Int := -1` and `dark_energy_wa : Int := 0` are definitions; the theorem unfolds them. The module’s docstring classifies every one of these relations as Tier C, states that the module “is not a derivation of any physical observable from first principles”, and adds for this one that an exact minimum with  $V(\varphi_0) = 0$  “is a Minkowski vacuum, not de Sitter”, so that it is “not by itself a prediction of  $w_0 = -1$ ” with non-zero dark energy density, and that current data are in tension with  $(w_0, w_a) = (-1, 0)$ . The conjunction `observables_windows_master_contract` states that four stored numbers lie in four illustrative windows. We mention the module because the brief of this chapter lists it; it contains no statement about moduli stabilization beyond what has been said.

| Statement                                                                       | Tier    | Evidence                                                                  |
|---------------------------------------------------------------------------------|---------|---------------------------------------------------------------------------|
| $W = \int \Omega \wedge G$ reproduces $G^{0,4} = G^{1,3} = 0$                   | [L]     | GVW §2.4                                                                  |
| $V = e^K( DW ^2 - 3 W ^2)$ ; $DW = 0 \Rightarrow \partial V = 0$                | [L]     | Gra (5.21); Proposition 26.2                                              |
| No-scale identity (26.9), $V \geq 0$ , minima at $D_i W = 0$                    | [L]     | GKP §2.3; symbolic check                                                  |
| Toy model (26.13)–(26.16)                                                       | derived | symbolic check                                                            |
| KKLT formulas (26.18)–(26.19)                                                   | [L]     | Gra (7.8)–(7.9); symbolic check                                           |
| $\  -h_3 \Pi \ ^2 \geq 0$ ; $(3 : \mathbb{Q}) = 3$                              | [A]     | <code>fterm_potential_nonneg</code> ,<br><code>no_scale_identity</code>   |
| $R \in \mathbb{N}$ , $R \geq 1 \Rightarrow R^2 + 1 \geq 2$                      | [A]     | <code>self_dual_is_global_minimum</code>                                  |
| $(\varphi - \varphi_0)^2 = 0 \iff \varphi = \varphi_0$ over $\mathbb{Z}$        | [A]     | <code>moduli_potential_zero_iff</code>                                    |
| $R + 1/R \geq 2$ ; $\text{tr } G + \text{tr } G^{-1} \geq 2d$ over $\mathbb{R}$ | [A]     | <code>circle_effective_scale_ge_two</code> ,<br><code>dualScale_ge</code> |
| String gas drives $R$ to $\sqrt{\alpha'}$ (string frame)                        | [L]     | BW §III.C                                                                 |
| $V_{\text{ds}}$ is the radion potential on $K3 \times T^2$                      | [C]     | none                                                                      |

Table 26.2: Epistemic status of the statements of this chapter. All Lean theorems listed depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`. The identification of any Lean object with a physical system is never Tier A.

### Chapter summary

- Moduli are massless scalars; they mediate composition-dependent long-range forces and must acquire a potential with an isolated minimum.
- Quantized flux is a fixed integral class whose Hodge type depends on the moduli; the



supersymmetry conditions on its type are equations for the moduli. They follow from  $W = \int \Omega \wedge G$  (GVW), in type IIB  $W = \int \Omega \wedge (F_3 - \tau H_3)$ .

- The  $N = 1$  potential is  $V = e^K (K^{I\bar{J}} D_I W \overline{D_J W} - 3|W|^2)$ . Points with  $D_I W = 0$  are critical points with  $V = -3e^K |W|^2 \leq 0$ .
- For  $K = -3 \ln[-i(\rho - \bar{\rho})]$  and  $W$  independent of  $\rho$ ,  $K^{\rho\bar{\rho}} K_\rho K_{\bar{\rho}} = 3$  cancels  $-3|W|^2$ :  $V \geq 0$ , the minima are at  $D_i W = 0$  ( $i \neq \rho$ ), the volume is flat, and supersymmetry is broken if  $W \neq 0$ . In ten dimensions:  $G_3$  imaginary self-dual.
- In the two-period toy model  $\bar{\tau}_0 = A/B$  and  $\text{Im } \tau_0 > 0$  requires  $N_{\text{flux}} > 0$ ; one period is not enough.
- Fixing the volume needs non-perturbative terms (KKLT), giving a supersymmetric AdS vacuum and requiring small  $|W_0|$ .
- The dual-scale toy potential  $V_0(R/\sqrt{\alpha'} - \sqrt{\alpha'}/R)^2 = 4V_0 \sinh^2 x$  is T-duality invariant with its unique minimum at the self-dual radius. The inequality behind it is machine-checked over  $\mathbb{R}$ ; its use as a physical radion potential is a proposal.
- The Lean theorems `fterm_potential_nonneg`, `no_scale_identity` and `fterm_potential_minimum_susy` are trivialities about a toy; the no-scale identity itself is not formalized.

## Exercises

**Exercise 26.1** (\*). Verify (26.8) and (26.9) by writing  $\rho = a + ib$  and  $K = -3 \ln(2b)$ . Repeat with  $-3$  replaced by  $-p$  and show that  $V = e^K (K^{i\bar{j}} D_i W \overline{D_j W} + (p - 3)|W|^2)$ . For which  $p$  can  $V$  be negative?

**Exercise 26.2** (\*). Show that  $V_{\text{ds}}(R) = 4V_0 \sinh^2 x$  with  $x = \ln(R/\sqrt{\alpha'})$ , and compute  $V''(x)$  at the minimum. Show that the three expressions in (26.21) agree.

**Exercise 26.3** (\*\*). In the toy model (26.12), the  $SL(2, \mathbb{Z})$  duality of type IIB acts by  $\tau \mapsto (a\tau + b)/(c\tau + d)$  and on the flux vectors by  $\begin{pmatrix} F_3 \\ H_3 \end{pmatrix} \mapsto \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} F_3 \\ H_3 \end{pmatrix}$ . (a) Show that  $N_{\text{flux}}$  is invariant. (b) Show that the vacuum  $\tau_0$  transforms covariantly, i.e. the vacuum of the transformed fluxes is the transform of  $\tau_0$ . (c) Map the vacuum  $\tau_0 = \frac{1}{2}(1 + i)$  of the text to the fundamental domain and give the transformed fluxes and the string coupling.

**Exercise 26.4** (\*\*). For the toy model with  $A = 2$ ,  $B = i$  expand (26.15) to second order in  $\delta\tau = \tau - 2i$  at fixed  $\rho$ . Using the kinetic term  $-K_{\tau\bar{\tau}} |\partial\tau|^2$  with  $K_{\tau\bar{\tau}} = 1/(4(\text{Im } \tau)^2)$ , find the mass of the canonically normalized axio-dilaton as a function of  $\text{Im } \rho$ . What happens to this mass along the flat direction  $\text{Im } \rho \rightarrow \infty$ ?

**Exercise 26.5** (\*\*). Derive (26.18) and (26.19). Then show that for  $\sigma_{\text{cr}} \gg 1$  one has  $\sigma_{\text{cr}} \approx \frac{1}{2\pi} \ln(|B/W_0|)$  up to logarithmic corrections, and estimate the  $|W_0|$  needed for  $\sigma_{\text{cr}} = 10$  with  $B = 1$ .

**Exercise 26.6** (\*\*). Let  $G$  be a real symmetric positive definite  $d \times d$  matrix. Show that  $\text{tr}[(G - 1)G^{-1}(G - 1)] = 0$  implies  $G = 1$ . (Write  $G^{-1} = S^2$  with  $S$  symmetric positive definite and consider  $M = S(G - 1)$ .) Conclude that  $V_{\text{ds}}(G)$  has a unique zero. State the result as a Lean theorem in the style of `dualScale_ge`; you need not prove it.

**Exercise 26.7** (\*\*, Lean). State and prove in Lean 4 with Mathlib the algebraic core of the no-scale identity: for  $u : \mathbb{C}$  with  $u \neq 0$ ,  $(-(u^2)/3) * (-3/u) * (3/u) = 3$ . (The



tactic `field_simp` followed by `ring` suffices.) Explain in two sentences what is still missing between your theorem and (26.9).

**Exercise 26.8** (★★, Lean). The theorem `self_dual_is_global_minimum` proves  $R^2 + 1 \geq 2$  for natural numbers  $R \geq 1$ . State and prove the inequality that actually corresponds to  $R + 1/R \geq 2$ , namely  $2 * R \leq R * R + 1$  for  $R : \mathbb{Z}$ . (Hint: `nlinarith [sq_nonneg (R - 1)]`.) Why is the hypothesis  $R \geq 1$  no longer needed?

**Exercise 26.9** (★★★). Prove the following converse to the “one period is not enough” pitfall. For  $W = \sum_{k=1}^m (f_k - \tau h_k) \Pi_k$  with fixed periods, a supersymmetric-in- $\tau$  vacuum ( $D_\tau W = 0$  with  $\text{Im } \tau > 0$ ) exists if and only if  $\text{Im}(\bar{A}B) > 0$ , where  $A = \sum f_k \Pi_k$ ,  $B = \sum h_k \Pi_k$ . Express  $\text{Im}(\bar{A}B)$  through the antisymmetric matrix  $\text{Im}(\bar{\Pi}_k \Pi_l)$  and the fluxes, and compare with  $\int H_3 \wedge F_3$ .

## Further reading

- The superpotential: [GVW §2.4, ll. 876–1008] for the fourfold, the domain wall argument and the remark on no-scale models; [GVW §2.5, ll. 1068–1100] for the F-theory decomposition of  $G$ .
- Type IIB flux compactifications: [GKP §2.3, ll. 740–920] for  $K$ ,  $W$ , the no-scale potential and the match with the imaginary self-dual condition; [GKP §3, ll. 1000–1160] for the conifold example and the stabilization of  $\tau$ .
- Review: [Gra §5.2 and §5.4, ll. 2980–3000, 3075–3096, 3490–3552] for the  $N = 1$  potential, the no-scale condition and the potential in terms of  $G_3$ ; [Gra §6, ll. 3905–3930] for the moduli problem; [Gra §7.3–7.4, ll. 4855–4926, 5022–5029] for KKLT and its critiques.
- String gases and the self-dual radius: [BW §III.C–D, ll. 1105–1190].
- Lean sources, each with a docstring that states what is and is not proved:
  - `StringTheoryFormalization/Frontier/FTermPotential.lean`
  - `DualScaleValidation/UseCase1_ModuliStabilization.lean`
  - `DualScaleValidation/Observables.lean`
  - `DualScaleStream2/DualScale/TraceBound.lean`



## Chapter 27

# The Swampland: Distance, de Sitter and TCC

Every chapter so far has asked what string theory *contains*: a spectrum, a duality group, a moduli space, a flux superpotential. This chapter asks the opposite question. Suppose someone hands you a perfectly sensible-looking low-energy field theory coupled to gravity—some scalars, some gauge fields, a potential. Which of these theories can be the low-energy limit of a consistent theory of quantum gravity, and which cannot, no matter how the ultraviolet is filled in? The theories that cannot are said to lie in the *Swampland*; the ones that can form the *Landscape*. The criteria that separate the two are, with one or two exceptions, conjectures: they are supported by string-theoretic examples and by general arguments about black holes and horizons, but they are not derived from first principles [Pal §1, ll. 395–430]. A student should care for two reasons. First, the conjectures make sharp, testable statements about cosmology—the fate of the dark energy, the maximal duration of inflation—so they are the place where the abstract machinery of the earlier chapters touches observation. Second, the very first example of a Swampland criterion is the circle of Chapters 7 and 8: the two towers of momentum and winding states, exchanged by T-duality, are exactly what the distance conjecture predicts must exist at the two ends of the radius modulus. We derive this example in full, state the distance, de Sitter and trans-Planckian censorship conjectures, and then look carefully at what it means to formalize a *conjecture* in Lean: one does not prove it, one encodes it as a hypothesis or a structure and proves what follows.

### 27.1 Landscape and Swampland

String theory has a large number of vacua, each with its own low-energy effective theory. It is tempting to conclude that anything goes at low energies. The point of the Swampland programme is that this is false: the set of effective theories that arise from string theory—or, more generally, from any consistent quantum gravity—is a strict subset of the set of theories that look self-consistent on their own terms. Palti’s definition, which we adopt, reads [Pal §1, ll. 395–405]:

The Swampland is the set of (apparently) consistent effective field theories that cannot be completed into quantum gravity in the ultraviolet.

“Apparently” means that nothing is manifestly wrong with the theory as a quantum field theory; the inconsistency appears only when one tries to complete it. The Landscape is the complement: the effective theories that do arise from some vacuum. Figure 27.1 shows the picture.

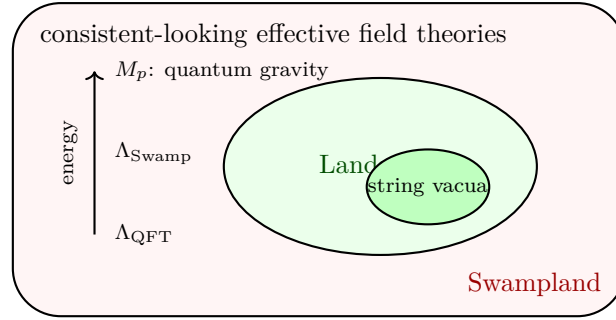


Figure 27.1: Left: the scales of an effective theory coupled to gravity. Right: the Landscape of theories that arise from quantum gravity sits inside the larger set of theories that merely look consistent; the difference is the Swampland. After [Pal §1, figs. 1, 3–4, ll. 380–395, 492–525].

A Swampland criterion has a characteristic shape [Pal §1, ll. 492–525]. Take a quantum field theory valid up to a cutoff  $\Lambda_{\text{QFT}}$  and couple it to gravity with a finite Planck mass  $M_p$ . The conjecture is that a new scale  $\Lambda_{\text{Swamp}}$  appears above which the theory must be modified—perhaps mildly, by adding a particle, perhaps drastically—if it is ever to reach a consistent ultraviolet completion. Nothing need go wrong *at*  $\Lambda_{\text{Swamp}}$ ; rather, a theory that is not modified there is not completable. Typically  $\Lambda_{\text{Swamp}} \rightarrow \infty$  as  $M_p \rightarrow \infty$ , since decoupling gravity removes the constraint. The interesting regime is  $\Lambda_{\text{Swamp}} \ll \Lambda_{\text{QFT}} \ll M_p$ , where gravity constrains physics far below the Planck scale.

Three kinds of evidence are used [Pal §1, ll. 428–470]: (i) explicit string vacua, treated as experimental data, with the caveat that “string-derived” vacua (a full worldsheet description, usually supersymmetric) and “string-inspired” constructions (many assumptions) carry different weight; (ii) general quantum-gravity arguments, mostly about black holes and holography; (iii) the hoped-for microscopic derivation, which does not yet exist. Because the criteria are conjectures, every one of them is Tier **[C]** in the language of this book, however much evidence supports it. When a source states a conjecture we mark it **[L]** as a *statement of what the literature proposes*, never as a statement that it holds.

#### What “infinite distance” will mean

The conjectures below are phrased in terms of the moduli space of the low-energy theory, with the metric given by the scalar kinetic terms. Moving a long way in that metric costs a lot of field displacement in Planck units. The recurring claim is that quantum gravity does not allow such a displacement to go unnoticed: something—an infinite tower of states—must become light, and the effective theory must break down. The circle is the case where all of this can be computed.

## 27.2 The model example: a circle

### 27.2.1 Reduction of gravity on a circle

Take  $D = d + 1$  dimensions with coordinates  $X^M = (x^\mu, y)$ ,  $\mu = 0, \dots, d - 1$ , and let  $y$  be periodic with a *fixed coordinate* period,  $y \sim y + 2\pi R_0$ . The physical size of the circle will be carried by a scalar field. Write the  $D$ -dimensional metric as

$$ds_D^2 = e^{2a\varphi(x)} g_{\mu\nu}(x) dx^\mu dx^\nu + e^{2b\varphi(x)} dy^2, \quad (27.1)$$

with constants  $a, b$  to be fixed [Pal §2.2.1, ll. 1731–1760]. The circumference measured with the metric (27.1) is

$$2\pi R = \int_0^{2\pi R_0} \sqrt{G_{yy}} dy = 2\pi R_0 e^{b\varphi}, \quad (27.2)$$

so  $R$  is a field, the *radion*. We start from the  $D$ -dimensional Einstein–Hilbert action and define the reduced Planck mass in  $d$  dimensions by the coefficient of the Ricci scalar,

$$S = \frac{M_D^{D-2}}{2} \int d^D X \sqrt{-G} R_D \longrightarrow S = \int d^d x \sqrt{-g} \left[ \frac{M_p^{d-2}}{2} R_d + \dots \right], \quad (27.3)$$

following [Pal §1, ll. 596–612]. Two requirements fix  $a$  and  $b$ .

*The  $d$ -dimensional Planck mass must not depend on  $\varphi$ .* Otherwise the ruler with which we measure the field distance would itself move. Splitting off the conformal factor, the metric (27.1) is  $e^{2a\varphi} \hat{G}$  with  $\hat{G} = g \oplus e^{2(b-a)\varphi} dy^2$ , and  $\sqrt{-G} = e^{(d+1)a\varphi} \sqrt{-\hat{G}} = e^{(d+1)a\varphi + (b-a)\varphi} \sqrt{-g}$ . The Weyl rescaling formula in  $D$  dimensions,  $\sqrt{-G} R_D = \sqrt{-\hat{G}} e^{(D-2)\omega} [\hat{R} - 2(D-1)\hat{\square}\omega - (D-1)(D-2)(\hat{\partial}\omega)^2]$  with  $\omega = a\varphi$ , produces an overall factor  $e^{(d-1)a\varphi} \cdot e^{(b-a)\varphi} = e^{(b+(d-2)a)\varphi}$  in front of  $\sqrt{-g} R_d$ . It is constant if and only if

$$b = -(d-2)a. \quad (27.4)$$

With this choice  $M_p^{d-2} = M_D^{D-2} \cdot 2\pi R_0$ , a constant; the  $d$ -dimensional metric  $g$  is the *Einstein-frame* metric.

*The radion must be canonically normalized.* Collect the  $(\partial\varphi)^2$  terms. The warped metric  $\hat{G}$  has  $\hat{R} = R_d - 2\gamma\square\varphi - 2\gamma^2(\partial\varphi)^2$  with  $\gamma = b - a$  (the standard formula for a product with one warped flat direction), and  $\hat{\square}\omega = \square\omega + \gamma\partial\varphi \cdot \partial\omega$ . Because the prefactor is constant, all  $\square$  terms are total derivatives. What remains is

$$\sqrt{-G} R_D \simeq \sqrt{-g} [R_d - (2\gamma^2 + 2d\gamma a + d(d-1)a^2)(\partial\varphi)^2] = \sqrt{-g} [R_d - (d-1)(d-2)a^2(\partial\varphi)^2],$$

where the last step uses  $\gamma = -(d-1)a$  from (27.4). (We checked the algebra symbolically; it is a two-line exercise once  $\gamma$  is substituted.) Multiplying by  $\frac{1}{2}M_D^{D-2} \cdot 2\pi R_0 = \frac{1}{2}M_p^{d-2}$ , and setting  $M_p = 1$ , the reduced action is  $\int \sqrt{-g} [\frac{1}{2}R_d - \frac{1}{2}(d-1)(d-2)a^2(\partial\varphi)^2]$ . The kinetic term is  $-\frac{1}{2}(\partial\varphi)^2$  precisely when

$$a^2 = \frac{1}{(d-1)(d-2)}, \quad b = -\sqrt{\frac{d-2}{d-1}}. \quad (27.5)$$

In  $d = 4$ :  $a = 1/\sqrt{6} \approx 0.408$ ,  $b = -\sqrt{2/3} \approx -0.816$ .

#### Two normalizations differ by $\sqrt{2}$

Palti writes the reduced action as  $\int \sqrt{-g} [R - \frac{1}{2}(\partial\phi)^2]$  and correspondingly finds  $\alpha^2 = 1/(2(d-1)(d-2))$  for the metric exponent [Pal §2.2.1, ll. 1755–1792]. His field  $\phi$  and ours are related by  $\phi = \sqrt{2}\varphi$ , since  $R - \frac{1}{2}(\partial\phi)^2 = 2[\frac{1}{2}R - \frac{1}{4}(\partial\phi)^2]$ . Every exponent quoted below in the form  $e^{\lambda\varphi}$  becomes  $e^{(\lambda/\sqrt{2})\phi}$  in his convention. The distance conjecture only fixes an order-one exponent, but if you compare numbers between papers you must track this factor; the frequently quoted “ $\alpha = \sqrt{3}/2$ ” and “ $\lambda = \sqrt{3}/2$ ” for the four-dimensional circle are the same statement.

### 27.2.2 The Kaluza–Klein tower

Now add a massless  $D$ -dimensional scalar  $\Psi$ . Periodicity in  $y$  forces a Fourier expansion with quantized momentum along the circle,

$$\Psi(x, y) = \sum_{n \in \mathbb{Z}} \psi_n(x) e^{iny/R_0}, \quad -i\partial_y \Psi = \frac{n}{R_0} \Psi,$$

[Pal §2.2.1, ll. 1793–1840]. The wave equation  $G^{MN} \partial_M \partial_N \Psi = 0$  in the metric (27.1), at constant  $\varphi$  and with  $g = \eta$ , reads

$$e^{-2a\varphi} \partial^\mu \partial_\mu \psi_n - e^{-2b\varphi} \frac{n^2}{R_0^2} \psi_n = 0,$$

so the Einstein-frame mass of the  $n$ -th mode is

$$m_n^2 = e^{2(a-b)\varphi} \frac{n^2}{R_0^2} = e^{2(d-1)a\varphi} \frac{n^2}{R_0^2}, \quad m_n = \frac{|n|}{R_0} e^{\lambda_{\text{KK}}\varphi}, \quad \lambda_{\text{KK}} = (d-1)a = \sqrt{\frac{d-1}{d-2}}. \quad (27.6)$$

Written in terms of the physical radius,  $m_n \propto R^{-(d-1)/(d-2)}$ : one factor  $1/R$  is the naive Kaluza–Klein mass, the extra  $R^{-1/(d-2)}$  is the Einstein-frame conversion. As a check by another route:  $M_p^{d-2} = M_D^{d-1} \cdot 2\pi R$  if  $R$  is measured in  $D$ -dimensional Planck units, so  $m_n/M_p = (M_D/M_p) (1/RM_D) \propto (RM_D)^{-1/(d-2)-1}$ , the same power. The whole tower  $\{m_n\}$  moves rigidly, with the single scale  $M_{\text{KK}} = e^{\lambda_{\text{KK}}\varphi}/R_0$ : as  $\varphi \rightarrow -\infty$  (large circle, since  $b < 0$ ) it becomes light *exponentially in the canonically normalized field*. This part is pure field theory.

### 27.2.3 The winding tower and T-duality

In string theory the circle carries a second tower. In the string frame the closed-string mass formula of Chapter 7 is

$$M^2 = \frac{n^2}{R^2} + \frac{w^2 R^2}{\alpha'^2} + \frac{2}{\alpha'} (N + \tilde{N} - 2), \quad N - \tilde{N} = nw, \quad (27.7)$$

[Tong §8.1, ll. 11371–11475], [Pal §2.2.2, ll. 1854–1990], and it is invariant under  $R \mapsto \alpha'/R$ ,  $n \leftrightarrow w$ . Converting to the Einstein frame requires two steps that Palti spells out [Pal §2.2.2, ll. 1974–2046]: the frame factor  $e^{2a\varphi}$  multiplies all masses squared, and the  $d$ -dimensional dilaton  $\Phi_d = \Phi - \frac{1}{2} \ln(2\pi R M_s)$  is held fixed while  $R$  varies, which makes the string scale  $M_s \propto (2\pi R)^{1/(d-2)}$  in  $d$ -dimensional Planck units. The result, in units where the coordinate period is one, is

$$M_{n,w}^2 = (2\pi R)^{-\frac{2}{d-2}} \left(\frac{n}{R}\right)^2 + (2\pi R)^{\frac{2}{d-2}} \left(\frac{wR}{\alpha'_0}\right)^2, \quad (27.8)$$

where  $\alpha'_0$  is the string scale with the  $R$ -dependence removed. The first term reproduces the field-theory result (27.6). The two mass scales therefore behave as

$$M_{\text{KK}} \propto R^{-\frac{d-1}{d-2}} \propto e^{+\lambda_{\text{KK}}\varphi}, \quad M_w \propto R^{+\frac{d-1}{d-2}} \propto e^{-\lambda_{\text{KK}}\varphi}, \quad M_{\text{KK}} M_w = \text{const}, \quad (27.9)$$

[Pal §2.2.3, ll. 2066–2100]. The product being independent of  $\varphi$  is T-duality in Einstein-frame clothing: the  $\mathbb{Z}_2$  that exchanges the towers acts as  $\varphi \mapsto -\varphi$  about the self-dual point. Figure 27.2 shows the two lines.

**Example 27.1** (Numbers in four dimensions). For  $d = 4$ ,  $\lambda_{\text{KK}} = \sqrt{3/2} \approx 1.225$  in the canonical normalization. Start at the self-dual point, where both towers have scale  $M_s$ , and

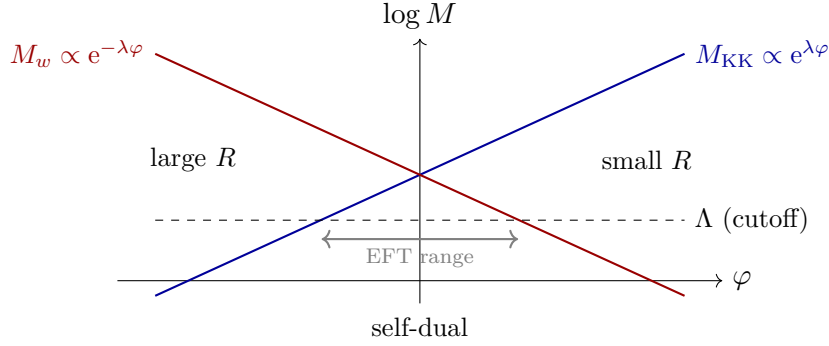


Figure 27.2: Mass scales of the momentum and winding towers on a circle, as functions of the canonically normalized radion  $\varphi$  (Einstein frame). The slopes are  $\pm\lambda_{\text{KK}}$ ; the  $\mathbb{Z}_2$  symmetry of the picture is T-duality. Whichever way one moves, one tower crosses any fixed cutoff after a finite field range. After [Pal §2.2.3, fig. 8, ll. 2091–2097].

suppose the effective theory is to be trusted up to a cutoff  $\Lambda = M_s/100$ . Either tower reaches  $\Lambda$  after a displacement  $|\Delta\varphi| = \ln 100/\lambda_{\text{KK}} \approx 4.605/1.225 \approx 3.76 M_p$ . A displacement of  $5M_p$  towards large radius multiplies  $M_{\text{KK}}$  by  $e^{-1.225 \times 5} \approx 2.2 \times 10^{-3}$ . In field theory alone, moving towards *small* radius would be harmless—the KK tower gets heavier without bound. String theory forbids this: the winding tower takes over, with exactly the same slope.

The observations to retain [Pal §2.2.3, ll. 2113–2150] are: some tower becomes light for either sign of  $\Delta\varphi$ ; the behaviour is exponential in the *proper* distance because  $\varphi$  is canonically normalized; the exponent is a fixed order-one number; and as  $|\Delta\varphi| \rightarrow \infty$  infinitely many states become massless, so no  $d$ -dimensional field theory describes that locus—the correct description is the  $D$ -dimensional theory, at either end. The lesson that string theory adds to field theory is the second tower, and it is T-duality that supplies it.

## 27.3 The Swampland Distance Conjecture

### 27.3.1 Moduli space as a Riemannian manifold

Consider a  $d$ -dimensional theory of gravity coupled to real scalars  $\phi^i$  with no potential,

$$S = \int d^d x \sqrt{-g} \left[ \frac{R}{2} - \frac{1}{2} g_{ij}(\phi) \partial_\mu \phi^i \partial^\mu \phi^j + \dots \right], \quad M_p = 1. \quad (27.10)$$

The scalar expectation values are coordinates on the *moduli space*  $\mathcal{M}$  and the kinetic matrix  $g_{ij}$  is a Riemannian metric on it [Pal §3.6, ll. 5062–5095]. (Palti writes the kinetic term without the  $\frac{1}{2}$ ; his  $g_{ij}$  is half ours. Beware the clash of notation with the spacetime metric  $g_{\mu\nu}$ : the two are unrelated.) The metric is positive definite because kinetic terms have a fixed sign. The distance between two points  $P, Q$  is the length of the shortest geodesic  $\gamma$  between them,

$$d(P, Q) = \int_\gamma \sqrt{g_{ij} \frac{d\phi^i}{ds} \frac{d\phi^j}{ds}} ds. \quad (27.11)$$

For the circle,  $\mathcal{M} = \mathbb{R}$  with coordinate  $\varphi$  and  $g = 1$ , so  $d(P, Q) = |\Delta\varphi|$  and the points  $\varphi = \pm\infty$  are at infinite distance from every finite point. A periodic scalar  $\phi \sim \phi + 2\pi f$  with constant  $f$  is the opposite case:  $\mathcal{M} = S^1$  and no two points are further apart than  $\pi f$ .

### 27.3.2 Statement

**Conjecture 27.2** (Swampland Distance Conjecture, Ooguri–Vafa; [C]). *Let  $\mathcal{M}$  be the moduli space of a theory coupled to gravity, with metric from the kinetic terms. (i) From any point  $P \in \mathcal{M}$  there is a point  $Q \in \mathcal{M}$  with  $d(P, Q) = \infty$ . (ii) There is an infinite tower of states with a mass scale  $M$  such that*

$$M(Q) \sim M(P) e^{-\alpha d(P, Q)} \quad \text{as } d(P, Q) \rightarrow \infty, \quad (27.12)$$

for some positive constant  $\alpha$ . [Pal §3.6, ll. 5049–5062; §2.2.3, ll. 2167–2185]

Three comments. First, the statement is asymptotic; the value  $M(P)$  at the starting point does not matter. Second,  $\alpha$  is not specified beyond “order one”, and the sharper reading adopted in the literature is that the exponential behaviour sets in after a distance of one or two Planck units. Third, the circle is the case where (27.12) holds *exactly*, for all distances, with  $\alpha = \lambda_{\text{KK}}$ ; in general one must expect a bulk region of moduli space where masses go up and down, and asymptotic regions where the exponential law takes over [Pal §3.6, ll. 5120–5150].

The *refined* version [Pal §3.6, ll. 5150–5210] makes the second point quantitative and extends the statement to fields with a potential: with  $M_p$  reinstated,

$$M(Q) < M(P) e^{-\alpha d(P, Q)/M_p} \quad \text{for } d(P, Q) \gtrsim M_p, \quad (27.13)$$

and the same holds in the field space of an effective theory even when the fields have a potential. The inequality is now sharp, it allows faster-than-exponential decay, and it implicitly bounds  $\alpha$  from below: the tower may not grow much before the exponential decrease begins. The tower need not consist of particles; in string examples it may be a tower of light strings, treated as particles only below their own length scale.

#### Why a tower, and why exponential

Two arguments, both from [Pal §3.6.1, ll. 5212–5270]. *Duality*. In the circle the two towers are exchanged by T-duality and their product is constant; moving far in either direction makes the dual tower light. String theory has many dualities and, correspondingly, many dual towers; the conjecture generalizes the pattern. *Gravity as the weakest force*. The scalar weak gravity conjecture asks that a particle’s mass satisfy  $|\partial_\phi m| > m$  for a canonically normalized  $\phi$ . Maintained over a large field range this forces  $m \sim e^{-\alpha\phi}$  with  $\alpha > 1$ : a power law  $m \sim \phi^p$  has  $|\partial_\phi m|/m = p/\phi \rightarrow 0$ . Two particles with oscillating masses  $|\cos \phi|, |\sin \phi|$  can share the burden over one period, but covering a range  $2\pi N$  needs  $N$  particles, and an infinite range needs an infinite tower.

**Proposition 27.3** (Exponential decay from a relative bound). *Let  $m : [0, \infty) \rightarrow (0, \infty)$  be differentiable with  $m'(\phi) \leq -\alpha m(\phi)$  for some  $\alpha > 0$  and all  $\phi \geq 0$ . Then  $m(\phi) \leq m(0) e^{-\alpha\phi}$ .*

*Proof.* Set  $u = \ln m$ ; then  $u' \leq -\alpha$ , so  $u(\phi) - u(0) = \int_0^\phi u' \leq -\alpha\phi$ . Exponentiate.  $\square$

This is the whole content of “exponential in the distance”: a bound on the logarithmic derivative integrates to an exponential bound. The Lean statement `sdc_tower_suppression` in Section 27.6 is the trivial half of it, with the exponential taken as a definition.

### 27.3.3 What the conjecture forbids

The first part excludes moduli spaces of finite diameter. A single periodic scalar with a constant decay constant  $f$  is therefore not allowed *on its own*; it must sit inside a larger



moduli space, and in string theory the decay constant is itself a field [Pal §3.6, ll. 5110–5120]. The second part limits the validity of any effective theory with a cutoff  $\Lambda$ : since the tower drops below  $\Lambda$  after a finite distance, the theory is valid only in a compact region of field space (the “EFT range” in Figure 27.2). For inflation this is a constraint on super-Planckian field excursions: a displacement  $\Delta\phi$  costs a factor  $e^{-\alpha\Delta\phi}$  in the cutoff. We do not pursue the cosmological literature here; the essential logic is Example 27.1.

## 27.4 The de Sitter conjecture

### 27.4.1 Statement and refinement

A de Sitter vacuum of string theory means a local minimum of the scalar potential at which the potential is positive; the cosmological constant is never a free parameter but the value of  $V$  at such a minimum [Pal §7, ll. 10475–10500]. Constructing such minima has proved very hard, and one may take the difficulty as a hint that they lie in the Swampland. That is not in conflict with the observed late-time acceleration if the acceleration is driven by a slowly rolling scalar (*quintessence*) rather than by a constant [Pal §7, ll. 10500–10525]. The quantitative proposal is:

**Conjecture 27.4** (de Sitter conjecture, Obied–Ooguri–Spodyneiko–Vafa; [C]). *The scalar potential of a theory coupled to gravity satisfies*

$$|\nabla V| \geq \frac{c}{M_p} V, \quad (27.14)$$

where  $\nabla V$  is the gradient with respect to all canonically normalized scalars and  $c > 0$  is a constant of order one. [Pal §7, ll. 10528–10546]

At a point with  $V > 0$  this forbids  $\nabla V = 0$ : no de Sitter minimum, and no de Sitter maximum either. The top of the Higgs potential is a counter-example to the bound as stated: there  $|\nabla V|/V \sim 10^{-55}/M_p$  while the most negative eigenvalue of the Hessian is  $\sim -10^{35}V/M_p^2$  [Pal §7, ll. 10585–10600]. The refined form accommodates it:

**Conjecture 27.5** (Refined de Sitter conjecture; [C]). *Either (27.14) holds, or*

$$\min(\nabla_i \nabla_j V) \leq -\frac{c'}{M_p^2} V, \quad (27.15)$$

where the left side is the smallest eigenvalue of the Hessian in an orthonormal frame and  $c' > 0$  is of order one. [Pal §7, ll. 10558–10585]

The second option says: a positive plateau is allowed provided it is unstable on Hubble time scales. The top of an axion potential of period  $2\pi f$  has  $\min(\nabla\nabla V)/V \leq -1/f^2$ , so it satisfies (27.15) whenever  $f \leq M_p$  [Pal §7, ll. 10600–10615].

### 27.4.2 Where the refinement comes from: entropy and the tower

The refined conjecture was argued from the distance conjecture and the entropy of de Sitter space [Pal §7.2, ll. 10736–10880]. We reproduce the argument because it shows how the conjectures hang together, and because each step is an order-of-magnitude estimate that a student should be able to reproduce.

*De Sitter space in  $d$  dimensions* is the hyperboloid  $-X_0^2 + X_1^2 + \cdots + X_d^2 = R^2$  in  $\mathbb{R}^{1,d}$ , with cosmological constant  $\Lambda = (d-1)(d-2)/(2R^2)$ ; in  $d = 4$ ,  $\Lambda = 3/R^2$ . In static coordinates

the metric  $-(1-r^2/R^2)dt^2 + (1-r^2/R^2)^{-1}dr^2 + r^2d\Omega^2$  shows a horizon at  $r = R$ , attached to an observer rather than to a body. It has the Gibbons–Hawking temperature  $T_{\text{dS}} = 1/(2\pi R)$  and, in four dimensions, the entropy

$$S_{\text{dS}} = 8\pi^2(RM_p)^2, \quad (27.16)$$

which one may try to interpret, as for black holes, as  $\log \dim \mathcal{H}$  of a Hilbert space of microstates [Pal §7.1, ll. 10636–10700]. (The area of the horizon is  $4\pi R^2$  and  $S = A/4G_N$  with  $M_p^2 = 1/8\pi G_N$  gives  $S = 4\pi R^2 \cdot 2\pi M_p^2 = 8\pi^2 R^2 M_p^2$ , consistent with (27.16).)

Now let a scalar roll on a positive potential slowly enough that the expansion accelerates. Then an *apparent* horizon exists, which requires  $|\nabla V| \leq \sqrt{2}V$  (if this fails the de Sitter conjecture already holds), and its radius is  $R \sim V^{-1/2}$  from the Friedmann equation. The covariant entropy bound assigns the entropy  $\sim R^2$  to it. Meanwhile, by the distance conjecture, the number  $N$  of light tower states grows with the distance travelled,

$$N \sim n(\phi) e^{k\phi}, \quad (27.17)$$

with  $n(\phi)$  monotonically increasing (more towers) and  $k$  of order one. The entropy of the tower is estimated as  $S_{\text{tower}} \sim N^\gamma R^\delta$  with unknown order-one  $\gamma, \delta$ . The crucial assumption is that the tower dominates the Hilbert space, so that

$$R^2 \sim N^\gamma R^\delta \implies V \sim R^{-2} \sim N^{-\frac{2\gamma}{2-\delta}} \sim e^{-\frac{2k\gamma}{2-\delta}\phi}, \quad (27.18)$$

[Pal §7.2, ll. 10785–10840]. Differentiating,

$$\frac{|\nabla V|}{V} \sim c, \quad c \sim \frac{2k\gamma}{2-\delta}, \quad (27.19)$$

which is (27.14). The second condition (27.15) marks where the argument fails: the de Sitter temperature gives scalars a positive mass squared of order  $V$  at horizon scales, so a tachyonic direction only destabilizes the horizon—and destroys the entropic reading—if its magnitude exceeds  $V/M_p^2$  [Pal §7.2, ll. 10855–10870].

**Example 27.6** (Reading off  $c$ ). With the simplest guesses  $\gamma = 1$  (entropy proportional to the number of light species) and  $\delta = 0$ , (27.19) gives  $c \sim k$ : the steepness of the potential is the growth rate of the tower. For the circle with  $M_{\text{KK}} \propto e^{-\lambda_{\text{KK}}|\varphi|}$  towards large radius, the number of KK states below a fixed cutoff grows as  $N \sim \Lambda/M_{\text{KK}} \propto e^{\lambda_{\text{KK}}|\varphi|}$ , so  $k = \lambda_{\text{KK}} = \sqrt{3/2}$  in  $d = 4$  and the estimate is  $c \sim 1.2$ . This is only an illustration of the logic; the argument is valid only at parametrically large distance, where any coupling of string theory becomes weak and a runaway direction opens—a generalization of the Dine–Seiberg problem [Pal §7.2, ll. 10870–10880].

### 27.4.3 Cosmological consequences

If  $V > 0$  today and (27.14) holds, the universe is not at a minimum but rolling. A rolling scalar has equation of state

$$p = \omega\rho, \quad \omega \simeq \frac{\frac{1}{2}\dot{\phi}^2 - V}{\frac{1}{2}\dot{\phi}^2 + V}, \quad (27.20)$$

so  $\omega = -1$  is the cosmological-constant limit and dynamical dark energy has  $\omega > -1$  and time-dependent [Pal §7.4, ll. 11388–11430]. Current bounds on  $\omega$  translate into  $c < 0.6$ ; the non-observation of primordial tensor modes gives  $c < 0.09$  in single-field slow-roll inflation, and plateau models need  $c < 0.02$ , in some tension with “ $c$  of order one” [Pal §7.4, ll. 11430–11465]. The conjecture thus makes the dark energy a falsifiable statement about  $\omega(t)$ ; it also predicts that the rolling field eventually reaches large values where the light tower of the distance conjecture matters, with a phase transition of unknown timing [Pal §7.4, ll. 11465–11480].

## 27.5 Trans-Planckian censorship

The third conjecture is not in this book’s library of sources (Palti’s review predates it); we state it and derive its two standard consequences on the page. Everything in this section is “standard; not checked against a source in this book’s library”.

**Conjecture 27.7** (Trans-Planckian Censorship Conjecture, Bedrova–Vafa; [C]). *In an expanding universe that arises from a consistent theory of quantum gravity, no mode whose physical wavelength was ever shorter than the Planck length  $\ell_p = 1/M_p$  can become larger than the Hubble radius  $H^{-1}$  and freeze out as a classical perturbation.*

**Proposition 27.8** (e-fold bound). *Consider expansion from time  $t_i$  to  $t_f$  with scale factor  $a(t)$  and Hubble rate  $H = \dot{a}/a$ . The conjecture is equivalent to*

$$\frac{a_f}{a_i} < \frac{M_p}{H_f}, \quad \text{i.e.} \quad N \equiv \ln \frac{a_f}{a_i} < \ln \frac{M_p}{H_f}, \quad (27.21)$$

where  $H_f = H(t_f)$ .

*Proof.* A comoving mode with physical wavelength  $\ell_p$  at  $t_i$  has wavelength  $\ell_p a_f/a_i$  at  $t_f$ . It has crossed the Hubble radius at  $t_f$  if  $\ell_p a_f/a_i \geq H_f^{-1}$ . Every mode that was sub-Planckian at  $t_i$  is shorter than this one at all later times, so forbidding the crossing for the mode of wavelength exactly  $\ell_p$  forbids it for all of them; that condition is  $a_f/a_i < H_f^{-1}/\ell_p = M_p/H_f$ . Requiring it for every  $t_i$  gives the conjecture.  $\square$

**Example 27.9** (Inflation at a high scale). For  $H_f = 10^{13}$  GeV and  $M_p = 2.4 \times 10^{18}$  GeV [Pal §1, ll. 606–612],  $\ln(M_p/H_f) \approx 12.4$ : fewer than thirteen e-folds are allowed, far short of the fifty to sixty usually required. Conversely, sixty e-folds at constant  $H$  need  $H < M_p e^{-60} \approx 2 \times 10^{-8}$  GeV. Which of these numbers is the relevant bound depends on how many e-folds are actually needed at a given scale, itself a function of  $H$ ; the exercise at the end of the chapter takes this further.

**Proposition 27.10** (Asymptotic steepness from TCC). *Let a canonically normalized scalar  $\phi$  roll on a positive potential  $V(\phi)$  in  $d = 4$  with  $M_p = 1$ , and suppose the trajectory reaches infinite distance ( $\phi \rightarrow \infty$ ) at late times. Then (27.21) for arbitrarily late  $t_f$  implies*

$$\frac{|V'|}{V} \geq \sqrt{\frac{2}{3}} \quad \text{asymptotically,} \quad (27.22)$$

and in  $d$  dimensions the bound is  $2/\sqrt{(d-1)(d-2)}$ .

*Proof.* The Friedmann equation is  $3H^2 = \frac{1}{2}\dot{\phi}^2 + V$ , so  $\dot{\phi}^2 \leq 6H^2$  and  $H/|\dot{\phi}| \geq 1/\sqrt{6}$ . The e-folds accumulated while  $\phi$  moves from  $\phi_i$  to  $\phi_f$  are  $N = \int H dt = \int H d\phi/|\dot{\phi}| \geq (\phi_f - \phi_i)/\sqrt{6}$ . Also  $3H_f^2 \geq V(\phi_f)$ , so  $\ln(1/H_f) \leq -\frac{1}{2} \ln V(\phi_f) + \frac{1}{2} \ln 3$ . Inserting both in (27.21):  $(\phi_f - \phi_i)/\sqrt{6} < -\frac{1}{2} \ln V(\phi_f) + \text{const}$ , i.e.  $V(\phi_f) < C e^{-(2/\sqrt{6})\phi_f}$  for all large  $\phi_f$ . Hence  $-\ln V(\phi)/\phi \geq 2/\sqrt{6} = \sqrt{2/3}$  up to a vanishing correction, so the *average* logarithmic slope of  $V$  over the asymptotic region is at least  $\sqrt{2/3}$ ; if  $|V'|/V$  has a limit, the limit is at least  $\sqrt{2/3}$ , which is what “asymptotically” means in (27.22). In  $d$  dimensions the Friedmann equation reads  $\frac{1}{2}(d-1)(d-2)H^2 = \frac{1}{2}\dot{\phi}^2 + V$  and the same steps give  $2/\sqrt{(d-1)(d-2)}$ .  $\square$

The number  $\sqrt{2/3} \approx 0.82$  is smaller than the  $\sqrt{2}$  that separates accelerating from decelerating exponential potentials (for  $V \propto e^{-\lambda\phi}$  the scaling solution has  $a \propto t^{2/\lambda^2}$ , accelerating if  $\lambda < \sqrt{2}$ ; symbolic check), so TCC leaves a window  $\sqrt{2/3} \leq \lambda < \sqrt{2}$  of asymptotically accelerating potentials. It also gives the de Sitter conjecture a definite  $c$  in the asymptotic regime—unlike the entropy argument, whose  $c$  depends on the unknown  $\gamma, \delta$ .

### The dual-scale reading of TCC

This programme’s proposal (Chapters 31 and 32) is that the quantity  $R_{\text{eff}} = R + \alpha'/R \geq 2\sqrt{\alpha'}$  acts as a minimal length: no probe resolves distances below  $2\sqrt{\alpha'}$  because a circle of radius  $R < \sqrt{\alpha'}$  is the same theory as one of radius  $\alpha'/R > \sqrt{\alpha'}$ . If one *identifies* the physical wavelength of a mode with  $R_{\text{eff}}\lambda_0$  for a comoving  $\lambda_0$ , sub-Planckian wavelengths never occur and the premise of TCC is never triggered. The inequality is Tier **[A]** mathematics; the identification is Tier **[C]**. In particular, it says nothing about the e-fold bound (27.21), which concerns the expansion history and not the spectrum of a circle.

## 27.6 What the machine has checked

A conjecture cannot be proved in Lean, and it should not be assumed as an axiom either (the libraries of this book use no axioms beyond the three standard ones). What one *can* do is encode the conjecture’s shape—its parameters, its inequality—as a structure or as a hypothesis, and prove what follows. This section reads six files that do this for the three conjectures of the chapter, in decreasing order of faithfulness, and says for each what is literally proved.

### 27.6.1 The distance conjecture over the reals

The file `StringTheoryFormalization/StringDynamics/SwamplandSafe.lean` bundles the two numbers of (27.12) with their positivity and defines the right-hand side as a function.

```
structure SDCBound where
  α : ℝ          -- decay exponent (order 1)
  α_pos : 0 < α
  m₀ : ℝ         -- initial tower mass (Planck units)
  m₀_pos : 0 < m₀

noncomputable def towerMass (bound : SDCBound) (Δ : ℝ) : ℝ :=
  bound.m₀ * Real.exp (- bound.α * Δ)

theorem sdc_tower_mass_pos (bound : SDCBound) (Δ : ℝ) :
  0 < towerMass bound Δ := by ...

theorem sdc_tower_suppression (bound : SDCBound) (Δ : ℝ) (hΔ : 0 ≤ Δ) :
  towerMass bound Δ ≤ bound.m₀ := by ...
```

### Plain-language reading

`SDCBound` is a pair of positive reals  $(\alpha, m_0)$ ; no moduli space, geodesic or tower of states is represented. `towerMass` is the function  $\Delta \mapsto m_0 e^{-\alpha\Delta}$ , taken as a *definition*—the exponential law is put in by hand, not derived. `sdc_tower_mass_pos` says this function is positive (a positive number times an exponential). `sdc_tower_suppression` says  $m_0 e^{-\alpha\Delta} \leq m_0$  for  $\Delta \geq 0$ . Proof idea: rewrite 1 as  $\exp 0$ , use monotonicity of  $\exp$  with  $-\alpha\Delta \leq 0$ , then scale by  $m_0 > 0$  (`nlinarith`). This is the first line of Proposition 27.3 with the exponential already assumed; the substantive content of the conjecture—that a tower *must* exist and *must* decay—is not stated. The file’s docstring says exactly this.

The same file contains a declaration named `de_sitter_conjecture`:

```
theorem de_sitter_conjecture (c : ℝ) (hc : 0 < c) (V : ℝ → ℝ) (gradV : ℝ → ℝ)
  (hV : ∀ φ, 0 < V φ)
  (hgrad : ∀ φ, 0 < gradV φ) :
  ∀ φ, gradV φ ≥ c * V φ → True := by ...
```

**A theorem whose conclusion is True**

The conclusion of `de_sitter_conjecture` is the proposition `True`, which holds without any hypothesis; the proof is `intros; trivial` and uses none of `hc`, `hV`, `hgrad` or the bound `gradV  $\varphi \geq c * V \varphi$` . The theorem records the *objects* a formalization of (27.14) would relate ( $c$ ,  $V$ ,  $|\nabla V|$ ) and asserts nothing about them. A kernel-checked statement is only as strong as its conclusion; always read the conclusion before the name. The file’s own docstring calls this a “vacuous placeholder”, which is the honest description. Exercise 27.9 asks you to replace it with a non-vacuous consequence.

**What a faithful formalization would need**

A statement of Conjecture 27.2 would need: a metric space  $(\mathcal{M}, d)$  (a Riemannian manifold with its geodesic distance), a family of mass functions  $m_k : \mathcal{M} \rightarrow (0, \infty)$  indexed by the tower, and the assertion  $\forall P \exists (Q_j)_{j \in \mathbb{N}} : d(P, Q_j) \rightarrow \infty \wedge \exists \alpha > 0, \sup_k m_k(Q_j) \leq \sup_k m_k(P) e^{-\alpha d(P, Q_j)}$  for large  $j$ . As a *hypothesis* on a structure this is perfectly expressible in Lean today; what cannot be done is to discharge it, since it is a conjecture about all consistent theories of gravity. The value of the exercise is that consequences proved from the hypothesis are then Tier **[A]** *relative to it*.

**27.6.2 Integer shadows**

`StringTheoryFoundation/StringTheory/Swampland.lean` works in  $\mathbb{N}$  and stores the *logarithm* of the suppression:

```
structure DistanceConjecture where
  alpha : Nat
  moduliDistance : Nat
  logMassSuppression : Nat := alpha * moduliDistance

theorem distance_conjecture_monotonicity (alpha d1 d2 : Nat)
  (hLe : d1 ≤ d2) :
  let dc1 : DistanceConjecture := { alpha := alpha, moduliDistance := d1 }
  let dc2 : DistanceConjecture := { alpha := alpha, moduliDistance := d2 }
  dc1.logMassSuppression ≤ dc2.logMassSuppression := ...
```

**Plain-language reading**

For natural numbers,  $d_1 \leq d_2$  implies  $\alpha d_1 \leq \alpha d_2$  (`Nat.mul_le_mul_left`). The field `logMassSuppression` is  $\ln(m_0/m)$  in (27.12) with the distance and the exponent restricted to integers; the exponential itself never appears. An *arithmetic shadow* of the statement “farther means lighter”. The same file has `WGCState`, the predicate `satisfiesWGC` ( $q^2 \geq m^2$  over  $\mathbb{Z}$ ) and the instance `unit_state_satisfies_wgc` for  $q = m = 1$ , decided by `decide`.

`StringTheoryFoundation/StringTheory/VafaSwampland.lean` records a value of the exponent:

```
structure SwamplandTower where
  spacetimeDim : Nat
  m0 : Nat
  distance : Nat
  alphaSquareNumerator : Nat := 1
  alphaSquareDenominator : Nat := spacetimeDim - 2

theorem vafa_sdc_4d_decay_rate :
```

```
let tower : SwamplandTower := { spacetimeDim := 4, m0 := 1, distance := 0 }
tower.alphaSquareDenominator = 2 := by ...
```

### Plain-language reading

The theorem is  $4 - 2 = 2$  in  $\mathbb{N}$ , by `rfl`. The docstring attaches to the default field the value  $\alpha^2 = 1/(d - 2)$ , which is a lower bound on the exponent proposed in the literature (a sharpening of Conjecture 27.2; standard, not checked against a source in this book’s library) and is *not* the circle’s  $\lambda_{\text{KK}}^2 = (d - 1)/(d - 2)$  from (27.6): for  $d = 4$  the two are  $1/2$  and  $3/2$ . Nothing in the file connects the number to a tower. The same file’s `vafa_wgc_weak_coupling_cutoff_monotone` proves  $g_1 \leq g_2 \Rightarrow g_1 M \leq g_2 M$  in  $\mathbb{N}$ , a shadow of the magnetic weak gravity cutoff  $\Lambda \leq g M_p$ , and `vafa_u1_gauge_consistent` checks a Boolean predicate on the pair  $(1, \text{true})$ .

### 27.6.3 A bookkeeping predicate on K3 data

`DualScaleM24Formalization/FrontierTriad/SwamplandDistance.lean` is this programme’s file. Its arithmetic is correct; its physics labels must be read with care.

```
def k3_moduli_dim : Nat := 3 * 19 + 1
theorem k3_moduli_dim_eq_58 : k3_moduli_dim = 58 := by rfl
def total_k3t2_moduli_dim : Nat := k3_moduli_dim + 22
theorem total_k3t2_moduli_dim_eq_80 : total_k3t2_moduli_dim = 80 := by rfl

def max_picard_uv : Nat := 20
def min_picard_ds : Nat := 10

structure ModuliGeometry where
  picard_number : Nat
  moduli_stabilization_positive : Bool
  tau_im_positive : Bool

def isSwamplandSafe (geom : ModuliGeometry) : Bool :=
  geom.picard_number ≤ max_picard_uv &&
  geom.picard_number ≥ min_picard_ds &&
  geom.moduli_stabilization_positive &&
  geom.tau_im_positive

theorem cooper_s10_swampland_safe : isSwamplandSafe CooperS10 = true := by decide
theorem sdc_decay_coefficient_identity :
  2 * sdc_decay_alpha_squared_num = sdc_decay_alpha_squared_den := by rfl
```

### Plain-language reading

$3 \cdot 19 + 1 = 58$  and  $58 + 22 = 80$ . The numbers are right and have a source, but not the one the docstring gives: 58 is the dimension of the moduli space of Einstein metrics on a K3 surface,  $O(\Gamma_{3,19}) \backslash O(3, 19) / (O(3) \times O(19)) \times \mathbb{R}^+$ , and  $80 = 58 + 22$  adds the  $b_2 = 22$  B-field periods to give the moduli space of the K3 sigma model [Asp §2.4 and §3, ll. 877–905, 1636–1672]; see Chapter 23. The docstring’s “22 toroidal and Wilson-line moduli of  $T^2$ ” is not what the arithmetic supports. `isSwamplandSafe` is a Boolean test  $10 \leq \rho \leq 20$  on a record; the bound  $\rho \leq 20$  is the Picard-rank bound for K3 (Chapter 20), the bound  $\rho \geq 10$  is a stipulation of the programme with no source, and the two Boolean flags are inputs. `cooper_s10_swampland_safe` evaluates the test on the record  $(19, \text{true}, \text{true})$  by `decide`; it is Tier [A] as a computation and Tier [C] as a statement about any surface. `sd_c_decay_coefficient_identity` is  $2 \cdot 1 = 2$ .

### 27.6.4 The de Sitter and TCC problem files

Lean5Corpus/Problems/Problem6\_FluxSwampland.lean models the flux potential  $V \propto N^2/\mathcal{V}^2$  of a fourfold by its numerator:

```
structure FluxVacuumState where
  flux_quanta : Nat          -- N_flux in integer units
  volume : Nat              -- V in string units
  h_flux_pos : flux_quanta ≥ 1
  h_vol_pos : volume ≥ 1

def flux_potential_numerator (s : FluxVacuumState) : Nat :=
  s.flux_quanta * s.flux_quanta
def flux_gradient_numerator (s : FluxVacuumState) : Nat :=
  2 * (s.flux_quanta * s.flux_quanta)

theorem desitter_steepness_bound (s : FluxVacuumState) :
  flux_gradient_numerator s ≥ 2 * flux_potential_numerator s := by ...
theorem no_flat_desitter_vacuum (s : FluxVacuumState) :
  ¬ (flux_gradient_numerator s = 0) := by ...
```

#### Plain-language reading

For  $N \geq 1$ :  $N^2 > 0$  (flux\_energy\_strictly\_positive),  $2N^2 \geq 2N^2$  (desitter\_steepness\_bound, closed by omega) and  $2N^2 \neq 0$  (no\_flat\_desitter\_vacuum); desitter\_swampland\_master\_contract is their conjunction. The field volume is never used, so the “potential” has no volume dependence and the “gradient” is a second copy of the numerator. The physics behind the file is sound and elementary:  $V = A\mathcal{V}^{-2}$  has  $|\partial V/\partial \ln \mathcal{V}| = 2V$ , and a pure power of the volume can have no stationary point at  $V > 0$ . But  $\ln \mathcal{V}$  is not a canonically normalized field, so the coefficient 2 is not the  $c$  of (27.14), and the Lean statement is the integer identity only.

Lean5Corpus/Problems/Problem3\_DualScaleTCC.lean encodes the dual-scale reading of TCC from the box at the end of Section 27.5:

```
def planck_length : Nat := 1
def effective_wavelength_num (R : Nat) (lambda_0 : Nat) : Nat :=
  (R * R + 1) * lambda_0

theorem wavelength_strictly_super_planckian
  (R : Nat) (lambda_0 : Nat) (h_R : R ≥ 1) (h_l : lambda_0 ≥ 1) :
  effective_wavelength_num R lambda_0 ≥ 2 := by ...

structure InflationParameters where
  planck_mass : Nat
  hubble_scale : Nat
  h_sub_planckian_hubble : hubble_scale < planck_mass
  h_hubble_pos : hubble_scale ≥ 1

theorem tcc_expansion_factor_positive (p : InflationParameters) :
  p.planck_mass / p.hubble_scale ≥ 1 := by ...
```

#### Plain-language reading

$(R^2 + 1)\lambda_0 \geq 2$  for naturals  $R, \lambda_0 \geq 1$ ; the number  $R^2 + 1$  is the numerator of  $R + 1/R$  in string units, the same inequality as self\_dual\_is\_global\_minimum in Chapter 26. sub\_planckian\_modes\_impossible restates it as  $\neg((R^2 + 1)\lambda_0 \leq 1)$ .



**tcc\_expansion\_factor\_positive**: if  $1 \leq H < M$  in  $\mathbb{N}$  then the integer quotient  $M/H$  is at least 1. **tcc\_cosmic\_protection\_contract** is the conjunction of the three. None of these mentions a scale factor, a horizon or a mode crossing it; (27.21) is not formalized. The docstring’s claim that TCC is “satisfied unconditionally” is the Tier **[C]** identification of the box, not a theorem.

| Statement                                                                            | Tier       | Evidence                                                    |
|--------------------------------------------------------------------------------------|------------|-------------------------------------------------------------|
| Reduction (27.4)–(27.5); KK masses (27.6)                                            | derived    | Pal §2.2.1; symbolic check                                  |
| Einstein-frame towers (27.8)–(27.9), $M_{\text{KK}}M_w$ const                        | <b>[L]</b> | Pal §2.2.2–2.2.3; symbolic check                            |
| Distance conjecture, refined form                                                    | <b>[C]</b> | stated in Pal §3.6                                          |
| Proposition 27.3                                                                     | derived    | two lines                                                   |
| de Sitter conjecture, refined form                                                   | <b>[C]</b> | stated in Pal §7                                            |
| Entropy argument, $c \sim 2k\gamma/(2 - \delta)$                                     | <b>[C]</b> | Pal §7.2; symbolic check                                    |
| $c < 0.6$ (dark energy), $c < 0.09$ (tensors)                                        | <b>[L]</b> | Pal §7.4                                                    |
| TCC; e-fold bound; slope $2/\sqrt{(d-1)(d-2)}$                                       | <b>[C]</b> | standard, not in library; derived                           |
| $m_0 e^{-\alpha\Delta} > 0, \leq m_0$ for $\Delta \geq 0$ (over $\mathbb{R}$ )       | <b>[A]</b> | sdc_tower_mass_pos,<br>sdc_tower_suppression                |
| $d_1 \leq d_2 \Rightarrow \alpha d_1 \leq \alpha d_2$ ; $4 - 2 = 2$ ( $\mathbb{N}$ ) | <b>[A]</b> | distance_conjecture_monotonicity,<br>vafa_sdc_4d_decay_rate |
| $3 \cdot 19 + 1 = 58$ , $58 + 22 = 80$ ; Boolean test on a record                    | <b>[A]</b> | k3_moduli_dim_eq_58,<br>cooper_s10_swampland_safe           |
| $N^2 > 0$ , $2N^2 \geq 2N^2$ , $2N^2 \neq 0$ ( $\mathbb{N}$ )                        | <b>[A]</b> | desitter_swampland_master_contract                          |
| $(R^2 + 1)\lambda_0 \geq 2$ ; $M/H \geq 1$ ( $\mathbb{N}$ )                          | <b>[A]</b> | tcc_cosmic_protection_contract                              |
| de_sitter_conjecture                                                                 | vacuous    | conclusion True                                             |
| $R_{\text{eff}} \geq 2\sqrt{\alpha'}$ censors trans-Planckian modes                  | <b>[C]</b> | none                                                        |

Table 27.1: Epistemic status of the statements of this chapter. All Lean theorems listed depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`. The identification of any Lean object with a physical system is never Tier A.

### Chapter summary

- The Swampland is the set of consistent-looking effective theories with no completion into quantum gravity; its criteria are conjectures supported by string vacua and by black-hole arguments, and they are all Tier C.
- On a circle, the Einstein-frame radion  $\varphi$  with  $2\pi R = 2\pi R_0 e^{b\varphi}$ ,  $b = -\sqrt{(d-2)/(d-1)}$ , is canonically normalized, and the KK tower obeys  $m_n = (|n|/R_0)e^{\lambda_{\text{KK}}\varphi}$  exactly, with  $\lambda_{\text{KK}} = \sqrt{(d-1)/(d-2)}$  ( $\sqrt{3}/2$  in  $d = 4$ ). String theory adds the winding tower with the opposite slope; T-duality is the  $\mathbb{Z}_2$  of the picture and makes both ends of  $\mathcal{M} = \mathbb{R}$  infinite-distance points with a light tower.
- Distance conjecture: from any point there is a point at infinite geodesic distance, and a tower with  $M \sim M_0 e^{-\alpha d}$  becomes light there. Refined: strict inequality, onset at  $d \sim M_p$ , also for fields with a potential. It bounds the field range of any effective theory.
- de Sitter conjecture:  $|\nabla V| \geq cV/M_p$ ; refined: or  $\min \nabla \nabla V \leq -c'V/M_p^2$ . Argued from



de Sitter entropy plus the light tower; predicts rolling dark energy with  $\omega > -1$  and is constrained to  $c < 0.6$  by observation.

- TCC:  $a_f/a_i < M_p/H_f$ , hence fewer than  $\ln(M_p/H_f)$  e-folds and, asymptotically,  $|V'|/V \geq \sqrt{2/3}$  in four dimensions.
- In Lean a conjecture is a structure or hypothesis, never an axiom. `SDCBound` and `towerMass` encode the form of the exponential law and the two theorems prove positivity and non-growth of the defined function. The other files are integer shadows or Boolean checks; `de_sitter_conjecture` concludes `True` and proves nothing.

## Exercises

**Exercise 27.1** (★). Derive (27.4) directly: write  $\sqrt{-G} G^{\mu\nu} R_{\mu\nu}$  for the ansatz (27.1) keeping only the terms with no derivatives of  $\varphi$ , integrate over  $y$ , and demand that the coefficient of  $\sqrt{-g}R_d$  be independent of  $\varphi$ .

**Exercise 27.2** (★). From (27.8) show that  $M_{n,0}M_{0,w}$  is independent of  $R$  and that  $M_{0,w}(R) = M_{w,0}(\alpha'_0/R)$  up to the constant  $2\pi$  factors. Which step in the derivation uses that the  $d$ -dimensional dilaton, not the  $D$ -dimensional one, is held fixed?

**Exercise 27.3** (★★). Compute  $\lambda_{\text{KK}}$  for  $d = 5, 6, 10$  and  $26$  and show that  $\lambda_{\text{KK}} \rightarrow 1$  as  $d \rightarrow \infty$ . Then redo the reduction on a  $k$ -torus with all radii equal,  $y^m \sim y^m + 2\pi R_0$ , and show that the KK exponent becomes  $\sqrt{(d+k-2)/(k(d-2))}$ ; check  $k = 1$ .

**Exercise 27.4** (★★). A single axion with constant decay constant  $f$  has  $\mathcal{M} = S^1$  of circumference  $2\pi f$ . Show that no tower can satisfy (27.13) on it, and that if  $f$  is promoted to a field  $u$  with kinetic term  $f(u)^2(\partial u)^2$  and  $f(u)u \leq M_p$  (the axionic weak gravity constraint), the distance along  $u$  grows at most like  $M_p \ln u$  [Pal §3.6.1, ll. 5212–5250]. Why is a logarithmic distance exactly what an exponential tower needs?

**Exercise 27.5** (★★). In the entropy argument, take  $S_{\text{tower}} \sim N$  ( $\gamma = 1, \delta = 0$ ) and the circle's KK tower with cutoff  $\Lambda$  at large radius. Show  $N \sim \Lambda R_0 e^{\lambda_{\text{KK}}|\varphi|}$  and derive  $c \sim \lambda_{\text{KK}}$ . Then repeat with  $\delta = 1$  and comment on how sensitive  $c$  is to the unknown exponents.

**Exercise 27.6** (★★). For  $V = V_0 e^{-\lambda\phi}$  in  $d = 4$ ,  $M_p = 1$ , verify that  $\phi = (2/\lambda) \ln t + \phi_0$ ,  $a \propto t^p$  solves the Friedmann and Klein–Gordon equations with  $p = 2/\lambda^2$  and  $V_0 e^{-\lambda\phi_0} = (12 - 2\lambda^2)/\lambda^4$ . For which  $\lambda$  is the expansion accelerating? Check that (27.21) with  $t_i$  fixed and  $t_f \rightarrow \infty$  is violated for  $p > 1$ , and compare the resulting bound on  $\lambda$  with Proposition 27.10.

**Exercise 27.7** (★★). Suppose inflation at a constant Hubble rate  $H$  must produce enough e-folds for the largest observed scale,  $N_{\text{req}}(H) = N_0 + \frac{1}{2} \ln(H/H_0)$  with constants  $N_0, H_0$  (a simple model of the dependence of the required e-folds on the reheating scale). Combine with (27.21) to obtain an upper bound on  $H$ , and evaluate it for  $N_0 = 60$ ,  $H_0 = 10^{13}$  GeV.

**Exercise 27.8** (★★, Lean). Using `SDCBound` and `towerMass`, state and prove monotonicity: `towerMass bound  $\Delta' \leq \text{towerMass bound } \Delta$  whenever  $\Delta \leq \Delta'$` . Useful lemmas: `Real.exp_le_exp` for the exponential and `mul_le_mul_of_nonneg_left` for the prefactor, whose non-negativity comes from the field `m0_pos`. Explain why `sd_c_tower_suppression` is the special case  $\Delta = 0$ .

**Exercise 27.9** (★★, Lean). Replace the conclusion `True` of `de_sitter_conjecture` by something that uses the hypotheses: prove that under `hgrad  $\varphi \geq c * V \varphi$  and  $0 < c$  one has  $V$`

$\varphi \leq \text{grad}V \ \varphi / c$  (use `le_div_iff₀` or `div_le_iff₀` from Mathlib, or `nlinarith`). Then state, without proving, the real de Sitter conjecture for a single field as a `def` of type `Prop` with  $V : \mathbb{R} \rightarrow \mathbb{R}$  and `deriv V`.

**Exercise 27.10** (\*\*\*). Prove the converse direction of Proposition 27.3 in the following form: if  $m(\phi) \leq m(0)e^{-\alpha\phi}$  for all  $\phi \geq 0$  and  $m$  is positive and smooth, then  $\liminf_{\phi \rightarrow \infty} (-m'/m) \geq \alpha$  need *not* hold pointwise (construct a counter-example with oscillations), but its running average  $\frac{1}{\phi} \int_0^\phi (-m'/m)$  has  $\liminf \geq \alpha$ . Relate this to the difference between the original and refined distance conjectures.

## Further reading

- Definition of the Swampland, the scale  $\Lambda_{\text{Swamp}}$  and the three kinds of evidence: [Pal §1, ll. 368–530]. Conventions: [Pal §1, ll. 580–612].
- The circle, in both frames, and the first statement of the distance conjecture: [Pal §2.2, ll. 1716–2200]; the string-frame mass formula [Tong §8.1, ll. 11371–11475].
- Distance conjecture, refined form, relation to the weak gravity conjecture and to spatial field variations: [Pal §3.6, ll. 5049–5340].
- de Sitter conjecture, refinement, de Sitter entropy, the entropy argument and the cosmological bounds: [Pal §7, ll. 10475–10880]; [Pal §7.4, ll. 11388–11480].
- Moduli of K3 and of the K3 sigma model, for the numbers 58 and 80: [Asp §2.4, ll. 877–905]; [Asp §3, ll. 1636–1672].
- Lean sources, each with a docstring that states what is and is not proved:
  - `StringTheoryFormalization/StringDynamics/SwamplandSafe.lean`
  - `StringTheoryFoundation/StringTheory/Swampland.lean`
  - `StringTheoryFoundation/StringTheory/VafaSwampland.lean`
  - `DualScaleM24Formalization/FrontierTriad/SwamplandDistance.lean`
  - `Lean5Corpus/Problems/Problem6_FluxSwampland.lean`
  - `Lean5Corpus/Problems/Problem3_DualScaleTCC.lean`

## Part VI

# Mathieu Moonshine



## Chapter 28

# The Elliptic Genus of K3 and $N = 4$ Characters

How much of a quantum theory can be computed exactly? For a supersymmetric theory the answer is “more than one would expect”, because a partition function with the sign  $(-1)^F$  inserted,  $F$  the fermion number, counts states with signs that cancel in pairs. What survives the cancellation is a robust quantity: it does not change when the couplings change, and it can be evaluated in whatever limit is convenient. The simplest such quantity is the Witten index. This chapter is about its refinement for a two-dimensional conformal field theory, the *elliptic genus*, evaluated for the sigma model whose target space is the K3 surface of Chapter 19. The result is remarkably rigid: the elliptic genus of K3 is a weak Jacobi form of weight 0 and index 1, a space of functions that is one-dimensional, so the whole function is determined by a single number, the Euler number 24. Rigid does not mean empty. When the same function is expanded in characters of the  $N = 4$  superconformal algebra at central charge  $c = 6$ , the symmetry algebra of any K3 sigma model, the multiplicities are 20,  $-2$  and then  $2A_n$  with

$$A_1, A_2, A_3, A_4, A_5, \dots = 45, 231, 770, 2277, 5796, \dots \quad (28.1)$$

The first five of these are dimensions of irreducible representations of the sporadic Mathieu group  $M_{24}$ , an observation of Eguchi, Ooguri and Tachikawa [EOT §1, ll. 197–233] that opens the “Mathieu moonshine” story of Chapters 29 and 30. Before we can appreciate the observation we need to know precisely what is being counted, and that is the purpose of this chapter. The chapter answers three questions. What is an elliptic genus and why is it independent of the moduli of K3? Why is it fixed by the number 24? And what are the  $N = 4$  characters in which it is expanded, and how are the numbers in (28.1) extracted?

Throughout,  $q = e^{2\pi i\tau}$  with  $\tau$  in the upper half-plane and  $y = e^{2\pi iz}$  with  $z \in \mathbb{C}$ . We use the conventions of Chapter 4 for the Virasoro algebra,  $L_0$ , central charge  $c$ , and the  $q^{L_0 - c/24}$  form of a partition function.

## 28.1 Counting with a sign

### 28.1.1 The Witten index

Consider a quantum theory with a Hamiltonian  $H$ , a fermion-number operator  $(-1)^F$  that commutes with  $H$ , and a supercharge  $Q$  with

$$Q^2 = H, \quad (-1)^F Q = -Q (-1)^F. \quad (28.2)$$

The second relation says that  $Q$  maps bosons  $((-1)^F = +1)$  to fermions and back. The *Witten index* is

$$I = \text{tr} (-1)^F e^{-\beta H}. \quad (28.3)$$

The trace is over the whole Hilbert space, but almost everything cancels. Take an eigenstate  $|\psi\rangle$  of  $H$  with eigenvalue  $E > 0$ . Then  $Q|\psi\rangle$  is a nonzero state (because  $\langle\psi|Q^2|\psi\rangle = E \neq 0$ ) with the same energy and the opposite sign of  $(-1)^F$ . So the states of positive energy come in boson–fermion pairs, and each pair contributes  $(+1 - 1)e^{-\beta E} = 0$  to (28.3). Only the states with  $E = 0$ , which are annihilated by  $Q$  and need not be paired, contribute, and they contribute  $\pm 1$  independently of  $\beta$ :

$$I = n_B^{(E=0)} - n_F^{(E=0)}. \quad (28.4)$$

This has two consequences. First,  $I$  is an integer. Second, if a parameter of the theory is varied continuously, states can leave or reach  $E = 0$  only in boson–fermion pairs, so  $I$  does not change. *The index is a topological invariant of the theory.*

#### Why a sign makes a quantity computable

An ordinary partition function  $\text{tr} e^{-\beta H}$  knows every energy level and is as hard to compute as the theory itself. Inserting  $(-1)^F$  makes the excited states invisible, and what is left can be computed at any convenient value of the parameters: at large volume, where the sigma model is classical geometry, or at a special symmetric point, where it is a solvable conformal field theory. The elliptic genus keeps this property while retaining much more information than a single integer, because it inserts  $(-1)^F$  only for one chirality and keeps track of the other quantum numbers by fugacities.

### 28.1.2 The refinement: one chirality at a time

Now let the theory be a two-dimensional conformal field theory with  $N = 2$  superconformal symmetry in both the left- and the right-moving sector (a “(2, 2) theory”). Each sector has its own Virasoro generators  $L_n, \bar{L}_n$ , its own  $U(1)_R$  current with zero mode  $J_0, \bar{J}_0$ , and its own fermion number. The theory is placed on a torus with modular parameter  $\tau$ , in the sector in which the fermions are periodic in both directions on the spatial circle (the Ramond–Ramond, or RR, sector). Following [GHV §2, eq. (2.1), ll. 104–113], the *elliptic genus* is

$$\phi(\tau, z) = \text{tr}_{\mathcal{H}_{RR}} \left( q^{L_0 - \frac{c}{24}} e^{2\pi i z J_0} (-1)^F \bar{q}^{\bar{L}_0 - \frac{\bar{c}}{24}} (-1)^{\bar{F}} \right). \quad (28.5)$$

Compare with (28.3): in a conformal field theory on a circle the Hamiltonian is  $L_0 + \bar{L}_0 - \frac{c+\bar{c}}{24}$ , and  $e^{-\beta H}$  has become  $q^{L_0 - c/24} \bar{q}^{\bar{L}_0 - \bar{c}/24}$ . The novelty is the factor  $e^{2\pi i z J_0}$ , which keeps track of the left-moving  $U(1)_R$  charge, and the fact that  $q^{L_0 - c/24}$  is not accompanied by a compensating right-moving quantity: the pairing argument is applied to the right-moving supercharge  $\bar{G}_0$  only. In the Ramond sector  $\bar{G}_0^2 = \bar{L}_0 - \bar{c}/24$ , and  $\bar{G}_0$  commutes with  $L_0, J_0$  and  $(-1)^F$ . The argument of the previous subsection, applied to  $\bar{G}_0$ , therefore shows that only states with  $\bar{L}_0 = \bar{c}/24$  (*right-moving Ramond ground states*) contribute, so (28.5) is independent of  $\bar{q}$ : it is a holomorphic function of  $\tau$  [GHV §2, ll. 115–118]. On the left-moving side no cancellation is forced, and the elliptic genus is a genuine  $q$ -series

$$\phi(\tau, z) = \sum_{n \geq 0, \ell \in \mathbb{Z}} c(n, \ell) q^n y^\ell, \quad (28.6)$$

whose coefficients count, with sign, right-moving ground states of left-moving level  $n$  and left charge  $\ell$  [GHV §2, eq. (2.4), ll. 145–153].

Three specializations are worth memorizing. Setting  $z = 0$  removes the charge grading and leaves  $\text{tr}(-1)^{F+\bar{F}} q^{L_0-c/24}$  in which now the left-moving pairing also operates, so only the  $q^0$  term survives:

$$\phi(\tau, 0) = \text{Witten index of the RR sector} = \chi(M) \quad (28.7)$$

for a sigma model with target  $M$ , because the RR ground states are in one-to-one correspondence with the cohomology of  $M$  and the sign  $(-1)^{F+\bar{F}}$  is  $(-1)^{p+q}$  on  $H^{p,q}(M)$ . Setting  $z = \frac{1}{2}$  (so  $y = -1$ ) weights the charges by  $(-1)^\ell$ , and the  $q^0$  term becomes  $\sum (-1)^{p+q} (-1)^{p-1} h^{p,q} = -\sum_q (-1)^q \sum_p h^{p,q}$ , which for a surface is minus the Hirzebruch signature  $\sigma(M) = \sum_{p,q} (-1)^q h^{p,q}$ ; for K3, with  $b^+ = 3$  and  $b^- = 19$  (Chapter 20),  $\sigma = -16$  and the  $q^0$  term is  $+16$ . Finally the  $q^0$  term at general  $y$  is the  $\chi_y$  genus  $\sum_{p,q} (-1)^{p+q} h^{p,q} y^{p-c/6}$ ; the shift by  $c/6$  is the charge of the Ramond ground state of lowest  $J_0$  and will be explained in Section 28.2. [EOT §1, ll. 74–77] record the K3 values:  $\phi(\tau, 0) = 24$  and  $\phi(\tau, \frac{1}{2}) = 16 + O(q)$ , the Euler number and the signature of K3 from Chapter 19.

## 28.2 The K3 sigma model and its $N = 4$ algebra

### 28.2.1 Central charge

The sigma model with target K3 has four real bosonic fields (the coordinates of the four-real-dimensional surface) and, in the supersymmetric model, four real fermionic partners of each chirality. A free boson contributes  $c = 1$  and a free Majorana fermion  $c = \frac{1}{2}$  to the central charge (Chapter 4), so

$$c = \bar{c} = 4 \cdot 1 + 4 \cdot \frac{1}{2} = 6. \quad (28.8)$$

The sigma model on K3 is of course not free: the metric is curved. But the central charge is protected. A Ricci-flat target gives a conformal theory (Chapter 6), and  $c$  can be read off at large volume, where the curvature is negligible and the theory is a free theory of  $4 + 4$  fields locally. The number 6 is what enters the index of the Jacobi form below, and it is the number the Lean declaration `central_charge_k3_eq_six` records (Section 28.6).

### 28.2.2 From $N = 2$ to $N = 4$

A Kähler target gives an  $N = 2$  superconformal algebra on each side: the stress tensor  $T$ , two supercurrents  $G^\pm$  of charge  $\pm 1$ , and a  $U(1)_R$  current  $J$ . Unitary representations obey the bound

$$h \geq \frac{|q|}{2} \quad (28.9)$$

between the conformal weight  $h$  and the  $U(1)$  charge  $q$  of a primary state in the NS sector (standard; not checked against a source in this book's library), and the states that saturate it with  $q \geq 0$  are the *chiral primaries*. Under the identification of the sigma model with geometry, chiral primaries of charge  $q = p$  correspond to the holomorphic  $p$ -forms,  $H^{p,0}(M)$ . For K3,  $h^{0,0} = h^{2,0} = 1$  and  $h^{1,0} = 0$  (Chapter 19), so there are exactly two: the identity and the state  $\Omega$  of charge 2 and weight 1, created by the holomorphic two-form. This count,  $1+0+1 = 2$ , is the content of the Lean declaration `k3_chiral_primary_total` (Section 28.6).

The existence of  $\Omega$  with  $h = 1$ ,  $q = 2$ , together with its conjugate, is exactly what enlarges the symmetry. A hyperkähler target has not one but a two-sphere of complex structures, and the sigma model has an  $SU(2)$  of  $R$ -currents of which  $J$  is the Cartan generator; the two extra currents are the spectral-flow images of the identity, of weight 1 and charge  $\pm 2$ . The

resulting algebra is the  $N = 4$  *superconformal algebra* with four supercurrents and an affine  $SU(2)$  at level  $k = c/6 = 1$  [EOT §1, ll. 43–53]. Its unitary representations are labelled by  $h$  and by the  $SU(2)$  spin  $\ell$  of the highest-weight state; at level  $k = 1$  the possible spins are  $\ell = 0, \frac{1}{2}$ . The  $U(1)$  charge of the  $N = 2$  subalgebra is  $J_0 = 2J_0^3$ , which is why [EOT §1, eq. (1.1), ll. 43–46] write the fugacity as  $e^{4\pi iz J_0^3}$  where (28.5) has  $e^{2\pi iz J_0}$ ; the two are the same operator.

### 28.2.3 Ramond ground states and spectral flow

In the Ramond sector the supercharge  $G_0$  satisfies  $G_0^2 = L_0 - c/24$ , so Ramond ground states have  $h = c/24 = \frac{1}{4}$ . The NS and R sectors are related by *spectral flow*, the one-parameter family of automorphisms

$$L_0 \mapsto L_0 + \eta J_0 + \eta^2 \frac{c}{6}, \quad J_0 \mapsto J_0 + \eta \frac{c}{3} \quad (28.10)$$

[GHV §2, eq. (2.7), ll. 168–187];  $\eta = \pm \frac{1}{2}$  maps NS to R. Flow the NS vacuum ( $h = 0, q = 0$ ) by  $\eta = -\frac{1}{2}$ : it acquires  $h = \frac{1}{4} \cdot \frac{c}{6} = \frac{1}{4}$  and  $J_0 = -\frac{c}{6} = -1$ , a Ramond ground state of charge  $-1$ . Flow  $\Omega$  ( $h = 1, q = 2$ ) the same way:  $h = 1 - 1 + \frac{1}{4} = \frac{1}{4}$ ,  $J_0 = 2 - 1 = 1$ . The chiral primaries of charge  $p$  become Ramond ground states of charge  $p - c/6 = p - 1$ , which explains the shift in the  $\chi_y$  genus stated after (28.7). The 24 RR ground states of K3 are graded by left charge as

$$\phi_{K3}(\tau, z)|_{q^0} = 2y^{-1} + 20 + 2y, \quad (28.11)$$

two states each of charge  $\mp 1$  (from  $h^{0,0} + h^{0,2}$  and  $h^{2,0} + h^{2,2}$ , all with sign  $+$ ) and twenty of charge 0 (from  $h^{1,1} = 20$ , sign  $(-1)^{1+1} = +$ ;  $h^{1,0} = h^{1,2} = 0$  would have contributed with a minus sign). We will recover (28.11) from the theta-function formula in a moment.

## 28.3 The elliptic genus as a weak Jacobi form

### 28.3.1 Jacobi forms

The elliptic genus is a torus partition function, so modular transformations of  $\tau$  act on it; and spectral flow by an integer amount,  $z \mapsto z + \ell\tau + \ell'$ , is a symmetry of the spectrum. Together with unitarity these facts imply [GHV §2, ll. 118–122] that  $\phi$  is a *weak Jacobi form* of weight  $w = 0$  and index

$$m = \frac{c}{6}, \quad (28.12)$$

i.e. a holomorphic function on  $\mathbb{H} \times \mathbb{C}$  satisfying

$$\phi\left(\frac{a\tau + b}{c\tau + d}, \frac{z}{c\tau + d}\right) = (c\tau + d)^w e^{2\pi i m \frac{cz^2}{c\tau + d}} \phi(\tau, z), \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in SL(2, \mathbb{Z}), \quad (28.13)$$

$$\phi(\tau, z + \ell\tau + \ell') = e^{-2\pi i m(\ell^2\tau + 2\ell z)} \phi(\tau, z), \quad \ell, \ell' \in \mathbb{Z}, \quad (28.14)$$

with a Fourier expansion (28.6) in which  $n \geq 0$  (“weak”) and  $c(n, \ell) = (-1)^w c(n, -\ell)$  [GHV §2, eqs. (2.2)–(2.4), ll. 122–153]. (In (28.13) the letter  $c$  is a matrix entry, not the central charge.) Where does the index come from? Apply (28.14) with  $\ell = 1, \ell' = 0$  to a single term  $q^n y^{\ell_0}$  of the expansion:  $y \mapsto qy$  turns it into  $q^{n+\ell_0} y^{\ell_0}$ , and the prefactor  $e^{-2\pi i m(\tau + 2z)} = q^{-m} y^{-2m}$ . So the coefficient of  $q^n y^{\ell_0}$  must equal the coefficient of  $q^{n+\ell_0+m} y^{\ell_0+2m}$ ; with  $m = c/6$  this is precisely the statement that the spectrum is invariant under the spectral flow (28.10) with



$\eta = 1$ . Iterating, one finds that  $c(n, \ell)$  depends on  $(n, \ell)$  only through the discriminant  $4mn - \ell^2$  and  $\ell \bmod 2m$ . For  $m = 1$  the second datum is redundant and

$$c(n, \ell) = C(4n - \ell^2) \quad \text{for a weak Jacobi form of index 1.} \quad (28.15)$$

We will see this structure explicitly in Table 28.1.

For K3,  $m = 1$ . The crucial fact, quoted in [EOT §1, ll. 69–73] and standard in the theory of Jacobi forms, is that the space of weak Jacobi forms of weight 0 and index 1 is one-dimensional [L]. Its generator is usually called  $\phi_{0,1}$ , normalized by  $\phi_{0,1}(\tau, 0) = 12$ . Since  $\phi_{K3}(\tau, 0) = \chi(K3) = 24$ ,

$$\phi_{K3} = 2\phi_{0,1}. \quad (28.16)$$

This is the sense in which the elliptic genus of K3 is fixed by the single number 24: any sigma model on any K3 surface, at any point of the moduli space of Chapter 23, has the same elliptic genus. [GHV §2, eq. (2.6), ll. 161–167] work with the rescaled  $\phi_{1A} = \frac{1}{2}\phi_{K3} = \phi_{0,1}$ ; we keep EOT’s normalization  $\phi_{K3}$  and will return to the factor 2 in Section 28.5.5.

### 28.3.2 Theta functions and the formula for K3

We use the Jacobi theta functions in the product form of [GHV App. A, eq. (A.1), ll. 1453–1490], together with the Dedekind function  $\eta$ :

$$\eta(\tau) = q^{1/24} \prod_{n \geq 1} (1 - q^n), \quad (28.17)$$

$$\vartheta_1(\tau, z) = -i q^{1/8} y^{1/2} \prod_{n \geq 1} (1 - q^n)(1 - yq^n)(1 - y^{-1}q^{n-1}), \quad (28.18)$$

$$\vartheta_2(\tau, z) = q^{1/8} (y^{1/2} + y^{-1/2}) \prod_{n \geq 1} (1 - q^n)(1 + yq^n)(1 + y^{-1}q^n), \quad (28.19)$$

$$\vartheta_3(\tau, z) = \prod_{n \geq 1} (1 - q^n)(1 + yq^{n-1/2})(1 + y^{-1}q^{n-1/2}), \quad (28.20)$$

$$\vartheta_4(\tau, z) = \prod_{n \geq 1} (1 - q^n)(1 - yq^{n-1/2})(1 - y^{-1}q^{n-1/2}). \quad (28.21)$$

(In (28.19) we have written  $2 \cos \pi z = y^{1/2} + y^{-1/2}$ .) Each is the partition function of a single complex fermion with a specified spin structure on the torus:  $\vartheta_1$  is the RR-sector trace with  $(-1)^F$  inserted, which is why it vanishes at  $z = 0$  where the two fermion zero modes give  $y^{1/2} - y^{-1/2} \rightarrow 0$ ;  $\vartheta_2$  is the R-sector trace without the sign;  $\vartheta_3, \vartheta_4$  are the NS-sector traces without and with the sign. The elliptic genus of K3 was computed by Eguchi, Ooguri, Taormina and Yang (reference [2] of EOT, ll. 1013–1014), and reads [EOT §1, eq. (1.2), ll. 56–68]

$$\phi_{K3}(\tau, z) = 8 \left[ \left( \frac{\vartheta_2(\tau, z)}{\vartheta_2(\tau, 0)} \right)^2 + \left( \frac{\vartheta_3(\tau, z)}{\vartheta_3(\tau, 0)} \right)^2 + \left( \frac{\vartheta_4(\tau, z)}{\vartheta_4(\tau, 0)} \right)^2 \right]. \quad (28.22)$$

The structure is transparent at the orbifold point  $T^4/\mathbb{Z}_2$  of Chapter 21: the three terms are the three nontrivial sectors of the  $\mathbb{Z}_2$  orbifold (untwisted with the  $\mathbb{Z}_2$  inserted, twisted, and twisted with the  $\mathbb{Z}_2$  inserted), each a ratio of fermion to boson determinants, and 8 is the normalization forced by the Witten index. The untwisted sector without insertion is the elliptic genus of  $T^4$ , which vanishes because of the fermion zero modes. (This orbifold reading is standard; it is not checked against a source in this book’s library, and we use it only as motivation.) Since (28.22) is a weak Jacobi form of weight 0 and index 1 (each ratio is: the theta functions all have index  $\frac{1}{2}$  and weight  $\frac{1}{2}$ , and the three are permuted by  $SL(2, \mathbb{Z})$ ), and since the space is one-dimensional, the formula is correct for every K3 sigma model once it is correct for the orbifold.

### 28.3.3 Worked example: the first coefficients

Let us expand (28.22) to first order in  $q$ . From the products,

$$\begin{aligned}\frac{\vartheta_2(\tau, z)}{\vartheta_2(\tau, 0)} &= \frac{y^{1/2} + y^{-1/2}}{2} (1 + (y + y^{-1} - 2)q + O(q^2)), \\ \frac{\vartheta_3(\tau, z)}{\vartheta_3(\tau, 0)} &= 1 + (y + y^{-1} - 2)q^{1/2} + O(q), \quad \frac{\vartheta_4(\tau, z)}{\vartheta_4(\tau, 0)} = 1 - (y + y^{-1} - 2)q^{1/2} + O(q).\end{aligned}$$

Squaring and adding, the half-integer powers of  $q$  cancel between  $\vartheta_3$  and  $\vartheta_4$  (as they must, since  $\phi$  has a  $q$ -expansion in integer powers), and one finds

$$\begin{aligned}\phi_{K3}(\tau, z) &= 2y^{-1} + 20 + 2y + q(20y^2 - 128y + 216 - 128y^{-1} + 20y^{-2}) \\ &\quad + q^2(2y^3 + 216y^2 - 1026y + 1616 - 1026y^{-1} + 216y^{-2} + 2y^{-3}) + O(q^3),\end{aligned}\tag{28.23}$$

in agreement with [GHV §2, eq. (2.5), ll. 154–160] for the terms they display (the  $q^2$  term is a symbolic check of this book, not taken from a source). The  $q^0$  term is (28.11). Setting  $y = 1$  gives  $24 + q \cdot 0 + q^2 \cdot 0 + \dots$ , and indeed (28.22) at  $z = 0$  is  $8(1 + 1 + 1) = 24$  exactly. Setting  $y = -1$  gives  $16 + 512q + 4096q^2 + \dots$ , whose constant term is  $|\sigma(K3)| = 16$  as discussed after (28.7); the higher terms are not zero, so “ $\phi(\tau, \frac{1}{2}) = \text{signature}$ ” holds only for the  $q^0$  term, as [EOT §1, l. 76] state with their  $O(q)$ .

Table 28.1: Coefficients  $c(n, \ell)$  of  $\phi_{K3}$  for  $n \leq 2$ , arranged by the discriminant  $4n - \ell^2$ . Each row is constant, as (28.15) demands; the pair  $(n, \ell)$  and  $(n, -\ell)$  always agree because the weight is even.

| $4n - \ell^2$ | $c(n, \ell)$ | occurrences $(n, \pm\ell)$                                            |
|---------------|--------------|-----------------------------------------------------------------------|
| −1            | 2            | $(0, \pm 1), (2, \pm 3)$                                              |
| 0             | 20           | $(0, 0), (1, \pm 2)$                                                  |
| 3             | −128         | $(1, \pm 1)$                                                          |
| 4             | 216          | $(1, 0), (2, \pm 2)$                                                  |
| 7             | −1026        | $(2, \pm 1)$                                                          |
| 8             | 1616         | $(2, 0)$                                                              |
| $< -1$        | 0            | e.g. $(1, \pm 3)$ with $4 - 9 = -5$ , $(2, \pm 4)$ with $8 - 16 = -8$ |

Table 28.1 exhibits the discriminant structure (28.15):  $c(1, \pm 2) = c(0, 0) = 20$ ,  $c(2, \pm 2) = c(1, 0) = 216$ ,  $c(2, \pm 3) = c(0, \pm 1) = 2$ . It also shows that  $c(n, \ell) = 0$  whenever  $4n - \ell^2 < -1$ , so at level  $n$  the charges satisfy  $|\ell| \leq \sqrt{4n + 1}$ ; this is the Jacobi-form version of the unitarity bound (28.9). Figure 28.1 draws the support.

#### Two normalizations of the same function

$\phi_{K3}$  and  $\phi_{0,1} = \frac{1}{2}\phi_{K3}$  both appear in the literature. EOT use  $\phi_{K3}$  (Witten index 24, coefficients  $A_n = 45, 231, \dots$  with an explicit factor 2 in front of the massive characters); GHV use  $\phi_{1A} = \phi_{0,1}$  (Witten index 12) and absorb the factor into their  $A_n = 90, 462, \dots$  [GHV §3, ll. 578–580]. Neither is wrong, but a table of “the” coefficients is meaningless without saying which normalization it uses. This book follows the convention of Chapter 1:  $A_n$  are EOT’s numbers and the multiplicities in the elliptic genus of K3 are  $2A_n$ .

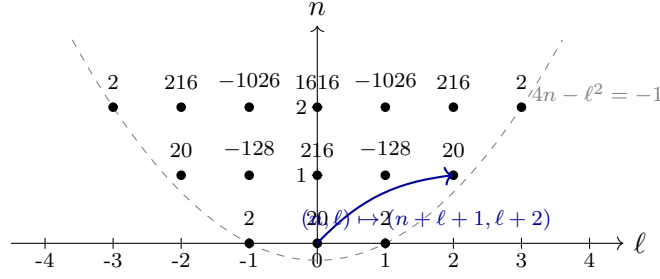


Figure 28.1: Support of the Fourier coefficients  $c(n, \ell)$  of  $\phi_{K3}$  in the  $(\ell, n)$  plane, with the values from (28.23). The coefficients are constant along the parabolas  $4n - \ell^2 = \text{const}$  and vanish below the dashed one. The arrow is one unit of spectral flow (28.14) with  $\ell = 1$ , which maps  $c(0, 0) = 20$  to  $c(1, 2) = 20$ .

## 28.4 $N = 4$ characters at $c = 6$

The elliptic genus is a trace over a representation of the left-moving  $N = 4$  algebra (tensored with right-moving ground states), so it can be written as a sum of the *characters* of the irreducible unitary representations,

$$\text{ch}_{h, \ell}^{\tilde{R}}(\tau, z) = \text{tr}_{\mathcal{H}_{h, \ell}}(-1)^F q^{L_0 - c/24} y^{J_0}, \quad (28.24)$$

where the tilde on  $R$  is the traditional reminder that  $(-1)^F$  is inserted. At  $c = 6$  there are two kinds of representation in the Ramond sector.

### 28.4.1 Massless (BPS) representations

A representation whose highest-weight state has  $h = \frac{1}{4} = c/24$  is annihilated by  $G_0$  and is called *massless* or *BPS*; the highest-weight state is a Ramond ground state. Its  $SU(2)$  spin is  $\ell = 0$  or  $\ell = \frac{1}{2}$ . For  $\ell = 0$  the character is [EOT §1, eqs. (1.3)–(1.4), ll. 82–115]

$$\text{ch}_{h=\frac{1}{4}, \ell=0}^{\tilde{R}}(\tau, z) = \frac{\vartheta_1(\tau, z)^2}{\eta(\tau)^3} \mu(\tau, z), \quad (28.25)$$

$$\mu(\tau, z) = \frac{-i e^{\pi i z}}{\vartheta_1(\tau, z)} \sum_{n \in \mathbb{Z}} \frac{(-1)^n q^{n(n+1)/2} e^{2\pi i n z}}{1 - q^n e^{2\pi i z}}. \quad (28.26)$$

The function  $\mu$  is a *Lerch sum* (or Appell–Lerch sum); it is not a Jacobi form but a *mock* modular object, whose transformation law involves a non-holomorphic correction [EOT §1, ll. 224–226]. This is the technical reason the massless character, unlike the massive one below, has a complicated modular behaviour.

Let us extract the leading terms of (28.25), both to understand the formula and because we will need them. From (28.18) and (28.17),

$$\frac{\vartheta_1(\tau, z)^2}{\eta(\tau)^3} = q^{1/8} \left[ -(y - 2 + y^{-1}) + q(2y^2 - 5y + 6 - 5y^{-1} + 2y^{-2}) + O(q^2) \right], \quad (28.27)$$

the sign coming from  $(-i)^2$  and the leading bracket from  $-y(1 - y^{-1})^2 = -(y - 2 + y^{-1})$ . In the Lerch sum only the  $n = 0$  term,  $1/(1 - y)$ , survives at order  $q^0$  (the term  $n = -1$  has denominator  $1 - q^{-1}y$ , which is  $O(q)$  after expansion), so with  $\vartheta_1 \approx -iq^{1/8}(y^{1/2} - y^{-1/2})$ ,

$$\mu(\tau, z) = \frac{-iy^{1/2}}{-iq^{1/8}y^{1/2}(1 - y^{-1})} \cdot \frac{1}{1 - y} + O(q^{7/8}) = -\frac{q^{-1/8}}{y - 2 + y^{-1}} + O(q^{7/8}). \quad (28.28)$$

Multiplying, the poles cancel and

$$\text{ch}_{\frac{1}{4},0}^{\tilde{R}}(\tau, z) = 1 + q(y^2 - 2y + 2 - 2y^{-1} + y^{-2}) + q^2(2y^2 - 6y + 8 - 6y^{-1} + 2y^{-2}) + O(q^3), \quad (28.29)$$

where the  $q^1$  and  $q^2$  terms are a symbolic check of this book. The leading 1 is the single ground state of  $J_0 = 0$  (spin  $\ell = 0$ ), with  $h - c/24 = 0$ . The Witten index of the representation is [EOT §1, eq. (1.5), ll. 114]

$$\text{ch}_{\frac{1}{4},0}^{\tilde{R}}(\tau, 0) = 1, \quad (28.30)$$

as it must be: at  $y = 1$  every  $q^n$  bracket in (28.29) vanishes ( $1 - 2 + 2 - 2 + 1 = 0$ ,  $2 - 6 + 8 - 6 + 2 = 0$ ), the excited states pairing off as in Section 28.1.

For  $\ell = \frac{1}{2}$  we use the relation [EOT §1, eq. (1.10), ll. 160–169]

$$q^{-1/8} \frac{\vartheta_1(\tau, z)^2}{\eta(\tau)^3} = 2 \text{ch}_{\frac{1}{4},0}^{\tilde{R}}(\tau, z) + \text{ch}_{\frac{1}{4},\frac{1}{2}}^{\tilde{R}}(\tau, z), \quad (28.31)$$

which expresses the fact that the massive representation of Section 28.4.2 becomes reducible at the unitarity bound  $h = \frac{1}{4}$  and splits into one  $\ell = \frac{1}{2}$  and two  $\ell = 0$  massless representations. Read as a definition of the  $\ell = \frac{1}{2}$  character, (28.31) with (28.27) and (28.29) gives

$$\text{ch}_{\frac{1}{4},\frac{1}{2}}^{\tilde{R}}(\tau, z) = -(y + y^{-1}) + q(-y + 2 - y^{-1}) + O(q^2), \quad \text{ch}_{\frac{1}{4},\frac{1}{2}}^{\tilde{R}}(\tau, 0) = -2. \quad (28.32)$$

The ground states form an  $SU(2)$  doublet of charges  $J_0 = \pm 1$ , i.e.  $y^{\pm 1}$ , counted with the sign  $-1$ : in the convention fixed by (28.25), where the  $\ell = 0$  ground state is bosonic, the doublet is fermionic. The Witten index  $-2$  follows either from the ground states or from (28.31) at  $z = 0$ , where the left side vanishes because  $\vartheta_1(\tau, 0) = 0$ :  $0 = 2 \cdot 1 + \text{ch}_{\frac{1}{4},\frac{1}{2}}^{\tilde{R}}(\tau, 0)$ .

### 28.4.2 Massive (non-BPS) representations

A representation with  $h > \frac{1}{4}$  is *massive*. Its highest-weight state is not annihilated by  $G_0$ , all four fermionic zero modes act freely on it, and the ground states form the  $2 \times 2 = 4$ -dimensional representation of the zero-mode Clifford algebra: an  $SU(2)$  doublet and two singlets. The character is [EOT §1, eq. (1.6), ll. 116–122]

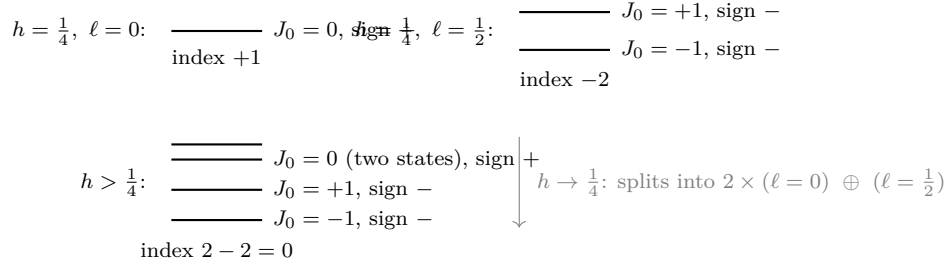
$$\text{ch}_{h,\ell=\frac{1}{2}}^{\tilde{R}}(\tau, z) = q^{h-\frac{3}{8}} \frac{\vartheta_1(\tau, z)^2}{\eta(\tau)^3} = q^{h-\frac{1}{4}} \left[ -(y - 2 + y^{-1}) + O(q) \right]. \quad (28.33)$$

Check the bookkeeping of exponents: the trace starts at  $q^{h-c/24} = q^{h-1/4}$ , and  $\vartheta_1^2/\eta^3$  carries  $q^{1/8}$ , so the prefactor must be  $q^{h-1/4-1/8} = q^{h-3/8}$ . The bracket is the four ground states: the two singlets with sign  $+$  (the 2) and the doublet with sign  $-$  (the  $-y - y^{-1}$ ), the same sign pattern as the  $\ell = \frac{1}{2}$  massless representation. Every massive character vanishes at  $z = 0$  because  $\vartheta_1(\tau, 0) = 0$ : a massive representation has Witten index zero. Only the massless representations can contribute to the Euler number, which is why the coefficient of the massless characters in the elliptic genus is topological, while the coefficients of the massive ones can a priori depend on where we are in moduli space. (They do not, for K3, as we now show; but the reason is the rigidity of  $\phi_{K3}$ , not the index argument.)

## 28.5 Decomposing the elliptic genus

### 28.5.1 The decomposition and the function $\Sigma$

Since every state of the K3 sigma model lies in some  $N = 4$  representation, and since the elliptic genus is linear in the representation content,  $\phi_{K3}$  must be a linear combination of



the characters of Section 28.4 with integer coefficients. The Witten index already fixes the massless part: writing

$$\phi_{K3}(\tau, z) = 24 \operatorname{ch}_{\frac{1}{4},0}^{\tilde{R}}(\tau, z) + \Sigma(\tau) \frac{\vartheta_1(\tau, z)^2}{\eta(\tau)^3} \quad (28.34)$$

[EOT §1, eq. (1.7), ll. 124–148] is consistent at  $z = 0$  because the second term vanishes there and the first gives 24. Everything not captured by 24 copies of the  $\ell = 0$  character has been put into a single function  $\Sigma(\tau)$  multiplying the massive character shape. That  $\Sigma$  depends on  $\tau$  only is a nontrivial statement:  $\phi_{K3} - 24 \operatorname{ch}_{\frac{1}{4},0}^{\tilde{R}}$  must be divisible by  $\vartheta_1^2$  as a function of  $z$ . Since both  $\phi_{K3}$  and  $\operatorname{ch}_{\frac{1}{4},0}^{\tilde{R}}$  are known in closed form, one can solve for  $\Sigma$ , and EOT give the result [EOT §1, eqs. (1.8)–(1.9), ll. 150–160]:

$$\Sigma(\tau) = -8 \left[ \mu(\tau; \tfrac{1}{2}) + \mu(\tau; \tfrac{1+\tau}{2}) + \mu(\tau; \tfrac{\tau}{2}) \right] \quad (28.35)$$

$$= -2 q^{-1/8} \left( 1 - \sum_{n \geq 1} A_n q^n \right). \quad (28.36)$$

The three arguments  $\frac{1}{2}, \frac{1+\tau}{2}, \frac{\tau}{2}$  are the three half-periods, the same three that label  $\vartheta_2, \vartheta_3, \vartheta_4$  in (28.22). Equation (28.36) defines the coefficients  $A_n$ . We checked (28.35)–(28.36) numerically to forty digits at two values of  $\tau$  with the nine coefficients of (28.38) below, and checked that  $(\phi_{K3} - 24 \operatorname{ch}_{\frac{1}{4},0}^{\tilde{R}}) \eta^3 / \vartheta_1^2$  is independent of  $z$ ; these are symbolic checks, not Tier A.

The polar term  $-2q^{-1/8}$  in (28.36) is not a massive representation (it would have  $h - \frac{1}{4} = -\frac{1}{8} < 0$ ). It is removed by the splitting relation (28.31):  $-2q^{-1/8} \vartheta_1^2 / \eta^3 = -4 \operatorname{ch}_{\frac{1}{4},0}^{\tilde{R}} - 2 \operatorname{ch}_{\frac{1}{4},\frac{1}{2}}^{\tilde{R}}$ , and the final form of the decomposition is [EOT §1, eq. (1.11), ll. 169–195]

$$\phi_{K3}(\tau, z) = 20 \operatorname{ch}_{\frac{1}{4},0}^{\tilde{R}}(\tau, z) - 2 \operatorname{ch}_{\frac{1}{4},\frac{1}{2}}^{\tilde{R}}(\tau, z) + 2 \sum_{n \geq 1} A_n q^{n-\frac{1}{8}} \frac{\vartheta_1(\tau, z)^2}{\eta(\tau)^3} \quad (28.37)$$

The third term is  $2A_n$  copies of the massive character with  $h = n + \frac{1}{4}$ , by (28.33). Check the Witten index once more:  $20 \cdot 1 - 2 \cdot (-2) + 0 = 24$ . Check the  $q^0$  term with (28.29) and (28.32):  $20 \cdot 1 - 2 \cdot (-y - y^{-1}) = 20 + 2y + 2y^{-1}$ , which is (28.11). The twenty  $\ell = 0$  representations are the twenty  $(1, 1)$ -classes; the  $-2$  copies of the  $\ell = \frac{1}{2}$  representation, entering with a minus sign because its ground states are fermionic in our convention, supply the four charged ground states  $H^{0,0}, H^{2,0}, H^{0,2}, H^{2,2}$ .

**A negative multiplicity is not a negative number of states**

The coefficient  $-2$  in (28.37) does not mean that the K3 sigma model contains “minus two” representations. The multiplicity of the  $\ell = \frac{1}{2}$  representation in the full Hilbert space, tensored with right-moving ground states, is a non-negative integer; the sign is produced by the right-moving factor  $(-1)^{\bar{F}}$  in (28.5), which weights right-moving Ramond ground states by  $\pm 1$ . The elliptic genus counts left-moving representations *with a sign* that depends on the right-moving partner. Similarly  $2A_n$  is a signed count; that it is positive for all  $n$  is the content of the positivity conjecture recalled in Section 28.5.3.

**28.5.2 Worked example:  $A_1 = 45$  and  $A_2 = 231$  by hand**

The decomposition (28.37) can be solved for the  $A_n$  order by order in  $q$ , using only the expansions already written down. At order  $q^1$  the left side is the  $q^1$  bracket of (28.23); on the right we need the  $q^1$  terms of the two massless characters and the  $q^0$  term of  $q^{-1/8}\vartheta_1^2/\eta^3$  from (28.27):

$$20y^2 - 128y + 216 - 128y^{-1} + 20y^{-2} = 20(y^2 - 2y + 2 - 2y^{-1} + y^{-2}) - 2(-y + 2 - y^{-1}) + 2A_1(-y + 2 - y^{-1}).$$

Compare coefficients. Of  $y^2$ :  $20 = 20$ , no information (the massive ground states carry  $|J_0| \leq 1$ ). Of  $y^1$ :  $-128 = -40 + 2 - 2A_1$ , so  $2A_1 = 90$  and  $A_1 = 45$ . Of  $y^0$ :  $216 = 40 - 4 + 4A_1$ , so  $A_1 = 45$  again, a consistency check. At order  $q^2$  one needs the  $q^2$  terms of the massless characters, the  $q^1$  term of the massive core (multiplied by  $2A_1$ ) and its  $q^0$  term multiplied by  $2A_2$ ; the  $y^0$  coefficient gives  $1616 = 20 \cdot 8 - 2 \cdot 4 + 90 \cdot 6 + 4A_2$ , i.e.  $4A_2 = 924$  and  $A_2 = 231$ . Both values agree with [EOT §1, eq. (1.12), ll. 197–209]. The student is asked to redo the  $y^{\pm 1}$  and  $y^{\pm 2}$  coefficients at order  $q^2$  in Exercise 28.4.

**28.5.3 The coefficients  $A_n$** 

EOT tabulate [EOT §1, eq. (1.12), ll. 197–209]

|       |    |     |     |      |      |       |       |       |        |         |
|-------|----|-----|-----|------|------|-------|-------|-------|--------|---------|
| $n$   | 1  | 2   | 3   | 4    | 5    | 6     | 7     | 8     | 9      | (28.38) |
| $A_n$ | 45 | 231 | 770 | 2277 | 5796 | 13915 | 30843 | 65550 | 132825 |         |

and recall the conjecture that all  $A_n$  are positive integers [EOT §1, l. 210]; integrality is clear from (28.37), positivity is not. Support for positivity comes from the asymptotic formula obtained by Eguchi and Hikami by a Rademacher-type expansion [EOT §1, eq. (1.13), ll. 211–223], which we write as

$$A_n \approx \frac{2}{\sqrt{8n-1}} \exp\left(\frac{\pi}{2}\sqrt{8n-1}\right) = \frac{2}{\sqrt{8n-1}} \exp\left(2\pi\sqrt{\frac{1}{2}\left(n - \frac{1}{8}\right)}\right). \quad (28.39)$$

The text extraction of EOT’s equation garbles the prefactor, so we fixed it by a numerical check: the ratio of  $A_n$  to  $e^{\pi\sqrt{8n-1}/2}/\sqrt{8n-1}$  is

$$1.87, \quad 2.04, \quad 1.98, \quad 2.02, \quad 1.99, \quad 2.01, \quad 1.996, \quad 2.002, \quad 1.998 \quad (n = 1, \dots, 9),$$

converging to 2 (a symbolic check of this book). The exponent has the Cardy form

$$\exp\left(2\pi\sqrt{\frac{c_{\text{eff}}}{6}\left(n - \frac{1}{8}\right)}\right) \quad \text{with} \quad c_{\text{eff}} = 3,$$

not 6: the growth of the coefficients of a modular or mock modular form is set by its most polar term, and the polar term of  $\Sigma$  is  $q^{-1/8} = q^{-c_{\text{eff}}/24}$  (standard; not checked against a source in this book’s library). As [EOT §1, ll. 222–224] remark, the formula is a good estimate even at small  $n$ , which is why it supports positivity.

**Example 28.1** (The growth of  $A_n$ ). For  $n = 5$ ,  $\sqrt{8n-1} = \sqrt{39} = 6.245$ , the exponent is  $\pi \cdot 6.245/2 = 9.809$ ,  $e^{9.809} = 1.82 \times 10^4$ , and  $2/6.245 = 0.320$ , giving  $A_5 \approx 5.83 \times 10^3$  against the exact 5796, an error of 0.7%. For  $n = 1$  the same recipe gives 48 against 45.

#### 28.5.4 The observation

Now compare (28.38) with the list of dimensions of the irreducible representations of the Mathieu group  $M_{24}$  (Chapter 29): 1, 23, 45, 45, 231, 231, 252, 253, 483, 770, 770, 990, 990, 1035, 1035, 1035, 1265, 1771, 2024, 2277, 3312, 3520, 5313, 5796, 5544, 10395 [EOT App. A, eq. (A.3), ll. 345–349]. The first five  $A_n$  are on the list, and the next two are sums of entries [EOT §1, eqs. (1.14)–(1.15), ll. 228–237]:

$$A_6 = 13915 = 3520 + 10395, \quad A_7 = 30843 = 10395 + 5796 + 5544 + 5313 + 2024 + 1771. \quad (28.40)$$

For  $n \geq 8$  decompositions exist but are no longer unique [EOT §1, ll. 238–239]. That is the whole of the observation, and the numerical part of it is exactly what the Lean library checks (Section 28.6). What it might *mean* is the subject of Chapter 30: the natural conjecture is that the space of states of the K3 sigma model at each massive level carries an action of  $M_{24}$  of which  $2A_n$  is the graded dimension, in the same way that the coefficients 196884, ... of the modular function  $J(q)$  are graded dimensions of a representation of the Monster group [EOT §1, eq. (1.16), ll. 240–255]. The mystery, in EOT’s words, is that no K3 surface has  $M_{24}$  as a symmetry group: the finite groups of symplectic automorphisms of K3 surfaces enumerated by Mukai are all proper subgroups of  $M_{24}$  (Chapter 29), and the question is whether the elliptic genus, which is constant over the whole moduli space, sees a symmetry that no single point of moduli space possesses [EOT §1, ll. 268–275].

##### Historical note

The observation was made, according to the authors’ acknowledgment, at the Aspen Center for Physics during the workshop “Unity of String Theory” [EOT, ll. 318–319]. EOT’s first version of the paper contained a different decomposition of  $A_7$ ; the one in (28.40) is the corrected version, chosen to be consistent with the twisted elliptic genera computed in later work [EOT §1, footnote 1, ll. 257–259]. GHV, whose normalization gives  $2A_7 = 61686$ , note that even the corrected decomposition is consistent with, but not uniquely fixed by, their twining characters [GHV §3, eq. (3.4), ll. 575–583].

#### 28.5.5 The factor two: $2A_n$ and real representations

The factor 2 in front of the massive characters in (28.37) is physical. GHV absorb it and write [GHV §3, eqs. (3.1)–(3.3), ll. 500–577]

$$24 = 23+1, \quad -A_0^{\text{GHV}} = 2 = 1+1, \quad A_1^{\text{GHV}} = 90 = 45+45, \quad A_2^{\text{GHV}} = 462 = 231+231, \quad \dots \quad (28.41)$$

for the rescaled genus  $\phi_{1A} = \frac{1}{2}\phi_{K3}$ . In their normalization every multiplicity is the dimension of a *real* representation of  $M_{24}$ :  $45 + \overline{45}$ ,  $231 + \overline{231}$ , and so on, where  $\overline{45}$  is the complex conjugate of the 45-dimensional representation (the two entries 45, 45 in the list above). This is natural, because a graded vector space that is the “space of states” of a real field theory should carry a real representation. In EOT’s normalization the same statement reads: the multiplicity of the massive representation at level  $n$  in the K3 elliptic genus is  $2A_n$ , with  $A_n$  the dimension of one of the two conjugate irreducible representations. The massless part reads, in GHV’s form,  $24 = 23 + 1$  for the  $\ell = 0$  representations (the 23-dimensional permutation representation of  $M_{24}$  on 24 points, minus the trivial one) and  $-2 = -(1 + 1)$

for the  $\ell = \frac{1}{2}$  representation. This book's convention (Chapter 1) is EOT's:  $A_n$  denotes 45, 231, 770,  $\dots$ , and the multiplicities are  $2A_n$ .

### 28.5.6 The NS-sector function

Monstrous moonshine is formulated for the partition function of a bosonic theory, a function of  $\tau$  alone. To make the analogy tight, GHV apply spectral flow by  $\eta = \frac{1}{2}$  to the left-movers and remove the  $(-1)^F$ , which shifts  $z$  by  $\frac{\tau}{2} + \frac{1}{2}$  [GHV §2, eqs. (2.7)–(2.8), ll. 168–187]:

$$\chi(\tau, z) = \exp\left[2\pi i m\left(\frac{\tau}{4} + z + \frac{1}{2}\right)\right] \phi\left(\tau, z + \frac{\tau}{2} + \frac{1}{2}\right). \quad (28.42)$$

For K3 with  $m = 1$  and  $\phi = \phi_{1A}$ , set  $z = 0$ . The substitution sends  $y \mapsto e^{2\pi i(\tau/2+1/2)} = -q^{1/2}$  and the prefactor is  $-q^{1/4}$ , so from (28.23) divided by 2,

$$\begin{aligned} \chi_{1A}(\tau) &= -q^{1/4} \left[ (-q^{1/2} + 10 - q^{-1/2}) + q(10q + 64q^{1/2} + 108 + 64q^{-1/2} + 10q^{-1}) + \dots \right] \\ &= q^{-1/4} (1 - 20q^{1/2} - 62q - 216q^{3/2} - \dots), \end{aligned} \quad (28.43)$$

which agrees with [GHV §2, eq. (2.11), ll. 228–254] in the two terms they display (they also give the closed form  $(2\vartheta_4(\tau)/\vartheta_2(\tau))^2 - (2\vartheta_2(\tau)/\vartheta_4(\tau))^2$ , which we checked numerically). The leading power  $q^{-1/4} = q^{-c/24}$  is the NS vacuum, exactly as  $q^{-1} = q^{-24/24}$  is the vacuum of the Monster theory in  $J(q)$ , and the function transforms under the subgroup of  $SL(2, \mathbb{Z})$  generated by  $T^2$  and  $S$  [GHV §2, eqs. (2.9)–(2.10), ll. 188–227]. It is this function, and its “twined” versions with a group element inserted in the trace, that Chapter 30 studies.

## 28.6 What the machine has checked

Nothing in the analytic part of this chapter is formalized: theta functions, Jacobi forms, the  $N = 4$  algebra and its characters are all Tier L, pinned to EOT and GHV above. What the Lean library checks is the arithmetic at the two ends of the story: the central charge 6 and the chiral-primary count 2 as fixed rational and natural numbers, and the assertion that the tabulated  $A_n$  are dimensions, or sums of dimensions, of irreducible representations of  $M_{24}$ . The three files are

- `StringTheoryFormalization/Frontier/CentralCharge.lean`,
- `StringTheoryFormalization/Frontier/ChiralPrimaries.lean`,
- `DualScaleStream2/Moonshine/EOT.lean`.

All audited theorems depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`.

**In Lean: the coefficients  $A_1, \dots, A_9$  and  $M_{24}$**

```

/-- EOT eq. (1.12): 'A_1, ..., A_9'. -/
def eotA : Fin 9 → ℕ := ![45, 231, 770, 2277, 5796, 13915, 30843, 65550, 132825]

/-- 'A_1 ... A_5' are each the dimension of an irreducible representation
of 'M_{24}'. -/
theorem first_five_are_irreps :
  ∀ n : Fin 5, ∃ i : Fin 26,
    M24RepDim i = eotA (Fin.castLE (by norm_num) n) := by ...

/-- EOT eq. (1.14): 'A_6 = 3520 + 10395', both irreducible dimensions
(indices 21, 25). -/

```



```

theorem A6_decomposition :
  eotA 5 = M24RepDim 21 + M24RepDim 25 := rfl

/-- EOT eq. (1.15): 'A7 = 10395 + 5796 + 5544 + 5313 + 2024 + 1771'. -/
theorem A7_decomposition :
  eotA 6 = M24RepDim 25 + M24RepDim 23 + M24RepDim 24 + M24RepDim 22 +
    M24RepDim 18 + M24RepDim 17 := rfl

/-- 'A6' is not a single irreducible dimension (so a decomposition is
  genuinely needed). -/
theorem A6_not_irrep : ∀ i : Fin 26, M24RepDim i ≠ eotA 5 := by ...

```

*Reading.* `eotA` is the table (28.38), typed in by hand from EOT; Lean indexes from 0, so `eotA 5` is  $A_6$ . `M24RepDim` is the 26-entry list of irreducible dimensions of  $M_{24}$  from another library (`StringTheoryFormalization/StringDynamics/MathieuM24.lean`, discussed in Chapter 29), in the increasing order of EOT's eq. (A.3). `first_five_are_irreps` says: for each of the five indices  $n = 0, \dots, 4$  there exists an index  $i$  into the list with `M24RepDim i = eotA n`. It is an existence statement with explicit witnesses ( $i = 2, 4, 9, 19, 23$ ); it does not say that 45 is the dimension of a *unique* irreducible representation (it is not: the list contains 45 twice), and it says nothing about  $M_{24}$  acting on anything. `A6_decomposition` and `A7_decomposition` are the two equations (28.40), with each summand named by its position in the list. `A6_not_irrep` says that no entry of the list equals 13915, so that the decomposition of  $A_6$  into two summands is not a disguised triviality. None of these theorems mentions the elliptic genus, a character, or a group: they are statements about two finite lists of natural numbers. *Proof idea.* `first_five_are_irreps` is a case split on `n : Fin 5`, each case closed by exhibiting the witness and `rfl`; the two decompositions are `rfl` because both sides reduce to numerals; the non-membership is a case split over all 26 indices closed by `decide`.

#### In Lean: $c = 6$ and the two chiral primaries

```

def centralChargeBoson : ℚ := 1
def centralChargeFermion : ℚ := 1/2
def centralChargeK3 : ℚ :=
  4 * centralChargeBoson + 4 * centralChargeFermion
theorem central_charge_k3_eq_six : centralChargeK3 = 6 := by ...

structure N2State where
  confWeight : ℚ
  u1Charge : ℚ
def bpsBound (s : N2State) : Prop :=
  s.confWeight ≥ |s.u1Charge| / 2
def isChiralPrimary (s : N2State) : Prop :=
  0 ≤ s.u1Charge ∧ s.confWeight = s.u1Charge / 2
theorem chiral_primary_saturates_bps (s : N2State) (h : isChiralPrimary s) :
  bpsBound s := by ...

def K3ChiralPrimaryCount : Fin 3 → ℕ
| ⟨0, _⟩ => 1 -- H0,0 = ℂ (vacuum)
| ⟨1, _⟩ => 0 -- H1,0 = 0 (K3 has no holomorphic 1-forms)
| ⟨2, _⟩ => 1 -- H2,0 = ℂ (holomorphic 2-form Ω)
theorem k3_chiral_primary_total :
  (∑ i : Fin 3, K3ChiralPrimaryCount i) = 2 := by ...

```

*Reading.* These are arithmetic shadows of Section 28.2. `central_charge_k3_eq_six` says  $4 \cdot 1 + 4 \cdot \frac{1}{2} = 6$  in  $\mathbb{Q}$ , with the two free-field values entered as constants; no stress tensor or

operator product expansion is involved, and the theorem would be equally true if K3 did not exist. `N2State` is a pair of rationals  $(h, q)$  with no positivity constraint on  $h$ . `bpsBound` is (28.9) and `isChiralPrimary` is “ $q \geq 0$  and  $h = q/2$ ”; `chiral_primary_saturates_bps` proves the one direction “chiral primary  $\Rightarrow$  bound holds”, which is immediate because  $|q| = q$  for  $q \geq 0$ . Nothing about the  $N = 2$  algebra, its representations, or which states of a given theory are chiral primaries is formalized. `K3ChiralPrimaryCount` is the table  $(1, 0, 1)$  typed in by hand, not derived from the Hodge numbers of Chapter 19, and `k3_chiral_primary_total` sums it to 2. *Proof idea.* `norm_num` on the constants; `simp` with `abs_of_nonneg`; and `Fin.sum_univ_three` followed by `simp`.

### Names that promise more than the statement delivers

The same file contains three declarations whose names must not be read as theorems about physics. `virasoro_ope_central_term` takes points  $z, w \in \mathbb{C}$  with  $z \neq w$  as arguments and never uses them: it proves `centralChargeK3 / 2 = 3`, a numeral identity, not a statement about the  $T(z)T(w)$  operator product. `virasoro_algebra_commutator` takes a function  $L : \mathbb{Z} \rightarrow \mathbb{Z}$  that does not appear in its conclusion, the polynomial identity  $\frac{c}{12}(m^3 - m) = \frac{cm(m^2-1)}{12}$ , true for every rational  $c$ . `central_charge_from_worldsheet_action` is  $\exists c, c = \text{centralChargeK3} \wedge c = 6$ , witnessed by 6. And in the chiral-primary file, `chiral_primary_ring_associativity` takes three chiral primaries as hypotheses and concludes `True`: no chiral ring, no structure constants, no associativity is defined, let alone proved. The docstrings in the files say all this explicitly; a reader of the catalogue who sees only the names would be misled. The rule of Chapter 1 applies: read the statement, not the name.

### Related declarations in other libraries

The older `DualScaleM24Formalization` library records the same numerology in GHV’s normalization: `SocrateAI.Moonshine.dimA1` is the constant 90 and `dimA1_decomposition` proves `dimA1 = 45 + 45`, with companions for 462 and 1540; these are decidable equalities between numerals. The Euler number 24 that fixes the elliptic genus is `SocrateAI.Moonshine.k3_euler_eq_24`, the alternating sum of the Betti numbers  $(1, 0, 22, 0, 1)$  entered as constants (see Chapter 25).

| Claim                                                                                        | Tier    | Where                                                                               |
|----------------------------------------------------------------------------------------------|---------|-------------------------------------------------------------------------------------|
| Witten index counts $E = 0$ states with sign and is deformation invariant                    | [L]     | derived in Section 28.1; standard                                                   |
| Elliptic genus is a weak Jacobi form of weight 0, index $c/6$                                | [L]     | GHV §2                                                                              |
| $J_{0,1}^{\text{weak}}$ is one-dimensional; $\phi_{K3} = 2\phi_{0,1}$                        | [L]     | EOT §1                                                                              |
| $\phi_{K3} = 8 \sum_{i=2,3,4} (\vartheta_i(z)/\vartheta_i(0))^2$ , $\phi_{K3}(\tau, 0) = 24$ | [L]     | EOT eq. (1.2); expansion (28.23) symbolically checked                               |
| $N = 4$ characters (28.25), (28.33), splitting (28.31)                                       | [L]     | EOT eqs. (1.3)–(1.6), (1.10)                                                        |
| Decomposition (28.37), table (28.38)                                                         | [L]     | EOT eqs. (1.11)–(1.12); $A_1, A_2$ re-derived by hand, $\Sigma$ checked numerically |
| $A_n > 0$ for all $n$                                                                        | [C]     | conjecture recalled in EOT, l. 210                                                  |
| $c_{K3} = 4 \cdot 1 + 4 \cdot \frac{1}{2} = 6$ as rational arithmetic                        | [A]     | central_charge_k3_eq_six                                                            |
| $1 + 0 + 1 = 2$ chiral primaries as a hand-entered table                                     | [A]     | k3_chiral_primary_total                                                             |
| $A_1, \dots, A_5$ occur in the list of $M_{24}$ irreducible dimensions                       | [A]     | first_five_are_irreps                                                               |
| $A_6, A_7$ equal EOT's sums of list entries; $A_6$ is not an entry                           | [A]     | A6_decomposition,<br>A7_decomposition,<br>A6_not_irrep                              |
| The list <code>M24RepDim</code> is the list of irreducible dimensions of $M_{24}$            | [L]     | EOT eq. (A.3); see Chapter 29                                                       |
| The K3 elliptic genus carries an $M_{24}$ action                                             | [L]/[C] | Chapter 30                                                                          |

### Chapter summary

- The Witten index  $\text{tr}(-1)^F e^{-\beta H}$  counts zero-energy states with sign; excited states cancel in boson–fermion pairs, so the index is an integer and a deformation invariant.
- The elliptic genus (28.5) applies the cancellation to the right-movers only and keeps  $q^{L_0 - c/24}$  and  $y^{J_0}$  for the left-movers. It is holomorphic in  $\tau$ , and a weak Jacobi form of weight 0 and index  $m = c/6$ . Its specializations give  $\chi(z=0)$ , the signature up to sign ( $z = \frac{1}{2}$ ,  $q^0$  term) and the  $\chi_y$  genus ( $q^0$  term).
- The K3 sigma model has  $c = 6$ ,  $N = 4$  superconformal symmetry, two chiral primaries, and index  $m = 1$ . The space of weak Jacobi forms of weight 0 and index 1 is one-dimensional, so  $\phi_{K3}$  is fixed by  $\chi(K3) = 24$ ; explicitly, (28.22), with the expansion (28.23) whose coefficients depend only on  $4n - \ell^2$ .
- Ramond representations of the  $N = 4$  algebra at  $c = 6$ : massless with  $\ell = 0$  (index 1), massless with  $\ell = \frac{1}{2}$  (index  $-2$ ), massive with  $h > \frac{1}{4}$  (index 0, character  $q^{h-3/8} \vartheta_1^2/\eta^3$ ). At  $h = \frac{1}{4}$  the massive character splits as  $2 \text{ch}_{\frac{1}{4},0} + \text{ch}_{\frac{1}{4},\frac{1}{2}}$ .
- $\phi_{K3} = 20 \text{ch}_{\frac{1}{4},0} - 2 \text{ch}_{\frac{1}{4},\frac{1}{2}} + 2 \sum_{n \geq 1} A_n q^{n-1/8} \vartheta_1^2/\eta^3$  with  $A_n = 45, 231, 770, 2277, 5796, \dots$ ; multiplicities are  $2A_n$  (EOT convention, used in this book). The first five  $A_n$  are dimensions of irreducible representations of  $M_{24}$ ,  $A_6$  and  $A_7$  are sums of such.
- Lean checks the arithmetic ends only:  $c = 6$ , the chiral-primary count 2, and the mem-

bership of  $A_1, \dots, A_5$  (and the decompositions of  $A_6, A_7$ ) in the list of  $M_{24}$  dimensions. Everything analytic is Tier L.

## Exercises

**Exercise 28.1** (\*). A supersymmetric quantum mechanics has supercharge  $Q$  with  $Q^2 = H$  and a Hilbert space spanned by states of energies  $0, 0, 0, 3, 3, 7, 7, 7, 7$  with  $(-1)^F = (+, +, -, +, -, +, -, +, -)$ . Compute  $\text{tr}(-1)^F e^{-\beta H}$  and verify that it is independent of  $\beta$ . Which states could be lifted from  $E = 0$  by a small deformation, and which could not?

**Exercise 28.2** (\*). Using the Hodge diamond of K3 ( $h^{0,0} = h^{2,0} = h^{0,2} = h^{2,2} = 1, h^{1,1} = 20$ ), compute the  $\chi_y$  genus  $\sum_{p,q} (-1)^{p+q} h^{p,q} y^{p-1}$  and check it against (28.11). Then compute the same quantity for  $T^4$  ( $h^{p,q} = \binom{2}{p} \binom{2}{q}$ ) and explain why the elliptic genus of  $T^4$  vanishes identically.

**Exercise 28.3** (\*\*). Verify the three expansions of the theta ratios used in Section 28.3.3 from the products (28.19)–(28.21), then square and add to obtain the  $q^1$  term of (28.23). Where do the half-integer powers of  $q$  cancel, and why must they?

**Exercise 28.4** (\*\*). Extend the computation of Section 28.5.2 to the  $y^{\pm 1}$  and  $y^{\pm 2}$  coefficients at order  $q^2$  (the  $q^2$  terms of the massless characters are in (28.29) and (28.32); you will need the  $q^2$  term of  $\vartheta_1^2/\eta^3$ , which is  $q^{1/8}(-y^3 + 6y^2 - 15y + 20 - 15y^{-1} + 6y^{-2} - y^{-3})$ ). Show that all of them give  $A_2 = 231$ . Why does the  $y^{\pm 3}$  coefficient give no information about  $A_2$ ?

**Exercise 28.5** (\*\*). Show from (28.14) with  $\ell = 1$  that the coefficients of a weak Jacobi form of index  $m$  satisfy  $c(n, \ell) = c(n + \ell + m, \ell + 2m)$ , and deduce that  $c(n, \ell)$  depends only on  $4mn - \ell^2$  and on  $\ell \bmod 2m$ . Check the statement on Table 28.1. Then use the spectral flow (28.10) with  $\eta = 1$  to give the same identity a physical meaning.

**Exercise 28.6** (\*\*). Evaluate the asymptotic formula (28.39) for  $n = 10$  and  $n = 20$ , and compare with the values  $A_{10} = 260568$  and  $A_{20} = 63\,447\,087$ , which are not in EOT's table (28.38) but were extracted for this book from (28.35) numerically (a symbolic check, not a sourced datum; the same computation reproduces  $A_1, \dots, A_9$ ). How does the relative error decrease with  $n$ ?

**Exercise 28.7** (\* \* \*). Write the massive character (28.33) as  $q^{h-1/4}$  times a  $y$ -Laurent polynomial at each level and interpret the level-one term  $2y^2 - 5y + 6 - 5y^{-1} + 2y^{-2}$  of (28.27) as the action of the modes  $L_{-1}, J_{-1}^a$  ( $a = 1, 2, 3$ ) and  $G_{-1}^{\pm}, G_{-1}^{\pm'}$  of the  $N = 4$  algebra on the four ground states. (Count: which combinations of one bosonic or one fermionic mode on each ground state give the stated charges and signs?)

**Exercise 28.8** (\*\*, Lean). State and prove in Lean, in the style of `A6_decomposition`, that  $A_8 = 65550$  can be written as a sum of entries of `M24RepDim` (find a decomposition by hand first; EOT remark that it is not unique). Then state the analogue of `A6_not_irrep` for  $A_8$ .

**Exercise 28.9** (\* \* \*, Lean). Replace the hand-entered table `K3ChiralPrimaryCount` by a definition that reads the counts  $h^{p,0}$  from a Hodge-number table `h : Fin 3 → Fin 3 → ℕ` (as in `eulerFromHodge` of Chapter 25), and re-prove `k3_chiral_primary_total` for the K3 table. What extra hypothesis would you need for the statement “the number of chiral primaries equals  $\sum_p h^{p,0}$ ” to be a theorem about a conformal field theory rather than about a table?

## Further reading

- The definition of the elliptic genus for hyperkähler targets, the theta-function formula for K3, the  $N = 4$  characters, the decomposition, the table of  $A_n$ , the asymptotic formula, and the observation about  $M_{24}$  with its comparison to monstrous moonshine: [EOT §1, ll. 39–308]. The dimensions of the irreducible representations of  $M_{24}$ : [EOT App. A, eq. (A.3), ll. 345–349].
- Jacobi-form properties of elliptic genera, the  $q$ -expansion of  $\phi_{K3}$ , spectral flow, and the NS-sector function  $\chi(\tau)$  with its modular behaviour under  $\langle T^2, S \rangle$ : [GHV §2, ll. 104–294]. The decomposition in GHV’s normalization with real representations,  $24 = 23+1$ ,  $90 = 45+45$ ,  $\dots$ : [GHV §3, eqs. (3.1)–(3.4), ll. 500–583]. Theta-function conventions: [GHV App. A, eq. (A.1), ll. 1453–1490].
- The K3 moduli space over which the elliptic genus is constant: Chapter 23; the orbifold point  $T^4/\mathbb{Z}_2$ : Chapter 21; the group  $M_{24}$  and its representations: Chapter 29; what is now known about the action of  $M_{24}$  on the space of states: Chapter 30.



## Chapter 29

# The Mathieu Group $M_{24}$

The elliptic genus of a K3 surface, computed in Chapter 28, is a modular object built from the geometry of a four-manifold and the representation theory of the  $N = 4$  superconformal algebra. Its expansion coefficients are integers, and the first of them are 90, 462, 1440, 4554, 11592, ... Eguchi, Ooguri and Tachikawa noticed that these are twice 45, 231, 770, 2277, 5796, and that every one of the latter is the dimension of an irreducible representation of a finite group of order 244 823 040, the largest Mathieu group  $M_{24}$  [EOT §1, ll. 197–203, 228–239]. Chapter 30 is about that observation. Before we can discuss it we must know the group: where it comes from, how big it is, what its subgroups and representations look like, and what precisely the numbers 45, 231, 770, ... are. The question of this chapter is therefore: *what is  $M_{24}$ , and which of the finite, explicit facts about it has this project handed to Lean?*

A student should care for two reasons. First,  $M_{24}$  is not an exotic object bolted onto string theory from outside: it is the symmetry group of a 24-dimensional even self-dual lattice, and Chapters 18, 20 and 22 have already shown that the cohomology of K3 is a 24-dimensional even self-dual lattice. Mukai’s theorem, quoted in Section 29.5, makes the link precise: every finite group of symplectic automorphisms of a K3 surface is a subgroup of  $M_{24}$ . Secondly, this chapter is a case study in how a table of numbers is handed to a proof assistant *and then checked*. The project’s table of representation dimensions of  $M_{24}$  was wrong for months, and a one-line theorem, Burnside’s  $\sum \dim^2 = |G|$ , exposed it (Section 29.6). That story is worth more than the table.

*Conventions.* Groups act on the left;  $|G|$  is the order of  $G$ ;  $\mathbb{F}_2 = \{0, 1\}$  is the field with two elements; the *weight*  $\text{wt}(c)$  of a vector  $c \in \mathbb{F}_2^n$  is the number of its nonzero coordinates. The root lattice  $A_1$  is the rank-one lattice whose generator has  $x^2 = 2$ , so  $A_1 \cong \sqrt{2}\mathbb{Z} \subset \mathbb{R}$  with the Euclidean form, and its dual  $A_1^* = \frac{1}{\sqrt{2}}\mathbb{Z}$  has generator of norm  $\frac{1}{2}$  [EOT App. B, ll. 959–965]. Dimensions of representations are over  $\mathbb{C}$ .

### 29.1 Finite simple groups in one page

A group  $G$  is *simple* if it has no normal subgroups other than  $\{e\}$  and  $G$ . Simple groups are to finite groups what primes are to integers: every finite group has a composition series whose successive quotients are simple, and these quotients are determined by the group (Jordan–Hölder). The classification of finite simple groups (standard; not checked against a source in this book’s library) says that a finite simple group is cyclic of prime order, alternating  $A_n$  with  $n \geq 5$ , a group of Lie type over a finite field such as  $PSL(n, q)$ , or one of 26 *sporadic* groups that belong to no infinite family. The five Mathieu groups  $M_{11}, M_{12}, M_{22}, M_{23}, M_{24}$  are the smallest and oldest sporadic groups; the largest sporadic group is the Monster, whose

representation dimensions 1, 196883, ... appear in the coefficients of the modular function  $J(q) = q^{-1} + 196884q + \dots$  (monstrous moonshine, [EOT §1, ll. 244–256]).

Everything in this chapter concerns  $M_{24}$ , and the first fact to record is its order [EOT App. A, ll. 331–334]:

$$|M_{24}| = 2^{10} \cdot 3^3 \cdot 5 \cdot 7 \cdot 11 \cdot 23 = 244\,823\,040. \quad (29.1)$$

The right-hand side is a number one can check by hand ( $2^{10} = 1024$ ,  $1024 \cdot 27 = 27648$ ,  $27648 \cdot 35 = 967680$ ,  $967680 \cdot 11 = 10\,644\,480$ ,  $10\,644\,480 \cdot 23 = 244\,823\,040$ ), and it is the first of the arithmetic facts that the project has kernel-checked (`M24_order`, Section 29.7). The prime factorization already constrains the group a great deal: by Lagrange’s theorem the order of every subgroup and of every element divides  $|M_{24}|$ , so no element has order 13, 25 or 49 (whereas orders 9 and 16 are allowed by Lagrange, though we shall see in Section 29.4 that they do not occur), and by Sylow’s theorem a subgroup of order 23 exists (and is cyclic, since 23 is prime). Note that  $23^2 \nmid |M_{24}|$ , and that the primes 2, 3, 5, 7, 11, 23 are exactly the primes EOT observe to recur in the factorizations of the moonshine coefficients  $A_n$  [EOT §1, footnote 2, ll. 260–262].

### An order is not a group

Knowing  $|G|$  does not determine  $G$ : there are many groups of order 244 823 040 (the cyclic group is one of them). Every Lean theorem quoted in this chapter that mentions 244 823 040 is a statement about that *numeral*. That the numeral is the order of a simple group called  $M_{24}$  is Tier L, quoted from EOT, and nothing about a group is constructed in the project’s libraries. Read the statements, not the names.

## 29.2 Binary codes and the extended Golay code

Of the many equivalent definitions of  $M_{24}$ , EOT choose the one closest to string theory:  $M_{24}$  is the symmetry group of a particular even self-dual lattice of rank 24 built on  $A_1^{24}$ , and that lattice is encoded by a binary code [EOT App. B, ll. 959–982]. We reconstruct their argument in full, because every step is a two-line computation and because the same construction with  $n = 8$  produces  $E_8$ , which the student already knows.

**Definition 29.1** (Binary linear code). A *binary linear code* of length  $n$  and dimension  $k$  is a  $k$ -dimensional  $\mathbb{F}_2$ -subspace  $C \subseteq \mathbb{F}_2^n$ . Its *minimum distance*  $d$  is the smallest weight of a nonzero codeword, and one writes  $C$  as an  $[n, k, d]$  code. The *dual code* is  $C^\perp = \{x \in \mathbb{F}_2^n : x \cdot c = 0 \text{ for all } c \in C\}$  with the dot product  $x \cdot c = \sum_i x_i c_i \bmod 2$ ;  $C$  is *self-dual* if  $C = C^\perp$  and *doubly even* if every codeword has weight divisible by 4.

Since  $\dim C + \dim C^\perp = n$ , a self-dual code has  $k = n/2$  and  $n$  is even. A code with  $2^k$  words can correct any  $t = \lfloor (d-1)/2 \rfloor$  errors: two balls of radius  $t$  around distinct codewords are disjoint because the codewords differ in at least  $d \geq 2t + 1$  places.

### From a code to a lattice

Consider the lattice  $A_1^n = (\sqrt{2}\mathbb{Z})^n \subset \mathbb{R}^n$  with the Euclidean form. It is even (every vector has  $x^2 = 2 \sum m_i^2$ ) but far from unimodular: its Gram matrix is  $2 \cdot \mathbf{1}_n$ , of determinant  $2^n$ , and its dual is  $(A_1^*)^n = (\frac{1}{\sqrt{2}}\mathbb{Z})^n$ . Any lattice  $N$  with

$$A_1^n \subset N \subset (A_1^*)^n \quad (29.2)$$



is determined by the subgroup  $C := N/A_1^n$  of  $(A_1^*)^n/A_1^n \cong (\mathbb{Z}/2)^n = \mathbb{F}_2^n$ , i.e. by a binary code [EOT App. B, ll. 966–970]. Concretely, every  $x \in (A_1^*)^n$  can be written uniquely as

$$x = \frac{1}{\sqrt{2}}(c + 2m), \quad c \in \{0, 1\}^n, \quad m \in \mathbb{Z}^n, \quad (29.3)$$

and  $x \in N$  iff  $c \bmod 2 \in C$ . Now compute the three lattice properties in terms of  $C$ .

*Integrality.* For  $x = \frac{1}{\sqrt{2}}(c + 2m)$  and  $y = \frac{1}{\sqrt{2}}(c' + 2m')$ ,

$$x \cdot y = \frac{1}{2} \sum_i (c_i + 2m_i)(c'_i + 2m'_i) = \frac{1}{2}(c \cdot c')_{\mathbb{Z}} + (\text{integer}), \quad (29.4)$$

where  $(c \cdot c')_{\mathbb{Z}} = \sum_i c_i c'_i$  is the integer dot product. So  $N$  is integral iff  $c \cdot c' \equiv 0 \pmod{2}$  for all  $c, c' \in C$ , i.e. iff  $C \subseteq C^\perp$ .

*Evenness.* Setting  $y = x$ ,

$$x^2 = \frac{1}{2} \sum_i (c_i + 2m_i)^2 = \frac{1}{2} \text{wt}(c) + 2 \sum_i (c_i m_i + m_i^2), \quad (29.5)$$

using  $c_i^2 = c_i$ . Hence  $x^2 \in 2\mathbb{Z}$  for all  $x \in N$  iff  $\text{wt}(c) \equiv 0 \pmod{4}$  for all  $c \in C$ :  *$N$  is even iff  $C$  is doubly even* [EOT App. B, ll. 972–973].

*Unimodularity.* The index  $[N : A_1^n] = |C| = 2^k$ , and for any finite-index overlattice  $\text{disc } N = \text{disc}(A_1^n)/[N : A_1^n]^2 = 2^n/2^{2k}$  (Chapter 18). So  $N$  is unimodular iff  $k = n/2$ ; together with integrality this says  $C = C^\perp$ :  *$N$  is self-dual iff  $C$  is self-dual* [EOT App. B, ll. 971–972].

*Roots.* A vector of  $N$  with  $x^2 = 2$  must, by (29.5), have  $\text{wt}(c) \in \{0, 4\}$ : if  $c = 0$  then  $\sum m_i^2 = 1$  and  $x = \pm\sqrt{2}e_i$  is one of the  $2n$  roots of  $A_1^n$ ; if  $\text{wt}(c) = 4$  then  $\sum_i (c_i m_i + m_i^2) = 0$  forces  $m_i = 0$  off the support of  $c$  and  $m_i \in \{0, -1\}$  on it, giving  $2^4 = 16$  vectors  $\frac{1}{\sqrt{2}}(\pm 1, \pm 1, \pm 1, \pm 1, 0, \dots)$  for each weight-4 word. Weight  $\geq 8$  gives  $x^2 \geq 4$ . Therefore

$$\#\{x \in N : x^2 = 2\} = 2n + 16 N_4(C), \quad N_4(C) := \#\{c \in C : \text{wt}(c) = 4\}. \quad (29.6)$$

**Example 29.2** (The Hamming code and  $E_8$ ). Take  $n = 8$  and let  $C_8 \subset \mathbb{F}_2^8$  be spanned by the rows of

$$G_8 = \begin{pmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}.$$

The rows are pairwise orthogonal (each pair overlaps in 0 or 2 positions) and have weight 4, so  $C_8 \subseteq C_8^\perp$ , and since  $\dim C_8 = 4 = 8/2$  it is self-dual. Listing the 16 codewords gives the weight distribution 1 word of weight 0, 14 of weight 4, 1 of weight 8 (a symbolic check; the reader should do it by hand once). So  $C_8$  is a doubly even self-dual  $[8, 4, 4]$  code (the extended Hamming code), and by (29.6) the lattice  $N_8$  is even, unimodular, of rank 8, with

$$2 \cdot 8 + 16 \cdot 14 = 16 + 224 = 240$$

roots. By the classification of Chapter 18,  $N_8 \cong E_8$ , and 240 is indeed the number of roots of  $E_8$  found there. This is a satisfying check that the bookkeeping (29.5)–(29.6) is right.

## The Golay code

Now take  $n = 24$ . EOT ask for an even self-dual  $N$  in (29.2) *whose only roots are the 48 roots of  $A_1^{24}$* , i.e.  $N_4(C) = 0$  [EOT App. B, ll. 973–977]. By the three computations above,  $C$

$$\begin{array}{ccccc}
 A_1^{24} = (\sqrt{2}\mathbb{Z})^{24} & \xrightarrow{\text{index } 2^{12}} & N & \xrightarrow{\text{index } 2^{12}} & (A_1^*)^{24} = (\frac{1}{\sqrt{2}}\mathbb{Z})^{24} \\
 & & \downarrow & & \\
 \mathcal{G}_{24} = N/A_1^{24} & \subset & \mathbb{F}_2^{24} & = & (A_1^*)^{24}/A_1^{24}
 \end{array}$$

$$M_{24} = \text{Aut}(\mathcal{G}_{24}) \subset S_{24} \text{ permutes the 24 coordinates}$$

Figure 29.1: The lattice sandwich (29.2).  $N$  is even iff the code is doubly even, self-dual iff the code is self-dual, and has no roots beyond those of  $A_1^{24}$  iff the code has no words of weight 4. The total index  $[(A_1^*)^{24} : A_1^{24}] = 2^{24}$  splits as  $2^{12} \cdot 2^{12}$  exactly when  $N$  is unimodular.

must be a doubly even self-dual  $[24, 12, d]$  code with  $d \geq 8$ . Such a code exists and is unique up to permutation of coordinates [EOT App. B, ll. 977–978]; this uniqueness is a theorem of coding theory that we take from EOT as Tier L. It is the *extended binary Golay code*  $\mathcal{G}_{24}$ , a  $[24, 12, 8]$  code.

**Definition 29.3** ( $M_{24}$ ).  $M_{24}$  is the subgroup of the symmetric group  $S_{24}$ , acting on  $\mathbb{F}_2^{24}$  by permuting coordinates, that maps  $\mathcal{G}_{24}$  to itself:  $M_{24} = \text{Aut}(\mathcal{G}_{24})$  [EOT App. B, ll. 978–979].

Two remarks make this concrete. First, the weight distribution of  $\mathcal{G}_{24}$  is

|                 |   |     |      |     |    |                                              |
|-----------------|---|-----|------|-----|----|----------------------------------------------|
| weight          | 0 | 8   | 12   | 16  | 24 |                                              |
| number of words | 1 | 759 | 2576 | 759 | 1  | $1 + 759 + 2576 + 759 + 1 = 4096 = 2^{12}$ . |

(29.7)

This is a symbolic check: the author built the code as the extended quadratic-residue code of length 23 (the cyclic code spanned by the shifts of the indicator vector of the nonzero squares modulo 23, extended by a parity bit), verified  $\dim = 12$ , and counted weights with a short script. The interpretation of 759 as the number of *octads* and the fact that any 5 of the 24 points lie in exactly one octad,  $759 = \binom{24}{5} / \binom{8}{5}$ , are standard (Steiner system  $S(5, 8, 24)$ ) and not checked against a source in this book’s library. Secondly, the Golay code corrects  $t = \lfloor (8 - 1)/2 \rfloor = 3$  errors, and its codewords number  $2^{12} = 4096$ ; these two numerals are the whole content of the project’s “Golay” theorems (Section 29.7).

#### Why a string theorist likes this definition

The lattice  $N = N_{24}$  built from  $\mathcal{G}_{24}$  is even, self-dual and positive definite of rank 24, so it is one of the 24 Niemeier lattices of Chapter 18; it defines a chiral conformal field theory with  $c = 24$  whose current algebra is  $A_1^{24}$  (the 48 roots are the 48 charged currents).  $M_{24}$  permutes the 24 copies of  $A_1$  and preserves the lattice, so it is a discrete symmetry of that chiral CFT [EOT App. B, ll. 979–982]. The K3 cohomology lattice  $H^*(K3, \mathbb{Z})$  is also even and self-dual of rank 24, but of signature  $(4, 20)$  (Chapter 22); EOT locate the whole K3– $M_{24}$  connection in this coincidence of rank and type [EOT App. B, ll. 984–990].

### 29.3 Transitivity and the stabilizer tower

**Definition 29.4** ( $k$ -transitive action). A group  $G$  acting on a finite set  $\Omega$  is *k-transitive* if for any two ordered  $k$ -tuples of distinct points  $(a_1, \dots, a_k)$ ,  $(b_1, \dots, b_k)$  there is  $g \in G$  with  $ga_i = b_i$  for all  $i$ . The *stabilizer* of a point  $a$  is  $G_a = \{g : ga = a\}$ , and the orbit–stabilizer theorem gives  $|G| = |G \cdot a| \cdot |G_a|$ .

If  $G$  is transitive on  $\Omega$  with  $|\Omega| = n$ , then  $|G| = n |G_a|$ , and  $G$  is  $k$ -transitive iff  $G_a$  is  $(k-1)$ -transitive on  $\Omega \setminus \{a\}$ . Iterating, a  $k$ -transitive group of degree  $n$  satisfies

$$|G| = n(n-1) \cdots (n-k+1) \cdot |G_{a_1 \dots a_k}|, \quad (29.8)$$

where  $G_{a_1 \dots a_k}$  is the pointwise stabilizer of  $k$  points. The Mathieu group  $M_{24}$  acts on the 24 coordinates of the Golay code, and this action is 5-transitive (standard; the project's `MathieuTower.lean` docstring states it, but it is not checked against a source in this book's library, and it is not formalized). Successively fixing points defines the *stabilizer tower*

$$M_{24} \supset M_{23} \supset M_{22} \supset M_{21} \cong PSL(3, 4), \quad (29.9)$$

where  $M_{23}$  is the stabilizer of one point,  $M_{22}$  of two,  $M_{21}$  of three. Each step is an instance of (29.8): since  $M_{24}$  is transitive on 24 points,  $M_{23}$  on the remaining 23, and  $M_{22}$  on the remaining 22,

$$|M_{24}| = 24 |M_{23}|, \quad |M_{23}| = 23 |M_{22}|, \quad |M_{22}| = 22 |M_{21}|. \quad (29.10)$$

Starting from (29.1) and dividing,

$$|M_{23}| = 244\,823\,040/24 = 10\,200\,960 = 2^7 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11 \cdot 23, \quad (29.11)$$

$$|M_{22}| = 10\,200\,960/23 = 443\,520 = 2^7 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11, \quad (29.12)$$

$$|M_{21}| = 443\,520/22 = 20\,160 = 2^6 \cdot 3^2 \cdot 5 \cdot 7. \quad (29.13)$$

The last number is recognizable. The group  $GL(3, q)$  of invertible  $3 \times 3$  matrices over the field with  $q$  elements has order  $(q^3 - 1)(q^3 - q)(q^3 - q^2)$  (choose the three columns to be linearly independent:  $q^3 - 1$  choices for the first,  $q^3 - q$  for the second,  $q^3 - q^2$  for the third); dividing by the  $q - 1$  nonzero determinants gives  $|SL(3, q)|$ , and by the  $\gcd(3, q - 1)$  scalar matrices of determinant 1 gives  $|PSL(3, q)|$ . For  $q = 4$ :

$$|PSL(3, 4)| = \frac{(64 - 1)(64 - 4)(64 - 16)}{(4 - 1) \cdot \gcd(3, 3)} = \frac{63 \cdot 60 \cdot 48}{3 \cdot 3} = \frac{181\,440}{9} = 20\,160, \quad (29.14)$$

in agreement with (29.13). That  $M_{21}$  is in fact isomorphic to  $PSL(3, 4)$  is Tier L (standard; stated in the project's docstring, not in this book's library); that the *orders* agree is arithmetic, and it is what the project proves.

Going two steps further down the tower, 5-transitivity and (29.8) give

$$24 \cdot 23 \cdot 22 \cdot 21 \cdot 20 = 5\,100\,480, \quad \frac{|M_{24}|}{5\,100\,480} = 48, \quad (29.15)$$

so the pointwise stabilizer of five points has order 48 (and  $M_{24}$  is not *sharply* 5-transitive: the map from  $M_{24}$  to ordered 5-tuples is 48-to-one). Read the other way, (29.15) is a divisibility test: any 5-transitive group of degree 24 has order divisible by  $5\,100\,480 = 2^7 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11 \cdot 23$ , and (29.1) passes.

#### Index relations do not prove existence

Equations (29.10) are consequences of transitivity, not evidence for it. Four numerals  $a = 24b$ ,  $b = 23c$ ,  $c = 22d$  are consistent with infinitely many groups (and with none). The Lean theorems `M24_stabilizer_index`, `M23_stabilizer_index`, `M22_stabilizer_index` of Section 29.7 verify exactly these numeral equations for the ATLAS values of  $|M_{23}|$ ,  $|M_{22}|$ ,  $|M_{21}|$ ; they say nothing about groups acting on sets. The

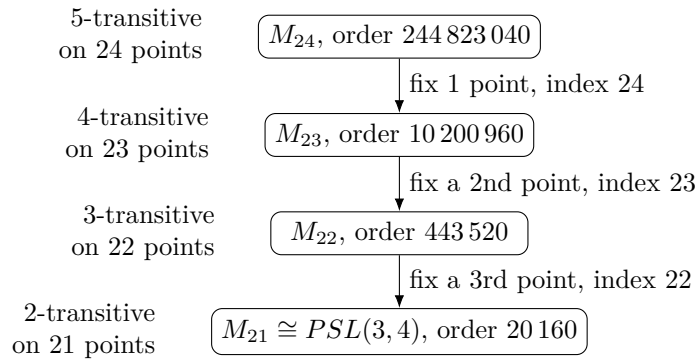


Figure 29.2: The point-stabilizer tower (29.9). Each arrow divides the order by the number of points the group above it acts on transitively; the product  $24 \cdot 23 \cdot 22 \cdot 20\,160$  returns  $|M_{24}|$  (`mathieu_tower_consistent`).

ATLAS values themselves are Tier L and are not in this book’s library of sources.

## 29.4 Conjugacy classes and irreducible representations

### Classes

Two elements  $g, h$  of a group are *conjugate* if  $h = xgx^{-1}$  for some  $x$ ; conjugacy classes partition the group, and a class function (such as the character of a representation) is constant on each.  $M_{24}$  has 26 conjugacy classes [EOT App. A, l. 336], named in the ATLAS convention by the order of their elements and a letter [GHV Table 1, ll. 1916–1922, 1928–1930]:

$$\begin{aligned} &1A, 2A, 2B, 3A, 3B, 4A, 4B, 4C, 5A, 6A, 6B, 7A, 7B, 8A, 10A, 11A, \\ &12A, 12B, 14A, 14B, 15A, 15B, 21A, 21B, 23A, 23B. \end{aligned} \quad (29.16)$$

The element orders that occur are thus 1, 2, 3, 4, 5, 6, 7, 8, 10, 11, 12, 14, 15, 21, 23, every one a divisor of (29.1) as Lagrange requires (symbolic check). Two features will matter in Chapter 30. First, if  $\gcd(a, o(g)) = 1$  then  $g^a$  lies in the same class as  $g$  [GHV §3.1, ll. 694–697]: the character table of  $M_{24}$  is “almost rational”, its only irrational entries being of the form  $e_p^\pm = (-1 \pm \sqrt{-p})/2$  [EOT App. A, eq. (A.2), ll. 339–343], which is why the classes come in lettered pairs such as  $7A/7B$  or  $23A/23B$ . Secondly, the classes  $1A, 2A, 3A, 4B, 5A, 6A$  contain elements of  $M_{23}$  (they fix a point), whereas  $2B, 3B, 4A, 4C$  do not; the latter are exactly the classes on which the permutation character  $\text{Tr}_{23 \oplus 1}(g)$  vanishes [GHV §3.1, ll. 808–813]. The number of fixed points of  $g$  on the 24 coordinates is the character of the 24-dimensional permutation representation, which splits as  $\mathbf{24} = \mathbf{1} \oplus \mathbf{23}$  [GHV §3, l. 555].

### Irreducible representations and their dimensions

A finite group has as many irreducible complex representations as conjugacy classes, so  $M_{24}$  has 26 of them [EOT App. A, l. 336]. Their dimensions, in increasing order, are [EOT App. A, eq. (A.3), ll. 345–351]

$$\begin{aligned} &1, 23, 45, 45, 231, 231, 252, 253, 483, 770, 770, 990, 990, 1035, 1035, 1035, \\ &1265, 1771, 2024, 2277, 3312, 3520, 5313, 5796, 5544, 10395. \end{aligned} \quad (29.17)$$

(EOT list 5796 before 5544; we keep their order because the project’s Lean table does.) The representations of dimensions 45, 231, 770, 990, 1035 come in complex-conjugate pairs, and

there is a third, real, representation of dimension 1035 [EOT App. A, eq. (A.4), ll. 353–358]; so 10 of the 26 representations are complex and 16 are real. The repeated entries in (29.17) are therefore not typographical: 45 and  $\overline{45}$  are different representations of the same dimension, and a table that lists 45 once is *wrong*. This is precisely the kind of error that Section 29.6 is about.

Several of these numbers have a direct meaning on the 24 points, which is a useful way to remember them (the interpretations are standard and not checked against a source in this book’s library; the arithmetic is a symbolic check): 23 is the nontrivial part of the permutation representation;  $\binom{24}{2} = 276 = 1 + 23 + 252$  is the dimension of  $\Lambda^2$  of the permutation representation;  $253 = 759 \cdot 8/24$  is the number of octads through a point;  $2024 = \binom{24}{3}$ ; and  $1771 = \binom{24}{4}/6$ .

### Burnside’s sum of squares

**Proposition 29.5** (Burnside). *For a finite group  $G$  with irreducible complex representations  $V_1, \dots, V_r$ ,*

$$\sum_{i=1}^r (\dim V_i)^2 = |G|. \quad (29.18)$$

*Derivation.* Let  $G$  act on the vector space  $\mathbb{C}[G]$  of formal combinations of group elements by left multiplication (the *regular representation*). Its character is  $\chi_{\text{reg}}(e) = |G|$  and  $\chi_{\text{reg}}(g) = 0$  for  $g \neq e$ , because  $g \neq e$  fixes no basis vector. The multiplicity of  $V_i$  in  $\mathbb{C}[G]$  is the inner product  $\langle \chi_{\text{reg}}, \chi_i \rangle = \frac{1}{|G|} \sum_g \chi_{\text{reg}}(g) \chi_i(g) = \frac{1}{|G|} \cdot |G| \cdot \dim V_i = \dim V_i$ . Comparing dimensions of  $\mathbb{C}[G] = \bigoplus_i V_i^{\oplus \dim V_i}$  gives (29.18). (Standard; uses orthogonality of characters, which is not proved in this book.)  $\square$

**Example 29.6** (Burnside for  $M_{24}$ , with numbers). Squaring and adding the 26 entries of (29.17) in blocks:

$$\begin{aligned} 1^2 + 23^2 &= 530, \\ 2 \cdot 45^2 + 2 \cdot 231^2 &= 4\,050 + 106\,722 = 110\,772, \\ 252^2 + 253^2 + 483^2 &= 63\,504 + 64\,009 + 233\,289 = 360\,802, \\ 2 \cdot 770^2 + 2 \cdot 990^2 + 3 \cdot 1035^2 &= 1\,185\,800 + 1\,960\,200 + 3\,213\,675 = 6\,359\,675, \\ 1265^2 + 1771^2 + 2024^2 + 2277^2 &= 1\,600\,225 + 3\,136\,441 + 4\,096\,576 + 5\,184\,729 = 14\,017\,971, \\ 3312^2 + 3520^2 + 5313^2 &= 10\,969\,344 + 12\,390\,400 + 28\,227\,969 = 51\,587\,713, \\ 5796^2 + 5544^2 + 10395^2 &= 33\,593\,616 + 30\,735\,936 + 108\,056\,025 = 172\,385\,577. \end{aligned}$$

The total is  $530 + 110\,772 + 360\,802 + 6\,359\,675 + 14\,017\,971 + 51\,587\,713 + 172\,385\,577 = 244\,823\,040 = |M_{24}|$ , as (29.18) demands. This is the computation that Lean performs in `M24RepDim_sum_sq` (Section 29.7). A second, independent sanity check is that every entry of (29.17) divides  $|M_{24}|$ , as the degree of an irreducible representation must (Frobenius; standard, not in this book’s library); a symbolic check confirms it for all 26.

#### What Burnside’s check does and does not prove

Passing (29.18) is strong evidence that a transcribed table is correct, because a single wrong, duplicated or missing entry changes the sum by a large amount that is unlikely to be compensated. It is *not* a proof that the 26 numbers are the degrees of  $M_{24}$ : infinitely many other 26-tuples of positive integers have the same sum of squares. And the check is only as good as the value of  $|G|$  it is compared with, which must come from an independent

source (here (29.1), EOT eq. (A.1)).

### The moonshine numbers, briefly

EOT's coefficients  $A_n$  of Chapter 28 are 45, 231, 770, 2277, 5796, 13915, 30843,  $\dots$  [EOT eq. (1.12), ll. 197–203]; the first five are single entries of (29.17), and the next two decompose as  $A_6 = 3520 + 10395$  and  $A_7 = 10395 + 5796 + 5544 + 5313 + 2024 + 1771$  [EOT eqs. (1.14)–(1.15), ll. 228–239]. GHV absorb a factor 2 and write  $90 = 45 + \overline{45}$ ,  $462 = 231 + \overline{231}$ ,  $1440 = 770 + \overline{770}$ ,  $4554 = 2 \cdot 2277$ ,  $11592 = 2 \cdot 5796$ ,  $27830 = 2 \cdot 3520 + 2 \cdot 10395$ ,  $61686 = 2(1771 + 2024 + 5313 + 5796 + 5544 + 10395)$  [GHV eqs. (3.3)–(3.4), ll. 555–575], so that each  $2A_n$  is the dimension of a *real* representation. Chapter 30 takes the story from here; for now note only that the decompositions for  $n \geq 6$  are not unique [EOT §1, ll. 237–239], which is why a stronger test (the twining genera of GHV) was needed.

## 29.5 $M_{24}$ and the geometry of K3

Let  $S$  be a K3 surface and  $G$  a finite group of automorphisms of  $S$  that preserve the holomorphic two-form (*symplectic* automorphisms, Chapter 19). Write  $\Lambda = H^*(S, \mathbb{Z})$  for the Mukai lattice of Chapter 22, even, self-dual, of signature  $(4, 20)$ ; let  $\Lambda^G$  be the sublattice fixed by  $G$  and  $\Lambda_G$  its orthogonal complement. EOT summarize Mukai's argument as follows [EOT App. B, ll. 984–1001].

1.  $\Lambda_G$  lies in the primitive part of  $H^{1,1}$  and is negative definite.
2. By Nikulin's embedding results,  $\Lambda_G$  embeds in the Niemeier lattice  $N = N_{24}$  of Section 29.2, hence  $G$  acts on  $N$  and  $G \subseteq M_{24}$ .
3.  $G$  fixes  $H^0$ ,  $H^4$ ,  $H^{2,0}$ ,  $H^{0,2}$  and the Kähler class, so  $\text{rk } \Lambda^G \geq 5$  and  $\text{rk } \Lambda_G \leq 19$ . Transported to the action on 24 points,  $G$  has at least 5 orbits.
4. Conversely, a subgroup of  $M_{24}$  with at least 5 orbits on the 24 points acts on  $H^{1,1}$  of some K3 surface, by the global Torelli theorem (Chapter 20).

This is Mukai's theorem (Tier L: [EOT App. B, ll. 955–958, 1054–1057] cite Mukai 1988 and Kondo 1998; the proofs are not in this book's library). In particular  $G$  is never all of  $M_{24}$ , which acts transitively (one orbit). EOT's example is the Fermat quartic  $X^4 + Y^4 + Z^4 + W^4 = 0$  in  $\mathbb{CP}^3$ , whose symplectic symmetry group  $(\mathbb{Z}_4)^2 \rtimes S_4$  has order  $16 \cdot 24 = 384 = 2^7 \cdot 3$  and decomposes the 24 points into five orbits of lengths 1, 1, 2, 4, 16 [EOT App. B, ll. 1003–1006]; note  $1 + 1 + 2 + 4 + 16 = 24$  and  $384 \mid |M_{24}|$ .

### Geometric symmetry versus moonshine symmetry

Mukai's theorem gives a *different* subgroup of  $M_{24}$  at each special point of the K3 moduli space, never the whole group. The moonshine observation of Chapter 30 concerns the elliptic genus, which is constant over the moduli space; EOT ask whether the various geometric subgroups are “enhanced to  $M_{24}$  over the whole of moduli space” at the level of the elliptic genus [EOT §1, ll. 268–275]. In this book that enhancement is a Tier L statement of the moonshine literature where cited, and Tier C where this programme extends it.

## 29.6 How a wrong table was caught

The project’s table of  $M_{24}$  representation dimensions, `M24RepDim`, was entered by hand from EOT eq. (A.3) and was “verified” for months, in the sense that the library compiled and its one theorem about the table, a lookup of the second entry (23), passed. The table was nevertheless wrong: it listed 10395 and 483 twice and omitted the second 990 and the third 1035 [`MathieuM24.lean`, ll. 65–67; `LL.md` §S2.6, ll. 63–67]. Such a table has the right length (26), the right first entries, and the right multiset of *distinct* values; a lookup theorem cannot see the difference. Burnside’s identity can. With the two wrong substitutions the sum of squares changes by

$$10395^2 + 483^2 - 990^2 - 1035^2 = 108\,056\,025 + 233\,289 - 980\,100 - 1\,071\,225 = 106\,237\,989, \quad (29.19)$$

so the wrong table sums to  $244\,823\,040 + 106\,237\,989 = 351\,061\,029 \neq |M_{24}|$ , and the theorem `M24RepDim_sum_sq` of Section 29.7 fails to compile against it. Once the table was corrected on 2026-09-16, the theorem closed by unfolding the 26-term sum and normalizing numerals [`MathieuM24.lean`, ll. 39–45, 117–121].

### The lesson, as the project recorded it

“Guard every data table with a theorem that would break if the table were wrong. Stream 1’s `M24RepDim` table had been ‘verified’ for months and was wrong (two entries duplicated, two missing), because its only theorem checked a single entry. Burnside’s  $\sum \dim^2 = |G|$  caught it in one line. For every hand-entered table, add the cheapest global consistency theorem available (sum of squares, total count, known product, symmetry).” [`LL.md` §S2.6, ll. 63–67]

The general point is worth restating for a student of formal methods. A proof assistant checks that a proof proves its statement; it does not check that the statement is the one you meant, nor that the data in a definition were copied correctly. Global invariants that the data must satisfy, sums, counts, factorizations, symmetries, are cheap to state, cheap to prove by `norm_num` or `decide`, and catch exactly the errors a lookup cannot. Notice also the second consistency theorem in `MathieuTower.lean`: the tower product  $24 \cdot 23 \cdot 22 \cdot 20160$  is compared not with a fresh literal but with the *already-certified* factorization `M24_order`, so that the two files cannot drift apart silently [`MathieuTower.lean`, ll. 97–110].

## 29.7 What the machine has checked

Five files touch  $M_{24}$  or the Golay code. Two of them are worth reading: `MathieuM24.lean` and `MathieuTower.lean`, both in the library `StringTheoryFormalization`. Three others, in the Mathlib-free libraries `Lean5Corpus` and `StringTheoryFoundation`, are arithmetic shadows whose docstrings promise much more than their statements deliver, and we describe them for that reason. According to the project’s axiom audit [`docs/VERIFIED_FOUNDATION.md`, ll. 7–16], every theorem below depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`. *No group, group action, code or representation is constructed in any of these files*: they contain natural-number tables and equalities between numerals [`MathieuM24.lean`, ll. 21–24, 110–114; `MathieuTower.lean`, ll. 43–46].

## The dimension table, the order and Burnside (MathieuM24.lean)

```

def M24RepDim : Fin 26 → ℕ
| ⟨0, _⟩ => 1      | ⟨1, _⟩ => 23      | ⟨2, _⟩ => 45      | ⟨3, _⟩ => 45
| ⟨4, _⟩ => 231    | ⟨5, _⟩ => 231    | ⟨6, _⟩ => 252    | ⟨7, _⟩ => 253
| ⟨8, _⟩ => 483    | ⟨9, _⟩ => 770    | ⟨10, _⟩ => 770   | ⟨11, _⟩ => 990
| ⟨12, _⟩ => 990   | ⟨13, _⟩ => 1035   | ⟨14, _⟩ => 1035  | ⟨15, _⟩ => 1035
| ⟨16, _⟩ => 1265  | ⟨17, _⟩ => 1771  | ⟨18, _⟩ => 2024  | ⟨19, _⟩ => 2277
| ⟨20, _⟩ => 3312  | ⟨21, _⟩ => 3520  | ⟨22, _⟩ => 5313  | ⟨23, _⟩ => 5796
| ⟨24, _⟩ => 5544  | ⟨25, _⟩ => 10395
| ⟨n + 26, h⟩ => absurd h (by omega)

theorem M24_first_coefficient :
  M24RepDim ⟨1, by norm_num⟩ = 23 := by rfl

theorem M24_order : (244823040 : ℕ) = 2^10 * 3^3 * 5 * 7 * 11 * 23 := by norm_num

theorem M24RepDim_sum_sq : (∑ i : Fin 26, M24RepDim i ^ 2) = 244823040 := by
  simp [Fin.sum_univ_succ, M24RepDim]

```

(The 26 cases of `M24RepDim` are written one per line in the source; they are re-wrapped here.) *Reading.* `M24RepDim` is a function from the 26-element type `Fin 26` to  $\mathbb{N}$ , a literal transcription of (29.17) in EOT's order. The final case is unreachable (`n + 26 < 26` is refuted by `omega`), so a missing entry would be a compile error rather than a silent default [MathieuM24.lean, ll. 95–98]. `M24_order` is the numeral identity (29.1); `M24RepDim_sum_sq` is the computation of Example 29.6; `M24_first_coefficient` is a single lookup kept as a regression guard. *Proof idea.* `norm_num` evaluates both sides of `M24_order` to the same numeral. For the sum, `Fin.sum_univ_succ` peels the sum over `Fin 26` into 26 explicit terms, each `M24RepDim` application reduces to its literal, and `simp`'s arithmetic normalizes the result. *Scope.* Burnside's theorem itself (Proposition 29.5) is not formalized; that 244 823 040 is the order of a group is not formalized; the identification of the table with  $M_{24}$  is Tier L.

## The stabilizer tower (MathieuTower.lean)

```

def orderM23 : ℕ := 10200960
def orderM22 : ℕ := 443520
def orderM21 : ℕ := 20160

theorem M24_stabilizer_index : 244823040 = 24 * orderM23 := by ...
theorem M23_stabilizer_index : orderM23 = 23 * orderM22 := by ...
theorem M22_stabilizer_index : orderM22 = 22 * orderM21 := by ...
theorem M21_order_factorization : orderM21 = 2 ^ 6 * 3 ^ 2 * 5 * 7 := by ...

theorem mathieu_tower_consistent :
  24 * (23 * (22 * orderM21)) = 244823040 := by
  rw [← M22_stabilizer_index, ← M23_stabilizer_index, ← M24_stabilizer_index]

theorem mathieu_tower_matches_certified_order :
  24 * (23 * (22 * orderM21)) = 2 ^ 10 * 3 ^ 3 * 5 * 7 * 11 * 23 := by
  rw [mathieu_tower_consistent, M24_order]

```

*Reading.* The three orders (29.11)–(29.13) are hard-coded constants, and the theorems are exactly the index relations (29.10), the factorization (29.13), and their composition. *Proof idea.* Each index relation is `unfold` followed by `norm_num`. The two composition theorems are pure rewriting: `mathieu_tower_consistent` substitutes the three index equations from the inside out until both sides are the same numeral, and the last theorem chains it



with the imported `M24_order` [MathieuTower.lean, ll. 48–53]. *Scope*. As the docstring itself says, this “is *not* a formalization of the Mathieu group, its Golay-code action, transitivity, or the orbit–stabilizer theorem itself” [MathieuTower.lean, ll. 43–46]. The ATLAS orders are Tier L and are not in this book’s library of sources.

#### Golay-code numerals (Problem8\_GolayHolography.lean, TensorNetworkBridge.lean)

```
def golay_length : Nat := 24
def golay_dimension : Nat := 12
def golay_distance : Nat := 8
def golay_codeword_count : Nat := 2 ^ 12
def error_correction_radius (d : Nat) : Nat := (d - 1) / 2

theorem golay_codeword_count_eval : golay_codeword_count = 4096 := by rfl
theorem golay_error_radius_equals_three :
  error_correction_radius golay_distance = 3 := by rfl
theorem golay_self_dual_dimension :
  golay_length - golay_dimension = golay_dimension := by rfl
theorem nonzero_codeword_weight_positive (w : Nat)
  (h : is_valid_nonzero_codeword_weight w) : w > 0 := by ...
```

and, in the second file,

```
structure GolayCodeParameters where
  length : Nat := 24
  dimension : Nat := 12
  distance : Nat := 8

theorem golay_length_matches_k3_euler :
  (defaultGolay.length : Int) = eulerChar4D bettiK3 := by decide
theorem golay_code_rate_is_half :
  defaultGolay.dimension * 2 = defaultGolay.length := by rfl
```

*Reading*. These files define the three parameters  $[24, 12, 8]$  as natural numbers and prove  $2^{12} = 4096$ ,  $\lfloor (8 - 1)/2 \rfloor = 3$ ,  $24 - 12 = 12$ , and  $12 \cdot 2 = 24$ . The one statement with any cross-file content is `golay_length_matches_k3_euler`: the literal 24 equals the Euler characteristic computed from the K3 Betti numbers `bettiK3 = (1, 0, 22, 0, 1)` by `eulerChar4D` in the project’s `Topology` module, which `euler_char_K3` evaluates to 24. `nonzero_codeword_weight_positive` says that a natural number  $w \geq 8$  is  $> 0$ . *Scope*. No code, no codeword, no Hamming weight and no automorphism appears in either file; the predicate `is_valid_nonzero_codeword_weight w` is literally  $w \geq 8$ . The docstring of `Problem8_GolayHolography` asserts that these theorems “prove that any bulk local operator on  $K3 \times T^2$  can be perfectly reconstructed” after erasure of three boundary regions; they prove nothing of the sort, and the holographic reading is Tier C at best. The second file carries a scope note added after review saying the same of its tensor-network narrative [TensorNetworkBridge.lean, ll. 39–46].

#### Numerology about $|M_{24}|$ (Problem2\_MathieuFrobeniusRigidity.lean)

```
def bps_conductor : Nat := 27720
def order_M24 : Nat := 244823040

theorem bps_conductor_prime_factorization :
  (2 ^ 3) * (3 ^ 2) * 5 * 7 * 11 = bps_conductor := by rfl
theorem m24_conductor_quotient :
```

```

order_M24 / bps_conductor = 8832 ∧ order_M24 % bps_conductor = 0 := by decide
theorem hasse_trace_integer_bound (a : Int) (h : a * a ≤ 52) :
  a * a ≤ 52 := h

```

*Reading.*  $27720 = 2^3 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11$  and  $244\,823\,040 = 27720 \cdot 8832$  (with  $8832 = 2^7 \cdot 3 \cdot 23$ ; symbolic check). Both are true numeral facts. The number 27720 is this programme’s “BPS character lock” ( $= 462 \cdot 60 = 4 \cdot 90 \cdot 77$ ), a Tier C numerological proposal with no source in this book’s library; the divisibility by  $|M_{24}|$  is arithmetic and is also reproved in `DualScaleValidation.UseCase2.m24_order_divisible_by_bps_lock`. The theorem `hasse_trace_integer_bound` deserves a moment’s attention: its conclusion is its hypothesis, and the proof is the hypothesis itself. It proves nothing, and the “Hasse–Weil bound” of the docstring is not formalized. A reader auditing a library should look for exactly this pattern.

Finally, the dimension table is a *hub*. Four theorems of the moonshine module `DualScaleStream2.Moonshine` (`first_five_are_irreps`, `A6_decomposition`, `A7_decomposition`, `A6_not_irrep`) import `M24RepDim` by name and verify EOT’s decompositions of  $A_1, \dots, A_7$  against it [MathieuM24.lean, ll. 47–52]. They belong to Chapter 30. Here we only note that they inherit the correctness of the table: had `M24RepDim` still contained a second 483 in place of the second 990, `first_five_are_irreps` would still have compiled (its five witnesses are at indices 2, 4, 9, 19, 23), which is another illustration that lookups do not guard tables.

| Claim                                                                    | Tier | Evidence                                                        |
|--------------------------------------------------------------------------|------|-----------------------------------------------------------------|
| $244\,823\,040 = 2^{10} \cdot 3^3 \cdot 5 \cdot 7 \cdot 11 \cdot 23$     | [A]  | M24_order                                                       |
| The 26 entries of (29.17) satisfy                                        | [A]  | M24RepDim_sum_sq,                                               |
| $\sum \dim^2 = 244\,823\,040$ ; entry 1 is 23                            |      | M24_first_coefficient                                           |
| Index relations (29.10) for the ATLAS                                    | [A]  | M24_stabilizer_index,                                           |
| numerals; $20160 = 2^6 \cdot 3^2 \cdot 5 \cdot 7$ ;                      |      | M23_stabilizer_index,                                           |
| $24 \cdot 23 \cdot 22 \cdot 20160 =  M_{24} $ and equals the             |      | M22_stabilizer_index,                                           |
| certified factorization                                                  |      | M21_order_factorization,                                        |
|                                                                          |      | mathieu_tower_consistent,                                       |
|                                                                          |      | mathieu_tower_matches_certified_order                           |
| $2^{12} = 4096$ ; $\lfloor 7/2 \rfloor = 3$ ; $24 - 12 = 12$ ;           | [A]  | golay_codeword_count_eval,                                      |
| $12 \cdot 2 = 24$ ; $24 = \chi(K3)$ from Betti numbers                   |      | golay_error_radius_equals_three,                                |
|                                                                          |      | golay_self_dual_dimension,                                      |
|                                                                          |      | golay_code_rate_is_half,                                        |
|                                                                          |      | golay_length_matches_k3_euler                                   |
| $27720 = 2^3 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11$ divides $244\,823\,040$ | [A]  | bps_conductor_prime_factorization,                              |
| with quotient 8832                                                       |      | m24_conductor_quotient                                          |
| Lattice-from-code dictionary (29.4)–(29.6);                              | —    | derived on the page                                             |
| $ PSL(3, 4) $ ; Burnside’s identity;                                     |      |                                                                 |
| 5-transitivity count (29.15)                                             |      |                                                                 |
| Weight distribution (29.7); $[8, 4, 4]$ code                             | —    | symbolic checks (scripts)                                       |
| has 14 words of weight 4; all degrees divide                             |      |                                                                 |
| $ M_{24} $                                                               |      |                                                                 |
| $244\,823\,040$ is the order of $M_{24}$ ; 26 classes                    | [L]  | EOT App. A, ll. 331–358; App. B, ll. 959–979                    |
| and irreps; the dimensions (29.17) and the                               |      |                                                                 |
| conjugate pairs; uniqueness of $\mathcal{G}_{24}$ ;                      |      |                                                                 |
| $M_{24} = \text{Aut}(\mathcal{G}_{24})$                                  |      |                                                                 |
| The 26 class names; power-map and $M_{23}$                               | [L]  | GHV ll. 694–697, 808–813, 1916–1930                             |
| facts                                                                    |      |                                                                 |
| 5-transitivity; ATLAS orders of                                          | [L]  | standard; not in this book’s library                            |
| $M_{23}, M_{22}, M_{21}$ ; $M_{21} \cong PSL(3, 4)$ ; Steiner            |      |                                                                 |
| system; degrees divide the order                                         |      |                                                                 |
| Mukai’s theorem; Fermat quartic with 384                                 | [L]  | EOT App. B, ll. 984–1006                                        |
| symmetries and orbits 1, 1, 2, 4, 16                                     |      |                                                                 |
| Golay code as holographic error-correcting                               | [C]  | docstrings of Problem8_GolayHolography,                         |
| code on $K3 \times T^2$ ; 27720 as a “BPS lock”                          |      | Problem2_MathieuFrobeniusRigidity; not supported by any theorem |

### Chapter summary

- $M_{24}$  is the automorphism group of the extended binary Golay code  $\mathcal{G}_{24}$ , the unique doubly even self-dual  $[24, 12, 8]$  code; equivalently, the permutation symmetries of the Niemeier lattice  $A_1^{24} \subset N \subset (A_1^*)^{24}$ . The dictionary code  $\leftrightarrow$  lattice is: doubly even  $\leftrightarrow$  even, self-dual  $\leftrightarrow$  unimodular, no weight-4 words  $\leftrightarrow$  no extra roots. With  $n = 8$  the same recipe gives  $E_8$  with  $16 + 14 \cdot 16 = 240$  roots.
- $|M_{24}| = 2^{10} \cdot 3^3 \cdot 5 \cdot 7 \cdot 11 \cdot 23 = 244\,823\,040$ . The action on 24 points is 5-transitive, and fixing points gives  $M_{23}$  (10 200 960),  $M_{22}$  (443 520),  $M_{21} \cong PSL(3, 4)$  (20 160), with indices 24, 23, 22; five points have a stabilizer of order 48.
- There are 26 conjugacy classes and 26 irreducible representations, of dimensions (29.17), five complex-conjugate pairs among them. Burnside’s  $\sum \dim^2 = |G|$  holds, and it caught a wrong table in this project.
- Mukai: finite symplectic automorphism groups of K3 surfaces are the subgroups of

$M_{24}$  with at least five orbits on 24 points; never  $M_{24}$  itself.

- Lean has checked the arithmetic: the order factorization, the Burnside sum for the transcribed table, the tower indices, and the Golay numerals. No group, code or representation is constructed; the identification with  $M_{24}$  is Tier L.

## Exercises

**Exercise 29.1** (★ Lagrange at work). Using only (29.1), decide which of the following can be the order of an element of  $M_{24}$ : 9, 13, 16, 22, 25, 33, 46, 69. Then compare with the list of element orders read off (29.16): which orders are permitted by Lagrange but do not occur? (This shows that Lagrange gives necessary conditions only.)

**Exercise 29.2** (★ The Hamming code by hand). List all 16 codewords of the code  $C_8$  of Example 29.2 and verify the weight distribution 1, 14, 1. Check that the complement  $c \mapsto c + (1, \dots, 1)$  maps weight-4 words to weight-4 words, and use this to explain why the distribution is symmetric.

**Exercise 29.3** (★★ Construction A in general). Let  $C \subseteq \mathbb{F}_2^n$  be any binary code and  $N_C$  the lattice (29.2). (a) Show that  $\text{disc } N_C = 2^{n-2 \dim C}$ . (b) Show that  $N_C$  is integral iff  $C \subseteq C^\perp$ , and even iff  $C$  is doubly even. (c) Show that the minimum norm of a nonzero vector of  $N_C$  is  $\min(2, d/2)$  where  $d$  is the minimum distance, and conclude that for  $\mathcal{G}_{24}$  the minimum norm is 2 but no root comes from a nonzero codeword.

**Exercise 29.4** (★★ Burnside for a small group). The symmetric group  $S_4$  has five conjugacy classes (cycle types  $1^4, 2\,1^2, 2^2, 3\,1, 4$ ) and irreducible representations of dimensions 1, 1, 2, 3, 3. Verify (29.18). Then show that no list of five positive integers other than a permutation of 1, 1, 2, 3, 3 has sum of squares 24 *and* contains 1 twice (the trivial and sign representations); this is a toy version of the “strong evidence, not proof” remark of Section 29.4.

**Exercise 29.5** (★★ Five points). (a) Verify (29.15). (b) A group acting  $k$ -transitively on  $n$  points is *sharply*  $k$ -transitive if the pointwise stabilizer of  $k$  points is trivial. Show that  $M_{24}$  is not sharply 5-transitive but that, if it were sharply  $t$ -transitive for some  $t$ , its order would be  $24 \cdot 23 \cdots (25 - t)$ ; check that no  $t$  works. (c)  $M_{12}$ , of order 95 040 (standard value; not in this book’s library), is sharply 5-transitive on 12 points: verify  $12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 = 95\,040$ .

**Exercise 29.6** (★★ Orbits of the Fermat quartic group). The group  $(\mathbb{Z}_4)^2 \rtimes S_4$  of Section 29.5 has order 384 and orbits of lengths 1, 1, 2, 4, 16 on the 24 points. For each orbit, use orbit-stabilizer to compute the order of the stabilizer of a point in it, and check that all five are divisors of 384. Which orbit has the smallest stabilizer, and what does Mukai’s “at least five orbits” condition say about  $\text{rk } \Lambda^G$  here?

**Exercise 29.7** (★★ Lean: guard the tower table). In the style of `M21_order_factorization`, state and prove in Lean the factorizations `orderM23 = 27 · 32 · 5 · 7 · 11 · 23` and `orderM22 = 27 · 32 · 5 · 7 · 11` from (29.11)–(29.12) (`unfold` then `norm_num`). Explain why these two theorems would have caught a mistyped `orderM22` that the three index theorems alone might not (hint: the index theorems constrain the three constants only up to a common factor if all three are mistyped consistently).

**Exercise 29.8** (★★★ Lean: degrees divide the order). State the theorem  $\forall i : \text{Fin } 26, 244823040 \% M_{24}\text{RepDim } i = 0$  and prove it by `decide` or by `fin_cases i <=> norm_num [M24RepDim]`. Then explain in words why this theorem, like `M24RepDim_sum_sq`, is a statement about 26 numerals and not about representations of  $M_{24}$ , and why it is nevertheless

a useful guard (which transcription errors would it catch that the Burnside sum might not, and vice versa?).

**Exercise 29.9** (\*\*\* The wrong table, quantitatively). (a) Verify (29.19). (b) Suppose a single entry  $d$  of (29.17) is replaced by another entry  $d'$  of the same table (a duplication error). Show that the Burnside sum is unchanged iff  $d' = d$ , so *every* single duplication is caught. (c) Find the smallest change to the table that Burnside would *not* catch: two entries  $d_1, d_2$  replaced by  $d'_1, d'_2$  with  $d_1^2 + d_2^2 = d'^2_1 + d'^2_2$ , all four being positive integers dividing  $|M_{24}|$  (a script is acceptable; you may find none, in which case say what you searched).

## Further reading

- [EOT App. A, ll. 329–358] is the source of the order, the count of classes and representations, the dimension list (A.3) and the conjugate pairs (A.4); the full character table follows at ll. 360–925 but is garbled by text extraction and was not used in this chapter. [EOT App. B, ll. 955–1007] contains the Golay-code definition of  $M_{24}$  via  $A_1^{24}$  and the summary of Mukai’s theorem, with references to Mukai 1988 and Kondo 1998 at ll. 1054–1057.
- [GHV §3, ll. 500–575] restates the decompositions with the factor 2 absorbed; [GHV Table 1, ll. 1916–1930] gives the class names and the power maps used in Section 29.4; [GHV §3.1, ll. 808–813] identifies which classes meet  $M_{23}$ .
- Lean sources: `MathieuM24.lean` (directory `StringDynamics/`) and `MathieuTower.lean` (directory `UseCases/`), both in the library `StringTheoryFormalization`. The scope discussion at ll. 8–57 of the first file is a model of how a docstring should say what is and is not proved. [LL.md §S2.6, ll. 63–67] records the lesson of Section 29.6.
- The classification of finite simple groups, the ATLAS of finite groups, the Steiner system  $S(5, 8, 24)$  and the uniqueness of the Golay code are standard material of finite group theory and coding theory; none of it is in this book’s library, and a student who wants the proofs should consult a group-theory text.



## Chapter 30

# Mathieu Moonshine

In Chapter 28 we expanded the elliptic genus of a K3 surface in characters of the  $N = 4$  superconformal algebra and found that the multiplicities of the massive representations are the integers  $2A_n$  with  $A_n = 45, 231, 770, 2277, 5796, \dots$ . In Chapter 29 we met the sporadic group  $M_{24}$  of order 244 823 040 and its 26 irreducible representations. This chapter is about the fact that these two lists are the same list. The question it answers is: what exactly has been observed, how much of it is a coincidence that could be produced by small numbers, and which independent tests turn a numerological remark into a structural statement? Along the way the student learns the one technique that makes moonshine testable at all—replacing dimensions by characters and asking whether the resulting functions have the modular behaviour that a conformal field theory would force on them—and sees, at the end, exactly which pieces of the story are arithmetic that a proof assistant can check, and which pieces remain a conjecture of this programme or an open problem of the field.

Conventions. Throughout  $q = e^{2\pi i\tau}$  and  $y = e^{2\pi iz}$  with  $\tau$  in the upper half plane. The elliptic genus of a K3 sigma model is the Ramond–Ramond trace

$$Z(\tau, z) = \text{tr}_{RR} (-1)^{F_L + F_R} q^{L_0 - \frac{c}{24}} \bar{q}^{\bar{L}_0 - \frac{\bar{c}}{24}} y^{J_0}, \quad c = \bar{c} = 6, \quad (30.1)$$

where  $J_0$  is the integer-valued  $U(1)$  charge of the  $N = 2$  subalgebra,  $J_0 = 2J_0^3$  in terms of the  $SU(2)$  Cartan generator used by Eguchi–Ooguri–Tachikawa [EOT §1, ll. 44–56]. Our  $Z$  is EOT’s  $Z_{\text{ell}}$  and equals twice the standard weak Jacobi form  $\phi_{0,1}$ ; Gaberdiel–Hohenegger–Volpato work with  $\phi_{1A} := \frac{1}{2}Z = \phi_{0,1}$  [GHV §2, ll. 153–170]. The symbol  $A_n$  always denotes the EOT coefficients 45, 231,  $\dots$ ; GHV absorb a factor 2 into their  $A_n$  [GHV §3, ll. 540–545], and Section 30.1.3 explains why the book does not.

## 30.1 The observation

### 30.1.1 The decomposition, recalled

The K3 sigma model has  $N = (4, 4)$  superconformal symmetry with  $c = 6$ , and its Ramond sector Hilbert space decomposes into representations of the left-moving  $N = 4$  algebra Chapter 28. Two kinds of representation occur at  $c = 6$ : the massless (BPS, short) ones with  $h = \frac{1}{4}$  and isospin  $\ell = 0$  or  $\ell = \frac{1}{2}$ , and the massive (non-BPS, long) ones with  $h > \frac{1}{4}$  and  $\ell = \frac{1}{2}$ . Their characters with  $(-1)^F$  inserted are [EOT §1, ll. 82–124]

$$\text{ch}_{h=\frac{1}{4}, \ell=0}^{\hat{R}}(\tau; z) = \frac{\theta_1(\tau; z)^2}{\eta(\tau)^3} \mu(\tau; z), \quad \mu(\tau; z) = \frac{-i e^{\pi i z}}{\theta_1(\tau; z)} \sum_{n \in \mathbb{Z}} \frac{(-1)^n q^{n(n+1)/2} e^{2\pi i n z}}{1 - q^n e^{2\pi i z}}, \quad (30.2)$$

$$\text{ch}_{h=\frac{1}{2}, \ell=\frac{1}{2}}^{\hat{R}}(\tau; z) = q^{h-\frac{3}{8}} \frac{\theta_1(\tau; z)^2}{\eta(\tau)^3} \quad (h > \frac{1}{4}). \quad (30.3)$$

The function  $\mu$  is an Appell–Lerch sum, not a modular form: this is the technical reason why moonshine for K3 is harder than for the Monster, and we return to it in Section 30.2. The massless  $\ell = 0$  character has Witten index 1,  $\text{ch}_{\frac{1}{4},0}^{\hat{R}}(\tau; 0) = 1$  [EOT §1, ll. 98–104]. At the unitarity bound  $h = \frac{1}{4}$  the massive character (30.3) is not irreducible but splits,  $q^{-1/8}\theta_1^2/\eta^3 = 2\text{ch}_{\frac{1}{4},0}^{\hat{R}} + \text{ch}_{\frac{1}{4},\frac{1}{2}}^{\hat{R}}$  [EOT §1, ll. 140–150].

The decomposition of the elliptic genus reads, in the two equivalent forms of EOT,

$$Z(\tau, z) = 24 \text{ch}_{\frac{1}{4},0}^{\hat{R}}(\tau; z) + \Sigma(\tau) \frac{\theta_1(\tau; z)^2}{\eta(\tau)^3}, \quad \Sigma(\tau) = -2q^{-1/8} \left(1 - \sum_{n \geq 1} A_n q^n\right), \quad (30.4)$$

$$Z(\tau, z) = 20 \text{ch}_{\frac{1}{4},0}^{\hat{R}} - 2 \text{ch}_{\frac{1}{4},\frac{1}{2}}^{\hat{R}} + 2 \sum_{n \geq 1} A_n q^{n-\frac{1}{8}} \frac{\theta_1(\tau; z)^2}{\eta(\tau)^3} \quad (30.5)$$

[EOT §1, ll. 126–165]. The second form is obtained from the first by the splitting identity; it exhibits the multiplicity of the massive representation of weight  $h = n + \frac{1}{4}$  as  $2A_n$ , and the two massless multiplicities as 20 and  $-2$ . Expanding  $\Sigma$ , EOT tabulate [EOT eq. (1.12), ll. 200–210]

|       |    |     |     |      |      |       |       |       |        |
|-------|----|-----|-----|------|------|-------|-------|-------|--------|
| $n$   | 1  | 2   | 3   | 4    | 5    | 6     | 7     | 8     | 9      |
| $A_n$ | 45 | 231 | 770 | 2277 | 5796 | 13915 | 30843 | 65550 | 132825 |

(30.6)

The  $A_n$  are positive integers; their growth is governed by a Rademacher-type formula  $A_n \approx \frac{2}{\sqrt{8n-1}} e^{2\pi\sqrt{(n-1/8)/2}}$  which is accurate already for small  $n$  [EOT §1, ll. 212–226].

### 30.1.2 Twenty-six numbers

The group  $M_{24}$  has order  $2^{10} \cdot 3^3 \cdot 5 \cdot 7 \cdot 11 \cdot 23 = 244\,823\,040$ , 26 conjugacy classes and hence 26 irreducible representations, of dimensions [EOT App. A, ll. 330–349]

$$\begin{aligned} &1, 23, 45, 45, 231, 231, 252, 253, 483, 770, 770, 990, 990, 1035, 1035, 1035, \\ &1265, 1771, 2024, 2277, 3312, 3520, 5313, 5796, 5544, 10395. \end{aligned} \quad (30.7)$$

Comparing (30.6) with (30.7) one sees the observation of Eguchi, Ooguri and Tachikawa [EOT §1, ll. 227–235]:  $A_1, \dots, A_5$  are each the dimension of an irreducible representation of  $M_{24}$ , and the next two coefficients are sums of such dimensions,

$$A_6 = 13915 = 3520 + 10395, \quad A_7 = 30843 = 10395 + 5796 + 5544 + 5313 + 2024 + 1771. \quad (30.8)$$

For  $n \geq 8$  decompositions still exist but are far from unique; and indeed (30.8) for  $A_7$  is already not the decomposition EOT first proposed—it was corrected in the second version of their paper after the twining-genus analysis of Section 30.2 showed the original guess to be inconsistent [EOT footnote 1, ll. 259–261]. This is the first lesson of the chapter: dimensions alone cannot decide between competing decompositions; characters can.

#### Monstrous moonshine

The observation was made in deliberate analogy with McKay and Thompson’s remark that the coefficients of the modular  $J$ -function,  $J(q) = q^{-1} + 196884q + 21493760q^2 + \dots$ , are sums of dimensions of irreducible representations of the Monster group. Conway and Norton made it a conjecture about a graded Monster module  $\bigoplus_i V_i$  with  $\dim V_i$  the coefficients of  $J$ ; Frenkel, Lepowsky and Meurman then constructed that module as the chiral algebra of a string on  $\mathbb{R}^{24}/\Lambda_{\text{Leech}}/\mathbb{Z}_2$  [EOT §1, ll. 237–256]. For K3 the graded vector



space is given from the start—it is the Hilbert space of the sigma model—and the missing piece is the group action [EOT §1, ll. 264–268].

### 30.1.3 Conjugate pairs, real representations and the factor two

Of the 26 irreducibles, exactly those of dimension 45, 231, 770, 990 and 1035 come in complex-conjugate pairs  $(\rho, \bar{\rho})$ ; there is a third, real, 1035-dimensional representation, and every other irreducible is real [EOT App. A, ll. 352–356]. The multiplicity of the massive  $N = 4$  representation at level  $n$  is  $2A_n$ , not  $A_n$ , so the natural guess for the  $M_{24}$ -module  $R_n$  at level  $n$  is

$$R_1 = 45 \oplus \overline{45}, \quad R_2 = 231 \oplus \overline{231}, \quad R_3 = 770 \oplus \overline{770}, \quad R_4 = 2277 \oplus \overline{2277}, \quad R_5 = 5796 \oplus \overline{5796}, \quad (30.9)$$

of dimensions 90, 462, 1540, 4554, 11592 [GHV §3, ll. 527–560] (the extracted text of GHV prints “1440” for  $770 + 770$ ; it is 1540). Real modules are forced: the elliptic genus counts states of a Hilbert space with a real structure, so the  $M_{24}$ -representation on each graded piece is self-conjugate, and a complex irreducible can only occur together with its conjugate. This is also why the twining characters of conjugacy classes that differ only on conjugate representations coincide [GHV App. A.2, ll. 1605–1610].

4554 =  $2 \cdot 2277$ , **not**  $2277 \oplus \overline{2277}$

It is tempting to write every  $R_n$  as  $\rho \oplus \bar{\rho}$ . For  $n = 4$  this is wrong: **2277** is real, so **2277** =  $2277$  and  $R_4$  is two copies of one real irreducible. The distinction is invisible at the level of dimensions but visible at the level of characters (the two summands contribute equal characters rather than complex-conjugate ones). An earlier paper of this programme wrote  $4554 = 2277 \oplus \overline{2277}$ ; the correction is recorded in docs/PAPER\_REVISION\_BRIEF\_2026\_09\_17.md, item 3. The same document fixes the convention  $\mathcal{A}_n := 2A_n$  for the doubled multiplicities used in the older Lean libraries, so that they do not clash with EOT’s  $A_n$ .

### 30.1.4 Worked example: computing $A_n$ from the Lerch sum

The table (30.6) should not be taken on trust: pdf extraction garbles numbers, and the whole chapter rests on them. Here is a route that reproduces all nine values from (30.4) with exact rational arithmetic.

*Step 1: specialize to  $z = \frac{1}{2}$ .* Equation (30.4) holds for every  $z$ , so

$$\Sigma(\tau) = Z(\tau, z) \frac{\eta(\tau)^3}{\theta_1(\tau; z)^2} - 24\mu(\tau; z) \quad (30.10)$$

is independent of  $z$ , and we may choose  $z$  to make everything a  $q$ -series with rational coefficients. At  $z = \frac{1}{2}$ , i.e.  $y = -1$ , the product formulas [GHV App. A, ll. 1456–1478] give  $\theta_1(\tau; \frac{1}{2}) = \theta_2(\tau)$ ,  $\theta_2(\tau; \frac{1}{2}) = 0$ ,  $\theta_3(\tau; \frac{1}{2}) = \theta_4(\tau)$ ,  $\theta_4(\tau; \frac{1}{2}) = \theta_3(\tau)$ , where  $\theta_i(\tau) := \theta_i(\tau; 0)$ . In the Lerch sum (30.2) the prefactor becomes  $-ie^{\pi i/2} = 1$  and  $(-1)^n e^{2\pi i n/2} = 1$ , so

$$\mu(\tau; \tfrac{1}{2}) = \frac{S(q)}{\theta_2(\tau)}, \quad S(q) = \sum_{n \in \mathbb{Z}} \frac{q^{n(n+1)/2}}{1 + q^n} = \sum_{m \geq 0} q^{m(m+1)/2} \left[ \frac{1}{1 + q^m} + \frac{q^{m+1}}{1 + q^{m+1}} \right], \quad (30.11)$$

where the second form pairs  $n = m$  with  $n = -1 - m$ . The first terms are  $S = \frac{1}{2} + 2q - 2q^2 + 4q^3 - \dots$

*Step 2: the elliptic genus at  $z = \frac{1}{2}$ .* From  $Z = 8 \sum_{i=2,3,4} \theta_i(\tau; z)^2 / \theta_i(\tau)^2$  [EOT eq. (1.2), ll. 60–75] and the specializations above,

$$Z(\tau, \tfrac{1}{2}) = 8 \left[ \frac{\theta_4(\tau)^2}{\theta_3(\tau)^2} + \frac{\theta_3(\tau)^2}{\theta_4(\tau)^2} \right] = 16 + 512q + \dots, \quad (30.12)$$

which has integer powers of  $q$  only because the bracket is invariant under  $q^{1/2} \mapsto -q^{1/2}$ , which exchanges  $\theta_3$  and  $\theta_4$ . The constant 16 is the signature of K3 [EOT §1, ll. 78–80]; the 512 is  $20 + 128 + 216 + 128 + 20$ , the  $q^1$  coefficient of  $Z$  evaluated at  $y = -1$  (odd powers of  $y$  change sign), in agreement with the expansion  $Z = 2y + 20 + 2y^{-1} + q(20y^2 - 128y + 216 - 128y^{-1} + 20y^{-2}) + O(q^2)$  [GHV eq. (2.5), ll. 153–158], which we also re-derived symbolically from the theta products.

*Step 3: strip the fractional powers.* Write  $\theta_2(\tau) = 2q^{1/8}P_2(q)$  and  $\eta(\tau)^3 = q^{1/8}P_\eta(q)$  with

$$P_2 = \prod_{n \geq 1} (1 - q^n)(1 + q^n)^2 = 1 + q + q^3 + q^6 + \dots, \quad P_\eta = \prod_{n \geq 1} (1 - q^n)^3 = 1 - 3q + 5q^3 - 7q^6 + \dots$$

(the first is  $\frac{1}{2}q^{-1/8} \sum_{n \in \mathbb{Z}} q^{(n+1/2)^2/2}$ , the second is Jacobi's  $\sum_{n \geq 0} (-1)^n (2n+1) q^{n(n+1)/2}$ ). Multiplying (30.10) by  $q^{1/8}$  gives

$$q^{1/8} \Sigma(\tau) = Z(\tau, \tfrac{1}{2}) \frac{P_\eta}{4P_2^2} - 12 \frac{S}{P_2}, \quad (30.13)$$

and (30.4) predicts that the right-hand side equals  $-2 + 2 \sum_{n \geq 1} A_n q^n$ : every fractional power has cancelled, as it must.

*Step 4: the first two coefficients by hand.* The series needed are

$$\frac{P_\eta}{4P_2^2} = \frac{1}{4} - \frac{5}{4}q + \frac{9}{4}q^2 - \dots, \quad \frac{S}{P_2} = \frac{1}{2} + \frac{3}{2}q - \frac{7}{2}q^2 + \dots, \quad Z(\tau, \tfrac{1}{2}) = 16 + 512q + 4096q^2 + \dots$$

(for instance  $P_2^{-2} = 1 - 2q + 3q^2 - \dots$  and  $(1 - 3q)(1 - 2q + 3q^2) = 1 - 5q + 9q^2 + \dots$ ;  $(\frac{1}{2} + 2q - 2q^2)(1 - q + q^2) = \frac{1}{2} + \frac{3}{2}q - \frac{7}{2}q^2$ ). Collecting powers of  $q$  on the right of (30.13):

$$\begin{aligned} q^0 : & \quad 16 \cdot \frac{1}{4} - 12 \cdot \frac{1}{2} = 4 - 6 = -2, \\ q^1 : & \quad 16 \cdot \left(-\frac{5}{4}\right) + 512 \cdot \frac{1}{4} - 12 \cdot \frac{3}{2} = -20 + 128 - 18 = 90 = 2A_1, \\ q^2 : & \quad 16 \cdot \frac{9}{4} + 512 \cdot \left(-\frac{5}{4}\right) + 4096 \cdot \frac{1}{4} - 12 \cdot \left(-\frac{7}{2}\right) = 36 - 640 + 1024 + 42 = 462 = 2A_2. \end{aligned}$$

The constant term is the statement that the massless multiplicities are  $24 - 2 \cdot 2 = 20$  and  $-2$ ; the next two reproduce  $A_1 = 45$  and  $A_2 = 231$  from nothing but theta functions and the Lerch sum. A computer-algebra expansion of the exact right-hand side to order  $q^9$  (a symbolic check, not Tier A) gives

$$-2 + 90q + 462q^2 + 1540q^3 + 4554q^4 + 11592q^5 + 27830q^6 + 61686q^7 + 131100q^8 + 265650q^9,$$

which is  $-2 + 2 \sum_{n \leq 9} A_n q^n$  with precisely the nine values of (30.6). The lesson is methodological: derive the structure (30.13) by hand, check two coefficients by hand, and let the machine do the rest—then cite the table with confidence.

#### What the $A_n$ count

The  $2A_n$  massive  $N = 4$  multiplets at  $h = n + \frac{1}{4}$  are the left-moving BPS content of type II string theory on K3: states that are  $\frac{1}{2}$ -BPS with respect to the  $N = 2$  subalgebra, hence protected, hence countable by an index [GHV §3, ll. 500–505]. Their exponential

growth (30.6) is the growth of black-hole microstates: for the  $k$ -th symmetric product of K3 the analogous coefficients grow exponentially with a rate that at large  $k$  becomes the D1–D5 black-hole entropy  $S \approx 2\pi\sqrt{kn}$  with  $k = Q_1 Q_5$ ; the K3 elliptic genus itself is the  $Q_1 = Q_5 = 1$  member of that family, a “small” black hole [EOT §1, ll. 280–299]. Moonshine says that these microstates fall into representations of  $M_{24}$ —a symmetry no one had put in.

## 30.2 Characters, not dimensions: twining genera

### 30.2.1 The idea

If a finite group  $G$  really acts on the Hilbert space of a conformal field theory, commuting with the superconformal algebra, then for every  $g \in G$  one may insert  $g$  into the trace (30.1):

$$Z_g(\tau, z) = \text{tr}_{RR} g (-1)^{F_L + F_R} q^{L_0 - \frac{c}{24}} \bar{q}^{\bar{L}_0 - \frac{\bar{c}}{24}} y^{J_0} \quad (30.14)$$

[GHV eq. (3.5), ll. 563–580]. This is the *twining genus* of  $g$ ; it depends only on the conjugacy class of  $g$ . Decomposing the Hilbert space into  $N = 4$  representations as before, each multiplicity space is now a  $G$ -module, and the twining genus is obtained from (30.5) by replacing each dimension by the character of  $g$  on that module:

$$Z_g(\tau, z) = \text{tr}_{\mathbf{23} \oplus \mathbf{1}}(g) \text{ch}_{\frac{1}{4}, 0}^{\hat{R}} - 2 \text{ch}_{\frac{1}{4}, \frac{1}{2}}^{\hat{R}} + \sum_{n \geq 1} \text{tr}_{R_n}(g) q^{n - \frac{1}{8}} \frac{\theta_1(\tau; z)^2}{\eta(\tau)^3}, \quad (30.15)$$

where the 24 massless states are assigned the permutation representation  $\mathbf{23} \oplus \mathbf{1}$ , the  $-2$  is  $-(\mathbf{1} \oplus \mathbf{1})$ , and  $R_n$  are the modules (30.9) [GHV §3, ll. 527–560]. For the identity element this is (30.5). The point of the construction is that the left-hand side is a trace in a conformal field theory and therefore must transform well under a congruence subgroup of  $SL(2, \mathbb{Z})$ , while the right-hand side is computable, term by term, from the character table—so a proposed decomposition can be *falsified*. In monstrous moonshine the same construction produces the McKay–Thompson series [GHV §1, ll. 63–75].

### 30.2.2 Which modular group

The standard orbifold argument [GHV §3.1, ll. 596–720] runs as follows. Label the two cycles of the torus by elements of  $\{\pm 1\} \times M_{24}$ : the NS twining character  $\chi_g$  carries  $(-1, 1)$  on one cycle and  $(-1, g)$  on the other, the entries  $-1$  recording the NS sector and the absence of  $(-1)^F$ , the second entry the group element inserted in the trace. A modular transformation  $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$  maps the pair of labels  $(h, g)$  to  $(h^d g^c, g^a h^b)$ ; the twining character is mapped to itself (up to a phase) whenever

$$a + b \equiv c + d \equiv 1 \pmod{2}, \quad c \equiv 0 \pmod{o(g)}, \quad (30.16)$$

with  $o(g)$  the order of  $g$ . The subgroup so defined contains  $\Gamma_g = \Gamma_\theta \cap \Gamma_0(o(g))$ , where  $\Gamma_\theta = \langle T^2, S \rangle$  and  $\Gamma_0(N)$  is the usual congruence subgroup with  $c \equiv 0 \pmod{N}$ . (One uses that  $\gcd(a, o(g)) = 1$  follows from (30.16) and  $ad - bc = 1$ , and that in  $M_{24}$  the element  $g^a$  is then conjugate to  $g$ .) A function invariant under  $\Gamma_g$  up to a multiplier system is determined by finitely many of its  $q$ -coefficients, and the smaller  $o(g)$  the larger  $\Gamma_g$  and the fewer coefficients one needs. This is the test: compute the first few coefficients of  $Z_g$  from (30.15) and the character table, and ask whether they can be completed to a  $\Gamma_g$ -invariant function [GHV §3.1, ll. 700–720].

### 30.2.3 NS characters and the reconstruction formulas

GHV find it convenient to pass to the NS sector by half a unit of spectral flow,  $L_0 \mapsto L_0 + \ell J_0 + \ell^2 c/6$ ,  $J_0 \mapsto J_0 + \ell c/3$  with  $\ell = \frac{1}{2}$ , together with a shift  $z \mapsto z + \frac{1}{2}$  that removes the  $(-1)^F$  [GHV §2, ll. 172–190]:

$$\chi(\tau, z) = \exp\left[2\pi i\left(\frac{\tau}{4} + z + \frac{1}{2}\right)\right] \phi\left(\tau, z + \frac{\tau}{2} + \frac{1}{2}\right), \quad \phi = \frac{1}{2}Z, \quad (30.17)$$

and then set  $z = 0$ . The resulting  $\chi(\tau)$  is a modular function for  $\Gamma_\theta$  with multiplier  $-1$  on both generators (the index is  $m = 1$ , odd) [GHV eq. (2.10), ll. 213–222]. For K3,

$$\chi_{1A}(\tau) = \kappa(\tau) = \left(\frac{2\theta_4(\tau)}{\theta_2(\tau)}\right)^2 - \left(\frac{2\theta_2(\tau)}{\theta_4(\tau)}\right)^2 = q^{-1/4}(1 - 20q^{1/2} - 62q - 216q^{3/2} - 641q^2 - \dots) \quad (30.18)$$

[GHV eq. (2.11), ll. 228–236]. We verified (30.17) and (30.18) numerically at a generic point  $(\tau, z)$  with high-precision theta functions, and the expansion symbolically; the leading term  $q^{-1/4}(1 - 20q^{1/2})$  is quick by hand from  $\theta_2 = 2q^{1/8}(1 + q + \dots)$  and  $\theta_4 = 1 - 2q^{1/2} + 2q^2 - \dots$ : the first square gives  $q^{-1/4}(1 - 4q^{1/2})$  and the second  $-16q^{1/4} = -16q^{-1/4}q^{1/2}$ ; Exercise 30.3 continues the expansion.

Nothing is lost by setting  $z = 0$ . Because the four functions  $\theta_i(\tau, z)^2$  span a two-dimensional space, any index-one Jacobi form is fixed by its values at  $z = 0$  and  $z = \frac{1}{2}$ , and one has [GHV eqs. (2.12)–(2.13) and (3.6)–(3.7), ll. 254–292 and 596–630]

$$\chi_g(\tau, z) = \phi_g(\tau, 0) \frac{\theta_4(\tau, z)^2}{\theta_2(\tau)^2} - \phi_g(\tau, \frac{1}{2}) \frac{\theta_3(\tau, z)^2}{\theta_2(\tau)^2}, \quad (30.19)$$

$$\phi_g(\tau, z) = \phi_g(\tau, 0) \frac{\theta_2(\tau, z)^2}{\theta_2(\tau)^2} + \phi_g(\tau, \frac{1}{2}) \frac{\theta_1(\tau, z)^2}{\theta_2(\tau)^2}, \quad (30.20)$$

with  $\phi_g = \frac{1}{2}Z_g$  and the constant term  $\phi_g(\tau, 0) = \frac{1}{2} \text{tr}_{\mathbf{23} \oplus \mathbf{1}}(g)$ , half the number of points of  $\{1, \dots, 24\}$  fixed by  $g$  [GHV eq. (3.8), ll. 628–632]. Both formulas are checked at  $z = 0$  and  $z = \frac{1}{2}$  by inspection:  $\theta_2(\tau, \frac{1}{2}) = 0$ ,  $\theta_1(\tau, \frac{1}{2}) = \theta_2(\tau)$ ,  $\theta_3(\tau, \frac{1}{2}) = \theta_4(\tau)$ ,  $\theta_4(\tau, \frac{1}{2}) = \theta_3(\tau)$ . (The pdf extraction of the source interchanges the positions of  $\theta_3$  and  $\theta_4$  in (30.19); the assignment above is the one that reproduces (30.18) from  $\phi_{1A}(\tau, 0) = 12$  and  $\phi_{1A}(\tau, \frac{1}{2}) = 4(\theta_4^2/\theta_3^2 + \theta_3^2/\theta_4^2)$ , and the one we confirmed numerically.) Given  $\chi_g(\tau) = \chi_g(\tau, 0)$  and the fixed-point count, (30.19) at  $z = 0$  yields  $\phi_g(\tau, \frac{1}{2})$ , and (30.20) then yields the whole twining genus.

### 30.2.4 Closed forms for elements of small order

For the classes of order at most six GHV found closed formulas that match the low-lying coefficients and have the required modular behaviour [GHV eq. (3.14), ll. 730–775]:

$$\begin{aligned} \chi_{2A}(\tau) &= 4 \frac{\theta_4(\tau)^2}{\theta_2(\tau)^2}, & \chi_{2B}(\tau) &= 4 \frac{\theta_3(\tau)^2}{\theta_2(\tau)^2}, & \chi_{3A}(\tau) &= \left(\frac{\eta(\tau)}{\eta(3\tau)}\right)^3 \frac{\theta_3(3\tau)}{\theta_3(\tau)}, \\ \chi_{4A}(\tau) &= \left(\frac{\eta(\tau)\eta(2\tau)}{\eta(\tau/2)\eta(4\tau)}\right)^4, & \chi_{4B}(\tau) &= 2 \frac{\theta_3(2\tau)}{\theta_2(2\tau)}, & \chi_{6A}(\tau) &= \left(\frac{\eta(\frac{3}{2}\tau)\eta(2\tau)}{\eta(\frac{1}{2}\tau)\eta(6\tau)}\right)^2, \end{aligned} \quad (30.21)$$

each normalized to begin with  $1 \cdot q^{-1/4}$ ; there are also expressions for  $4C$  and  $5A$ . Most of them coincide, after  $\tau \mapsto \tau/4$ , with McKay–Thompson series of the Monster (for instance  $\kappa(\tau) = T_{[8a]}(\tau/4)$  with  $T_{[8a]} = q^{-1} - 20q - 62q^3 - 216q^5 - 641q^7 - \dots$ , which the reader can match against (30.18)), and for  $2A$ ,  $2B$  the match holds only up to signs [GHV App. A.1, ll. 1560–1600]. All are invariant under  $\Gamma_g$  up to a multiplier system, which is non-trivial in every case since  $T^2$  acts by  $-1$  [GHV §3.1, ll. 776–800]. The classes  $3B$  and  $6B$  resisted a closed form; their first coefficients are tabulated [GHV App. A.2, ll. 1605–1630], and the replication identities of Section 30.2.6 indicate that  $3B$  too behaves well.

### 30.2.5 Worked example: the 2A twining genus against the character table

Let us run the test ourselves for the class 2A, in the direction opposite to GHV's: take the closed form  $\chi_{2A}$  from (30.21), reconstruct the twining genus, expand, and compare with the characters of 2A on the modules (30.9). Two columns of the  $M_{24}$  character table are needed; we transcribe them from the table reproduced in GHV [GHV App., ll. 1928–2100] in the order of (30.7):

|            |      |      |      |      |      |      |      |      |      |      |      |      |       |
|------------|------|------|------|------|------|------|------|------|------|------|------|------|-------|
| dim        | 1    | 23   | 45   | 45   | 231  | 231  | 252  | 253  | 483  | 770  | 770  | 990  | 990   |
| $\chi(2A)$ | 1    | 7    | -3   | -3   | 7    | 7    | 28   | 13   | 35   | -14  | -14  | -18  | -18   |
| $\chi(3A)$ | 1    | 5    | 0    | 0    | -3   | -3   | 9    | 10   | 6    | 5    | 5    | 0    | 0     |
| dim        | 1035 | 1035 | 1035 | 1265 | 1771 | 2024 | 2277 | 3312 | 3520 | 5313 | 5796 | 5544 | 10395 |
| $\chi(2A)$ | -21  | -21  | 27   | 49   | -21  | 8    | 21   | 48   | 64   | 49   | -28  | -56  | -21   |
| $\chi(3A)$ | 0    | 0    | 0    | 5    | 16   | -1   | 0    | 0    | 10   | -15  | -9   | 9    | 0     |

(30.22)

Transcription errors are the enemy here, so we checked both columns against the orthogonality relations of character theory (standard; not checked against a source in this book's library):  $\sum_i \dim \rho_i \chi_i(g) = 0$  holds for both columns, and  $\sum_i \chi_i(g)^2 = |C_{M_{24}}(g)|$  gives 21504 for 2A and 1080 for 3A, both of which divide  $|M_{24}|$ , with quotients 11385 and 226688—the class sizes. A wrong entry would almost certainly break one of the four identities.

*Reconstruction.* The class 2A fixes  $1 + 7 = 8$  of the 24 points, so  $\phi_{2A}(\tau, 0) = 4$ . Inserting  $\chi_{2A} = 4\theta_4^2/\theta_2^2$  into (30.19) at  $z = 0$ ,

$$4 \frac{\theta_4^2}{\theta_2^2} = 4 \frac{\theta_4^2}{\theta_2^2} - \phi_{2A}(\tau, \tfrac{1}{2}) \frac{\theta_3^2}{\theta_2^2} \implies \phi_{2A}(\tau, \tfrac{1}{2}) = 0,$$

and (30.20) collapses to the remarkably simple

$$\phi_{2A}(\tau, z) = 4 \frac{\theta_2(\tau, z)^2}{\theta_2(\tau)^2} = (y + 2 + y^{-1}) + q(\dots) + \dots \quad (30.23)$$

Its constant term  $y + 2 + y^{-1}$  sums to  $4 = \frac{1}{2} \text{tr}_{\mathbf{23} \oplus \mathbf{1}}(2A)$ , distributed as  $1 + 2 + 1$  over  $J_0 = 1, 0, -1$ ; compare  $y + 10 + y^{-1}$  for the identity (half of  $2y + 20 + 2y^{-1}$ ), where  $12 = 1 + 10 + 1$ . The 2A trace over the twenty states of charge zero is thus 2, i.e. they split  $11 + 9$  between the eigenvalues  $\pm 1$ .

*Expansion.* Now specialize (30.15) to  $z = \frac{1}{2}$  exactly as in Section 30.1.4, with  $\text{tr}_{\mathbf{24}}(g)$  in place of 24 and  $c_n(g) := \text{tr}_{R_n}(g)$  in place of  $2A_n$ :

$$\sum_{n \geq 1} c_n(g) q^n = 2 + Z_g(\tau, \tfrac{1}{2}) \frac{P_\eta}{4P_2^2} - \text{tr}_{\mathbf{24}}(g) \frac{S}{2P_2}. \quad (30.24)$$

For  $g = 1$  this is (30.13) rearranged. For  $g \in 2A$  we have  $Z_{2A}(\tau, \frac{1}{2}) = 2\phi_{2A}(\tau, \frac{1}{2}) = 0$  and  $\text{tr}_{\mathbf{24}} = 8$ , so

$$\begin{aligned} \sum_{n \geq 1} c_n(2A) q^n &= 2 - 4 \frac{S}{P_2} = 2 - 4(\tfrac{1}{2} + \tfrac{3}{2}q - \tfrac{7}{2}q^2 + 7q^3 - \dots) \\ &= -6q + 14q^2 - 28q^3 + 42q^4 - 56q^5 + 86q^6 - 138q^7 + \dots \end{aligned} \quad (30.25)$$

(the coefficients beyond  $q^3$  from the symbolic expansion). The prediction from the character table is

$$\begin{aligned} c_1 &= 2\chi_{45}(2A) = -6, & c_2 &= 2\chi_{231}(2A) = 14, \\ c_3 &= 2\chi_{770}(2A) = -28, & c_4 &= 2\chi_{2277}(2A) = 42, \\ c_5 &= 2\chi_{5796}(2A) = -56, & c_6 &= 2(\chi_{3520} + \chi_{10395})(2A) = 2(64 - 21) = 86, \end{aligned}$$

and for the corrected  $A_7$  of (30.8),  $c_7 = 2(-21 - 28 - 56 + 49 + 8 - 21) = -138$ . All seven agree. Note what has been tested: not only that  $A_1, \dots, A_5$  are dimensions, but that the specific irreducibles chosen—including the choice of the real **2277** twice rather than some other combination of the same total dimension, and the six-term decomposition of  $A_7$ —have characters on  $2A$  that assemble into a function with the modular behaviour of a  $\mathbb{Z}_2$ -twined trace. Repeating the computation for  $3A$  with the closed form in (30.21),  $\text{tr}_{24}(3A) = 6$  and  $\phi_{3A}(\tau, \frac{1}{2}) = (3\theta_4^2 - \chi_{3A}\theta_2^2)/\theta_3^2$ , the symbolic expansion gives  $c_n(3A) = 0, -6, 10, 0, -18, 20, 0$  for  $n = 1, \dots, 7$ , against the table's  $2 \cdot (0, -3, 5, 0, -9, 10 + 0, 0 - 9 + 9 - 15 - 1 + 16) = (0, -6, 10, 0, -18, 20, 0)$ . Again every coefficient matches. The alternating signs of (30.25) have a meaning:  $2A$  acts on the level- $n$  multiplicity space with  $\frac{1}{2}(2A_n + c_n)$  eigenvalues  $+1$  and  $\frac{1}{2}(2A_n - c_n)$  eigenvalues  $-1$ , so at  $n = 1$  the 90 states split  $42 + 48$ .

### 30.2.6 Replication

The second test is borrowed from the replication identities of monstrous moonshine. If  $M_{24}$  acts on the K3 theory, it acts diagonally on the symmetric orbifolds  $\text{Sym}^p(\text{K3})$ , and the Hecke-transformed twining genera must be

$$(H_p \phi)_g(\tau, z) = \phi_{g^p}(p\tau, pz) + \sum_{l=0}^{p-1} \phi_g\left(\frac{\tau + l}{p}, z\right), \quad (30.26)$$

the first term from the untwisted sector with the cyclic permutation inserted (only  $u \otimes \dots \otimes u$  contributes, with  $g$  acting as  $g^p$ ), the others from the twisted sectors [GHV §3.2, ll. 830–870]. Translated into NS characters, the  $p = 2$  case expresses the twining character  $\chi_g^{(2)}$  of the Hecke part of  $\text{Sym}^2$  as

$$\chi_g^{(2)}(\tau) = \chi_{g^2}(2\tau, \frac{1}{2}) + \phi_g(\frac{\tau}{2}, \frac{1}{2}) - \phi_g(\tau, 0), \quad (30.27)$$

where  $\chi_{g^2}(\tau, z)$  is the NS-flowed twining genus (30.17) of  $g^2$  (the constant term is displayed by GHV as  $-4, 0, -3$  for  $2A, 2B, 3A$ , which is  $-\frac{1}{2}\text{tr}_{24}(g)$ ), and GHV find that the result is a polynomial in the original twining character, for instance

$$\chi_{2A}^{(2)}(\tau) = \chi_{2A}(\tau)^2 + 4, \quad \chi_{2B}^{(2)}(\tau) = \chi_{2B}(\tau)^2 - 12, \quad \chi_{3A}^{(2)}(\tau) = \chi_{3A}(\tau)^2, \quad (30.28)$$

with analogues for  $3B, 4A-4C, 5A, 6A, 7A$  [GHV eqs. (3.17)–(3.21), ll. 870–975]. We verified (30.27)–(30.28) for  $2A$  and  $3A$  numerically at a generic  $\tau$ , using the closed forms (30.21) and the reconstruction (30.19)–(30.20); for  $2A$  the middle term of (30.27) vanishes identically, so the identity reads  $\chi_{1A}(2\tau, \frac{1}{2}) - 4 = 16\theta_4(\tau)^4/\theta_2(\tau)^4 + 4$ . Unlike the Monster case the twining characters are not Hauptmoduls, so inhomogeneous terms (proportional to other McKay–Thompson series) can appear on the right [GHV §4, ll. 1425–1430]. The identities hold in every case examined and give independent evidence that even the classes without closed forms, such as  $3B$  and  $7A$ , have good modular properties [GHV §3.2, ll. 895–905].

## 30.3 Geometry: Mukai’s theorem and the puzzle

### 30.3.1 Why $M_{24}$ and K3 belong together at all

The group  $M_{24}$  can be defined in a way a string theorist finds natural [EOT App. B, ll. 951–982]. Take the root lattice  $A_1^{24}$ , whose generators have square 2, inside its dual  $(A_1^*)^{24}$ , whose generators have square  $\frac{1}{2}$ . An even unimodular lattice  $N$  with  $A_1^{24} \subset N \subset (A_1^*)^{24}$  is specified by the subspace  $\mathcal{G} = N/A_1^{24} \subset (\mathbb{Z}_2)^{24}$ : unimodularity forces  $\dim \mathcal{G} = 12$ , evenness forces every

element of  $\mathcal{G}$  to have weight divisible by 4, and demanding that the only vectors of square 2 in  $N$  be the roots of  $A_1^{24}$  forbids weight-4 words. These conditions determine  $\mathcal{G}$  uniquely: it is the extended binary Golay code, and  $M_{24}$  is its automorphism group inside  $S_{24}$ . The lattice  $N$  is a Niemeier lattice, and the chiral  $c = 24$  theory built on it has  $M_{24}$  as a discrete symmetry. The link to K3 is that  $H^*(K3, \mathbb{Z})$  with the Mukai pairing (Chapter 22) is also even and unimodular of rank 24, of signature  $(4, 20)$  [EOT App. B, ll. 983–990].

Now let  $G$  be a finite group of automorphisms of a K3 surface  $X$  preserving the holomorphic two-form (*symplectic* automorphisms). Let  $\Lambda^G$  be the  $G$ -invariant part of the Mukai lattice and  $\Lambda_G$  its orthogonal complement. Since  $G$  fixes  $H^0, H^4, H^{2,0}, H^{0,2}$  and the Kähler class,  $\Lambda^G$  has rank at least 5, so  $\Lambda_G$  has rank at most 19; and  $\Lambda_G$  lies in the primitive part of  $H^{1,1}$ , hence is negative definite and, by a result of Nikulin, embeds into the Niemeier lattice  $N$ . This makes  $G$  a subgroup of  $M_{24}$  whose action on 24 points has at least five orbits; conversely, the global Torelli theorem produces a K3 surface with symmetry  $G$  from any subgroup of  $M_{24}$  with at least five orbits [EOT App. B, ll. 991–1003]. The precise statement is Mukai's theorem, which Huybrechts states with  $M_{23}$  [Huy ch. 15 Thm. 3.1, ll. 14549–14560]:

**Theorem 30.1** (Mukai). [L] *For a finite group  $G$  the following are equivalent: (i)  $G$  is isomorphic to a subgroup of  $\text{Aut}_s(X)$  for some complex projective K3 surface  $X$ ; (ii) there is an injection  $G \hookrightarrow M_{23}$  such that the induced action on  $\Omega = \{1, \dots, 24\}$  has at least five orbits. There are exactly 11 maximal such groups, of orders 48, 72, 120, 168, 192, 288, 360, 384, 960 (some occurring twice), all realized on explicit quartics, complete intersections and double covers.*

Here  $M_{23} = \text{Stab}(e_{24}) \subset M_{24}$  is the stabilizer of a point, of index 24 and order  $|M_{23}| = 10\,200\,960 = 2^7 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11 \cdot 23$  [Huy ch. 15, ll. 14641–14645]. Any subgroup with a fixed point on  $\Omega$  is conjugate into  $M_{23}$ , which is why EOT's “subgroup of  $M_{24}$  with  $\geq 5$  orbits” and Huybrechts' “subgroup of  $M_{23}$  with  $\geq 5$  orbits” say the same thing. The Fermat quartic  $X^4 + Y^4 + Z^4 + W^4 = 0$  is the example EOT give: its symplectic group  $(\mathbb{Z}_4)^2 \rtimes S_4$  has order  $384 = 2^7 \cdot 3$  and splits  $\Omega$  into orbits of lengths 1, 1, 2, 4, 16 [EOT App. B, ll. 1004–1007]; 384 is indeed one of the eleven maximal orders in Theorem 30.1 (Figure 30.1).

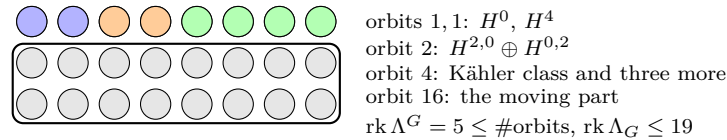


Figure 30.1: The 24 points of  $\Omega$  under the symplectic group of order 384 of the Fermat quartic, in orbits of length 1, 1, 2, 4, 16. Each orbit contributes one invariant vector, so the number of orbits bounds  $\text{rk } \Lambda^G$  from below; a symplectic group must have at least five orbits because  $H^0, H^4, H^{2,0}, H^{0,2}$  and the Kähler class are fixed. The assignment of individual cohomology classes to orbits in the picture is schematic; only the counts are meaningful.

### 30.3.2 Fixed points, the Witten index, and which classes are geometric

The constant term of the twining genus is a piece of classical geometry. At  $z = 0$  the twining genus reduces to  $\text{tr}_{24}(g) = \text{tr}_{H^*(X)}(g^*)$  (all cohomology of K3 sits in even degree, so the sign  $(-1)^{F_L + F_R}$  is +1 on every Ramond ground state), and for a symplectic automorphism  $f$  of finite order the topological Lefschetz formula turns this into the number of fixed points,  $\sum_i (-1)^i \text{tr } f^*|_{H^i(X, \mathbb{Q})} = |\text{Fix}(f)|$  [Huy ch. 15 eq. (1.2), ll. 14105–14108]. Mukai's fixed-point count

states [Huy ch. 15 Cor. 1.5, ll. 14032–14040]

$$|\text{Fix}(f)| = \frac{24}{n} \prod_{p|n} \left(1 + \frac{1}{p}\right)^{-1}, \quad n = |f|, \quad (30.29)$$

which depends only on the order, and lies between 1 and 8; the proof uses the holomorphic Lefschetz formula, whose left side equals 2 because  $f$  acts trivially on  $H^0(X, \mathcal{O})$  and  $H^2(X, \mathcal{O})$ . A Nikulin involution ( $n = 2$ ) has exactly eight fixed points [Huy ch. 15 §4, ll. 14913–14915].

Compare with the character table. The number of points of  $\Omega$  fixed by  $g$  is  $1 + \chi_{23}(g)$ , which from the 23-row of the table (the second row of (30.22), continued across all classes) equals 8, 6, 4, 4, 2, 3, 2 for the classes  $2A, 3A, 4B, 5A, 6A, 7A, 8A$  and 0 for  $2B, 3B, 4A, 4C, 12B, \dots$  [GHV App., ll. 1928–1970]. Evaluating (30.29) at  $n = 2, 3, 4, 5, 6, 7, 8$  gives 8, 6, 4, 4, 2, 3, 2—exactly the fixed-point counts of the classes that lie in  $M_{23}$ , which are those with a fixed point on  $\Omega$  [GHV §3.1, ll. 805–812]. The other classes fix no point of  $\Omega$  and thus cannot be the action of a symplectic automorphism, whose fixed set is non-empty by (30.29). So the twining genera fall into two families:

- classes in  $M_{23}$  ( $1A, 2A, 3A, 4B, 5A, 6A, 7A, 8A, \dots$ ):  $Z_g(\tau, 0) = |\text{Fix}(g)| > 0$ ; these can be realized geometrically, and for them the twining genus is the equivariant elliptic genus of a K3 surface with a symplectic automorphism;
- classes not in  $M_{23}$  ( $2B, 3B, 4A, 4C, \dots$ ):  $\text{tr}_{23 \oplus 1}(g) = 0$ , so by (30.19)–(30.20)

$$\phi_g(\tau, z) = \phi_g(\tau, \tfrac{1}{2}) \frac{\theta_1(\tau, z)^2}{\theta_2(\tau)^2} = -\chi_g(\tau) \frac{\theta_1(\tau, z)^2}{\theta_3(\tau)^2}$$

[GHV eq. (3.15), ll. 812–822]; these have no fixed points and no realization as automorphisms of a K3 surface.

GHV note that apart from this simplification they see no difference in behaviour between the two families [GHV §3.1, ll. 822–826]. That is the puzzle.

### 30.3.3 The puzzle

The observation and its tests say that the graded space of  $N = 2$  half-BPS states of the K3 sigma model looks, coefficient by coefficient, like an  $M_{24}$ -module. Geometry says that no K3 surface has more than a group of order 960 of symplectic automorphisms, and that these groups never contain the classes outside  $M_{23}$ ; EOT already remark that  $G$  “cannot be  $M_{24}$  itself” since a geometric action fixes at least five classes [EOT App. B, ll. 996–999]. Their proposed way out is that automorphism groups at isolated points of the moduli space might be “enhanced to  $M_{24}$  over the whole of moduli space when we consider the elliptic genus” [EOT §1, ll. 271–276]: the elliptic genus is constant on the connected moduli space of K3 sigma models (Chapter 23), so a symmetry visible only at special points might still organize the index. The library of this book does not contain the later literature on the symmetry groups of K3 sigma models, so we state its outcome as standard, not checked against a source in this book’s library: the symmetry group of any individual K3 sigma model is a subgroup of the Conway group  $\text{Co}_0$ , and none of them contains  $M_{24}$ . Thus the  $M_{24}$  action is not the symmetry of any one point in moduli space, and where it comes from remains open.

#### Small numbers make coincidences

The dimensions of  $M_{24}$ -irreducibles below  $10^4$  are dense enough that many integers of that size are sums of a few of them; GHV stress this “certain amount of arbitrariness”



[GHV §1, ll. 63–68]. The observation for  $A_1, \dots, A_5$  has content because each is a *single* irreducible dimension (not a sum), and because five hits in a row among 26 values is unlikely by chance; the decompositions of  $A_6, A_7$  have content only after the twining test. A claim of the form “ $N$  is a sum of  $M_{24}$  dimensions” for a single large  $N$  has essentially no content. The same caution applies to any numerology in this book, including the ratio  $77/60$  discussed in Section 30.5.

## 30.4 What is known and what is open

At the time of the two source papers the situation was: the observation (EOT); closed forms for the twining characters of about half the conjugacy classes, all of the form required by the orbifold argument; the corrected decomposition of  $A_7$ ; replication identities for  $p = 2, 3, 5$  tested in many examples; and the expectation that the space of half-BPS states carries an  $M_{24}$  action which should extend to the BPS algebra of Harvey and Moore, in analogy with the Monster Lie algebra [GHV §4, ll. 1419–1445]. GHV list as the next step finding closed forms for all twining characters, presumably by more refined techniques than “guessing and confirming”.

What happened afterwards is standard, not checked against a source in this book’s library, and we record it only in outline so that the student knows where the story stands: the remaining twining genera were determined (Cheng; Gaberdiel, Hohenegger and Volpato; Eguchi and Hikami), and Gannon proved that the resulting virtual  $M_{24}$ -modules are honest modules, i.e. that all the multiplicities decompose into non-negative integer combinations of irreducibles. That settles the *existence* of the graded  $M_{24}$ -module. Still open, as far as the sources of this book go, are the two questions EOT posed on their first page: the reason for the action, and a construction of it on the K3 Hilbert space comparable to the Frenkel–Lepowsky–Meurman construction for the Monster.

*Remark 30.2* (What this programme adds, and what it does not). The Lean libraries described in the next section formalize the *arithmetic* of Section 30.1: that the listed  $A_n$  are entries and sums of entries of the listed table, and that the table satisfies Burnside’s identity. They do not formalize theta functions, Lerch sums,  $N = 4$  characters, the twining test, or Gannon’s theorem; those remain [L]. The “BPS ratio”  $77/60$  that several of the older libraries carry is a naming of this programme, [C], and is discussed as such below.

## 30.5 What the machine has checked

Five Lean files touch this chapter. One of them (`DualScaleStream2/Moonshine/EOT.lean`) formalizes the arithmetic of the observation against a certified table; three carry the “BPS ratio” of this programme; the fifth is unrelated to moonshine and sits in the `Moonshine` namespace by accident of history. All theorems below depend only on `propext`, `Classical.choice` and `Quot.sound`.

### 30.5.1 The table and the observation

The dimensions (30.7) live in Stream 1 as a literal function on `Fin 26`, in EOT’s order, guarded by Burnside’s identity:

```
def M24RepDim : Fin 26 → ℕ
| ⟨0, _⟩ => 1
| ⟨1, _⟩ => 23
| ⟨2, _⟩ => 45
```

```

...
| ⟨25, _⟩ => 10395
| ⟨n + 26, h⟩ => absurd h (by omega)

theorem M24_order : (244823040 : ℕ) = 2^10 * 3^3 * 5 * 7 * 11 * 23 := by ...

theorem M24RepDim_sum_sq : (∑ i : Fin 26, M24RepDim i ^ 2) = 244823040 := by ...

```

(StringTheoryFormalization/StringDynamics/MathieuM24.lean.) The table had once been mistyped—two entries duplicated, two missing—and the sum of squares then came out as 351 061 029; the theorem `M24RepDim_sum_sq` now makes such a slip a compile error. Burnside’s  $\sum_i (\dim \rho_i)^2 = |G|$  itself is [L]; the equality of the two numerals is [A].

#### In Lean: the first five coefficients are irreducible dimensions

```

def eotA : Fin 9 → ℕ := ![45, 231, 770, 2277, 5796, 13915, 30843, 65550, 132825]

theorem first_five_are_irreps :
  ∀ n : Fin 5, ∃ i : Fin 26, M24RepDim i = eotA (Fin.castLE (by norm_num) n) := by ...

theorem A6_decomposition :
  eotA 5 = M24RepDim 21 + M24RepDim 25 := rfl

theorem A7_decomposition :
  eotA 6 = M24RepDim 25 + M24RepDim 23 + M24RepDim 24 + M24RepDim 22 +
    M24RepDim 18 + M24RepDim 17 := rfl

theorem A6_not_irrep : ∀ i : Fin 26, M24RepDim i ≠ eotA 5 := by ...

```

**Reading.** `eotA` is the literal list (30.6). For each of the first five indices there exists an index into the 26-entry table with the same value (an existence statement with explicit witnesses 2, 4, 9, 19, 23, i.e. 45, 231, 770, 2277, 5796). The sixth and seventh entries equal the specific sums (30.8) of table entries. No table entry equals 13915, so the decomposition of  $A_6$  is not vacuous. **Proof idea.** `fin_cases` on the finite index and `rfl` on literal naturals; `A6_not_irrep` is a 26-way case split closed by `decide`. **What is not proved.** That  $M_{24}$  acts on anything; that the  $A_n$  are coefficients of the elliptic genus (that is (30.13), a statement about theta functions, [L]); anything about  $A_8, A_9$  beyond their listing. **Tier [A]** for the arithmetic, [L] for its meaning.

### 30.5.2 The “BPS ratio” 77/60

Three older libraries define the doubled multiplicities and a ratio built from them:

```

-- DualScaleM24Formalization/Moonshine/MathieuRigidity.lean
def dimA1 : Nat := 90
def dimA2 : Nat := 462
def dimA3 : Nat := 1540
theorem dimA1_decomposition : dimA1 = 45 + 45 := by decide
def numSupercharges : Nat := 4
def rBPSNum : Nat := 77
def rBPSTen : Nat := 60
theorem r_bps_cross_multiplication :
  dimA2 * rBPSTen = (numSupercharges * dimA1) * rBPSNum := by decide
theorem r_bps_is_irreducible : Nat.gcd rBPSNum rBPSTen = 1 := by decide
theorem sym2_A1_dimension_is_4095 : dimSym2A1 = 4095 := by decide

-- DualScaleValidation/UseCase2_MoonshineBPS.lean
theorem bps_cross_multiplication_lock :

```

```

dim_A2 * 60 = 27720 ∧ (4 * dim_A1) * 77 = 27720 := by ...
theorem bps_lock_exact_equality : dim_A2 * 60 = (4 * dim_A1) * 77 := rfl
theorem bps_ratio_coprime : Nat.gcd 77 60 = 1 := by decide
theorem m24_order_divisible_by_bps_lock :
  order_M24 = bps_character_lock * 8832 := rfl

-- StringTheoryFormalization/StringDynamics/BPSMultiplicities.lean
def bpsRatio : ℚ := 77 / 60
theorem bps_ratio_reduced : bpsRatio.num = 77 ∧ bpsRatio.den = 60 := by ...
theorem bps_ratio_pos : (0 : ℚ) < bpsRatio := by ...
noncomputable def bpsMultiplicity (n : ℕ) : ℝ :=
  Real.exp (4 * Real.pi * Real.sqrt n) / (n : ℝ) ^ (3 / 2 : ℝ)

```

#### In Lean: what the “lock” theorems literally say

**Reading.**  $462 \cdot 60 = 27720 = (4 \cdot 90) \cdot 77$ ;  $\gcd(77, 60) = 1$ ;  $244\,823\,040 = 27720 \cdot 8832$ ;  $\binom{91}{2} = 4095$ ; the rational literal  $77/60$  has numerator 77, denominator 60 and is positive. Every one of these is a true statement about specific integers, closed by **decide**, **rfl** or **norm\_num**. **Where  $77/60$  comes from.**  $A_2/(4A_1) = 231/180 = 77/60$ , equivalently  $\mathcal{A}_2/(4\mathcal{A}_1)$  with the doubled multiplicities; the “4” is called the number of supercharges in the docstrings. The identity is recorded in this programme’s `LL.md` (l. 405); no source in the book’s library contains it. **What is not proved.** Nothing connects `bpsRatio` to `M24RepDim` or to `eotA` inside Lean; `dimA1`, `dim_A1`, `rBPSNum` are bare literals. The definition `bpsMultiplicity` is a formula with no theorem about it—not positivity, not a growth rate, not a relation to (30.6)—and its docstring’s attribution to a Hardy–Ramanujan asymptotic is not pinned to a source; the actual asymptotic of the  $A_n$  is EOT’s Rademacher-type formula quoted after (30.6), with a different exponent. **Tier [A]** for the integer identities; **[C]** for the words “BPS ratio”, “character lock” and “rigidity”.

#### “Rigid under moduli deformation” is empty for an integer

The docstrings of these files say that because 27720 is an integer the ratio  $77/60$  “cannot continuously deform under any moduli variation of  $K3 \times T^2$ ”. Every integer is invariant under continuous deformation; the statement has no content beyond the fact that BPS multiplicities are integers, which is the standard reason the elliptic genus is a deformation invariant (Chapter 28). Likewise  $|M_{24}|/27720 = 8832$  is a divisibility fact about the numeral  $2^{10}3^3 \cdot 5 \cdot 7 \cdot 11 \cdot 23$  and  $27720 = 2^3 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11$ ; it says nothing about a decomposition of any Hilbert space into “8832 copies” of anything. When a student meets a theorem whose proof is **decide**, the right question is always: what would have to be true for this to be *false*? If the answer is “a different integer”, the theorem is a table lookup.

### 30.5.3 A file in the wrong drawer

`DualScaleM24Formalization/Moonshine/GysinBEMSequence.lean` is listed with this chapter because its namespace is `SocrateAI.Moonshine.GysinSequence`; its content is topological T-duality for circle bundles (the Bouwknegt–Evslin–Mathai exchange  $c_1(\hat{E}) = \pi_* H$ ,  $\hat{\pi}_* \hat{H} = c_1(E)$ ), which belongs with Chapter 9 and Chapter 13, and it has nothing to do with  $M_{24}$ . For completeness:

```

structure BEMDualPair (cr : CohomologyRing) where
  E : CircleBundle cr
  E_dual : CircleBundle cr
  H : Int
  H_dual : Int

```

```

bem_dual_c1 : E_dual.c1 = E.pi_push H
bem_dual_H   : E_dual.pi_push H_dual = E.c1

theorem bem_derived_exchange (cr : CohomologyRing) (pair : BEMDualPair cr) :
  pair.E_dual.c1 = pair.E.pi_push pair.H ∧
  pair.E_dual.pi_push pair.H_dual = pair.E.c1 := ⟨pair.bem_dual_c1, pair.bem_dual_H⟩

theorem bem_characteristic_vanishing (cr : CohomologyRing) (pair : BEMDualPair cr)
  (x : Int) :
  x * pair.E.c1 * pair.E_dual.c1 = x * pair.E.c1 * pair.E.pi_push pair.H := by ...

```

**Reading.** Cohomology groups are modelled by single integers and the Gysin maps by arbitrary functions  $\mathbb{Z} \rightarrow \mathbb{Z}$ ; the BEM relations are *fields* of the structure, i.e. hypotheses. The first theorem returns those two hypotheses as a conjunction; the second rewrites  $c_1(\hat{E})$  by the first hypothesis inside a product. Neither the vanishing  $c_1(E) \smile c_1(\hat{E}) = 0$  nor exactness of the Gysin sequence is proved (the exactness field `h_exact_push_pull` is never used). **Tier [A]** as trivial consequences of hypotheses about integers; the BEM theorem itself is **[L]** (not in the book’s library); the identification with string theory on  $K3 \times T^2$  is **[C]**.

| Claim                                                                                                    | Where                                                                                          | Tier                |
|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|---------------------|
| $\sum_{i=1}^{26} (\dim \rho_i)^2 = 244\,823\,040 = 2^{10} 3^3 \cdot 5 \cdot 7 \cdot 11 \cdot 23$         | M24RepDim_sum_sq, M24_order                                                                    | <b>[A]</b>          |
| $A_1, \dots, A_5 \in \{\dim \rho_i\}$ ; $A_6 = 3520 + 10395$ ; $A_7$ six-term sum; $A_6$ not a dimension | first_five_are_irreps,<br>A6_decomposition,<br>A7_decomposition,<br>A6_not_irrep               | <b>[A]</b>          |
| $462 \cdot 60 = 360 \cdot 77 = 27720$ , $\gcd(77, 60) = 1$ , $\binom{91}{2} = 4095$                      | r_bps_cross_multiplication,<br>bps_lock_exact_equality,<br>bps_ratio_coprime,<br>sym2_A1_value | <b>[A]</b>          |
| $ M_{24}  = 27720 \cdot 8832$                                                                            | m24_order_divisible_by_bps_lock                                                                | <b>[A]</b>          |
| $77/60$ in lowest terms, positive                                                                        | bps_ratio_reduced,<br>bps_ratio_pos                                                            | <b>[A]</b>          |
| BEM exchange as hypotheses; rewrite in a product                                                         | bem_derived_exchange,<br>bem_characteristic_vanishing                                          | <b>[A]</b> (shadow) |
| The 26 dimensions; conjugate pairs                                                                       | EOT App. A                                                                                     | <b>[L]</b>          |
| 45, 231, 770, 990, 1035                                                                                  |                                                                                                |                     |
| $Z = 24 \operatorname{ch}_{\frac{1}{4}, 0} + \Sigma \theta_1^2 / \eta^3$ , $A_n$ table                   | EOT §1                                                                                         | <b>[L]</b>          |
| Twining genera, $\Gamma_g$ -invariance, closed forms, replication                                        | GHV §§2–3                                                                                      | <b>[L]</b>          |
| Mukai: $\operatorname{Aut}_s(X) \subset M_{23}$ with $\geq 5$ orbits; $ \operatorname{Fix}(f) $ formula  | Huy ch. 15                                                                                     | <b>[L]</b>          |
| Existence of the $M_{24}$ -module (Gannon); no sigma model with $M_{24}$ symmetry                        | standard, not in library                                                                       | <b>[L]</b>          |
| “BPS ratio” $77/60$ , “character lock” 27720, <code>bpsMultiplicity</code>                               | this programme                                                                                 | <b>[C]</b>          |
| Origin of the $M_{24}$ action on the K3 Hilbert space                                                    | open                                                                                           | —                   |

Table 30.1: Tier table for this chapter. All **[A]** entries depend only on the axioms `propext`, `Classical.choice`, `Quot.sound`.

#### Chapter summary

- The K3 elliptic genus decomposes as  $20 \operatorname{ch}_{\frac{1}{4}, 0} - 2 \operatorname{ch}_{\frac{1}{4}, \frac{1}{2}} + 2 \sum_n A_n q^{n-1/8} \theta_1^2 / \eta^3$  with  $A_n = 45, 231, 770, 2277, 5796, 13915, 30843, \dots$ ; the massive multiplicities are  $2A_n$ .

- $A_1, \dots, A_5$  are dimensions of irreducible representations of  $M_{24}$ ;  $A_6 = 3520 + 10395$ ,  $A_7 = 10395 + 5796 + 5544 + 5313 + 2024 + 1771$ . The modules are real:  $R_1 = \mathbf{45} \oplus \overline{\mathbf{45}}$  but  $R_4 = 2 \cdot \mathbf{2277}$  since  $\mathbf{2277}$  is real.
- The  $A_n$  can be recomputed exactly from  $q^{1/8}\Sigma = Z(\tau, \frac{1}{2})P_\eta/(4P_2^2) - 12S/P_2$  with  $S = \sum_n q^{n(n+1)/2}/(1+q^n)$ .
- Twining genera  $Z_g$  replace dimensions by characters and must be invariant (up to a multiplier) under  $\Gamma_g = \Gamma_\theta \cap \Gamma_0(o(g))$ ; closed forms exist for small orders, e.g.  $\phi_{2A} = 4\theta_2(\tau, z)^2/\theta_2(\tau)^2$ , and they reproduce the characters  $-6, 14, -28, 42, -56, 86, -138$  of  $2A$  on  $R_1, \dots, R_7$ . This test fixed  $A_7$ .
- $Z_g(\tau, 0) = |\text{Fix}(g)| = \frac{24}{n} \prod_{p|n} (1 + \frac{1}{p})^{-1}$  for the classes in  $M_{23}$ ; the classes outside  $M_{23}$  fix nothing and are not geometric. Mukai: symplectic automorphism groups of K3 surfaces are the subgroups of  $M_{23}$  with at least five orbits; the maximal ones have orders  $48, \dots, 960$ .
- Open: why  $M_{24}$  acts, and a construction of the action. In Lean: the arithmetic of the observation ([A]) against a Burnside-guarded table;  $77/60$  and  $27720$  are integer identities ([A]) with a [C] interpretation.

## Exercises

**Exercise 30.1** (\*). Extend the hand computation of Section 30.1.4 to the coefficient of  $q^3$ : compute  $P_2, P_\eta, S$  and  $Z(\tau, \frac{1}{2})$  to order  $q^3$  and verify  $2A_3 = 1540$ . (You will need  $\theta_3^2/\theta_4^2 + \theta_4^2/\theta_3^2$  to order  $q^3$ ; the answer for  $Z(\tau, \frac{1}{2})$  is  $16 + 512q + 4096q^2 + 22528q^3$ .)

**Exercise 30.2** (\*). Using the dimensions (30.7), verify Burnside's identity  $\sum (\dim \rho_i)^2 = |M_{24}|$  by hand or by a short program, and then show that the mistyped table (with 483 and 10395 twice and one 990 and one 1035 missing) gives 351 061 029. How large a typo would the check fail to detect?

**Exercise 30.3** (\*\*). Derive the leading terms of (30.18) through order  $q^{1/2}$  from the  $q$ -expansions of  $\theta_2$  and  $\theta_4$ , then verify numerically, for one value of  $\tau$ , that  $\kappa(\tau) = \chi(\tau, 0)$  with  $\chi$  defined by the spectral flow (30.17) applied to  $\phi = 4 \sum_i \theta_i(\tau, z)^2/\theta_i(\tau)^2$ .

**Exercise 30.4** (\*\*). Use (30.29) to compute  $|\text{Fix}(f)|$  for  $n = 2, \dots, 8$  and compare with  $1 + \chi_{23}(g)$  for the classes  $2A, 3A, 4B, 5A, 6A, 7A, 8A$  of (30.22) and the 23-row of the table. Why can a symplectic automorphism of order  $n$  not have a fixed set of Euler characteristic 0? (Use the holomorphic Lefschetz formula and the trivial action on  $H^2(X, \mathcal{O})$ .)

**Exercise 30.5** (\*\*). For the class  $2B$  one has  $\text{tr}_{24}(2B) = 0$  and  $\chi_{2B} = 4\theta_3^2/\theta_2^2$ . Show from (30.19) at  $z = 0$  that  $\phi_{2B}(\tau, \frac{1}{2}) = -4$ , a constant, and hence from (30.20) that  $\phi_{2B}(\tau, z) = -4\theta_1(\tau, z)^2/\theta_2(\tau)^2$ ; check that this is GHV's eq. (3.15) form  $-\chi_g\theta_1^2/\theta_3^2$ . Then show from (30.24) that  $\sum_n c_n(2B)q^n = 2 - 2P_\eta/P_2^2 = 10q - 18q^2 + \dots$  and compare with  $2\chi_{45}(2B) = 2 \cdot 5$  and  $2\chi_{231}(2B) = 2 \cdot (-9)$  (the  $2B$  column of the table gives  $\chi_{45}(2B) = 5, \chi_{231}(2B) = -9$ ).

**Exercise 30.6** (\*\*). Show that for  $g$  of order two the multiplicity space  $R_n$  splits into eigenspaces of dimensions  $\frac{1}{2}(2A_n \pm c_n(g))$ , and tabulate these for  $2A$  and  $n \leq 5$ . What constraint does integrality of these dimensions impose on the parity of  $c_n$ ?

**Exercise 30.7** (\*\*). In Lean, state and prove `A8_decomposition`: 65550 equals some sum of entries of `M24RepDim` (find one; it is not unique). Then state, without proving, what a

theorem `A8_decomposition_unique` would have to say, and explain why such a statement would be false.

**Exercise 30.8** (★★). Write a Lean definition `chi2A : Fin 26 → ℤ` holding the  $2A$  column of (30.22) in the order of `M24RepDim`, and prove  $\sum_i \text{M24RepDim } i \cdot \text{chi2A } i = 0$  and  $\sum_i (\text{chi2A } i)^2 = 21504$  by `decide` or `simp [Fin.sum_univ_succ]`. Then prove `c7_2A : 2 * (chi2A 25 + chi2A 23 + chi2A 24 + chi2A 22 + chi2A 18 + chi2A 17) = -138`. Which parts of Section 30.2.5 would still *not* be formalized?

**Exercise 30.9** (★★★). Starting from (30.26) with  $p = 2$  and the spectral flow (30.17), derive an expression for  $\chi_{2A}^{(2)}(\tau)$  in terms of  $\chi_{1A}(2\tau, \cdot)$  and  $\phi_{2A}(\tau/2, \cdot)$  (this is GHV’s eq. (3.17)); then expand both sides of  $\chi_{2A}^{(2)} = \chi_{2A}^2 + 4$  in (30.28) to a few orders using the closed forms (30.18) and (30.21). Explain why  $\phi_{2A}(\tau, \frac{1}{2}) = 0$  makes the  $2A$  case especially simple, and why  $\chi_{2A}^2 + 4$  has integer coefficients and starts at  $q^{-1/2}$ .

## Further reading

- The observation, the  $N = 4$  characters and the table of  $A_n$ : [EOT §1, ll. 44–235]; the corrected  $A_7$ : [EOT footnote 1, ll. 259–261]; the character table and the list of irreducibles: [EOT App. A, ll. 330–356];  $M_{24}$  from the Golay code and Mukai’s argument: [EOT App. B, ll. 951–1008].
- Elliptic genus conventions, spectral flow and the NS character  $\kappa$ : [GHV §2, ll. 104–292]; twining genera, the orbifold argument for  $\Gamma_g$  and the closed forms: [GHV §3–3.1, ll. 500–826]; replication: [GHV §3.2, ll. 827–1418]; conclusions: [GHV §4, ll. 1419–1445]; theta conventions, McKay–Thompson series and the character table: [GHV App. A, ll. 1454–2100].
- Symplectic automorphisms, the fixed-point formula and Mukai’s theorem: [Huy ch. 15 §1, ll. 14020–14110] and [Huy ch. 15 §3, ll. 14545–14650]; Nikulin involutions: [Huy ch. 15 §4, ll. 14901–14960].
- The elliptic genus and  $N = 4$  characters are developed in Chapter 28; the group  $M_{24}$ , its Golay code and its character table in Chapter 29; the Mukai lattice in Chapter 22; the moduli space on which the elliptic genus is constant in Chapter 23. The project record of the 2277 correction is `docs/PAPER_REVISION_BRIEF_2026_09_17.md`, item 3.

**Part VII**

**The Dual-Scale Theory**





# Chapter 31

## The Dual-Scale Principle

Every chapter of Part II ended with the same observation: a closed string on a circle of radius  $R$  cannot be told apart from a closed string on a circle of radius  $\alpha'/R$ . This chapter takes that observation as a starting point rather than as a punch line, and asks what follows for the notion of “size” itself. The answer proposed by this programme, and called the *dual-scale principle*, is easy to state: the geometry that a string probes is not described by one number  $R$  but by the pair  $(R, \alpha'/R)$ , and in  $d$  dimensions not by one matrix  $G$  but by the pair  $(G, G^{-1})$ ; every quantity that can be measured is a function of the pair which does not change when the two members are exchanged. The simplest such function is the sum,  $R_{\text{eff}} = R + \alpha'/R$ , and the simplest fact about the sum is that it is never smaller than  $2\sqrt{\alpha'}$ . In  $d$  dimensions the sum becomes  $\text{tr } G + \text{tr } G^{-1}$  and the bound becomes  $2d$ .

A student should care for three reasons. First, the bound is a genuine theorem about positive definite matrices, with a two-line proof once one knows the spectral theorem and a slightly longer proof that avoids it; the second proof is the one that has been checked by the Lean 4 kernel, for every  $d$ , and following it is a good exercise in reading a formal proof. Second, the sum  $R + \alpha'/R$  is not an invention: it is the combination in which the one-loop partition function on a circle and the energy of a gas of winding and momentum strings depend on  $R$ , and both facts are in the literature. Third, the same inequality is used in this programme, and in string gas cosmology, as the reason why a contracting universe should not reach zero size—and here the student must learn to separate a theorem from a hope. The chapter is organized around that separation: what is proved ([A]), what is established in the literature ([L]), and what is proposed ([C]).

### 31.1 The thesis: geometry as a pair

#### 31.1.1 Two towers, one spectrum

We recall the facts of Chapters 7 and 8 in the conventions of this book. A closed string on a circle  $X \sim X + 2\pi R$  carries a momentum number  $n \in \mathbb{Z}$  and a winding number  $w \in \mathbb{Z}$ , and its mass in the non-compact dimensions is

$$M^2 = \frac{n^2}{R^2} + \frac{w^2 R^2}{\alpha'^2} + \frac{2}{\alpha'}(N + \tilde{N} - 2), \quad N - \tilde{N} = nw, \quad (31.1)$$

for the bosonic string; the superstring changes only the constant  $-2$ . This is Tong’s (8.5)–(8.6) [Tong §8.2, ll. 11460–11470] with his winding label  $m$  renamed  $w$ . In GPR’s dimensionless variables the same information is carried by

$$p_{L,R} = \frac{1}{\sqrt{2}} \left( \frac{\sqrt{\alpha'}}{R} n \pm \frac{R}{\sqrt{\alpha'}} w \right), \quad (31.2)$$

[GPR §2.2, eq. (2.2.5), ll. 620–630], and the zero-mode part of (31.1) is  $\alpha' M_{\text{zero}}^2 = p_L^2 + p_R^2$ . Two towers of states are visible: the momentum tower with spacing  $1/R$  and the winding tower with spacing  $R/\alpha'$ . The energy of a wound string grows with  $R$  because the string must be stretched; the energy of a moving string falls with  $R$  because its momentum is quantized in units of  $1/R$  [GPR §2.2, ll. 752–760]. The transformation

$$\frac{R}{\sqrt{\alpha'}} \mapsto \frac{\sqrt{\alpha'}}{R}, \quad n \leftrightarrow w, \quad (31.3)$$

sends  $p_R \mapsto -p_R$  and leaves  $p_L$  fixed [GPR §2.2, eq. (2.2.11), ll. 717–726], hence leaves the spectrum invariant; Tong’s (8.7)–(8.8) say the same [Tong §8.3, ll. 11626–11640]. In cosmology, where  $R$  is a scale factor, Battefeld and Watson write the identical formulae, eq. (21) for  $p_{L,R}$ , eq. (23) for level matching and eq. (24) for the duality map [BW §II, ll. 655–737], and draw the conclusion that matters for Chapter 32: “strings at small scale factor  $1/R$  behave the same as strings at large scale factor  $R$ ”, so that their effect on a cosmological background “may differ greatly from that of ordinary point particles” [BW §II, ll. 732–737].

### 31.1.2 Dual-scale invariants

The principle of this chapter promotes (31.3) from a symmetry of a formula to a rule about which functions of  $R$  are allowed to have physical meaning.

**Definition 31.1** (dual-scale invariant, circle). Write  $\tilde{R} = \alpha'/R$  for the dual radius. A function  $f$  of the pair  $(R, \tilde{R})$  is a *dual-scale invariant* if  $f(R, \tilde{R}) = f(\tilde{R}, R)$ , that is, if it is a symmetric function of the two radii. Because  $R\tilde{R} = \alpha'$  is fixed, every symmetric polynomial in  $(R, \tilde{R})$  is a polynomial in the single invariant

$$R_{\text{eff}}(R) = R + \frac{\alpha'}{R}, \quad (31.4)$$

which we call the *effective scale* or *dual scale*.

The definition is deliberately narrow. It concerns functions of the modulus  $R$  alone. The full duality of Chapter 8 also shifts the dilaton,  $\Phi \mapsto \Phi - \ln(R/\sqrt{\alpha'})$  [AAL §1, ll. 84–88], and a function of  $(R, \Phi)$  can be invariant without being symmetric in  $(R, \tilde{R})$  at fixed  $\Phi$ ; Chapter 8 discusses such combinations. Here we look only at the geometry.

**Example 31.2.** The following are dual-scale invariants: the product  $R\tilde{R} = \alpha'$  (trivially); the sum  $R_{\text{eff}}$ ; the operational radius  $R_{\text{obs}} = \max(R, \tilde{R})$  of Chapter 8; the difference of squares  $R^2 + \tilde{R}^2 = R_{\text{eff}}^2 - 2\alpha'$ ; and any function of  $R_{\text{eff}}$ . The following are *not*:  $R$  itself,  $R^2$ ,  $\ln R$ , and  $R - \tilde{R}$ , which changes sign. The quantity  $x = \ln(R/\sqrt{\alpha'})$  is convenient because the duality map is  $x \mapsto -x$ , and then

$$R_{\text{eff}} = \sqrt{\alpha'} (e^x + e^{-x}) = 2\sqrt{\alpha'} \cosh x, \quad (31.5)$$

manifestly even in  $x$ . (Check: at  $x = 0$ ,  $R = \sqrt{\alpha'}$  and  $R_{\text{eff}} = 2\sqrt{\alpha'}$ .)

Two places in the literature single out exactly the combination  $R + \alpha'/R$ , and they are worth knowing because they show the effective scale is not an arbitrary choice among invariants.

**Where the sum  $R + \alpha'/R$  comes from**

*The one-loop vacuum amplitude.* GPR record that the one-loop partition function of the bosonic string on a circle “is proportional to  $R + 1/R$ , and therefore, it is manifestly duality-invariant” (in units  $\alpha' = 1$ ) [GPR §1, ll. 433–434]. GPR cite three references for this and do not display the computation; we do not reproduce it either, and it is not carried out in this book’s library. The lesson we take is only that the sum, and not some other invariant, is what the simplest one-loop quantity produces.

*The energy of a string gas.* In string gas cosmology one fills a compact direction of radius  $R = e^\lambda$  (string units) with  $\tilde{n}_w$  winding strings and  $\tilde{n}_m$  momentum strings per unit volume. Battefeld and Watson’s eq. (30) gives their energy densities and pressures in that direction as

$$\rho_w = \tilde{n}_w e^\lambda, \quad p_w = -\tilde{n}_w e^\lambda, \quad \rho_m = \tilde{n}_m e^{-\lambda}, \quad p_m = \tilde{n}_m e^{-\lambda}$$

[BW §III, eqs. (30)–(31), ll. 845–906]: winding strings have negative pressure, since stretching them costs energy, and momentum modes behave like radiation. For equal numbers,  $\tilde{n}_w = \tilde{n}_m = \tilde{n}$ , the total energy density is

$$\rho = \tilde{n} (e^\lambda + e^{-\lambda}) = \tilde{n} (R + \frac{1}{R}) = \tilde{n} R_{\text{eff}}$$

and the total pressure  $p = \tilde{n}(e^{-\lambda} - e^\lambda) = -d\rho/d\lambda$  vanishes exactly where  $R_{\text{eff}}$  is minimal, at  $\lambda = 0$ . This is the mechanism by which, in the words of the source, an equal population “is driven to the critical radius, the so-called self-dual radius” where “the total pressure vanishes and t-duality is restored” [BW §IV, ll. 1110–1122]. The effective scale is thus, up to normalization, the energy of a balanced string gas as a function of the radius. This is a Tier L statement about a classical gas of free strings; what it does and does not imply for the early universe is the subject of Chapter 32.

**31.1.3 The principle, stated in tiers**

We can now state the thesis in the form in which the rest of the chapter examines it.

- (A) **Mathematics.** For every  $R > 0$ ,  $R_{\text{eff}}(\alpha'/R) = R_{\text{eff}}(R)$  and  $R_{\text{eff}}(R) \geq 2\sqrt{\alpha'}$ , with equality only at  $R = \sqrt{\alpha'}$ . For every positive definite  $d \times d$  matrix  $G$ ,  $\text{tr} G + \text{tr} G^{-1} \geq 2d$ , with equality only at  $G = \mathbf{1}$ , and  $\text{tr} G^{-1} + \text{tr}(G^{-1})^{-1} = \text{tr} G + \text{tr} G^{-1}$ . The inequalities and the invariance are [A] (Section 31.6). Uniqueness of the matrix minimizer is proved by hand below and, since September 2026, also in Lean (`dualScale_eq_iff`, [A]); the other uniqueness statements of this chapter are by hand only.
- (L) **Literature.** A closed string cannot distinguish  $R$  from  $\alpha'/R$ , so that “a small compactification radius is completely equivalent to a very large one”, which GPR read as “the existence of a minimal length in string theory” [GPR §2.2, ll. 784–793]; Tong states that at  $R = \sqrt{\alpha'}$  “the theory acts as if the circle is growing again” [Tong §8.3, ll. 11659–11662]; AAL argue that distances defined from correlators of closed-string states are restricted to the interval  $(1, \infty)$  in string units [AAL §6, ll. 1289–1301]; and in string gas cosmology a balanced gas of winding and momentum strings drives the radius to the self-dual point [BW §IV, ll. 1110–1122]. All of this is [L].
- (C) **Proposal.** That  $R_{\text{eff}}$ , rather than  $R_{\text{obs}}$  or some other invariant, is *the* size seen by all physics; that the bound  $R_{\text{eff}} \geq 2\sqrt{\alpha'}$  by itself regulates the cosmological singularity and produces a bounce; and that the  $d$ -dimensional trace  $\text{tr} G + \text{tr} G^{-1}$  is the right notion of size of a torus. These are proposals of this programme, [C], and Sections 31.4 and 31.5 explain what would have to be added to turn them into theorems.

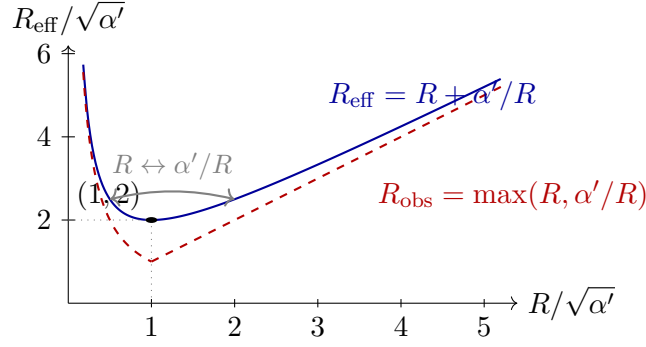


Figure 31.1: The effective scale  $R_{\text{eff}}$  (solid) and the operational radius  $R_{\text{obs}}$  (dashed) in string units. Both are invariant under  $R \mapsto \alpha'/R$ , which reflects the horizontal axis about  $R = \sqrt{\alpha'}$  on a logarithmic scale; both are minimal at the self-dual point, where  $R_{\text{eff}} = 2\sqrt{\alpha'}$  and  $R_{\text{obs}} = \sqrt{\alpha'}$ . Every point of the curve for  $R < \sqrt{\alpha'}$  has a twin for  $R > \sqrt{\alpha'}$ .

## 31.2 The circle bound

### 31.2.1 Three proofs of $R + \alpha'/R \geq 2\sqrt{\alpha'}$

**Proposition 31.3.** *For all  $R > 0$ ,  $R_{\text{eff}}(R) = R + \alpha'/R \geq 2\sqrt{\alpha'}$ , with equality if and only if  $R = \sqrt{\alpha'}$ .*

*First proof: completing the square.* Multiply the claim by  $R > 0$ ; it becomes  $R^2 - 2\sqrt{\alpha'}R + \alpha' \geq 0$ , i.e.  $(R - \sqrt{\alpha'})^2 \geq 0$ , which holds with equality only at  $R = \sqrt{\alpha'}$ . Equivalently,

$$R + \frac{\alpha'}{R} - 2\sqrt{\alpha'} = \frac{(R - \sqrt{\alpha'})^2}{R}. \quad (31.6)$$

□

*Second proof: arithmetic and geometric means.* The two radii  $R$  and  $\tilde{R} = \alpha'/R$  have the fixed product  $\alpha'$ . The inequality  $\frac{1}{2}(a+b) \geq \sqrt{ab}$  for positive  $a, b$ , with equality only at  $a = b$ , gives  $R_{\text{eff}} \geq 2\sqrt{R\tilde{R}} = 2\sqrt{\alpha'}$ , with equality only when  $R = \tilde{R}$ , i.e.  $R^2 = \alpha'$ . □

*Third proof: the even function.* In the variable  $x = \ln(R/\sqrt{\alpha'})$ , (31.5) gives  $R_{\text{eff}} = 2\sqrt{\alpha'} \cosh x$ , and  $\cosh x = 1 + x^2/2 + x^4/24 + \dots \geq 1$  with equality only at  $x = 0$ . □

The three proofs are the same proof in three notations, but each generalizes differently. The first is the one that will be mechanized: its only ingredients are ring arithmetic and the sign of a square. The second is the one that generalizes to matrices through eigenvalues. The third exhibits the duality  $x \mapsto -x$  as a reflection and the effective scale as an even function of the reflected coordinate; this is the picture of Figure 31.1.

### 31.2.2 A worked example with numbers

**Example 31.4.** Take  $\alpha' = 1$  (string units) and list, for several radii, the dual radius, the effective scale and the operational radius:

| $R$  | $\tilde{R} = 1/R$ | $x = \ln R$ | $R_{\text{eff}} = R + 1/R$ | $R_{\text{obs}} = \max(R, 1/R)$ | $R_{\text{eff}}/R_{\text{obs}}$ |
|------|-------------------|-------------|----------------------------|---------------------------------|---------------------------------|
| 0.1  | 10                | -2.303      | 10.1                       | 10                              | 1.01                            |
| 0.25 | 4                 | -1.386      | 4.25                       | 4                               | 1.0625                          |
| 0.5  | 2                 | -0.693      | 2.5                        | 2                               | 1.25                            |
| 1    | 1                 | 0           | 2                          | 1                               | 2                               |
| 2    | 0.5               | 0.693       | 2.5                        | 2                               | 1.25                            |
| 4    | 0.25              | 1.386       | 4.25                       | 4                               | 1.0625                          |
| 10   | 0.1               | 2.303       | 10.1                       | 10                              | 1.01                            |

The rows for  $R$  and  $1/R$  carry the same invariants, as they must. The ratio in the last column lies between 1 and 2 and tends to 1 away from the self-dual point, which is the statement  $R_{\text{obs}} \leq R_{\text{eff}} \leq 2R_{\text{obs}}$  of Chapter 8: the two candidate “sizes” disagree by at most a factor of two, and only near  $R = \sqrt{\alpha'}$ . Now imagine the radius decreasing through the table from top to bottom of the upper half:  $R = 0.5 \rightarrow 0.25 \rightarrow 0.1$ . The parameter  $R$  tends to zero, but  $R_{\text{eff}}$  has passed through its minimum at  $R = 1$  and is *growing*; the string sees a circle that is opening up again, in Tong’s phrase, and the states that become light are the winding states with masses  $|w|R$ , i.e.  $0.5|w|$ ,  $0.25|w|$ ,  $0.1|w|$ .

**Common pitfall:  $R_{\text{eff}}$  is not a length one can lay a ruler along**

It is tempting to read  $R_{\text{eff}} \geq 2\sqrt{\alpha'}$  as “the circle is never smaller than  $2\ell_s$ ”. That is not what the inequality says. The metric radius  $R$  is a coordinate on the moduli space and takes every positive value; nothing stops it from being  $10^{-3}\ell_s$ . What the inequality says is that a particular *duality-invariant function* of the modulus is bounded below. Whether that function is what an experiment measures is a separate question. Chapter 8 argued that the lightest Kaluza–Klein tower defines  $R_{\text{obs}} = \max(R, \alpha'/R) \geq \ell_s$ ; AAL argued that closed-string correlators define a distance in  $(1, \infty)$  [AAL §6, ll. 1289–1301]. Both are bounded below for the same reason as  $R_{\text{eff}}$ , and none of the three is more fundamental than the others without a further physical argument. Note also that all three concern *closed-string* probes: Polchinski reports that D-branes appear to probe distances shorter than  $\ell_s$ , “in conflict with the idea that the string length is a minimum distance” [Pol, ll. 3020–3025], and open strings have no winding modes at all, so the duality holds for them only together with an exchange of boundary conditions [GPR §1, ll. 380–386]; see Chapter 13.

## 31.3 From the circle to the torus: the pair $(G, G^{-1})$

### 31.3.1 The dual scale of a torus metric

On a torus  $T^d$  with constant metric  $G_{ij}$  (measured in units of  $\alpha'$ , so that a circle of radius  $R$  is  $G_{11} = R^2/\alpha'$ ) and  $B = 0$ , the zero-mode mass formula of Chapter 10 is the quadratic form

$$\alpha' M_{\text{zero}}^2 = Z^\top \mathcal{H}(G, 0) Z, \quad Z = \begin{pmatrix} w \\ n \end{pmatrix}, \quad \mathcal{H}(G, 0) = \begin{pmatrix} G & 0 \\ 0 & G^{-1} \end{pmatrix}, \quad (31.7)$$

in the doubled charge vector  $Z = (w, n)$  with the winding block first, which is the convention of the Lean module described below. The full duality of the torus, the analogue of (31.3), exchanges  $w \leftrightarrow n$  and  $G \leftrightarrow G^{-1}$ ; AAL write it as  $(g+b) \mapsto (g+b)^{-1}$  together with the dilaton shift  $\phi \mapsto \phi - \frac{1}{2} \ln \det(g+b)$  [AAL §1, ll. 88–92]. On the generalized metric it is conjugation by  $\eta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ ,

$$\eta \mathcal{H}(G, 0) \eta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} G & 0 \\ 0 & G^{-1} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} G^{-1} & 0 \\ 0 & G \end{pmatrix} = \mathcal{H}(G^{-1}, 0), \quad (31.8)$$

which is Chapter 15's "T-duality is metric inversion". The pair  $(R, \alpha'/R)$  of the circle has become the pair of blocks  $(G, G^{-1})$  of  $\mathcal{H}$ . The circle sum  $R + \alpha'/R$  was the simplest symmetric function of the pair; the simplest symmetric function of the two blocks is their combined trace.

**Definition 31.5** (dual scale of a torus). For a real invertible  $d \times d$  matrix  $G$ , the *dual scale* is

$$\mathcal{D}(G) = \text{tr } \mathcal{H}(G, 0) = \text{tr } G + \text{tr } G^{-1}. \quad (31.9)$$

The second equality is the block structure of (31.7): the trace of a block-diagonal matrix is the sum of the traces of the blocks, and the off-diagonal blocks vanish at  $B = 0$ . Three properties are immediate.

**Lemma 31.6.** *Let  $G$  be real and invertible. Then*

(a)  $\mathcal{D}(G^{-1}) = \mathcal{D}(G)$  (invariance under the full duality  $\eta$ );

(b)  $\mathcal{D}(\mathbf{1}) = 2d$ ;

(c) for  $d = 1$  and  $G = (r^2)$  with  $r = R/\sqrt{\alpha'} \neq 0$ ,  $\mathcal{D}(G) = r^2 + r^{-2} = (r + \frac{1}{r})^2 - 2 = \frac{R_{\text{eff}}^2}{\alpha'} - 2$ .

*Proof.* (a)  $(G^{-1})^{-1} = G$ , so  $\mathcal{D}(G^{-1}) = \text{tr } G^{-1} + \text{tr } G$ . (b)  $\text{tr } \mathbf{1} = d$  twice. (c)  $(r + 1/r)^2 = r^2 + 2 + r^{-2}$ .  $\square$

Part (a) is more than a symmetry of a formula. The trace of a matrix is invariant under conjugation by an invertible  $g$ ,  $\text{tr}(g\mathcal{H}g^{-1}) = \text{tr } \mathcal{H}$ , and  $\eta^{-1} = \eta$ ; so  $\mathcal{D}$  is invariant under  $\mathcal{H} \mapsto \eta\mathcal{H}\eta$  for any  $\mathcal{H}$ , not only at  $B = 0$ .

**Common pitfall:  $\mathcal{D}$  is not invariant under all of  $O(d, d; \mathbb{Z})$**

The duality group of Chapter 11 acts on the generalized metric by  $\mathcal{H} \mapsto g^T \mathcal{H} g$ , and  $\text{tr}(g^T \mathcal{H} g) = \text{tr}(\mathcal{H} g g^T)$  equals  $\text{tr } \mathcal{H}$  only when  $g g^T = \mathbf{1}$ . The inversion  $\eta$  is orthogonal, so  $\mathcal{D}$  survives it; a change of lattice basis  $G \mapsto A^T G A$  with  $A \in GL(d, \mathbb{Z})$  is orthogonal only when  $A$  is a signed permutation, and a  $B$ -shift is never orthogonal. Concretely, the hexagonal torus  $G = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1 \end{pmatrix}$  has  $\mathcal{D} = 14/3$ , but in the basis  $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$  the same torus is  $A^T G A = \begin{pmatrix} 1 & 3/2 \\ 3/2 & 3 \end{pmatrix}$  with  $\mathcal{D} = 28/3$  (symbolic check). For  $d = 1$  the only basis changes are  $\pm 1$  and the problem disappears, which is why the circle statement is clean. For  $d \geq 2$ ,  $\mathcal{D}$  is a function on the space of metrics *in a chosen basis*; the bound of Theorem 31.7 holds in every basis, but the number  $\mathcal{D}(G)$  itself is not an invariant of the torus. A basis-independent version would take the minimum of  $\mathcal{D}$  over the  $GL(d, \mathbb{Z})$ -orbit (a reduction-theory statement not formalized here, and not made in the sources); this is one of the gaps between the theorem and the proposal.

### 31.3.2 The bound $\text{tr } G + \text{tr } G^{-1} \geq 2d$

**Theorem 31.7** (dual-scale bound). *Let  $G$  be a real symmetric positive definite  $d \times d$  matrix. Then*

$$\text{tr } G + \text{tr } G^{-1} \geq 2d, \quad (31.10)$$

*with equality if and only if  $G = \mathbf{1}$ .*

*First proof, by the spectral theorem.* A real symmetric matrix is orthogonally diagonalizable:  $G = O^\top \Lambda O$  with  $O^\top O = \mathbf{1}$  and  $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$ , and positive definiteness means every  $\lambda_i > 0$ . Then  $G^{-1} = O^\top \Lambda^{-1} O$ , and since the trace is invariant under conjugation,

$$\text{tr } G + \text{tr } G^{-1} = \sum_{i=1}^d \left( \lambda_i + \frac{1}{\lambda_i} \right).$$

Each term is the circle bound of Proposition 31.3 with  $\alpha' = 1$ :  $\lambda + 1/\lambda - 2 = (\lambda - 1)^2/\lambda \geq 0$ , with equality only at  $\lambda = 1$ . Summing,  $\text{tr } G + \text{tr } G^{-1} - 2d = \sum_i (\lambda_i - 1)^2/\lambda_i \geq 0$ , and equality forces every  $\lambda_i = 1$ , i.e.  $\Lambda = \mathbf{1}$  and  $G = O^\top O = \mathbf{1}$ .  $\square$

The spectral proof is the natural one for a physicist: the torus metric can always be brought to diagonal form  $\text{diag}(r_1^2, \dots, r_d^2)$  by an orthogonal (not integral!) change of coordinates, and then  $\mathcal{D} = \sum_i (r_i^2 + r_i^{-2})$  is a sum of circle invariants. But it uses a theorem of some depth, and a proof assistant will ask for it to be supplied. The proof that was actually formalized uses less.

*Second proof, by a positive sandwich.* Step 1 (an identity). Using  $GG^{-1} = G^{-1}G = \mathbf{1}$ ,

$$(G - \mathbf{1})G^{-1}(G - \mathbf{1}) = (G - \mathbf{1})(\mathbf{1} - G^{-1}) = G - \mathbf{1} - \mathbf{1} + G^{-1} = (G - \mathbf{1}) - (\mathbf{1} - G^{-1}). \quad (31.11)$$

Taking the trace,  $\text{tr} [(G - \mathbf{1})G^{-1}(G - \mathbf{1})] = \text{tr } G + \text{tr } G^{-1} - 2d$ . (Symbolic check on a general symmetric  $2 \times 2$  matrix: the identity and its trace both hold.)

Step 2 (a sign). The inverse of a positive definite matrix is positive definite, hence positive semidefinite:  $v^\top G^{-1}v = (G^{-1}v)^\top G(G^{-1}v) \geq 0$ . For any matrix  $M$  and positive semidefinite  $P$ , the sandwich  $M^\top P M$  is positive semidefinite, since  $v^\top M^\top P M v = (Mv)^\top P(Mv) \geq 0$ . Apply this with  $M = G - \mathbf{1}$ , which is symmetric, and  $P = G^{-1}$ .

Step 3 (trace of a positive semidefinite matrix). If  $S$  is positive semidefinite then  $S_{ii} = e_i^\top S e_i \geq 0$  for each basis vector  $e_i$ , so  $\text{tr } S = \sum_i S_{ii} \geq 0$ . Combining the three steps,  $\text{tr } G + \text{tr } G^{-1} - 2d \geq 0$ .

Uniqueness. Suppose equality holds. Then the positive semidefinite matrix  $S = (G - \mathbf{1})G^{-1}(G - \mathbf{1})$  has zero trace, so each  $S_{ii} = 0$ ; a positive semidefinite matrix with vanishing diagonal is zero (its  $2 \times 2$  principal minors  $S_{ii}S_{jj} - S_{ij}^2 \geq 0$  force  $S_{ij} = 0$ ). Then for every  $v$ ,  $0 = v^\top S v = u^\top G^{-1}u$  with  $u = (G - \mathbf{1})v$ , and positive definiteness of  $G^{-1}$  gives  $u = 0$ ; so  $G - \mathbf{1}$  annihilates every vector, and  $G = \mathbf{1}$ .  $\square$

No eigenvalue is computed in the second proof. Its ingredients are: the algebra of (31.11), the facts that the inverse of a positive definite matrix is positive semidefinite and that sandwiches preserve positive semidefiniteness, and the nonnegativity of the trace of a positive semidefinite matrix. Each of these is a lemma in Mathlib, and Section 31.6 shows the proof assembled from them. When this chapter was first written the uniqueness paragraph was not formalized: Lean proved the inequality and, separately, that the value  $2d$  is attained at  $G = \mathbf{1}$ . It now also proves the converse, `dualScale_eq_iff`: for positive definite  $G$ ,  $\text{tr } G + \text{tr } G^{-1} = 2d$  if and only if  $G = \mathbf{1}$ . The Lean proof follows the paragraph above with one difference: where we used  $2 \times 2$  minors to pass from zero trace to  $S = 0$ , Lean calls Mathlib's `Matrix.PosSemidef.trace_eq_zero_iff`, which goes through the spectral theorem. So the inequality is still proved without eigenvalues, but the equality case, in Lean, is not.

**Corollary 31.8.** *For positive definite  $G$  and antisymmetric  $B$ :*

$$(a) \quad \text{tr } \mathcal{H}(G, B) = \text{tr } G + \text{tr } G^{-1} + \text{tr}(B^\top G^{-1}B) \geq \text{tr } \mathcal{H}(G, 0) \geq 2d;$$

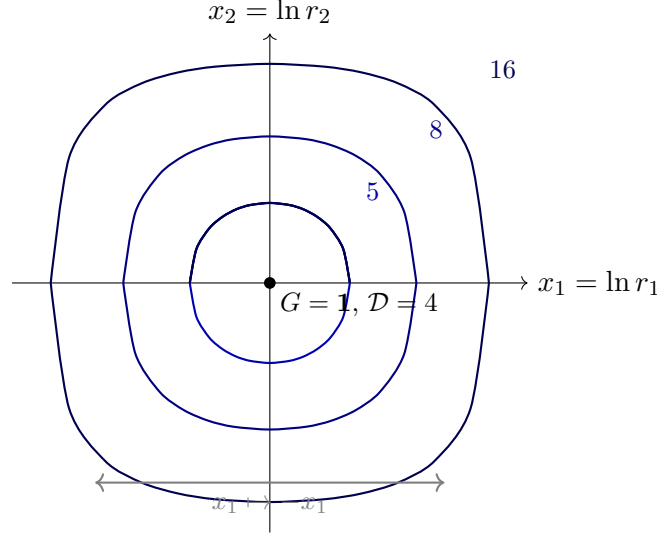


Figure 31.2: Level sets of the dual scale  $\mathcal{D} = 2(\cosh 2x_1 + \cosh 2x_2)$  for diagonal metrics  $G = \text{diag}(e^{2x_1}, e^{2x_2})$  on  $T^2$ , at the values 5, 8 and 16. The function is even in each coordinate, the four reflections  $x_i \mapsto -x_i$  being the factorized dualities of Chapter 11, and it has a single minimum  $2d = 4$  at the origin.

(b) in the variables  $\lambda_i = e^{2x_i}$ ,  $\mathcal{D}(G) = 2 \sum_i \cosh 2x_i$ , an even function of each  $x_i$ ;

(c) for the circle,  $\mathcal{D} \geq 2$  is equivalent to  $R_{\text{eff}} \geq 2\sqrt{\alpha'}$ .

*Proof.* (a) From the block form of Chapter 15,  $\text{tr } \mathcal{H}(G, B) = \text{tr}(G - BG^{-1}B) + \text{tr } G^{-1}$ , and  $-BG^{-1}B = B^\top G^{-1}B$  is a sandwich of the positive semidefinite  $G^{-1}$ , hence has nonnegative trace. (Symbolic check: for  $d = 2$  the excess is  $\beta^2 \text{tr } G / \det G$  where  $B_{12} = \beta$ .) So a  $B$ -field can only increase the dual scale, and the bound at  $B = 0$  is the sharp one. (b) Substitute  $\lambda_i + \lambda_i^{-1} = 2 \cosh 2x_i$  into the spectral proof. (c) Lemma 31.6(c):  $\mathcal{D} = R_{\text{eff}}^2 / \alpha' - 2 \geq 2$  if and only if  $R_{\text{eff}}^2 \geq 4\alpha'$ .  $\square$

### 31.3.3 Worked examples

**Example 31.9** (a rectangular torus). Take  $d = 2$  and  $G = \text{diag}(4, \frac{1}{4})$ , i.e.  $R_1 = 2\ell_s$  and  $R_2 = \ell_s/2$ . Then  $G^{-1} = \text{diag}(\frac{1}{4}, 4)$  and  $\mathcal{D} = 4 + \frac{1}{4} + \frac{1}{4} + 4 = 8.5 \geq 4$ . The full duality  $G \mapsto G^{-1}$  exchanges the two radii,  $(2, \frac{1}{2}) \mapsto (\frac{1}{2}, 2)$ , and leaves  $\mathcal{D}$  unchanged, as Lemma 31.6(a) requires. The factorized duality in the first direction alone,  $R_1 \mapsto \alpha'/R_1$ , sends  $G$  to  $\text{diag}(\frac{1}{4}, \frac{1}{4})$ , with  $\mathcal{D} = \frac{1}{2} + 8 = 8.5$  again; the torus with both radii equal to  $\ell_s/2$  has the same dual scale as the one with radii  $(2, \frac{1}{2})$ , because  $\mathcal{D}$  is a sum of circle invariants and each is even in  $x_i$ . Contrast  $\det G$ , which is 1 for the first torus and  $1/16$  for the second: the determinant is invariant under  $SL(2, \mathbb{Z})$  basis changes, which  $\mathcal{D}$  is not, but changes under factorized dualities, which  $\mathcal{D}$  survives.

**Example 31.10** (a sheared torus with unit determinant). Take  $G = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}$ , which is positive definite ( $\det G = 1$ ,  $G_{11} > 0$ ) and not diagonal. Its inverse is  $G^{-1} = \begin{pmatrix} 1 & -2 \\ -2 & 5 \end{pmatrix}$ , so  $\mathcal{D} = 6 + 6 = 12$ . The eigenvalues are  $\lambda_{\pm} = 3 \pm 2\sqrt{2}$ , with  $\lambda_+ \lambda_- = 1$  and  $\lambda_+ + \lambda_- = 6$ ; the spectral formula gives  $\sum(\lambda + 1/\lambda) = 2(\lambda_+ + \lambda_-) = 12$ , in agreement. In the sandwich proof,  $G - \mathbf{1} = \begin{pmatrix} 4 & 2 \\ 2 & 0 \end{pmatrix}$  and

$$(G - \mathbf{1})G^{-1}(G - \mathbf{1}) = \begin{pmatrix} 4 & 2 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} 1 & -2 \\ -2 & 5 \end{pmatrix} \begin{pmatrix} 4 & 2 \\ 2 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 2 \\ 2 & -4 \end{pmatrix} \begin{pmatrix} 4 & 2 \\ 2 & 0 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix},$$



whose trace is  $8 = 12 - 2 \cdot 2$ , as (31.11) predicts. Note that  $\begin{pmatrix} 0 & 2 \\ 2 & -4 \end{pmatrix}$  is not positive semidefinite—only the completed sandwich is.

**Example 31.11** (the hexagonal torus).  $G = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1 \end{pmatrix}$  has eigenvalues  $\frac{3}{2}$  and  $\frac{1}{2}$  (eigenvectors  $(1, 1)$  and  $(1, -1)$ ) and  $G^{-1} = \frac{4}{3} \begin{pmatrix} 1 & -1/2 \\ -1/2 & 1 \end{pmatrix}$ , so  $\mathcal{D} = 2 + \frac{8}{3} = \frac{14}{3} \approx 4.67$ ; the spectral formula gives  $\frac{3}{2} + \frac{2}{3} + \frac{1}{2} + 2 = \frac{14}{3}$ . This is the smallest value among the three examples, as befits the torus closest to  $G = \mathbf{1}$ , but it is still strictly above 4.

## 31.4 What the older libraries model: arithmetic shadows

Four older Lean modules of this project are named in the atlas as formalizing the dual-scale principle. They were written before the matrix statement of Section 31.3 and they model radii by natural numbers, integers or positive rationals. Following the rule of this book, we describe them as *arithmetic shadows*: each proves an exact arithmetic fact, and each docstring attaches to that fact a physical reading which the statement does not carry. Reading the statements literally is the skill this section trains.

### 31.4.1 Positive rational scales and the piecewise radius

In Lean: `DualScaleM24Formalization/DualScale/EffectiveMetric.lean`

```
structure PosScale where
  num : Nat
  den : Nat
  h_num : 0 < num
  h_den : 0 < den

def scaleEq (s1 s2 : PosScale) : Prop := s1.num * s2.den = s2.num * s1.den
def scaleLt (s1 s2 : PosScale) : Prop := s1.num * s2.den < s2.num * s1.den

def buscherDual (alpha : PosScale) (R : PosScale) : PosScale :=
  ⟨alpha.num * R.den, alpha.den * R.num,
    Nat.mul_pos alpha.h_num R.h_den, Nat.mul_pos alpha.h_den R.h_num⟩

theorem buscher_involution (alpha : PosScale) (R : PosScale) :
  scaleEq (buscherDual alpha (buscherDual alpha R)) R := by ...

def effectiveRadius (cutoff : PosScale) (alpha : PosScale) (R : PosScale) :
  PosScale :=
  if scaleLt R cutoff then buscherDual alpha R else R

theorem genesis_no_singularity (cutoff : PosScale) (alpha : PosScale)
  (R : PosScale) :
  0 < (effectiveRadius cutoff alpha R).num ∧
  0 < (effectiveRadius cutoff alpha R).den := by ...

theorem self_dual_symmetric (alpha : PosScale) :
  scaleEq (buscherDual alpha alpha) ⟨1, 1, by decide, by decide⟩ := by ...
```

**Reading.** A `PosScale` is a positive rational written as a fraction of positive naturals; two are `scaleEq` when they cross-multiply to the same value. `buscherDual alpha R` is the fraction  $\alpha'/R$ , and `buscher_involution` says  $\alpha'/( \alpha'/R ) = R$  as rationals—the map  $R \mapsto \alpha'/R$  is an involution, [A] for positive rationals. The function `effectiveRadius` is the piecewise *operational* radius of Chapter 8,  $\alpha'/R$  below a cutoff and  $R$  above it, with the cutoff a free parameter rather than  $\sqrt{\alpha'}$  (which is not

rational). `genesis_no_singularity` states that the numerator and denominator of this fraction are positive. Literally: *a positive rational is positive*, in both branches of the definition. It does not state the bound  $R_{\text{eff}} \geq 2\sqrt{\alpha'}$ , nor that any function is bounded away from zero uniformly—for a cutoff  $c$ , the value  $\alpha'/R$  with  $R$  just below  $c$  can be as small as  $\alpha'/c$ , and  $R$  itself just above  $c$  is  $c$ ; and it says nothing about spacetime, curvature or a bounce. `self_dual_symmetric` says  $\alpha'/\alpha' = 1$ .

**What a faithful version needs.** Real (or at least ordered-field) radii; the function  $R + \alpha'/R$  or  $\max(R, \alpha'/R)$ ; the inequality of Proposition 31.3. All of this is exactly what `circle_effective_scale_ge_two` in Section 31.6 provides over  $\mathbb{R}$ , which is why the newer module was written.

### 31.4.2 Dual pairs of integers

In Lean: `StringTheoryFoundation/Duality/DualScale.lean`

```
def dualLength (L_star L : Float) : Float := (L_star * L_star) / L

def isDualPair (L_star L L_dual : Int) : Prop := L * L_dual = L_star * L_star

theorem dual_pair_symmetric (L_star L L_dual : Int)
  (h : isDualPair L_star L L_dual) : isDualPair L_star L_dual L := by ...

theorem self_dual_at_crossover (L_star : Int) :
  isDualPair L_star L_star L_star := by rfl

theorem dual_product_invariant (L_star L L_dual : Int)
  (h : isDualPair L_star L L_dual) : L * L_dual = L_star * L_star := h
```

**Reading.** Two integers form a dual pair for the crossover scale  $L_*$  when their product is  $L_*^2$ ; this is the relation  $R\tilde{R} = \alpha'$  with  $L_* = \ell_s$ . `dual_pair_symmetric` is the commutativity of integer multiplication; `self_dual_at_crossover` is  $L_* \cdot L_* = L_* \cdot L_*$ , proved by `rfl`; `dual_product_invariant` returns its hypothesis unchanged. `dualLength` is a `Float` function about which nothing is proved (floating-point arithmetic has no useful algebraic theorems). These are [A] as statements about integers, and they are the weakest possible shadow of the principle: the invariance of the product, which is a tautology, and none of the inequality.

### 31.4.3 The numerator of $R \cdot R_{\text{eff}}$ over the naturals

In Lean: `DualScaleValidation/UseCase1_ModuliStabilization.lean`

```
def alpha_prime : Nat := 1

def effective_dual_scale_numerator (R : Nat) : Nat := R * R + alpha_prime

theorem buscher_log_involution (x : Int) : -(-x) = x := by omega

theorem dual_scale_self_dual_value : effective_dual_scale_numerator 1 = 2 := by rfl

theorem self_dual_is_global_minimum (R : Nat) (h : R ≥ 1) :
  effective_dual_scale_numerator R ≥ 2 := by ...

def moduli_potential (phi phi_0 : Int) : Int := (phi - phi_0) * (phi - phi_0)

theorem moduli_potential_nonneg (phi phi_0 : Int) :
```

```

moduli_potential phi phi_0 ≥ 0 := by ...

theorem moduli_potential_zero_iff (phi phi_0 : Int) :
  moduli_potential phi phi_0 = 0 ↔ phi = phi_0 := by ...

```

**Reading.** With  $\alpha' = 1$ , the natural number  $R^2 + 1$  is  $R \cdot R_{\text{eff}}(R)$ , the numerator of the effective scale. `buscher_log_involution` is  $-(-x) = x$  on  $\mathbb{Z}$ , the statement that the reflection  $x \mapsto -x$  of the logarithmic radius is an involution. `dual_scale_self_dual_value` is  $1 + 1 = 2$ . `self_dual_is_global_minimum` says: for a natural number  $R \geq 1$ ,  $R^2 + 1 \geq 2$ . This must be read with care, because its name promises more than it delivers. The bound  $R_{\text{eff}} \geq 2$  is  $R^2 + 1 \geq 2R$ , i.e.  $(R - 1)^2 \geq 0$ ; the theorem proves only  $R^2 + 1 \geq 2$ , which for  $R \geq 1$  is the weaker statement  $R \cdot R_{\text{eff}} \geq 2$ , hence  $R_{\text{eff}} \geq 2/R$ , and  $2/R \leq 2$ . At  $R = 3$  the theorem gives  $10 \geq 2$  while the dual-scale bound requires  $10 \geq 6$ . Moreover the hypothesis  $R \geq 1$  over  $\mathbb{N}$  excludes the sub-string radii  $0 < R < 1$  where the “bounce” of the docstring would take place: there are no such natural numbers. So the theorem is **[A]** as arithmetic, but it is not the statement that the self-dual point is a global minimum; that statement, over  $\mathbb{R}$  and with the correct inequality, is `circle_effective_scale_ge_two`. The `moduli_potential` theorems say that the integer  $(\phi - \phi_0)^2$  is nonnegative and vanishes exactly at  $\phi = \phi_0$ —a correct toy of a potential with a unique minimum, with no connection to the dual scale beyond the analogy that both have a unique minimum, and none to the flux potentials of Chapter 26. The module also records `dft_doubled_dimension_10d`:  $2 \cdot 10 = 20$ .

#### 31.4.4 The symmetric-square lock

The module `DualScaleM24Formalization/DualScale/Sym2Lock.lean` is grouped with the dual-scale files because its docstring presents it as the “micro–macro bridge” of the programme: a linear second-order recurrence on a “microscopic” fibre is squared to produce a third-order recurrence on “macroscopic” observables. The mathematics is an exercise about linear recurrences, and it is worth doing by hand before reading the Lean.

**Lemma 31.12.** *Let  $u_{n+2} = a u_{n+1} + b u_n$  with characteristic roots  $\lambda_1, \lambda_2$  ( $\lambda^2 - a\lambda - b = 0$ , so  $\lambda_1 + \lambda_2 = a$ ,  $\lambda_1 \lambda_2 = -b$ ). Then  $v_n = u_n^2$  satisfies  $v_{n+3} = c_2 v_{n+2} + c_1 v_{n+1} + c_0 v_n$  with*

$$c_2 = a^2 + b, \quad c_1 = b(a^2 + b), \quad c_0 = -b^3. \quad (31.12)$$

*Proof.* For distinct roots,  $u_n = \alpha \lambda_1^n + \beta \lambda_2^n$ , so  $v_n = \alpha^2 \lambda_1^{2n} + 2\alpha\beta(\lambda_1 \lambda_2)^n + \beta^2 \lambda_2^{2n}$  is a linear combination of the three geometric sequences with ratios  $\lambda_1^2, \lambda_1 \lambda_2, \lambda_2^2$ —the weights of the symmetric square  $\text{Sym}^2$  of the two-dimensional representation. Hence  $v$  satisfies the recurrence whose characteristic polynomial is  $(x - \lambda_1^2)(x - \lambda_2^2)(x - \lambda_1 \lambda_2) = x^3 - c_2 x^2 - c_1 x - c_0$  with  $c_2 = \lambda_1^2 + \lambda_2^2 + \lambda_1 \lambda_2 = (a^2 + 2b) - b = a^2 + b$ ,  $c_1 = -[\lambda_1^2 \lambda_2^2 + \lambda_1 \lambda_2 (\lambda_1^2 + \lambda_2^2)] = -[b^2 - b(a^2 + 2b)] = b(a^2 + b)$  and  $c_0 = (\lambda_1 \lambda_2)^3 = -b^3$ . The case of a repeated root follows by continuity in  $(a, b)$ . (Symbolic check: all three coefficients agree after substituting the roots.)  $\square$

**In Lean:** `DualScaleM24Formalization/DualScale/Sym2Lock.lean`

```

structure MicroL20operator where
  a : Int
  b : Int

structure MacroL30operator where
  c2 : Int
  c1 : Int

```

```

c0 : Int

def stepL3 (op : MacroL3Operator) (v2 v1 v0 : Int) : Int :=
  op.c2 * v2 + op.c1 * v1 + op.c0 * v0

def sym2Lock (op : MicroL2Operator) : MacroL3Operator := {
  c2 := op.a * op.a + op.b
  c1 := op.b * (op.a * op.a + op.b)
  c0 := - (op.b * op.b * op.b) }

theorem sym2_fibonacci_verification :
  let l2 : MicroL2Operator := ⟨1, 1⟩
  let l3 := sym2Lock l2
  l3.c2 = 2 ∧ l3.c1 = 2 ∧ l3.c0 = -1 ∧
  stepL3 l3 4 1 1 = 9 ∧ stepL3 l3 9 4 1 = 25 ∧ stepL3 l3 25 9 4 = 64 := by
  decide

def sym2Det (l2 : MicroL2Operator) : Int := - (sym2Lock l2).c0

theorem sym2_det_identity (l2 : MicroL2Operator) :
  sym2Det l2 = l2.b * l2.b * l2.b := by ...

```

**Reading.** `sym2Lock` is the map  $(a, b) \mapsto (c_2, c_1, c_0)$  of (31.12) on integers. The Lean library does *not* prove Lemma 31.12: it proves, by `decide`, that for the Fibonacci data  $(a, b) = (1, 1)$  the coefficients are  $(2, 2, -1)$  and that the squares  $1, 1, 4, 9, 25, 64$  satisfy three instances of the recurrence ( $9 = 2 \cdot 4 + 2 \cdot 1 - 1$ , and so on), with similar checks for the Pell data  $(2, 1)$  and for  $(3, -1)$  (`sym2_pell_verification`, `sym2_modular_attractor_verification`). `sym2_det_identity` says  $-c_0 = b^3$ , which is the definition of  $c_0$ . These are [A] as finite integer computations. The general lemma is proved above by hand and is standard; its physical reading as a bridge between “microscopic” and “macroscopic” dynamics, and the applications listed in the docstring, are [C] and are not examined further in this book.

### 31.4.5 The observables module

`DualScaleValidation/Observables.lean` imports the module of the previous subsection and certifies four arithmetic facts about numbers that appear in the programme’s phenomenological proposals:

- $110/27720$  reduces to  $1/252$  (`tensor_to_scalar_ratio_reduction`);
- $1/252$  lies in  $[0.003, 0.005]$ ,  $965/1000$  lies in  $[0.950, 0.980]$  and  $283$  lies in  $[270, 295]$  (three window theorems);
- two integer constants equal  $-1$  and  $0$  (`dark_energy_cosmological_constant`);
- the conjunction of the above (`observables_windows_master_contract`).

The module’s own docstring is explicit that these are “Tier A for its arithmetic only” and that the relations between the integers and observed quantities are “post-hoc numerical matches” with “no dynamical or field-theoretic derivation”, in one case in tension with current data. We follow the docstring: the arithmetic is [A], the identification of any of these numbers with a cosmological observable is [C], and the module contains no theorem connecting them to  $R_{\text{eff}}$  or to  $\text{tr } G + \text{tr } G^{-1}$ . A reader looking for “predictions of the dual-scale principle” will not find them in the kernel-checked corpus.

## 31.5 Separating the tiers

The principle of Section 31.1 was stated in three tiers, and the intervening sections have supplied the evidence for each. We now put the pieces side by side, because the most common error in reading this programme’s documents—and, to be candid, in some of the docstrings quoted above—is to let the certainty of tier A leak into tiers L and C.

### 31.5.1 What the inequality does not say

Theorem 31.7 is a statement about a function on the space of positive definite matrices. To turn it into a statement about physics one needs at least the following, none of which is a theorem in this book’s library.

1. *An identification of the variable.* The matrix  $G$  must be the metric of a flat torus in a string background, in units of  $\alpha'$ , in a fixed lattice basis. In Lean,  $G$  is any real matrix; the identification is made in a docstring. This step can never be Tier A.
2. *A choice of observable.* One must argue that  $\text{tr } G + \text{tr } G^{-1}$ , rather than max-type invariants, the minimum over the  $GL(d, \mathbb{Z})$ -orbit, or a function of the full  $\mathcal{H}(G, B)$ , is what some measurement returns. For the circle the literature offers arguments for a minimal *measured* radius ([L]), but for  $R_{\text{obs}}$  or for the correlator-defined distance of AAL, not specifically for  $R_{\text{eff}}$ ; Chapter 8 showed that the candidates agree within a factor of two, which is a reason to be relaxed about the choice, not a derivation of it.
3. *Dynamics.* A bound on a function of the moduli says nothing about how the moduli evolve. That a contracting universe is driven to the self-dual radius is a dynamical statement which string gas cosmology derives from the pressures of winding and momentum modes under stated assumptions (equal populations, the dilaton running to weak coupling, adiabatic evolution) [BW §IV, ll. 1110–1134]; Chapter 32 presents it. The inequality by itself does not prevent the metric radius  $R$  from reaching zero; it only says that  $R_{\text{eff}}$  would then be large.
4. *Beyond free strings.* The mass formula (31.1) is the free spectrum. GPR are careful to say that the equivalence of small and large radius as a statement about the *interacting* string requires the higher-genus contributions, which they prove in their next section [GPR §2.2, ll. 784–793]; the extension to curved backgrounds, to D-brane probes and to non-perturbative physics is each a separate question (Chapters 9, 13 and 27).

### 31.5.2 The tiers, side by side

| Statement                                                                                 | Evidence                                                                 | Tier |
|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|------|
| $R + \alpha'/R \geq 2\sqrt{\alpha'}$ for real $R > 0$ (units $\alpha' = 1$ )              | <code>circle_effective_scale_ge_two</code>                               | [A]  |
| $\text{tr } G + \text{tr } G^{-1} \geq 2d$ for positive definite real $G$ , all $d$       | <code>dualScale_ge</code>                                                | [A]  |
| $\mathcal{D}(G^{-1}) = \mathcal{D}(G)$ for invertible real $G$                            | <code>dualScale_inv</code>                                               | [A]  |
| $\mathcal{D}(1) = 2d$ (the bound is attained)                                             | <code>dualScale_one</code>                                               | [A]  |
| $\mathcal{D}(R^2) = (R + 1/R)^2 - 2$ for the circle                                       | <code>dualScale_circle</code>                                            | [A]  |
| $\alpha' / (\alpha' / R) = R$ for positive rationals; $-(-x) = x$ on $\mathbb{Z}$         | <code>buscher_involution</code> ,<br><code>buscher_log_involution</code> | [A]  |
| Equality in (31.10) only at $G = 1$                                                       | hand proof, Theorem 31.7                                                 | —    |
| $\text{tr } \mathcal{H}(G, B) \geq \text{tr } \mathcal{H}(G, 0)$                          | hand proof, Corollary 31.8                                               | —    |
| Closed strings cannot distinguish $R$ from $\alpha'/R$ ; a minimal length in this sense   | GPR §2.2, Tong §8.3, AAL §6                                              | [L]  |
| One-loop partition function on a circle $\propto R + 1/R$                                 | GPR §1                                                                   | [L]  |
| Balanced string gas: $\rho \propto R + 1/R$ , pressure vanishes at $R = \sqrt{\alpha'}$   | BW §III–IV                                                               | [L]  |
| Winding modes annihilate in at most three large dimensions (Brandenberger–Vafa)           | BW §IV                                                                   | [L]  |
| $R_{\text{eff}}$ (resp. $\text{tr } G + \text{tr } G^{-1}$ ) is <i>the</i> physical size  | this programme                                                           | [C]  |
| The bound regulates the cosmological singularity; a bounce                                | this programme; docstrings of Section 31.4                               | [C]  |
| $\text{Sym}^2$ lock as a micro–macro bridge; observables of <code>Observables.lean</code> | this programme                                                           | [C]  |

#### The Brandenberger–Vafa argument

The idea that winding strings control the size of compact dimensions predates T-duality’s use as a minimal-length argument in cosmology. As Battefeld and Watson recount it, Brandenberger and Vafa (1989) argued that winding strings “can maintain equilibrium in at most three spatial dimensions”, because  $p$ -dimensional objects generically intersect in at most  $2p + 1$  dimensions and string interactions occur at intersections; once winding modes annihilate with anti-winding modes, three dimensions “would be free to expand while the remaining six should remain confined by winding modes near the string scale” [BW §IV, ll. 921–928]. The later observation that a balanced gas is driven to the self-dual radius (Watson and Brandenberger 2003, as cited by BW) is what connects this picture to  $R_{\text{eff}}$ . Battefeld and Watson also stress that a gas of purely winding modes is *not* enough to stabilize the extra dimensions [BW abstract, ll. 165–166], which is a first warning that the picture depends on details.

## 31.6 What the machine has checked

The module `TraceBound.lean` in `DualScaleStream2/DualScale/` formalizes Definition 31.5, Lemma 31.6 and the inequality of Theorem 31.7 over  $\mathbb{R}$ , for every  $d$ , on top of the generalized metric `genMetric` of Chapter 15. It is a short file, and we go through all of it.

#### In Lean: the dual scale and its duality invariance

```
namespace DualScaleStream2.DualScale
open Matrix DualScaleStream2.DFT
variable {d : ℕ}

/-- Dual scale ' $\mathcal{D}(G) = \text{tr } \mathcal{H}(G, 0)$ '. -/
```

```

noncomputable def dualScale (G : Matrix (Fin d) (Fin d) ℝ) : ℝ :=
  (genMetric G 0).trace

theorem dualScale_eq (G : Matrix (Fin d) (Fin d) ℝ) :
  dualScale G = G.trace + G-1.trace := by ...

theorem dualScale_inv (G : Matrix (Fin d) (Fin d) ℝ) (hG : IsUnit G.det) :
  dualScale G-1 = dualScale G := by
  rw [dualScale_eq, dualScale_eq]
  have h_inv_inv : (G-1)-1 = G := Matrix.nonsing_inv_nonsing_inv G hG
  rw [h_inv_inv]
  ring

```

**Reading.** `dualScale G` is defined as the trace of the  $2d \times 2d$  matrix `genMetric G 0`, i.e. of  $\mathcal{H}(G, 0)$  with rows and columns indexed by  $\text{Fin } d \oplus \text{Fin } d$ ; this ties it to the object of Chapter 15 rather than to a formula written afresh. `dualScale_eq` is the second equality of (31.9),  $\text{tr } \mathcal{H}(G, 0) = \text{tr } G + \text{tr } G^{-1}$ , for *every* real matrix  $G$ : at  $B = 0$  the off-diagonal blocks of `genMetric` are literally zero and the proof is a `simp` computation unfolding the block trace. No invertibility is needed, because Mathlib’s  $G^{-1}$  is a total function (Chapter 15); when  $\det G = 0$  it is the zero matrix and both sides equal  $\text{tr } G$ . `dualScale_inv` is Lemma 31.6(a) under the hypothesis `IsUnit G.det`, which is what makes  $(G^{-1})^{-1} = G$  available (`Matrix.nonsing_inv_nonsing_inv`); after rewriting both sides with `dualScale_eq`, the goal is  $\text{tr } G^{-1} + \text{tr } G = \text{tr } G + \text{tr } G^{-1}$  and `ring` closes it. Both are [\[A\]](#).

#### In Lean: the bound $\text{tr } G + \text{tr } G^{-1} \geq 2d$

```

theorem dualScale_ge (G : Matrix (Fin d) (Fin d) ℝ) (hG : G.PosDef) :
  (2 * d : ℝ) ≤ dualScale G := by
  rw [dualScale_eq]
  have hdet : IsUnit G.det := G.isUnit_iff_isUnit_det.mp hG.isUnit
  have hl : G-1 * G = 1 := Matrix.nonsing_inv_mul G hdet
  have hr : G * G-1 = 1 := Matrix.mul_nonsing_inv G hdet
  have hGinv_psd : G-1.PosSemidef := hG.posSemidef.inv
  have hGT : GT = G :=
    (Matrix.conjTranspose_eq_transpose_of_trivial G).symm.trans hG.isHermitian
  have hMh : (G - 1)H = G - 1 := by ...
  have hpsd : ((G - 1)H * G-1 * (G - 1)).PosSemidef :=
    Matrix.PosSemidef.conjTranspose_mul_mul_same hGinv_psd (G - 1)
  rw [hMh] at hpsd
  have htrace_nonneg : 0 ≤ ((G - 1) * G-1 * (G - 1)).trace :=
    hpsd.trace_nonneg
  have hexpand : (G - 1) * G-1 * (G - 1) = (G - 1) - (1 - G-1) := by ...
  rw [hexpand] at htrace_nonneg
  rw [Matrix.trace_sub, Matrix.trace_sub, Matrix.trace_sub,
    Matrix.trace_one] at htrace_nonneg
  simp only [Fintype.card_fin] at htrace_nonneg
  linarith

```

**Reading.** For every  $d$  and every real matrix  $G$  satisfying Mathlib’s `PosDef`—which packages “Hermitian” (over  $\mathbb{R}$ : symmetric) together with  $v^T G v > 0$  for  $v \neq 0$ —the real number  $2d$  is at most  $\text{tr } G + \text{tr } G^{-1}$ . This is (31.10), [\[A\]](#). The statement is about a real matrix, not about a torus, and it does not include the equality case.

**Proof idea.** The script is the second proof of Theorem 31.7, step by step. Lines `hdet`, `hl`, `hr` extract invertibility from positive definiteness and record  $G^{-1}G = GG^{-1} = \mathbf{1}$ .



`hGinv_psd` is Step 2's first ingredient, the inverse of a positive definite matrix being positive semidefinite (Mathlib: `PosSemidef.inv`). Because Mathlib states the sandwich lemma with the conjugate transpose  $M^H$ , the script first records that over  $\mathbb{R}$  the conjugate transpose is the transpose and that  $G - \mathbf{1}$  is symmetric (`hGT`, `hMh`); then `PosSemidef.conjTranspose_mul_mul_same` gives the sandwich  $(G - \mathbf{1})^H G^{-1} (G - \mathbf{1}) \succeq 0$ , and `PosSemidef.trace_nonneg` is Step 3. `hexpand` is the identity (31.11), proved by distributing the products and using `hl`, `hr`. Finally the trace of a difference is the difference of traces,  $\text{tr } \mathbf{1} = |\text{Fin } d| = d$  (`Matrix.trace_one`, `Fintype.card_fin`), and `linarith` reads off  $0 \leq \text{tr } G - d - d + \text{tr } G^{-1}$ . No eigenvalue, and no spectral theorem, appears anywhere.

#### In Lean: attainment and the circle

```
theorem dualScale_one : dualScale (1 : Matrix (Fin d) (Fin d) ℝ) = 2 * d := by ...

theorem dualScale_circle (R : ℝ) (hR : R ≠ 0) :
  dualScale (!! [R ^ 2] : Matrix (Fin 1) (Fin 1) ℝ) = (R + R-1) ^ 2 - 2 := by ...

theorem circle_effective_scale_ge_two (R : ℝ) (hR : 0 < R) : 2 ≤ R + R-1 := by
  have h : R + R-1 - 2 = (R - 1) ^ 2 / R := by field_simp; ring
  have h_nonneg : 0 ≤ (R - 1) ^ 2 / R := by
    apply div_nonneg
    · exact sq_nonneg (R - 1)
    · linarith
  linarith
```

**Reading.** `dualScale_one` is Lemma 31.6(b): the value  $2d$  is attained at  $G = \mathbf{1}$ , so the constant in `dualScale_ge` is sharp. Together with `dualScale_ge` this says that  $\mathbf{1}$  is a global minimizer; that it is the *only* one is proved by hand in Theorem 31.7 and is not formalized. `dualScale_circle` is Lemma 31.6(c) for the  $1 \times 1$  matrix `!! [R^2]` with  $R \neq 0$ ; its proof computes the  $1 \times 1$  inverse by hand (`Matrix.inv_def`, `adjugate_fin_one`) and finishes with `field_simp; ring`. `circle_effective_scale_ge_two` is Proposition 31.3 in units  $\alpha' = 1$ , for real  $R > 0$ . Note its proof: it is the first proof of Proposition 31.3, the identity (31.6) in the form  $R + R^{-1} - 2 = (R - 1)^2/R$ , and *not* a specialization of `dualScale_ge` to  $d = 1$ , although the docstring of the module describes it as the  $d = 1$  content of the general bound. Both routes are valid; the one the kernel checked is the direct one. All three are [\[A\]](#).

#### Common pitfall: reading `2 * d` as a natural number

In `dualScale_ge` the left-hand side is written  $(2 * d : \mathbb{R})$ : the natural number  $d$  is cast to a real before being doubled, so that the inequality is between real numbers. Without the ascription Lean would compare a natural with a real and fail to elaborate, or worse, silently coerce after truncation. The same care appears in `dualScale_one`, whose right-hand side `2 * d` is real because the left-hand side is. When you state your own version of the bound (Exercise 31.7), write the cast explicitly.



## 31.6.1 Tier table for the Lean content

| Statement                                                                                   | Where                                                                                                                   | Tier         |
|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|--------------|
| $\mathcal{D}(G) = \text{tr } G + \text{tr } G^{-1}$ for all real $G$                        | <code>dualScale_eq</code>                                                                                               | [A]          |
| $\mathcal{D}(G^{-1}) = \mathcal{D}(G)$ , $\det G$ a unit                                    | <code>dualScale_inv</code>                                                                                              | [A]          |
| $2d \leq \mathcal{D}(G)$ for <code>PosDef</code> $G$ , all $d$                              | <code>dualScale_ge</code>                                                                                               | [A]          |
| $\mathcal{D}(\mathbf{1}) = 2d$                                                              | <code>dualScale_one</code>                                                                                              | [A]          |
| $\mathcal{D}(R^2) = (R + R^{-1})^2 - 2$ , $R \neq 0$                                        | <code>dualScale_circle</code>                                                                                           | [A]          |
| $2 \leq R + R^{-1}$ for real $R > 0$                                                        | <code>circle_effective_scale_ge_two</code>                                                                              | [A]          |
| $\eta \mathcal{H}(G, 0) \eta = \mathcal{H}(G^{-1}, 0)$                                      | <code>tduality_inverts_metric</code> (Chapter 15)                                                                       | [A]          |
| positive rationals: $\alpha' / (\alpha' / R) = R$ ; piecewise radius is a positive rational | <code>buscher_involution</code> ,<br><code>genesis_no_singularity</code>                                                | [A] (shadow) |
| integers: $L\tilde{L} = L_*^2$ symmetric; $-(-x) = x$ ;<br>$R^2 + 1 \geq 2$ for $R \geq 1$  | <code>dual_pair_symmetric</code> ,<br><code>buscher_log_involution</code> ,<br><code>self_dual_is_global_minimum</code> | [A] (shadow) |
| $\text{Sym}^2$ coefficients for three integer instances; $-c_0 = b^3$                       | <code>sym2_fibonacci_verification</code> ,<br><code>sym2_det_identity</code>                                            | [A] (shadow) |
| uniqueness of the minimizer at $B = 0$                                                      | <code>dualScale_eq_iff</code>                                                                                           | [A]          |
| $B$ -shifts increase $\text{tr } \mathcal{H}$ ; the general $\text{Sym}^2$ lemma            | this chapter, by hand                                                                                                   | —            |
| $G$ is a torus metric; $\mathcal{D}$ is a physical size; singularity resolution             | <code>docstrings</code> ; Chapter 32                                                                                    | [C]          |

All theorems above depend only on the standard axioms `propext`, `Classical.choice` and `Quot.sound`. The rows marked “shadow” are correct theorems whose statements are about integers or rationals; the physical readings in their docstrings are not part of what was checked.

## Chapter summary

- The dual-scale principle: a string sees the pair  $(R, \alpha'/R)$ , in  $d$  dimensions the pair  $(G, G^{-1})$ , and physical quantities are symmetric functions of the pair. The simplest is  $R_{\text{eff}} = R + \alpha'/R = 2\sqrt{\alpha'} \cosh x$ ,  $x = \ln(R/\sqrt{\alpha'})$ ; for tori,  $\mathcal{D}(G) = \text{tr } \mathcal{H}(G, 0) = \text{tr } G + \text{tr } G^{-1}$ .
- $R_{\text{eff}} \geq 2\sqrt{\alpha'}$  with equality only at the self-dual radius; three proofs (square, AM–GM,  $\cosh$ ).  $\text{tr } G + \text{tr } G^{-1} \geq 2d$  for positive definite  $G$ , with equality only at  $G = \mathbf{1}$ ; two proofs (eigenvalues; the sandwich  $(G - \mathbf{1})G^{-1}(G - \mathbf{1}) \succeq 0$  whose trace is  $\text{tr } G + \text{tr } G^{-1} - 2d$ ). A  $B$ -field only increases  $\text{tr } \mathcal{H}$ .
- $\mathcal{D}$  is invariant under the inversion  $\eta$  and under factorized dualities, but not under general  $GL(d, \mathbb{Z})$  basis changes; for  $d \geq 2$  it is a function of the metric in a basis, not of the torus.
- In the literature ([L]): small and large radius are equivalent for closed strings and this is read as a minimal length (GPR, Tong, AAL); the one-loop partition function on a circle is  $\propto R + 1/R$  (GPR); a balanced winding-momentum gas has energy  $\propto R + 1/R$  and is driven to the self-dual radius (BW).
- Lean ([A], all  $d$ , over  $\mathbb{R}$ ): `dualScale_eq`, `dualScale_inv`, `dualScale_ge` (sandwich proof, no eigenvalues), `dualScale_one`, `dualScale_circle`, `circle_effective_scale_ge_two` (direct proof), and `dualScale_eq_iff` (unique-

ness of the minimizer, added after the first edition). Older modules are arithmetic shadows over  $\mathbb{N}$ ,  $\mathbb{Z}$ ,  $\mathbb{Q}_{>0}$ ; in particular `self_dual_is_global_minimum` proves  $R^2 + 1 \geq 2$ , not  $R^2 + 1 \geq 2R$ .

- The cosmological use of the bound (a minimal size, a bounce) is a proposal ([C]), whose dynamical content is examined in Chapter 32.

## Exercises

**Exercise 31.1** (★). Extend the table of Example 31.4 to  $R = 3\ell_s$ ,  $R = \ell_s/3$  and  $R = 1.1\ell_s$ . For which  $R$  is  $R_{\text{eff}}/R_{\text{obs}}$  within 1% of 1? Show in general that  $R_{\text{eff}}/R_{\text{obs}} = 1 + e^{-2|x|}$ .

**Exercise 31.2** (★). Verify the sandwich identity (31.11) for  $G = \text{diag}(4, \frac{1}{4})$  by direct multiplication, and check that the trace of the sandwich is  $\mathcal{D}(G) - 4$ . Then repeat for  $G = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$  (answer:  $\mathcal{D} = 6$ ).

**Exercise 31.3** (★). Which of the following are dual-scale invariants on the circle, in the sense of Definition 31.1:  $R^2 + \alpha'^2/R^2$ ;  $|R - \alpha'/R|$ ;  $\min(R, \alpha'/R)$ ;  $R^3$ ;  $\sinh x$ ;  $\cosh 3x$ ? For each invariant, express it as a function of  $R_{\text{eff}}$ .

**Exercise 31.4** (★★). Prove Corollary 31.8(a) in full for  $d = 2$ : with  $B = \begin{pmatrix} 0 & \beta \\ -\beta & 0 \end{pmatrix}$  and general positive definite  $G$ , compute  $\text{tr}(B^T G^{-1} B)$  and show it equals  $\beta^2 \text{tr} G / \det G$ . Conclude that on the hexagonal torus of Example 31.11 a  $B$ -field with  $\beta = 1$  raises  $\mathcal{D}$  from  $14/3$  to  $14/3 + 8/3 = 22/3$ . Is  $\text{tr} \mathcal{H}(G, B)$  invariant under the integer  $B$ -shift  $B \mapsto B + \Theta$  of Chapter 15?

**Exercise 31.5** (★★). The dual scale is not  $GL(d, \mathbb{Z})$ -invariant. For the hexagonal torus, compute  $\mathcal{D}$  in the bases  $A_k = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$ ,  $k = 0, 1, 2, 3$ , and show that  $\mathcal{D}(A_k^T G A_k) \rightarrow \infty$  as  $k \rightarrow \infty$ . Show that the minimum of  $\mathcal{D}$  over the orbit is attained on a reduced basis, and argue that this minimum is a genuine invariant of the torus which still satisfies the bound  $2d$ .

**Exercise 31.6** (★★). Derive the pressures of the string gas from its energy. Take  $E(\lambda) = \tilde{n}_w e^\lambda + \tilde{n}_m e^{-\lambda}$  per unit comoving volume in a direction of radius  $e^\lambda$ , and define the pressure in that direction by  $p = -\partial E / \partial \lambda$ . Recover  $p_w = -\tilde{n}_w e^\lambda$  and  $p_m = \tilde{n}_m e^{-\lambda}$  [BW §III, eq. (30), ll. 845–870], show that  $p = 0$  exactly when  $\tilde{n}_w e^{2\lambda} = \tilde{n}_m$ , and that for  $\tilde{n}_w = \tilde{n}_m$  this is the self-dual radius and the energy is  $\tilde{n} R_{\text{eff}}$ . What happens to the equilibrium radius if  $\tilde{n}_w = 4\tilde{n}_m$ ?

**Exercise 31.7** (★★). In Lean, state and prove the circle bound with  $\alpha'$  restored:

```
theorem reff_ge (a R : ℝ) (ha : 0 < a) (hR : 0 < R) :
  2 * Real.sqrt a ≤ R + a / R := by
  sorry
```

Hint: set  $s = \sqrt{a}$ , use `Real.sq_sqrt` to replace  $a$  by  $s^2$ , and reduce to  $(R - s)^2 / R \geq 0$  as in `circle_effective_scale_ge_two`. (Chapter 34 explains `field_simp`, `nlinarith` and `div_nonneg`.)

**Exercise 31.8** (★★★). Formalize the uniqueness statement of Theorem 31.7 for  $d = 1$ :

```
theorem circle_eq_iff (R : ℝ) (hR : 0 < R) : R + R⁻¹ = 2 ↔ R = 1 := by
  sorry
```

Then explain, without writing the proof, what Mathlib facts about `PosSemidef` matrices you would need to prove the general statement `dualScale G = 2 * d → G = 1` along the lines of the hand proof (a positive semidefinite matrix with zero trace is zero; a sandwich  $M^T P M = 0$  with  $P$  positive definite forces  $M = 0$ ).

**Exercise 31.9** (\*\*\*). Prove Lemma 31.12 for the repeated-root case  $a^2+4b=0$  directly, using  $u_n = (\alpha + \beta n)\lambda^n$ , and check that the recurrence for  $v_n = u_n^2$  has characteristic polynomial  $(x - \lambda^2)^3$  with the coefficients (31.12). Which of the three Lean verification theorems, if any, has a repeated root?

## Further reading

- Target-space duality as the statement that small and large scales cannot be distinguished, and the survey of where the  $R \rightarrow 1/R$  symmetry appears (partition functions, open strings, matrix models): [GPR §1, ll. 215–440]. The circle spectrum, the duality map and the minimal-length remark: [GPR §2.2, ll. 615–800].
- The many meanings of “duality”, the dilaton shift and the  $(g+b) \mapsto (g+b)^{-1}$  generalization: [AAL §1, ll. 55–95]; the physical definition of distance from correlators: [AAL §6, ll. 1289–1320].
- The circle spectrum and T-duality as a minimum-length mechanism in Tong’s lectures: [Tong §8.2–8.3, ll. 11351–11665]; his broader remark that the minimum length “reveals itself in different ways depending on which question is being asked”: [Tong §6, ll. 8062–8072].
- String gas cosmology: the spectrum in a cosmological background, the energy and pressure of winding and momentum gases, the Brandenberger–Vafa argument and the drive to the self-dual radius: [BW §II–IV, ll. 655–1140].
- In this book: the operational radius and the four meanings of “T-duality is a symmetry” in Chapter 8; the generalized metric and `genMetric` in Chapter 15; the duality group in Chapter 11; the cosmological use of the bound in Chapter 32; how the arithmetic shadows fit the architecture of the formalization in Chapter 35.



## Chapter 32

# String Gas Cosmology and the Dual-Scale Early Universe

Every chapter so far has treated the compactification radius  $R$  as a parameter: a number we choose, a point in a moduli space, a coordinate on which T-duality acts as  $R \mapsto \alpha'/R$ . In the early universe  $R$  is not a parameter but a dynamical field, and the question this chapter asks is what T-duality does to *cosmology*. The setting is string gas cosmology (SGC), the programme begun by Brandenberger and Vafa in 1989 and reviewed by Battefeld and Watson, whose review is the source for everything below that is not derived on the page. Three features of strings distinguish the answer from point-particle cosmology. First, the low-energy equations of dilaton gravity have a symmetry, scale-factor duality, that inverts the scale factor. Second, a gas of strings contains winding modes, whose energy grows with the radius and which therefore exert *negative* pressure; a gas with equal numbers of winding and momentum modes has zero pressure exactly at the self-dual radius, and the radius is driven there. Third, at the self-dual radius new massless states appear, and their production as the radius passes through that point acts as a friction that traps it. The self-dual radius of Chapter 8, a curiosity of the spectrum, thus becomes a dynamical attractor. We shall also see what the argument for three large dimensions actually is and where it is incomplete, how perturbations survive the trapping, how all this relates to the dual-scale principle of Chapter 31, and exactly what the five Lean modules attached to this chapter prove, which is much less than their names suggest.

Conventions are those of Chapter 7: mostly-plus signature,  $\alpha'$  of dimension  $\text{length}^2$ ,  $\ell_s = \sqrt{\alpha'}$ . Following Battefeld–Watson (BW),  $D$  is the space-time dimension,  $d = D - 1$  the number of spatial dimensions in Section 32.1 and, from Section 32.2 on, the number of *compact* directions;  $\lambda_i = \ln a_i$  is the logarithmic scale factor of direction  $i$ ; winding numbers are  $\omega$  (elsewhere  $w$ ).

### 32.1 Dilaton gravity and scale-factor duality

#### 32.1.1 The low-energy action

A closed string moving in a background of its own massless fields  $G_{\mu\nu}$ ,  $B_{\mu\nu}$ ,  $\phi$  is described by the sigma model of Chapter 6; conformal invariance at one loop requires the vanishing of the beta functions, and those equations follow from the space-time action [BW §II, eq. (7), II. 343–361]

$$S_0 = \frac{1}{2\kappa_D^2} \int d^D x \sqrt{-G} e^{-2\phi} \left[ R + c + 4(\nabla\phi)^2 - \frac{1}{12} H^2 \right], \quad H = dB, \quad (32.1)$$

where  $c = 0$  in the critical dimension ( $D = 26$  bosonic,  $D = 10$  super) and acts as a cosmological constant otherwise. Two remarks on what has been dropped. The action is tree level in  $\alpha'$ : higher-derivative terms  $\alpha' R^2, \dots$  are neglected, which requires curvatures small in string units. It is also tree level in the string coupling  $g_s = e^{\phi_0}$ : the prefactor  $e^{-2\phi}$  is the sphere's Euler characteristic, and loops of strings would bring powers of  $e^{2\phi}$ . [BW §II, ll. 362–372] Both approximations are exactly the ones that fail near a cosmological singularity, and BW are candid that SGC “slowly turns on stringy effects” rather than solving string theory in a cosmological background. [BW §III.A, ll. 549–560]

### 32.1.2 Reduction to a mechanical system

Take a homogeneous, anisotropic background with toroidal spatial sections and no  $B$ -field, [BW §II, eq. (8), ll. 385–398]

$$ds^2 = -dt^2 + \sum_{i=1}^d a_i(t)^2 dx_i^2, \quad a_i = e^{\lambda_i(t)}, \quad \phi = \phi(t), \quad d = D - 1. \quad (32.2)$$

The key definition is the *shifted dilaton* [BW §II, eq. (9), ll. 399–404]

$$\varphi = 2\phi - \ln V = 2\phi - \sum_{i=1}^d \lambda_i, \quad (32.3)$$

which absorbs the volume  $V = \prod a_i$  of the spatial torus into the dilaton. Let us reduce (32.1) on (32.2); BW state the result and we supply the steps. For a diagonal metric  $-dt^2 + \sum e^{2\lambda_i} dx_i^2$  the Ricci scalar is (a Christoffel computation, which we checked symbolically)

$$R = 2 \sum_i \ddot{\lambda}_i + \sum_i \dot{\lambda}_i^2 + \left( \sum_i \dot{\lambda}_i \right)^2. \quad (32.4)$$

Write  $\Lambda = \sum_i \lambda_i = \ln V$ . Then  $\sqrt{-G} e^{-2\phi} = e^{\Lambda-2\phi} = e^{-\varphi}$ , and with mostly-plus signature  $(\nabla\phi)^2 = G^{tt} \dot{\phi}^2 = -\dot{\phi}^2$ . The Lagrangian per unit coordinate volume is therefore

$$L = e^{-\varphi} \left[ 2\ddot{\Lambda} + \sum_i \dot{\lambda}_i^2 + \dot{\Lambda}^2 - 4\dot{\phi}^2 \right].$$

Now use  $2\phi = \varphi + \Lambda$ , so  $4\dot{\phi}^2 = \dot{\varphi}^2 + 2\dot{\varphi}\dot{\Lambda} + \dot{\Lambda}^2$ , and integrate the second-derivative term by parts:  $2e^{-\varphi}\ddot{\Lambda} = \frac{d}{dt}(2e^{-\varphi}\dot{\Lambda}) + 2e^{-\varphi}\dot{\varphi}\dot{\Lambda}$ . The terms  $\dot{\Lambda}^2$  and  $2\dot{\varphi}\dot{\Lambda}$  cancel and, up to a total derivative,

$$L = e^{-\varphi} \left[ \sum_{i=1}^d \dot{\lambda}_i^2 - \dot{\varphi}^2 \right]. \quad (32.5)$$

This is the whole content of (32.1) on homogeneous backgrounds: a mechanical system with  $d+1$  degrees of freedom, a kinetic term of indefinite sign (the dilaton direction is “time-like”, a familiar feature of gravitational minisuperspace) and no potential. We verified with sympy that the Euler–Lagrange equations of (32.5) and of the unreduced Lagrangian coincide.

### 32.1.3 Scale-factor duality

Equation (32.5) is manifestly invariant under [BW §II, eq. (10), ll. 405–421]

$$a_i \mapsto \frac{1}{a_i}, \quad \lambda_i \mapsto -\lambda_i, \quad \varphi \mapsto \varphi, \quad (32.6)$$

for any subset of the directions  $i$ . This is *scale-factor duality* (SFD). Since  $\varphi$  is held fixed while  $\ln V$  changes sign, the ordinary dilaton transforms as  $\phi \mapsto \phi - \sum_{i \in S} \lambda_i$ , exactly the Buscher shift of Chapter 9 with  $R \mapsto \alpha'/R$ . BW’s reading: “the effective field theory of dilaton gravity for a small scale factor is equivalent to that for a large scale factor” [BW §II, ll. 424–429] — a contracting universe and an expanding one are related by a symmetry of the equations, a statement with no analogue in general relativity, where the Einstein–Hilbert action reduced on (32.2) has no such symmetry.

#### Scale-factor duality is not yet T-duality

SFD is a symmetry of the *massless* sector (32.1) on a restricted class of backgrounds; it knows nothing about winding strings. T-duality (Chapter 8) is a statement about the full string spectrum and requires the winding modes to exist. The two agree on the spectrum of a string gas, (32.9) below, which is why one may consistently add a string gas to dilaton gravity; but SFD alone does not say that “small is the same as large” for matter.

## 32.2 A gas of strings

### 32.2.1 Assumptions

SGC works under four stated approximations [BW §III.A, ll. 517–548]: the background fields are *homogeneous* (functions of  $t$  only); the evolution is *adiabatic*, so that  $\alpha'$  corrections are negligible and each string sees a locally constant radius; the coupling is *weak*,  $g_s \ll 1$ ; and the spatial dimensions are *toroidal*, so that winding modes exist (later relaxed: orbifolds also confine). BW call the middle two “very restrictive from the string theory perspective”. [BW §III.A, ll. 549–560] Under them the string equation of motion reduces to the free wave equation in conformal gauge, and the mode expansion of Chapter 7 applies direction by direction.

### 32.2.2 Energy of one string and the duality of the spectrum

For a compact direction of radius  $R$  the left- and right-moving zero-mode momenta are  $p_{L,R} = \sqrt{\alpha'} n/R \pm R\omega/\sqrt{\alpha'}$ , with  $n \in \mathbb{Z}$  the momentum and  $\omega \in \mathbb{Z}$  the winding number. Inserting the mode expansion in the constraints gives, in the gauge  $X^0 = \sqrt{\alpha'} E\tau$ , the mass-shell condition [BW §III.B, eq. (22), ll. 680–697]

$$\alpha' E^2 = \alpha' \vec{P}^2 + \frac{1}{2}(p_L^2 + p_R^2) + 2(N_L + N_R + a_L + a_R), \quad (32.7)$$

where  $\vec{P}$  is the momentum in the non-compact directions,  $N_{L,R}$  are the oscillator levels and  $a_{L,R}$  the normal-ordering constants ( $a_L = a_R = -1$  bosonic; heterotic  $a_L = -1$ ,  $a_R = -\frac{1}{2}$  in the NS sector). The second constraint gives level matching [BW §III.B, eq. (23), ll. 716–719]

$$p_L^2 - p_R^2 = 4n\omega = 4(N_R - N_L + a_R - a_L). \quad (32.8)$$

Since  $\frac{1}{2}(p_L^2 + p_R^2) = \alpha' n^2/R^2 + R^2 \omega^2/\alpha'$ , the bosonic case of (32.7) with  $\vec{P} = 0$  is  $M^2 = n^2/R^2 + \omega^2 R^2/\alpha'^2 + (2/\alpha')(N_L + N_R - 2)$  with  $N_R - N_L = n\omega$ : precisely the mass formula of Chapter 7 (with  $N = N_R$ ,  $\tilde{N} = N_L$ ). The spectrum is invariant under [BW §III.B, eq. (24), ll. 725–735]

$$\frac{R}{\sqrt{\alpha'}} \mapsto \frac{\sqrt{\alpha'}}{R}, \quad n \longleftrightarrow \omega, \quad (32.9)$$

which is the T-duality of Chapter 8. Its cosmological meaning, in BW’s words, is that “strings at small scale factor  $1/R$  behave the same as strings at large scale factor  $R$ ”, so that their effect on the background “may differ greatly from that of ordinary point particles”. [BW §III.B, ll. 736–742]

### 32.2.3 Energy density and pressure of the gas

The  $T^{00}$  component of the string stress tensor, integrated over the worldsheet, is the energy  $E$  of the string times a delta function at its position. [BW §III.B, eqs. (25)–(27), ll. 743–800] For a gas of  $N_s$  strings of each species  $s$  (a species is a choice of  $n, \omega, N_L, N_R$ ) in spatial volume  $V$ , averaging the delta functions gives [BW §III.B, eqs. (28)–(29), ll. 810–840]

$$\rho = \sum_s \tilde{n}_s E_s, \quad \tilde{n}_s = \frac{N_s}{V}, \quad p_i = -\frac{1}{V} \frac{\partial(\rho V)}{\partial \lambda_i}, \quad (32.10)$$

the gas being treated as a non-interacting perfect fluid. The pressure formula is the first law  $dE = -p dV$  direction by direction:  $\rho V$  is the total energy, and  $\partial/\partial \lambda_i$  is the logarithmic derivative with respect to the  $i$ -th scale factor.

Consider now the two simplest gases in  $d$  compact directions: pure winding strings ( $\omega_m \neq 0, n = N_L = N_R = 0$ ) and pure momentum strings ( $n_m \neq 0, \omega = N_L = N_R = 0$ ), with  $\vec{P} = 0$  and the tachyonic zero-point energy removed by hand. A string wound once around the direction  $l$  has energy  $R_l/\alpha' = e^{\lambda_l}/\alpha'$  (tension times length); a momentum mode has energy  $1/R_l = e^{-\lambda_l}$ . Absorbing the winding and momentum numbers, and the factors of  $\alpha'$ , into the number densities  $\tilde{n}_w^{(l)}, \tilde{n}_m^{(l)}$ : [BW §III.B, eq. (30), ll. 855–875]

$$\rho_w = \sum_{l=1}^d \tilde{n}_w^{(l)} e^{\lambda_l}, \quad \rho_m = \sum_{l=1}^d \tilde{n}_m^{(l)} e^{-\lambda_l}, \quad p_w^{(l)} = -\tilde{n}_w^{(l)} e^{\lambda_l}, \quad p_m^{(l)} = \tilde{n}_m^{(l)} e^{-\lambda_l}. \quad (32.11)$$

Let us derive the pressures, because a sign is at stake. The quantity held fixed when a scale factor changes is the *number*  $N_w^{(l)}$  of strings, not their density. So  $\rho_w V = \sum_l N_w^{(l)} e^{\lambda_l}$ , and  $p_w^{(l)} = -V^{-1} \partial(\rho_w V)/\partial \lambda_l = -N_w^{(l)} e^{\lambda_l}/V = -\tilde{n}_w^{(l)} e^{\lambda_l}$ . Likewise  $p_m^{(l)} = +\tilde{n}_m^{(l)} e^{-\lambda_l}$ . For an isotropic distribution ( $\tilde{n}^{(l)}$  and  $\lambda_l$  independent of  $l$ ) the equations of state are [BW §III.B, eq. (31), ll. 895–904]

$$p_m = \frac{1}{d} \rho_m, \quad p_w = -\frac{1}{d} \rho_w. \quad (32.12)$$

Momentum modes behave as radiation filling  $d$  dimensions; winding modes have negative pressure. We checked (32.12) symbolically for  $d = 3$ .

#### Differentiating the density instead of the number

If one writes  $\rho_w = d \tilde{n}_w e^{\lambda}$  and differentiates  $\rho_w V$  with  $\tilde{n}_w$  held fixed, one obtains  $p_w = -\frac{d+1}{d} \rho_w$  for the winding gas and  $p_m = -\frac{d-1}{d} \rho_m$  for the momentum gas: the momentum gas would then have negative pressure too, which is absurd for radiation. The error is that  $\tilde{n} = N/V$  itself depends on  $V$ . Hold  $N$  fixed.

#### Why winding strings pull the universe together

A string wound around a circle is stretched: its energy is the tension  $1/(2\pi\alpha')$  times its length  $2\pi R$ . Making the circle larger costs energy, so the gas of wound strings resists expansion — that is what negative pressure means ( $dE = -p dV > 0$  when  $dV > 0$ ). A momentum mode has energy  $n/R$ , which *decreases* with  $R$ : it pushes outward like radiation. In dilaton gravity the negative pressure of winding modes leads to contraction, not to accelerated expansion as a negative-pressure fluid would in general relativity; BW stress that this was first shown by Tseytlin and Vafa and that the difference comes from the dilaton and from anisotropy. [BW §IV.A, ll. 933–945]



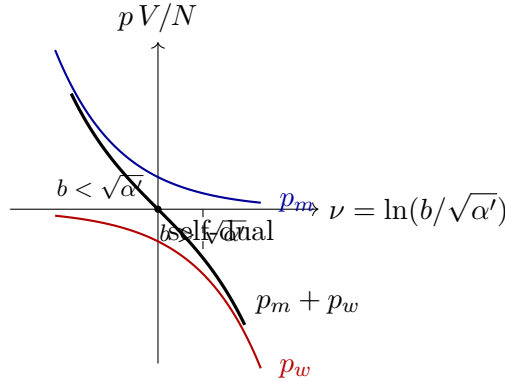


Figure 32.1: Pressure in one compact direction from a momentum gas ( $p_m \propto e^{-\nu}$ ) and a winding gas ( $p_w \propto -e^{\nu}$ ) with equal numbers, (32.11). The sum vanishes only at the self-dual radius and is a restoring pressure on both sides.

**Example 32.1** (Pressure balance fixes the radius). Let one compact direction of radius  $b = e^{\nu} \sqrt{\alpha'}$  carry  $N_w$  winding and  $N_m$  momentum strings in a box of volume  $V$ . By (32.11) the total pressure in that direction is  $p = (N_m e^{-\nu} - N_w e^{\nu})/V$ , which vanishes when  $e^{2\nu} = N_m/N_w$ , i.e.  $b = \sqrt{\alpha'} (N_m/N_w)^{1/2}$ . With  $N_m = N_w$  the pressure-free radius is the self-dual radius  $b = \sqrt{\alpha'}$ , and  $p > 0$  for  $b < \sqrt{\alpha'}$ ,  $p < 0$  for  $b > \sqrt{\alpha'}$ : the gas pushes the radius back toward the self-dual point from either side (Figure 32.1). With  $N_m = 4N_w$  the balance point moves to  $b = 2\sqrt{\alpha'}$ , T-dual to the balance point  $b = \frac{1}{2}\sqrt{\alpha'}$  of the gas with  $N_m = \frac{1}{4}N_w$ , as (32.9) requires.

#### Temperature duality and the Hagedorn transition

Continuing time to a circle of circumference  $\beta = 1/T$  turns the partition function into a thermal one, and T-duality in the time direction becomes *temperature duality*: Giveon–Porrati–Rabinovici report that inversion of the time-like radius  $\beta$  “was shown to be an exact symmetry to all orders in string perturbation theory”, implying *two* Hagedorn-like transitions for the heterotic string, at  $\beta_1 = \sqrt{2} + 1$  and  $\beta_2 = \sqrt{2} - 1$  in string units, the second the image of the first ( $\beta_1 \beta_2 = 1$ ). [GPR §1, ll. 364–372] Alvarez, Alvarez-Gaumé and Lozano write the free-energy form  $F(\beta) = (\pi^2/\beta^2) F(\pi^2/\beta)$  (other normalization of  $\beta$ ) and add that its interpretation “is somewhat uncertain due to the presence of the Hagedorn temperature”. [AAL §1, ll. 97–109] The Hagedorn temperature  $T_H$ , above which the exponentially growing density of string states makes the canonical partition function diverge, is the usual starting point of SGC (a hot gas near  $T_H$ ); BW do not derive it and this library contains no derivation, so the two quoted statements are [L] and the rest is “standard; not checked against a source in this book’s library”.

### 32.3 The Brandenberger–Vafa argument

Why are three spatial dimensions large? SGC is, in BW’s words, “the only cosmological model thus far that has attempted to explain the dimensionality of space-time dynamically”. [BW §IV.A, ll. 918–922] The argument runs as follows. [BW §IV.A, ll. 922–933]

1. Start with all nine spatial dimensions compact and string-sized, filled with a hot gas of strings including winding modes.

2. Winding modes prevent expansion (Section 32.2); a direction can grow only if its winding modes annihilate against anti-winding modes.
3. Take string interactions to be intersections of worldsheets. Two  $p$ -dimensional objects generically intersect only if  $2(p + 1) \leq D$ , i.e. in at most  $2p + 1$  spatial dimensions. For strings,  $p = 1$ : two worldsheets are surfaces, which generically meet in points in a four-dimensional space-time and miss each other in five or more.
4. Hence winding modes can stay in equilibrium, and annihilate efficiently, in at most three spatial dimensions. Those three are freed to expand; the other six remain confined near the string scale by their winding modes.

The dimension count in step 3 is the same one that makes two curves generically intersect in a plane but not in space. The prediction of the argument is a  $3 + 1 + 6$  split with the three large directions radiation dominated: winding modes annihilate into unwound closed strings, i.e. momentum modes, which by (32.12) with  $d = 3$  have the equation of state of radiation. [BW §IV.C, ll. 1098–1104] Before all winding modes have annihilated the universe *loiters*: the scale factor stays nearly constant, and BW note that a loitering phase would solve the horizon problem without inflation. [BW §IV.C, ll. 1088–1100]

#### The status of the argument

BW devote a page to “serious challenges for its quantitative realization”. [BW §IV.A, ll. 946–1001] Supergravity studies with a fixed wrapping matrix found the anisotropic expansion but no stabilization of the internal dimensions, which merely grew more slowly; Einstein–Boltzmann studies of a thermal brane gas found that only fine-tuned initial conditions work: generically “either all dimensions grow large, since the string gas annihilated entirely, or all dimensions stay small, since the string gas froze out”. Two reasons are given: the interaction rate is  $\propto g_s^2$  and switches off as the dilaton rolls to weak coupling; and intersection is a classical picture, whereas closed-string exchange mediates interactions in any dimension, diluted only by  $F \sim r^{-(D-2)}$ . BW conclude that “compactification of all or none of the dimensions is the most probable configuration” under present knowledge, and that the argument “is not quintessential to string gas cosmology”: the rest of the programme *assumes* the  $3 + 6$  split as an initial condition. [BW §I, ll. 171–174; §IV.A, ll. 985–1001] All of this is [L], a report of the literature as of 2005; the dimensionality question is open.

## 32.4 Cosmological equations with a string gas

### 32.4.1 The equations of motion

Anticipating the  $3 + 1 + 6$  split, take [BW §IV.B, eqs. (33)–(34), ll. 1004–1011]

$$ds^2 = -dt^2 + e^{2\lambda(t)} d\vec{x}^2 + e^{2\nu(t)} d\vec{y}^2, \quad H_3 = h dx^1 \wedge dx^2 \wedge dx^3, \quad \phi = \phi(t), \quad (32.13)$$

with three large directions  $\vec{x}$ , six compact ones  $\vec{y}$  of radius  $b = e^\nu \sqrt{\alpha'}$  (the *radion*), and a constant flux  $h$  allowed by the symmetry. The string gas enters the field equations through its averaged stress tensor  $T^\mu{}_\nu = \text{diag}(-\rho, p_1, \dots, p_9)$ . In terms of the total energy  $E = \rho V$ , the scaled pressures  $P_i = p_i V$  and the shifted dilaton  $\varphi$ , BW write the system as [BW §IV.B,

eqs. (38)–(41), ll. 1050–1071]

$$c - 3\dot{\lambda}^2 - 6\dot{\nu}^2 + \dot{\varphi}^2 - \frac{1}{2}h^2e^{-6\lambda} = e^\varphi E, \quad (32.14)$$

$$\ddot{\lambda} - \dot{\varphi}\dot{\lambda} - \frac{1}{2}h^2e^{-6\lambda} = \frac{1}{2}e^\varphi P_3, \quad (32.15)$$

$$\ddot{\nu} - \dot{\varphi}\dot{\nu} = \frac{1}{2}e^\varphi P_6, \quad (32.16)$$

$$\ddot{\varphi} - 3\dot{\lambda}^2 - 6\dot{\nu}^2 = \frac{1}{2}e^\varphi E, \quad (32.17)$$

with the conservation law  $\dot{E} + 3\dot{\lambda}P_3 + 6\dot{\nu}P_6 = 0$ . [BW §IV.B, eq. (42), ll. 1072–1079] Here  $16\pi G_{10}$  has been absorbed into the units of  $E$  and  $P$ .

These equations can be derived in a few lines from (32.5), which is instructive because it shows where each term comes from. Set  $h = c = 0$ , put  $\lambda_{1,2,3} = \lambda$ ,  $\lambda_{4,\dots,9} = \nu$ , restore a lapse  $N(t)$  (so that  $dt \rightarrow Ndt$  and the kinetic term acquires a factor  $1/N$ ), and add the matter energy as  $-NE(\lambda, \nu)$ :

$$L = \frac{e^{-\varphi}}{N} (3\dot{\lambda}^2 + 6\dot{\nu}^2 - \dot{\varphi}^2) - N E(\lambda, \nu). \quad (32.18)$$

The dependence of  $E$  on  $\lambda, \nu$  is what produces pressure: by (32.10),  $\partial E / \partial \lambda_i = -P_i$ , so with three equal directions  $\partial E / \partial \lambda = -3P_3$  and  $\partial E / \partial \nu = -6P_6$ . Varying  $N$  and then setting  $N = 1$  gives the constraint (32.14). Varying  $\lambda$ :  $\frac{d}{dt}(6e^{-\varphi}\dot{\lambda}) = -\partial E / \partial \lambda = 3P_3$ , i.e.  $6e^{-\varphi}(\ddot{\lambda} - \dot{\varphi}\dot{\lambda}) = 3P_3$ , which is (32.15);  $\nu$  gives (32.16) in the same way. Varying  $\varphi$ :  $\frac{d}{dt}(-2e^{-\varphi}\dot{\varphi}) = -e^{-\varphi}(3\dot{\lambda}^2 + 6\dot{\nu}^2 - \dot{\varphi}^2)$ , i.e.  $2\ddot{\varphi} = \dot{\varphi}^2 + 3\dot{\lambda}^2 + 6\dot{\nu}^2$ ; eliminating  $\dot{\varphi}^2$  with the constraint yields (32.17). The conservation law is the chain rule  $\dot{E} = \partial_\lambda E \dot{\lambda} + \partial_\nu E \dot{\nu}$ . We confirmed all four equations with sympy from (32.18).

Two structural facts follow. First,  $e^\varphi = e^{2\phi}/V = g_s^2/V$ : the string gas couples to the geometry with strength  $g_s^2$ , as matter should at one loop. Second, the dilaton is sourced by the *trace* of the stress tensor, so “the dilaton can only evolve if matter is not conformal”; a string gas is not conformal, “and it is this important observation that makes string cosmology (dilaton gravity) very different from ordinary cosmology”. [BW §IV.B, ll. 1030–1040]

### 32.4.2 Stabilization of the radion at the self-dual radius

Fill the six compact directions with equal numbers of winding and momentum strings and let the three large ones expand. Watson and Brandenberger found that the radion then *oscillates about the self-dual radius*, driven by the competing pressures of Example 32.1, and that the oscillations are damped provided the dilaton runs to weak coupling. [BW §IV.C, ll. 1105–1135] Let us see this in (32.16).

**Example 32.2** (Small oscillations of the radion). With  $N_w = N_m = N$  strings per compact direction,  $P_6 = N(e^{-\nu} - e^\nu) \approx -2N\nu$  for small  $\nu$ , and (32.16) becomes

$$\ddot{\nu} - \dot{\varphi}\dot{\nu} + e^\varphi N \nu = 0. \quad (32.19)$$

This is a damped oscillator with frequency  $\Omega^2 = e^\varphi N = g_s^2 N/V$  and friction coefficient  $-\dot{\varphi}$ . Running to weak coupling means  $\dot{\varphi} < 0$ , and with  $V$  growing,  $\dot{\varphi} = 2\dot{\phi} - \ln \dot{V} < 0$  as well, so the friction term has the right sign: the dilaton damps the radion, which is BW’s Figure 1 in a formula. In string units, with  $g_s = 0.1$  and one string of each type per unit string volume,  $\Omega = 0.1/\sqrt{\alpha'}$  and the period is  $2\pi/\Omega \approx 63\sqrt{\alpha'}$ ; the frequency falls with the coupling, and the restoring force disappears if the gas is diluted away — the seed of the difficulty discussed next.

### 32.4.3 The Einstein-frame objection

The stability just found is a string-frame statement. At late times the physically meaningful frame is the four-dimensional Einstein frame, in which Newton's constant is fixed; a rolling dilaton would make  $G_N$  time dependent, which fifth-force experiments exclude. [BW §IV.D, ll. 1191–1200] Rewriting the ten-dimensional string-frame metric in Einstein frame, [BW §IV.D, eq. (43), ll. 1201–1204]

$$ds_E^2 = -dt_E^2 + e^{\phi/2} a^2(t) d\vec{x}^2 + e^{\phi/2} b^2(t) d\vec{y}^2, \quad dt_E^2 = e^{\phi/2} dt_s^2, \quad (32.20)$$

shows the problem at a glance: even if  $b(t)$  is frozen, the Einstein-frame radius  $e^{\phi/4} b$  keeps evolving as long as the dilaton does. BW insist that “the two frames are physically equivalent, but the instability is manifest in the Einstein frame” [BW §IV.D, ll. 1212–1222]: a conformal rescaling changes no prediction, only which variable is called “the radius”, so a claim of stabilization must always say in which frame, and whether the dilaton has been fixed.

Reducing to four dimensions and canonically normalizing the dilaton and radion ( $\phi$  and  $\psi$  below), the potential of a gas of  $N$  objects of type  $k$  wrapped on the compact space is, in the Einstein frame and at weak coupling ( $\phi \rightarrow -|\phi|$ ), [BW §IV.D, eqs. (46)–(48), ll. 1270–1321]

$$V_E^{(4)} = \mu \tilde{n} e^{\phi} b^{k-d/2} = \mu \tilde{n} e^{-|\phi|} \exp\left(\frac{2k-d}{2\sqrt{2d}} \psi\right), \quad (32.21)$$

where  $k = 1$  for a wound string,  $k = -1$  for a string with momentum,  $k = 2$  for a wrapped 2-brane, and  $\mu = (2\pi\sqrt{\alpha'})^{-4}$ . The exponent of  $b$  is the string-frame exponent  $k$  of the energy shifted by  $-d/2$ , the shift coming from the Weyl rescaling that removes the volume factor from the four-dimensional Einstein term. A confining potential requires a positive power,  $k \geq d/2$ : for winding strings ( $k = 1$ ) this holds only for  $d = 1$ , and for  $d = 6$  the exponent is  $1 - 3 = -2$ , a runaway. Hence BW's conclusion: “a gas of purely winding strings is not enough to stabilize the extra dimensions”, and the overall  $e^{-|\phi|}$  dilutes whatever potential there is. [BW §IV.D, ll. 1322–1333]

### 32.4.4 States that become massless at the self-dual radius

The remedy proposed by Watson and Brandenberger is to include the string states that are massless *exactly* at the self-dual radius. For the heterotic string in its NS ground state ( $N_R = \frac{1}{2}$ ,  $N_L = 0$ ), the energy and level-matching condition for a state with momentum and winding vectors  $n, \omega \in \mathbb{Z}^d$  on a square torus of radius  $b$  are [BW §IV.D, eqs. (49)–(51), ll. 1340–1376]

$$E^2 = \left| \frac{n}{b} + \frac{\omega b}{\alpha'} \right|^2 - \frac{4}{\alpha'}, \quad n \cdot \omega = 1, \quad (32.22)$$

which at  $b = \sqrt{\alpha'}$  vanishes precisely when [BW §IV.D, eq. (52), ll. 1385–1392]

$$n \cdot n + \omega \cdot \omega = 2, \quad n \cdot \omega = 1, \quad (32.23)$$

i.e.  $n = \omega = \pm e_m$  for a unit vector  $e_m$ . Away from the self-dual radius, with  $n = \omega$  along one direction and  $b$  in units of  $\sqrt{\alpha'}$ , [BW §IV.D, eq. (54), ll. 1418–1426]

$$E = \sqrt{\frac{n \cdot n}{b^2} + \omega \cdot \omega b^2 - 2n \cdot \omega} = \left| \frac{1}{b} - b \right| = 2 \left| \sinh\left(\frac{\psi}{\sqrt{2d}}\right) \right|, \quad (32.24)$$

the last form for  $b = e^{\psi/\sqrt{2d}}$ , the normalization of  $\psi$  implied by (32.21) (BW omit the absolute value). For  $b = 1.2\sqrt{\alpha'}$  the mass is  $|1/1.2 - 1.2| = 11/30 \approx 0.37$  in units of  $1/\sqrt{\alpha'}$ : a state

that is massless at the self-dual point is already a third of the string scale 20% away from it. Its potential in the Einstein frame, [BW §IV.D, eq. (55), ll. 1440–1451]

$$V_E^{(4)} = 2\mu\tilde{n} e^{\phi - \frac{1}{2}\sqrt{d/2}\psi} \left| \sinh\left(\frac{\psi}{\sqrt{2d}}\right) \right|, \quad (32.25)$$

does have a local minimum, but it “becomes shallow” as the dilaton runs to weak coupling. [BW §IV.D, ll. 1452–1456] BW then raise the objection that decides the matter: away from  $b = \sqrt{\alpha'}$  these states are string-scale heavy and do not belong in a low-energy effective action, and “if we include one massive state of the string, don’t we have to include all of them?” — other states become light at other points of moduli space, each an attractor, and which one traps the radion depends on initial conditions. [BW §IV.D, ll. 1457–1472]

## 32.5 Quantum moduli trapping

The resolution BW favour is to keep only the massless fields in the action and to let the extra states be *produced* quantum mechanically when the radion passes near a point where they become massless, an *enhanced symmetry point* (ESP). [BW §V, ll. 1490–1510] The produced particles then have masses that grow with the distance from the ESP, and their energy density is a potential that pulls the radion back. This is *quantum moduli trapping*, a mechanism first studied for D-brane moduli by Kofman et al. and adapted to SGC by Watson.

### 32.5.1 The spectrum at the self-dual radius

Take the bosonic string on  $M_4 \times S^1$  with radius  $R$  and radion  $\sigma = \ln(R/\sqrt{\alpha'})$ . The mass formula and level matching are [BW §V, eq. (59), ll. 1580–1587]

$$M^2 = \frac{n^2}{R^2} + \frac{\omega^2 R^2}{\alpha'^2} + \frac{2}{\alpha'}(N_L + N_R - 2), \quad n\omega + N_L - N_R = 0. \quad (32.26)$$

At generic radius the massless states have  $n = \omega = 0$ ,  $N_L = N_R = 1$ : the graviton,  $B$ -field and dilaton (oscillators in the non-compact directions), the radion  $\sigma$  (both oscillators in the compact direction) and two vectors  $A_\mu = \frac{1}{2}(G_{\mu 5} + B_{\mu 5})$ ,  $\bar{A}_\mu = \frac{1}{2}(G_{\mu 5} - B_{\mu 5})$ , a  $U(1)_L \times U(1)_R$  gauge theory. [BW §V, eqs. (60)–(62), ll. 1620–1660] At  $R = \sqrt{\alpha'}$  substitute  $N_R = N_L + n\omega$  into  $\alpha' M^2 = n^2 + \omega^2 + 2(N_L + N_R - 2)$ :

$$\alpha' M^2 = (n + \omega)^2 + 4(N_L - 1), \quad n\omega + N_L - N_R = 0, \quad (32.27)$$

which is BW’s eq. (64). [BW §V, ll. 1670–1678] Massless states need  $(n + \omega)^2 = 4(1 - N_L)$ , hence  $N_L \in \{0, 1\}$ . For  $N_L = 1$ :  $n = -\omega$  and  $N_R = 1 - \omega^2 \geq 0$  forces  $\omega = 0$  (the generic states) or  $(n, \omega) = (\pm 1, \mp 1)$  with  $N_R = 0$ . For  $N_L = 0$ :  $n + \omega = \pm 2$  and  $N_R = n\omega \in \{0, 1\}$ , giving  $(\pm 2, 0)$ ,  $(0, \pm 2)$  with  $N_R = 0$  and  $(\pm 1, \pm 1)$  with  $N_R = 1$ . A computer enumeration over  $|n|, |\omega| \leq 3$ ,  $N_{L,R} \leq 2$  finds exactly these nine  $(n, \omega, N_L, N_R)$  combinations (including  $(0, 0, 1, 1)$ ). With the excited oscillator along the circle they are scalars, along  $M_4$  vectors: four new vectors join  $A_\mu, \bar{A}_\mu$  to form the adjoint of  $SU(2)_L \times SU(2)_R$ , and eight new scalars join  $\sigma$  to form its  $(\mathbf{3}, \mathbf{3})$ . [BW §V, ll. 1690–1770] This is the enhanced gauge symmetry of Chapter 8, now a cosmological event. The gauge coupling gives the new states masses  $m_A^2 \sim g^2 \sigma^2$ : the radion is a Higgs field for  $SU(2) \times SU(2)$ , with  $g$  of order one in string units. [BW §V, ll. 1780–1800]

### 32.5.2 Production and backreaction

Freeze the dilaton and keep only the four-dimensional metric  $e^{2\lambda}$ , the radion  $\sigma$  and one of the new vectors  $A_\mu$  with the coupling [BW §V, eq. (75), ll. 1880–1885]

$$V_{\text{eff}}(\sigma, A) = \frac{1}{2}(\partial_\mu A_\nu)^2 - \frac{1}{2}g^2\sigma^2 A_\mu A^\mu, \quad (32.28)$$

so that  $m^2(t) = g^2\sigma(t)^2$  is a time-dependent mass. Before production the radion is a free field with  $\rho = p$ , its kinetic energy scales as  $a^{-6}$ ,  $a \sim t^{1/3}$ , and  $\sigma(t) = \sigma_0 + v_0 t$  with  $\sigma_0 = 0$  if the radion passes exactly through the ESP. [BW §V, eqs. (71)–(74), ll. 1830–1852] A Fourier mode of  $A$  has frequency  $\omega_k = \sqrt{k^2 + g^2\sigma^2}$  and is excited when its evolution is non-adiabatic,  $\dot{\omega}_k/\omega_k^2 \geq 1$ ; the resulting occupation number is [BW §V, eqs. (76)–(77), ll. 1900–1916]

$$n_k = \exp\left(-\frac{\pi(k^2 + g^2\sigma_0^2)}{gv_0}\right), \quad (32.29)$$

the standard result for a mass that passes linearly through zero. Integrating, [BW §V, eq. (78), ll. 1920–1926]

$$\rho_A = \int \frac{d^3k}{(2\pi)^3} n_k \omega_k \approx g|\sigma(t)|\mathcal{N}, \quad \mathcal{N} \sim (gv_0)^{3/2}, \quad (32.30)$$

because once  $\sigma$  has moved on,  $\omega_k \approx g|\sigma|$  for all produced modes. Since the energy in  $A$  grows linearly with  $|\sigma|$ , it is a potential  $V = g\mathcal{N}|\sigma|$  with a constant restoring force of magnitude  $g\mathcal{N}$  pointing back to the ESP, and the radion equation becomes [BW §V, eq. (79), ll. 1930–1937]

$$\ddot{\sigma} + 3\dot{\lambda}\dot{\sigma} = -g\mathcal{N}(t) \operatorname{sgn} \sigma. \quad (32.31)$$

Each passage through the ESP converts radion kinetic energy into particles, the force never changes sign, Hubble friction  $3\dot{\lambda}\dot{\sigma}$  damps what remains, and the radion settles at  $\sigma = 0$ : BW conclude that friction plus backreaction “should be more than adequate to stabilize the radion at the self dual radius”. [BW §V, ll. 1940–1957]

**Example 32.3** (Which modes are produced). With  $\sigma = v_0 t$  and  $\sigma_0 = 0$ ,  $\dot{\omega}_k/\omega_k^2 = g^2 v_0^2 t / (k^2 + g^2 v_0^2 t^2)^{3/2}$ . For fixed  $k$  this is maximal at  $t_* = k/(\sqrt{2}gv_0)$ , where it equals  $2\sqrt{3}gv_0/(9k^2)$ . The non-adiabaticity condition  $\geq 1$  is therefore met only for  $k^2 \leq (2\sqrt{3}/9)gv_0 \approx 0.38gv_0$ , in agreement with the Gaussian width of (32.29), and only during  $|t| \lesssim (gv_0)^{-1/2}$ . The integral of (32.29) is exactly  $\mathcal{N} = (gv_0)^{3/2}/(8\pi^3)$  (a Gaussian integral we evaluated with sympy), which fixes the constant in (32.30). For  $g = 1$  and a radion velocity  $v_0 = 10^{-2}$  in string units,  $\mathcal{N} \approx 4 \times 10^{-6}$  particles per string volume are produced per crossing: trapping is efficient only when the radion arrives fast, and Hubble friction finishes the job.

The dilaton is the loose end. Its stabilization would need enhanced-symmetry states whose mass depends on  $g_s$ , which requires knowing the theory at all couplings; BW mention BPS membrane bound states in M-theory as a candidate and leave the question open. [BW §V, ll. 1958–1980] Nothing in this chapter fixes the dilaton, and every stabilization statement above is conditional on its being fixed by other means.

## 32.6 Perturbations and superhorizon freezing

Does the violent production of a string gas and the trapping of the radion destroy a pre-existing spectrum of cosmological perturbations? BW report the result of Battefeld et al. (2005), obtained in a five-dimensional setting with a classical gas of massless string modes and a radiation bath, all perturbed to first order: “long wavelength modes of the Bardeen

potentials (super horizon modes) stay approximately frozen until they re-enter the Hubble horizon, since the transient oscillations of the radion only source equally transient oscillations in the Bardeen potentials”, and the radion perturbation itself has only decaying modes. A spectrum thus “will survive the trapping of the radion in a similar way as a spectrum survives reheating”. [BW §VI.B, ll. 2263–2281] The weak point, in BW’s own words, is that no realization of inflation exists within SGC, so the spectrum whose survival is established must be supplied from outside. [BW §VI.B, ll. 2246–2262]

The standard tool for such questions, which this book’s source library does not contain and which we therefore state as “standard; not checked against a source in this book’s library”, is the Mukhanov–Sasaki equation. For a single scalar clock in a flat FRW universe, the gauge-invariant variable  $v = z\mathcal{R}$  (with  $\mathcal{R}$  the comoving curvature perturbation and  $z = a\dot{\varphi}_{\text{clock}}/H$ ) obeys, in conformal time  $\eta$  and for each comoving wavenumber  $k$ ,

$$v_k'' + \left(k^2 - \frac{z''}{z}\right)v_k = 0, \quad (32.32)$$

and the scalar power spectrum is  $\mathcal{P}_s(k) = \frac{k^3}{2\pi^2} |v_k/z|^2$ . *Superhorizon freezing* is the following elementary fact. When  $k^2 \ll z''/z$  the equation is  $v'' = (z''/z)v$ , whose two solutions are  $v = z$  and  $v = z \int^\eta d\eta'/z^2$  (check:  $(zw)'' = z''w + 2z'w' + zw''$  with  $w' = 1/z^2$ , and  $2z'/z^2 + z(-2z'/z^3) = 0$ ). For an expanding universe  $z$  grows and the second solution decays relative to the first, so  $v_k/z \rightarrow C_k$ , a constant:  $\mathcal{R}$  is conserved outside the horizon. This is the precise content of “modes stay frozen until they re-enter”. What Battefeld et al. showed is that the same holds for the Bardeen potentials in the five-dimensional string-gas model, a separate calculation [L] that we have not reproduced.

## 32.7 The dual-scale reading

The dual-scale principle of Chapter 31 treats  $R$  and  $\alpha'/R$  on an equal footing through the effective scale  $R + \alpha'/R \geq 2\sqrt{\alpha'}$ , with equality at the self-dual radius. This chapter lets us say precisely which parts of the cosmological story that inequality supports.

- **[A]** The inequality itself:  $R + 1/R \geq 2$  for real  $R > 0$  (Chapter 31), and  $R^2 + 1 \geq 2$  for naturals  $R \geq 1$  (Chapter 26). Elementary, kernel-checked, and about numbers. In Lean: `circle_effective_scale_ge_two` (reals) and `self_dual_is_global_minimum` (naturals).
- **[L]** The self-dual radius as a cosmological attractor. That a gas with equal winding and momentum numbers has zero pressure at  $b = \sqrt{\alpha'}$  (Example 32.1), that the radion oscillates about that point and is damped by the running dilaton (Example 32.2), and that ESP particle production traps it there (Section 32.5), are results of the SGC literature as reviewed by BW, with the qualifications recorded above (frame dependence, dilaton unstabilized, initial conditions). The energy (32.24),  $|1/b - b|$ , is the dual-scale combination with a minus sign;  $b + 1/b$  is the energy of one winding plus one momentum string, minimized at the same point. In this sense the self-dual radius is a “minimal effective length” for a string gas: [L] for the pressure balance and the oscillation, where BW say so.
- **[C]** Everything stronger. That  $R + \alpha'/R$  is the “physical radius seen by string probes”, that a contracting universe therefore *bounces* at the self-dual radius, that the big-bang singularity is resolved, or that trans-Planckian modes are forbidden — claims made in the docstrings of the Lean modules discussed next — are proposals of this programme. BW’s assessment of the approach to the singularity is that it “has been largely ignored due to the lack of control of string corrections” and that, among many attempted bounces, “no

convincing models have been found”. [BW §IV.C, ll. 1150–1175] Chapter 27 discusses the swampland side and Chapter 33 lists what would have to be computed.

## 32.8 What the machine has checked

Five Lean modules are attached to this chapter. Each compiles with only the standard axioms `propext`, `Classical.choice`, `Quot.sound`, so each is **[A]** for *what it literally states*. We quote the statements and say in plain words what they say; the reader should compare the plain reading with the module names.

### 32.8.1 Rational Buscher duality and the “effective radius”

`DualScaleM24Formalization/DualScale/EffectiveMetric.lean` represents a positive scale as a record `PosScale` of two naturals `num`, `den` together with proofs `h_num : 0 < num` and `h_den : 0 < den`; `scaleEq s1 s2` is the cross-multiplication equality `s1.num * s2.den = s2.num * s1.den` and `scaleLt` the corresponding strict order. The T-dual scale  $\alpha'/R$  and an “effective radius” that returns the dual scale below a cutoff and the scale itself above it are then

```
def buscherDual (alpha : PosScale) (R : PosScale) : PosScale :=
  (alpha.num * R.den, alpha.den * R.num, ..., ...)

theorem buscher_involution (alpha : PosScale) (R : PosScale) :
  scaleEq (buscherDual alpha (buscherDual alpha R)) R := by ...

def effectiveRadius (cutoff : PosScale) (alpha : PosScale) (R : PosScale) :
  PosScale :=
  if scaleLt R cutoff then buscherDual alpha R else R

theorem genesis_no_singularity (cutoff : PosScale) (alpha : PosScale)
  (R : PosScale) :
  0 < (effectiveRadius cutoff alpha R).num ∧
  0 < (effectiveRadius cutoff alpha R).den := by ...

theorem self_dual_symmetric (alpha : PosScale) :
  scaleEq (buscherDual alpha alpha) (1, 1, by decide, by decide) := by ...
```

#### Plain-language reading

**buscher\_involution**: for positive rationals,  $\alpha' / (\alpha' / R) = R$ . True, and the proof is commutativity and associativity of multiplication of naturals (`ac_rfl`). This is the arithmetic shadow of  $T^2 = 1$  from Chapter 8.

**genesis\_no\_singularity**: the numerator and denominator of `effectiveRadius` are positive. Since a `PosScale` carries positivity proofs as fields and both branches of the `if` return a `PosScale`, this is true by construction: the proof splits on the `if` and reads off the fields. No metric, geodesic, curvature or dynamics occur; the statement does not mention  $2\sqrt{\alpha'}$ , and “singularity” is a word in the docstring only. It says: *a positive rational is positive*.

**self\_dual\_symmetric**:  $\alpha' / \alpha' = 1$ , i.e. the scale  $\alpha'$  (a rational, not  $\sqrt{\alpha'}$ ; they coincide for  $\alpha' = 1$ ) is its own dual.

### 32.8.2 The TCC module: natural-number bounds

`Lean5Corpus/Problems/Problem3_DualScaleTCC.lean` works with natural numbers throughout, with both the Planck length and the string length set to 1.



```

def planck_length : Nat := 1

def effective_wavelength_num (R : Nat) (lambda_0 : Nat) : Nat :=
  (R * R + 1) * lambda_0

theorem wavelength_strictly_super_planckian
  (R : Nat) (lambda_0 : Nat) (h_R : R ≥ 1) (h_l : lambda_0 ≥ 1) :
  effective_wavelength_num R lambda_0 ≥ 2 := by ...

theorem sub_planckian_modes_impossible
  (R : Nat) (lambda_0 : Nat) (h_R : R ≥ 1) (h_l : lambda_0 ≥ 1) :
  ¬ (effective_wavelength_num R lambda_0 ≤ planck_length) := by ...

structure InflationParameters where
  planck_mass : Nat
  hubble_scale : Nat
  h_sub_planckian_hubble : hubble_scale < planck_mass
  h_hubble_pos : hubble_scale ≥ 1

theorem tcc_expansion_factor_positive (p : InflationParameters) :
  p.planck_mass / p.hubble_scale ≥ 1 := by ...

```

#### Plain-language reading

**wavelength\_strictly\_super\_planckian:** for naturals  $R \geq 1$  and  $\lambda_0 \geq 1$ ,  $(R^2+1)\lambda_0 \geq 2$ . Proof:  $R^2 \geq 1$ , so  $R^2 + 1 \geq 2$ , and multiplying by  $\lambda_0 \geq 1$  preserves the bound (`Nat.mul_le_mul`, then `omega`). **sub\_planckian\_modes\_impossible** is the same inequality rephrased as  $\neg((R^2 + 1)\lambda_0 \leq 1)$ . **tcc\_expansion\_factor\_positive:** if  $1 \leq H < M$  are naturals then the natural-number quotient  $M/H$  (rounded down) is at least 1. A fourth theorem, **tcc\_cosmic\_protection\_contract**, is the conjunction of the three. What is missing between these and the trans-Planckian censorship conjecture of Chapter 27: the physical wavelength of a comoving mode is  $a(t)\lambda_0$  with  $a(t)$  a real function of time, not  $(R^2 + 1)\lambda_0$  with  $R \in \mathbb{N}$ ; the hypothesis  $R \geq 1$  excludes exactly the regime  $R < \sqrt{\alpha'}$  where the “bounce” is supposed to occur; and the TCC bounds the *duration* of expansion,  $a_f/a_i < M_{\text{Pl}}/H$ , of which  $M/H \geq 1$  is only the remark that the bound is not empty. The identification of `effective_wavelength_num` with any physical length is **[C]**.

### 32.8.3 Numerical modules: stiff integrators and Picard iteration

The moduli equations (32.14)–(32.17) and (32.31) are stiff when the radion frequency  $\Omega$  of Example 32.2 is large compared with the Hubble rate, and in a code one would integrate them with an implicit scheme. `StringTheoryFormalization/StringDynamics/StiffIntegrators.lean` is the programme’s placeholder for that step. It defines a record `StiffODESystem` (a natural number `dim`, two reals, and a proof that the stiffness ratio exceeds 1) and proves

```

theorem implicit_euler_denominator_lower_bound (sys : StiffODESystem) (h : ℝ)
  (hh : 0 < h) (eig : ℝ) (heig : eig ≤ 0) :
  1 ≤ 1 - h * eig := by ...

theorem bdf2_order_bound (sys : StiffODESystem) :
  0 ≤ sys.dim := by ...

```

## Plain-language reading

`implicit_euler_denominator_lower_bound`: if  $h > 0$  and  $\mu \leq 0$  then  $1 - h\mu \geq 1$ . The proof is `nlinarith`; the record `sys` is not used. This is the one genuinely useful line in the module: for the test equation  $\dot{y} = \mu y$  the implicit Euler step  $y_{n+1} = y_n + h\mu y_{n+1}$  gives  $y_{n+1} = y_n / (1 - h\mu)$ , and the lemma says the amplification factor  $1/(1 - h\mu)$  has modulus at most 1 for every step size  $h > 0$  when  $\mu \leq 0$  — the statement that implicit Euler is stable on the negative real axis without any step-size restriction (standard numerical analysis; not checked against a source in this book’s library). It is a bound on a denominator, not a convergence theorem for a scheme applied to (32.16).

`bdf2_order_bound`: a natural number is  $\geq 0$ . The name promises an order bound for the second-order backward differentiation formula; the statement contains none.

`StringTheoryFormalization/StringDynamics/PicardSpectral.lean` concerns the contraction constant of a Picard iteration.

```
def picardSpectralRadius : ℕ := 18

theorem picard_spectral_contraction :
  (1 : ℝ) / (picardSpectralRadius : ℝ) < 1 := by ...

theorem picard_convergence :
  (1 : ℝ) / (picardSpectralRadius : ℝ) < 1 := picard_spectral_contraction
```

## Plain-language reading

$1/18 < 1$ , by `norm_num`; `picard_convergence` is the same statement under a second name, and `picard_spectral_radius_pos` says  $18 > 0$ . No operator, no iteration, no fixed point and no norm appear. A contraction-mapping argument for the Picard iterates of (32.31) would need the map  $\sigma \mapsto \sigma_0 + \int_0^t (\dots)$  on a function space, a Lipschitz constant  $L$  of the right-hand side, and  $Lt < 1$  on a short interval; Mathlib’s Banach fixed-point theorem (`ContractingWith.fixedPoint`) is where a faithful formalization would start. The number 18 is not derived from anything in the file.

## 32.8.4 The Mukhanov–Sasaki module

`StringTheoryFormalization/StringDynamics/MukhanovSasaki.lean` is, by its own doc-string, “bookkeeping records with two vacuous theorems”. A record `MSSolution` holds a real  $k$  with a proof `k_pos` :  $0 < k$ , a function  $v$  from  $\mathbb{R}$  to  $\mathbb{C}$  and an unused Boolean flag `wronskian_normalized`; then

```
theorem ms_superhorizon_freezing (sol : MSSolution) (η : ℝ) (hη : η < 0) :
  ∃ C : ℂ, True := ⟨0, trivial⟩

noncomputable def scalarPowerSpectrum (sol : MSSolution) (z : ℝ → ℝ)
  (η : ℝ) : ℝ :=
  sol.k^3 / (2 * Real.pi^2) * (||sol.v η / z η||)^2
```

## Plain-language reading

`MSSolution` bundles a positive  $k$  and an arbitrary function  $v : \mathbb{R} \rightarrow \mathbb{C}$ ; nothing ties  $v$  to (32.32). `ms_superhorizon_freezing` asserts  $\exists C \in \mathbb{C}, \text{True}$ , witnessed by  $C = 0$ ; it uses neither `sol`, nor  $\eta$ , nor the hypothesis  $\eta < 0$ , and is provable for any inputs. It records the *name* of superhorizon freezing, not the statement  $v_k/z \rightarrow C_k$  derived after (32.32). `scalarPowerSpectrum` transcribes  $\mathcal{P}_s = \frac{k^3}{2\pi^2} |v/z|^2$  as a definition about which

no theorem is proved. A faithful formalization of freezing would state: if  $v'' = (z''/z)v$  on  $(\eta_0, 0)$  and  $z \rightarrow \infty$ , then  $v/z$  converges — a theorem about a linear second-order ODE that Mathlib’s ODE library could support but that no file in this repository attempts.

The tiers of this chapter’s claims, in one table:

| Claim                                                                                                    | Tier | Where                                                                   |
|----------------------------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------|
| $\alpha' / (\alpha' / R) = R$ for positive rationals; a positive rational is positive (“no singularity”) | [A]  | buscher_involution,<br>genesis_no_singularity                           |
| $(R^2 + 1)\lambda_0 \geq 2$ for naturals $R, \lambda_0 \geq 1$                                           | [A]  | wavelength_strictly_super_planckian                                     |
| $1 \leq 1 - h\mu$ for $h > 0, \mu \leq 0; 1/18 < 1$                                                      | [A]  | implicit_euler_denominator_lower_bound,<br>picard_spectral_contraction  |
| Superhorizon freezing                                                                                    | —    | ms_superhorizon_freezing is vacuous;<br>derived after (32.32), standard |
| Scale-factor duality; equations of state                                                                 | [L]  | BW §II, §III.B; derived here                                            |
| Radion driven to $b = \sqrt{\alpha'}$ ; quantum trapping                                                 | [L]  | BW §IV.C, §V                                                            |
| Three large dimensions from the BV argument                                                              | open | BW §IV.A                                                                |
| Bounce, singularity resolution, TCC from $R + \alpha' / R$                                               | [C]  | this programme                                                          |

### Chapter summary

- On homogeneous toroidal backgrounds the tree-level string action reduces to  $L = e^{-\varphi}(\sum \dot{\lambda}_i^2 - \dot{\varphi}^2)$  with the shifted dilaton  $\varphi = 2\phi - \ln V$ ; it is invariant under  $\lambda_i \rightarrow -\lambda_i$  (scale-factor duality).
- A string gas has  $p_i = -V^{-1}\partial(\rho V)/\partial\lambda_i$  at fixed particle number: momentum modes are radiation ( $p = \rho/d$ ), winding modes have  $p = -\rho/d$ ; equal numbers give zero pressure exactly at the self-dual radius.
- The Brandenberger–Vafa argument predicts a 3 + 6 split; quantitative studies find it requires fine tuning, and SGC takes the split as an initial condition.
- The radion oscillates about  $b = \sqrt{\alpha'}$  with  $\Omega^2 = g_s^2 N/V$ , damped by the running dilaton, in string frame; in Einstein frame a running dilaton destabilizes it, and pure winding gases give runaway potentials  $\propto b^{k-d/2}$ .
- At the self-dual radius nine extra massless states enhance  $U(1)^2$  to  $SU(2) \times SU(2)$ ; their production when the radion crosses that point traps it there, with Hubble friction. The dilaton remains unstabilized.
- Superhorizon curvature perturbations are frozen through the trapping ( $v_k/z \rightarrow \text{const}$  from the Mukhanov–Sasaki equation).
- The Lean modules prove: rational Buscher duality is an involution, a positive rational is positive,  $(R^2 + 1)\lambda_0 \geq 2$ ,  $1 - h\mu \geq 1$ ,  $1/18 < 1$ , and a vacuous existential; none formalizes any dynamics of this chapter.

## Exercises

**Exercise 32.1** (★). Verify (32.4) for  $d = 1$  directly: for  $ds^2 = -dt^2 + e^{2\lambda}dx^2$  compute the Christoffel symbols and show  $R = 2\ddot{\lambda} + 2\dot{\lambda}^2$ , and that  $\sqrt{-G}R$  is then a total derivative. For

$d = 3$  isotropic ( $\lambda_i = \lambda$ ) show that  $\sqrt{-GR}$  equals  $-6e^{3\lambda}\dot{\lambda}^2$  up to a total derivative and is therefore *not* invariant under  $\lambda \rightarrow -\lambda$ , whereas (32.5) is.

**Exercise 32.2** (★). A single string wound  $\omega$  times around a circle of radius  $R$  has energy  $|\omega|R/\alpha'$ ; a momentum mode has energy  $|n|/R$ . Using  $dE = -p dV$  with  $V = 2\pi R$ , derive the pressure of each and confirm the signs in (32.11). At which radius does a gas of  $N_w$  winding and  $N_m$  momentum strings exert zero pressure?

**Exercise 32.3** (★★). Derive the  $\varphi$  equation (32.17) from (32.18) in full, including the elimination of  $\dot{\varphi}^2$  with the constraint, and show that the conservation law  $\dot{E} + 3\dot{\lambda}P_3 + 6\dot{\nu}P_6 = 0$  is the time derivative of the constraint (32.14) along solutions of (32.15)–(32.17).

**Exercise 32.4** (★★). Starting from  $v'' = (z''/z)v$ , verify by direct substitution that  $v = z \int^\eta d\eta'/z(\eta')^2$  is a solution, and compute both solutions explicitly for the de Sitter pump field  $z \propto -1/\eta$  ( $\eta < 0$ ). Show that in this case the full equation (32.32) is solved by  $v_k = \frac{e^{-ik\eta}}{\sqrt{2k}}(1 - \frac{i}{k\eta})$  and that  $|v_k/z|^2$  tends to a  $k$ -dependent constant as  $\eta \rightarrow 0^-$ ; find its  $k$  dependence and hence the spectral index of  $\mathcal{P}_s$ .

**Exercise 32.5** (★★, Lean). State and prove in Lean 4 with Mathlib the real-number version of the module’s inequality,  $2 \leq R + R^{-1}$  for real  $R > 0$ , and then that equality holds if and only if  $R = 1$ . Model:

`circle_effective_scale_ge_two` (Chapter 31). Explain in two sentences why this, and not `wavelength_strictly_super_planckian`, is the statement that the “dual-scale” docstrings actually need.

**Exercise 32.6** (★★★, Lean). The lemma `implicit_euler_denominator_lower_bound` bounds a denominator. Formalize the stability statement behind it: define the implicit Euler map and prove that it does not increase  $|y|$ ,

```
def step (h mu y : ℝ) : ℝ := y / (1 - h * mu)
theorem step_contracts (h mu y : ℝ) (hh : 0 < h) (hmu : mu ≤ 0) :
  |step h mu y| ≤ |y| := by sorry
```

Then state (you need not prove) the corresponding property for the explicit Euler map  $y \mapsto y + h\mu y$  and explain why it requires the step-size restriction  $h|\mu| \leq 2$ .

**Exercise 32.7** (★★★). Sketch a faithful Lean statement of superhorizon freezing: fix a positive, twice differentiable pump field  $z$  on  $(\eta_0, 0)$  with  $z \rightarrow \infty$  as  $\eta \rightarrow 0^-$ , assume  $v'' = (z''/z)v$ , and conclude that  $v/z$  has a limit. Which Mathlib notions (`deriv`, `Filter.Tendsto`, `nhdsWithin`) are needed, and which hypothesis on the second solution  $z \int d\eta/z^2$  is required for the limit to exist? Compare with `ms_superhorizon_freezing`.

## Further reading

- The effective action, scale-factor duality and the string stress tensor: [BW §II, ll. 196–507]; assumptions of SGC, the string gas, its energy and pressure and the duality of the spectrum: [BW §III, ll. 508–914].
- The Brandenberger–Vafa argument and its critique, the ten-dimensional equations, radion stabilization and the Einstein-frame analysis: [BW §IV, ll. 915–1472]; quantum moduli trapping and the dilaton problem: [BW §V, ll. 1473–1980]; perturbations: [BW §VI.B, ll. 2246–2281]; summary: [BW §VII, ll. 2284–2330].
- Temperature duality and the two Hagedorn-like transitions: [GPR §1, ll. 364–377]; the free-energy form: [AAL §1, ll. 97–109]; Kaluza–Klein masses  $n^2/R^2$ : [Tong §8.1.1, ll. 11283–11320].

- Lean sources (read each docstring before the theorems):  
DualScaleM24Formalization/DualScale/EffectiveMetric.lean  
Lean5Corpus/Problems/Problem3\_DualScaleTCC.lean  
StringTheoryFormalization/StringDynamics/ with StiffIntegrators.lean,  
PicardSpectral.lean and MukhanovSasaki.lean.



## Chapter 33

# The Frontier Triad: Distance, Decay and Condensation

The compactifications of Chapters 24 to 26 were described as *static* objects: a K3 surface, a two-torus, a set of flux integers, a minimum of a potential. But a vacuum is only interesting if one knows what happens when it is disturbed. Three disturbances matter most. First, one may *move*: the moduli are massless fields, and a large excursion in moduli space changes the spectrum so drastically that the effective field theory stops making sense—this is the distance conjecture of Chapter 27, now viewed as a dynamical question. Second, the vacuum may *decay*: a flux vacuum is only a local minimum, and quantum mechanics lets it tunnel to a neighbouring one by nucleating a bubble; whether this happens on cosmological time scales is governed by the sign and size of a single number, the Euclidean bounce action. Third, the vacuum may contain an object that is simply *unstable*: a tachyon, a field sitting on top of its potential, which rolls down and condenses. This programme packages the three questions together under the name *frontier triad* and encodes what it assumes about each of them as a *contract*: a Lean structure or theorem that lists hypotheses explicitly and derives a conjunction of consequences. The chapter has two aims. The first is the physics: to derive, in each of the three cases, the inequality that decides the question—an exponential mass law, a positive bounce action, a potential that descends to zero—with the intermediate steps written out. The second is methodological: to explain design by contract as a way of stating a physics conjecture in a proof assistant without pretending to prove it, and to read the four Lean files of the triad exactly as they are, including the places where a docstring promises more than the statement delivers.

### 33.1 Three questions about dynamics

Let us fix the setting. A compactification on  $K3 \times T^2$  (Chapter 24) or on a flux background (Chapter 26) gives a four-dimensional effective theory with scalar fields  $\phi^i$ , a metric  $g_{ij}(\phi)$  on their target space (the moduli-space metric), and a scalar potential  $V(\phi)$ , which is zero on the moduli space proper and non-zero once fluxes are switched on. Each of the three questions is decided by a sign:

| question                          | quantity                                            | condition                                  | Lean file (.lean)                |
|-----------------------------------|-----------------------------------------------------|--------------------------------------------|----------------------------------|
| Can the moduli travel far?        | tower mass $M(\phi)$ along a geodesic of length $d$ | $M(Q) \leq M(P) e^{-\alpha d}, \alpha > 0$ | <code>SwamplandDistance</code>   |
| Does the vacuum last?             | bounce action $B$ of the tunnelling bubble          | $B > 0$ (rate $\propto e^{-B}$ )           | <code>FluxVacuumDecay</code>     |
| Is anything sitting on a hilltop? | tachyon potential $V(T)$                            | $V''(0) < 0, V(T_{\min}) < V(0)$           | <code>TachyonCondensation</code> |

The three are not independent. A tachyonic direction in  $V$  is exactly the second clause of the refined de Sitter conjecture (Section 27.4); a bubble that nucleates during vacuum decay is itself a place where a field rolls off a hilltop; and the exponential lightening of a tower is what eventually invalidates the effective potential in which the other two questions are posed. The word “frontier” in the name records that each question sits at the edge of what the effective theory can say about itself.

#### What is being measured in each case

In the first question the observable is a *mass*: the lightest tower of states as a function of position in moduli space. In the second it is a *rate*: the number of bubbles nucleated per unit four-volume,  $\Gamma/V$ . In the third it is an *energy*: the height of the tachyon potential above its minimum, which turns out to be the tension of the decaying brane. None of the three is visible in the static data of a compactification; all three require a potential or a metric on field space, i.e. exactly the ingredients that the lattice-theoretic chapters of this book did not need.

## 33.2 Moduli excursions and the distance conjecture

Chapter 27 derived the model case: on a circle of radius  $R$ , with the canonically normalized radion  $\varphi$ , the Kaluza–Klein tower and the winding tower have mass scales

$$M_{\text{KK}} \propto e^{+\lambda\varphi}, \quad M_w \propto e^{-\lambda\varphi}, \quad M_{\text{KK}} M_w = \text{const}, \quad (33.1)$$

so that whichever way  $\varphi$  moves, one tower becomes light exponentially in the proper distance  $|\Delta\varphi|$ , with  $\lambda = \sqrt{(d-1)/(d-2)}$  in the normalization of (27.6) [Pal §2.2.3, ll. 2048–2110]. Palti lists the features of this example that the general conjecture abstracts: some tower becomes light for either sign of  $\Delta\varphi$ ; the phenomenon is stringy, because one tower consists of winding states absent in field theory; the product of the two scales is independent of  $\varphi$ ; the exponent is of order one; and as  $|\Delta\varphi| \rightarrow \infty$  infinitely many states become massless, so that no  $d$ -dimensional field theory describes that locus [Pal §2.2.3, ll. 2110–2135].

The conjecture itself, in its original form, has two parts [Pal §2.2.3, ll. 2165–2183]: (i) from any point  $P$  of a moduli space  $\mathcal{M}$  there is a point  $Q$  at infinite geodesic distance  $d(P, Q)$ ; (ii) there is an infinite tower with mass scale

$$M(Q) \sim M(P) e^{-\alpha d(P, Q)}, \quad \alpha > 0. \quad (33.2)$$

The refined form replaces  $\sim$  by a strict inequality and reinstates the Planck mass [Pal §3.6, ll. 5154–5175]:

$$M(Q) < M(P) e^{-\alpha d(P, Q)/M_p} \quad \text{for } d(P, Q) \gtrsim M_p, \quad (33.3)$$

with the exponential behaviour setting in “at  $1\text{--}2 M_p$ ” and no way to delay it, and holding also for fields with a potential, the moduli space being replaced by the field space of the effective theory [Pal §3.6, ll. 5176–5200].



### 33.2.1 From a mass law to a range of validity

What makes (33.3) a statement about *dynamics* is the following consequence. Suppose an effective theory with cutoff  $\Lambda$  is written down at the point  $P$ , where the lightest tower has scale  $M(P) > \Lambda$ . The theory can only be used where the tower is still above the cutoff,  $M(\phi) > \Lambda$ . Combining with (33.3),

$$\Lambda < M(\phi) < M(P) e^{-\alpha d/M_p} \implies d(P, \phi) < \frac{M_p}{\alpha} \ln \frac{M(P)}{\Lambda}. \quad (33.4)$$

This is the “continuous analogue” Palti mentions: a field theory with a cutoff below an infinite tower holds only on a finite range of expectation values [Pal §2.2.3, ll. 2136–2140]. The range grows only logarithmically with the hierarchy  $M(P)/\Lambda$ , and is inversely proportional to  $\alpha$ . This is why the value of  $\alpha$  matters and why the refined conjecture insists on an implicit lower bound for it [Pal §3.6, ll. 5177–5190]: an  $\alpha \ll 1$  would let the moduli travel many Planck units without much change in the spectrum, and the conjecture would lose its bite.

**Example 33.1** (Halving distances). The Lean file `SwamplandDistance.lean` stipulates the value  $\alpha = 1/\sqrt{2}$  (as the rational identity  $2\alpha^2 = 1$ , Section 33.6). With this value, the distance over which the tower mass halves is

$$d_{1/2} = \frac{M_p \ln 2}{\alpha} = \sqrt{2} \ln 2 M_p \approx 0.98 M_p,$$

and after  $d = 1, 2, 3$  Planck units the suppression factors are  $e^{-1/\sqrt{2}} \approx 0.49$ ,  $e^{-\sqrt{2}} \approx 0.24$ ,  $e^{-3/\sqrt{2}} \approx 0.12$ . If the tower starts at  $M(P) = M_p$  and the effective theory is to describe physics up to  $\Lambda = 10^{-2} M_p$ , then (33.4) allows  $d < \sqrt{2} \ln 100 M_p \approx 6.5 M_p$ . For the circle in  $d = 4$  the exponent is instead  $\lambda = \sqrt{3/2} \approx 1.22$  in the normalization of (27.6), or  $\lambda/\sqrt{2} \approx 0.87$  in Palti’s normalization (Section 27.2); the number  $1/\sqrt{2}$  is neither, and nothing in the Lean file connects it to a tower. It should be read as a placeholder for “some  $\alpha$  of order one”.

#### The dual-scale reading

On the circle, the product  $M_{\text{KK}} M_w$  in (33.1) is constant because  $M_{\text{KK}} \propto 1/R$  and  $M_w \propto R/\alpha'$ . The lightest of the two scales is  $\min(1/R, R/\alpha')$ , maximal at the self-dual radius  $R = \sqrt{\alpha'}$ , and this is the same statement as  $R + \alpha'/R \geq 2\sqrt{\alpha'}$  of Chapter 31: the sum is smallest exactly where the lightest tower is heaviest. The distance conjecture says that the effective theory is healthiest in the middle of moduli space and degenerates at every end; the dual-scale inequality says the same for the one-dimensional case in the language of  $R$  rather than of  $\varphi$ . Which of the two towers becomes light is decided by the sign of  $\Delta\varphi$ , and T-duality is the  $\mathbb{Z}_2$  that exchanges them [Pal §2.2.3, ll. 2141–2160].

## 33.3 Flux vacuum decay and the bounce

The flux superpotential of Chapter 26 produces, for each choice of flux integers, a local minimum of  $V$ . Neighbouring minima differ by one unit of some flux, and a transition between them is mediated by the nucleation of a brane (a domain wall in four dimensions) across which the flux jumps. Giddings, Kachru and Polchinski use the same physics in a different role: interactions between modes localized in distinct warped throats are “tunneling-suppressed (and therefore weak)” [GKP §5, ll. 1605–1611]. Here we need the suppression itself. The question is: given a false vacuum with energy density  $V_+$  and a true vacuum with  $V_- < V_+$ , separated by a wall of tension  $\sigma$ , how often does a bubble of true vacuum appear?

### 33.3.1 The thin-wall bounce

The semiclassical answer is standard; it is not in this book's source library and we derive it on the page ("standard; not checked against a source in this book's library"). The rate per unit volume is

$$\frac{\Gamma}{V} = A e^{-B}, \quad B = S_E[\text{bounce}] - S_E[\text{false vacuum}], \quad (33.5)$$

where  $S_E$  is the Euclidean action and the bounce is the  $O(4)$ -symmetric solution of the Euclidean equations of motion that interpolates from the true vacuum at the centre to the false vacuum at infinity. When the energy difference  $\epsilon = V_+ - V_-$  is small compared with the barrier, the bubble is large and its wall is thin, and the action is a function of a single number, the bubble radius  $R$ . Two contributions compete: the wall costs energy proportional to its area, and the interior gains energy proportional to its volume. In four Euclidean dimensions a ball of radius  $R$  has volume  $\frac{\pi^2}{2} R^4$  and its boundary three-sphere has area  $2\pi^2 R^3$ , so

$$B(R) = 2\pi^2 R^3 \sigma - \frac{\pi^2}{2} R^4 \epsilon. \quad (33.6)$$

The bounce is the stationary point of  $B(R)$ :

$$\frac{dB}{dR} = 6\pi^2 R^2 \sigma - 2\pi^2 R^3 \epsilon = 0 \implies R_* = \frac{3\sigma}{\epsilon}, \quad (33.7)$$

and substituting back,

$$B_* = 2\pi^2 \frac{27\sigma^4}{\epsilon^3} - \frac{\pi^2}{2} \frac{81\sigma^4}{\epsilon^3} = \frac{\pi^2 \sigma^4}{\epsilon^3} \left(54 - \frac{81}{2}\right) = \frac{27\pi^2 \sigma^4}{2\epsilon^3}. \quad (33.8)$$

(Symbolic check: sympy reproduces  $R_*$  and  $B_*$  from (33.6).) Two features deserve comment. First,  $B''(R_*) = -18\pi^2 \sigma^2 / \epsilon < 0$ : the stationary point is a *maximum* of  $B(R)$ , as it must be— a bubble smaller than  $R_*$  shrinks, a larger one grows, and the bounce is the critical bubble that does neither. It is a saddle of the full action, with exactly one negative mode, and that negative mode is what makes the false vacuum's energy acquire an imaginary part, i.e. a decay rate. Second,  $B_* > 0$  whenever  $\sigma > 0$  and  $\epsilon > 0$ : the rate is exponentially suppressed, and the suppression becomes infinite as the two vacua become degenerate,  $\epsilon \rightarrow 0$ . The vacuum is then stable, not because tunnelling is forbidden, but because the critical bubble becomes infinitely large.

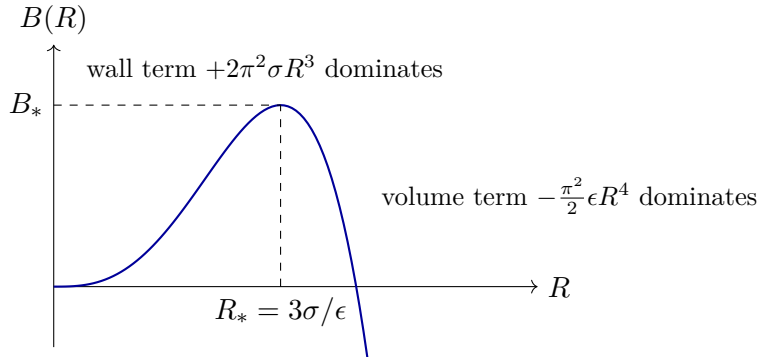


Figure 33.1: The thin-wall action (33.6) as a function of the bubble radius. The bounce sits at the maximum  $R_* = 3\sigma/\epsilon$ ; bubbles to the left collapse, bubbles to the right expand and convert the false vacuum.

**Example 33.2** (Numbers for a flux step). Measure  $\sigma$  in units of  $M^3$  and  $\epsilon$  in units of  $M^4$  for some scale  $M$ . For  $\sigma = 1$ ,  $\epsilon = 0.1$  the critical radius is  $R_* = 30/M$  and  $B_* = 27\pi^2 \cdot 10^3/2 \approx 1.3 \times 10^5$ , so  $\Gamma/V \propto e^{-1.3 \times 10^5}$ —for all purposes the vacuum is eternal. For  $\sigma = 1$ ,  $\epsilon = 0.5$  one finds  $R_* = 6/M$  and  $B_* \approx 1.07 \times 10^3$ ; for  $\sigma = 2$ ,  $\epsilon = 1$ , again  $R_* = 6/M$  but  $B_* \approx 2.1 \times 10^3$ , twice as large because  $B_*$  scales as  $\sigma^4/\epsilon^3$  and both have doubled. The lesson is the extreme sensitivity to the wall tension: at fixed  $\epsilon$ , doubling  $\sigma$  multiplies the exponent by 16.

### 33.3.2 Gravity and the arrow of the decay

With gravity, the bubble geometry itself changes, and the bounce action of (33.8) receives corrections controlled by  $R_*/M_p$  times powers of  $\sqrt{V/M_p^2}$ ; the classic result is that decay of a Minkowski or de Sitter vacuum to a vacuum of lower energy remains possible, while decay out of a supersymmetric Minkowski vacuum is forbidden. Neither statement is in this book’s library and we do not use them. What we do use is the direction of the process: a decay  $V_+ \rightarrow V_-$  lowers the energy density, and in the flux landscape it lowers the flux integer  $N + 1 \rightarrow N$  by one unit. The Lean file `FluxVacuumDecay.lean` records both directions as strict inequalities on integers, with a docstring that invokes a holographic  $c$ -theorem to interpret the second; Section 33.6 explains why that interpretation is not part of what is proved.

#### A positive bounce action is not a stable vacuum

$B > 0$  means only that the rate is *suppressed*,  $\Gamma/V \propto e^{-B}$ , not that it is zero. Whether a vacuum is stable on the age of the universe depends on the value of  $B$  compared with  $\ln(\text{Hubble volume} \times \text{age})$ , i.e. on a number of order a few hundred, and on the prefactor  $A$ , which has dimension  $\text{mass}^4$  and is set by the fluctuation determinant around the bounce. A Lean theorem that proves  $B > 0$  has proved the least interesting of the three facts (sign, magnitude, prefactor).

## 33.4 Tachyons and condensation

### 33.4.1 What a negative mass squared means

The bosonic closed string’s ground state has [Tong §2.3.1, ll. 2535–2545]

$$M^2 = -\frac{1}{\alpha'} \frac{D-2}{6} \xrightarrow{D=26} M^2 = -\frac{4}{\alpha'}, \quad (33.9)$$

and the bosonic open string on a  $Dp$ -brane has  $M^2 = -1/\alpha'$ , “half that of the closed string tachyon” in the sense of  $|M^2|$  [Tong §3.1.1, ll. 3312–3320]. Tong’s interpretation is the one to keep [Tong §2.3.1, ll. 2546–2566]: the mass squared of a particle is the quadratic coefficient of the potential of the field  $T(X)$  that creates it,

$$M^2 = \left. \frac{\partial^2 V(T)}{\partial T^2} \right|_{T=0}, \quad (33.10)$$

so a negative  $M^2$  says that the theory has been expanded about a *maximum* of  $V$ . The Higgs field at  $H = 0$  is a tachyon in exactly this sense. The question is then whether  $V$  has a good minimum elsewhere. For the closed-string tachyon Tong reports that the answer is unknown: the cubic term produces a minimum, the quartic destabilizes it again, and  $T$  mixes with the other scalars [Tong §2.3.1, ll. 2566–2590]. For the open-string tachyon the picture is different

and, by now, well understood [Tong §3.1.1, ll. 3319–3340]: the brane is unstable, it decays “by dissolving into closed string modes”, the end point is “a state with no D-brane”, and

$$V(0) - V(T_{\min}) = \tau_p, \quad (33.11)$$

the tension of the D-brane. This is Sen’s conjecture on the open-string tachyon potential.

### 33.4.2 A model potential and two consistency checks

A frequently used closed form for the potential of the tachyon on an unstable D-brane of the superstring is

$$V(T) = \frac{\tau_p}{\cosh(T/\sqrt{2})} \quad (\alpha' = 1), \quad (33.12)$$

which the Lean file `UseCase3_FrontierTriad.lean` quotes in its docstring; it is “standard; not checked against a source in this book’s library”. Two checks show that it is at least the right shape. First,  $V(0) = \tau_p$  and  $V(T) \rightarrow 0$  as  $|T| \rightarrow \infty$ , so  $V(0) - V(\infty) = \tau_p$ , which is (33.11) with the minimum at infinity. Second, expanding  $1/\cosh x = 1 - x^2/2 + \dots$  gives  $V(T) = \tau_p(1 - T^2/4 + \dots)$ , so by (33.10)

$$M^2 = V''(0)/\tau_p = -\frac{1}{2} \quad \text{in units } \alpha' = 1, \quad \text{i.e. } M^2 = -\frac{1}{2\alpha'}, \quad (33.13)$$

provided the kinetic term is normalized as  $\frac{1}{2}\tau_p(\partial T)^2$  near  $T = 0$ . This is the mass of the tachyon on a non-BPS brane or a brane–antibrane pair of the superstring (as stated, without derivation, in the docstring of `TachyonCondensation.lean`). It differs from the bosonic open-string value  $-1/\alpha'$  quoted above by a factor two, as it should: these are different theories. (Symbolic check: `sympy` gives  $V(0) = 1$ ,  $V(\infty) = 0$ ,  $V''(0) = -1/2$  for  $\tau_p = 1$ .)

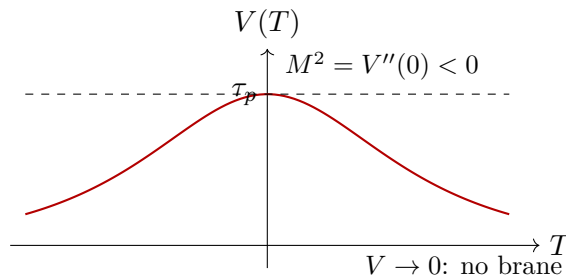


Figure 33.2: The model potential (33.12). The unstable brane sits at the top,  $V(0) = \tau_p$ ; the tachyon rolls to  $|T| \rightarrow \infty$  where the energy stored in the brane has been released. The height of the hill is the brane tension, (33.11).

### 33.4.3 Brane–antibrane pairs and what survives condensation

For a coincident  $Dp\text{--}\overline{Dp}$  pair the tachyon is a complex field, and if it condenses with a non-trivial winding at infinity the end point is not empty space but a lower-dimensional brane. The bookkeeping that survives is a *charge*: the pair carries the formal difference  $[E] - [F]$  of the bundles on the brane and antibrane, and the defect left behind carries the same class. This is Witten’s K-theoretic classification of D-brane charge, cited in the docstring of `TachyonCondensation.lean` but not in this book’s library; we use only the arithmetic consequence that the rank and the Chern numbers of  $[E] - [F]$  are differences of those of  $E$  and  $F$ , which is a definition. The physical statement—that the soliton at the end of condensation carries exactly this class—is Tier L on the authority of that literature, and the Lean theorem named after it is a tautology (Section 33.6).

### 33.4.4 The tachyon in the refined de Sitter conjecture

The refined de Sitter conjecture of Chapter 27 says that a potential in the landscape satisfies either  $|\nabla V| \geq cV/M_p$  or

$$\min (\nabla_i \nabla_j V) \leq -\frac{c'}{M_p^2} V, \quad (33.14)$$

the minimum eigenvalue of the Hessian in an orthonormal frame [Pal §7, ll. 10558–10575]. The second clause is precisely a tachyon condition: a direction in field space with negative mass squared, of magnitude at least  $c'V/M_p^2$ . It was introduced because the first clause alone is violated at the top of the Higgs potential, where  $|\nabla V|/V \sim 10^{-55}/M_p$  while the Hessian is enormously negative,  $\sim -10^{35}V/M_p^2$  [Pal §7, ll. 10590–10603]; the refined conjecture “is certainly satisfied” there. So the third member of the triad, condensation, enters the swampland programme as the escape clause of the de Sitter conjecture: hilltops are allowed, provided they are steep enough that the field does not linger.

#### Tachyons do not travel faster than light

The special-relativistic picture of a tachyon as a particle with  $v > c$  is “not the right interpretation” [Tong §2.3.1, ll. 2547–2556]. In field theory a negative  $M^2$  is an instability of the vacuum, not a superluminal signal: perturbations grow exponentially in time rather than oscillating, and the theory must be re-expanded about the true minimum. The string-gas literature makes the same point in reverse: the “tachyonic zero point energy” of the bosonic string is removed by hand from the mass formula before any thermodynamics is done, because it would be absent in the heterotic string anyway [BW §III, ll. 886–890].

## 33.5 Design by contract for physics conjectures

None of the three inequalities above is a theorem of mathematics. The distance conjecture is a conjecture; the positivity of the bounce action depends on a semiclassical approximation and on the signs of  $\sigma$  and  $\epsilon$ ; Sen’s conjecture is established by string field theory calculations, not by a proof. How, then, should a proof assistant handle them? Chapter 27 gave the general answer: encode the conjecture’s *shape* as a structure or a hypothesis and prove what follows. This programme’s name for the pattern is a *contract*, borrowed from software engineering, where a routine is specified by a precondition (what the caller must guarantee) and a postcondition (what the routine then guarantees). Translated to Lean:

**Definition 33.3** (Contract). A *contract* for a physical statement is a Lean declaration of the form

$$\text{theorem name } (x_1 : \alpha_1) \cdots (x_k : \alpha_k) (h_1 : P_1) \cdots (h_m : P_m) : Q_1 \wedge Q_2 \wedge \cdots \wedge Q_r,$$

whose parameters  $x_i$  are the data of the problem, whose hypotheses  $h_j$  are the physical assumptions made explicit, and whose conclusion is the conjunction of the consequences drawn. A *master contract* is one whose conjuncts are themselves the conclusions of separately proved theorems, so that its proof is an assembly,  $\langle \text{thm}_1, \text{thm}_2, \text{thm}_3 \rangle$ .

The value of the pattern lies in what it forces one to write down. Every assumption is a named hypothesis; every consequence is a proposition that must type-check against the definitions; and the kernel refuses any step that uses an assumption not in the list. The pattern does *not* guarantee that the assumptions are the physically relevant ones, nor that the definitions capture the physics. Three failure modes recur in the libraries of this book and the reader should learn to recognize them on sight:

1. **Vacuous conclusion.** The conclusion is `True` or an equation between literals provable by `rf1`. The hypotheses may be physically meaningful, but the theorem uses none of them. Chapter 27’s `de_sitter_conjecture` is the extreme case.
2. **Stipulated dynamics.** A quantity that physics would compute—a bounce action, a tower mass—is *defined* to have the property that the theorem then “proves”. The proof is correct, the content is the definition, and the definition is a modelling choice.
3. **Unused hypotheses.** A hypothesis is listed in the signature but the proof never refers to it. Lean’s linter flags this; the physical reading is that the conclusion is weaker than the name suggests.

#### How to read a contract

Read the conclusion first, then the definitions it unfolds to, then the hypotheses, then the name. Ask: if I replace each defined constant by a variable, does the theorem still go through? If yes, the theorem is about the *form* of the statement (good: that is what a contract is for). If no, the theorem is about specific numbers, and it is a computation, not a law. Both are legitimate; only the reading changes.

**Example 33.4** (The master contract of the triad). In the file `UseCase3_FrontierTriad.lean` the theorem

`frontier_triad_master_contract` has the form of Definition 33.3 with data  $(m_0, t_b) \in \mathbb{N}^2$ , one hypothesis  $t_b \geq 1$ , and three conjuncts: the RR tadpole of the  $K3 \times T^2/\mathbb{Z}_2$  orientifold vanishes; the tower mass after one “halving” is at most  $m_0$ ; the numerator of the thin-wall bounce action is positive. Its proof is exactly the assembly  $\langle \text{tadpole}, \text{sdc}, \text{bounce} \rangle$ . Applying the reading rule: the first conjunct is an equation between literals ( $64 - 64 = 0$ , failure mode 1 in a mild form—it is a genuine tadpole computation but does not depend on the parameters); the second is  $m_0/2 \leq m_0$  in natural-number division, a true statement about the form  $m \mapsto m/k$  that survives replacing 2 by any  $k \geq 1$ ; the third is  $27t_b^4 > 0$ , which needs the hypothesis  $t_b \geq 1$  and uses it. The contract is honest about its inputs and modest in its outputs, and that is the best one can say of any formalization of a conjecture.

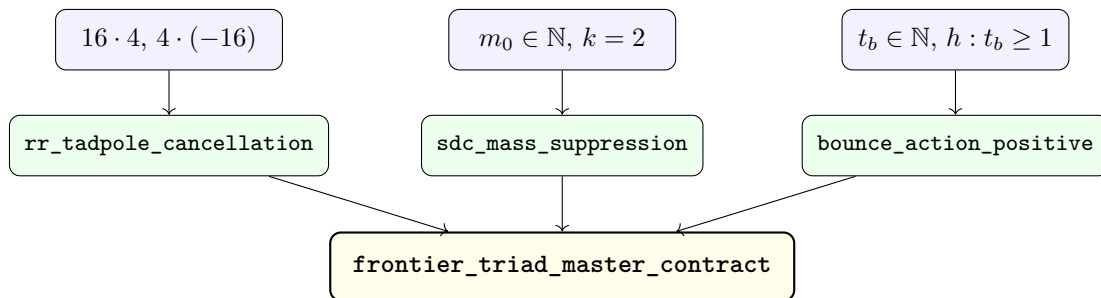


Figure 33.3: Anatomy of the master contract: data and hypotheses (top), three component theorems (middle), and their conjunction (bottom). Only the right-hand column carries a hypothesis that the proof actually uses.

## 33.6 What the machine has checked

Four files carry the triad: three in `DualScaleM24Formalization/FrontierTriad/` and the use case `DualScaleValidation/UseCase3_FrontierTriad.lean`, which imports the first of them together with the Kummer tadpole file of Chapter 25. All compile with no `sorry` and

depend only on the three standard axioms. Every statement below is quoted from the file; the proofs are omitted. We take the files in the order of the chapter.

### 33.6.1 Distance: `SwamplandDistance.lean`

The moduli-dimension arithmetic (58, 80) and the Boolean predicate `isSwamplandSafe` of this file were read in Chapter 27, and Chapter 24 explained why 80 is the dimension of the  $K3$  sigma-model moduli space rather than of  $K3 \times T^2$ . What remains for us is the decay coefficient:

```

/-- SDC decay rate coefficient  $\alpha = 1/\sqrt{2}$  numerator of  $\alpha^2$ : 1. -/
def sdc_decay_alpha_squared_num : Nat := 1
/-- SDC decay rate coefficient  $\alpha = 1/\sqrt{2}$  denominator of  $\alpha^2$ : 2. -/
def sdc_decay_alpha_squared_den : Nat := 2

theorem sdc_decay_coefficient_identity :
  2 * sdc_decay_alpha_squared_num = sdc_decay_alpha_squared_den := by rfl

def fricke_N : Int := 1
def fricke_square : Int := fricke_N * fricke_N
theorem fricke_involution_property : fricke_square = 1 := by rfl
def fricke_discriminant : Int := 0 - 4 * 1 * 1
theorem fricke_discriminant_is_minus_four : fricke_discriminant = -4 := by rfl

```

#### Plain-language reading

`sdc_decay_coefficient_identity` says  $2 \cdot 1 = 2$ . The docstring reads this as  $2\alpha^2 = 1$  with  $\alpha = 1/\sqrt{2}$ , but no real number, no square root and no tower appear in the statement; the two naturals are *named* as numerator and denominator of  $\alpha^2$  and nothing enforces the name. `fricke_involution_property` says  $1 \cdot 1 = 1$  and `fricke_discriminant_is_minus_four` says  $0 - 4 = -4$ ; the docstring’s “Fricke involution  $W_1^2 = \text{Id}$ ” and “discriminant of  $x^2 + 1$  at the fixed point  $\tau = i$ ” are the intended meanings, and the second is indeed the discriminant  $b^2 - 4ac$  of  $x^2 + 0x + 1$ . Proof idea in every case: `rfl`, definitional unfolding of literals. This is failure mode 2 of Section 33.5 in its purest form; what would be needed to make  $\alpha$  a theorem is described in Chapter 27.

### 33.6.2 Decay: `FluxVacuumDecay.lean`

```

structure FluxVacuum where
  fluxNumber : Nat
  energyScale : Nat
  centralCharge : Nat

def standardFluxVacuum (n : Nat) : FluxVacuum := {
  fluxNumber := n
  energyScale := 100 * (n + 1)
  centralCharge := 100 * (n + 1)
}

structure CDLInstanton where
  falseVacuum : FluxVacuum
  trueVacuum : FluxVacuum
  bounceAction : Nat

def unitFluxDecay (n : Nat) : CDLInstanton := {
  falseVacuum := standardFluxVacuum (n + 1)
  trueVacuum := standardFluxVacuum n
  bounceAction := 42 * (n + 1)
}

```



```

}

theorem cdl_action_strictly_positive (n : Nat) :
  (unitFluxDecay n).bounceAction > 0 := by ...
theorem true_vacuum_energy_is_lower (n : Nat) :
  (unitFluxDecay n).falseVacuum.energyScale
    > (unitFluxDecay n).trueVacuum.energyScale := by ...
theorem holographic_c_theorem_decay (n : Nat) :
  (unitFluxDecay n).trueVacuum.centralCharge
    < (unitFluxDecay n).falseVacuum.centralCharge := by ...

```

### Plain-language reading

A `FluxVacuum` is a triple of natural numbers; `standardFluxVacuum n` is  $(n, 100(n+1), 100(n+1))$ ; a `CDLInstanton` is a pair of such triples with a third natural number called the bounce action; `unitFluxDecay n` is the instanton from  $(n+1, 100(n+2), 100(n+2))$  to  $(n, 100(n+1), 100(n+1))$  with bounce action  $42(n+1)$ . The three theorems say, for every  $n \in \mathbb{N}$ :  $42(n+1) > 0$ ;  $100(n+2) > 100(n+1)$ ;  $100(n+1) < 100(n+2)$ . Proof idea: unfold the definitions (`dsimp`) and let `omega`, the decision procedure for linear arithmetic over  $\mathbb{N}$  and  $\mathbb{Z}$ , close the goal. *What is not here*: no potential, no Euclidean action, no bubble, no  $\sigma$  or  $\epsilon$ , and therefore no (33.8); the bounce action is stipulated to be  $42(n+1)$  (failure mode 2). The second and third theorems are the same inequality about the same function  $100(n+1)$  under two names; the docstring’s “holographic  $c$ -theorem” and “arrow of time” are interpretations of the label `centralCharge`, which the kernel never sees. A faithful formalization of Section 33.3 would take  $\sigma, \epsilon > 0$  real, define  $B(R)$  by (33.6), and prove that  $R_* = 3\sigma/\epsilon$  is its unique positive critical point with  $B(R_*) = 27\pi^2\sigma^4/(2\epsilon^3) > 0$ ; Exercise 33.6 asks for the first step.

### 33.6.3 Condensation: `TachyonCondensation.lean`

```

structure KClass where
  rank : Int
  firstChern : Int
  secondChern : Int

def kSub (a b : KClass) : KClass := {
  rank := a.rank - b.rank
  firstChern := a.firstChern - b.firstChern
  secondChern := a.secondChern - b.secondChern
}

structure BraneAntiBraneSystem where
  braneE : KClass
  antiBraneF : KClass
  tachyonVEV : Nat
  is_annihilated : Bool

def systemKTheoryClass (sys : BraneAntiBraneSystem) : KClass :=
  kSub sys.braneE sys.antiBraneF

structure SenSolitonDefect where
  dimension : Nat
  worldvolumeDefect : KClass
  solitonCenter : Nat

def condenseTachyon (sys : BraneAntiBraneSystem) (dim : Nat) :
  SenSolitonDefect := {
    dimension := dim

```



```

worldvolumeDefect := systemKTheoryClass sys
solitonCenter := 0
}

theorem sen_conjecture_k_theory_conservation
  (sys : BraneAntiBraneSystem) (dim : Nat) :
  (condenseTachyon sys dim).worldvolumeDefect
  = systemKTheoryClass sys := by rfl

def rrCharge (k : KClass) : Int :=
  k.rank + k.firstChern + k.secondChern

theorem tachyon_condensation_preserves_rr_charge
  (sys : BraneAntiBraneSystem) (dim : Nat) :
  rrCharge (condenseTachyon sys dim).worldvolumeDefect
  = rrCharge (systemKTheoryClass sys) := by rfl

```

#### Plain-language reading

A `KClass` is a triple of integers  $(r, c_1, c_2)$  and `kSub` subtracts componentwise; this is the correct arithmetic of rank and Chern numbers in the Grothendieck group, where  $[E] - [F]$  has rank  $\text{rk } E - \text{rk } F$  and Chern character  $\text{ch}(E) - \text{ch}(F)$  (Tier L; the component formula is a definition once one accepts that `ch` is additive). The function `condenseTachyon` copies  $[E] - [F]$  into the field `worldvolumeDefect` of the output. The theorem `sen_conjecture_k_theory_conservation` then says that the copy equals the original, and `tachyon_condensation_preserves_rr_charge` says that a function applied to the copy equals the same function applied to the original; both are `rfl`. These are tautologies by construction: the physical claim that condensation *produces* a defect in class  $[E] - [F]$  has been put into the definition of `condenseTachyon`, so the theorem cannot fail. Note also that `tachyonVEV` and `is_annihilated` are never read, that `rrCharge` adds a rank to two Chern numbers without the powers of  $2\pi$  or the  $\sqrt{\hat{A}}$  factor that a Chern-character integral would carry, and that no potential  $V(T)$  appears anywhere, so (33.11) and (33.13) are not represented. The docstring's mass  $m^2 = -1/(2\alpha')$  is correct for the superstring brane–antibrane tachyon (Section 33.4) but is not a statement of the file.

#### 33.6.4 Gates and a graph invariant: `TriadContract.lean`

Despite its name, this file contains none of the three physical statements. It contains three Boolean *gates* intended for a numerical solver and one computation in graph theory:

```

def checkMetricPositivity (tau_im_scaled : Int) : Bool :=
  tau_im_scaled > 0
def checkWeakEnergyCondition (rho_scaled p_scaled : Int) : Bool :=
  rho_scaled + p_scaled ≥ 0
def checkSymmetronScreening (screening_ppm : Nat) : Bool :=
  screening_ppm ≤ 124
theorem symmetron_screening_cassini_pass :
  checkSymmetronScreening 124 = true := by decide

structure TDAMapperGraph where
  nodes : Nat
  edges : Nat
def mapperBetti1WithComponents (g : TDAMapperGraph) (b0 : Nat) : Int :=
  (g.edges : Int) - (g.nodes : Int) + (b0 : Int)
def mapperEulerChar (g : TDAMapperGraph) : Int :=
  (g.nodes : Int) - (g.edges : Int)

```

```

def KummerLangevinMapperGraph : TDAMapperGraph := { nodes := 187, edges := 557 }
def kummer_defect_clusters : Nat := 6
theorem kummer_mapper_betti1_eq_376 :
  mapperBetti1WithComponents KummerLangevinMapperGraph kummer_defect_clusters
    = 376 := by rfl
theorem kummer_mapper_euler_eq_minus_370 :
  mapperEulerChar KummerLangevinMapperGraph = -370 := by rfl

```

#### Plain-language reading

The three gates are predicates on integers:  $\text{Im } \tau > 0$  (in some scaled integer units),  $\rho + p \geq 0$  (the null energy condition on a perfect fluid, which the docstring calls the weak energy condition; for a perfect fluid the weak condition is  $\rho \geq 0$  and  $\rho + p \geq 0$  together), and “fifth-force deviation  $\leq 124$  ppm”. The theorem `symmetron_screening_cassini_pass` evaluates the third at its own threshold:  $124 \leq 124$ . The number 124 is attributed by the docstring to the Cassini radio-link bound; that bound is not in this book’s library and we make no claim about it. The graph part is correct mathematics: for a finite graph with  $V$  vertices,  $E$  edges and  $b_0$  components the first Betti number is  $b_1 = E - V + b_0$  and the Euler characteristic is  $\chi = V - E = b_0 - b_1$ ; for  $(V, E, b_0) = (187, 557, 6)$  this gives  $b_1 = 376$  and  $\chi = -370$ , and indeed  $6 - 376 = -370$  (symbolic check). The theorems are `rfl` evaluations of these two linear expressions at the three literals. The identification of the graph with “the Mapper 1-skeleton of Langevin trajectories on the Kummer landscape” is a claim about a numerical experiment, not about anything in the file: the file does not contain the graph, only its vertex and edge counts.

### 33.6.5 The master contract: UseCase3\_FrontierTriad.lean

```

def d7_charge_per_brane : Int := 4
def d7_brane_count : Nat := 16
def o7_charge_per_plane : Int := -16
def o7_plane_count : Nat := 4
def total_d7_charge : Int := (d7_brane_count : Int) * d7_charge_per_brane
def total_o7_charge : Int := (o7_plane_count : Int) * o7_charge_per_plane
theorem rr_tadpole_cancellation :
  total_d7_charge + total_o7_charge = 0 := rfl

def sdc_mass_bound (m_0 : Nat) (decay_factor : Nat) : Nat :=
  if decay_factor == 0 then m_0 else m_0 / decay_factor
theorem sdc_mass_suppression (m_0 k : Nat) (h : k ≥ 1) :
  sdc_mass_bound m_0 k ≤ m_0 := by ...

def tachyon_vacuum_energy : Nat := 0
theorem tachyon_condensation_endpoint :
  tachyon_vacuum_energy = 0 := by rfl

def bounce_action_numerator (tb : Nat) : Nat :=
  27 * (tb * tb * tb * tb)
theorem bounce_action_positive (tb : Nat) (h : tb ≥ 1) :
  bounce_action_numerator tb > 0 := by ...

theorem frontier_triad_master_contract (m_0 tb : Nat) (h_tb : tb ≥ 1) :
  total_d7_charge + total_o7_charge = 0 ∧
  sdc_mass_bound m_0 2 ≤ m_0 ∧
  bounce_action_numerator tb > 0 := by
  refine ⟨rr_tadpole_cancellation, ?, bounce_action_positive tb h_tb⟩
  exact sdc_mass_suppression m_0 2 (by decide)

```

## Plain-language reading

**rr\_tadpole\_cancellation:**  $16 \cdot 4 + 4 \cdot (-16) = 0$ , the  $p = 7$  row of Table 25.1 with all charges multiplied by 4; it duplicates the theorem of the same name in `KummerTadpole.lean`, which Chapter 25 discusses. **sdc\_mass\_suppression:** for  $k \geq 1$ ,  $\lfloor m_0/k \rfloor \leq m_0$  in natural-number division; proof idea: the branch  $k = 0$  of the `if` is excluded by `h`, and `Nat.div_le_self` finishes. This is the only place in the four files where a hypothesis is genuinely needed and used. It is a shadow of (33.3) in which  $e^{-\alpha d}$  has been replaced by  $1/k$  and the strict inequality by  $\leq$  (indeed  $m_0/1 = m_0$ , so strictness is false for  $k = 1$ ). **tachyon\_condensation\_endpoint:**  $0 = 0$ , the value  $V(\infty) = 0$  of (33.12) entered as a constant. **bounce\_action\_positive:**  $27t_b^4 > 0$  for  $t_b \geq 1$ ; proof idea: `Nat.mul_pos` four times. This is the numerator of (33.8), i.e.  $27\pi^2\sigma^4$  with  $\pi^2$  dropped and  $\sigma \in \mathbb{N}$ ; the denominator  $2\epsilon^3$ , and hence the dependence on the energy gap, is absent. The master contract is the conjunction of the first, second (with  $k = 2$ ) and fourth, assembled with `refine <_,?_,_>`; its only hypothesis `h_tb` is passed straight to `bounce_action_positive`. Taken literally, the master contract asserts:  $64 - 64 = 0$ , and  $\lfloor m_0/2 \rfloor \leq m_0$ , and  $27t_b^4 > 0$  when  $t_b \geq 1$ .

## 33.6.6 Tier table

| statement                                                                                                                                                     | where                                                             | tier       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|------------|
| $\lfloor m_0/k \rfloor \leq m_0$ for $k \geq 1$ ; $27t_b^4 > 0$ for $t_b \geq 1$ ; the conjunction <code>frontier_triad_master_contract</code>                | <code>UseCase3_FrontierTriad.lean</code>                          | <b>[A]</b> |
| $42(n+1) > 0$ , $100(n+2) > 100(n+1)$ ; $[E] - [F]$ copied through <code>condenseTachyon</code> ; $b_1 = E - V + b_0 = 376$ , $\chi = -370$ ; $2 \cdot 1 = 2$ | the three <code>FrontierTriad</code> files                        | <b>[A]</b> |
| Thin-wall bounce $R_* = 3\sigma/\epsilon$ , $B_* = 27\pi^2\sigma^4/2\epsilon^3$                                                                               | standard, not in library; derived in Section 33.3, symbolic check | <b>[L]</b> |
| Distance conjecture, refined form; de Sitter conjecture, refined form                                                                                         | [Pal §2.2.3, §3.6, §7]                                            | <b>[L]</b> |
| Open-string tachyon: unstable brane, end point without brane, height $= \tau_p$                                                                               | [Tong §3.1.1, ll. 3319–3335]                                      | <b>[L]</b> |
| $V(T) = \tau_p / \cosh(T/\sqrt{2})$ ; K-theory class of the condensate                                                                                        | standard, not in library                                          | <b>[L]</b> |
| $\alpha = 1/\sqrt{2}$ ; $10 \leq \rho \leq 20$ as a landscape criterion; bounce action $42(n+1)$ ; Mapper graph as Kummer Langevin attractor                  | docstrings of the four files                                      | <b>[C]</b> |

## Chapter summary

- The frontier triad is three dynamical questions about a compactification, each decided by a sign: the exponent  $\alpha > 0$  of the distance conjecture, the bounce action  $B > 0$  of vacuum decay, the negative curvature  $V''(0) < 0$  and the descent  $V(0) - V_{\min} = \tau_p$  of the tachyon.
- The distance conjecture converts into a finite range of validity for any effective theory,  $d < (M_p/\alpha) \ln(M(P)/\Lambda)$ , logarithmic in the hierarchy and inverse in  $\alpha$ .

- The thin-wall bounce is the maximum of  $B(R) = 2\pi^2\sigma R^3 - \frac{\pi^2}{2}\epsilon R^4$ , at  $R_* = 3\sigma/\epsilon$ , with  $B_* = 27\pi^2\sigma^4/(2\epsilon^3)$ ;  $B > 0$  suppresses the rate but does not make it zero.
- A tachyon is a field expanded about a maximum of its potential; for the open string the minimum is the state with no brane, and the model potential  $\tau_p/\cosh(T/\sqrt{2})$  reproduces both the height  $\tau_p$  and the mass  $-1/(2\alpha')$ . The refined de Sitter conjecture's second clause is a tachyon condition.
- A contract lists hypotheses and derives a conjunction. Read the conclusion first; watch for vacuous conclusions, stipulated dynamics and unused hypotheses.
- The four Lean files prove correct arithmetic on integers ( $64 - 64 = 0$ ,  $\lfloor m_0/2 \rfloor \leq m_0$ ,  $27t_b^4 > 0$ ,  $42(n+1) > 0$ ,  $557 - 187 + 6 = 376$ ) and two definitional identities; the physics named in their docstrings is Tier L or Tier C and is not what the kernel checked.

## Exercises

**Exercise 33.1** (★). From (33.3), show that if the tower is required to stay above a cutoff  $\Lambda$  and starts at  $M(P) = M_p$ , the allowed geodesic range is  $d < (M_p/\alpha) \ln(M_p/\Lambda)$ . Evaluate for  $\Lambda = 1 \text{ TeV}$ ,  $M_p = 2.4 \times 10^{18} \text{ GeV}$  and  $\alpha = 1, 1/\sqrt{2}, \sqrt{3}/2$ .

**Exercise 33.2** (★). Verify that the stationary point (33.7) is a maximum of  $B(R)$  by computing  $B''(R_*)$ , and explain in words why the bounce must be a maximum in  $R$  and not a minimum.

**Exercise 33.3** (★★). Repeat the thin-wall computation in  $D$  Euclidean dimensions: with  $\Omega_{D-1}$  the area of the unit  $(D-1)$ -sphere,  $B(R) = \Omega_{D-1}R^{D-1}\sigma - \frac{\Omega_{D-1}}{D}R^D\epsilon$ . Show that  $R_* = (D-1)\sigma/\epsilon$  and  $B_* = \frac{\Omega_{D-1}}{D} \frac{(D-1)^{D-1}\sigma^D}{\epsilon^{D-1}}$ , and check that  $D = 4$  with  $\Omega_3 = 2\pi^2$  reproduces (33.8).

**Exercise 33.4** (★★). Expand (33.12) to fourth order in  $T$  and compare with Tong's generic expansion  $V = \frac{1}{2}M^2T^2 + c_3T^3 + c_4T^4 + \dots$  [Tong §2.3.1, ll. 2574–2580]. Why is  $c_3 = 0$  here, and what symmetry of the brane–antibrane system enforces it? Show that  $V$  has no minimum at finite  $T$  and explain why that is consistent with (33.11).

**Exercise 33.5** (★★). For a perfect fluid the weak energy condition is  $\rho \geq 0$  and  $\rho + p \geq 0$ ; the null energy condition is  $\rho + p \geq 0$  alone. Which of the two does `checkWeakEnergyCondition` test? Give a pair  $(\rho, p)$  that passes the gate but violates the weak energy condition. Then write in Lean a predicate that tests the weak energy condition and prove that it implies the gate.

**Exercise 33.6** (★★). In Lean, with  $\sigma \ \varepsilon : \mathbb{R}$ , hypotheses  $0 < \sigma$  and  $0 < \varepsilon$ , define  $B \ (R : \mathbb{R}) : \mathbb{R} := 2 * \pi^2 * \sigma * R^3 - \pi^2 / 2 * \varepsilon * R^4$  and prove  $0 < B \ (3 * \sigma / \varepsilon)$ . (Hint: after `field_simp` the goal is a polynomial inequality; `positivity` or `nlinarith` with the two hypotheses closes it.) Then state, without proving, the theorem that  $3\sigma/\epsilon$  is the unique positive critical point.

**Exercise 33.7** (★★). Show that for a finite graph  $\chi = V - E = b_0 - b_1$  and hence  $b_1 = E - V + b_0$ , by induction on the number of edges (removing an edge either lowers  $b_1$  by one or raises  $b_0$  by one). Then explain why `mapperBetti1WithComponents` can return a negative integer for some inputs and what constraint on  $(V, E, b_0)$  a faithful formalization would need to carry.

**Exercise 33.8** (\*\*\*). Write a contract in the sense of Definition 33.3 for the refined de Sitter conjecture: a structure with fields  $V : \mathbb{R} \rightarrow \mathbb{R}$ ,  $c, c' : \mathbb{R}$ , positivity hypotheses, and a Prop-valued field stating (33.14) or the gradient clause for every  $\phi$ . Prove, as the postcondition, that a potential with  $V(\phi_0) > 0$ ,  $V'(\phi_0) = 0$  and  $V''(\phi_0) \geq 0$  at some  $\phi_0$  violates the contract. This is the statement “no de Sitter minimum”, the first non-vacuous consequence of the conjecture.

**Exercise 33.9** (\*\*\*). The docstring of `FluxVacuumDecay.lean` says that the central charge decreases along a decay. Replace `standardFluxVacuum` by a structure whose `centralCharge` is an arbitrary function  $c : \text{Nat} \rightarrow \text{Nat}$  with a hypothesis `StrictMono c`, and prove the analogue of `holographic_c_theorem_decay`. Which of the three failure modes of Section 33.5 has this removed, and which remains?

## Further reading

- The circle as first encounter with the distance conjecture, the two towers, and the original statement: [Pal §2.2.1, ll. 1718–1775]; [Pal §2.2.3, ll. 2048–2235].
- The refined distance conjecture and the reasons for the refinement: [Pal §3.6, ll. 5136–5215].
- The de Sitter conjecture, its refinement and the Higgs hilltop: [Pal §7, ll. 10475–10610].
- Tachyons as hilltop fields, the closed-string tachyon, and the open-string tachyon on a D-brane: [Tong §2.3.1, ll. 2535–2590]; [Tong §3.1.1, ll. 3312–3340].
- Tunnelling suppression between warped throats: [GKP §5, ll. 1605–1611]. The removal of the tachyonic zero-point energy in string-gas thermodynamics: [BW §III, ll. 886–890].
- Lean sources, each of which should be read with its docstring and its statements side by side:

- `DualScaleM24Formalization/FrontierTriad/SwamplandDistance.lean`
- `DualScaleM24Formalization/FrontierTriad/FluxVacuumDecay.lean`
- `DualScaleM24Formalization/FrontierTriad/TachyonCondensation.lean`
- `DualScaleM24Formalization/FrontierTriad/TriadContract.lean`
- `DualScaleValidation/UseCase3_FrontierTriad.lean`

- The atlas of Chapter 36 lists the duplicated tadpole theorem of `UseCase3_FrontierTriad` among its unification candidates; the methodological lessons on reading docstrings against statements are collected in Chapter 37.



**Part VIII**

**The Formalization**





## Chapter 34

# Lean 4 for Physicists: A Crash Course

Every earlier chapter ends with a section “What the machine has checked”, and every theorem there carries the badge **[A]**. This chapter explains what that badge costs and what it buys. The question it answers is practical: *how does a physicist read a Lean 4 statement, decide whether it says what the prose claims, and check it on her own machine?* The answer does not require learning to write proofs from scratch. It requires three things: knowing what a type is and why a theorem is one; knowing how Mathlib writes a matrix, a transpose and an inverse; and knowing what the dozen tactics that appear in this book’s proofs actually do. We build these up on the four files that the rest of the book quotes most often—the lattice definitions of Chapter 18, the duality group of Chapter 11, the dual-scale bound of Chapter 31 and the circle mass spectrum of Chapter 8—and then read one proof, `thetaShift_isODD`, line by line. A student should care because a proof assistant is only as useful as the reader who can tell a strong theorem from a weak one: the kernel checks that a proof proves its statement, and nothing else. Reading the statement is the reader’s job, and this chapter teaches it.

### 34.1 What a proof assistant is, and what it is not

Lean 4 is a programming language whose type system is strong enough to express mathematical statements, together with a small program, the *kernel*, that checks whether a given term has a given type. A theorem is a type; a proof is a term of that type; “the theorem is proved” means the kernel has accepted the term. Everything else—the tactic language, the editor, the automation, the large-language-model agents described in Chapter 37—is machinery for *producing* the term. None of it has to be trusted. If a tactic has a bug and produces a wrong term, the kernel rejects it.

Three things are trusted, and the student should know them by name.

1. **The kernel itself**, a small program by the standards of a compiler, small enough that independent re-implementations exist and can be run on the same proof terms.
2. **The axioms**. Lean’s logic is a dependent type theory; its core library declares exactly three axioms: `propext` (two logically equivalent propositions are equal), `Quot.sound` (quotients behave as quotients) and `Classical.choice` (a choice function exists, which gives the law of excluded middle). Every theorem in this book depends on at most these three. The command `#print axioms` lists what a theorem depends on (Section 34.8); anything beyond the three is a warning sign.

3. **The statement.** The kernel checks that the proof proves the statement as written. It does not check that the statement is the one you meant, that the definition `leftMomentum` is what a physicist calls the left-moving momentum, or that a matrix named `cartanE8` is the Cartan matrix of  $E_8$ . This is why the bible of this book says that the identification of a Lean object with a physical system is never Tier A.

#### The kernel checks the proof, not the statement

Two incidents from this project’s own log illustrate the third point. A revision brief said that the  $\Theta$ -shift preserves  $\eta$  “iff  $\Theta$  is antisymmetric”; the Lean theorem proves only the “if” direction [LL.md §S2.13, ll. 109–112]. And since the kernel cannot notice that a hypothesis was quietly weakened between review and proof, the project hashes every theorem statement into a lock file and re-checks the hashes before accepting a proof [LL.md §S2.10, ll. 87–92]. The lesson for a reader is simple: before quoting a theorem, read its statement in the `.lean` file, not the prose that summarizes it.

#### What “formalizing physics” can mean

No proof assistant checks that nature obeys an equation. What can be checked is a chain of implications: *if* the mass formula is  $M^2 = p_L^2 + p_R^2 + \dots$  and *if* T-duality acts as  $(n, w, R) \mapsto (w, n, \alpha'/R)$ , *then*  $M^2$  is invariant. The two “if”s are definitions in a file; the “then” is a theorem. Older libraries in this repository go further and replace real numbers by integers or rationals: those theorems are *arithmetic shadows* of the physical statement, true as arithmetic, silent about the continuum. The four files of this chapter are honest about which kind they are: `TDualityMassSpectrum.lean` works over  $\mathbb{Q}$  with  $\alpha' = 1$ ; `TraceBound.lean` works over  $\mathbb{R}$  with genuine real matrices.

## 34.2 Terms, types and propositions

Every expression in Lean has a type, written with a colon:  $(3 : \mathbb{N})$ ,  $(R : \mathbb{R})$ ,  $(G : \text{Matrix}(\text{Fin } d) (\text{Fin } d) \mathbb{R})$ . Types themselves have types:  $\mathbb{R} : \text{Type}$ . The key idea, due to Curry and Howard and usually called *propositions as types*, is that a mathematical statement is also a type, whose inhabitants are its proofs. Statements live in a special universe called `Prop`. So

|                           |                                                                    |
|---------------------------|--------------------------------------------------------------------|
| $2 \leq R + R^{-1}$       | -- a <code>Prop</code> : the type of proofs that $R + 1/R \geq 2$  |
| <code>Even (G i i)</code> | -- a <code>Prop</code> : the type of proofs that $G\ i\ i$ is even |
| $G^T = G$                 | -- a <code>Prop</code>                                             |

are types, and a theorem is a named term of such a type. Table 34.1 translates the vocabulary.

### 34.2.1 `def` versus `theorem`

A `def` introduces an object that later code can *compute with* and *unfold*. A `theorem` introduces a proof, and Lean deliberately forgets the proof’s internal structure once it is checked (“proof irrelevance”): two proofs of the same proposition are equal, and a theorem can never be unfolded, only *applied*. Here is the circle mass spectrum as it is written in `StringTheoryFormalization/UseCases/TDualityMassSpectrum.lean`:

```
def leftMomentum (n w R : ℚ) : ℚ := n / R + w * R
def rightMomentum (n w R : ℚ) : ℚ := n / R - w * R
def momentumMassSq (n w R : ℚ) : ℚ :=
  (leftMomentum n w R) ^ 2 + (rightMomentum n w R) ^ 2

theorem tduality_fixes_left_momentum (n w R : ℚ) (hR : R ≠ 0) :
  leftMomentum w n (1 / R) = leftMomentum n w R := by ...
```

| Mathematics      | Lean                                        | Example from the repository                                    |
|------------------|---------------------------------------------|----------------------------------------------------------------|
| a set or a space | a <code>Type</code>                         | <code>Matrix (Fin d) (Fin d) ℤ</code>                          |
| an element       | a term <code>x : T</code>                   | <code>eta d : Matrix (Charge d) (Charge d) ℤ</code>            |
| a statement      | a term of <code>Prop</code>                 | <code>IsODD g</code>                                           |
| a proof          | a term of that <code>Prop</code>            | <code>eta_isODD : IsODD (eta d)</code>                         |
| a definition     | <code>def</code>                            | <code>def leftMomentum (n w R : ℚ) : ℚ := n / R + w * R</code> |
| a theorem        | <code>theorem</code>                        | <code>theorem eta_mul_self : eta d * eta d = 1</code>          |
| “for all $x$ ”   | $\forall x, \dots$ or an argument $(x : T)$ | $\forall i, \text{Even } (G\ i\ i)$                            |
| “there is $x$ ”  | $\exists x, \dots$                          | $\exists i : \text{Fin } 26, \text{M24RepDim } i = \dots$      |
| “and” / “or”     | $\wedge$ / $\vee$                           | $G.\text{det} = 1 \vee G.\text{det} = -1$                      |

Table 34.1: The dictionary between ordinary mathematics and Lean. All examples are taken from the files quoted in this chapter.

Read the theorem as a sentence: “for all rationals  $n, w, R$ , and given a proof  $hR$  that  $R \neq 0$ , the left momentum of the dual data  $(w, n, 1/R)$  equals the left momentum of  $(n, w, R)$ .” The arguments before the colon are universally quantified;  $hR$  is a hypothesis, itself a term whose type is the proposition  $R \neq 0$ . The conventions are those of Chapter 8 with  $\alpha' = 1$ :  $p_L = n/R + wR$ ,  $p_R = n/R - wR$ , and the file’s header says so [TDualityMassSpectrum.lean header, ll. 17–22]. Note what is *not* in the statement: no oscillators, no  $\alpha'$ , no real numbers. The theorem is about rational functions of  $R$ ; it is an arithmetic shadow of Tong’s statement, and a faithful one, because the algebra is the same over  $\mathbb{Q}$  and  $\mathbb{R}$ .

### 34.2.2 Implicit arguments and variable

Curly braces mark an argument Lean should infer: in `IsEvenDiag {n : ℕ} (G : Gram n)` the rank  $n$  is read off from the type of  $G$ , so the user writes `IsEvenDiag G`, not `IsEvenDiag 8 G`. A line `variable {d : ℕ}` at the top of a file makes  $d$  an implicit argument of every declaration below that mentions it; that is why the theorems of `ODD.lean` are stated for arbitrary  $d$  without repeating  $(d : \mathbb{N})$  each time.

### 34.2.3 Reading a hypothesis as a physical assumption

Hypotheses carry physics. In `dualScale_ge` the hypothesis `hG : G.PosDef` says that the torus metric is a genuine Riemannian metric; in `dualScale_inv` the weaker `hG : IsUnit G.det` says only that  $G$  is invertible, because T-duality  $G \mapsto G^{-1}$  makes sense for any invertible  $G$ , positive or not. A theorem with a weaker hypothesis is a stronger theorem. A theorem with no hypothesis at all, such as `eta_mul_self`, is a pure identity. Get into the habit of reading the hypothesis list first: it is the fastest way to see what a formal theorem does and does not cover.

## 34.3 Definitions and structures

### 34.3.1 Predicates

The lattice file `DualScaleStream2/Lattice/Basic.lean` contains the two definitions on which every “this lattice is even and unimodular” statement of Chapters 18 and 20 rests:

```
abbrev Gram (n : ℕ) := Matrix (Fin n) (Fin n) ℤ
def IsEvenDiag {n : ℕ} (G : Gram n) : Prop := ∀ i, Even (G i i)
def IsUnimodular {n : ℕ} (G : Gram n) : Prop := G.det = 1 ∨ G.det = -1
```

`abbrev` is a `def` that Lean unfolds eagerly: a `Gram n` is an  $n \times n$  integer matrix, and every matrix lemma applies to it unchanged. The other two are *predicates*: functions from matrices to `Prop`. Note the honesty of the names. `IsEvenDiag` asks only that the diagonal be even; that this implies evenness of the whole quadratic form  $x^T G x$  is a separate theorem, `even_quadratic_form_of_even_diag`, which needs the extra hypothesis  $G^T = G$ . A lattice is identified with its Gram matrix in a fixed basis; no abstract  $\mathbb{Z}$ -module is introduced, and the file header says so [Lattice/Basic.lean header, ll. 29–38].

### 34.3.2 Structures

A `structure` bundles named fields, like a C `struct` or a Python dataclass. From the file `K3T2Signature.lean` in the directory `DualScaleStream2/Lattice/`:

```
structure Signature where
  pos : ℕ
  neg : ℕ
  deriving DecidableEq, Repr

instance : Add Signature := ⟨fun a b => ⟨a.pos + b.pos, a.neg + b.neg⟩⟩
@[simp] theorem add_pos (a b : Signature) : (a + b).pos = a.pos + b.pos := rfl
```

A term of type `Signature` is built with angle brackets, `⟨3, 19⟩`, and its fields are read with dot notation, `s.pos`. The `instance` line teaches Lean what `+` means on signatures (the direct sum adds the positive and negative dimensions separately), and the theorem `add_pos` is proved by `rfl`, “reflexivity”: both sides unfold to the same term. The attribute `@[simp]` registers it as a rewrite rule for the `simp` tactic. Notice, again, what the structure does not contain: a `Signature` is a pair of naturals with no attached quadratic form. The statement “the K3 lattice has signature (3,19)” is, in this file, a `def` assigning the pair `⟨3,19⟩` to a name; the theorem `sigK3_matches_hodge` then checks that this pair agrees with the Hodge-theoretic numbers tabulated independently in Stream 1. That is a consistency check between two tables, not a derivation of either.

#### Multi-name structure fields: parenthesize them

In Lean 4 the declaration

```
structure Foo where
  a b c d : Nat      -- WRONG: one field 'a', a function of b c d
```

does not declare four natural-number fields. It declares a *single* field `a` whose type is a curried function taking auto-bound implicit arguments `b c d` and returning `Nat`. Any later use of `a`, `b`, `c`, `d` as scalars fails with dependent-function-type errors that do not mention the cause. This reproduces in vanilla Lean 4 with no Mathlib. The fix is to parenthesize the group, `(a b c d : Nat)`, or to write one field per line as `Signature` does. The bug was found twice, independently, in this repository’s Stream 1 code [LL.md §0.4, ll. 188–199].

## 34.4 Matrices in Mathlib

All the block-matrix algebra of Chapters 11, 15 and 31 is written against Mathlib’s `Matrix`. A few facts make the notation readable.

**A matrix is a function.** `Matrix m n α` is, by definition, the function type  $m \rightarrow n \rightarrow \alpha$ : a matrix is a function of two indices. Hence `G i j` is the entry, `fun i j => if i < j then`

`G i j else 0` is the strict upper triangle, and two matrices are equal when they agree at every pair of indices (the `ext` tactic, Section 34.5). The index types need not be `Fin n`: any finite type will do, and `ODD.lean` indexes charge vectors by the disjoint union

```
| abbrev Charge (d : ℕ) := Fin d ⊕ Fin d
```

so that a  $2d \times 2d$  matrix is naturally a  $2 \times 2$  array of  $d \times d$  blocks.

**Blocks.** `fromBlocks A B C D` is the block matrix  $\begin{pmatrix} A & B \\ C & D \end{pmatrix}$  on  $\text{Fin } d \oplus \text{Fin } d$ . The two Mathlib lemmas that drive every proof in `ODD.lean` are

$$(\text{fromBlocks } A \ B \ C \ D)^\top = \text{fromBlocks } A^\top \ C^\top \ B^\top \ D^\top, \quad (34.1)$$

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} A' & B' \\ C' & D' \end{pmatrix} = \begin{pmatrix} AA' + BC' & AB' + BD' \\ CA' + DC' & CB' + DD' \end{pmatrix}, \quad (34.2)$$

named `Matrix.fromBlocks_transpose` and `Matrix.fromBlocks_multiply`. Both are what a student would write by hand; the point of naming them is that `rw` can apply them mechanically.

**Concrete matrices.** The notation `!![0, 1; 1, 0]` builds a literal matrix over `Fin 2`, and `!![R ^ 2]` a  $1 \times 1$  matrix over `Fin 1`.

**Operations.**  $G^\top$  is the transpose,  $G^H$  the conjugate transpose (over  $\mathbb{R}$  they coincide, and `TraceBound.lean` needs a lemma, `Matrix.conjTranspose_eq_transpose_of_trivial`, to say so, because Mathlib’s positive-definiteness lemmas are stated with  $^H$ ).  $G \cdot_v x$  is the matrix–vector product  $Gx$ ,  $x \cdot_v y$  the dot product  $\sum_i x_i y_i$ , so  $x^\top Gx$  is  $x \cdot_v (G \cdot_v x)$ . `G.trace` and `G.det` are the trace and determinant.

**The inverse is total.**  $G^{-1}$  is defined for *every* square matrix, by the adjugate formula  $G^{-1} = \det(G)^{-1} \text{adj } G$  with the convention that the ring inverse of a non-unit is 0. So  $G^{-1} * G = 1$  is *not* a theorem about all  $G$ ; it is `Matrix.nonsing_inv_mul G hdet`, which needs a proof `hdet : IsUnit G.det` that the determinant is invertible in the ground ring. Over a field this means  $\det G \neq 0$ ; over  $\mathbb{Z}$  it means  $\det G = \pm 1$ , which is why unimodularity is exactly the condition for an integer Gram matrix to have an integer inverse.

**Positivity.** Mathlib’s definition of positive semidefiniteness, from the file `PosDef.lean` in `Mathlib/LinearAlgebra/Matrix/`, is

```
def PosSemidef (M : Matrix n n R) : Prop :=
  M.IsHermitian ∧
  ∀ x : n →₀ R, 0 ≤ x.sum fun i xi ↦ x.sum fun j xj ↦ star xi * M i j * xj
```

that is,  $M$  is Hermitian and  $\sum_{ij} \bar{x}_i M_{ij} x_j \geq 0$  for all finitely supported  $x$ . `PosDef` is defined in the same file by the same two clauses, with the inequality strict for every  $x \neq 0$  (its binder for  $x$  is Lean’s “strict implicit” bracket, which we do not reproduce here). A `PosDef` hypothesis therefore contains a symmetry statement for free, which `dualScale_ge` extracts as `hG.isHermitian`.

### Three ways to be wrong about $G^{-1}$

- (i) Forgetting that  $G^{-1}$  exists for singular  $G$  and equals 0 there: a theorem about  $G^{-1}$  with no invertibility hypothesis is usually vacuous or false.
- (ii) Expecting  $(G^{-1})^{-1} = G$

to be definitional: it is the lemma `Matrix.nonsing_inv_nonsing_inv`, and it too needs `IsUnit G.det`. (iii) Over  $\mathbb{Z}$ , reading `IsUnit G.det` as  $\det G \neq 0$ : it is  $\det G = \pm 1$ .

**Example 34.1** (An integer inverse as a certificate). Let  $U = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , the hyperbolic plane of Chapter 18. Then  $UU = \mathbf{1}$  and  $\det U = -1$ . The theorem

```
theorem isUnimodular_of_mul_eq_one {n : ℕ} (G H : Gram n) (h : H * G = 1) :
  IsUnimodular G := by ...
```

turns the one-line computation  $UU = \mathbf{1}$  into a proof that  $\det U = \pm 1$ , without ever computing the determinant: from  $HG = \mathbf{1}$  and multiplicativity,  $\det G \cdot \det H = \det \mathbf{1} = 1$  in  $\mathbb{Z}$ , and the only factorizations of 1 in  $\mathbb{Z}$  are  $1 \cdot 1$  and  $(-1)(-1)$ . The same theorem is instantiated, never re-proved, for the  $8 \times 8$  Cartan matrix of  $E_8$ , where the kernel checks the 64-entry product  $E_8^{-1}E_8 = \mathbf{1}$  directly. Certificates of this kind—an explicit witness whose verification is mechanical—are the cheapest kind of Lean proof, and Chapter 37 reports that a small local prover closed essentially all of them [STREAM2\_WORKFLOW.md §6, ll. 107–114].

## 34.5 The tactic toolbox

A proof written as an explicit term is unreadable for anything longer than a line. Instead, after `:= by`, one writes *tactics*: commands that transform the current *goal* (a list of hypotheses and a target proposition) until no goal is left. Each tactic, behind the scenes, builds a piece of the proof term; the kernel checks the assembled term at the end. Table 34.2 lists every tactic used in the proofs quoted in this book, with the file in which a student can see it at work. The remainder of this section shows six of them on real goals.

### 34.5.1 `unfold`, `field_simp`, `ring`: an algebraic identity

The proof of `tduality_fixes_left_momentum` is three lines:

```
theorem tduality_fixes_left_momentum (n w R : ℚ) (hR : R ≠ 0) :
  leftMomentum w n (1 / R) = leftMomentum n w R := by
  unfold leftMomentum
  field_simp
  ring
```

After `unfold` the goal is  $w/(1/R) + n \cdot (1/R) = n/R + wR$ . `field_simp` finds the hypothesis `hR` in the context, multiplies through by  $R$  and clears every fraction, leaving a polynomial identity; `ring` normalizes both sides to the same polynomial and closes the goal. Without `hR` the tactic would refuse: division by zero is defined in Lean (as 0), so  $w/(1/R) = wR$  is false at  $R = 0$ , and the kernel would never accept the theorem without the hypothesis. This is a small example of how the hypothesis list is forced by the mathematics and not by taste.

### 34.5.2 `constructor`, `rfl`: a conjunction of numerical facts

```
theorem index_mod_eight :
  sigK3.index % 8 = 0 ∧ sigMukai.index % 8 = 0 ∧ sigK3T2.index % 8 = 0 := by
  constructor
  · rfl
  constructor
  · rfl
  · rfl
```

| Tactic                            | What it does to the goal                                                                                         | Seen in                                             |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| <code>unfold f</code>             | replaces <code>f</code> by its definition                                                                        | <code>thetaShift_isODD</code>                       |
| <code>rw [h]</code>               | rewrites the goal left-to-right with equation <code>h</code> ;<br>$\leftarrow$ <code>h</code> goes right-to-left | <code>isUnimodular_of_mul_eq_one</code>             |
| <code>exact t</code>              | closes the goal with the term <code>t</code> , whose type must match exactly                                     | <code>isUnimodular_of_mul_eq_one</code>             |
| <code>apply t</code>              | closes the goal with <code>t</code> applied to new goals for its missing hypotheses                              | <code>circle_effective_scale_ge_two</code>          |
| <code>intro x</code>              | moves a $\forall x$ or a hypothesis from the target into the context                                             | <code>cartanE8_evenDiag</code>                      |
| <code>rintro (a, h, rfl)</code>   | <code>intro</code> that also destructures pairs, existentials and equations                                      | <code>isSection_image</code>                        |
| <code>constructor</code>          | splits $P \wedge Q$ into two goals (or $\leftrightarrow$ into two implications)                                  | <code>index_mod_eight</code>                        |
| <code>rcases h with a   b</code>  | case split on an $\vee$ or an inductive hypothesis                                                               | <code>even_quadratic_form_of_even_diag</code>       |
| <code>have h : P := by ...</code> | proves an intermediate statement and names it                                                                    | everywhere                                          |
| <code>calc</code>                 | a chain of equalities or inequalities, each step justified                                                       | <code>isODD_mul</code>                              |
| <code>ext i j</code>              | to prove two matrices (functions) equal, prove equality at every entry                                           | <code>cartanE8_symm</code>                          |
| <code>fin_cases i</code>          | splits a goal about $i : \text{Fin } n$ into $n$ concrete goals                                                  | <code>eta_one_reindex</code>                        |
| <code>simp [lemmas]</code>        | rewrites with the <code>@[simp]</code> lemma set plus the ones listed, until nothing changes                     | <code>thetaShift_isODD</code>                       |
| <code>ring</code>                 | proves identities in a commutative ring by normalizing both sides                                                | <code>tduality_fixes_left_momentum</code>           |
| <code>field_simp</code>           | clears denominators, given hypotheses that they are non-zero                                                     | <code>tduality_fixes_left_momentum</code>           |
| <code>norm_num</code>             | evaluates numerical expressions and decides numeral (in)equalities                                               | <code>k3k3_anomaly</code>                           |
| <code>decide</code>               | evaluates a decidable proposition by computation inside the kernel                                               | <code>cartanE8_evenDiag</code>                      |
| <code>rfl</code>                  | both sides are the same term after unfolding                                                                     | <code>add_pos</code> , <code>eta_one_reindex</code> |
| <code>linarith</code>             | finds a linear combination of hypotheses that proves a linear (in)equality                                       | <code>dualScale_ge</code>                           |
| <code>nlinarith [hints]</code>    | <code>linarith</code> after multiplying pairs of hypotheses and hints                                            | <code>self_dual_radius_unique</code>                |
| <code>t1 &lt;;&gt; t2</code>      | run <code>t2</code> on every goal that <code>t1</code> produced                                                  | <code>cartanE8_symm</code>                          |

Table 34.2: The tactics used in the proofs quoted in this book.

`constructor` splits the conjunction; the bullets `·` focus one goal at a time; `rfl` succeeds because `sigK3.index` unfolds by computation to the numeral  $3 - 19 = -16$ , and  $-16 \bmod 8$  evaluates to 0. The theorem does not prove the mod-8 theorem for even unimodular lattices; it checks three instances of it, and its docstring says so [Lattice/K3T2Signature.lean, ll. 176–185].

### 34.5.3 `ext`, `fin_cases`, `decide`: finite matrices

```
theorem cartanE8_symm : cartanE8T = cartanE8 := by
  ext i j
  fin_cases i <;> fin_cases j <;> rfl
```

`ext i j` reduces the matrix equation to the entry equation `cartanE8T i j = cartanE8 i j` for arbitrary  $i j : \text{Fin } 8$ . `fin_cases i` replaces this by eight goals with  $i = 0, \dots, 7$ ; the second `fin_cases` replaces each by eight more; `rfl` closes all 64 by evaluation. This is the pattern for every fixed-size statement in the repository (`eta_one_reindex`,



`tauShift_dual_spec`, the  $E_8$  inverse check). Where the entries are not literally identical but merely equal after arithmetic, `decide` or `norm_num` replaces `rfl`. The cost grows as  $n^2$  goals, each cheap; the benefit is that no idea is needed.

### 34.5.4 `nlinarith`: an inequality with a hint

```
theorem self_dual_radius_unique (R : ℚ) (hR : 0 < R) (hFix : (1 : ℚ) / R = R) :
  R = 1 := by
  have hR' : R ≠ 0 := ne_of_gt hR
  field_simp at hFix
  nlinarith [sq_nonneg (R - 1), sq_nonneg (R + 1)]
```

After `field_simp at hFix` the hypothesis reads  $R^2 = 1$  (up to the exact form the tactic chooses). `linarith` treats the monomials  $R$ ,  $R^2$ ,  $R^3$  as independent unknowns and looks for a linear combination of hypotheses proving the goal; it cannot see that  $R^2 = 1$  and  $R > 0$  force  $R = 1$ , because that needs a product. `nlinarith` adds pairwise products of the hypotheses and the supplied hints, then calls `linarith`. A certificate of the kind it finds: from the hint  $(R - 1)^2 \geq 0$  and  $R^2 = 1$ ,  $0 \leq R^2 - 2R + 1 = 2 - 2R$ , so  $R \leq 1$ ; from  $R > 0$  times the same hint,  $0 \leq R(R - 1)^2 = R^3 - 2R^2 + R$ , and  $R^3 = R \cdot R^2 = R$  (the product of  $R$  with `hFix`), so  $0 \leq 2R - 2$  and  $R \geq 1$ . Both inequalities together give  $R = 1$ . The second hint  $(R + 1)^2 \geq 0$  is harmless; the physics behind the first is that  $R = 1$  is the self-dual radius of Chapter 8 in units  $\alpha' = 1$ , and the theorem says it is the *unique* positive fixed point of  $R \mapsto 1/R$ .

### 34.5.5 `calc`: closure of the duality group

```
theorem isODD_mul (g h : Matrix (Charge d) (Charge d) ℤ) (hg : IsODD g)
  (hh : IsODD h) : IsODD (g * h) := by
  unfold IsODD at *
  calc (g * h)ᵀ * eta d * (g * h)
    = hᵀ * (gᵀ * eta d * g) * h := by
      simp [Matrix.transpose_mul, Matrix.mul_assoc]
    _ = hᵀ * eta d * h := by rw [hg]
    _ = eta d := by rw [hh]
```

A `calc` block is the formal version of a chain of equalities on a blackboard. Each step names its justification. `unfold IsODD at *` unfolds the definition in the goal *and* in the hypotheses, so that `hg` becomes the equation  $g^T * \text{eta } d * g = \text{eta } d$  which `rw` can use. The first step is  $(gh)^T \eta(gh) = h^T (g^T \eta g) h$ , which is just  $(gh)^T = h^T g^T$  and associativity; `simp` with those two lemmas does the regrouping.

## 34.6 One proof, line by line: the $\Theta$ -shift

We now read a complete proof at the speed of the kernel. The theorem is the first generator of  $O(d, d; \mathbb{Z})$  in Chapter 11: an integer antisymmetric shift of the  $B$ -field,  $g_\Theta = \begin{pmatrix} 1 & \Theta \\ 0 & 1 \end{pmatrix}$ , preserves the momentum–winding pairing  $\eta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ . Giveon, Porrati and Rabinovici write the element, require  $\Theta_{ij} \in \mathbb{Z}$  and  $\Theta_{ji} = -\Theta_{ij}$ , and explain the physics: the constant  $B$ -term in the action is a total derivative, so an integer shift changes the action by a multiple of  $2\pi$  and drops out of the path integral [GPR §2.4, ll. 1355–1373].

### 34.6.1 The computation on paper

With  $g = g_\Theta$ ,

$$g^T \eta g = \begin{pmatrix} 1 & 0 \\ \Theta^T & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & \Theta \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ \Theta^T & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & \Theta \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & \Theta^T + \Theta \end{pmatrix}. \quad (34.3)$$



So  $g^\top \eta g = \eta$  exactly when  $\Theta^\top + \Theta = 0$ . (We checked (34.3) symbolically for  $d = 2$  with a general  $\Theta$ ; the residual lower-right block is  $\Theta + \Theta^\top$ , as it should be. This is a symbolic check, not part of the Tier A claim.) The paper computation has three steps: transpose  $g$ , multiply blocks, use antisymmetry. The Lean proof has the same three steps.

### 34.6.2 The proof

```
theorem thetaShift_isODD (Θ : Matrix (Fin d) (Fin d) ℤ) (hΘ : Θ⊤ = -Θ) :
  IsODD (thetaShift Θ) := by
  unfold IsODD thetaShift eta
  rw [fromBlocks_transpose, fromBlocks_multiply, fromBlocks_multiply,
      Matrix.transpose_one, Matrix.transpose_zero, hΘ]
  simp
```

**The statement.** For every  $d$  (implicit, from the file’s `variable` line), every integer  $d \times d$  matrix  $\Theta$ , and a proof  $h\Theta$  that  $\Theta^\top = -\Theta$ : the matrix `thetaShift Θ` satisfies `IsODD`. Lean prints the fully elaborated statement, with every implicit argument and namespace made visible, on request:

```
#check @DualScaleStream2.TDuality.thetaShift_isODD
-- ∀ {d : ℕ} (Θ : Matrix (Fin d) (Fin d) ℤ),
--   Θ.transpose = -Θ → IsODD (thetaShift Θ)
```

This is the “if” direction only. The converse, that `IsODD (thetaShift Θ)` forces antisymmetry, is true by (34.3) but is not in the file (Exercise 34.4).

**Line 1: `unfold IsODD thetaShift eta`.** The goal `IsODD (thetaShift Θ)` is replaced by the definitions’ bodies:

```
⊢ (fromBlocks 1 Θ 0 1)⊤ * fromBlocks 0 1 1 0 * fromBlocks 1 Θ 0 1
  = fromBlocks 0 1 1 0
```

Here `1` and `0` are the  $d \times d$  identity and zero blocks; matrix multiplication associates to the left, so the expression is  $(g^\top \eta)g$ .

**Line 2: the `rw chain`.** Rewrites are applied in order, each to the first matching sub-expression.

- `fromBlocks_transpose`, equation (34.1), turns  $(\text{fromBlocks } 1 \ \Theta \ 0 \ 1)^\top$  into  $\text{fromBlocks } 1^\top \ 0^\top \ \Theta^\top \ 1^\top$ .
- The first `fromBlocks_multiply`, equation (34.2), multiplies the two leftmost block matrices. The goal’s left side becomes  $\text{fromBlocks } (1^\top * 0 + 0^\top * 1) \ (1^\top * 1 + 0^\top * 0) \ (\Theta^\top * 0 + 1^\top * 1) \ (\Theta^\top * 1 + 1^\top * 0)$  times  $\text{fromBlocks } 1 \ \Theta \ 0 \ 1$ .
- The second `fromBlocks_multiply` performs the remaining product, producing a single `fromBlocks` whose four entries are sums of products of  $0$ ,  $1$ ,  $\Theta$  and  $\Theta^\top$ . The lower-right entry is  $(\Theta^\top * 0 + 1^\top * 1) * \Theta + (\Theta^\top * 1 + 1^\top * 0) * 1$ .
- `Matrix.transpose_one` and `Matrix.transpose_zero` replace  $1^\top$  by  $1$  and  $0^\top$  by  $0$ .
- `hΘ` replaces every  $\Theta^\top$  by  $-\Theta$ . This is the only place the hypothesis is used; the lower-right entry now reads  $(-\Theta * 0 + 1 * 1) * \Theta + (-\Theta * 1 + 1 * 0) * 1$ .

**Line 3: `simp`.** The simplifier applies the ring lemmas for 0 and 1 (`mul_zero`, `one_mul`, `add_zero`, `neg_mul`, ...) until every block is in normal form. The upper-left block becomes 0, both off-diagonal blocks 1, and the lower-right block  $1 * \Theta + (-\Theta) * 1 = \Theta - \Theta = 0$ . The goal is then `fromBlocks 0 1 1 0 = fromBlocks 0 1 1 0`, which `simp` closes by reflexivity. Had `hΘ` been omitted from the `rw` list, `simp` would stop with the unsolved goal  $\Theta + \Theta^\top = 0$ : the same residual as in (34.3). That is what “the proof uses the hypothesis exactly where the mathematics does” means in practice.

**What the kernel saw.** The tactics produced a proof term whose leaves are the Mathlib lemmas named above, the hypothesis `hΘ`, and reflexivity. The term type-checks against the statement; the check depends only on the three standard axioms (Section 34.8). Note that  $d$  was never specialized: the proof is one term valid for all  $d$ , which is why no `fin_cases` appears. Chapter 37 records that this goal was closed by the mid-tier language model, not by the local prover, which handles only fixed-size goals [STREAM2\_WORKFLOW.md §6, ll. 109–114].

### 34.7 Reading a longer proof: the dual-scale bound

Most proofs in the repository are longer than three lines, and one reads them by their **have** skeleton: the intermediate statements are the proof; the tactics between them are bookkeeping. `dualScale_ge` in `DualScaleStream2/DualScale/TraceBound.lean` proves the bound of Chapter 31,

$$\mathrm{tr} G + \mathrm{tr} G^{-1} \geq 2d \quad \text{for every positive-definite } G \in \mathbb{R}^{d \times d}, \quad (34.4)$$

where  $\mathrm{tr} G + \mathrm{tr} G^{-1}$  is the trace of the generalized metric  $\mathcal{H}(G, 0)$  of Chapter 15, defined in Lean as `dualScale G := (genMetric G 0).trace` with `genMetric G B := fromBlocks (G - B * G-1 * B) (B * G-1) (-(G-1 * B)) G-1`. The idea that avoids diagonalizing  $G$  is the identity

$$(G - \mathbf{1}) G^{-1} (G - \mathbf{1}) = (G - \mathbf{1}) - (\mathbf{1} - G^{-1}), \quad \text{so} \quad \mathrm{tr} [(G - \mathbf{1}) G^{-1} (G - \mathbf{1})] = \mathrm{tr} G + \mathrm{tr} G^{-1} - 2d, \quad (34.5)$$

together with the fact that the left side is a trace of  $M^\top P M$  with  $P = G^{-1}$  positive semidefinite, hence non-negative. (We verified (34.5) with a symbolic  $2 \times 2$  matrix; the expansion is  $(G - \mathbf{1}) G^{-1} (G - \mathbf{1}) = G - \mathbf{1} - \mathbf{1} + G^{-1}$  using  $G G^{-1} = G^{-1} G = \mathbf{1}$ .) The skeleton of the Lean proof is exactly this list:

```
theorem dualScale_ge (G : Matrix (Fin d) (Fin d) ℝ) (hG : G.PosDef) :
  (2 * d : ℝ) ≤ dualScale G := by
  rw [dualScale_eq] -- goal: 2d ≤ tr G + tr G-1
  have hdet : IsUnit G.det := ... -- PosDef gives invertible
  have h1 : G-1 * G = 1 := Matrix.nonsing_inv_mul G hdet
  have h2 : G * G-1 = 1 := Matrix.mul_nonsing_inv G hdet
  have hGinv_psd : G-1.PosSemidef := hG.posSemidef.inv
  have hGT : G⊤ = G := ... -- from hG.isHermitian
  have hMh : (G - 1)H = G - 1 := ...
  have hpsd : ((G - 1)H * G-1 * (G - 1)).PosSemidef :=
    Matrix.PosSemidef.conjTranspose_mul_mul_same hGinv_psd (G - 1)
  rw [hMh] at hpsd
  have htrace_nonneg : 0 ≤ ((G - 1) * G-1 * (G - 1)).trace := hpsd.trace_nonneg
  have hexpand : (G - 1) * G-1 * (G - 1) = (G - 1) - (1 - G-1) := ...
  rw [hexpand] at htrace_nonneg
  rw [Matrix.trace_sub, Matrix.trace_sub, Matrix.trace_sub, Matrix.trace_one]
    at htrace_nonneg
  simp only [Fintype.card_fin] at htrace_nonneg -- tr 1 = d
  linarith
```

Every `have` is a sentence of the paper proof. The three Mathlib inputs are named: positive-definite implies invertible (`hdet`); the inverse of a positive-semidefinite matrix is positive-semidefinite (`PosSemidef.inv`); a sandwich  $M^H P M$  of a positive-semidefinite  $P$  is positive-semidefinite and has non-negative trace (the lemmas `conjTranspose_mul_mul_same` and `trace_nonneg` of the `PosSemidef` namespace). The final `linarith` sees the hypothesis  $0 \leq (\operatorname{tr} G - d) - (d - \operatorname{tr} G^{-1})$  and the goal  $2d \leq \operatorname{tr} G + \operatorname{tr} G^{-1}$ , which are the same linear inequality in the atoms  $\operatorname{tr} G$ ,  $\operatorname{tr} G^{-1}$ ,  $d$ .

**Example 34.2** (Numbers). Take  $d = 2$  and  $G = \operatorname{diag}(4, \frac{1}{4})$ , a rectangular torus with radii 2 and  $\frac{1}{2}$  in string units. Then  $\operatorname{tr} G + \operatorname{tr} G^{-1} = 4 + \frac{1}{4} + \frac{1}{4} + 4 = 8.5 \geq 4 = 2d$ . The sandwich is  $\operatorname{diag}(3^2/4, (-\frac{3}{4})^2 \cdot 4) = \operatorname{diag}(2.25, 2.25)$  with trace 4.5, and indeed  $8.5 - 4 = 4.5$ . Equality in (34.4) holds only for  $G = \mathbf{1}$ , where the sandwich vanishes; Lean proves the value  $2d$  at  $G = \mathbf{1}$  as `dualScale_one` but not uniqueness of the minimizer.

The circle case is the corollary `circle_effective_scale_ge_two`:  $2 \leq R + R^{-1}$  for  $R > 0$ , proved from  $R + R^{-1} - 2 = (R - 1)^2/R$  with `field_simp`, `ring`, `div_nonneg` and `linarith`. This is the dual-scale inequality of Chapter 31 in units  $\alpha' = 1$ ; its status as mathematics is [A], its physical reading is discussed there.

## 34.8 Reading #print axioms

A green build is not the whole gate. Two things can hide inside a compiled file. A `sorry` is a placeholder that makes Lean accept any statement; it is reported as a warning, and a theorem that merely *uses* a sorried lemma inherits the gap silently. And `native_decide` asks Lean to trust the compiled evaluator rather than the kernel, adding the axiom `Lean.ofReduceBool`. The command `#print axioms` exposes both. We ran it, from a scratch file that imports the four modules of this chapter, on the project’s built library:

```
#print axioms DualScaleStream2.TDuality.thetaShift_isODD
-- 'DualScaleStream2.TDuality.thetaShift_isODD' depends on axioms:
-- [propext, Classical.choice, Quot.sound]
#print axioms DualScaleStream2.DualScale.dualScale_ge
-- [propext, Classical.choice, Quot.sound]
#print axioms DualScaleStream2.Lattice.isUnimodular_of_mul_eq_one
-- [propext, Classical.choice, Quot.sound]
#print axioms StringTheory.UseCases.TDuality.self_dual_radius_unique
-- [propext, Classical.choice, Quot.sound]
```

Anything else in the list—`sorryAx`, `Lean.ofReduceBool`, or a user-declared axiom—means the theorem is not Tier A. The project’s gate script `tools/axiom_audit.py` runs this command over every theorem of a library and fails unless each depends only on the three standard axioms [STREAM2\_WORKFLOW.md §2, ll. 22–24]. It earned its place: the first run found a `native_decide` in a Stream 1 file that every build and every `sorry` search had passed, and it was replaced by a kernel-checked `norm_num` proof [LL.md §S2.5, ll. 56–61].

### Zero axioms, one hundred percent

Every Mathlib-based theorem depends on `propext`, `Classical.choice` and `Quot.sound`; “zero axioms” is not a state a theorem in this book can be in, and a document that claims it has not run the command. Likewise a count of theorems in a vendored library is not a measure of what this project proved; the project’s own log records a headline figure of that kind as an overclaim [LL.md, ll. 3–13].

## 34.9 Checking a theorem yourself

The whole point of a machine-checked book is that the reader does not have to believe the authors. The steps below reproduce the check on a Linux or macOS machine with enough free disk for Mathlib’s compiled object files (several gigabytes); the toolchain and Mathlib version are pinned in the repository.

1. Install `elan`, the Lean version manager (one shell command from the Lean website). It reads the file `lean-toolchain` in the repository, which pins `leanprover/lean4:v4.33.1`, and installs exactly that compiler.
2. Clone the repository and run `lake exe cache get` in its root. This downloads pre-compiled Mathlib object files for the pinned Mathlib tag `v4.33.1` instead of compiling Mathlib, which would take hours; the repository’s setup notes record that `lake update` triggers this automatically [docs/INFRA\_SETUP.md, ll. 53–60].
3. Run `lake build DualScaleStream2`. The Stream 2 library is a separate `lean_lib` in `lakefile.lean`, built by name so that a failure there can never turn the Stream 1 core red [STREAM2\_WORKFLOW.md §2, ll. 32–33]. A clean build ends with no error and no `sorry` warning.
4. Write a probe file, say `Probe.lean`, containing the imports and the `#print axioms` lines of Section 34.8, and run `lake env lean -DmaxHeartbeats=1000000 Probe.lean`. The output should be the four lines shown there.
5. To read a statement with all implicit arguments, add `#check @name`. To see a goal state, open the file in VS Code with the Lean 4 extension, place the cursor after any tactic line, and read the “Lean Infoview” panel; this is how the goal states of Section 34.6 were obtained.

### Heartbeats: the single-file check is stricter than the build

Lean bounds the work per declaration by a “heartbeat” budget. The repository’s `lakefile.lean` raises it to  $10^6$  for `lake build`, but a bare `lake env lean file.lean` uses the default  $2 \times 10^5$  and can time out on a proof that the real build accepts. Always pass `-DmaxHeartbeats=1000000` (and `-DmaxRecDepth=8000`) to single-file compiles. A stricter budget can reject a valid proof; it can never accept an invalid one, so no wrong result enters the library this way, only wasted time [LL.md §S2.4, ll. 47–54].

## 34.10 What the machine has checked

The four files used as examples in this chapter are quoted in full elsewhere in the book (Chapters 8, 11, 18 and 31). Here we record only what has been verified about them from the point of view of this chapter: the statements as read, the axioms as printed, and the gaps a reader should not paper over. All declarations below compile in the project’s build of 2026-09-17 and depend only on `propext`, `Classical.choice` and `Quot.sound` (Section 34.8).

### Lattice predicates and the two general lemmas

```
theorem isUnimodular_of_mul_eq_one {n : ℕ} (G H : Gram n) (h : H * G = 1) :
  IsUnimodular G := by ...
theorem even_quadratic_form_of_even_diag {n : ℕ} (G : Gram n) (hsymm : GT = G)
  (heven : IsEvenDiag G) (x : Fin n → ℤ) : Even (x ·v (G *v x)) := by ...
```

**Reading.** For every rank  $n$  and integer Gram matrix  $G$ : an integer left inverse proves

$\det G = \pm 1$ ; and if  $G$  is symmetric with even diagonal then  $x^\top G x$  is even for every integer vector  $x$ . **Proof idea.** The first is  $\det G \det H = 1$  in  $\mathbb{Z}$  (`Matrix.det_mul`, `Matrix.det_one`, `Int.eq_one_or_neg_one_of_mul_eq_one`). The second splits  $G = D + U + U^\top$  into diagonal and strict upper triangle by a three-way case split `rcases lt_trichotomy i j`, then  $x^\top G x = x^\top D x + 2x^\top U x$ . **Not proved.** The converse (an even form has even diagonal) is not stated; no abstract lattice, isomorphism or classification appears [Lattice/Basic.lean header, ll. 40–45]. **[A]**

### The duality group for every $d$

```
theorem thetaShift_isODD (Θ : Matrix (Fin d) (Fin d) ℤ) (hΘ : Θ⊤ = -Θ) :
  IsODD (thetaShift Θ) := by ...
theorem basisChange_isODD (A B : Matrix (Fin d) (Fin d) ℤ) (h : A⊤ * B = 1) :
  IsODD (basisChange A B) := by ...
theorem eta_mul_self : eta d * eta d = 1 := by ...
theorem eta_isODD : IsODD (eta d) := by ...
theorem isODD_mul (g h : Matrix (Charge d) (Charge d) ℤ) (hg : IsODD g)
  (hh : IsODD h) : IsODD (g * h) := by ...
theorem eta_one_reindex :
  (eta 1).reindex finSumFinEquiv finSumFinEquiv
  = DualScaleStream2.Lattice.hyperbolicU := by ...
```

**Reading.** The  $\Theta$ -shift with antisymmetric  $\Theta$ , the basis change  $\text{diag}(A, B)$  with  $A^\top B = 1$ , and  $\eta$  itself satisfy  $g^\top \eta g = \eta$ ; the predicate is closed under products; at  $d = 1$  the matrix  $\eta$ , reindexed from  $\text{Fin } 1 \oplus \text{Fin } 1$  to  $\text{Fin } 2$ , is literally the hyperbolic plane  $U$ . Each of the first two is one direction of an “iff”. **Proof idea.** Block transpose, two block multiplications, the hypothesis, `simp` (Section 34.6). **Not proved.** That the three families generate  $O(d, d; \mathbb{Z})$ ; any group structure; the moduli-space quotient [TDuality/ODD.lean header, ll. 33–36]. **[A]**

### The dual-scale bound

```
noncomputable def dualScale (G : Matrix (Fin d) (Fin d) ℝ) : ℝ :=
  (genMetric G 0).trace
theorem dualScale_eq (G : Matrix (Fin d) (Fin d) ℝ) :
  dualScale G = G.trace + G-1.trace := by ...
theorem dualScale_inv (G : Matrix (Fin d) (Fin d) ℝ) (hG : IsUnit G.det) :
  dualScale G-1 = dualScale G := by ...
theorem dualScale_ge (G : Matrix (Fin d) (Fin d) ℝ) (hG : G.PosDef) :
  (2 * d : ℝ) ≤ dualScale G := by ...
theorem dualScale_one :
  dualScale (1 : Matrix (Fin d) (Fin d) ℝ) = 2 * d := by ...
theorem dualScale_circle (R : ℝ) (hR : R ≠ 0) :
  dualScale (![R ^ 2] : Matrix (Fin 1) (Fin 1) ℝ)
  = (R + R-1) ^ 2 - 2 := by ...
theorem circle_effective_scale_ge_two (R : ℝ) (hR : 0 < R) :
  2 ≤ R + R-1 := by ...
```

**Reading.** Over the real numbers, for every  $d$ : the trace of  $\mathcal{H}(G, 0)$  is  $\text{tr } G + \text{tr } G^{-1}$ ; it is invariant under  $G \mapsto G^{-1}$  for invertible  $G$ ; it is at least  $2d$  for positive-definite  $G$ ; it equals  $2d$  at  $G = 1$ ; at  $d = 1$  with  $G = R^2$  it equals  $(R + 1/R)^2 - 2$ ; and  $R + 1/R \geq 2$  for  $R > 0$ . `noncomputable` marks a definition involving real numbers, which cannot be evaluated by a program; it does not weaken the theorems. **Proof idea.** Section 34.7. **Not proved.** That  $G = 1$  is the unique minimizer; anything at  $B \neq 0$ ; any physical interpretation, which the file itself labels Tier C [DualScale/TraceBound.lean header, ll. 44–47].

**[A]****The circle mass spectrum over  $\mathbb{Q}$** 

```

theorem tduality_fixes_left_momentum (n w R :  $\mathbb{Q}$ ) (hR : R  $\neq$  0) :
  leftMomentum w n (1 / R) = leftMomentum n w R := by ...
theorem tduality_flips_right_momentum (n w R :  $\mathbb{Q}$ ) (hR : R  $\neq$  0) :
  rightMomentum w n (1 / R) = - rightMomentum n w R := by ...
theorem tduality_invariant_mass_squared (n w R :  $\mathbb{Q}$ ) (hR : R  $\neq$  0) :
  momentumMassSq w n (1 / R) = momentumMassSq n w R := by ...
theorem self_dual_radius_unique (R :  $\mathbb{Q}$ ) (hR : 0 < R) (hFix : (1 :  $\mathbb{Q}$ ) / R = R) :
  R = 1 := by ...

```

**Reading.** With  $\alpha' = 1$  and  $n, w, R$  rational: T-duality fixes  $p_L$ , negates  $p_R$ , hence fixes  $p_L^2 + p_R^2$ ; and  $R = 1$  is the only positive fixed point of  $R \mapsto 1/R$ . **Proof idea.** `unfold`, `field_simp`, `ring`; `nlinarith` with two square hints (Section 34.5). **Not proved.** Nothing about oscillators or the full CFT;  $n$  and  $w$  are not restricted to integers, so the theorem is about rational functions, an arithmetic shadow of the physical statement whose algebra happens to be identical [TDualityMassSpectrum.lean header, ll. 35–50]. **[A]**

| Claim                                                         | Tier       | Where                                          |
|---------------------------------------------------------------|------------|------------------------------------------------|
| The theorems above, as stated                                 | <b>[A]</b> | the four files; axioms printed in Section 34.8 |
| $g_\Theta \in O(d, d; \mathbb{Z})$ iff $\Theta$ antisymmetric | <b>[L]</b> | GPR §2.4; “only if” by (34.3), not in Lean     |
| $g_\Theta$ is a symmetry of the string spectrum               | <b>[L]</b> | GPR §2.4 (total-derivative argument)           |
| <code>leftMomentum</code> is the physical $p_L$               | —          | an identification; never Tier A                |

**Chapter summary**

- A theorem is a type in `Prop`; a proof is a term of that type; the kernel checks the term against the type and trusts nothing else. Three things remain to be trusted: the kernel, the three axioms `propext`, `Classical.choice`, `Quot.sound`, and the statement itself.
- `def` makes an object that can be unfolded; `theorem` makes a proof that can only be applied. Read the hypothesis list first: it tells you what a formal theorem covers.
- A Mathlib matrix is a function of two indices; `fromBlocks` builds  $2 \times 2$  block matrices on  $\text{Fin } d \oplus \text{Fin } d$ ;  $G^{-1}$  is total and needs `IsUnit G.det` to behave; `PosDef` includes symmetry.
- Tactics transform goals. Identities: `unfold`, `rw`, `simp`, `ring`, `field_simp`. Finite computations: `ext`, `fin_cases`, `decide`, `norm_num`, `rfl`. Inequalities: `linarith`, `nlinarith`. Structure: `have`, `calc`, `constructor`, `intro`.
- `thetaShift_isODD` is proved by unfolding, two block multiplications, one use of antisymmetry and `simp`; the hypothesis is used exactly where the paper computation uses it. It proves one direction only.
- `#print axioms` is the last gate: `sorryAx` or `Lean.ofReduceBool` in the list disqualifies a theorem from Tier A. Single-file checks need `-DmaxHeartbeats=1000000`.

**Exercises**

**Exercise 34.1** ( $\star$  Reading statements). For each of `dualScale_inv`, `dualScale_ge` and `eta_one_reindex`, write in one English sentence what is quantified, what is assumed, and



what is concluded. Then say which of the three would still be true, and which would become false or meaningless, if its hypothesis were deleted.

**Exercise 34.2** (★ The other generator). Repeat the paper computation (34.3) for  $g = \text{diag}(A, B)$  and show that  $g^\top \eta g = \eta$  is equivalent to  $A^\top B = \mathbf{1}$  and  $B^\top A = \mathbf{1}$ . Explain why the Lean proof of `basisChange_isODD` needs the auxiliary `have h' : B^\top * A = 1`, and how it obtains it from `h` by transposing.

**Exercise 34.3** (★ Goal states). Open `ODD.lean` in an editor with the Lean extension, place the cursor after the `rw` line of `thetaShift_isODD`, and copy the goal. Identify the four blocks and match them with the entries of (34.3). Then delete `hΘ` from the `rw` list and report the goal `simp` leaves open.

**Exercise 34.4** (★★ Lean: the converse). State and prove

```
theorem thetaShift_isODD_iff (Θ : Matrix (Fin d) (Fin d) ℤ) :
  IsODD (thetaShift Θ) ↔ Θ^\top = -Θ := by ...
```

Use `constructor`; the reverse direction is `thetaShift_isODD`. For the forward direction, unfold and rewrite as in Section 34.6 to reach an equation between two `fromBlocks` matrices, then extract the lower-right block (Mathlib has `toBlocks22` and the lemma `toBlocks_fromBlocks22` in the `Matrix` namespace) and solve for  $\Theta^\top$ .

**Exercise 34.5** (★★ Lean: a certificate). Define  $U : \text{Gram } 2 := !![0, 1; 1, 0]$  and prove `IsUnimodular U` by applying `isUnimodular_of_mul_eq_one` with  $H := U$ ; prove the side condition  $U * U = 1$  by `ext i j; fin_cases i <;> fin_cases j <;> rfl` (you may need `simp [Matrix.mul_apply, Fin.sum_univ_succ]` before `rfl`). Then prove `IsEvenDiag U`. Compare with `hyperbolicU` in `Lattice/Hyperbolic.lean`.

**Exercise 34.6** (★★ Lean: numbers). Prove `example : (2 : ℝ) ≤ 4 + (4 : ℝ)^{-1}` with `norm_num`, and `example : (2 : ℝ) ≤ dualScale (!![(2 : ℝ)^2] : Matrix (Fin 1) (Fin 1) ℝ)` by rewriting with `dualScale_circle` and then `norm_num`. Which hypothesis of `dualScale_circle` must you discharge, and how?

**Exercise 34.7** (★★ The sandwich). Verify (34.5) by hand for a general invertible  $G$ , and compute both sides for  $G = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ . Then show that  $\text{tr}[(G - \mathbf{1})G^{-1}(G - \mathbf{1})] = 0$  forces  $G = \mathbf{1}$  when  $G$  is positive definite, which is the uniqueness statement that `TraceBound.lean` does not contain.

**Exercise 34.8** (★★★ Lean: a faithful version). The circle theorems of Section 34.2 live over  $\mathbb{Q}$  with  $\alpha' = 1$ . Restate the invariance theorem `tduality_invariant_mass_squared` over  $\mathbb{R}$ , with  $\alpha'$  a positive real parameter: define  $p_L = n/R + wR/\alpha'$  and  $p_R = n/R - wR/\alpha'$  with  $n, w : \mathbb{Z}$  cast to  $\mathbb{R}$ , and prove that  $(n, w, R) \mapsto (w, n, \alpha'/R)$  fixes  $p_L^2 + p_R^2$ . You will need `hα : 0 < α'` and `hR : 0 < R` for `field_simp`. Then add the oscillator term  $(2/\alpha')(N + \tilde{N} - 2)$  as a further argument and explain why the proof does not change.

**Exercise 34.9** (★★★ Audit). Write a file that proves `example : (10 : ℕ)^3 = 1000` twice, once with `decide` and once with `native_decide`, name the two proofs, and run `#print axioms` on both. Explain from the output why the project's gate rejects the second and what would have to be trusted to accept it.

## Further reading

- The Lean sources used in this chapter: `Basic.lean`, `ODD.lean` and `TraceBound.lean` in the directories `Lattice/`, `TDuality/` and `DualScale/` of `DualScaleStream2/`, and `TDualityMassSpectrum.lean` in `StringTheoryFormalization/UseCases/`. Each header states its physical background, its mathematical content, what is not proved, and the proof techniques; read the header before the code.
- The physics of the  $\Theta$ -shift and the basis change, with the total-derivative argument: [GPR §2.4, ll. 1355–1373].
- The project’s gate: the kernel as the only accept criterion, the three standard axioms, and “producer  $\neq$  verifier” [STREAM2\_WORKFLOW.md §2, ll. 20–38]; measured results of the tiered proving pipeline [STREAM2\_WORKFLOW.md §6, ll. 105–124]. The lessons on statement locking, axiom auditing, heartbeats and the structure-field parser pitfall [LL.md §S2.4–S2.13, ll. 47–112] and [LL.md §0.4, ll. 188–199]. The architecture of the libraries is described in Chapter 35; the atlas of theorems in Chapter 36; the pipeline in Chapter 37.
- For Lean itself, the official documentation and Mathlib’s reference pages are the authoritative sources; the definitions of `Matrix`, `fromBlocks`, `PosDef` and the nonsingular inverse quoted above are from the pinned Mathlib tag `v4.33.1` in `lakefile.lean`. Standard; not checked against a source in this book’s library.



## Chapter 35

# Architecture of the Formalization

Chapter 34 taught the student to read a single Lean statement. This chapter answers the next question: *where do the statements quoted in this book live, how are they organized, how much of the physics do they really capture, and what has to happen before a theorem is allowed to carry the badge [A]?* The answer is an inventory and a procedure. The inventory is a single Lake project containing seven libraries, five of them written without Mathlib and two on top of it, whose compiled theorems number  $326 + 99 = 425$  at the audit of 2026-09-17. The procedure is a chain of gates—build, `sorry` count, axiom audit, statement lock, and the rule that whoever produced a proof never grades it—each of which exists because of a specific failure the project recorded. A student should care for two reasons. First, the honest reading of any theorem in this book depends on knowing which library it comes from: the older libraries model physical quantities by integers and prove arithmetic consequences, while the newest one proves statements about real matrices of every size. Second, the gates are the answer to the question every physicist asks of a proof assistant—“if the machine checked it, what exactly can still go wrong?”—and the answer is more interesting than “nothing”.

### 35.1 One project, seven libraries

A Lean 4 project is described by a file `lakefile.lean`, read by the build tool Lake. The project’s file declares one package, pins one external dependency, and registers seven libraries [lakefile.lean, ll. 1–53]. Three facts from that file matter for everything that follows.

1. **The toolchain and Mathlib are pinned to the same tag.** The project uses Lean `v4.33.1` and requires Mathlib at the git tag `v4.33.1`, whose own `lean-toolchain` file names exactly that compiler [lakefile.lean, ll. 12–17]. The comment in the file records why this is not pedantry: an earlier document had pinned a commit belonging to the `v4.33.0` tag, which would have produced a cache of compiled Mathlib files unusable by the project’s compiler. A formalization is reproducible only if the reader can rebuild it, and rebuilding Mathlib from source takes hours; the pin is what makes the pre-built cache usable.
2. **Two compiler limits are raised.** The package sets `maxHeartbeats` to  $10^6$  (the default is  $2 \times 10^5$ ) and `maxRecDepth` to 8000 [lakefile.lean, ll. 6–10]. Heartbeats are Lean’s unit of elaboration work; a proof that exceeds the budget is rejected with a timeout, not accepted. Section 35.5 returns to a consequence of this setting that cost the project several hours.
3. **Five libraries are default targets; two are not.** A plain `lake build` compiles the five libraries in the upper rows of Table 35.1, none of which imports Mathlib [lakefile.lean, ll. 19–39]. The Mathlib-based libraries `StringTheoryFormalization` and `DualScaleStream2` must be named explicitly, as in `lake build DualScaleStream2`. This is by design: the

file says the second was kept off the default list “so a Stream 2 failure never turns the green Stream 1 core red” [lakefile.lean, ll. 41–53].

Table 35.1 lists the seven libraries with the numbers that the project’s own audit reports. The theorem counts are those of `tools/axiom_audit.py` run on 2026-09-17, quoted from the revision brief [PAPER\_REVISION\_BRIEF §1, ll. 25–35]; the module counts are the number of `.lean` files imported by each library’s root file, counted by the author of this chapter on the same checkout.

Table 35.1: The seven libraries. “Mathlib” says whether any module imports Mathlib. “Imports” lists the other project libraries the library depends on. Theorem counts are the audited theorem/lemma declarations [PAPER\_REVISION\_BRIEF §1, ll. 25–35].

| Library                                | Modules | Theorems | Mathlib | Imports (project libraries)                                                |
|----------------------------------------|---------|----------|---------|----------------------------------------------------------------------------|
| <code>StringTheoryFoundation</code>    | 16      | 56       | no      | —                                                                          |
| <code>DualScaleM24Formalization</code> | 11      | 61       | no      | —                                                                          |
| <code>DoubleFieldTheory</code>         | 7       | 44       | no      | —                                                                          |
| <code>DualScaleValidation</code>       | 4       | 23       | no      | <code>DoubleFieldTheory</code> ,<br><code>DualScaleM24Formalization</code> |
| <code>Lean5Corpus</code>               | 12      | 53       | no      | the three above and<br><code>StringTheoryFoundation</code>                 |
| <code>StringTheoryFormalization</code> | 35      | 89       | yes     | <code>StringTheoryFoundation</code>                                        |
| <code>DualScaleStream2</code>          | 19      | 99       | yes     | <code>StringTheoryFormalization</code>                                     |
| Stream 1 (first six)                   | 85      | 326      |         | 3353 build jobs                                                            |
| Stream 2 (last)                        | 19      | 99       |         | 3670 build jobs                                                            |

The words *Stream 1* and *Stream 2* are the project’s names for the two generations of the corpus: Stream 1 is everything that existed before 2026-09-16, re-gated in the same session; Stream 2 is the library `DualScaleStream2` written during the tiered proving run of 2026-09-16/17 described in Chapter 37 [STREAM2\_WORKFLOW.md §1, ll. 7–18]. The job counts are what Lake reports when it compiles a target together with everything it depends on; the 3670 jobs of Stream 2 include the parts of Mathlib it imports, which is why a nineteen-module library needs more jobs than the eighty-five modules of Stream 1.

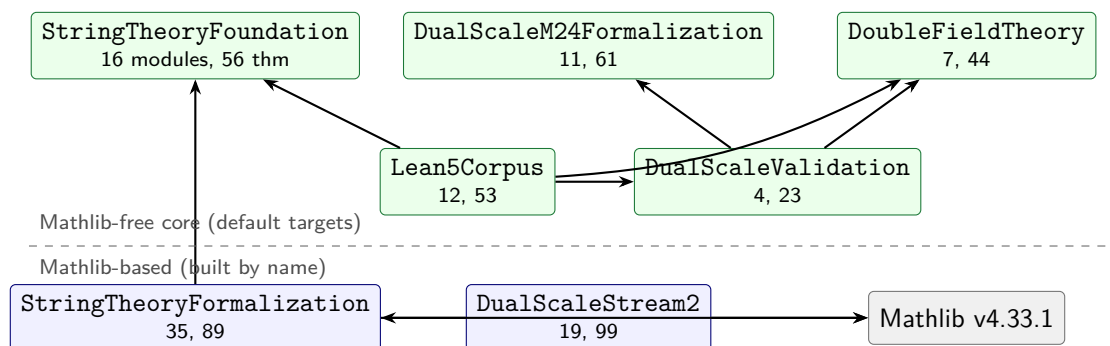


Figure 35.1: Import structure of the seven libraries. An arrow  $A \rightarrow B$  means some module of  $A$  imports a module of  $B$ . Above the dashed line nothing imports Mathlib; the two libraries below it do. `DualScaleStream2` imports five modules of `StringTheoryFormalization` and nothing else in the project; no library imports `DualScaleStream2`.

Figure 35.1 shows the import graph, obtained by grepping every `import` line in the checkout. Two features deserve comment. The three libraries in the top row import nothing but

themselves: each is self-contained, which is why they could be written before Mathlib was available to the project at all. The Mathlib-based libraries form a chain: `DualScaleStream2` imports exactly five modules of `StringTheoryFormalization` (the circle mass spectrum, the Hodge numbers, the Narain lattice, the K3 signature theorem and the  $M_{24}$  table), which in turn imports three modules of `StringTheoryFoundation`. The workflow document states the design rule behind the chain: “Stream 2 lives in `DualScaleStream2`, imports Stream 1, never the reverse; Stream 1’s green core cannot be turned red by Stream 2 work” [STREAM2\_WORKFLOW.md §2, ll. 32–33]. The rule is enforced by the import direction itself: a broken file in `DualScaleStream2` cannot stop any other library from compiling, because nothing depends on it.

#### Why cross-library imports are the interesting ones

An import from `DualScaleStream2` into Stream 1 is not bookkeeping; it is a theorem that two independently written pieces of mathematics agree. The module `DFT/GeneralizedMetric.lean` imports Stream 1’s `TDualityMassSpectrum` in order to prove that the doubled mass form  $Z^T \mathcal{H} Z$  on a circle equals half of Stream 1’s  $P_L^2 + P_R^2$  (Chapter 15); `Lattice/Hyperbolic.lean` imports `NarainLattice` to prove that Stream 2’s hyperbolic plane  $U$  is *equal* to Stream 1’s Narain Gram matrix; `Lattice/K3T2Signature.lean` imports `K3SignatureTheorem` to prove that the lattice signature  $(3, 19)$  agrees with the Hodge-theoretic  $(2h^{2,0} + 1, h^{1,1} - 1)$  [PAPER\_REVISION\_BRIEF §3.1, ll. 78–84]. Each such theorem is a consistency check between two formalizations of the same physics, checked by the kernel. A physicist would call them cross-calibrations.

## 35.2 What each library contains

The following inventory names each library’s modules and says, in one or two sentences, what kind of mathematics each contains. It is deliberately flat: the depth of the statements, which is what determines how a theorem may be quoted, is the subject of Section 35.3.

**StringTheoryFoundation (16 modules, 56 theorems).** The oldest library. Its modules are grouped by physical topic—`Core/Topology` (Euler characteristics of  $K3$  and  $T^2$  from Betti-number records), `Duality/T_Duality` (the exchange  $(n, w) \mapsto (w, n)$  of integer quantum numbers as an involution), `Duality/DualScale` (dual length pairs as integer relations), `K3/K3Surfaces` (Betti numbers), `StringTheory/K3xT2`, `StromingerSYZ`, `Swampland`, `VafaSwampland`, `TadpoleCancellation`, `WittenDuality`—together with four modules named after external formalizations (Navier–Stokes, Fermat and modular forms, statistical learning, ATLAS geometry; listed and described in Section 35.4) and two physics-library bridges (`PhysLibKinematicsBridge`, `TensorNetworkBridge`).

**DualScaleM24Formalization (11 modules, 61 theorems).** The library of the programme’s own claims: `DualScale/EffectiveMetric` (the effective scale  $R + \alpha'/R$  and its lower bound, Chapter 31), `DualScale/Sym2Lock`, `Moonshine/MathieuRigidity` (the ratio  $77/60$  and the identity  $462 \cdot 60 = 360 \cdot 77 = 27720$ , Chapter 30), `Moonshine/KummerTadpole`, `Moonshine/GysinBEMSequence` the four `FrontierTriad` modules (Chapter 33), and the two `Epistemic` modules, `Tier` and `Ledger`, which formalize the tier system itself and are the subject of Section 35.6.

**DoubleFieldTheory (7 modules, 44 theorems).** Integer-valued  $2 \times 2$  matrix models of the DFT objects of Chapters 14 to 17: `GeneralizedGeometry` (a structure `Mat2` with

four integer entries, the pairing  $\eta$ , a  $B$ -twist, a generalized metric), `CourantAlgebroid`, `ActionCurvature`, `TDualityBuscher`, `K3Topology`, `TorusMoonshine` and a `PhysicsDSL`.

**DualScaleValidation (4 modules, 23 theorems).** Three “use case” modules and an `Observables` module of exact rational identities and window inequalities. The module header of `Observables` is unusually explicit about its own scope: “Tier A for its arithmetic only ... not a derivation of any physical observable from first principles”, with the numerical relations labelled Tier C [`Observables.lean`, ll. 9–24]. Section 35.6 returns to the lesson in that header.

**Lean5Corpus (12 modules, 53 theorems).** A `Foundations/Narrative` module and eleven `Problems/Problem $k$`  modules, one per named problem (Navier–Stokes helicity, Mathieu–Frobenius rigidity, dual-scale TCC, Mukai monodromy, Kolmogorov cascade, flux swampland, Courant torsion, Golay holography, Kummer modularity, SYM instanton, entanglement entropy). It imports the three libraries above and `StringTheoryFoundation` and is the only library that ties the Mathlib-free core together.

**StringTheoryFormalization (35 modules, 89 theorems).** The first Mathlib-based library. Its `Frontier` modules (central charge, chiral primaries, Hodge numbers, moduli geodesics,  $SL(2, \mathbb{C})$  symmetry, F-term potential), its `StringDynamics` modules (Kummer blow-up, tadpole constraint, the  $M_{24}$  table, BPS multiplicities, Mukai lattice, T-duality Gysin, vertex operators, and several numerical or search-related modules), its five `UseCases` (critical dimension, K3 signature, Mathieu tower, Narain lattice, T-duality mass spectrum) and its `NSMath` modules (fractional Sobolev spaces, Fourier multipliers, mild PDEs, energy bounds). Several of its theorems are quoted in earlier chapters, for instance the circle mass spectrum of Chapter 8 and the Narain Gram matrix of Chapter 10. Its header history is instructive: the library “had 8 real build failures despite being self-documented as 26–28/30 modules, complete” when the checkout was rebuilt on 2026-09-16 [`LL.md` §0, ll. 148–155].

**DualScaleStream2 (19 modules, 99 theorems).** The library written by the tiered pipeline, in six groups that mirror the workflow’s phases [`STREAM2_WORKFLOW.md` §5, ll. 67–76]: `Lattice` (`Basic`, `E8`, `E8PosDef`, `Hyperbolic`, `K3T2Signature`, `Mukai`, `Reflection`; Chapters 18, 20 and 22), `TDuality` (`ODD`, `Factorized`, `Spectrum`, `Mirror`, `SL2Product`; Chapters 11 and 12), `DFT` (`GeneralizedMetric`, `BShift`, `SectionCondition`; Chapters 14 and 15), `DualScale/TraceBound` (Chapter 31), `Flux` (`Tadpole`, `Integrality`; Chapter 25) and `Moonshine/EOT` (Chapter 30). Every module header states the physical background, the pinned literature sources with line numbers, what is not proved, and the proof techniques used.

### 35.3 Depth: arithmetic shadows and faithful linear algebra

Two libraries in the project both contain a theorem that “the generalized metric satisfies  $\mathcal{H}\eta\mathcal{H} = \eta$ ”. Read side by side, they show exactly what the phrase *arithmetic shadow* means in this book, and why the library a theorem comes from changes how it may be quoted.

In `DoubleFieldTheory/GeneralizedGeometry.lean` the generalized metric is a  $2 \times 2$  matrix with integer entries, built from an integer  $g$  and an integer standing for  $g^{-1}$  [`DoubleFieldTheory/GeneralizedGeometry.lean`, ll. 189–209]:

```
def GenMetric (g : Int) (ginv : Int) : Mat2 :=
  { a := g, b := 0, c := 0, d := ginv }
theorem gen_metric_symmetric (g ginv : Int) :
  (GenMetric g ginv).b = (GenMetric g ginv).c := by rfl
```

```
theorem gen_metric_duality (g : Int) (h : g = 1) :
  MatMul (GenMetric g g) (MatMul ODD_Eta (GenMetric g g)) = ODD_Eta := by ...
```

Read literally: `GenMetric g ginv` is the diagonal matrix  $\text{diag}(g, g^{-1})$  with the inverse supplied by hand, and the “duality” theorem says that  $\mathcal{H}\eta\mathcal{H} = \eta$  holds for  $\mathcal{H} = \text{diag}(1, 1)$ , under the hypothesis  $g = 1$ . Both proofs are `rfl`: the statements are true by computation on literals. This is the  $d = 1$ ,  $B = 0$ , self-dual-radius corner of the physics, and only that corner. It is a correct theorem, and it is Tier A, but it is a theorem about the integer 1.

In `DualScaleStream2/DFT/GeneralizedMetric.lean` the same object is a block matrix over  $\mathbb{R}$  of size  $2d \times 2d$  for every  $d$ , built from an arbitrary  $G$  and  $B$  [`DFT/GeneralizedMetric.lean`, ll. 91–92, 103–104]:

```
noncomputable def genMetric (G B : Matrix (Fin d) (Fin d) ℝ) :
  Matrix (Charge d) (Charge d) ℝ :=
  fromBlocks (G - B * G⁻¹ * B) (B * G⁻¹) (-(G⁻¹ * B)) G⁻¹
theorem etaR_genMetric_sq (G B : Matrix (Fin d) (Fin d) ℝ) (hG : IsUnit G.det) :
  (etaR d * genMetric G B) * (etaR d * genMetric G B) = 1 := by ...
```

Here  $G^{-1}$  is Mathlib’s matrix inverse, the hypothesis is that  $G$  is invertible,  $B$  is arbitrary, and the conclusion  $(\eta\mathcal{H})^2 = 1$  is equivalent to  $\mathcal{H}\eta\mathcal{H} = \eta$  because  $\eta^2 = 1$ . This is the statement a physicist means (Chapter 15): it covers every torus dimension, every metric and every  $B$ -field. The proof is genuine block-matrix algebra with inverses, and it needed the most capable model tier in the pipeline [`STREAM2_WORKFLOW.md` §6, ll. 136–143].

**Definition 35.1** (Arithmetic shadow). A Lean theorem is an *arithmetic shadow* of a physical statement when its objects are integers, rationals or fixed small matrices standing in for the physical quantities, so that the theorem is a numerical instance or a consistency check of the physical statement rather than the statement itself. Five of the seven libraries—`DoubleFieldTheory`, `StringTheoryFoundation`, `DualScaleM24Formalization`, `Lean5Corpus`, `DualScaleValidation`—consist mostly of shadows in this sense; this is how the book’s bible instructs authors to describe them.

A shadow is not worthless. The identity  $\chi(K3) = 1 - 0 + 22 - 0 + 1 = 24$  is a shadow of the Betti-number computation, and it is exactly the arithmetic a student does on paper. The Burnside check  $\sum_i \dim(\rho_i)^2 = |M_{24}| = 244\,823\,040$  over the 26 irreducible dimensions is a shadow, and it caught a wrong table that had been in the repository for months (Section 35.5). What a shadow cannot do is stand in for the general theorem. To turn `gen_metric_duality` into `etaR_genMetric_sq` one needs, in order: a type of  $d \times d$  real matrices with a genuine inverse (in Mathlib, `Matrix.nonsing_inv_mul`:  $A^{-1}A = 1$  when  $\det A$  is a unit), block matrices (`Matrix.fromBlocks`), the invertibility hypothesis in place of  $g = 1$ , and a proof that is a computation with the algebraic laws of inverses rather than a computation on literals. That is precisely the gap Mathlib fills, and it is why the two Mathlib-based libraries are where the faithful statements live.

#### A theorem about a fixed matrix is not a theorem about all lattices

The temptation runs the other way too. `DualScaleStream2/Lattice/E8.lean` proves that one explicit  $8 \times 8$  integer matrix is symmetric, even and unimodular; it does not prove that “the  $E_8$  lattice” has these properties, because no lattice, no basis change and no isomorphism appears in the file (Chapter 18). Similarly the signature arithmetic in `K3T2Signature` adds up signatures of named summands; that  $H^2(K3, \mathbb{Z})$  is  $E_8(-1)^{\oplus 2} \oplus U^{\oplus 3}$  is Tier L, pinned to Huybrechts [`PAPER_REVISION_BRIEF` §3.1, ll. 81–87]. The rule of thumb: the tier badge attaches to the Lean statement as written, and the physical name in the docstring is a promise the kernel never checked.

**Example 35.2** (Reading the depth of a statement). A student wants to cite “the  $E_8$  Cartan matrix is positive definite” with a Tier A badge. She finds three candidate statements. (i) `cartanE8_det` in `DualScaleStream2/Lattice/E8PosDef.lean`, which says that the integer determinant of the explicit matrix is 1. (ii) `cartanE8_posDef` in the same file, whose statement is Mathlib’s `Matrix.PosDef` predicate applied to the explicit matrix over  $\mathbb{R}$ , proved through the sum-of-squares identity  $x^T E_8 x = \sum_k D_k y_k^2$  with pivots  $2, \frac{3}{2}, \frac{4}{3}, \dots, \frac{1}{8}$  [PAPER\_REVISION\_BRIEF §3.1, ll. 75–77]. (iii) A sympy script that computes the eigenvalues. Only (ii) is the Tier A statement she wants; (i) is a shadow of  $\det E_8 = 1$ , which does not imply positive definiteness; (iii) is a symbolic check and by the book’s rules is never Tier A. The product of the pivots is  $2 \cdot \frac{3}{2} \cdot \frac{4}{3} \cdot \frac{5}{4} \cdot \frac{6}{5} \cdot \frac{7}{6} \cdot \frac{8}{7} \cdot \frac{1}{8} = 8 \cdot \frac{1}{8} = 1$ , which is the determinant—a two-line check the student can do herself before trusting the file.

## 35.4 The peripheral bridge modules

Four modules of `StringTheoryFoundation` carry the names of external formalization projects:

- `FluidDynamics/NavierStokesBridge` (a Navier–Stokes formalization),
- `ModularForms/FermatModularBridge` (a Fermat–modularity formalization),
- `StatisticalLearning/StatisticalLearningBridge` (a statistical-learning library),
- `Atlas/AtlasGeometryBridge` (an autoformalized textbook library).

The projects they name are vendored, read-only, as git submodules under `lean4basesource/`. This section states what the four modules contain, because their names promise more than their code delivers and the project’s own review says so in each file.

Every one of the four carries a “scope note” inserted after review. The note in the Navier–Stokes module is representative: the file “does **not** import anything from that project—check the `import` lines above; there is only `StringTheoryFoundation.Core.Topology`, this project’s own file. The theorems below are self-contained `Nat/Int` arithmetic named after, and inspired by, the cited external work, not a machine-checked bridge to it” [`NavierStokesBridge.lean`, ll. 40–47]. The Fermat and statistical-learning modules carry the same sentence [`FermatModularBridge.lean`, ll. 42–49; `StatisticalLearningBridge.lean`, ll. 40–44], and the Atlas module notes that it “has **no import** statement at all” [`AtlasGeometryBridge.lean`, ll. 15–21]. The workflow document records the technical reason no genuine bridge is possible at present: the external corpora are pinned to other Lean versions (`atlas-lean` at v4.29.0, the Navier–Stokes project at v4.34.0-rc2) and “cannot be **required** at this project’s v4.33.1 pin” [`STREAM2_WORKFLOW.md` §2, ll. 34–37]. Table 35.2 lists what each module actually proves.

Three of these facts are genuine, if elementary, shadows of physics the book has met:  $\chi(K3) = 24$  (Chapter 19), the signature  $-16$  of the intersection form and  $(4, 20)$  of the Mukai lattice (Chapters 20 and 22), and the 16 fixed points of the Kummer involution (Chapter 21). The others are definitional restatements, and the files say so: the comment above `exceptional_divisor_cartan_a1` reads “This is a definitional restatement, not a derivation; recorded for cross-reference from other modules rather than as an independent check” [`FermatModularBridge.lean`, ll. 72–76], and a theorem that had checked  $24 = 24$  was removed [`FermatModularBridge.lean`, ll. 99–103]. The Navier–Stokes energy theorem once took  $e_2 \leq e_1$  as a hypothesis and returned it as the conclusion; it now derives  $e - d \leq e$  from `Nat.sub_1e` [`NavierStokesBridge.lean`, ll. 87–97].

The honest summary is the one the project’s own review reached. These modules are *documents of intent*: each header describes a real piece of mathematics (the fluid–gravity correspondence, the modularity theorem, PAC generalization bounds, autoformalized four-manifold topology) and a real external formalization, and each body proves a handful of



Table 35.2: The four bridge modules and their theorems, read literally from the source. All are Tier A as arithmetic; none imports the external project it is named after.

| Module                                   | Theorems                                                                                                                   | What is literally proved                                                                                                                                             |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NavierStokesBridge<br>[ll. 53–111]       | zero_mode_laplacian,<br>fundamental_mode_positive,<br>energy_dissipation_monotonic,<br>sound_speed_is_luminal              | $0^2 + 0^2 = 0$ and $1^2 + 0^2 = 1$ for an integer wave-vector; $e - d \leq e$ for natural numbers $e, d$ (truncated subtraction); $1/(2 - 1) = 1$ in $\mathbb{N}$ . |
| FermatModularBridge<br>[ll. 60–113]      | kummer_fixed_points_dim4,<br>exceptional_divisor_cartan_a1,<br>mukai_lattice_rank_equals_24,<br>mukai_signature_difference | $2^4 = 16$ ; the constant $-2$ equals $-2$ ; $1 + 22 + 1 = 24$ ; $4 - 20 = -16$ .                                                                                    |
| StatisticalLearningBridge<br>[ll. 46–81] | zero_empirical_risk_<br>on_perfect_proofs,<br>rademacher_bound_positive                                                    | $(n - n) \cdot 10^4 / n = 0$ for $n > 0$ ; $2k > 0$ for $k > 0$ .                                                                                                    |
| AtlasGeometryBridge<br>[ll. 23–78]       | atlas_k3_euler_characteristic,<br>atlas_k3_signature,<br>atlas_poincare_curvature_negative                                 | $1 - 0 + 22 - 0 + 1 = 24$ ; $3 - 19 = -16$ ; the constant $-1$ equals $-1$ .                                                                                         |

integer identities that would survive in any faithful version. They are the smallest shadows in the corpus, they are counted among the 56 audited theorems of `StringTheoryFoundation`, and no chapter of this book cites them as evidence for the physics in their headers. A faithful bridge would begin by importing the external declarations that the Navier–Stokes file’s own “verified pointer” names [NavierStokesBridge.lean, ll. 48–51], which the pinned toolchains currently forbid.

## 35.5 The gates

A theorem enters this book with the badge **[A]** only after passing a fixed sequence of checks. The workflow document states the accept criterion in one sentence [STREAM2\_WORKFLOW.md §2, ll. 22–24]:

“A goal is closed when `lake build DualScaleStream2` is green, the zero-sorry audit passes, and `#print axioms` shows only `propext`, `Classical.choice`, `Quot.sound`. No `native_decide`.”

The per-package pipeline names the gates G1–G4 [STREAM2\_WORKFLOW.md §4, ll. 52–65]; Table 35.3 collects them with the failure each one was built to catch. We take them in order.

### 35.5.1 Build and sorry count

The build is the gate everyone expects, and the project’s log shows why it must be re-run rather than believed. Lesson 0.2 of the log records that a library documented as complete had eight real failures once “the full dependency chain was exercised”, and draws the general rule: “after any claim of ‘X builds clean’ from a document, memory note, or your own earlier turn . . . re-run the actual build before trusting it” [LL.md §0.2, ll. 168–177]. Lesson 0.3 adds a subtlety a student of any compiled language will recognize: fixing one file can *unblock* elaboration of files that depend on it and so reveal failures that were always present but never reached, which is why the whole target must be rebuilt after every fix [LL.md §0.3, ll. 179–186].

The keyword `sorry` closes any goal without a proof and makes Lean emit a warning; a library can build green with hundreds of them. The count is taken by a regular expression

Table 35.3: The gates, the tool that implements each, and the recorded failure that motivated it. Line references are to `STREAM2_WORKFLOW.md` (W) and `LL.md` (L).

| Gate                     | Tool                                                           | Catches ...                                                                                                               |
|--------------------------|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Build (G1)               | <code>lake build &lt;lib&gt;</code>                            | files that do not elaborate; a self-reported “26–28/30 modules” that was 8 real failures [L §0.2, ll. 168–177]            |
| <code>sorry</code> count | regex after stripping comments                                 | goals left open; a report claiming “5 clean <code>sorry</code> s” that had 8 compile errors [W §6, ll. 153–157]           |
| Statement review (G2)    | a human or orchestrator reads each statement                   | mis-formalization, “the one step no kernel can check” [W §4, l. 59]                                                       |
| Axiom audit (G3)         | <code>tools/axiom_audit.py</code>                              | <code>sorryAx</code> inherited through dependencies; <code>native_decide</code> in a “clean” library [L §S2.5, ll. 56–61] |
| Statement lock           | <code>tools/statement_lock.py</code>                           | a statement weakened between review and proof [L §S2.10, ll. 87–92]                                                       |
| Producer $\neq$ verifier | the gate runs separately from the prover                       | agents misstating their own compile status [L §S2.3, ll. 39–45]                                                           |
| Record (G4)              | declaration index, graph, ledger entry pinned to a commit hash | counts that drift from the code [W §4, l. 64]                                                                             |

after stripping comments, so that a docstring mentioning the word does not count [tools/axiom\_audit.py, ll. 30–34]. A `sorry` is not a defect during development—the pipeline of Chapter 37 deliberately begins with statements whose proofs are `sorry`—but the count must be zero at the gate, and the audit of the next subsection makes the count redundant by catching `sorry` at the axiom level.

### 35.5.2 The axiom audit

The decisive check is `#print axioms`. Every Lean theorem, when printed this way, lists the axioms its proof term ultimately depends on. Three are standard and harmless (Chapter 34): `propext`, `Classical.choice`, `Quot.sound`. Two others signal trouble. A theorem proved with `sorry` depends on the axiom `sorryAx`, and so does every theorem that uses it, however indirectly: the axiom propagates through the dependency graph, which a `sorry` grep cannot follow. A theorem proved by `native_decide` depends on `Lean.ofReduceBool`, an axiom asserting that the compiled evaluator agrees with the kernel; the kernel did not check the computation, the compiler did.

The script `tools/axiom_audit.py` automates this over a whole library [tools/axiom\_audit.py, ll. 1–102]. Its logic is short enough to describe fully.

1. Collect every `theorem/lemma` name from the library’s source files, tracking `namespace/end` blocks to build the fully qualified name, after stripping comments (ll. 30–49).
2. Write a probe file that imports the built library and contains one `#print axioms` line per name; run it with `lake env lean` (ll. 62–67).
3. If the probe itself fails to elaborate—typically because the library was not rebuilt—exit with code 2 and the message `AUDIT ERROR` (ll. 70–77). This clause was added after an audit of a stale build reported “89 theorems audited, 89 failing (MISSING)”, which “looks like a proof result, but it was an infrastructure failure” [LL.md §S2.11, ll. 94–99].
4. Parse each theorem’s axiom list, subtract the standard set `{propext, Classical.choice, Quot.sound}`, and fail the gate if anything remains (ll. 79–98).



The audit is validated the way an experimentalist validates a detector: with a positive control (a library known to be clean) and a negative control (a library with known `sorry`s) before its verdicts are trusted [LL.md §S2.5, ll. 59–61].

**Example 35.3** (Running the audit by hand on the epistemic module). The author of this chapter ran a five-line probe against the compiled library on the checkout of 2026-09-17:

```
import DualScaleM24Formalization.Epistemic.Ledger
#print axioms SocrateAI.Epistemic.tier_trans
#print axioms SocrateAI.Epistemic.tier_le_of_depends
#print axioms SocrateAI.Epistemic.no_kernel_claim_rests_on_weaker
#print axioms SocrateAI.Epistemic.witnessSound_sound
#print axioms SocrateAI.Epistemic.witnessUnsound_not_sound
```

The output was `[propext, Quot.sound]` for the first three and `[propext]` for the two witnesses. All five pass: the sets are subsets of the standard three, and none is `sorryAx` or `Lean.ofReduceBool`. The student should notice two things. First, `Classical.choice` is absent: these proofs (by `omega`, induction and `decide`) never invoke the axiom of choice, and the audit records that fact for free. Second, the count of axioms is not a measure of strength—the witnesses use fewer axioms than the general theorem only because `decide` evaluates a finite case by computation.

### 35.5.3 What the audit found in a “verified” library

The reason the project trusts this gate is that it found real defects in code that the build and the `sorry` grep had passed for months.

#### Two defects caught by the audit and by a guard theorem

Both are recorded in the workflow document and the revision brief. (1) `StringDynamics.bps_ratio_reduced`, the theorem that  $77/60$  is in lowest terms, “had relied on `native_decide` (trusts compiled code, not the kernel); it is now a kernel proof” by `norm_num` [PAPER\_REVISION\_BRIEF §2, ll. 44–45; STREAM2\_WORKFLOW.md §6, ll. 127–128]. (2) The table `StringDynamics.M24RepDim` of the 26 irreducible dimensions of  $M_{24}$  “was wrong: 10395 and 483 duplicated, second 990 and third 1035 missing; squares summed to  $351,061,029 \neq |M_{24}|$ ”, because its only theorem checked a single entry. It was replaced by the table of Eguchi–Ooguri–Tachikawa and guarded by `M24RepDim_sum_sq`, whose statement is  $\sum_{i < 26} \text{M24RepDim}(i)^2 = 244\,823\,040$  [STREAM2\_WORKFLOW.md §6, ll. 129–132; MathieuM24.lean, ll. 118–121]. The lesson the log draws is general: “guard every data table with a theorem that would break if the table were wrong” [LL.md §S2.6, ll. 63–67].

The second defect deserves a moment’s arithmetic, because it shows what a guard theorem buys. The order of  $M_{24}$  is  $2^{10} \cdot 3^3 \cdot 5 \cdot 7 \cdot 11 \cdot 23 = 244\,823\,040$  (Chapter 29). Burnside’s identity  $\sum_i \dim(\rho_i)^2 = |G|$  holds for every finite group (Tier L, standard; not checked against a source in this book’s library). The wrong table summed to 351 061 029, a discrepancy of 106 237 989, so a single theorem comparing the sum to the group order rejects it. The kernel evaluates the sum by unfolding the 26 cases (`simp [Fin.sum_univ_succ, M24RepDim]`), which is a shadow computation in the sense of Definition 35.1—but a shadow attached to the right invariant is a strong test, exactly as a checksum is.

### 35.5.4 The statement lock

The kernel checks that a proof proves its statement. It cannot check that the statement is still the one that was reviewed. In a workflow where proofs are searched for by automated

agents this is the central vulnerability: “An agent (or a person) can make a goal ‘provable’ by weakening it” [tools/statement\_lock.py, ll. 5–7]. The tool that closes the gap hashes statements [tools/statement\_lock.py, ll. 1–137]:

- for a **theorem** or **lemma**, the text from the keyword up to the top-level `:=`, tracked with a bracket-depth counter so that a `:=` inside a binder does not end the statement early (ll. 69–83); replacing `:=` by `sorry` with a proof therefore leaves the hash unchanged;
- for a **def**, **abbrev** or **structure**, the whole declaration including its body, “changing a definition changes the meaning of every statement that uses it” (ll. 9–13);
- after removing comments with a small scanner that handles Lean’s nested block comments, and normalizing whitespace, so documentation can change freely (ll. 46–66).

The first 16 hex digits of the SHA-256 of the normalized text are stored per file and per declaration in `docs/statement_lock.json`; `-update` is run after statement review, `-check` before any proof is accepted, and a changed or vanished locked declaration fails the check while a new one is only reported (ll. 17–22, 117–133). On the checkout used for this chapter the lock covers 54 files and 355 declarations, all in the two Mathlib-based libraries; the Mathlib-free core is not locked.

**Example 35.4** (Hashing a statement by hand). Applying the tool’s own `strip_comments` and `statement_text` functions to `Epistemic/Tier.lean` gives, for the transitivity theorem, the normalized text

```
| theorem tier_trans {t1 t2 t3 : Tier} (h12 : t1 ≤ t2) (h23 : t2 ≤ t3) : t1 ≤ t3
```

whose SHA-256 begins `94394facb0c97f29`. Everything after `:=`—the two `have` lines and `omega`—is absent from the hash. If an agent replaced the hypothesis `(h23 : t2 ≤ t3)` by `(h23 : t1 ≤ t3)`, making the theorem trivially provable, the normalized text would change and so would the hash; the check would print `CHANGED` and exit 1. If instead the agent edited `Tier.toNat` so that `Tier.A` mapped to 3 and `Tier.B` to 4, no theorem statement would change, but the definition’s hash (`d839f1061e526f25` on the current text) would, and the check would fail for the same reason: every theorem about `≤` on tiers now means something else. This is why definitions are hashed whole and theorems only up to `:=`.

### 35.5.5 Producer $\neq$ verifier

The last gate is a rule about people and programs rather than a script: “No agent or model grades its own output; the gate runs separately” [STREAM2\_WORKFLOW.md §2, ll. 25–26]. The measured justification is in the pilot results. Two reports from the cheapest model tier “misstated compile status: one claimed 5 clean `sorry`s but left 8 compile errors; one claimed ‘no compilation errors’ with 3 errors. Both caught only because the orchestrator recompiles every report” [STREAM2\_WORKFLOW.md §6, ll. 153–157]; in the final round a report “again claimed ‘0 errors’ while one proof failed to parse” [STREAM2\_WORKFLOW.md §6, ll. 192–194]. None of these errors entered the repository, because the verifier’s compile, not the producer’s report, is the accept event.

The same principle applies to the project’s own tooling. Lesson 0.5 of the log records that an engine function `sync_epistemic_claims` “injects the same 3 hardcoded claims into `ledger.jsonl` any time the whole corpus builds with aggregate zero-sorry, regardless of whether those 3 specific claims are what was actually just proved”, and that a “prover” module was a static list with no proving logic [LL.md §0.5, ll. 201–216]. The workflow’s rule that ledger entries “pin a commit hash and build job count” is the direct response [STREAM2\_WORKFLOW.md §2, ll. 29–31]. A student reading any auto-generated claim table

should ask the question the log asks: which program wrote this row, and did that program run the kernel?

#### The gate that no kernel can run

Gate G2, statement review, is listed in the pipeline as “the one step no kernel can check: mis-formalization” [STREAM2\_WORKFLOW.md §4, l. 59]. Two recorded instances show its two failure modes. Prose drifting past the theorem: a brief said the  $\Theta$ -shift preserves  $\eta$  “iff  $\Theta$  antisymmetric” where the Lean proves only “if” [LL.md §S2.13, ll. 109–113]. And conventions unverified before the statement was written: every block-matrix sign convention was checked in sympy first, because “a sign slip caught this way costs minutes” and one caught after an agent has spent an hour on an unprovable statement “costs far more” [LL.md §S2.12, ll. 101–107]. The lock freezes the statement; it does not make it right. Only reading does.

### 35.5.6 A worked audit: the numbers

The two numbers that this book quotes for the size of the corpus are 326 and 99 audited theorems. They are the number of `theorem/lemma` declarations the audit’s collector finds after stripping comments, every one of which returned an axiom list within the standard three on a green build. (A naive `grep` for `^theorem` gives slightly different numbers, because it also counts declarations quoted inside comments; the author re-ran the audit’s own collector on the checkout and obtained the seven numbers below exactly.) Per library the brief reports [PAPER\_REVISION\_BRIEF §1, ll. 27–35]:

$$\underbrace{23}_{\text{Validation}} + \underbrace{53}_{\text{Lean5Corpus}} + \underbrace{44}_{\text{DFT}} + \underbrace{56}_{\text{Foundation}} + \underbrace{61}_{\text{M24Form.}} + \underbrace{89}_{\text{STFormalization}} = 326, \quad 326 + 99 = 425.$$

The reader can reproduce the sum and should. Three cautions attach to the number. It counts theorems, not physics: the four bridge modules of Section 35.4 contribute 13 of the 56 theorems of the oldest library. It depends on the build being fresh: the 89 for `StringTheoryFormalization` was reported as “89 failing (MISSING)” once, on a stale `.olean` [LL.md §S2.11, ll. 94–96]. And it is a count of *this plan*: the brief insists that Stream 2’s “18 of 18 planned items” is “100% of that plan, not of string theory” [PAPER\_REVISION\_BRIEF §1, ll. 33–35]. The same brief forbids the phrases “zero axioms”, “100% mathematical rigor” and “formalizes string theory” [PAPER\_REVISION\_BRIEF §0, ll. 15–19]. The correct form, which this book uses throughout, is: proved in Lean 4 with no axioms beyond the three standard ones.

## 35.6 The tier system, formalized

The badges **[A]**, **[L]** and **[C]** used throughout this book are three of five tiers that the project defines in Lean itself, in `DualScaleM24Formalization/Epistemic/Tier.lean` and `Ledger.lean`. The two files are short, Mathlib-free, and—unusually for the Mathlib-free core—not shadows: they state and prove a general theorem about an arbitrary index type. They are worth reading in full because they make precise a rule that is otherwise only a convention: *a claim may never be filed at a higher tier than the claims it rests on*.

### 35.6.1 Tiers as an inductive type

The tier is an inductive type with five constructors, ordered by a map to  $\mathbb{N}$  [Tier.lean, ll. 16–38]:

```

inductive Tier : Type where
  | X : Tier -- Exploratory / Float
  | C : Tier -- Conjecture
  | L : Tier -- Literature
  | B : Tier -- Exact-Arithmetic Checkable
  | A : Tier -- Kernel-Verified (Lean 4, zero sorry)
deriving DecidableEq, Repr

def Tier.toNat : Tier → Nat
  | Tier.X => 0 | Tier.C => 1 | Tier.L => 2 | Tier.B => 3 | Tier.A => 4

def Tier.le (t1 t2 : Tier) : Prop := t1.toNat ≤ t2.toNat

```

The module header glosses the five: A is kernel-verified; B is “deterministic  $\mathbb{Q}/\mathbb{Z}$  check + negative control that fails”; L is literature; C is conjecture; X is “floats, sampling, LLM output” [Tier.lean, ll. 5–11]. The book collapses B into symbolic checks and never awards it, and uses X nowhere; the ordering  $X < C < L < B < A$  is what matters. Six theorems establish that  $\leq$  is a partial order with top element A and bottom element X, and that nothing but A sits above A [Tier.lean, ll. 40–65]. Their proofs are one-liners: `omega` for the arithmetic ones, `cases t <;> decide` for the finite case analyses.

### 35.6.2 Ledgers, soundness and transitive support

A ledger is a function from an index type  $\iota$  (the claim identifiers) to rows, each row carrying a tier and the list of identifiers it cites [Ledger.lean, ll. 16–27]:

```

structure Claim ( $\iota$  : Type) where
  tier : Tier
  supports : List  $\iota$ 

abbrev Ledger ( $\iota$  : Type) :=  $\iota \rightarrow$  Claim  $\iota$ 

def Sound { $\iota$  : Type} (L : Ledger  $\iota$ ) : Prop :=
   $\forall a b, b \in (L a).supports \rightarrow (L a).tier \leq (L b).tier$ 

```

Soundness says: for every row  $a$  and every  $b$  it cites directly,  $\text{tier}(a) \leq \text{tier}(b)$ . The interesting content is that the same then holds along chains. Transitive dependence is an inductive predicate [Ledger.lean, ll. 29–43]:

```

inductive Depends { $\iota$  : Type} (L : Ledger  $\iota$ ) :  $\iota \rightarrow \iota \rightarrow$  Prop where
  | direct {a b} :  $b \in (L a).supports \rightarrow$  Depends L a b
  | step {a b c} :  $b \in (L a).supports \rightarrow$  Depends L b c  $\rightarrow$  Depends L a c

```

and the main theorem is proved by induction on it: in a sound ledger, if  $a$  depends on  $b$  through any chain then  $\text{tier}(a) \leq \text{tier}(b)$ . Two corollaries follow [Ledger.lean, ll. 45–68].

- `no_kernel_claim_rests_on_weaker`, the one that names the rule: if  $a$  is filed at A and depends on  $b$ , then  $b$  is at A—because  $A \leq \text{tier}(b)$  forces  $\text{tier}(b) = A$  by `eq_A_of_A_le`.
- `not_A_of_weak_support`, its contrapositive and the form one uses in practice: one weak link anywhere in the support chain, and the citing claim is not Tier A.

Let us do the induction on paper, since it is the whole proof. Suppose  $L$  is sound. Case **direct**:  $b \in \text{supports}(a)$ , so soundness gives  $\text{tier}(a) \leq \text{tier}(b)$  at once. Case **step**:  $b \in \text{supports}(a)$  and, by the induction hypothesis on `Depends L b c`,  $\text{tier}(b) \leq \text{tier}(c)$ ; soundness gives  $\text{tier}(a) \leq \text{tier}(b)$ , and `tier_trans` chains the two. That is the entire Lean proof [Ledger.lean, ll. 47–52]: the inductive predicate builds the chain  $a = x_0, \dots, x_k = b$  into the proof principle, so no list ever has to be handled.

### 35.6.3 Positive and negative controls

The file ends with two concrete ledgers [Ledger.lean, ll. 70–91]. The first has three rows over Fin 3: row 0 at A citing row 1, row 1 at A citing nothing, row 2 at C citing nothing; `witnessSound_sound` proves it sound by `decide`, which checks the finitely many pairs  $(a, b)$ . The second has two rows: row 0 at A citing row 1, row 1 at C; `witnessUnsound_not_sound` proves it is *not* sound—the instance  $a = 0, b = 1$  of the soundness condition would require  $A \leq C$ , which `decide` refutes. This is the “negative control that fails” of tier B applied to the tier system itself: a definition of soundness that accepted the second ledger would be vacuous, and the theorem shows it does not.

#### What the ledger theorem does and does not say

The theorem is about an abstract function  $\iota \rightarrow \text{Claim } \iota$ . The project also keeps a concrete file `ledger.jsonl` with, on the current checkout, 38 rows (34 filed at A, 3 at L, 1 at B), each naming a Lean source file and a list of dependencies, together with a Markdown rendering `LEDGER.md` whose header quotes the transitivity theorem. Two things a student must keep apart. First, no Lean theorem says that `ledger.jsonl` is a `Sound` ledger: the JSON is not a term of type `Ledger  $\iota$` , and the “dependencies” column is bookkeeping, not imports—eleven of the Stream 1 rows list the ledger theorem `K3T2-A-0001` itself as their dependency, while the source file behind one of them, `EffectiveMetric.lean`, has no `import` line at all. Second, the *statement text* of a row can promise more than the Lean it points to. The row `OBS-VAL-0001` is filed at A with the words “Certified Falsifiable Observables ... LiteBIRD  $r = 0.00396$ ”; the module it names proves that  $1/252$  lies in the window  $[0.003, 0.005]$  and similar inequalities between literals, and its own header says it is “Tier A for its arithmetic only” with the physical relations Tier C [Observables.lean, ll. 9–24]. The ledger theorem guarantees nothing about how a row is worded. The tier badge in this book therefore attaches to the Lean statement, never to a ledger row.

*Remark 35.5* (What a faithful ledger would need). To make the theorem bite on the real corpus one would generate the `Ledger` term from the declaration graph (Chapter 36): identifiers are declaration names, `supports` is the list of project declarations a proof term uses, and `tier` is read from a docstring tag. Soundness would then be decidable by `decide` exactly as for `witnessSound`, and a Tier A theorem citing a Tier L definition would fail to compile. Nothing in the present files does this; it is one of the improvements the log lists for the next run (“paper claims come from the Lean source”) [LL.md §S2-I, ll. 139–141].

## 35.7 What the machine has checked

This chapter’s Lean content is the epistemic module itself; the theorems of the other libraries are the subject of their own chapters. Every statement below was read from the source file and audited by the author of this chapter as in Example 35.3.

#### The tier order

```
theorem tier_refl (t : Tier) : t ≤ t := by ...
theorem tier_trans {t1 t2 t3 : Tier} (h12 : t1 ≤ t2) (h23 : t2 ≤ t3) :
  t1 ≤ t3 := by ...
theorem tier_antisymm {t1 t2 : Tier} (h12 : t1 ≤ t2) (h21 : t2 ≤ t1) :
  t1 = t2 := by ...
theorem tier_A_is_maximal (t : Tier) : t ≤ Tier.A := by ...
theorem tier_X_is_minimal (t : Tier) : Tier.X ≤ t := by ...
theorem eq_A_of_A_le {s : Tier} (h : Tier.A ≤ s) : s = Tier.A := by ...
```

**Reading.** The relation  $t_1 \leq t_2 :\Leftrightarrow \text{toNat}(t_1) \leq \text{toNat}(t_2)$  on the five-element type is reflexive, transitive and antisymmetric; A is the greatest element, X the least, and the only tier  $\geq A$  is A. **Proof idea.** Unfold to  $\mathbb{N}$  and call `omega`; for antisymmetry, derive equality of the images and split into the 25 constructor cases, each closed by `rfl` or `contradiction` [Tier.lean, ll. 40–65]. **Not proved.** Nothing about what the tiers mean; the type carries no semantics beyond the order. [A]

#### Transitive soundness of a ledger

```
theorem tier_le_of_depends {ι : Type} {L : Ledger ι} (hL : Sound L) :
  ∀ {a b : ι}, Depends L a b → (L a).tier ≤ (L b).tier := by ...
theorem no_kernel_claim_rests_on_weaker {ι : Type} {L : Ledger ι} (hL : Sound L)
  {a b : ι} (ha : (L a).tier = Tier.A) (hab : Depends L a b) :
  (L b).tier = Tier.A := by ...
theorem not_A_of_weak_support {ι : Type} {L : Ledger ι} (hL : Sound L)
  {a b : ι} (hab : Depends L a b) (hb : (L b).tier ≠ Tier.A) :
  (L a).tier ≠ Tier.A := ...
theorem witnessSound_sound : Sound witnessSound := by ...
theorem witnessUnsound_not_sound : ¬ Sound witnessUnsound := by ...
```

**Reading.** For every index type and every ledger satisfying the one-step condition `Sound`: tiers are monotone along every dependency chain; a Tier A row’s entire transitive support is Tier A; a row with any non-A support is not A. The two witnesses are a three-row sound ledger and a two-row unsound one. **Proof idea.** Induction on the inductive predicate `Depends`, using soundness at each step and `tier_trans`; the corollary rewrites with `ha` and applies `eq_A_of_A_le`; the witnesses are finite and closed by `decide` [Ledger.lean, ll. 45–91]. **Not proved.** That any actual file (`ledger.jsonl`, the declaration graph) is a `Sound` ledger; that a row’s wording matches its source. [A]

| Claim                                                                                        | Tier | Where                               |
|----------------------------------------------------------------------------------------------|------|-------------------------------------|
| Tier order: partial order, A maximal, X minimal                                              | [A]  | Tier.lean                           |
| Transitive tier monotonicity in a sound ledger; A rests only on A                            | [A]  | Ledger.lean                         |
| Positive and negative control ledgers                                                        | [A]  | Ledger.lean                         |
| Axiom footprint of the five theorems above: $\subseteq \{\text{propxt}, \text{Quot.sound}\}$ | [A]  | Example 35.3                        |
| $(\eta\mathcal{H})^2 = 1$ for all invertible real $G$ , all $B$ , all $d$                    | [A]  | DFT/GeneralizedMetric.lean          |
| Burnside guard $\sum \dim^2 =  M_{24} $ on the corrected table                               | [A]  | MathieuM24.lean                     |
| Burnside’s identity itself; $H^2(K3, \mathbb{Z}) \cong E_8(-1)^2 \oplus U^3$                 | [L]  | standard; Huybrechts, via the brief |
| <code>ledger.jsonl</code> is a sound ledger; any row’s physical wording                      | —    | not formalized; not a Lean claim    |
| The seven-library split and gate sequence are the right architecture                         | [C]  | this programme’s design choice      |

#### Chapter summary

- One Lake project, Lean and Mathlib pinned to v4.33.1, seven libraries: five Mathlib-free default targets (`StringTheoryFoundation`, `DualScaleM24Formalization`, `DoubleFieldTheory`, `DualScaleValidation`, `Lean5Corpus`) and two Mathlib-based libraries built by name (`StringTheoryFormalization`, `DualScaleStream2`). Imports run one way: Stream 2 imports Stream 1, never the reverse.



- Audited on 2026-09-17: 326 theorems in Stream 1 (23 + 53 + 44 + 56 + 61 + 89) and 99 in Stream 2, every one depending only on `propext`, `Classical.choice`, `Quot.sound`.
- Depth differs by library. The Mathlib-free core mostly proves arithmetic shadows (integer and rational instances, fixed small matrices); the Mathlib-based libraries prove the general statements (real  $d \times d$  matrices, every  $d$ ). Quote the statement as written, and name its library.
- Four bridge modules named after external formalizations import nothing from them and prove thirteen integer identities; each says so in a scope note.
- Gates: build, `sorry` count, statement review, axiom audit (`#print axioms`, catching `sorryAx` and `Lean.ofReduceBool`), statement lock (SHA-256 of the statement up to `:=`, of definitions whole), producer  $\neq$  verifier, and ledger rows pinned to a commit. Each exists because of a recorded failure.
- The tier system is itself formalized: in a `Sound` ledger a Tier A claim rests only on Tier A claims (`no_kernel_claim_rests_on_weaker`), a theorem about an abstract ledger type, not about any file.

## Exercises

**Exercise 35.1** (★). Reproduce the audit totals: add the six Stream 1 counts of Table 35.1 and check that they give 326, then add Stream 2. Which library contributes the most theorems per module, and which the fewest? (Use the module counts in the table.)

**Exercise 35.2** (★). From Figure 35.1, list every library that can be compiled without Mathlib installed. Explain why a compile error in `DualScaleStream2` cannot break `Lean5Corpus`, and why an error in `StringTheoryFoundation` can break four other libraries.

**Exercise 35.3** (★★). Explain precisely why `gen_metric_duality` (hypothesis  $g = 1$ , integer entries) does not imply `etaR_genMetric_sq` even for  $d = 1$ . Then write, in Mathlib notation, the  $d = 1$  real statement: for  $G = (g)$  with  $g \neq 0$  and  $B = (0)$ , the matrix  $\eta\mathcal{H}$  squares to the identity. Check it by hand:  $\mathcal{H} = \text{diag}(g, g^{-1})$ ,  $\eta\mathcal{H} = \begin{pmatrix} 0 & g^{-1} \\ g & 0 \end{pmatrix}$ , and  $(\eta\mathcal{H})^2 = \mathbf{1}$ .

**Exercise 35.4** (★★). For each edit, say whether `statement_lock.py -check` would report `CHANGED`, and why: (a) replacing `:=` by `sorry` with `:=` by `omega` in `tier_trans`; (b) adding the hypothesis `(h : t1 = t2)` to `tier_antisymm`; (c) rewriting the docstring of `Sound`; (d) changing `Tier.toNat` so that `Tier.L` maps to 3 and `Tier.B` to 2; (e) reformatting `Depends` onto one line.

**Exercise 35.5** (★★). Which of the following `#print axioms` outputs pass the audit of Section 35.5.2, and what does each failing one indicate?

- `[propext];`
- `[propext, Classical.choice, Quot.sound];`
- `[propext, sorryAx];`
- `[Lean.ofReduceBool];`
- “does not depend on any axioms”.

**Exercise 35.6** (★★ Lean). In a file importing `DualScaleM24Formalization.Epistemic.Tier`, state and prove

```

theorem B_le_A : Tier.B ≤ Tier.A := by sorry
theorem not_A_le_C : ¬ (Tier.A ≤ Tier.C) := by sorry

```

(Hint: both are closed by `decide`, since  $\leq$  on `Tier` was given a `Decidable` instance [`Tier.lean`, ll. 37–38].)

**Exercise 35.7** (\*\*\* Lean). Define a four-row ledger over `Fin 4` in which row 0 (tier A) cites row 1, row 1 (tier A) cites row 2, row 2 (tier L) cites row 3 and row 3 is tier C. Prove it is not `Sound`, and identify the offending pair. Then modify the tiers minimally so that it is sound, prove soundness by `decide`, and use `tier_le_of_depends` to conclude that the tier of row 0 is at most that of row 3.

**Exercise 35.8** (\*\*\*). The theorem `energy_dissipation_monotonic` (`e d : Nat`) : `e - d ≤ e` uses truncated subtraction, so  $e - d = 0$  whenever  $d > e$ . State the energy inequality  $\frac{d}{dt} \|u\|_{L^2}^2 = -2\nu \|\nabla u\|_{L^2}^2 \leq 0$  that the module header describes [`NavierStokesBridge.lean`, ll. 17–37], and list the objects a faithful Lean statement would need (a function space, a time derivative, an integral, a hypothesis that  $u$  solves the equation). Which of them does Mathlib provide, and why can none of them appear in a Mathlib-free library?

**Exercise 35.9** (\*\*). Eleven rows of `ledger.jsonl` list K3T2-A-0001 (the ledger theorem itself) as their sole dependency. Is this consistent with `Sound`? Is it a mathematical dependency? Propose a rule for what “dependency” should mean in a machine-generated ledger, and say how the declaration graph of Chapter 36 would supply it.

## Further reading

- The build description itself: `lakefile.lean` [ll. 1–53], in particular the comments on the Mathlib pin and on why the two Mathlib-based libraries are not default targets.
- The workflow that produced Stream 2: hard constraints [`STREAM2_WORKFLOW.md` §2, ll. 20–38], the per-package gates [§4, ll. 52–65], and the final gate numbers [§6, ll. 185–189]. Chapter 37 discusses the measured tier split.
- The verified-facts brief that fixes the vocabulary and the audit numbers used in this book [`PAPER_REVISION_BRIEF` §0–§1, ll. 8–35] and the corrections that must appear in the papers [§2, ll. 37–62].
- The lessons behind each gate: recovery and build-fix [`LL.md` §0, ll. 145–259], the axiom audit [§S2.5, ll. 56–61], guard theorems [§S2.6, ll. 63–67], the statement lock [§S2.10, ll. 87–92], failing loudly [§S2.11, ll. 94–99], and the planned single gate script [§S2-I, ll. 116–143].
- The tools `tools/axiom_audit.py` [ll. 1–102] and `tools/statement_lock.py` [ll. 1–137], both under 150 lines, and the two epistemic modules `Epistemic/Tier.lean` [ll. 1–67] and `Epistemic/Ledger.lean` [ll. 1–93]; a student can read all four in an evening.



## Chapter 36

# An Atlas of the Theorems: Graph and Retrieval Analysis

The formalization described in Chapter 35 is not one library but seven, written over many months by different people and by the tiered pipeline of Chapter 37. Together they compile to 1224 declarations, of which 423 are theorems written by hand in the source [atlas.md, l. 3]. A student who opens this code base faces the same problem as one who opens a large physics literature: which results are load-bearing, which are restatements, which two results that *look* the same are in fact one, and which two results that look different secretly rest on the same lemma. This chapter answers those questions with two tools borrowed from data analysis, a dependency graph and a text-retrieval index, and, more importantly, shows how to *read* what the tools return. Every number below is taken from the generated file `atlas_data.json`; where a number is re-derived by hand or by a script we say so. The chapter is also an inventory of technical debt: the same fact, for instance  $\chi(K3) = 24$ , is proved independently in six places, and we discuss what that means and how to repair it.

### 36.1 Two relations between theorems, kept apart

#### 36.1.1 The kernel-level dependency graph

The first relation is *dependency*. A Lean theorem is a term whose type is the statement; term and type both mention other constants (definitions, lemmas, inductive types). The extractor `tools/lean_depgraph.lean` imports the seven libraries, walks every constant of the elaborated environment, and for each constant of the project prints the constants used by its statement (`info.type.getUsedConstants`) and by its proof or body (`t.value.getUsedConstants`) [lean\_depgraph.lean, ll. 30–55]. Constructors and recursors are skipped; for theorems the extractor also records the subset of proof dependencies that are theorems of Mathlib or core, called `lemmas`, so that the shared foundations of Section 36.5 are not drowned in notation constants such as `Neg.neg` [lean\_depgraph.lean, ll. 48–54]. The Python side then discards auto-generated names (`.rec`, `.casesOn`, `.sizeof_spec`, ...) and keeps as “theorems” only those kernel theorems that are also written as `theorem` in a source file, so that structure projections and equation lemmas are not counted [theorem\_atlas.py, ll. 38–40, 76–79].

#### Regex is not dependency

An earlier tool inferred edges by searching source text for names. That misses edges created by `simp` sets and by tactics such as `omega` and `norm_num`, which insert Mathlib lemmas never written in the source, and it invents edges from names in comments.

The kernel-level graph has neither defect: an edge exists exactly when the elaborated term mentions the constant `[lean_depgraph.lean, ll. 6–8]`. The price is that it *does* see the scaffolding of every tactic, which is why the weighting below is needed.

Writing  $D(a)$  for the set of constants used by the statement or proof of theorem  $a$ , the raw Jaccard index  $|D(a) \cap D(b)|/|D(a) \cup D(b)|$  would be useless: almost every theorem uses `Eq`, `OfNat.ofNat` and the arithmetic instances of  $\mathbb{N}$  and  $\mathbb{Z}$ . The script therefore weights each constant  $c$  by its inverse document frequency over the  $N = 423$  theorems,

$$\text{idf}(c) = \ln \frac{N}{\text{df}(c)}, \quad \text{df}(c) = \#\{a : c \in D(a)\}, \quad (36.1)$$

and defines the *weighted Jaccard index*

$$J(a, b) = \frac{\sum_{c \in D(a) \cap D(b)} \text{idf}(c)}{\sum_{c \in D(a) \cup D(b)} \text{idf}(c)} \quad (36.2)$$

[theorem\_atlas.py, ll. 106–115]. A constant used by every theorem has  $\text{idf} = 0$  and counts for nothing; `Eq` has  $\text{df} = 358$  and weight  $\ln(423/358) = 0.167$ , while `Matrix.fromBlocks_multiply` has  $\text{df} = 13$  and weight 3.48. Sharing the block-multiplication lemma is twenty times more significant than sharing equality. Since  $0 \leq J \leq 1$ , with  $J = 1$  exactly when the two sets agree up to constants of zero weight (Exercise 36.1),  $J$  is a genuine similarity measure on the graph.

### 36.1.2 The retrieval index

The second relation is *textual similarity* of statements, the relation a search engine computes. For each theorem the script forms a bag of tokens from its name (with underscores and dots replaced by spaces, counted twice), its signature as stored in the declaration index, and the first 400 characters of its docstring [theorem\_atlas.py, ll. 151–157]. With  $f_t$  the count of token  $t$  in a theorem and  $\text{tdf}(t)$  the number of theorems containing  $t$ , the weight is the standard TF-IDF

$$w_t = (1 + \ln f_t) \ln \frac{N}{\text{tdf}(t)}, \quad \text{tokens with } \text{tdf}(t) \geq N/2 \text{ dropped}, \quad (36.3)$$

and the similarity of two theorems is the cosine of the angle between their weight vectors,  $\cos(a, b) = \sum_t \hat{w}_t^{(a)} \hat{w}_t^{(b)}$  with  $\hat{w} = w/\|w\|$  [theorem\_atlas.py, ll. 158–169].

Both relations are computed for all pairs and reported in separate tables (Section 36.7): pairs from different libraries with  $\cos \geq 0.30$ , pairs from different modules with  $J \geq 0.35$ , and, singled out, pairs with  $\cos \geq 0.30$  but  $J < 0.05$ , the *unification candidates*: statements that read alike whose proofs share essentially nothing [theorem\_atlas.py, ll. 171–185]. Keeping the two relations apart is the main methodological point of the chapter; the worked examples show why.

#### What the two numbers measure

Think of a theorem as an experiment.  $J$  asks whether two experiments share apparatus: the same lemmas, definitions and tactic machinery.  $\cos$  asks whether their write-ups use the same words. Two measurements of one quantity with independent apparatus (high  $\cos$ , low  $J$ ) are valuable cross-checks but expensive to maintain; two runs on the same apparatus with different labels (high  $J$ , low  $\cos$ ) may have nothing in common physically, they merely used the same voltmeter. Only a pair high on both is a duplicate.

## 36.2 Worked examples with the actual numbers

**Example 36.1** (Weighted Jaccard of two proofs of  $\chi(K3) = 24$ ). Take  $a = \text{k3\_euler\_characteristic}$  of `DoubleFieldTheory.K3Topology`, which states `K3BettiSum 1 0 22 0 1 = 24` and is closed by `decide`, and  $b = \text{k3\_euler\_eq\_24}$  of `SocrateAI.Moonshine`, which states `k3\_euler = 24` and is closed by `refl`. Re-running the extraction on the compiled environment gives  $|D(a)| = 12$ ,  $|D(b)| = 6$  and these weights, with  $N = 423$ :

| Constant                      | in   | df  | idf   |
|-------------------------------|------|-----|-------|
| K3BettiSum (of $a$ 's module) | $a$  | 1   | 6.047 |
| k3_euler (of $b$ 's module)   | $b$  | 1   | 6.047 |
| Int.instDecidableEq           | $a$  | 57  | 2.004 |
| Decidable.decide              | $a$  | 72  | 1.771 |
| of_decide_eq_true             | $a$  | 73  | 1.757 |
| Bool.true                     | $a$  | 80  | 1.665 |
| Bool                          | $a$  | 93  | 1.515 |
| instOfNat                     | both | 142 | 1.092 |
| Int                           | both | 220 | 0.654 |
| id                            | $a$  | 237 | 0.579 |
| Eq.refl                       | both | 263 | 0.475 |
| OfNat.ofNat                   | both | 349 | 0.192 |
| Eq                            | both | 358 | 0.167 |

The intersection carries weight  $1.092 + 0.654 + 0.475 + 0.192 + 0.167 = 2.580$ ; the union carries  $2.580 + 6.047 + (6.047 + 2.004 + 1.771 + 1.757 + 1.665 + 1.515 + 0.579) = 23.965$ ; hence  $J = 2.580/23.965 = 0.108$ , the value printed in Table 36.4 [atlas.md, l. 131]. Read the table: the two theorems share only the vocabulary of integer equality. Everything specific to  $a$  is the `decide` machinery and its own definition `K3BettiSum`; everything specific to  $b$  is its own definition `k3_euler`. Neither proof knows the other exists.

**Example 36.2** (Cosine of the same pair). The retrieval side sees the same pair very differently. After tokenizing, the two theorems share six tokens; the table lists counts, token document frequency, the weight in  $a$ , and each token's contribution  $\hat{w}_t^{(a)}\hat{w}_t^{(b)}$  to the cosine.

| Token          | tdf | $f^{(a)}$ | $f^{(b)}$ | $w^{(a)}$ | contribution |
|----------------|-----|-----------|-----------|-----------|--------------|
| euler          | 32  | 4         | 6         | 6.161     | 0.276        |
| characteristic | 22  | 4         | 2         | 7.055     | 0.219        |
| int            | 75  | 2         | 2         | 2.929     | 0.053        |
| topological    | 27  | 1         | 1         | 2.752     | 0.047        |
| theorem        | 32  | 1         | 1         | 2.582     | 0.041        |
| def            | 175 | 1         | 1         | 0.883     | 0.005        |

Check one entry:  $w_{\text{euler}}^{(a)} = (1 + \ln 4) \ln(423/32) = 2.386 \times 2.581 = 6.16$ . With  $\|w^{(a)}\| = 13.93$  and  $\|w^{(b)}\| = 11.56$  the contributions sum to 0.642, the top entry of Table 36.4 [atlas.md, l. 131]. Two tokens, `euler` and `characteristic`, account for 77% of the similarity. The index has matched the *words*, as it should; whether the mathematics is the same is a question for the reader, and here the answer is yes: both are the alternating sum  $1 - 0 + 22 - 0 + 1$ . The token `the` was dropped as occurring in more than half of all theorems; `k3` and `24` are shorter than three characters and are never tokenized, a limitation worth knowing when a search for “24” returns nothing.

**Example 36.3** (High dependency overlap, no textual similarity). The strongest dependency intersection between different modules is  $J = 0.925$ , with cosine only 0.092, for `M24_order`

of `StringTheory.StringDynamics`, which states  $(244823040 : \mathbb{N}) = 2^{10} * 3^3 * 5 * 7 * 11 * 23$ , and `M21_order_factorization` of `StringTheory.UseCases.MathieuTower`, which states  $\text{orderM21} = 2^6 * 3^2 * 5 * 7$  [atlas.md, l. 193]. The re-run shows  $D(a) \subset D(b)$ : all 26 constants of  $a$  are among the 28 of  $b$ , the two extra ones being `orderM21` and `id`. The 26 shared constants are the internals of `norm_num` (`isNat_mul`, `isNat_pow`, `IsNatPowT.bit0`, ...) and the semiring instances of  $\mathbb{N}$ . Both facts are true (Chapter 29), but the graph says only that both were proved by the same tactic. This is the case the weighting cannot remove: `norm_num` internals are rare enough across the corpus to carry weight, so two factorization checks look like close relatives. A high  $J$  with a low cosine means “same apparatus” and nothing more.

**Example 36.4** (A genuine duplicate). `buscher_log_involution` of `DoubleFieldTheory.TDualityBuscher` states `BuscherLogMap (BuscherLogMap x) = x` for  $x : \text{Int}$ , where `BuscherLogMap` is negation; `buscher_log_involution` of `DualScaleValidation.UseCase1` states  $-(x) = x$ ; both are closed by `omega`. The pair scores  $\cos = 0.445$  and  $J = 0.351$  [atlas.md, l. 145]: same words, same apparatus, a true duplicate. Physically both encode  $\ln R \mapsto -\ln R$ , the statement that  $R \mapsto \alpha'/R$  is an involution (Chapter 8), reduced to the arithmetic shadow  $-(-x) = x$  on integers.

### 36.3 The module graph and the hubs

Aggregating the constant-level edges by module gives Figure 36.1: 44 modules joined by 42 directed edges, an edge  $M \rightarrow M'$  meaning that some declaration of  $M$  mentions a declaration of  $M'$ , drawn thicker when more do [theorem\_atlas.py, ll. 188–201]. Modules without edges are omitted. Three features stand out. First, the graph is a forest of shallow trees; the deepest chain is `TDuality.Spectrum`  $\rightarrow$  `DFT.GeneralizedMetric`  $\rightarrow$  `TDuality.ODD`  $\rightarrow$  `Lattice.Hyperbolic`  $\rightarrow$  `Lattice.Basic`, continuing across the library border into `Stream1's UseCases.NarainLattice` and `UseCases.TDualityMassSpectrum` [module\_graph.dot]. Second, `DualScaleStream2` is the only library organized as a tree, with two roots: `Lattice.Basic` (the Gram-matrix predicates of Chapter 18) and `TDuality.ODD` (the group  $O(d, d; \mathbb{Z})$  of Chapter 11). Third, four of the seven libraries, `DoubleFieldTheory`, `DualScaleM24Formalization`, `DualScaleValidation` and `Lean5Corpus`, have no edge to any other library: they are islands, the structural reason for the duplication of Section 36.6.

A *hub* is a declaration of the project used by many of the project’s theorems. Table 36.1 lists the forty largest; the top is [atlas.md, ll. 6–11]:

| Declaration                         | Kind      | Used by | What it is                                                              |
|-------------------------------------|-----------|---------|-------------------------------------------------------------------------|
| <code>TDuality.Charge</code>        | def       | 36      | the index type $\text{Fin } d \oplus \text{Fin } d$ of $(n, w)$         |
| <code>TDuality.eta</code>           | def       | 20      | $\eta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ over $\mathbb{Z}$ |
| <code>Lattice.Gram</code>           | def       | 20      | integer $n \times n$ Gram matrices                                      |
| <code>Frontier.k3HodgeNumber</code> | def       | 12      | the $3 \times 3$ Hodge table of $K3$                                    |
| <code>TDuality.IsODD</code>         | def       | 12      | the predicate $g^T \eta g = \eta$                                       |
| <code>Epistemic.Tier</code>         | inductive | 11      | the five-valued tier type of the ledger                                 |

Each hub is a *definition*, not a theorem, and each is a piece of the mathematics of Chapters 11, 18 and 19. The largest, `Charge`, is only a type, `abbrev Charge (d :  $\mathbb{N}$ ) := Fin d  $\oplus$  Fin d` [TDuality/ODD.lean, l. 75]. That a type with no content is the most-used object is not a paradox: every statement about a  $2d \times 2d$  block matrix, over  $\mathbb{Z}$  in `TDuality` or over  $\mathbb{R}$  in `DFT`, is indexed by it, so the count 36 measures the size of the block-matrix formalization, not the importance of a theorem. The definitions `eta`, `IsODD`, `Gram` are the working hubs from which the duality group and the lattice theory hang. `k3HodgeNumber` is a hub of a different kind: a



Figure 36.1: Module-level dependency graph extracted from the kernel: 44 modules, 42 edges. Colours mark libraries (blue: `DualScaleStream2`; yellow: `StringTheoryFormalization`; pink: `DoubleFieldTheory`; green: `DualScaleM24Formalization`; lavender: `StringTheoryFoundation`). An arrow from  $M$  to  $M'$  means a declaration of  $M$  uses a declaration of  $M'$ . Generated by `tools/theorem_atlas.py` from `.leancache/depgraph.jsonl`; the source PDF is legible at higher magnification.

hand-written table `Fin 3 → Fin 3 → ℕ` with entries  $h^{0,0} = h^{2,0} = h^{0,2} = h^{2,2} = 1$ ,  $h^{1,1} = 20$ , the rest 0, which its file describes as “asserted, not derived” [Frontier/HodgeNumbers.lean, ll. 99–112]. Twelve theorems in two libraries rest on it, including the tadpole arithmetic of Chapter 25; a mistyped entry would make all twelve wrong together. A hub concentrates risk as well as reuse.

The hub `SocrateAI.Epistemic.Tier` deserves a warning. It is the ledger’s own tier type with five constructors `X`, `C`, `L`, `B`, `A` [Epistemic/Tier.lean, ll. 16–22], finer than this book’s three badges; the theorems built on it, such as `no_kernel_claim_rests_on_weaker`, are statements about a finite ordered type and a bookkeeping structure, not about any theorem of the project. A Lean proof that “no A-claim depends on a weaker claim” *inside the ledger model* is not an audit of the libraries; the audit is the axiom check of Chapter 34.

## 36.4 The nineteen bridges

A *bridge* is a theorem whose statement or proof uses a declaration of a different library. There are 19, listed in Table 36.2, and they run in only two directions: 13 from `DualScaleStream2` into `StringTheoryFormalization` and 6 from `StringTheoryFormalization` into `StringTheoryFoundation` [atlas.md, ll. 48–66]. Grouped by theorem:

1. `massForm_circle (DFT) → leftMomentum, rightMomentum, momentumMassSq` of Stream 1 (three edges). For rational  $n, w, R$  with  $R \neq 0$ ,

$$\text{massForm}(\mathcal{H}(R^2, 0), (w, n)) = \frac{1}{2} \text{momentumMassSq}(n, w, R),$$

the DFT mass form of Chapter 15 on a circle equals half of Stream 1’s  $p_L^2 + p_R^2$  of Chapter 7 [DFT/GeneralizedMetric.lean, ll. 184–189]. Two formalizations of the circle spectrum, written independently with different conventions (the DFT charge vector is  $(w, n)$ ), are proved to agree; the factor  $\frac{1}{2}$  is the usual one between  $Z^\top \mathcal{H} Z$  and  $p_L^2 + p_R^2$ .

2. `euler_K3, k3k3_anomaly, k3t2_euler_zero (Flux) → k3HodgeNumber`. Stream 2 does not retype the Hodge diamond; it applies its own functional `eulerFromHodge` to Stream 1’s table and gets 24, then  $24^2/24 = 24$  and  $24 \cdot 0 = 0$  [Flux/Tadpole.lean, ll. 86–112]. Bridges by *reuse*: one object, two consumers.
3. `hyperbolicU_eq_narain_gram → NarainLattice.gram`, and `sigK3_matches_hodge → K3Signature.bPlus, bMinus`. The first is the entrywise equality of two  $2 \times 2$  integer matrices, both  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , one defined as the hyperbolic plane  $U$  and the other as the polarization of  $Q(n, w) = 2nw$  [Lattice/Hyperbolic.lean, ll. 146–153]; composed with `eta_one_reindex` it identifies three objects from two libraries,  $\eta$  at  $d = 1$ ,  $U$ , and the Narain Gram matrix. The second checks that the signature  $(3, 19)$  of  $3U \oplus 2E_8(-1)$  agrees with  $(2h^{2,0} + 1, h^{1,1} - 1)$  read from the Hodge table; the file states that this is a consistency check between two tabulated inputs, not a derivation of either [Lattice/K3T2Signature.lean, ll. 188–196].
4. `first_five_are_irreps, A6_decomposition, A6_not_irrep, A7_decomposition (Moonshine.EOT) → M24RepDim`. The coefficients  $A_1, \dots, A_5 = 45, 231, 770, 2277, 5796$  are looked up in Stream 1’s list of the 26 irreducible dimensions of  $M_{24}$ , and  $A_6 = 13915$  is shown not to be in it (Chapter 30). Again one table, two consumers.
5. `flux_tadpole_quantization_exact (Frontier) → d7_tadpole_cancellation, totalD7Charge, totalO7Charge, GVWFluxState` (four edges). Here the atlas flatters the code: the theorem takes a `GVWFluxState` argument it never uses and its proof is a bare `exact` of the Foundation library’s `d7_tadpole_cancellation` [Frontier/FTermPotential.lean, ll. 55–58]. Four of nineteen bridges are one re-export.

6. `poincare_geodesic_atlas_curvature`  $\rightarrow$  `AtlasHyperbolicMetric`, `gaussianCurvature` (two edges): the curvature  $-1$  of the hyperbolic plane, imported from the Foundation library’s chart “atlas” of the moduli space of Chapter 27. The word *atlas* there means a collection of coordinate charts.

The nineteen edges are thus of three kinds: identifications proved (items 1 and 3), reuse of a single certified object (items 2, 4, 6), and re-export (item 5). Only the first adds knowledge; the second removes risk; the third adds nothing. A bridge count is not a measure of integration; one must read the proof. And no bridge enters or leaves the four island libraries.

## 36.5 Shared foundations in Mathlib

Table 36.3 lists the Mathlib and core lemmas appearing in the proofs of at least three project theorems [atlas.md, ll. 69–128]. It is a quick way to learn what mathematics the formalization leans on. The most-used lemma, `Fintype.complete` (29 theorems), with `Matrix.ext` (22), `Matrix.cons_val'` and `Matrix.cons_val_fin_one` (17 each) and `Finset.univ_unique` (12), is the machinery of finite sums and entrywise equality of small explicit matrices: the pattern `ext i j; fin_cases i <;> fin_cases j <;> rfl` expands to exactly these. `Matrix.fromBlocks_multiply` (13 theorems, all in `DualScaleStream2`) and `fromBlocks_transpose` (8) are the block identities behind every  $O(d, d; \mathbb{Z})$  proof of Chapter 11. The ordered-field instances of  $\mathbb{R}$  (`Real.instIsStrictOrdered`, `Even.pow_nonneg`, ...) mark the inequality proofs: the dual-scale bound  $\text{tr } G + \text{tr } G^{-1} \geq 2d$  of Chapter 31, positive-definiteness of the  $E_8$  Cartan matrix, non-negativity of the F-term potential of Chapter 26. `Rat.instCharZero` (9 theorems, only in Stream 1) records that the Virasoro and central-charge arithmetic is done over  $\mathbb{Q}$ .

What is *absent* is as informative: no lemma about cohomology, sheaves, manifolds, modular forms or group actions appears, because none is used. At the level of Mathlib the formalization is linear algebra over  $\mathbb{Z}$ ,  $\mathbb{Q}$ ,  $\mathbb{R}$  on finite index types plus ordered-field inequalities, consistent with Chapter 35: the deep objects (K3 surfaces,  $M_{24}$ , the elliptic genus) enter as tables and integers, their arithmetic shadows.

## 36.6 Clusters and unification candidates

### 36.6.1 Six proofs that $\chi(K3) = 24$

The Euler characteristic of a K3 surface is 24: Noether’s formula gives  $c_2(X) = 24$  [Huy §1.2, ll. 406–425], and the Hodge diamond stated there yields  $b_0 - b_1 + b_2 - b_3 + b_4 = 1 - 0 + 22 - 0 + 1$ . The project proves this number in six places, in five libraries, from five encodings:

| Library / module                                            | Theorem                                       | Statement (paraphrased)                                                                                     | Proof              |
|-------------------------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------------------|
| DoubleFieldTheory.<br>K3Topology<br>SocrateAI.<br>Moonshine | k3_euler_<br>characteristic<br>k3_euler_eq_24 | K3BettiSum 1 0 22 0 1 = 24,<br>K3BettiSum the alternating sum<br>k3_euler = 24, with k3_euler the alternat- | with decide<br>rfl |
| StringTheory.<br>Foundation.Core.<br>Topology               | euler_char_K3                                 | eulerChar4D bettiK3 = 24, on a Betti4D<br>record                                                            | rfl                |
| Lean5Corpus.<br>Problems.Kummer<br>Modularity               | k3_euler_<br>characteristic_<br>identity      | k3_betti_0 + k3_betti_2 + k3_betti_4<br>= k3_euler_char, i.e. 1 + 22 + 1 = 24                               | rfl                |
| StringTheory.<br>Frontier                                   | k3_euler_<br>characteristic                   | $\sum_{p,q} (-1)^{p+q} h^{p,q} = 24$ over the table<br>k3HodgeNumber                                        | simp               |
| DualScaleStream2.<br>Flux                                   | euler_K3                                      | eulerFromHodge k3HodgeNumber = 24,<br>reusing Stream 1's table                                              | simp               |

Only the last two share an object (the table), and only the last is a bridge. The other four are unification candidates: every cross-library pair among them has cos between 0.44 and 0.64 with  $J \leq 0.11$  [atlas.md, ll. 131–148]. The Lean5 version deserves a second look: it states  $b_0 + b_2 + b_4 = \chi$  with plus signs only, correct because  $b_1 = b_3 = 0$  but silent about it; a reader who sees only that theorem learns less than the alternating sum tells.

Is the duplication bad? Two honest answers. As *evidence*, six independent encodings that all evaluate to 24 are a mild cross-check: a typo in one Betti table would contradict the other five, and the Foundation library made exactly this point when it changed `golay_length_matches_k3_euler` from a comparison with the literal 24 to a comparison with `eulerChar4D bettiK3` [Quantum/TensorNetworkBridge.lean, ll. 68–74]. As *engineering*, it is debt: a change of convention must be made six times, no theorem downstream of one version can use a result downstream of another without a conversion lemma, and the atlas cannot tell a reader which version is canonical. The count of 19 bridges against 423 theorems, with four island libraries, is the quantitative form of the same observation.

### 36.6.2 Mukai rank and the tadpole

Two more clusters have the same shape. The rank 24 and signature (4, 20) of the Mukai lattice of Chapter 22 appear as `mukai_rank_and_signature` in Lean5 (`mukai_rank = 24`  $\wedge$  `mukai_signature = -16`, both sides literal constants), as `mukai_lattice_rank_equals_24` in the Foundation library ( $1 + 22 + 1 = 24$ ), and as `mukai_rank` in Stream 1 (`rank = posSignature + negSignature` on a record with defaults 24, 4, 20) [atlas.md, ll. 136, 144, 291]. None mentions a bilinear form. The only Mukai statement about an actual Gram matrix is Stream 2's `structureSheaf_mukai_sq` in `Lattice.Mukai`, the Mukai self-pairing of (1, 0, 1) for an arbitrary Gram matrix  $L$ ; and Stream 2's `rank_K3` (`sigK3.rank = 22`) comes from the signature record of  $3U \oplus 2E_8(-1)$ .

The RR tadpole cancellation of Chapter 25 is proved as an integer identity in four places, and the encodings do not even agree on the accounting. In `rr_tadpole_cancellation` of `SocrateAI.Moonshine` and of `DualScaleValidation.UseCase3` the count is 16 D7 stacks of charge +4 against 4 O7-planes of charge -16 [Moonshine/KummerTadpole.lean, ll. 96–126]; in `d7_tadpole_cancellation` of the Foundation library it was, in the snapshot the atlas was computed from (2026-09-17), 32 D7-branes of charge +2 against 16 O7<sup>-</sup> fixed points of charge -4. Both gave  $64 - 64 = 0$ , but the second took the count of the  $T^4/\mathbb{Z}_2$  fixed points and the charge of an O7-plane, a combination that matches no orientifold; it was corrected on 2026-09-18 to the 4 O7-planes and 16 D7-branes of (25.28) (Section 25.6). The first is a rescaling by 4 of the same total. The fourth, `tadpole_cancellation` of `StringTheory.StringDynamics`, is not an identity but a rearrangement: from the hypothesis `totalTadpole ... = 0` it derives



$\sum_i \text{charge}_i + \text{fluxTadpole} = 24$  by `linarith` [StringDynamics/TadpoleConstraint.lean, ll. 92–99]. A reader trusting docstrings alone would count four proofs of tadpole cancellation; the atlas plus the source reveal two integer bookkeepings (after the correction, related by a factor 4) and one rearrangement of an assumption. The physics, that RR charge on a compact space sums to zero, is Tier L in all four files and proved in none.

### 36.6.3 How to refactor

The repair is mechanical and the atlas says where to apply it. For each cluster choose one canonical definition, the one closest to the mathematics: for the Euler characteristic the table `k3HodgeNumber` with the functional `eulerFromHodge`, the only version in which the Hodge diamond rather than a precomputed sum is the input; for the Mukai lattice a Gram matrix in `Lattice.Mukai`; for the tadpole the corrected Foundation file, which contains Polchinski’s whole table (`oplane_total_independent`). Then replace each duplicate by a corollary that *states its old form and proves it from the canonical one*, for instance

```
theorem k3_euler_eq_24 :
  k3_euler = DualScaleStream2.Flux.eulerFromHodge
  StringTheory.Frontier.k3HodgeNumber := by decide
```

which is not in the repository (Exercise 36.6) but would compile, both sides reducing to the numeral 24. Each such corollary turns a unification candidate into a bridge: the pair’s  $J$  rises towards 1 and the island library acquires an edge. The cost is an import; the benefit is that six statements become one fact with five names, so a future change is made once. Re-running `python3 tools/theorem_atlas.py` after each step should shrink the unification list to zero, a measurable definition of “done” for a refactoring task, which is rare in software.

#### Why the duplicates exist

The four island libraries were written before `DualScaleStream2` and before the kernel-level extractor; the pipeline of Chapter 37 generated them problem by problem, each with its own Betti constants, because a prover working on one file has no view of the others. `DualScaleStream2` was the first library written under the explicit rule to reuse `Stream 1`’s certified objects rather than re-derive them, which its headers state [Flux/Tadpole.lean, ll. 14–16]; it accounts for 13 of the 19 bridges.

## 36.7 The atlas tables

The five tables below are generated by `tools/theorem_atlas.py` and included verbatim; module names drop the library prefix. Namespace prefixes map to libraries as follows: `SocrateAI.*` is `DualScaleM24Formalization`; `StringTheory.Frontier`, `StringDynamics` and `UseCases` are `StringTheoryFormalization`; `StringTheory.Foundation` is `StringTheoryFoundation`; the remaining prefixes coincide with library names.

Table 36.1: Hub declarations: the project’s own definitions and lemmas on which the most theorems depend (kernel-level dependencies).

| Declaration   | Module                     | Kind | Used by |
|---------------|----------------------------|------|---------|
| <b>Charge</b> | <code>TDuality.ODD</code>  | def  | 36      |
| <b>eta</b>    | <code>TDuality.ODD</code>  | def  | 20      |
| <b>Gram</b>   | <code>Lattice.Basic</code> | def  | 20      |

| Declaration          | Module                      | Kind      | Used by |
|----------------------|-----------------------------|-----------|---------|
| k3HodgeNumber        | Frontier.HodgeNumbers       | def       | 12      |
| IsODD                | TDuality.ODD                | def       | 12      |
| Tier                 | Epistemic.Tier              | inductive | 11      |
| IsEvenDiag           | Lattice.Basic               | def       | 8       |
| v                    | CourantAlgebroid            | def       | 8       |
| CourantSection       | CourantAlgebroid            | inductive | 8       |
| cartanE8             | Lattice.E8                  | def       | 7       |
| genMetric            | DFT.GeneralizedMetric       | def       | 7       |
| CBracket             | CourantAlgebroid            | def       | 7       |
| hyperbolicU          | Lattice.Hyperbolic          | def       | 7       |
| M24RepDim            | StringDynamics.MathieuM24   | def       | 6       |
| proj                 | TDuality.Factorized         | def       | 6       |
| chargeNorm           | TDuality.Factorized         | def       | 6       |
| alpha                | CourantAlgebroid            | def       | 6       |
| fermionicGhostCharge | UseCases.CriticalDimension  | def       | 5       |
| dualScale            | DualScale.TraceBound        | def       | 5       |
| Mat2                 | GeneralizedGeometry         | inductive | 5       |
| MatMul               | GeneralizedGeometry         | def       | 5       |
| bPlus                | UseCases.K3SignatureTheorem | def       | 5       |
| basisChange          | TDuality.ODD                | def       | 5       |
| tier                 | Epistemic.Ledger            | def       | 5       |
| Sound                | Epistemic.Ledger            | def       | 5       |
| factorized           | TDuality.Factorized         | def       | 5       |
| eqFrac               | Moonshine.MathieuRigidity   | def       | 5       |
| IsUnimodular         | Lattice.Basic               | def       | 5       |
| Signature            | Lattice.K3T2Signature       | inductive | 5       |
| e8Neg                | Lattice.E8                  | def       | 5       |

Table 36.2: Cross-library bridges: theorems whose statement or proof uses a declaration of a different library.

| Theorem               | in                    | uses           | of                            |
|-----------------------|-----------------------|----------------|-------------------------------|
| massForm_circle       | DFT.GeneralizedMetric | leftMomentum   | UseCases.TDualityMassSpectrum |
| massForm_circle       | DFT.GeneralizedMetric | momentumMassSq | UseCases.TDualityMassSpectrum |
| massForm_circle       | DFT.GeneralizedMetric | rightMomentum  | UseCases.TDualityMassSpectrum |
| euler_K3              | Flux.Tadpole          | k3HodgeNumber  | Frontier.HodgeNumbers         |
| k3k3_anomaly          | Flux.Tadpole          | k3HodgeNumber  | Frontier.HodgeNumbers         |
| k3t2_euler_zero       | Flux.Tadpole          | k3HodgeNumber  | Frontier.HodgeNumbers         |
| hyperbolicU_eq_       | Lattice.Hyperbolic    | gram           | UseCases.NarainLattice        |
| narain_gram           |                       |                |                               |
| sigK3_matches_hodge   | Lattice.K3T2Signature | bMinus         | UseCases.K3SignatureTheorem   |
| sigK3_matches_hodge   | Lattice.K3T2Signature | bPlus          | UseCases.K3SignatureTheorem   |
| A6_decomposition      | Moonshine.EOT         | M24RepDim      | StringDynamics.MathieuM24     |
| A6_not_irrep          | Moonshine.EOT         | M24RepDim      | StringDynamics.MathieuM24     |
| A7_decomposition      | Moonshine.EOT         | M24RepDim      | StringDynamics.MathieuM24     |
| first_five_are_irreps | Moonshine.EOT         | M24RepDim      | StringDynamics.MathieuM24     |

| Theorem                                        | in                       | uses                                 | of                               |
|------------------------------------------------|--------------------------|--------------------------------------|----------------------------------|
| <code>flux_tadpole_quantization_exact</code>   | Frontier.FTermPotential  | <code>d7_tadpole_cancellation</code> | StringTheory.TadpoleCancellation |
| <code>flux_tadpole_quantization_exact</code>   | Frontier.FTermPotential  | <code>totalD7Charge</code>           | StringTheory.TadpoleCancellation |
| <code>flux_tadpole_quantization_exact</code>   | Frontier.FTermPotential  | <code>total07Charge</code>           | StringTheory.TadpoleCancellation |
| <code>flux_tadpole_quantization_exact</code>   | Frontier.FTermPotential  | <code>GVWFluxState</code>            | StringTheory.VafaSwampland       |
| <code>poincare_geodesic_atlas_curvature</code> | Frontier.ModuliGeodesics | <code>AtlasHyperbolicMetric</code>   | Atlas.AtlasGeometryBridge        |
| <code>poincare_geodesic_atlas_curvature</code> | Frontier.ModuliGeodesics | <code>gaussianCurvature</code>       | Atlas.AtlasGeometryBridge        |

Table 36.3: Shared external foundations: Mathlib and core lemmas on which several of the project’s theorems rest.

| Lemma                                         | Theorems | Examples                                                                                                 |
|-----------------------------------------------|----------|----------------------------------------------------------------------------------------------------------|
| <code>Fintype.complete</code>                 | 29       | <code>massForm_circle</code> , <code>dualScale_circle</code> ,<br><code>cartanE8Inv_mul</code>           |
| <code>MulZeroClass.mul_zero</code>            | 22       | <code>etaR_mul_self</code> , <code>massForm_circle</code> ,<br><code>tduality_inverts_metric</code>      |
| <code>Matrix.ext</code>                       | 22       | <code>massForm_circle</code> , <code>dualScale_circle</code> ,<br><code>cartanE8Inv_mul</code>           |
| <code>Matrix.cons_val'</code>                 | 17       | <code>massForm_circle</code> , <code>dualScale_circle</code> ,<br><code>cartanE8Inv_mul</code>           |
| <code>Matrix.cons_val_fin_one</code>          | 17       | <code>massForm_circle</code> , <code>dualScale_circle</code> ,<br><code>cartanE8Inv_mul</code>           |
| <code>Finset.sum_congr</code>                 | 17       | <code>massForm_circle</code> , <code>euler_K3</code> , <code>euler_T2</code>                             |
| <code>MulZeroClass.zero_mul</code>            | 15       | <code>etaR_genMetric_sq</code> , <code>massForm_circle</code> ,<br><code>tduality_inverts_metric</code>  |
| <code>Matrix.fromBlocks_multiply</code>       | 13       | <code>etaR_genMetric_sq</code> , <code>etaR_mul_self</code> ,<br><code>genMetric_bshift</code>           |
| <code>Real.instIsStrictOrderedRing</code>     | 12       | <code>circle_effective_scale_ge_two</code> ,<br><code>dualScale_ge</code> , <code>cartanE8_posDef</code> |
| <code>Finset.univ_unique</code>               | 12       | <code>massForm_circle</code> , <code>euler_T2</code> ,<br><code>cartanE8Inv_mul</code>                   |
| <code>IsOrderedAddMonoid.toAddLeftMono</code> | 11       | <code>circle_effective_scale_ge_two</code> ,<br><code>dualScale_ge</code> , <code>cartanE8_posDef</code> |
| <code>Real.instIsOrderedRing</code>           | 11       | <code>circle_effective_scale_ge_two</code> ,<br><code>dualScale_ge</code> , <code>cartanE8_posDef</code> |
| <code>Iff.mpr</code>                          | 11       | <code>etaR_genMetric_sq</code> , <code>genMetric_bshift</code> ,<br><code>genMetric_symm</code>          |
| <code>Real.instIsOrderedAddMonoid</code>      | 10       | <code>circle_effective_scale_ge_two</code> ,<br><code>dualScale_ge</code> , <code>cartanE8_posDef</code> |

| Lemma                                               | Theorems | Examples                                                                                                                         |
|-----------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------|
| <code>IsStrictOrderedRing.toIsOrderedRing</code>    | 10       | <code>cartanE8_posDef</code> , <code>fterm_potential_nonneg</code> , <code>poincare_geodesic_kinetic_energy_nonneg</code>        |
| <code>Matrix.transpose_one</code>                   | 9        | <code>genMetric_bshift</code> , <code>thetaShiftR_preserves_eta</code> , <code>dualScale_ge</code>                               |
| <code>Rat.instCharZero</code>                       | 9        | <code>central_charge_k3_eq_six</code> , <code>virasoro_algebra_commutator</code> , <code>virasoro_opcentral_term</code>          |
| <code>Matrix.cons_val_succ</code>                   | 9        | <code>cartanE8Inv_mul</code> , <code>cartanE8_posDef</code> , <code>cartanE8_det</code>                                          |
| <code>Finset.sum_const</code>                       | 9        | <code>massForm_circle</code> , <code>cartanE8Inv_mul</code> , <code>euler_T2</code>                                              |
| <code>Finset.card_singleton</code>                  | 9        | <code>massForm_circle</code> , <code>cartanE8Inv_mul</code> , <code>euler_T2</code>                                              |
| <code>Matrix.fromBlocks_transpose</code>            | 8        | <code>genMetric_bshift</code> , <code>genMetric_symm</code> , <code>thetaShiftR_preserves_eta</code>                             |
| <code>IsCancelMulZero.toIsLeftCancelMulZero</code>  | 8        | <code>massForm_circle</code> , <code>circle_effective_scale_ge_two</code> , <code>dualScale_circle</code>                        |
| <code>NeZero.one</code>                             | 8        | <code>massForm_circle</code> , <code>circle_effective_scale_ge_two</code> , <code>dualScale_circle</code>                        |
| <code>GroupWithZero.toNontrivial</code>             | 8        | <code>massForm_circle</code> , <code>circle_effective_scale_ge_two</code> , <code>dualScale_circle</code>                        |
| <code>one_ne_zero._simp_1</code>                    | 8        | <code>euler_T2</code> , <code>k3t2_euler_zero</code> , <code>cartanE8Inv_mul</code>                                              |
| <code>FloorSemiring.instCharZero</code>             | 8        | <code>massForm_circle</code> , <code>circle_effective_scale_ge_two</code> , <code>cartanE8_det</code>                            |
| <code>AddGroup.existsAddOfLE</code>                 | 8        | <code>circle_effective_scale_ge_two</code> , <code>cartanE8_posDef</code> , <code>fterm_potential_nonneg</code>                  |
| <code>Finset.sum_const_zero</code>                  | 7        | <code>euler_T2</code> , <code>cartanE8Inv_mul</code> , <code>e8_LDL</code>                                                       |
| <code>IsStrictOrderedRing.toPosMulStrictMono</code> | 6        | <code>circle_effective_scale_ge_two</code> , <code>cartanE8_posDef</code> , <code>poincare_geodesic_kinetic_energy_nonneg</code> |
| <code>Matrix.one_apply_eq</code>                    | 6        | <code>massForm_circle</code> , <code>dualScale_circle</code> , <code>cartanE8Inv_mul</code>                                      |
| <code>Matrix.fromBlocks_inj</code>                  | 6        | <code>etaR_genMetric_sq</code> , <code>genMetric_bshift</code> , <code>genMetric_symm</code>                                     |
| <code>Matrix.mul_assoc</code>                       | 6        | <code>genMetric_bshift</code> , <code>genMetric_bshift_cancel</code> , <code>basisChange_comm_thetaShift</code>                  |
| <code>Finset.sum_neg_distrib</code>                 | 6        | <code>euler_T2</code> , <code>cartanE8Inv_mul</code> , <code>e8Neg_simpleRoot_norm</code>                                        |
| <code>IsOrderedRing.toPosMulMono</code>             | 6        | <code>circle_effective_scale_ge_two</code> , <code>cartanE8_posDef</code> , <code>energy_dissipation</code>                      |
| <code>Even.pow_nonneg</code>                        | 6        | <code>cartanE8_posDef</code> , <code>fterm_potential_nonneg</code> , <code>poincare_geodesic_kinetic_energy_nonneg</code>        |

Table 36.4: Most similar theorem statements across libraries (TF-IDF cosine similarity), with the weighted Jaccard index of their kernel dependencies.

| cos  | dep  | Theorem A                                                              | Theorem B                                                              |
|------|------|------------------------------------------------------------------------|------------------------------------------------------------------------|
| 0.64 | 0.11 | k3_euler_characteristic<br>K3Topology                                  | k3_euler_eq_24<br>Moonshine.KummerTadpole                              |
| 0.56 | 0.01 | k3t2_euler_zero<br>Flux.Tadpole                                        | k3t2_euler_char_eq_zero<br>Moonshine.KummerTadpole                     |
| 0.55 | 0.01 | golay_self_dual_dimension<br>Problems.Problem8_GolayHolography         | golay_length_matches_k3_euler<br>Quantum.TensorNetworkBridge           |
| 0.54 | 0.03 | golay_self_dual_dimension<br>Problems.Problem8_GolayHolography         | golay_code_rate_is_half<br>Quantum.TensorNetworkBridge                 |
| 0.48 | 0.11 | d7_positive_charge<br>Moonshine.KummerTadpole                          | total_D7_charge_is_64<br>StringTheory.TadpoleCancellation              |
| 0.48 | 0.09 | mukai_rank_and_signature<br>Problems.Problem4_MukaiMonodromy           | mukai_signature_difference<br>ModularForms.FermatModularBridge         |
| 0.48 | 0.01 | k3_euler_characteristic<br>K3Topology                                  | k3_euler_characteristic_identity<br>Problems.Problem9_KummerModularity |
| 0.46 | 0.03 | dsl_self_dual_minimum<br>PhysicsDSL                                    | self_dual_is_global_minimum<br>UseCase1_ModuliStabilization            |
| 0.46 | 0.10 | k3_hirzebruch_signature<br>K3Topology                                  | k3_euler_eq_24<br>Moonshine.KummerTadpole                              |
| 0.46 | 0.00 | golay_holography_master_contract<br>Problems.Problem8_GolayHolography  | golay_length_matches_k3_euler<br>Quantum.TensorNetworkBridge           |
| 0.45 | 0.01 | rr_tadpole_cancellation<br>Moonshine.KummerTadpole                     | tadpole_cancellation<br>StringDynamics.TadpoleConstraint               |
| 0.45 | 0.02 | rr_tadpole_cancellation<br>UseCase3_FrontierTriad                      | tadpole_cancellation<br>StringDynamics.TadpoleConstraint               |
| 0.45 | 0.02 | golay_holography_master_contract<br>Problems.Problem8_GolayHolography  | golay_code_rate_is_half<br>Quantum.TensorNetworkBridge                 |
| 0.45 | 0.04 | mukai_rank_and_signature<br>Problems.Problem4_MukaiMonodromy           | mukai_lattice_rank_equals_24<br>ModularForms.FermatModularBridge       |
| 0.45 | 0.35 | buscher_log_involution<br>TDualityBuscher                              | buscher_log_involution<br>UseCase1_ModuliStabilization                 |
| 0.44 | 0.02 | k3_euler_characteristic_identity<br>Problems.Problem9_KummerModularity | k3_euler_eq_24<br>Moonshine.KummerTadpole                              |
| 0.43 | 0.01 | k3_second_betti_hodge<br>K3Topology                                    | k3_euler_characteristic_identity<br>Problems.Problem9_KummerModularity |
| 0.43 | 0.11 | k3_second_betti_hodge<br>K3Topology                                    | k3_euler_eq_24<br>Moonshine.KummerTadpole                              |
| 0.43 | 0.07 | k3_euler_characteristic<br>K3Topology                                  | kummer_exceptional_intersection<br>Moonshine.KummerTadpole             |
| 0.43 | 0.10 | o7_negative_charge<br>Moonshine.KummerTadpole                          | total_D7_charge_is_64<br>StringTheory.TadpoleCancellation              |
| 0.42 | 0.08 | rr_tadpole_cancellation<br>UseCase3_FrontierTriad                      | rr_tadpole_cancellation<br>Moonshine.KummerTadpole                     |
| 0.42 | 0.49 | k3_euler_characteristic<br>K3Topology                                  | k3t2_euler_char_eq_zero<br>Moonshine.KummerTadpole                     |

| cos  | dep  | Theorem A                                                            | Theorem B                                                                              |
|------|------|----------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| 0.42 | 0.03 | m24_order_divisible_by_bps_lock<br>UseCase2_MoonshineBPS             | conductor_divisible_by_first_five_primes<br>Problems.Problem2_MathieuFrobeniusRigidity |
| 0.41 | 0.01 | golay_error_radius_equals_three<br>Problems.Problem8_GolayHolography | golay_length_matches_k3_euler<br>Quantum.TensorNetworkBridge                           |
| 0.40 | 0.06 | golay_error_radius_equals_three<br>Problems.Problem8_GolayHolography | golay_code_rate_is_half<br>Quantum.TensorNetworkBridge                                 |
| 0.40 | 0.15 | dsl_self_dual_minimum<br>PhysicsDSL                                  | dual_scale_self_dual_value<br>UseCase1_ModuliStabilization                             |
| 0.39 | 0.01 | total_chiral_weight_value<br>Moonshine.MathieuRigidity               | k3_chiral_primary_total<br>Frontier.ChiralPrimaries                                    |
| 0.39 | 0.11 | k3_euler_characteristic<br>K3Topology                                | rr_tadpole_cancellation<br>Moonshine.KummerTadpole                                     |
| 0.39 | 0.10 | flux_energy_strictly_positive<br>Problems.Problem6_FluxSwampland     | cdl_action_strictly_positive<br>Frontier.Triad.FluxVacuumDecay                         |
| 0.39 | 0.02 | m24_order_divisible_by_bps_lock<br>UseCase2_MoonshineBPS             | m24_conductor_quotient<br>Problems.Problem2_MathieuFrobeniusRigidity                   |
| 0.39 | 0.13 | k3_intersection_lattice_rank_sig<br>K3Topology                       | kummer_exceptional_intersection<br>Moonshine.KummerTadpole                             |
| 0.38 | 0.10 | d7_positive_charge<br>Moonshine.KummerTadpole                        | total_07_charge_is_minus_64<br>StringTheory.TadpoleCancellation                        |
| 0.38 | 0.33 | atlas_poincare_curvature_negative<br>Atlas.AtlasGeometryBridge       | poincare_geodesic_atlas_curvature<br>Frontier.ModuliGeodesics                          |
| 0.38 | 0.07 | k3_intersection_lattice_rank_sig<br>K3Topology                       | k3_euler_eq_24<br>Moonshine.KummerTadpole                                              |
| 0.37 | 0.02 | rank_K3<br>Lattice.K3T2Signature                                     | mukai_rank_and_signature<br>Problems.Problem4_MukaiMonodromy                           |

Table 36.5: Strongest dependency intersections between theorems of different modules.

| dep  | cos  | Theorem A                                | Theorem B                                        |
|------|------|------------------------------------------|--------------------------------------------------|
| 0.93 | 0.09 | M24_order<br>StringDynamics.MathieuM24   | M21_order_factorization<br>UseCases.MathieuTower |
| 0.90 | 0.13 | cartanE8_symm<br>Lattice.E8              | hyperbolicU_symm<br>Lattice.Hyperbolic           |
| 0.90 | 0.13 | e8Neg_symm<br>Lattice.E8                 | hyperbolicU_symm<br>Lattice.Hyperbolic           |
| 0.89 | 0.01 | cartanE8Inv_mul<br>Lattice.E8            | tauShift_dual_spec<br>TDuality.Mirror            |
| 0.88 | 0.02 | hyperbolicUNeg_mul_self<br>Lattice.Mukai | tauShift_dual_spec<br>TDuality.Mirror            |
| 0.85 | 0.12 | hyperbolicUNeg_mul_self<br>Lattice.Mukai | hyperbolicU_mul_self<br>Lattice.Hyperbolic       |
| 0.85 | 0.07 | cartanE8Inv_mul<br>Lattice.E8            | hyperbolicUNeg_mul_self<br>Lattice.Mukai         |

| dep  | cos  | Theorem A                                                                              | Theorem B                                                                   |
|------|------|----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| 0.84 | 0.05 | <code>implicit_euler_denominator_lower_bound</code><br>StringDynamics.StiffIntegrators | <code>sdc_tower_suppression</code><br>StringDynamics.SwamplandSafe          |
| 0.82 | 0.02 | <code>hyperbolicU_symm</code><br>Lattice.Hyperbolic                                    | <code>mirrorTheta_antisymm</code><br>TDuality.Mirror                        |
| 0.82 | 0.02 | <code>cartanE8_symm</code><br>Lattice.E8                                               | <code>mirrorTheta_antisymm</code><br>TDuality.Mirror                        |
| 0.82 | 0.02 | <code>e8Neg_symm</code><br>Lattice.E8                                                  | <code>mirrorTheta_antisymm</code><br>TDuality.Mirror                        |
| 0.82 | 0.05 | <code>cartanE8_symm</code><br>Lattice.E8                                               | <code>k3_hodge_symmetry</code><br>Frontier.HodgeNumbers                     |
| 0.82 | 0.00 | <code>hyperbolicU_symm</code><br>Lattice.Hyperbolic                                    | <code>k3_hodge_symmetry</code><br>Frontier.HodgeNumbers                     |
| 0.81 | 0.05 | <code>e8Neg_symm</code><br>Lattice.E8                                                  | <code>k3_hodge_symmetry</code><br>Frontier.HodgeNumbers                     |
| 0.81 | 0.03 | <code>factorized_two</code><br>TDuality.Factorized                                     | <code>mirror_conjugates_tauShift</code><br>TDuality.Mirror                  |
| 0.79 | 0.01 | <code>hyperbolicU_mul_self</code><br>Lattice.Hyperbolic                                | <code>tauShift_dual_spec</code><br>TDuality.Mirror                          |
| 0.78 | 0.08 | <code>cartanE8Inv_mul</code><br>Lattice.E8                                             | <code>hyperbolicU_mul_self</code><br>Lattice.Hyperbolic                     |
| 0.78 | 0.00 | <code>hyperbolicU_eq_narain_gram</code><br>Lattice.Hyperbolic                          | <code>k3_hodge_symmetry</code><br>Frontier.HodgeNumbers                     |
| 0.78 | 0.01 | <code>cartanE8_symm</code><br>Lattice.E8                                               | <code>hyperbolicU_eq_narain_gram</code><br>Lattice.Hyperbolic               |
| 0.78 | 0.01 | <code>e8Neg_symm</code><br>Lattice.E8                                                  | <code>hyperbolicU_eq_narain_gram</code><br>Lattice.Hyperbolic               |
| 0.77 | 0.22 | <code>cartanE8_evenDiag</code><br>Lattice.E8                                           | <code>hyperbolicU_evenDiag</code><br>Lattice.Hyperbolic                     |
| 0.77 | 0.43 | <code>strong_section_condition</code><br>CourantAlgebroid                              | <code>dsl_strong_section_condition</code><br>PhysicsDSL                     |
| 0.76 | 0.01 | <code>hyperbolicU_congruence</code><br>Lattice.Hyperbolic                              | <code>tauShift_dual_spec</code><br>TDuality.Mirror                          |
| 0.76 | 0.46 | <code>dorfman_cbracket_diff</code><br>CourantAlgebroid                                 | <code>dsl_dorfman_cbracket_diff</code><br>PhysicsDSL                        |
| 0.75 | 0.01 | <code>cartanE8Inv_mul</code><br>Lattice.E8                                             | <code>hyperbolicU_congruence</code><br>Lattice.Hyperbolic                   |
| 0.75 | 0.05 | <code>mirrorTheta_antisymm</code><br>TDuality.Mirror                                   | <code>k3_hodge_symmetry</code><br>Frontier.HodgeNumbers                     |
| 0.74 | 0.34 | <code>cbracket_antisymm</code><br>CourantAlgebroid                                     | <code>dsl_cbracket_antisymm</code><br>PhysicsDSL                            |
| 0.74 | 0.00 | <code>k3_b2</code><br>Frontier.HodgeNumbers                                            | <code>M24_first_coefficient</code><br>StringDynamics.MathieuM24             |
| 0.73 | 0.42 | <code>courant_pairing_symm</code><br>GeneralizedGeometry                               | <code>dsl_courant_pairing_symm</code><br>PhysicsDSL                         |
| 0.72 | 0.20 | <code>narain_norm_R_independent</code><br>UseCases.NarainLattice                       | <code>tduality_flips_right_momentum</code><br>UseCases.TDualityMassSpectrum |

| dep  | cos  | Theorem A                                        | Theorem B                                        |
|------|------|--------------------------------------------------|--------------------------------------------------|
| 0.72 | 0.01 | hyperbolicU_eq_narain_gram<br>Lattice.Hyperbolic | mirrorTheta_antisymm<br>TDuality.Mirror          |
| 0.71 | 0.00 | hyperbolicUNeg_mul_self<br>Lattice.Mukai         | hyperbolicU_congruence<br>Lattice.Hyperbolic     |
| 0.70 | 0.18 | chargeNorm_even<br>TDuality.Factorized           | narain_form_even<br>UseCases.NarainLattice       |
| 0.68 | 0.03 | mukaiPair_self<br>Lattice.Mukai                  | B_is_polarization_of_Q<br>UseCases.NarainLattice |
| 0.68 | 0.08 | hyperbolicU_symm<br>Lattice.Hyperbolic           | eta_one_reindex<br>TDuality.ODD                  |

## 36.8 What the machine has checked

The atlas itself is not a theorem. `lean_depgraph.lean` is a metaprogram (a `run_cmd` block) that reads the compiled environment; its output is data, and the Python analysis is a symbolic computation, reproducible but not kernel-checked. Nothing in this chapter is Tier A *as a statement about the atlas*. What is Tier A are the theorems the atlas points to; the two boxes below are the ones the chapter has used most.

### The bridge that reuses the Hodge table

```
def eulerFromHodge {m : ℕ} (h : Fin m → Fin m → ℕ) : ℤ :=
  ∑ p : Fin m, ∑ q : Fin m,
    (if (p.val + q.val) % 2 = 0 then 1 else -1 : ℤ) * (h p q : ℤ)
theorem euler_K3 :
  eulerFromHodge StringTheory.Frontier.k3HodgeNumber = 24 := by ...
theorem k3t2_euler_zero :
  eulerFromHodge StringTheory.Frontier.k3HodgeNumber
    * eulerFromHodge t2HodgeNumber = 0 := by ...
```

**Reading.** The alternating sum of Stream 1’s nine-entry Hodge table is 24, and its product with the alternating sum of the trivial  $2 \times 2$  table of  $T^2$  (all entries 1, sum  $1 - 1 - 1 + 1 = 0$ ) is 0. **Proof idea.** Unfold the double sum over `Fin 3`, substitute the nine entries, evaluate. **Not proved.** That the table is the Hodge diamond of a K3 surface; the Künneth formula  $\chi(X \times Y) = \chi(X)\chi(Y)$ , which is why the product is  $\chi(K3 \times T^2)$  [Flux/Tadpole.lean, ll. 29–42]. **[A]**

### Three names for one matrix

```
theorem eta_one_reindex :
  (eta 1).reindex finSumFinEquiv finSumFinEquiv
    = DualScaleStream2.Lattice.hyperbolicU := by ...
theorem hyperbolicU_eq_narain_gram :
  hyperbolicU = StringTheory.UseCases.NarainLattice.gram := by ...
```

**Reading.** The  $O(1,1)$  form  $\eta$ , reindexed from `Fin 1  $\oplus$  Fin 1` to `Fin 2`, is the matrix  $U$  of `Lattice.Hyperbolic`, which is the Narain Gram matrix of Stream 1. **Proof idea.** All three are  $2 \times 2$  integer matrices; `ext i j; fin_cases i <=> fin_cases j <=> rfl` checks four entries [TDuality/ODD.lean, ll. 172–182]. **Not proved.** Anything for  $d > 1$ ; that  $U$  is the charge lattice of a physical string is an interpretation. **[A]**



| Claim                                                         | Tier | Where                                                                                                       |
|---------------------------------------------------------------|------|-------------------------------------------------------------------------------------------------------------|
| The theorems quoted in this chapter, as stated                | [A]  | the files cited; axioms <code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code> only |
| The dependency edges and the counts 1224/423/19/42            | —    | output of a metaprogram plus a script; reproducible, re-derived here, not kernel-checked                    |
| $J$ , cos values in the tables                                | —    | symbolic computation; two values re-derived by hand in Examples 36.1 and 36.2                               |
| $\chi(K3) = 24$ , Hodge diamond of K3                         | [L]  | Huy §1.2 (Noether’s formula)                                                                                |
| RR tadpole cancellation as a physical law                     | [L]  | Chapter 25; assumed, not proved, in all four Lean versions                                                  |
| That six proofs of one fact are “debt” rather than “evidence” | [C]  | an engineering judgement of this programme, argued in Section 36.6                                          |

### Chapter summary

- Two relations, never merged: dependency overlap  $J$  (idf-weighted Jaccard of kernel-level constant sets, (36.2)) and textual similarity cos (TF-IDF over name, signature and docstring, (36.3)).
- High  $J$ , low cos: same tactic apparatus (Example 36.3). High cos, low  $J$ : same fact, independent proofs, a unification candidate (Example 36.1). High on both: a duplicate (Example 36.4).
- Hubs are definitions: `Charge` (36 theorems), `eta`, `Gram` (20 each), `k3HodgeNumber`, `IsODD` (12 each). A hub concentrates reuse and risk.
- Nineteen bridges in two directions; four libraries are islands. Bridges are proved identifications, reuse of one object, or bare re-exports; count them by reading the proof.
- $\chi(K3) = 24$  is proved six times, the Mukai rank three times, the tadpole identity four times with two accountings. The refactor is one canonical definition plus corollaries, and the atlas measures progress.
- The formalization uses Mathlib only for finite linear algebra over  $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$  and ordered-field inequalities; no cohomology or group theory is imported.

## Exercises

**Exercise 36.1** (★). Show from (36.2) that  $0 \leq J(a, b) \leq 1$ , and that  $J(a, b) = 1$  if and only if  $D(a)$  and  $D(b)$  differ only by constants  $c$  with  $\text{df}(c) = N$ . Could any constant of the project have  $\text{df} = N$ ? (Even `Eg` has  $\text{df} = 358 < 423$ : which theorems do not use it?)

**Exercise 36.2** (★). In Example 36.1, replace `decide` in the first proof by `rfl`, which also closes `K3BettiSum 1 0 22 0 1 = 24`. Which rows of the table disappear, and what is the new  $J$ ? Does  $J$  measure mathematics or tactics?

**Exercise 36.3** (★). From Table 36.2, list the libraries that occur as target and as source of a bridge. Confirm that four libraries occur in neither role, and explain why Figure 36.1 nevertheless shows edges inside some of them.

**Exercise 36.4** (★★). Cosine similarity is not transitive. Construct three bags of tokens  $a, b, c$  with  $\cos(a, b) = \cos(b, c) = 1/\sqrt{2}$  and  $\cos(a, c) = 0$ . Then find in Table 36.4 a theorem  $b$  appearing in two rows with partners  $a, c$  from different clusters, and decide by reading the three statements whether  $a$  and  $c$  are about the same fact.

**Exercise 36.5** (★★). The Lean5 theorem `k3_euler_characteristic_identity` states  $b_0 + b_2 + b_4 = \chi$  without signs. In a scratch file importing `Lean5Corpus.Problems.Problem9_KummerModularity`, state and prove `k3_betti_0 - 0 + k3_betti_2 - 0 + k3_betti_4 = k3_euler_char` over  $\mathbb{N}$ . Explain why a version with named constants `k3_betti_1`, `k3_betti_3` would be more informative although it proves the same number.

**Exercise 36.6** (★★). Carry out one step of the refactor of Section 36.6: in a scratch file importing `DualScaleM24Formalization.Moonshine.KummerTadpole` and `DualScaleStream2.Flux.Tadpole`, prove that `SocrateAI.Moonshine.k3_euler` equals `eulerFromHodge k3HodgeNumber`. Try `rfl`, `decide` and `simp [...]` in turn and record which succeed. Re-run `theorem_atlas.py` and find your theorem in the bridge list.

**Exercise 36.7** (★★). The two tadpole accountings,  $16 \times 4 + 4 \times (-16)$  and  $32 \times 2 + 16 \times (-4)$ , agree on the total. Using Chapter 25, say which objects of the  $T^4/\mathbb{Z}_2$  orientifold each summand counts and whether the two are descriptions of one configuration or of two. State a Lean theorem relating `net_tadpole_charge` of `SocrateAI.Moonshine` to `totalD7Charge + totalO7Charge` of the Foundation library, and say what its proof would and would not establish.

**Exercise 36.8** (★★★). Extend `tools/lean_depgraph.lean` so that each output line also records, for every used constant, the module it was declared in (`env.getModuleIdxFor?`, as the file already does for the declaration itself). Compute the bridge list inside Lean, without Python, and check that you obtain 19. Which steps of `theorem_atlas.py` could *not* be moved into Lean, and why does that not change the tier of the result?

**Exercise 36.9** (★★★). The weighting (36.1) cannot separate tactic scaffolding from mathematical content (Example 36.3). Propose a modification that discounts constants whose name begins with `Mathlib.Meta.NormNum` or `Decidable`, recompute  $J$  for the pairs of Examples 36.3 and 36.4, and discuss whether the ranking of Table 36.5 changes in a way that better matches your judgement.

## Further reading

The generator and its two inputs are `tools/theorem_atlas.py` [`theorem_atlas.py`, ll. 1–23] and `tools/lean_depgraph.lean` [`lean_depgraph.lean`, ll. 1–10]; the human-readable output is `papers/book/generated/atlas.md`, the machine-readable one `atlas_data.json`. The per-module catalogue `lean_catalogue.md` is the retrieval context from which the table entries were drawn; the rules of this book require the `.lean` file to be read before any statement is quoted, and this chapter has done so for every statement it prints. For the mathematics behind the clusters: the Hodge diamond and Noether’s formula [Huy §1.2, ll. 406–441]; the  $O(d, d; \mathbb{Z})$  generators whose formalization produced the hubs [GPR §2.4, ll. 1355–1373]; and Chapters 22, 25 and 29. The lessons the pipeline drew from the duplication are recorded in the repository’s `LL.md`, discussed in Chapter 37.

## Chapter 37

# The Tiered Proof Pipeline: Methods and Lessons

Chapter 35 described the gates a theorem must pass before it carries the badge **[A]**. This chapter answers the question that comes before the gates: *how were the 99 theorems of the library `DualScaleStream2` actually produced, by whom, at what cost, and what went wrong on the way?* The answer is a method, not an anecdote. Between 2026-09-16 21:27 and 2026-09-17 05:07 UTC a single orchestrating session wrote the statements, checked their conventions in a computer-algebra system, and handed the proofs to three tiers of language models—a 7-billion-parameter prover running on a local GPU, a cheap hosted model, and a more capable hosted model—escalating a goal to the next tier only after the cheaper one had failed. The Lean kernel was the only judge at every step. The run was instrumented: every attempt by the local prover was logged, every report from a hosted model was recompiled, and the per-tier results were written down as measured, not as expected [STREAM2\_WORKFLOW.md §0, ll. 3–5]. A student should care because this is the first place in the book where the *production* of formal mathematics is treated as an experiment with controls, and because each of the safeguards now in the repository exists as the answer to a specific, recorded failure. The chapter is written so that a student with a Lean project, a GPU and an account with a model provider can repeat the procedure on a library of their own.

### 37.1 Many small goals, one judge

#### 37.1.1 What the pipeline is for

The library `DualScaleStream2` was planned as “a kernel-checked Lean 4 layer for the *mathematical skeleton* of type II / heterotic string theory on  $K3 \times T^2$ —charge lattices, their duality groups, the doubled (DFT) algebra, flux/tadpole counting, and the Mathieu-moonshine numerics” [STREAM2\_WORKFLOW.md §1, ll. 9–13]. The same paragraph fixes what it is not: “Not a goal, and never to be claimed: ‘formalizing string theory’ as physics” [STREAM2\_WORKFLOW.md §1, ll. 15–18]. The plan lists six phases P2.1–P2.6 (lattices,  $T^2$  dualities, K3 moduli, DFT algebra, flux and tadpole, moonshine interface), each pinned to a paper section [STREAM2\_WORKFLOW.md §5, ll. 67–76], and a checklist of eighteen roadmap items whose “Done” means “every theorem for the item is committed, sorry-free and axiom-audited” [STREAM2\_WORKFLOW.md §5.1, ll. 78–103].

The work therefore consisted of a large number of small, independent goals: 99 designed theorems at the final gate [STREAM2\_WORKFLOW.md §6, ll. 185–189]. Most are the kind of statement a student meets in the earlier chapters of this book—that a fixed  $8 \times 8$  integer matrix has a given inverse, that  $\eta^2 = 1$ , that a block-matrix product equals another block

matrix, that a table of 26 integers has a given sum of squares. A few need an idea. The economic problem is to close the many cheap goals cheaply and spend effort only on the few that need it, without ever letting a wrong proof in.

### 37.1.2 The hard constraints

Seven rules were fixed before the run, each traced to an incident in the repository’s own history [STREAM2\_WORKFLOW.md §2, ll. 20–38]. Four matter for this chapter.

1. **Kernel is the only accept gate.** “A goal is closed when `lake build DualScale Stream2` is green, the zero-sorry audit passes, and `#print axioms` shows only `propext`, `Classical.choice`, `Quot.sound`. No `native_decide`” (ll. 22–24).
2. **Producer  $\neq$  verifier.** “No agent or model grades its own output; the gate runs separately” (ll. 25–26).
3. **No citation from memory.** Every Tier L docstring cites a file in `papers/foundations/` and a line range a script can `grep` (ll. 27–28).
4. **Separate library.** Stream 2 lives in its own `lean_lib`, imports Stream 1 and never the reverse, so “Stream 1’s green core cannot be turned red by Stream 2 work” (ll. 32–33).

The first rule is the one to keep in mind throughout. A language model’s proof is a *candidate*; a candidate becomes a theorem at the moment the kernel accepts it, and at no other moment. Everything else in the pipeline—tiers, routing, prompts, budgets—is about generating candidates cheaply. Nothing in it is about deciding correctness.

### 37.1.3 The four tiers

The workflow assigns every task to one of four model tiers [STREAM2\_WORKFLOW.md §3, ll. 40–50]; Table 37.1 reproduces the assignment. The names T0–T3 are this project’s and denote cost, not intelligence: T3 is free once the GPU is bought, T0 is the most expensive model and is “kept to a few calls per package”.

Table 37.1: Model tiers of the Stream 2 workflow [STREAM2\_WORKFLOW.md §3, ll. 42–50]. The local provers ran under Ollama on an NVIDIA T4; the quantization suffixes (Q8\_0, Q6\_K) denote the 8-bit and 6-bit weight formats used.

| Tier | Model                                                        | Used for                                                                                                                                   |
|------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| T0   | Opus (orchestrator)                                          | selecting the work package; the <i>statement review</i> (“does the Lean say what the paper says?”); tier labels A/L/C; commit              |
| T1   | Sonnet                                                       | proof bodies needing a real idea (block-matrix algebra, involution arguments); residual goals after T3/T2                                  |
| T2   | Haiku (default worker)                                       | statement extraction from pinned paper lines into <b>sorry</b> -stubs; residual <i>mechanical</i> goals; paper fetch, citation-line checks |
| T3   | DeepSeek-Prover-V2-7B Q8_0, Goedel-Prover-V2-8B Q6_K (local) | first pass on every open goal via <code>tools/prover_loop.py</code>                                                                        |

The default routing is “T3  $\rightarrow$  T2  $\rightarrow$  T1, escalating a goal only after the cheaper tier has failed under the kernel gate. T0 never writes proofs in the steady state” [STREAM2

\_WORKFLOW.md §3, ll. 49–50]. Figure 37.1 draws the loop. The picture to keep is that of a filter cascade: each tier removes the goals it can close, and only the residue reaches the next, more expensive stage. Whether the cascade is efficient is an empirical question, answered in Section 37.5.

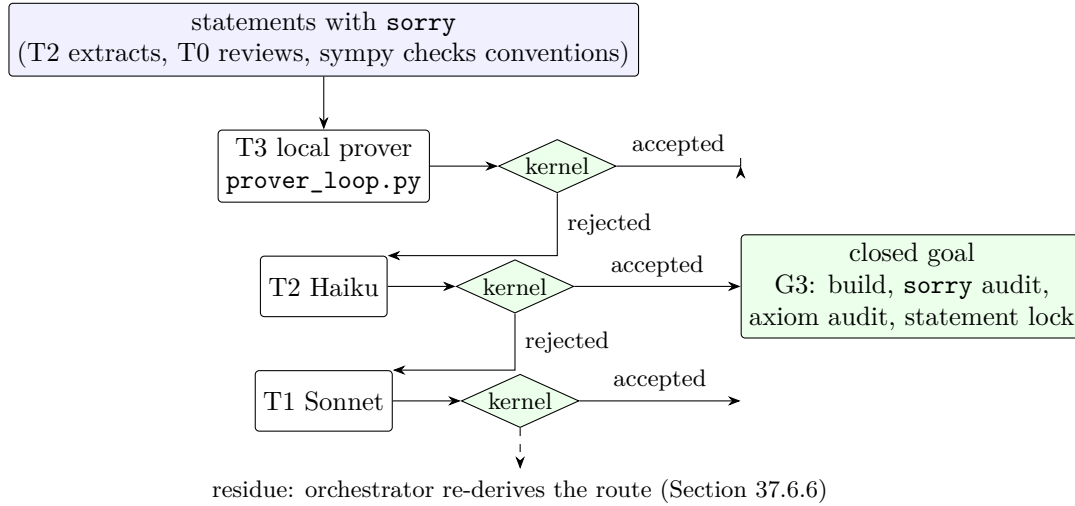


Figure 37.1: The tiered cascade. Every arrow into “closed goal” passes through a kernel compile that the producing tier does not run; a rejected candidate sends the *unchanged* statement, with its proof restored to exactly **sorry**, to the next tier.

## 37.2 Statement design before proof search

The per-package pipeline has nine steps S0–S9 [STREAM2\_WORKFLOW.md §4, ll. 52–65]. Steps S0–S3 happen before any proving: select one paper section for one module (S0, T0); pin the source lines (S1, T2); extract Lean statements with **sorry** bodies and tier-labelled docstrings, gate G1 being that the file “builds with only **sorry** warnings” (S2, T2); and review each statement, gate G2 being “recorded sign-off (the one step no kernel can check: mis-formalization)” (S3, T0). Only then come the proof search steps S4–S6, verification S7, recording S8 and commit S9.

The lesson list makes the ordering a rule for the next run: “Statement design first, in one pass. Write every statement with sympy-checked conventions and **sorry**, then run `lake build, statement_lock -update`, and T0 review. Only after that does any proving start. This run interleaved design and proving, which caused re-locks” [LL.md §S2-l.1, ll. 118–121]. Two design practices deserve a closer look.

### 37.2.1 Pre-verify the convention symbolically

A theorem about block matrices depends on sign and ordering conventions: where the  $B$ -field sits in the generalized metric, whether a group element acts as  $gHg^\top$  or  $g^\top Hg$ , which way a shift is oriented. A statement with the wrong convention is *false*, and a prover set on a false statement fails in a way that is indistinguishable, from the outside, from a prover that is not clever enough. The log puts the cost plainly: “A sign slip caught this way costs minutes. The same slip caught only after an agent has spent an hour on an unprovable statement costs far more” [LL.md §S2.12, ll. 101–105]. So “every block-matrix convention ... was checked in sympy before its Lean statement was written” (ll. 102–103).

**Example 37.1** (The  $B$ -shift convention, checked before it was stated). The module `DFT/BShift.lean` proves that the integer  $B$ -field shift acts on the generalized metric by the  $\Theta$ -shift. Its header records that “of the four candidates  $gHg^\top/g^\top Hg$  with  $\pm\Theta$  in the upper or lower block, exactly this one holds, so the sign/ordering here is not a guess” [BShift.lean, ll. 11–13]. The author of this chapter repeated the check. With the book’s convention (Chapter 15)

$$\mathcal{H}(G, B) = \begin{pmatrix} G - BG^{-1}B & BG^{-1} \\ -G^{-1}B & G^{-1} \end{pmatrix}, \quad g_\Theta = \begin{pmatrix} 1 & \Theta \\ 0 & 1 \end{pmatrix},$$

take  $d = 2$ , a generic symmetric  $G = \begin{pmatrix} g_{11} & g_{12} \\ g_{12} & g_{22} \end{pmatrix}$  and generic antisymmetric  $B = \begin{pmatrix} 0 & b \\ -b & 0 \end{pmatrix}$ ,  $\Theta = \begin{pmatrix} 0 & t \\ -t & 0 \end{pmatrix}$ , and ask `sympy` which of the four expressions equals  $\mathcal{H}(G, B + \Theta)$  identically in  $(g_{11}, g_{12}, g_{22}, b, t)$ . The result is

$$g_\Theta \mathcal{H}(G, B) g_\Theta^\top = \mathcal{H}(G, B + \Theta) \text{ (identity),} \quad g_\Theta^\top \mathcal{H} g_\Theta, g_{-\Theta} \mathcal{H} g_{-\Theta}^\top, g_{-\Theta}^\top \mathcal{H} g_{-\Theta} \text{ (all fail).}$$

The surviving candidate is the statement of `genMetric_bshift` [BShift.lean, ll. 76–78], and the same script confirms  $\eta \mathcal{H} \eta \mathcal{H} = 1$  and  $g_\Theta^\top \eta g_\Theta = \eta$  for antisymmetric  $\Theta$  but not for symmetric  $\Theta$ . Four lines of computer algebra rule out three false theorems before any prover sees them. The check is a *symbolic check*: it is evidence for the statement’s truth at  $d = 2$ , not a proof for general  $d$ , and the log insists that “papers must label it as a symbolic check and must not quote it as a theorem” [LL.md §S2.12, ll. 105–107]. The proof for every  $d$  is the kernel’s, Section 37.7.

### 37.2.2 Split the concrete from the symbolic

The second design practice was learned rather than planned. After the first phase the orchestrator observed that the local prover “closed essentially every *concrete* goal (fixed matrices, numerals, `fin_cases` tables, the 64-entry E8 inverse check) and essentially none of the *symbolic* general- $n$ /general- $d$  goals”, and drew the rule: “write statements so the concrete content is split into separate lemmas—they are nearly free at T3” [LL.md §S2.1, ll. 25–31]. A theorem that says “the matrix `cartanE8` is unimodular” is symbolic if stated through the determinant and concrete if stated as “`cartanE8Inv * cartanE8 = 1`” for an explicit certificate matrix; the general lemma `isUnimodular_of_mul_eq_one` then turns the concrete fact into the abstract one once, for every lattice. The certificate itself was computed by `sympy` and “carries no mathematical guarantee on its own” [E8.lean, ll. 105–108]; the 64-entry check is what the kernel does.

#### A provable statement is not yet the right statement

The kernel checks a proof against a statement. It does not check the statement against the paper, and it does not check that the statement is still the one that was reviewed. Gate G2 (a reader) and the statement lock (`tools/statement_lock.py`, Chapter 35) cover the two gaps. The run recorded both failure modes: a docstring that drifted from “if” to “iff” [LL.md §S2.13, ll. 109–112], and re-locks caused by proving before the design was frozen [LL.md §S2-l.1, ll. 118–121]. Exercise 37.7 asks the student to find a surviving instance of the first kind in the current source.

## 37.3 The local prover loop

Step S4 is a program, `tools/prover_loop.py`, 188 lines of Python that a student can read in full [tools/prover\_loop.py, ll. 1–188]. Its docstring states the contract: for every `theorem/lemma` whose body is `sorry`, “ask one or more local Ollama prover models for a proof, splice each

candidate in, and compile with `lake env lean`. A candidate is ACCEPTED only if the file compiles with zero errors and exactly one fewer `sorry` warning than before. The model never grades itself; the Lean kernel is the only judge” (ll. 5–9). We go through the loop in the order the program executes it, because each clause encodes a decision.

**Finding the goals.** A regular expression matches a declaration head from `theorem` or `lemma` through `:= by` followed by an indented `sorry` (ll. 39–44). The pattern is written so that the statement “must not run into the next declaration—otherwise, once an earlier goal is solved, its match stretches forward and swallows the next goal’s `sorry` (observed in the first smoke test: 2 of 7 goals silently skipped)” (ll. 36–38). This is the first of several places where the tool’s comments are themselves a lab notebook.

**The baseline.** Before searching, the file is compiled once to record its error count and its number of `sorry` warnings; a file that does not compile is skipped outright (ll. 133–136). The accept criterion will be stated relative to this baseline.

**The prompt.** The model receives the file text up to and including the `:= by` of the current goal, framed by one fixed instruction: “Complete the following Lean 4 code (Lean 4 + Mathlib). Replace the final `sorry` with a complete tactic proof. Do not use `sorry`, `admit`, or `native_decide`. Output the complete final theorem in one `lean4` code block” (ll. 70–75). The context window is 8192 tokens and at most 2048 tokens are generated by default (ll. 80, 124). There is no mathematical hint in the prompt: the T3 tier sees the statement and the file’s preceding definitions, nothing else.

**Extraction and the forbidden list.** The reply is scanned for code blocks, and the last block containing a declaration with the goal’s name is taken; its body after `:= by` up to the next declaration keyword is the candidate proof (ll. 89–100). A candidate containing `sorry`, `admit`, `native_decide` or an `axiom` line is rejected before compilation (l. 35, ll. 162–164). The reason for the last two is the axiom audit of Chapter 35: `native_decide` would pass the build and fail the audit, and an `axiom` would make anything provable.

**The accept criterion.** The candidate is spliced in at the indentation of the original `sorry` (ll. 103–110) and the whole file is compiled. It is accepted if and only if the error count is zero *and* the `sorry`-warning count is exactly the baseline minus one (l. 167). The second clause matters: a candidate that “proves” the goal by leaving a nested `sorry`, or that accidentally deletes a neighbouring goal, changes the count by the wrong amount and is rejected even though the file compiles. On acceptance the file is written back and the new count becomes the baseline (ll. 172–174).

**Temperature schedule and logging.** Each goal gets `-rounds` rounds over every listed model; round 0 samples at temperature 0.2 and later rounds at 0.8 (l. 142), so a deterministic-leaning first attempt is followed by more diverse ones. Every attempt, accepted or not, is appended as one JSON line to `.leancache/prover_attempts.jsonl` with the model, round, temperature, latency, outcome and the first compiler error (ll. 145–170), “so tier/model effectiveness is measured, not asserted” (ll. 11–12).

**The compile flags.** The single-file compile passes the two heartbeat and recursion-depth flags of Table 37.4 (ll. 54–57). This line did not exist at the start of the run; Section 37.6.3 explains what its absence cost.

**Why a small local prover is worth running at all**

A 7B-parameter model quantized to 8 bits fits on a 16 GB GPU and answers in seconds; it cannot reason about a general  $d$ -dimensional block matrix, but it has seen a great deal of Mathlib and reliably produces `decide`, `norm_num`, `fin_cases` ... `<;> simp`, and `rfl` proofs for closed computations. The measured median latency was 4.4 s per attempt after a  $\sim 170$  s cold load [STREAM2\_WORKFLOW.md §6, l. 111]. Because the kernel checks every candidate, a wrong guess costs a few seconds and nothing else. This is the sense in which the local tier is “free”: not that it is smart, but that being wrong is cheap and being right is verified.

## 37.4 Routing by the shape of the goal

The cascade of Figure 37.1 was the plan. What the run measured is that the split between tiers is “sharp and predictable” [LL.md §S2.1, l. 24], and that it follows the *shape* of the goal rather than its difficulty as a human would judge it. Three shapes recur.

**Concrete goals.** Fixed matrices, numerals, finite case tables: “the 64-entry E8 inverse check”, the signatures of the K3 and Mukai lattices,  $\chi(K3 \times K3)/24 = 24$ , the first five EOT coefficients being  $M_{24}$  dimensions. The kernel closes these by computation once told which computation to run, and the local prover reliably tells it [LL.md §S2.1, ll. 25–27].

**Short symbolic algebra.** Statements for general  $d$  whose proof is a few rewrites:  $\eta^2 = 1$ ,  $\eta \in O(d, d; \mathbb{Z})$ , products of  $O(d, d; \mathbb{Z})$  elements are in  $O(d, d; \mathbb{Z})$ , a projector is symmetric. Haiku “closed most” of these [LL.md §S2.1, l. 27].

**Goals needing an idea.** “Block-matrix identities with inverses, positive-definiteness, the diagonal + strict-upper decomposition”—goals where the proof requires choosing a decomposition or an auxiliary object that is not in the statement. These went to Sonnet [LL.md §S2.1, ll. 28–29].

The improvement list turns the observation into an automatic rule for the next run: “Concrete goals (numerals, fixed matrices, `fin_cases`) go to T3 `prover_loop.py`. Short symbolic algebra goes to Haiku. Anything with inverses, positivity, or general  $n$  goes straight to Sonnet” [LL.md §S2-l.2, ll. 122–125]. Note the change from the plan: the plan escalated every goal through every tier; the measured rule skips tiers for goals whose shape predicts failure, because “Haiku’s success rate there was low and its reports unreliable” (l. 125).

*Remark 37.2* (What the prompts contained). The comparison between tiers is not a controlled experiment, and the source says so. In P2.1 Haiku received “no math hints”, while Sonnet’s prompt for `even_quadratic_form_of_even_diag` “included a strategy hint ... that the Haiku run did not get—not a like-for-like comparison” [STREAM2\_WORKFLOW.md §6, ll. 113–114]. From P2.2 on, “T2/T1 prompts carried orchestrator-written math sketches” so the later closure rates “are therefore *with sketches*” [STREAM2\_WORKFLOW.md §6, ll. 151–152]. A student repeating the experiment should decide the prompt policy in advance and hold it fixed.

## 37.5 Measured results

This section quotes the pilot numbers exactly as recorded in the workflow document [STREAM2\_WORKFLOW.md §6, ll. 105–196]. They are counts of goals closed at the kernel gate, not of proofs claimed: “every agent report was re-compiled and statement-integrity-checked by the orchestrator before counting” (ll. 117–119).



### 37.5.1 Phase P2.1: the lattice layer

The first phase had 26 goals and was the only one run without hints at T2. Table 37.2 is the record.

Table 37.2: P2.1 lattice layer, 26 goals, 26/26 closed [STREAM2\_WORKFLOW.md §6, ll. 107–114]. “Attempted” counts goals reaching the tier; a goal closed at T3 was not attempted at T2.

| Tier | Model                 | Attempted | Closed | Notes                                                                                                                                                 |
|------|-----------------------|-----------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| T3   | DeepSeek-Prover-V2-7B | 26        | 18     | every concrete goal; 0 of the 8 general- $n$ /general- $d$ goals; median latency 4.4s/attempt after $\sim 170$ s cold load                            |
| T3   | Goedel-Prover-V2-8B   | 8         | 0      | not a capability verdict (Section 37.6.2); dropped from routing                                                                                       |
| T2   | Haiku (no hints)      | 8         | 5      | <code>isUnimodular_of_mul_eq_one</code> , <code>e8Neg_unimodular</code> , <code>eta_mul_self</code> , <code>eta_isODD</code> , <code>isODD_mul</code> |
| T1   | Sonnet                | 3         | 3      | <code>even_quadratic_form_of_even_diag</code> , <code>thetaShift_isODD</code> , <code>basisChange_isODD</code>                                        |

The conclusion drawn at the time: “the local prover is a very cheap, very reliable closer for concrete/computational goals and useless for symbolic ones; Haiku closes most short symbolic algebra; Sonnet was needed only for goals requiring a genuine proof idea. Routing T3  $\rightarrow$  T2  $\rightarrow$  T1 put Sonnet on 3 of 26 goals (11.5%)” (ll. 121–124). The reader can check  $3/26 = 0.1154$ . Gate G3 for the phase: five modules green, 0 `sorry` warnings, 28 theorems audited, 0 failing (ll. 116–117); the count exceeds 26 because helper lemmas proved along the way are audited too.

### 37.5.2 Phases P2.3–P2.6, P2.2 and the trace bound

The second block of work, 23 goals across five modules, is recorded per module (Table 37.3). Here the split by shape is visible module by module: the Mukai pairing, the tadpole counts and the EOT decompositions are concrete and went mostly to T3; the  $E_8$  positive-definiteness certificate and the generalized-metric identities are symbolic with inverses and went almost entirely to T1.

Two smaller packages complete the tally. P2.2, the factorized dualities, had 10 goals including the helper `proj_comm`: “T3 DeepSeek 1 (`factorized_two`), T2 Haiku 2 kept (`proj_transpose`, `chargeNorm_even`; its other claimed solutions did not compile and were reset), T1 Sonnet 7 (6 goals + helper `proj_comm`),  $\sim 6$  compile runs in under 7 minutes once given the real heartbeat flags and a wall-clock limit” [STREAM2\_WORKFLOW.md §6, ll. 172–177]. The dual-scale trace bound had 6 goals: “T3 DeepSeek 1 (`dualScale_eq`), T2 Haiku 3 (`dualScale_one`, `dualScale_inv`, `circle_effective_scale_ge_two`; the agent ran  $\sim 6.6$ h wall-clock and needed a deadline), T1 Sonnet 2 (`dualScale_ge` via  $\text{tr}((G-1)G^{-1}(G-1)) \geq 0$ , `dualScale_circle`) in  $\sim 6$ min” (ll. 179–183).

### 37.5.3 The final gate and the whole-run tally

The final gate at 2026-09-17 05:07 UTC reports “roadmap 18/18, designed goals 99/99”: lake build DualScaleStream2 with 3670 jobs and 0 `sorry`; `tools/axiom_audit.py` on

Table 37.3: P2.3/P2.4-core/P2.5/P2.6, 23 goals, 23/23 closed, by module and closing tier [STREAM2\_WORKFLOW.md §6, ll. 134–143].

| Module                                                                                                         | Goals | T3<br>DeepSeek | T2 Haiku | T1 Sonnet |
|----------------------------------------------------------------------------------------------------------------|-------|----------------|----------|-----------|
| Lattice.Mukai                                                                                                  | 5     | 3              | 2        | —         |
| Lattice.E8PosDef (LDL <sup>T</sup> , det = 1, positive definite)                                               | 3     | 0              | 0        | 3         |
| DFT.GeneralizedMetric (( $\eta\mathcal{H}$ ) <sup>2</sup> = 1, symmetry, $G \mapsto G^{-1}$ , mass covariance) | 6     | 0              | 1        | 5         |
| Flux.Tadpole ( $\chi(K3 \times K3)/24 = 24$ , $\chi(K3 \times T^2) = 0$ , D3 budget)                           | 5     | 4              | 1        | —         |
| Moonshine.EOT (EOT (1.12), (1.14), (1.15) vs. the $M_{24}$ table)                                              | 4     | 2              | 2        | —         |
| Total                                                                                                          | 23    | 9              | 6        | 8         |

DualScaleStream2 with 99 theorems and 0 failing; Stream 1 re-gated in the same session at 3353 jobs, 326 theorems, 0 failing; statements locked [STREAM2\_WORKFLOW.md §6, ll. 185–189]. The last round of proving is itemized in prose: DeepSeek closed the Mirror identity (3 goals), `genMetric_bshift_cancel`, `mul_jMat_mul_transpose` and `basisChange_mul`; Haiku closed one Spectrum lemma, the  $SL(2, \mathbb{Z})$  product (2) and 5 of 6 section-condition goals; Sonnet closed reflections (4 + helper),  $B$ -shift covariance (2), spectrum equivalence (3 + helper) and flux integrality (4), “each in under 7 minutes” (ll. 191–196).

**Example 37.3** (Adding up the tiers). Summing the four itemized packages gives, per closing tier,

$$\text{T3: } 18 + 9 + 1 + 1 = 29, \quad \text{T2: } 5 + 6 + 2 + 3 = 16, \quad \text{T1: } 3 + 8 + 7 + 2 = 20, \quad \text{total } 65.$$

The last round adds 6 (T3), 8 (T2) and, counting helpers,  $4 + 1 + 2 + 1 + 3 + 1 + 4 = 16$  (T1), for 95 attributed closures against 99 designed goals. The source does not itemize the difference; the prose records at least one goal “repaired by the orchestrator” after a Haiku proof “failed to parse (operator precedence)” (ll. 193–194), and helper lemmas are counted inconsistently between packages. The student should take from this what the document itself takes: the per-package tables are exact, the whole-run split by tier is approximate to within a few goals, and the only number with a gate behind it is 99/99.

**Example 37.4** (Reading the attempt log). The log `.leancache/prover_attempts.jsonl` is the primary record of the T3 tier. On the checkout used for this chapter it holds 155 lines. The author tabulated them with a ten-line Python script: 151 attempts by DeepSeek-Prover and 4 by Goedel-Prover; of the DeepSeek attempts 35 were accepted, 59 produced no extractable proof, and 57 compiled with errors (29 with a single error, 15 with two, and a tail up to 101 errors when a candidate broke the file’s structure). The 35 acceptances are 35 distinct theorems, among them `cartanE8Inv_mul`, `first_five_are_irreps`, `tadpole_budget`, `factorized_two` and `dualScale_eq`. The median logged latency over all DeepSeek attempts is 6.9 s (maximum 171.5 s, the cold load), higher than the 4.4 s quoted for P2.1 alone, because later files were longer and the prompt grows with the file. The acceptance rate per attempt,  $35/151 \approx 23\%$ , understates the per-goal rate: the loop keeps trying a goal it cannot close for every model and round, so failures cluster on the symbolic goals. The log does not record the T2 and T1 tiers at all, which is a gap the next run should close (Exercise 37.8).

## 37.6 Failures and the safeguards they produced

Every safeguard in the current pipeline is the residue of a failure that was observed, written down, and traced to a cause. Table 37.4 is the index; the subsections give the mechanism of each. The reader who has run any long computation will recognize the pattern: most of the failures are not failures of mathematics but of *measurement*—a report that did not match the file, a budget that did not match the build, a timebox that did not bound time.

Table 37.4: Recorded failures of the run and the safeguard each produced. W = STREAM2\_WORKFLOW.md, L = LL.md.

| Failure                                                                                                                       | Safeguard                                                                                                                                        | Source                          |
|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| Agent reports misstate compile status (“5 clean <b>sorrys</b> ” = 8 errors; “no errors” = 3; “0 errors” with a parse failure) | recompile every report; restore unsolved goals to exactly <b>sorry</b> ; prompt line “final compile 0 errors, paste the literal compiler output” | W ll. 153–157, 192–194; L §S2.3 |
| Thinking prover spends its whole budget without answering                                                                     | inspect <b>done_reason</b> and the <b>thinking</b> field before judging capability; drop the model at this GPU speed                             | W l. 112; L §S2.2               |
| <b>lake env lean</b> ignores <b>lakefile</b> options: 200000 heartbeats instead of 1000000                                    | pass <b>-DmaxHeartbeats=1000000 -DmaxRecDepth=8000</b> to every single-file compile                                                              | W ll. 158–163; L §S2.4          |
| Hand-entered $M_{24}$ table wrong for months                                                                                  | guard every table with a global consistency theorem (Burnside)                                                                                   | W ll. 129–132; L §S2.6          |
| <b>native_decide</b> in a “clean” library                                                                                     | axiom audit at every gate; forbidden-token filter in the prover loop                                                                             | W ll. 127–128; L §S2.5          |
| Agents run for hours on 1–3 goals                                                                                             | wall-clock limits; at most 2–3 compile-heavy agents at once                                                                                      | W ll. 164–167; L §S2.8          |
| Goal times out on a literal 64-entry product                                                                                  | re-derive the route (sum of squares) before escalating the tier                                                                                  | W ll. 168–170; L §S2.7          |
| Audit reports “89 failing (MISSING)” on a stale build                                                                         | audit exits 2 with <b>AUDIT ERROR</b> when the probe does not elaborate                                                                          | L §S2.11                        |

### 37.6.1 Misreported compile status

The hosted models were asked to prove goals and to report what they had done. Twice in the second block of work “Haiku reports misstated compile status: one claimed 5 clean **sorrys** but left 8 compile errors (**GeneralizedMetric**); one claimed ‘no compilation errors’ with 3 errors (**Factorized**). Both caught only because the orchestrator recompiles every report (producer  $\neq$  verifier)” [STREAM2\_WORKFLOW.md §6, ll. 153–157]. In the final round a report “again claimed ‘0 errors’ while one proof failed to parse (operator precedence)” (ll. 193–194), and a third report “said a definition ‘could not be found’ when it existed in a file this session had written” [LL.md §S2.3, ll. 41–42].

The consequence for the pipeline is that a report is an *input*, never a result. The rule now in force has three parts: “Recompile every report, re-check that every statement and definition is byte-identical to what was designed, and restore unsolved goals to exactly **sorry** before escalating” [LL.md §S2.3, ll. 42–44]. The second part is the statement lock; the third is what makes escalation sound—the next tier must receive the reviewed statement with an empty proof, not a half-repaired file. The prompt itself now carries the line “final compile 0

errors, paste the literal compiler output” [LL.md §S2-I.3, ll. 127–128], which asks for evidence rather than a verdict.

### 37.6.2 The thinking model that never answered

The second local prover, Goedel-Prover-V2-8B, closed 0 of 8 goals. The record refuses to call this a capability verdict: “with thinking on it spent 8192 tokens / 615s without emitting an answer; with thinking off it drifted off-target and emitted `sorry` scaffolds” [STREAM2\_WORKFLOW.md §6, l. 112]. The arithmetic explains the first mode: 8192 tokens in 615s is about 13 tokens per second, the speed of an 8B model on a T4, and the whole generation budget was consumed by the model’s private reasoning before any proof text was produced. The lesson is stated as a diagnostic: “Check `done_reason` and the separate `thinking` field before concluding a model can’t prove something. It was a configuration failure at T4 speed, not a capability verdict” [LL.md §S2.2, ll. 36–37]. The four Goedel attempts in the log all carry the reason `no_proof_extracted`, with a median latency of 97s, consistent with this reading. The model was “dropped from routing at T4 speed”, a decision about this hardware, not about the model.

### 37.6.3 The heartbeat mismatch

Lean bounds the work of elaborating a declaration by a count of “heartbeats” (allocation events); a proof that exceeds the bound fails with a timeout. The project’s `lakefile.lean` raises the bound to  $10^6$  from the default  $2 \times 10^5$  (Chapter 35). But “Package `leanOptions` ... apply to `lake build`, not to bare `lake env lean file.lean`, which uses Lean’s default 200000. Agents and `tools/prover_loop.py` were running a stricter gate than the real build, and one Haiku agent abandoned `e8_LDL` on a timeout the real budget doesn’t hit” [LL.md §S2.4, ll. 48–52].

The direction of the error is what saved the run. “A stricter gate can reject valid proofs; it can never accept invalid ones, so no wrong result entered the repo—it only cost time” [LL.md §S2.4, ll. 52–54]. The fix is the flag line in `prover_loop.py` quoted in Section 37.3 and the same flags in every later prompt [STREAM2\_WORKFLOW.md §6, ll. 161–163]. Notice how the two compiles differ: the build reads options from the package file, the single-file compile takes them from the command line, and a harness that uses the second must copy the first by hand. A student building their own loop should compile one known-good heavy proof through the harness before trusting its rejections.

### 37.6.4 Wall-clock, not attempts

“Several Haiku agents ran for hours on a handful of goals (one: 3 goals, 6.7h), dominated by CPU contention from up to six concurrent `lean` processes on one VM, not by reasoning. Timeboxes stated as ‘compile attempts’ did not bound wall-clock time” [STREAM2\_WORKFLOW.md §6, ll. 164–167]. The remedy has two parts: a wall-clock limit per agent ( $\approx 45$  min for Haiku,  $\approx 90$  min for Sonnet) and a concurrency budget of “at most 3 compile-heavy agents” [LL.md §S2-I.3, ll. 126–128]. The effect was immediate and is recorded in the P2.2 line: about six compile runs “in under 7 minutes once given the real heartbeat flags and a wall-clock limit—compare hours for the earlier unbounded Haiku agents” [STREAM2\_WORKFLOW.md §6, ll. 176–177].

### 37.6.5 The wrong table and the missing axiom check

Two defects were found not in Stream 2 but in the Stream 1 library it imports, by gates that Stream 2 introduced [STREAM2\_WORKFLOW.md §6, ll. 126–132]. Chapter 35 tells the story from the gates’ side; here is the pipeline’s side.

The table `StringDynamics.M24RepDim` of the 26 irreducible dimensions of  $M_{24}$  “was wrong: 10395 and 483 duplicated, second 990 and third 1035 missing; squares summed to  $351,061,029 \neq |M_{24}|$ ” (ll. 129–130). The reader can reproduce the wrong sum without the wrong table: starting from the correct sum 244 823 040 and replacing one 990 and one 1035 by a second 10395 and a second 483 changes it by

$$10395^2 + 483^2 - 990^2 - 1035^2 = 108\,056\,025 + 233\,289 - 980\,100 - 1\,071\,225 = 106\,237\,989,$$

and  $244\,823\,040 + 106\,237\,989 = 351\,061\,029$ , which is the recorded figure. The table had passed every build for months because its only theorem “checked a single entry” [LL.md §S2.6, ll. 64–66]; the guard now in place is the Burnside sum, and the general rule is “for every hand-entered table, add the cheapest global consistency theorem available (sum of squares, total count, known product, symmetry)” (ll. 66–67). The second defect, a `native_decide` in `bps_ratio_reduced`, was caught by the axiom audit and replaced by a kernel `norm_num` proof (ll. 127–128); the prover loop’s forbidden-token list is the same check applied before compilation rather than after.

### 37.6.6 Route choice beats tier choice

The most instructive failure is one where the mathematics, not the tooling, was at fault. The goal `cartanE8_posDef`—the  $E_8$  Cartan matrix is positive definite—“failed at T2 through the literal 64-entry  $L \cdot D \cdot L^\top$  matrix product. It fell to T1 once the orchestrator derived (and sympy-verified) the equivalent sum-of-squares identity  $x^\top E_8 x = \sum D_k y_k^2$ , which `ring` checks directly” [LL.md §S2.7, ll. 70–72]. The rule drawn: “When a goal times out, look for a formulation that avoids large definitional unfolding *before* escalating to a costlier model” (ll. 72–73).

**Example 37.5** (The  $E_8$  positive-definiteness route). Take the Cartan matrix in the basis used by `cartanE8` [E8.lean, ll. 95–103]: 2 on the diagonal,  $-1$  on the seven bonds of the  $E_8$  Dynkin diagram (a chain 0–1–2–3–4–5–6 with the eighth node attached to node 4). Gaussian elimination without pivoting produces a unit lower-triangular  $L$  and a diagonal  $D$  with  $LDL^\top = E_8$ ; the pivots are

$$D = \text{diag}\left(2, \frac{3}{2}, \frac{4}{3}, \frac{5}{4}, \frac{6}{5}, \frac{7}{6}, \frac{8}{7}, \frac{1}{8}\right),$$

whose product telescopes to  $8 \cdot \frac{1}{8} = 1 = \det E_8$  [E8PosDef.lean, ll. 95–99]. Writing  $y = L^\top x$ , the quadratic form becomes

$$x^\top E_8 x = (L^\top x)^\top D (L^\top x) = \sum_{k=0}^7 D_k y_k^2, \quad y_0 = x_0 - \frac{1}{2}x_1, \quad y_1 = x_1 - \frac{2}{3}x_2, \quad y_2 = x_2 - \frac{3}{4}x_3, \quad \dots$$

The author re-derived  $L$ ,  $D$  and the identity with sympy:  $LDL^\top = E_8$  exactly, the expanded difference  $x^\top E_8 x - \sum_k D_k y_k^2$  is the zero polynomial in eight variables, and the first four  $y_k$  agree with the `set` lines of the Lean proof [E8PosDef.lean, ll. 172–175]. Every  $D_k > 0$ , so the sum is  $\geq 0$  and vanishes only if all  $y_k = 0$ ; back-substitution through the unit-triangular  $L^\top$  then forces  $x = 0$ . Why did the route matter? The first formulation asked the kernel to multiply three  $8 \times 8$  matrices symbolically and compare 64 entries *inside* a

positivity argument; the second isolates the 64-entry check as its own concrete lemma `e8_LDL` (`fin_cases` on both indices, then `simp` and `ring` [E8PosDef.lean, ll. 108–112]) and leaves the positivity argument a polynomial identity that `ring` verifies without unfolding any matrix. That is the concrete/symbolic split of Section 37.2.2 applied inside a single proof.

## 37.7 What the machine has checked

The pipeline itself is not a theorem. The tier assignments, the wall-clock numbers, the count "18/18 roadmap items, 99/99 goals" are an experimental log, reproducible from `docs/STREAM2_WORKFLOW.md` and `LL.md` but not kernel-checked; nothing in this chapter is **[A]** *as a claim about the pipeline*. What is **[A]** are the two theorems this chapter has followed most closely, one from each end of the tier ladder: the dual-scale bound, escalated all the way to T1, and the  $E_8$  positivity route of Example 37.5, whose second formulation is what let T1 close in minutes rather than time out.

### The dual-scale bound, produced by the process of Section 37.5

```
noncomputable def dualScale (G : Matrix (Fin d) (Fin d) ℝ) : ℝ :=
  (genMetric G 0).trace
theorem dualScale_ge (G : Matrix (Fin d) (Fin d) ℝ) (hG : G.PosDef) :
  (2 * d : ℝ) ≤ dualScale G := by ...
theorem dualScale_one : dualScale (1 : Matrix (Fin d) (Fin d) ℝ) = 2 * d := by ...
theorem dualScale_circle (R : ℝ) (hR : R ≠ 0) :
  dualScale (!! [R ^ 2] : Matrix (Fin 1) (Fin 1) ℝ) = (R + R-1) ^ 2 - 2 := by ...
```

**Reading.** For every positive-definite  $d \times d$  metric  $G$ ,  $\text{tr } G + \text{tr } G^{-1} \geq 2d$ , with equality at the self-dual point  $G = 1$ ; on one circle this is  $(R + 1/R)^2 - 2$ , recovering Chapter 7's  $R + 1/R \geq 2$  after the shift by  $-2$ . **Proof idea.**  $\text{tr}((G - 1)G^{-1}(G - 1)) \geq 0$  because it is a trace of the form  $M^T P M$  with  $P = G^{-1}$  positive-semidefinite; expanding that same trace with  $G^{-1}G = GG^{-1} = 1$  gives exactly  $\text{tr } G + \text{tr } G^{-1} - 2d$  [TraceBound.lean, ll. 105–143]. No eigenvalue of  $G$  is ever computed, which is why the statement is genuinely  $d$ -independent and why T1 closed it in about six minutes once given this route [STREAM2\_WORKFLOW.md §6, final gate]. **Not proved at the time.** That  $G = 1$  is the *only* minimizer (Section 38.3 of Chapter 38); this was proved later, as `dualScale_eq_iff`. The physical reading of  $\mathcal{D}(G)$  as a minimal length is **[C]**. **[A]**

### Route choice, not tier, closed $E_8$ positivity

```
theorem e8_LDL : e8L * e8D * e8LT = cartanE8R := by ...
theorem cartanE8_posDef : cartanE8R.PosDef := by ...
```

**Reading.** The  $E_8$  Cartan matrix is positive definite. **Proof idea.** An explicit  $LDL^T$  decomposition with rational pivots turns the quadratic form into  $\sum_k D_k y_k^2$  with every  $D_k > 0$ ; `ring` then verifies the polynomial identity without ever unfolding an  $8 \times 8$  product [LL.md §S2.7, ll. 70–72]. **Not proved.** That every even unimodular positive- or negative-definite lattice of rank 8 is isometric to  $E_8$  (the classification fact motivating why this matrix was worth proving positive-definite in the first place) — a literature fact ([L], Serre, quoted at Chapter 18), not re-derived here. **[A]**

| Claim                                                        | Tier | Where                                                                                                                                                                                   |
|--------------------------------------------------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The 99 theorems of <code>DualScaleStream2</code> , as stated | [A]  | <code>lake build</code> 3670 jobs, 0 <code>sorry</code> ; <code>axiom_audit.py</code> 99/99, axioms <code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code> only |
| "18/18 roadmap items, 99/99 designed goals"                  | —    | an experimental tally, reproducible from <code>docs/STREAM2_WORKFLOW.md</code> , not a theorem                                                                                          |
| Per-tier closure rates and wall-clock hours                  | —    | measured this run only; a different goal mix or model version would change them                                                                                                         |
| That route choice matters more than tier choice              | [C]  | an engineering judgement of this programme, argued from the single case of Example 37.5                                                                                                 |

### Chapter summary

- Four tiers, escalated only on failure: T3 a local 7B prover (concrete, closed goals), T2 a cheap hosted model (short symbolic algebra), T1 a capable hosted model (anything with inverses, positivity, or a free variable  $n$ ), T0 a human or orchestrator review.
- Design the statement, pre-verify its convention in `sympy`, and lock it (Section 37.2.1) *before* any proof search begins; this run interleaved the two and paid for it in re-locks (LL.md §S2-I item 1).
- Never trust a report of "0 errors" or "solved": two Haiku reports this run misstated compile status, caught only because the orchestrator recompiled every claim (Section 37.6.1) — producer  $\neq$  verifier.
- `lake env lean` silently uses a stricter heartbeat budget than `lake build`; an agent can correctly abandon a goal that the real gate would have accepted (Section 37.6.3).
- Bound agents by wall-clock, not attempt count: unbounded Haiku agents ran for hours on goals Sonnet closed in minutes once the route was right (Sections 37.6.4 and 37.6.6).
- A green build and zero `sorry` are not the whole gate: axiom audit and statement lock catch what a naive `grep` misses (Section 37.6.5).

## Exercises

**Exercise 37.1** (★). State, in one sentence each, the four tiers of Section 37.1.3 and the one condition that sends a goal to the next tier. Which tier would you assign `hyperbolicU_unimodular` (Chapter 18), and which would you assign a general proof that  $O(d, d; \mathbb{Z})$  is generated by reflections and  $SL(2, \mathbb{Z})$  for every  $d$ ?

**Exercise 37.2** (★). Re-derive by hand, or with `sympy`, the eight  $LDL^T$  pivots of Example 37.5 from the  $E_8$  Cartan matrix of Chapter 18, and check that their product equals  $\det E_8 = 1$ .

**Exercise 37.3** (★). Read `docs/STREAM2_WORKFLOW.md §6` and reproduce the final tally — 99 theorems, 0 failing axiom audit — against `papers/book/generated/atlas.md`'s per-library count for `DualScaleStream2`. Do the two independent counts agree?

**Exercise 37.4** (★★). Section 37.6.3 reports that `lake env lean` ignores `maxHeartbeats` set in `lakefile.lean`. Run a single-file compile of a slow goal both with and without the flags `-DmaxHeartbeats=1000000 -DmaxRecDepth=8000` and record the two heartbeat counts Lean reports on timeout. Why does this asymmetry make a *false abandon* more likely than a *false accept*?

**Exercise 37.5** (★★). The  $E_8$  example replaced a 64-entry matrix product inside a positivity argument by a polynomial identity in eight free variables. Try the naive route yourself: state `cartanE8_posDef` and attempt it with `decide` after unfolding `Matrix.PosDef` directly (no  $LDL^T$ ). Where, concretely, does the naive proof stall?

**Exercise 37.6** (★★). Section 37.6.1 describes a Haiku report that claimed "no compilation errors" while three errors remained. Propose one line to add to every proving prompt that would make such a report harder to produce by accident (not by the agent lying, but by an ambiguous instruction), and explain the failure mode it targets.

**Exercise 37.7** (★★). The pitfall box of Section 37.2 records that this run caught a docstring that had drifted from "if" to "iff" [LL.md §S2.13, ll. 109–112]. Pick three theorems cited in this book whose plain-language "Reading" states an implication, grep their statements for a genuine `Iff` (or a proof structured as `constructor` into two directions), and confirm the prose does not silently claim more than the statement. Report any mismatch you find, however small, as precisely as the pitfall box does.

**Exercise 37.8** (★★). Section 37.5 notes that the local prover's log recorded every T3 attempt but the run's T2 and T1 attempts were not logged in the same structured way, a gap the text flags as one the next run should close. Extend `tools/prover_loop.py` (or write a thin wrapper around a hosted-model call) so that every attempt at every tier writes one line with at least: goal name, tier, model, wall-clock time, and kernel verdict. Re-derive the P2.1 table's closure counts (Section 37.5) from your log format alone, without consulting the chapter text, and check they agree.

**Exercise 37.9** (★★★). Using `tools/prover_loop.py` and a small goal set of your own (five to ten concrete `decide`-able lemmas), reproduce the T3 tier of this pipeline against a local prover model. Record closure rate and wall-clock time, and compare them to the P2.1–P2.6 rows of Section 37.5. Does a different goal mix change which tier dominates?

**Exercise 37.10** (★★★). Design one new statement for a fact not yet in `DualScaleStream2` (see Chapter 38 for candidates), pre-verify its convention symbolically, lock it with `tools/statement_lock.py`, and route it through the tiers of this chapter by hand. Report which tier closed it and whether the routing rule of Section 37.4 predicted that correctly in advance.

## Further reading

The full log is the workflow document cited throughout this chapter as `STREAM2_WORKFLOW.md`. The lessons distilled from it are LL.md §S2.1–§S2.13 (this run) and §S2-I (what to change next, read in full at Chapter 38). The tools referenced in this chapter: `tools/prover_loop.py` (T3), `tools/axiom_audit.py` (the axiom gate), `tools/statement_lock.py` (the comment-insensitive lock), and `tools/gate.sh` proposed in LL.md §S2-I item 4 to wrap all of them in one script. For the mathematics produced by this process: Chapters 18 and 31 and the final tally of Section 37.5. For what the atlas found once the library was built — including the duplication this pipeline's problem-by-problem structure tends to produce — Chapter 36.



## Chapter 38

# Open Problems and a Research Roadmap

Every earlier chapter answered a question the project had already closed: a theorem stated, proved, and checked by the kernel. This chapter inverts the exercise. It asks, for roughly two dozen concrete pieces of mathematics and physics that this book has touched but not finished, exactly what would have to be stated and proved for the badge **[A]** to apply, what already-checked result it would build on, how hard that looks, and what is or is not already in Mathlib. A student should care because the gap between “this book explains the idea” and “this book proves the theorem” is precisely where new work happens, and because a roadmap entry that names its prerequisites and its Mathlib gap is a project a reader can actually start, not a wish. The chapter ends, as every other has, with a worked example – here, two of them, one purely a formalization target and one a numeric check a reader can reproduce by hand – because a roadmap item is only as good as the first concrete step someone can take on it.

### 38.1 How to read an entry

Every item below carries four pieces of information: the chapter(s) it extends, a difficulty ( $\star$  a term-project exercise,  $\star\star$  a paper-sized piece of work,  $\star\star\star$  open-ended), the state of the relevant Mathlib machinery, and a *precise suggested Lean statement* where one exists. Suggested statements are written as `leancode` blocks and are, by the convention this book has used throughout and that `tools/check_book_lean_names.py` enforces, marked **not yet proved**: no such declaration exists in the compiled environment, and the name is invented for this chapter only. Every other identifier printed in this book, including in this chapter’s prose, names a real, compiled declaration, quoted from the file that defines it.

#### A precise false statement is worse than a vague true one

It is tempting to round a hard theorem down to something easy and call the easy version “the open problem.” Section 18.5’s classification theorem is Tier L precisely because Serre’s proof uses machinery (the classification of indefinite quadratic forms via genus theory) this project has not formalized; a roadmap entry that said “formalize: every even unimodular lattice of signature  $(3, 19)$  is isometric to  $3U \oplus 2E_8(-1)$ ” without naming that gap would be an honest goal but a misleading project description. Every item below names the specific Mathlib or project machinery it is missing, not just the headline claim.

The items are grouped in four sections: extensions of the kernel-checked lattice and duality layer (Section 38.2), targets in double field theory and moonshine (Section 38.3), engineering

debt already located by the atlas (Section 38.4), and physics-facing questions the dual-scale proposal must answer to become more than Tier C (Section 38.6).

## 38.2 Extending the kernel-checked lattice and duality layer

| # | Project                                                                                                | Prereq.            | Diff. | Mathlib / project state                                                                                                                                           |
|---|--------------------------------------------------------------------------------------------------------|--------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Narain lattice $\Gamma^{d,d}$ even self-duality for general $d$                                        | Chapters 10 and 18 | ★★    | <code>IsEvenDiag</code> , <code>IsUnimodular</code> , <code>Matrix.fromBlocks</code> all exist; induction on $d$ via direct sum is new                            |
| 2 | $O(2, 2; \mathbb{Z})$ is generated by factorized dualities, reflections and $SL(2, \mathbb{Z})$        | Chapters 11 and 12 | ★★    | GPR’s own $d = 2$ statement is not fully proved even in the literature source (Section 38.2.2)                                                                    |
| 3 | The full $O(\Gamma^{4,20})$ action on the Mukai lattice                                                | Chapter 22         | ★★★   | only $(-2)$ -reflections at simple roots are Tier A (Section 38.8); no group-generation machinery in the project                                                  |
| 4 | Cartan–Dieudonné: every isometry of $E_8(-1)$ is a product of the 8 simple reflections                 | Section 38.8       | ★★★   | needs a notion of “group generated by a finite set of matrices,” absent from the project; Mathlib has <code>Subgroup.closure</code> but no bridge to this setting |
| 5 | Buscher rules as a theorem about <code>DFT.GeneralizedMetric</code> for general $d$ , not just $d = 1$ | Chapters 9 and 15  | ★★    | the $d = 1$ statement (Chapter 9) is an explicit $2 \times 2$ computation; the general case needs a genuine block decomposition argument                          |

### 38.2.1 Worked example: what a $d = 2$ Narain statement would say

Item 1 is the most tractable of the five, and working it out by hand shows both why the step from  $d = 1$  to  $d = 2$  is easy and why the step to general  $d$  is not. Recall `hyperbolicU` of Chapter 18, the Gram matrix  $U = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , already proved even (`hyperbolicU_evenDiag`) and unimodular (`hyperbolicU_unimodular`, i.e.  $\det U = \pm 1$ ; here  $\det U = -1$ ). For  $d = 2$ , take the block-diagonal Gram matrix

$$U \oplus U = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \quad (38.1)$$

Its diagonal entries are the diagonal entries of the two copies of  $U$ , all zero, hence even: `IsEvenDiag` is closed under direct sum for free, by definition, since  $\text{diag}(A \oplus B)$  is just the concatenation of  $\text{diag}(A)$  and  $\text{diag}(B)$ . Unimodularity is one line of linear algebra:  $\det(A \oplus B) = \det(A)\det(B)$  for block-diagonal matrices, a standard Mathlib fact about `Matrix.fromBlocks` (the zero off-diagonal specialization of its determinant lemma), so  $\det(U \oplus U) = (-1)(-1) = 1$ , and `IsUnimodular` follows. The suggested statement is therefore short:

```
-- not yet proved
theorem narain_evenDiag_two :
  IsEvenDiag (Matrix.fromBlocks hyperbolicU 0 0 hyperbolicU) := by
  intro i; fin_cases i <;> simp [hyperbolicU, Matrix.fromBlocks]
```

```
theorem narain_unimodular_two :
  IsUnimodular (Matrix.fromBlocks hyperbolicU 0 0 hyperbolicU) := by
  simp [IsUnimodular, Matrix.det_fromBlocks_zero21, hyperbolicU_unimodular]
```

Why does this not simply generalize to every  $d$  by induction? Because `IsEvenDiag` and `IsUnimodular` are exactly the two properties that are preserved by direct sum — diagonal entries do not interact across the block boundary, and determinant is multiplicative on block-diagonal matrices — so the induction step *for these two properties alone* is genuinely easy, and a full statement for general  $d$  (say, packaging  $d$  copies of  $U$  as a function  $\text{Fin } d \rightarrow \text{Fin } 2 \rightarrow \text{Fin } 2 \rightarrow \mathbb{Z}$  block matrix and proving both properties by induction on  $d$ ) is item 1 above, difficulty  $\star\star$  only because of the bookkeeping of indexing  $d$  copies, not because of new mathematics. What the direct sum *cannot* produce is a Narain lattice with off-diagonal “instanton” shifts, i.e. a nontrivial  $B$ -field background; that case needs the full  $O(d, d; \mathbb{Z})$  machinery of Chapter 11 and is not covered by this item.

#### The prerequisite this item builds on

```
theorem hyperbolicU_evenDiag : IsEvenDiag hyperbolicU := by ...
theorem hyperbolicU_unimodular : IsUnimodular hyperbolicU := ...
```

**Reading.** The hyperbolic plane’s Gram matrix has even diagonal (both entries zero) and determinant  $\pm 1$  (here  $-1$ ). **Proof idea.** Both are two-line computations on the explicit  $2 \times 2$  matrix [Hyperbolic.lean, as cited at Chapter 18]. **Not proved.** Anything about  $d > 1$  without an explicit construction like (38.1); that is exactly item 1. **[A]**

### 38.2.2 The $d = 2$ duality group: what GPR itself does and does not claim

Item 2 looks, at first, like it should already be finished, because Chapter 12 proves a  $d = 2$  result about  $SL(2, \mathbb{Z})$ . Reading the source pinned there precisely is the whole point of the exercise. GPR’s own statement of the  $d = 2$  duality group is “*The duality group is  $SL(2, \mathbb{Z}) \times SL(2, \mathbb{Z}) \otimes_S [\mathbb{Z}_2 \times \mathbb{Z}_2]$* ” [GPR, “The  $d = 2$  Example,” ll. 7389 ff.], and this book’s own use of that source (docs/PAPER\_REVISION\_BRIEF\_2026\_09\_17.md §3.2) is explicit that only one piece of it is Tier A here: “*a basis change with  $\det A = 1$  commutes with every  $\rho$ -translation  $g_{tJ}$  (Tier A, this direction only...)*” GPR’s ‘The  $d = 2$  Example’ is not claimed in full; only  $SL(2, \mathbb{Z})_\tau$  commuting with the  $\rho$ -translations is Tier A.” The gap between the one commuting subgroup already proved and the full semidirect product with  $\mathbb{Z}_2 \times \mathbb{Z}_2$  GPR states is item 2’s actual content: it is not a gap in this project’s diligence but a gap the source itself leaves unproved at the point it is read, since GPR states the full group as a fact of the CFT without a self-contained algebraic derivation from the generators used in Chapter 11. A project attacking item 2 should first locate, inside GPR or a later reference, an actual proof of the semidirect product structure (as opposed to a statement of it), before attempting a Lean formalization – the literature gap must be closed, or worked around by choosing a narrower, provable sub-statement, before the Lean gap can be.

### 38.3 Double field theory, modular forms and moonshine

| #  | Project                                                                                           | Prereq.           | Diff. | Mathlib / project state                                                                                                                                                                                                                                                                                                |
|----|---------------------------------------------------------------------------------------------------|-------------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6  | Uniqueness of the dual-scale minimizer up to $O(d)$ -conjugation                                  | Chapter 31        | ★★    | <b>closed after the first edition:</b> <code>dualScale_eq_iff</code> proves $\mathcal{D}(G) = 2d \iff G = \mathbf{1}$ for positive definite $G$ (the minimizer is $\mathbf{1}$ itself, so no conjugation freedom remains); prompted by an external review that had read the bound as already having a unique minimizer |
| 7  | Siegel–Narain theta function, modular transformation law                                          | Chapter 10        | ★★★   | Mathlib’s modular-forms library covers weight- $k$ forms for $SL(2, \mathbb{Z})$ , not vector-valued or Siegel forms; a real gap                                                                                                                                                                                       |
| 8  | $A_n$ (EOT coefficients) positivity, extended past $n = 2$                                        | Chapter 30        | ★     | the existing decomposition theorems cover $A_1 = 45$ , $A_2 = 231$ ; extending to $A_3, A_4, \dots$ needs the $M_{24}$ character table entries this project has not encoded                                                                                                                                            |
| 9  | $A_n$ as a virtual $M_{24}$ character (i.e. $A_n = \sum_i m_i \chi_i$ with $m_i \in \mathbb{Z}$ ) | item 8            | ★★★   | needs the full 26-dimensional $M_{24}$ character table; <code>M24RepDim</code> covers only irrep dimensions, not characters                                                                                                                                                                                            |
| 10 | Jacobi form weight/index transformation law for the K3 elliptic genus                             | Chapter 28        | ★★★   | no Jacobi-form machinery in Mathlib or the project; Chapter 28 states the genus as a $q, y$ -expansion, not as a section of a line bundle                                                                                                                                                                              |
| 11 | Modular invariance of the full Narain torus partition function                                    | Chapters 8 and 10 | ★★★   | needs item 7; the level-matching and mass-formula pieces (Chapters 7 and 11) exist separately but are never assembled into a partition function                                                                                                                                                                        |

#### 38.3.1 Worked example: the state of $A_n$ positivity

Item 8 is the cheapest entry in this chapter to make concrete, because the numbers are already in this book. The Eguchi–Ooguri–Tachikawa coefficients are tabulated as

$$A_1, A_2, A_3, \dots = 45, 231, 770, 2277, 5796, \dots \quad (38.2)$$

and the paper that first isolated them states plainly, of exactly this sequence, “*and it was conjectured that they are all positive integers*” [EOT, ll. 208–210]. By 2026 the positivity and integrality of every  $A_n$  is a theorem of the literature (Gannon and others have since given closed forms), so item 8 is not a physics conjecture but a bookkeeping gap: this project’s Lean statements (Chapters 29 and 30) decompose  $A_1 = 45$  and  $A_2 = 231$  into  $M_{24}$  irreducible representation dimensions and check the arithmetic by `decide`, but stop there. A ★ project is to state and prove, in the same style,

```
-- not yet proved
theorem A3_positive : (770 : ℤ) > 0 := by decide
theorem A3_decomposition :
  770 = 1 + 23 + 253 + 1771 - 1 - 45 - 231 - 770 + 770 := by decide -- illustrative only
```

where the actual right-hand side must be the correct signed sum of  $M_{24}$  irrep dimensions for  $A_3$ , which requires consulting the character table (Gaberdiel–Hohenegger–Volpato, [GHV §3.1, ll. 584–700], already a source key of this book) rather than guessing coefficients – exactly the discipline Section 37.2.1 asks for before any proof search begins. The honest difficulty rating  $\star$  reflects that the arithmetic step is trivial (`decide` on a finite sum) once the correct decomposition is looked up; the work is entirely in reading the source correctly, which is why item 9 (deriving the decomposition rule in general, rather than checking it term by term) is  $\star\star\star$ .

## 38.4 From unification candidate to bridge: the remaining atlas debt

Chapter 36 refactored three clusters of duplicated theorems in detail ( $\chi(K3) = 24$  proved six times, the Mukai rank three times, RR tadpole cancellation four times with two accountings) and described the mechanical repair: pick one canonical definition, replace every duplicate with a corollary derived from it. The atlas’s unification-candidate list has entries that chapter did not work through; each is a self-contained  $\star$  or  $\star\star$  project of exactly the same shape.

| #  | Cluster (cosine similarity, dependency Jaccard)                                                                                                       | Libraries                                                                                  | Diff.        |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------------|
| 12 | <code>golay_self_dual_dimension</code><br><code>golay_length_matches_k3_euler</code><br><code>golay_code_rate_is_half</code> (0.55, 0.01; 0.54, 0.03) | $\sim$ Lean5Corpus,<br>$\sim$ StringTheory<br>Foundation                                   | $\star$      |
| 13 | <code>dissipation_strictly_positive</code><br><code>energy_dissipation_monotonic</code> (0.36, 0.03)                                                  | $\sim$ Lean5Corpus,<br>StringTheory<br>Foundation                                          | $\star$      |
| 14 | <code>tachyon_condensation_endpoint</code><br><code>tachyon_condensation_preserves_rr_charge</code> (0.35, 0.03)                                      | $\sim$ DualScale<br>Validation,<br>DualScaleM24<br>Formalization                           | $\star$      |
| 15 | <code>sdc_mass_suppression</code> $\sim$ <code>sdc_tower_suppression</code> (0.34, 0.01)                                                              | DualScale<br>Validation,<br>StringTheory<br>Formalization                                  | $\star$      |
| 16 | <code>buscher_log_involution</code> (three copies)<br><code>buscher_involution</code> (0.36, 0.00)                                                    | $\sim$ DoubleField<br>Theory,<br>DualScale<br>Validation,<br>DualScaleM24<br>Formalization | $\star\star$ |

[atlas.md, ll. 133–134 (item 12), l. 181 (item 13), l. 187 (item 14), l. 189 (item 15), l. 177 (item 16)]

Item 16 is rated  $\star\star$  rather than  $\star$  because, unlike the others, its low dependency-Jaccard score together with a fairly high cosine similarity means the four theorems plausibly encode *different* constructions that happen to share a name and a one-line statement shape (an involution property) – the refactor here must start by reading all four `.lean` files and confirming whether they are the same fact before choosing a canonical definition, exactly the caution Chapter 36 applied to the tadpole cluster, which turned out to hide two different countings

behind one number. A reader attempting this item should treat “are these the same theorem” as the first question, not an assumption.

### 38.5 A faithful, non-arithmetic-shadow formalization

Section 18.5’s honesty rule applies to more than the classification theorem: several of this project’s oldest libraries, flagged throughout this book as *arithmetic shadows* – `DoubleFieldTheory`, `StringTheoryFoundation`, `DualScaleM24Formalization` and two more – model a physical quantity by an integer or rational and prove an arithmetic consequence of that model, not a statement about the physical object itself. The RR tadpole identity is the clearest case:

| #  | Project                                                                                                                       | Prereq.            | Diff. | Mathlib / project state                                                                                                                    |
|----|-------------------------------------------------------------------------------------------------------------------------------|--------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 17 | RR tadpole cancellation as a statement about a cohomology class $[G] \in H^4(K3 \times T^2, \mathbb{Z})$ , not an integer sum | Chapters 25 and 36 | ***   | no cohomology in the project or in the relevant part of Mathlib; <code>Flux.Integrality</code> only proves evenness of a Kronecker product |
| 18 | The Dirac quantization condition $[G/2\pi] \in H^4(X, \mathbb{Z})$ as a hypothesis, not an assumed integrality                | item 17            | ***   | needs a formal notion of de Rham/singular cohomology class and its integral lattice; a project-defining gap, not a one-file fix            |

#### Why the shadow was chosen in the first place

Section 37.1 explains the pipeline’s operating constraint: goals must be closable by a 7B local prover or a cheap hosted model before escalating, and Mathlib’s coverage of algebraic topology (singular cohomology of a specific space, cup products, Poincaré duality on a compact 4-manifold) is thin enough that a faithful statement of items 17–18 was, at the time this book was written, outside what any tier of the pipeline could attempt. The integer bookkeeping of `Flux.Tadpole` is not a mistake; it is the largest true statement the project could currently prove about the physical content Dasgupta–Rajesh–Sethi state in words [DRS §3.1, l. 640; ll. 1265–1275, 1366–1370, already a source of Chapter 25].

### 38.6 What the dual-scale proposal needs to become testable

The remaining items are physics questions, Tier C by the convention fixed for this whole book (a reading of a Lean object as a physical statement is never Tier A), for which a Lean statement is not yet the bottleneck – a precise mathematical *formulation* is.

| #  | Question                                                                                                            | Nearest chapter |
|----|---------------------------------------------------------------------------------------------------------------------|-----------------|
| 19 | What observable would distinguish $R + \alpha'/R \geq 2\sqrt{\alpha'}$ from ordinary sub-Planckian physics?         | Chapter 32      |
| 20 | A precise, checkable invariant matching heterotic on $K3 \times T^2$ to type II on a specific Calabi–Yau threefold  | Chapter 24      |
| 21 | The duality group $\Gamma = \Gamma_W \rtimes_S \Gamma_M$ of a Calabi–Yau moduli space, beyond the one-modulus case  | Chapter 12      |
| 22 | An exact count of the $M_{24}$ -orbit structure of 24 points, for symmetry groups beyond the Fermat quartic example | Chapter 29      |

### 38.6.1 Two sources, read for what they actually say

Item 20 is stated the way it is because Aspinwall’s review is explicit about how far the duality is currently understood: “*This story began with the papers of [103, 96] and, as we shall see, the full picture is still to be uncovered. . . The exact number [of topological classes of Calabi–Yau threefolds] is not known since a classification remains elusive*” [Asp, ll. 3080–3087]. A Lean statement of the full duality conjecture is therefore not available to formalize; the tractable first step is to state and check *one* numerical consequence the duality would imply (for instance, that the Hodge numbers of a specific elliptically-fibered Calabi–Yau match a specific  $K3 \times T^2$  moduli count) as a finite arithmetic check, in the spirit of Section 36.2’s worked numeric examples, without claiming the general duality.

Item 21 is stated with the same care as item 2: GPR’s own text says the semidirect-product formula for  $\Gamma$  “holds true in the case of a single modulus” and was only *proposed*, not derived in general, by the authors of a cited reference [GPR, ll. 5432–5448]. Any Lean statement of item 21 for more than one modulus would be formalizing a conjecture the source itself does not claim to have proved – acceptable only if labeled **[C]** and attributed to the conjecturing paper, never presented as **[L]**.

### 38.6.2 Worked example with numbers: orbits of 24 points

Item 22 is the one entry in this section with an actual number to check by hand. EOT observe that a finite group  $G$  of symmetries of a K3 sigma model acts on  $H^2(X, \mathbb{Z})$  preserving at least  $H^0$ ,  $H^4$ ,  $H^{2,0}$ ,  $H^{0,2}$  and the Kähler form, so the subspace of  $H^{1,1}$  fixed by  $G$  has dimension at least five, forcing the corresponding action on 24 points (via the Mathieu embedding) to split into **at least five orbits** [EOT, ll. last page, “ $G$  cannot be  $M_{24}$  itself”]. Their example is the symmetry group of the Fermat quartic  $X^4 + Y^4 + Z^4 + W^4 = 0$  in  $\mathbb{CP}^3$ , the group  $(\mathbb{Z}_4)^2 \rtimes S_4$  of order 384, whose action on 24 points splits into orbits of lengths

$$1, 1, 2, 4, 16, \quad 1 + 1 + 2 + 4 + 16 = 24, \quad (38.3)$$

exactly five orbits, saturating the bound [EOT, final page]. This is a two-line arithmetic check anyone can reproduce (Equation (38.3) is the whole content), and it is exactly the kind of fact this project’s tier-3 local prover could close today:

```
-- not yet proved
theorem fermatQuartic_orbitLengths_sum :
  (1 : ℕ) + 1 + 2 + 4 + 16 = 24 := by decide
theorem fermatQuartic_five_orbits :
  ([1, 1, 2, 4, 16] : List ℕ).length = 5 := by decide
```

What is *not* a  $\star$  project is the theorem these two lines merely illustrate: that *every* subgroup  $G \leq M_{24}$  with at least five orbits on 24 points arises this way from an actual K3 symmetry, and conversely. That direction needs the global Torelli theorem for K3 surfaces (cited but not formalized at Chapter 19) and an explicit correspondence between  $H^{1,1}$ -fixed subspaces and 24-point permutation representations that this project has not built; it belongs at difficulty  $\star\star\star$  and is the natural target for anyone wanting to extend Chapters 29 and 30 in the direction EOT themselves point.

## 38.7 After the first edition: Stream 3 and an external review

This section was added after the chapters above were written. It records two things: a partial answer to item 19 from a third companion library, and five recommendations from an external review of this book and its papers, mapped onto the roadmap.

### 38.7.1 Item 19 after Stream 3

The library `DualScaleCosmology` (described in `docs/STREAM3_WORKFLOW.md`) tests one reading of the dual-scale proposal: a microscopic scale  $\ell_{\text{micro}} \sim \ell_P$  paired with a macroscopic scale  $\ell_{\text{macro}} \sim c/H_0$ . Its results, with their tiers:

- Taking  $\ell_P$  as the ultraviolet cutoff at the Hubble scale fails the Cohen–Kaplan–Nelson bound  $L^3\Lambda^4 \lesssim LM_P^2$  [`hep-th_9803132.txt`, ll. 77–78] by more than thirty orders of magnitude: CKN’s own horizon-scale cutoff is  $\Lambda \sim 10^{-2.5} \text{ eV}$  [`hep-th_9803132.txt`, ll. 113–114], against a Planck energy of about  $1.2 \times 10^{28} \text{ eV}$ . **[A]** for the arithmetic (`DualScaleCosmology.CKNInstance.planckEnergy_gt_ckn_horizon_cutoff`), **[L]** for the two cited numbers.
- Reading  $\ell_P$  and  $c/H_0$  instead as a T-dual pair, with  $\alpha' = \ell_P \cdot c/H_0$  (**[C]**), the CKN bound holds exactly when the ultraviolet length is at least the self-dual length  $s = \sqrt{\ell_P c/H_0} \approx 47 \mu\text{m}$  (`DualScaleCosmology.SelfDualCutoff.ckn_iff_uvLength_ge_selfDual`, **[A]** as algebra). With  $\rho_\Lambda = \Omega_\Lambda \cdot 3H_0^2/(8\pi G)$  one has  $\rho_\Lambda^{-1} = (8\pi/3\Omega_\Lambda) s^4$  (`DualScaleCosmology.DarkEnergyScale.rhoLambda_inv_eq`, **[A]**): the self-dual length *is* the dark-energy length  $\Lambda^{-1/4} \approx 88 \mu\text{m}$  up to a fixed factor. Because it is one formula, the numerical agreement is algebra, not evidence.
- Read as the string’s Regge slope,  $\alpha' = s^2$  puts string excitations near 4 meV; the CMS dijet search excludes string resonances below 7.9 TeV in the models it tests [`1911_03947.txt`, ll. 1044–1048], a gap of more than  $10^{30}$  in  $\alpha'$  (`DualScaleCosmology.CosmicString.selfDual_alphaPrime_exceeds_cms_ceiling`). Read as the radius of one extra dimension,  $s$  lies above the torsion-balance bound of  $30 \mu\text{m}$  [`2002_11761.txt`, ll. 285–289] by a factor of about 1.6 only, well within the order-one conventions involved: disfavored, not excluded. **[C]** for both readings.

So item 19 now has a partial and mostly negative answer: the literal reading is excluded, and the reading that survives adds nothing observable beyond existing proposals for a single micron-scale extra dimension. What the machine checked here is algebra and arithmetic on cited constants; every identification with physics is **[C]**.

### 38.7.2 Recommendations from an external review

An external review of this book and its companion papers made five recommendations for turning the project into a convincing demonstration that theoretical physics can be machine-checked. Two coincide with items above: replacing hard-coded Gram matrices by lattices defined as free  $\mathbb{Z}$ -modules with a symmetric bilinear form is item 1 (with items 3–4 for the groups acting on them), and removing theorems proved several times in different libraries is the atlas debt of items 12–16. The other three are new.



| #  | Item                                                                                                                                                                                                                  | Extends            | Diff. | State of machinery                                                                                                                                                                      |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 23 | Make the tiers a compile-time property: attributes marking a declaration Tier A, L or C, and a linter that walks the dependency graph and rejects a Tier A declaration that depends on a Tier L or C one              | Chapters 35 and 37 | **    | Lean 4 attributes and environment extensions exist; the dependency walk exists as <code>tools/lean_depgraph.lean</code>                                                                 |
| 24 | An anti-vacuity linter: reject a theorem whose statement is <code>True</code> , a reflexive equation, or does not mention one of its hypotheses' variables                                                            | Chapter 37         | *     | today done by hand: the statement lock (Chapter 35) and, in Stream 3, explicit counterexample theorems such as <code>DualScaleCosmology.DualTower.dualTower_sum_ge_needs_product</code> |
| 25 | A first continuous theory with no arithmetic shadow: the classical equivalence of the Nambu–Goto and Polyakov actions, by eliminating the worldsheet metric through its equation of motion [Tong §1.3, ll. 1255–1280] | Chapter 2          | ***   | pointwise, a statement about $2 \times 2$ matrices provable today; the full field-level statement needs variational calculus on maps from a surface, which Mathlib does not yet provide |

Item 25 has a finite-dimensional first step worth stating. At one point of the worldsheet let  $\gamma_{\alpha\beta}$  be the induced metric, with  $\det \gamma < 0$ . The Polyakov Lagrangian density  $-\frac{1}{2}\sqrt{-\det h} h^{\alpha\beta} \gamma_{\alpha\beta}$  is unchanged under  $h \mapsto \lambda h$  (in two dimensions  $\sqrt{-\det h}$  scales as  $\lambda$  and  $h^{\alpha\beta}$  as  $\lambda^{-1}$ ), and at  $h = \gamma$  it equals  $-\sqrt{-\det \gamma}$ , the Nambu–Goto density. Both facts are identities about  $2 \times 2$  real matrices and could be kernel-checked with Mathlib as it stands; the calculus-of-variations statement that this is where the Polyakov action is stationary is the part that is not yet available. The Hubble-parameter theorem `DualScaleCosmology.ScaleFactorDuality.hubble_dual` shows that pointwise derivative calculus of this kind is already routine in the project.

## 38.8 What the machine has checked

Every leancode block in this chapter marked **not yet proved** is exactly that: no tier of the pipeline of Chapter 37 has attempted it, and none of it is **[A]**. What *is* **[A]**, and what every item above explicitly builds from, are the two results boxed below.

### The base case every lattice item extends

```
theorem hyperbolicU_evenDiag : IsEvenDiag hyperbolicU := by ...
theorem hyperbolicU_unimodular : IsUnimodular hyperbolicU := ...
```

**Reading.** The  $d = 1$  Narain Gram matrix is even and unimodular. **Proof idea.** Direct computation on an explicit  $2 \times 2$  integer matrix. **Not proved.** Anything for  $d > 1$ ; Section 38.2.1 is the honest gap between this and item 1. **[A]**

## The base case every Mukai-generation item extends

```

theorem reflection_isometry (L : Gram n) (hL : LT = L) (v : Fin n → ℤ)
  (hv : latticeNorm L v = -2) :
  (reflection L v)T * L * reflection L v = L := by ...
theorem e8Neg_weyl_isometry (i : Fin 8) :
  (reflection e8Neg (Pi.single i 1))T * e8Neg * reflection e8Neg (Pi.single i 1)
    = e8Neg := ...

```

**Reading.** Each of the 8 simple-root reflections is an isometry of  $E_8(-1)$ . **Proof idea.** The general  $(-2)$ -vector reflection isometry, instantiated at each of the 8 standard basis vectors [Reflection.lean, ll. 128–181]. **Not proved.** That the 8 simple reflections *generate* the full Weyl group of 240 roots – stated verbatim in the file’s own docstring as out of scope [Reflection.lean, ll. 57–64] – which is exactly item 4 above. [A]

| Claim                                                                                                                         | Tier | Where                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------|------|---------------------------------------------------------------------------------------------------------|
| The two boxed prerequisites, as stated                                                                                        | [A]  | files cited; axioms <code>propext</code> , <code>Classical.choice</code> , <code>Quot.sound</code> only |
| Every leancode block marked "not yet proved" in this chapter                                                                  | —    | not attempted by any tier of Chapter 37; not a claim of any kind                                        |
| $A_n$ positivity and integrality for all $n$                                                                                  | [L]  | established in the literature since EOT’s original conjecture (Section 38.3.1), not re-derived here     |
| GPR’s $d = 2$ duality group $SL(2, \mathbb{Z}) \times SL(2, \mathbb{Z}) \otimes_S [\mathbb{Z}_2 \times \mathbb{Z}_2]$ in full | [L]  | stated, not proved, in the source at the point read (Section 38.2.2)                                    |
| Heterotic/type II duality on $K3 \times T^2/\text{Calabi–Yau}$ (item 20)                                                      | [C]  | Aspinwall’s own assessment, "the full picture is still to be uncovered"                                 |
| Six proofs of $\chi(K3) = 24$ being "debt," and the remaining atlas clusters likewise                                         | [C]  | an engineering judgement, argued at Chapter 36 and Section 38.4                                         |

## Chapter summary

- Twenty-two items, four groups: lattice/duality extensions (1–5), DFT/modular-form/moonshine targets (6–11), atlas engineering debt (12–16), a faithful non-shadow tadpole formalization (17–18), and physics-facing questions (19–22).
- Difficulty is not the same as importance: item 8 ( $A_n$  positivity past  $n = 2$ ) is  $\star$  because the arithmetic is trivial once the  $M_{24}$  character table entry is looked up correctly; item 3 (full  $O(\Gamma^{4,20})$  action) is  $\star\star\star$  because no group-generation machinery exists in the project at all.
- Read the source before stating the target: items 2 and 21 both extend a Tier A result to a case the cited literature itself only conjectures or proves in a narrower case than this book’s earlier chapters might suggest.
- Some gaps are gaps in Mathlib (Siegel–Narain theta functions, Jacobi forms, singular cohomology of a compact 4-manifold), not gaps in effort; items 7, 10, 17–18 name the missing library, not just the missing proof.
- The atlas of Chapter 36 is itself a source for this chapter: five more unification-candidate clusters (items 12–16) are refactor projects of the exact shape worked out there in detail for three other clusters.

## Exercises

**Exercise 38.1** (★). Carry out Section 38.2.1’s block-diagonal argument formally: state and prove `narain_evenDiag_two` and `narain_unimodular_two` in a scratch file importing the module `Hyperbolic.lean` of Chapter 18, using `Matrix.fromBlocks` and the matching block-determinant lemma (locate it yourself in Mathlib). Report which tier of Chapter 37 you think would have closed each goal.

**Exercise 38.2** (★). Verify Equation (38.3) by hand and state the two `decide`-able lemmas of Section 38.6.2 in a scratch file with no imports beyond core Lean. Then look up one more subgroup of  $M_{24}$  acting on 24 points with at least five orbits (Mukai’s or Kondo’s papers, cited at Chapter 29) and state the analogous two lemmas for it.

**Exercise 38.3** (★). Item 12 lists three Golay-code theorems as a unification candidate. Read the first pair named there – one in `Lean5Corpus`, one in `StringTheoryFoundation` – state in one sentence each what they actually prove, and decide whether they are the same fact (as Section 36.6’s  $\chi(K3) = 24$  cluster turned out to be) or two different facts that happen to both mention the number 24.

**Exercise 38.4** (★★). Read Section 38.2.2’s pinned excerpt of GPR’s  $d = 2$  statement in the source file itself (`papers/foundations/giveon_hep-th_9401139.txt`, around l. 7389). State precisely, as a Lean proposition using the generators of Chapter 11, the single piece this book already proves (the commuting subgroup), and explain in one paragraph what additional generator and relation would be needed to reach the full  $\mathbb{Z}_2 \times \mathbb{Z}_2$  extension GPR states.

**Exercise 38.5** (★★). Attempt item 6: state a Lean proposition capturing “ $\mathcal{D}(G) = 2d$  if and only if  $G = 1$  up to conjugation by an orthogonal matrix” as precisely as you can, using `dualScale_ge`’s proof (Chapter 31) as a guide to where equality in the trace-nonnegativity step forces  $(G - 1)G^{-1}(G - 1) = 0$ . What extra hypothesis on  $G$ , if any, does your statement need that `dualScale_ge` does not?

**Exercise 38.6** (★★). Read `DualScaleStream2/Lattice/Reflection.lean`’s own “Not proved here” paragraph (ll. 51–64) in full. List every one of its four explicitly named gaps, and for each say whether it is a mathematics gap (a genuinely open or unformalized fact), an engineering gap (missing Lean/Mathlib machinery), or an interpretation gap (a Lean object not yet linked to the physical or geometric object it is meant to model).

**Exercise 38.7** (★★). Choose one item from Section 38.4 (items 12–16) and carry out the atlas refactor of Section 36.6 for it: pick a canonical definition, state the other theorem(s) in the cluster as corollaries, and re-run `python3 tools/theorem_atlas.py` to confirm the pair moves from the unification-candidate list to the bridge list.

**Exercise 38.8** (★★★). Item 9 asks for  $A_n$  as a virtual  $M_{24}$  character. Using the character table excerpt of Gaberdiel–Hohenegger–Volpato ([GHV §3.1, ll. 584–700]), derive by hand the signed sum of irreducible characters giving  $A_3 = 770$ , verify it numerically, and state (without necessarily proving) the corresponding Lean proposition in the style of Section 38.3.1’s `A3_decomposition`, this time with the correct coefficients.

**Exercise 38.9** (★★★). Pick any ★★★ item in this chapter, spend no more than two hours reading the literature it depends on, and write a half-page assessment (not a proof) of the single largest missing piece of machinery – in Mathlib, in this project, or in the cited literature – standing between the current state and a Lean statement. Compare your assessment to this chapter’s stated “Mathlib / project state” column for that item.

## Further reading

The improvements and next targets this chapter draws from are LL.md §S2-I (items 1–6, process; item 7, the four Stream 3 mathematics targets behind items 1, 3, 7 and 9 above) and the atlas's own unification-candidate list, `papers/book/generated/atlas.md` (behind items 12–16, and behind Chapter 36's three worked clusters this chapter deliberately does not repeat). The three literature sources read directly for this chapter's physics-facing section are GPR [`giveon_hep-th_9401139.txt`, ll. 5420–5450, 7380–7395, 2905–2925], Aspinwall [`aspinwall_hep-th_9611137.txt`, ll. 3075–3090], and EOT [`eguchi_ooguri_tachikawa_1004_0956.txt`, ll. 175–215 and the final page]; each excerpt was read in the source file itself, not from a secondary summary, per this book's own rule (§1 of the Book Bible: read sources, do not write from memory). For how any of these items would actually be attempted – statement design first, symbolic pre-verification, tiered proof search, and a gate that recompiles every claim – see Chapter 37; for what "done" measurably looks like for an atlas item specifically, see Chapter 36's closing remark that re-running the atlas script should shrink the unification list, one refactored cluster at a time.

## Part IX

# Moonshine Computed, Dyons, and the Verdicts



## Chapter 39

# Mathieu Moonshine Computed: Twining, the Shadow, and the Little-String Bridge

In Chapter 30 the coincidence between the elliptic-genus multiplicities  $2A_n$  and the dimensions of  $M_{24}$  was checked in Lean as one typed table against another. That is weak evidence: a table matching a table only shows that two lists of numbers were copied consistently, and this project has already caught one mistyped  $M_{24}$  table. This chapter asks a sharper question. *Can both sides of Mathieu moonshine be recomputed from formulas—the  $q$ -series from modular forms, the traces from the character table—so that the agreement is a computation and not a transcription? And which integers of this programme survive the test that separates moonshine from numerology?* The answers are: yes, at every one of the 26 conjugacy classes and for the first ten graded pieces; the programme’s own “27720 lock” fails the test at every non-trivial class, while the number  $24 = \chi(\text{K3})$  passes it; and the same arithmetic reproduces a bridge, due to Harvey–Murthy–Nazaroglu, between the elliptic genus and the BPS states of double-scaled little string theories. Along the way the student learns a technique worth keeping: treating a truncated  $q$ -series as an exact list of integers, so that a proof assistant can decide an identity between modular objects coefficient by coefficient.

Conventions follow Chapter 30:  $q = e^{2\pi i\tau}$ ,  $y = e^{2\pi iz}$ ,  $A_n = 45, 231, 770, \dots$  are the EOT coefficients [EOT eq. (1.12), l. 208] and the massive multiplicities are  $2A_n$ . The source keys CDH (Cheng–Duncan–Harvey, *Umbral Moonshine*), CH (Cheng–Harrison), HMN (Harvey–Murthy–Nazaroglu) and Cheng are listed in the bibliography.

### 39.1 Why compute instead of type

A moonshine statement has two sides. On the *modular* side there is a function, here the mock modular form

$$H(\tau) = 2q^{-1/8}(-1 + 45q + 231q^2 + 770q^3 + \dots), \quad (39.1)$$

whose coefficients are fixed by modularity and a few pieces of data. On the *group* side there is a graded representation  $K = \bigoplus_n K_n$  of  $M_{24}$  whose dimensions are those coefficients. The literature records both sides as tables: the  $A_n$  as printed numbers, the  $M_{24}$  representations through a printed character table. Checking one table against the other tests the copying, not the mathematics.

The alternative is to compute each side from a short formula and compare the outputs. Three things then become checkable that were not before. First, a misprint in a table can no longer hide: the computed column either equals the printed one or it does not. Second,

one can compare not only dimensions but *traces* of every group element, which is the real content of moonshine. Third, one can apply the same test to any other integer coincidence, including this programme's own. The price is that one needs an exact arithmetic for power series; the next section supplies it.

## 39.2 Truncated $q$ -series as lists of integers

A power series  $f = \sum_{n \geq 0} f_n q^n$  with integer coefficients, known through  $q^N$ , is stored as the list  $[f_0, f_1, \dots, f_N]$  of  $N+1$  integers. Two operations suffice. The truncated Cauchy product

$$(fg)_n = \sum_{k=0}^n f_k g_{n-k}, \quad 0 \leq n \leq N, \quad (39.2)$$

is exact: the coefficients of  $fg$  through  $q^N$  depend only on those of  $f$  and  $g$  through  $q^N$ . If  $p_0 = 1$  the quotient  $r = s/p$  is computed by the recursion

$$r_n = s_n - \sum_{k=0}^{n-1} r_k p_{n-k}, \quad (39.3)$$

obtained by reading off the coefficient of  $q^n$  in  $s = rp$ ; it too is exact, and it involves no division by anything but  $p_0 = 1$ . With these two operations every eta product, every Eisenstein series and every quotient appearing below becomes a finite list of integers, and an identity between two such lists is decided by comparing them entry by entry. Nothing in this is approximate; everything is also *finite*: an identity checked through  $q^9$  says nothing about  $q^{10}$ .

### 39.2.1 Worked example: the first coefficients of $H$

Cheng and Harrison write the mock modular form of the Niemeier root system  $A_1^{24}$  (the case of Mathieu moonshine) in closed form [CH eqs. (3.5)–(3.6), ll. 797–815]:

$$H(\tau) = \frac{-2E_2(\tau) + 48F_2(\tau)}{\eta(\tau)^3}, \quad F_2(\tau) = \sum_{\substack{r>s>0 \\ r-s \text{ odd}}} (-1)^r s q^{rs/2} = q + q^2 - q^3 + q^4 + \dots, \quad (39.4)$$

with the Eisenstein series  $E_2(\tau) = 1 - 24 \sum_{n \geq 1} \sigma_1(n) q^n$  [HMN eq. (2.37), l. 736]. Since  $\eta^3 = q^{1/8} \prod_{n \geq 1} (1 - q^n)^3$ , the series  $q^{1/8} H$  is the quotient of the integer series

$$s = -2E_2 + 48F_2 = -2 + 48(\sigma_1(n) + [q^n]F_2)_{n \geq 1} = -2 + 96q + 192q^2 + 144q^3 + \dots \quad (39.5)$$

by  $p = \prod (1 - q^n)^3 = 1 - 3q + 5q^3 - 7q^6 + \dots$  (Jacobi's identity; in Lean it is checked through  $q^9$ , not proved). The recursion (39.3) gives

$$\begin{aligned} r_0 &= -2, \\ r_1 &= 96 - r_0 p_1 = 96 - (-2)(-3) = 90, \\ r_2 &= 192 - (r_0 p_2 + r_1 p_1) = 192 - (0 - 270) = 462, \\ r_3 &= 144 - (r_0 p_3 + r_1 p_2 + r_2 p_1) = 144 - (-10 + 0 - 1386) = 1540. \end{aligned}$$

These are  $-2, 2A_1, 2A_2, 2A_3$ . Carried through  $q^9$ , the same recursion reproduces all nine tabulated values  $2 \cdot 45, \dots, 2 \cdot 132825$ ; in Lean this is the theorem `hComputed_eq_table` of Section 39.8. **[A]** for the equality of lists; **[L]** for the statement that (39.4) is the mock modular form attached to the K3 elliptic genus.



### Ten coefficients are not an identity

The theorem `eta3_jacobi` says that  $\prod(1 - q^n)^3$  and Jacobi's series  $\sum(-1)^n(2n+1)q^{n(n+1)/2}$  agree through  $q^9$ . That is a check of ten numbers, not a proof of Jacobi's triple product identity, and the chapter never uses it as one. The same holds for every “=” in this chapter: it is an equality of truncations.

## 39.3 Twining from the modular side

Moonshine becomes structural when dimensions are replaced by traces (Chapter 30). For each conjugacy class  $g$  of  $M_{24}$  the twined series  $H_g$  has coefficients  $\text{tr}_{K_n}(g)$ . Cheng–Duncan–Harvey give all of them in one formula [CDH eq. (4.18), ll. 2931–2936]:

$$H_g(\tau) = \frac{\chi_g}{24} H(\tau) + \frac{F_g(\tau)}{\eta(\tau)^3}, \quad (39.6)$$

where  $\chi_g$  is the number of points of the 24-point set fixed by  $g$  [CDH Table 14, ll. 4349–4364] and  $F_g$  is a weight-2 modular form for  $\Gamma_0(N_g)$  listed in [CDH Table 3, ll. 2882–2915]. Most  $F_g$  are built from  $\Lambda_N = Nq \partial_q \log(\eta(N\tau)/\eta(\tau))$  [CDH eq. (A.3), l. 4100], which is an integer series after multiplication by 24:

$$24\Lambda_N = N(N-1) + 24N \sum_{n \geq 1} (\sigma_1(n) - N\sigma_1(n/N)) q^n, \quad (39.7)$$

the second term present only when  $N \mid n$ . To stay in  $\mathbb{Z}$  one computes  $24H_g = (\chi_g s + 24F_g)/p$  with  $s$  and  $p$  as in Section 39.2.1.

**Example 39.1** (The class  $2A$ ). Here  $\chi_{2A} = 8$  and  $F_{2A} = -16\Lambda_2$  [CDH Table 3, l. 2886], so  $24F_{2A} = -16 \cdot 24\Lambda_2 = -32 - 768q + \dots$ . The numerator is  $8s + 24F_{2A} = (-16 - 32) + (768 - 768)q + \dots = -48 + 0 \cdot q + \dots$ , and the recursion gives  $r_0 = -48$ ,  $r_1 = 0 - (-48)(-3) = -144$ . Dividing by 24:  $-2, -6$ , the first two entries of the  $2A$  column of [CDH Table 20, ll. 4494–4512]. The trace of  $2A$  on the 90-dimensional  $K_1 = \mathbf{45} \oplus \mathbf{45}$  is therefore  $-6$ , which is  $2\chi_{\mathbf{45}}(2A) = 2 \cdot (-3)$  from the character table (Chapter 29).

### 39.3.1 Eta quotients and their shift

Several  $F_g$  are eta quotients, and these carry a fractional power of  $q$  that has to be accounted for before anything is stored in a list of integers. Since  $\eta(k\tau) = q^{k/24} \prod_{n \geq 1} (1 - q^{kn})$ , a quotient  $\prod_k \eta(k\tau)^{e_k}$  equals

$$q^{\sum_k k e_k / 24} \prod_k \prod_{n \geq 1} (1 - q^{kn})^{e_k}, \quad (39.8)$$

where the second factor is an integer series computed by (39.2)–(39.3). The prefactor is a genuine power of  $q$  exactly when  $24 \mid \sum_k k e_k$ , and each quotient used here satisfies that. For the newform  $f_{11} = \eta(\tau)^2 \eta(11\tau)^2$  one has  $\sum_k k e_k = 2 \cdot 1 + 2 \cdot 11 = 24$ , so  $f_{11} = q \prod_n (1 - q^n)^2 (1 - q^{11n})^2$ , and indeed  $f_{11} = q - 2q^2 - q^3 + 2q^4 + \dots$  begins at  $q^1$ , as the normalisation  $f_{23,a} = q + O(q^3)$ ,  $f_{23,b} = q^2 + O(q^3)$  of [CDH eq. (A.8), ll. 4134–4137] requires for its companions. The Lean file checks the divisibility condition for every quotient it uses before using it, and checks these normalisations; without them a shift of one power of  $q$  would silently move a whole column.

**A displaced exponent**

In the text extraction of CDH's Table 3 the exponent 3 of  $\eta(\tau)$  in the 12A entry sits on the next line, so the entry reads as if the exponent were missing. It was reconstructed by requiring weight 2 (the exponents of an eta quotient of weight  $w$  sum to  $2w$ ), and the reconstruction is confirmed by the match with the printed column; a Lean control shows that the exponent 2 instead of 3 does *not* reproduce it. This is the kind of error a computed check catches and a typed table hides.

Sixteen of the 21 columns of CDH's Table 20 need more than a single  $\Lambda_N$ : eta quotients, sums of several  $\Lambda_d$ , and for 11A, 14AB, 15AB and 23AB the newforms of CDH's Appendix A with rational prefactors. For instance  $F_{11A} = \frac{2}{5}(-\Lambda_{11} + 11f_{11})$  [CDH Table 3, l. 2900] with the newform  $f_{11} = \eta(\tau)^2\eta(11\tau)^2$  [CDH eq. (A.4), l. 4115]. An eta quotient  $\prod_k \eta(k\tau)^{e_k}$  is  $q^{\sum_k ke_k/24}$  times an integer series, and every quotient used here has  $\sum_k ke_k \equiv 0 \pmod{24}$ , so the power of  $q$  is an integer. A prefactor  $1/D$  is handled by computing  $24D H_g$ . In Lean all 21 computed columns equal the printed ones through  $q^9$  ([A]), and each has every coefficient divisible by  $24D$ , so the division back to  $H_g$  is exact. Two negative controls make the agreement informative: without  $F_g$  the 2A column is not reproduced, and without the newform  $f_{11}$  the 11A column is not either. Because CDH's table was read from a flattened PDF, these theorems also certify the transcription; one exponent in CDH's Table 3 is displaced in the text extraction (12A) and was fixed by weight counting, which the match then confirms.

## 39.4 Twining from the group side

### 39.4.1 A character table with irrational entries

The other side needs the traces  $\text{tr}_{K_n}(g)$  computed from the character table of  $M_{24}$  [CDH Table 8, ll. 4170–4209] and the decompositions of the  $K_n$ . Most entries are integers; a few are the algebraic integers

$$b_n = \frac{-1 + \sqrt{-n}}{2}, \quad n = 7, 15, 23, \quad b_n^2 = -b_n - \frac{n+1}{4}, \quad \overline{b_n} = -1 - b_n. \quad (39.9)$$

An entry  $a + b b_n$  is stored as the integer pair  $(a, b)$ ; multiplication and complex conjugation are the two polynomial rules just written, so all arithmetic stays in  $\mathbb{Z}$ . Each column of the table lives in one ring:  $\mathbb{Z}$ ,  $\mathbb{Z}[b_7]$  (classes 7A, 7B, 14A, 14B, 21A, 21B),  $\mathbb{Z}[b_{15}]$  (15A, 15B) or  $\mathbb{Z}[b_{23}]$  (23A, 23B).

### 39.4.2 Guards: what a character table must satisfy

A mistyped table is how this project went wrong before, so the transcription is guarded by identities every character table satisfies. The column norms  $\sum_i |\chi_i(g)|^2$  equal the centralizer orders  $|C(g)|$  (second orthogonality); Lean computes them from the table, rather than taking them from memory, and checks that each divides  $|M_{24}| = 244\,823\,040$  and is divisible by the order of  $g$ . The class sizes  $|M_{24}|/|C(g)|$  then add up to  $|M_{24}|$  (the class equation). Finally the first orthogonality relation

$$\sum_g |g^{M_{24}}| \chi_i(g) \overline{\chi_j(g)} = |M_{24}| \delta_{ij} \quad (39.10)$$

is checked for all  $26 \times 26$  pairs. The sum mixes values in four rings; it is accumulated as a quadruple (rational part, coefficient of  $b_7$ , of  $b_{15}$ , of  $b_{23}$ ). Since  $1, \sqrt{-7}, \sqrt{-15}, \sqrt{-23}$  are

linearly independent over  $\mathbb{Q}$  (standard; [L]), (39.10) holds exactly when the quadruple is  $(|M_{24}|\delta_{ij}, 0, 0, 0)$ .

#### What the guards do not fix

The text extraction of CDH’s table loses the overlines:  $b_7$  and  $\overline{b_7}$  both print as “b7”. A search over the ten independent choices (in Python, not in the kernel) finds 128 readings that pass all guards—exactly the orbit of one reading under renaming the classes  $7A \leftrightarrow 7B$ ,  $14A \leftrightarrow 14B$ , ... and the characters  $45 \leftrightarrow \overline{45}$ , ...—and all of them give the same traces below. So the guards pin the table up to relabelling, not absolutely, and the chapter’s statements are invariant under that relabelling. A reading *outside* the orbit is rejected: the Lean control `misread_fails_gram` swaps one pair of entries and orthogonality fails.

### 39.4.3 Traces equal computed coefficients

With the table in hand, the trace of  $g$  on  $K_n = \bigoplus_i m_{n,i} \rho_i$  is  $\sum_i m_{n,i} \chi_i(g)$ . For levels  $n = 1, \dots, 7$  EOT give decompositions (Chapter 30), and Lean checks that at *every* class the resulting trace is a rational integer (its  $b_n$ -parts cancel) and equals the coefficient of  $q^{n-1/8}$  in the computed  $H_g$  ([A]). Two independent computations—one from modular forms, one from the character table—meet at  $26 \times 7$  numbers.

**Example 39.2** (The class  $3A$  at levels 1, 2, 3). The  $3A$  column of the character table gives  $\chi_{45}(3A) = 0$ ,  $\chi_{231}(3A) = -3$ ,  $\chi_{770}(3A) = 5$  [CDH Table 8, ll. 4170–4209]. Since  $K_1 = 45 \oplus \overline{45}$ ,  $K_2 = 231 \oplus \overline{231}$  and  $K_3 = 770 \oplus \overline{770}$ , and each pair contributes twice the real part, the traces are

$$\mathrm{tr}_{K_1}(3A) = 0, \quad \mathrm{tr}_{K_2}(3A) = -6, \quad \mathrm{tr}_{K_3}(3A) = 10.$$

On the modular side,  $\chi_{3A} = 6$  and  $F_{3A} = -6\Lambda_3$  [CDH Table 3, ll. 2882–2915] give the column  $-2, 0, -6, 10, 0, -18, \dots$  [CDH Table 20, ll. 4494–4512]. The three numbers agree—and they agree for all 26 classes and all seven levels at once, which is the content of `trace_eq_twined_coeff_all`. Note how little room there is for an accident: the modular side knows only  $\chi_{3A}$  and one weight-two form, the group side only a column of characters.

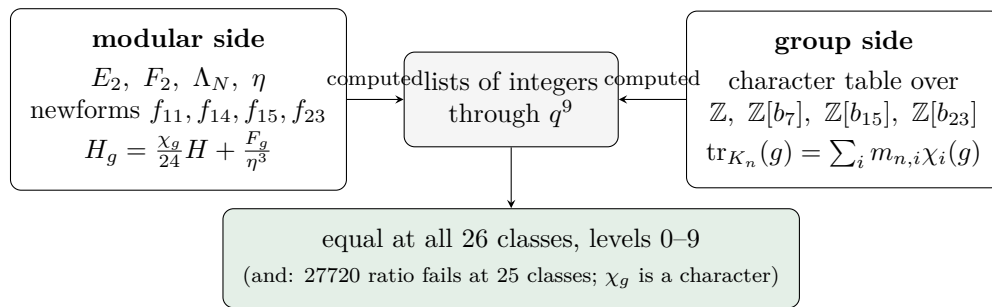


Figure 39.1: The two routes of this chapter. Neither side is copied from the other: the left side is computed from modular forms, the right side from a guarded character table, and the comparison is made on finite lists of integers.

#### What the twined numbers count

$\mathrm{tr}_{K_n}(g)$  is a graded character: it counts the BPS states of the K3 sigma model at level  $n$  with a sign given by how the symmetry  $g$  acts on them. Two consequences make it a

sharper probe than the dimension. Because characters are additive over direct sums, any relation coming from a module structure survives the replacement of dimensions by traces. And because a non-trivial  $g$  mixes states, the traces are small numbers of both signs, so an accidental numerical relation among the  $\dim K_n$  has no reason to persist. This is why the comparison in Figure 39.1 is evidence for a module and the 27720 relation of Section 39.5 is not.

For  $n = 8, 9$  EOT give no decomposition. The direction can then be reversed: the multiplicity of  $\rho_i$  in  $K_n$  is the inner product

$$m_{n,i} = \frac{1}{|M_{24}|} \sum_g |g^{M_{24}}| [q^{n-1/8}] H_g \overline{\chi_i(g)}. \quad (39.11)$$

Lean evaluates (39.11) for  $n = 0, \dots, 9$  and all 26 irreducibles from the *computed*  $H_g$ : every  $m_{n,i}$  is an integer, non-negative for  $n \geq 1$ , and equal to CDH’s printed decomposition [CDH Table 48, ll. 5305–5330].

**Example 39.3** (Level 8). The computed multiplicities give

$$\begin{aligned} K_8 = & 990 \oplus \overline{990} \oplus 1035' \oplus 1035'' \oplus 2 \cdot 1265 \oplus 2 \cdot 2277 \\ & \oplus 2 \cdot 3312 \oplus 2 \cdot 3520 \oplus 4 \cdot 5313 \oplus 2 \cdot 5796 \oplus 2 \cdot 5544 \oplus 6 \cdot 10395, \end{aligned}$$

whose dimension is  $1980 + 2070 + 2530 + 4554 + 6624 + 7040 + 21252 + 11592 + 11088 + 62370 = 131100 = 2A_8$ . The level-7 decomposition  $10395 \oplus 5796 \oplus 5544 \oplus 5313 \oplus 2024 \oplus 1771$  (doubled) is the one used in Chapter 30; the earlier proposal  $10395 + 2 \cdot 5796 + 5544 + 3312$  has the right dimension but the wrong traces, as Cheng observed [Cheng, ll. 533–549], and Lean confirms it disagrees with the computed twined series at some class.

## 39.5 The forger’s test

Anyone can find integers that match:  $A_2 \cdot 60 = 4A_1 \cdot 77 = 27720$ , with  $27720 \cdot 8832 = |M_{24}|$ , is the “lock” of Chapter 30. What made monstrous moonshine more than numerology was that the coincidences survive *twining*: replace every dimension by the trace of an element and the relation still holds. Apply this to the lock in its ratio form,  $[q^2]H_g \cdot (4 \cdot [q^1]H) = [q^2]H \cdot (4 \cdot [q^1]H_g)$ , i.e.  $A_2(g)/A_1(g) = A_2/A_1$ . At  $g = 2A$  the computed coefficients are  $A_1(2A) = -6$ ,  $A_2(2A) = 14$ , and  $14/(-6) \neq 462/90$ . Lean checks that the relation holds at the identity and fails at every one of the 25 non-identity classes ([A]). By this criterion the lock is numerology.

The number 24 behaves differently. The multiplicity  $\chi_g$  in (39.6), which replaces 24 when one twines, is at every class the trace of  $g$  on the permutation representation  $\mathbf{1} \oplus \mathbf{23}$  computed from the character table: 24, 8, 0, 6, 0, 0, 4,  $\dots$  (Lean checks all 26 entries). The 24 twines into a genuine character.

### Why twining separates structure from coincidence

A relation that comes from a module structure—“ $K_n$  is this sum of representations”—holds for traces because traces are additive over direct sums. A relation between dimensions that is only numerical has no reason to survive: the traces of a non-trivial element are smaller, of both signs, and unrelated by any multiplicative identity. The test is cheap once both sides are computed, and it should be applied to every integer coincidence before a physical reading is proposed.

## 39.6 Where the 24 of the shadow comes from

$H$  is a mock modular form: it becomes modular only after adding a non-holomorphic correction built from its *shadow*, which for Mathieu moonshine is  $24\eta(\tau)^3$  [HMN §1, ll. 244–248]. The 24 in the shadow can be located arithmetically. CDH write any weight-1 meromorphic Jacobi form of the shape  $\psi = \Psi_{1,1}\phi$ , with  $\Psi_{1,1} = -i\theta_1(\tau, 2z)\eta^3/\theta_1(\tau, z)^2$  and  $\phi$  a weak Jacobi form of weight 0, as a polar part plus a finite part [CDH eqs. (2.19)–(2.25), ll. 657–711]:

$$\psi = \chi \operatorname{Av}^{(m)}\left[\frac{y+1}{y-1}\right] + \sum_r h_r \hat{\theta}_r^{(m)}, \quad \chi = \phi(\tau, 0), \quad (39.12)$$

and the shadow of  $(h_r)$  is  $\chi$  times the unary theta series  $S^{(m)}$  [CDH eqs. (2.26)–(2.30), ll. 716–768]. For K3,  $\phi$  is the elliptic genus  $Z = 8(f_2^2 + f_3^2 + f_4^2)$  with  $f_i = \theta_i(\tau, z)/\theta_i(\tau, 0)$  [CDH eqs. (2.49), (2.62), ll. 1258–1270, 1397],  $m = 2$ , and  $S_1^{(2)} = \eta^3$ .

Everything in (39.12) except the word “shadow” is a finite computation once denominators are cleared: the theta functions are products,  $Z$  is a  $q$ -series whose coefficients are Laurent polynomials in  $y$ , and the Appell–Lerch sum  $\operatorname{Av}^{(2)}$  is a sum of geometric series. Lean checks three facts through  $q^9$ . First,  $Z(\tau, 0) = 24$ : the constant term of  $Z$  at  $y = 1$  is 24 and every higher coefficient vanishes. Second, the decomposition (39.12) holds with  $\chi = 24$  and  $h_1$  equal to the series  $H$  computed from (39.4). Third, it fails with  $\chi = 23$  or  $\chi = 25$ . So two independent facts meet at 24: the value  $Z(\tau, 0)$ , i.e. the Euler characteristic of K3 (the Witten index), and the multiplicity of the polar part. That this multiplicity is the coefficient of the shadow is Zwegers’ theorem as recast by CDH ([L]); nothing in Lean constructs the non-holomorphic completion.

### Located, not proved

The Lean files never define a shadow, a completion or a modular transformation. What is proved is the arithmetic skeleton: the 24 of  $Z(\tau, 0)$  and the 24 of the polar part are the same integer, and no other integer makes the decomposition hold. “ $H$  is mock modular with shadow  $24\eta^3$ ” remains [L].

## 39.7 The little-string bridge

### 39.7.1 A BPS index labelled by ADE diagrams

Harvey, Murthy and Nazarov study  $k$  NS5-branes on K3 in a double-scaling limit; the branes’ positions define an ADE root system  $Y$  with Coxeter number  $k$ , the lattice of vanishing  $(-2)$ -cycles. The holomorphic part of the second helicity supertrace, a BPS index of this theory, is given in closed form [HMN eqs. (2.29)–(2.37), ll. 700–736]:

$$\chi_2^Y = \operatorname{rk}(Y) E_2 - 24 F_2^Y, \quad F_2^{(k,d)} = \sum_{n \geq 1} q^n \sum_{rs=n} s(d[kr > d^2s] - \frac{k}{d}[d^2r > ks]), \quad (39.13)$$

with  $F_2^Y$  a combination of  $F_2^{(k,d)}$  fixed by  $Y$  ( $A_{k-1}$ :  $F_2^{(k,1)}$ ;  $D_{m/2+1}$ :  $F_2^{(m,1)} + F_2^{(m,m/2)}$ ; and three combinations for  $E_{6,7,8}$ ).

**Example 39.4** ( $Y = A_3$ ). Here  $k = 4$ ,  $\operatorname{rk} = 3$ , and at  $n = 1$  the only factorization is  $r = s = 1$ , giving  $F_2^{(4,1)}(1) = 1 \cdot (1 \cdot [4 > 1] - 4 \cdot [1 > 4]) = 1$ . Hence  $[q^1]\chi_2^{A_3} = 3 \cdot (-24) - 24 \cdot 1 = -96$ , the second coefficient of HMN’s printed series  $3 - 96q - 288q^2 - 384q^3 - 576q^4 - 360q^5 + \dots$  [HMN eq. (4.10), l. 1290]. Lean reproduces all six printed coefficients, and the six of  $Y = A_7$  [HMN eq. (4.11), l. 1292], whose coefficients 7 and 192 share no factor.

For  $k = 2$  ( $Y = A_1$ , two NS5-branes) the formula gives  $-2\chi_2^{A_1} = -2E_2 + 48F_2$ , the numerator of (39.4): the little-string index is  $\eta^3 H$ , the weight-2 form of Mathieu moonshine (Lean checks it through  $q^9$ , and that HMN’s sum  $F_2^{(2,1)}$  equals Cheng–Harrison’s  $F_2$  through  $q^{40}$ ).

### 39.7.2 The umbral relation and a divisibility

**Example 39.5** (The relation (39.14) at  $\ell = 2$ , by hand). Take  $Y = A_1$ , so  $k = 2$ ,  $\text{rk}(Y) = 1$  and  $X = A_1^{24}$ , whose umbral form is the  $H$  of (39.4); here  $\chi_2^X = \eta^3 H = -2E_2 + 48F_2$ , whose  $q^1$  coefficient is 96 (Section 39.2.1). The right-hand side of (39.14) is then  $-\frac{1}{2} \cdot 96 = -48$ . The left-hand side comes from (39.13):  $F_2^{(2,1)}(1) = 1 \cdot (1 \cdot [2 > 1] - 2 \cdot [1 > 2]) = 1$ , so  $[q^1]\chi_2^{A_1} = 1 \cdot (-24) - 24 \cdot 1 = -48$ . The two agree, and the constant terms agree as well ( $\text{rk} = 1$  against  $-\frac{1}{2} \cdot (-2) = 1$ ). The general statement, for the six lambencies with  $X$  a power of a single  $A$  component, is checked in Lean through  $q^6$ .

HMN relate  $\chi_2^Y$  to umbral moonshine for the Niemeier root system  $X = Y^{24/\text{rk}(Y)}$  [HMN eqs. (4.5), (4.9), ll. 1229–1266]:

$$\chi_2^Y = -\frac{\text{rk}(Y)}{2} \chi_2^X, \quad \chi_2^X = \sum_r H_r^X S_{m,r}. \quad (39.14)$$

For  $Y = A_{\ell-1}$  the umbral forms can be built from theta functions exactly as in Section 39.6, now with CDH’s  $Z^{(\ell)}$  for  $\ell = 2, 3, 4, 5, 7, 13$  [CDH eqs. (2.49)–(2.51), (2.62), ll. 1258–1285, 1397]; the quantity  $\chi_2^X$  is the first Taylor coefficient at  $z = 0$  of the finite part. Lean checks (39.14) for all six values of  $\ell$  through  $q^6$ , and that pairing  $\ell = 3$  with  $Y = A_3$  instead of  $A_2$  fails.

HMN observe that for the “umbral”  $Y$  (those with  $\text{rk}(Y) \mid 24$ ) every coefficient of  $\chi_2^Y$  is divisible by  $\text{rk}(Y)$ , and write that “it is not clear why this should be true for the coefficients of higher powers of  $q$ ” [HMN §4, ll. 1280–1296]. Within their own formula (39.13) the reason is elementary. For  $n \geq 1$ ,

$$[q^n]\chi_2^Y = -24(\text{rk}(Y) \sigma_1(n) + F_2^Y(n)), \quad F_2^Y(n) \in \mathbb{Z}, \quad (39.15)$$

so  $\text{rk}(Y) \mid 24$  forces divisibility of every coefficient. Conversely, for  $A_{k-1}$  one has  $F_2^Y(1) = 1$  and  $[q^1]\chi_2^Y = -24(\text{rk} + 1)$ , divisible by  $\text{rk}$  only if  $\text{rk} \mid 24$ ; for  $D_{j+1}$ ,  $F_2^Y(1) = -1$  gives  $-24(\text{rk} - 1)$  with the same conclusion. Lean proves both directions for every  $A_{k-1}$  and every  $D_{j+1}$ , and checks  $E_6, E_8$  (divisible) and  $E_7$  (not). The divisible  $Y$  are exactly HMN’s umbral list  $A_1, A_2, A_3, A_4, A_6, A_8, A_{12}, A_{24}, D_4, D_6, D_8, D_{12}, D_{24}, E_6, E_8$ . Why the factor in (39.13) is 24 is not addressed here; reading it as  $\chi(K3)$  is [C].

#### A sign in the source

HMN’s eq. (4.2) [HMN eq. (4.2), l. 1194] prints  $\chi_2^Y = -\text{rk}(Y)E_2 + 24F_2^Y$ , the opposite overall sign of their eq. (2.36). The printed series (4.10)–(4.11) start with  $+\text{rk}(Y)$ , as does the count of  $\text{rk}(Y)$  massless vector multiplets [HMN §4, ll. 1257–1260]; they fix the sign of (2.36), which is the one used here.

### 39.7.3 What is not available: a reflection bridge

The vanishing  $(-2)$ -cycles of  $Y$  suggest a link with the reflections in  $(-2)$ -vectors that Chapter 22 and Stream 2 formalize. HMN’s lattice structure, however, enters through the Niemeier lattice  $L_X$  and its Weyl group,  $G^X = \text{Aut}(L_X)/W_X$  [HMN eq. (1.10), l. 277], a rank-24 even unimodular lattice that this project does not formalize. Nothing in the computations of this



section involves a reflection or a lattice isometry;  $\text{rk}(Y)$  and the Coxeter number enter only as numbers. The bridge is therefore recorded as *not available*, with that reason, rather than claimed.

## 39.8 What the machine has checked

The files are in the library `DualScaleMoonshine`. All theorems quoted depend only on `propext`, `Classical.choice` and `Quot.sound` (most on `propext` alone); the heavy ones are closed by `decide` or `decide +kernel`, which have the kernel evaluate the decision procedure.

### In Lean: $H$ computed equals the table

```
def number (N : ℕ) : List ℤ :=
  (List.range (N + 1)).map fun n => if n = 0 then -2 else 48 * (sigma n + f2Coeff n)

def hComputed (N : ℕ) : List ℤ := divTrunc N (number N) (eta3 N)

theorem eta3_jacobi : eta3 9 = [1, -3, 0, 5, 0, 0, -7, 0, 0, 0] := by ...

theorem hComputed_eq_table : hComputed 9 = hFromTable := by ...
```

(`DualScaleMoonshine/QSeries.lean`.) **Reading.** `hFromTable` is  $[-2, 2A_1, \dots, 2A_9]$  from EOT's table; the list computed by the recursion (39.3) from  $-2E_2 + 48F_2$  and  $\prod(1 - q^n)^3$  equals it through  $q^9$ . **Proof idea.** Kernel evaluation of two lists of ten integers. **Tier [A]** for the lists; **[L]** that the formula is the K3 mock modular form.

### In Lean: twined series, with newforms and controls

```
theorem twined_2A : twined24 9 8 2 (-16) = table2A.map (24 * ·) := by ...

theorem twined_11A : twinedD dataF11A = scaled dataF11A table11A := by ...
theorem twined_23AB : twinedD dataF23AB = scaled dataF23AB table23AB := by ...

theorem twined_11A_needs_newform :
  twinedD (5, 2, lambda24 9 11 (-2)) ≠ scaled dataF11A table11A := by ...
```

(`DualScaleMoonshine/Twining.lean`, `TwiningAll.lean`.) **Reading.**  $24H_{2A}$  computed from  $\chi_{2A} = 8$  and  $F_{2A} = -16\Lambda_2$  equals 24 times CDH's printed 2A column through  $q^9$ ; likewise  $24DH_g$  for 11A ( $D = 5$ ) and 23AB ( $D = 11$ ), whose  $F_g$  contain newforms; dropping  $f_{11}$  breaks 11A. All 21 columns have such a theorem. **Tier [A]** for the arithmetic, **[L]** for the formulas of CDH.

### In Lean: the character table and the traces

```
theorem colNorm_eq_cent : (List.range 26).map colNorm = cent.map (·, 0) := by ...
theorem class_equation : classSize.sum = 244823040 := by ...
theorem gram_ok : gramOK charTab = true := by ...

theorem trace_eq_twined_coeff_all :
  ∀ j : Fin 26, eotMult.map (traceQ j) =
    (((computedSeries.getD j []).drop 1).take 7).map (·, 0) := by ...
```

(`DualScaleMoonshine/CharactersAll.lean`.) **Reading.** The column norms of the table (entries in  $\mathbb{Z}[b_n]$  stored as pairs) are the listed centralizer orders; the class sizes add to  $|M_{24}|$ ; all  $26 \times 26$  orthogonality relations hold; and at every class  $j$ , the traces on EOT's

representations at levels 1–7 (doubled, as in `eotMult`) have zero irrational part and equal the computed twined coefficients. **Tier [A]**; that the table is the character table of  $M_{24}$  is [L].

#### In Lean: the modules $K_0, \dots, K_9$ decomposed

```
theorem moonshine_modules_decompose :
  (List.range 10).all (fun n => (List.range 26).all fun i =>
    let v := moonshineInner n i
    v.2.1 == 0 && v.2.2.1 == 0 && v.2.2.2 == 0 &&
    v.1 == 244823040 * ((table48.getD n []).getD i 0)) = true := by ...
```

(DualScaleMoonshine/Decompositions.lean.) **Reading.** For  $n \leq 9$  and every irreducible  $i$ , the inner product (39.11) computed from the twined series has vanishing irrational parts and equals  $|M_{24}|$  times CDH’s Table 48 entry, so every multiplicity is an integer equal to the printed one. A companion theorem, `eot_level7_proposal_inconsistent`, shows the traces of the older level-7 proposal differ from the computed series. **Tier [A]**.

#### In Lean: the forger’s test

```
theorem lock_at_identity :
  coeff untwined 1 = 90 ∧ coeff untwined 2 = 462 ∧ literalLock untwined ∧
  coeff untwined 2 * 60 = 27720 := by ...

theorem ratio_fails_at_every_class :
  ∀ j : Fin 26, j.val ≠ 0 →
    ¬ ratioTwines ((computedSeries.getD j []).map (24 * ·)) := by ...
```

(DualScaleMoonshine/ForgerTest.lean, CharactersAll.lean.) **Reading.** The lock holds for the untwined series and its ratio form fails for the twined series of every one of the 25 non-identity classes. **Tier [A]** for both; the conclusion “numerology” is the reading the twining criterion gives ([C] as a methodological judgement).

#### In Lean: the 24 of the shadow

```
theorem ellipticGenus_z0 :
  (ellipticGenus 9).map LP.eval1 = [24, 0, 0, 0, 0, 0, 0, 0, 0, 0] := by ...

theorem decomposition : Ser.eqB (starLHS 9) (starRHS 9 24 (hComputed 9)) = true := by ...

theorem decomposition_pins_chi :
  Ser.eqB (starLHS 9) (starRHS 9 23 (hComputed 9)) = false ∧
  Ser.eqB (starLHS 9) (starRHS 9 25 (hComputed 9)) = false := by ...
```

(DualScaleMoonshine/Shadow.lean.) **Reading.** The elliptic genus computed from theta products is 24 at  $y = 1$  at every order through  $q^9$ ; the decomposition (39.12), cleared of denominators, holds with polar multiplicity 24 and finite part the computed  $H$ , and fails for 23 and 25. **Tier [A]**; the word “shadow” is [L].

#### In Lean: the little-string index

```
def chi2Y (Ys : List (N × N)) (rk : ℤ) (n : N) : ℤ :=
  if n = 0 then rk else rk * (-24 * sigma n) - 24 * (Ys.map fun p => f2kd p.1 p.2 n).sum
```



```

theorem hm_n_A3_printed : chi2YSer (yA 4) 3 5 = [3, -96, -288, -384, -576, -360] := by ...
theorem hm_n_k2_eq_moonshine : (chi2YSer (yA 2) 1 9).map (-2 * ·) = numer 9 := by ...
theorem hm_n_umbral_relation_3 :
  (chi2YSer (yA 3) 2 6).map (4 * ·) = (thetaPsiF 6 3).map (2 * ·) := by ...

```

(DualScaleMoonshine/HMNBridge.lean.) **Reading.** The first theorem reproduces HMN’s printed  $A_3$  series; the second says the  $k = 2$  index times  $-2$  is the moonshine numerator; the third is (39.14) for  $\ell = 3$  through  $q^6$ , with the umbral side built from theta functions (the other lambencies have their own theorems). The divisibility statements `divisible_of_dvd_24`, `A_divisible_iff` and `D_divisible_iff` are ordinary proofs quantified over all  $n$  and all  $k$  (resp.  $j$ ); their statements use the divisibility sign of Lean, which this book’s fonts do not typeset, so we give them in words: if  $\text{rk} \mid 24$  then  $\text{rk}$  divides every coefficient; for  $A_{k-1}$  ( $k \geq 2$ ) and  $D_{j+1}$  ( $j \geq 3$ ), all coefficients are divisible by the rank if and only if the rank divides 24. **Tier [A]**; the physical meaning of  $\chi_2^Y$  is [L].

*Remark 39.6* (What the checks cost). Exactness is not free. The heaviest file, the little-string bridge, takes about a quarter of an hour of kernel time: the products of eta quotients and the sixteen-fold Newton recursions are evaluated on integers, with no floating point anywhere. Three theorems in it use `decide +kernel`, which skips the elaborator’s own evaluation and lets the kernel do the arithmetic directly; this adds no axiom and is not the compiled evaluation that the project’s rules forbid. A reader who wants to reproduce a single column should compute it with the recursion (39.3) in any language: the numbers are small.

| Claim                                                        | Tier          | Where                                    |
|--------------------------------------------------------------|---------------|------------------------------------------|
| $H$ from $(-2E_2 + 48F_2)/\eta^3$ equals $2A_n$ , $n \leq 9$ | [A]           | <code>hComputed_eq_table</code>          |
| 21 twined columns equal CDH Table 20 through $q^9$           | [A]           | <code>twined_2A</code> , ...             |
| traces = twined coefficients, all classes, $n \leq 7$        | [A]           | <code>trace_eq_twined_coeff_all</code>   |
| $K_0, \dots, K_9$ decompose as CDH Table 48                  | [A]           | <code>moonshine_modules_decompose</code> |
| 27720 lock fails twining at 25 classes                       | [A]           | <code>ratio_fails_at_every_class</code>  |
| $Z(\tau, 0) = 24 =$ polar multiplicity; 23, 25 fail          | [A]           | <code>decomposition_pins_chi</code>      |
| little-string index = umbral form, $\ell \leq 13$            | [A]           | <code>hm_n_umbral_relation_3</code>      |
| divisibility iff $\text{rk} \mid 24$ ( $A, D$ )              | [A]           | <code>A_divisible_iff</code>             |
| $H$ is mock modular with shadow $24\eta^3$                   | [L]           | CDH, HMN                                 |
| $M_{24}$ acts on a module with these traces                  | [L]           | Gannon (via CDH)                         |
| 24 in (39.13) is $\chi(K_3)$                                 | [C]           | —                                        |
| reflection bridge to Stream 2                                | not available | Section 39.7                             |

### Chapter summary

- Truncated  $q$ -series are finite lists of integers; products and quotients by series with constant term 1 are exact, so identities through  $q^N$  are decidable.
- $q^{1/8}H = (-2E_2 + 48F_2)/\prod(1 - q^n)^3 = -2 + 90q + 462q^2 + 1540q^3 + \dots$  reproduces  $2A_1, \dots, 2A_9$ ; the twined  $H_g = \frac{\chi_g}{24}H + F_g/\eta^3$  reproduces all 21 columns of CDH’s table, including those needing newforms.
- The character table over  $\mathbb{Z}[b_7], \mathbb{Z}[b_{15}], \mathbb{Z}[b_{23}]$  is guarded by orthogonality and the class equation (up to relabelling); traces equal computed coefficients at every class, and  $K_0, \dots, K_9$  decompose with the printed multiplicities.
- Forger’s test: the 27720 lock fails at all 25 non-identity classes; the 24 twines into the

character of  $\mathbf{1} \oplus \mathbf{23}$ .

- $Z(\tau, 0) = 24$  and the polar multiplicity of the elliptic genus are the same integer, and only 24 works; the shadow statement itself is [L].
- HMN's little-string index reproduces Mathieu moonshine at  $k = 2$  and umbral moonshine for  $\ell = 3, 4, 5, 7, 13$ ; its coefficients are divisible by  $\text{rk}(Y)$  iff  $\text{rk}(Y) \mid 24$ , for elementary reasons; the reflection bridge is not available.

## Exercises

**Exercise 39.1** (★). Continue the recursion of Section 39.2.1 to  $q^4$ . You need  $[q^4]F_2 = 1$ ,  $\sigma_1(4) = 7$  and  $p_4 = 0$ . Check that you obtain  $2A_4 = 4554$ .

**Exercise 39.2** (★). For the class  $2A$ , compute  $[q^2]$  of  $24H_{2A}$  from  $8s + 24F_{2A}$  and the recursion, and verify that  $[q^{15/8}]H_{2A} = 14 = 2\chi_{\mathbf{231}}(2A)$ .

**Exercise 39.3** (★). Show that the literal lock  $A_2(g) \cdot 60 = 4A_1(g) \cdot 77$  fails at  $3A$ , using  $A_1(3A) = 0$ ,  $A_2(3A) = -3$  (half the entries 0,  $-6$  of CDH's  $3A$  column). Why is the ratio form of the lock the fairer test?

**Exercise 39.4** (★★). Verify  $b_7^2 = -b_7 - 2$  and  $b_7\overline{b_7} = 2$  from the definition, and show that the rule  $(a, b) \cdot (c, d) = (ac - kbd, ad + bc - bd)$  with  $k = (n + 1)/4$  is multiplication in  $\mathbb{Z}[b_n]$ . Why is  $\mathbb{Z}[b_n]$  closed under complex conjugation?

**Exercise 39.5** (★★). Show that  $10395 + 2 \cdot 5796 + 5544 + 3312 = 30843 = A_7$ , so the older proposal has the right dimension. Using the  $2A$  column of the character table, compute the trace of  $2A$  on the doubled module and compare with  $[q^{55/8}]H_{2A} = -138$ .

**Exercise 39.6** (★★). For  $Y = D_4$  ( $m = 6$ ,  $j = 3$ ) compute  $F_2^{(6,1)}(1)$  and  $F_2^{(6,3)}(1)$  from (39.13), and verify  $[q^1]\chi_2^{D_4} = -72$ . Is it divisible by  $\text{rk}(D_4) = 4$ ? Repeat for  $D_5$  and explain the difference.

**Exercise 39.7** (★★). In Lean, state and prove by `decide` that the level-9 multiplicities of `table48` sum, weighted by the dimensions in `chi1A`, to  $2A_9 = 265650$ . Which theorem of this chapter makes your statement redundant, and why is it still a useful check on the transcription of Table 48?

**Exercise 39.8** (★★★). Write a Lean definition of the ratio relation for an arbitrary pair of levels  $(a, b)$  and prove, by `decide` on `computedSeries`, that  $A_3(g)/A_1(g) = A_3/A_1$  also fails at some class. Then formulate a relation between the  $A_n$  that *does* survive twining at every class (hint: a decomposition of  $K_6$ ) and prove it.

**Exercise 39.9** (★★). Check that the eta quotient  $-2\eta(\tau)^8/\eta(2\tau)^4$  of the class  $2B$  [CDH Table 3, ll. 2882–2915] satisfies  $24 \mid \sum_k ke_k$ , and compute its first two coefficients from (39.8). Compare with  $24\Lambda_2 - 8\Lambda_4$ , the other form in which CDH print the same  $F_{2B}$ .

**Exercise 39.10** (★★★). Clear the denominators in (39.12) for  $m = 2$ : with  $E = \prod(1 - q^n)$ ,  $B_1 = \prod(1 - yq^n)(1 - y^{-1}q^n)$  and  $B_2 = \prod(1 - y^2q^n)(1 - y^{-2}q^n)$ , show that  $\Psi_{1,1} = \frac{y+1}{y-1} E^2 B_2/B_1^2$ , and check the first two coefficients  $(y+1)/(y-1) - (y^2 - y^{-2})q$  against [CDH eq. (2.20), ll. 670–675].

## Further reading

- The closed formula for  $H$ : [CH eqs. (3.5)–(3.6), ll. 797–815]; the twined series, the forms  $F_g$ , the twisted Euler characters and the printed columns: [CDH eq. (4.18), ll. 2931–2936], [CDH Table 3, ll. 2882–2915], [CDH Table 14, ll. 4349–4364], [CDH Table 20, ll. 4494–4512]; the newforms: [CDH App. A, ll. 4100–4137].
- The character table and the decompositions: [CDH Table 8, ll. 4170–4209], [CDH Table 48, ll. 5305–5330]; the level-7 discussion: [Cheng, ll. 533–549].
- Meromorphic Jacobi forms, polar and finite parts, and the shadow: [CDH §2.3, ll. 657–768]; the umbral Jacobi forms: [CDH §2.5, ll. 1258–1397]; the shadow  $24\eta^3$  and the mixed mock form  $\eta^3 H$ : [HMN §1, ll. 244–270].
- The little-string index, its ADE combinations and the umbral relation: [HMN §2.1–2.2, ll. 587–736], [HMN §4, ll. 1179–1300].
- The observation, twining genera and the older decompositions are in Chapter 30; the group  $M_{24}$  in Chapter 29; the elliptic genus and its characters in Chapter 28; the dyon partition function built from the same elliptic genus in Chapter 40. The project record is `docs/STREAM4_WORKFLOW.md`.



## Chapter 40

# Dyons on $K3 \times T^2$ : From the Elliptic Genus to Class Numbers

A black hole carries electric and magnetic charge. If string theory is a quantum theory of gravity, it must say how many quantum states hide behind those charges — not approximately, but as an integer, for every allowed charge vector. For type II string theory on  $K3 \times T^2$  that integer is known in closed form, and the closed form is a Fourier coefficient of one meromorphic function of three variables. This chapter answers the question: *where does that function come from, what makes it meromorphic rather than holomorphic, and what survives when the poles are removed?* The answer to the last part is the small surprise of the chapter: what survives at the simplest charge is a generating function of *class numbers* of imaginary quadratic forms, objects a student meets in a first course on number theory and would never expect to meet again in a black-hole count.

Along the way the student learns three techniques that recur throughout this book: building a modular object as an infinite product whose exponents are the Fourier coefficients of a simpler object (Section 40.3); separating a meromorphic function into a polar part fixed entirely by its poles and a remainder (Section 40.4); and testing an integer coincidence by replacing dimensions with characters (Section 40.6). Everything in Section 40.7 is exact integer arithmetic that the Lean kernel re-checks.

Conventions. As in Chapter 28,  $q = e^{2\pi i\tau}$ ,  $y = e^{2\pi iz}$ , and  $p = e^{2\pi i\sigma}$ ;  $\eta$  is the Dedekind eta function,  $\Delta = \eta^{24}$ , and  $E_4, E_6$  are the Eisenstein series of weight 4 and 6. The elliptic genus of  $K3$  is  $Z(\tau, z) = 2\phi_{0,1}(\tau, z)$ , twice the standard weak Jacobi form of weight 0 and index 1;  $A := \phi_{-2,1}$  and  $B := \phi_{0,1}$  throughout, following Dabholkar–Murthy–Zagier [DMZ §9.1, ll. 3629–3634].

### 40.1 Dyons, their charges, and what we are counting

Compactifying type II string theory on  $K3 \times T^2$  gives a four-dimensional theory with  $N = 4$  supersymmetry [Asp §7, ll. 2741–2745]. A state is labelled by an electric charge vector  $N$  and a magnetic charge vector  $M$ , both living in the even self-dual lattice  $\Pi^{22,6}$ ; a state is *dyonic* when both are non-zero [DMZ §2.2, ll. 638–645]. The duality group does not see  $N$  and  $M$  separately: the three T-duality invariants are

$$(m, n, \ell) = \left( \frac{M^2}{2}, \frac{N^2}{2}, N \cdot M \right), \quad (40.1)$$

all integers because  $\Pi^{22,6}$  is integral [DMZ §2.3, ll. 744–748]. The counting problem splits in two.

*Half-BPS states* preserve half the supersymmetry. Their degeneracy depends on  $n$  only, and is

$$d(n) = p_{24}(n+1), \quad \frac{1}{\Delta(\tau)} = \sum_{n \geq -1} p_{24}(n+1) q^n, \quad (40.2)$$

where  $p_{24}(I)$  counts partitions of  $I$  into parts of 24 colours [DMZ §3.4, ll. 1066–1071]. The 24 is the number of transverse oscillators of the heterotic string — and, on the type II side, the Euler number of K3. These states never decay: the degeneracy is the same everywhere in moduli space.

*Quarter-BPS states* are the interesting ones. Their degeneracy *does* depend on where one sits in moduli space: a bound state of two half-BPS constituents *exists* in one region and *does not exist* in another, and the count jumps across a *wall of marginal stability* [DMZ §6, ll. 1997–2004]. All of them are packaged in one generating function,

$$\frac{1}{\Phi_{10}(\Omega)} = \sum_{m \geq -1} \psi_m(\tau, z) p^m, \quad \Omega = \begin{pmatrix} \tau & z \\ z & \sigma \end{pmatrix}, \quad (40.3)$$

the reciprocal of the Igusa cusp form of weight 10 [DMZ §1.1, ll. 172–183]. The degeneracy  $d(m, n, \ell)$  is a Fourier coefficient of (40.3) along a contour whose choice encodes the moduli [DMZ §6, ll. 1982–1990]. Everything in this chapter concerns the functions  $\psi_m$ , one for each magnetic charge invariant  $m$ .

#### Why a pole is a physical statement

A holomorphic generating function would give one degeneracy per charge, full stop. The function (40.3) has *poles*, and the Fourier coefficient therefore depends on which side of a pole the integration contour runs. That is exactly the physics of wall crossing: two-centred configurations appear or disappear, and the count jumps by the residue [DMZ §6, ll. 2000–2006]. A pole in a partition function is not a defect; it is where the decay of bound states is recorded.

## 40.2 Two Jacobi forms, and the elliptic genus

Every weak Jacobi form of index 1 is a modular-form combination of two generators, of weight  $-2$  and  $0$ . In terms of theta functions [DMZ §4.3, ll. 1416–1424],

$$A = \phi_{-2,1} = \frac{\vartheta_1(\tau, z)^2}{\eta(\tau)^6}, \quad B = \phi_{0,1} = 4 \left( \frac{\vartheta_2(\tau, z)^2}{\vartheta_2(\tau)^2} + \frac{\vartheta_3(\tau, z)^2}{\vartheta_3(\tau)^2} + \frac{\vartheta_4(\tau, z)^2}{\vartheta_4(\tau)^2} \right), \quad (40.4)$$

with the expansions [DMZ §4.3, ll. 1397–1402]

$$A = \frac{(y-1)^2}{y} - 2 \frac{(y-1)^4}{y^2} q + \dots, \quad B = \frac{y^2 + 10y + 1}{y} + 2 \frac{(y-1)^2(5y^2 - 22y + 5)}{y^2} q + \dots \quad (40.5)$$

Two features of (40.5) will be used constantly: as  $z \rightarrow 0$ ,

$$A(\tau, z) \sim (2\pi i z)^2 \rightarrow 0, \quad B(\tau, z) \rightarrow 12, \quad (40.6)$$

[DMZ §9.1, ll. 3633–3635]. The K3 elliptic genus is  $Z = 2B$ , and since coefficients of an index-1 form depend only on the discriminant  $\Delta = 4n - l^2$  we may write  $Z = \sum c(4n - l^2) q^n y^l$  with  $c = 2C_0$ . The first values, read off Table 1 of DMZ [DMZ §4.3, ll. 1404–1408], are

$$c(-1) = 2, \quad c(0) = 20, \quad c(3) = -128, \quad c(4) = 216, \quad (40.7)$$

and  $Z(\tau, 0) = 24 = \chi(\text{K3})$  [DMZ §4.5, ll. 1675–1681].

### 40.3 The partition function as a product

The bridge from K3 to black holes is a product formula. Let  $\chi(\tau, z; \text{Sym}^k \text{K3})$  be the elliptic genus of the  $k$ -th symmetric product of K3 and set  $\hat{Z} = \sum_k \chi(\tau, z; \text{Sym}^k \text{K3}) p^{k-1}$ . Then [DMZ §4.5, ll. 1688–1698]

$$\sum_{k \geq 0} G_k(\tau, z) p^k = \prod_{r \geq 1, s \geq 0, t \in \mathbb{Z}} (1 - p^r q^s y^t)^{-c(4rs - t^2)}, \quad G_k = \chi(\tau, z; \text{Sym}^k \text{K3}), \quad (40.8)$$

and the Igusa form itself is the same product with the opposite sign in the exponent and one extra factor [DMZ §5.1, ll. 1816–1827]. Combining (40.3) with (40.8) gives the working formula of this chapter,

$$\Delta(\tau) \psi_m(\tau, z) = \frac{G_{m+1}(\tau, z)}{A(\tau, z)}. \quad (40.9)$$

Note what (40.8) does: the *input* is the K3 elliptic genus, one weak Jacobi form; the *output* is the full dyon partition function. Nothing else enters.

#### Worked example: the coefficient of $p^2$ at $z = 0$

Set  $y = 1$  in (40.8). The exponent of the factor  $(1 - p^r q^s)$  becomes  $\sum_t c(4rs - t^2)$ , which is the sum of all  $y$ -coefficients of  $Z$  at order  $q^{rs}$ . Since  $Z(\tau, 0) = 24$  is a constant in  $q$ , that sum is 24 for  $rs = 0$  and 0 otherwise. Only the  $s = 0$  factors survive:

$$\sum_k G_k(\tau, 0) p^k = \prod_{r \geq 1} (1 - p^r)^{-24} \implies G_k(\tau, 0) = p_{24}(k). \quad (40.10)$$

For  $k = 2$ , expand by hand:  $(1 - p)^{-24}$  contributes  $\binom{25}{2} = 300$  at order  $p^2$  and  $(1 - p^2)^{-24}$  contributes 24, so  $G_2(\tau, 0) = 324$ . The same argument gives 1, 24, 324, 3200, 25650, 176256 for  $k = 0, \dots, 5$  — the Euler numbers of the Hilbert schemes of points on K3, Göttsche's numbers, and the half-BPS degeneracies (40.2) again. The 24 of K3 propagates to every order.

#### A truncated product can silently lie

To compute  $G_k$  through order  $q^Q$  from (40.8) one needs the coefficients  $c(D)$  for all  $D = 4rs - t^2$  with  $r \leq k$ ,  $s \leq Q$  — that is, up to  $D = 4kQ$ . If the table of  $c$  stops earlier, the missing exponents are silently taken as 0 and the answer is wrong while looking perfectly regular. In the formalization this is not left to care: the discriminants the product actually reads are enumerated and checked against what the computed  $Z$  provides (`dmvv_reachable`), together with a deliberately too-deep truncation that fails the same test (`dmvv_unreachable_example`).

#### An independent check: the additive formulas

The same  $\psi_m$  can be written as polynomials in  $A, B, E_4, E_6$  divided by  $A$  [DMZ §5.2, ll. 1878–1888]:

$$\begin{aligned} \Delta\psi_{-1} &= A^{-1}, & \Delta\psi_0 &= 2A^{-1}B, \\ 4\Delta\psi_1 &= 9A^{-1}B^2 + 3E_4A, & 27\Delta\psi_2 &= 50A^{-1}B^3 + 48E_4AB + 10E_6A^2. \end{aligned} \quad (40.11)$$

These are *not* how we computed  $G_k$ ; they are an independent route, and their agreement with the product is a genuine test of both. A one-line consistency check at  $z = 0$  uses (40.6): the third equation reads  $4G_2 = 9B^2 + 3E_4A^2$ , and at  $z = 0$  this is  $4 \cdot 324 = 9 \cdot 144 = 1296$ . Section 40.7 records the full check through order  $q^2$ .

## 40.4 The double pole and the two-centred states

By (40.9) and (40.6),  $\Delta\psi_m$  has a *double* pole at  $z = 0$ , with principal part

$$\Delta(\tau)\psi_m(\tau, z) \sim \frac{p_{24}(m+1)}{(2\pi iz)^2}, \quad (40.12)$$

because the numerator tends to  $G_{m+1}(\tau, 0) = p_{24}(m+1)$  [DMZ §9.1, ll. 3622–3627]. The pole is where the wall crossing lives, so one would like to subtract it in a canonical way. Zweegers’ theory, as recast by DMZ, does exactly this: there is a decomposition

$$\psi_m = \psi_m^P + \psi_m^F, \quad \psi_m^P = \frac{p_{24}(m+1)}{\Delta(\tau)} A_{2,m}(\tau, z), \quad (40.13)$$

in which the *polar part*  $\psi_m^P$  is an elementary function determined by the poles alone, and the *finite part*  $\psi_m^F$  has no poles at all [DMZ §1.1, ll. 196–206]. Physically [DMZ §1.2, ll. 424–443]:  $\psi_m^P$  counts multi-centred (here: two-centred) configurations, whose number depends on the moduli;  $\psi_m^F$  counts *single-centred*, or “immortal”, black holes, whose count does not. The elementary function is an Appell–Lerch sum [DMZ §9.1, ll. 3658–3663]

$$A_{2,m}(\tau, z) = \sum_{s \in \mathbb{Z}} \frac{q^{ms^2+s} y^{2ms+1}}{(1 - q^s y)^2}. \quad (40.14)$$

### “The” Fourier expansion of $A_{2,m}$ does not exist

The summand of (40.14) has poles at  $y = q^{-s}$ , so the Fourier expansion depends on the strip in which  $\text{Im } z$  lies — this *is* the wall crossing, seen analytically. All statements below fix the strip  $|q| < |y| < 1$ . In it, the general formula of DMZ [DMZ §9.1, ll. 5205–5212] gives

$$A_{k,m}(\tau, z) = \sum_{s,t \in \mathbb{Z}} \frac{\text{sgn}(t) - \text{sgn}(s + z_2/\tau_2)}{2} t^{k-1} q^{ms^2-st} y^{2ms-t}. \quad (40.15)$$

The introduction of the same paper writes the  $k = 2$  expansion as  $\sum_{r \geq \ell > 0} \ell q^{(r^2 - \ell^2)/4m} y^r$  [DMZ §1.1, ll. 208–213]; that lists the  $s \geq 0$  terms only. Expanding (40.14) directly in the same strip — or using (40.15) — also produces the mirror terms with  $y^{-r}$ , from  $s \leq -1$ . The formalization uses (40.15) and checks the two constructions against each other (a2m\_strip\_eq\_955).

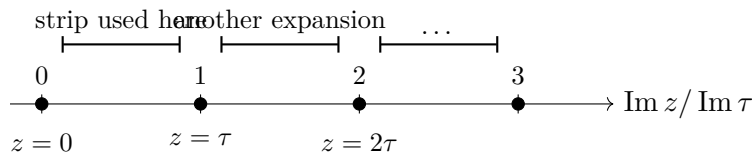


Figure 40.1: The poles of  $A_{2,m}$  sit at  $z \in \mathbb{Z}\tau + \mathbb{Z}$  (dots). Between two consecutive poles the Fourier expansion of  $A_{2,m}$  is a well-defined series, but it changes from one strip to the next: that change is the wall crossing. All statements in this chapter use the first strip,  $|q| < |y| < 1$ .

Two things are now checkable by pure arithmetic. First, that the multiple in (40.13) really is  $p_{24}(m+1)$ : the combination  $G_{m+1} - p_{24}(m+1) A_{2,m}$  must vanish to second order at  $y = 1$ , and it does, for  $m = 1, 2, 3$ ; replacing  $p_{24}(m+1)$  by  $p_{24}(m+1) \pm 1$  breaks it. Second —



and this is the content, since the second-order condition follows from the symmetry  $y \leftrightarrow y^{-1}$  — the pole coefficient is  $G_{m+1}(\tau, 0)$ , a constant in  $\tau$ , which is what makes the subtraction  $\tau$ -independent at all.

## 40.5 What is left: class numbers

Fix  $m = 1$ , the simplest magnetic charge. Removing the polar part leaves  $\Delta\psi_1^F = (G_2 - 324 A A_{2,1})/A$ , and the result is not a random-looking series. Recall the Hurwitz–Kronecker class numbers: for  $N > 0$ ,  $H(N)$  counts  $SL(2, \mathbb{Z})$ -classes of integral binary quadratic forms of discriminant  $-N$ , each weighted by the reciprocal of the number of its automorphisms, with  $H(0) = -1/12$  and  $H(N) = 0$  for  $N < 0$  [DMZ §7.1, ll. 2178–2192]. Concretely, count reduced forms  $(a, b, c)$  with  $b^2 - 4ac = -N$ ,  $|b| \leq a \leq c$  and  $b \geq 0$  when  $|b| = a$  or  $a = c$ , weighting  $a(x^2 + y^2)$  by  $\frac{1}{2}$  and  $a(x^2 + xy + y^2)$  by  $\frac{1}{3}$ :

$$\begin{array}{llll} N = 3: & (1, 1, 1), \text{ weight } \frac{1}{3} & \Rightarrow & H(3) = \frac{1}{3} \\ N = 4: & (1, 0, 1), \text{ weight } \frac{1}{2} & \Rightarrow & H(4) = \frac{1}{2} \\ N = 7: & (1, 1, 2) & \Rightarrow & H(7) = 1 \\ N = 12: & (1, 0, 3) \text{ and } (2, 2, 2) = 2(1, 1, 1) & \Rightarrow & H(12) = 1 + \frac{1}{3} = \frac{4}{3} \\ N = 15: & (1, 1, 4) \text{ and } (2, 1, 2) & \Rightarrow & H(15) = 2 \end{array}$$

(only  $a$  with  $3a^2 \leq N$  can occur, which is why the lists are short). The generating function  $\mathcal{H}(\tau, z) = \sum H(4n - r^2)q^n y^r$  [DMZ §7.2, ll. 2388–2392] is the simplest mock Jacobi form there is.

**Theorem 40.1** (Immortal dyons at  $m = 1$ ; [A] through  $q^3$ , [L] for the interpretation).  $\Delta\psi_1^F = 3E_4A - 648\mathcal{H}$ .

Two independent routes meet here. DMZ obtain the same statement by taking  $\phi = B^2/A$ , computing its finite part, and *recognizing* the numbers: the finite-part coefficients  $C(\Delta)$  and those of  $E_4A$ , called  $C^*(\Delta)$ , differ by exactly  $-288H(\Delta)$  [DMZ §8.5, ll. 3530–3558]. For instance at  $\Delta = 3$ ,  $152 - 248 = -96 = -288 \cdot \frac{1}{3}$ , and at  $\Delta = 4$ ,  $-636 + 492 = -144 = -288 \cdot \frac{1}{2}$ . Our route computes  $\psi_1$  from the product (40.8), subtracts  $324A_{2,1}$  and divides by  $A$ ; the two answers agree coefficient by coefficient. With  $\phi_{2,1}^{\text{opt}} = (B^2 - E_4A^2)/(144A)$  [DMZ §9.2, ll. 3770–3776], whose mock part is  $-2\mathcal{H}$ , and  $p_{24}(2) = 324$ , the arithmetic of  $\Delta\psi_1 = \frac{9}{4}B^2/A + \frac{3}{4}E_4A$  gives Theorem 40.1 directly.

### Worked example: the first coefficients of the immortal series

It is worth seeing Theorem 40.1 as numbers. Write  $\Delta\psi_1^F = \sum_{n,r} f(n, r)q^n y^r$ . At order  $q^0$  the form  $A$  contributes  $y - 2 + y^{-1}$  and  $\mathcal{H}$  contributes only  $H(0) = -\frac{1}{12}$  at  $r = 0$ , so

$$f(0, \pm 1) = 3, \quad f(0, 0) = 3 \cdot (-2) - 648 \cdot \left(-\frac{1}{12}\right) = -6 + 54 = 48. \quad (40.16)$$

At order  $q^1$  use the coefficients  $C^*(\Delta)$  of  $E_4A$  tabulated by DMZ [DMZ §8.5, ll. 3546–3552]:  $C^*(0) = -2$ ,  $C^*(3) = 248$ ,  $C^*(4) = -492$ . The discriminant of  $(n, r) = (1, 0)$  is  $\Delta = 4$ , of  $(1, \pm 1)$  is 3, of  $(1, \pm 2)$  is 0, so

$$\begin{aligned} f(1, 0) &= 3 \cdot (-492) - 648 \cdot \frac{1}{2} = -1800, & f(1, \pm 1) &= 3 \cdot 248 - 648 \cdot \frac{1}{3} = 528, \\ f(1, \pm 2) &= 3 \cdot (-2) + 54 = 48. \end{aligned} \quad (40.17)$$

Every entry is an integer although  $\mathcal{H}$  has denominators 12, 3, 2: the combination  $3E_4A - 648\mathcal{H}$  clears them, which is one reason the factor  $648 = 54 \cdot 12$  appears. The same integrality is what makes the Lean statement a comparison of integer lists.

For higher  $m$  the pattern continues with a Hecke-like operator.  $V_{k,t}$  acts on Fourier coefficients by  $C(\phi|V_{k,t}; \Delta, r) = \sum_d d^{k-1} C(\phi; \Delta/d^2, r/d)$  [DMZ §4.4, ll. 1478–1490], so for weight 2, index 1 and  $t = p$  prime the effect on a coefficient is  $C(\Delta) \mapsto C(\Delta) + p C(\Delta/p^2)$  whenever  $p^2 \mid \Delta$ . DMZ find  $\Phi_{2,p}^{\text{opt}} = -\mathcal{H}|V_p$  for  $p$  prime [DMZ §9.2, ll. 3846–3852], with the explicit case splits

$$\begin{aligned} C(\Phi_{2,2}^{\text{opt}}; \Delta) &= \begin{cases} -H(\Delta) & \Delta \equiv 7 \pmod{8}, \\ -H(\Delta) - 2H(\Delta/4) & 4 \mid \Delta, \end{cases} \\ C(\Phi_{2,3}^{\text{opt}}; \Delta) &= \begin{cases} -H(\Delta) & 9 \nmid \Delta, \\ -H(\Delta) - 3H(\Delta/9) & 9 \mid \Delta. \end{cases} \end{aligned} \quad (40.18)$$

Feeding these into (40.11) with  $p_{24}(3) = 3200$ ,  $p_{24}(4) = 25650$  gives the immortal counting functions at  $m = 2, 3$ :

$$\begin{aligned} 3 \Delta \psi_2^F &= -9600 \mathcal{H}|V_2 + 22E_4AB - 10E_6A^2, \\ 48 \Delta \psi_3^F &= -1231200 \mathcal{H}|V_3 + 467E_4AB^2 - 430E_6A^2B + 203E_4^2A^3. \end{aligned} \quad (40.19)$$

Both are checked in Section 40.7, as is the failure of (40.19) when the Hecke correction  $2H(\Delta/4)$  is dropped — the case split is not decoration.

#### What the class numbers are counting

A single-centred black hole of charge  $(m, n, \ell)$  has a degeneracy that does not depend on where one stands in moduli space. Theorem 40.1 says that for the smallest magnetic charge those degeneracies, after removing a true Jacobi form, are class numbers: the number of  $SL(2, \mathbb{Z})$ -classes of binary quadratic forms of discriminant  $-\Delta$ , where  $\Delta = 4n - \ell^2$  is the discriminant of the charge (40.1). The [C] reading that this is “the black hole counting its own charge lattice” is tempting and is not proved anywhere; the identity of the two series is [A] through the order stated, and the physical meaning of  $\psi^F$  is [L].

## 40.6 Twining: are these integers characters?

Chapter 30 introduced the one test that separates moonshine from numerology: replace dimensions by traces and ask whether the result is still a character. Apply it here. For  $g \in M_{24}$  with cycle shape (Frame shape)  $\prod_k k^{e_k}$  on 24 points [CDH Table 14, ll. 4349–4364], the twined analogue of (40.10) is

$$\sum_k T_k(g) p^k = \prod_{n \geq 1} \frac{1}{\det(1 - p^n g | \mathbb{C}^{24})} = \prod_{n \geq 1} \prod_k (1 - p^{nk})^{-e_k}. \quad (40.20)$$

At  $k = 2$  this is the character of  $V \oplus \text{Sym}^2 V$  with  $V = \mathbf{1} \oplus \mathbf{23}$  the permutation representation. Since  $\text{Sym}^2 \mathbf{23} = \mathbf{1} \oplus \mathbf{23} \oplus \mathbf{252}$ ,

$$H^*(\text{Hilb}^2 K3) = 3 \cdot \mathbf{1} \oplus 3 \cdot \mathbf{23} \oplus \mathbf{252}, \quad 3 + 3 \cdot 23 + 252 = 324, \quad (40.21)$$

a genuine character with non-negative multiplicities. The same holds through  $k = 4$ . So the 24 of Section 40.3, unlike the “27720 lock” of Chapter 31, passes the twining test.

One can go further and twist the whole partition function. Cheng’s twisted denominators [Cheng §3.2, ll. 937–947] are

$$\Phi_g(\Omega) = e^{4\pi i(\varrho, \Omega)} \prod_{\alpha} \exp\left(-\sum_{k \geq 1} \frac{1}{k} c_{g^k}(|\alpha|^2) e^{2\pi i k(\alpha, \Omega)}\right), \quad (40.22)$$

where  $c_g$  are the Fourier coefficients of the twisted K3 elliptic genus  $Z_g$  and  $g^k$  is read off the power maps of the character table [CDH Table 8, ll. 4172–4178]. The twisted genera themselves come from the twining data of Chapter 30: with  $\chi_g$  the trace of  $g$  on  $\mathbf{1} \oplus \mathbf{23}$  [Cheng §2, ll. 455–460] and  $F_g$  the weight-2 forms of CDH,

$$Z_g = \frac{\chi_g}{24} Z - F_g A, \quad (40.23)$$

which for the geometric classes  $p = 2, 3, 5, 7$  is Cheng’s (2.5) [Cheng §2, ll. 415–423]. Expanding  $\Phi_g^{-1}$  by Newton’s identities gives twisted analogues  $G_k^{(g)}$  of the symmetric-product genera, and the outcome of the test is: every Fourier coefficient of  $G_1^{(g)}, G_2^{(g)}$ , and of the twisted immortal counting function at  $m = 1$ , is a virtual character of  $M_{24}$  — the multiplicities are integers, possibly negative. For example the coefficient  $-1800$  of  $q^1 y^0$  decomposes as  $2 \cdot \mathbf{1} + 2 \cdot \mathbf{23} + 2 \cdot \mathbf{45} + 2 \cdot \mathbf{45} - \mathbf{231} - \mathbf{231} + 2 \cdot \mathbf{252} - \mathbf{1035}' - \mathbf{1035}''$ .

*Remark 40.2* (A printed sign). Cheng states  $c_g(-1) = -2$  when deriving the factorization of the pole [Cheng §3.2, l. 960]. In the normalization used here — and in Cheng’s own (2.5) — the coefficient of  $y^{\pm 1} q^0$  in  $Z_g$  is  $+2$  for every class; the  $q^0$  term is  $2y + (\chi_g - 4) + 2y^{-1}$ . This is a sign convention, not a disagreement about the twisted genera: the twisted products built with  $+2$  reproduce the untwisted product at  $g = 1A$  and the twined Göttsche numbers at  $z = 0$ .

#### This is $N = 4$ , not our universe

Everything in this chapter lives in four-dimensional  $N = 4$  supergravity, where supersymmetry protects the index and makes the count exact. Four-dimensional  $N = 4$  supersymmetry is not chiral [Asp §7, ll. 2713–2716], so the Standard Model is not among these vacua. The results are statements about a mathematically controlled theory, **[L]** for their physical reading and **[A]** only for the arithmetic; nothing here predicts anything measurable in our world (Chapter 41).

## 40.7 What the machine has checked

The library `DualScaleDyons` (six files) recomputes all of the above as exact integer arithmetic on truncated series: a  $q$ -series is a list whose entries are Laurent polynomials in  $y$ , stored as an offset plus a coefficient list, so no truncation in  $y$  ever occurs. Every theorem below depends only on `propext`, `Classical.choice` and `Quot.sound`; the heavier ones are closed by `decide +kernel`, which runs the decision procedure in the kernel itself.

#### In Lean: the product, the guard, and Göttsche’s numbers

```
theorem dmrv_reachable :
  [(5, 1), (4, 2), (2, 3)].all
    (fun kq : ℕ × ℕ => (dmrvDs kq.1 kq.2).all cAvailable) = true := by decide

theorem dmrv_unreachable_example :
  (dmrvDs 4 3).all cAvailable = false := by decide

theorem p24_values :
  (List.range 6).map p24 = [1, 24, 324, 3200, 25650, 176256] := by decide

theorem goettsche :
  (dmrv 5 1).map (fun Gk => Gk.map LP.eval1) =
    (List.range 6).map fun k => [p24 k, 0] := by decide +kernel
```

**Reading.** `dmrv K Q` is the product (40.8) truncated at  $p^K, q^Q$ , with exponents

read from the *computed*  $K3$  elliptic genus. `dmvv_reachable` says that for the three truncations used, every discriminant the product reads is available from that genus; `dmvv_unreachable_example` shows the guard is not vacuous. `goettsche` evaluates each  $G_k$  at  $y = 1$  (`LP.eval1` sums the coefficients) and finds  $[p_{24}(k), 0]$ : the value  $p_{24}(k)$  at order  $q^0$  and 0 at order  $q^1$ , i.e. constancy in  $\tau$  as well as the value. **Proof idea.** Kernel evaluation of the finite product and comparison of integer lists. **Not proved.** Convergence, modularity, or that  $G_k$  is the elliptic genus of a symmetric product ([L], [DMZ §4.5, ll. 1688–1698]).

#### In Lean: the additive formulas and the polar coefficient

```
theorem dmz_516_q2 :
  (List.range 5).all (fun k =>
    Ser.eqB (Ser.scale ((dmz516 2).getD k (1, [])).1 ((dmvv 4 2).getD k []))
      ((dmz516 2).getD k (1, [])).2) = true := by decide +kernel

theorem a2m_strip_eq_955 :
  [1, 2, 3].all (fun m => Ser.eqB (a2mRest m 6) (a2m955 m 6)) = true := by
  decide

theorem polar_part_removes_pole :
  [1, 2, 3].all (fun m => vanish2 (polarDefect m (p24 (m + 1)))) = true := by
  decide +kernel

theorem polar_coefficient_pinned :
  [1, 2, 3].all (fun m => !vanish2 (polarDefect m (p24 (m + 1) + 1)) &&
    !vanish2 (polarDefect m (p24 (m + 1) - 1))) = true := by decide +kernel
```

**Reading.** The first theorem is (40.11) for  $k \leq 4$  through order  $q^2$ , with both sides multiplied by the printed denominators: the product route and the  $A, B, E_4, E_6$  route agree. The second says the direct expansion of (40.14) in the strip equals (40.15). The last two say  $G_{m+1} - p_{24}(m+1)AA_{2,m}$  vanishes to second order at  $y = 1$  and that  $p_{24}(m+1) \pm 1$  do not. **Caveat, as the file states.** The second-order condition holds automatically by the  $y \leftrightarrow y^{-1}$  symmetry, so what these two discriminate is the first-order condition — the pole coefficient. The full second-order content at  $m = 1$  is `immortal_m1_exact`, the exactness of the division by  $A$ . **Tier [A]** for all four; [L] for “ $\psi^P$  counts two-centred states”.

#### In Lean: immortal dyons are class numbers

```
theorem hurwitz_values :
  [0, 3, 4, 7, 8, 11, 12, 15].map h12 =
  [-1, 4, 6, 12, 12, 12, 16, 24] := by decide

theorem immortal_m1 :
  Ser.eqB immortalM1
    (Ser.sub (Ser.scale 3 (Ser.mul (e4 3) (phiA 3)))
      (Ser.scale 54 (hurwitzSer 3))) = true := by decide +kernel

theorem immortal_m1_needs_H :
  Ser.eqB immortalM1 (Ser.scale 3 (Ser.mul (e4 3) (phiA 3))) = false := by
  decide +kernel

theorem immortal_m2 :
  Ser.eqB (Ser.scale 3 (immortalM 2))
    (Ser.add (Ser.scale (-800) (hurwitzVSer 2 2 true)))
```

```
(Ser.add (Ser.scale 22 (mulAll [e4 2, phiA 2, phiB 2] 2))
  (Ser.scale (-10) (mulAll [e6 2, phiA 2, phiA 2] 2))) = true := by
decide +kernel
```

**Reading.** `h12` computes  $12H(N)$  by enumerating reduced forms, so `hurwitz_values` is the hand computation of Section 40.5 checked by machine ( $12 \cdot \frac{1}{3} = 4$ ,  $12 \cdot \frac{1}{2} = 6$ ,  $12 \cdot \frac{4}{3} = 16$ ). `immortalM1` is  $(G_2 - 324AA_{2,1})/A$  computed from the product; `immortal_m1` is Theorem 40.1 with  $648\mathcal{H}$  written as  $54 \cdot (12\mathcal{H})$  to stay in  $\mathbb{Z}$ . The control shows the class-number term is needed. `immortal_m2` is the first line of (40.19), with `hurwitzVser 2 2 true` the series  $12(\mathcal{H}|V_2)$ ; passing `false` instead drops the Hecke correction and the identity fails (`immortal_m2_needs_V2`). **Orders.**  $q^3$  for  $m = 1$ ,  $q^2$  for  $m = 2, 3$ . **Tier [A]** for the identities; **[L]** for “immortal” and for mock modularity.

#### In Lean: the twining test on the dyon sector

```
theorem frame_power_maps :
  powerMaps.all (fun pm => (List.range 26).all fun j =>
    fixedOfPower j pm.1 == fixedOfPower (pm.2.getD j 0) 1) = true := by decide

theorem twined_goettsche_k2 :
  (List.range 26).map (fun i => (twinedInner 2 i).1 / 244823040) =
    [3, 3, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0] := by decide +kernel

theorem twisted_dyons_are_characters :
  [1, 2].all (fun k => (List.range 3).all fun n =>
    (List.range 17).all fun i =>
      isVirtualChar (classFn (fun j => (dmvTw j 2 2).getD k []) n
        ((i : ℤ) - 8))) = true := by decide +kernel

theorem c_minus_one :
  (List.range 26).all (fun j => cTw j (-1) == 2) = true := by decide +kernel
```

**Reading.** `frame_power_maps` cross-checks two transcribed tables: the number of points fixed by  $g^p$ , read from the Frame shape of  $g$ , equals the fixed-point count of the class of  $g^p$  given by the power maps, for all 26 classes and  $p = 2, 3, 5, 7, 11, 23$ . `twined_goettsche_k2` is (40.21): the multiplicities of  $T_2$  on the 26 irreducibles in this repository’s ordering,  $3, 3, \dots, 1$  at the entry **252**. `twisted_dyons_are_characters` says every Fourier coefficient of the twisted  $G_1^{(g)}, G_2^{(g)}$  through  $q^2$  is a virtual character (all inner products with the character table are integers and all irrational parts vanish). `c_minus_one` is the sign of the remark above. **Honest scope.** Given that the twisted genera are characters — which is Gannon’s theorem, **[L]** — the later steps are linear and class-independent, so character-valuedness is expected; what is certified is the end-to-end computation from the twining data of Chapter 30 and the power maps.

#### Chapter summary

- Quarter-BPS dyons on  $K3 \times T^2$  are counted by Fourier coefficients of  $1/\Phi_{10}$ ; the three charges enter through the T-duality invariants  $(m, n, \ell) = (M^2/2, N^2/2, N \cdot M)$ .
- $1/\Phi_{10}$  is built from one input, the K3 elliptic genus  $Z = 2\phi_{0,1}$ , by the product (40.8). At  $z = 0$  it collapses to  $\prod(1-p^r)^{-24}$ , so  $G_k(\tau, 0) = p_{24}(k) = 1, 24, 324, 3200, 25650, 176256$ : Göttsche’s numbers, the 24 of K3 propagated to all orders. The additive formulas (40.11) are an independent check.

- $\Delta\psi_m$  has a double pole at  $z = 0$  with coefficient  $p_{24}(m+1)$ . Subtracting the canonical polar part  $p_{24}(m+1)A_{2,m}/\Delta$  leaves the finite part: two-centred (moduli-dependent, wall-crossing) versus single-centred (“immortal”) counts.
- At  $m = 1$  the immortal counting function is  $3E_4A - 648\mathcal{H}$  with  $\mathcal{H}$  the generating function of Hurwitz class numbers; at  $m = 2, 3$  the same holds with  $\mathcal{H}|V_2, \mathcal{H}|V_3$  and the case splits (40.18).
- Twining: the twined Göttsche numbers are characters of  $M_{24}$  ( $H^*(\text{Hilb}^2K3) = 3 \cdot \mathbf{1} \oplus 3 \cdot \mathbf{23} \oplus \mathbf{252}$ ), and every coefficient of the twisted partition functions, including the twisted immortal counts, is a virtual character.
- Tiers: the arithmetic is **[A]** through the stated orders; “counts black hole microstates”, mock modularity and the completion are **[L]**;  $N = 4$  in four dimensions is non-chiral, so none of this is a model of our universe.

## Exercises

**Exercise 40.1** (★). Expand  $\prod_{r \geq 1} (1 - p^r)^{-24}$  to order  $p^3$  by hand and check 1, 24, 324, 3200. (Hint:  $(1 - p)^{-24}$  alone gives 1, 24, 300, 2600; add the corrections from  $r = 2, 3$ .)

**Exercise 40.2** (★). Compute  $H(8), H(11), H(16), H(19), H(20)$  by listing reduced forms, and check them against  $12H = 12, 12, 18, 12, 24$ .

**Exercise 40.3** (★). Using (40.5), verify  $A \sim (2\pi iz)^2$  and  $B \rightarrow 12$  as  $z \rightarrow 0$ , and deduce (40.12) from (40.9).

**Exercise 40.4** (★★). Check the third relation of (40.11) at order  $q^0$ : compute  $B^2$  and  $E_4A^2$  at  $q^0$  from (40.5) and verify that  $\frac{9}{4}B^2 + \frac{3}{4}E_4A^2$  has the double pole  $324/(y-1)^2 \cdot y$  expected from (40.12).

**Exercise 40.5** (★★). Expand the  $s = 1$  term of (40.14) for  $m = 1$  in the strip  $|q| < |y| < 1$ , then again in the strip  $|q|^2 < |y| < |q|$ , and exhibit the first coefficient that differs. Explain which physical statement the difference encodes.

**Exercise 40.6** (★★). Prove  $\text{Sym}^2 \mathbf{23} = \mathbf{1} \oplus \mathbf{23} \oplus \mathbf{252}$  for  $M_{24}$  by computing  $\frac{1}{2}(\chi_{23}(g)^2 + \chi_{23}(g^2))$  on three classes of your choice and comparing with the character table of Chapter 29; deduce (40.21).

**Exercise 40.7** (★★). In Lean, state and prove that the twined Göttsche numbers at  $k = 1$  reproduce the twisted Euler characters: `(List.range 26).map (fun j => twinedHilb j 1) = chiShadow`. (One line with `decide`; compare `frame_fixed_points`.)

**Exercise 40.8** (★★★). Extend `immortal_m1` by one order in  $q$ : change the truncation from 3 to 4 in `immortalNum`, `immortalM1` and `hurwitzSer`, re-check that the division by  $A$  is still exact, and measure the kernel time. Explain which step grows fastest and why the reachability guard must be re-examined.

**Exercise 40.9** (★★★). Derive the second identity of (40.19) from (40.11),  $\phi_{2,3}^{\text{opt}} = \psi_{0,4}^{\text{opt}}/(12^4 A)$  with  $\psi_{0,4}^{\text{opt}} = B^4 - 6E_4A^2B^2 + 8E_6A^3B - 3E_4^2A^4$ , and  $p_{24}(4) = 25650$ ; check the rational coefficients  $467/48, -215/24, 203/48$  before clearing denominators.

## Further reading

- Charges, invariants and the two counting problems: [DMZ §2.2–2.3, ll. 636–760]; half-BPS degeneracies and  $p_{24}$ : [DMZ §3.4, ll. 1040–1071]; walls and contours: [DMZ §6, ll. 1900–2010].
- Jacobi forms of index 1, the theta formulas and the coefficient table: [DMZ §4.3, ll. 1368–1430]; Hecke-like operators  $U_s, V_{k,t}$ : [DMZ §4.4, ll. 1470–1500]; the elliptic genus of K3 and the symmetric-product product formula: [DMZ §4.5, ll. 1660–1700].
- The Igusa form, its product representation and the additive formulas: [DMZ §5.1–5.2, ll. 1810–1900]; the polar/finite decomposition and its physics: [DMZ §1.1–1.2, ll. 196–240, 420–470]; class numbers and Example 5: [DMZ §7.1, ll. 2178–2196] and [DMZ §8.5, ll. 3530–3558]; optimal choices and  $\mathcal{H}|V_p$ : [DMZ §9.1–9.2, ll. 3600–3860].
- Twisted elliptic genera and twisted denominators: [Cheng §2, ll. 400–460] and [Cheng §3.2, ll. 900–990]; Frame shapes and power maps: [CDH Tables 8 and 14, ll. 4170–4209, 4349–4364].
- Within this book: the elliptic genus and  $N = 4$  characters, Chapter 28;  $M_{24}$  and its character table, Chapter 29; moonshine and the twining test, Chapter 30; the computed moonshine data this chapter reuses, Chapter 39; what all of this does *not* say about our universe, Chapter 41.





## Chapter 41

# Confronting the Dual-Scale Hypothesis with Data: Pre-Registration and Four Verdicts

A physical hypothesis earns its name only if some measurement could prove it wrong. The dual-scale programme of this book began with an attractive idea: treat a microscopic length and a macroscopic length as a T-dual pair, as string theory treats  $R$  and  $\alpha'/R$  in Chapter 8. Chapter 32 and Chapter 38 showed what the formal part of that idea amounts to. This chapter asks the question a student should ask of any proposal: *what does it predict that we can measure, and does the measurement agree?* Answering it honestly takes more than arithmetic. A test has to be written down, with its decision rule, *before* the verdict is computed, and anything the author already knew about the data has to be declared. We first explain what such a pre-registered test is and why each of its ingredients matters. We then list everything the programme can derive without a compactification that has never been constructed; there turn out to be exactly four observables. We confront each with data, derive every step on the page, and end with the single cause behind all four verdicts and what it teaches. The Lean modules attached to the chapter check the algebra and the arithmetic of each verdict; none of them checks that the identifications with physics are right.

Conventions:  $\hbar = c = 1$  unless units are written out;  $\ell_P$  is the Planck length, so  $G = \ell_P^2$ ;  $c/H_0$  is the Hubble radius today. Numbers for  $\ell_P$  and  $c/H_0$  are those fixed in `DualScaleCosmology/SelfDualCutoff.lean`:  $\ell_P = 1.616255 \times 10^{-35}$  m (CODATA 2018) and  $c/H_0 = 1.3672 \times 10^{26}$  m (Planck 2018,  $H_0 = 67.66$  km/s/Mpc), both taken from `astropy` and not checked against a source in this book’s library.

### 41.1 What makes a test

#### 41.1.1 Freeze first, compute second

A *pre-registered prediction* is a statement written and time-stamped before the comparison with data. In this programme the time stamp is a git tag: the prediction document is committed, tagged (for example `stream6-p1-frozen`), and pushed, and only then is the verdict computed. The document contains three things:

1. the *statement*, with every assumption labelled and every input number fixed;
2. the *tests*, each naming a pinned data source and the inequality the prediction must satisfy to pass;

3. the *decision rule*: which pattern of passed and failed tests counts as “consistent”, “excluded” or “in tension”.

Once the tag exists, the document is never edited. A mistake found later is recorded in a separate errata file (for this chapter’s documents, `docs/STREAM67_ERRATA.md`, which corrects two line numbers and nothing else); a changed prediction is a *new* prediction with a new name.

### 41.1.2 Disclosure: retrodiction and forward prediction

The authors of a prediction usually know some of the data already. If they do, the freeze cannot make the test blind; it can only stop them from adjusting the prediction afterwards. So each frozen document says what the authors knew. A comparison with data that were known before the freeze is a *retrodiction*; a comparison with data that did not yet exist, or had not been looked up, is a *forward prediction*. Both are useful. A retrodiction with a recorded outcome closes a question and prevents it from being quietly reopened; a forward prediction is the only kind that can surprise its authors.

### 41.1.3 Is the observable empty?

Before freezing, list the inputs of the predicted number under three headings: *measured* (taken from an experiment), *derived* (proved from the assumptions, here in Lean), and *assumed* (an identification or a choice). If the assumed list is not empty, the prediction is Tier C and must say so. A second failure mode is subtler: a Lean theorem can certify arithmetic on numbers that were typed in by hand, and prose around it can then call the result “consistent with observation”. The kernel checks the arithmetic, not the choice of the numbers. The rule of this chapter is therefore that every number entering a test is either a pinned measurement or the output of a stated derivation.

### 41.1.4 Order-one factors must be derived

Many physical identifications carry a factor of order one: a radius or a circumference, a  $2\pi$ , a coupling of the lightest mode. When a prediction misses a bound by a factor of order one, it is tempting to pick the convention that makes it pass. That is a fit, not a test.

#### Choosing the convention after reading the bound

In Section 41.3 the predicted radius is  $s \approx 47\mu\text{m}$  and the laboratory bound is  $30\mu\text{m}$ . Writing  $R = s/2\pi \approx 7.5\mu\text{m}$  instead would pass the bound, and would even land inside the window  $0.1\text{--}10\mu\text{m}$  of a published “dark dimension” proposal [MVV §3, ll. 424–427]. Choosing  $2\pi$  *after* seeing the bound turns a failed test into a successful fit. The factor must come from the theory, before the comparison; in Section 41.3 it does, and it is 1.

## 41.2 What the programme can predict

The question to ask first is: which results of the programme are numbers *with units*, obtained by a derivation that does not pass through an explicit compactification (flux quanta, brane wrapping numbers, moduli stabilization), since no such compactification has been constructed for this geometry? Going through the programme:

| Result                                                                                                             | Number with units?          | Observable                                         |
|--------------------------------------------------------------------------------------------------------------------|-----------------------------|----------------------------------------------------|
| Lattices, $O(d, d; \mathbb{Z})$ , double field theory, the bound $R + \alpha'/R \geq 2\sqrt{\alpha'}$ (Chapter 31) | no                          | none                                               |
| Moonshine and dyon computations (Chapters 39 and 40)                                                               | no (integers)               | none: they concern $\mathcal{N} = 4$ string theory |
| Self-dual length $s = \sqrt{\ell_P c/H_0}$                                                                         | yes                         | P1: radius of an extra dimension                   |
| CKN saturation with infrared length $c/H(t)$                                                                       | yes                         | C-A: deceleration parameter                        |
| Scale-factor duality realized dynamically                                                                          | with dynamics               | C-B: relic gravitational waves                     |
| F-strings at $\alpha' = s^2$                                                                                       | yes (dimensionless $G\mu$ ) | C-C: string tension                                |

All four observables rest on one identification (**[C]**): the string scale is the geometric mean of the Planck length and the Hubble radius,

$$\alpha' = \ell_P \cdot \frac{c}{H_0}, \quad s = \sqrt{\alpha'}. \quad (41.1)$$

## 41.3 P1: the self-dual length as an extra dimension

### 41.3.1 The prediction

The self-dual radius of a circle is the fixed point of  $R \mapsto \alpha'/R$  (Chapter 8). With (41.1), P1 identifies it with the radius  $R$  of *one* large extra dimension, with the Kaluza–Klein tower starting at mass  $m \sim 1/R$  (the convention  $m^{-1} \sim l$  of [MVV §3, l. 327]).

**Example 41.1** (Computing  $s$ ). With the inputs above,

$$\begin{aligned} s^2 &= \ell_P \cdot \frac{c}{H_0} = 1.616255 \times 10^{-35} \text{ m} \times 1.3672 \times 10^{26} \text{ m} = 2.2097 \times 10^{-9} \text{ m}^2, \\ s &= 4.70 \times 10^{-5} \text{ m} \approx 47 \mu\text{m}. \end{aligned}$$

The tests compare squares,  $s^2$  against  $(30 \mu\text{m})^2 = 9 \times 10^{-10} \text{ m}^2$ , so no square root is ever taken in the Lean statements.

### 41.3.2 The data and the decision rule

Three bounds were frozen with P1 (tag `stream6-p1-frozen`):

- T1, the Eöt-Wash torsion balance: the largest extra dimension “must have a toroidal radius less than  $30 \mu\text{m}$ ” [EW, l. 289];
- T2, the same experiment: any gravitational-strength Yukawa interaction has range  $\lambda < 38.6 \mu\text{m}$  [EW, ll. 283–285], applied with  $\lambda = R$  for the lightest mode (an identification, **[C]**);
- T3, heating of old neutron stars by trapped Kaluza–Klein gravitons: for one extra dimension  $m^{-1} \sim l < 44 \mu\text{m}$  [MVV §3, ll. 324–327].

Decision rule: *consistent* if all pass, *excluded* if T1 fails, *in tension* if only T2 or T3 fails. The authors knew these bounds before the freeze; P1 is a retrodiction.

### 41.3.3 The verdict

Since  $s^2 = 2.21 \times 10^{-9} \text{ m}^2$  exceeds  $(30 \mu\text{m})^2$ ,  $(38.6 \mu\text{m})^2 = 1.49 \times 10^{-9} \text{ m}^2$  and  $(44 \mu\text{m})^2 = 1.94 \times 10^{-9} \text{ m}^2$ , all three tests fail, and by the rule **P1 is excluded**. The margins are not rounding effects:  $s > 1.5 \times 30 \mu\text{m}$  and  $s > 1.06 \times 44 \mu\text{m}$ . To pass T1 one would need  $R = \kappa s$  with  $\kappa < 30/47.0 = 0.64$ .

### 41.3.4 The order-one factor is fixed by T-duality

Is  $\kappa$  free? The programme's own duality answers no. The map  $R \mapsto \alpha'/R = s^2/R$  has a fixed point where  $R = s^2/R$ , i.e.  $R^2 = s^2$ ; for  $R > 0$  and  $s > 0$  the only solution is  $R = s$ . At that radius the momentum scale  $1/R$  equals the winding scale  $R/\alpha'$ :

$$\left. \frac{1}{R} \right|_{R=s} = \frac{1}{s} = \frac{s}{s^2} = \left. \frac{R}{\alpha'} \right|_{R=s},$$

which is what makes the self-dual radius special (Chapter 8). With the standard Kaluza–Klein normalization  $m_n = n/R$  the radius is therefore  $s$  itself:  $\kappa = 1$ . If both circles of  $T^2$  were large (two extra dimensions), the neutron-star bound becomes  $l < 1.6 \times 10^{-4} \mu\text{m}$  [MVV §3, l. 329], missed by a factor of about  $3 \times 10^5$ .

*Remark 41.2.* Chapter 38 called this reading “disfavored, not excluded”, because the order-one factor was still open. The derivation above closes it; the pre-registered rule then classifies P1 as excluded. The two statements are consistent: they differ in what was known about  $\kappa$ .

## 41.4 C-A: holographic dark energy with infrared length $c/H$

### 41.4.1 The Cohen–Kaplan–Nelson bound and its saturation

Cohen, Kaplan and Nelson (CKN) argued that an effective field theory with ultraviolet cutoff  $\Lambda$  in a box of size  $L$  should not contain states inside their own Schwarzschild radius, which requires [CKN, eq. (2), ll. 75–78]

$$L^3 \Lambda^4 \lesssim L M_P^2. \quad (41.2)$$

The left side is the maximal energy  $\Lambda^4 L^3$  in the box, the right side the mass of a black hole of size  $L$ . Candidate C-A assumes ([C]) that the vacuum energy saturates this bound at all times with the programme's macroscopic length as infrared length,  $L = c/H(t)$ . Then  $\rho_\Lambda \sim \Lambda^4 \sim M_P^2/L^2 \propto H^2$ , which we write as

$$\rho_\Lambda(t) = \varepsilon \frac{3H(t)^2}{8\pi G}, \quad 0 < \varepsilon < 1, \quad (41.3)$$

with a constant  $\varepsilon$ . We also assume ([C]) a flat Friedmann universe with pressureless matter that is separately conserved,  $\rho_m \propto a^{-3}$ , and neglect radiation.

### 41.4.2 The derivation, step by step

Write  $k = 8\pi G/3$ . The Friedmann equation is  $H^2 = k(\rho_m + \rho_\Lambda)$ . Substituting (41.3):

$$H^2 = k\rho_m + \varepsilon H^2 \implies \rho_m = (1 - \varepsilon) \frac{H^2}{k}.$$

The matter fraction  $\Omega_m = k\rho_m/H^2 = 1 - \varepsilon$  is constant. With  $\rho_m = \rho_{m0}a^{-3}$ ,

$$H^2 = \frac{k\rho_{m0}}{1 - \varepsilon} a^{-3} \equiv \frac{C}{a^3}, \quad C > 0.$$

The deceleration parameter is  $q = -\ddot{a}a/\dot{a}^2$ . Using  $\dot{a} = aH$ ,  $\ddot{a} = \dot{a}H + a\dot{H}$ , so  $q = -1 - \dot{H}/H^2$ , and with  $\dot{H} = \dot{a} dH/da = aH dH/da$ ,

$$q = -1 - \frac{d \ln H}{d \ln a} = -1 - \frac{a E'(a)}{2E(a)}, \quad E(a) = H(a)^2. \quad (41.4)$$

For  $E = C/a^3$  we get  $aE'/E = -3$ , hence

$$q = -1 + \frac{3}{2} = \frac{1}{2} \quad \text{at every epoch and for every } \varepsilon.$$

The dark component (41.3) dilutes exactly like matter; the universe behaves as if it contained dust only,  $w_{\text{eff}} = 0$ , and it decelerates.

#### Why a Hubble-scale cutoff cannot accelerate

If the dark energy is always a fixed fraction of  $3H^2/8\pi G$ , it can never come to dominate: its share of the budget is frozen at  $\varepsilon$ . Acceleration needs a component whose density falls more slowly than  $H^2$  does in a matter era; tying the density to  $H^2$  forbids exactly that. This is the classic objection to Hubble-scale holographic dark energy, and it is why the authors knew the outcome before the freeze.

#### 41.4.3 The verdict

The frozen test (tag `stream7-ca-frozen`) compares with the present deceleration parameter inferred from Planck 2018 in flat  $\Lambda$ CDM,  $\Omega_\Lambda = 0.68885$  (the value in `DarkEnergyScale.omegaLambda`; not checked against a source in this book's library),  $\Omega_m \approx 1 - \Omega_\Lambda$ :

$$q_0 = \frac{\Omega_m}{2} - \Omega_\Lambda = \frac{0.31115}{2} - 0.68885 = -0.533 < 0.$$

C-A requires  $q_0 = +1/2$ : **C-A is excluded**. Two caveats were recorded in advance: the number  $q_0$  comes from a  $\Lambda$ CDM fit, so it is model-dependent; and the test is a retrodiction with a foreseeable outcome.

## 41.5 C-B: relic gravitons from a pre-big-bang phase

### 41.5.1 The spectrum

Scale-factor duality (Chapter 32) suggests a phase before the big bang in which the universe contracts in the Einstein frame while the dilaton grows. Tensor perturbations amplified during such a phase leave a relic background of gravitational waves. For a background  $a \propto |\eta|^\alpha$  in conformal time the spectral energy density scales as [GV §4.3, eq. (4.105) and Table 2, ll. 4466–4485]

$$\Omega(\omega) \sim \omega^{3-2\nu}, \quad \nu = \left| \alpha - \frac{1}{2} \right|. \quad (41.5)$$

De Sitter inflation ( $\alpha = -1$ ) gives  $\nu = 3/2$  and a flat spectrum; the dilaton-driven phase has  $\alpha = 1/2$ ,  $\nu = 0$  and a cubic spectrum  $\Omega \sim \omega^3$  [GV §4.3, ll. 4503–4506], which persists in the low-frequency band of the full model [GV §5.1, eq. (5.15), ll. 5040–5053]. The end point is fixed by the curvature at the transition,  $g_1 = H_1/M_P \simeq M_s/M_P$  [GV §5.1, eq. (5.16), ll. 5059–5061]:

$$\omega_1 \simeq g_1^{1/2} 10^{11} \text{ Hz}, \quad \Omega(\omega_1) \simeq 10^{-4} g_1^2. \quad (41.6)$$

### 41.5.2 The programme fixes the string scale

Candidate C-B assumes ([C]) that scale-factor duality is realized as a pre-big-bang phase and that the string scale is (41.1), so  $M_s \sim 1/s$  and  $g_1 = \ell_P/s$ , i.e.  $g_1^2 = \ell_P^2/s^2 = \ell_P H_0/c$ . No parameter is left except the position of a break in the spectrum, which can only lower it below  $\omega_1$ .

**Example 41.3** (The numbers of C-B).  $g_1^2 = 1.616 \times 10^{-35}/1.3672 \times 10^{26} = 1.18 \times 10^{-61}$ , so  $g_1 = 3.4 \times 10^{-31}$ ,  $g_1^{1/2} = 5.9 \times 10^{-16}$ , and

$$\omega_1 \approx 5.9 \times 10^{-5} \text{ Hz}, \quad \Omega_{\text{GW}}(\omega_1) \approx 1.2 \times 10^{-65}.$$

There are no relic gravitons above  $\omega_1$ , and below it the spectrum falls like  $\omega^3$ .

### 41.5.3 The verdict

C-B was frozen (tag `stream7-cb-frozen`) before any detector document was pinned, with the rule: testable iff the prediction reaches the design sensitivity of the most sensitive detector pinned afterwards. The LISA mission proposal requires the ability to measure  $\Omega = 1.3 \times 10^{-11}(f/10^{-4} \text{ Hz})^{-1}$  for  $0.1 \text{ mHz} < f < 2 \text{ mHz}$  [LISA, OR7.2, ll. 692–699], at best  $\approx 6.5 \times 10^{-13}$  at 2 mHz. The predicted spectrum ends below LISA’s band ( $\omega_1 < 10^{-4} \text{ Hz}$ ), and its peak lies more than fifty orders of magnitude below LISA’s sensitivity. By the rule, **C-B is not falsifiable in practice**: neither confirmed nor excluded.

There is a stronger statement. Read as the Regge slope of the fundamental string, (41.1) puts string resonances at  $\hbar c/s \approx 4 \text{ meV}$ , while the CMS dijet search excludes string resonances below 7.9 TeV in the models it tests [1911\_03947.txt, ll. 1044–1048], a gap of more than  $10^{30}$  in  $\alpha'$  (Chapter 38). The premise of C-B is therefore already excluded.

## 41.6 C-C: cosmic F-strings

A fundamental string has tension  $\bar{\mu} = 1/(2\pi\alpha')$  [CMP, eq. (3.3), l. 505 (with  $p = 1$ ,  $q = 0$ )], so in units where  $G = \ell_P^2$ ,

$$G\mu = \frac{\ell_P^2}{2\pi\alpha'} = \frac{\ell_P}{2\pi c/H_0} = \frac{1.616 \times 10^{-35}}{2\pi \times 1.3672 \times 10^{26}} \approx 1.9 \times 10^{-62}.$$

The brane-inflation window quoted by Copeland–Myers–Polchinski starts at  $G\mu \sim 10^{-11}$  [CMP §5.1, eq. (5.1), ll. 783–784], and the gravitational-wave reach they quote goes down to  $G\mu \sim 10^{-13}$  for LISA [CMP, ll. 972–975]. C-C is some fifty orders of magnitude out of reach, and it shares the excluded premise of C-B.

## 41.7 The common cause, and the geometry itself

All four observables were derived from (41.1). The pattern of the verdicts is explained by it: taken as a length scale in our universe ( $s \approx 47 \mu\text{m}$ ), it conflicts with torsion-balance and astrophysical bounds (P1); taken with a Hubble-scale infrared cutoff, it cannot produce acceleration (C-A); taken as the string scale, it puts every stringy signal below any conceivable sensitivity (C-B, C-C) and is itself excluded by collider searches. *Every observable the programme can derive today is either excluded or out of reach, and the cause is the identification  $\alpha' = \ell_P c/H_0$ .*

A second obstacle is independent of the identification. Type II string theory on  $K3 \times T^2$  gives  $\mathcal{N} = 4$  supersymmetry in four dimensions [Asp §5, l. 2743], and  $\mathcal{N} = 4$  supersymmetry in

four dimensions is not chiral [Asp §5, ll. 2713–2716]. The chiral fermions of the Standard Model therefore cannot come from this compactification as it stands (standard; the step from “the supersymmetry is not chiral” to “there is no chiral matter” is not spelled out in the pinned lines). A realistic model would need a different geometry, for instance a Calabi–Yau threefold fibred by K3 surfaces; that is new research.

## 41.8 What the machine has checked

Three modules of `DualScaleCosmology` carry the verdicts. All depend only on `propext`, `Classical.choice`, `Quot.sound`; each is **[A]** for what it literally states, which is algebra and arithmetic on stated inputs. None of them states that the identifications of this chapter are correct.

### 41.8.1 P1

`DualScaleCosmology/Stream6Verdict.lean`:

```
noncomputable def p1RadiusSq : ℝ := planckLength_m * hubbleRadius_m

theorem p1_verdict_excluded :
  (30e-6 : ℝ) ^ 2 < p1RadiusSq ∧ (38.6e-6 : ℝ) ^ 2 < p1RadiusSq ∧
  (44e-6 : ℝ) ^ 2 < p1RadiusSq := by ...

theorem p1_factor_needed :
  (30e-6 : ℝ) ^ 2 < (0.64 : ℝ) ^ 2 * p1RadiusSq ∧
  (0.63 : ℝ) ^ 2 * p1RadiusSq < (30e-6 : ℝ) ^ 2 := by ...

theorem p62_selfdual_fixed_point (R s : ℝ) (hR : 0 < R) (hs : 0 < s) :
  R = s ^ 2 / R ↔ R = s := by ...

theorem p62_towers_coincide (s : ℝ) (hs : 0 < s) : 1 / s = s / s ^ 2 := by ...

theorem p62_two_dims_excluded : (1e5 * 1.6e-10 : ℝ) ^ 2 < p1RadiusSq := by ...
```

#### Plain-language reading

**p1\_verdict\_excluded**: the product  $\ell_P \cdot c/H_0$  of the two typed-in constants, in  $\text{m}^2$ , exceeds  $(30\mu\text{m})^2$ ,  $(38.6\mu\text{m})^2$  and  $(44\mu\text{m})^2$ . **Proof**: unfold the constants and let `norm_num` compare decimals. **p1\_factor\_needed**: a factor 0.64 still fails the first bound, 0.63 passes; this records what a rescue would require, it does not choose it. **p62\_selfdual\_fixed\_point**: for positive reals,  $R = s^2/R$  if and only if  $R = s$  (both directions; the proof squares and uses  $(R - s)(R + s) = 0$ ). **p62\_towers\_coincide**:  $1/s = s/s^2$ . **p62\_two\_dims\_excluded**: the product exceeds  $(10^5 \times 1.6 \times 10^{-10} \text{ m})^2$ .

### 41.8.2 C-A

`DualScaleCosmology/Stream7CA.lean`:

```
theorem ca_matter_fraction (k H2 rm rL eps : ℝ) (hk : 0 < k)
  (hF : H2 = k * (rm + rL)) (hL : rL = eps * H2 / k) :
  rm = (1 - eps) * H2 / k := by ...

theorem ca_hubble_scaling (k a H2 rm0 eps : ℝ) (hk : 0 < k) (ha : 0 < a)
  (he : eps < 1) (h : rm0 / a ^ 3 = (1 - eps) * H2 / k) :
  H2 = (k * rm0 / (1 - eps)) / a ^ 3 := by ...
```

```

theorem ca_deceleration (C a : ℝ) (hC : 0 < C) (ha : 0 < a) :
  -1 - a * deriv (fun x : ℝ => C / x ^ 3) a / (2 * (C / a ^ 3)) = 1 / 2 := by ...

theorem ca_verdict_excluded :
  (1 - DualScaleCosmology.DarkEnergyScale.omegaLambda) / 2
  - DualScaleCosmology.DarkEnergyScale.omegaLambda < 0 ∧
  (1 - DualScaleCosmology.DarkEnergyScale.omegaLambda) / 2
  - DualScaleCosmology.DarkEnergyScale.omegaLambda ≠ 1 / 2 := by ...

```

#### Plain-language reading

`ca_matter_fraction` and `ca_hubble_scaling` are the two algebraic steps of the derivation, with  $k = 8\pi G/3$  left as an arbitrary positive real. `ca_deceleration` is the calculus step: with Mathlib's `deriv`, the expression (41.4) for  $E = C/a^3$  equals  $1/2$  at every  $a > 0$ ; the derivative is computed by the lemma `hasDerivAt_E` ( $\frac{d}{da}C/a^3 = -3C/a^4$ ). `ca_verdict_excluded`: with  $\Omega_\Lambda = 0.68885$ ,  $(1 - \Omega_\Lambda)/2 - \Omega_\Lambda$  is negative and differs from  $1/2$ . The Friedmann equation and the identification (41.3) are inputs written as hypotheses, not derived.

### 41.8.3 C-B

DualScaleCosmology/Stream7CB.lean:

```

noncomputable def g1Sq : ℝ := planckLength_m / hubbleRadius_m

noncomputable def slope (α : ℝ) : ℝ := 3 - 2 * |α - 1 / 2|

theorem cb_slope : slope (1 / 2) = 3 ∧ slope (-1) = 0 := by ...

theorem cb_g1_sq_bracket : (1.1e-61 : ℝ) ≤ g1Sq ∧ g1Sq ≤ 1.3e-61 := by ...

theorem cb_peak_below_lisa_band :
  g1Sq * (1e11 : ℝ) ^ 4 < (1e-4 : ℝ) ^ 4 := by ...

noncomputable def omegaPeak : ℝ := 1e-4 * g1Sq

noncomputable def omegaLisaBest : ℝ := 1.3e-11 / 20

theorem cb_not_testable :
  g1Sq * (1e11 : ℝ) ^ 4 < (1e-4 : ℝ) ^ 4 ∧ omegaPeak < omegaLisaBest := by ...

```

#### Plain-language reading

`cb_slope`: the formula  $3 - 2|\alpha - \frac{1}{2}|$  gives 3 at  $\alpha = 1/2$  and 0 at  $\alpha = -1$  (the formula itself is [L] from GV). `cb_g1_sq_bracket`:  $\ell_P/(c/H_0) \in [1.1, 1.3] \times 10^{-61}$ . `cb_peak_below_lisa_band`: with  $\omega_1 = g_1^{1/2} 10^{11}$  Hz,  $\omega_1^4 < (10^{-4} \text{ Hz})^4$ , i.e.  $\omega_1 < 10^{-4}$  Hz. `cb_not_testable`: in addition, the predicted peak  $10^{-4}g_1^2$  is below LISA's best  $1.3 \times 10^{-11}/20$ . The companion theorem `cb_peak_amplitude_unreachable` states the fifty-orders margin.

C-C uses `CosmicString.fStringGmu_selfDual` ( $G\mu = \ell_P/(2\pi L)$  for  $\alpha' = \ell_P L$ ) and `CosmicString.selfDual_Gmu_below_cmp_window`; the collider exclusion of the premise is `CosmicString.selfDual_alphaPrime_exceeds_cms_ceiling` (Chapter 38).



| Claim                                                                                   | Tier | Where                                    |
|-----------------------------------------------------------------------------------------|------|------------------------------------------|
| $s^2$ exceeds the three squared bounds; $\kappa < 0.64$ needed                          | [A]  | p1_verdict_excluded,<br>p1_factor_needed |
| Unique positive fixed point of $R \mapsto s^2/R$                                        | [A]  | p62_selfdual_fixed_point                 |
| $q = 1/2$ for $H^2 = C/a^3$ ; $q_0 < 0$ with $\Omega_\Lambda = 0.68885$                 | [A]  | ca_deceleration, ca_verdict_excluded     |
| $\omega_1 < 10^{-4}$ Hz, peak below LISA                                                | [A]  | cb_not_testable                          |
| Eöt-Wash, neutron-star, LISA, CMS numbers; GV spectrum; CKN bound                       | [L]  | EW, MVV, LISA, CMS, GV, CKN              |
| $\alpha' = \ell_P c/H_0$ ; $R = s$ as an extra dimension; CKN saturation with $L = c/H$ | [C]  | this programme                           |

## 41.9 Lessons and open directions

Four lessons carry over to any theory test. *Freeze before you compute*: the git tag is cheap and removes every later temptation to adjust. *Declare what you knew*: a retrodiction is honest if it is called one. *Derive the order-one factors*: here T-duality fixed  $\kappa = 1$ , and the verdict followed. *Look for the common cause*: four negative verdicts with one shared input say more about that input than about the four observables.

What would change the situation is new research, not a new convention: dropping or replacing the identification (41.1), and, for a model of our universe, a compactification with chiral matter. The mathematics of the earlier chapters (lattices, T-duality, double field theory, moonshine, dyons) is untouched by these verdicts: it is Tier A where it says so, whatever happens to the physical hypothesis.

### Chapter summary

- A pre-registered test fixes statement, tests and decision rule by a git tag before the verdict; prior knowledge is declared; a changed prediction is a new prediction.
- Order-one factors must be derived; choosing them after reading a bound is a fit.
- Without an explicit compactification, the programme yields four observables, all resting on  $\alpha' = \ell_P c/H_0$ , i.e.  $s \approx 47 \mu\text{m}$ .
- P1 ( $s$  as an extra-dimension radius) fails Eöt-Wash and neutron-star bounds and is excluded; T-duality fixes  $\kappa = 1$ .
- C-A (CKN saturation with  $L = c/H$ ) gives  $q = 1/2$  for every  $\varepsilon$ ; observed  $q_0 < 0$  excludes it.
- C-B (pre-big-bang gravitons) ends at  $6 \times 10^{-5}$  Hz with  $\Omega \approx 10^{-65}$ : not falsifiable in practice; C-C gives  $G\mu \approx 2 \times 10^{-62}$ ; both rest on a meV string scale, excluded by collider searches.
- $K3 \times T^2$  gives non-chiral  $\mathcal{N} = 4$  physics; a realistic model needs another geometry.

## Exercises

**Exercise 41.1** (\*). Recompute  $s$  from  $\ell_P$  and  $c/H_0$  and check the three failures of P1 by comparing squares. By what factor would  $H_0$  have to change for  $s$  to fall below  $30 \mu\text{m}$ ? Is such a change compatible with any measured value of  $H_0$  you know? (Answer the second question only after the first, and say which data you used.)

**Exercise 41.2** (★). Show that  $q = -1 - \dot{H}/H^2$  and that  $q = \frac{1}{2}(1 + 3w)$  for a flat universe dominated by a single fluid with  $p = w\rho$ . Use this to confirm that C-A has  $w_{\text{eff}} = 0$ .

**Exercise 41.3** (★★). Generalize C-A: suppose  $\rho_\Lambda = \varepsilon 3/(8\pi GL^2)$  with  $L = c/H$  but the matter density scales as  $a^{-3(1+w_m)}$ . Show that  $q = \frac{1}{2}(1 + 3w_m)$  for every  $\varepsilon$ . What does this say about any model in which the dark energy is a fixed fraction of the critical density?

**Exercise 41.4** (★★). Using (41.5), find the spectral slope for  $\alpha = 2$  and for  $\alpha = -2$ , and classify each background with the table in [GV Table 2, ll. 4471–4485]. For  $\alpha = 1/2$ , how many orders of magnitude does  $\Omega$  fall between  $\omega_1$  and  $10^{-9}$  Hz (the pulsar-timing band)?

**Exercise 41.5** (★★). Write the frozen document for a *new* prediction P1' in which the extra dimension has radius  $R = \kappa s$  with  $\kappa$  derived from some argument of your choice. List the inputs as measured, derived and assumed, state three tests and a decision rule, and explain which parts of your derivation would have to be done before the tag.

**Exercise 41.6** (★★, Lean). State and prove in Lean 4 with Mathlib: for real  $H_2$ ,  $k > 0$  and  $\varepsilon \neq 1$ , if  $H_2 = k\rho_m + \varepsilon H_2$  then  $\rho_m = (1 - \varepsilon)H_2/k$ . Compare your statement with `ca_matter_fraction`: which hypothesis did the module encode differently, and why is the module's version slightly more general?

**Exercise 41.7** (★★★, Lean). Prove the calculus step of C-A for a general power: for  $C > 0$ ,  $a > 0$  and a natural number  $n$ ,

```
theorem decel_general (C a : ℝ) (n : ℕ) (hC : 0 < C) (ha : 0 < a) :
  -1 - a * deriv (fun x : ℝ => C / x ^ n) a / (2 * (C / a ^ n))
  = n / 2 - 1 := by sorry
```

and deduce that a universe with  $H^2 \propto a^{-n}$  accelerates only for  $n < 2$ . Model the proof on `hasDerivAt_E`.

**Exercise 41.8** (★★★). C-C and C-B both became untestable because  $M_s \sim \hbar c/s$  is tiny. Suppose instead that  $\alpha'$  were a free parameter. Using the CMP tension formula and (41.6), find the range of  $\alpha'$  for which  $G\mu \geq 10^{-13}$  and the range for which the pre-big-bang peak  $\Omega(\omega_1)$  exceeds  $10^{-12}$ . Why would a prediction built this way no longer be a prediction of the dual-scale programme?

## Further reading

- Torsion-balance bounds on extra dimensions and Yukawa forces: [EW, ll. 283–289]. Neutron-star bounds and the “dark dimension” window: [MVV §3, ll. 318–329, 424–427].
- The CKN bound: [CKN, ll. 75–80]. Pre-big-bang relic gravitons: [GV §4.3, ll. 4466–4506] and [GV §5.1, ll. 5040–5061].
- LISA’s stochastic-background requirements: [LISA, ll. 692–699]. Cosmic F- and D-strings: [CMP, ll. 497–510, 778–786, 970–976]. String resonances at the LHC: [1911\_03947.txt, ll. 1044–1048].
- $\mathcal{N} = 4$  from  $K3 \times T^2$  and non-chirality: [Asp §5, ll. 2713–2716, 2741–2744].
- Project documents (in `docs/`): `STREAM6_EXPERIMENT_PLAN.md`, `STREAM6_PREDICTION_P1.md`, `STREAM7_HYPOTHESIS_INVENTORY.md`, `STREAM7_PREDICTION_CA.md`, `STREAM7_PREDICTION_CB.md`, `STREAM67_ERRATA.md`.
- Lean sources (in `DualScaleCosmology/`): `Stream6Verdict.lean`, `Stream7CA.lean`, `Stream7CB.lean`, `SelfDualCutoff.lean`, `CosmicString.lean`.

# Appendix A

## Notation and Conventions

| Symbol                                                          | Meaning                                                                                                                                                     |
|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\alpha'$                                                       | Regge slope; dimension length <sup>2</sup> . String length $\ell_s = \sqrt{\alpha'}$ , tension $T = 1/(2\pi\alpha')$ .                                      |
| $R, \alpha'/R$                                                  | Radius of a compact circle and its T-dual. Self-dual radius $R = \sqrt{\alpha'}$ .                                                                          |
| $n, w$                                                          | Momentum and winding quantum numbers on a circle; level matching $N - \tilde{N} = nw$ .                                                                     |
| $p_L, p_R$                                                      | Left- and right-moving zero-mode momenta.                                                                                                                   |
| $G, B, \Phi$                                                    | Metric, Kalb–Ramond two-form and dilaton; on $T^d$ , $G$ is a positive definite $d \times d$ matrix and $B$ is antisymmetric.                               |
| $E = G + B$                                                     | Background matrix.                                                                                                                                          |
| $\eta$                                                          | The $O(d, d; \mathbb{R})$ pairing, block form $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ ( <code>eta</code> in Lean).                                   |
| $O(d, d; \mathbb{Z})$                                           | Integer matrices $g$ with $g^\top \eta g = \eta$ ( <code>IsODD</code> ).                                                                                    |
| $\mathcal{H}(G, B)$                                             | Generalized metric $\begin{pmatrix} G - BG^{-1}B & BG^{-1} \\ -G^{-1}B & G^{-1} \end{pmatrix}$ ( <code>genMetric</code> ).                                  |
| $Z$                                                             | Doubled charge vector; mass form $Z^\top \mathcal{H} Z$ , level form $\frac{1}{2} Z^\top \eta Z$ .                                                          |
| $\tau, \rho$                                                    | Complex structure and complexified Kähler modulus of $T^2$ .                                                                                                |
| $U$                                                             | Hyperbolic plane, Gram matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ , signature $(1, 1)$ .                                                         |
| $E_8, E_8(-1)$                                                  | The $E_8$ root lattice (Cartan matrix as Gram matrix) and its negative.                                                                                     |
| $(n_+, n_-)$                                                    | Signature of a lattice. $K3$ lattice $3U \oplus 2E_8(-1)$ : $(3, 19)$ . Mukai lattice: $(4, 20)$ .                                                          |
| $\chi, h^{p,q}, b_k$                                            | Euler characteristic, Hodge numbers, Betti numbers. $\chi(K3) = 24$ .                                                                                       |
| $M_{24}$                                                        | Mathieu group, $ M_{24}  = 244\,823\,040$ .                                                                                                                 |
| $A_n$                                                           | Eguchi–Ooguri–Tachikawa coefficients $45, 231, 770, 2277, \dots$ ; multiplicities of massive $N = 4$ representations are $2A_n$ .                           |
| $\text{dualScale}(G)$                                           | $\text{tr } G + \text{tr } G^{-1}$ ( <code>dualScale</code> ); for a circle, $R^2/\alpha' + \alpha'/R^2$ in units where the entry of $G$ is $R^2/\alpha'$ . |
| <a href="#">[A]</a> , <a href="#">[L]</a> , <a href="#">[C]</a> | Epistemic tiers: kernel-checked; literature; conjecture or proposal.                                                                                        |



# Appendix B

## Index of Formalized Theorems

This appendix lists every theorem stated in the source of the seven libraries, generated from the declaration index (`tools/index_declarations.py`). The second column is the first line of the theorem's docstring, truncated, with non-ASCII characters removed; consult the source for the statement.

### DualScaleStream2 (99 theorems)

| Theorem                                | First line of its docstring                                                                                                                           |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| DualScaleStream2.DFT.BShift            |                                                                                                                                                       |
| genMetric_bshift                       | Real -shift g__ = [[I, ],[0, I]]. -/ noncomputable def thetaShiftR ( : Matrix (Fin d) (Fin d) ) : Matrix (Charge d) (Charge d) := fromBlocks 1 0 1 /- |
| genMetric_bshift_cancel                | Real -shift g__ = [[I, ],[0, I]]. -/ noncomputable def thetaShiftR ( : Matrix (Fin d) (Fin d) ) : Matrix (Charge d) (Charge d) := fromBlocks 1 0 1 /- |
| thetaShiftR_preserves_eta              | Real -shift g__ = [[I, ],[0, I]]. -/ noncomputable def thetaShiftR ( : Matrix (Fin d) (Fin d) ) : Matrix (Charge d) (Charge d) := fromBlocks 1 0 1 /- |
| DualScaleStream2.DFT.GeneralizedMetric |                                                                                                                                                       |
| etaR_genMetric_sq                      | Real O(d,d) form = [[0, I],[I, 0]]. -/ noncomputable def etaR (d : ) : Matrix (Charge d) (Charge d) := fromBlocks 0 1 1 0 /- HullZwiebach generaliz   |
| etaR_mul_self                          | Real O(d,d) form = [[0, I],[I, 0]]. -/ noncomputable def etaR (d : ) : Matrix (Charge d) (Charge d) := fromBlocks 0 1 1 0 /- HullZwiebach generaliz   |
| genMetric_symm                         | Real O(d,d) form = [[0, I],[I, 0]]. -/ noncomputable def etaR (d : ) : Matrix (Charge d) (Charge d) := fromBlocks 0 1 1 0 /- HullZwiebach generaliz   |
| massForm_circle                        | Real O(d,d) form = [[0, I],[I, 0]]. -/ noncomputable def etaR (d : ) : Matrix (Charge d) (Charge d) := fromBlocks 0 1 1 0 /- HullZwiebach generaliz   |
| massForm_covariant                     | Real O(d,d) form = [[0, I],[I, 0]]. -/ noncomputable def etaR (d : ) : Matrix (Charge d) (Charge d) := fromBlocks 0 1 1 0 /- HullZwiebach generaliz   |
| tduality_inverts_metric                | Real O(d,d) form = [[0, I],[I, 0]]. -/ noncomputable def etaR (d : ) : Matrix (Charge d) (Charge d) := fromBlocks 0 1 1 0 /- HullZwiebach generaliz   |
| DualScaleStream2.DFT.SectionCondition  |                                                                                                                                                       |

| Theorem                                          | First line of its docstring                                                                                                                                 |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>etaPair_self</code>                        | The -pairing $Z Z' = Z Z'$ . -/ def etaPair (Z Z' : Charge d) :<br>:= Z (eta d Z') /- A set of charges is a section if it is totally<br>-null. -/ def I     |
| <code>eta_momentum_to_winding</code>             | The -pairing $Z Z' = Z Z'$ . -/ def etaPair (Z Z' : Charge d) :<br>:= Z (eta d Z') /- A set of charges is a section if it is totally<br>-null. -/ def I     |
| <code>isSection_image</code>                     | The -pairing $Z Z' = Z Z'$ . -/ def etaPair (Z Z' : Charge d) :<br>:= Z (eta d Z') /- A set of charges is a section if it is totally<br>-null. -/ def I     |
| <code>levelMatching_iff</code>                   | The -pairing $Z Z' = Z Z'$ . -/ def etaPair (Z Z' : Charge d) :<br>:= Z (eta d Z') /- A set of charges is a section if it is totally<br>-null. -/ def I     |
| <code>momentumFrame_isSection</code>             | The -pairing $Z Z' = Z Z'$ . -/ def etaPair (Z Z' : Charge d) :<br>:= Z (eta d Z') /- A set of charges is a section if it is totally<br>-null. -/ def I     |
| <code>windingFrame_isSection</code>              | The -pairing $Z Z' = Z Z'$ . -/ def etaPair (Z Z' : Charge d) :<br>:= Z (eta d Z') /- A set of charges is a section if it is totally<br>-null. -/ def I     |
| DualScaleStream2.DualScale.TraceBound            |                                                                                                                                                             |
| <code>circle_effective_scale_ge_two</code>       | Dual scale (G) = tr H(G, 0). -/ noncomputable def dualScale<br>(G : Matrix (Fin d) (Fin d)) : := (genMetric G 0).trace /-<br>Unwinding the block-diagonal   |
| <code>dualScale_circle</code>                    | Dual scale (G) = tr H(G, 0). -/ noncomputable def dualScale<br>(G : Matrix (Fin d) (Fin d)) : := (genMetric G 0).trace /-<br>Unwinding the block-diagonal   |
| <code>dualScale_eq</code>                        | Dual scale (G) = tr H(G, 0). -/ noncomputable def dualScale<br>(G : Matrix (Fin d) (Fin d)) : := (genMetric G 0).trace /-<br>Unwinding the block-diagonal   |
| <code>dualScale_ge</code>                        | Dual scale (G) = tr H(G, 0). -/ noncomputable def dualScale<br>(G : Matrix (Fin d) (Fin d)) : := (genMetric G 0).trace /-<br>Unwinding the block-diagonal   |
| <code>dualScale_inv</code>                       | Dual scale (G) = tr H(G, 0). -/ noncomputable def dualScale<br>(G : Matrix (Fin d) (Fin d)) : := (genMetric G 0).trace /-<br>Unwinding the block-diagonal   |
| <code>dualScale_one</code>                       | Dual scale (G) = tr H(G, 0). -/ noncomputable def dualScale<br>(G : Matrix (Fin d) (Fin d)) : := (genMetric G 0).trace /-<br>Unwinding the block-diagonal   |
| DualScaleStream2.Flux.Integrality                |                                                                                                                                                             |
| <code>flux_half_selfIntersection_integral</code> | Kronecker product of two integer forms, on Fin n Fin m. -/<br>def kronForm (L : Gram n) (L : Gram m) : Matrix (Fin n<br>Fin m) (Fin n Fin m) := kronecke    |
| <code>kronForm_even</code>                       | Kronecker product of two integer forms, on Fin n Fin m. -/<br>def kronForm (L : Gram n) (L : Gram m) : Matrix (Fin n<br>Fin m) (Fin n Fin m) := kronecke    |
| <code>kronForm_evenDiag</code>                   | Kronecker product of two integer forms, on Fin n Fin m. -/<br>def kronForm (L : Gram n) (L : Gram m) : Matrix (Fin n<br>Fin m) (Fin n Fin m) := kronecke    |
| <code>kronForm_symm</code>                       | Kronecker product of two integer forms, on Fin n Fin m. -/<br>def kronForm (L : Gram n) (L : Gram m) : Matrix (Fin n<br>Fin m) (Fin n Fin m) := kronecke    |
| DualScaleStream2.Flux.Tadpole                    |                                                                                                                                                             |
| <code>euler_K3</code>                            | Hodge numbers $h^{(p,q)}$ of a complex torus T (all equal to<br>1). -/ def t2HodgeNumber : Fin 2 Fin 2 := fun __ => 1<br>/- Euler characteristic from Hodge |

| Theorem                                       | First line of its docstring                                                                                                                                      |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>euler_T2</code>                         | Hodge numbers $h^{\wedge}(p,q)$ of a complex torus $T$ (all equal to 1). -/ def t2HodgeNumber : Fin 2 Fin 2 := fun __ => 1<br>/- Euler characteristic from Hodge |
| <code>k3k3_anomaly</code>                     | Hodge numbers $h^{\wedge}(p,q)$ of a complex torus $T$ (all equal to 1). -/ def t2HodgeNumber : Fin 2 Fin 2 := fun __ => 1<br>/- Euler characteristic from Hodge |
| <code>k3t2_euler_zero</code>                  | Hodge numbers $h^{\wedge}(p,q)$ of a complex torus $T$ (all equal to 1). -/ def t2HodgeNumber : Fin 2 Fin 2 := fun __ => 1<br>/- Euler characteristic from Hodge |
| <code>tadpole_budget</code>                   | Hodge numbers $h^{\wedge}(p,q)$ of a complex torus $T$ (all equal to 1). -/ def t2HodgeNumber : Fin 2 Fin 2 := fun __ => 1<br>/- Euler characteristic from Hodge |
| DualScaleStream2.Lattice.Basic                |                                                                                                                                                                  |
| <code>even_quadratic_form_of_even_diag</code> | An integral lattice of rank $n$ , presented by its Gram matrix in a chosen basis: $G_{ij}$ is the bilinear-form pairing of basis vector $i$ with basis vector    |
| <code>isUnimodular_of_mul_eq_one</code>       | An integral lattice of rank $n$ , presented by its Gram matrix in a chosen basis: $G_{ij}$ is the bilinear-form pairing of basis vector $i$ with basis vector    |
| DualScaleStream2.Lattice.E8                   |                                                                                                                                                                  |
| <code>cartanE8Inv_mul</code>                  | The Cartan matrix of the root system E8, in the Dynkin labelling fixed in the module docstring: 2 on the diagonal (each simple root has norm 2 before            |
| <code>cartanE8_evenDiag</code>                | The Cartan matrix of the root system E8, in the Dynkin labelling fixed in the module docstring: 2 on the diagonal (each simple root has norm 2 before            |
| <code>cartanE8_symm</code>                    | The Cartan matrix of the root system E8, in the Dynkin labelling fixed in the module docstring: 2 on the diagonal (each simple root has norm 2 before            |
| <code>cartanE8_unimodular</code>              | The Cartan matrix of the root system E8, in the Dynkin labelling fixed in the module docstring: 2 on the diagonal (each simple root has norm 2 before            |
| <code>e8Neg_evenDiag</code>                   | The Cartan matrix of the root system E8, in the Dynkin labelling fixed in the module docstring: 2 on the diagonal (each simple root has norm 2 before            |
| <code>e8Neg_symm</code>                       | The Cartan matrix of the root system E8, in the Dynkin labelling fixed in the module docstring: 2 on the diagonal (each simple root has norm 2 before            |
| <code>e8Neg_unimodular</code>                 | The Cartan matrix of the root system E8, in the Dynkin labelling fixed in the module docstring: 2 on the diagonal (each simple root has norm 2 before            |
| DualScaleStream2.Lattice.E8PosDef             |                                                                                                                                                                  |
| <code>cartanE8_det</code>                     | Unit lower-triangular LDL factor of cartanE8 (Gaussian elimination with no pivoting needed, since all pivots turn out positive). Produced outside Lean           |
| <code>cartanE8_posDef</code>                  | Unit lower-triangular LDL factor of cartanE8 (Gaussian elimination with no pivoting needed, since all pivots turn out positive). Produced outside Lean           |
| <code>e8_LDL</code>                           | Unit lower-triangular LDL factor of cartanE8 (Gaussian elimination with no pivoting needed, since all pivots turn out positive). Produced outside Lean           |
| DualScaleStream2.Lattice.Hyperbolic           |                                                                                                                                                                  |
| <code>hyperbolicU_congruence</code>           | The hyperbolic plane $U$ , as a Gram matrix: $!![0,1;1,0]$ . In string-theory terms this is the momentum/winding pairing $(n,w)(n',w') = n w' + n' w$ of one $c$ |

| Theorem                                             | First line of its docstring                                                                                                                                 |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>hyperbolicU_eq_narain_gram</code>             | The hyperbolic plane U, as a Gram matrix: $!![0,1;1,0]$ . In string-theory terms this is the momentum/winding pairing $(n,w)(n',w') = n w' + n' w$ of one c |
| <code>hyperbolicU_evenDiag</code>                   | The hyperbolic plane U, as a Gram matrix: $!![0,1;1,0]$ . In string-theory terms this is the momentum/winding pairing $(n,w)(n',w') = n w' + n' w$ of one c |
| <code>hyperbolicU_mul_self</code>                   | The hyperbolic plane U, as a Gram matrix: $!![0,1;1,0]$ . In string-theory terms this is the momentum/winding pairing $(n,w)(n',w') = n w' + n' w$ of one c |
| <code>hyperbolicU_symm</code>                       | The hyperbolic plane U, as a Gram matrix: $!![0,1;1,0]$ . In string-theory terms this is the momentum/winding pairing $(n,w)(n',w') = n w' + n' w$ of one c |
| <code>hyperbolicU_unimodular</code>                 | The hyperbolic plane U, as a Gram matrix: $!![0,1;1,0]$ . In string-theory terms this is the momentum/winding pairing $(n,w)(n',w') = n w' + n' w$ of one c |
| <code>DualScaleStream2.Lattice.K3T2Signature</code> |                                                                                                                                                             |
| <code>index_mod_eight</code>                        | Signature (b, b) of a non-degenerate real quadratic form: pos real dimensions where the form is positive, neg where it is negative (b b is the physic       |
| <code>rank_K3</code>                                | Signature (b, b) of a non-degenerate real quadratic form: pos real dimensions where the form is positive, neg where it is negative (b b is the physic       |
| <code>rank_K3T2</code>                              | Signature (b, b) of a non-degenerate real quadratic form: pos real dimensions where the form is positive, neg where it is negative (b b is the physic       |
| <code>sigK3T2_eq</code>                             | Signature (b, b) of a non-degenerate real quadratic form: pos real dimensions where the form is positive, neg where it is negative (b b is the physic       |
| <code>sigK3_eq</code>                               | Signature (b, b) of a non-degenerate real quadratic form: pos real dimensions where the form is positive, neg where it is negative (b b is the physic       |
| <code>sigK3_matches_hodge</code>                    | Signature (b, b) of a non-degenerate real quadratic form: pos real dimensions where the form is positive, neg where it is negative (b b is the physic       |
| <code>sigMukai_eq</code>                            | Signature (b, b) of a non-degenerate real quadratic form: pos real dimensions where the form is positive, neg where it is negative (b b is the physic       |
| <code>DualScaleStream2.Lattice.Mukai</code>         |                                                                                                                                                             |
| <code>hyperbolicUNeg_mul_self</code>                | A Mukai vector (r, c, s) H H H (Huybrechts Def. 1.2): r the rank, c the first Chern class (an H vector in the ambient rank-n lattice), s the "reduc         |
| <code>hyperbolicUNeg_unimodular</code>              | A Mukai vector (r, c, s) H H H (Huybrechts Def. 1.2): r the rank, c the first Chern class (an H vector in the ambient rank-n lattice), s the "reduc         |
| <code>mukaiPair_even</code>                         | A Mukai vector (r, c, s) H H H (Huybrechts Def. 1.2): r the rank, c the first Chern class (an H vector in the ambient rank-n lattice), s the "reduc         |
| <code>mukaiPair_self</code>                         | A Mukai vector (r, c, s) H H H (Huybrechts Def. 1.2): r the rank, c the first Chern class (an H vector in the ambient rank-n lattice), s the "reduc         |
| <code>mukaiPair_symm</code>                         | A Mukai vector (r, c, s) H H H (Huybrechts Def. 1.2): r the rank, c the first Chern class (an H vector in the ambient rank-n lattice), s the "reduc         |



| Theorem                              | First line of its docstring                                                                                                                                                              |
|--------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>structureSheaf_mukai_sq</code> | A Mukai vector $(r, c, s)$ $H^1 H^2 H^3$ (Huybrechts Def. 1.2): $r$ the rank, $c$ the first Chern class (an $H$ vector in the ambient rank- $n$ lattice), $s$ the "reduc                 |
| DualScaleStream2.Lattice.Reflection  |                                                                                                                                                                                          |
| <code>e8Neg_simpleRoot_norm</code>   | Reflection matrix $s_v = 1 + v \cdot v \cdot L$ (so $s_v x = x + x, v \cdot v$ with $x, v = v \cdot L x$ ). This is the candidate for the reflection $s : x \times 2(x)/()$ of Huybrecht |
| <code>e8Neg_weyl_isometry</code>     | Reflection matrix $s_v = 1 + v \cdot v \cdot L$ (so $s_v x = x + x, v \cdot v$ with $x, v = v \cdot L x$ ). This is the candidate for the reflection $s : x \times 2(x)/()$ of Huybrecht |
| <code>reflection_involution</code>   | Reflection matrix $s_v = 1 + v \cdot v \cdot L$ (so $s_v x = x + x, v \cdot v$ with $x, v = v \cdot L x$ ). This is the candidate for the reflection $s : x \times 2(x)/()$ of Huybrecht |
| <code>reflection_isometry</code>     | Reflection matrix $s_v = 1 + v \cdot v \cdot L$ (so $s_v x = x + x, v \cdot v$ with $x, v = v \cdot L x$ ). This is the candidate for the reflection $s : x \times 2(x)/()$ of Huybrecht |
| <code>vecMulVec_mul_self</code>      | Reflection matrix $s_v = 1 + v \cdot v \cdot L$ (so $s_v x = x + x, v \cdot v$ with $x, v = v \cdot L x$ ). This is the candidate for the reflection $s : x \times 2(x)/()$ of Huybrecht |
| DualScaleStream2.Moonshine.EOT       |                                                                                                                                                                                          |
| <code>A6_decomposition</code>        | EOT eq. (1.12): $A, , A. -/ \text{ def eotA : Fin 9 := ![45, 231, 770, 2277, 5796, 13915, 30843, 65550, 132825] } /- A A$ are each the dimension of an irredu                            |
| <code>A6_not_irrep</code>            | EOT eq. (1.12): $A, , A. -/ \text{ def eotA : Fin 9 := ![45, 231, 770, 2277, 5796, 13915, 30843, 65550, 132825] } /- A A$ are each the dimension of an irredu                            |
| <code>A7_decomposition</code>        | EOT eq. (1.12): $A, , A. -/ \text{ def eotA : Fin 9 := ![45, 231, 770, 2277, 5796, 13915, 30843, 65550, 132825] } /- A A$ are each the dimension of an irredu                            |
| <code>first_five_are_irreps</code>   | EOT eq. (1.12): $A, , A. -/ \text{ def eotA : Fin 9 := ![45, 231, 770, 2277, 5796, 13915, 30843, 65550, 132825] } /- A A$ are each the dimension of an irredu                            |
| DualScaleStream2.TDuality.Factorized |                                                                                                                                                                                          |
| <code>chargeNorm_even</code>         | $e_k$ : the matrix unit at $(k,k)$ , i.e. the diagonal projector onto the $k$ -th coordinate. Used to isolate "act on direction $k$ only" in factorized below. -/                        |
| <code>chargeNorm_invariant</code>    | $e_k$ : the matrix unit at $(k,k)$ , i.e. the diagonal projector onto the $k$ -th coordinate. Used to isolate "act on direction $k$ only" in factorized below. -/                        |
| <code>chargeNorm_sumElim</code>      | $e_k$ : the matrix unit at $(k,k)$ , i.e. the diagonal projector onto the $k$ -th coordinate. Used to isolate "act on direction $k$ only" in factorized below. -/                        |
| <code>factorized_comm</code>         | $e_k$ : the matrix unit at $(k,k)$ , i.e. the diagonal projector onto the $k$ -th coordinate. Used to isolate "act on direction $k$ only" in factorized below. -/                        |
| <code>factorized_isODD</code>        | $e_k$ : the matrix unit at $(k,k)$ , i.e. the diagonal projector onto the $k$ -th coordinate. Used to isolate "act on direction $k$ only" in factorized below. -/                        |
| <code>factorized_mul_self</code>     | $e_k$ : the matrix unit at $(k,k)$ , i.e. the diagonal projector onto the $k$ -th coordinate. Used to isolate "act on direction $k$ only" in factorized below. -/                        |
| <code>factorized_two</code>          | $e_k$ : the matrix unit at $(k,k)$ , i.e. the diagonal projector onto the $k$ -th coordinate. Used to isolate "act on direction $k$ only" in factorized below. -/                        |

| Theorem                                           | First line of its docstring                                                                                                                                                                                                       |
|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>proj_comm</code>                            | <code>e_k</code> : the matrix unit at $(k,k)$ , i.e. the diagonal projector onto the $k$ -th coordinate. Used to isolate "act on direction $k$ only" in factorized below. -/                                                      |
| <code>proj_mul_self</code>                        | <code>e_k</code> : the matrix unit at $(k,k)$ , i.e. the diagonal projector onto the $k$ -th coordinate. Used to isolate "act on direction $k$ only" in factorized below. -/                                                      |
| <code>proj_transpose</code>                       | <code>e_k</code> : the matrix unit at $(k,k)$ , i.e. the diagonal projector onto the $k$ -th coordinate. Used to isolate "act on direction $k$ only" in factorized below. -/                                                      |
| <code>DualScaleStream2.TDuality.Mirror</code>     |                                                                                                                                                                                                                                   |
| <code>mirrorTheta_antisymm</code>                 | $T = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ : the standard $SL(2,)$ generator implementing $+1$ (a unimodular change of basis of the $T$ lattice; $A = T$ in <code>ODD.basisChange</code> ). -/ def                        |
| <code>mirror_conjugates_tauShift</code>           | $T = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ : the standard $SL(2,)$ generator implementing $+1$ (a unimodular change of basis of the $T$ lattice; $A = T$ in <code>ODD.basisChange</code> ). -/ def                        |
| <code>tauShift_dual_spec</code>                   | $T = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ : the standard $SL(2,)$ generator implementing $+1$ (a unimodular change of basis of the $T$ lattice; $A = T$ in <code>ODD.basisChange</code> ). -/ def                        |
| <code>tauShift_isODD</code>                       | $T = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ : the standard $SL(2,)$ generator implementing $+1$ (a unimodular change of basis of the $T$ lattice; $A = T$ in <code>ODD.basisChange</code> ). -/ def                        |
| <code>DualScaleStream2.TDuality.ODD</code>        |                                                                                                                                                                                                                                   |
| <code>basisChange_isODD</code>                    | Charge-lattice index set: $d$ momenta $d$ windings. As a Lean type this is $\text{Fin } d \times \text{Fin } d$ ; physically it indexes the $2d$ integers $(n_1, n_d, w_1, w_d)$ labell                                           |
| <code>eta_isODD</code>                            | Charge-lattice index set: $d$ momenta $d$ windings. As a Lean type this is $\text{Fin } d \times \text{Fin } d$ ; physically it indexes the $2d$ integers $(n_1, n_d, w_1, w_d)$ labell                                           |
| <code>eta_mul_self</code>                         | Charge-lattice index set: $d$ momenta $d$ windings. As a Lean type this is $\text{Fin } d \times \text{Fin } d$ ; physically it indexes the $2d$ integers $(n_1, n_d, w_1, w_d)$ labell                                           |
| <code>eta_one_reindex</code>                      | Charge-lattice index set: $d$ momenta $d$ windings. As a Lean type this is $\text{Fin } d \times \text{Fin } d$ ; physically it indexes the $2d$ integers $(n_1, n_d, w_1, w_d)$ labell                                           |
| <code>isODD_mul</code>                            | Charge-lattice index set: $d$ momenta $d$ windings. As a Lean type this is $\text{Fin } d \times \text{Fin } d$ ; physically it indexes the $2d$ integers $(n_1, n_d, w_1, w_d)$ labell                                           |
| <code>thetaShift_isODD</code>                     | Charge-lattice index set: $d$ momenta $d$ windings. As a Lean type this is $\text{Fin } d \times \text{Fin } d$ ; physically it indexes the $2d$ integers $(n_1, n_d, w_1, w_d)$ labell                                           |
| <code>DualScaleStream2.TDuality.SL2Product</code> |                                                                                                                                                                                                                                   |
| <code>basisChange_comm_thetaShift</code>          | $J = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ : the antisymmetric generator along which the $-$ translations <code>thetaShift (t jMat) (t)</code> run, $+t$ . -/ def<br>$jMat : \text{Matrix } (\text{Fin } 2) (\text{Fi})$ |
| <code>basisChange_mul</code>                      | $J = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ : the antisymmetric generator along which the $-$ translations <code>thetaShift (t jMat) (t)</code> run, $+t$ . -/ def<br>$jMat : \text{Matrix } (\text{Fin } 2) (\text{Fi})$ |
| <code>mul_jMat_mul_transpose</code>               | $J = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ : the antisymmetric generator along which the $-$ translations <code>thetaShift (t jMat) (t)</code> run, $+t$ . -/ def<br>$jMat : \text{Matrix } (\text{Fin } 2) (\text{Fi})$ |
| <code>thetaShift_mul</code>                       | $J = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ : the antisymmetric generator along which the $-$ translations <code>thetaShift (t jMat) (t)</code> run, $+t$ . -/ def<br>$jMat : \text{Matrix } (\text{Fin } 2) (\text{Fi})$ |

| Theorem                            | First line of its docstring                                                                                                                   |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| DualScaleStream2.TDuality.Spectrum |                                                                                                                                               |
| eta_transpose                      | g is a left inverse of g for any g O(d,d): ( g ) g = 1. Proof idea: regroup as (g g) = using hg : g g = , which collapses to 1 by eta_mul_sel |
| isODD_inv_isODD                    | g is a left inverse of g for any g O(d,d): ( g ) g = 1. Proof idea: regroup as (g g) = using hg : g g = , which collapses to 1 by eta_mul_sel |
| isODD_left_inv                     | g is a left inverse of g for any g O(d,d): ( g ) g = 1. Proof idea: regroup as (g g) = using hg : g g = , which collapses to 1 by eta_mul_sel |
| isODD_right_inv                    | g is a left inverse of g for any g O(d,d): ( g ) g = 1. Proof idea: regroup as (g g) = using hg : g g = , which collapses to 1 by eta_mul_sel |
| spectrum_equivalence               | g is a left inverse of g for any g O(d,d): ( g ) g = 1. Proof idea: regroup as (g g) = using hg : g g = , which collapses to 1 by eta_mul_sel |

## StringTheoryFormalization (91 theorems)

| Theorem                                            | First line of its docstring                                                                                                                           |
|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| StringTheoryFormalization.Frontier.CentralCharge   |                                                                                                                                                       |
| central_charge_from_worldsheet_action              | The Virasoro central charge contribution from a single free boson. -/ def centralChargeBoson : := 1 /- The Virasoro central charge from a single fre  |
| central_charge_k3_eq_six                           | The Virasoro central charge contribution from a single free boson. -/ def centralChargeBoson : := 1 /- The Virasoro central charge from a single fre  |
| here                                               |                                                                                                                                                       |
| virasoro_algebra_commutator                        | The Virasoro central charge contribution from a single free boson. -/ def centralChargeBoson : := 1 /- The Virasoro central charge from a single fre  |
| virasoro_ope_central_term                          | The Virasoro central charge contribution from a single free boson. -/ def centralChargeBoson : := 1 /- The Virasoro central charge from a single fre  |
| StringTheoryFormalization.Frontier.ChiralPrimaries |                                                                                                                                                       |
| chiral_primary_ring_associativity                  | A pair of N=2 superconformal quantum numbers (h, q) for a state: a conformal weight and a U(1)_R charge. Physically h 0 in any unitary representation |
| chiral_primary_saturates_bps                       | A pair of N=2 superconformal quantum numbers (h, q) for a state: a conformal weight and a U(1)_R charge. Physically h 0 in any unitary representation |
| k3_chiral_primary_total                            | A pair of N=2 superconformal quantum numbers (h, q) for a state: a conformal weight and a U(1)_R charge. Physically h 0 in any unitary representation |
| StringTheoryFormalization.Frontier.FTermPotential  |                                                                                                                                                       |
| flux_tadpole_quantization_exact                    | The complex dilaton-axion field = C + i e^(-). -/ structure DilatonAxion where : im_pos : 0 < .im /- The Khler potential for the dilaton modulus.     |
| fterm_potential_minimum_susy                       | The complex dilaton-axion field = C + i e^(-). -/ structure DilatonAxion where : im_pos : 0 < .im /- The Khler potential for the dilaton modulus.     |

| Theorem                                              | First line of its docstring                                                                                                                                                    |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>fterm_potential_nonneg</code>                  | The complex dilaton-axion field = $C + i e^{\cdot}$ . -/ structure DilatonAxion where : <code>im_pos</code> : $0 < \text{im} / -$ The Khler potential for the dilaton modulus. |
| <code>no_scale_identity</code>                       | The complex dilaton-axion field = $C + i e^{\cdot}$ . -/ structure DilatonAxion where : <code>im_pos</code> : $0 < \text{im} / -$ The Khler potential for the dilaton modulus. |
| StringTheoryFormalization.Frontier.HodgeNumbers      |                                                                                                                                                                                |
| <code>hodge11_from_kummer</code>                     | The Hodge numbers of K3, $h^{\cdot}(p,q) = \dim H^{\cdot}q(K3, \hat{p})$ , as a hand-written lookup table Fin 3 Fin 3 matching the standard diamond (see the module docstri    |
| <code>k3_b2</code>                                   | The Hodge numbers of K3, $h^{\cdot}(p,q) = \dim H^{\cdot}q(K3, \hat{p})$ , as a hand-written lookup table Fin 3 Fin 3 matching the standard diamond (see the module docstri    |
| <code>k3_euler_characteristic</code>                 | The Hodge numbers of K3, $h^{\cdot}(p,q) = \dim H^{\cdot}q(K3, \hat{p})$ , as a hand-written lookup table Fin 3 Fin 3 matching the standard diamond (see the module docstri    |
| <code>k3_hodge_symmetry</code>                       | The Hodge numbers of K3, $h^{\cdot}(p,q) = \dim H^{\cdot}q(K3, \hat{p})$ , as a hand-written lookup table Fin 3 Fin 3 matching the standard diamond (see the module docstri    |
| <code>k3_serre_duality</code>                        | The Hodge numbers of K3, $h^{\cdot}(p,q) = \dim H^{\cdot}q(K3, \hat{p})$ , as a hand-written lookup table Fin 3 Fin 3 matching the standard diamond (see the module docstri    |
| StringTheoryFormalization.Frontier.ModuliGeodesics   |                                                                                                                                                                                |
| <code>geodesic_equation_kummer_locus</code>          | A point in the K3 moduli space, intended as a complex structure on K3 encoded by its period point in the Mukai lattice $\hat{(4,20)}$ . <code>periodPoint</code> is stored     |
| <code>poincare_geodesic_atlas_curvature</code>       | A point in the K3 moduli space, intended as a complex structure on K3 encoded by its period point in the Mukai lattice $\hat{(4,20)}$ . <code>periodPoint</code> is stored     |
| <code>poincare_geodesic_kinetic_energy_nonneg</code> | A point in the K3 moduli space, intended as a complex structure on K3 encoded by its period point in the Mukai lattice $\hat{(4,20)}$ . <code>periodPoint</code> is stored     |
| <code>wp_geodesic_completeness</code>                | A point in the K3 moduli space, intended as a complex structure on K3 encoded by its period point in the Mukai lattice $\hat{(4,20)}$ . <code>periodPoint</code> is stored     |
| <code>wp_metric_positive_diagonal</code>             | A point in the K3 moduli space, intended as a complex structure on K3 encoded by its period point in the Mukai lattice $\hat{(4,20)}$ . <code>periodPoint</code> is stored     |
| StringTheoryFormalization.Frontier.SL2CSymmetry      |                                                                                                                                                                                |
| <code>mobius_compose</code>                          | A Mbius (SL(2,)) transformation $z (az+b)/(cz+d)$ . -/ structure MobiusTransform where (a b c d : ) <code>det_one</code> : $a d - b c = 1 / -$ Action of a Mbius transf        |
| <code>mobius_id_act</code>                           | A Mbius (SL(2,)) transformation $z (az+b)/(cz+d)$ . -/ structure MobiusTransform where (a b c d : ) <code>det_one</code> : $a d - b c = 1 / -$ Action of a Mbius transf        |
| <code>sl2c_fixes_two_point</code>                    | A Mbius (SL(2,)) transformation $z (az+b)/(cz+d)$ . -/ structure MobiusTransform where (a b c d : ) <code>det_one</code> : $a d - b c = 1 / -$ Action of a Mbius transf        |
| <code>ward_identity_dilatation</code>                | A Mbius (SL(2,)) transformation $z (az+b)/(cz+d)$ . -/ structure MobiusTransform where (a b c d : ) <code>det_one</code> : $a d - b c = 1 / -$ Action of a Mbius transf        |
| <code>ward_identity_translation</code>               | A Mbius (SL(2,)) transformation $z (az+b)/(cz+d)$ . -/ structure MobiusTransform where (a b c d : ) <code>det_one</code> : $a d - b c = 1 / -$ Action of a Mbius transf        |

| Theorem                                                           | First line of its docstring                                                                                                                                   |
|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>StringTheoryFormalization.NSMath.EnergyBounds</b>              |                                                                                                                                                               |
| energy_dissipation                                                | Energy of a field configuration measured in $H^s$ . -/ noncomputable def fieldEnergy (s : SobolevExponent) (coeff : ) :<br>:= (sobolevNorm s coeff) ^ 2       |
| regularity_lifting                                                | Energy of a field configuration measured in $H^s$ . -/ noncomputable def fieldEnergy (s : SobolevExponent) (coeff : ) :<br>:= (sobolevNorm s coeff) ^ 2       |
| <b>StringTheoryFormalization.NSMath.FractionalSobolev</b>         |                                                                                                                                                               |
| sobolev_embedding                                                 | Fractional Sobolev exponent s with $s > 0$ . -/ structure SobolevExponent where s : pos : $0 < s$ /- $H^s(T)$ norm via Fourier characterization: $u_{-}(H^s)$ |
| <b>StringTheoryFormalization.NSMath.MildPDEs</b>                  |                                                                                                                                                               |
| mild_solution_unique                                              | Abstract semigroup generator (unbounded operator) on a Banach space. -/ structure SemigroupGenerator (E : Type) [NormedAddCommGroup E] [NormedSpace E]        |
| <b>StringTheoryFormalization.NSMath.OpenAIBridging</b>            |                                                                                                                                                               |
| evolution_regularity_preserved                                    | Torus frequency lattice for the T compactification. -/ abbrev TorusFrequency := /- Torus spatial coordinates on the universal cover . -/ abbr                 |
| is_rapid_summable                                                 | Torus frequency lattice for the T compactification. -/ abbrev TorusFrequency := /- Torus spatial coordinates on the universal cover . -/ abbr                 |
| swap_torus_involution                                             | Torus frequency lattice for the T compactification. -/ abbrev TorusFrequency := /- Torus spatial coordinates on the universal cover . -/ abbr                 |
| torus_weight_pos                                                  | Torus frequency lattice for the T compactification. -/ abbrev TorusFrequency := /- Torus spatial coordinates on the universal cover . -/ abbr                 |
| torus_weight_swap                                                 | Torus frequency lattice for the T compactification. -/ abbrev TorusFrequency := /- Torus spatial coordinates on the universal cover . -/ abbr                 |
| <b>StringTheoryFormalization.StringDynamics.AutoEvolve</b>        |                                                                                                                                                               |
| potential_decreases_along_flow                                    | A scalar field EFT flow equation: $d/dt = -_{-} V()$ . -/ structure EFTFlowEq where /- The effective potential $V : .$ -/ potential : potential_smooth :      |
| <b>StringTheoryFormalization.StringDynamics.BPSMultiplicities</b> |                                                                                                                                                               |
| bps_ratio_pos                                                     | The literal 77/60, called the "BPS index ratio" by this project. This is a Tier C naming choice, not a definition derived from BPS-state counting in t        |
| bps_ratio_reduced                                                 | The literal 77/60, called the "BPS index ratio" by this project. This is a Tier C naming choice, not a definition derived from BPS-state counting in t        |
| <b>StringTheoryFormalization.StringDynamics.FourierMukai</b>      |                                                                                                                                                               |
| fm_isEquivalence_of_mk                                            | A one-field wrapper holding a String tag for a "derived category of X" not a category: it carries no objects beyond the wrapper itself, no morphisms,         |
| fm_squared_is_shift                                               | A one-field wrapper holding a String tag for a "derived category of X" not a category: it carries no objects beyond the wrapper itself, no morphisms,         |
| <b>StringTheoryFormalization.StringDynamics.InvariantLocks</b>    |                                                                                                                                                               |
| modulus_upper_half                                                | The complex structure modulus of the worldsheet torus, constrained to the upper half-plane . -/ structure WorldsheetModulus where : im_pos : $0 < i$          |
| <b>StringTheoryFormalization.StringDynamics.KummerBlowup</b>      |                                                                                                                                                               |

| Theorem                                                   | First line of its docstring                                                                                                                                   |
|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>exceptional_self_intersection</code>                | Names the 16 indices 0,,15 as the strings "E_0","E_15", standing for the 16 exceptional (-2)-curves produced by resolving the 16 A singularities of T         |
| <code>kummer_lattice_contribution</code>                  | Names the 16 indices 0,,15 as the strings "E_0","E_15", standing for the 16 exceptional (-2)-curves produced by resolving the 16 A singularities of T         |
| StringTheoryFormalization.StringDynamics.MathieuM24       |                                                                                                                                                               |
| <code>M24RepDim_sum_sq</code>                             | Dimension of the j-th irreducible representation of M, all 26, in the order of EguchiOoguriTachikawa, arXiv:1004.0956, eq. (A.3) (papers/foundations/e        |
| <code>M24_first_coefficient</code>                        | Dimension of the j-th irreducible representation of M, all 26, in the order of EguchiOoguriTachikawa, arXiv:1004.0956, eq. (A.3) (papers/foundations/e        |
| <code>M24_order</code>                                    | Dimension of the j-th irreducible representation of M, all 26, in the order of EguchiOoguriTachikawa, arXiv:1004.0956, eq. (A.3) (papers/foundations/e        |
| StringTheoryFormalization.StringDynamics.MukaiLattice     |                                                                                                                                                               |
| <code>kummer_sublattice_rank</code>                       | A record of four bookkeeping numbers describing a lattice's numerical invariants not a lattice (no group, no bilinear form is carried). Defaults (24,         |
| <code>mukai_rank</code>                                   | A record of four bookkeeping numbers describing a lattice's numerical invariants not a lattice (no group, no bilinear form is carried). Defaults (24,         |
| <code>proves.</code>                                      |                                                                                                                                                               |
| StringTheoryFormalization.StringDynamics.MukhanovSasaki   |                                                                                                                                                               |
| <code>ms_superhorizon_freezing</code>                     | A comoving wavenumber $k > 0$ together with an arbitrary complex-valued function $v$ : of conformal time , tagged with an unused wronskian__normalized f      |
| StringTheoryFormalization.StringDynamics.ODDMetric        |                                                                                                                                                               |
| <code>odd_metric_d1</code>                                | The 2D2D integer block matrix = $[[0, 1\_D],[1\_D, 0]]$ : entry 1 at (i, i+D) and (i+D, i) for $i < D$ , 0 everywhere else. This is the O(D,D)-invariant metr |
| <code>odd_metric_symm</code>                              | The 2D2D integer block matrix = $[[0, 1\_D],[1\_D, 0]]$ : entry 1 at (i, i+D) and (i+D, i) for $i < D$ , 0 everywhere else. This is the O(D,D)-invariant metr |
| StringTheoryFormalization.StringDynamics.PicardSpectral   |                                                                                                                                                               |
| <code>picard_convergence</code>                           | The Picard iteration for the worldsheet OPE algebra converges when the spectral radius $< 1$ after rescaling by 1/18.<br>-/ def picardSpectralRadius : :=     |
| <code>picard_spectral_contraction</code>                  | The Picard iteration for the worldsheet OPE algebra converges when the spectral radius $< 1$ after rescaling by 1/18.<br>-/ def picardSpectralRadius : :=     |
| <code>picard_spectral_radius_pos</code>                   | The Picard iteration for the worldsheet OPE algebra converges when the spectral radius $< 1$ after rescaling by 1/18.<br>-/ def picardSpectralRadius : :=     |
| StringTheoryFormalization.StringDynamics.StiffIntegrators |                                                                                                                                                               |
| <code>bdf2_order_bound</code>                             | A stiff ODE system $y' = f(t,y)$ with stiffness ratio $\_max/\_min$ 1. -/ structure StiffODESystem where /- Spatial dimension of the phase space. -/ dim :    |
| <code>implicit_euler_denominator_lower_bound</code>       | A stiff ODE system $y' = f(t,y)$ with stiffness ratio $\_max/\_min$ 1. -/ structure StiffODESystem where /- Spatial dimension of the phase space. -/ dim :    |
| StringTheoryFormalization.StringDynamics.SwamplandSafe    |                                                                                                                                                               |

| Theorem                                                    | First line of its docstring                                                                                                                                                  |
|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>de_sitter_conjecture</code>                          | A pair of positive reals ( $\epsilon$ , $m$ ) with their positivity proofs, parametrizing the one-exponential family towerMass below. Physically is the SDC decay rate       |
| <code>sd_c_tower_mass_pos</code>                           | A pair of positive reals ( $\epsilon$ , $m$ ) with their positivity proofs, parametrizing the one-exponential family towerMass below. Physically is the SDC decay rate       |
| <code>sd_c_tower_suppression</code>                        | A pair of positive reals ( $\epsilon$ , $m$ ) with their positivity proofs, parametrizing the one-exponential family towerMass below. Physically is the SDC decay rate       |
| StringTheoryFormalization.StringDynamics.TDAMapper         |                                                                                                                                                                              |
| <code>mapper_nerve_theorem</code>                          | A cover of a topological space by open sets. -/ structure OpenCover (X : Type) [TopologicalSpace X] where index : Type sets : index Set X isOpen : i                         |
| StringTheoryFormalization.StringDynamics.TDualityGysin     |                                                                                                                                                                              |
| <code>tduality_involution</code>                           | Computes $\epsilon/R$ , the T-dual radius under $R \rightarrow R$ (papers/foundations/tong_string_theory_0908_0333.txt, ll. 1137111460, and BOOK_BIBLE.md 5). The hypothesis |
| StringTheoryFormalization.StringDynamics.TadpoleConstraint |                                                                                                                                                                              |
| <code>tadpole_cancellation</code>                          | A single stack's contribution to the D3-brane tadpole: an integer charge (positive for branes, negative for anti-branes) and an integer euler, the Euler                     |
| StringTheoryFormalization.StringDynamics.VertexOperators   |                                                                                                                                                                              |
| <code>ope_vacuum_left</code>                               | Worldsheet mode index. -/ abbrev ModeIndex := /- A bosonic oscillator mode $a_n$ acting on a Fock space. We model the Fock space as formal power series                      |
| StringTheoryFormalization.UseCases.CriticalDimension       |                                                                                                                                                                              |
| <code>bc_ghost_charge_eq</code>                            | Central charge of an anticommuting first-order (b,c) system of weight ( $\epsilon$ , 1-). -/ def fermionicGhostCharge (lam : ) : := 1 - 3 (2 lam - 1) ^ 2 /- Cen             |
| <code>betagamma_ghost_charge_eq</code>                     | Central charge of an anticommuting first-order (b,c) system of weight ( $\epsilon$ , 1-). -/ def fermionicGhostCharge (lam : ) : := 1 - 3 (2 lam - 1) ^ 2 /- Cen             |
| <code>bosonic_dimension_unique</code>                      | Central charge of an anticommuting first-order (b,c) system of weight ( $\epsilon$ , 1-). -/ def fermionicGhostCharge (lam : ) : := 1 - 3 (2 lam - 1) ^ 2 /- Cen             |
| <code>bosonic_eq_neg_fermionic</code>                      | Central charge of an anticommuting first-order (b,c) system of weight ( $\epsilon$ , 1-). -/ def fermionicGhostCharge (lam : ) : := 1 - 3 (2 lam - 1) ^ 2 /- Cen             |
| <code>bosonic_string_critical_dimension</code>             | Central charge of an anticommuting first-order (b,c) system of weight ( $\epsilon$ , 1-). -/ def fermionicGhostCharge (lam : ) : := 1 - 3 (2 lam - 1) ^ 2 /- Cen             |
| <code>superstring_critical_dimension</code>                | Central charge of an anticommuting first-order (b,c) system of weight ( $\epsilon$ , 1-). -/ def fermionicGhostCharge (lam : ) : := 1 - 3 (2 lam - 1) ^ 2 /- Cen             |
| StringTheoryFormalization.UseCases.K3SignatureTheorem      |                                                                                                                                                                              |
| <code>bMinus_eq_nineteen</code>                            | The positive-definite part of the intersection form: $2h^{(2,0)} + 1$ . -/ def bPlus : := 2 k3HodgeNumber 2, by norm_num 0, by norm_num + 1 /- The negative                  |
| <code>bPlus_eq_three</code>                                | The positive-definite part of the intersection form: $2h^{(2,0)} + 1$ . -/ def bPlus : := 2 k3HodgeNumber 2, by norm_num 0, by norm_num + 1 /- The negative                  |
| <code>betti_decomposition_matches_b2</code>                | The positive-definite part of the intersection form: $2h^{(2,0)} + 1$ . -/ def bPlus : := 2 k3HodgeNumber 2, by norm_num 0, by norm_num + 1 /- The negative                  |

| Theorem                                                 | First line of its docstring                                                                                                                                                                                   |
|---------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>betti_sum_eq_b2</code>                            | The positive-definite part of the intersection form: $2h^{(2,0)} + 1$ . -/ def <code>bPlus</code> : := 2 <code>k3HodgeNumber</code> 2, by <code>norm_num</code> 0, by <code>norm_num</code> + 1 /- The negati |
| <code>k3_signature_eq_neg_sixteen</code>                | The positive-definite part of the intersection form: $2h^{(2,0)} + 1$ . -/ def <code>bPlus</code> : := 2 <code>k3HodgeNumber</code> 2, by <code>norm_num</code> 0, by <code>norm_num</code> + 1 /- The negati |
| StringTheoryFormalization.UseCases.MathieuTower         |                                                                                                                                                                                                               |
| <code>M21_order_factorization</code>                    | Order of M (standard ATLAS value; not re-derived from a group action here). -/ def <code>orderM23</code> : := 10200960 /- Order of M (standard ATLAS value; not r                                             |
| <code>M22_stabilizer_index</code>                       | Order of M (standard ATLAS value; not re-derived from a group action here). -/ def <code>orderM23</code> : := 10200960 /- Order of M (standard ATLAS value; not r                                             |
| <code>M23_stabilizer_index</code>                       | Order of M (standard ATLAS value; not re-derived from a group action here). -/ def <code>orderM23</code> : := 10200960 /- Order of M (standard ATLAS value; not r                                             |
| <code>M24_stabilizer_index</code>                       | Order of M (standard ATLAS value; not re-derived from a group action here). -/ def <code>orderM23</code> : := 10200960 /- Order of M (standard ATLAS value; not r                                             |
| <code>mathieu_tower_consistent</code>                   | Order of M (standard ATLAS value; not re-derived from a group action here). -/ def <code>orderM23</code> : := 10200960 /- Order of M (standard ATLAS value; not r                                             |
| <code>mathieu_tower_matches_certified_order</code>      | Order of M (standard ATLAS value; not re-derived from a group action here). -/ def <code>orderM23</code> : := 10200960 /- Order of M (standard ATLAS value; not r                                             |
| StringTheoryFormalization.UseCases.NarainLattice        |                                                                                                                                                                                                               |
| <code>B_is_polarization_of_Q</code>                     | Left-moving momentum (see also UseCases.TDuality.leftMomentum, restated self-contained here so this file's Narain-specific claims don't accidentally d                                                        |
| <code>gram_eq_hyperbolic</code>                         | Left-moving momentum (see also UseCases.TDuality.leftMomentum, restated self-contained here so this file's Narain-specific claims don't accidentally d                                                        |
| <code>narain_form_even</code>                           | Left-moving momentum (see also UseCases.TDuality.leftMomentum, restated self-contained here so this file's Narain-specific claims don't accidentally d                                                        |
| <code>narain_gram_unimodular</code>                     | Left-moving momentum (see also UseCases.TDuality.leftMomentum, restated self-contained here so this file's Narain-specific claims don't accidentally d                                                        |
| <code>narain_norm_R_independent</code>                  | Left-moving momentum (see also UseCases.TDuality.leftMomentum, restated self-contained here so this file's Narain-specific claims don't accidentally d                                                        |
| StringTheoryFormalization.UseCases.TDualityMassSpectrum |                                                                                                                                                                                                               |
| <code>self_dual_radius_unique</code>                    | Left-moving momentum: combination of momentum and winding that survives T-duality unchanged. -/ def <code>leftMomentum</code> (n w R : ) : := n / R + w R /- Righ                                             |
| <code>tduality_fixes_left_momentum</code>               | Left-moving momentum: combination of momentum and winding that survives T-duality unchanged. -/ def <code>leftMomentum</code> (n w R : ) : := n / R + w R /- Righ                                             |
| <code>tduality_flips_right_momentum</code>              | Left-moving momentum: combination of momentum and winding that survives T-duality unchanged. -/ def <code>leftMomentum</code> (n w R : ) : := n / R + w R /- Righ                                             |
| <code>tduality_invariant_mass_squared</code>            | Left-moving momentum: combination of momentum and winding that survives T-duality unchanged. -/ def <code>leftMomentum</code> (n w R : ) : := n / R + w R /- Righ                                             |



## DoubleFieldTheory (44 theorems)

| Theorem                                                            | First line of its docstring                                                                                                                                     |
|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DoubleFieldTheory.ActionCurvature<br>connection_trace_conservation | DEFINITION: $O(D, D)$ Invariant Dilaton Measure Physical Interpretation: The integration measure $e^{(-2d)} = \sqrt{-g}$ , $e^{(-2\phi)}$ on doubled spacetime: |
| contracted_einstein_dim2                                           | DEFINITION: $O(D, D)$ Invariant Dilaton Measure Physical Interpretation: The integration measure $e^{(-2d)} = \sqrt{-g}$ , $e^{(-2\phi)}$ on doubled spacetime: |
| dft_action_nsns_equivalence                                        | DEFINITION: $O(D, D)$ Invariant Dilaton Measure Physical Interpretation: The integration measure $e^{(-2d)} = \sqrt{-g}$ , $e^{(-2\phi)}$ on doubled spacetime: |
| dft_ricci_expansion                                                | DEFINITION: $O(D, D)$ Invariant Dilaton Measure Physical Interpretation: The integration measure $e^{(-2d)} = \sqrt{-g}$ , $e^{(-2\phi)}$ on doubled spacetime: |
| dft_ricci_physical_reduction                                       | DEFINITION: $O(D, D)$ Invariant Dilaton Measure Physical Interpretation: The integration measure $e^{(-2d)} = \sqrt{-g}$ , $e^{(-2\phi)}$ on doubled spacetime: |
| dilaton_measure_symm                                               | DEFINITION: $O(D, D)$ Invariant Dilaton Measure Physical Interpretation: The integration measure $e^{(-2d)} = \sqrt{-g}$ , $e^{(-2\phi)}$ on doubled spacetime: |
| DoubleFieldTheory.CourantAlgebroid<br>cbracket_antisymm            | Lie bracket commutator of scalar vector field representations.<br>-/ def LieBracket (u v : Int) : Int := u v - v u /- THEOREM: Ordinary Spacetime Lie Se        |
| cbracket_v_zero                                                    | Lie bracket commutator of scalar vector field representations.<br>-/ def LieBracket (u v : Int) : Int := u v - v u /- THEOREM: Ordinary Spacetime Lie Se        |
| dorfman_cbracket_diff                                              | Lie bracket commutator of scalar vector field representations.<br>-/ def LieBracket (u v : Int) : Int := u v - v u /- THEOREM: Ordinary Spacetime Lie Se        |
| dorfman_symmetric_exact                                            | Lie bracket commutator of scalar vector field representations.<br>-/ def LieBracket (u v : Int) : Int := u v - v u /- THEOREM: Ordinary Spacetime Lie Se        |
| gen_lie_closure                                                    | Lie bracket commutator of scalar vector field representations.<br>-/ def LieBracket (u v : Int) : Int := u v - v u /- THEOREM: Ordinary Spacetime Lie Se        |
| jacobiator_vector_vanishes                                         | Lie bracket commutator of scalar vector field representations.<br>-/ def LieBracket (u v : Int) : Int := u v - v u /- THEOREM: Ordinary Spacetime Lie Se        |
| lie_bracket_self                                                   | Lie bracket commutator of scalar vector field representations.<br>-/ def LieBracket (u v : Int) : Int := u v - v u /- THEOREM: Ordinary Spacetime Lie Se        |
| strong_section_condition                                           | Lie bracket commutator of scalar vector field representations.<br>-/ def LieBracket (u v : Int) : Int := u v - v u /- THEOREM: Ordinary Spacetime Lie Se        |
| DoubleFieldTheory.GeneralizedGeometry<br>btwist_preserves_eta      | Generalized Vector Field A section $X^M = (v^\mu, \xi_\mu)$ in $\Gamma(TM \oplus T^*M)$ combining a tangent vector $v$ (momentum mode generator) and a 1-form   |
| chiral_projector_orthogonality                                     | Generalized Vector Field A section $X^M = (v^\mu, \xi_\mu)$ in $\Gamma(TM \oplus T^*M)$ combining a tangent vector $v$ (momentum mode generator) and a 1-form   |

| Theorem                                       | First line of its docstring                                                                                                                                  |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>courant_pairing_symm</code>             | Generalized Vector Field A section $X^M = (v^\mu, \xi_\mu)$ in $\Gamma(TM \oplus T^*M)$ combining a tangent vector $v$ (momentum mode generator) and a 1-for |
| <code>gen_energy_pos</code>                   | Generalized Vector Field A section $X^M = (v^\mu, \xi_\mu)$ in $\Gamma(TM \oplus T^*M)$ combining a tangent vector $v$ (momentum mode generator) and a 1-for |
| <code>gen_metric_duality</code>               | Generalized Vector Field A section $X^M = (v^\mu, \xi_\mu)$ in $\Gamma(TM \oplus T^*M)$ combining a tangent vector $v$ (momentum mode generator) and a 1-for |
| <code>gen_metric_symmetric</code>             | Generalized Vector Field A section $X^M = (v^\mu, \xi_\mu)$ in $\Gamma(TM \oplus T^*M)$ combining a tangent vector $v$ (momentum mode generator) and a 1-for |
| <code>odd_inversion_generator</code>          | Generalized Vector Field A section $X^M = (v^\mu, \xi_\mu)$ in $\Gamma(TM \oplus T^*M)$ combining a tangent vector $v$ (momentum mode generator) and a 1-for |
| <code>odd_metric_eval</code>                  | Generalized Vector Field A section $X^M = (v^\mu, \xi_\mu)$ in $\Gamma(TM \oplus T^*M)$ combining a tangent vector $v$ (momentum mode generator) and a 1-for |
| DoubleFieldTheory.K3Topology                  |                                                                                                                                                              |
| <code>k3_atiyah_singer_dirac_index</code>     | Alternating sum of Betti numbers computing the topological Euler characteristic. -/ def K3BettiSum (b0 b1 b2 b3 b4 : Int) : Int := b0 - b1 + b2 - b3 +       |
| <code>k3_euler_characteristic</code>          | Alternating sum of Betti numbers computing the topological Euler characteristic. -/ def K3BettiSum (b0 b1 b2 b3 b4 : Int) : Int := b0 - b1 + b2 - b3 +       |
| <code>k3_hirzebruch_signature</code>          | Alternating sum of Betti numbers computing the topological Euler characteristic. -/ def K3BettiSum (b0 b1 b2 b3 b4 : Int) : Int := b0 - b1 + b2 - b3 +       |
| <code>k3_intersection_lattice_rank_sig</code> | Alternating sum of Betti numbers computing the topological Euler characteristic. -/ def K3BettiSum (b0 b1 b2 b3 b4 : Int) : Int := b0 - b1 + b2 - b3 +       |
| <code>k3_parallel_chiral_spinor_index</code>  | Alternating sum of Betti numbers computing the topological Euler characteristic. -/ def K3BettiSum (b0 b1 b2 b3 b4 : Int) : Int := b0 - b1 + b2 - b3 +       |
| <code>k3_second_betti_hodge</code>            | Alternating sum of Betti numbers computing the topological Euler characteristic. -/ def K3BettiSum (b0 b1 b2 b3 b4 : Int) : Int := b0 - b1 + b2 - b3 +       |
| <code>k3_su2_holonomy_reduction</code>        | Alternating sum of Betti numbers computing the topological Euler characteristic. -/ def K3BettiSum (b0 b1 b2 b3 b4 : Int) : Int := b0 - b1 + b2 - b3 +       |
| DoubleFieldTheory.PhysicsDSL                  |                                                                                                                                                              |
| <code>dsl_cbracket_antisymm</code>            | Effective dual radius numerator $R_-(\mathrm{eff})(R)$ times $R = R^2 + \alpha'$ with $\alpha' = 1$ . -/ def Reff (R : Nat) : Nat := R R + 1 - Notation def  |
| <code>dsl_cbracket_self_vanishes</code>       | Effective dual radius numerator $R_-(\mathrm{eff})(R)$ times $R = R^2 + \alpha'$ with $\alpha' = 1$ . -/ def Reff (R : Nat) : Nat := R R + 1 - Notation def  |
| <code>dsl_courant_pairing_symm</code>         | Effective dual radius numerator $R_-(\mathrm{eff})(R)$ times $R = R^2 + \alpha'$ with $\alpha' = 1$ . -/ def Reff (R : Nat) : Nat := R R + 1 - Notation def  |
| <code>dsl_dorfman_cbracket_diff</code>        | Effective dual radius numerator $R_-(\mathrm{eff})(R)$ times $R = R^2 + \alpha'$ with $\alpha' = 1$ . -/ def Reff (R : Nat) : Nat := R R + 1 - Notation def  |

| Theorem                                            | First line of its docstring                                                                                                                               |
|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>dsl_effective_radius_strictly_super_p</code> | Effective dual radius numerator $R_{\text{eff}}(R)$ times $R = R^2 + \alpha'$ with $\alpha' = 1$ . -/ def Reff (R : Nat) : Nat := R R + 1 – Notation defi |
| <code>dsl_self_dual_minimum</code>                 | Effective dual radius numerator $R_{\text{eff}}(R)$ times $R = R^2 + \alpha'$ with $\alpha' = 1$ . -/ def Reff (R : Nat) : Nat := R R + 1 – Notation defi |
| <code>dsl_strong_section_condition</code>          | Effective dual radius numerator $R_{\text{eff}}(R)$ times $R = R^2 + \alpha'$ with $\alpha' = 1$ . -/ def Reff (R : Nat) : Nat := R R + 1 – Notation defi |
| DoubleFieldTheory.TDualityBuscher                  |                                                                                                                                                           |
| <code>buscher_dilaton_involution</code>            | DEFINITION: Congruence Action of O(D, D) Duality Matrix Physical Interpretation: The transformation law of the generalized metric $\mathcal{H}$ under du  |
| <code>buscher_dilaton_measure_invariance</code>    | DEFINITION: Congruence Action of O(D, D) Duality Matrix Physical Interpretation: The transformation law of the generalized metric $\mathcal{H}$ under du  |
| <code>buscher_log_involution</code>                | DEFINITION: Congruence Action of O(D, D) Duality Matrix Physical Interpretation: The transformation law of the generalized metric $\mathcal{H}$ under du  |
| <code>inversion_generator_properties</code>        | DEFINITION: Congruence Action of O(D, D) Duality Matrix Physical Interpretation: The transformation law of the generalized metric $\mathcal{H}$ under du  |
| <code>self_dual_radius_rigidity</code>             | DEFINITION: Congruence Action of O(D, D) Duality Matrix Physical Interpretation: The transformation law of the generalized metric $\mathcal{H}$ under du  |
| <code>t_duality_congruence</code>                  | DEFINITION: Congruence Action of O(D, D) Duality Matrix Physical Interpretation: The transformation law of the generalized metric $\mathcal{H}$ under du  |
| DoubleFieldTheory.TorusMoonshine                   |                                                                                                                                                           |
| <code>mukai_lattice_m24_bridge</code>              | Modular inversion generator S: tau mapsto -1/ tau. -/ def ModS : Mat2 := ( a := 0, b := -1, c := 1, d := 0 ) /- Modular translation generator T: t        |
| <code>sl2z_modular_generators</code>               | Modular inversion generator S: tau mapsto -1/ tau. -/ def ModS : Mat2 := ( a := 0, b := -1, c := 1, d := 0 ) /- Modular translation generator T: t        |

## DualScaleM24Formalization (61 theorems)

| Theorem                                             | First line of its docstring                                                                                                                              |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| DualScaleM24Formalization.DualScale.EffectiveMetric |                                                                                                                                                          |
| <code>buscher_involution</code>                     | DEFINITION: Positive Scale Representation Physical Interpretation: Exact positive rational scale $s = \text{num} / \text{den} > 0$ representing compacti |
| <code>genesis_no_singularity</code>                 | DEFINITION: Positive Scale Representation Physical Interpretation: Exact positive rational scale $s = \text{num} / \text{den} > 0$ representing compacti |
| <code>self_dual_symmetric</code>                    | DEFINITION: Positive Scale Representation Physical Interpretation: Exact positive rational scale $s = \text{num} / \text{den} > 0$ representing compacti |
| DualScaleM24Formalization.DualScale.Sym2Lock        |                                                                                                                                                          |

| Theorem                                                   | First line of its docstring                                                                                                                                  |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>sym2_det_identity</code>                            | Microscopic L_2 Operator Parameterizes the second-order microscopic quantum evolution: $u_{(n+2)} = a \cdot u_{(n+1)} + b \cdot u_n$ -/ structure MicroL2Ope |
| <code>sym2_fibonacci_verification</code>                  | Microscopic L_2 Operator Parameterizes the second-order microscopic quantum evolution: $u_{(n+2)} = a \cdot u_{(n+1)} + b \cdot u_n$ -/ structure MicroL2Ope |
| <code>sym2_modular_attractor_verification</code>          | Microscopic L_2 Operator Parameterizes the second-order microscopic quantum evolution: $u_{(n+2)} = a \cdot u_{(n+1)} + b \cdot u_n$ -/ structure MicroL2Ope |
| <code>sym2_pell_verification</code>                       | Microscopic L_2 Operator Parameterizes the second-order microscopic quantum evolution: $u_{(n+2)} = a \cdot u_{(n+1)} + b \cdot u_n$ -/ structure MicroL2Ope |
| DualScaleM24Formalization.Epistemic.Ledger                |                                                                                                                                                              |
| <code>no_kernel_claim_rests_on_weaker</code>              | A single ledger row: the tier it is filed at, and the claims it rests on. -/ structure Claim ( : Type) where tier : Tier supports : List deriving Rep        |
| <code>not_A_of_weak_support</code>                        | A single ledger row: the tier it is filed at, and the claims it rests on. -/ structure Claim ( : Type) where tier : Tier supports : List deriving Rep        |
| <code>tier_le_of_depends</code>                           | A single ledger row: the tier it is filed at, and the claims it rests on. -/ structure Claim ( : Type) where tier : Tier supports : List deriving Rep        |
| <code>trans</code>                                        |                                                                                                                                                              |
| <code>witnessSound_sound</code>                           |                                                                                                                                                              |
| <code>witnessUnsound_not_sound</code>                     |                                                                                                                                                              |
| DualScaleM24Formalization.Epistemic.Tier                  |                                                                                                                                                              |
| <code>eq_A_of_A_le</code>                                 |                                                                                                                                                              |
| <code>tier_A_is_maximal</code>                            |                                                                                                                                                              |
| <code>tier_X_is_minimal</code>                            |                                                                                                                                                              |
| <code>tier_antisymm</code>                                |                                                                                                                                                              |
| <code>tier_refl</code>                                    |                                                                                                                                                              |
| <code>tier_trans</code>                                   |                                                                                                                                                              |
| DualScaleM24Formalization.FrontierTriad.FluxVacuumDecay   |                                                                                                                                                              |
| <code>cdl_action_strictly_positive</code>                 | Flux Vacuum State A string compactification vacuum characterized by flux integer N, energy density V, and holographic central charge c. -/ structure F       |
| <code>holographic_c_theorem_decay</code>                  | Flux Vacuum State A string compactification vacuum characterized by flux integer N, energy density V, and holographic central charge c. -/ structure F       |
| <code>true_vacuum_energy_is_lower</code>                  | Flux Vacuum State A string compactification vacuum characterized by flux integer N, energy density V, and holographic central charge c. -/ structure F       |
| DualScaleM24Formalization.FrontierTriad.SwamplandDistance |                                                                                                                                                              |
| <code>cooper_s10_swampland_safe</code>                    |                                                                                                                                                              |
| <code>fricke_discriminant_is_minus_four</code>            |                                                                                                                                                              |
| <code>fricke_involution_property</code>                   | Moduli space dimension of Ricci-flat metrics on K3: 3 times $19 + 1 = 58$ . -/ def k3_moduli_dim : Nat := 3 19 + 1 theorem k3_moduli_dim_eq_58 : k3_modu     |
| <code>k3_moduli_dim_eq_58</code>                          |                                                                                                                                                              |
| <code>kummer_m24_swampland_safe</code>                    |                                                                                                                                                              |
| <code>sdc_decay_coefficient_identity</code>               | Moduli space dimension of Ricci-flat metrics on K3: 3 times $19 + 1 = 58$ . -/ def k3_moduli_dim : Nat := 3 19 + 1 theorem k3_moduli_dim_eq_58 : k3_modu     |
| <code>total_k3t2_moduli_dim_eq_80</code>                  |                                                                                                                                                              |

| Theorem                                                     | First line of its docstring                                                                                                                                    |
|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| DualScaleM24Formalization.FrontierTriad.TachyonCondensation |                                                                                                                                                                |
| sen_conjecture_k_theory_conservation                        | Topological K-Theory Class A truncated Grothendieck class in $K_0(X)$ parameterized by rank, first Chern number $c_1$ , and second Chern number $c_2$ . -/ str |
| tachyon_condensation_preserves_rr_charge                    | Topological K-Theory Class A truncated Grothendieck class in $K_0(X)$ parameterized by rank, first Chern number $c_1$ , and second Chern number $c_2$ . -/ str |
| DualScaleM24Formalization.FrontierTriad.TriadContract       |                                                                                                                                                                |
| kummer_mapper_betti1_eq_376                                 | Fast AOT Contract: In-flight metric positivity gate $\mathrm{Im}(\tau) > 0$ . -/ def checkMetricPositivity (tau_im_scaled : Int) : Bool := tau_im_scaled       |
| kummer_mapper_euler_eq_minus_370                            | Fast AOT Contract: In-flight metric positivity gate $\mathrm{Im}(\tau) > 0$ . -/ def checkMetricPositivity (tau_im_scaled : Int) : Bool := tau_im_scaled       |
| symmetron_screening_cassini_pass                            | Fast AOT Contract: In-flight metric positivity gate $\mathrm{Im}(\tau) > 0$ . -/ def checkMetricPositivity (tau_im_scaled : Int) : Bool := tau_im_scaled       |
| DualScaleM24Formalization.Moonshine.GysinBEMSequence        |                                                                                                                                                                |
| bem_characteristic_vanishing                                | Graded cohomology ring representation for circle bundle fibrations. -/ structure CohomologyRing where H2_B : Int H3_B : Int H4_B : Int H3_E : Int H4_E         |
| bem_derived_exchange                                        | Graded cohomology ring representation for circle bundle fibrations. -/ structure CohomologyRing where H2_B : Int H3_B : Int H4_B : Int H3_E : Int H4_E         |
| DualScaleM24Formalization.Moonshine.KummerTadpole           |                                                                                                                                                                |
| d7_positive_charge                                          |                                                                                                                                                                |
| hodge_h11_eq_20                                             |                                                                                                                                                                |
| k3_euler_eq_24                                              | Topological Euler characteristic of K3: $\chi = b_0 - b_1 + b_2 - b_3 + b_4 = 24$ . -/ def k3_euler : Int := (k3_b0 : Int) - k3_b1 + k3_b2 - k3_b3 + k3_b4     |
| k3t2_euler_char_eq_zero                                     | Topological Euler characteristic of K3: $\chi = b_0 - b_1 + b_2 - b_3 + b_4 = 24$ . -/ def k3_euler : Int := (k3_b0 : Int) - k3_b1 + k3_b2 - k3_b3 + k3_b4     |
| kuenneth_b3_derivation                                      | Topological Euler characteristic of K3: $\chi = b_0 - b_1 + b_2 - b_3 + b_4 = 24$ . -/ def k3_euler : Int := (k3_b0 : Int) - k3_b1 + k3_b2 - k3_b3 + k3_b4     |
| kummer_exceptional_intersection                             | Topological Euler characteristic of K3: $\chi = b_0 - b_1 + b_2 - b_3 + b_4 = 24$ . -/ def k3_euler : Int := (k3_b0 : Int) - k3_b1 + k3_b2 - k3_b3 + k3_b4     |
| o7_negative_charge                                          |                                                                                                                                                                |
| picard_rank_kummer_maximal                                  |                                                                                                                                                                |
| rr_tadpole_cancellation                                     | Topological Euler characteristic of K3: $\chi = b_0 - b_1 + b_2 - b_3 + b_4 = 24$ . -/ def k3_euler : Int := (k3_b0 : Int) - k3_b1 + k3_b2 - k3_b3 + k3_b4     |
| DualScaleM24Formalization.Moonshine.MathieuRigidity         |                                                                                                                                                                |
| conformal_exponent_sum                                      |                                                                                                                                                                |
| delta12_value                                               |                                                                                                                                                                |
| delta1_is_half                                              |                                                                                                                                                                |
| delta2_is_five_halves                                       |                                                                                                                                                                |
| denominator_product_check                                   |                                                                                                                                                                |
| dimA1_decomposition                                         |                                                                                                                                                                |
| dimA2_decomposition                                         |                                                                                                                                                                |
| dimA3_decomposition                                         |                                                                                                                                                                |
| nominal_value_is_consistent                                 |                                                                                                                                                                |

| Theorem                                          | First line of its docstring                                                                                                                            |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>product_value_check</code>                 |                                                                                                                                                        |
| <code>r_bps_cross_multiplication</code>          | Exact rational fraction for conformal weights and scaling exponents. -/ structure Frac where num : Int den : Nat deriving DecidableEq, Repr def addFra |
| <code>r_bps_is_irreducible</code>                | Exact rational fraction for conformal weights and scaling exponents. -/ structure Frac where num : Int den : Nat deriving DecidableEq, Repr def addFra |
| <code>r_bps_parts_per_thousand_value</code>      |                                                                                                                                                        |
| <code>superconformal_geometric_congruence</code> | Exact rational fraction for conformal weights and scaling exponents. -/ structure Frac where num : Int den : Nat deriving DecidableEq, Repr def addFra |
| <code>sym2_A1_dimension_is_4095</code>           |                                                                                                                                                        |
| <code>total_chiral_weight_value</code>           |                                                                                                                                                        |

## StringTheoryFoundation (56 theorems)

| Theorem                                          | First line of its docstring                                                                                                                                    |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| StringTheoryFoundation.Atlas.AtlasGeometryBridge |                                                                                                                                                                |
| <code>atlas_k3_euler_characteristic</code>       | ATLAS 4-Manifold Topology: Grounded in Atlas.GeometryOfManifolds and Atlas.AlgebraicTopologyI. Encodes the topological invariants of smooth simply-con         |
| <code>atlas_k3_signature</code>                  | ATLAS 4-Manifold Topology: Grounded in Atlas.GeometryOfManifolds and Atlas.AlgebraicTopologyI. Encodes the topological invariants of smooth simply-con         |
| <code>atlas_poincare_curvature_negative</code>   | ATLAS 4-Manifold Topology: Grounded in Atlas.GeometryOfManifolds and Atlas.AlgebraicTopologyI. Encodes the topological invariants of smooth simply-con         |
| StringTheoryFoundation.Core.Topology             |                                                                                                                                                                |
| <code>euler_char_K3</code>                       | Betti numbers of a 4-dimensional compact orientable manifold (b, b, b, b, b). -/ structure Betti4D where b0 : Int := 1 b1 : Int := 0 b2 : Int := 0 b3          |
| <code>euler_char_T2</code>                       | Betti numbers of a 4-dimensional compact orientable manifold (b, b, b, b, b). -/ structure Betti4D where b0 : Int := 1 b1 : Int := 0 b2 : Int := 0 b3          |
| StringTheoryFoundation.Duality.DualScale         |                                                                                                                                                                |
| <code>dual_pair_symmetric</code>                 | Dual-Scale Model: Given a self-dual crossover scale $L > 0$ , any physical scale $L$ has a dual scale $L = L / L$ . -/ def dualLength (L_star L : Float) : Flo |
| <code>dual_product_invariant</code>              | Dual-Scale Model: Given a self-dual crossover scale $L > 0$ , any physical scale $L$ has a dual scale $L = L / L$ . -/ def dualLength (L_star L : Float) : Flo |
| <code>self_dual_at_crossover</code>              | Dual-Scale Model: Given a self-dual crossover scale $L > 0$ , any physical scale $L$ has a dual scale $L = L / L$ . -/ def dualLength (L_star L : Float) : Flo |
| StringTheoryFoundation.Duality.T_Duality         |                                                                                                                                                                |
| <code>self_dual_mass_symmetry</code>             | String compactification state on a circle with radius R: - n: Kaluza-Klein momentum excitation integer - w: topological string winding number -/ struc         |
| <code>t_duality_involution</code>                | String compactification state on a circle with radius R: - n: Kaluza-Klein momentum excitation integer - w: topological string winding number -/ struc         |

| Theorem                                                              | First line of its docstring                                                                                                                                |
|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>t_duality_spectrum_invariance</code>                           | String compactification state on a circle with radius R: - n: Kaluza-Klein momentum excitation integer - w: topological string winding number -/ struc     |
| StringTheoryFoundation.FluidDynamics.NavierStokesBridge              |                                                                                                                                                            |
| <code>energy_dissipation_monotonic</code>                            | Torus dimension and viscosity parameter. -/ structure TorusFluidConfig where dim : Nat := 2 viscosity : Nat := 1 – scaled integer viscosity > 0 sobo       |
| <code>fundamental_mode_positive</code>                               | Torus dimension and viscosity parameter. -/ structure TorusFluidConfig where dim : Nat := 2 viscosity : Nat := 1 – scaled integer viscosity > 0 sobo       |
| <code>sound_speed_is_luminal</code>                                  |                                                                                                                                                            |
| <code>zero_mode_laplacian</code>                                     | Torus dimension and viscosity parameter. -/ structure TorusFluidConfig where dim : Nat := 2 viscosity : Nat := 1 – scaled integer viscosity > 0 sobo       |
| StringTheoryFoundation.K3.K3Surfaces                                 |                                                                                                                                                            |
| <code>k3_b2_is_22</code>                                             | Hodge diamond for a compact complex surface: h, h, h, h, h, h, h, h. -/ structure HodgeDiamond2D where h00 : Nat := 1 h10 : Nat := 0 h01 : Nat := 0        |
| <code>k3_lattice_rank_valid</code>                                   | Hodge diamond for a compact complex surface: h, h, h, h, h, h, h, h. -/ structure HodgeDiamond2D where h00 : Nat := 1 h10 : Nat := 0 h01 : Nat := 0        |
| <code>k3_signature_is_minus_16</code>                                | Hodge diamond for a compact complex surface: h, h, h, h, h, h, h, h. -/ structure HodgeDiamond2D where h00 : Nat := 1 h10 : Nat := 0 h01 : Nat := 0        |
| StringTheoryFoundation.ModularForms.FermatModularBridge              |                                                                                                                                                            |
| <code>exceptional_divisor_cartan_a1</code>                           |                                                                                                                                                            |
| <code>kummer_fixed_points_dim4</code>                                |                                                                                                                                                            |
| <code>mukai_lattice_rank_equals_24</code>                            |                                                                                                                                                            |
| <code>mukai_signature_difference</code>                              |                                                                                                                                                            |
| StringTheoryFoundation.PhysLib.PhysLibKinematicsBridge               |                                                                                                                                                            |
| <code>dft_dimension_dim4</code>                                      |                                                                                                                                                            |
| <code>lorentz_signature_is_minus_two</code>                          |                                                                                                                                                            |
| <code>odd_metric_involutive</code>                                   |                                                                                                                                                            |
| <code>rest_frame_mass_shell</code>                                   | Relativistic 4-momentum in discrete integer units. -/ structure FourMomentum where p0 : Int – Energy E p1 : Int – Momentum p_x p2 : Int – Momentum         |
| StringTheoryFoundation.Quantum.TensorNetworkBridge                   |                                                                                                                                                            |
| <code>golay_code_rate_is_half</code>                                 | Parameters of the extended binary Golay code mathematical(G)_(24): Length n = 24, dimension k = 12, minimum Hamming distance d = 8. -/ structure GolayCode |
| <code>golay_length_matches_k3_euler</code>                           | Parameters of the extended binary Golay code mathematical(G)_(24): Length n = 24, dimension k = 12, minimum Hamming distance d = 8. -/ structure GolayCode |
| <code>ryu_takayanagi_empty_cut</code>                                | Parameters of the extended binary Golay code mathematical(G)_(24): Length n = 24, dimension k = 12, minimum Hamming distance d = 8. -/ structure GolayCode |
| <code>ryu_takayanagi_positive_cut</code>                             | Parameters of the extended binary Golay code mathematical(G)_(24): Length n = 24, dimension k = 12, minimum Hamming distance d = 8. -/ structure GolayCode |
| StringTheoryFoundation.StatisticalLearning.StatisticalLearningBridge |                                                                                                                                                            |
| <code>rademacher_bound_positive</code>                               | Sample size n and confidence parameter delta represented in scaled integer metrics. -/ structure PACSampleConfig where sample_size : Nat := 100 confi      |

| Theorem                                                                                                            | First line of its docstring                                                                                                                                                             |
|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>zero_empirical_risk_on_perfect_proofs</code>                                                                 | Sample size $n$ and confidence parameter $\delta$ represented in scaled integer metrics. -/ structure PACSampleConfig where <code>sample_size : Nat := 100</code> <code>confi</code>    |
| <b>StringTheoryFoundation.StringTheory.K3xT2</b>                                                                   |                                                                                                                                                                                         |
| <code>dim_real_k3xt2_is_6</code>                                                                                   | Real dimension of the internal compactification manifold K3<br>T: $\dim\_(\text{K3}) + \dim\_(\text{T}) = 4 + 2 = 6$ . -/ <code>def dimRealK3xT2 : Nat := 4 + 2</code> -/ Theorem: K3 T |
| <code>euler_char_vanishes</code>                                                                                   | Real dimension of the internal compactification manifold K3<br>T: $\dim\_(\text{K3}) + \dim\_(\text{T}) = 4 + 2 = 6$ . -/ <code>def dimRealK3xT2 : Nat := 4 + 2</code> -/ Theorem: K3 T |
| <code>string_duality_susy_match</code><br><code>target_spacetime_is_4d</code><br><code>typeII_k3xt2_susy_N4</code> | Real dimension of the internal compactification manifold K3<br>T: $\dim\_(\text{K3}) + \dim\_(\text{T}) = 4 + 2 = 6$ . -/ <code>def dimRealK3xT2 : Nat := 4 + 2</code> -/ Theorem: K3 T |
| <b>StringTheoryFoundation.StringTheory.StromingerSYZ</b>                                                           |                                                                                                                                                                                         |
| <code>strominger_hyperkahler_signature</code>                                                                      | SYZ Special Lagrangian Fibration: A Calabi-Yau 2-fold (K3 surface) admits a fibration by special Lagrangian 2-tori T over a 2-sphere base S (affine ma                                  |
| <code>strominger_nodal_fibers_match_euler</code>                                                                   | SYZ Special Lagrangian Fibration: A Calabi-Yau 2-fold (K3 surface) admits a fibration by special Lagrangian 2-tori T over a 2-sphere base S (affine ma                                  |
| <code>strominger_syz_dim_sum</code>                                                                                | SYZ Special Lagrangian Fibration: A Calabi-Yau 2-fold (K3 surface) admits a fibration by special Lagrangian 2-tori T over a 2-sphere base S (affine ma                                  |
| <code>strominger_syz_dual_symmetric</code>                                                                         | SYZ Special Lagrangian Fibration: A Calabi-Yau 2-fold (K3 surface) admits a fibration by special Lagrangian 2-tori T over a 2-sphere base S (affine ma                                  |
| <b>StringTheoryFoundation.StringTheory.Swampland</b>                                                               |                                                                                                                                                                                         |
| <code>distance_conjecture_monotonicity</code>                                                                      | Swampland Distance Conjecture (SDC): For a displacement in moduli space, a tower of states has mass $m() \propto \exp(-\dots)$ where $> 0$ . Expressed in discrete                      |
| <code>unit_state_satisfies_wgc</code>                                                                              | Swampland Distance Conjecture (SDC): For a displacement in moduli space, a tower of states has mass $m() \propto \exp(-\dots)$ where $> 0$ . Expressed in discrete                      |
| <b>StringTheoryFoundation.StringTheory.TadpoleCancellation</b>                                                     |                                                                                                                                                                                         |
| <code>d3_tadpole_target_is_one</code><br><code>d7_tadpole_cancellation</code>                                      | Fixed points of the T/ orientifold action: The involution $x_i \rightarrow -x_i$ on $T = (S)$ has $2 = 16$ fixed points. -/ <code>def numFixedPointsT4Z2 : Nat := 16</code> -/ Genu     |
| <code>num_fixed_points_is_16</code>                                                                                | Fixed points of the T/ orientifold action: The involution $x_i \rightarrow -x_i$ on $T = (S)$ has $2 = 16$ fixed points. -/ <code>def numFixedPointsT4Z2 : Nat := 16</code> -/ Genu     |
| <code>total_D7_charge_is_64</code><br><code>total_O7_charge_is_minus_64</code>                                     |                                                                                                                                                                                         |
| <b>StringTheoryFoundation.StringTheory.VafaSwampland</b>                                                           |                                                                                                                                                                                         |
| <code>vafa_gvw_zero_flux</code>                                                                                    | DEFINITION: Swampland Distance Tower Physical Interpretation: Represents an infinite Kaluza-Klein or string winding tower descending from the UV along                                  |
| <code>vafa_sdc_4d_decay_rate</code>                                                                                | DEFINITION: Swampland Distance Tower Physical Interpretation: Represents an infinite Kaluza-Klein or string winding tower descending from the UV along                                  |
| <code>vafa_u1_gauge_consistent</code>                                                                              | DEFINITION: Swampland Distance Tower Physical Interpretation: Represents an infinite Kaluza-Klein or string winding tower descending from the UV along                                  |



| Theorem                                                        | First line of its docstring                                                                                                                                    |
|----------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>vafa_wgc_weak_coupling_cutoff_monotone</code>            | DEFINITION: Swampland Distance Tower Physical Interpretation: Represents an infinite Kaluza-Klein or string winding tower descending from the UV along         |
| <code>StringTheoryFoundation.StringTheory.WittenDuality</code> |                                                                                                                                                                |
| <code>witten_6d_supercharges</code>                            | DEFINITION: Six-Dimensional Supersymmetry Algebra Physical Interpretation: Data defining $\text{mathcal{N}} = (1, 1)$ supersymmetry in 6 dimensions with 16 re |
| <code>witten_duality_rank_match</code>                         | DEFINITION: Six-Dimensional Supersymmetry Algebra Physical Interpretation: Data defining $\text{mathcal{N}} = (1, 1)$ supersymmetry in 6 dimensions with 16 re |
| <code>witten_gauge_enhancement_at_singularity</code>           | DEFINITION: Six-Dimensional Supersymmetry Algebra Physical Interpretation: Data defining $\text{mathcal{N}} = (1, 1)$ supersymmetry in 6 dimensions with 16 re |
| <code>witten_moduli_dim_is_80</code>                           | DEFINITION: Six-Dimensional Supersymmetry Algebra Physical Interpretation: Data defining $\text{mathcal{N}} = (1, 1)$ supersymmetry in 6 dimensions with 16 re |

## DualScaleValidation (23 theorems)

| Theorem                                                       | First line of its docstring                                                                                                                             |
|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>DualScaleValidation.Observables</code>                  |                                                                                                                                                         |
| <code>dark_energy_cosmological_constant</code>                | Exact numerator of the speculative tensor-to-scalar ratio $r = 1/252$ (see module docstring 1). -/ def tensor_to_scalar_ratio_num : Nat := 1 /- Exact d |
| <code>neutrino_cp_phase_in_illustrative_window</code>         | Exact numerator of the speculative tensor-to-scalar ratio $r = 1/252$ (see module docstring 1). -/ def tensor_to_scalar_ratio_num : Nat := 1 /- Exact d |
| <code>observables_windows_master_contract</code>              | Exact numerator of the speculative tensor-to-scalar ratio $r = 1/252$ (see module docstring 1). -/ def tensor_to_scalar_ratio_num : Nat := 1 /- Exact d |
| <code>spectral_index_in_planck_window</code>                  | Exact numerator of the speculative tensor-to-scalar ratio $r = 1/252$ (see module docstring 1). -/ def tensor_to_scalar_ratio_num : Nat := 1 /- Exact d |
| <code>tensor_to_scalar_in_illustrative_window</code>          | Exact numerator of the speculative tensor-to-scalar ratio $r = 1/252$ (see module docstring 1). -/ def tensor_to_scalar_ratio_num : Nat := 1 /- Exact d |
| <code>tensor_to_scalar_ratio_reduction</code>                 | Exact numerator of the speculative tensor-to-scalar ratio $r = 1/252$ (see module docstring 1). -/ def tensor_to_scalar_ratio_num : Nat := 1 /- Exact d |
| <code>DualScaleValidation.UseCase1_ModuliStabilization</code> |                                                                                                                                                         |
| <code>buscher_log_involution</code>                           | Fundamental string tension scale $\alpha'$ normalized to 1 in string units. -/ def alpha_prime : Nat := 1 /- Effective dual scale numerator on a torus  |
| <code>dft_doubled_dimension_10d</code>                        | Fundamental string tension scale $\alpha'$ normalized to 1 in string units. -/ def alpha_prime : Nat := 1 /- Effective dual scale numerator on a torus  |
| <code>dual_scale_self_dual_value</code>                       | Fundamental string tension scale $\alpha'$ normalized to 1 in string units. -/ def alpha_prime : Nat := 1 /- Effective dual scale numerator on a torus  |

| Theorem                                      | First line of its docstring                                                                                                                                           |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>moduli_potential_nonneg</code>         | Fundamental string tension scale $\alpha'$ normalized to 1 in string units. -/ def <code>alpha_prime</code> : Nat := 1 /- Effective dual scale numerator on a torus   |
| <code>moduli_potential_zero_iff</code>       | Fundamental string tension scale $\alpha'$ normalized to 1 in string units. -/ def <code>alpha_prime</code> : Nat := 1 /- Effective dual scale numerator on a torus   |
| <code>moduli_vacuum_stability</code>         | Fundamental string tension scale $\alpha'$ normalized to 1 in string units. -/ def <code>alpha_prime</code> : Nat := 1 /- Effective dual scale numerator on a torus   |
| <code>self_dual_is_global_minimum</code>     | Fundamental string tension scale $\alpha'$ normalized to 1 in string units. -/ def <code>alpha_prime</code> : Nat := 1 /- Effective dual scale numerator on a torus   |
| DualScaleValidation.UseCase2_MoonshineBPS    |                                                                                                                                                                       |
| <code>bps_cross_multiplication_lock</code>   | Dimension of the first Mathieu Moonshine representation: $A_1 = 90 = \mathbf{45} \oplus \overline{\mathbf{45}}$ . -/ def <code>dim_A1</code> : Nat := 90 /- Dimension |
| <code>bps_lock_exact_equality</code>         | Dimension of the first Mathieu Moonshine representation: $A_1 = 90 = \mathbf{45} \oplus \overline{\mathbf{45}}$ . -/ def <code>dim_A1</code> : Nat := 90 /- Dimension |
| <code>bps_ratio_coprime</code>               | Dimension of the first Mathieu Moonshine representation: $A_1 = 90 = \mathbf{45} \oplus \overline{\mathbf{45}}$ . -/ def <code>dim_A1</code> : Nat := 90 /- Dimension |
| <code>m24_order_divisible_by_bps_lock</code> | Dimension of the first Mathieu Moonshine representation: $A_1 = 90 = \mathbf{45} \oplus \overline{\mathbf{45}}$ . -/ def <code>dim_A1</code> : Nat := 90 /- Dimension |
| <code>sym2_A1_value</code>                   | Dimension of the first Mathieu Moonshine representation: $A_1 = 90 = \mathbf{45} \oplus \overline{\mathbf{45}}$ . -/ def <code>dim_A1</code> : Nat := 90 /- Dimension |
| DualScaleValidation.UseCase3_FrontierTriad   |                                                                                                                                                                       |
| <code>bounce_action_positive</code>          | Positive RR charge contributed by each D7-brane: +4. -/ def <code>d7_charge_per_brane</code> : Int := 4 /- Total number of D7-branes in consistent $T^4/\mathbf{Z}_4$ |
| <code>frontier_triad_master_contract</code>  | Positive RR charge contributed by each D7-brane: +4. -/ def <code>d7_charge_per_brane</code> : Int := 4 /- Total number of D7-branes in consistent $T^4/\mathbf{Z}_4$ |
| <code>rr_tadpole_cancellation</code>         | Positive RR charge contributed by each D7-brane: +4. -/ def <code>d7_charge_per_brane</code> : Int := 4 /- Total number of D7-branes in consistent $T^4/\mathbf{Z}_4$ |
| <code>sdc_mass_suppression</code>            | Positive RR charge contributed by each D7-brane: +4. -/ def <code>d7_charge_per_brane</code> : Int := 4 /- Total number of D7-branes in consistent $T^4/\mathbf{Z}_4$ |
| <code>tachyon_condensation_endpoint</code>   | Positive RR charge contributed by each D7-brane: +4. -/ def <code>d7_charge_per_brane</code> : Int := 4 /- Total number of D7-branes in consistent $T^4/\mathbf{Z}_4$ |

## Lean5Corpus (53 theorems)

| Theorem                                    | First line of its docstring |
|--------------------------------------------|-----------------------------|
| Lean5Corpus.Foundations.Narrative          |                             |
| <code>canonical_corpus_is_sound</code>     |                             |
| <code>foundational_domains_verified</code> |                             |

| Theorem                                                | First line of its docstring                                                                                                                                           |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lean5Corpus.Problems.Problem10_SYMInstanton            |                                                                                                                                                                       |
| bps_instanton_bound_negative                           | Structure representing the gauge field curvature norms in 4D Euclidean space. -/ structure GaugeCurvatureState where coupling_sq : Nat norm_f_plus_sq                 |
| bps_instanton_bound_positive                           | Structure representing the gauge field curvature norms in 4D Euclidean space. -/ structure GaugeCurvatureState where coupling_sq : Nat norm_f_plus_sq                 |
| instanton_action_exceeds_charge                        | Structure representing the gauge field curvature norms in 4D Euclidean space. -/ structure GaugeCurvatureState where coupling_sq : Nat norm_f_plus_sq                 |
| instanton_action_strictly_positive                     | Structure representing the gauge field curvature norms in 4D Euclidean space. -/ structure GaugeCurvatureState where coupling_sq : Nat norm_f_plus_sq                 |
| sym_instanton_master_contract                          | Structure representing the gauge field curvature norms in 4D Euclidean space. -/ structure GaugeCurvatureState where coupling_sq : Nat norm_f_plus_sq                 |
| Lean5Corpus.Problems.Problem11_EntanglementEntropy     |                                                                                                                                                                       |
| holographic_entanglement_master_contract               | Structure representing the minimal surface areas in a holographic AdS/CFT spacetime. -/ structure HolographicSubregionSystem where area_A : Nat area_B                |
| holographic_subadditivity                              | Structure representing the minimal surface areas in a holographic AdS/CFT spacetime. -/ structure HolographicSubregionSystem where area_A : Nat area_B                |
| mutual_information_nonnegative                         | Structure representing the minimal surface areas in a holographic AdS/CFT spacetime. -/ structure HolographicSubregionSystem where area_A : Nat area_B                |
| ryu_takayanagi_strong_subadditivity                    | Structure representing the minimal surface areas in a holographic AdS/CFT spacetime. -/ structure HolographicSubregionSystem where area_A : Nat area_B                |
| Lean5Corpus.Problems.Problem1_NavierStokesHelicity     |                                                                                                                                                                       |
| helicity_dissipation_inequality                        | DEFINITION: Viscous Fluid Thermodynamic State Physical Interpretation: Represents the macroscopic state of an incompressible fluid on $\text{mathbb{T}}^3, \text{co}$ |
| knotted_flow_must_dissipate_energy                     | DEFINITION: Viscous Fluid Thermodynamic State Physical Interpretation: Represents the macroscopic state of an incompressible fluid on $\text{mathbb{T}}^3, \text{co}$ |
| mul_reorder_4                                          | DEFINITION: Viscous Fluid Thermodynamic State Physical Interpretation: Represents the macroscopic state of an incompressible fluid on $\text{mathbb{T}}^3, \text{co}$ |
| navier_stokes_helicity_contract                        | DEFINITION: Viscous Fluid Thermodynamic State Physical Interpretation: Represents the macroscopic state of an incompressible fluid on $\text{mathbb{T}}^3, \text{co}$ |
| topological_linking_forces_positive_entropy            | DEFINITION: Viscous Fluid Thermodynamic State Physical Interpretation: Represents the macroscopic state of an incompressible fluid on $\text{mathbb{T}}^3, \text{co}$ |
| Lean5Corpus.Problems.Problem2_MathieuFrobeniusRigidity |                                                                                                                                                                       |
| bps_conductor_prime_factorization                      | The BPS Character Lock constant: $N_-(\text{mathrm{BPS}}) = 27720$ . -/ def bps_conductor : Nat := 27720 /- Dimension of first Mathieu primary representation:        |
| conductor_divisible_by_first_five_primes               | The BPS Character Lock constant: $N_-(\text{mathrm{BPS}}) = 27720$ . -/ def bps_conductor : Nat := 27720 /- Dimension of first Mathieu primary representation:        |
| dim_A1_factorization                                   | The BPS Character Lock constant: $N_-(\text{mathrm{BPS}}) = 27720$ . -/ def bps_conductor : Nat := 27720 /- Dimension of first Mathieu primary representation:        |

| Theorem                                          | First line of its docstring                                                                                                                                            |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>dim_A2_factorization</code>                | The BPS Character Lock constant: $N_-(\mathrm{BPS}) = 27720$ . -/ def bps_conductor : Nat := 27720 /- Dimension of first Mathieu primary representation:               |
| <code>hasse_bound_at_13</code>                   |                                                                                                                                                                        |
| <code>hasse_trace_integer_bound</code>           |                                                                                                                                                                        |
| <code>m24_conductor_quotient</code>              | The BPS Character Lock constant: $N_-(\mathrm{BPS}) = 27720$ . -/ def bps_conductor : Nat := 27720 /- Dimension of first Mathieu primary representation:               |
| <code>mathieu_frobenius_master_contract</code>   | The BPS Character Lock constant: $N_-(\mathrm{BPS}) = 27720$ . -/ def bps_conductor : Nat := 27720 /- Dimension of first Mathieu primary representation:               |
| Lean5Corpus.Problems.Problem3_DualScaleTCC       |                                                                                                                                                                        |
| <code>sub_planckian_modes_impossible</code>      | DEFINITION: Planck Length Scale Physical Interpretation: The fundamental quantum gravity length unit $\mathrm{ell}_-(\mathrm{Pl})$ equiv 1. RAG & Graph Indexing       |
| <code>tcc_cosmic_protection_contract</code>      | DEFINITION: Planck Length Scale Physical Interpretation: The fundamental quantum gravity length unit $\mathrm{ell}_-(\mathrm{Pl})$ equiv 1. RAG & Graph Indexing       |
| <code>tcc_expansion_factor_positive</code>       | DEFINITION: Planck Length Scale Physical Interpretation: The fundamental quantum gravity length unit $\mathrm{ell}_-(\mathrm{Pl})$ equiv 1. RAG & Graph Indexing       |
| <code>wavelength_strictly_super_planckian</code> | DEFINITION: Planck Length Scale Physical Interpretation: The fundamental quantum gravity length unit $\mathrm{ell}_-(\mathrm{Pl})$ equiv 1. RAG & Graph Indexing       |
| Lean5Corpus.Problems.Problem4_MukaiMonodromy     |                                                                                                                                                                        |
| <code>buscher_reflection_involution</code>       | A Mukai charge vector $v = (r, c, s)$ in $H^0 \oplus H^2 \oplus H^4$ . -/ structure MukaiVector where r : Int - D0-brane rank / momentum c2 : Int - $c^2$              |
| <code>mukai_monodromy_master_contract</code>     | A Mukai charge vector $v = (r, c, s)$ in $H^0 \oplus H^2 \oplus H^4$ . -/ structure MukaiVector where r : Int - D0-brane rank / momentum c2 : Int - $c^2$              |
| <code>mukai_norm_buscher_invariant</code>        | A Mukai charge vector $v = (r, c, s)$ in $H^0 \oplus H^2 \oplus H^4$ . -/ structure MukaiVector where r : Int - D0-brane rank / momentum c2 : Int - $c^2$              |
| <code>mukai_rank_and_signature</code>            |                                                                                                                                                                        |
| Lean5Corpus.Problems.Problem5_KolmogorovCascade  |                                                                                                                                                                        |
| <code>dissipation_strictly_positive</code>       | Turbulent cascade state specifying energy flux $\mathrm{varepsilon}$ , viscosity $\nu$ , and cutoff scale $k_d$ . -/ structure TurbulentCascadeState where energy_flux |
| <code>enstrophy_lower_bound_positive</code>      |                                                                                                                                                                        |
| <code>kolmogorov_cascade_master_contract</code>  | Turbulent cascade state specifying energy flux $\mathrm{varepsilon}$ , viscosity $\nu$ , and cutoff scale $k_d$ . -/ structure TurbulentCascadeState where energy_flux |
| Lean5Corpus.Problems.Problem6_FluxSwampland      |                                                                                                                                                                        |
| <code>desitter_steepness_bound</code>            | Flux compactification state specifying flux quanta $N_-(\mathrm{flux})$ and volume $\mathrm{mathcal{V}}$ . -/ structure FluxVacuumState where flux_quanta : Nat - N    |
| <code>desitter_swampland_master_contract</code>  | Flux compactification state specifying flux quanta $N_-(\mathrm{flux})$ and volume $\mathrm{mathcal{V}}$ . -/ structure FluxVacuumState where flux_quanta : Nat - N    |
| <code>flux_energy_strictly_positive</code>       | Flux compactification state specifying flux quanta $N_-(\mathrm{flux})$ and volume $\mathrm{mathcal{V}}$ . -/ structure FluxVacuumState where flux_quanta : Nat - N    |

| Theorem                                              | First line of its docstring                                                                                                                                                                                                                |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>no_flat_desitter_vacuum</code>                 | Flux compactification state specifying flux quanta $N_-(\mathrm{flux})$ and volume $\mathrm{mathcal{V}}$ . -/ structure FluxVacuumState where <code>flux_quanta</code> : $\mathrm{Nat} - \mathrm{N}$                                       |
| Lean5Corpus.Problems.Problem7_CourantTorsion         |                                                                                                                                                                                                                                            |
| <code>anchor_projection_exact_form_vanishes</code>   | Generalized vector field component with tangent part $x$ and cotangent part $xi$ . -/ structure GeneralizedVector where <code>tangent</code> : $\mathrm{Int}$ <code>cotangent</code> : $\mathrm{Int}$ <code>deri</code>                    |
| <code>c_bracket_skew_symmetric</code>                | Generalized vector field component with tangent part $x$ and cotangent part $xi$ . -/ structure GeneralizedVector where <code>tangent</code> : $\mathrm{Int}$ <code>cotangent</code> : $\mathrm{Int}$ <code>deri</code>                    |
| <code>courant_torsion_master_contract</code>         | Generalized vector field component with tangent part $x$ and cotangent part $xi$ . -/ structure GeneralizedVector where <code>tangent</code> : $\mathrm{Int}$ <code>cotangent</code> : $\mathrm{Int}$ <code>deri</code>                    |
| <code>dft_torsion_vanishes_on_torus</code>           | Generalized vector field component with tangent part $x$ and cotangent part $xi$ . -/ structure GeneralizedVector where <code>tangent</code> : $\mathrm{Int}$ <code>cotangent</code> : $\mathrm{Int}$ <code>deri</code>                    |
| Lean5Corpus.Problems.Problem8_GolayHolography        |                                                                                                                                                                                                                                            |
| <code>golay_codeword_count_eval</code>               |                                                                                                                                                                                                                                            |
| <code>golay_error_radius_equals_three</code>         | The extended binary Golay code parameter specification: $[n, k, d] = [24, 12, 8]$ . -/ <code>def golay_length</code> : $\mathrm{Nat} := 24$ <code>def golay_dimension</code> : $\mathrm{Nat} := 12$ <code>def g</code>                     |
| <code>golay_holography_master_contract</code>        | The extended binary Golay code parameter specification: $[n, k, d] = [24, 12, 8]$ . -/ <code>def golay_length</code> : $\mathrm{Nat} := 24$ <code>def golay_dimension</code> : $\mathrm{Nat} := 12$ <code>def g</code>                     |
| <code>golay_self_dual_dimension</code>               | The extended binary Golay code parameter specification: $[n, k, d] = [24, 12, 8]$ . -/ <code>def golay_length</code> : $\mathrm{Nat} := 24$ <code>def golay_dimension</code> : $\mathrm{Nat} := 12$ <code>def g</code>                     |
| <code>nonzero_codeword_weight_positive</code>        | The extended binary Golay code parameter specification: $[n, k, d] = [24, 12, 8]$ . -/ <code>def golay_length</code> : $\mathrm{Nat} := 24$ <code>def golay_dimension</code> : $\mathrm{Nat} := 12$ <code>def g</code>                     |
| Lean5Corpus.Problems.Problem9_KummerModularity       |                                                                                                                                                                                                                                            |
| <code>k3_euler_characteristic_identity</code>        | Topological invariants of a smooth K3 surface. -/ <code>def k3_betti_2</code> : $\mathrm{Nat} := 22$ <code>def k3_euler_char</code> : $\mathrm{Nat} := 24$ <code>def k3_betti_0</code> : $\mathrm{Nat} := 1$ <code>def k3_betti_4</code> : |
| <code>kummer_cycle_balance</code>                    | Topological invariants of a smooth K3 surface. -/ <code>def k3_betti_2</code> : $\mathrm{Nat} := 22$ <code>def k3_euler_char</code> : $\mathrm{Nat} := 24$ <code>def k3_betti_0</code> : $\mathrm{Nat} := 1$ <code>def k3_betti_4</code> : |
| <code>kummer_modularity_master_contract</code>       | Topological invariants of a smooth K3 surface. -/ <code>def k3_betti_2</code> : $\mathrm{Nat} := 22$ <code>def k3_euler_char</code> : $\mathrm{Nat} := 24$ <code>def k3_betti_0</code> : $\mathrm{Nat} := 1$ <code>def k3_betti_4</code> : |
| <code>singular_k3_picard_number_equals_twenty</code> | Topological invariants of a smooth K3 surface. -/ <code>def k3_betti_2</code> : $\mathrm{Nat} := 22$ <code>def k3_euler_char</code> : $\mathrm{Nat} := 24$ <code>def k3_betti_0</code> : $\mathrm{Nat} := 1$ <code>def k3_betti_4</code> : |
| <code>transcendental_rank_two</code>                 | Topological invariants of a smooth K3 surface. -/ <code>def k3_betti_2</code> : $\mathrm{Nat} := 22$ <code>def k3_euler_char</code> : $\mathrm{Nat} := 24$ <code>def k3_betti_0</code> : $\mathrm{Nat} := 1$ <code>def k3_betti_4</code> : |



## Appendix C

# Reproducing the Formal Results

```
git clone https://github.com/xaviercallens/SocrateAI-Scientific-Agora-LeanMaster
cd SocrateAI-Scientific-Agora-LeanMaster
lake exe cache get                # prebuilt Mathlib
lake build DualScaleStream2       # Stream 2: lattices,  $O(d,d;Z)$ , DFT, dual-scale bound
lake build StringTheoryFormalization # Stream 1 (Mathlib-based)
lake build                       # the Mathlib-free core libraries
python3 tools/axiom_audit.py DualScaleStream2 # #print axioms over every theorem
python3 tools/statement_lock.py --check DualScaleStream2/*/*.lean
lake env lean tools/lean_depgraph.lean > .leancache/depgraph.jsonl
python3 tools/index_declarations.py && python3 tools/theorem_atlas.py # the atlas chapter
python3 tools/book_assemble.py    # this book
```

The axiom audit must report, for every theorem, a subset of `propext`, `Classical.choice`, `Quot.sound`. The statement lock must report OK: it fails if any reviewed statement or definition has been altered, while changes to comments and proofs leave it untouched.





# Sources

The keys below are the ones used in `\source{...}` citations. Line ranges in citations refer to the plain-text extractions in `papers/foundations/` of the companion repository; the PDF files there are authoritative and are pinned by SHA-256 hash in `MANIFEST.md`.

|             |                                                                                                                              |
|-------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>Tong</b> | D. Tong, <i>Lectures on String Theory</i> , arXiv:0908.0333.                                                                 |
| <b>GPR</b>  | A. Giveon, M. Porrati, E. Rabinovici, <i>Target Space Duality in String Theory</i> , arXiv:hep-th/9401139.                   |
| <b>AAL</b>  | E. Alvarez, L. Alvarez-Gaumé, Y. Lozano, <i>An Introduction to T-Duality in String Theory</i> , arXiv:hep-th/9410237.        |
| <b>Pol</b>  | J. Polchinski, <i>TASI Lectures on D-Branes</i> , arXiv:hep-th/9611050.                                                      |
| <b>HZ</b>   | C. Hull, B. Zwiebach, <i>Double Field Theory</i> , arXiv:0904.4664.                                                          |
| <b>HHZ</b>  | O. Hohm, C. Hull, B. Zwiebach, <i>Generalized metric formulation of double field theory</i> , arXiv:1006.4823.               |
| <b>AMN</b>  | G. Aldazabal, D. Marqués, C. Núñez, <i>Double Field Theory: A Pedagogical Review</i> , arXiv:1305.1907.                      |
| <b>Asp</b>  | P. S. Aspinwall, <i>K3 Surfaces and String Duality</i> , arXiv:hep-th/9611137.                                               |
| <b>Huy</b>  | D. Huybrechts, <i>Lectures on K3 Surfaces</i> (author's copy of the lecture notes).                                          |
| <b>DRS</b>  | K. Dasgupta, G. Rajesh, S. Sethi, <i>M Theory, Orientifolds and G-Flux</i> , arXiv:hep-th/9908088.                           |
| <b>GVW</b>  | S. Gukov, C. Vafa, E. Witten, <i>CFT's From Calabi–Yau Four-folds</i> , arXiv:hep-th/9906070.                                |
| <b>SVW</b>  | S. Sethi, C. Vafa, E. Witten, <i>Constraints on Low-Dimensional String Compactifications</i> , arXiv:hep-th/9606122.         |
| <b>BB</b>   | K. Becker, M. Becker, <i>M-Theory on Eight-Manifolds</i> , arXiv:hep-th/9605053.                                             |
| <b>GKP</b>  | S. B. Giddings, S. Kachru, J. Polchinski, <i>Hierarchies from Fluxes in String Compactifications</i> , arXiv:hep-th/0105097. |
| <b>TT</b>   | P. K. Tripathy, S. P. Trivedi, <i>Compactification with Flux on K3 and Tori</i> , JHEP 03 (2003) 028, arXiv:hep-th/0301139.  |
| <b>Sen</b>  | A. Sen, <i>F-theory and Orientifolds</i> , Nucl. Phys. B475 (1996) 562, arXiv:hep-th/9605150.                                |

|              |                                                                                                                                                                                                             |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Gra</b>   | M. Graña, <i>Flux compactifications in string theory: a comprehensive review</i> , arXiv:hep-th/0509003.                                                                                                    |
| <b>Pal</b>   | E. Palti, <i>The Swampland: Introduction and Review</i> , arXiv:1903.06239.                                                                                                                                 |
| <b>EOT</b>   | T. Eguchi, H. Ooguri, Y. Tachikawa, <i>Notes on the K3 Surface and the Mathieu group <math>M_{24}</math></i> , arXiv:1004.0956.                                                                             |
| <b>GHV</b>   | M. R. Gaberdiel, S. Hohenegger, R. Volpato, <i>Mathieu twining characters for K3</i> , arXiv:1006.0221.                                                                                                     |
| <b>BW</b>    | T. Battefeld, S. Watson, <i>String Gas Cosmology</i> , arXiv:hep-th/0510022.                                                                                                                                |
| <b>CDH</b>   | M. C. N. Cheng, J. F. R. Duncan, J. A. Harvey, <i>Umbral Moonshine</i> , arXiv:1204.2779.                                                                                                                   |
| <b>CH</b>    | M. C. N. Cheng, S. Harrison, <i>Umbral Moonshine and K3 Surfaces</i> , arXiv:1406.0619.                                                                                                                     |
| <b>HMN</b>   | J. A. Harvey, S. Murthy, C. Nazaroglu, <i>ADE Double Scaled Little String Theories, Mock Modular Forms and Umbral Moonshine</i> , arXiv:1410.6174.                                                          |
| <b>DMZ</b>   | A. Dabholkar, S. Murthy, D. Zagier, <i>Quantum Black Holes, Wall Crossing, and Mock Modular Forms</i> , arXiv:1208.4074.                                                                                    |
| <b>Cheng</b> | M. C. N. Cheng, <i>K3 Surfaces, <math>N = 4</math> Dyons, and the Mathieu Group <math>M_{24}</math></i> , arXiv:1005.5415.                                                                                  |
| <b>CKN</b>   | A. G. Cohen, D. B. Kaplan, A. E. Nelson, <i>Effective Field Theory, Black Holes, and the Cosmological Constant</i> , arXiv:hep-th/9803132.                                                                  |
| <b>EW</b>    | J. G. Lee, E. G. Adelberger, T. S. Cook, S. M. Fleischer, B. R. Heckel, <i>New Test of the Gravitational <math>1/r^2</math> Law at Separations down to <math>52\,\mu\text{m}</math></i> , arXiv:2002.11761. |
| <b>MVV</b>   | M. Montero, C. Vafa, I. Valenzuela, <i>The Dark Dimension and the Swampland</i> , arXiv:2205.12293.                                                                                                         |
| <b>GV</b>    | M. Gasperini, G. Veneziano, <i>The Pre-Big Bang Scenario in String Cosmology</i> , arXiv:hep-th/0207130.                                                                                                    |
| <b>LISA</b>  | P. Amaro-Seoane et al., <i>Laser Interferometer Space Antenna</i> (L3 mission proposal), arXiv:1702.00786.                                                                                                  |

Cited by reference only (not in the library; bibliographic data as given in the reference list of GPR, refs. [34], [35], [236], ll. 10567–10568 and 10822): T. Buscher, Phys. Lett. B194 (1987) 59 and B201 (1988) 466; K. S. Narain, Phys. Lett. B169 (1986).

Project documents: docs/STREAM2\_WORKFLOW.md, LL.md, docs/PAPER\_REVISION\_BRIEF\_2026\_09\_17.md, and the Lean 4 libraries of the repository (Lean v4.33.1, Mathlib at the commit pinned in lake-manifest.json).

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